

Forbidden pc-Minors for Corner-Peelable Ample Partial Cubes

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Abstract

We study the forbidden partial-cube minors of corner-peelable ample concept classes. The nonample members are exactly the cubes with one antipodal pair removed. For the ample members, we give exact recognition criteria and partial structural constructions. Using CCMW's equivalence between corner peelings and acyclic unique-sink orientations, we interpret rooted factors as prescribed-sink obstructions and separator gluing as compatibility of outgoing boundary directions. Every ample obstruction has a unique forbidden graph three-core, and finite labelled square attachment data reconstruct all extensions of that core. Obstructions with a cut vertex have exactly two rooted factors; separating edges admit an exact decomposition into two or three pieces. A finite-certificate construction realizes three pieces in a 1237-concept obstruction whose nonempty-coordinate reductions all peel. Product attachments convert peeling down to a prescribed ample subfamily into ordinary peeling and transfer minimal relative failures to forbidden pc-minors. Replacing one coordinate by a block, with its forbidden signs repeated across the block, preserves forbiddenness and constructs obstructions with arbitrarily many maximal reduction descent supports. Compressing repeated signed-clause columns gives a canonical reduced forbidden class with positive integer multiplicities on its coordinates. Graph pruning and column compression alternate canonically; their inverse stages recover every obstruction from a forbidden base of minimum degree at least three with distinct columns. Every nonempty corner-peelable ample family occurs as a coordinate face of such a base, even with isometric dimension minus VC dimension fixed at nine. A further construction replicates resolving additions to a fixed forbidden seed over a layer cube. For cornerless seeds, its terminal members are parameterized by partitions of the signed layer facets. Explicit terminal examples show that this construction is not exhaustive, and another infinite family rules out generation solely by corner deletions from cubes. The least cardinality of an ample obstruction lies between 64 and 267. A complete intrinsic generative classification remains open: the missing ingredients lie in the intrinsic construction and admissibility of these reduced forbidden bases and their factors.

Keywords. Partial cube, pc-minor, ample class, corner peeling, unique-sink orientation, representation map, Yang complex, simplicial ball, shellability, Alexander duality, linear quotients.

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1 Introduction

A family of binary words *shatters* a set of coordinates when its projection onto those coordinates contains every binary pattern. The family is *ample* when each shattered set of coordinates also supports a full cube inside the family. These classes, also called lopsided systems, were introduced by Lawrence and developed from combinatorial and metric viewpoints [Law83, BCDK06]. Join two words when they differ in one coordinate. The resulting graph is a partial cube: its graph distances are exactly the Hamming distances between the words.

A vertex is a *corner* if it belongs to a unique maximal cube. A *corner peeling* successively deletes corners until the family is empty. Reversing such a deletion order gives an ordering whose nonempty prefixes are ample [CCMW22, Proposition 4.2]. We write $\mathcal{D}_{\text{amp}}$ for the corner-peelable ample families. The term *dismantlable* is also used for these families; it refers here to corner deletion, rather than domination of vertices in an arbitrary graph. Hall’s maximum class of VC dimension three has no corner and shows that an ample class need not belong to $\mathcal{D}_{\text{amp}}$ [CCMW22, Theorem 4.5].

1.1 The classification problem

Partial-cube minors, or *pc-minors*, combine convex restrictions with contractions of coordinate classes [KM20]. For $X \subseteq Q_E = \{0, 1\}^E$, let $\rho_e^i X$ be the section with coordinate e fixed to i , and let $\pi_e X$ be the projection deleting coordinate e . Constant coordinates are removed from sections. The class $\mathcal{D}_{\text{amp}}$ is pc-minor closed, so it is determined by the basis

$$\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) = \{O : O \notin \mathcal{D}_{\text{amp}}, \text{ every proper pc-minor of } O \text{ lies in } \mathcal{D}_{\text{amp}}\}.$$

Our aim is to understand the structure of this basis and give explicit constructions of its members.

The nonample part is completely classified: Proposition 2.8 identifies it as $\{Q_n \setminus \{x, \bar{x}\} : n \geq 3\}$, where $\bar{x}$ is the antipode of x . For the ample part, pc-minor closure gives the exact recognition criterion

$$O \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff \begin{aligned} &O \notin \mathcal{D}_{\text{amp}}, \\ &\rho_e^0 O, \rho_e^1 O, \pi_e O \in \mathcal{D}_{\text{amp}} \quad (e \in E), \end{aligned} \tag{1.1}$$

for an ample family embedded with every coordinate active, meaning that it occurs on an edge. Equivalent algebraic and topological forms use linear quotients and Yang-complex shellability (Theorem 3.3 and Corollary 3.4). Signed minimal nonfaces give a Boolean formulation (Theorem 5.3).

An intrinsic generative description asks for more: finite structural parameters, conditions specifying which parameters are admissible, a construction that always produces an obstruction, and recovery of every obstruction from such data. The parameters may have arbitrary finite size, and recursive constructions are allowed. A bounded catalogue is unnecessary. For example, a down-set of arbitrary size can serve as a structural parameter, as in the [companion classification for daisy cubes](#). For the present problem, a construction that assumes an arbitrary forbidden seed still leaves that seed to be classified. Likewise, replacing the tests in (1.1) by equivalent shelling or ordering conditions gives a recognition theorem without supplying the desired structural inputs.

1.2 Acyclic USOs and the organization of the results

CCMW identify corner-peelable ample classes with those admitting an acyclic unique-sink orientation (USO) [CCMW22, Proposition 6.12]. For an ample class a USO requires both a unique sink on each contained cube and containment of the cube spanned by each vertex’s outgoing directions. Its

out-map records those directions and is a representation map [CCMW22, Theorems 6.1 and 6.8]. Section 2.2 gives the precise definitions.

This organizes the structural results by the information they preserve. Canonical graph and signed-column reductions act on the support itself. Rooted factors and separator constructions instead prescribe boundary data on acyclic USOs of positive pieces. Peeling to a root means realizing it as a sink. Peeling to a positive subfamily means extending its acyclic USO with all boundary edges pointing inward (Proposition 2.1). At a separating cube, a boundary vertex can have outgoing exclusive directions in at most one piece, because two would require an absent mixed square (Proposition 2.20). This single condition explains the separator bound and the five boundary conditions used at an edge.

Negative certificates must exclude every acyclic USO. Protected missing cubes do so by forcing a directed cycle in every possible USO (Proposition 2.9). One cyclic orientation alone does not certify nonpeelability, and the existence of acyclic USOs on proper minors does not make arbitrary inherited maps acyclic. For retained targets that are themselves nonpeelable, relative peeling requires only acyclicity outside the target; its full scope is preserved in the deletion statements below.

1.3 Canonical reduction and explicit constructions

A unique core and all its square attachments. The strongest reduction applying to every ample obstruction is Theorem 2.53. Repeatedly delete vertices of degree less than three. Each deletion is a degree-two corner deletion that preserves forbidden-minor status. The result is the unique graph three-core K , the largest induced subgraph of minimum degree at least three. It is nonempty, ample, and forbidden, and has the same active coordinates as the original family.

The inverse operation is explicit. Given an ample family, choose a missing word and two coordinates such that the other three vertices of its square are present and the coordinate pair is not shattered. Adjoining that word is an ample square completion. Over a forbidden base, it preserves forbidden-minor status. More generally, it preserves the possibility of peeling to every retained subset of the base (Proposition 2.13). To see the operation at the smallest scale, $\{00, 01, 10\}$ acquires its missing word 11 using the coordinate pair $\{1, 2\}$; the three present words specify the entire addition.

Theorem 2.60 turns a sequence of these operations into finite structural data: abstract new vertices, a coordinate pair and three targets for each vertex, and an acyclic dependency digraph. Coordinate consistency and injectivity determine the new words; distinct pairs absent from the core's shattered sets ensure that every addition is valid. The data are recovered uniquely from the resulting family, up to renaming the abstract new vertices. Thus construction and recovery of *all attachments* are settled. The unresolved input is precisely which ample three-cores are forbidden. Such a core can itself have corners.

Rooted factors and gluing. Say that A *peels to* a when it has a corner peeling ending at a . A rooted pair (A, a) has the factor property needed here exactly when

1. A is ample and corner-peelable;
2. no acyclic USO of A has sink a ;
3. every proper pc-minor obtained while retaining the root, or its image under contraction, admits an acyclic USO with that sink.

These are the *rooted factors* used below. Theorem 2.19 says that gluing any two such factors at their roots gives a forbidden pc-minor. Every ample obstruction with a cut vertex arises this way. It has

exactly two blocks and a unique cut vertex, so the factors are recovered uniquely up to interchange. This is a complete decomposition of that branch, with an explicitly unresolved rooted-factor input.

The same inputs yield two-connected examples. Put rooted factors Y, C on disjoint coordinate sets, with common root 0, choose an edge $\{0, v\}$ of Y , and choose a nonempty down-set $L \subseteq C$. Here down-set means that replacing any ones by zeros keeps a word in L . The family

$$X = Y \cup C \cup (v + L) \tag{1.2}$$

is forbidden, and is two-connected exactly when $|L| > 1$ (Theorem 2.47). All unused coordinates are zero, and $v + L$ is the translated copy of L across the chosen edge. The marked coordinate blocks and edge recover Y, C, L . The down-set is a structural parameter; the rooted factors still satisfy the three recognition conditions above.

There is also exact recovery at a separating edge, meaning an edge whose two endpoints form a vertex cut. Its union with each component outside the edge gives a canonical convex piece. There are two or three pieces, and Theorem 2.32 gives necessary and sufficient gluing conditions. These use five attainable out-map conditions: an inward boundary at the edge, either endpoint as sink, and either endpoint having just the common edge as its outgoing edge. Equivalently, these specify retention of the edge, its possible final endpoints, and its possible first endpoint deletions as leaves. Proposition 2.37 realizes the boundary profile $(0, 1, 1, 0, 0)$ from rooted inputs: both endpoint sinks are attainable, but neither leaf exit nor edge retention is possible. Thus sink feasibility alone cannot replace the five conditions. Theorem 2.42 reduces existence of a three-piece forbidden union exactly to existence of a positive piece failing edge retention but admitting both an endpoint sink and an endpoint leaf exit. No minor-minimality assumption on that witness is needed; both boundary properties survive edge-retaining minors. Proposition 2.43 first gives a uniform construction from a rooted factor, one repairing addition, and a proper face containing the acquired cube. If that face omits a root neighbour, the output is another rooted factor, with all proper rooted minor tests automatically satisfied. A 308-concept example results. The operation has a recoverable shore presentation but is not closed under direct iteration; Remark 2.44 distinguishes this limitation from coverage by the earlier ample-addition construction. Theorem 2.45 supplies the other outcome with profile $(0, 1, 0, 1, 0)$ on 330 concepts, certified by explicit orders. Gluing it, its reversal, and the 581-concept piece gives a forbidden 1237-concept class with three components outside the edge. Thus the bound of three is attained. All its nonempty-coordinate reductions peel, so this example also belongs to the seed side of reduction descent. A complementary backward step is given by Proposition 2.72: an ordinarily positive nonroot-corner deletion followed by rooted minor minimalization reaches a smaller rooted factor. If a proper minor is needed, it preserves an endpoint repair on a face of the deleted corner cube. Iteration ends at factors whose root is the unique source in every acyclic USO; whether these can have other geometric corners remains a separate question. Corollary 2.73 bypasses that question by allowing a rooted deletion to enter the ordinary nonpeelable branch. Its terminal inputs are rooted factors with a unique geometric corner and cornerless ample forbidden pc-minors. Negative-minor selection and admissible inverse expansions still prevent an intrinsic forward generation theorem. Compatibility between two pieces matters: even in a forbidden three-core, an elementary minor can peel through their interaction although one changed piece still fails to peel to the shared edge (Remark 2.48).

The separator branch is also detectable from shattered pairs. Its crossing graph has one vertex per active coordinate and joins the shattered pairs. Disconnection detects cut vertices; after ruling those out, a cut vertex of the crossing graph detects separating edges in that direction. Contracting the direction recovers all such edges. If deleting that crossing-graph vertex leaves two components, there is one separating edge and a unique pair of convex factors (Proposition 2.33

and Corollary 2.34). Thus the remaining branch has a two-connected crossing graph; the peeling properties of the recovered factors still require intrinsic criteria.

In the separator branch, descent through nonpeelable reductions also has a canonical endpoint: there is at most one descent coordinate, and after that one step all further reductions peel (Theorem 5.34). A nonpeelable carrier must retain every remaining coordinate, so its direction is adjacent to all others in the crossing graph. Failure of two-connectivity permits at most one such direction. Whether its carrier actually peels is still a separate condition. Square attachments leave every nonpeelable reduction unchanged as an embedded family (Proposition 5.35). Hence the same conclusion holds whenever the canonical three-core has a separator, even if the full obstruction has none.

Replacing coordinates by blocks. There is also a construction that applies to every ample forbidden input, without choosing resolving additions. Replace a coordinate z by a nonempty block E . Keep its two sections at the constant words $0_E, 1_E$, and put their union in every nonconstant section. Theorem 5.36 proves that the result is forbidden exactly when the original family is forbidden. On signed minimal nonfaces, the rule simply replaces z by all of E , repeating its forbidden sign. The two constant sections recover the input. For $|E| \geq 2$, the output is its own graph three-core and has a two-connected crossing graph. If the input has no nonpeelable proper reduction, the maximal descent supports are exactly $E \setminus \{e\}$, $e \in E$, and every endpoint is the original family with one coordinate renamed (Corollary 5.37). Thus descent supports need not be unique even though these examples share one terminal isomorphism type.

The operation also has canonical unmarked recovery. Group coordinates whose columns in the signed minimal-nonface presentation agree up to sign, and compress each group to one coordinate. Ampleness forces the section pattern required by the construction, and the reduced family is again forbidden. Theorem 5.38 proves that this gives an exact parametrization by reduced forbidden inputs with distinct columns up to sign and arbitrary positive coordinate multiplicities. The description of these reduced inputs remains open.

The two canonical reductions also fit together. Repeated columns force minimum degree at least three, so graph pruning and column compression are never available simultaneously in a forbidden class. Exhaust one type, then the other, until neither applies. Theorem 5.39 proves that the resulting base and all intermediate phases are canonical. There are at most $2d - 5$ phases for VC dimension d , and the difference between isometric dimension and VC dimension is unchanged. Conversely, alternate nonempty square attachment data and nontrivial coordinate multiplicities over any forbidden base with minimum degree at least three and distinct columns. These are exactly the recovered construction stages; no further compatibility tests are required. Intrinsic admissibility of the final forbidden bases remains the unresolved input.

This input does not become finite at a fixed value of the dimension difference. Theorem 6.14 constructs, from any nonempty ample family K , a nonpeelable family X_K with minimum degree at least four, distinct columns, complete crossing graph, and dimension difference nine, such that

$$X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff K \in \mathcal{D}_{\text{amp}}.$$

The input K is recovered as a proper coordinate face. Thus all corner-peelable ample families still occur inside reduced forbidden bases. Taking K to be a cube supplies an explicit infinite family of these bases; it does not classify all of them. The construction also preserves every input clause change at fixed shadow. Proposition 6.15 uses this to show that changing one sign bit can destroy pc-minimality while both outputs remain nonpeelable and satisfy the reduction stopping conditions. Intrinsic sign admissibility for ampleness therefore does not supply the missing minimality criterion.

Construction	What construction and recovery settle	Remaining input or coverage
Alternating graph and column reductions	Canonical reduced base; all square attachment and coordinate-block stages have intrinsic parameters and unique recovery.	Forbidden bases of minimum degree at least three with distinct signed columns.
Gluing at a cut vertex or separating edge	Canonical factors and exact gluing conditions for every obstruction with the specified separator.	Attainable sink and boundary out-map conditions of the factors; coverage outside these separator classes.
Down-set completion	Uniform obstruction transfer and recovery of the marked data in (1.2).	The two rooted factors.
Signed-facet assembly	All terminal layer data over a cornerless seed with eligible tips.	The seed and resolving tips; this construction is not exhaustive.

Table 1: Scope of the structural results. Each row distinguishes the parameters already constructed intrinsically from the remaining problem.

Replicated repairs and signed-facet partitions. Let $S \subseteq Q_F$ be an ample forbidden seed. Choose a set T of distinct missing words a , such that every $S \cup \{a\}$ is corner-peelable and their acquired shattered supports P_a are pairwise distinct. On a disjoint nonempty coordinate set E , choose nonempty proper ample families $R_a \subsetneq Q_E$. Form

$$X = (S \times Q_E) \cup \bigcup_{a \in T} (\{a\} \times R_a). \quad (1.3)$$

Theorem 6.8 says that X is forbidden exactly when each R_a is corner-peelable and every signed facet $\{t : t_e = i\}$ of the layer cube is contained in at least one R_a . Fixing all layer coordinates and intersecting the resulting sections recovers S ; the other old words and their occurrences recover the tips and their families.

Call an obstruction *terminal* when no corner deletion leaves another forbidden pc-minor. For a cornerless seed, the terminal members of (1.3) have a completely structural description of their layer data. Choose a surjection

$$\lambda : E \times \{0, 1\} \longrightarrow T, \quad \lambda(e, 0) \neq \lambda(e, 1),$$

and set

$$R_a = \bigcup_{\lambda(e,i)=a} \{t \in Q_E : t_e = i\}.$$

These conditions suffice, every terminal member in this class has this form, and the marked sections recover λ (Theorem 6.22). With two tips, the two families are cubes punctured at antipodal words. This gives an explicit infinite family of order $(|S| + 2)2^{|E|} - 2$ over every eligible two-tip seed. The seed's forbiddenness and the resolving property of its tips remain input conditions.

1.4 What remains to be classified

Table 1 separates two outstanding tasks. For the separator branches, the precise question is which boundary out-maps can be realized by acyclic USOs of the factors, and how to construct and recover supports for which the required conditions fail. This does not replace structural generation by a search over orientations. First, the primitive cores, rooted factors, and resolving tips need structural

admissibility conditions that imply their required peeling properties. Second, the construction classes need to cover all forbidden cores. The all-obstruction three-core reduction settles the attachment part of the second task, but its core input is still the original obstruction problem with a minimum-degree restriction.

Several examples delimit these tasks. A terminal obstruction with 1158 concepts and a cut vertex lies outside the replicated-repair class; another terminal example of the same order is two-connected and also has no such presentation (Theorem 2.66 and Proposition 6.38). Thus the missing coverage cannot be supplied merely by allowing cut vertices or by restricting attention to two-connected terminal families. Nor can all obstructions be generated by corner deletions from cubes: Corollary 6.18 gives an infinite family with nonpeelable cube complements, which exclude such a descent. Changing signs while preserving the shadow cannot always produce a cornerless core either: a two-connected 1162-concept three-core has a shadow forcing at least three corners in every ample realization (Corollary 2.56). These examples restrict proposed constructions; they do not establish impossibility of an intrinsic classification using other finite parameters.

Nor can the one-hole construction of cornerless ample classes be converted to a forbidden-minor construction merely by selecting its pc-minimal outputs. A 580-concept cornerless graph three-core has a corner in every nonempty proper pc-minor, yet its unique forbidden minor has 292 concepts and one corner (Corollary 5.23). The proper-minor peeling conditions therefore retain information that corner existence, even in all proper minors, does not supply.

The contrast with daisy cubes also concerns decomposition. Their peripheral coordinates expose a down-set structure from which a generative description can be recovered. Every ample obstruction here is periphery-free (Proposition 2.52), so that starting decomposition is unavailable. A complete description needs a structural replacement for the primitive cores and factors.

1.5 Bounds, verification, and a guide to the proofs

Let m_{amp} be the least number of concepts in an ample forbidden pc-minor. The current bounds are

$$64 \leq m_{\text{amp}} \leq 267 \quad (\text{Theorem 7.159}).$$

The upper bound comes from an explicit family of order 267 (Proposition 7.55), obtained by iterating the contract-and-lift step that turns Hall’s class into the 291-concept family of Proposition 7.45; the lower bound combines structural reductions with finite exclusions. Each computationally supported result identifies its construction or proof certificate and its checking procedure. Uniform construction theorems are proved symbolically; checks of fixed inputs or examples are stated with their actual scope. These bounds are a separate extremal result and do not imply that the structural classification is nearly complete.

For the generative results, start with the USO and relative-extension dictionary in Section 2.2, then the basic operations and local extension results in Section 2.4. Continue through the cut-vertex and separating-edge theorems to Section 2.7, which gives the full three-core and square-attachment proof chain before the specialized rooted constructions. Their dependency is explicit: the cut-vertex theorem excludes degree-two cut vertices in an obstruction; periphery-freeness excludes leaves; and square-deletion invariance preserves forbidden-minor status. These facts yield the canonical core, and the square attachment data invert every such deletion. The down-set construction extends the rooted-factor examples, while the replicated-repair and signed-facet theorems in Section 6 describe the separate layered construction.

For recognition, Section 3 gives the algebraic and Yang-complex criteria. Section 4 describes critical-core extensions, and Section 5 develops signed minimal nonfaces and one-concept additions. Section 7 contains the fixed witnesses and the minimum-order argument. Appendix A supplies the

relative peeling theorem used in the lower bound. The detailed proofs and finite certificates remain available independently of this overview; the complete intrinsic generative classification remains open.

2 Basic operations and structural decompositions

2.1 Definitions and elementary operations

All graphs are finite partial cubes. A cube embedding is *irredundant* if every coordinate occurs on an edge; its number of coordinates is the *isometric dimension*. We identify each graph with an irredundant isometric family $X \subseteq Q_E = \{0, 1\}^E$. For $e \in E$ and $b \in \{0, 1\}$, write

$$\rho_e^b X = \{x|_{E \setminus \{e\}} : x \in X, x_e = b\}$$

for the restriction to one e -semicube, with the constant coordinate removed, and write

$$\pi_e X = \rho_e^0 X \cup \rho_e^1 X$$

for contraction of the complete e -coordinate class. Graphically, the contraction identifies both endpoints of every edge in that Θ -class. A *pc-minor* is obtained by a finite sequence of nonempty convex restrictions and coordinate contractions [KM20, Section 2.1]; it is *proper* if it is not isomorphic to the original partial cube.

For $A \subseteq E$, the *reduction* along A is

$$X^A = \{z \in Q_{E \setminus A} : \{z\} \times Q_A \subseteq X\}.$$

In particular, $X^{\{e\}} = \rho_e^0 X \cap \rho_e^1 X$. Unlike a restriction or contraction, a reduction need not be a pc-minor.

CCMW call ordinary coordinate projection a *restriction* and the full-fibre operation above a *reduction*. We use reduction throughout. For pc-minors we retain the graph terms *convex restriction* and *coordinate contraction*: ρ_e^b is a coordinate-face intersection, whereas π_e is CCMW's restriction. Older source labels and verifier filenames using the historical name refer to this same operation X^A .

For $S \subseteq E$, the family X *shatters* S if its projection onto S is all of Q_S . It *strongly shatters* S if X contains a full S -dimensional cube obtained by fixing all coordinates outside S . The family is *ample* if its shattered and strongly shattered coordinate sets coincide [BCDK06]. The Sandwich theorem states that every finite cube family satisfies

$$|\text{SSh}(X)| \leq |X| \leq |\text{Sh}(X)|,$$

where $\text{SSh}(X)$ and $\text{Sh}(X)$ denote the strongly shattered and shattered sets, respectively. Equality in either inequality holds exactly when X is ample [BCDK06, Corollary 1 and Theorem 2]. We refer to these cardinality statements as the *Sandwich inequalities* and to the ample case as the *Sandwich equality*.

A vertex is a *corner* if it belongs to a unique maximal cube. A *corner peeling*, written in deletion order as $x_1, \dots, x_m$, is an ordering in which x_i is a corner of $X \setminus \{x_1, \dots, x_{i-1}\}$ for every i . Its reversal is the prefix-building convention used for ample and isometric orders in [CCMW22, Proposition 4.2]. We use $\mathcal{D}_{\text{amp}}$ for the ample families admitting such an ordering.

For a vertex x and $A \subseteq E$, write x^A for the vertex obtained by flipping the coordinates in A . Put

$$I_X(x) = \{e \in E : x^{\{e\}} \in X\}, \quad \partial_e^b X = \{x \in X : x_e = b, x^{\{e\}} \notin X\}.$$

Thus $I_X(x)$ is the set of edge directions incident with x , and $\partial_e^b X$ is the signed e -boundary on side b . Write $\text{Cor}(X)$ for the set of corners of X . If x is a corner, $Q_X(x)$ denotes its unique maximal cube. Finally,

$$Q_n^{--} = Q_n \setminus \{x, \bar{x}\}$$

denotes an n -cube with one antipodal pair removed; its isomorphism type is independent of the choice of x .

2.2 Representation maps and boundary conditions

The orientation formulation identifies which information a gluing argument must preserve. An orientation D of the one-inclusion graph of an ample $H \subseteq Q_F$ has *out-map*

$$r_D(x) = \{f \in F : x^{\{f\}} \in H, x \longrightarrow x^{\{f\}}\}.$$

Following Chalopin–Chepoi–Moran–Warmuth (CCMW), it is a *unique-sink orientation* (USO) if both conditions hold:

(C1) $\{x^J : J \subseteq r_D(x)\} \subseteq H$ for every $x \in H$;

(C2) every contained cube has exactly one sink in its induced orientation.

A *representation map* is a bijection $r : H \rightarrow \text{Sh}(H)$ such that distinct x, y differ on at least one coordinate of $r(x) \cup r(y)$. CCMW’s Theorems 6.1 and 6.8 identify representation maps with the out-maps of USOs. Their Proposition 6.12 identifies $H \in \mathcal{D}_{\text{amp}}$ with existence of an acyclic USO [CCMW22]. Every topological order of that orientation is a corner deletion order. We use these established results throughout.

The two USO conditions must be kept together. On the path $\{00, 01, 11\}$, directing both edges out of 01 satisfies (C2), since only vertices and edges are contained cubes, but violates (C1): its outgoing square would require 10. Directing the path $00 \rightarrow 01 \rightarrow 11$ satisfies both and realizes sink 11.

Write $\mathcal{A}(H)$ for the set of acyclic USOs of H . For an ample family, forbiddenness is the condition $\mathcal{A}(H) = \emptyset$ while $\mathcal{A}(M) \neq \emptyset$ for every nonempty proper pc-minor M . The witnesses on different minors are chosen independently. Positivity of a minor does not require that every representation map on it be acyclic. Existence of a representation map on every ample class is not assumed.

For nonempty ample $T \subseteq H$, say that H *peels to* T when corners outside T can be deleted until exactly T remains. An *inward boundary at* T means that every edge between $H \setminus T$ and T is directed into T .

Proposition 2.1 (Relative peeling and extension of a representation map). *Suppose a USO D_T on T is given. Then H peels to T if and only if D_T extends to a USO on H with inward boundary at T and an acyclic induced orientation on $H \setminus T$. If $T \in \mathcal{D}_{\text{amp}}$, relative peeling is equivalent to existence of an acyclic USO on H with inward boundary at T . Every prescribed acyclic USO of T then extends.*

Proof. Reverse the relative peeling, adding each corner as a source. Old outgoing cubes are unchanged and the new source has its corner cube. If a contained cube B contains the new source v and has more than one vertex, its old part $B \setminus \{v\}$ is a coordinate-face intersection of the previous ample family. Its induced orientation is a USO by CCMW Lemma 6.9 and has one sink: by Theorem 6.8 its representation map takes the value $\emptyset$ exactly once. Adding the source preserves this sink, so (C2) holds as well. The added vertices are acyclic and the boundary enters T .

Conversely, take a source of the remaining outside acyclic digraph. The inward boundary makes it a source of the whole remaining USO. By CCMW Proposition A.2, deleting this singleton source component leaves an ample class with the restricted representation map; the source is a corner by (C1). Repeating deletes precisely the outside vertices. If D_T is acyclic, so is the whole extension, since a cycle cannot enter T and leave. $\square$

For $T = \{a\}$, this is existence of an acyclic USO with prescribed sink a . For a nonpeelable retained target, the extension may have cycles inside T . Thus relative peeling must not be replaced unconditionally by existence of an acyclic USO on the whole family. The rank-two invariance theorem below also covers targets with no supplied representation map and the empty target; its deletion formulation preserves that full scope.

Proposition 2.2 (A quadratic form of metric elimination). *Let H be a nonempty isometric cube family, let $f_2(H)$ be its number of contained squares, and let O be an acyclic orientation of its one-inclusion graph. Write $d_O^+(v)$ for the outdegree of v and set*

$$\varepsilon_H(O) = \sum_{v \in H} \binom{d_O^+(v)}{2} - f_2(H).$$

Then $\varepsilon_H(O) \geq 0$. Equality holds exactly when a topological order of O is a corner peeling; in that case every topological order is a corner peeling and H is ample. In particular, for ample H ,

$$H \in \mathcal{D}_{\text{amp}} \iff \min_{O \text{ acyclic}} \varepsilon_H(O) = 0.$$

The minimum over acyclic orientations in which a is a sink is zero exactly when H peels to a .

Proof. An outgoing pair at v consists of two edges $v \rightarrow u, v \rightarrow w$. Either their other square vertex $z = u \oplus v \oplus w$ is absent, or the pair is a source at v of a contained square. Every acyclically oriented square has one or two sources. Consequently $\varepsilon_H(O)$ is exactly the number of outgoing pairs with absent other square vertex, plus the number of squares with two sources. This proves nonnegativity.

Take a topological order, in deletion order. If the score is zero, any two remaining neighbours u, w of the next vertex v have their other square vertex z in H . Moreover z has not been deleted: otherwise both v and z would precede u, w , giving two sources in that square. Thus deleting v preserves every distance, by replacing any two-edge segment through v by the surviving two-edge segment through z . Every suffix is isometric. Proposition 5.62 identifies this order with a corner peeling, and also proves ampleness without assuming it initially. This reasoning applies to every topological order. Conversely a corner order supplies a surviving z for every outgoing pair, so neither counted defect occurs and the score is zero. Finally an acyclic orientation with sink a has a topological order ending at a ; the preceding equivalence proves the rooted assertion. $\square$

The proposition is a scalar reformulation of the existing metric-elimination criterion, not a new characterization of its generable inputs. Its outgoing pair count is the source version of the two-frame objective used in the abstract-objective-function literature [JKK02, Theorems 1 and 5]. The cited two-face theorem can be applied to each contained cube after acyclicity and (C1) have been imposed. The metric argument above is stronger for this purpose: zero score already implies (C1), so no separate higher-dimensional outgoing-cube test is required. Acyclicity remains essential; a directed square has score -1 .

For the fixed rooted-negative 292-vertex control, the stored order ending at its root has score exactly one. Its only defect is the outgoing pair

$$1116 \longrightarrow 1628, \quad 1116 \longrightarrow 92, \quad 604 = 1116 \oplus 1628 \oplus 92 \notin H.$$

Every contained square has one source. The existing exhaustive certificate that no corner order ends at the root excludes score zero, so the rooted minimum is exactly one. The earlier relaxed order had score 17; this was the value of that order, not a lower bound. The rooted-positive 294-vertex control has minimum zero, witnessed by its existing order. `research/quadratic_orientation_score.py/json` replays both bounds, the defect count, and the old rooted-negative certificate. These examples distinguish the two rooted problems numerically; they do not imply that every rooted factor has minimum one, that local score descent finds a minimum, or that a score-one orientation alone certifies nonpeelability. The lower bound here still uses the earlier negative proof.

Remark 2.3 (The score can increase under last-fibre contraction). An order on a cube family induces an acyclic orientation by directing each edge toward its later endpoint. Under contraction of coordinate j , order the image vertices by the positions of their last preimages. This operation does not preserve an upper bound of one on the score. Identify the vertices of Q_4 with the integers $0, \dots, 15$ in binary, with coordinate 0 the least significant bit. The order

$$14, 6, 10, 12, 8, 4, 9, 0, 1, 11, 3, 7, 2, 15, 5, 13$$

has score one: among its 24 squares, precisely $\{2, 3, 6, 7\}$ has two sources (namely 6 and 3). Contracting coordinate 0, and retaining the zero bit in the image labels, gives the last-fibre order

$$8, 0, 10, 6, 2, 14, 4, 12.$$

This is an order on Q_3 . Two of its six squares have two sources: $\{0, 2, 4, 6\}$, with sources 0, 6, and $\{2, 6, 10, 14\}$, with sources 6, 10. The score is therefore two. Equivalently, the outgoing-pair totals are 25 and 8, respectively. All other squares have one source, and there are no missing diagonals in either full cube.

Both cubes admit corner orders ending at the displayed final vertex, so their optimized rooted scores are both zero. Thus this example refutes nonincrease for the transported order, not for the minimum over all orders. The zero-score transport supplied by corner-peeling minor closure is unaffected. The orders and the independent square-source counts are replayed in `research/quadratic_score_transport.py/json`.

2.3 Closure and elementary constructions

A coordinate e is *peripheral* if, after orienting it, $\rho_e^1 X \subseteq \rho_e^0 X$. Equivalently,

$$X = (B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B,$$

where $B = \pi_e X$ is the contraction base and S is the duplicated expansion site. The family is *periphery-free* if it has no peripheral coordinate, and *Cartesian-prime* if it is not a Cartesian product of two nontrivial cube families. We write

$$\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) = \{O : O \notin \mathcal{D}_{\text{amp}} : \text{every proper pc-minor of } O \text{ belongs to } \mathcal{D}_{\text{amp}}\}.$$

We use that $\mathcal{D}_{\text{amp}}$ is pc-minor closed and that Cartesian products satisfy $A \square B \in \mathcal{D}_{\text{amp}}$ if and only if $A, B \in \mathcal{D}_{\text{amp}}$; the product statement is proved in Proposition 2.6.

Proposition 2.4 (Pc-minor closure). *The class $\mathcal{D}_{\text{amp}}$ is closed under pc-minors.*

Proof. Reverse a corner-peeling order to obtain an ample prefix order $x_1, \dots, x_m$, by [CCMW22, Proposition 4.2], and put $X_i = \{x_1, \dots, x_i\}$. Intersecting these prefixes with a coordinate semicube,

omitting empty sets and repetitions, gives an ample prefix order of the nonempty restriction: every retained set is an intersection of an ample class with a cube.

For contraction along e , list distinct projected concepts in order of their first occurrence. If the j -th new projected concept first occurs at index i_j , the first j concepts of the new order form exactly

$$\pi_e X_{i_j} = \rho_e^0 X_{i_j} \cup \rho_e^1 X_{i_j}.$$

This is the projection of an ample prefix and is ample by [CCMW22, Section 3.1]. It is not the reduction $X_{i_j}^{\{e\}}$, which uses the intersection of the sections. Thus both operations give ample prefix orders. Reversing them gives corner peelings of the elementary pc-minors. Nonempty convex restrictions are intersections of coordinate semicubes, and restrictions and contractions generate the pc-minor relation, proving the claim. $\square$

Proposition 2.5 (Reduction closure). *If $X \in \mathcal{D}_{\text{amp}}$, then for every $A \subseteq E$, the reduction X^A is either empty or belongs to $\mathcal{D}_{\text{amp}}$.*

Proof. If $X^A = \emptyset$, there is nothing to prove. Otherwise, reverse a corner peeling of X to obtain an ample prefix order $x_1, \dots, x_m$, and put $X_i = \{x_1, \dots, x_i\}$. For $z \in X^A$, define its completion time to be the largest index of a concept in the full fibre $\{z\} \times Q_A$. Order the members of X^A by increasing completion time. Distinct full fibers are disjoint, so their completion times are distinct. At every completion time i , the resulting prefix is exactly X_i^A . Reductions of ample families are ample [CCMW22, Section 3.1], so this is an ample prefix order of X^A . Reversing it gives a corner peeling. $\square$

Proposition 2.6 (Exact peripheral-expansion and product tests). *Let $B, S \subseteq Q_F$ be nonempty ample families with $S \subseteq B$, and let e be a new coordinate. Then*

$$Y = (B \times \{0\}) \cup (S \times \{1\}) \quad \text{satisfies} \quad Y \in \mathcal{D}_{\text{amp}} \iff B, S \in \mathcal{D}_{\text{amp}}. \quad (\text{A4g1a})$$

If A and B are nonempty ample families on disjoint coordinate sets, then

$$A \sqcup B \in \mathcal{D}_{\text{amp}} \iff A, B \in \mathcal{D}_{\text{amp}}. \quad (\text{A4g1b})$$

In particular, peripheral expansions and Cartesian products preserve non-corner-peelability as soon as one section or factor is non-corner-peelable.

Proof. A set of old coordinates is shattered by Y exactly when it is shattered by B , while a set containing e is shattered exactly when its old-coordinate part is shattered by S . Hence

$$|\text{Sh}(Y)| = |\text{Sh}(B)| + |\text{Sh}(S)| = |B| + |S| = |Y|,$$

so Y is ample. If $Y \in \mathcal{D}_{\text{amp}}$, its two signed e -restrictions show, by pc-minor closure, that $B, S \in \mathcal{D}_{\text{amp}}$. Conversely, take a prefix-isometric order of $B \times \{0\}$, followed by the lift to the 1-side of a prefix-isometric order of S . The old layer is isometric. Two vertices in a partial new layer have a geodesic supplied by the corresponding prefix of S , while a geodesic from the old layer to a new vertex $(s, 1)$ may first reach $(s, 0)$ inside $B \times \{0\}$ and then use the e -edge. Thus every prefix is isometric, proving (A4g1a).

The product of two ample families is ample. If $A \sqcup B \in \mathcal{D}_{\text{amp}}$, fixing a vertex in either factor realizes the other factor as a face restriction, so both factors lie in $\mathcal{D}_{\text{amp}}$. Conversely, let $a_1, \dots, a_p$ and $b_1, \dots, b_q$ be prefix-isometric orders of A and B , respectively, and order the product fiber by fiber:

$$(a_1, b_1), \dots, (a_1, b_q), (a_2, b_1), \dots, (a_2, b_q), \dots, (a_p, b_1), \dots, (a_p, b_q).$$

A prefix ending in the j -th vertex of the i -th fiber is

$$(\{a_1, \dots, a_{i-1}\} \square B) \cup (\{a_i\} \square \{b_1, \dots, b_j\}).$$

Two vertices in the last partial fiber have a geodesic there. For a vertex in a completed fiber and one in the partial fiber, first move inside the completed B -fiber to the second vertex's B -coordinate, and then follow an A -geodesic with that coordinate fixed. This is a product geodesic contained in the prefix. Hence the displayed order is prefix-isometric and proves (A4g1b). $\square$

Lemma 2.7 (Cartesian primality). *Every member of $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ is Cartesian-prime.*

Proof. If $O = A \square B$ with both factors nontrivial, then A and B are proper convex subgraphs of O and hence belong to $\mathcal{D}_{\text{amp}}$. The product equivalence then gives $O \in \mathcal{D}_{\text{amp}}$, a contradiction. $\square$

Proposition 2.8 (The nonample layer). *The nonample part of the forbidden basis is*

$$\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \setminus \mathcal{A} = \{Q_n^{--} : n \geq 3\}.$$

Proof. The elementary restrictions of Q_n^{--} are singly punctured cubes and its elementary contractions are hypercubes. These families admit corner peelings, whereas Q_n^{--} is not ample, so Q_n^{--} is an obstruction. Conversely, the forbidden-pc-minor characterization of ampleness [KM20, Corollary 7.4] gives such a pc-minor in every nonample family. Minimality therefore forces $O \cong Q_n^{--}$. $\square$

Proposition 2.9 (Protected missing cubes force cycles). *Let $H \subseteq Q_F$ be ample and let $S \subseteq H$ with $|S| \geq 2$. Suppose for each $x \in S$ there is $J_x \subseteq F$ such that $x^{\{e\}} \in S$ for every $e \in J_x$, but $x^{J_x} \notin H$. Every USO of H , if one exists, has a directed cycle contained in S . In particular, $H \notin \mathcal{D}_{\text{amp}}$.*

Proof. Not all directions in J_x can be outgoing at x , by (C1). Thus every vertex of the finite digraph induced on S has an incoming neighbour in S . Following predecessors until a vertex repeats yields a directed cycle. CCMW's acyclic-USO characterization gives the conclusion. $\square$

This is also a first-deletion certificate: no vertex of S can be the first one removed while all its specified neighbours remain. The statement quantifies over every USO but does not assert a fixed common cycle, existence of any USO, or that S is a source strongly connected component. CCMW Proposition A.2 deletes source components of a chosen representation map; that map-dependent operation is not a pc-minor operation.

2.4 One-concept additions and square deletions

We collect the elementary steps used in the core-and-attachment construction. The local test identifies which missing concepts may be adjoined. Over a forbidden family every allowed addition either restores peelability or gives another forbidden family. Additions of support size at most two always take the second alternative, and their inverse corner deletions preserve forbiddenness as well.

For words $x, y \in Q_F$, let $\text{Diff}(x, y) = \{f \in F : x_f \neq y_f\}$. The shattered supports form a simplicial complex. A *minimal nonface* is a support outside that complex whose every proper subset belongs to it. The unique new shattered support in the next lemma is called the *acquired support* of the addition.

We use the following standard facts about ample families: their restrictions, coordinate contractions, reductions, and cube complements are ample, and every nonempty ample family is isometric [CCMW22, Section 3.1 and Theorems 3.1–3.2]. The *ample corner-deletion criterion* says that deleting a concept from an ample family leaves an ample family exactly when the concept is a corner

[CCMW22, Lemma 4.1(iii)]. A reduction to a retained subset T successively deletes corners outside T until precisely T remains; the empty sequence is allowed. Reduction to $\{r\}$, followed by deletion of r , is a peeling with prescribed endpoint r .

Lemma 2.10 (Local test for an ample one-concept extension). *Let $S \subseteq Q_F$ be a nonempty ample family, let $a \in Q_F \setminus S$, and put*

$$I_S(a) = \{f \in F : a^{\{f\}} \in S\}.$$

Then $S \cup \{a\}$ is ample if and only if

$$a^U \in S \quad \text{for every nonempty } U \subseteq I_S(a), \quad (2.1)$$

$$I_S(a) \notin \text{Sh}(S). \quad (2.2)$$

When these conditions hold, a is a corner of $S \cup \{a\}$, its unique maximal cube has support $I_S(a)$, and

$$\text{Sh}(S \cup \{a\}) = \text{Sh}(S) \dot{\cup} \{I_S(a)\}. \quad (2.3)$$

Moreover, $I_S(a)$ is a minimal nonface of the shattered-set complex $\text{Sh}(S)$. In fact, (2.1) holds for every S, a in the hypotheses. Thus the extension is ample exactly when it is isometric, equivalently when

$$\text{Diff}(a, x) \cap I_S(a) \neq \emptyset \quad \text{for every } x \in S. \quad (2.4)$$

Proof. Put $A = S \cup \{a\}$ and $I = I_S(a)$. We first establish the asserted automatic punctured cube. There is nothing to show when $I = \emptyset$. Otherwise restrict S to the face agreeing with a outside I . This is an ample family containing every I -neighbor of a , but not a . Its complement in that face is ample, hence connected, and contains a with none of its neighbors. The complement is therefore $\{a\}$, proving (2.1). All I -traces other than $a|_I$ are consequently present in S . Condition (2.2) is therefore equivalent to (2.4), including when $I = \emptyset$, when both are false. A cube geodesic from a to $x \in S$ must start along a direction in this intersection. Conversely, any such first edge can be followed by a geodesic in the isometric family S , giving a path of Hamming length from a to x . Thus the displayed condition is equivalent to isometry of A .

Suppose first that A is ample. Since $S = A \setminus \{a\}$ is also ample, the ample corner-deletion criterion makes a a corner of A . Its unique maximal cube contains every edge of A incident with a , and hence has support I . Its other vertices are precisely the a^U with $\emptyset \neq U \subseteq I$, which proves (2.1).

Deleting a preserves every cube avoiding a . It also preserves a cube on each proper subset $T \subsetneq I$: if $f \in I \setminus T$, then the T -face through $a^{\{f\}}$ lies in S . Thus deletion can remove only I from the strongly shattered sets. Since the two ample families have sizes differing by one, it must remove I , so $I \notin \text{Sh}(S)$ and (2.2) and (2.3) follow.

Conversely, suppose that (2.1) and (2.2) hold. The family A strongly shatters every member of $\text{Sh}(S)$, because S is ample, and it strongly shatters I by (2.1). Hence

$$|\text{SSh}(A)| \geq |\text{Sh}(S)| + 1 = |S| + 1 = |A|.$$

The Sandwich inequality gives the reverse inequality; equality in its lower bound characterizes ample families. Therefore A is ample. The displayed punctured cube contains all neighbors of a , so it is the unique maximal cube through a , and the first part of the proof gives (2.3).

Finally, $I \notin \text{Sh}(S)$, while the parallel face through $a^{\{f\}}$ used above shows that every proper subset $T \subsetneq I$ is strongly shattered by S . Thus I is a minimal nonface of $\text{Sh}(S)$. $\square$

Definition 2.11 (Punctured-face parametrization). For a nonempty ample family $S \subseteq Q_F$, define

$$\mathcal{P}(S) = \left\{ (a, I) : \begin{array}{l} a \in Q_F \setminus S, \quad I = I_S(a), \\ a^U \in S \text{ for every } \emptyset \neq U \subseteq I, \quad I \notin \text{Sh}(S) \end{array} \right\}. \quad (2.5)$$

Thus an element of $\mathcal{P}(S)$ is an oriented cube face whose only missing vertex is a , and whose support is not shattered by S . The support alone is not the datum: different missing patterns on the same coordinate set give different oriented punctured faces.

Proposition 2.12 (One-concept extensions of a forbidden family). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be irredundantly embedded in Q_F . Then*

$$(a, I) \mapsto S \cup \{a\} \quad (2.6)$$

is a bijection from $\mathcal{P}(S)$ onto the ample one-concept extensions of S . Every such extension belongs to exactly one of $\mathcal{D}_{\text{amp}}$ and $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$. Consequently

$$\mathcal{P}(S) = \mathcal{P}_{\text{peel}}(S) \dot{\cup} \mathcal{P}_{\text{obs}}(S), \quad (2.7)$$

where the first part consists of the punctured faces whose completed family is corner-peelable, and every element of the second part completes to another forbidden pc-minor.

Proof. Lemma 2.10 proves the asserted bijection. Let $E = S \cup \{a\}$ be one of these ample extensions. The same lemma makes a a corner of E .

Let μ be an elementary restriction or contraction of E . Since the embedding of S is irredundant, $\mu(S)$ is a proper pc-minor of S , and hence belongs to $\mathcal{D}_{\text{amp}}$. Applying μ either erases the new concept, identifies it with an old image, or retains one new concept:

$$\mu(E) = \mu(S) \quad \text{or} \quad \mu(E) = \mu(S) \cup \{a_\mu\}. \quad (2.8)$$

In the second case both displayed families are ample, because ampleness is pc-minor closed. The ample corner-deletion criterion makes a_μ a corner of $\mu(E)$. Peeling a_μ first and then $\mu(S)$ shows that $\mu(E) \in \mathcal{D}_{\text{amp}}$. Thus all elementary pc-minors of E belong to $\mathcal{D}_{\text{amp}}$, and pc-minor closure gives the same conclusion for every proper pc-minor.

If E is corner-peelable, it belongs to the first part of (2.7). Otherwise the preceding paragraph proves $E \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$, which is the second part. The two alternatives are disjoint by definition. $\square$

The next invariance concerns every retained target. For targets admitting a representation map, Proposition 2.1 interprets it as preservation of inward-boundary extension possibilities. Its deletion formulation also covers targets for which no map is supplied.

Proposition 2.13 (Additions of support size at most two). *Let $S \subseteq Q_F$ be nonempty and ample, and suppose that $H = S \cup \{a\}$ is an ample one-concept extension with acquired support $P = I_S(a)$. If $|P| \leq 2$, then for every $T \subseteq S$, including $T = \emptyset$,*

$$H \text{ reduces to } T \iff S \text{ reduces to } T. \quad (2.9)$$

Both reductions use only corner deletions of concepts outside T ; a sequence of length zero is allowed. In particular,

$$H \in \mathcal{D}_{\text{amp}} \iff S \in \mathcal{D}_{\text{amp}},$$

and every old prescribed endpoint is preserved. Thus a one-concept addition that makes any previously impossible retained target reachable must acquire a support of size at least three.

Proof. The added concept a is a corner of H , by the ample corner-deletion criterion. Any sequence from S to T can therefore be prefixed by deletion of a .

For the converse, suppose that H reduces to T but S does not, and choose a counterexample with $|S|$ least. One has $|S| \geq 2$, since a singleton reduces to each of its subsets. Let z be the first deleted concept in the chosen sequence from H . It lies outside T and is not a , since deleting a first would leave a sequence from S to T .

If z is also a corner of S , put $S' = S \setminus \{z\}$ and $H' = H \setminus \{z\}$. Both are ample, S' is nonempty and contains T , and $H' = S' \cup \{a\}$ reduces to T . Its acquired support is $I_{S'}(a) \subseteq P$, by Lemma 2.10. Minimality gives a sequence from S' to T . Prefixing it by z contradicts the choice of S .

Suppose instead that z is not a corner of S . The unique maximal cube through z in H must contain a : if it avoided a , it would still contain every cube through z in S , making z a corner there. Since a is also a corner of H , this cube is precisely the completed P -cube through a . The concepts a, z cannot be adjacent. Removing an adjacent vertex from that cube would leave at z the full face on its remaining incident directions, again making z a corner of S .

It follows that $|P| = 2$ and a, z are opposite vertices of the completed square. Each has exactly the other two square vertices as neighbors. Hence exchanging a, z and fixing every other vertex is a graph automorphism of H . It fixes T pointwise, because neither a nor z belongs to T , and identifies $H \setminus \{z\}$ with $S = H \setminus \{a\}$. Applying it to the remaining deletion sequence gives a sequence from S to T , the final contradiction.

Taking $T = \emptyset$ gives ordinary peelability. Taking $T = \{r\}$ and then deleting r gives the assertion for each prescribed endpoint $r \in S$. $\square$

The proof also transports a supplied deletion sequence without searching for another one. If its first concept is a , remove that first entry. If the first concept z is a corner of both families, retain it and apply the same procedure to the remaining sequence. In the opposite-square case, exchange a, z in the remaining sequence. Each recursive step reduces the family size.

The bounded checker `verify_damp_relative_low_support.py` independently computes reachable retained subsets for all 336 eligible one-concept extensions in Q_3 . It checks 5,616 extension–target pairs and replays 4,056 transported deletion orders, exercising all three cases of the proof. It also checks ordinary peelings and complete negative relative-deletion graphs for fixed ample families of orders 294, 295, 296, with a retained vertex and a retained incident edge. The respective negative graphs have 1, 2, 4 states for each target. These tests, recorded in `damp_relative_low_support_verification.json`, verify the stated finite instances; the proposition is proved uniformly above.

Corollary 2.14 (Descent at corners of degree at most two). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor, and let a be a corner of H of degree at most two. Then $S = H \setminus \{a\}$ is again an irredundantly embedded ample forbidden pc-minor. Consequently, if no corner deletion of H leaves an ample forbidden pc-minor, every corner of H has degree at least three.*

Proof. The corner-deletion criterion makes S ample, and Proposition 2.13 makes it nonpeelable. Its embedding is irredundant: if a coordinate became constant, S would be a proper signed restriction of H , which must peel. Put $I = I_S(a)$, so $|I| \leq 2$.

For an elementary map μ , the family μH peels, and either $\mu H = \mu S$ or μH is an ample one-concept extension of the ample μS . In the latter case its acquired support still has size at most two. Indeed, a restriction in coordinate e retaining a leaves exactly the incident directions $I \setminus \{e\}$. A contraction in a direction $e \in I$ identifies a with its old neighbor. If $e \notin I$, contraction retains precisely the shattered sets avoiding e ; since $\text{Sh}(H) = \text{Sh}(S) \dot{\cup} \{I\}$, the acquired support of the contracted extension is still I . Proposition 2.13 therefore makes μS peelable as well; empty

images are already peelable. All elementary pc-minors of S peel, so S is forbidden. The terminality assertion follows immediately. $\square$

2.5 Product attachments and relative obstructions

A relative peeling of an ample family A to a nonempty subfamily K deletes only corners outside K and leaves exactly K ; write $A \searrow K$. The next construction turns this question into ordinary peelability and transfers minimal relative failures to forbidden pc-minors. No convexity or gatedness of K in A is required.

We use the elementary incident cube criterion: a vertex is a corner exactly when the cube spanned by all its incident directions is contained in the family. Every incident edge extends to a maximal cube, which proves the criterion. Intersecting that cube with a coordinate face also proves that a peeling restricts to a peeling of every coordinate section.

Fix a finite ample family $R \subseteq Q_F$ containing 0_F , which peels but has no peeling ending at 0_F . The construction uses only these two peeling properties. A fixed example is the 289-concept family of Remark 2.67, whose only corner is its root. For nonempty ample families $K \subseteq A \subseteq Q_E$, with E, F disjoint, put

$$P_R(A, K) = (A \times \{0_F\}) \cup (K \times R). \quad (2.10)$$

Thus a copy of R is attached over each vertex of K , and the copies share all cubes supplied by K . This differs from attaching independent copies at the individual vertices: the mixed cubes are essential.

Theorem 2.15 (Testing a relative peeling). *The family $P_R(A, K)$ is ample, has $|A| + (|R| - 1)|K|$ vertices, and satisfies*

$$P_R(A, K) \text{ peels} \iff K \text{ peels and } A \searrow K.$$

In particular, a fixed choice of R converts relative peeling to ordinary peeling with a constant factor increase in the number of vertices.

Proof. Write $T \subseteq E$ and $I \subseteq F$ for the two parts of a shattered support. If I is empty, ampleness of A supplies its cube. If I is nonempty, a nonzero trace on I can occur only in $K \times R$. Realizing every trace on T at that fixed nonzero trace shows that K shatters T . Projection onto F also shows that R shatters I . Their witnessing cubes have a product cube in $K \times R$. This proves ampleness; the cardinality follows by subtracting the overlap $K \times \{0\}$.

Suppose $P_R(A, K)$ peels. Fix $k \in K$. Its section obtained by fixing all E coordinates to k is a copy of R , so the restricted peeling is a peeling of R . At the deletion of $(k, 0)$ there must be a remaining neighbor in this copy outside its root: otherwise the root could be postponed to last, contradicting the choice of R . Let $f \in F$ be its direction. If a remaining $(a, 0)$ with $a \in A \setminus K$ were adjacent to $(k, 0)$, its direction $e \in E$ and f would span a missing mixed square, since $(a, 0^{\{f\}})$ is absent. The incident cube criterion excludes this.

Restrict the peeling to $A \times \{0\}$. Every $k \in K$ is therefore deleted without a remaining neighbor outside K . Retain all these vertices and run only the deletions in $A \setminus K$. A retained vertex introduces no new neighbor at a later outside deletion, so the original corner cube and all incident directions at that deletion are unchanged. This gives $A \searrow K$. Finally choose any $r \in R \setminus \{0\}$. The section with F coordinates fixed to r is K , so K peels by restriction to this coordinate face.

Conversely, perform a relative peeling of A to K . Vertices of $A \setminus K$ have no neighbors with nonzero F coordinates, so these are also valid deletions in the attachment. What remains is $K \times R$. Given peelings of R and K , delete the R layers in the first order, using the second order within each layer. At each step the incident cube is the product of the current corner cubes: the current

layer contains its remaining part of K , and all later layers still contain all of K . This verifies every deletion by the incident cube criterion and finishes the proof. $\square$

Call $(R, 0)$ a *minimal rooted failure* when R peels, $R \not\searrow \{0\}$, and every elementary root-preserving image $\rho_f^0 R$ and $\pi_f R$ peels to its root. Assume all coordinates of A and R vary. Set $P_R(A, \emptyset) = A$.

Theorem 2.16 (Transfer of minimality). *Let $(R, 0)$ be a minimal rooted failure and let $K \subseteq A$ be nonempty ample peelable families. Then $P_R(A, K)$ is a forbidden pc-minor if and only if*

1. $A \not\searrow K$;
2. for every elementary operation $\mu \in \{\rho_e^0, \rho_e^1, \pi_e\}$ on a coordinate of A with $\mu K \neq \emptyset$, one has $\mu A \searrow \mu K$.

Proof. For an old coordinate $e \in E$, direct application of the operation gives

$$\mu P_R(A, K) = P_R(\mu A, \mu K).$$

For an added coordinate $f \in F$, the three identities are

$$\begin{aligned} \rho_f^0 P_R(A, K) &= P_{\rho_f^0 R}(A, K), & \pi_f P_R(A, K) &= P_{\pi_f R}(A, K), \\ \rho_f^1 P_R(A, K) &= K \times \rho_f^1 R. \end{aligned}$$

The first two added-coordinate images peel: peel their rooted factor down to the root, layer by layer over K as in Theorem 2.15, leaving A , and then peel A . The third peels by the product construction and pc-minor closure of peelability for R .

An old-coordinate image with empty μK is simply μA , which peels. If μK is nonempty, it is ample and peelable by pc-minor closure; Theorem 2.15 says that the image peels exactly when $\mu A \searrow \mu K$. The same theorem says the original attachment is nonpeelable exactly when $A \not\searrow K$. Since every active coordinate operation produces a proper image, and elementary images suffice by pc-minor closure, these statements prove both directions. $\square$

This gives a uniform construction of forbidden minors from minimal relative failures of pairs (A, K) . Its input admissibility still uses relative peelings. It therefore transfers a classification problem; it does not supply intrinsic admissibility for those pairs or exhaust all forbidden minors by such attachments.

Remark 2.17 (Verification scope). The checker `verify_gated_relative_tests.py` replays an ordinary peeling and all 24 elementary root-preserving orders of the fixed 289-concept factor, and checks that its root is its only corner. It constructs specified boundary-record orders, including a nonconvex attachment site, and verifies the coordinate-minor identities on that attachment. A 577-concept positive attachment has a replayed order; a 577-concept negative attachment is cornerless. The record is `gated_relative_tests_verification.json`. The universal statements above follow from their proofs; no exhaustive classification of relative pairs is claimed.

2.6 Gluing at vertices and separating cubes

Gluing at a vertex. A *rooted family* is a pair (A, a) with $a \in A$. We say that A *peels to* a if it has a corner peeling whose last concept is a . A *root-preserving pc-minor* is obtained by coordinate contractions, carrying the root to its image, and convex restrictions that contain the current root. Peeling to the root is preserved by these operations: reverse the peeling to obtain an ample prefix

order starting at the root, restrict the prefixes, or order projected concepts by their first occurrence. The resulting ample prefix order still starts at the image of the root. Here an ample prefix order means an ordering whose every nonempty initial segment is ample; reversing such an order gives a corner peeling.

Proposition 2.18 (Peeling after gluing at one vertex). *Let $A_1, \dots, A_m$ be nonempty ample families, each containing 0, on pairwise disjoint coordinate sets. Embed them in the product cube by setting all other coordinates to zero, and put*

$$X = A_1 \vee \dots \vee A_m = \bigcup_{i=1}^m A_i.$$

Then X is ample. It peels to 0 if and only if every A_i peels to 0. Moreover,

$$X \in \mathcal{D}_{\text{amp}} \iff \begin{cases} A_i \in \mathcal{D}_{\text{amp}} \text{ for every } i, \\ \text{at most one } A_i \text{ fails to peel to 0.} \end{cases} \quad (2.11)$$

Proof. A pair of coordinates from different blocks is not shattered, because the trace 11 is absent. The shadow of X is therefore the union of the block shadows, with the empty support shared. Hence

$$|\text{Sh}(X)| = 1 + \sum_i (|\text{Sh}(A_i)| - 1) = 1 + \sum_i (|A_i| - 1) = |X|,$$

which proves ampleness.

For the peeling assertion, use Proposition 2.1 with retained singleton $\{0\}$. A USO on the union restricts to each piece. Its outgoing directions at 0 can lie in at most one exclusive block, by (C1), since mixed squares are absent. Thus in every other piece 0 is a sink, and that piece peels to 0. A sink at 0 in the union requires this in every piece.

Conversely, choose acyclic USOs with sink 0 in all but a possible exceptional piece, and any acyclic USO in that piece. At 0, all outgoing edges belong to the exceptional piece, so (C1) holds; elsewhere it holds within the unique piece. Every cube lies in a piece, giving (C2). A simple cycle cannot cross between pieces through their only common vertex. The union is therefore acyclic. Choosing sink 0 in every piece also proves the prescribed-sink assertion. $\square$

Recall that a *cut vertex* disconnects a connected graph when deleted. Its *blocks* are its maximal two-connected subgraphs and its bridges, with each bridge regarded as a two-vertex block.

Theorem 2.19 (Forbidden pc-minors with a cut vertex). *An ample family with a cut vertex belongs to $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ if and only if it is obtained by gluing two rooted ample families (A_1, a_1) , (A_2, a_2) at their roots, where each pair satisfies all three conditions:*

1. $A_i \in \mathcal{D}_{\text{amp}}$;
2. no acyclic USO of A_i has sink a_i ;
3. every proper root-preserving pc-minor of (A_i, a_i) admits an acyclic USO with sink at the image of the root.

By Proposition 2.1, conditions 2 and 3 are exactly failure and proper-minor restoration of peeling to the root. Each A_i has no cut vertex. The glued graph has exactly two blocks and one cut vertex, so the two rooted factors are recovered uniquely up to interchange.

Proof. First note how a cut vertex a of a partial cube separates its coordinates. No coordinate can differ from its value at a in two components of the graph with a removed. Otherwise choose vertices x, y in those components whose bits in that coordinate differ from the bit at a . Every path between them passes through a , so its length is at least $d_H(x, a) + d_H(a, y) > d_H(x, y)$, where d_H is Hamming distance, contradicting isometry. Thus, after translating a to zero, the induced families on each component together with a occupy disjoint coordinate blocks. They and any union of them are convex: a shortest path cannot leave a component through a and return through the same vertex. They therefore give the vertex-gluing presentation above.

Let $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ have a cut vertex. Each rooted component family is a proper convex restriction and belongs to $\mathcal{D}_{\text{amp}}$. By (2.11), at least two of them fail to peel to the cut vertex. If there were any additional component, the union of two failing ones would be a proper convex nonpeelable restriction. Hence there are exactly two component families A_1, A_2 , and both satisfy conditions 1 and 2.

An elementary root-preserving restriction or contraction in A_1 produces a proper minor of X obtained by gluing the resulting rooted family to the unchanged A_2 . That minor peels, whereas A_2 does not peel to its root. The vertex-gluing test forces the changed family to peel to its root. The same reasoning applies to A_2 . Since peeling to the root is preserved by subsequent root-preserving pc-minors, condition 3 follows in full.

Conversely, glue two pairs satisfying the three conditions, using disjoint coordinate sets and identifying their roots with 0. The result is ample and nonpeelable by the vertex-gluing test. For a coordinate of A_1 , contraction or restriction to the semicube containing 0 replaces A_1 by a proper root-preserving minor. Peel that minor to 0, then peel A_2 . Restriction to the opposite semicube discards A_2 entirely and leaves a pc-minor of the peelable A_1 . All three elementary images therefore peel. The argument for a coordinate of A_2 is identical, proving forbidden-minor status.

It remains to show that a rooted factor (A, a) satisfying the conditions has no cut vertex. If its root is a cut vertex, the rooted part of Proposition 2.18 forces one component family to fail to peel to the root. That family is a proper root-preserving convex restriction, contrary to condition 3. If another vertex $b \neq a$ is a cut vertex, let U consist of b and the component containing a , and let V_j be the other component families, rooted at b . The same deletion argument shows that A peels to a exactly when U peels to a and every V_j peels to b : before deleting b , all the other branches must disappear, because the branch containing the retained a still has an edge at b . The pair (U, a) is a proper root-preserving convex restriction, so it peels to its root. Each (V_j, b) is also a proper root-preserving pc-minor of (A, a) : contract all coordinates outside its block, which sends a to b . Condition 3 makes all these pairs peel to their roots, a contradiction. Thus the factors have no cut vertices. Condition 2 also excludes a singleton or a single edge, since either peels to any chosen root. Thus the factors are two-connected, and the asserted block decomposition and uniqueness follow. $\square$

The theorem completely describes the block decomposition of an ample forbidden pc-minor with a cut vertex. Its conditions still use peeling and rooted minor minimality. An intrinsic classification must therefore also identify these rooted factors without those recognition tests; the theorem is not such an admissibility criterion.

Separating coordinate faces. A nonempty coordinate-face intersection D is *separating* when deleting all its vertices disconnects the graph. In particular, a separating edge means that its two endpoints form a vertex cut; it does not mean that the edge is a bridge. For $D \subseteq A$, a peeling of A to D deletes corners outside D until precisely D remains.

Proposition 2.20 (Ample assembly and USO compatibility at a coordinate face). *Let $\emptyset \neq D \subseteq Q_E$ and $A_i \subseteq Q_{E \cup F_i}$ be ample, with pairwise disjoint exclusive blocks F_i and $A_i \cap (Q_E \times \{0\}^{F_i}) = D$. Embed the pieces by putting zero in the other blocks, and set $X = \bigcup_{i=1}^m A_i$. Then X is ample if and only if*

$$\text{Sh}(A_i) \cap \text{Sh}(A_j) = \text{Sh}(D) \quad (i \neq j). \quad (2.12)$$

This condition is automatic when $D = Q_E$. When it holds, USOs on the pieces glue to a USO of X exactly when they agree on D and, at each $d \in D$, at most one piece has an outgoing edge from d into $A_i \setminus D$. The glued USO is acyclic exactly when its union orientation is acyclic. For $|D| \leq 2$, acyclicity of the piece USOs already implies acyclicity of their union.

Proof. The pieces intersect pairwise in D , so $|X| = \sum_i |A_i| - (m-1)|D|$. Every contained cube belongs to a piece: a cube varying coordinates in two exclusive blocks would contain a word nonzero in both blocks, which is absent; a cube with a fixed nonzero exclusive coordinate stays in that piece. Cubes with all exclusive coordinates fixed to zero lie in D . Consequently

$$\text{SSh}(X) = \bigcup_i \text{Sh}(A_i),$$

where SSh denotes the strongly shattered supports. Each piece shadow contains $\text{Sh}(D)$, of cardinality $|D|$. Thus this union has cardinality $|X|$ exactly when the supports outside $\text{Sh}(D)$ occur in only one piece, which is (2.12). Equality in the lower Sandwich inequality characterizes ampleness, proving both directions. For $D = Q_E$, any support common to two piece shadows lies in E and already belongs to $\text{Sh}(D) = 2^E$, so the test is automatic.

Every contained cube belongs to a piece, so (C2) holds for the union whenever the orientations agree. Outside D , all neighbours lie in one piece. At $d \in D$, outgoing directions into two exclusive blocks violate (C1) because their square is absent. If at most one piece supplies outgoing exclusive directions, all outgoing directions have their cube in that piece. This proves necessity and sufficiency. Conversely, the pieces inherit USOs from the union by CCMW Lemma 6.9, as they are coordinate-face intersections.

A simple directed cycle crossing a singleton intersection is impossible. If $D = \{s, t\}$ is oriented $s \rightarrow t$, a simple directed cycle visiting multiple pieces would contain a directed path $t \rightarrow s$ within one piece, with interior outside D . This path and the common edge contradict acyclicity in that piece. For larger faces the union acyclicity condition is retained; agreement alone is not being asserted sufficient. $\square$

Corollary 2.21 (Finite boundary data for acyclic gluing). *Use the hypotheses and notation of Proposition 2.20, assume its shadow-intersection condition, and put $k = |D|$. For each piece A_i and each total order σ of D , let $\mathcal{S}_i(\sigma)$ consist of the sets $S \subseteq D$ for which there is an acyclic USO of A_i with a topological order restricting to σ on D , and with*

$$S = \{d \in D : d \text{ has an outgoing neighbour in } A_i \setminus D\}.$$

Then X is corner-peelable if and only if there are one order σ and pairwise disjoint sets $S_i \in \mathcal{S}_i(\sigma)$. One may replace every $\mathcal{S}_i(\sigma)$ by its inclusion-minimal members without changing this criterion. Thus a piece has at most $k!2^k$ possible boundary records before this reduction, independently of the number of its interior vertices.

Proof. Restrict an acyclic USO of X to its coordinate-face pieces and restrict one global topological order to D . The preceding proposition gives the disjointness of the resulting sets S_i , proving necessity.

Conversely, choose witnessing piece orientations and topological orders. They orient each edge of D from its earlier to its later endpoint in σ , so they agree on D . Disjointness of the S_i gives a USO of X by the preceding proposition. To see that it is acyclic, split each witnessing order into blocks of outside vertices before the first vertex of σ , between successive vertices of σ , and after the last vertex. Concatenate the pieces' blocks within each interval, and insert each boundary vertex once. The resulting order restricts to the chosen order on each piece. Every edge belongs to a piece, so this is a topological order of the union. CCMW Proposition 6.12 makes it a corner peeling.

If disjoint records have been chosen, replace each S_i by a minimal member of $\mathcal{S}_i(\sigma)$ contained in it. Such a member exists by finiteness, and disjointness persists. The converse is immediate. There are $k!$ orders and 2^k subsets, giving the stated bound. $\square$

This eliminates a test on the full union digraph from the compatibility rule, once the attainable boundary records of the pieces are supplied. It does not intrinsically characterize those records or generate their realizing pieces. In particular, the finite bound is for a fixed separator; it is not a bound on the primitive inputs to the forbidden-minor problem. Unlike the original cube case, this statement also applies to an ample coordinate-face intersection that is not a full cube. In particular, it applies to all eight canonical faces in Theorem 2.86: their recovered pieces satisfy the shadow test because their union is ample. In forward assembly, the shadow test must be imposed explicitly.

For example, let $D = \{0, 1, 3\} \subseteq Q_2$ and take two copies of $Q_3 \setminus \{2\}$, identifying their zero third-bit sections with D and using different exclusive coordinates. Both pieces are ample, but their shadows intersect in $2^{\{0,1\}}$, strictly larger than $\text{Sh}(D) = \{\emptyset, \{0\}, \{1\}\}$. Their union has 11 vertices, 10 strongly shattered supports and 12 shattered supports, so it is not ample. Replacing the second piece by $D \square Q_1$ gives an ample union of order 10. The direct checks in `research/face_assembly_shadow.py/json` cover these controls and all 1032 assemblies obtained from nonempty ample $D \subseteq Q_2$ and two ample pieces in three coordinates with zero exclusive section D .

Proposition 2.22 (Transport of boundary records). *Let $D = Q_E \times \{0\}^F \subseteq A \subseteq Q_{E \cup F}$ be ample, and let a corner order of A realize a boundary record (σ, S) as in Corollary 2.21. For $f \in F$, identify D with its image after deleting coordinate f .*

1. *Restriction to $f = 0$ realizes a record (σ, T) with $T \subseteq S$.*
2. *Contraction in f , with the order transported by last-fibre positions, realizes a record (τ, T) with $T \subseteq S$. Moreover, if $x \in D \setminus S$ precedes $y \in D$ in σ , then x precedes y in τ .*

In particular, contraction preserves the boundary order whenever S is empty.

Proof. Both transported orders are corner orders by pc-minor closure and its order-level proof. Under restriction, the boundary vertices keep their positions relative to one another, and every remaining outward edge was already an outward edge with the same orientation. This proves the first assertion.

For contraction, the representative of a double fibre is its later endpoint, equivalently its sink. CCMW Proposition 6.14 gives the outgoing directions of the image vertex from this representative, with f erased. If a boundary vertex d has its outside mate d^f as representative, the original edge $d \rightarrow d^f$ puts d in S . If its representative is d itself, any outgoing exclusive direction in the image was already such a direction at d . In either case, an outward vertex of the contracted record belongs to S .

Write $p(v)$ for the position in the original order. The new position used to order a boundary vertex d is $\max\{p(v) : \pi_f(v) = d\}$, which is at least $p(d)$. For $d \notin S$, a present mate d^f precedes d , so this maximum is exactly $p(d)$. Thus $p(x) < p(y)$ with $x \notin S$ implies that the new position of x is smaller than that of y . This proves the second assertion and its final consequence. $\square$

The statement concerns a transported witness. It does not assert that the original order σ is unattainable in the minor when $\tau \neq \sigma$. For example, on the square $A = \{0, 1, 2, 3\}$ with boundary edge $D = \{0, 1\}$, the corner order $0, 1, 3, 2$ has $\sigma = (0, 1)$ and $S = D$. Contracting bit 1 gives the last-fibre boundary order $(1, 0)$ and empty outward set, but the minor edge also admits $(0, 1)$. The transport argument does not preserve the order of every attainable record, even if the minor orientation may be chosen anew, as the next proposition shows. The verifier `research/boundary_record_transport.py/json` checks the stated law, not that stronger existential assertion. If the minor peels to D , same-order attainability does hold: restrict the original corner order to D to obtain σ , and append σ to a relative peeling of the minor to D . This realizes $(\sigma, \emptyset)$. Consequently, a counterexample to same-order attainability must have a minor that itself fails relative peeling to D ; relative-minimal pieces cannot supply such a counterexample in their exclusive proper minors.

Proposition 2.23 (An attainable boundary order can disappear under contraction). *Preservation of attainable records with the same boundary order and a smaller outward set fails under exclusive-coordinate contraction, already for an edge boundary. It fails even when only inclusion-minimal outward sets are recorded for each boundary order.*

Proof. Use the positive 289-concept class $C \subseteq Q_{12}$ of Remark 2.67, translated so that its unique corner is 0. It contains 2 and 4, the neighbours of 0 in bits 1 and 2. Put $D = \{0, 2\}$ and, for a fresh coordinate f , form the peripheral expansion

$$A = (D \times \{0\}) \cup (C \times \{1\}).$$

Both A and C are ample and corner-peelable. Deleting $(2, 0)$, then $(0, 0)$, then following any corner order of the upper copy of C realizes the lower-boundary order $\sigma = (2, 0)$ with outward set D .

In fact this is the only attainable outward set for that boundary order. Every corner of the upper layer would have to be a corner of C : a corner cube in A restricts to a corner cube in that layer. The only candidate is $(0, 1)$, which is not a corner because its neighbours $(0, 0)$ and $(4, 1)$ have missing square vertex $(4, 0)$. Thus initially the only corners of A are its two lower vertices. After deleting $(2, 0)$, the same argument still blocks $(0, 1)$, and the only corner is $(0, 0)$. Any order realizing σ must therefore start with these two deletions. Both have their upper mates still present, so both belong to the outward set. Hence D is inclusion-minimal, indeed the unique record for σ .

Contracting f returns C and preserves the marked edge D . Every corner order of C starts at its unique corner 0, so no order can restrict to $(2, 0)$ on D , regardless of its outward set. By the acyclic-USO correspondence, changing the minor orientation cannot repair this failure. The saved class, orders, and forced first corners are replayed in `research/boundary_attainability_failure.py/json`. $\square$

This does not contradict minor closure of positive gluing. The lost record has outward set all of D , so any other piece compatible with that record must have empty outward set and hence peel to D . Such pieces can be peeled to D first, leaving the positive contracted piece C . Consequently this lost record alone does not distinguish positive gluing contexts. Inclusion-minimality at each fixed boundary order is insufficient to make the raw record families monotone under minors.

Theorem 2.24 (Finite gluing algebra and comparison by contexts). *Fix a nonempty corner-peelable ample family $D \subseteq Q_E$. A piece is an ample family $A \subseteq Q_{E \cup F}$ whose zero F -section is exactly D . Private coordinate sets are renamed to be disjoint before gluing. Put*

$$U_D = 2^E \setminus \text{Sh}(D), \quad T_A = (\text{Sh}(A) \cap 2^E) \setminus \text{Sh}(D),$$

and let Σ_D be the set of corner orders of D . For each piece define its record family

$$\mathcal{R}(A) = \{(T_A, \sigma, S) : \sigma \in \Sigma_D, S \in \mathcal{S}_A(\sigma)\}.$$

Here $\mathcal{S}_A(\sigma)$ is defined in Corollary 2.21. In particular, $\mathcal{R}(A) = \emptyset$ if A is not corner-peelable. Set

$$\Omega_D = 2^{U_D} \times \Sigma_D \times 2^D.$$

For arbitrary $P, Q \subseteq \Omega_D$ define

$$P * Q = \{(T \cup T', \sigma, S \cup S') : (T, \sigma, S) \in P, (T', \sigma, S') \in Q, T \cap T' = \emptyset, S \cap S' = \emptyset\}. \quad (2.13)$$

Then the following statements hold.

1. This product is associative and commutative, with zero $\emptyset$ and identity $I_D = \{(\emptyset, \sigma, \emptyset) : \sigma \in \Sigma_D\}$. For two pieces glued along D ,

$$\mathcal{R}(A \cup_D B) = \mathcal{R}(A) * \mathcal{R}(B),$$

where the left side is defined to be empty if the union is not ample. Thus the union is ample and corner-peelable exactly when the product is nonempty.

2. Let $\mathfrak{R}_D$ consist of the record families realized by pieces, together with zero. It is a finite monoid. On it define

$$P \preceq Q \iff \text{for every } R \in \mathfrak{R}_D, P * R \neq \emptyset \implies Q * R \neq \emptyset.$$

Mutual comparison is an equivalence relation compatible with product. Its classes form a partially ordered commutative monoid with at most $2^{|\Omega_D|}$ elements.

3. If A' is obtained from A by contractions of private coordinates and restrictions of private coordinates to zero, and its zero remaining-private-coordinate section is still exactly D , then

$$\mathcal{R}(A) \preceq \mathcal{R}(A').$$

If $X = A \cup_D B$ is an ample forbidden pc-minor and the corresponding operation on X is proper, this comparison is strict modulo mutual comparison.

Proof. The first coordinate T_A records exactly the supports that can prevent ample gluing. Indeed, private coordinate sets are disjoint, so the common shattered supports of two pieces lie in 2^E . Proposition 2.20 says that their union is ample exactly when $T_A \cap T_B = \emptyset$. If it is ample, its shattered supports are the union of the piece shadows, as shown in that proposition's proof. Its first record coordinate is consequently $T_A \cup T_B$.

An order on a piece restricts to a corner order on D , by coordinate face restriction. In an acyclic USO of the union, restriction to the two pieces supplies a common boundary order and disjoint outward sets; the outward set of the union is their union. Conversely the proof of Corollary 2.21 combines any two records with a common order and disjoint outward sets into an acyclic USO, with exactly that union of outward sets. This proves the record identity when the union is ample. If the shadows conflict, every record pair is excluded by its first coordinates. If either piece is nonpeelable, the union cannot be ample and peelable, since that piece is a coordinate face of the union. Thus the identity and positivity test also cover all zero cases.

In a product of three formal record families, a resulting record is specified by one common order, three pairwise disjoint first-coordinate sets, and three pairwise disjoint outward sets. These conditions and the two unions are independent of parenthesization, proving associativity. Commutativity, the zero, and the identity follow directly from (2.13). The identity is realized by D itself. The record identity shows that a product of realized families is either realized by their ample union or is zero. Hence $\mathfrak{R}_D$ is a submonoid of the finite monoid 2^{Ω_D} .

The relation $\preceq$ is reflexive and transitive. If $P \preceq Q$ and $H \in \mathfrak{R}_D$, apply its definition to $H * R$, for each $R \in \mathfrak{R}_D$, to obtain $P * H \preceq Q * H$. It follows that mutual comparison is a product congruence and the induced relation on equivalence classes is a partial order. Finiteness and the stated bound follow from the ambient monoid.

For the minor assertion, choose any piece B in fresh private coordinates. Every allowed operation on $A \cup_D B$ fixes B and induces the specified operation on A . Thus $A' \cup_D B$ is a pc-minor of $A \cup_D B$. Ample corner-peelable families are closed under these minors, proving the implication defining $\preceq$ for every realized context; the zero context imposes no condition. Finally, if X is forbidden and the resulting minor is proper, then

$$\mathcal{R}(A) * \mathcal{R}(B) = \emptyset, \quad \mathcal{R}(A') * \mathcal{R}(B) \neq \emptyset.$$

The context B therefore excludes the reverse comparison, proving strictness. $\square$

This comparison keeps the effect of a piece on every gluing context, rather than individual boundary orders. It is monotone even in Proposition 2.23, where a raw order disappears. For a cube boundary U_D is empty, so the first record coordinate is unnecessary, and every private-coordinate operation in item 3 preserves the marked face. For a noncube boundary the first coordinate enforces the shadow condition and must be retained. Moreover, a private contraction can enlarge the zero private-coordinate section; item 3 applies only when the resulting section is still D . No comparison between different marked faces is asserted.

The algebra is finite and explicit, but its realizable submonoid $\mathfrak{R}_D$ has not been intrinsically characterized. Nor does the finite number of comparison classes bound the sizes of pieces minimal for a class: proper minors may immediately change class, however large the original piece is. Thus the theorem gives a minor-monotone comparison and a necessary strictness test for forbidden assemblies, not an exhaustive piece generator. The new content beyond the boundary compatibility criterion is its associative composition and its comparison under all realizable contexts; no preservation of individual records is asserted.

Remark 2.25 (A square already permits a cyclic compatible union). The acyclicity condition in Proposition 2.20 cannot be dropped for a square. Use binary integer labels and let $D = \{0, 1, 2, 3\}$. Orient a first three-cube $A = \{0, \dots, 7\}$ by the corner order

$$4, 0, 1, 5, 7, 6, 2, 3.$$

Orient a second copy by the corner order $0, 2, 4, 6, 7, 5, 1, 3$, then replace its bit 2 by a fresh bit 3. The two orientations agree on the diamond $0 \rightarrow 1, 0 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 3$ in D . Their boundary vertices with outgoing exclusive edges are respectively $\{1\}$ and $\{0, 2\}$, which are disjoint. Their union is therefore a USO, but contains the directed cycle

$$1 \rightarrow 5 \rightarrow 7 \rightarrow 6 \rightarrow 2 \rightarrow 10 \rightarrow 11 \rightarrow 9 \rightarrow 1.$$

The first piece forces 1 before 2 in every topological order, whereas the second forces 2 before 1. They cannot supply the same boundary order in Corollary 2.21. The union family is $Q_2 \square P_3$ and is

corner-peelable; the failure concerns these prescribed orientations, not the support. The verifier `research/cube_separator_cycle.py/json` replays both piece orders and independently checks (C1), (C2), and the directed cycle in the union. Thus the automatic acyclicity for singleton and edge intersections is sharp at the next cube size.

Theorem 2.26 (Pieces at a separating cube). *Let $D = Q_E$, and let $A_1, \dots, A_m$ be ample families containing D , on coordinate sets $E \dot{\cup} F_i$, where the sets F_i are pairwise disjoint. Embed each family by putting zero in the other blocks, and put $X = \bigcup_i A_i$. Then X is ample. If X is corner-peelable, at most $|D|$ of the pieces A_i fail to peel to D .*

Now suppose X is an ample forbidden pc-minor and D is a separating k -cube. The components K_i of $X \setminus D$ give a presentation of the preceding form with $A_i = D \cup K_i$. Each piece is a proper convex restriction, belongs to $\mathcal{D}_{\text{amp}}$, and fails to peel to D . Moreover,

$$2 \leq m \leq |D| + 1 = 2^k + 1. \quad (2.14)$$

The pieces and their exclusive coordinate sets are uniquely recovered from the specified separating cube, up to permutation of the pieces.

Proof. Ampleness follows from Proposition 2.20. If $X \in \mathcal{D}_{\text{amp}}$, choose an acyclic USO and restrict it to each piece. A piece failing to peel to D must have an outgoing edge from some $d \in D$ to its exclusive part, by Proposition 2.1. By the compatibility law, different such pieces use disjoint nonempty sets of boundary vertices. There are at most $|D|$ of them. This does not require $A_i \setminus D$ to be connected.

For the recovery assertion, normalize the separating cube by making its fixed coordinates zero, and let E be its varying coordinates. A coordinate $f \notin E$ cannot take value one in two different components of $X \setminus D$. A path between such concepts must pass through D , where $f = 0$, and hence must change f twice, contrary to the existence of a cube geodesic. Assign each coordinate outside E to its unique component. The piece $D \cup K_i$ is exactly the restriction setting all other exclusive coordinates to zero. The same holds for the union of any selected pieces. These are convex restrictions and have the claimed coordinate form.

Each piece belongs to $\mathcal{D}_{\text{amp}}$ by proper-minor minimality. If one peeled to D , its outside concepts could first be removed from X , followed by a peeling of the proper convex restriction formed by the other pieces. This would peel X , a contradiction. Thus every piece fails to peel to D . If there were more than $|D| + 1$ pieces, the union of any $|D| + 1$ of them would be a proper convex restriction which cannot peel, by the first part. This proves (2.14). The component construction also proves uniqueness for the marked cube. $\square$

Proposition 2.27 (Pieces at a coordinate face). *Let X be ample and let D be a nonempty coordinate-face intersection of X . Normalize the fixed coordinates of D to zero, and let E be its free coordinate set. The components $W_1, \dots, W_m$ of $X \setminus D$ have disjoint exclusive coordinate blocks F_i outside E . Each $A_i = D \cup W_i$, and each union of selected such pieces, is a coordinate restriction of X . If $X \in \mathcal{D}_{\text{amp}}$, at most $|D|$ of the A_i fail to peel to D . If X is forbidden and $m \geq 2$, every A_i is positive, none peels to D , and $2 \leq m \leq |D| + 1$.*

Proof. This extends the recovery and counting argument of Theorem 2.26; D need not itself be a cube. If a fixed coordinate j takes value one in two outside components, a geodesic between the corresponding vertices stays in the $j = 1$ semicube and avoids D , contradicting their separation. Thus each active fixed coordinate belongs to one component. Every outside vertex has a nonzero fixed coordinate, and all its nonzero fixed coordinates belong to its component. Setting the other blocks to zero consequently recovers $D \cup W_i$, and likewise recovers any selected union of pieces.

Suppose X is positive and restrict an acyclic USO to each of these coordinate faces. If no edge at a vertex of D points into W_i , a topological order can put all of W_i before D , proving relative peeling. Thus each piece failing relative peeling must use a boundary vertex with an outgoing exclusive edge. Two different pieces cannot use the same vertex: the square in their two exclusive directions is absent, contradicting (C1). This gives the bound $|D|$.

If X is forbidden and $m \geq 2$, every piece is a proper restriction and hence positive. A piece peeling to D could be removed first from X , since its outside vertices have no neighbours in the other pieces. The remaining proper restriction is positive, a contradiction. If $m > |D| + 1$, the union of any $|D| + 1$ pieces is a proper positive restriction, contradicting the preceding count. This proves the assertion. $\square$

Proposition 2.28 (Projections of pieces outside a coordinate face). *In Proposition 2.27, assume $X \neq D$, let p project onto the free coordinates E , put $B = p(X)$ and $C_i = p(A_i)$, and identify D with $p(D)$. Then B and the C_i are ample,*

$$B = \bigcup_i C_i, \quad C_i \cap C_j = D \quad (i \neq j),$$

and no edge joins $C_i \setminus D$ to $C_j \setminus D$ for $i \neq j$. Thus each $C_i \setminus D$ is a union of components of $B \setminus D$. If every coordinate in E varies on D , then D is gated in A_i if and only if $C_i = D$. Consequently the number of pieces in which D is not gated is at most the number of components of $B \setminus D$.

Proof. Ampleness follows by coordinate contraction. Suppose distinct pieces have projected vertices $u, v \notin D$ that are equal or adjacent. Choose preimages in their outside components. A geodesic between them in X has E -trace constantly u if $u = v$, and otherwise only u or v : an isometric cube path never changes a coordinate on which its endpoints agree. This geodesic therefore avoids D , contradicting that its endpoints lie in different components of $X \setminus D$. This proves the intersection and edge assertions, and hence the component statement.

Recall that a gate of x in D is a vertex $g \in D$ satisfying $d(x, z) = d(x, g) + d(g, z)$ for every $z \in D$; D is gated when each vertex has a gate. If $p(x) \in D$, the Hamming-distance identity makes $p(x)$ a gate of x . Thus $C_i = D$ implies gatedness. Conversely, let g be a gate of $x \in A_i$. If g and x differed in some coordinate of E , choose $z \in D$ with the bit of x there, using variation on D . The route through g would change this coordinate twice, contradicting the gate equality in the isometric cube metric. Thus $g = p(x)$, proving $p(x) \in D$ and the converse. Each nongated piece uses at least one of the disjoint components already identified, giving the bound. $\square$

For an edge, the bound leaves at most two pieces that can fail relative peeling in a peelable union. Whether those two pieces are compatible depends on the order in which the endpoints are removed.

Definition 2.29 (Acyclic USO boundary conditions at an edge). Let $D = \{u, v\}$ be an edge in direction e of an ample family A . Define five Boolean values, each quantifying over acyclic USOs of A with out-map r :

- $r_D(A) = 1$ if some such map satisfies $r(u), r(v) \subseteq \{e\}$;
- $t_s(A) = 1$, for $s \in D$, if some such map satisfies $r(s) = \emptyset$;
- $\ell_s(A) = 1$, for $s \in D$, if some such map satisfies $r(s) = \{e\}$.

For two such families, write $\chi_D(A, B) = 1$ if some ordering (s, t) of the endpoints satisfies

$$(t_t(A) = 1 \text{ and } \ell_s(B) = 1) \quad \text{or} \quad (t_t(B) = 1 \text{ and } \ell_s(A) = 1). \quad (2.15)$$

Otherwise put $\chi_D(A, B) = 0$.

These retain the original deletion interpretations. By Proposition 2.1, r_D means peeling to D , and t_s means peeling to s . In a topological deletion order the remaining neighbours of a deleted vertex are precisely its outgoing neighbours. Thus ℓ_s means a complete peeling in which s is the first endpoint deleted and is then a leaf adjacent to the other endpoint. Conversely each such peeling induces the required out-map value.

These five values record attainable boundary conditions on acyclic USOs. The following theorem makes their gluing effect exact; intrinsic structural tests for attainability remain open.

Theorem 2.30 (Peeling after gluing along an edge). *Let $A_1, \dots, A_m \in \mathcal{D}_{\text{amp}}$ have the coordinate-block presentation of Theorem 2.26, with common edge D . Put $I = \{i : r_D(A_i) = 0\}$. Then*

$$\bigcup_i A_i \in \mathcal{D}_{\text{amp}} \iff |I| \leq 1 \quad \text{or} \quad (I = \{i, j\} \text{ and } \chi_D(A_i, A_j) = 1). \quad (2.16)$$

Proof. Suppose the union has an acyclic USO, and orient its common edge $s \rightarrow t$. Each piece in I must have outgoing exclusive directions at some endpoint. Proposition 2.20 permits at most one such piece at each endpoint, so $|I| \leq 2$. If $I = \{i, j\}$, the two pieces must use different endpoints. The piece using s has no outgoing edge at t , so realizes $t_t = 1$. The piece using t has just the common outgoing edge at s , so realizes $\ell_s = 1$. This is $\chi_D(A_i, A_j) = 1$.

Conversely, every piece outside I can extend either prescribed orientation of D with inward boundary, by Proposition 2.1. If $|I| \leq 1$, choose an acyclic USO of that possible exceptional piece and extend its edge orientation to the others; if I is empty, choose either edge orientation. The compatibility law gives an acyclic USO of the union. If $t_t(A_i) = \ell_s(A_j) = 1$, the witnessing maps both orient $s \rightarrow t$. Only A_i can have outgoing exclusive directions at s , and only A_j can have them at t . Extend the common edge orientation to all other pieces with inward boundary. Again Proposition 2.20 gives an acyclic USO, hence a peeling of the union. $\square$

Corollary 2.31 (Six contexts determine every edge-gluing type). *Fix an ordered edge $D = \{u, v\}$ and use the context comparison of Theorem 2.24. A piece has one of the following types:*

- **0** if it is not corner-peelable;
- **1** if it peels to D ;
- $[p]$, where $p = (t_u, t_v, \ell_u, \ell_v) \in \{0, 1\}^4$, if it is corner-peelable but does not peel to D .

Two pieces have the same context type if and only if they have the same entry in this list. In particular, there are at most eighteen types; this does not assert that all sixteen vectors p are realized. Among realized vector types,

$$[p] \preceq [q] \iff p \leq q \text{ coordinatewise}, \quad \mathbf{0} \prec [p] \prec \mathbf{1}.$$

Write $Z = [(0, 0, 0, 0)]$. The multiplication law is

$$[p] * [q] = \begin{cases} Z, & p_{t_u} q_{\ell_v} \vee p_{t_v} q_{\ell_u} \vee q_{t_u} p_{\ell_v} \vee q_{t_v} p_{\ell_u} = 1, \\ \mathbf{0}, & \text{otherwise,} \end{cases} \quad (2.17)$$

with identity $\mathbf{1}$ and absorbing element $\mathbf{0}$. Every product of three vector types is $\mathbf{0}$.

Six fixed positive pieces suffice to distinguish all the types and test the comparison: the edge D , a piece of type Z , and four pieces realizing the four unit vectors in $\{0, 1\}^4$. These pieces are supplied by Remark 2.36 and its positive union, together with endpoint reversals.

Proof. The edge-gluing theorem makes a piece retaining D neutral for ordinary positivity, and a nonpeelable piece makes every union nonpeelable by coordinate-face closure. For two positive pieces failing retention, that theorem is exactly the Boolean test in (2.17).

Suppose such a pair has a positive union. Its union cannot retain D , since relative peeling restricts to either piece. We show that all four boundary values of the union vanish. If an acyclic USO made u a sink, its restrictions would make u a sink in both pieces. Each piece must have an outward edge at some boundary vertex, or it would retain D . Both would therefore have outward edges at v , violating the gluing condition. The same argument excludes a sink at v . If a map instead had out-map $\{e\}$ at u , where e is the edge direction, neither piece could have an outward edge at u . Both would again require outward edges at v , a contradiction. Interchanging endpoints excludes the fourth boundary value. Thus the positive product has type Z . The displayed multiplication law follows, and multiplication of Z by any vector type gives $\mathbf{0}$, proving the three-factor assertion.

The two pieces in Remark 2.36 have vector types $(0, 1, 0, 0)$ and $(0, 0, 1, 0)$; reversing endpoints supplies $(1, 0, 0, 0)$ and $(0, 0, 0, 1)$. Their known positive union has type Z by the preceding paragraph. Gluing a vector-type piece to these four unit-vector pieces tests, respectively, its four dual Boolean entries: t_u is tested by the ℓ_v unit, t_v by the ℓ_u unit, and conversely. Hence different vectors are distinguished by realized contexts, and a failed coordinatewise comparison is likewise witnessed by one of these four contexts.

Coordinatewise inclusion is sufficient for comparison against every context, by the displayed Boolean formula and the two exceptional types. The edge distinguishes $\mathbf{0}$ from every positive type. The realized piece Z distinguishes $\mathbf{1}$ from every vector type: it glues positively to the former and negatively to the latter. These observations also establish the strict comparisons and show that the six stated contexts determine both equivalence and comparison for all pieces. $\square$

Thus edge context comparison needs no unknown collection of test pieces. The required sink and leaf values themselves still express attainability of acyclic USO boundary conditions. The corollary neither supplies intrinsic tests for those values nor classifies their minor-minimal realizing pieces. It sharpens the finite gluing algebra at an edge using the existing profile theorem and existing witnesses; it does not enlarge the known forbidden families. The elementary algebra is checked independently in `research/edge_context_algebra.py/json`, allowing all sixteen formal vectors without claiming their realizability.

Theorem 2.32 (Forbidden minors with a separating edge). *Let $m \geq 2$, and let $A_1, \dots, A_m$ be irredundantly embedded ample families whose coordinate sets intersect pairwise in precisely the coordinate of $D = \{u, v\}$, and whose common edge is D . Normalize $u = 0$, and suppose each $A_i \setminus D$ is nonempty and connected. Their union X is an ample forbidden pc-minor if and only if all the following conditions hold.*

1. $m \in \{2, 3\}$, every $A_i \in \mathcal{D}_{\text{amp}}$, and $r_D(A_i) = 0$ for every i .
2. If $m = 2$, then $\chi_D(A_1, A_2) = 0$. If $m = 3$, then $\chi_D(A_i, A_j) = 1$ for every pair of distinct indices.
3. For each exclusive coordinate f of A_i and each $\mu \in \{\rho_f^0, \pi_f\}$, the following holds: when $m = 2$, writing $j \neq i$ for the other index,

$$r_D(\mu A_i) = 1 \quad \text{or} \quad \chi_D(\mu A_i, A_j) = 1; \tag{2.18}$$

when $m = 3$, one has $r_D(\mu A_i) = 1$.

4. For each of the two restrictions and the contraction in the shared coordinate, at most one image μA_i fails to peel to the singleton μD .

Every ample forbidden pc-minor with a specified separating edge arises this way. Its pieces are recovered uniquely as that edge united with each component of its complement, up to permutation. The union is two-connected if and only if all its pieces are two-connected.

Proof. The union is ample by Theorem 2.26. Its components outside D are exactly the stated $A_i \setminus D$.

Suppose X is forbidden. The cube-separator theorem gives item 1. For two pieces, nonpeelability and the exact gluing law give $\chi_D(A_1, A_2) = 0$. For three pieces, each two-piece union is a proper convex restriction and peels; the same law gives all three compatibilities in item 2. An elementary exclusive-coordinate restriction to zero or contraction changes only A_i and retains D . If there are two pieces, the exact gluing law is precisely (2.18). If there are three, the other two pieces still fail relative peeling; thus the changed piece must peel to D . Finally, an elementary operation in the shared coordinate changes the intersection to a singleton, and Proposition 2.18 gives exactly item 4.

Conversely, items 1 and 2 make X nonpeelable by Theorem 2.30. A restriction to one in an exclusive coordinate discards every other piece and is a pc-minor of the peelable A_i . The two other elementary operations in that coordinate give a peelable union by item 3 and the edge-gluing law; in the three-piece case the unchanged pair is compatible by item 2. The shared-coordinate images peel by item 4 and the vertex-gluing law. Every elementary pc-minor therefore peels, so X is forbidden.

For an arbitrary obstruction with a separating edge, the cube-separator theorem recovers the coordinate blocks and their convex pieces. These meet the presentation hypotheses, and the first implication proves their conditions. Their recovery is canonical for the specified edge.

If every piece is two-connected, deleting any vertex leaves connected residual pieces sharing at least one endpoint of D , so the union is two-connected. Conversely, if deleting a vertex disconnects a piece, the surviving endpoint or edge D lies in one component, and any other component has no path to the rest of X . That vertex is then a cut vertex of X . This proves the final equivalence. $\square$

This is an exact decomposition of the separating-edge branch. Its five boundary values are still peeling predicates. The three-piece case is realized by Theorem 2.45 below, so the bound of three pieces is sharp. The three-piece case forces every exclusive elementary image to peel to the edge. In the two-piece case, compatibility with the unchanged piece is essential: Remark 2.48 gives a forbidden two-connected graph three-core for which the changed factor still fails relative peeling while the full proper image peels.

The preceding decomposition assumes that a separating edge has been specified. Shattered pairs detect exactly which coordinate directions can contain such an edge, and coordinate contractions recover the edges themselves. For an irredundant partial cube $H \subseteq Q_F$, write $\text{Cr}(H)$ for its *crossing graph*: its vertices are F , and distinct f, g are adjacent when $\{f, g\} \in \text{Sh}(H)$. This is the usual crossing graph, since shattering a pair means that all four intersections of its coordinate semicubes are nonempty. We use the usual convention that a two-connected graph has at least three vertices.

Proposition 2.33 (Detecting and recovering separating edges). *Let $H \subseteq Q_F$ be an irredundant partial cube with $|F| \geq 3$. Then H has a cut vertex if and only if $\text{Cr}(H)$ is disconnected. If H is two-connected, the following assertions hold for each $e \in F$.*

1. The separating edges of direction e are exactly the fibres $\pi_e^{-1}(z) \cap H$, where z ranges over the cut vertices of $\pi_e H$. Components outside an edge correspond to components outside its image by projection and inverse image.

2. Such an edge exists if and only if $\text{Cr}(H) - e$ is disconnected.
3. If $\text{Cr}(H) - e$ has exactly two components, with vertex sets F_1, F_2 , there is exactly one separating e -edge D . The two pieces obtained by adjoining D to the components of $H \setminus D$ have exclusive coordinate sets F_1, F_2 , and are convex coordinate restrictions.

Consequently an ample forbidden pc-minor has neither a cut vertex nor a separating edge exactly when its crossing graph is two-connected.

Proof. The cut-vertex assertion is [KM02, Theorem 3.4]. The dimension assumption excludes the one-edge degeneracy, both for H and for its one-coordinate contractions. We derive the edge assertions by applying this theorem after contraction and then lifting the resulting cut vertices.

Projection preserves shattering of every remaining coordinate pair, so

$$\text{Cr}(\pi_e H) = \text{Cr}(H) - e.$$

Each fibre of π_e is a singleton or an e -edge, and is therefore connected. Moreover every edge of the projected graph lifts to an edge between its two fibres. Indeed, choose one preimage of each endpoint. They differ either in the projected edge direction alone or in that direction and e . In the latter case a length-two geodesic in H supplies the required edge. The same argument applies after deleting any whole fibre: all edges between surviving fibres still lift. Thus the connected components of $H \setminus \pi_e^{-1}(z)$ correspond bijectively to those of $(\pi_e H) \setminus \{z\}$. A cut vertex z cannot have a singleton fibre, since that singleton would be a cut vertex of H . This proves item 1. The contraction is again an irredundant partial cube, and the cited theorem now proves item 2.

For item 3, recall why the components of the crossing graph of a partial cube are exactly the coordinate sets of its blocks, including bridge blocks. Here a block is a maximal two-connected subgraph or a bridge with its endpoints. Each block is convex: a path leaving a block can return to it only through the same cut vertex. A coordinate cannot occur in two different blocks. Otherwise choose a cut vertex separating the two blocks and, in each of them, an endpoint of an edge of that coordinate on the side opposite the cut vertex. A geodesic between these endpoints would have to change that coordinate twice. Also, two coordinates occurring in different blocks cannot cross: a cut vertex separating the blocks prevents their simultaneously taking values opposite its values. Finally, the crossing graph of a two-connected block is connected by the cited theorem, and that of a bridge block is a singleton. These facts identify the blocks with the crossing-graph components.

It follows that $\pi_e H$ has precisely two blocks, meeting at one cut vertex z . Item 1 gives a unique separating e -edge D , and exactly two components outside it. Normalize $D = \{0, \mathbf{1}_{\{e\}}\}$. A coordinate $f \neq e$ cannot take value one in both components: a geodesic between such vertices would pass through D , where $f = 0$, and repeat direction f . Assign f to its unique component. Adjoining D to a component is therefore exactly the restriction setting all coordinates assigned to the other component to zero. Its exclusive coordinates are those of the corresponding quotient block, namely F_1 or F_2 . This proves item 3.

An ample forbidden pc-minor has VC dimension at least three, since ample classes of VC dimension at most two are corner-peelable [CCMW22, Proposition 4.11]. It therefore satisfies the dimension assumption. The cut-vertex assertion and item 2 say that absence of both kinds of separator is equivalent to connectivity of the crossing graph and absence of a cut vertex in it, as required. $\square$

Corollary 2.34 (Construction and recovery when a shadow splits in two). *Let $\Sigma \subseteq 2^F$ be a simplicial complex containing every singleton, and let its graph of two-element faces be connected.*

Suppose deleting a vertex e from that graph leaves exactly two components, with vertex sets F_1, F_2 . Put

$$\Sigma_i = \Sigma \cap 2^{F_i \cup \{e\}} \quad (i = 1, 2).$$

All ample realizations of Σ are obtained as follows: choose ample realizations $H_i \subseteq Q_{F_i \cup \{e\}}$ of Σ_i , choose an oriented e -edge in each, translate both chosen edges to $D = \{0, \mathbf{1}_{\{e\}}\}$, put zero in the other exclusive coordinates, and take any cube translate of $H_1 \cup H_2$. Every output has a unique separating edge of direction e , and that edge recovers the two convex pieces, up to their order and a simultaneous reversal of the chosen edge orientation.

Within this construction, the forbidden pc-minors are exactly the pairs satisfying Theorem 2.32 with $m = 2$.

Proof. Translation preserves shadows, ampleness, and separators, so it suffices to check the normalized union. No face of Σ meets both F_1 and F_2 , since such a face would contain a pair joining the two components. Hence

$$\Sigma = \Sigma_1 \cup \Sigma_2, \quad \Sigma_1 \cap \Sigma_2 = \{\emptyset, \{e\}\}.$$

The coordinate-block union has shadow Σ by the argument of Theorem 2.26, and

$$|H_1 \cup H_2| = |H_1| + |H_2| - 2 = |\Sigma|.$$

It is therefore ample. Its crossing graph is connected, so it is two-connected. Proposition 2.33 gives uniqueness of its separating e -edge.

We check that the chosen edge is this separating edge. The graph of Σ_i is connected: any path from a vertex of F_i to e stays in F_i until its last step. Thus H_i is two-connected. If $|F_i| \geq 2$, its crossing graph remains connected after deletion of e , so the proposition shows that $H_i \setminus D$ is connected. If $|F_i| = 1$, connectivity of the graph forces Σ_i to be the full two-coordinate simplex, and $H_i = Q_2$; deleting D leaves the opposite edge. In both cases the two nonempty sets $H_i \setminus D$ are exactly the components outside the chosen edge in the union.

Conversely, let H be any ample realization of Σ . The proposition recovers the unique separating e -edge and its two convex pieces with exclusive coordinate sets F_1, F_2 . Normalize this edge to D . Projection onto $F_i \cup \{e\}$ gives precisely the corresponding piece, since the other piece projects into D . Projection preserves shattering on these coordinates, so that piece has shadow Σ_i . Convex coordinate restrictions preserve ampleness. Choosing an endpoint of the recovered edge also recovers the final translation. Thus these are exactly the construction data. The pieces meet all presentation hypotheses of Theorem 2.32, which proves the last assertion. $\square$

Remark 2.35 (What the shadow determines). The shadow determines separating-edge directions, but in general it does not determine how many separating edges occur in one direction. For example, let the first factor have coordinate e , and assign three distinct coordinates to the edges of either four-vertex tree. The ample partial cubes $Q_1 \square P_4$ and $Q_1 \square K_{1,3}$ have the same shadow: besides the empty face and singletons, it contains exactly the three pairs $\{e, f\}$. They have respectively two and one separating e -edges, as deleting such an edge amounts to deleting a vertex of the tree. The contraction, rather than its shadow alone, recovers their positions and multiplicities.

For the 1162-concept example in Remark 2.48, the crossing graph has just one cut vertex, coordinate 10. Deleting it gives the sets $\{0, \dots, 13\} \setminus \{10\}$ and $\{14, \dots, 25\}$. The corresponding pieces have orders 871 and 293. The corollary therefore forces a unique separating edge and these piece orders in every ample realization of that shadow. In the stored realization the edge is $\{0, 1024\}$. By contrast, the fixed 291-concept family G in Remark 2.50 has crossing graph K_{12} , so every ample realization of its shadow has neither a cut vertex nor a separating edge.

These statements identify and recover the separator branch without assuming a marked separator. They do not decide the five peeling properties of its factors, or generate the remaining forbidden cores whose crossing graphs are two-connected. The checker `verify_damp_crossing_separators.py` verifies the fixed examples and all full-dimensional ample families in Q_3, Q_4 , as well as two nonample partial-cube controls; the general conclusions follow from the proofs above.

Remark 2.36 (Two relative failures can have a peelable union). Use the translated 289-concept family C from Remark 2.67, with edge $D = \{u, v\} = \{0, 2\}$, and set

$$A = C, \quad B = \{0, 8192\} \cup \{2 + 2^{12}x : x \in C\}.$$

The old coordinates of the second copy are disjoint from those of A . The four words $0, 2, 8192, 8194$ form a square attached along an edge of this second copy. Both pieces are ample and two-connected. Their five properties, in the orientation (u, v) , are

	r_D	t_u	t_v	ℓ_u	ℓ_v
A	0	0	1	0	0
B	0	0	0	1	0

The first family has only corner u , of degree three, and a stored peeling ending at v . The second has initial corners $0, 8192$. Keeping both endpoints forces deletion of 8192 and then stops. Keeping either endpoint as the final concept also fails; the complete negative state graphs have two and four states. Deleting 8192 , then the leaf u , then peeling the remaining copy of C gives $\ell_u(B) = 1$.

Their 578-concept union peels explicitly: delete 8192 , peel A to v with its last v omitted, and then peel the second copy of C . Thus the vertex-gluing rule “at most one piece fails relative peeling” is false for edges. The endpoint compatibility in (2.15) is necessary.

The checker `verify_damp_edge_separator_profiles.py` validates these profiles and this complete order, a cornerless 576-concept incompatible double, and a nonminimal 865-concept three-piece union. It also recovers the pieces of orders 289, 871 in Remark 6.39. Both recovered pieces have the first displayed profile. All 48 exclusive elementary factor images peel to the shared edge, and one factor peels to its root in each of the three shared-coordinate images. These are the conditions of Theorem 2.32 checked on the recovered pieces. The source endpoint order is recorded in `computations/round807_edge_endpoint_order.json`, and the full report is `damp_edge_separator_profiles_verification.json`.

Proposition 2.37 (Transferring rooted failure to an edge boundary). *Let $0 \in C \subseteq A, B \subseteq Q_F$ be ample, with $A \cap B = C$, $A \cup B$ ample and $\text{Sh}(A) \cap \text{Sh}(B) = \text{Sh}(C)$. Suppose $C \in \mathcal{D}_{\text{amp}}$, both A and B peel to C , and both admit acyclic USOs with sink 0 . For a fresh coordinate e , set*

$$L = (A \times \{0\}) \cup (B \times \{1\}), \quad D = \{u, v\} = \{(0, 0), (0, 1)\}.$$

Then L is ample. If $p = 1$ when C admits an acyclic USO with sink 0 , and $p = 0$ otherwise, its five boundary values are exactly

$$(r_D(L), t_u(L), t_v(L), \ell_u(L), \ell_v(L)) = (p, 1, 1, p, p).$$

In particular, when $p = 0$, either endpoint is an attainable sink but neither leaf exit nor retention of the edge is possible.

Proof. The supports avoiding e are those shattered by $A \cup B$, and the supports containing e are $I \cup \{e\}$ with $I \in \text{Sh}(A) \cap \text{Sh}(B) = \text{Sh}(C)$. Their number is $|A \cup B| + |C| = |L|$, proving ampleness.

To end at u , delete $B \setminus C$ in the upper layer by its relative order, then the whole upper copy of C by an ordinary core order, and finish with an order of A to 0. The first deletions have no vertical mates. During the core deletions the lower A contains every needed mate cube. Thus every deletion is a corner deletion. Interchanging the layers ends at v .

If $p = 1$, clear both wings and peel $C \times \{0, 1\}$ down to D using an order of C ending at 0. Either orientation of the remaining edge gives both leaf exits. Conversely, record a core word whenever its full two-point fibre first breaks during a peeling. While a full fibre exists, every neighbour in the current reduction lifts to a neighbour of the deleted vertex, whose mate is still present. Its corner cube projects to a corner cube in that reduction. These events therefore form a corner deletion sequence starting from C .

If the peeling retains D , this sequence ends at $\{0\}$ without deleting 0. If instead an endpoint s is the first one deleted and is then a leaf, stop immediately before that deletion. Its mate is still present and it has no old-coordinate neighbour. Hence 0 is isolated in the current reduction. That reduction is ample, nonempty and connected, so it is $\{0\}$. Again the recorded events peel C to 0. This proves that each of r_D, ℓ_u, ℓ_v implies $p = 1$ and finishes all cases. $\square$

This reuses the two-layer contraction and endpoint schedules of the paired root construction (Proposition 2.63 and its [relative-wing extension](#)). The additional conclusion is the exact five-value boundary profile. It refutes the implication that two attainable endpoint sinks force edge retention; its obstruction is the prescribed sink of the reduction, not absence of ordinary positive orders on the layers.

Theorem 2.38 (Relative obstructions at cube boundaries of arbitrary dimension). *Take the ample inputs C, A, B and root 0 of Proposition 2.37. Thus C is positive, A and B peel both to C and to 0, and their intersection and shadow intersection are those of C . For $k \geq 1$, let E be a fresh k -coordinate set and put*

$$R_A = Q_E \setminus \{1^E\}, \quad R_B = Q_E \setminus \{0^E\}, \quad D_k = \{0\} \times Q_E,$$

$$Y_k = (C \times Q_E) \cup ((A \setminus C) \times R_A) \cup ((B \setminus C) \times R_B).$$

Then Y_k is ample and positive, and it peels to every nonempty proper face of D_k . It peels to D_k if and only if C peels to 0. If $(C, 0)$ is a rooted factor using every old coordinate, every proper pc-minor of the pair (Y_k, D_k) with nonempty target peels to its target image, although Y_k does not peel to D_k . Here a pc-minor of the pair applies the same coordinate operations to both entries and discards operations whose target image is empty.

The certified inputs $|C| = 289$, $|A| = 291$, $|B| = 290$ give, for every $k \geq 1$, such a relative obstruction with

$$|Y_k| = 292 \cdot 2^k - 3, \quad \text{idim}(Y_k) = 12 + k.$$

Thus even the simultaneous ability to retain every proper face of a cube does not imply the ability to retain that cube.

Proof. Write $U = A \cup B$. Inclusion and the shadow-intersection hypothesis give $\text{Sh}(U) = \text{Sh}(A) \cup \text{Sh}(B)$: the union on the right has cardinality $|A| + |B| - |C| = |U| = |\text{Sh}(U)|$. For an old support I and a new support $J \subsetneq E$, the combined support $I \cup J$ is shattered in Y_k exactly when I is shattered in U . For sufficiency choose the wing whose shadow contains I ; its punctured E -cube contains a J -face. Necessity follows by projection. If $J = E$, the layers 0^E and 1^E force I to be shattered in both A and B , equivalently in C . Conversely $C \times Q_E$ contains all the required cubes. Consequently

$$|\text{Sh}(Y_k)| = (2^k - 1)|U| + |C| = |Y_k|,$$

so Y_k is ample.

We describe the corner orders used below. For a relative order from A to C , delete the copies of each successive old vertex by a corner order of R_A . Do the same for B and R_B . The punctured cubes are positive, for example after a translation they are downward-closed. There are no old-coordinate edges from $A \setminus C$ to $B \setminus C$: such an edge would give a missing-diagonal square in the ample two-layer family of Proposition 2.37. At each deletion, therefore, the old neighbour cube is the one in the relative wing order; the new neighbour cube is the one in its row order. Their product is present, with core neighbours having every new-coordinate mate. This verifies every corner deletion and clears both wings to $C \times Q_E$. That product is positive, proving ordinary positivity of Y_k . If C peels to 0, the same procedure followed by its product relative order peels Y_k to D_k .

Conversely, suppose a peeling retains D_k . Record an old core vertex when its complete E -fibre first loses a vertex. These first-loss events are corner deletions in the current E -reduction: at a full fibre, every remaining reduction neighbour has all its mates, and the corner cube of the deleted vertex projects to the requisite reduction cube. Initially that reduction is C , since neither punctured wing has a full fibre. At the end it is $\{0\}$, whose fibre was never touched. The recorded sequence would peel C to 0. This is the relative form of the reduction-closure argument, proving the equivalence.

Let F be a nonempty proper face of Q_E . It omits at least one of $0^E, 1^E$, so it lies in R_M for one wing $M \in \{A, B\}$. A punctured cube $Q_E \setminus \{v\}$ peels to any face avoiding v : choose a coordinate fixed on the face opposite to v , clear the punctured opposite semicube using its full mates, then peel the remaining full cube to the desired face. A full cube also peels to any of its faces, by successively clearing opposite semicubes.

Clear the other wing completely. For each vertex in a relative order of M to C , delete only its copies in $R_M \setminus F$, using the punctured-cube relative order just described. The same product corner check applies. The remainder is

$$(C \times Q_E) \cup ((M \setminus C) \times F).$$

For each vertex in an ordinary order of C , clear its copies outside F using a full-cube relative order to F . Old neighbours at these copies come only from the surviving core; all needed mates in F remain. The remainder is $M \times F$, which peels to $\{0\} \times F$ using the given root order of M . This proves retention of every proper face of D_k , including all its vertices.

Assume now that $(C, 0)$ is a rooted factor using every old coordinate. An old-coordinate operation with nonempty target is a root-preserving restriction or contraction. The resulting C' peels to 0. The corresponding wings A', B' peel to C' : relative corner orders pass to their coordinate images by omitting repeated families, as in Corollary 2.39. The same corollary's no-diagonal argument gives $A' \cap B' = C'$. The image of Y_k consequently has the same three-part form; clear its wings, then peel $C' \times Q_E$ to the target cube. A restriction in a new coordinate inherits a peeling to its target facet from the proper-face result just proved. Contraction in any new coordinate sends both punctured rows onto the full smaller cube, giving $U \times Q_{E \setminus \{e\}}$. The family U peels to 0: clear $B \setminus C$ first, then use the root order of A . Thus this contraction also peels to its target. These cover every elementary operation with nonempty target; relative minor closure covers subsequent operations. Rooted failure of C excludes retention in the original pair.

The finite inputs are those already used for the 581-vertex edge transfer, with $A \setminus C = \{128, 160\}$ and $B \setminus C = \{290\}$. Substituting $|U| = 292$ and $|C| = 289$ gives the stated sizes. At $k = 2$, `research/cube_boundary_retention.py/json` checks exact shadows, an ordinary corner order and its nonclashing out-map, all four boundary-edge relative orders, and all 30 elementary proper relative-minor orders. Failure at the full square follows from the rooted-core proof, not a negative search. $\square$

For $k = 1$ this is the existing edge transfer. The opposite punctured rows also occur in the ordinary common-seed constructions, for example Proposition 6.36. The new conclusion is their relative-boundary behaviour with a positive rooted-critical core: ordinary positivity and every proper-face retention coexist with a minimal failure to retain the entire cube. This rules out any fixed proper-face dimension cutoff as a universal test for cube retention. It does not generate the rooted input factors or classify all relative obstructions.

Corollary 2.39 (An edge compatibility test and a forbidden double). *Assume $p = 0$ in Proposition 2.37. For a positive ample piece K on coordinates disjoint from F except for the common edge coordinate, $L \cup_D K$ is positive exactly when*

$$r_D(K) = 1 \quad \text{or} \quad \ell_u(K) = 1 \quad \text{or} \quad \ell_v(K) = 1.$$

If moreover $(C, 0)$ is a rooted factor and uses every coordinate of F , then two copies of L , on disjoint old coordinate sets and with D identified, form a forbidden pc-minor of order $2(|A| + |B|) - 2$.

Proof. The compatibility test is Theorem 2.30 with the displayed five values substituted. In particular, two copies of L have a nonpeelable union.

Consider an elementary old-coordinate operation retaining D in one copy. It replaces C by a proper root-preserving minor C' that peels to 0. Images of relative deletion chains, with repetitions omitted, are relative deletion chains: every intermediate image is ample and each step loses at most one word outside the retained image. Thus the new sections A', B' peel to C' . Their intersection is C' . For face intersections this is immediate; under contraction a new common word outside C' would come from opposite vertices of an old-coordinate and e square with both intermediate vertices absent, contradicting isometry. Consequently the changed copy peels to D , by clearing its wings and then its core product. Its union with the unchanged positive copy peels by the edge-gluing theorem. An old-coordinate face intersection not containing D discards the other copy and leaves a positive minor of L .

For either face intersection in e , or contraction of e , both copies have positive images with the common root as an attainable sink: take the root-preserving image of a peeling of L ending at the corresponding endpoint. The vertex-gluing theorem makes their union positive. These are all elementary pc-minors, so the double is forbidden. $\square$

The reduction of the forbidden double in e is $C \vee C$, already a forbidden class by the rooted-factor theorem. Thus this double belongs to the established inverse-construction branch above negative reductions; it does not generate a previously uncovered reduction-terminal seed.

The corollary is an application of the existing separator classification, not intrinsic generation of the rooted input C . For a stored example, use the first control in `relative_wing_square.json`: $|C| = 289$, $A = C \cup \{128, 160\}$ and $B = C \cup \{290\}$. The root 0 is the only corner of C , so $p = 0$. The resulting L has 581 words and profile $(0, 1, 1, 0, 0)$; its forbidden double has 1160 words. The input rooted-factor certificates are the previously established ones. The checker `research/verify_edge_boundary_transfer.py` replays both endpoint orders and the relative wing orders and verifies the unique-corner certificate of the core. The profile and double's minimality follow from the uniform proofs above. This does not settle existence of forbidden three-piece edge unions: it shows why attainable sinks alone cannot exclude them, and why the leaf conditions in their classification must be retained.

Proposition 2.40 (Rooted factors inside one edge context type). *Let $C \subseteq Q_F$ be ample and corner-peelable, with marked vertex a , and use a fresh coordinate e to form*

$$P = C \times Q_{\{e\}}, \quad D = \{a\} \times Q_{\{e\}}.$$

If $h = 1$ when C peels to a and $h = 0$ otherwise, then

$$(r_D(P), t_{(a,0)}(P), t_{(a,1)}(P), \ell_{(a,0)}(P), \ell_{(a,1)}(P)) = (h, h, h, h, h).$$

Consequently (C, a) is a rooted factor if and only if P has the positive zero-vector context type Z and every proper elementary private-coordinate minor retaining D peels to D . Here C is embedded on its active coordinates, and a private restriction retaining D fixes its coordinate to the bit of a .

Moreover, every rooted factor occurs in this way as a piece of an ample forbidden pc-minor with a separating edge. One may glue P to the fixed 581-vertex piece L of Corollary 2.39, identifying its marked edge with D and renaming all private coordinates. The resulting forbidden family has $2|C| + 579$ vertices.

Proof. Products of ample corner-peelable families are ample and corner-peelable. If C peels to a , deleting its other vertex fibres in that order peels P to D . Indeed, at each fibre deletion the old corner cube is multiplied by the remaining part of its edge fibre. Relative peeling to D implies all four sink and leaf values, by finishing with the appropriate order on the edge.

Conversely, a relative peeling to D restricts to a peeling of either e -section ending at a . A sink at (a, i) restricts to a sink at a in that section. Likewise an out-map equal to $\{e\}$ at (a, i) has empty out-map after restriction to its e -section, so gives an acyclic USO of C with sink a . Thus each of the five values implies that C peels to a , proving the displayed profile.

A private root-containing restriction or contraction replaces the prism by $(\mu C) \times Q_{\{e\}}$ and carries D to the edge over μa . The profile identity says that the image retains that edge exactly when μC peels to μa . These are precisely the elementary rooted minimality conditions, and ordinary positivity of C is assumed. This proves the equivalence and also recovers C and its root from either section of the marked prism.

For the last assertion translate a to zero. The fixed L is positive, fails retention of its edge, attains both endpoints as sinks, and every proper elementary private minor retaining its edge retains that edge, by Corollary 2.39 and its proof. Its two shared-coordinate restrictions and its shared-coordinate contraction peel to the image of an endpoint, by rooted minor closure. The union in the statement is exactly the product attachment $P_C(L, D)$ of Theorem 2.16. All old-coordinate images of (L, D) with nonempty image of D therefore peel relatively to that image, whereas L itself fails retention. That theorem proves forbiddenness. The attachment formula gives $|L| + (|C| - 1)|D| = 581 + 2(|C| - 1)$. Both pieces have vertices outside D and no edges between their private parts, so the common edge is a separator. $\square$

Thus classifying minor-minimal realizers even of the single edge type Z contains the entire rooted-factor input problem, already within marked prisms that actually occur in forbidden assemblies. The proposition is an application of the earlier product-attachment and edge-transfer results, not a new mechanism for generating rooted factors. In particular, settling which Boolean types are realizable would not by itself settle their support generation.

Lemma 2.41 (Transport of edge boundary conditions). *Let $D = \{u, v\}$ be an edge of an ample class K . A pc-minor operation retains D if it is a coordinate-face intersection containing both endpoints or a contraction in a coordinate other than the direction of D . Each of $r_D, t_u, t_v, \ell_u, \ell_v$, when true, remains true after such an operation, with the endpoints replaced by their images.*

Proof. Apply the operation to the ample residual families of a witnessing peeling. A step loses at most one image word; omit repetitions. This gives a peeling of the image. For a face intersection the resulting order is the filtered order. For a contraction it records a word when its last lift is deleted. Retained endpoints persist, proving transport of r_D and the two sink conditions.

For ℓ_s , consider the moment when s is deleted as a leaf adjacent to the other endpoint t . A face intersection containing D preserves this leaf and its neighbour. Suppose instead coordinate f is contracted, where f differs from the edge direction. The f -mate of s is absent, so this deletion removes the last lift of its image. No new neighbour of that image can appear outside the direction of D : if a quotient neighbour had only a lift in the opposite f -layer, that lift would have distance two from s with both intermediate vertices absent (its matching-layer lift and the f -mate of s). This contradicts isometry of the current ample family. Thus the image of s is still a leaf. The image of t remains, so it is still the first endpoint deleted. The image order witnesses ℓ_s . Iterating proves the claim for sequences of these operations. $\square$

Theorem 2.42 (An exact existence criterion for the three-piece branch). *The following statements are equivalent.*

1. *There exists an ample forbidden pc-minor with an edge whose deletion leaves exactly three connected components.*
2. *There exists a positive ample class K with an edge $D = \{u, v\}$ such that*

$$r_D(K) = 0, \quad (t_u(K) \vee t_v(K)) = (\ell_u(K) \vee \ell_v(K)) = 1.$$

In item 2 the sink and leaf exit may be witnessed by different acyclic USOs, and their endpoints may agree or differ. No minor-minimality hypothesis or connectivity assumption on $K \setminus D$ is needed.

Proof. For necessity, the separator classification gives three positive pieces, all failing edge retention and pairwise compatible. Each compatible pair has a witness consisting of a sink value in one piece and a leaf-exit value in the other. If no piece possessed both kinds of value, pieces with a sink value and pieces with a leaf-exit value would form disjoint groups, and compatibility could occur only across the groups. Three pieces cannot all be pairwise compatible in this way. Thus one piece satisfies item 2.

For sufficiency, pass to a minor K_0 retaining D that still fails to peel to D and is minimal among such minors. All the positive boundary values in item 2 survive by Lemma 2.41. Every proper elementary operation in an exclusive coordinate retaining D now makes K_0 peel to D . Moreover $K_0 \setminus D$ is connected. Otherwise the canonical pieces at D are proper convex minors. They all peel to D by minimality, and their outside deletion orders concatenate to peel K_0 to D , a contradiction.

First suppose $t_u(K_0) = \ell_v(K_0) = 1$, interchanging endpoint names if necessary. Glue three copies of K_0 along D on disjoint exclusive coordinate sets. Each pair is compatible, while all three fail retention; thus their union is nonpeelable. An exclusive elementary operation retaining D changes one piece to a relative-positive piece and leaves a compatible pair. An exclusive face intersection not containing D leaves a positive minor of a single piece. For the face image through u , all three pieces peel to the common root, by restricting the sink- u orders. The same is true for the face through v : immediately before the leaf deletion of v in a witnessing order, its own coordinate shore is a singleton. Indeed isometry forces every other residual vertex into the shore of its only neighbour u . Filtering the preceding order to the v -shore peels it to v . Finally, contraction of the shared direction sends the sink- u orders to orders ending at the common root. The vertex-gluing law therefore makes all three shared-coordinate images positive. This proves forbiddenness.

Otherwise item 2 supplies a same-endpoint pair, say $t_u(K_0) = \ell_u(K_0) = 1$. Use a fixed relative-critical piece L with profile $(0, 1, 1, 0, 0)$ from Corollary 2.39, chosen with $L \setminus D$ connected. Such a piece exists by the certified 581-word instance above. Glue L , K_0 , and an endpoint-reversed copy of K_0 . Every pair is compatible: L supplies either sink and the other pieces supply their

respective leaf exits; the two copies of K_0 supply a sink and the opposite leaf exit to each other. All three fail retention, so the union is negative. Each exclusive elementary image retaining D peels to D and leaves a compatible pair; an image discarding D lies in a single positive piece. In either shared-coordinate face image at most one piece can fail at the common root: L attains both endpoints, and one of the two copies attains the chosen endpoint. In the contraction all three images attain the common root. The vertex-gluing law proves positivity of these images. Hence this union is forbidden as well. In both constructions the three complements of D are connected, occupy disjoint exclusive blocks, and have no cross edges, so deleting D leaves exactly three components. $\square$

The equivalence reduces existence to joint attainability on one positive piece. The preceding $(0, 1, 1, 0, 0)$ construction realizes only the sink conditions, and the older $(0, 0, 0, 1, 0)$ collar realizes only a leaf exit. The next construction realizes both on the same piece and therefore settles existence of the three-piece branch.

Proposition 2.43 (Rooted factors from a repair in a proper face). *Let $(C, 0)$ be a rooted factor on its active coordinate set F , and suppose that 0 is its only corner. Let $c \notin C$ be an ample one-concept addition with acquired support P , such that $C \cup \{c\}$ peels to 0 . Choose a proper coordinate face $Q_J(0)$ containing c and its entire acquired cube. Put*

$$B = C \cap Q_J(0), \quad T = B \cup \{c\}, \quad K = (T \times \{0\}) \cup (C \times \{1\}), \quad D = \{u, v\} = \{(0, 0), (0, 1)\}.$$

Then K is ample and ordinarily peelable, with

$$r_D(K) = t_v(K) = \ell_v(K) = 0, \quad \ell_u(K) = 1.$$

Every proper root-preserving pc-minor of (K, u) peels to its root. Every proper old-coordinate elementary image retaining D peels to D . Consequently either (K, u) is a rooted factor, or K has the mixed profile $(0, 1, 0, 1, 0)$.

If $Q_J(0)$ omits a neighbour of 0 in C , then (K, u) is a rooted factor. This last conclusion requires no search for a peeling of K .

Proof. The punctured acquired cube lies in B , so T is ample and $\text{Sh}(T) = \text{Sh}(B) \dot{\cup} \{P\}$. Since $P \notin \text{Sh}(C)$, the section shadows intersect exactly in $\text{Sh}(B)$. The section union is $C \cup \{c\}$; shadow counting proves ampleness of K .

As a proper root-containing face of C , the class B peels to 0 . Delete $(c, 0)$ and then the lower copy of $B \setminus \{0\}$ using such an order. The untouched upper C supplies every vertical mate cube. Now u is a leaf at v ; delete it and peel C ordinarily. This proves ordinary positivity and $\ell_u = 1$. The upper shore cannot peel to its root, so neither edge retention nor a sink at v is possible. A first-endpoint leaf exit at v would also peel its own shore to v : just before that exit its shore is a singleton by isometry, and the preceding order restricts to a peeling of that shore. Thus $\ell_v = 0$.

Let μ be an old-coordinate operation retaining D . Both μC and μB peel to their roots, by rooted minimality and rooted minor closure respectively. The lower difference $\mu T \setminus \mu B$ has at most one word. If it survives, it has no upper mate: a contraction identifying c with an old word would identify it with a neighbour in its acquired cube, which already belongs to B . Deleting this lower word leaves the ample nested expansion with sections $\mu B \subseteq \mu C$, so the word is a corner. Peel the lower μB to its root and then the upper μC to its root, retaining D . This proves all the stated old-coordinate tests. The root-containing face in the new coordinate is T , which peels to 0 by deleting c and then peeling B . The contraction is $C \cup \{c\}$, which peels to 0 by hypothesis. These are every elementary root-preserving operation. Closure under further such operations proves the assertion about all proper rooted minors and hence the dichotomy.

Suppose finally that a root neighbour h of C is omitted by the chosen face. In a hypothetical peeling ending at u , consider its first deletion from the upper C . All upper words are still present, so the old directions of that corner would make its projection a corner of C . Its projection must therefore be 0. But v still has neighbours u and $(h, 1)$, whereas $(h, 0)$ is absent. Its incident cube is incomplete. This excludes the first upper deletion and proves that u is unattainable as a sink. $\square$

The first-deletion argument reuses the missing-root-neighbour mechanism of Theorem 2.101. The additional conclusion is a one-coordinate construction that supplies every rooted minor test; the earlier square construction used two new coordinates and three repairs. For the 289-concept core and repair $c = 290$, choose $J = \{1, 5, 6, 7, 8\}$. Then $|B| = 18$, $|T| = 19$, and $h = 4$ is omitted. The output is a 308-concept rooted factor of dimension 13. The checker `research/repair_face_rooted_factor.py` verifies its shadow, ordinary order, and all 26 proper rooted elementary orders; its negative certificate is the uniform argument above. Adding coordinate 2 to J gives the face used in the next theorem. Containing every root neighbour removes this particular obstruction; it is not asserted to be sufficient for a sink in general.

Remark 2.44 (Recovering a repaired-face presentation and its limits). The construction has a necessary condition visible without its marked new coordinate. Normalize the root of a candidate H to zero. For each edge at the root, with direction e , project its root shore and opposite shore to T_e and C_e . A repaired-face presentation requires

$$|T_e \setminus C_e| = 1, \quad T_e \cap C_e = C_e \cap Q_J(0),$$

for a proper face containing T_e . The singleton is the recovered repair; C_e is the recovered input. The least possible J is the union of the supports of the words of T_e . Any larger face realizing the same data has the same intersection with C_e . The acquired-cube and rooted-input hypotheses of Proposition 2.43 must still be checked; the singleton test alone is not sufficient.

On the 308- and 330-concept examples this recovers exactly one coordinate, namely the added coordinate, and recovers $c = 290$ and the stated input C . For the established 292-concept noncorner-root factor, the root-edge directions are 1, 2, 5, 6, 8. The respective root-shore exclusive counts are 75, 35, 59, 97, 90. Thus it has no repaired-face presentation, even before any peeling condition is tested. This does not make it a new irreducible input: the established corner deletions 32, 96, 64 recover the same 289-concept factor, so the earlier ample-addition construction already covers it. These recovery checks are recorded in `research/repair_face_recovery.json`.

Finally every output of the repaired-face operation has the nonroot corner $(c, 0)$: it has no vertical mate and its old neighbours span its acquired cube. Therefore its output cannot directly be reused as the unique-corner input required by this operation. These are limits of this particular construction, not of a combination with ample additions or other proved growth operations. No exhaustive recovery assertion follows from it.

Theorem 2.45 (A forbidden three-piece edge union, with finite certificates). *Let $C \subseteq Q_{\{0, \dots, 11\}}$ be the translated 289-concept rooted factor used in Proposition 2.37. Use integer words with coordinate i represented by bit 2^i , and put*

$$c = 290, \quad J = \{1, 2, 5, 6, 7, 8\}, \quad T = (C \cup \{c\}) \cap Q_J(0), \quad B = C \cap Q_J(0).$$

For a fresh coordinate e , define

$$K = (T \times \{0\}) \cup (C \times \{1\}), \quad D = \{u, v\} = \{(0, 0), (0, 1)\}.$$

Then $|T| = 41$, $|K| = 330$, and K is ample with exact profile

$$(r_D, t_u, t_v, \ell_u, \ell_v) = (0, 1, 0, 1, 0).$$

Every proper elementary pc-minor operation in an old coordinate retaining D makes K peel to D , and $K \setminus D$ is connected.

Glue K , an endpoint-reversed copy of K , and the 581-concept piece L of Corollary 2.39 along D , using disjoint old coordinate sets. The result is an ample forbidden pc-minor with 1237 concepts and isometric dimension 37. Deleting D leaves exactly three components. All its nonempty-coordinate reductions are corner-peelable.

Proof. The input has acquired support $P = \{5, 6, 7, 8\}$, and the chosen proper face contains its entire acquired cube. Proposition 2.43 therefore proves ampleness, ordinary positivity, $r_D = t_v = \ell_v = 0$, and $\ell_u = 1$, as well as every exclusive relative-minor condition. It remains to certify the sink and apply the separator construction.

The remaining assertion $t_u = 1$ is certified by the explicit 330-word order `SINK_ORDER` in `research/three_piece_edge_witness.py`. The checker reconstructs K from the stored input, checks that this list contains every word exactly once and ends at u , and verifies every corner deletion. Independently, it forms the outgoing coordinate sets of the resulting acyclic orientation, checks that they biject onto $\text{Sh}(K)$, and checks nonclashing for every pair of distinct words. CCMW's representation-map characterization therefore also certifies the acyclic USO with sink u . The source words, both positive orders, and the computed checks are recorded in `research/three_piece_edge_witness.json`. This finite replay, rather than a general assertion about lifting endpoint repairs, supplies $t_u = 1$.

The checker also replays all 24 relative minor orders furnished by the preceding proposition. If $K \setminus D$ were disconnected, its proper convex pieces at D would all peel to D by this minimality, so their orders would concatenate to peel K to D . Thus it is connected; this is also checked directly.

The three profiles in the proposed union are $(0, 1, 0, 1, 0)$, $(0, 0, 1, 0, 1)$, and $(0, 1, 1, 0, 0)$. The same-endpoint construction in the proof of Theorem 2.42 applies without further minimalization: the preceding relative-minor checks supply its hypotheses for K , and Corollary 2.39 supplies them for L . It proves nonpeelability and positivity of every proper pc-minor. The three connected complements of D are its three components. Counting the common edge gives $330 + 330 + 581 - 4 = 1237$ concepts and $12 + 12 + 12 + 1 = 37$ coordinates.

Finally, an exclusive-coordinate reduction equals the reduction of the corresponding positive piece and is positive by reduction closure. Reduction in e is $B \vee B \vee C$ at the common root. Both copies of B peel to their roots, and C is ordinarily peelable, so the vertex-gluing theorem makes this reduction positive. Further reductions remain positive. This also agrees with the previously proved separator descent theorem (Theorem 5.34). $\square$

Corollary 2.46 (Trace partitions and minimality in the three-piece example). *Let X be the 1237-concept forbidden class in Theorem 2.45. Put its shared edge in coordinate 0, and the three private coordinate blocks in positions $1, \dots, 12$, $13, \dots, 24$, and $25, \dots, 36$, respectively. Let $A = \pi_0 X$ and $C = X^0$, retaining a zero in the erased bit position. Then $|A| = 870$, $|C| = 367$, and both are corner-peelable. Order the components of $A \setminus C$ by their least integer word; their sizes are $249, 1, 249, 1, 2, 1$. For $t \in \{0, 1\}^6$, assign component i to layer t_i , retaining both layers over C , and denote the resulting ample lift by X_t . Then*

$$X_{001100} \in \mathcal{D}_{\text{amp}}, \quad X_{100000}, X_{001000} \notin \mathcal{D}_{\text{amp}}.$$

Consequently the positive label set cannot be a union of equality spaces of partitions having at most one nonsingleton block, even though (A, C) is recovered from an ample forbidden class. Nevertheless, forbiddenness within this fixed pair has the exact selector

$$X_t \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff t_0 \neq t_1, \quad t_2 \neq t_3, \quad t_4 \neq t_5, \quad t_1 \neq t_3. \quad (2.19)$$

Its four outputs are all isomorphic to X .

Proof. Positivity of A follows from minor closure; positivity of C is part of Theorem 2.45. The stated component counts follow directly from its three pieces and are also checked by `research/obstruction_trace_partition_counterexample.py`.

For 001100, each of the first two pieces has constant component labels. It is a peripheral expansion of the 290-concept repaired class along its 40-concept face. Both classes peel to their common root, so first peeling the smaller layer to that root and then the larger layer to the root retains the shared edge. Remove these two pieces outside the edge. The remaining piece has constant labels on its two components, hence is a peripheral expansion of its positive 292-concept contraction along the positive 289-concept reduction. It is positive. The cited replay checks the resulting complete order on all 1237 vertices.

For each of 100000 and 001000, the accompanying JSON record supplies a set S of 155 vertices and, for each $x \in S$, two distinct directions i, j such that

$$x^i, x^j \in S, \quad x^{\{i,j\}} \notin X_t.$$

The replay checks these inclusions and exclusions against the reconstructed family. In any putative peeling, the first deleted vertex of S would retain both neighbours but lack the required square vertex. It could not be a corner. This proves both negative assertions without assumptions about which peeling choices are made.

We prove the selector without classifying all positive labelings. If $t_0 = t_1$ or $t_2 = t_3$, the corresponding 330-concept piece peels to the common edge by the same argument as above. The full union is positive if and only if the union of the other two pieces is positive: one direction is restriction closure, and the other prefixes the relative peeling. That smaller union is a proper coordinate face. Thus the full union is either positive or has a proper negative minor, and cannot be forbidden.

Suppose now both pairs are mixed. The shore at the singleton-component label t_1 of the first piece is its repaired face T ; the other shore is the 289-concept class with unique corner zero. The analogous labels for the second piece are t_3 and $1 - t_3$. If $t_1 = t_3$, restriction to the other shore therefore contains two copies of the 289-concept class joined at zero. Neither copy peels to zero, since zero is its unique initial corner. Proposition 2.18 makes this proper section negative. Consequently $t_1 \neq t_3$ is necessary.

If in addition $t_4 = t_5$, the third piece is a peripheral expansion whose thin shore, at $1 - t_4$, is also the 289-concept class. Since $t_1 \neq t_3$, exactly one of the first two pieces has that same rooted-failing shore. This proper section again has two rooted failures at its common vertex and is negative. Thus $t_4 \neq t_5$ is necessary.

Conversely, the four labelings satisfying all four inequalities are obtained from the original labeling 100101 by complementing coordinate zero and exchanging the first two private coordinate blocks. These are cube isometries of the recovered pair, so all four outputs are forbidden by Theorem 2.45. The replay `research/six_component_forbidden_selector.py` verifies the four isometries, the relative orders, and the rooted-failing sections. It classifies 48 labels by a relative-positive piece, 12 by a negative proper section, and the remaining four by these isometries. This proves (2.19).

An equality space with one nonsingleton block B accepts 001100 only if B lies in $\{0, 1, 4, 5\}$ or in $\{2, 3\}$. In the first case it also accepts the negative label 001000; in the second it also accepts 100000. A discrete partition accepts every label. Therefore no such space can both contain 001100 and be contained in the positive label set, which proves the claimed failure of a union representation. $\square$

The coordinate pair is inherited from a forbidden class, but its other labelings need not be forbidden. Thus minimality of one labeling does not justify simplifying the full positive-label set of the uncoloured pair. The counterexample concerns a proposed restriction on trace partitions, not the completeness of the partition-certificate theorem or the validity of the existing three-piece construction.

The construction shows that the three-piece alternative in the separator classification is necessary, including among forbidden classes with all proper reductions positive. It is a new realization of that alternative, using the earlier rooted factor and repair as inputs; it does not supply an intrinsic generator of all rooted factors or all reduced forbidden bases.

Theorem 2.47 (Down-set completions of two rooted blocks). *Let F, G be disjoint finite coordinate sets, and let $Y \subseteq Q_F$ and $C \subseteq Q_G$ be irredundantly embedded ample families containing 0, with $|C| > 1$. Suppose $e \in F$ and $v = \mathbf{1}_{\{e\}} \in Y$. Let $K \subseteq C$ be a nonempty down-set: whenever $x \in K$, every word whose support is contained in the support of x also belongs to K . Embed the families by setting all other coordinates to zero, and put*

$$B = C \cup (v + K), \quad X = Y \cup C \cup (v + K), \quad D = \{0, v\}. \quad (2.20)$$

Here $v + x$ sets coordinate e to one and retains the G -coordinates of x . Then X is ample and

$$X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff (Y, 0), (C, 0) \text{ are rooted factors}. \quad (2.21)$$

Here the rooted-factor conditions are those of Theorem 2.19. For fixed F, G, e , the parameters are recovered uniquely: $Y = X \cap Q_F$, $C = X \cap Q_G$, and $K = \{x \in Q_G : v + x \in X\}$.

When these equivalent obstruction conditions hold, X is two-connected exactly when $|K| > 1$. If Y, C have minimum degree at least three and K is a cube of dimension at least two containing 0, then X also has minimum degree at least three and is its own graph three-core. Its order is $|Y| + |C| + |K| - 2$.

Proof. The down-set K is ample and peels to 0: delete its nonzero words in decreasing order of support size. At each deletion the remaining neighbors of the deleted word are exactly its downward neighbors and span its full lower cube. The family B is the peripheral expansion of C with site K , so it is ample by Proposition 2.6. The edge-amalgam shadow count in Theorem 2.26 makes $X = Y \cup B$ ample as well. The recovery formulas and the order formula follow from $Y \cap B = D$ and $Y \cap C = \{0\}$.

We first prove the ordinary peeling identity

$$X \in \mathcal{D}_{\text{amp}} \iff Y, C \in \mathcal{D}_{\text{amp}} \text{ and at least one of } Y, C \text{ peels to } 0. \quad (2.22)$$

The families Y, C are convex restrictions of X , so their ordinary peelability is necessary. Delete the words $v + x$, $x \in K \setminus \{0\}$, in decreasing support size. Each is a corner: its downward cube in the upper layer and the matching lower cube are present. These deletions leave $Y \cup C$. Thus if one factor peels to 0, the vertex-gluing law gives a complete peeling of X .

To prove the remaining implication, suppose both factors peel ordinarily but neither peels to 0. In the notation of Definition 2.29, the ordered-edge profile of B is exactly

$$(r_D(B), t_0(B), t_v(B), \ell_0(B), \ell_v(B)) = (0, 0, 0, 0, 1). \quad (2.23)$$

Indeed, an order ending at either endpoint would contract in coordinate e to an order of C ending at 0. Peeling to D would also give such an endpoint order. The upper-layer deletion order just given leaves v a leaf; delete it and then peel C , proving $\ell_v(B) = 1$. If 0 could be the first deleted endpoint and then a leaf, its only neighbor would be the retained v . Isometry forces every other residual word onto the $e = 1$ side: a geodesic from 0 must start with its sole incident edge. Restricting the preceding deletion sequence to $e = 0$ would therefore peel C to 0, a contradiction. This proves $\ell_0(B) = 0$. Since $t_0(Y) = 0$, also $r_D(Y) = 0$, and (2.23) gives $\chi_D(Y, B) = t_0(Y) = 0$. The exact edge-gluing law proves that X cannot peel. This establishes (2.22).

Suppose now that X is forbidden. The restrictions Y, C are proper, and the identity just proved shows that both are ordinarily peelable and fail to peel to 0. For a coordinate $f \in F \setminus \{e\}$, a root-preserving restriction or contraction replaces Y by its image and leaves C, K unchanged. That proper image of X peels, so (2.22) forces the image of Y to peel to 0. For $f = e$, either root-preserving operation yields the vertex union of its image of Y with C ; the vertex-gluing law forces the same conclusion. For a coordinate of G , either root-preserving operation replaces C, K by their images. The image of K is again a nonempty down-set, so the peeling identity forces the image of C to peel to 0. If that image is a singleton the conclusion holds directly. Root-preserving minor closure now gives all three rooted-factor conditions for both inputs.

Conversely, assume those rooted-factor conditions. The peeling identity makes X nonpeelable. A root-preserving elementary operation in $F \setminus \{e\}$ makes the image of Y peel to 0, so the same identity makes the full image peel. A root-preserving operation in G makes the image of C peel to 0, with the same consequence. The root restriction and contraction in e give vertex unions of C with root-positive images of Y , and hence peel. A restriction opposite the root in $F \setminus \{e\}$ discards B and leaves a minor of Y . One in G discards Y and leaves a peripheral expansion of a minor of C along a down-set, or just its nonempty base if the site is empty. Both peel by Proposition 2.6. Finally, the opposite restriction in e gives the vertex union of Y_e^1 and K . The first family peels ordinarily and the second peels to their common 0, so their union peels. Thus every proper elementary minor of X peels.

Both rooted factors have no cut vertex by Theorem 2.19. If $K = \{0\}$, then $X = Y \cup C$ has the cut vertex 0. If $|K| > 1$, the upper copy of K is connected and has at least two matching edges to the two-connected C . Deleting a lower vertex leaves that upper copy attached through another matching edge. Deleting an upper vertex leaves every surviving upper vertex attached to C . Hence B is two-connected, and gluing Y and B along D preserves two-connectivity. Old vertices lose no neighbors. If K is an r -cube, every new vertex $v + x$, $x \neq 0$, has its r upper cube neighbors and its matching lower neighbor. For $r \geq 2$ this proves the asserted degree bound. $\square$

Remark 2.48 (Factor-minor compatibility is essential in a three-core). Let $C \subseteq Q_{12}$ and $Y_0 \subseteq Q_{14}$ be, respectively, the 289-concept core and the 870-concept rooted factor in Remark 2.67. Use binary integer notation with coordinates numbered from zero, and define

$$v = 1024, \quad Y = Y_0 \cup \{v\}, \quad C' = 2^{14}C, \quad K' = 2^{14}\{0, 2, 4, 6\}, \quad B = C' \cup (v + K').$$

The added word v has acquired support 1027 in Y_0 . The resulting $(Y, 0)$ is a rooted factor: delete v and ordinarily peel Y_0 ; its complete root-retaining deletion graph has eight states and never reaches $\{0\}$. Its 28 proper elementary root-preserving images peel to 0. Their orders are obtained from the stored orders for Y_0 by first deleting the image of v when that image is new. All these orders are checked directly.

Theorem 2.47 now gives an ample forbidden pc-minor

$$X = Y \cup B, \quad |X| = 1162, \quad \dim X = 26, \quad \text{VC}(X) = 4.$$

It is two-connected, has minimum degree three, and is its own graph three-core. Deleting the edge endpoints $D = \{0, 1024\}$ leaves two components, whose canonical pieces are precisely Y and B , of orders 871 and 293.

Consider the restriction $\mu = \rho_0^0$, exclusive to Y . Its factor image is

$$H = \mu Y = \{0, 1024\} \cup \{4x + 2 : x \in C \cup \{128\}\}, \quad |H| = 292.$$

The complete corner-deletion graph retaining D has five states and never reaches D , so $r_D(H) = 0$. Nevertheless H peels to 0: delete 1024, follow a stored root-ending order of $C \cup \{128\}$ in the upper layer, and finish at 0. On the unchanged side, (2.23) gives $r_D(B) = 0$ and $\ell_v(B) = 1$. Thus

$$r_D(H) = r_D(B) = 0, \quad \chi_D(H, B) = 1, \quad \mu X = H \cup B \in \mathcal{D}_{\text{amp}}.$$

In particular, the compatibility alternative in (2.18) cannot be replaced by relative positivity of the changed factor, even for two-connected forbidden graph three-cores.

The checker `verify_damp_coupled_edge_minors.py` replays all 52 rooted input minor orders and all 78 output elementary-minor orders. It checks ampleness through exact shadows, the canonical pieces, the graph degrees and connectivity, and the complete 32-state negative deletion graph of X . The explicit positive order of $H \cup B$ has 583 concepts. This example is not terminal: deleting the corner $99328 = v + 2^{14} \cdot 6$ replaces the square site by the down-set $\{0, 2, 4\}$, so the theorem still gives forbiddenness. The down-set parameter is intrinsic, but the two rooted-factor inputs remain unclassified; this theorem does not supply a global generative classification.

Remark 2.49 (A forbidden pc-minor with two blocks). Let P be the 294-concept peelable family of Remark 6.47, translated by its only corner 1894 so that this corner becomes 0. The fixed checker `verify_damp_cut_vertex_obstruction.py` verifies peelings ending at 0 for all 24 elementary root-preserving pc-minors of P . The pair $(P, 0)$ therefore satisfies the theorem: it peels ordinarily, while its unique corner cannot be left last. Consequently

$$X = (P \times \{0\}^{12}) \cup (\{0\}^{12} \times P) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}).$$

It has 587 concepts, isometric dimension 24, VC dimension three, no corners, and the unique cut vertex 0. Deleting that vertex leaves two components of order 293. The checker independently constructs and validates complete peelings of all 72 elementary pc-minors of X ; the report `damp_cut_vertex_587_verification.json` records every order. Both copies of P are gated in X , with gate 0 for vertices in the other copy: for each such vertex x and every $y \in P$, the equality $d(x, y) = d(x, 0) + d(0, y)$ holds. Thus $\mathcal{D}_{\text{amp}}$ is not closed under vertex gluing, even when both pieces are gated and peelable.

Remark 2.50 (A minimal rooted failure whose root is not a corner). The root of a factor in Theorem 2.19 need not be a corner. Let G be the fixed cornerless 291-concept family recorded as `base` in `damp_fixed_291_repair_creation.json`, and put

$$H = G \cup \{91\}, \quad a = 29.$$

Here integers denote their twelve-bit cube words in the stored embedding. The family H is ample and peelable, with VC dimension four and initial corners 61, 91, 93, 125; in particular, a is not a corner. Deleting 91 first leaves the cornerless G . The complete corner calculation with 29 retained has eight states. It shows that every complete peeling begins with one of

$$(61, 125, 93), \quad (93, 125, 61), \quad (125, 61, 93), \quad (125, 93, 61),$$

and then deletes 29 fourth. Thus H cannot peel to 29, although all 24 elementary root-preserving pc-minors do peel to the image of 29.

The same rooted minimality holds after deleting any one of 61, 93, 125, after deleting 61, 125, and after deleting all three. Each remaining family also has a verified ordinary peeling. In particular,

$$B = H \setminus \{61, 125, 93\}$$

is a 289-concept ample peelable family of VC dimension three, with 29 as its only corner, and $(B, 29)$ satisfies the three conditions of Theorem 2.19. After translating 29 to 0, the vertex doubles $H \vee H$ and $B \vee B$ are therefore ample forbidden pc-minors of orders 583 and 577, respectively, both in dimension 24. The first has VC dimension four and eight corners; the second has VC dimension three and no corners. Both have one cut vertex. Deleting 61, 125, 93 in the first block, then in the second block, gives six obstruction-preserving corner deletions from the first double to the second. These labels refer to the original words within the corresponding block before translation.

The independent checker `verify_damp_noncorner_rooted_obstruction.py` validates all 144 rooted elementary peelings of the six displayed factors. For each endpoint double it also validates all 72 elementary pc-minor peelings and a complete negative deletion certificate, with 64 states for the 583-concept family and one state for the cornerless 577-concept family. The families, orders, and corner-descent chain are recorded in `damp_noncorner_rooted_obstruction_verification.json`.

Corollary 2.51 (Corner descent in the cut-vertex branch). *Let $X = A_1 \vee A_2$ satisfy Theorem 2.19, with common root a . For $x \in \text{Cor}(A_i) \setminus \{a\}$, the family $X \setminus \{x\}$ is a forbidden pc-minor if and only if $(A_i \setminus \{x\}, a)$ still satisfies the three rooted factor conditions in that theorem.*

Call an ample forbidden pc-minor corner-descent terminal if none of its corner deletions is another forbidden pc-minor. A cornered terminal ample forbidden pc-minor with a cut vertex exists if and only if there is a rooted factor satisfying those three conditions that has a nonroot corner, but no nonroot corner deletion preserving the conditions.

Proof. The corners of X are exactly the nonroot corners of its two factors: the root has edges in both coordinate blocks and no mixed square. Deleting a nonroot corner leaves an ample family, still with the root as a cut vertex. If the result is forbidden, the cut-vertex theorem shows that the changed branch is its other rooted block and satisfies the three conditions. Conversely, if those conditions hold, the theorem applies directly to the two remaining factors. Notice that failure to peel to the root is automatic in the changed factor: otherwise prefixing its rooted peeling by x would peel the original factor to a .

If a cornered terminal X exists, at least one of its rooted factors has a nonroot corner, and the first assertion excludes every condition-preserving deletion in that factor. In the other direction, glue a factor with the stated properties to $(B, 29)$ from Remark 2.50. This second factor's only corner is its root. All corners of the union therefore come from the first factor, and the first assertion makes every one of their deletions fail forbidden-minor status. The union is the required cornered terminal obstruction. $\square$

2.7 Canonical three-cores and square attachments

The cut-vertex decomposition (Theorem 2.19) rules out degree-two cut vertices in an ample forbidden minor. Together with periphery-freeness and the small-corner descent proved in Section 2.4, this yields the unique graph three-core. We then prove the inverse construction: all square attachments are specified by coordinate pairs and targets in an acyclic dependency digraph. This entire construction and recovery argument uses the elementary results already established above; the unresolved input is the choice of the forbidden core.

Proposition 2.52 (Every dismantlable-ample obstruction is periphery-free). *If $O \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$, then O has no peripheral coordinate. Consequently, every member of*

$$\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \setminus \{Q_n^{--} : n \geq 3\}$$

is ample, periphery-free, and Cartesian-prime.

Proof. Suppose that e is peripheral. After orienting e , write

$$O = (B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B.$$

The contraction base $B = O/e$ and duplicated site S are proper pc-minors of O , so both belong to $\mathcal{D}_{\text{amp}}$. The exact peripheral-expansion theorem for $\mathcal{D}_{\text{amp}}$ says that the displayed expansion belongs to $\mathcal{D}_{\text{amp}}$ if and only if both B and S do. This contradicts $O \notin \mathcal{D}_{\text{amp}}$. The remaining claims follow from Proposition 2.8 and Lemma 2.7. $\square$

Theorem 2.53 (The graph three-core of an ample forbidden minor). *Let O be an ample forbidden pc-minor for $\mathcal{D}_{\text{amp}}$. Every degree-two vertex of its graph is a corner. Repeatedly delete vertices of degree less than three, in any order. Every deletion is a degree-two corner deletion, and the final family K is a nonempty ample forbidden pc-minor. It is the graph's three-core: the unique largest induced subgraph of minimum degree at least three. In particular, K is independent of every deletion choice.*

The families K and O have the same active coordinates, the same shattered supports of size at least three, and the same VC dimension. Conversely, every ample forbidden minor is obtained from its three-core by a sequence of ample one-concept square completions on those coordinates. Starting from any ample forbidden minor of minimum degree at least three, every such sequence produces an ample forbidden minor with that same three-core.

Proof. We first record a local fact for any ample family A . A vertex of degree two is either a corner or a cut vertex. Translate it to 0, and let its neighbors be $0^{\{p\}}, 0^{\{q\}}$. Every geodesic from 0 starts in one of these directions and never changes that coordinate again. Thus 0 is the only concept with trace 00 on $\{p, q\}$. If any concept realizes 11, this pair is shattered. Ampleness makes it strongly shattered, so a full $\{p, q\}$ -square exists. Its 00-vertex must be 0, making 0 a corner. If no concept realizes 11, the remaining concepts split into the nonempty 10- and 01-sections. Each section is connected by isometry, and no edge joins the sections. Hence 0 is a cut vertex.

In O , a cut vertex has degree at least four: by Theorem 2.19, it belongs to two two-connected rooted factors, each contributing at least two edges. Consequently every degree-two vertex of O is a corner. There is no vertex of degree zero, since O is connected and is not a singleton. There is no leaf either: isometry would make its edge coordinate peripheral, contrary to Proposition 2.52.

Corollary 2.14 preserves ample forbidden-minor status after every degree-two deletion. The preceding paragraph therefore applies at each step. The process cannot reach the empty family, since every intermediate family is forbidden. It terminates at a nonempty K of minimum degree at least three. If L is any induced subgraph of O of minimum degree at least three, none of its vertices can be deleted: as long as L remains, each has at least three surviving neighbors. Thus L is contained in every possible terminal K . This proves the largest-subgraph property and uniqueness.

Deleting a degree-two corner removes precisely its two-coordinate support from the shadow. It removes no singleton support and no support of size at least three. Active coordinates are therefore unchanged. An ample forbidden family has VC dimension at least three, since every ample family of VC dimension at most two is corner-peelable [CCMW22, Proposition 4.11]. Thus VC dimension is unchanged as well. Reversing the deletions gives the asserted square completions.

For the last assertion, Proposition 2.13 preserves nonpeelability at each square addition, and Proposition 2.12 preserves forbidden-minor status. In reverse addition order, each new vertex has degree two when it is removed. The original family of minimum degree at least three remains. The uniqueness just proved identifies it as the three-core of the final family. $\square$

This is a canonical reduction of the full classification to ample forbidden families of minimum degree at least three. The inverse steps have intrinsic tests: choose a missing concept and two coordinates whose other three square vertices are present, and require that their support is not shattered. The one-concept extension criterion in Lemma 2.10 then gives ampleness. No peeling decision is needed for these steps. Recognizing the possible three-cores intrinsically remains open; a three-core need not be cornerless.

Lemma 2.54 (A support pattern forcing corners). *Let $H \subseteq Q_F$ be ample with every coordinate active, and let Γ be the graph on F in which f, g are adjacent exactly when $\{f, g\} \in \text{Sh}(H)$. Suppose that*

$$F = \{e\} \dot{\cup} I \dot{\cup} A \dot{\cup} B, \quad I, A, B \neq \emptyset, \quad P = \{e\} \cup I \in \text{Sh}(H),$$

that the induced graphs $\Gamma[A]$ and $\Gamma[B]$ are connected, and that

$$\{a, i\} \notin \text{Sh}(H) \quad (a \in A, i \in I), \quad \{b, e\} \notin \text{Sh}(H) \quad (b \in B). \quad (2.24)$$

Then the unique cube of H with support P contains at least $2^{|I|} - 1$ corners of H . All hypotheses depend only on the shattered supports, so the same conclusion holds for every ample family with the same shadow.

Proof. The missing pairs prevent any support $P \cup \{f\}$, $f \notin P$, from being shattered. Ampleness supplies a cube C of support P . It is unique: two such cubes would differ on some coordinate $f \notin P$, and their projections would shatter $P \cup \{f\}$.

We first locate every edge whose direction is outside P . If $\{f, g\}$ is not shattered, all edges of direction f have the same value of coordinate g . Otherwise, one such edge at $g = 0$ and one at $g = 1$ would realize all four traces on $\{f, g\}$. Every active direction has an edge, so these fixed values are defined.

It follows that all edges of each direction $a \in A$ have a fixed trace $\alpha_a \in Q_I$. If a, a' are adjacent in $\Gamma[A]$, ampleness supplies an $\{a, a'\}$ -square. Its two edge directions have the same trace on I , so $\alpha_a = \alpha_{a'}$. Connectivity gives one common $\alpha \in Q_I$ for all directions in A . For each $b \in B$, all edges of direction b have a fixed value β_b of coordinate e . An $\{b, b'\}$ -square gives $\beta_b = \beta_{b'}$ whenever b, b' are adjacent in $\Gamma[B]$. Connectivity therefore gives one common value $\beta \in \{0, 1\}$.

Choose any vertex $x \in C$ such that

$$x_e \neq \beta, \quad x|_I \neq \alpha.$$

It has no neighbor in a direction of A , because such an edge would have trace α on I . It has no neighbor in a direction of B , because such an edge would have value β on e . Its incident directions are thus exactly P , and all their combinations occur in C . Every cube through x is contained in C , making x a corner. The cube C contains exactly $2^{|I|} - 1$ vertices satisfying the two displayed conditions, which proves the bound. $\square$

Remark 2.55 (Sharpness of the support bound). The bound is attained on the distinguished cube for every $k = |I| \geq 1$. Take $A = \{a\}$, $B = \{b\}$, and let H be the union of the coordinate cubes with supports $\{e\} \cup I$, $\{e, a\}$, and $I \cup \{b\}$, all based at the zero word. This is a down-set and hence ample. On the first cube, an a -edge occurs exactly when the I -trace is zero, and a b -edge occurs

exactly when $x_e = 0$. Such vertices are not corners: their neighbor in the outside direction and a neighbor in a direction of the first cube lack the fourth vertex of their square. All other vertices of the first cube are corners, so their number is exactly $2^k - 1$.

Corollary 2.56 (A forbidden three-core whose shadow forces corners). *There is a two-connected ample forbidden pc-minor X of minimum degree three such that every ample $H \subseteq Q_{26}$ with $\text{Sh}(H) = \text{Sh}(X)$ has at least three corners. In particular, preserving the full shadow cannot transform every ample forbidden three-core into a cornerless ample family. This remains impossible even if intermediate families need not be forbidden pc-minors.*

Proof. Take the 1162-concept three-core of Remark 2.48, with its stored coordinate numbering, and put

$$e = 10, \quad I = \{15, 16\}, \quad A = \{0, \dots, 13\} \setminus \{10\}, \quad B = \{14, \dots, 25\} \setminus \{15, 16\}.$$

The duplicated square supplies the $P = \{10, 15, 16\}$ -cube. The shadow of the down-set completion has no pairs between A and I , and no pairs between B and $\{10\}$. The induced graph on A contains the star with center 0, and the one on B contains the star with center 14. These support assertions are checked from the complete shadow by `verify_damp_shadow_forced_corners.py`. They depend on the shadow alone, so Lemma 2.54 applies to every ample realization and gives $2^2 - 1 = 3$ corners.

The verifier independently checks the shadow, both spanning stars, and all missing pairs. On the original realization, the three forced corners are 33792, 66560, 99328; their unique cube has support P . It also replays all 78 proper elementary-minor peelings and the complete 32-state nonpeelability certificate, and checks six translated sharpness examples with $|I| = 1, 2, 3$. The record is `damp_shadow_forced_corners_verification.json`. The universal same-shadow conclusion is the symbolic lemma, not an enumeration of the possible signs. $\square$

Theorem 2.57 (Relative peeling when only small supports are added). *Let $\emptyset \neq B \subseteq A \subseteq Q_F$ be ample, and assume*

$$|I| \leq 2 \quad \text{for every } I \in \text{Sh}(A) \setminus \text{Sh}(B). \quad (2.25)$$

Then A can be reduced to B by deleting corners of degree at most two, without deleting any concept of B . More precisely, starting from B , repeatedly adding a concept of A with the largest number of neighbors in the current family always gives an ample family and eventually reaches A . Every addition acquires a support of size at most two.

For every $T \subseteq B$, including the empty target, A can be reduced to T by corner deletions if and only if B can be so reduced. In particular, $A \in \mathcal{D}_{\text{amp}}$ if and only if $B \in \mathcal{D}_{\text{amp}}$, and for every $b \in B$, peeling A to b is possible if and only if peeling B to b is possible. If the active coordinates of A and B agree, every displayed deletion has degree exactly two.

Proof. We adapt the maximum-neighbor construction in [CCMW22, Proposition 4.11]. Its dimension bound is needed only for cubes not already contained in the starting family.

First, every cube of A of dimension at least three lies in B . To see this, extend it to a maximal cube Q of A , with support I . The reduction A^I has one vertex for each full I -cube and is connected, by [CCMW22, Theorem 3.2]. The vertex corresponding to Q is isolated: an incident edge would enlarge Q by one coordinate. Hence Q is the only I -cube of A . Condition (2.25) puts I in $\text{Sh}(B)$. Ampleness supplies an I -cube in B , which must therefore be Q . The same argument shows that any maximal square of A is the only square of its support in A .

Suppose that $B \subseteq S \subsetneq A$ is ample, and choose $x \in A \setminus S$ maximizing its number of neighbors in S . Since A is connected and S is nonempty, this maximum is positive. Put

$$P = \{e \in F : x^{\{e\}} \in S\}.$$

By [CCMW22, Lemma 4.1(i)], the punctured P -cube at x is contained in S . Thus $|P| \leq 2$: otherwise its completion in A would be a cube of dimension at least three containing $x \notin B$.

If $|P| = 2$, the completed square is maximal in A , since no larger cube can contain x . It is the unique square of that support. Hence $P \notin \text{Sh}(S)$: strong shattering in the ample S would give a second such square. Lemma 2.10 now makes $S \cup \{x\}$ ample, with acquired support P .

If $|P| = 1$, every concept of $A \setminus S$ has at most one neighbor in S . Therefore S is locally convex in A : every concept of A at distance one from two concepts of S belongs to S . Connected local convexity implies convexity in an ample family by [CCMW22, Lemma 4.10]. Write $P = \{p\}$ and $u = x^{\{p\}} \in S$. If some $z \in S$ had $z_p = x_p$, then x would lie on a shortest u, z -path in A , contrary to convexity. Thus p is constant on S , so P is not shattered there. The one-concept extension test again gives ampleness.

These arguments apply after each addition. They produce the claimed growth order, whose reverse is the required corner-deletion prefix. For a fixed $T \subseteq B$, Proposition 2.13 preserves the existence of a reduction to T at every addition. Iteration proves the asserted equivalence for every retained target; the empty target and singleton targets give the stated special cases. If the active coordinate sets agree, every intermediate S already realizes both values of each active coordinate. The case $|P| = 1$ is then impossible, proving the last assertion. $\square$

Theorem 2.58 (A canonical partial order for square attachments). *Under the hypotheses of Theorem 2.57, suppose also that A and B have the same active coordinates, and put $D = A \setminus B$. There is a unique orientation of the edges with at least one endpoint in D such that*

1. *every edge between D and B points toward B ;*
2. *every vertex of D has exactly two outgoing edges;*
3. *the orientation induced on D is acyclic.*

Let $P_{A,B}$ be the partial order on D in which $x < y$ means that a directed path runs from x to y . The relative corner-deletion orders from A to B are exactly its linear extensions. Moreover,

$$C \text{ is ample} \iff A \setminus C \text{ is an order ideal of } P_{A,B} \quad (B \subseteq C \subseteq A). \quad (2.26)$$

Thus the ample intermediate families form a distributive lattice under union and intersection.

If I_x is the pair of coordinate directions of the two outgoing edges at x , then

$$x \mapsto I_x \text{ is a bijection } D \longrightarrow \text{Sh}(A) \setminus \text{Sh}(B).$$

In particular, the support removed with each concept is independent of the deletion order.

Proof. Take a relative deletion order supplied by the preceding theorem. Orient each edge toward its later endpoint, regarding all of B as retained after D . Each deleted vertex has degree two, so this gives all three properties.

Suppose two orientations had these properties. Their boundary edges agree. At each vertex of D , the number of internal outgoing edges is fixed: it is two minus the number of its neighbors in B . On the edges where the orientations differ, orient according to the first orientation. Each vertex then has equal indegree and outdegree in this difference graph. If it were nonempty, following outgoing edges would give a directed cycle, contradicting acyclicity. Hence the orientations agree.

Any relative peeling loses only supports in the shadow difference. All singleton supports remain in B , so every deletion loses a pair and has degree two. Every relative deletion order therefore induces this orientation and is a linear extension of $P_{A,B}$. Conversely, any two linear extensions of a

finite partial order are connected by swaps of consecutive incomparable elements: move the first element of the desired order to the front and continue inductively. Here incomparable elements are nonadjacent in the graph. Consecutive nonadjacent corner deletions commute. Indeed, deleting the first vertex leaves the second's neighbor set unchanged, so the second was already a corner. Its unique maximal cube avoids the first vertex. The first vertex's unique maximal cube must likewise avoid the second, since otherwise the two cubes would coincide. Either deletion therefore preserves the other's corner cube. Starting with the known relative peeling, the swaps realize every linear extension.

If C is ample with $B \subseteq C \subseteq A$, both nested pairs $C \subseteq A$ and $B \subseteq C$ satisfy the preceding theorem and have the same active coordinates. Concatenate the resulting relative peelings. The removed set $A \setminus C$ is an initial segment of a linear extension, and hence an order ideal. Conversely, every order ideal is the initial segment of a linear extension; its corner deletions leave an ample intermediate family. This proves (2.26). Order ideals are closed under union and intersection, so the complementary intermediate families form the asserted distributive lattice.

Finally, each corner deletion removes exactly the support of its two outgoing directions from the shadow. These supports are distinct and exhaust the shadow difference. The orientation is unique, proving the claimed bijection and order independence. $\square$

Apply these results with $B = K$, an ample forbidden minor of minimum degree at least three. Any ample $A \supseteq K$ on the same active coordinates, with no new shattered support of size at least three, is obtained by square additions and is therefore forbidden. Its graph three-core is exactly K . This gives an admissibility test for the extension without supplying an addition order, and a canonical description of every ample intermediate family. Intrinsic admissibility of the possible cores themselves remains open.

For the 579-concept example over its 577-concept core, the partial order is the chain $8198 < 8194$, giving exactly three ample intermediate families. For the nine-square completion, the partial order is a nine-element antichain, giving 512 ample intermediate families. All these intermediate families are forbidden.

The checker `verify_damp_small_support_intervals.py` verifies the greedy construction and the canonical partial orders on the 579- and 586-concept examples over their 577-concept core, and on a small example of VC dimension three. It checks every intermediate subset in those three intervals against (2.26). A further example with a new coordinate checks the degree-one step. Its report is `damp_small_support_intervals_verification.json`.

The preceding criterion recognizes an extension when its upper family is already given. We next specify finite data that construct the new concepts from the lower family. The data record which three existing concepts complete each square; the coordinates of the added concepts are derived from these incidences.

Definition 2.59 (Square attachment data). Fix a nonempty ample family $K \subseteq Q_F$, and fix an order on F . Take a finite set D of abstract symbols disjoint from K . For each $x \in D$, specify two coordinates $p_x < q_x$ and three targets

$$u_x, v_x, w_x \in K \dot{\cup} D.$$

Write $I_x = \{p_x, q_x\}$. The data are *admissible* when the following conditions hold.

1. The digraph on D with arrows from x to each of its targets lying in D is acyclic.
2. The pairs I_x are distinct, and none belongs to $\text{Sh}(K)$.

3. Set $\phi(k) = k$ for $k \in K$ and define $\phi(x) = \phi(u_x)^{\{p_x\}}$. This recursion terminates by item 1. The resulting map on $K \dot{\cup} D$ is injective and satisfies

$$\phi(v_x) = \phi(x)^{\{q_x\}}, \quad \phi(w_x) = \phi(x)^{\{p_x, q_x\}} \quad (x \in D). \quad (2.27)$$

Their output is $A = K \cup \phi(D)$.

All checks in this definition concern a finite labelled digraph, coordinate flips, equality of derived words, and the shattered pairs of the given base. Neither words for the new concepts nor a total order are supplied; there is no ampleness or peeling test on the output. The value of $\phi(x)$ can be computed simply by following successive u -targets to a base concept and applying the recorded coordinate flips.

Theorem 2.60 (Construction and recovery from square attachment data). *Admissible square attachment data over K produce an ample family with the same active coordinates as K , and*

$$\text{Sh}(A) = \text{Sh}(K) \dot{\cup} \{I_x : x \in D\}. \quad (2.28)$$

Every order of the symbols in which all targets precede their source constructs A by ample square additions. Conversely, every ample $A \supseteq K$ with the same active coordinates and with no new shattered support of size at least three arises this way. Its data are unique up to renaming the symbols of D .

The reachability order of the dependency digraph is exactly the partial order $P_{A,K}$ of Theorem 2.58, under the identification by ϕ . In particular,

$$|D| \leq \binom{|F|}{2} - |\{I \in \text{Sh}(K) : |I| = 2\}|.$$

If K is an ample forbidden pc-minor of minimum graph degree at least three, all outputs are ample forbidden pc-minors with graph three-core K . Conversely, every ample forbidden pc-minor is obtained from its canonical graph three-core by its uniquely recovered attachment data.

Proof. Choose an order with every target before its source, which exists by acyclicity. We prove the shadow formula throughout the construction. Suppose the concepts preceding x have already been added and form an ample family S . Injectivity makes $\phi(x)$ a new concept, and the three targets are precisely the other vertices of its I_x -square. By induction and the distinctness condition, $I_x \notin \text{Sh}(S)$.

There cannot be an additional neighbor of $\phi(x)$ in an outside direction $e \notin I_x$. Indeed, the punctured-cube assertion of [CCMW22, Lemma 4.1(i)] would put the full I_x -square based at $\phi(x)^{\{e\}}$ inside S . That would shatter I_x , a contradiction. Thus the incident support of the new concept is exactly I_x . The one-concept criterion, Lemma 2.10, proves that the addition is ample and acquires precisely I_x . Its two directions were already active in the punctured square. All other coordinate values already occurred at its targets. Induction proves the first assertions and (2.28) for any allowed order.

For recovery, take an upper family satisfying the stated hypotheses. Theorem 2.58 gives its unique acyclic orientation toward K , with two outgoing edges at each concept of $A \setminus K$. Use those concepts as temporary symbols. Let $p_x < q_x$ be their two outgoing directions and let u_x, v_x be the corresponding targets. Set $w_x = x^{\{p_x, q_x\}}$. In any relative peeling this is the opposite vertex of the square deleted with x , so it is present and is deleted later, or lies in K . All three dependency arrows therefore point forward in that peeling. The pairs are distinct and absent from $\text{Sh}(K)$, by the support–concept bijection of the same theorem. Finally, $\phi(x) = x$ satisfies the recursion and (2.27). The recovered data are admissible and reconstruct A .

Conversely, reversing any construction order from admissible data gives a relative peeling. At the deletion of x , its two neighbors are exactly its prescribed targets u_x, v_x . Hence the data induce the unique orientation just used for recovery. This determines their coordinate pair and both edge targets; the opposite vertex then determines w_x . Injectivity identifies its symbol uniquely. This proves uniqueness up to renaming.

For the assertion about partial orders, the two edge-target arrows are precisely the oriented graph edges incident to the new concepts. Every relative peeling also retains w_x when it deletes x . Thus, when $w_x \in D$, x precedes w_x in every linear extension of $P_{A,K}$. In a finite partial order this means $x < w_x$: otherwise a linear extension with w_x before x exists. Adding the diagonal-target arrows therefore does not change reachability. The cardinality bound follows from the distinct pairs. Finally, Theorem 2.53 supplies both forbidden soundness for square additions over a forbidden core and the canonical core reduction of every ample forbidden pc-minor. The recovery just proved applies to that reduction. $\square$

This is an intrinsic construction of all the square attachments over a given core. The unresolved part of the full generative classification is still intrinsic admissibility of the cores themselves, which may have corners.

2.8 Rooted constructions and endpoint dependence

The boundary datum in this section is a prescribed sink. By Proposition 2.1, every assertion that a family peels to a means that it admits an acyclic USO with sink a . The constructions below transport this constraint, which ordinary positivity does not guarantee. We retain deletion orders where they give explicit certificates and preserve the stronger relative statements.

Proposition 2.61 (Operations with a prescribed endpoint). *The following statements hold for nonempty ample families.*

1. Let $S \subseteq B \subseteq Q_F$ and

$$Y = (B \times \{0\}) \cup (S \times \{1\}).$$

For $b \in B$, the family Y peels to $(b, 0)$ if and only if B peels to b and $S \in \mathcal{D}_{\text{amp}}$. For $s \in S$, it peels to $(s, 1)$ if and only if both B and S peel to s .

2. The product $A \square B$ peels to (a, b) if and only if A peels to a and B peels to b .
3. Suppose $H = S \cup \{c\}$ is an ample one-concept extension with acquired support of size at most two. For every $r \in S$, the family H peels to r if and only if S peels to r .

In particular, an ample addition that makes peeling to an old concept possible must acquire a support of size at least three.

Proof. For the necessity in item 1, contract the new coordinate and restrict to the upper layer. Contraction preserves the root and gives the assertion for B . An upper root is also preserved by the upper restriction; for a lower root, ordinary pc-minor closure gives only the required ordinary peeling of S .

For a lower endpoint, first delete the entire upper layer following an ordinary peeling of S , and then peel B to b . Each upper deletion is a corner deletion: its incident support consists of its current directions in S and the new coordinate, and the whole corresponding lower cube is present in B . For an upper endpoint $(s, 1)$, first peel the upper layer to that vertex, retaining it. Then follow a peeling of B to s , retaining $(s, 0)$. The single remaining upper vertex affects only the lower vertex

$(s, 0)$, so all other lower deletions remain corners. Finish by deleting $(s, 0)$ and then $(s, 1)$. This proves sufficiency in both cases.

In item 2, root-preserving face restrictions prove necessity. For sufficiency, use the product construction of Proposition 2.6 with ample prefix orders starting at a and b . Its first product concept is (a, b) ; reversing the order gives the required endpoint.

For item 3, apply Proposition 2.13 with retained target $\{r\}$. Reducing to this singleton and then deleting it is exactly a peeling ending at r . $\square$

Proposition 2.62 (Unrestricted sections at a degree-two corner). *Let $A, D, C \subseteq Q_F$ be ample families containing zero, and put*

$$U = A \cup D \cup C, \quad Y = \{(0, 00)\} \cup (A \times \{01\}) \cup (D \times \{10\}) \cup (C \times \{11\}).$$

No containment or relative-peeling assumptions are imposed. Then Y is ample if and only if all four conditions hold:

$$\begin{aligned} \text{Sh}(U) &= \text{Sh}(A) \cup \text{Sh}(D) \cup \text{Sh}(C), \\ \text{Sh}(A \cup C) &= \text{Sh}(A) \cup \text{Sh}(C), \\ \text{Sh}(D \cup C) &= \text{Sh}(D) \cup \text{Sh}(C), \\ \text{Sh}(A) \cap \text{Sh}(D) &\subseteq \text{Sh}(C). \end{aligned} \tag{2.29}$$

Every ample family with a marked degree-two corner has this presentation, uniquely up to exchanging the corner's two directions and the corresponding side sections, after translating the corner to zero. In particular, the criterion includes every rooted factor whose distinguished root is such a corner. It does not impose rooted failure or rooted minimality.

Proof. Write $\mathcal{A} = \text{Sh}(A)$, $\mathcal{D} = \text{Sh}(D)$, and $\mathcal{C} = \text{Sh}(C)$. Sorting shattered supports according to their two new coordinates gives the exact identity

$$|\text{Sh}(Y)| = |\text{Sh}(U)| + |\mathcal{A} \cap \text{Sh}(D \cup C)| + |\mathcal{D} \cap \text{Sh}(A \cup C)| + 1. \tag{2.30}$$

For example, a support using the first new coordinate and not the second has its old part shattered by both projected shores A and $D \cup C$. A support using both new coordinates cannot contain an old coordinate, because the 00 section is a singleton. The support consisting of the two new coordinates is shattered, since all sections contain zero.

Suppose first that Y is ample. Its two coordinate faces with sections (A, C) and (D, C) are ample. The first has shadow size $|\text{Sh}(A \cup C)| + |\mathcal{A} \cap \mathcal{C}|$ and cardinality $|A| + |C| = |\mathcal{A}| + |\mathcal{C}|$. Equality forces $\text{Sh}(A \cup C) = \mathcal{A} \cup \mathcal{C}$, since shadows always contain the union of the individual shadows. The other face gives the analogous equality for $D \cup C$. With these two identities, subtracting $|Y| = 1 + |\mathcal{A}| + |\mathcal{D}| + |\mathcal{C}|$ from (2.30) yields

$$|\text{Sh}(Y)| - |Y| = |\text{Sh}(U)| - |\mathcal{A} \cup \mathcal{D} \cup \mathcal{C}| + |(\mathcal{A} \cap \mathcal{D}) \setminus \mathcal{C}|. \tag{2.31}$$

Both terms on the right are nonnegative. Their sum is zero, so both vanish, proving the remaining two conditions.

Conversely, the middle two conditions give (2.31), and the first and last make its right side zero. Equality in the Sandwich bound therefore makes Y ample. Its marked vertex has exactly its two new neighbours, and their full square is present, so it is a degree-two corner.

For recovery, translate a marked degree-two corner to zero and call its incident directions the two new coordinates. In the coordinate face fixing both to zero, the marked vertex is isolated, since

it has no remaining incident direction. This face is ample and hence connected, so it consists of that vertex alone. The corner square supplies zero in each of the other three sections. Those sections are ample by face closure and are uniquely determined by the marked vertex and its two directions. The criterion already proved applies to them. Exchanging the two directions only exchanges A and D . $\square$

This gives support admissibility for the full marked degree-two-corner class, including non-contained opposite sections. In the paired special case below, the containment and shadow hypotheses imply these four conditions. The unrestricted criterion does not make the three sections relatively peelable to a common core or decide peeling to the marked root. The verification `research/degree_two_section_shadows.py/json` checks all triples of ample two-coordinate sections containing zero and the archived 870, 872, and 879 presentations. These are checks of the shadow criterion, not a rooted-factor census.

Proposition 2.63 (Exact endpoint transfer through a square). *Let $C \subseteq A, D \subseteq Q_F$ be ample families containing 0, with $|A \setminus C|, |D \setminus C| \leq 1$ and $A \cap D = C$. Suppose that $A \cup D$ is ample and $\text{Sh}(A) \cap \text{Sh}(D) = \text{Sh}(C)$. For two fresh coordinates q, r , put*

$$Y = \{(0, 00)\} \cup (A \times \{01\}) \cup (D \times \{10\}) \cup (C \times \{11\}), \quad a = (0, 00). \quad (2.32)$$

Then Y is ample, a is a degree-two corner, and

$$Y \in \mathcal{D}_{\text{amp}} \iff C \in \mathcal{D}_{\text{amp}}, \quad (2.33)$$

$$Y \text{ peels to } a \iff C \text{ peels to } 0. \quad (2.34)$$

If C uses every coordinate of F , then (Y, a) is a rooted factor of Theorem 2.19 if and only if $(C, 0)$ is a rooted factor and both A, D peel to 0. No corner condition on the root of C is required.

If $C \in \mathcal{D}_{\text{amp}}$, the endpoints in all three other sections satisfy

$$\begin{aligned} Y \text{ peels to } (x, 01) &\iff A \text{ peels to } x && (x \in A), \\ Y \text{ peels to } (x, 10) &\iff D \text{ peels to } x && (x \in D), \\ Y \text{ peels to } (x, 11) &\iff C \text{ peels to } x && (x \in C). \end{aligned} \quad (2.35)$$

Consequently, whenever (Y, a) is a rooted factor, a is its only vertex z for which (Y, z) is a rooted factor. Moreover, its attainable sink set is not isometric: it contains $(0, 01), (0, 10)$ but neither of their common cube neighbours $(0, 00), (0, 11)$.

For $|C| > 1$, the corners of Y are a , the at most two concepts outside the three copies of C , and the members of the following two sets:

$$\begin{aligned} \{(x, 01) : x \in \text{Cor}(C) \setminus \{0\}, d_H(x, z) \neq 1 \text{ for all } z \in A \setminus C\}, \\ \{(x, 10) : x \in \text{Cor}(C) \setminus \{0\}, d_H(x, z) \neq 1 \text{ for all } z \in D \setminus C\}. \end{aligned} \quad (2.36)$$

Proof. We first compute the shadow and give the forward peeling constructions. The converse for the prescribed endpoint uses the first deletion among the three nonroot copies of old 0. At that moment one of two overlaps has already been peeled to 0.

A support using neither new coordinate is shattered exactly when it belongs to $\text{Sh}(A \cup D)$. The old-coordinate projections of the two q -sections are A, D ; those of the two r -sections are D, A . The 00-fibre is a singleton. Consequently

$$\text{Sh}(Y) = \text{Sh}(A \cup D) \dot{\cup} \{I \cup \{q\} : I \in \text{Sh}(C)\} \dot{\cup} \{I \cup \{r\} : I \in \text{Sh}(C)\} \dot{\cup} \{\{q, r\}\}. \quad (2.37)$$

Its size is $|A \cup D| + 2|C| + 1 = |Y|$, proving ampleness. The root has exactly its two new-coordinate neighbors and their square. Each extra concept in the 01- or 10-layer has no neighbor in another layer and is a corner of its ample one-concept extension.

Necessity in (2.33) follows by restricting to $q = r = 1$. For sufficiency, delete a and both extra concepts that are present, then peel $C \times \{01, 10, 11\}$. Given an ordinary peeling of C , delete each old word's copies in the order 10, 11, 01. For the reverse implication in (2.34), delete the extra concepts first, retain a , and use a peeling of C to 0. Delete the copies of each nonzero old word in that same order, then finish the square at old 0 in the order 11, 01, 10, 00. Every deletion has its full incident product cube.

Suppose conversely that Y peels to a , and stop just before the first deletion of a vertex $(0, t)$ with $t \in \{01, 10, 11\}$. Until this time the full square at old 0 is retained. The corner $(0, t)$ has both new directions. It cannot have an old-coordinate neighbor: its corner cube would also require that neighbor in the singleton 00-fibre. Its current old-coordinate section is ample and connected, so that section is the singleton $\{0\}$.

If $t = 01$ or 11 , restrict the preceding deletion sequence to $r = 1$ and strongly project q . Initially the result is $A \cap C = C$. If $t = 10$, restrict to $q = 1$ and strongly project r , obtaining $D \cap C = C$ initially. In each case the projected family just before the selected deletion is $\{0\}$, because one of the two sections has become a singleton. Throughout the preceding sequence the full two-point fibre of 0 remains. Record an old word when its full two-point fibre first disappears, omitting repeated projected families. At such a deletion the deleted vertex has an edge to its mate. Each incident direction of its word in the overlap lifts to an incident direction of that vertex; the full corner cube together with its mate layer therefore projects to a complete corner cube in the overlap. Thus every recorded word is a corner of the current overlap. The sequence ends at $\{0\}$ without deleting 0. Appending 0 proves (2.34).

Assume now that C uses every old coordinate. A root-preserving restriction or contraction μ in an old coordinate gives the same square construction with $C' = \mu C$, $A' = \mu A$, and $D' = \mu D$. The side differences still have size at most one. They cannot share a word outside C' : its 01- and 10-copies would have distance two with neither intermediate word present, contrary to isometry of the ample minor. The union of the three old sections is ample by contraction, and the shadow count above forces $\text{Sh}(A') \cap \text{Sh}(D') = \text{Sh}(C')$. Thus the endpoint equivalence applies to every old-coordinate image, even if C' has constant coordinates or is a singleton.

If (Y, a) is a rooted factor, the two equivalences make C ordinarily peelable and unable to peel to 0. Every elementary root-preserving image of C peels to 0, by applying the endpoint equivalence to the corresponding proper image of Y . Hence $(C, 0)$ is a rooted factor. The $q = 0$ restriction of Y , followed by contraction of r , gives $(A, 0)$; the $r = 0$ restriction followed by contraction of q gives $(D, 0)$. Rooted minimality makes both endpoints possible.

Conversely, suppose that $(C, 0)$ is a rooted factor and that A, D peel to 0. The equivalences give ordinary peelability and failure at a for Y , and all old-coordinate rooted minors peel to their roots. The $q = 0$ restriction is A with a leaf attached at 0; follow a peeling of A to 0 and delete the leaf root last. The $r = 0$ restriction uses D . Contracting q gives D at $r = 0$ and A at $r = 1$. Delete the upper extra concept, if present, then its copy of C using an ordinary peeling; all its corner cubes lift into the lower D -layer. Finish with D to 0. Contracting r interchanges A, D . All elementary rooted minors peel, and rooted pc-minor closure proves minimality.

We next prove (2.35), assuming $C \in \mathcal{D}_{\text{amp}}$. Necessity follows by restricting a root-ending order to the indicated coordinate section. For an endpoint $(x, 01)$, first delete the extra concept in the 10 section, if present, and delete a . Both are corners and they are nonadjacent, so these two deletions are valid in this order. The 10 and 11 sections are now both C , with no 00 section. Delete the entire 10 copy by an ordinary order of C , then the entire 11 copy by an ordinary order of C . At

each step its old corner cube has all its mates in the untouched adjacent layer: first the 11 copy of C , then the 01 copy of $A \supseteq C$. No other layer supplies a neighbour. These are corner deletions, leaving exactly A at 01. Finish with an order ending at x . Interchanging the two side sections gives the 10 assertion. For an endpoint $(x, 11)$, delete both possible extras and a . The remaining family is $C \sqcap \{01, 10, 11\}$. The second factor is a three-vertex path and peels to its middle vertex 11. The product endpoint rule in Proposition 2.61 gives sufficiency.

If (Y, a) is a rooted factor, the equivalence already proved makes both side endpoints attainable and the core endpoint unattainable. The root a is unattainable as well. This proves the missing-common-neighbours assertion. Finally, for any other unattainable z , the corresponding equivalence in (2.35) locates its failure in its own coordinate section. This is a proper root-preserving minor of Y , so (Y, z) is not a rooted factor. Attainable vertices cannot be roots of rooted factors by definition. This proves uniqueness.

Finally suppose $|C| > 1$. Every nonzero $(x, 11)$ has both new directions but lacks $(x, 00)$, so is not a corner. A vertex $(x, 01)$, with $x \in C \setminus \{0\}$, has its q -mate. Its corner cube would force x to be a corner of C ; if the extra concept of A is adjacent to x , that cube would also require the absent copy of the extra concept at 11. Conversely, when x is a corner of C and has no such extra neighbor, its old corner cube times the q -edge is complete. This proves the first line of (2.36); interchanging q, r proves the second. Each nonroot copy of old 0 has both new directions and an old neighbor, since C is nontrivial and connected. Its cube would require that neighbor in the 00-fibre, excluding all three copies. The root and extra concepts were checked at the start of the proof. $\square$

Theorem 2.64 (Rooted transfer through the two section overlaps). *Use the presentation and the four shadow conditions of Proposition 2.62. Put*

$$R = A \cap C, \quad S = D \cap C,$$

and suppose that $R, S \in \mathcal{D}_{\text{amp}}$ and

$$A \searrow R, \quad C \searrow R, \quad D \searrow S, \quad C \searrow S.$$

Here $H \searrow K$ denotes a corner peeling retaining K . Then $Y \in \mathcal{D}_{\text{amp}}$ and, for $a = (0, 00)$,

$$\begin{aligned} Y \text{ peels to } a &\iff (R \text{ peels to } 0 \text{ and } D \text{ peels to } 0) \\ &\quad \text{or } (S \text{ peels to } 0 \text{ and } A \text{ peels to } 0). \end{aligned} \tag{2.38}$$

Every endpoint in another section is attainable in Y exactly when it is attainable in that section alone.

Assume every coordinate of F is active in $A \cup D \cup C$. Then (Y, a) is a rooted factor exactly when all the following hold:

1. A and D peel to zero;
2. neither R nor S peels to zero;
3. for every old coordinate $f \in F$ and each root-preserving restriction or contraction μ in f , at least one of $\mu R, \mu S$ peels to the image of zero.

The singleton a is then the only critical root of Y . In particular, if $(R, 0)$ and $(S, 0)$ are rooted factors each using every coordinate of F , item 3 holds automatically. No containment of C in either side is required.

Proof. The shadow criterion makes Y ample. Isometry implies $A \cap D \subseteq C$: a nonzero word in $(A \cap D) \setminus C$ would give two vertices in opposite side sections with neither common neighbour present. Similarly there are no edges between $A \setminus C$ and $C \setminus A$ in their union: such an edge would produce a missing-intermediate diagonal square in the ample face with sections A, C . The corresponding assertion holds for D, C . Thus relative orders in the side sections also give

$$A \cup C \searrow C, \quad D \cup C \searrow C.$$

For example, while deleting $A \setminus R$ no neighbour in $C \setminus A$ is introduced, so each corner cube remains valid.

All three sections are positive by the relative orders and positivity of R, S . Delete a first. To retain A , clear $D \setminus S$, delete its remaining copy of S with mates in C , clear $C \setminus R$ in 11, and delete that copy of R with mates in A . Interchanging the sides gives a relative order retaining D . To retain C , clear $A \setminus R$ and $D \setminus S$, then delete the side copies of R and S , both with mates in C . In each relative step the deleted word has no mate in the adjacent layer; in each whole-layer step its corner cube has all its mates in an untouched containing layer. These are corner orders. Appending a root-ending order in the retained section proves ordinary positivity and sufficiency for every non-singleton-section endpoint. Necessity is coordinate-face restriction.

Suppose an order ends at a . The two root-containing faces with sections $\{0\}, A$ and $\{0\}, D$ force both side endpoints to be attainable. Stop just before its first deletion of one of $(0, 01), (0, 10), (0, 11)$. The root square is still present. Its old-coordinate section must be a singleton, since an old neighbour in its corner cube would require an absent old neighbour in the 00 section. If the first deletion is at 01 or 11, record the first destruction of each edge fibre in the face with sections A, C ; if it is at 10, use the face with sections D, C . As in Proposition 2.63, a broken fibre's corner cube projects to a corner cube in the remaining reduction. The zero fibre survives until the stopping time. The resulting order peels R or S , respectively, to zero. This proves necessity in (2.38).

For sufficiency, suppose R and D peel to zero. Relatively clear $A \setminus R$ in 01, then peel its copy of R to zero with mates in C . Delete $(0, 01)$, now a corner of its root square. Clear $C \setminus S$ in 11, and delete the remaining copy of S by an ordinary order with mates in D . The remaining family is D with a leaf a attached at zero; peel D to zero and finish at a . The other alternative follows by interchanging the two sides.

Relative orders pass to coordinate images: their residual images are ample and lose at most one word at each step; omitting repetitions gives a relative order to the image target. For every old-coordinate operation one also has

$$\mu R = \mu A \cap \mu C, \quad \mu S = \mu D \cap \mu C.$$

Restrictions are immediate. A new intersection under contraction would give a missing-intermediate diagonal square in the original ample adjacent face. Consequently all old-coordinate rooted images satisfy the same hypotheses, including positivity of the image overlaps; their shadow conditions hold because the images of Y are ample.

If (Y, a) is a rooted factor, its two new-coordinate root restrictions force item 1. The endpoint equivalence then makes root failure exactly item 2. Applying that equivalence to each old-coordinate rooted image, and using endpoint minor closure for A, D , gives item 3. These images are proper by the active-coordinate assumption.

Conversely, items 1–3 give positivity of every old-coordinate rooted image by (2.38). The two new-coordinate root restrictions are positive leaves on A, D . One new contraction has sections $A, D \cup C$ and root in A . Use $D \cup C \searrow C \searrow R$, delete the remaining R layer with mates in A , and finish at zero in A . Here $(D \cup C) \cap A = R$, so relative deletions outside R have no mates in

the retained layer. The other contraction similarly uses $A \cup C \searrow C \searrow S$ and finishes in D . Thus all elementary rooted images are positive. Ordinary positivity and root failure, already proved, give rooted minimality. Every other unattainable vertex fails in its own proper section by the full endpoint formula, so cannot be a critical root. The stated sufficient condition for item 3 is immediate from the definition of the two rooted factors. $\square$

If $C \subseteq D$, then $S = C$. Since $C \searrow R$, an endpoint order of R extends to C , including after every coordinate operation. The endpoint condition reduces to attainability of zero in A, C . If all old coordinates are active in C , the rooted-factor criterion reduces to $(C, 0)$ being a rooted factor and A, D peeling to zero. This is the one-containing-side rule, including the 872 example; the contained relative-wing construction is its special case $R = C$.

Both containments can fail. In the partial-tip construction with old core B and a tip γ occurring only in the opposite section, put $A = B \cup \{128\}$, $D = B \cup \{290\}$, and $C = B \cup \{\gamma\}$. Then $R = S = B$. For the archived tips 64 and 604, all four relative orders remove a single tip. The theorem covers both 871-concept rooted factors, including the case $\gamma = 604$ where C itself peels to zero. Thus the obstruction at the singleton is controlled by the two overlaps, not by the opposite section alone.

The verification `research/two_overlap_square.py/json` checks these two presentations, their clearing orders and the two new-coordinate contraction orders. The earlier one-sided schedules remain in `research/noncontained_relative_square.py/json`. Intrinsic selection of the jointly failing overlaps in items 2–3 and exhaustive recovery of the four relative orders are still open; the theorem does not remove those requirements by calling them a generator.

Remark 2.65 (Recovering the paired extensions). Let Z be an ample family with a corner root of degree two. Translate the root to 0, and let q, r be its incident coordinates. The $q = r = 0$ section is $\{0\}$: the root is isolated in that section, which must be connected. The other three sections recover A, D, C in positions 01, 10, 11, respectively. Suppose that $C \subseteq A, D$, that both side differences have size at most one, and that C uses every remaining coordinate. These are intrinsic tests on the recovered sections.

Isometry forces $A \cap D = C$, by the missing-intermediate-word argument in the proof. All three sections and their union are ample. The exact shadow formula, with $\text{Sh}(A) \cap \text{Sh}(D)$ in the two one-new-coordinate terms, and $|\text{Sh}(Z)| = |Z|$, forces that intersection to equal $\text{Sh}(C)$. Thus Proposition 2.63 applies and recovers all parameters uniquely, up to interchanging q, r . Within this type, rooted minimality is exactly rooted minimality of C and the two repairing endpoint conditions. This reduces the admissibility problem to smaller inputs; it does not replace those remaining recognition conditions by intrinsic ones, or assert that every degree-two-root factor has this section pattern.

The root of the recovered core need not be a corner. Take the 292-concept family H , rooted at 29, from Remark 2.50, and translate its root to zero to obtain C . The extensions by 128 and 290 both peel to zero and satisfy the shadow hypotheses. The resulting Y is a rooted factor with 879 concepts, dimension 14, VC dimension five, and eleven corners. The replay `verify_damp_paired_root_transfer.py` checks all 24 rooted minor orders of C , all 28 of Y , and its complete 704-state negative certificate. Four positive controls over $C = \{0, 1, 3, 7\}$, with optional tips 2, 5, test every one of 321 reachable states before the first root-square deletion. All 231 possible exits from these states induce a core peeling ending at zero. The data and the recovered corner sets are recorded in `damp_paired_root_transfer_verification.json`.

Theorem 2.66 (Two endpoint repairs give a terminal cut-vertex obstruction). *Let $(C, 0)$, irredundantly embedded in Q_F , satisfy the three rooted-factor conditions of Theorem 2.19, and suppose that 0 is its only corner. Let $A = C \cup \{\alpha\}$, $D = C \cup \{\delta\}$, where $\alpha, \delta \notin C$ are distinct. Assume that*

$A, D, A \cup D$ are ample, that both A, D peel to 0, and that their shadows intersect in $\text{Sh}(C)$. Then the square construction (2.32) is a rooted factor with corners exactly

$$a = (0, 00), \quad \alpha^* = (\alpha, 01), \quad \delta^* = (\delta, 10).$$

Neither nonroot corner deletion preserves rooted minimality. Gluing (Y, a) to a copy of $(C, 0)$ on disjoint coordinates gives a corner-descent-terminal ample forbidden pc-minor $X = Y \vee C$, with a cut vertex, exactly two corners, and order $4|C| + 2$. It has no nontrivial replicated common-seed presentation with resolving tips.

Proof. Proposition 2.63 gives the rooted-factor conditions and, for every $R \subseteq \{\alpha^*, \delta^*\}$,

$$\text{Cor}(Y \setminus R) = \{a\} \cup (\{\alpha^*, \delta^*\} \setminus R). \quad (2.39)$$

After deleting α^* , the same transfer criterion fails because its 01-section is now C , which cannot peel to its root. Explicitly, restrict to $q = 0$ and contract r to expose the proper bad rooted minor $(C, 0)$. After deleting δ^* , restrict to $r = 0$ and contract q to get the same witness.

The cut-vertex theorem makes $X = Y \vee C$ ample and forbidden. The root ceases to be a corner, so its corners are exactly the two tips. The preceding restriction and contraction, with the second block retained, give the proper minor $C \vee C$ of either corner deletion. This minor is nonpeelable by Proposition 2.18, proving terminality. Every nontrivial replicated common-seed construction with resolving tips is two-connected by Proposition 2.113; the cut vertex of X therefore excludes such a presentation. $\square$

Remark 2.67 (A fixed terminal example with 1158 concepts). Let $B = (G \cup \{91\}) \setminus \{61, 125, 93\}$ be the 289-concept rooted factor from Remark 2.50. Translate its root 29 to zero, obtaining $C = \{x \text{ xor } 29 : x \in B\}$. Use the two tips $\alpha = 128$ and $\delta = 290$, corresponding to the original words 157 and 319. Their extensions peel to zero and acquire the distinct supports

$$\{1, 2, 7, 8\} \quad \text{and} \quad \{5, 6, 7, 8\}, \quad \text{with bitmasks } 390, 480.$$

Their union is ample and its shadow is $\text{Sh}(C)$ together with these two supports. Thus the theorem applies. The resulting Y has 870 concepts in dimension 14. With new bits q, r in positions 0, 1, its nonroot corners have words 514, 1161. Gluing a second C in bit positions 14, $\dots$, 25 produces X of order 1158, isometric dimension 26, and VC dimension four. Both corners have degree four; the root has degree five. The minimum degree is four, so X is already its graph three-core. Deleting the cut vertex leaves components of orders 869 and 288. Each corner deletion has the cornerless 577-concept double $C \vee C$ as a proper minor.

The independent replay `verify_damp_terminal_cut_vertex.py` checks the input shadows, both repairing endpoint orders, all 24 rooted elementary orders of C , all 28 of Y , and all 78 ordinary elementary orders of X . It verifies the complete four-state negative certificates for both Y with its root retained and X , as well as both terminality witnesses. The report is `damp_terminal_cut_vertex_verification.json`; the new endpoint orders are stored in `computations/round790_paired_endpoint_repairs.json`.

This example refutes universal preserving nonroot-corner deletion and universal existence of a common-seed layer set for cornered terminal obstructions. Proposition 6.36 below uses its two resolving extensions to refute cornerlessness of the largest recovered seed as well. Intrinsic admissibility of the rooted factors and their endpoint-repairing extensions remains open.

Lemma 2.68 (Rooted minimality along ample additions). *Let (H, a) be a rooted factor, irredundantly embedded in Q_F , and let $K = H \cup \{c\} \subseteq Q_F$ be ample, with $c \notin H$. Then either K peels to a , or (K, a) is a rooted factor. More generally, suppose*

$$H = H_0 \subset H_1 \subset \cdots \subset H_m \subseteq Q_F, \quad |H_{i+1} \setminus H_i| = 1,$$

and every H_i is ample. If H_m does not peel to a , then every (H_i, a) is a rooted factor.

Proof. The new concept c is a corner of K by Lemma 2.10. Deleting it and following an ordinary peeling of H proves $K \in \mathcal{D}_{\text{amp}}$. Let μ be an elementary root-preserving restriction or contraction. Every coordinate is active in H , so μH is a proper rooted minor and peels to μa . Both μH and μK are ample; their difference has size at most one. If the difference is nonempty, its concept is a corner of μK by the same lemma and is distinct from μa . Delete it first, then peel μH to μa . If the difference is empty, use that peeling directly. Thus all elementary rooted minors of K peel to their roots. Closure under further root-preserving pc-minors gives the third rooted-factor condition of Theorem 2.19. If K does not peel to a , all three conditions hold.

Peeling to a persists under an ample one-concept addition: delete the new corner first. Hence no H_i in the displayed chain can peel to a when H_m cannot. Apply the first assertion successively. The fixed coordinate set is essential to this argument: it makes every elementary image of the initial factor a proper minor. $\square$

Proposition 2.69 (A smaller rooted input and a second root). *Let $C \subseteq Q_{12}$ be the 289-concept rooted factor, normalized at zero, used in Proposition 2.37. In its integer encoding put $b = 256$ and $H = C \setminus \{0\}$. Then $|H| = 288$ and all three pairs*

$$(H, b), \quad (C, 0), \quad (C, b)$$

are rooted factors. The first two have corner roots of degree three; the third has a noncorner root of degree four. The concepts attainable as the final vertex of a peeling of C are exactly $C \setminus \{0, b\}$. These same 287 concepts are the attainable final vertices of H .

Pairwise gluing of these three rooted inputs, allowing repetition, gives six pairwise nonisomorphic ample forbidden pc-minors, all of isometric dimension 24. Their orders and numbers of corners are

<i>rooted factors</i>	<i>order</i>	<i>corners</i>
$(H, b), (H, b)$	575	0
$(H, b), (C, 0)$	576	0
$(H, b), (C, b)$	576	1
$(C, 0), (C, 0)$	577	0
$(C, 0), (C, b)$	577	1
$(C, b), (C, b)$	577	2

Proof. Direct corner checks give

$$\text{Cor}(C) = \{0\}, \quad \text{Cor}(H) = \{b\}.$$

Thus every peeling of C begins with $0, b$. The previously certified ordinary order supplies an ordinary peeling of H after its first vertex is removed. Corner deletion preserves ampleness, and every coordinate remains active in H .

The new rooted-minimality assertion is certified explicitly, rather than inferred from the forced prefix. The replay `research/forced_prefix_rooted_inputs.py` reconstructs C from `research/relative_wing_square.json`. For H it checks an ordinary peeling, its unique corner b , and

complete orders ending at the root image for all 24 elementary root-preserving restrictions and contractions. Rooted minor closure then covers every proper root-preserving pc-minor. Since b is the only initial corner and $|H| > 1$, H cannot peel to b . Consequently (H, b) is a rooted factor.

The pair $(C, 0)$ is the established input. Since C is an ample one-concept extension of H , Lemma 2.68 applies at the old root b . The forced prefix excludes a peeling of C ending at b , so (C, b) is also a rooted factor. The replay separately checks its 24 rooted elementary orders using the constructive proof of that lemma. It also checks the three stated root degrees.

For each $a \in C \setminus \{0, b\}$ the accompanying JSON record supplies a complete peeling of C ending at a . All 287 orders are replayed. Their forced initial 0 may be removed to obtain orders of H with the same endpoints. The unique corners exclude the remaining endpoints in C and H , proving both assertions about attainable final vertices.

Apply Theorem 2.19 to each unordered pair of rooted inputs. It proves forbiddenness and dimension 24. The orders follow by identifying the two roots. A nonroot corner is unaffected by vertex gluing, while the common root is not a corner, since incident directions from different blocks lack their mixed square. The rooted inputs have respectively zero, zero, and one nonroot corner, which gives the displayed corner counts. The six order/corner-count pairs are distinct, proving pairwise nonisomorphism. As an independent finite check, the replay constructs and verifies complete orders for all 72 elementary pc-minors of each output, 432 orders in total. $\square$

The 577-concept double of $(C, 0)$ is reused here. The additional input comes from deleting its first forced vertex and changing the distinguished root, followed by a new rooted-minor certification. The proposition does not assert that this operation preserves rooted minimality in general. It therefore expands the collection of certified inputs and outputs, without providing an exhaustive intrinsic rooted-factor generator.

Proposition 2.70 (Attainable sinks need not form an ample family). *For a corner-peelable ample family A , put*

$$\text{Sink}(A) = \{a \in A : A \text{ has a corner peeling ending at } a\}.$$

There is a 658-concept corner-peelable ample family H of isometric dimension 25 such that the graph induced by $\text{Sink}(H)$ is disconnected. In particular, $\text{Sink}(H)$ is not ample. For every positive integer m , a corner-peelable ample family exists whose attainable-sink graph has at least 2^m connected components.

Proof. Take two copies K_1, K_2 of the 330-concept piece from Theorem 2.45, sharing the edge $D = \{u, v\}$ and with disjoint private coordinates. Reverse the endpoints in the second copy. Their exact profiles are

$$(r_D, t_u, t_v, \ell_u, \ell_v)(K_1) = (0, 1, 0, 1, 0), \quad (r_D, t_u, t_v, \ell_u, \ell_v)(K_2) = (0, 0, 1, 0, 1).$$

Their union H is ample by cube-face gluing, has $330+330-2 = 658$ concepts, and uses $12+12+1 = 25$ coordinates. Both compatibility witnesses $t_u(K_1)\ell_v(K_2)$ and $t_v(K_2)\ell_u(K_1)$ hold, so H is corner-peelable.

The first witness constructs a peeling ending in $K_2 \setminus D$. Choose a peeling of K_1 ending at u and a peeling of K_2 in which v is the first boundary vertex removed and is then a leaf adjacent to u . Both orders restrict to v, u on D . Merge their outside blocks before, between, and after these two boundary vertices, as in Corollary 2.21. The first order has no vertex after u . The second must have a vertex after u : otherwise it would end at u , contradicting $t_u(K_2) = 0$. Thus the merged order ends in $K_2 \setminus D$. Using the other compatibility witness interchanges the roles and gives an attainable sink in $K_1 \setminus D$.

Corollary 2.31 shows that a positive union of two pieces failing retention has all four boundary values zero. Hence neither u nor v belongs to $\text{Sink}(H)$. There are no edges between $K_1 \setminus D$ and $K_2 \setminus D$. The attainable-sink graph meets both sets and is therefore disconnected. Every nonempty ample family has connected one-inclusion graph, proving that this sink set is not ample. The replay `research/disconnected_attainable_sinks.py/json` constructs and checks the two complete orders; it does not enumerate all attainable sinks.

Finally, for positive ample families A, B one has

$$\text{Sink}(A \times B) = \text{Sink}(A) \times \text{Sink}(B).$$

Necessity follows by restricting a root-ending order to the coordinate sections through its final vertex. For sufficiency, take root-ending orders in both factors and delete the A -fibres in the B order, using the A order within each fibre. The incident cube at each step is the product of the two relevant corner cubes; all later fibres remain complete. This ends at the prescribed pair of roots. The one-inclusion graph on the displayed sink set is the Cartesian product of the two sink graphs. Its connected components are exactly the products of their components, since paths project to paths and paths within factors lift with the other coordinate fixed. Applying this to H^m gives at least 2^m components. $\square$

The ample attainable-sink set in Proposition 2.69 is therefore a property of that input, not a general recovery principle. The counterexample uses two existing edge pieces and the existing gluing criterion; it supplies no new forbidden-minor family. It excludes treating the attainable sinks of arbitrary positive ample families as one ample residual class, or as a union with a universally bounded number of connected components. The next proposition shows that rooted minimality does not restore ampleness of the sink set, although it makes no claim about disconnectedness.

Proposition 2.71 (Rooted minimality does not make the sink set ample). *There is an ample rooted factor $(S, 0)$ with 870 concepts and isometric dimension 14 such that $\text{Sink}(S)$ is not isometric and therefore not ample. More precisely, in the integer encoding below, $1, 2 \in \text{Sink}(S)$ but $0, 3 \notin \text{Sink}(S)$.*

Proof. Use the established 870-concept rooted factor from Remark 2.67. In its stored coordinates, let C be the normalized 289-concept core, put $A = C \cup \{128\}$ and $B = C \cup \{290\}$, and write

$$S = \{0\} \cup \{4x + 2 : x \in A\} \cup \{4x + 1 : x \in B\} \cup \{4x + 3 : x \in C\}.$$

The pair $(S, 0)$ is a rooted factor by the earlier construction. In particular, 0 cannot be an attainable sink. The coordinate face with both low bits equal to one is the copy $\{4x + 3 : x \in C\}$. An order ending at 3 would restrict to an order of C ending at zero, which is impossible because zero is the only initial corner of C . Thus 3 is also unattainable.

Let $c = (0, c_2, \dots, c_{289})$ be the stored ordinary order of C , and let $a = (a_1, \dots, a_{290})$ and $b = (b_1, \dots, b_{290})$ be the stored orders of A and B ending at zero. The following concatenated lists are complete corner orders of S :

$$\begin{aligned} (0, 1, 3, 1161), & \quad (4c_i + 1)_{i=2}^{289}, \quad (4c_i + 3)_{i=2}^{289}, \quad (4a_i + 2)_{i=1}^{290}; \\ (0, 514, 2, 3), & \quad (4c_i + 2)_{i=2}^{289}, \quad (4c_i + 3)_{i=2}^{289}, \quad (4b_i + 1)_{i=1}^{290}. \end{aligned}$$

They end at 2 and 1, respectively. The prefixes remove the root, the other two copies of core zero, and the unused repair. The next two blocks clear copies of $C \setminus \{0\}$ while their corner cubes have mates in an untouched adjacent layer. The last block uses the repair kept in the final shore, which makes zero attainable there.

For completeness, the finite prefix assertions and every subsequent corner deletion are replayed by `research/rooted_nonample_sink_set.py`. It reconstructs these lists from the earlier core and repair orders, checks the complete four-state negative deletion graph protecting zero, and replays all 28 rooted elementary-minor orders. Hence the input's rooted minimality and the two new endpoint assertions are checked independently of any greedy search. The exact lists and source hashes are in the accompanying JSON record.

The two vertices 1 and 2 differ in their two low coordinates, so their only common neighbours in the ambient cube are 0 and 3. Both are absent from $\text{Sink}(S)$. Its induced graph therefore has no path of length two between two vertices at Hamming distance two. It is not isometric and, since nonempty ample families are isometric, is not ample. $\square$

This strengthens the failure of the ample-sink-set principle to the actual rooted-factor domain. It does not assert that the sink set of this factor is disconnected, or classify all its attainable sinks. The two previously unsuccessful greedy endpoint tests are resolved by retaining the repair needed in the final shore; a greedy stall was not a negative certificate.

Proposition 2.72 (Corner deletion followed by rooted minor descent). *Let (H, a) be a rooted factor on its active coordinate set F . Suppose $x \neq a$ is a corner and $S = H \setminus \{x\}$ is ordinarily peelable. There is a root-preserving pc-minor map μ for which $(C, b) = (\mu S, \mu a)$ is a rooted factor with $|C| < |H|$.*

If (S, a) is already a rooted factor, take μ to be the identity. Otherwise μ can be chosen proper, and

$$\mu H = C \cup \{d\}, \quad d = \mu x \notin C,$$

is an ample one-concept addition that peels to b . Write P for the incident support of x in H . In a normal form for μ , let R be the restricted coordinates and Q the contracted coordinates. Then

$$Q \cap P = \emptyset, \quad x_i = a_i \ (i \in R), \quad \text{the acquired support at } d \text{ is } P \setminus R, \quad |P \setminus R| \geq 3.$$

Thus the repair retains a face of the original corner cube; it is not an arbitrary new repairing addition.

Iterating these steps ends at a rooted factor (K, r) in which every complete peeling starts at r . Equivalently, r is the unique source in every acyclic USO of K .

Proof. The class S is ample by corner deletion. It cannot peel to a , since prefixing such an order by x would peel H to a . Choose a root-preserving minor (C, b) of (S, a) minimal among those failing at their roots. Ordinary pc-minor closure makes C peelable; minimality supplies every proper rooted minor condition. Hence it is a rooted factor, and its size is at most $|S| < |H|$.

When S is not already minimal, the chosen map uses at least one proper coordinate operation. Applied to H it gives a proper root-preserving minor, which peels to b . Images of H and S differ by at most one word. They cannot coincide, because C fails at b . Their difference is therefore exactly $d = \mu x$, which is distinct from $b \in C$.

Restrictions and contractions on distinct coordinates commute, so write μ with disjoint sets R, Q and with every restriction taking the root's bit. A restriction discarding x would identify the two images; thus $x_i = a_i$ for every $i \in R$. Contracting a coordinate in P would also identify them, since the neighbour $x^{\{i\}}$ lies in S . Consequently $Q \cap P = \emptyset$.

First restrict in R . The corner cube at x becomes its retained face of support $P \setminus R$, and x remains a corner. The shadow difference between these two restricted ample families is exactly that support. Now contract in Q . A contraction retains precisely the shattered supports avoiding its coordinate; all members of Q avoid $P \setminus R$. The shadow difference therefore remains $\{P \setminus R\}$,

and the ample one-concept addition at d has this acquired support. A support of size at most two cannot change peeling to an old prescribed endpoint, by Proposition 2.61. Since adding d does change it here, $|P \setminus R| \geq 3$.

Whenever a peeling of the current rooted factor begins at a nonroot vertex x , its suffix proves that deleting x leaves an ordinarily peelable class, so the preceding step applies. Size strictly decreases, and iteration must stop. At a terminal factor no peeling begins away from the root. CCMW's correspondence identifies peelings with topological orders of acyclic USOs. A nonroot source in any such orientation can be chosen first in a topological order; conversely the first word of a peeling is a source. Thus the terminal property is exactly that the root is the unique source in every acyclic USO. $\square$

For a degree-three repair, the inequality in the proposition forbids restrictions in its support, agreeing with the more specialized Proposition 2.106. The additional statement here is the combined descent for arbitrary corner degree, including the retained-face description when restrictions lower that degree. It does not assert that the terminal factor has only one geometric corner: other corners could all leave nonpeelable classes when deleted. Nor is this an intrinsic forward generator, since ordinary peelability, rooted minimalization, and the inverse expansion data remain to be selected structurally.

The two recent controls illustrate distinct steps. In the 292-concept factor, deleting 32 leaves a rooted factor, as follows from its ample addition chain above the 289-concept factor and Lemma 2.68. For the 308-concept factor, delete the repair (290, 0) and contract the fresh coordinate. This recovers the 289-concept input; applying the same contraction before deletion recovers its endpoint repair $C \cup \{290\}$, of support $\{5, 6, 7, 8\}$. Thus allowing the minor step handles a failure of direct rooted minimality already present in the repaired-face construction.

Corollary 2.73 (Joint descent for rooted factors and ordinary obstructions). *Start with either a rooted factor (H, a) or an ample forbidden pc-minor H . The following procedure strictly decreases cardinality at every step and ends at one of these two types:*

1. *a rooted factor whose root is its only geometric corner;*
2. *a cornerless ample forbidden pc-minor.*

At a rooted factor with a nonroot corner x , delete x . If the result is ordinarily peelable, take a minimal root-preserving minor failing at its root, as in Proposition 2.72. If it is not ordinarily peelable, forget the root and take an ordinary forbidden pc-minor of the deletion. At an ordinary forbidden class with a corner, delete that corner and again take an ordinary forbidden pc-minor of the deletion. Once the procedure enters the ordinary branch it stays there.

Proof. At a rooted factor, deleting a nonroot corner preserves ampleness and failure to peel to the root. If the deletion is ordinarily positive, the preceding proposition gives a smaller rooted factor. Otherwise finite minor minimalization gives an ordinary forbidden pc-minor; it is ample because ampleness is pc-minor closed. At an ordinary nonpeelable ample family, a corner deletion remains nonpeelable: any peeling of the remainder could be prefixed by that corner. Thus ordinary minor minimalization is available at each later step as well. Each step begins with a deletion and then takes a minor, so size strictly decreases. In the rooted branch termination means that no nonroot corner is present. Ordinary positivity supplies at least one corner, necessarily the root. In the ordinary branch termination means cornerlessness. Both branches retain their stated minimality conditions by construction. $\square$

The ordinary part reuses the corner-deletion mechanism of Theorem 6.70; taking a minimal negative pc-minor after each deletion additionally keeps its states forbidden. The purpose of the joint statement is to allow ordinary nonpeelability after a rooted deletion, instead of treating it as a barrier. For example, the 1158-concept corner-descent-terminal class already constructed in Theorem 2.66 is not terminal for this procedure. After either corner deletion, its recorded restriction and contraction give the cornerless forbidden double $C \vee C$ of order 577. This is reuse of that theorem's witness, not a new example.

This is a backward reduction, not an intrinsic forward generator. It still uses negative-minor selection; reversing a step requires admissible expansions and corner additions that recover minimality. It proves neither that the terminal input is unique nor that the two input types have been generated. It does, however, remove the need to identify the source-based terminal condition in Proposition 2.72 with a unique geometric corner. That stronger source question can be considered separately.

Corollary 2.74 (Lifting a repairing addition through one contraction). *Let $\emptyset \neq D \subseteq C \subseteq Q_F$ be ample, let $K_1, \dots, K_k$ be the components of $C \setminus D$, and let $d \notin C$ with $A = C \cup \{d\}$ ample. For a fresh coordinate e , labels $t_i \in \{0, 1\}$, and $b \in \{0, 1\}$, put*

$$S_t = (D \times \{0, 1\}) \cup \bigcup_i (K_i \times \{t_i\}), \quad H_{t,b} = S_t \cup \{(d, b)\},$$

and define

$$I(d) = \{i : d \text{ has a neighbour in } K_i\}.$$

Then $H_{t,b}$ is ample if and only if $t_i = b$ for every $i \in I(d)$. In that case (d, b) is a corner, and the acquired support of its addition is the acquired support of d over C .

Suppose additionally that $A, D \in \mathcal{D}_{\text{amp}}$. If an ample $H_{t,b}$ is not in $\mathcal{D}_{\text{amp}}$, then some $j \notin I(d)$ has $t_j = 1 - b$. In particular, if $I(d) = \{1, \dots, k\}$, no labeling gives an ample forbidden pc-minor. This last conclusion requires positivity of D ; it does not discard the branch with a negative reduction.

Proof. The contraction of $H_{t,b}$ in e is A and its reduction is D . In the graph induced by $A \setminus D$, adjoining d merges precisely the components K_i with $i \in I(d)$ together with d ; all other components remain unchanged. Theorem 5.4 therefore says exactly that their labels must equal the label b of d . It also proves sufficiency. Both S_t and $H_{t,b}$ are then ample, so the one-concept extension criterion makes (d, b) a corner. There is no neighbour in direction e . Every old neighbour of d is either in D , which has both lifts, or in a component indexed by $I(d)$, which has label b . Thus its incident support is unchanged, as asserted.

If there is no $j \notin I(d)$ of label $1 - b$, all component labels equal b . Consequently

$$H_{t,b} = (A \times \{b\}) \cup (D \times \{1 - b\}).$$

This is a peripheral expansion of the positive base A at the positive site D , hence is positive by Proposition 2.6. Taking the contrapositive proves the final assertions. $\square$

This is an application of the existing component-label and peripheral positivity theorems, rather than a new parametrization of ample classes. It supplies a local constraint for reversing a contraction in the joint descent: a repair must leave an untouched component in the opposite layer if the repaired contraction and the reduction are positive. For example, take $C = \{00, 01, 10\}$, $D = \{00\}$, and $d = 11$. The two components of $C \setminus D$ both meet d , so lifting the addition forces both labels to agree and gives a positive peripheral expansion. Opposite labels cannot even give an ample addition. This small example illustrates the geometric constraint, not a forbidden-minor witness. The condition is necessary, not sufficient, for forbiddenness: it supplies neither failure of an acyclic USO nor positive USOs on the other proper pc-minors. Those remain the substantive forward-admissibility problem.

Remark 2.75 (Singleton reductions and a positive separated repair). The vertex-gluing criterion already settles the singleton-reduction case of Corollary 2.74. More generally, any ample expansion X with contraction A and reduction $\{a\}$ satisfies

$$X \in \mathcal{D}_{\text{amp}} \iff A \in \mathcal{D}_{\text{amp}}.$$

Indeed, the components of $A - a$, with a restored, occupy disjoint coordinate blocks, as proved in Theorem 2.19. In X , these pieces attach to the two ends of the sole edge in the new direction, according to their component labels. If A peels, all its pieces peel and at most one fails to peel to a , by Proposition 2.18. Peel every other piece to its attachment vertex. What remains is the exceptional piece with a pendant edge, or just the edge if there is no exceptional piece. Delete the pendant endpoint and peel the remaining piece, or peel the edge. Conversely, positivity of X implies positivity of its contraction A . Thus no active singleton-reduction expansion is an ample forbidden pc-minor.

This also gives a control inside the inverse-step problem, with a negative starting contraction. Take two copies $(B_1, 0), (B_2, 0)$ of the 289-concept rooted factor in Remark 2.67, on disjoint coordinates. Let $C = B_1 \vee B_2$, let $D = \{0\}$, and add the repairing tip $d = 290$ in the first block. The family C is the 577-concept forbidden double. The repaired family $A = (B_1 \cup \{d\}) \vee B_2$ peels: peel the first block to zero, then peel the second ordinarily. In the notation of Corollary 2.74, give $B_1 - 0$ label zero, $B_2 - 0$ label one, and lift d at zero. These are precisely the two components, since rooted factors have no cut vertex. The repair touches only the first component, so the second is the required untouched opposite-layer component. Nevertheless the 579-concept output peels: peel the repaired first block to its root, delete that root as a leaf across the new edge, and peel the second block. All its proper pc-minors therefore peel as well.

This reuses the known rooted factor and its certified endpoint repair; it is not a new forbidden family. It shows why component separation alone does not close forward admissibility, even with a forbidden input C and a positive repaired contraction. The following theorem strengthens this exclusion from singleton reductions to all convex reductions and supplies a lift of the prescribed USO.

Theorem 2.76 (Lifting acyclic USOs through a convex reduction). *Let X be ample, let e be active, and write $A = \pi_e X$ and $D = X^{\{e\}}$ for its contraction and reduction. If D is convex in A , every acyclic USO of A lifts to an acyclic USO of X whose orientation on every edge in an old direction is the given orientation on its projection. Consequently*

$$X \in \mathcal{D}_{\text{amp}} \iff A \in \mathcal{D}_{\text{amp}}.$$

In particular, an ample forbidden pc-minor has no coordinate whose reduction is convex in its contraction. This excludes every cubical reduction, in every dimension.

Proof. Fix an acyclic USO $\mathcal{O}$ of A . Write K_i for the components of $A \setminus D$ and t_i for their lift labels, using Theorem 5.4. Convexity in a partial cube means that D is obtained by fixing coordinates. Thus, for every cube $Q \subseteq A$, the intersection $Q \cap D$ is a face, possibly empty or all of Q . Its complement in Q , when nonempty, is connected and hence lies in a single K_i .

Orient every old-direction edge of X as its projection in $\mathcal{O}$. At $d \in D$, the outgoing cube Q_d of $\mathcal{O}$ is contained in A by (C1). If d has any outgoing neighbour outside D , all such neighbours belong to a single component K_i , by the preceding face observation. Orient the vertical edge over d towards layer t_i . If there is no such neighbour, choose either direction independently. We verify that this orientation is a USO.

For (C1), first take a vertex outside the doubled reduction. Its outgoing cube in A lifts wholly beside it: all vertices of that cube outside D belong to its own component, and all vertices in D have both lifts. Next take a lift of $d \in D$ with no outgoing vertical edge. Its outgoing old directions are a subset of those of Q_d . If outgoing neighbours outside D occur in that layer, all of $Q_d \setminus D$ has that layer's label, so the requisite cube lifts. If none occur, the outgoing directions are along the face $Q_d \cap D$, which has both lifts. Finally, at the source of the vertical edge, our choice ensures that no outgoing old neighbour outside D is present. Its old outgoing cube is a face contained in $Q_d \cap D$. The product of this face with the vertical edge is contained in X , proving (C1) there as well.

For (C2), a cube not using e projects to a cube of A with precisely its given USO. A cube using e is $Q \times \{0, 1\}$ for a cube $Q \subseteq D$. Let z be the unique sink of $\mathcal{O}$ on Q . Both lifts of every other vertex have an outgoing old edge within this cube. Its only sink is therefore the lift of z at the head of its vertical edge. This proves (C2), regardless of the independent choices of vertical directions.

A directed cycle containing an old-direction edge would project, after omitting vertical steps, to a nonempty directed closed walk in the acyclic orientation $\mathcal{O}$, which is impossible. A cycle using only vertical edges is also impossible, since those edges form a matching. Thus the lift is acyclic. CCMW's correspondence gives the forward positivity implication; the reverse follows from closure under contraction. For a forbidden X , the proper contraction A is positive, so convexity of D would contradict the conclusion. A contained cube is convex, which proves the last assertion. $\square$

Corollary 2.77 (A seven-vertex cube in every direction). *Let X be an ample forbidden pc-minor. For every active coordinate e , there are two other coordinates f, g and a coordinate three-face whose intersection with X has exactly seven vertices. In the two-dimensional projection of this face along e , exactly three vertices have both lifts. In particular, every nonempty single-coordinate reduction has at least three vertices and is not a cube.*

Proof. Put $A = \pi_e X$ and $D = X^{\{e\}}$. The preceding theorem makes D nonconvex in A . Both are ample, and D is nonempty and connected. CCMW Lemma 4.10 characterizes convexity of a connected subclass by containment of all ambient intervals between its vertices at distance two [CCMW22]. Since D is isometric, it contains at least one of the two possible intermediate vertices for any such pair. Failure of this criterion therefore gives a square of A , in directions f, g , with exactly three vertices in D . In X , those three vertices have both lifts, while the fourth has exactly one. Fixing all other coordinates gives the asserted seven-vertex intersection. $\square$

The horizontal construction and its free-bit count already appear, for representation maps without an acyclicity hypothesis, in the sibling CompressionSchemes manuscript, in its Round 717 theorem on restricting orientations and its coordinate-conditioning corollary, in `drafts/round717_restricting_orientations.tex`. The additions here are the projected-cycle proof of acyclicity, necessity for preserving every horizontal edge, and the resulting constraints on forbidden pc-minors. The construction and count are reused, not new results.

Unlike the earlier singleton argument, the lifting theorem handles every convex reduction and gives an orientation extending prescribed horizontal data. It uses the existing component-label parametrization and CCMW's USO criterion; the local-convexity assertion used in the corollary is CCMW's result. Nonconvex reductions require a different argument: at a vertical source, the old outgoing neighbours inside the reduction can fail to span a cube inside it. Thus an edge reduction does not require a new square-compatibility theory here. The star theorem below handles the three-vertex path and all larger stars without requiring horizontal preservation. The punctured-cube condition is necessary, not a sufficient generator.

Theorem 2.78 (All lifts preserving the horizontal orientation). *Let X be ample, e active, $A = \pi_e X$, and $D = X^{\{e\}}$. Fix an acyclic USO $\mathcal{O}$ of A . There is an acyclic USO of X whose every old-direction edge has the orientation of its projection in $\mathcal{O}$ if and only if the induced orientation $\mathcal{O}|_D$ is a USO of D .*

When this condition holds, let s be the number of $d \in D$ with no outgoing neighbour in $A \setminus D$. There are exactly 2^s such lifts. At each remaining d , the vertical edge is forced towards the component label of its outgoing neighbours outside D ; at the s specified vertices, either vertical direction may be chosen independently.

Proof. Suppose first that a lift exists. At the source of the vertical edge over d , all outgoing edges of $\mathcal{O}|_D$ are still outgoing. Condition (C1) on the lift requires their whole cube, together with its vertical translate, to lie in X . Their projected cube thus lies in D . This proves (C1) for $\mathcal{O}|_D$. Condition (C2) holds because every cube of D is a cube of A with the same orientation. Acyclicity is inherited as well.

Conversely, assume $\mathcal{O}|_D$ is a USO. For any cube Q of A , $Q \cap D$ is ample, and its complement in Q is ample. If nonempty, that complement is connected. This is the cube-complement fact used in Theorem 5.4; unlike the convex case, the intersection need not be a face. Hence $Q \setminus D$ lies in a single component of $A \setminus D$. In particular, the outgoing neighbours outside D of a fixed $d \in D$ all have one component label. Point the vertical edge towards that label, or choose either direction when there are no such neighbours.

At a vertex outside the doubled reduction, its whole outgoing cube lifts beside it by the preceding component observation. At the head of a vertical edge, if there are outgoing neighbours outside D , their entire outgoing cube lifts beside them by the same observation. If there are none, the outgoing cube lies in D by (C1) for $\mathcal{O}|_D$. At the tail of a vertical edge, no outgoing external neighbour is present; its old outgoing directions are exactly those of $\mathcal{O}|_D$. Their cube lies in D , so its product with the vertical edge lies in X . Thus (C1) holds everywhere. The proofs of (C2) and acyclicity from Theorem 2.76 apply without convexity: each doubled cube has its only sink over its base sink, and a directed cycle would project to a directed closed walk in $\mathcal{O}$.

Finally, if an external outgoing neighbour has label b , pointing the vertical edge away from b would give its tail two outgoing directions whose square lacks the neighbour's opposite lift. This violates (C1). All other vertical directions were arbitrary in the construction, proving both exhaustiveness and the count. $\square$

This criterion concerns preservation of every horizontal edge direction. It is more restrictive than allowing arbitrary acyclic USOs on X , or prescribing only the representation maps induced by CCMW's minor transport. It gives a direct special case of the general completion problem, with independent vertical choices instead of a compatibility search. If one acyclic USO of A induces a USO on D , every component labeling of this same pair gives a positive expansion. The theorem does not assert the converse of this last implication.

Corollary 2.79 (A forced orientation at a path reduction). *If an ample forbidden pc-minor X has a reduction $D = \{u, c, v\}$ inducing the path $u - c - v$, then every acyclic USO of $A = \pi_e X$ directs both edges away from c . Equivalently, every peeling of A deletes c before both u and v .*

Proof. The contraction A is positive. On the three-vertex path, (C2) is automatic, and the only failure of (C1) is directing both edges away from its middle vertex. Any other induced orientation would lift by the theorem, making X positive. The peeling formulation follows from the CCMW correspondence. $\square$

Failure for one prescribed orientation is not a negative certificate. For example, take $A = \{0, 1, 2, 3\}$, $D = \{0, 1, 2\}$, and orient the square A from 0 towards 3. The induced path orientation fails (C1), so no lift preserves these horizontal directions. Nevertheless the peripheral expansion $X = \{0, 1, 2, 3, 4, 5, 6\}$ peels in the order $(5, 6, 4, 3, 2, 1, 0)$. Here the new bit has value 4. The two initial deletions are corners of squares, the third is a leaf, and the remaining four vertices form a square. The universal quantifier in Corollary 2.79 is therefore essential.

Remark 2.80 (Positive expansions with no horizontally preserved USO). The existence condition in Theorem 2.78 is not necessary for positivity even after varying the contraction USO. Take the 289-concept factor A from Remark 2.67, with root zero, and put $D = \{0, 2, 4\}$. The words 2, 4 are neighbours of zero, and zero is the only geometric corner of A . Every acyclic USO of A therefore has zero as its unique source: sources are corners, and a finite acyclic orientation has a source. Both path edges point out of zero, so no such USO induces a USO on D .

Nevertheless every component expansion of (A, D) peels. The fixed connectivity check in `research/path_reduction_eligibility.py` shows that $A \setminus D$ is connected. Thus the only two labelings give

$$X = (A \times \{0\}) \cup (D \times \{1\}) \quad \text{and its reflection in the new coordinate.}$$

Each has 292 concepts. In X , delete $(2, 1)$ and $(4, 1)$ as square corners, then $(0, 1)$ as a leaf, and follow the recorded peeling of A . The verifier replays both complete orders. Thus neither positive expansion has an acyclic USO preserving all horizontal directions of any acyclic USO of its contraction. This strengthens the preceding fixed-orientation example by quantifying over every contraction orientation; the input factor and its corner certificate are reused.

The same bounded audit checks every three-vertex path in six stored positive supports: the 289, 292, 308, 330-concept inputs and the two repairing extensions of orders 291, 290. None of the 22316 paths has a disconnected complement. Hence these supports give only peripheral expansions over a path, and cannot supply a forbidden example by this construction. The result concerns these six inputs only. It neither excludes all path reductions of forbidden classes nor extends the convex-reduction theorem to arbitrary nonconvex reductions. The detailed inputs, counts, hashes, and positive orders are stored in the verifier's JSON report.

Theorem 2.81 (Exact label criterion for a path reduction). *Let $A \in \mathcal{D}_{\text{amp}}$ contain the ample path $D = \{u, c, v\}$ with edges uc, cv . Let $K_1, \dots, K_k$ be the components of $A \setminus D$, and form the ample expansion*

$$X_t = (D \times \{0, 1\}) \cup \bigcup_i (K_i \times \{t_i\}), \quad t \in \{0, 1\}^k.$$

For an acyclic USO $\mathcal{O}$ of A , let $S_{\mathcal{O}}$ be the vertices reachable from at least one vertex of D , including D itself. For $z \in \{u, v\}$, define the set of component indices

$$N_{\mathcal{O}}(z) = \{i : z \text{ has a neighbour in } K_i \cap S_{\mathcal{O}}\}.$$

Then

$$X_t \in \mathcal{D}_{\text{amp}} \iff \exists \mathcal{O} \exists z \in \{u, v\} : |\{t_i : i \in N_{\mathcal{O}}(z)\}| \leq 1.$$

Equivalently, the negative labelings are exactly the two-colourings for which every member of the hyperedge family

$$\mathcal{H}(A, D) = \{N_{\mathcal{O}}(u), N_{\mathcal{O}}(v) : \mathcal{O} \text{ an acyclic USO of } A\}$$

contains both colours. Empty or singleton hyperedges make such a colouring impossible.

Proof. We use two consequences of the component description. An exclusive vertex is a corner of the expansion exactly when its projection is a corner of the contraction: all its old neighbours lift beside it, and its corner cube does so by the cube-complement argument. At a doubled vertex z , a lift in layer b is a corner exactly when z is a corner of the current reduction and all its external neighbours have label $1 - b$. Indeed an external neighbour beside z would require its absent opposite lift in the corner cube containing the vertical direction; when none is present, the corner cube is the product of the reduction corner cube with that edge. These are the first-deletion rules from the [component-partition writeup](#).

Fix $\mathcal{O}$ and an endpoint z satisfying the displayed test. The set $S_{\mathcal{O}}$ has no outgoing edge to its complement. Choose a topological order deleting $A \setminus S_{\mathcal{O}}$ first. It is a peeling of A by CCMW, and its remaining family $S_{\mathcal{O}}$ is ample and positive. All deleted vertices are outside D , so the exclusive-corner rule lifts this prefix to X_t . In the remainder, all external neighbours of z have one label b , or there are none. Delete $(z, 1 - b)$, choosing either b in the empty case. The doubled-corner rule makes this a valid corner deletion. The contraction is still $S_{\mathcal{O}}$, whereas the reduction is the edge $D \setminus \{z\}$. This edge is convex in the contraction. Theorem 2.76 therefore peels the remainder, proving sufficiency.

Conversely, choose a peeling of X_t and consider its first deletion of a lift of a vertex in D . Until then, the reduction is still D . Its middle vertex c is not a corner, so the doubled-corner rule makes this first vertex a lift of an endpoint $z \in \{u, v\}$. All its external neighbours in the remaining expansion have one label, or none remain. The earlier exclusive deletions project to a valid corner-deletion prefix of A . The remaining contraction A' is positive, since the remaining expansion is positive and positivity is contraction-closed. Extend this prefix by a peeling of A' and let $\mathcal{O}$ be its acyclic USO. Every vertex deleted in the prefix precedes all of D ; no directed path from D can reach it. Hence $S_{\mathcal{O}} \subseteq A'$. The external neighbours of z in $S_{\mathcal{O}}$ are therefore a subset of the monochromatic set just identified. This gives the required test. Negating the existential condition proves the hypergraph assertion. $\square$

The general component-partition theorem permits several equality blocks and successive changes of the reduction. Here each positive certificate requires only one set of labels to agree: after the first reduction-vertex deletion, completion is automatic. The criterion allows the two horizontal copies of an edge to have different directions. In particular it accepts the positive example of Remark 2.80: there is only one complementary component, so every hyperedge has size at most one. This is an exact classification of labelings over a fixed positive contraction and path reduction, not an intrinsic generator of all forbidden classes. Constructing $\mathcal{H}(A, D)$ still quantifies over acyclic USOs of A , and forbidden-minor minimality is a further condition.

Theorem 2.82 (Expansions over stars preserve positivity). *Let X be ample and let e be active. Suppose the one-inclusion graph of $D = X^{\{e\}}$ is a star, including a singleton or an edge. Then*

$$X \in \mathcal{D}_{\text{amp}} \iff \pi_e X \in \mathcal{D}_{\text{amp}}.$$

Consequently no active coordinate reduction of an ample forbidden pc-minor is a star. Every coordinate edge class has at least four edges; if it has exactly four, its reduction is a four-vertex path.

Proof. Put $A = \pi_e X$ and assume A positive. We induct on the number r of edges of the star D . For $r \leq 1$, the reduction is a singleton or an edge, so Theorem 2.76 applies. Assume $r \geq 2$. We will remove one leaf lift, after clearing suitable exclusive vertices, and then apply induction to the smaller star.

Choose a peeling of A and lift its deletions up to, but not including, its first vertex in D . These are exclusive corner deletions, by the corner rules used in Theorem 2.81. Write A' for the remaining

positive contraction. Component labels below are inherited from the original expansion. If the next vertex is a leaf z , it is a corner of A' . Its incident cube minus D is connected when nonempty, by the connected-exterior lemma used in Theorem 2.78. Thus all its external neighbours have one component label. Delete the opposite lift of z (or either lift if there is no external neighbour). It is a corner: z is a leaf of D and this lift has no external neighbour. The reduction becomes $D \setminus \{z\}$ and the contraction remains A' . Induction completes the peeling.

It remains to treat the case where the centre c is a corner of A' . Translate c to zero and write

$$D = \{0\} \cup \{z_i = 0^{\{i\}} : i \in E\}, \quad |E| = r.$$

All other coordinates are zero on D . Cornerhood puts the entire E -cube at zero in A' . Its vertices of E -weight at least two form a connected set: each can reach the all-one E -vertex by adding bits. Let K_* be their component in $A' \setminus D$. Every neighbour of c outside D also belongs to K_* . Indeed, for its direction $j \notin E$ and distinct $i, k \in E$, the corner cube supplies the path $0^{\{j\}}, 0^{\{j,i\}}, 0^{\{j,i,k\}}, 0^{\{i,k\}}$ entirely outside D .

Every vertex of A' with E -weight at least two belongs to K_* : join it by a geodesic to the vertex of the E -cube having the same E -trace. The geodesic keeps that trace and avoids D . Every vertex with zero E -trace, other than c , also belongs to K_* : a geodesic from c keeps that trace, starts at an external neighbour, and cannot meet D again. Hence each other component K has only unit E -traces. Connectedness forces its trace to be one fixed unit vector, since two distinct unit vectors differ in two coordinates. It meets the rest of the graph only at the corresponding leaf z_i .

These are one-vertex attachments on disjoint coordinate blocks. To see this precisely, a coordinate outside E cannot take value one in two components of $A' \setminus D$. A geodesic between such vertices would keep that coordinate one, whereas any path between the components passes through D , where it is zero. For each component K , let F_K be the set of outside- E coordinates taking value one somewhere on K . These sets are pairwise disjoint. Every vertex of $K \neq K_*$ has a nonzero coordinate in F_K , since its unit trace alone is a vertex of D . Thus the core $B = D \cup K_*$ is the coordinate-face intersection obtained by setting all F_K , $K \neq K_*$, to zero. Each attached piece $K \cup \{z_i\}$ is likewise obtained by fixing its E -trace and setting all other outside- E blocks to zero. All these pieces are positive.

Restrict an acyclic USO of A' to the pieces. At most one leaf is a sink in the induced USO of B : an ample USO has exactly one global sink, since its out-map is a bijection onto its shattered sets and only one vertex receives the empty set, by CCMW's representation-map theorem. As $r \geq 2$, choose a leaf z with an outgoing edge in B . No attached piece at z can have an outgoing edge from z , because that edge and the chosen outgoing core edge would span a missing mixed square, violating (C1). Its restricted acyclic USO therefore has sink z , so the piece peels to z by CCMW Proposition 6.12.

Peel every attached piece at z to z . Their other vertices have no neighbours outside their piece, so these are valid deletions in the whole contraction and lift to exclusive corner deletions in the expansion. The resulting contraction is the coordinate-face intersection obtained by fixing the deleted attachment blocks to zero; hence it is positive. Every remaining external neighbour of z lies in K_* and has its single label, or no such neighbour remains. Delete the opposite lift of z , or either lift in the latter case. It is a corner by the same leaf rule as above. The reduction is the smaller star $D \setminus \{z\}$, so induction completes the proof of positivity.

The converse is contraction closure. For a forbidden class the proper contraction is positive, so its reduction cannot be a star. Active reductions are nonempty, and every ample family of at most three vertices is a star. An ample family with four vertices is either a square or a tree: if it has no square, ampleness gives VC dimension at most one, and its connected one-inclusion graph is a tree.

The square is excluded by convex-reduction lifting, and the three-leaf star by this theorem, leaving only the four-vertex path. Reduction vertices correspond bijectively to edges of the coordinate class, proving the conclusions. $\square$

The theorem extends the three-vertex path case to arbitrarily many leaves. In particular the hypergraph in Theorem 2.81 has no proper two-colouring for a positive contraction. Its exact criterion remains valid, but cannot produce a negative expansion. The proof uses a full cube at the first central corner to separate all remaining attachments, and the unique global sink of a CCMW USO to choose a leaf that can be cleared.

For comparison, the sibling CompressionSchemes manuscript already proves representation-map lifting over *every* tree reduction, in Theorem `round736-tree-overlap-lifting` of `drafts/round736_tree_overlap_lifting.tex`. Its outgoing-cube labels and quotient-tree sink select a lower root and give a representation map for every component colouring. That theorem does not assert acyclicity of the lifted orientation. The star theorem above proves corner-peeling closure for its stated subfamily; it does not require a prescribed upper map to survive. For arbitrary trees this positivity implication is false, as Theorem 2.85 below shows. A branch of a longer tree can meet a complementary component at several vertices, so the one-vertex attachment argument does not establish that extension. The first possible reduction shape has four vertices and is a path; the theorem below proves that it occurs in the ample forbidden basis.

Remark 2.83 (Tree lifting does not preserve acyclicity for every choice). The acyclicity qualification above is necessary even for a four-vertex path inside a cube. Encode vertices by binary masks, with coordinate weights 1, 2, 4, and take

$$U = \{0, \dots, 7\}, \quad H = \{1, 3, 7, 6\}, \quad (r(0), \dots, r(7)) = (7, 6, 1, 4, 3, 0, 5, 2).$$

Supports also use masks. The map r is an acyclic USO: the order $(0, 1, 4, 6, 2, 3, 7, 5)$ is a corner peeling and its outgoing supports are precisely the displayed values. The reduction is the path $1 - 3 - 7 - 6$. Its complement in U is connected.

In the Round 736 construction the only vertex whose outgoing cube is contained in H is 3. The labelled blocks are $\{1\}$ and $\{6, 7\}$, and the free singleton block is $\{3\}$; the quotient is directed toward $\{6, 7\}$. Choose lower root 6, give the complementary component bit zero, and choose the free vertical-source bit at 3 to be zero. The other three vertical-source bits are one. With new coordinate weight 8, the expansion and lifted map are

v	0	1	2	3	4	5	6	7	9	11	14	15
$R(v)$	7	6	1	12	3	0	5	2	10	4	8	9.

This is a representation map by the cited tree-lifting theorem, but its orientation contains the directed cycle

$$3 \longrightarrow 11 \longrightarrow 15 \longrightarrow 14 \longrightarrow 6 \longrightarrow 2 \longrightarrow 3.$$

Each arrow follows by testing its coordinate in the displayed out-map. Changing just the free bit at 3 exchanges its two assignments: $R(3) = 4$ and $R(11) = 12$. All vertical edges now point from the upper copy of H to the lower cube. The upper tree and lower cube are acyclic, so this modified map is acyclic. Independently, the expansion peels in the order $(9, 11, 15, 14, 0, 1, 4, 6, 2, 3, 7, 5)$. Thus the example refutes preservation for arbitrary certified choices; it does not refute existence of an acyclic choice or tree-reduction closure. The script `research/tree_lift_acyclicity.py` also checks the maps by nonclashing, outgoing cubes, and unique sinks on every contained cube.

Proposition 2.84 (Canonical free vertical directions in a tree lift). *Let $\emptyset \neq H \subseteq U$ be ample, with $G(H)$ a tree, and fix a representation map r of U . Put*

$$Q_h = Q(h, r(h)), \quad Z = \{h \in H : Q_h \subseteq H\}.$$

For $h \notin Z$, let W_h be the component of $U \setminus H$ containing $Q_h \setminus H$. Contract the edges of H whose endpoints lie outside Z and have the same label W_h , retaining every vertex of Z as a singleton. Let K_ be the sink block of the quotient oriented by r , as supplied by the Round 736 tree-lifting theorem. Fix $q \in K_*$, root H at q , and let $p(h)$ be the parent of $h \neq q$. Write s_q for the out-map of this rooted tree.*

Fix component bits t_W and use a fresh coordinate x to form

$$X_t = (H \times \{0, 1\}) \cup \bigcup_W (W \times \{t_W\}).$$

The cited theorem supplies a representation map R_b for every choice of source bits

$$b(h) = 1 - t_{W_h} \quad (h \notin Z), \quad b(h) \in \{0, 1\} \quad (h \in Z),$$

by the assignments

$$\begin{aligned} R_b(u, t_W) &= r(u) & (u \in W), \\ R_b(h, 1 - b(h)) &= r(h) & (h \in H), \\ R_b(h, b(h)) &= s_q(h) \cup \{x\} & (h \in H). \end{aligned}$$

Define a canonical choice b_ by retaining the forced bits, setting $b_*(q) = 0$ if $q \in Z$, and setting*

$$b_*(h) = b_*(p(h)) \quad (h \in Z \setminus \{q\}),$$

in increasing distance from q . Then

$$\text{some } R_b \text{ is acyclic} \iff R_{b_*} \text{ is acyclic.}$$

In particular, for fixed r and t , existence of an acyclic map within this construction requires at most $|K_|$ cycle tests, one for each root; there is no need to search the $2^{|Z|}$ free-bit assignments. Moreover, moving a successful root along an outgoing r -edge inside K_* preserves success. Thus it suffices to test the sinks of r restricted to K_* .*

Proof. The tree-lifting theorem states that r and s_q agree on every uncontracted tree edge. For $h \in Z$, the outgoing cube of r lies in a tree, so its support is empty or a singleton. Every edge incident to this singleton block is uncontracted. It follows that

$$r(h) = s_q(h) = \{\text{direction of } hp(h)\} \quad (h \in Z \setminus \{q\}),$$

and $r(q) = s_q(q) = \emptyset$ when $q \in Z$. Consequently changing a free bit changes only that vertical edge: all horizontal directions stay fixed. At either lift of a free nonroot vertex, the only outgoing horizontal edge goes to the corresponding lift of its parent.

Consider a free $h \neq q$ whose bit differs from its parent's. Put $a = b(p(h))$ and change $b(h)$ to a . Suppose this creates a directed cycle in a previously acyclic orientation. That cycle must use the new vertical edge, and its next edge is forced. Thus it contains

$$(h, a) \longrightarrow (h, 1 - a) \longrightarrow (p(h), 1 - a).$$

Replace this segment by

$$(h, a) \longrightarrow (p(h), a) \longrightarrow (p(h), 1 - a).$$

Both replacement edges existed before the change: the first is the fixed outgoing horizontal edge at h , and the second is the parent's vertical edge with source bit a . Every other edge of the cycle also existed before the change. The replacement therefore gives a nonempty directed closed walk in the old orientation, which contains a directed cycle, a contradiction. Matching a free bit to its parent's thus preserves acyclicity. Representation-map validity throughout is supplied by the original tree-lifting theorem.

If $q \in Z$, its two lifts have no outgoing horizontal edge. Reversing its vertical edge cannot create a cycle, since the new head has no outgoing edge. Set that bit to zero first. Then process the remaining free vertices in increasing distance from q , matching each bit to its parent's already fixed bit. Each step preserves acyclicity and the final assignment is b_* . Hence an acyclic R_b implies an acyclic R_{b_*} . The reverse implication is immediate, since b_* is an allowed choice. Repeating this single canonical test for each $q \in K_*$ proves the first bound on the number of tests.

To prove the root reduction, the case $|K_*| = 1$ is immediate. Otherwise K_* has one exterior label W , so its source bits all equal $a = 1 - t_W$. The canonical bits do not depend on the root within K_* : outside that subtree, paths toward any such root have the same initial segment up to their first vertex in K_* , where the prescribed bit is always a . Suppose $q \rightarrow q'$ is an r -edge inside K_* and move the root from q to q' . This reverses only the lower edge over qq' , from $(q', a) \rightarrow (q, a)$ to $(q, a) \rightarrow (q', a)$. All other edges stay fixed. If a cycle is created, it uses this new edge. The new lower root (q', a) has only its vertical edge outgoing, so the cycle contains

$$(q, a) \longrightarrow (q', a) \longrightarrow (q', 1 - a).$$

Replace this segment by

$$(q, a) \longrightarrow (q, 1 - a) \longrightarrow (q', 1 - a).$$

The latter edges existed before the root move: the first is vertical, and the second is the upper edge oriented by r . The old orientation would therefore contain a directed closed walk, a contradiction. Starting with any successful root, follow outgoing r -edges in K_* . Since its underlying graph is a finite tree, this ends at a sink, and every root along the path remains successful. This proves the final claim. $\square$

This normalization removes the vertical-choice part of the tree problem; it does not prove that a successful root always exists. It strengthens the earlier example's one-bit repair to a uniform operation on every tree lift. Round 736 supplies the maps and their quotient-tree structure; the additional argument here is that the indicated free-edge reversals cannot create cycles. The earlier general orientation-gluing criterion tests prescribed upper-layer assignments, whereas this proposition removes those free assignments for the certified tree roots. An acyclic r need not have any successful canonical root: Theorem 2.85 below gives a negative expansion with positive contraction, and hence refutes that existence claim.

Theorem 2.85 (A path reduction does not preserve corner peeling). *There is an ample forbidden pc-minor X for $\mathcal{D}_{\text{amp}}$ with 909 vertices in 37 active coordinates and an active coordinate e such that*

$$X \notin \mathcal{D}_{\text{amp}}, \quad \pi_e X \in \mathcal{D}_{\text{amp}}, \quad X_{e=0}, X_{e=1} \in \mathcal{D}_{\text{amp}},$$

while $X^{\{e\}}$ is the four-vertex path $2 - 0 - 4 - 5$ in binary-mask notation. In particular, expansions over tree reductions need not preserve corner peeling. The class X itself has a coordinate edge class of exactly four edges, so the bound in Theorem 2.82 is sharp.

Proof. We first describe the construction in terms of three pieces and their boundary conditions. The obstruction is the first attempted deletion from the doubled path. We then specify the finite inputs and prove positivity of every elementary pc-minor, grouping the checks by coordinate block.

The three inputs. Let B be positive ample with unique corner b , containing the induced path $H = a - b - c - d$. Denote its successive directions by f, g, h . Assume B has an acyclic USO with sink a and out-map $r_B(d) = \{h\}$. Let (A, a) be positive ample but not peelable to a , using a fresh coordinate block outside its attachment a . Let P be positive ample, meeting B exactly in $\{c, d\}$ and using a fresh coordinate block apart from the shared direction h . Require

$$t_c(P) = 1, \quad \ell_d(P) = 0.$$

The exclusive coordinate blocks of A and P are disjoint. Put

$$U = B \cup A \cup P, \quad S = H \cup A \cup P, \quad X = (B \times \{0\}) \cup (S \times \{1\}).$$

Thus $\pi_e X = U$ and $X^{\{e\}} = B \cap S = H$.

The coordinate-block gluing results make U and S ample. They are also positive. For U , glue an arbitrary acyclic USO of A to the prescribed map of B , which has sink a at their shared vertex. The resulting map still has out-map $\{h\}$ at d . A map of P with sink c then glues along cd , by Theorem 2.30. For S , first peel P to c , delete c and then b , and finish with any peeling of A . The lower section B is positive by hypothesis.

To check ampleness of X , use the section shadow identity. Within the coordinates of B , the shadow of S contains only the empty support and $\{f\}, \{g\}, \{h\}$: the first two come from the path $a - b - c$, and the third from the shared edge in P . All other coordinates of A and P are exclusive. Consequently

$$\text{Sh}(B) \cap \text{Sh}(S) = \text{Sh}(H) = \{\emptyset, \{f\}, \{g\}, \{h\}\}.$$

The old-coordinate shadow of X is $\text{Sh}(U)$, and its supports containing e are obtained by adjoining e to this intersection. Since U is ample, $|\text{Sh}(X)| = |U| + 4 = |X|$, proving ampleness.

Why no peeling can start deleting the doubled path. Suppose X has a complete peeling and consider its first deletion from $H \times \{0, 1\}$. No vertex of the lower copy of B has previously been deleted. Indeed, the first such deletion would restrict to a corner deletion of B , whose only corner is b , already in the doubled path. Both internal path vertices have noncorner projections in H , so neither of their lifts can be the first doubled-path corner. Each endpoint a, d has a neighbour in $B \setminus H$: otherwise its only neighbour in B would be its path neighbour, making it a corner of B , contrary to uniqueness of b . These neighbours are still present on the lower shore and have no upper mates. They prevent the lower lifts of a, d from being corners while their vertical mates remain.

If $(a, 1)$ is first, it can have no remaining neighbour in $A \setminus \{a\}$, since the lower mate of such a neighbour is absent. Restrict the preceding deletions to the upper copy of A . They are corner deletions retaining a ; the remaining class is ample and connected. With a isolated, it is the singleton $\{a\}$, contradicting the failure of peeling A to a . Finally, if $(d, 1)$ is first, it has no remaining exclusive neighbour in P . Restrict the complete peeling to the upper copy of P . It deletes d before c , and d then has exactly the neighbour c . This witnesses $\ell_d(P) = 1$, the other contradiction. These exhaust all possible first doubled-path deletions, proving $X \notin \mathcal{D}_{\text{amp}}$.

Explicit realization. Take for B the 289-vertex class C of Theorem 2.45, and put

$$(a, b, c, d) = (2, 0, 4, 5), \quad (f, g, h) = (2, 4, 1)$$

as coordinate masks. Its only corner is zero. The certificate in `research/four_vertex_path_obstruction.json` is a complete corner order ending at 2, in which 5 is deleted with sole remaining

neighbour 4. Take A to be a disjoint-coordinate copy of $(C, 0)$ attached at 2. Take P to be the 330-vertex piece of Theorem 2.45, identifying its lower endpoint with 4 and its upper endpoint with 5. Its established profile $(0, 1, 0, 1, 0)$ gives exactly $t_c(P) = 1$ and $\ell_d(P) = 0$. Thus all hypotheses of the construction hold.

More explicitly, keep the twelve bits of B , embed A by $v \mapsto 2 \oplus (v \ll 12)$, and embed P by

$$v \mapsto 4 \oplus ((v \bmod 4096) \ll 24) \oplus \lfloor v/4096 \rfloor.$$

Here $\oplus$ is bitwise exclusive-or and $v \ll k = 2^k v$. Use bit 2^{36} for the new coordinate. The resulting sizes are $|U| = 289 + 289 + 330 - 3 = 905$, $|S| = 620$, and $|X| = 289 + 620 = 909$. The verifier replays the core, contraction and section orders and checks the contraction map independently by support bijection and nonclashing. It also checks the exact section shadows. Negativity uses the first-deletion argument above and the previously proved profile, not an unsuccessful search.

Every proper minor is positive. It suffices to check contractions and the two sections of each active coordinate, since $\mathcal{D}_{\text{amp}}$ is pc-minor closed. We use four coordinate groups. The new coordinate e gives the three positive classes U, B, S already certified above.

For any of the three path directions f, g, h , the resulting reduction along e is a proper minor of H , hence a star or empty. To justify this, reduction along e commutes with operations on another coordinate j in an ample class. Restrictions commute immediately. For contraction, the only nontrivial inclusion arises when the projected e -mates are represented on opposite j -sides. These representatives have Hamming distance two; isometry supplies an intermediate vertex of their square, and hence an actual e -pair on one j -side. Thus $(\pi_j X)^{\{e\}} = \pi_j(X^{\{e\}})$. If e remains active, its contraction is a positive minor of U , so Theorem 2.82 applies. If e is constant, the class is a minor of the corresponding positive section of X . An empty reduction cannot occur with both sections nonempty: an ample class is connected and would then have an e -edge.

Consider a coordinate exclusive to A . Its section away from the attachment lies in A and is positive. Contraction or the section retaining a replaces A by a proper rooted minor A' , which peels to a by the rooted-factor property of $(C, 0)$. Perform that relative peeling on the upper shore. Each deleted vertex is outside the attachment and has no neighbour in the rest of the class, so its corner property is preserved. Then $(a, 1)$ is a corner: its incident directions are f, e , and their square is present. Delete it. The remaining reduction is $H \setminus \{a\}$, a three-vertex path, and the contraction is $B \cup P$. This contraction is positive by gluing the prescribed map of B , with leaf exit at d , to the sink- c map of P . The star theorem finishes the peeling.

For a coordinate exclusive to P , the section away from cd is a positive section of P . Contraction or the section retaining cd replaces P by a piece P' which peels to that edge. These are precisely the 24 relative orders verified in Theorem 2.45. Peel its upper exclusive vertices, then delete $(d, 1)$, whose two incident directions h, e span a present square. The remaining reduction is $H \setminus \{d\}$; the contraction $B \cup A$ is positive by vertex gluing, since B has a map with sink a . Again the star theorem applies.

Finally, a coordinate exclusive to B is one of bits $3, \dots, 11$. Its section away from H lies in B and is positive. For each contraction and each section retaining H , the resulting B' admits a relative peeling to H . The 18 explicit orders are stored and replayed in `research/four_vertex_path_minors.json`; their completed path orders also pass independent out-map bijectivity and nonclashing checks. Perform such a relative peeling on the lower shore. Its deleted vertices have no vertical mates or neighbours in the upper shore, so these are also corners of the full class. The remaining lower path can then be deleted in the order a, b, c, d . At each step its endpoint and vertical edge span a square, or only its vertical edge remains. The upper shore is now S , whose positive order was already established.

This covers all 37 coordinates and all 111 elementary operations. The replay additionally supplies complete orders for 109 of these operations; only contraction of bits 1 and 2 uses the star theorem without an explicitly stored full order. The structural argument above covers those two as well. Thus every proper pc-minor is positive, while X is negative. Its reduction H has four vertices, so its e -class has exactly four edges, proving sharpness with this explicit forbidden class. $\square$

This also refutes successful-root existence for the prescribed tree lift, rather than only the arbitrary-choice rule of Remark 2.83. For the certified contraction map above, the exterior labels on a, b, c, d are respectively the A piece, the B component at b , the same B component at c , and the P piece. The quotient sink block is the singleton $\{a\}$, and no vertex is free. The unique certified lift is cyclic; the verifier records an explicit cycle and checks its representation map. More fundamentally, no alternative acyclic map can exist on this expansion, by the theorem. The normalization proposition remains valid, but its remaining existence hypothesis is false.

The example does not imply that every four-edge coordinate class has the same three-piece presentation. There is, however, a canonical bounded separator for the entire four-edge branch. A coordinate face here means its intersection with the class, and may be a proper subset of its ambient cube.

Theorem 2.86 (Eight face types for a four-edge coordinate class). *Let X be an ample forbidden pc-minor with a coordinate e having exactly four edges. Normalize its reduction to $H = \{0, 1, 3, 7\} \subseteq Q_F$, where F consists of the three successive path directions, and make all other coordinates of H zero. Let D be the coordinate-face intersection of X obtained by fixing all coordinates outside $F \cup \{e\}$ to zero. Then D separates X , and $9 \leq |D| \leq 12$. Up to reversing the path and interchanging the e -sections, it is one of the following eight types. The last column bounds the number of pieces A_i in which D is not gated:*

$$D = ((H \cup L) \times \{0\}) \cup ((H \cup R) \times \{1\}).$$

L	R	$ D $	nongated pieces, at most
$\{2\}$	$\emptyset$	9	2
$\{2, 6\}$	$\emptyset$	10	1
$\{2, 5\}$	$\emptyset$	10	1
$\{2\}$	$\{5\}$	10	1
$\{2, 4, 6\}$	$\emptyset$	11	1
$\{2, 4, 5\}$	$\emptyset$	11	1
$\{2, 6\}$	$\{5\}$	11	0
$\{2, 4, 5, 6\}$	$\emptyset$	12	0

Every component W_i of $X \setminus D$ lies in one e -section. Both sections contain such components. The pieces $A_i = D \cup W_i$ are proper positive coordinate restrictions, use disjoint exclusive blocks, and fail to peel to D . If m_t is the number in section t , then

$$1 \leq m_t \leq |D_{e=t}|, \quad 2 \leq m_0 + m_1 \leq |D| \leq 12.$$

The specified coordinate determines D and its pieces uniquely, apart from the displayed normalization and permutation of the pieces.

Proof. The star-reduction theorem makes the four-vertex reduction a path rather than a star. A tree in a partial cube uses distinct coordinates on its edges, giving the normalization above. Write $U = \pi_e X$ and let K be its intersection with the F -face containing H . Then K is ample, $H \subseteq K \subseteq Q_F$, and D has contraction K and reduction H . In particular $|D| = |K| + 4 \leq 12$.

We first record the elementary geometry of this three-dimensional face. The complement of H in Q_F is the induced geodesic path

$$J = 2 - 6 - 4 - 5.$$

For an ample T with $H \subseteq T \subseteq Q_F$, complement duality says that $Q_F \setminus T$ is ample. If nonempty it is connected, so it is an interval of J . Conversely every interval of J is ample, since it is a geodesic path; its cube complement is therefore ample. There are exactly eleven possibilities, including the empty complement. Moreover every such T peels to H : enlarge its complementary interval one endpoint at a time until it is all of J . If the complement is initially empty, start at any vertex of J . Every intermediate complement, and hence every remaining T , is ample, so the successive single deletions are corner deletions.

Every vertex outside D has a nonzero coordinate outside $F \cup \{e\}$. No e -edge has an endpoint outside D , since all its projected pairs belong to H . Thus each outside component stays in one section. Suppose section t has no outside vertices. It is then an ample three-dimensional family containing H , and the preceding interval argument peels it to H . These relative deletions also work in X : the deleted vertices have no e -mates, so their full neighbourhoods are unchanged. The remaining class is a peripheral expansion of the other section at H . That section is positive by forbidden-minor minimality, and H is positive, so Proposition 2.6 makes the remainder positive. This contradicts nonpeelability of X . Both sections therefore have outside components, and D separates X .

Apply Proposition 2.27. It gives the canonical disjoint blocks, positivity of the pieces, and their failures to peel to D . For a component in section t , this failure is equivalent to the failure of its section piece $D_{e=t} \cup W_i$ to peel to $D_{e=t}$: each vertex deleted in a relative peeling is outside D , and has no vertical mate, so the corner tests in the two families coincide throughout. The whole section $X_{e=t}$ is positive. Applying the positive part of that proposition to its coordinate face $D_{e=t}$ gives $m_t \leq |D_{e=t}|$, and summing gives the stronger bound $m_0 + m_1 \leq |D|$.

If $K = H$, then H is a coordinate face of U , hence convex. Since U is positive, Theorem 2.76 would make X positive. Therefore $K \neq H$ and $|D| \geq 9$. To list D , its omitted set $Q_F \setminus K$ is an interval of J , or empty. The components of $K \setminus H$ are consequently the prefix and suffix of J left by that interval, omitting empty ones; when the interval is empty, all of J is one component. Each component has a single e -label by the component-expansion theorem (Theorem 5.4). Conversely that theorem makes every such labeling ample.

There are 26 nonprism labeled faces: 4 with one added vertex, 8 with two, 12 with three, and 2 with four. Reversing J and swapping the two labels leaves respectively one, three, three, and one types, which are exactly the table. Thus the list is exhaustive; it is a list of necessary face types, not a claim that every type occurs in a forbidden class. The literal 81 assignments of absent, lower, or upper status to the four vertices of J are also checked in `research/four_edge_face_types.py/json`. Finally the face and graph components were recovered without choices, proving uniqueness.

It remains to prove the last column. Project X onto $F \cup \{e\}$, obtaining an ample family V containing D . Its e -reduction is still H . Indeed, a new e -edge over $u \notin H$ would have preimages in X with the same F -trace u and opposite e -bits. A geodesic between them keeps that F -trace and crosses an e -edge of X , whose trace would have to belong to H , a contradiction. Put $K' = \pi_e V$. The complements $I = Q_F \setminus K$ and $I' = Q_F \setminus K'$ are intervals of J , possibly empty, with $I' \subseteq I$.

The vertices of $V \setminus D$ project bijectively onto $I \setminus I'$: no extra lift can be added over $K \setminus H$, since that would enlarge the e -reduction. Each component of $K' \setminus H$ has a single e -label, so the induced graphs on $V \setminus D$ and $I \setminus I'$ are isomorphic. The difference of two nested intervals has at most two components; when $|I| \leq 2$, it has at most one. Since $|I| = 12 - |D|$, Proposition 2.28 gives the first six bounds and the bound zero for $|D| = 12$. For the seventh row, $|I| = 1$ and the two components

of $K \setminus H$ already have different e -labels. If I' were empty, $K' = Q_F$ and $K' \setminus H = J$ would be connected and require a single label. Thus $I' = I$, $V = D$, and every piece has D gated. $\square$

For the 909-vertex class, this face is the first type and has nine vertices. Its three outside components have sizes 284, 288, 328. Their projections outside the canonical face contain respectively three, zero, and zero vertices. Thus precisely the first of these pieces fails to have D gated. The finite projection check in `research/four_edge_face_projections.py/json` verifies this statement and all containments between the 26 labelled nonprism faces. The bound and recovery apply to every forbidden class with a four-edge coordinate, whereas that example establishes realization of only one type. The pieces still carry relative-peeling conditions. These cannot in general be replaced by separate attainable-root and retained-edge tests, even for the smallest face type. Indeed, take Y_2 from Theorem 2.38, with its retained square, and attach a leaf at one square vertex in a fresh coordinate. Call the result T and the square together with this leaf K . Then K has the five-vertex shape $H \cup \{2\}$ in the first row of the table. The family T is positive and peels to each vertex and edge of K , but not to K : retaining K would retain the square in Y_2 . For a square edge, delete the leaf and use its certified retention; for the new leaf edge, peel Y_2 to its attaching root. Vertex retention follows from these orders. The peripheral expansion $(T \times \{0\}) \cup (H \times \{1\})$ is a positive one-sided piece with common face $(K \times \{0\}) \cup (H \times \{1\})$. It fails relative peeling to that face, and its outside part is connected for the displayed input. Its proper exclusive-coordinate images do peel to the face, by the relative-minor assertion of that theorem. This is an admissible local piece, not a claim that it occurs in a forbidden assembly. Generating their attainable USO data intrinsically, and proving compatible forward assembly, remain necessary before this decomposition becomes a generator of even this branch. Forbidden classes whose coordinate classes all have more than four edges are not covered by the eight-type assertion.

The separate source question is whether a rooted factor whose every peeling starts at its root can have another geometric corner. Low-rank extension invariance forces every such nonroot corner to have degree at least three. Moreover a degree-three corner cannot be adjacent to another corner: deleting both leaves an ample class whose two individual additions have the same support of size two, and square-pair invariance makes that class positive. Either single deletion is then positive, contradicting the required first vertex. These observations apply the earlier invariance results; they do not exclude higher-degree or nonadjacent additional corners.

The twelve stored positive one-concept extensions of the 291-concept obstruction do not supply a counterexample: each has complete peelings starting at two different vertices. The explicit orders are replayed by `research/unique_source_controls.py` and stored in its JSON report. This is a test of that fixed family only; the general source question remains open. It is not a prerequisite for the joint descent, whereas intrinsic forward generation remains open.

Theorem 2.87 (Endpoint transfer for a partially replicated tip). *Let $(B, 0)$ be a rooted factor using every coordinate of F . Let $\alpha, \delta, \gamma \in Q_F \setminus B$ be distinct ample one-concept additions with pairwise distinct acquired supports $P_\alpha, P_\delta, P_\gamma$. Suppose that $A = B \cup \{\alpha\}$ and $D = B \cup \{\delta\}$ both peel to zero. For two fresh coordinates q, r , put*

$$L = \{01, 10, 11\}, \quad S = \{(0, 00)\} \cup (A \times \{01\}) \cup (D \times \{10\}) \cup (B \times \{11\}),$$

and, for $R \subseteq L$, define

$$S_R = S \cup (\{\gamma\} \times R), \quad C = B \cup \{\gamma\}.$$

With $\text{Sh}(\emptyset) = \emptyset$, one has

$$\text{Sh}(S_R) = \text{Sh}(S) \dot{\cup} \{P_\gamma \cup J : J \in \text{Sh}(R)\}. \quad (2.40)$$

Thus S_R is ample exactly when $R \neq \{01, 10\}$. For every such R , the family S_R is ordinarily peelable, all its proper root-preserving pc-minors peel to their roots, and

$$S_R \text{ peels to } (0, 00) \iff |R| \geq 2 \text{ and } C \text{ peels to } 0. \quad (2.41)$$

Consequently $(S_R, (0, 00))$ is a rooted factor precisely when the right-hand side is false. The root is always a degree-two corner.

Proof. The compatible-extension union formula (6.20) applies to every subset of the three old-coordinate tips. They are pairwise nonadjacent by Lemma 6.7. For old supports, adding a nonempty row acquires exactly P_γ . A support containing q and not r is shattered when its old part is shattered by both projected q -sections. The only possible new old part is P_γ , and it occurs on both sides exactly when R shatters $\{q\}$. Interchanging q, r gives the other one-coordinate term. A support using both new coordinates has empty old part, since the 00 section is $\{0\}$; it is already shattered by S . The row R never shatters $\{q, r\}$. These observations prove (2.40), including the empty row.

The shadow of a singleton has size one; that of either edge of L has size two; and both L and its diagonal pair $\{01, 10\}$ have the three supports $\emptyset, \{q\}, \{r\}$. Since $|S_R| = |S| + |R|$, the Sandwich equality proves the ampleness assertion. Each allowable row can be built from the empty row by ample one-concept additions. It can also be enlarged to L by such additions. For an edge, add either endpoint first; for L , use a singleton, an incident edge, and then its third vertex.

We prove the endpoint equivalence next. If $|R| \leq 1$, the intersections of the 11 section with each adjacent section are both exactly B . Suppose that a peeling ends at $(0, 00)$, and consider the first deletion among $(0, 01), (0, 10), (0, 11)$. The full square at old zero is still present. As in the proof of Proposition 2.63, the selected corner cannot have any old-coordinate neighbor: its corner cube would require that neighbor in the singleton 00 section. Its current ample section is connected, and is therefore $\{0\}$. Choose the overlap along its edge to 11, or either edge if the selected vertex is $(0, 11)$. Initially this overlap is B . Recording the first destruction of each full edge fibre gives corner deletions of the overlap; the fibre of zero remains intact until the selected step. This would peel B to zero, a contradiction. This argument uses the two overlap equalities and does not require section containment.

The full-row family S_L is the paired construction over C , with sides $C \cup \{\alpha\}$ and $C \cup \{\delta\}$. Proposition 2.63 gives its endpoint equivalence with C . If C fails at zero, no allowable S_R can peel to its root: prefacing such a peeling by the reverse of an ample addition chain from S_R to S_L would give a peeling of S_L to its root.

It remains to prove sufficiency for an edge row when C peels to zero. Let $R = \{10, 11\}$. Delete $(\delta, 10)$, then use a peeling of C to zero to delete all nonzero words in its 10 copy. Every corner cube lifts to the untouched 11 copy of C , and no deleted word has a 00 mate. Delete $(\gamma, 11)$; it is now a corner with no mate in either adjacent section. Next delete $(0, 10)$, whose remaining incident cube is the square at old zero. The 11 section is now B . Peel it by an ordinary peeling of B ; its corner cubes lift into A at 01. What remains is A at 01 with the root attached at old zero. Peel A to zero and finish at $(0, 00)$. For $R = \{01, 11\}$, interchange q, r, A, D , and α, δ throughout this construction. The required endpoint order is then that of D . This proves (2.41) in every case.

The initial S is a rooted factor by Proposition 2.63. Every allowable S_R lies above it in an ample one-concept chain on the same active coordinates. Reversing the additions proves ordinary peelability. If the endpoint test is false, Lemma 2.68 proves rooted-factor status; if it is true, root-preserving pc-minor closure supplies the stated positive orders in all proper rooted minors. Finally, the root's two incident edges and their square are unchanged. $\square$

Remark 2.88 (Two individually ineffective additions restore the endpoint). Take the 289-concept core B and tips $\alpha = 128$, $\delta = 290$ from Remark 2.67. A third ample addition $\gamma = 604$, corresponding to the original tip 577, acquires support $\{5, 6, 9, 10\}$, with bitmask 1632, and peels to zero. Its endpoint order is stored in `computations/round793_third_endpoint_repair.json`. For the resulting 870-concept rooted factor S , put

$$u = (\gamma, 10), \quad v = (\gamma, 11).$$

Both $S \cup \{u\}$ and $S \cup \{v\}$ are rooted factors with 871 concepts. Yet $S \cup \{u, v\}$ peels to the original root. In the encoding $(x, qr) \mapsto 4x + q + 2r$, the new words are 2417, 2419. Each individual addition acquires support 6528, of size four; adding the second word after the first acquires 6530, of size five. The pair is adjacent, and the two initial supports coincide. This does not establish cooperation for additions with distinct initial supports.

The checker `verify_damp_partial_tip_endpoints.py` tests all eight occurrence rows for this repairing tip and for the nonrepairing tip 64 from Remark 2.90. It checks the shadow formula, the two nonample diagonal rows, complete negative graphs for the other negative cases, and explicit endpoint orders for both edge rows and the full row at 604. It also replays all 28 rooted elementary orders of S and all 84 orders of its three 871-concept extensions. The report is `damp_partial_tip_endpoints_verification.json`. Thus individual endpoint statuses do not suffice for simultaneous ample additions, even over a rooted factor. The theorem supplies the exact occurrence-row condition in this construction; the intrinsic criterion for whether $B \cup \{\gamma\}$ repairs zero remains open.

Proposition 2.89 (The opposite section need not be contained in both sides). *Let $(B, 0)$ be a rooted factor using every coordinate of F . Let $x, \alpha, \delta \in Q_F \setminus B$ be distinct concepts whose one-concept extensions are ample and acquire pairwise distinct shattered supports P_x, P_α, P_δ . Suppose that $B \cup \{x\}$ does not peel to zero, whereas $B \cup \{\alpha\}$ and $B \cup \{\delta\}$ do. Then, on two fresh coordinates q, r , the family*

$$\begin{aligned} Z = \{(0, 00)\} \cup ((B \cup \{\alpha\}) \times \{01\}) \\ \cup ((B \cup \{x, \delta\}) \times \{10\}) \cup ((B \cup \{x\}) \times \{11\}) \end{aligned} \quad (2.42)$$

is a rooted factor with a degree-two corner root $(0, 00)$. Its opposite section $C = B \cup \{x\}$ satisfies

$$C \setminus A = \{x\}, \quad C \subseteq D, \quad A \cap D = B, \quad A \setminus C = \{\alpha\}, \quad D \setminus C = \{\delta\},$$

where A, D denote its 01, 10 sections. Deleting $(x, 11)$ and then $(x, 10)$ preserves rooted-factor status and leaves the paired-extension construction over B .

Proof. Apply Theorem 2.87 with $\gamma = x$ and $R = \{10, 11\}$. Since $B \cup \{x\}$ does not peel to zero, the theorem makes this family, its singleton-row predecessor, and its empty-row predecessor rooted factors. Reversing the two row additions gives the stated corner deletions. The section differences follow directly from distinctness of the three tips and their exclusion from B . $\square$

Remark 2.90 (A noncontained section in a root-protected graph three-core). Take the translated 289-concept factor of Remark 2.67 as B , and use

$$x = 64, \quad \alpha = 128, \quad \delta = 290, \quad (P_x, P_\alpha, P_\delta) = (70, 390, 480)$$

in bitmask notation. The new support 70 is $\{1, 2, 6\}$. The family $B \cup \{64\}$ has only the corner 64; deleting it leaves B , whose only corner is zero. It therefore cannot peel to zero. The two repairing endpoints were verified in that earlier remark, so Proposition 2.89 applies.

The resulting Z has 872 concepts, dimension 14, and VC dimension four. Encode (v, qr) by $4v + q + 2r$. Its corners are precisely

$$0, \quad 257, \quad 259, \quad 514, \quad 1161.$$

The root has degree two and the four other corners have degree four. Every nonroot vertex has degree at least four. Thus the graph is unchanged by repeatedly deleting nonroot vertices of degree at most two, with the root protected. Nevertheless its 11 section contains the word 64, absent from its 01 section. Rooted minimality therefore does not force the containment required in Remark 2.65, even after this graph reduction and even when both side differences from the opposite section are singletons.

The fixed replay `verify_damp_opposite_section_noncontainment.py` checks all eight compatible tip unions and the ample chain of orders 870, 871, 872, 873. Its successive shadow gains have bitmasks 280, 282, 281. Complete root-protected negative graphs have 4, 8, 16, 28 states, respectively. It also checks ordinary peelings, all 24 rooted elementary orders of B , and all 28 of Z . The data are stored in `damp_opposite_section_noncontainment_verification.json`. The two preserving deletions 259, 257 return Z to the known 870-concept factor. Thus this example refutes automatic section containment but does not establish another terminal assembly type. In fact the full sectionwise endpoint law (2.35) still holds for this Z , despite its failure of containment. To verify this, let c be the stored ordinary order of B . The following prefixes, followed by the indicated complete copies of c , give relative orders:

prefix	layers cleared in order	retained family
259, 257, 1161, 0	$4B + 1, 4B + 3$	$4A + 2$
259, 514, 0	$4B + 2, 4B + 3$	$4D + 1$
259, 257, 514, 1161, 0	$4B + 1, 4B + 2$	$4B + 3$

Here $4F + t = \{4x + t : x \in F\}$. Each order is checked corner by corner in `research/noncontained_endpoint_transfer.py/json`. The first two orders prove sufficiency for every endpoint attainable in its side section. For the opposite section $C = B \cup \{64\}$, every peeling starts at its unique corner 64, so its attainable endpoints are exactly those of B . The third order therefore proves sufficiency there too. Necessity follows in each case by coordinate-face restriction.

Every unattainable vertex other than zero consequently fails in its own proper section; zero is the only critical root of Z . Thus the endpoint law and critical-root uniqueness do not imply the paired presentation's containment hypotheses. At the critical root, the two incident coordinates fix its four sections, so choosing another presentation cannot repair this failure. Intrinsic rooted-factor and endpoint-repair admissibility, and the full forbidden-pc-minor classification, remain open.

Proposition 2.91 (An antipodal corner pair does not restore the endpoint). *Let (B, a) be a rooted factor using every coordinate of F , and suppose that a is the only corner of B . Let $H = B \cup \{c\} \subseteq Q_F$ be an ample one-concept extension, with acquired support P and completed cube $Q = \{c^U : U \subseteq P\}$. Define*

$$E = \{v \in Q \setminus \{a, c\} : v^{\{e\}} \notin H \text{ for every } e \in F \setminus P\}. \quad (2.43)$$

Then $E = \text{Cor}(H) \setminus \{a, c\}$, and every member of E has Hamming distance at least two from c . Every peeling of H ending at a must start in E . If $E = \emptyset$ or $E = \{c^P\}$, then (H, a) is another rooted factor. In either case, every ordinary peeling of H deletes a among its first two concepts. Consequently, an endpoint-restoring addition must have some $v \in E$ with $2 \leq d_H(c, v) < |P|$.

Proof. The added concept is a corner whose unique maximal cube is Q . Any old concept that becomes a corner must have a corner cube containing c , since otherwise that cube already made it a

corner of B . Its cube is therefore Q . A vertex $v \in Q$ is a corner exactly when it has no neighbor outside Q . Indeed, all directions of P are incident with v . An additional direction e in its corner cube would force $c^{\{e\}} \in H$, contrary to the incident support of c . With no additional direction, Q is its full incident cube. Since a is the only old corner, this proves the formula for E .

If $v \in E$ were adjacent to c , deleting c would leave v as a corner of the opposite face of Q . This would be a corner of B different from a , a contradiction. A peeling ending at a cannot start at a ; it cannot start at c , since that leaves B , which fails at a . Its first concept must therefore belong to E .

If E is empty, this already excludes the endpoint. Suppose now that $E = \{d\}$, where $d = c^P$. The distance assertion gives $|P| \geq 2$. The only possible nonroot first deletions are c, d . Deleting c leaves only the corner a . We claim that deleting d likewise leaves no corner other than possibly a . The vertex c retains all directions of P , since it is not adjacent to d , but their cube is missing d . Thus c is no longer a corner. Any newly created corner must be a neighbor of d : deletion of a nonneighbor changes no incident direction and cannot fill a missing cube.

All neighbors of d lie in Q . Write one different from a as $w = d^{\{e\}}$, with $e \in P$. It is neither c nor d , and is not in E , so it has an outside neighbor $w^{\{f\}} \in H$, with $f \notin P$. After deleting d , it retains that neighbor and every direction in $P \setminus \{e\}$. Because c, d are antipodes, $c = w^{P \setminus \{e\}}$. A corner cube on these retained directions would therefore contain $c^{\{f\}}$, which is absent from H . Hence w cannot be a corner. This proves the claim, including every possible new nonroot corner.

In both cases, a first deletion different from a forces a to be the second deletion in any ordinary peeling. The endpoint a is therefore impossible. The family H does peel ordinarily, by deleting c and then peeling B . Lemma 2.68 now makes (H, a) a rooted factor. All tests in (2.43) are incident-edge tests on the completed cube; they do not ask for an endpoint order. Finally, if H peels to a , the set E is neither empty nor the singleton antipode. Its members all have distance at least two from c , and only the antipode has distance $|P|$. This proves the last assertion. $\square$

Remark 2.92 (Creating an old nonroot corner is insufficient). Use the normalized 289-concept rooted factor B from Remark 2.67. Its only corner is zero, of degree three. The addition $c = 167$, corresponding to the original word 186, is ample and acquires

$$P = \{0, 1, 6, 8\}, \quad \text{with bitmask } 323.$$

The completed four-cube has exactly two vertices without outside neighbors: $c = 167$ and its antipode $d = 484$. The latter belongs to B and becomes a new corner. All other fourteen cube vertices have outside neighbors, and zero lies outside this cube. Thus $E = \{484\}$, and the proposition applies. The 290-concept extension H is a rooted factor in dimension twelve and VC dimension four, with corners 0, 167, 484. Deleting either nonroot corner leaves zero as the only corner. In particular, exposing an old nonroot corner is necessary but not sufficient to restore the endpoint.

Compare the restoring tip 604 from Remark 2.88. Both tips have support size four, Hamming distance five from zero, and two tip-root disagreements inside their acquired support. A coordinate permutation identifies their punctured cubes together with the marked root and missing tip. Their exterior attachments differ: the restoring completion has three available old nonroot corners, 1084, 1116, 1148, whereas the nonrestoring completion has only the antipode 484. Thus even the marked punctured cube does not determine the endpoint without its exterior attachments.

The checker `verify_damp_rooted_clause_pivots.py` reconstructs both fixed extensions, verifies all fourteen exterior-neighbor witnesses and the marked-cube permutation, and replays the positive endpoint order. It checks the complete three-state root-protected negative graph of H , an ordinary peeling, and all 24 rooted elementary minor orders of H , as well as the 24 core orders. The report is `damp_rooted_clause_pivots_verification.json`. This gives an intrinsic sufficient con-

dition for a nonrestoring addition over a known unique-corner rooted factor. It does not characterize all endpoint repairs or settle intrinsic factor admissibility or the full classification.

Theorem 2.93 (A single clause can change an arbitrarily remote endpoint). *For every $j \geq 0$, there are two ordinarily corner-peelable ample families $B_j, K_j \subseteq Q_{12+2j}$, both containing zero, with the following properties.*

1. *$(B_j, 0)$ is a rooted factor, whereas K_j peels to zero. Every proper root-preserving pc-minor of either family peels to its root.*
2. *Their shattered-set complexes coincide. Their signed minimal-nonface presentations differ on exactly one four-element support, whose omitted traces are antipodal.*
3. *They contain exactly the same concepts at Hamming distance at most $j + 4$ from zero. The nearest difference has distance $j + 5$.*

Their common order, dimension, and VC dimension are

$$|B_j| = |K_j| = \frac{581 \cdot 3^j - 3}{2}, \quad \dim B_j = 12 + 2j, \quad \text{VCdim}(B_j) = 3 + j. \quad (2.44)$$

Their roots have degree three for $j = 0$, and degree two for $j \geq 1$. In particular, no fixed-radius neighborhood of the root, even together with the entire shattered-set complex, determines prescribed-root peelability or rooted-factor status among these families.

Proof. We give explicit recursive families. First we show that the paired square construction transports its own two endpoint repairs, so that it can be iterated without further endpoint searches.

Let C be ordinarily peelable, and suppose that $A = C \cup \{\alpha\}$, $D = C \cup \{\delta\}$ are compatible ample additions with distinct acquired supports P_α, P_δ , both peeling to zero. Write $\mathcal{S}(C; \alpha, \delta)$ for the square construction (2.32). Its two missing concepts $(\alpha, 11)$, $(\delta, 11)$ are ample one-concept additions with acquired supports

$$P_\alpha \cup \{q\}, \quad P_\delta \cup \{r\}, \quad (2.45)$$

respectively. For the first, its old punctured cube is present at 11, its q -mate and that entire cube are present at 01, and it has no r -mate. Its acquired support is absent from (2.37), so the local one-concept test proves ampleness. Interchanging q, r and α, δ proves the second assertion. These two supports are distinct, so the compatible-extension union formula also applies to their joint addition.

Both additions repair the square root. After adding $(\alpha, 11)$, peel the 01 copy of A down to zero, using its endpoint order; every deleted corner cube lifts into the 11 copy of A . Delete $(\alpha, 11)$ and then $(0, 01)$. The remaining 11 section is C ; peel it ordinarily, lifting its corner cubes into D at 10. Finally peel D to zero and delete the root last. For the addition $(\delta, 11)$, interchange the two new coordinates and the two side extensions in this sequence. This proves the repair propagation independently of whether C itself peels to zero.

Let B_0 be the normalized 289-concept factor used in Remark 2.92, and set

$$K_0 = (B_0 \setminus \{1084\}) \cup \{604\}, \quad \alpha_0 = 128, \quad \delta_0 = 290.$$

The verified peeling of $B_0 \cup \{604\}$ starts at 1084. Its suffix peels K_0 to zero. That first corner has support $I_0 = \{5, 6, 9, 10\}$, with bitmask 1632, so its deletion removes the newly acquired support and gives $\text{Sh}(K_0) = \text{Sh}(B_0)$. By Proposition 5.102, only the omitted trace on I_0 changes, from bitmask 576 to 1056. Both 604 and 1084 have weight five. The two additions by α_0, δ_0 are compatible and

repairing for both bases; these finite input conditions and all four endpoint orders are checked in the replay below.

Inductively define

$$\begin{aligned} B_{j+1} &= \mathcal{S}(B_j; \alpha_j, \delta_j), & K_{j+1} &= \mathcal{S}(K_j; \alpha_j, \delta_j), \\ \alpha_{j+1} &= (\alpha_j, 11), & \delta_{j+1} &= (\delta_j, 11). \end{aligned} \tag{2.46}$$

The propagation just proved maintains the two compatible repairs on both sides. Proposition 2.63 therefore makes every $(B_j, 0)$ a rooted factor and every K_j root-peelable. The two cores and their repairs have identical shadows at every step, by induction using (2.37) and (2.45). Ordinary peelability and the assertions about proper rooted minors follow from the same transfer theorem and root-preserving pc-minor closure. The order recurrence is $n_{j+1} = 3n_j + 3$, with $n_0 = 289$. Each step adds two active coordinates. The initial VC dimension is three and the initial repair supports have size four; the two shadow formulas increase these quantities by one at each step. This proves (2.44) and the root degrees.

Put $L = \{01, 10, 11\}$. Since the repair tips are the same for both families and the differing old words are nonzero, their differences are exactly

$$B_j \setminus K_j = \{1084\} \times L^j, \quad K_j \setminus B_j = \{604\} \times L^j.$$

Each word of L has weight at least one, and weight one is attained. This proves the asserted agreement through radius $j + 4$ and the first difference at radius $j + 5$.

The support I_0 , in the original twelve coordinates, remains a minimal nonface. For $j \geq 1$, the projection of B_j onto those coordinates is $B_0 \cup \{\alpha_0, \delta_0\}$, and the corresponding projection of K_j uses K_0 . The two compatible repairs do not fill I_0 , so its omitted traces remain 576 and 1056. Every other minimal nonface M misses some coordinate $e \in I_0$. The e -neighbor of each differing word belongs to both families: the original neighbors of 604 and 1084 lie in their common punctured four-cube and are nonzero, so all their L^j copies persist. That neighbor has the same M -trace as the differing word. Thus the two M -projections coincide. Only the omitted trace on I_0 changes, proving item 2.

Given any fixed radius, choose j large enough that it is at most $j + 4$. The two inputs then have the same root neighborhood and the same full shadow but opposite endpoint status. This proves the final assertion; no bound on the radius was assumed in advance. $\square$

The replay `verify_damp_rooted_clause_pivots.py` checks the first three levels, of orders 289, 870, 2613, including the common shadows, all signed minimal-nonface traces, explicit propagated repair orders, and the positive endpoint orders. It checks all 24, 28, 32 rooted elementary orders of the negative families. The first differences occur at distances 5, 6, 7, respectively. The uniform proof establishes all levels; the finite replay is stored in `damp_rooted_clause_pivots_verification.json`. This rules out fixed-radius root tests, even augmented by the shadow. It does not exclude a global criterion using the signed clauses and their attachment incidences, which remains the intrinsic admissibility problem.

Corollary 2.94 (A remote clause change can remove a forbidden pc-minor). *For every $j \geq 0$, there are two two-connected ample families $X_j, Y_j \subseteq Q_{24+2j}$, containing zero, such that $X_j \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ and $Y_j \in \mathcal{D}_{\text{amp}}$. They have the same shattered-set complex, and their signed minimal-nonface presentations differ on exactly one four-element support, with antipodal omitted traces. They agree on every concept at Hamming distance at most $j + 4$ from zero, and their nearest difference has distance $j + 5$. Their common order and VC dimension are*

$$|X_j| = |Y_j| = \frac{581 \cdot 3^j + 575}{2}, \quad \text{VCdim}(X_j) = \text{VCdim}(Y_j) = 3 + j.$$

Every proper pc-minor of either family belongs to $\mathcal{D}_{\text{amp}}$. Thus, even within this two-connected class with all proper minors peelable, a fixed-radius neighborhood of a marked vertex together with the full shadow does not determine forbidden-minor membership.

Proof. Take B_j, K_j from Theorem 2.93. Let T be a copy of B_0 on twelve fresh coordinates, with root zero, and first form

$$X'_j = B_j \vee T, \quad Y'_j = K_j \vee T.$$

Theorem 2.19 gives $X'_j \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$. Since K_j peels to zero and T peels ordinarily, Proposition 2.18 gives $Y'_j \in \mathcal{D}_{\text{amp}}$: peel K_j down to zero and then peel T ordinarily.

Write F_j, G for the two coordinate blocks. The shadow of each union is the union of its two block shadows. Since every coordinate is active, its minimal nonfaces are exactly the block minimal nonfaces and the pairs $\{e, f\}$, where $e \in F_j, f \in G$. Every such mixed pair omits 11. Old projections are unchanged, because the zero trace supplied by the other block was already present. Consequently the only changed clause is the old four-element clause; all mixed signed clauses coincide. The concept differences are also unchanged by gluing. The common order at this stage is $(581 \cdot 3^j + 573)/2$.

Choose a common root direction e in B_j, K_j and a root direction f in T . Such an e exists because the two root neighborhoods coincide. Complete the same square in both:

$$X_j = X'_j \cup \{0^{\{e, f\}}\}, \quad Y_j = Y'_j \cup \{0^{\{e, f\}}\}.$$

Theorem 2.114 makes X_j a two-connected ample forbidden pc-minor. The added concept in Y_j is a degree-two corner acquiring $\{e, f\}$. Deleting it and peeling Y'_j proves $Y_j \in \mathcal{D}_{\text{amp}}$.

For completeness, Y_j is also two-connected. The finite input K_0 is two-connected, as checked in the replay below. For $j \geq 1$, the three copies of K_{j-1} in the paired square induce its Cartesian product with the path $\{01, 11, 10\}$. A product of two nontrivial connected graphs is two-connected: after deleting (v, t) , choose another second-coordinate vertex t' . A surviving (w, s) with $w \neq v$ reaches the intact t' -layer along its column. If $w = v$, then $s \neq t$, so the intact s -layer first reaches a different column and then the t' -layer. Thus the whole surviving graph is connected. Adding the square root, which has two distinct neighbors, and each repairing tip, which has at least four neighbors in that product, preserves two-connectivity. Hence every K_j is two-connected. The graph of T is two-connected by the rooted-factor theorem. After deleting the common root in Y_j , the two blocks remain connected and the new square tip joins them. Deleting any other old vertex leaves the blocks connected through zero and the square tip with a surviving neighbor. Deleting the tip leaves Y'_j connected. These cases prove two-connectivity of Y_j .

The completion adds the same support $\{e, f\}$ to both shadows and removes precisely its common mixed clause. No larger mixed support can become a minimal nonface: any such support contains another mixed pair, which is still a nonface. The single changed four-element clause therefore persists. Both concept sets gain the same word, so the distance assertions persist as well. The shadow and order formulas give the stated parameters. All proper minors of X_j peel by forbidden-minor status; those of Y_j peel by pc-minor closure. $\square$

The replay `verify_damp_forbidden_clause_pivots.py` checks $j = 0, 1$, including the intermediate pairs of orders 577, 1158 and their two-connected completions of orders 578, 1159. It verifies both full signed presentations, all 600 elementary minor orders across the eight families, four positive peeling orders, the complete negative graphs, and two-connectivity of both completed families in each pair. Its report is `damp_forbidden_clause_pivots_verification.json`. The negative intermediate families are the previously constructed double and terminal cut-vertex example. The completions have a preserving degree-two corner deletion, so no new terminal type is asserted. This

gives opposite-status inputs for a global signed-clause criterion; it does not supply that criterion or the full classification.

Proposition 2.95 (Geometric realizations allow every final concept). *Call an ample family $H \subseteq Q_F$ geometrically realizable if it is the set of chamber sign vectors of finitely many affine hyperplanes, indexed by F , on a nonempty open convex polyhedron. This is realizability of its associated complex of oriented matroids. If H is geometrically realizable, then it peels to every prescribed $a \in H$. In particular, no rooted factor in Theorem 2.19 is geometrically realizable.*

Proof. We keep a point of the prescribed chamber while applying the geometric corner sweep of Knauer–Marc. The local step in their proof of [KM23, Proposition 5.5] is the following: translating a generic bounding halfspace into a realizing polyhedron, through its first nontrivial cell event, removes a COM corner and leaves a realizable COM. Here a COM corner may in general contain several topes. For a lopsided system it consists of a single ordinary corner, by [KM23, Proposition 5.3]. Consequently, in the ample case this step deletes one corner and leaves an ample geometric realization. We explain why the halfspace can always be chosen to preserve the prescribed chamber; this is an additional conclusion from the sweeping proof, rather than the statement of Proposition 5.5.

Let P realize H , and fix a point $p \in P$ with sign vector a . We may make P bounded without changing its COM: choose one witness for each realized sign vector, including those with zero entries, and intersect with a sufficiently large open polytope containing all these finitely many witnesses and p . Bounding facets can be made generic while retaining these witnesses and keeping the new polyhedron inside the old one: first move the facets a little inward, and then perturb them by less than the resulting margin. Thus no sign vector is added or lost. This is also the generic-position choice used in the cited sweep.

If $H = \{a\}$, there is nothing to prove. Otherwise choose $b \in H \setminus \{a\}$ and a coordinate on which a and b differ. Let ℓ be the affine functional of that hyperplane, oriented and scaled so that $\ell(p) = 1$; the chamber of b lies in $\ell < 0$. On the compact closure of P , choose a generic affine functional h with

$$\sup_{x \in \overline{P}} |h(x) - \ell(x)| < \frac{1}{4}.$$

Such choices are available arbitrarily close to ℓ , because only finitely many nongeneric incidences must be avoided. In particular, $h(p) > 3/4$, whereas $h(x) < 1/4$ whenever $x \in P$ and $\ell(x) \leq 0$.

Introduce the halfspace $h > s$, initially with $s < \min_{\overline{P}} h$, so that it is redundant, and translate its boundary by increasing s toward $1/2$. The point p stays strictly inside throughout. By the final position the chamber of b has disappeared, so there is a first nontrivial event before that position. Apply the local sweeping step and stop just after this event. It deletes one corner different from a , and the remaining ample class is realized on a bounded open polyhedron still containing p . Repeating this argument inductively on the number of concepts peels H to a . A rooted factor fails to peel to its specified root, so the last assertion follows by contraposition. $\square$

Geometric realizability is inherited by coordinate faces: intersect the realizing polyhedron with the open sign halfspaces of the fixed coordinates, then forget their hyperplanes. Therefore both families X_j, Y_j of Corollary 2.94, and both intermediate families X'_j, Y'_j in its proof, fail geometric realizability. Indeed, fixing every coordinate of F_j to zero recovers exactly the rooted factor $T = B_0$; the added square tip has a nonzero F_j -coordinate and is removed by this restriction. In particular, the 289-concept class B_0 , and the ordinarily peelable classes Y'_j, Y_j , show that ordinary peelability does not imply geometric realizability. The latter property cannot separate these opposite-status pairs.

Remark 2.96 (The affine oriented-matroid restriction is narrower). An affine oriented-matroid tope class is the projection of the $g = +$ topes of an oriented matroid onto its other coordinates, where g is a nonloop. Its shattered-set complex must be a matroid independence complex. To see this, let $\mathcal{T}$ be the full antipodal tope set and let M be the underlying matroid. A set S is shattered by $\mathcal{T}$ exactly when it is independent in M : restriction to an independent set is a free oriented matroid and has every sign vector as a tope; a dependent set contains a signed circuit, whose sign pattern cannot be the restriction of a tope by circuit–covector orthogonality. This is the restriction form of the rank–VC correspondence; see also [Mar24, Claim 2.2]. Antipodality gives, for the affine class C ,

$$I \in \text{Sh}(C) \iff I \cup \{g\} \in \text{Sh}(\mathcal{T}) \iff I \in \text{Ind}(M/g).$$

The first equivalence holds because every pattern on I in the positive half supplies every pattern in the negative half by negation. Thus $\text{Sh}(C)$ satisfies augmentation: if A, B are faces and $|A| < |B|$, then some $e \in B \setminus A$ has $A \cup \{e\}$ a face.

Already the common shadow of B_0, K_0 violates this condition: $A = \{1, 2\}$ and $B = \{1, 5, 6\}$ are shattered, but neither $\{1, 2, 5\}$ nor $\{1, 2, 6\}$ is shattered. In fact their common projection onto $\{1, 2, 5, 6\}$ has thirteen concepts and peels to zero. Thus failure of augmentation can occur in a positive proper minor. For every higher paired-square level, the fresh root pair $\{q, r\}$ is a maximal shattered set of size two, whereas $\{q, e, f\}$ is shattered for any shattered old pair $\{e, f\}$. This also violates augmentation. Hence none of the fixed rooted pairs is itself a full affine oriented-matroid tope class.

This restriction concerns the full positive half of one oriented matroid. It does not exclude fibers defined by several sign constraints or a representation using auxiliary elements. Likewise, the vertex-shellings in [HW25, Theorems 1.1–1.2] order cocircuits and the face lattices of tope cells; they do not assert an ample-class corner peeling to a prescribed concept. The mutation preservation lemma in [Wil25, Lemma 21] requires a Euclidean uniform oriented-matroid program and a mutation containing the objective element but excluding the distinguished affine element. A signed-clause exchange in an ample family does not by itself provide those data. No such auxiliary encoding or converse to geometric peeling is claimed.

The fixed replay `verify_damp_geometric_imports.py` checks the unique corner and ordinary peeling of B_0 , recovers its coordinate face in all eight ordinary families at $j = 0, 1$, and replays the four positive ordinary orders. It checks fourteen augmentation witnesses and explicit root-ending orders of the thirteen-concept projection and the five-concept projection $Q_2 \vee K_2$. Nonrealizability follows from Proposition 2.95; it is not an output of a numerical realizability test. The report is `damp_geometric_imports_verification.json`.

Proposition 2.97 (The ample completion of a single-clause exchange). *Let $H_s, H_t \subseteq Q_F$ be ample families with common shadow Σ , whose signed presentations differ only at $I \in \mathcal{N}(\Sigma)$. Write $s \neq t \in Q_I$ for their respective omitted traces. Let R be the family satisfying all the unchanged clauses, and put*

$$\Sigma^+ = \{J \subseteq F : N \not\subseteq J \text{ for every } N \in \mathcal{N}(\Sigma) \setminus \{I\}\}.$$

For $u \in Q_I$, write $R_u = \{y \in Q_{F \setminus I} : (u, y) \in R\}$. Then $R = H_s \cup H_t$ is ample with shadow Σ^+ , and

$$A := R_s = R_t = \bigcap_{u \in Q_I} R_u$$

is a nonempty ample family. In particular, $Q_I \times A \subseteq R$, and the two differences are precisely

$$H_t \setminus H_s = \{s\} \times A, \quad H_s \setminus H_t = \{t\} \times A.$$

The new shadow faces are

$$\Sigma^+ \setminus \Sigma = \{I \cup J : J \in \text{Sh}(A)\}.$$

All admissible choices for the omitted I -trace, with the other clauses fixed, are recovered by the intrinsic slice condition

$$P = \{u \in Q_I : R_u = A\}.$$

Conversely, every ample R and nonempty coordinate set I with nonempty $A = \bigcap_u R_u$ and at least two members of P produce such a single-clause family: its common shadow is $\text{Sh}(R) \setminus \{J : I \subseteq J\}$. Thus the construction and the slice recovery describe all single-clause exchanges with fixed shadow. For $u \in P$, the corresponding class is $H_u = R \setminus (\{u\} \times A)$. Deleting just $\{u\} \times A$ by corner deletions is possible exactly when $A \in \mathcal{D}_{\text{amp}}$: the corner orders of this slice are exactly the corner orders of A , with first coordinate u . If $|I| \geq 2$ and some H_u is peelable or is an ample forbidden pc-minor, then $A \in \mathcal{D}_{\text{amp}}$ automatically.

Proof. A word satisfying the unchanged clauses cannot violate both distinct I -clauses, so $R = H_s \cup H_t$. For every $i \in I$, the projections of H_s, H_t obtained by forgetting i coincide: their shadows coincide, and the signed presentations of these projections retain exactly the clauses supported away from i , which are unchanged. This uses Theorem 5.3 on the ample projections.

Take $(t, y) \in H_s \setminus H_t$. Equality of these projections forces $(t^{\{i\}}, y) \in H_t$ for every $i \in I$. The coordinate face $(H_t)_y \subseteq Q_I$ is ample, omits t , and contains every neighbor of t . In its signed presentation, a clause excluding t must involve every coordinate of I : otherwise the neighbor obtained by changing an omitted coordinate would violate that clause too. Thus I is a minimal nonface of the shadow of this face. All its proper subsets are shattered, and ampleness gives $(H_t)_y = Q_I \setminus \{t\}$. In particular $(s, y) \in H_t$. Interchanging the two classes now gives $(H_s)_y = Q_I \setminus \{s\}$. Consequently the two difference slices have the same outside assignments A , and $R_y = Q_I$ for every $y \in A$. Conversely any full I -cube in R has its s -vertex in $H_t \setminus H_s$, so its outside assignment belongs to A . This proves the slice and intersection formulas. The family A is nonempty and ample, since it is the s -coordinate face of H_t .

The old strongly shattered faces Σ , together with $I \cup J$ for $J \in \text{Sh}(A)$, are strongly shattered in R . The latter assertion follows from $Q_I \times A \subseteq R$ and ampleness of A . These two collections are disjoint and have total size $|\Sigma| + |A| = |R|$. The lower Sandwich inequality therefore forces R to be ample, with exactly these shadow faces. To identify this shadow with Σ^+ , a support containing an unchanged clause is not shattered. On a support containing no unchanged clause, the two projected families either are already full cubes, or omit only the extensions of s and t , respectively. Their union projects onto the full cube in either case. This proves $\text{Sh}(R) = \Sigma^+$.

For an arbitrary $u \in Q_I$, imposing the I -clause with omitted trace u gives $R \setminus (\{u\} \times R_u)$. Since $A \subseteq R_u$, its cardinality is $|\Sigma|$ exactly when $R_u = A$. The fixed-shadow signing theorem makes this equality equivalent to admissibility, proving the complete description of P .

For the converse construction, let R, I, A, P have the stated properties. Strong coordinate reduction preserves ampleness, so A is ample. Moreover

$$\text{Sh}(A) = \{J \subseteq F \setminus I : I \cup J \in \text{Sh}(R)\} :$$

ampleness identifies shattered supports with cube supports, and a cube on either side is exactly the corresponding full product cube on the other side. For $u \in P$, deleting $\{u\} \times A$ leaves every projection forgetting some $i \in I$ unchanged, since each deleted vertex has its i -neighbor in another slice of $Q_I \times A$. It removes all shattered supports containing I and retains every other shattered support. The number of lost supports is $|\text{Sh}(A)| = |A|$, exactly the number of deleted vertices. Hence H_u is ample with the asserted common shadow. The set I is a minimal nonface of that

shadow; every other minimal nonface avoids some $i \in I$, so its projection and omitted trace are unchanged. The presentations therefore differ only at I . Any two distinct deletions have union R , completing recovery.

Suppose $u \in P$, and only the vertices $\{u\} \times D$ have so far been deleted, where $D \subseteq A$. At a surviving (u, y) , the incident directions are precisely all directions of I , together with the neighbor directions of y in $A \setminus D$. They span a cube in the remaining family exactly when the latter directions span a cube at y in $A \setminus D$: the u -slice imposes this condition, and every other slice contains A . Thus the corner tests coincide at every step, proving the relative peeling assertion.

Finally, P contains at least s, t . Given $u \in P$, choose $v \in P \setminus \{u\}$. The v -coordinate face of H_u is A . When $|I| \geq 2$, this is a proper pc-minor, since H_u contains $(Q_I \setminus \{u\}) \times A$ and therefore has at least $3|A|$ concepts. Either positive pc-minor closure or forbidden-minor minimality gives $A \in \mathcal{D}_{\text{amp}}$. $\square$

Corollary 2.98 (All admissible signs in the fixed clause exchange). *Fix all clauses except the distinguished four-coordinate clause in Theorem 2.93, at any level j . There are exactly four admissible signs. On the old coordinates $(5, 6, 9, 10)$, in that order, their endpoint behavior is*

omitted trace	deleted vertex in the base completion	peels to zero
0110	604	no
1001	1084	yes
0101	1116	yes
1101	1148	yes

The outside family A in Proposition 2.97 is a translate of $\{01, 11, 10\}^j$, so it has 3^j concepts. The common ample completion has $(583 \cdot 3^j - 3)/2$ concepts. The same four choices exhaust the admissible signs for both ordinary pairs in the proof of Corollary 2.94, before and after square completion. In each case exactly the first sign gives a forbidden pc-minor; the other three classes peel, and every proper pc-minor of all four classes peels. All four square-completed classes are two-connected.

Proof. At level zero the common completion is $R_0 = B_0 \cup \{604\}$. The replay below checks that the only I -slices equal to the full-cube outside family $A_0 = \{28\}$, where I has bitmask 1632, are the four listed slices. Their traces have bitmasks 576, 1056, 1088, 1120, respectively. Proposition 2.97 therefore proves that the list is exhaustive. Deleting 604 returns the rooted factor B_0 . The stored peeling of R_0 to zero begins 1084, 1148, 1116. The other two prefixes 1116, 1148, 1084 and 1148, 1084, 1116 also consist of corner deletions and reach exactly the same remaining class. Appending the same suffix gives the three asserted positive endpoint orders.

In the paired-square construction, the outside family becomes $A_j = A_{j-1} \times \{01, 11, 10\}$, with the inherited fixed coordinates. Every old slice strictly larger than A_{j-1} remains strictly larger after this product. The additional root and the two repairing tips have old I -traces 0, 0, 32, all outside the four listed traces. Thus they do not change the four minimal slices, and no other slice can become minimal. This proves exhaustion at every level. The order formula adds 3^j to the order of B_j .

For each of the three positive choices the two repairing tips are still corners: at the base level their completed cubes are disjoint from the four exchanged vertices, and this separation persists in the paired square. Delete those tips, then lift a root-ending order of the old class through the three-layer path, finishing with 11, 01, 10, 00. This is exactly the positive-order construction in the proof of Theorem 2.93, and proves the three positive assertions by induction. The first choice remains B_j .

Gluing the fixed factor T adds concepts only in old I -trace zero. The common square tip also has old I -trace zero. These additions leave the four minimal slices unchanged, so the exhaustion

statement persists. The gluing and square-completion theorems give the first sign's forbidden status. For each other sign, peel its factor to zero and then T ordinarily, prefixing the square tip when it is present. Positive pc-minor closure gives all proper minors on these three sides; the first side already has forbidden-minor status. The three positive base factors are two-connected, as checked below. The product-core and attachment argument in the proof of Corollary 2.94 applies to each one and proves two-connectivity after square completion. $\square$

The replay `verify_damp_single_clause_completions.py` checks the three rooted levels and both ordinary stages at levels zero and one. It recovers the common completion, its full shadow, the outside family, and all admissible signs directly from the fixed clauses and slices. It checks the three base endpoint prefixes and their propagated orders, as well as the relative slice deletions. Its report is `damp_single_clause_completions_verification.json`. The proposition replaces a general single-clause exchange by a recovered ample completion and its unattached slices. The corollary settles all choices for these fixed exterior attachments; it does not characterize endpoint behavior for arbitrary completions. That remaining condition must retain the other slices of R , rather than only their common intersection or the graph of the admissible traces.

Proposition 2.99 (A cube layer can change adjacent endpoint choices). *Let $S \subseteq Q_F$ be ample and peel to $a \in S$. Suppose that $I \subseteq F$ is nonempty and that $C = Q_I \times \{y\} \subseteq S$, where $y \in Q_{F \setminus I}$. For $u \in Q_I$, put*

$$S_u = \{z \in Q_{F \setminus I} : (u, z) \in S\}, \quad P_0 = \{u : S_u = \{y\}\}.$$

Introduce a new coordinate f , and define

$$R = (S \times \{0\}) \cup (C \times \{1\}), \quad J = I \cup \{f\}.$$

Then R is an ample single-clause completion on J , with outside family $\{y\}$. Its admissible omitted traces are exactly

$$P = (P_0 \times \{0\}) \cup (Q_I \times \{1\}).$$

Write $X_{u,b} = R \setminus \{(u, y, b)\}$ for the member corresponding to $(u, b) \in P$. Their common shadow is

$$\text{Sh}(S) \cup \{T \cup \{f\} : T \subsetneq I\}.$$

For every $u \in Q_I$, the upper omission $X_{u,1}$ peels to $(a, 0)$. For $u \in P_0$ with $(u, y) \neq a$,

$$X_{u,0} \text{ peels to } (a, 0) \iff S \setminus \{(u, y)\} \text{ peels to } a.$$

After discarding any omission that removes $(a, 0)$, the graph on the remaining admissible traces, with edges at Hamming distance one, is connected. Nevertheless, whenever a nonroot corner deletion in S destroys the endpoint and its corner cube is C , two adjacent retained choices in this graph have opposite endpoint behavior.

Proof. The family R is the peripheral expansion of S along the cube C , so it is ample by Proposition 2.6. Its shadow is

$$\text{Sh}(R) = \text{Sh}(S) \cup \{T \cup \{f\} : T \subseteq I\}.$$

Its $(u, 0)$ -slice outside J is S_u , whereas every $(u, 1)$ -slice is $\{y\}$. Their intersection is therefore $\{y\}$, and their minimal slices are precisely the displayed set P . Since $|Q_I| \geq 2$, Proposition 2.97 applies. It gives all the admissible choices, shows that their presentations differ only at J , and obtains their common shadow by removing J from $\text{Sh}(R)$.

We use an explicit peeling of a punctured cube. If $v \in C$, delete the concepts of $C \setminus \{v\}$ in increasing Hamming distance from v , with arbitrary ordering at equal distance. At each step, all neighbors closer to v have disappeared, and all concepts farther from v remain. The remaining incident directions span the full cube increasing that distance, so every deletion is a corner deletion. Whenever a lower layer contains a remaining upper layer, an upper corner has the same incident cube with the additional direction f . Consequently an ordinary peeling of the upper layer deletes it completely while leaving the lower layer fixed.

In $X_{u,1}$, the upper layer is $C \setminus \{(u, y)\}$ and the lower layer is S . Delete the upper punctured cube by the preceding order, then peel S to a . This proves every upper choice positive. In $X_{u,0}$, the upper concept $(u, y, 1)$ is a corner: its I -cube is present and its lower neighbor is absent. Delete it first, then delete the remaining upper punctured cube while retaining $S \setminus \{(u, y)\}$ below. This proves the reverse implication of the endpoint equivalence. For the forward implication, restrict a peeling to the lower coordinate face. Deleting an upper concept leaves that face unchanged; a deleted lower corner restricts to a corner of the surviving face, since its incident cube restricts to the cube on the remaining lower directions. The restricted order ends at a , as required.

The whole upper trace cube remains among the root-retaining choices. Every remaining lower trace has its vertical neighbor in that cube, so the graph is connected. If $c = (u, y) \neq a$ is a corner of S with unique cube C , then $u \in P_0$. Indeed, S_u is an ample coordinate face containing y ; if it contained another concept, its connected graph would give a neighbor of c outside C . Thus, if deleting c destroys the endpoint, $X_{u,0}$ is negative and $X_{u,1}$ is positive. Their omitted traces differ only in f , proving the final assertion. $\square$

For a fixed instance, take $S = B_0 \cup \{604\}$ from Corollary 2.98, with root zero, I of bitmask 1632, and $y = 28$. The four elements of P_0 have bitmasks 576, 1056, 1088, 1120. Use bit $2^{12} = 4096$ for f . Then R has 306 concepts, and its twenty admissible omissions give 305-concept classes of dimension thirteen and VC dimension four. Their graph is connected, and every omission retains zero, since zero lies outside C . All sixteen upper choices peel to zero. Among the four lower choices, exactly the trace 576 fails, by the preceding corollary and the endpoint equivalence. In particular, the adjacent traces 576 and $4672 = 576 + 4096$, deleting concepts 604 and 4700, respectively, have opposite endpoint behavior.

The replay `verify_damp_adjacent_clause_endpoint_switch.py` checks all thirty-two possible signs, the twenty admissible presentations with their common full shadow, the connected trace graph, and nineteen explicit root-ending orders. The negative member has the proper rooted face B_0 , whose only corner is zero. Its ordinary peeling is checked as well. The report is `damp_adjacent_clause_endpoint_switch_verification.json`. Thus endpoint status need not be constant on a connected component of the root-retaining admissible graph. The negative member is not a rooted factor, since its proper face B_0 already fails at its root. Theorem 2.101 below gives an adjacent endpoint change whose negative member is itself a rooted factor.

Proposition 2.100 (Reducing an adjacent endpoint change to a rooted factor). *Let $H_s, H_t \subseteq Q_F$ be ample families with the same shadow, differing only at a clause on $I \subseteq F$, whose omitted traces $s, t \in Q_I$ differ in exactly one coordinate f . Let $a \in H_s \cap H_t$, suppose that $H_s \in \mathcal{D}_{\text{amp}}$, and suppose that H_t peels to a whereas H_s does not. If the root-containing face*

$$H_s^{a,f} := \{x \in H_s : x_f = a_f\}$$

peels to a , then there is a common sequence μ of root-preserving pc-minor operations, none using f , such that $(\mu H_s, \mu a)$ is a rooted factor and μH_t peels to μa . The two resulting families still have the same shadow and differ at just one clause, whose omitted traces differ exactly at f . Its

support is a subset of I , and its outside family is a nonempty pc-minor of the outside family A in Proposition 2.97.

Consequently, the following two assertions are equivalent:

1. no adjacent single-clause change retaining the root turns a rooted factor into a class that peels to its root;
2. whenever $H_s \in \mathcal{D}_{\text{amp}}$ and H_t peels to a common root in such an exchange, H_s peels to that root if and only if its root-containing f -face does.

Both assertions are false by Theorem 2.101. An adjacent endpoint change is impossible when $|I| \leq 2$, and also when A is a singleton and $|I| \leq 3$.

Proof. Put $R = H_s \cup H_t$. The completion theorem gives

$$R_s = R_t = A, \quad Q_I \times A \subseteq R, \quad H_u = R \setminus (\{u\} \times A) \quad (u = s, t).$$

We first describe how a common elementary root-preserving operation acts on this presentation. At every stage the image of H_t peels to the image of a , by root-preserving pc-minor closure. Thus, while retaining endpoint failure on the other side, we cannot use an operation that identifies the two families.

Contraction of a coordinate in I identifies the two images: each removed concept has its neighbor in that coordinate in both projections, so both images equal the corresponding projection of R . Contraction of $e \notin I$ instead preserves the presentation, replacing R, A by $\pi_e R, \pi_e A$. Indeed, the two displayed minimal slices both project to $\pi_e A$, and every other projected slice contains $\pi_e A$. The converse part of the completion theorem gives the same-shadow single-clause exchange on I .

For a root-containing restriction in a coordinate $e \notin I$, restrict R and A to the root's value of e . If the resulting outside family is empty, the differences disappear and the images coincide. Otherwise the two minimal slices remain equal to that restricted family, so the completion theorem again preserves the adjacent exchange. For $e \in I \setminus \{f\}$, the two omitted traces agree at e . If their common bit differs from a_e , both differences are discarded and the images coincide. If it equals a_e , the restricted union has two minimal slices equal to A , now indexed by the distinct adjacent traces $s|_{I \setminus \{e\}}, t|_{I \setminus \{e\}}$. The completion theorem preserves the exchange, on support $I \setminus \{e\}$.

Only restriction in f could destroy this presentation without identifying the two sides. It cannot preserve endpoint failure here. Any sequence of operations in the other coordinates commutes with the root-containing f -restriction. The resulting face of the negative side is therefore a root-preserving minor of the initially positive $H_s^{a,f}$, and peels to its root. This excludes an f -restriction at every subsequent stage as well.

Now choose successive proper elementary root-preserving minors that still fail at their roots, as long as any exist. If a proper descendant fails, its first elementary step must also fail, by root-preserving pc-minor closure. Apply the same operations to the positive partner. The preceding cases show inductively that f survives and that the adjacent exchange is retained. The changed support can only lose coordinates, and the outside family can only undergo restrictions and contractions, remaining nonempty. The order of the negative family strictly decreases, so the process terminates. Its final family still peels ordinarily, by pc-minor closure from H_s ; every proper root-preserving minor peels to its root. It is therefore a rooted factor. This proves the reduction.

If assertion 1 holds, a failure of the reverse implication in assertion 2 would give a rooted-factor counterexample by this reduction. The forward implication in assertion 2 is face restriction closure. Conversely, in a rooted factor the root-containing f -face is proper and peels to its root, so assertion

2 would exclude a positive adjacent partner. The face is proper because the changed coordinate is active. Thus the assertions are equivalent; their equivalence does not establish either one.

For the small-support statements, $|I| = 1$ gives disjoint members and hence no common root. If $I = \{f, k\}$, the common root has its k -bit opposite to $s_k = t_k$. The shared face

$$G = R \cap \{x : x_k = a_k\}$$

contains the root. Each member is a peripheral expansion of G , along one of its two copies of A distinguished by the f -bit. Proposition 2.61 makes either endpoint possible exactly when G peels to a and $A \in \mathcal{D}_{\text{amp}}$. Their endpoint statuses are therefore equal.

Finally suppose that $A = \{y\}$ and $|I| \leq 3$. The two concepts $(s, y), (t, y)$ are adjacent corners of the same full I -cube in R . Deleting either leaves the other a corner on support $I \setminus \{f\}$. Hence their simultaneous deletion leaves an ample family G containing a , and each of H_s, H_t is an ample one-concept extension of G with acquired support of size at most two. Item 3 of Proposition 2.61 makes their endpoint statuses equal. $\square$

To obtain a rooted-minimal counterexample, it therefore suffices to find an ordinarily peelable counterexample whose root-containing f -face peels to the root. No assumptions on its other proper rooted minors are required. This reduction concerns the adjacency question; it does not supply intrinsic admissibility for the full classification.

For the 305-concept pair above, this hypothesis fails at exactly the expected face. The root-containing restriction in coordinate 12 gives B_0 on the negative side and $B_0 \cup \{604\}$ on the positive side. All other twenty-five root-preserving elementary images of the negative member peel to zero. Among all twenty-six operations, ten identify the two sides, fifteen preserve an adjacent single-clause exchange, and only that restriction loses the exchange while retaining endpoint failure. The replay `verify_damp_adjacent_exchange_minor_reduction.py` checks these images, fifty-one endpoint orders, and a two-coordinate control with a three-concept outside family. Its report is `damp_adjacent_exchange_minor_reduction_verification.json`.

Theorem 2.101 (An enlarged root section permits an adjacent endpoint change). *Let $(B, 0)$ be a rooted factor using every coordinate of F , and suppose that zero is its only corner. Let $\alpha, \delta, \gamma \notin B$ be three distinct ample one-concept additions, with pairwise distinct acquired supports $P_\alpha, P_\delta, P_\gamma$, such that all three enlarged classes peel to zero. Write*

$$C_\gamma = \{\gamma^J : J \subseteq P_\gamma\}.$$

Suppose that a coordinate face D of B contains zero and $C_\gamma \setminus \{\gamma\}$, but omits a neighbor $h = 0^{\{e\}} \in B$ of zero. On two fresh coordinates q, r , define, for $R \subseteq Q_2$,

$$\begin{aligned} Y_R = & (D \times \{00\}) \cup ((B \cup \{\alpha\}) \times \{01\}) \\ & \cup ((B \cup \{\delta\}) \times \{10\}) \cup (B \times \{11\}) \cup (\{\gamma\} \times R). \end{aligned}$$

The family Y_R is ample exactly when R is ample, equivalently when R is neither diagonal pair of the square. For an ample row, with $\text{Sh}(\emptyset) = \emptyset$,

$$\text{Sh}(Y_R) = \text{Sh}(Y_\emptyset) \dot{\cup} \{P_\gamma \cup J : J \in \text{Sh}(R)\}.$$

Every such Y_R is two-connected, peels ordinarily, and has all proper root-preserving pc-minors peelable to their roots. Its endpoint condition is the intrinsic row test

$$Y_R \text{ peels to } (0, 00) \iff \{01, 11\} \subseteq R \text{ or } \{10, 11\} \subseteq R.$$

Thus eight of the fourteen ample rows give rooted factors and six give positive endpoints.

In particular, the four three-vertex rows $R = Q_2 \setminus \{z\}$ give classes with the same shadow and a single changing clause on $P_\gamma \cup \{q, r\}$. Their omitted traces have fixed old part $\gamma|_{P_\gamma}$ and new part z . Exactly the omission $z = 11$ gives a rooted factor; the other three classes peel to the root. Consequently an adjacent clause change can restore the endpoint even when every proper rooted minor of the negative member is positive.

Proof. We first construct each ample row above the known paired rooted factor. For the negative rows, the two overlaps opposite the root section then prevent any first deletion in a copy of the old core.

Let S be the paired construction with root section $\{0\}$, the same 01, 10, 11 sections, and no γ -copies. Proposition 2.63 makes S a rooted factor. The face D is proper, since $h \notin D$, so it peels to zero. Reverse such a peeling to add its nonzero concepts one by one in section 00. If an addition to the current old-coordinate face acquires support J , the corresponding addition in the full family acquires $J \cup \{q, r\}$: its old corner cube lifts through the other three copies of B . That support was not shattered before the addition, because its 00-section did not shatter J . The one-concept extension criterion therefore proves ampleness at every step and reaches $Y_\emptyset$.

The punctured cube $C_\gamma \setminus \{\gamma\}$ is now present in every section. Distinct acquired supports make α, δ, γ pairwise nonadjacent, by Lemma 6.7. Thus a row addition at γ , acquiring row support J , has precisely the combined incident support $P_\gamma \cup J$. Its punctured cube is present. Moreover, among the old-coordinate words in $Y_\emptyset$, none realizes $\gamma|_{P_\gamma}$: that trace is absent from B , and neither of the other two additions acquires P_γ . Consequently the combined support is new exactly when J is new in the row. Every ample subfamily of a square has a corner peeling, so reversing it constructs Y_R by ample one-concept additions and proves the shadow formula. For either diagonal row, its three shattered supports give three new shattered supports joined to P_γ , although only two concepts have been added. Thus the shadow has more supports than the family has concepts, excluding ampleness. This proves all the ampleness claims.

All additions took place on the active coordinate set of S . Lemma 2.68 therefore proves ordinary peelability and positivity of all proper rooted minors for every ample row. It remains to decide the endpoint itself.

Suppose first that neither displayed edge is contained in R . The intersections of sections 01, 11 and of sections 10, 11 are both exactly B . Consider a hypothetical peeling ending at $(0, 00)$, and its first deletion of a concept in $B \times \{01, 10, 11\}$. All three old core copies are intact up to this step. Choose one of the two displayed edges incident to the deleted concept's layer, and write b for its old word. The corner has its mate on that edge and all the neighbors of b in B . Its corner cube would therefore give the full cube on these old directions in both sections. Since their intersection is still B , the word b would be a corner of B . Hence $b = 0$.

All four copies of zero are still present: the original root is protected, and the other three belong to the undeleted old core copies. The candidate corner therefore has both fresh-coordinate directions and the old e -direction toward h , whose three nonroot copies also remain. Its corner cube would contain $(h, 00)$. That concept is absent, since $h \notin D$ and $\gamma \notin B$. This contradiction excludes the first deletion in any nonroot old core copy, and hence excludes a peeling to the root. Ordinary peelability and proper-minor positivity already proved above make Y_R a rooted factor.

Now suppose that R contains one of the displayed edges. If $00 \in R$, then 00 is a corner of the row: the row is either the full square or a three-vertex path with 00 as an endpoint. Its copy $(\gamma, 00)$ is therefore a corner of Y_R , and may be deleted first. Next delete $(D \setminus \{0\}) \times \{00\}$ by the chosen peeling of D to zero. At each step its corner cube lifts into the untouched copies of B in the other three layers. The remaining family is the old partially replicated-tip construction with row $R \setminus \{00\}$,

which is an edge or $\{01, 10, 11\}$. Theorem 2.87 peels it to the root because $B \cup \{\gamma\}$ does. This proves the positive direction, and exhausts the fourteen ample rows.

The starting factor S is two-connected. Each root-section addition has its two fresh-coordinate neighbors in the preceding family. Each γ -addition has at least three old-coordinate neighbors: a repairing support has size at least three, by Proposition 2.61. Adjoining a vertex with at least two neighbors preserves two-connectivity, since deleting an old vertex leaves at least one attachment and deleting the new vertex leaves the old graph. Thus every ample Y_R is two-connected.

Finally take $R = Q_2 \setminus \{z\}$. These four rows have the same shadow. Their common completion is Y_{Q_2} , and each differs from it by the single concept (γ, z) . That concept has unique cube $C_\gamma \times Q_2$. Proposition 2.97 therefore gives the common shadow and the asserted single clause on $P_\gamma \cup \{q, r\}$. The row criterion makes exactly $z = 11$ negative; its neighbors 01, 10 are positive. $\square$

For the fixed core $B = B_0$, take $\alpha = 128$, $\delta = 290$, $\gamma = 604$, and P_γ of bitmask 1632. Put

$$D = \{v \in B_0 : v_e = 0 \text{ for every } e \notin \{2, 3, 4, 5, 6, 9, 10\}\}.$$

This face has 28 concepts, contains the punctured repair cube, and omits the old root neighbor $h = 2$. The family $Y_\emptyset$ has 897 concepts. In the encoding $(v, qr) \mapsto 4v + q + 2r$, the four three-vertex rows give the following 900-concept classes:

omitted row trace	deleted concept in Y_{Q_2}	peels to zero
00	2416	yes
10	2417	yes
01	2418	yes
11	2419	no

They have dimension fourteen and VC dimension five. The completion has 901 concepts; the changed support has bitmask 6531, and its outside family is $\{112\}$. The negative omitted trace 2307 is adjacent to the positive trace 2306, differing in coordinate q . The negative member is a rooted factor, so its root-containing q -face also peels to zero. This refutes both equivalent assertions in Proposition 2.100.

The replay `verify_damp_root_section_square_pivots.py` checks the sixteen rows, all ample addition chains, the eight two-overlap obstruction certificates, six endpoint orders, and the four complete signed presentations. For the negative 900-concept member it replays all twenty-eight rooted elementary orders. Its report is `damp_root_section_square_pivots_verification.json`.

Corollary 2.102 (Adjacent signs can change forbidden-pc-minor status). *There are four ample classes of order 1188 and dimension twenty-six with a common shadow whose signed presentations vary on a square of adjacent signs at a single clause. Exactly one is a forbidden pc-minor; the other three peel. Every proper pc-minor of all four peels. The same conclusions hold for four two-connected classes of order 1189, obtained by completing a common root square. All eight classes have VC dimension five.*

Proof. Glue each of the four 900-concept classes above to a fixed copy of $(B_0, 0)$ on twelve fresh coordinates. The negative choice glues two rooted factors, so Theorem 2.19 makes it forbidden. Each other choice peels to the shared root and then peels the fixed factor ordinarily. The gluing adds the same cross-coordinate clauses to all four presentations, so their single varying clause and its four signs persist.

The first factor has the common root neighbor in coordinate 0; the second has one in coordinate 15. Add the common square tip $2^0 + 2^{15} = 32769$. This adds only the support $\{0, 15\}$ to the shadow

and leaves the old clause signs unchanged. Theorem 2.114 makes the negative choice forbidden; on a positive choice, delete the square tip first. All root factors and all three positive partners are two-connected, by the preceding theorem, so the square joins their two blocks after removing the root and makes every completed graph two-connected. Forbidden-minor minimality and positive pc-minor closure give all the claimed proper-minor conclusions. $\square$

The checker `verify_damp_forbidden_adjacent_sign_pivots.py` replays the eight signed presentations, six ordinary positive orders, and all 156 elementary orders of the two negative classes. Its report is `damp_forbidden_adjacent_sign_pivots_verification.json`. The negative root factor descends by corner deletions to the existing paired factor, and its glued obstruction descends to the existing 1158-concept construction. These examples refute adjacency and component endpoint invariance under rooted minimality; they do not establish another corner-descent terminal type or the full intrinsic classification.

Corollary 2.103 (Irreducibility and small corner deletions in rooted factors). *Let (A, a) satisfy the three rooted-factor conditions of Theorem 2.19. Then A is periphery-free and Cartesian-prime. Deleting a nonroot corner of degree at most two preserves all three conditions. If two nonroot corners are adjacent and have degree three, deleting either or both also preserves the conditions.*

Consequently, a rooted factor with a nonroot corner but no preserving nonroot corner deletion has all its nonroot corners of degree at least three. A degree-three nonroot corner has no nonroot corner as a neighbor.

Proof. If A had a peripheral coordinate, write it as Y in item 1 of Proposition 2.61. For a lower root, its contraction base is a proper root-preserving pc-minor and peels to the projected root; the upper section peels ordinarily because $A \in \mathcal{D}_{\text{amp}}$. For an upper root, both the contraction base and the upper section are proper root-preserving minors and peel to their roots. Either case makes A peel to its root, a contradiction. If A were a product of two nontrivial factors, the two root-preserving factor restrictions would peel to their roots, and item 2 would give the same contradiction.

For the deletion assertions, glue (A, a) to the fixed rooted factor of Remark 2.50. The result is forbidden by Theorem 2.19. Every nonroot corner of A is a corner of the glued family with the same degree. Apply Corollary 2.14 for a corner of degree at most two, and Corollary 5.106 for an adjacent degree-three pair. The resulting forbidden families still have a cut vertex, so their changed rooted factors satisfy all three conditions by the cut-vertex theorem. Finally, adjacent corners share their unique maximal cube and hence have the same degree. This proves the last assertion as well. $\square$

Remark 2.104 (An adjacent corner can change the prescribed endpoint). Let B be the 289-concept factor rooted at 29 from Remark 2.50, and put

$$M = B \cup \{157\}.$$

The addition is ample and acquires the support $\{1, 2, 7, 8\}$, with bitmask 390 in the stored embedding. Both 29 and 157 are corners of the completed four-cube; they are adjacent in coordinate 7. The 290-concept family M has verified peelings ending at either vertex. In contrast, B cannot peel to 29. Thus deleting a nonroot corner can destroy peeling to the root even when the root is also a corner and is adjacent to the deleted corner. This is a support-four change of endpoint behavior, whereas item 3 of Proposition 2.61 excludes such a change for additions of support size at most two.

The checker `verify_damp_rooted_operations.py` replays both endpoint orders and checks the full-cube addition. It also validates the peripheral and product constructions in the proof and checks periphery-freeness and all 2047 nontrivial coordinate bipartitions of each of the seven stored rooted factors. The report is `damp_rooted_operations_verification.json`. This single repair does not

itself give a terminal cut-vertex obstruction. Theorem 2.66 combines it with a second repair of the same core to construct one.

Proposition 2.105 (Proper images after deleting a degree-three corner). *Let (A, a) satisfy the rooted-factor conditions of Theorem 2.19, in an irredundant embedding $A \subseteq Q_F$. Let $c \neq a$ be a corner with three-coordinate incident support P , and put $B = A \setminus \{c\}$. For $e \in F$, let π_e forget coordinate e , and write*

$$R_e C = \pi_e \{x \in C : x_e = a_e\}, \quad a'_e = \pi_e a.$$

Then B is ample and cannot peel to a . If $B \in \mathcal{D}_{\text{amp}}$, the pair (B, a) is a rooted factor if and only if

$$\begin{aligned} \pi_e B \text{ peels to } a'_e & \quad (e \in F \setminus P), \\ R_e B \text{ peels to } a'_e & \quad (e \in F \setminus P, c_e = a_e). \end{aligned} \tag{2.47}$$

If one of these tests fails, the corresponding image of A is obtained from the image of B by adding the nonroot concept $\pi_e c$, acquiring exactly P ; this addition makes peeling to a'_e possible.

If a is itself a corner adjacent to c , the assumption $B \in \mathcal{D}_{\text{amp}}$ holds automatically. In this case all root-containing restrictions outside P occur in the displayed test.

Proof. Corner deletion preserves ampleness. A peeling of B to a , prefixed by c , would peel A to a , which is impossible. The punctured three-cube at c retains every direction in P . It also supplies a surviving concept with the same outside coordinates as c , so deleting c cannot make any outside coordinate constant. Thus B still uses every coordinate of F .

Every elementary root-preserving image μA peels to its root. We compare it with μB . If $\mu = \pi_e$ and $e \in P$, the removed concept is identified with its neighbor $c^{\{e\}}$ in B , so the two images coincide. For $\mu = R_e$ with $c_e \neq a_e$, the removed concept is discarded and again they coincide. If $\mu = R_e$, $e \in P$, and $c_e = a_e$, the image of A adds $\pi_e c$ on the two-coordinate support $P \setminus \{e\}$. Indeed, the incident cube at that concept is precisely the retained face of the original corner cube. Both images are ample, and the root is an old concept of the smaller image. Item 3 of Proposition 2.61 therefore makes μB peel to its root.

Only the operations in (2.47) remain. In an outside contraction, c has no neighbor in the contracted direction, so its image is a new concept. Contraction retains exactly the shattered supports avoiding e ; the identity $\text{Sh}(A) = \text{Sh}(B) \dot{\cup} \{P\}$ therefore preserves the support gain P . In an outside restriction retaining c , the entire corner cube survives and its incident support is still P . The added concept differs from the image of the root: equality under contraction would make a an outside neighbor of c , and a retained restriction is injective. This proves the asserted description of every failed test.

If all displayed tests pass, every elementary root-preserving image of B peels to its root. The same is true for every proper root-preserving pc-minor, by closure under further root-preserving operations. Together with $B \in \mathcal{D}_{\text{amp}}$ and the failure to peel to a , this gives exactly the three factor conditions. Their necessity gives the converse.

Finally suppose that a, c are adjacent corners. They share their unique maximal three-cube. Write $a = c^{\{f\}}$ and $I = P \setminus \{f\}$. Deleting c leaves a as a corner of its I -face, so $G = A \setminus \{a, c\}$ is ample. Each individual addition of a or c to G acquires the same two-coordinate support I . Proposition 5.104, applied to $A = G \cup \{a, c\} \in \mathcal{D}_{\text{amp}}$, gives $G \in \mathcal{D}_{\text{amp}}$. Deleting the corner a first in B , and then peeling G , gives $B \in \mathcal{D}_{\text{amp}}$. The two adjacent concepts agree outside P , proving the final assertion about the tests. $\square$

This proposition separates ordinary peelability from rooted minimality when the deleted corner is adjacent to a corner root. It does not show that its outside-coordinate tests always pass. The

fixed checks in `verify_damp_rooted_corner_images.py` validate the image and support identities and replay stored endpoint orders for three degree-three preserving deletions. Their report is `damp_rooted_corner_images_verification.json`.

Proposition 2.106 (Minimal sources of rank-three endpoint repairs). *Let $S \subseteq Q_F$ be ample and ordinarily peelable, let $a \in S$, and suppose that $H = S \cup \{c\}$ is an ample one-concept extension with acquired support P , where $|P| = 3$. Assume that H peels to a and S does not. There is a root-preserving pc-minor map μ , using only coordinates in $F \setminus P$, such that*

$$(T, t) = (\mu S, \mu a)$$

is a rooted factor of Theorem 2.19. Moreover, $\mu H = T \cup \{t\}$ is still an ample one-concept extension with acquired support P , and it peels to t . The whole punctured P -cube at the added concept is retained.

If a, c are adjacent corners of H , they remain adjacent corners of μH , both of degree three. In that case t is a corner of T of degree two.

Proof. First consider an elementary root-preserving operation in a coordinate $p \in P$. Contraction identifies c with its existing neighbor, so the images of H and S coincide. A root-containing restriction that discards c also gives equal images. If the restriction retains c , its image adds that concept on the two-coordinate support $P \setminus \{p\}$. The root is an old concept of the smaller image. The image of H peels to the root, by root-preserving minor closure; item 3 of Proposition 2.61 gives the same conclusion for the image of S . Thus every elementary operation in P makes peeling to the root possible.

Choose a pc-minor (T, t) of (S, a) , minimal among those that cannot peel to their roots. It is ample and ordinarily peelable by ordinary pc-minor closure. Minimality supplies the third rooted factor condition. Coordinate restrictions and contractions on distinct coordinates commute: both simply retain the words with specified restricted bits and then forget the contracted bits. Every minor has such a form with disjoint restricted and contracted coordinate sets. If the map to T used a coordinate of P , that operation could therefore be performed first. The preceding paragraph and root-preserving minor closure would make T peel to t , a contradiction. Hence only coordinates outside P are used.

Apply these operations simultaneously to S and H . At every intermediate stage the smaller image fails at its root, since its descendant T does; the larger image peels to its root. Consequently the images cannot coincide. Their difference remains one concept, so no restriction discards the added concept and no contraction identifies it with an old image. An outside restriction retaining the concept retains its entire P -cube. An outside contraction preserves the shadow difference $\{P\}$, because contraction keeps precisely the shattered supports avoiding its coordinate. By the ample one-concept extension criterion, the acquired support remains P at every stage. This proves the first assertion and the retained punctured-cube statement.

Suppose now that a, c are adjacent corners of H . They share their unique maximal cube, of support P . A root-containing restriction outside P retains this cube and both corners. An outside contraction also preserves either corner: its fibre is a singleton, and projecting the ample family obtained by deleting that corner leaves a one-concept difference of support P . The extension criterion again makes its image a corner of degree three. Induction over the operations preserves both corners and their edge, whose coordinate belongs to P . Deleting μc from μH leaves t in the opposite face of that three-cube, with exactly two incident directions. Thus t is a degree-two corner of T . $\square$

Corollary 2.107 (The remaining adjacent-corner endpoint test). *The following statements are equivalent.*

1. *There is an ample family H with adjacent degree-three corners a, c such that H peels to a , but $H \setminus \{c\}$ does not peel to a .*
2. *There is a rooted factor (T, t) whose root is a degree-two corner, and an ample one-concept addition adjacent to t , whose acquired support has size three, that makes peeling to t possible.*

If these equivalent profiles do not exist, deleting a degree-three nonroot corner adjacent to a corner root preserves the rooted-factor conditions.

Proof. For item 1, put $G = H \setminus \{a, c\}$. Deleting either corner leaves the other as a degree-two corner, so G is ample and both individual additions to G have the same two-coordinate support. Proposition 5.104 makes G ordinarily peelable, and Proposition 2.13 gives this property to $H \setminus \{c\}$. The preceding proposition therefore produces item 2. Conversely, in item 2 the added concept is a degree-three corner. Its cube contains the neighboring root; the root's two old incident directions and its new edge are exactly the directions of that cube. Thus both are degree-three corners, giving item 1.

For the last assertion, apply Proposition 2.105. Ordinary peelability after deletion is already proved there. In every remaining outside-coordinate image, the root and the deleted corner remain adjacent degree-three corners by the corner transport argument above. Failure of an endpoint test would give item 1. If that profile is absent, all tests pass and the deletion preserves the factor conditions. $\square$

This equivalence specifies a minimal minor source for a possible failure; it does not assert that such a source exists or that rooted factors cannot have degree-two roots. Its factor conditions are still recognition data, not intrinsic admissibility.

The fixed checker `verify_damp_minimal_endpoint_repair.py` validates the corner and support transport in all 18 outside images of a 71-concept example, obtained by 219 corner deletions from the 290-concept family. Both this family and its 70-concept corner deletion peel to 29, so it is a positive control, not a repair of endpoint failure. The checker also replays all 48 rooted elementary orders and the existing peripheral example with a degree-two corner root that fails as an endpoint. That example has an incident-coordinate contraction already failing at its root, so it is not a rooted factor. The report is `damp_minimal_endpoint_repair_verification.json`.

Proposition 2.108 (A prescribed endpoint can be lost under reduction). *Let B and $M = B \cup \{157\}$ be the families in Remark 2.104, and put*

$$Y = (M \times \{0\}) \cup (B \times \{1\}), \quad r = (29, 0).$$

Then Y is ample, has 579 concepts, dimension 13, and VC dimension four. It peels to r , but its reduction

$$Y^{\{e\}} = \{x : (x, 0), (x, 1) \in Y\} = B$$

in the new coordinate e cannot peel to the projected root 29. The full fibre through the root is present. The only corners of Y are $(157, 0)$ and $(29, 1)$, and every peeling ending at r starts by deleting $(29, 1)$. VC dimension four is the least possible for such a failure under a nonempty reduction.

Proof. The peripheral expansion is ample. Its shadow is

$$\text{Sh}(Y) = \text{Sh}(M) \dot{\cup} \{I \cup \{e\} : I \in \text{Sh}(B)\}.$$

Since M, B have orders 290, 289 and VC dimensions four and three, respectively, this gives the asserted size and VC dimension. Both use all twelve old coordinates, and the new coordinate is active. By Proposition 2.61, Y peels to r : first peel the upper copy of B ordinarily, and then peel M to 29. Its reduction is exactly B , whose only corner is 29 and which has more than one concept. That corner must be deleted first, so it cannot be an endpoint.

An upper vertex is a corner of Y exactly when its projection is a corner of B ; thus the only upper corner is $(29, 1)$. The lower vertex $(157, 0)$ has no upper mate and is a corner because 157 is a corner of M . Any other lower corner would have an upper mate, and its incident cube would force its projection to be a corner of B . The only candidate is $(29, 0)$, but its neighbors $(157, 0)$ and $(29, 1)$ have the missing opposite vertex $(157, 1)$, excluding that corner. This proves the two-corner assertion.

Deleting $(157, 0)$ leaves $B \sqcap Q_1$. This product cannot peel to r , since its root-containing lower restriction is B , which cannot peel to 29. Hence a peeling ending at r must start with $(29, 1)$.

For the dimension threshold, if Z^D shatters I , then Z shatters $I \cup D$: every trace on I in the reduction comes from a full D -fibre. Thus $\text{VC}(Z^D) \leq \text{VC}(Z) - |D|$. If Z is ample of VC dimension at most three and $D \neq \emptyset$, its nonempty reduction is ample of VC dimension at most two. Such a family peels to any prescribed concept, by the VC-two endpoint result recalled in Remark 6.47. This excludes a smaller VC dimension. The checker `verify_damp_rooted_strong_projection.py` independently checks the shadow, the two corners, and explicit ordinary and endpoint orders; its report is `damp_rooted_strong_projection_verification.json`. $\square$

Ordinary corner peelings do pass to reductions by ordering full fibres according to their first deleted member. In this example the fibre containing the retained root is destroyed first, so the induced peeling starts at its projected root. Thus the endpoint condition is lost even though the ordinary projection argument applies. The outside-coordinate tests in Proposition 2.105 use contractions and root-containing restrictions; reduction cannot replace their endpoint compatibility requirement.

Proposition 2.109 (Peeling the section opposite a prescribed endpoint). *Let*

$$X = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, with both sections nonempty, and put $C = A_0 \cap A_1$. Suppose that A_1 admits a relative corner peeling to C : a sequence of corner deletions removing exactly $A_1 \setminus C$ and retaining all of C . For $a \in A_0$, one then has

$$X \text{ peels to } (a, 0) \iff A_0 \text{ peels to } a \text{ and } C \in \mathcal{D}_{\text{amp}}. \quad (2.48)$$

Consequently, in a rooted factor satisfying Theorem 2.19, the section opposite the root in every active coordinate has no relative corner peeling to the overlap. Each such section has VC dimension at least three and uses at least six coordinates. It contains an ample family D strictly containing the overlap C , with every corner of D lying in C . In particular, every rooted factor uses at least seven coordinates.

Proof. The sections and their overlap are ample. The overlap is nonempty because an ample family is connected and an edge between the two sections projects into C . If X peels to $(a, 0)$, its root-containing lower restriction peels to a . The overlap peels ordinarily: order its full two-vertex fibres by their first deleted member in a peeling of X . Equivalently, reverse the peeling and order full fibres by completion time; the resulting prefixes are reductions of ample prefixes and hence are ample. This establishes necessity without prescribing an endpoint in C .

Conversely, first carry out the given relative peeling in the upper section of X . Every deleted vertex lies outside C and has no lower mate, so its incident corner cube is unchanged by the presence of the lower section. The remaining family is $(A_0 \times \{0\}) \cup (C \times \{1\})$. The lower-root peripheral test in Proposition 2.61 now gives a peeling to $(a, 0)$, proving the equivalence.

Apply this to a rooted factor, orienting any active coordinate so that the root lies in the lower section. That section is a proper root-preserving restriction and peels to its root. The overlap belongs to $\mathcal{D}_{\text{amp}}$, by the ordinary full-fibre argument above applied to an ordinary peeling of the factor. A relative peeling of the opposite section would therefore contradict its failure to peel to the root. The opposite section must have VC dimension at least three by Lemma A.16, and it must use at least six coordinates by Theorem A.137. Since the split coordinate is absent there, the whole factor uses at least seven coordinates.

Finally, keep deleting corners of the opposite section outside C for as long as possible. The process cannot reach C , since that would give a relative peeling. It stops at an ample family $D \supsetneq C$ all of whose corners lie in C , as claimed. No ordinary peeling of this terminal D is asserted. $\square$

The pairs $C \subsetneq D$ are upper-root-locked pairs in the sense of Definition 5.45; their one-hole description is given by Theorem 5.49. They give necessary local certificates in every opposite section. Satisfying these conditions in every coordinate does not imply failure at the root: Proposition 2.111 gives endpoint-positive counterexamples. A sufficient compatibility condition remains open. The checker `verify_damp_rooted_section_obstructions.py` validates all 84 section/overlap pairs of the seven stored rooted factors, including their complete relative deletion graphs and ordinary overlap orders. Its report is `damp_rooted_section_obstructions_verification.json`.

Lemma 2.110 (Corner cubes and initial section obstructions). *Let $A \subseteq Q_F$ be ample and let $a \in A$. In each active coordinate, orient the two sections so that the root is lower. Every corner of the opposite section lies in its overlap with the root-containing section, in every coordinate, if and only if*

$$a \in Q_{I_A(c)}(c) \quad \text{for every corner } c \text{ of } A. \quad (2.49)$$

Here $I_A(c)$ is the incident support of the corner c , and $Q_{I_A(c)}(c)$ is its unique maximal cube.

Proof. A corner of an opposite section outside the overlap has no mate in the root-containing section. Its lift c is therefore a corner of A with the same incident support, and the split coordinate belongs to $\text{Diff}(c, a) \setminus I_A(c)$. Conversely, if a corner c has such a coordinate, its whole corner cube lies in the opposite section and it has no mate across that coordinate. Its projection is a corner outside the overlap. The absence of all these cases is exactly $\text{Diff}(c, a) \subseteq I_A(c)$, which is the displayed cube condition. $\square$

Proposition 2.111 (Endpoint peeling need not remove a coordinate section first). *There is an ample family P of VC dimension three and a concept $a \in P$ such that P peels to a , but, for every active coordinate e , there is no relative corner peeling of P to its root-containing section*

$$P_e(a) = \{x \in P : x_e = a_e\}.$$

Furthermore, in every coordinate the opposite section has no corner outside its overlap with $P_e(a)$. The family is two-connected, periphery-free, and Cartesian-prime.

VC dimension three is the least possible for the failure of this section-first schedule in a nontrivial ample family. The failure can also occur when a is itself a corner, in VC dimension four.

Proof with fixed certificates. Take the 294-concept family P from Remark 6.47 and set $a = 1895$, in its stored twelve-coordinate embedding. A complete peeling ending at 1895 is recorded in `computations/round776_noncorner_roots.json`. The same underlying family, rooted at its unique corner 1894, is the rooted factor of Remark 2.49; Corollary 2.103 and Theorem 2.19 give the three asserted irreducibility properties. The checker `verify_damp_rooted_section_counterexample.py` replays the endpoint order and the relative certificates below.

The only corner of P is 1894, and 1895 is its neighbor in coordinate 0. For every $e \neq 0$, the unique initial corner lies in $P_e(a)$, so no relative peeling to that section can start. Coordinate 0 has exactly 153 concepts opposite the odd root 1895. Its complete relative deletion certificate has 13 states. The possible deletions follow the chain

$$1894, 1638, 1634, 1632, 1600, 3648,$$

with 3942 insertable anywhere after 1894. The checker verifies that these are exactly all allowed next deletions in every state while the odd section remains fixed. All six maximal orders therefore remove the same seven concepts and then stop, leaving 146 concepts of the opposite section. This proves failure in coordinate 0 as well.

The root 1895 lies in the unique corner cube at 1894. Lemma 2.110 proves the asserted initial overlap condition in every coordinate. Those opposite sections are strictly larger than their overlaps, since P is periphery-free. Thus all the local relative-obstruction conditions of Proposition 2.109 can hold in an endpoint-positive family.

For the corner-root example use $M = B \cup \{157\}$, rooted at 29, from Remark 2.104. All five corners have the same maximal four-cube, which contains 29. Its twelve root-containing-section intervals have complete negative certificates with 34 states in total. The same checker replays them, its endpoint peeling, and all 24 rooted elementary minor orders of each positive example. It also directly checks two-connectivity, periphery-freeness, and Cartesian primality. The report is `damp_rooted_section_counterexample_verification.json`.

Finally, if a nontrivial ample family has VC dimension at most two, choose any active coordinate. Its root-containing section is nonempty and ample. Lemma A.16 gives a relative peeling to that section, which then peels to the prescribed root by the same lemma applied to a singleton. Hence the failure cannot occur below VC dimension three. $\square$

The obstruction in an opposite section concerns which concepts may be deleted while that section's overlap is retained. A different possible test uses every missing cube of the original family, in every rank. Even this test loses information when a deletion removes a vertex of a cube that was originally complete.

Proposition 2.112 (Pruning with original missing cubes is incomplete). *Let $H \subseteq Q_F$ be finite and let $A \subseteq H$ be a set of protected concepts, possibly empty. Starting with $U = H$, repeatedly delete any $v \in U \setminus A$ for which*

$$Q_{I_U(v)}(v) \subseteq H, \quad I_U(v) = \{e \in F : v^{\{e\}} \in U\}. \quad (2.50)$$

Cube containment is tested in the original H , not in the current U . The final set, denoted $K_A(H)$, is independent of the choices. It is the greatest set U , with $A \subseteq U \subseteq H$, such that every $v \in U \setminus A$ has an original missing-cube witness

$$P \subseteq I_U(v), \quad v^P \notin H. \quad (2.51)$$

The witness may be chosen so that all v^J , $J \subsetneq P$, belong to H ; thus it is a punctured cube already present in the original family.

If H is ample and admits a corner peeling to A retaining all of A , then $K_A(H) = A$. The converse fails both for A a singleton and for $A = \emptyset$. Specifically, there is a 292-concept rooted factor $(B, 0)$ with $K_{\{0\}}(B) = \{0\}$, and a 583-concept ample forbidden pc-minor X with $K_{\emptyset}(X) = \emptyset$. Their dimensions are twelve and twenty-four, respectively, and both have VC dimension four.

Proof. If a concept satisfies (2.50), it continues to satisfy it after other concepts are deleted: its incident support only shrinks. More directly, every set satisfying (2.51) survives any allowed deletion sequence. At the first deletion from such a set, all the witnessing neighbors would still be present, contradicting (2.50). The set left when pruning stops itself satisfies (2.51). It is therefore the greatest such set, proving uniqueness and independence of the deletion choices. Choose an inclusion-minimal P in (2.51); all its proper subsets give vertices in H , as asserted.

Suppose that a corner peeling retaining A exists and that $K_A(H) \setminus A \neq \emptyset$. Just before its first deletion of a member v of that difference, every witnessing neighbor from (2.51) remains. The corner cube would contain v^P , which was absent originally and is still absent. This is impossible, proving necessity.

For the counterexamples, normalize the family in Remark 2.50 by putting

$$B = \{x \oplus 29 : x \in G \cup \{91\}\}, \quad X = B \cup \{2^{12}x : x \in B\},$$

where $\oplus$ is bitwise exclusive-or and G is the fixed 291-concept seed used there. That remark establishes the rooted-factor and forbidden-minor assertions, with the stated dimensions. The replay `verify_damp_original_cube_pruning.py` checks pruning sequences of lengths 291 and 583, leaving $\{0\}$ and $\emptyset$, respectively. Uniqueness of the final set gives the claimed values of K_A .

The first discrepancy is explicit. The corners of B are 32, 64, 70, 96. The checked pruning begins 32, 64. After deleting 32, the incident support at 64 still has bitmask 102. Its full four-cube belongs to B , so the pruning rule permits deleting 64; but that cube's vertex 32 has just disappeared. Hence 64 is not a corner of $B \setminus \{32\}$. The rule has forgotten a newly created hole in an originally complete cube. Already in the original square $\{0, 32, 64, 96\}$, these first two deletions remove both common neighbors of the opposite surviving concepts 0, 96. This is exactly the full-square obstruction in the metric criterion of Theorem 5.71.

The same replay independently checks the complete negative deletion graphs, with eight rooted states and sixty-four ordinary states, an ordinary peeling of B , and all $24 + 72$ elementary minor orders. These establish failure of actual peeling independently of the pruning computation. Its report is `damp_original_cube_pruning_verification.json`. $\square$

Thus no protected-set certificate using only (2.51), even with all original punctured cubes available, can detect these two failures. The sufficiency of a nontrivial protected set remains valid. A complete construction must also control deletions inside originally complete cubes; the pruning rule does not supply intrinsic rooted-factor or repair admissibility.

Proposition 2.113 (Two-connectivity of replicated extensions). *Let $S \subseteq Q_F$ be connected with $|S| \geq 2$, let $E \neq \emptyset$ be disjoint from F , and let $\mathcal{A} \subseteq Q_F \setminus S$. Suppose every $a \in \mathcal{A}$ has at least two neighbors in S . For arbitrary rows $R_a \subseteq Q_E$, the graph of*

$$X = (S \times Q_E) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times R_a)$$

is two-connected. In particular, this holds when S is ample and nonpeelable and every $S \cup \{a\}$ is ample and peelable.

Proof. The core $S \times Q_E$ is two-connected. Indeed, after deleting (s, t) , its layers indexed by $Q_E \setminus \{t\}$ form a connected graph: a nontrivial cube with one vertex removed is connected. Every surviving vertex in the t -layer has a neighbor in one of these other layers. The core has at least four vertices, so this proves the assertion also when $|E| = 1$.

Every added vertex (a, t) has at least two distinct neighbors in this core. After deleting any one vertex of X , the surviving core is connected, and every surviving added vertex still has a neighbor in it. The whole remaining graph is therefore connected. For the last assertion, a resolving ample one-concept extension has at least three neighbors in its nonpeelable base by Proposition 2.13. $\square$

The terminal cut-vertex obstruction of Theorem 2.66 therefore has no nontrivial replicated common-seed presentation. Its two degree-four corners admit no obstruction-preserving deletion. Thus a separate terminal assembly type is necessary; intrinsic rooted-factor admissibility remains open.

Theorem 2.114 (Square completion across rooted blocks). *Let $(A_1, 0)$ and $(A_2, 0)$ be rooted factors satisfying the three conditions of Theorem 2.19, on disjoint coordinate sets F_1, F_2 . Put*

$$N_i = \{e \in F_i : 0^{\{e\}} \in A_i\}, \quad X = A_1 \vee A_2,$$

and choose a nonempty set $\Gamma \subseteq N_1 \times N_2$. Then

$$X_\Gamma = X \cup \{0^{\{e,f\}} : (e, f) \in \Gamma\} \tag{2.52}$$

is a two-connected ample forbidden pc-minor. Its shadow is

$$\text{Sh}(X_\Gamma) = \text{Sh}(A_1) \cup \text{Sh}(A_2) \cup \{\{e, f\} : (e, f) \in \Gamma\},$$

with the empty support shared by the two block shadows. Every added concept is a degree-two corner. If each A_i has its root as its only corner, these are all the corners of X_Γ .

Conversely, let O be a two-connected ample forbidden pc-minor, and let c be a degree-two corner such that $O \setminus \{c\}$ has a cut vertex. Then (O, c) has the displayed construction with $|\Gamma| = 1$. The two rooted factors and their chosen root neighbors are recovered uniquely up to interchange.

Proof. The cut-vertex theorem makes X forbidden. For $(e, f) \in \Gamma$, the concept $c_{ef} = 0^{\{e,f\}}$ has exactly the two neighbors $0^{\{e\}}$ and $0^{\{f\}}$ in the enlarged family. Indeed, the old concepts use coordinates of only one block, and distinct added concepts have even Hamming distance. The opposite vertex of their square is 0. Until c_{ef} is added, the trace 11 on $\{e, f\}$ is absent: no other added concept realizes it. Thus each addition, in any order, is ample with acquired support $\{e, f\}$, by the one-concept extension criterion. This proves the shadow formula.

At every step ordinary nonpeelability is preserved by Proposition 2.13. Proposition 2.12 then makes each enlarged family forbidden. Induction proves forbidden-minor status for X_Γ .

Each A_i is two-connected. After deleting 0, the two remaining block graphs are connected, and any one added concept joins them through its two neighbors. Every other added concept also has neighbors in these connected blocks. If an old concept other than 0 is deleted, the surviving old graph remains connected through 0, and each added concept retains at least one neighbor. If an added concept is deleted, the connected old graph X remains, with every other added concept attached to it. These cases prove two-connectivity.

The added concepts have their displayed squares as unique maximal cubes. Suppose that each factor has only its root as a corner. Then X is cornerless by Proposition 2.18. Any old concept becoming a corner would have a maximal cube containing an added concept, hence that cube would

be one of the displayed squares. The old vertices of such a square are $0, 0^{\{e\}}, 0^{\{f\}}$. The root has degree at least four in X , since each factor is two-connected. Either other vertex has degree at least two in its old block and gains a new neighbor, so has degree at least three. None can be a corner of a square. Therefore the new concepts are exactly the corners.

For the converse, put $S = O \setminus \{c\}$. Corollary 2.14 makes S an ample forbidden pc-minor. Let u, v be the neighbors of c , and let w be the opposite vertex of its square. If r is a cut vertex of S , the graph $O \setminus \{r\}$ can be connected only if $r \notin \{u, v\}$ and u, v lie in different components of $S \setminus \{r\}$: the added vertex c has no other neighbors. Thus r separates u from v in S . The path u, w, v lies in S , so $r = w$. The cut-vertex theorem recovers the two rooted factors of S at w ; the neighbors u, v belong to different blocks. Translate w to zero. Their edge coordinates are in the two disjoint block coordinate sets, and $c = 0^{\{e, f\}}$. This is the construction with one selected pair. Its recovered root, blocks, and neighbors are determined by (O, c) , proving uniqueness up to interchange. $\square$

The theorem gives a construction and recovery rule for two-connected obstructions whose deletion of a marked degree-two corner creates a cut vertex. The rooted factors still require intrinsic admissibility. Every displayed completion has an obstruction-preserving corner descent back to X , so these examples are not corner-descent terminal.

Remark 2.115 (A two-connected obstruction with one degree-two corner). Use the 289-concept factor $(B, 29)$ from Remark 2.50, translated to root zero, and take two copies in coordinate blocks $0, \dots, 11$ and $12, \dots, 23$. The cornerless double X has 577 concepts. The root directions in either factor are 1, 2, 8. Completing just the square on directions 1, 13 adds the concept

$$2^1 + 2^{13} = 8194.$$

The result has 578 concepts, isometric dimension 24, VC dimension three, and exactly one corner, of degree two. It is two-connected, and deleting that corner returns the cut-vertex obstruction X .

More generally, any nonempty subset of the nine pairs between $\{1, 2, 8\}$ and $\{13, 14, 20\}$ gives a two-connected forbidden pc-minor. A subset of size k gives $577 + k$ concepts and exactly k degree-two corners. This describes 511 distinct families in the fixed labelled embedding; no count of their isomorphism classes is asserted.

Degree two is the least possible corner degree in an ample forbidden pc-minor. Such a family is connected and is not a singleton. If it had a leaf, every geodesic from that leaf would begin with its unique incident coordinate. Isometry forces all other concepts to have the opposite value in that coordinate, making the coordinate peripheral. This contradicts Proposition 2.52.

The checker `verify_damp_square_completed_blocks.py` verifies the one-square and nine-square examples. It checks the exact shadows using block projections and mixed supports, two-connectivity, marked recovery, all 144 elementary minor peelings, and complete nonpeelability certificates with 2 and 512 states. The report is `damp_square_completed_blocks_verification.json`.

2.9 Rooted reductions and attachment examples

Proposition 2.116 (A three-core reduction retaining the root and repair cube). *Let (A, a) be a rooted factor of Theorem 2.19. Repeatedly delete nonroot vertices of degree less than three. Each deletion is a degree-two corner deletion preserving the rooted-factor conditions. The final family is the unique largest induced subgraph containing a in which every nonroot vertex has degree at least three.*

If a is a degree-two corner and an adjacent ample support-three addition restores peeling to a , the same addition restores peeling in this final factor. Its whole punctured three-cube survives, and the

root is the final factor's only degree-two vertex. Thus the equivalent repair profiles of Corollary 2.107, if they exist, can be realized with this additional degree condition.

Proof. A rooted factor is two-connected. The local degree-two fact in the preceding proof therefore makes every degree-two vertex a corner; no vertex has smaller degree. Deleting a nonroot such corner preserves the factor conditions by Corollary 2.103. Induction justifies the entire process. Its uniqueness follows by the same containment argument as for the ordinary three-core, keeping the root throughout.

For the repair assertion, write $H = A \cup \{c\}$, and let C be the completed three-cube at the adjacent corners a, c . We claim that a nonroot degree-two vertex x of A lies outside C . The only vertices of $C \setminus \{c\}$ with fewer than three neighbors inside this punctured cube are the three neighbors of c . One is a . If x were one of the other two, then c, x would be adjacent degree-three corners of H . Put $G = H \setminus \{c, x\}$. Both individual additions of c, x to G complete squares of the same support. Attach the fixed rooted factor of Remark 2.50 at a , on fresh coordinates. The attachment makes ordinary peelability equivalent to peeling to a , by Proposition 2.18. It changes neither of the two square additions, since a is distinct from their tips. Proposition 5.104 therefore implies that G peels to a , since H does. Adding back the corner x would make A peel to a , a contradiction. This proves the claim.

For a degree-two vertex x outside C , the addition of c gives it no new neighbor. Its square in A is still its unique maximal cube in H , so it is a degree-two corner there as well. Deleting x preserves peeling of H to a , by item 3 of Proposition 2.61. In A , the deletion preserves the rooted-factor conditions, and it leaves all of $C \setminus \{c\}$ intact. The support of the addition remains the same three-coordinate support, since its punctured cube remains and deleting a concept cannot shatter that support. Repeating this argument proves the assertion at every stage and in the final factor. The retained square at a keeps its degree equal to two. $\square$

For a concrete two-stage reduction, let X be the 577-concept double in Remark 2.115. Add $8194 = 2^1 + 2^{13}$ and then $8198 = 2^1 + 2^2 + 2^{13}$. These complete squares on $\{1, 13\}$ and $\{2, 13\}$, respectively. The resulting 579-concept family is two-connected and has just one corner, 8198, of degree two. Deleting it exposes 8194 as the next degree-two corner; deleting that concept gives the three-core X . The first deletion alone still leaves a two-connected family. Thus the three-core reduction covers examples beyond the marked single-deletion recovery in Theorem 2.114. The checker `verify_damp_three_core.py` verifies this example, its 72 elementary minor orders, and the canonical reductions of the earlier one-square and nine-square examples. It also checks the existing nonresolving extension of the 291-concept family by 186: this is a 292-concept three-core with two degree-four corners, so three-cores need not be cornerless. Its report is `damp_three_core_verification.json`.

The attachment targets cannot be replaced by the list of acquired pairs. For example, use the 577-concept core K from Remark 2.115. The following admissible data give a four-element diamond; integers in the target columns denote base concepts, and letters denote abstract new concepts.

x	I_x	u_x	v_x	w_x	$\phi(x)$
a	$\{1, 13\}$	8192	2	0	8194
b	$\{2, 13\}$	a	6	2	8198
d	$\{1, 14\}$	24576	a	8192	24578
t	$\{2, 14\}$	d	b	a	24582

Its relative deletion order satisfies $t < b < a$ and $t < d < a$, with b, d incomparable, so there are six ample intermediate families. Using the same four pairs but taking $(u_x, v_x, w_x) = (2^{q_x}, 2^{p_x}, 0)$ in

every row instead gives the four concepts 8194, 8196, 16386, 16388. Its partial order is an antichain, giving sixteen ample intermediates. Both outputs have 581 concepts, VC dimension three, the same full shattered-set complex, and the same labelled three-core. They are not even graph isomorphic: the first has one corner and the second has four.

The checker `verify_damp_square_attachment_data.py` constructs both families from these abstract data, independently recovers their data from their orientations, and checks all 32 intermediate subsets. It replays 144 new elementary minor orders and verifies complete negative certificates with six and sixteen states, respectively. The report is `damp_square_attachment_data_verification.json`.

The remaining adjacent rank-three repair can be reduced further using the square attachment construction. The key point is to exclude a peeling that first removes the corner opposite the retained root.

Lemma 2.117 (Two opposite corner edges of a three-cube). *Let H be ample and contain a three-cube of support $\{p, q, f\}$. Write its vertices as*

$$c, \quad a = c^{\{f\}}, \quad z = c^{\{p,q\}}, \quad w = c^{\{p,q,f\}}, \quad u_i = c^{\{p\} \cup F_i}, \quad v_i = c^{\{q\} \cup F_i} \quad (i = 0, 1),$$

where $F_0 = \emptyset$ and $F_1 = \{f\}$. If c, a, z, w are all degree-three corners of H , then

$$H \text{ peels to } a \iff H \setminus \{c\} \text{ peels to } a.$$

Proof. Normalize $c = 0$. In the restriction $(p, q) = (0, 0)$, the edge $\{c, a\}$ is an isolated component: its two vertices have no neighbors outside the displayed cube. Ample restrictions are connected, so this restriction consists of that edge alone. The restriction $(p, q) = (1, 1)$ consists of $\{z, w\}$ for the same reason. Let U, V be the projections of the other two restrictions, with $(p, q) = (1, 0), (0, 1)$, respectively. They are ample and contain the same edge $E = \{e_0, e_1\}$, whose endpoints have f -bits zero and one and all other remaining bits zero.

We prove the following explicit criterion for the asserted endpoint:

$$U \text{ and } V \text{ peel to } e_1, \quad \text{at least one of } U, V \text{ peels relatively to } E. \quad (2.53)$$

Suppose first that H peels to a . Its root-containing restriction $q = 0$ is the peripheral expansion of U with duplicated site E , with the root in that site. Root-preserving minor closure and Proposition 2.61 give a peeling of U to e_1 . Restricting $p = 0$ gives the corresponding statement for V .

Follow the peeling until its first deletion from $W = \{u_0, u_1, v_0, v_1\}$. Before this time, deletions outside the displayed cube induce corner deletions in U or V retaining E : such vertices have no neighbors in either of the two other restrictions. Both residual families are therefore ample. Among c, a, z, w , the vertex a is retained. With all of W still present, the only possible sets of deleted vertices from $\{c, z, w\}$ are

$$\emptyset, \quad \{c\}, \quad \{z\}, \quad \{w\}, \quad \{z, w\}.$$

Indeed, deleting c leaves neither z nor w a corner; deleting z permits only w next among these vertices, and deleting w permits only z . After both are gone, c still has three neighbors but its incident cube is incomplete.

Consequently a lower vertex of W still has a neighbor in $\{c, z\}$, and an upper vertex still has the neighbor a . These neighboring vertices have no outside neighbors. Thus the first corner deleted from W has no neighbor in its own restriction outside E : an outside edge would require a missing square through the retained neighboring corner.

Suppose this first boundary deletion is u_i ; the case of v_i is obtained by interchanging p, q and U, V . Let U' be the residual family just before this deletion. Its $f = i$ section is connected and e_i

is isolated there, so that section is the singleton $\{e_i\}$. The remaining global family still peels to a . Restrict it to $q = 0$. If $i = 0$, this is the upper section of U' , joined to a at u_1 , with possibly the leaf c also attached to a . Contracting these attachment edges gives that upper section rooted at e_1 , so root-preserving minor closure makes it peel to e_1 . If $i = 1$, the lower section remains, joined to a by the path u_0, c, a . Here c must still be present, since otherwise this nonempty restriction would be disconnected. Contracting that path makes the lower section peel to e_0 . In either case, reattaching the deleted singleton as a leaf at the opposite endpoint allows that section peeling while retaining both members of E . Prepending the earlier deletions in U gives a relative peeling of U to E . This proves (2.53).

Conversely, suppose (2.53) holds, and first assume that U peels relatively to E . In $H \setminus \{c\}$, perform those deletions in its $(1, 0)$ restriction. Then delete

$$u_0, z, w, u_1$$

in that order. The first three deletions are square corners and the last is a leaf. The remaining graph is V with a attached as a leaf at v_1 . Peel V to v_1 , then delete v_1 , retaining a . If instead V supplies the relative peeling, interchange the two restrictions. Thus the criterion implies that $H \setminus \{c\}$ peels to a . Prefixing c also peels H to a , proving the equivalence. $\square$

Theorem 2.118 (Reduction to a switch between two square attachments). *Failure of the adjacent-corner endpoint property in Corollary 2.107 is equivalent to the existence of the following data. Take an ample family G , distinct coordinates p, q, r , and $b \in G$. Put*

$$a = b^{\{r\}}, \quad c = a^{\{p\}}, \quad z = a^{\{q\}}, \quad C = Q_{\{p, q, r\}}(a).$$

Require

$$G \cap C = C \setminus \{a, c, z\}, \quad \{p, r\}, \{q, r\} \notin \text{Sh}(G). \quad (2.54)$$

Require in addition that the two families

$$S = G \cup \{z, a\}, \quad T = G \cup \{c, a\}$$

have different endpoint behavior: T peels to a and S does not.

For all data satisfying (2.54), both S and T are ample square extensions of G , adding the same two supports in opposite orders, and

$$\text{Sh}(S) = \text{Sh}(T) = \text{Sh}(G) \cup \{\{p, r\}, \{q, r\}\}.$$

Their union H is ample; a, c, z are degree-three corners of its completed cube. In an endpoint-changing example, G is ordinarily peelable. One may choose such an example with S a rooted factor whose root a is its only degree-two vertex. The vertex z is then a noncorner degree-three neighbor of a in S .

Proof. First check the construction. The five supplied cube vertices give punctured squares at c and z , on supports $\{q, r\}$ and $\{p, r\}$, respectively. The distinct-pair construction of Theorem 2.60 makes $G + c$, $G + z$, and $G + c + z$ ample. It also gives the two square addition sequences z, a and c, a , with respective support sequences

$$\{p, r\}, \{q, r\} \quad \text{and} \quad \{q, r\}, \{p, r\}.$$

These have exactly the prescribed punctured squares, so the construction theorem proves the claims about S, T and their shadows. Adding a to $G + c + z$ completes its three-cube. Its support $\{p, q, r\}$

is not yet shattered, and the punctured-cube lemma excludes any further incident direction, as in that construction theorem. The one-concept criterion makes H ample. Its three displayed vertices have exactly their three cube neighbors, so they are corners. If T peels to a , prefixing z peels H to a . Failure in $S = H - c$ is therefore an adjacent rank-three repair. Ordinary rank-two invariance also makes G peelable.

For the other direction, choose an adjacent rank-three repair $H, S = H - c$ of minimum cardinality, with retained root a , and let C be the common three-cube at a, c . A successful peeling cannot start at a , which is retained, or at c , which would leave the bad side S . It cannot start outside C : such a first corner is also a corner of S , and deleting it leaves a smaller repair with the whole cube intact. Nor can it start at a neighbor $x \neq a$ of c in C . In that case c, x are adjacent degree-three corners, and their simultaneous removal leaves an ample family to which their individual additions are squares on the same support. Attach the fixed bad rooted factor at a . Propositions 2.18 and 5.104 imply that the smaller family peels to a ; adding back x would make S do so.

Write the cube directions as $\{p', q', f\}$, where $a = c^{\{f\}}$, and put $z' = c^{\{p', q'\}}$, $w' = z'^{\{f\}}$. We claim that a successful peeling can start at one of $a^{\{p'\}}, a^{\{q'\}}$. The only other possible first vertices are z' and w' . Suppose first that it starts at z' , the antipode of a . A second deletion outside C could be moved to the front: z' has no outside neighbor, deleting a nonneighbor cannot create a corner, and the cube of z' survives that outside deletion. This would give the already excluded smaller repair. Inside C , a second corner must neighbor z' , since a nonneighbor retains all three cube directions while their cube is incomplete.

The options are $c^{\{p'\}}, c^{\{q'\}}, w'$. For either of the first two, its remaining corner square contains c , which has no outside neighbor. Thus it has no outside neighbor either and was already a degree-three corner of H . Interchanging the first two adjacent corner deletions gives a peeling starting at a neighbor of c , which was excluded above. For w' , its remaining corner square contains a ; the same reasoning makes it a degree-three corner of H . Now c, a, z', w' satisfy Lemma 2.117, contradicting the repair. Thus z' cannot be first.

If the first vertex is w' , a second outside deletion can again move to the front. A second inside deletion must be one of its neighbors $a^{\{p'\}}, a^{\{q'\}}, z'$. Each one's remaining corner square contains c , so it has no outside neighbors and was already a corner of H . Interchanging the first two deletions makes that vertex first. The option z' was just excluded; either other option proves the claim.

Let $z \neq c$ be this successful first corner adjacent to a . Name the directions of ac, az as p, q , and the third as r . Put $G = H \setminus \{a, c, z\}$. Deleting a first leaves c, z as square corners; deleting both therefore makes G ample. These three deletions remove exactly the triple support and the two distinct pairs $\{q, r\}, \{p, r\}$. Thus (2.54) holds with $b = a^{\{r\}}$. The successful first deletion of z makes $T = H - z$ peel to a , whereas $S = H - c$ does not.

Finally, S is ordinarily peelable by the adjacent square-pair argument in Corollary 2.107. Proposition 2.106 would produce a smaller repair if S were not a rooted factor, and Proposition 2.116 would do so if it had another degree-two vertex. Minimality excludes both possibilities. The vertex z is not adjacent to c , so it retains all three incident directions in S , but its cube now lacks c . It is therefore a noncorner of degree three adjacent to the root. $\square$

The endpoint condition at this two-square switch remains open. The construction conditions are intrinsic, but deciding whether the switch changes the endpoint is precisely the unresolved question, not a completed admissibility test. In particular, the result does not exclude all degree-two-root factors.

The checker `verify_damp_two_square_endpoint_switch.py` validates the opposite-edge criterion on two positive connectors and a connector with two bad relative edges. The latter has complete negative endpoint certificates with 25 and 125 states before and after restoring the missing cube

corner. It also checks the five possible sets of removed cube vertices and all 20 first-boundary corner exclusions. For the switch itself, the existing 71-concept example gives $a = 29, c = 157, z = 25$, a 68-concept base, and two square supports with bitmasks 6, 130. Both 70-concept sides peel to 29, so this is a positive control. All 20 stored endpoint orders, including the two fixed edge tests, pass. The report is `damp_two_square_endpoint_switch_verification.json`.

Theorem 2.119 (A two-square switch with only small lower supports). *Assume the geometric switch conditions of (2.54). Flip coordinate values so that $a = 0$, and identify words with their sets of nonzero coordinates. Delete coordinate r from the two sections of G , writing*

$$G = (D \times \{0\}) \cup (U \times \{1\}), \quad E = D \cap U, \quad B = D \cup U, \quad d = \{p, q\}.$$

Then D, U, E, B are ample, $d \in E$, every word of D has $p = q = 1$, and U contains the square $\{0, \{p\}, \{q\}, d\}$. Moreover,

$$\text{Sh}(E) = \text{Sh}(D) \cap \text{Sh}(U), \quad \text{Sh}(B) = \text{Sh}(D) \cup \text{Sh}(U). \quad (2.55)$$

If

$$|I| \leq 2 \quad \text{for every } I \in \text{Sh}(D) \setminus \text{Sh}(E), \quad (2.56)$$

then the two switch sides S, T have the same endpoint behavior:

$$\begin{aligned} S \text{ peels to } a &\iff T \text{ peels to } a \\ &\iff U \text{ peels to } 0 \text{ and } E \text{ peels to } d \\ &\iff B \text{ peels to } 0 \text{ and } D \text{ peels to } d. \end{aligned} \quad (2.57)$$

In particular, a switch with $D \subseteq U$ cannot change the endpoint. Every endpoint-changing switch requires a support of size at least three in $\text{Sh}(D) \setminus \text{Sh}(U)$. Such a support avoids all three coordinates p, q, r . If the bad side (S, a) is a rooted factor, there must also be a support of size at least three in $\text{Sh}(U) \setminus \text{Sh}(D)$.

Proof. The five specified vertices of G realize the three traces 01, 11, 10 on each of $\{p, r\}$ and $\{q, r\}$. Since neither pair is shattered, the trace 00 is absent globally on each pair. Thus $r = 0$ forces $p = q = 1$. The same five vertices give $d \in E$ and the stated square in U . Sections, contractions, and reductions of an ample family are ample, so D, U, B, E are ample.

For completeness, a support avoiding r is shattered by G exactly when it is shattered by B . A support $I \cup \{r\}$ is shattered by G exactly when both sections shatter I . Strong shattering in G then supplies a full $I \cup \{r\}$ -cube, whose projection is an I -cube in E . The reverse implication follows from $E \subseteq D, U$. Hence the first equality in (2.55) holds and

$$\text{Sh}(G) = \text{Sh}(B) \dot{\cup} \{I \cup \{r\} : I \in \text{Sh}(E)\}.$$

An I -cube in G for a support avoiding r lies entirely in one section. Ampleness therefore also gives the second equality in (2.55).

Assume (2.56). By Theorem 2.57, delete $D \setminus E$ from D by corners of degree at most two. These deletions lift to both S and T , with the same degrees. Indeed, each deleted word has no upper mate, and the two added lower vertices form the path $0, \{q\}, d$ in S , or $0, \{p\}, d$ in T , attached to D only at $d \in E$. Thus no deleted word gains a neighbor from either the upper section or the path. Its entire incident cube remains in the current lower section. Proposition 2.61 preserves the old endpoint a at every deletion.

Write $h = q$ for S , or $h = p$ for T . The remaining family is

$$((E \cup \{0, \{h\}\}) \times \{0\}) \cup (U \times \{1\}).$$

Its lower section is contained in its upper section. The upper-root case of the peripheral criterion in Proposition 2.61, with the two levels interchanged, makes its endpoint at $(0, 0)$ equivalent to U peeling to 0 and $E \cup \{0, \{h\}\}$ peeling to 0. The latter is equivalent to E peeling to d : necessity follows by contracting the two edges of the attached path, and sufficiency follows by peeling E to d , then deleting $d, \{h\}$ and retaining 0. This proves the first two equivalences in (2.57) for both choices of h .

Theorem 2.57 also equates the endpoint d in D and E . By (2.55),

$$\text{Sh}(B) \setminus \text{Sh}(U) = \text{Sh}(D) \setminus \text{Sh}(E).$$

The same theorem applied to $U \subseteq B$ equates their endpoint 0, proving the last equivalence. When $D \subseteq U$, the support difference is empty. In every other endpoint-changing case, the contrapositive supplies a support of size at least three in that difference. Coordinates p, q are constant on D , and coordinate r has been removed, so this support avoids all three.

Finally suppose (S, a) is a rooted factor. Its section opposite $r = 0$ is U , and the overlap is $E \cup \{0, \{q\}\}$. The attached path uses only the coordinates p, q , which are constant on E , so

$$\text{Sh}(E \cup \{0, \{q\}\}) = \text{Sh}(E) \cup \{\{p\}, \{q\}\}.$$

If every support in $\text{Sh}(U) \setminus \text{Sh}(E)$ had size at most two, then Theorem 2.57 would peel U relatively to this overlap. This contradicts Proposition 2.109. Hence there is a support of size at least three in $\text{Sh}(U) \setminus \text{Sh}(E)$, which equals $\text{Sh}(U) \setminus \text{Sh}(D)$ by (2.55). $\square$

The existing 68-concept switch base has section sizes $|D| = |E| = 11$ and $|U| = 57$, so its equal positive endpoints are explained by the peripheral case. The checker `verify_damp_switch_section_supports.py` also treats two fixed nonperipheral switches with a three-concept lower square attachment. One has positive endpoints on both sides. In the other, both sides have a root-preserving contraction onto the bad rooted factor from Remark 2.50, and both endpoints fail. These controls check the two possible signs of the criterion; neither is an endpoint-changing switch. The remaining case requires supports of size at least three exclusive to each section, with the lower support outside the switch cube. Their compatibility with rooted minimality remains open.

Lemma 2.120 (Four corners along a three-cube path). *Let H be ample and contain a three-cube C . Suppose that c, a, z, v are four degree-three corners whose three consecutive edges form a shortest path between opposite vertices of C . Then*

$$H \text{ peels to } a \iff H \setminus \{c\} \text{ peels to } a.$$

Proof. The reverse implication follows by deleting c first. For the forward implication, suppose there is a counterexample of minimum cardinality. Normalize $a = 0$ and write

$$c = \{p\}, \quad z = \{q\}, \quad v = \{q, r\}, \quad b = \{r\}, \quad u = \{p, r\}, \quad d = \{p, q\}, \quad w = \{p, q, r\}.$$

The displayed words have zero entries outside $P = \{p, q, r\}$. The four assumed corners have no neighbors outside C . We first restrict the possible deletions inside this cube, and then use a peripheral restriction to contradict minimality.

Neither d nor u can be a corner of H . Indeed, if $x \neq a$ is a corner adjacent to c , then both c, x have degree three and $H \setminus \{c, x\}$ is ample. The two individual additions are squares on the same support. Glue the fixed bad rooted factor from Remark 2.50 at a . The vertex-gluing criterion and Proposition 5.104 make peeling H to a equivalent to peeling $H \setminus \{c, x\}$ to a . Adding back x preserves this endpoint by Proposition 2.61, contradicting the failure in $H - c$. Nor can w be a corner of H : the two opposite edges $\{c, a\}, \{w, v\}$ would then satisfy Lemma 2.117.

A successful first deletion outside C would leave a smaller counterexample with all four marked corners intact. Such a corner is also a corner of $H - c$, so root failure survives its deletion. The first deletion is therefore z, v , or b . If it is v , the second cannot lie outside C , since it would commute to the front. Here and below, consecutive nonadjacent corner deletions commute: the second is already a corner before the first, and it cannot lie diagonally in the first corner's cube, since that missing first vertex would prevent it from being a corner afterward. Inside $C - v$, a second corner is one of z, b, w . Its remaining square contains c , so it has no outside neighbor and was already a corner of H . The option w is excluded; interchanging the first two adjacent cube-corner deletions gives a successful first deletion at z or b . Interchanging q, r if needed, take this first vertex to be z . The vertex v is unchanged by this interchange and remains a degree-three corner of H .

The second deletion is now v . An outside deletion would again commute to the front. A corner of $C - z$ other than the retained root must be d or v . At d , the remaining square contains c , so cornerhood would make d a degree-three corner of H , already excluded. Thus the first two deletions are z, v .

A third deletion outside C would commute past both of these vertices, which have no outside neighbors in H . The nonroot corners of the remaining cube can only be b, d, w . Their squares contain a or c , so each would have no outside neighbor and be a corner of H . The exclusions of d, w force the third deletion to be b . In particular, b is also a degree-three corner of H .

The four vertices a, z, v, b form the entire restriction $p = 0$. They form a square with no outside neighbor in that restriction; connectedness of an ample restriction excludes any other component. Writing Q for their projected $\{q, r\}$ -square, we have

$$H = (Q \times \{0\}) \cup (V \times \{1\}), \quad Q \subseteq V,$$

where the displayed levels now refer to coordinate p . The image of c is 0, a degree-two corner of V . The peripheral endpoint criterion gives a peeling of V to 0. Its first corner cannot lie in Q : it is not the retained root, and any of the other three vertices has its entire square through 0, hence no outside neighbor. It would consequently make d, u , or w a corner of H , contrary to the exclusions.

Let $x \notin Q$ be this first corner of V , regarded as a vertex in the $p = 1$ section of H . It has no mate in the other section, so it is also a corner of H and $H - c$. The peripheral criterion applied to $V - x$ makes $H - x$ peel to a . But $H - c - x$ still fails at a , since otherwise prefixing x would peel $H - c$ to a . The four marked corners and their cube survive, giving the final contradiction to minimality. $\square$

Corollary 2.121 (The first two upper cubes in a smallest repair). *Suppose an adjacent rank-three endpoint repair exists, and choose $H, S = H - c$ of minimum cardinality, rooted at a . Use the switch coordinates of Theorem 2.118, normalized as $a = 0, c = \{p\}, z = \{q\}$, and let U be its upper r -section as in Theorem 2.119. A successful peeling of H can be chosen to begin*

$$z, \quad v = \{q, r\}, \quad x = \{q, r, e\}.$$

There are nonempty sets J, F , disjoint from $P = \{p, q, r\}$, with $e \in J$ and $F \cap J = \emptyset$, such that the corner supports of these second and third deletions are respectively

$$R = \{p, q\} \cup J, \quad R' = (\{p, q\} \cup (J \setminus \{e\})) \cup F. \quad (2.58)$$

The two corner cubes project to full cubes in the original U . Their intersection is the facet of the first cube with $e = 1$. In particular, U has a shattered support of size at least three containing both p, q , whereas the lower section has an exclusive support of size at least three avoiding all of P .

Proof. Put $C = Q_P(0)$ and write

$$b = \{r\}, \quad u = \{p, r\}, \quad d = \{p, q\}, \quad w = \{p, q, r\}.$$

Choose the successful first corner z using Theorem 2.118. The nonroot neighbors d, u of c cannot be corners of H , by the pointed square-pair argument in the preceding proof. After deleting z , an outside first corner would commute ahead of z , contradicting minimality. Inside the cube, the only nonroot possibilities are d and v . At d , the remaining square contains c ; hence it would have no outside neighbor and would already be a corner of H . This excludes d , so the second vertex is v .

In $H - z$, the corner at v contains the upper $\{p, q\}$ -square. It has no r -neighbor, so its support has the form $R = \{p, q\} \cup J$, with $J \cap P = \emptyset$. If J were empty, restoring z would make v a degree-three corner of H . Then c, a, z, v would satisfy Lemma 2.120, contradicting the repair. Thus $J \neq \emptyset$.

After deleting z, v , no nonroot vertex inside C can be the next corner. The vertices c, u retain all three cube directions while their cube is incomplete. Cornerhood at d would again force it to be a corner of H . For every $j \in J$, the upper cube at v supplies the outside neighbors $b^{\{j\}}, w^{\{j\}}$. The vertex b also retains its neighbor a , which has no outside neighbor, so the corresponding square at b is missing. The vertex w retains directions q, r ; a corner cube at w containing j would require $c^{\{j\}}$, also absent. Thus neither b nor w is a corner.

The third vertex x is therefore outside C . Unless it neighbors v , its deletion commutes past v and z , giving a successful first deletion outside C and a smaller repair. Hence $x = v^{\{e\}}$ for some $e \in J$. Its corner cube contains the full $R \setminus \{e\}$ -face opposite v . Direction e is absent after deleting v , and direction r is absent because the lower section has $p = q = 1$ apart from the three added vertices at outside word zero. Its support therefore has the form R' in (2.58), allowing for now $F = \emptyset$.

If F were empty, x would already be a corner of $H - z$ on the same cube as v , and also a corner of H , since z lies in the opposite r -section and has no outside neighbors. The deletions z, v, x could be reordered as x, z, v : deleting x leaves the cube at z intact, and after deleting z, x , the remaining face at v is a corner cube. This again contradicts minimality. Thus $F \neq \emptyset$. Both cubes are in the upper section. The second lies entirely in $e = 1$, so it did not use the removed vertex v and was already present in U . Their supports and positions give exactly the stated intersection. Finally, R is exclusive to U because p, q are constant on the lower section; the lower exclusive support follows from Theorem 2.119. $\square$

The four-corner path lemma excludes this configuration uniformly, without assumptions on the supports of the remaining family. The checker `verify_damp_four_corner_path.py` tests it on two nonperipheral controls with exclusive supports of size three in both sections: the 36-concept example and its corner deletion have positive endpoints, whereas the 317-concept example and its corner deletion contract to the known bad rooted factor. It also checks every local corner exclusion in the proof.

The new upper-cube pattern itself is realizable: a 23-concept ample family has a successful prefix with corner supports $\{p, q, r\}, \{p, q, e\}, \{p, q, f\}$, where $e \neq f$. Both 22-concept switch sides still peel to the root. Thus the required successive cubes do not alone imply failure. Whether this pattern can occur in an endpoint-changing switch with a rooted minimal bad side remains open.

2.10 Relative peeling and quantitative reductions

Proposition 2.52 lets us replace an absolute cardinality question by a smaller relative one. Define the *relative corner threshold*

$$\rho_{\mathcal{D}_{\text{amp}}} = \min\{|A \setminus B| : \emptyset \neq B \subsetneq A, A, B \in \mathcal{D}_{\text{amp}}, \text{Cor}(A) \subseteq B\}, \quad (2.59)$$

with the value $+\infty$ if there is no such pair. Thus a pair counted by $\rho_{\mathcal{D}_{\text{amp}}}$ consists of two dismantlable ample families, but every corner of the upper family is trapped in the lower one.

Proposition 2.122 (Two-wing reduction to a relative corner threshold). *Let H have minimum cardinality among all nonempty cornerless ample families. Then*

$$|H| \geq 2\rho_{\mathcal{D}_{\text{amp}}} + 2. \quad (2.60)$$

Consequently, $\rho_{\mathcal{D}_{\text{amp}}} \geq 22$ would exclude every such family of order less than 46 in one step.

Proof. Delete the constant coordinates of H and fix an active coordinate e . Write

$$A_b = \rho_e^b H \quad (b \in \{0, 1\}), \quad B = A_0 \cap A_1, \quad D_b = A_b \setminus B.$$

The two sections A_0, A_1 are ample, and the two-layer intersection lemma makes B ample. Isometry of H gives an e -edge, so B is nonempty. Every one of these three families has order smaller than $|H|$; hence Lemma 7.8 puts all of them in $\mathcal{D}_{\text{amp}}$.

The family H is a forbidden pc-minor for $\mathcal{D}_{\text{amp}}$: every proper pc-minor is ample and has smaller order, and therefore belongs to $\mathcal{D}_{\text{amp}}$ by the same lemma. Proposition 2.52 now shows that e is nonperipheral. Equivalently, both D_0 and D_1 are nonempty.

If $d \in D_b$ were a corner of A_b , then (d, b) would have no incident e -edge in H , and all its other incident cubes would be exactly the incident cubes of d in A_b . Thus (d, b) would be a corner of H , a contradiction. Hence

$$\text{Cor}(A_b) \subseteq B \quad (b \in \{0, 1\}).$$

Both nested pairs $B \subsetneq A_b$ occur in the minimum defining $\rho_{\mathcal{D}_{\text{amp}}}$, so $|D_b| \geq \rho_{\mathcal{D}_{\text{amp}}}$. Finally,

$$|H| = |A_0| + |A_1| = |D_0| + |D_1| + 2|B| \geq 2\rho_{\mathcal{D}_{\text{amp}}} + 2,$$

because B is nonempty. □

Proposition 2.123 (A twelve-concept trapped interval). *The relative corner threshold satisfies*

$$\rho_{\mathcal{D}_{\text{amp}}} \leq 12. \quad (2.61)$$

More precisely, the smaller section $Q \subseteq Q_{11}$ of the 291-concept obstruction in Proposition 7.45 contains a family $B \in \mathcal{D}_{\text{amp}}$ such that

$$|Q| = 85, \quad |B| = 73, \quad \text{Cor}(Q) \subseteq B,$$

and there is no ample family strictly between B and Q .

Computer-assisted proof. Let $K = P \cap Q$ be the 59-concept overlap in the construction of Proposition 7.45. In the decimal encoding of eleven-bit words, put

$$(w_1, \dots, w_{14}) = (2038, 898, 910, 398, 386, 1024, 1958, 1926, \\ 1974, 902, 390, 1942, 1152, 1408)$$

and $B_i = K \cup \{w_1, \dots, w_i\}$. The exact one-concept colon test in Lemma A.1 verifies that every B_i is ample. The family K belongs to $\mathcal{D}_{\text{amp}}$, and w_i is a corner of B_i , because deleting it leaves the ample family B_{i-1} . Reversing these fourteen additions and then peeling K therefore proves $B = B_{14} \in \mathcal{D}_{\text{amp}}$. The section Q also belongs to $\mathcal{D}_{\text{amp}}$.

The section-corner calculation gives $\text{Cor}(Q) \subseteq K \subseteq B$, and $|Q \setminus B| = 85 - (59 + 14) = 12$. Thus $B \subsetneq Q$ occurs in the minimum defining $\rho_{\mathcal{D}_{\text{amp}}}$. As an independent minimality check, the twelve acquired shattered supports are

$$14, 22, 26, 28, 518, 536, 1030, 1048, 1538, 1540, 1544, 1552.$$

Testing all $2^{12} - 2 = 4094$ proper nonempty subsets of the twelve remaining concepts against these supports finds no ample intermediate family. The reconstruction, the fourteen colon certificates, the terminal blocking vertices, and the interval scan are recorded in `damp_hall_relative_leaf_core.json` and reproduced by `analyze_damp_hall_relative_leaf_core.py`. The mathematical digest is

$$\text{e019f8c6ec0679ba930d06731c11988c98793fa5e26aba330739d8b503d721a0}.$$

□

Proposition 2.124 (The sharp interval has an octahedral support sphere). *For the twelve-concept interval $B \subsetneq Q$ in Proposition 2.123, let*

$$Y_0 = \{1, 2, 3, 4, 9, 10\}, \quad P_1 = \{1, 2\}, \quad P_2 = \{3, 4\}, \quad P_3 = \{9, 10\}.$$

All twelve acquired supports have order three, and their family is

$$\mathcal{H}_0 = \{T \in \binom{Y_0}{3} : P_i \subseteq T \text{ for some } i \in \{1, 2, 3\}\}. \quad (2.62)$$

Consequently, the pure complex whose facets are the rank-three supports in $\binom{Y_0}{3} \setminus \mathcal{H}_0$ is

$$\partial 2^{P_1} * \partial 2^{P_2} * \partial 2^{P_3} \cong S^0 * S^0 * S^0 \cong S^2, \quad (2.63)$$

the boundary complex of the three-dimensional cross-polytope. The three opposite vertex pairs of this octahedral sphere are precisely the direction pairs of the three vertex-disjoint squares in the relative graph.

Proof. Decode the twelve masks displayed in the proof of Proposition 2.123. They become

$$\begin{aligned} &\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \\ &\{1, 2, 9\}, \{3, 4, 9\}, \{1, 2, 10\}, \{3, 4, 10\}, \\ &\{1, 9, 10\}, \{2, 9, 10\}, \{3, 9, 10\}, \{4, 9, 10\}. \end{aligned}$$

These are exactly the four triples containing each of P_1, P_2, P_3 , and no triple contains two of the disjoint pairs. This proves (2.62). A three-subset of Y_0 contains none of the P_i exactly when it chooses one element from each pair. The eight such transversals are the facets of the join of the three zero-spheres, proving (2.63). The residual-edge reconstruction gives square directions $\{1, 2\}, \{3, 4\}, \{9, 10\}$; it is recorded independently in `damp_terminal_artinian_owner_paths_d12.json`. □

Remark 2.125 (The owner-cycle sphere target). Proposition 2.124 suggests a topological route to the two-connected theorem. It would be enough to prove that the lower and upper gate cycles of Corollary 2.142 force a rank-three support cycle containing a join of three nonempty zero-dimensional

parts. Such a join uses at least six active coordinates. The open-ear inequality (2.67) would then give $|D| \geq 12$ immediately. Proposition 2.183 obtains the same coordinate conclusion more directly from the positioned shore concepts and forced shielding owners, with one exact five-coordinate completion lemma. Thus the support sphere is not needed for the orders-ten-and-eleven exclusion, although the sharp interval still realizes the expected equality geometry by the octahedral sphere in (2.63).

Lemma 2.126 (Local carrier rules through residual degree three). *Let $K \subsetneq A \subseteq Q_E$ be ample, put*

$$R = A \setminus K, \quad \mathcal{H} = \text{Sh}(A) \setminus \text{Sh}(K),$$

and suppose that $c \in R$ is doubly blocked: neither $K \cup \{c\}$ nor $A \setminus \{c\}$ is ample. Then the following statements hold.

1. *Every compatible lower carrier of an inclusion-minimal support $S \in \mathcal{H}$ and upper carrier of an inclusion-maximal support $T \in \mathcal{H}$ meet in a full cube contained in R , whose free coordinate set is $T \setminus S$.*
2. *The concept c belongs to at least two lower carriers of minimal supports and at least two upper carriers of maximal supports.*
3. *If c has residual degree two, it marks an isolated support $S_c \in \mathcal{H}$, meaning that S_c is both minimal and maximal, and exactly one of the two residual edge coordinates at c belongs to S_c . More precisely, if*

$$P_c = \{e : c^{\{e\}} \in K\}, \quad W_c = \{e : c^{\{e\}} \in R\},$$

then

$$S_c = P_c \dot{\cup} \{\lambda_c\} \quad \text{for one } \lambda_c \in W_c. \quad (2.64)$$

If adjacent degree-two concepts c, d have edge coordinate e , then $e \in S_c \triangle S_d$.

4. *If c has residual degree three, there are a minimal support S^- and a maximal support S^+ such that*

$$S^- \subseteq S^+, \quad |S^+ \setminus S^-| \leq 1, \quad c \in C_{S^-}^- \cap C_{S^+}^+. \quad (\text{A4g1c})$$

Thus c marks either an isolated support or a minimal-maximal support cover.

Proof. Put $B = Q_E \setminus A$, which is ample by complement duality. For $S \in \mathcal{H}$, the two trace-defect classes

$$U_S = Q_S \setminus K|_S, \quad V_S = Q_{E \setminus S} \setminus B|_{E \setminus S}$$

are nonempty and ample, and $U_S \square V_S \subseteq R$. Sauer equality in the two projections shows that U_S is a singleton exactly when S is minimal in $\mathcal{H}$, and V_S is a singleton exactly when S is maximal. If u_S and v_T denote these singleton traces, their prescriptions agree on their common domain precisely when

$$u_S|_{S \setminus T} = v_T|_{S \setminus T}.$$

When they agree, all common extensions lie in R and form the full $(T \setminus S)$ -cube. This proves the first assertion.

Adjoining c to K shatters exactly the minimal relative supports whose lower missing trace is $c|_S$. The number of these supports is one exactly when $K \cup \{c\}$ is ample. Applying the complementary statement to $B \cup \{c\} = Q_E \setminus (A \setminus \{c\})$ gives the upper assertion. Double blocking therefore gives carrier depth at least two on both shores.

Translate c to the zero word and partition its coordinate directions as

$$P = \{e : c^{\{e\}} \in K\}, \quad Q = \{e : c^{\{e\}} \in B\}, \quad W = \{e : c^{\{e\}} \in R\}.$$

The commuting existential and universal operations for lopsided systems give two ample classes on W ,

$$L = \exists_Q \forall_P K, \quad U = \exists_P \forall_Q B.$$

The full-collar form of the blocker formula shows that L, U are nonempty, disjoint, avoid the zero word, and contain respectively every lower and upper obstruction signature. When $|W| = 2$, the three nonzero words form a path, so L and U are disjoint intervals in that path. Their order selects opposite residual directions. The two corresponding minimal transversals give the isolated support

$$S_c = P \dot{\cup} \{\lambda_c\}$$

for one $\lambda_c \in W$, proving (2.64). Projecting the coordinate of an edge between two such markers, and applying the same collision argument to the complementary pair, gives the stated symmetric-difference law. This is the degree-two isolated-support argument in a form that retains every coordinate constant on R .

Suppose finally that $|W| = 3$. An exact local calculation in Q_3 shows that any two nonempty disjoint ample classes avoiding zero have disjoint transversals of their inclusion-minimal words. There are 63 such ample classes and 672 ordered disjoint pairs. Shrinking the two transversals to minimal transversals of the actual lower and upper obstruction clutters preserves disjointness. They are both nonempty, so their union leaves zero or one of the three coordinates. The blocker formula now gives $S^- \subseteq S^+$, with difference equal to that unused set, and puts c in both displayed carriers. This proves (A4g1c). The calculation is reproduced by `verify_damp_degree_three_two_shore_dichotomy.py`; its digest is

51a1cbfec7834c283c3a73e23c8df9f3cf7ce22852d0e9e8102665934cf86fc0.

□

Proposition 2.127 (Component-size amplification in a terminal interval). *Let $K \subsetneq A$ be ample families, put $D = A \setminus K$, and suppose that*

$$K \cup \{d\} \quad \text{is nonample for every } d \in D.$$

Then every connected component of the graph induced by D has at least nine vertices. In particular, if $|D| \leq 17$, then this graph is connected.

Proof. Let $D_1, \dots, D_s$ be the vertex sets of the connected components of the graph induced by D , and let $\mathcal{Q}(A, K)$ denote the squarefree Alexander-dual module of the relative Yang complex for (A, K) . This module has a linear resolution because both endpoints are ample. The relative linear-presentation theorem [Bou26] says that its degree-one relations are of two types: an annihilator supported on one generator, coming from a neighbour in K , or an edge relation between two adjacent

members of D . Every such relation is supported inside one component D_i . Since these relations generate the full presentation, one obtains a direct-sum decomposition

$$\mathcal{Q}(A, K) \cong \bigoplus_{i=1}^s \mathcal{Q}(K \cup D_i, K). \quad (\text{A4g1d})$$

Each summand therefore has a linear resolution, so its relative Yang module is Cohen–Macaulay. The exact sequence from that relative module to the Yang modules of $K \cup D_i$ and K , together with ampleness of K , makes $K \cup D_i$ ample.

If $|D_i| \leq 8$, the arbitrary-upper relative linear-quotient theorem through eight facets [Bou26] supplies an ordering of the concepts of D_i whose first concept, say d , can be adjoined amply to K . This contradicts the hypothesis. Hence $|D_i| \geq 9$ for every i . Two components would consequently have total order at least eighteen, which proves the final assertion. $\square$

Proposition 2.128 (Small cut layers have one block shape). *Let $K \subsetneq A \subseteq Q_X$ be ample, put $D = A \setminus K$, and suppose that every $d \in D$ is doubly blocked:*

$$K \cup \{d\} \quad \text{and} \quad A \setminus \{d\} \quad \text{are nonample.}$$

If the graph induced by D has a cut vertex and $|D| \leq 11$, then $|D| = 11$. Moreover, its block–cut tree consists of two six-vertex dominoes sharing one articulation vertex, and the articulation has degree two inside each domino.

Proof. Proposition 2.127 makes the residual graph connected. It has minimum degree at least two. Indeed, an isolated residual concept teaches K , by taking the first edge of an A -geodesic to each member of K . At a residual leaf, the relative leaf-deletion dichotomy gives either the same teaching conclusion or a Cohen–Macaulay one-concept deletion. The latter makes $A \setminus \{d\}$ ample, since K is ample. Both alternatives contradict double blocking.

The block–cut tree has at least two leaf blocks. None is a bridge, since its nonarticulation endpoint would have residual degree one. Choose two leaf blocks L_1, L_2 . They are two-connected and

$$|D| \geq |L_1| + |L_2| - 1. \quad (2.65)$$

Thus one of them has at most six vertices.

We first classify a two-connected induced-cube graph on at most six vertices. It is bipartite and has minimum degree two. At order four the only possibility is C_4 . There is no possibility at order five: the bipartition would have sizes two and three and the graph would be $K_{2,3}$, whereas two cube vertices have at most two common neighbours. At order six, a bipartition of sizes two and four similarly gives the impossible graph $K_{2,4}$. With bipartition sizes three and three, the missing edges form a matching. Deleting zero or one edge from $K_{3,3}$ again leaves a $K_{2,3}$. Deleting two or three independent edges gives, respectively, the domino and C_6 . Hence the only block shapes are

$$C_4, \quad C_6, \quad \text{and the domino.}$$

It remains important where the articulation lies in the domino. Every nonarticulation vertex of residual degree two marks an isolated support by Lemma 2.126. Such a support contains exactly one of the two incident block directions; adjacent markers reverse ownership of their common direction. Compatible lower and upper carriers meet in a full residual cube. Applying only these necessary rules excludes a leaf C_4 , a leaf C_6 , and a domino rooted at a degree-three vertex. A domino rooted at a degree-two vertex is the sole surviving rooted shape.

For completeness, the finite check keeps coordinates outside the block symbolically through inclusion of their passive support parts. A carrier cube through a nonarticulation vertex may not use a passive direction, whereas a cube meeting the block only at its root is deliberately allowed to continue inward. The four first-witness searches visit respectively 5, 11, 11, 4 nodes. Exhausting the active masks in the last search gives exactly the four patterns

$$(2, 1, 4), \quad (3, 6, 5), \quad (4, 1, 2), \quad (5, 6, 3)$$

on its three degree-two nonroot vertices. Thus the check is a relaxation of the full carrier system, not a conditioning that discards passive boundary data. It is reproduced by `verify_damp_small_leaf_block_carriers.py` and `damp_small_leaf_block_carriers.json`; its digest is

$$3af61b6a1eeae8680d6d442d897558b5937a53543855c04b66f6bab3565083d0.$$

Consequently the smaller leaf block is a six-vertex domino rooted at a degree-two vertex. Inequality (2.65) now makes the other leaf block have order at most six, so it has the same rooted shape. The two blocks already account for eleven vertices. Equality throughout forces $|D| = 11$, makes their roots the same articulation, and leaves no further block in the block-cut tree. $\square$

Proposition 2.129 (Terminal cut layers through eleven are impossible). *Let $K \subsetneq A$ be ample families, put $D = A \setminus K$, and suppose that every $d \in D$ is doubly blocked:*

$$K \cup \{d\} \quad \text{and} \quad A \setminus \{d\} \quad \text{are nonample.}$$

If $|D| \leq 11$, then the graph induced by D is two-connected.

Computer-assisted proof. Proposition 2.127 makes the residual graph connected, and the proof of Proposition 2.128 gives it minimum degree at least two. It remains to exclude a cut vertex. If one exists, that proposition forces $|D| = 11$ and writes the graph as two six-vertex dominoes sharing an articulation which has degree two in each domino.

We first reduce the coordinate labels without a graph catalogue. Translate the common root to zero. A rooted domino has three distinct directions a, b, c , where a, c are incident with the root, and vertex set

$$\{0, a, a + b, c, a + c, a + b + c\} \subseteq \mathbb{F}_2^E. \quad (2.66)$$

The four root directions of the two dominoes are distinct, since the blocks meet only at their root. The middle direction of either domino cannot be a root direction of the other: if, for example, $b = a'$, then the second root neighbour b is adjacent to the first-block vertex $a + b$, producing an edge between the two blocks away from the root. Inducedness gives a contradiction. Thus the two middle directions are either equal or distinct. These are the only two coordinate profiles: five active coordinates in the *shared-middle* profile and six in the *disjoint* profile.

For each profile, instantiate the passive-coordinate carrier formula from the proof of Proposition 2.184 with all eleven relative supports. In addition to its support order, lower and upper carrier anchors, carrier depths, cross-carrier cubes, degree-two markers, and degree-three dichotomy, impose adjacent-marker reversal and one of the four active patterns certified above on each domino. Eleven formal passive coordinates suffice: replacing actual passive parts $P_1, \dots, P_{11}$ by

$$\widehat{P}_i = \{j : P_j \subseteq P_i\}$$

preserves exactly every passive inclusion and equality used by the formula. The one-concept deletion co-pairs are deliberately omitted. Consequently both formulas are relaxations of every genuine doubly blocked interval, rather than conditionings which suppress passive boundary data.

Both formulas are unsatisfiable. Their DIMACS sizes and the statistics of the independently checked proof cores are

profile	variables	clauses	core lemmas	resolution steps
shared middle	12,126	130,526	311,417	20,710,244
disjoint	13,476	145,285	749,165	51,150,761

The shared-middle proof uses no RAT lemma; the disjoint proof uses 26,857. The generator, manifest, proof cores, and aggregate checker are

```
solve_damp_outer_domino_wedge_carrier_cnf.py,
damp_outer_domino_wedge_carrier_cnf_manifest.json,
damp_outer_domino_wedge*_core.drat,
verify_damp_outer_domino_wedge_carrier_proofs.py.
```

The aggregate mathematical digest is

962dba99701919e0be0c25c98956e0373ff7147392124d319d272eeb3fd47942.

The two coordinate profiles exhaust the cut-vertex conclusion of Proposition 2.128, and both are impossible. Hence the connected residual graph has no cut vertex and is two-connected. $\square$

The connected conclusion still leaves many possible residual graphs. The next elementary bound shows that their two-connected parts cannot use many coordinates unless they also use many vertices. Here a coordinate is *active* on a cube subgraph if it labels one of its edges.

Lemma 2.130 (Coordinate cost of a two-connected cube graph). *Let G be a two-connected subgraph of a Boolean cube, and let Y be the set of coordinates active on G . Then*

$$|V(G)| \geq 2|Y|. \quad (2.67)$$

The subgraph need not be induced or isometric.

Proof. We first recall a short construction of an open-ear decomposition. Start with a cycle G_0 of G . Suppose that a subgraph G_i with an open-ear decomposition has already been constructed. If a component of $G - V(G_i)$ remains, it has at least two distinct neighbours in G_i : otherwise its unique attachment vertex would be a cut vertex of G . A shortest path through the component between two such attachment vertices is an open ear. After every vertex has been added, add the remaining edges as ears of length one. This constructs

$$G_0 \subseteq G_1 \subseteq \cdots \subseteq G_s = G,$$

where G_0 is a cycle and $G_i - G_{i-1}$ is the interior of a path whose distinct endpoints belong to G_{i-1} .

Every coordinate used on the initial cycle occurs a positive even number of times, and hence at least twice. If r_0 coordinates occur there, then

$$|V(G_0)| = |E(G_0)| \geq 2r_0.$$

Now consider an ear with ℓ edges, and suppose that it introduces k coordinates not used in the preceding connected subgraph. All old vertices have the same value in each such coordinate. Since the two endpoints of the ear are old, every new coordinate must therefore occur an even positive number of times on the ear, accounting for at least $2k$ edges. At least one further edge has an old coordinate. Indeed, otherwise the parity condition would make the two endpoints agree in every cube coordinate, contrary to their being distinct vertices. Thus

$$\ell \geq 2k + 1.$$

The ear contributes $\ell - 1 \geq 2k$ new vertices. Summing this inequality over the ears, including the length-one ears for which $k = 0$, gives (2.67). $\square$

The next localization is useful beyond the two-connected argument. It is the one-concept form of the minimal-nonample structure theorem from the Yang-complex dictionary. We state both directions so that all later uses of antipodal blockers refer to one source-of-truth result.

Proposition 2.131 (Antipodal-pair localization of failed augmentation). *Let $K \subseteq Q_X$ be ample and let $d \in Q_X \setminus K$. Then $K \cup \{d\}$ is nonample if and only if there are a coordinate set $T \subseteq X$, with $|T| \geq 2$, and $h = d^{\oplus T} \in K$ such that*

$$(K \cup \{d\}) \cap Q_T(d) = \{d, h\}, \quad K \cap Q_T(d) = \{h\}. \quad (2.68)$$

Here $Q_T(d)$ is the face obtained by fixing every coordinate outside T as in d . In particular, d and h are the antipodal vertices of this face.

Proof. If the displayed face exists, it is a nonample conditioning of $K \cup \{d\}$: two antipodal vertices do not form an ample family in a cube of dimension at least two. This proves the reverse implication.

Starting with the nonample class $K \cup \{d\}$, condition coordinates while a nonempty nonample conditioning remains, and stop at a conditioning-minimal nonample class H . Every nonample conditioning encountered contains d , since otherwise it would be a conditioning of the ample class K . Consequently

$$H \setminus \{d\}$$

is a nonempty conditioning of K , and is ample.

We claim that a minimal nonample class H with $H \setminus \{d\}$ ample has VC dimension one. Let n be its number of free coordinates and $k = \text{VC}(H)$. The structure theorem for minimal nonample classes [Bou26] gives

$$|H| = 2 \sum_{i=0}^{k-1} \binom{n-1}{i}, \quad \text{SSh}(H) = \{P \subseteq [n] : |P| \leq k-1\}.$$

Deleting a concept cannot create a strongly shattered support. Since $H \setminus \{d\}$ is ample, its shattered and strongly shattered supports coincide, and hence its VC dimension is at most $k-1$. Sauer's inequality therefore gives

$$2 \sum_{i=0}^{k-1} \binom{n-1}{i} - 1 = |H| - 1 \leq \sum_{i=0}^{k-1} \binom{n}{i}.$$

Pascal's identity reduces this inequality to

$$\binom{n-1}{k-1} \leq 1.$$

The same structure theorem gives $1 \leq k \leq n-1$, so necessarily $k=1$. It now gives $|H|=2$, and its antipodality conclusion says that the two concepts are antipodes on all n free coordinates.

Let T be those free coordinates and restore the fixed assignment of the conditioning. Then $H = \{d, h\}$, where $h = d^{\oplus T} \in K$, and the definition of the conditioning gives (2.68). $\square$

Lemma 2.132 (Antipodal deletion blockers descend along their faces). *Let $A \subseteq Q_X$, let $v \in A$, and suppose that for some $J \subseteq X$, with $|J| \geq 2$, one has*

$$A \cap Q_J(v) = Q_J(v) \setminus \{h\}, \quad h = v^{\oplus J}. \quad (2.69)$$

For every proper $S \subset J$, put $w = v^{\oplus S}$. Then

$$A \cap Q_{J \setminus S}(w) = Q_{J \setminus S}(w) \setminus \{h\}. \quad (2.70)$$

In particular, if $|J \setminus S| \geq 2$, then $A \setminus \{w\}$ is nonample.

Proof. Every member of $Q_{J \setminus S}(w)$ has the form

$$v^{\oplus(S \cup T)} \quad (T \subseteq J \setminus S).$$

This is a member of the J -face through v , and it equals h exactly when $T = J \setminus S$. Since $S \neq J$, all other displayed vertices belong to A by (2.69). This proves (2.70).

If $|J \setminus S| \geq 2$, the complement of $A \setminus \{w\}$ has, on this face, exactly the antipodal pair $\{w, h\}$. This conditioning is nonample. Conditioning and complement duality preserve ampleness, so $A \setminus \{w\}$ itself is nonample. $\square$

We can now replace every two-connected residual graph of order ten or eleven by one common boundary-ownership problem. For a family $L \subseteq Q_Y$ and $d \in Q_Y \setminus L$, write

$$N_L(d) = \{y \in Y : d^{\oplus y} \in L\}.$$

Proposition 2.133 (Five-coordinate normalization of the two-connected terminal frontier). *Let $K \subsetneq A \subseteq Q_X$ be ample, put $D = A \setminus K$, and suppose that*

$$K \cup \{d\} \text{ is nonample for every } d \in D. \quad (2.71)$$

Suppose moreover that the graph induced by D is two-connected and that $|D| \leq 11$. Let Y be its active coordinate set and put $Z = X \setminus Y$. Then the following statements hold.

1. *One has $|Y| \leq 5$, and all members of D have a common restriction $\sigma \in Q_Z$.*
2. *Identifying D with its projection to Y , put*

$$L = \{k|_Y : k \in K, k|_Z = \sigma\}, \quad B = L \dot{\cup} D.$$

Both L and B are ample, with the empty family allowed for L , and the set

$$R = \{d \in D : L \cup \{d\} \text{ is ample}\}$$

is nonempty.

3. *For every $d \in R$, there are a passive coordinate $z \in Z$ and a concept $c \in D$ such that*

$$c^{\oplus z} \in K, \quad d^{\oplus z} \notin K, \quad N_L(d) \cap \text{Diff}(d, c) = \emptyset. \quad (2.72)$$

In particular, if

$$S_z = \{c \in D : c^{\oplus z} \in K\} \quad (z \in Z),$$

then the nonempty passive boundary types S_z collectively block every active teaching root $d \in R$, in the precise sense of (2.72). Thus no enumeration of two-connected graphs of order ten or eleven is needed: every such terminal interval has the same five-coordinate passive-owner normal form.

Proof. The graph induced by D is connected, so every coordinate not in Y is constant on D ; call the common restriction σ . Lemma 2.130 gives

$$2|Y| \leq |D| \leq 11,$$

and hence $|Y| \leq 5$. Conditioning A and K on the assignment σ , and then deleting the constant coordinates, gives respectively B and L . Conditioning preserves ampleness, so both families are ample whenever they are nonempty.

If $L = \emptyset$, every singleton $\{d\}$ is ample and hence $R = D \neq \emptyset$. Suppose that $L \neq \emptyset$. Apply the five-coordinate relative-peeling theorem, Theorem A.137, to $L \subsetneq B$. In the resulting peeling, all members of D are removed before the members of L . Immediately before the last member of D is removed, the current ample family is $L \cup \{d\}$ for some $d \in D$. Thus $d \in R$, proving the second assertion.

Fix $d \in R$. The one-concept colon criterion says that d teaches L : for every $\ell \in L$,

$$N_L(d) \cap \text{Diff}(d, \ell) \neq \emptyset. \quad (2.73)$$

On the other hand, (2.71) and the same criterion give $h \in K$ such that

$$N_K(d) \cap \text{Diff}(d, h) = \emptyset, \quad N_K(d) = \{x \in X : d^{\oplus x} \in K\}. \quad (2.74)$$

Because A and K are ample, their relative Alexander-dual module has a linear resolution and in particular is linearly presented. The interval exit lemma therefore supplies a path from d inside $D \cap [d, h]$, followed by an edge cg with

$$c \in D \cap [d, h], \quad g \in K \cap [d, h].$$

The coordinate of this last edge cannot belong to Y . Indeed, in that case $g|_Z = \sigma$, so $g|_Y \in L$. Formula (2.73) would then give a coordinate in

$$N_L(d) \cap \text{Diff}(d, g) \subseteq N_K(d) \cap \text{Diff}(d, h),$$

contradicting (2.74).

Let $z \in Z$ be the coordinate of cg . Since every member of D has restriction σ , one has $g = c^{\oplus z} \in K$. Moreover, $c \in [d, h]$ and (2.74) give

$$N_L(d) \cap \text{Diff}(d, c) = \emptyset.$$

Finally $z \in \text{Diff}(d, h)$. If $d^{\oplus z} \in K$, then $z \in N_K(d) \cap \text{Diff}(d, h)$, another contradiction. Thus $d^{\oplus z} \notin K$, and all three conclusions in (2.72) follow. $\square$

The antipodal blocker lemma upgrades the last part of the proposition from one boundary edge to an entire peripheral cell.

Lemma 2.134 (The isolated blocker is a peripheral active cell). *Use the hypotheses and notation of Proposition 2.133, and let $d \in R$. Then there are a passive coordinate $z \in Z$, a nonempty set*

$$W \subseteq Y \setminus N_L(d),$$

and an ample family $C \subseteq D \cap Q_W(d)$ such that

$$d, d^{\oplus W} \in C, \quad (d^{\oplus W})^{\oplus z} \in K, \quad (2.75)$$

with

$$d \notin S_z, \quad d^{\oplus W} \in S_z, \quad C \cap S_z = \{d^{\oplus W}\},$$

and the conditioning of A to the face $Q_{W \cup \{z\}}(d)$ is the peripheral expansion

$$A \cap Q_{W \cup \{z\}}(d) = C \dot{\cup} \{(d^{\oplus W})^{\oplus z}\}, \quad (2.76)$$

whose duplicated site is the singleton $\{d^{\oplus W}\} \subseteq C$. The corresponding conditioning of K is the singleton $\{(d^{\oplus W})^{\oplus z}\}$. Moreover, C lies in the unique minimal active carrier of d .

Proof. Apply Proposition 2.131 to $K \cup \{d\}$, and write its isolated face as $Q_T(d)$, with antipodal member $h = d^{\oplus T} \in K$. The relative module of the ample pair $K \subsetneq A$ is linearly presented. The interval-exit lemma gives a path from d inside $D \cap [d, h]$, followed by an edge ch into $K \cap [d, h]$. By (2.68), the endpoint in K is necessarily h .

All members of D have passive restriction σ . The edge ch cannot have an active coordinate. Otherwise h would also have passive restriction σ , and hence $h|_Y \in L$. Since d teaches L , some coordinate of $N_L(d) \cap \text{Diff}(d, h)$ would give a member $d^{\oplus y} \in K \cap Q_T(d)$. It is different from h , since equality would make $T = \{y\}$ and the minimal nonample conditioning the ample one-cube. This contradicts (2.68). Thus ch has a passive coordinate z . Since c has passive restriction σ and h is adjacent to c , this is the only passive coordinate in T . Put

$$W = T \setminus \{z\} \subseteq Y.$$

The set W is nonempty: an antipodal pair on one free coordinate is the full one-cube and is ample, whereas the conditioning which produced $\{d, h\}$ is nonample. We have

$$c = h^{\oplus z} = d^{\oplus W} \in D.$$

If $y \in W \cap N_L(d)$, then $d^{\oplus y} \in L \subseteq K$ would be a member of $Q_T(d)$ different from h , again contradicting (2.68). Hence $W \cap N_L(d) = \emptyset$.

Put

$$C = A \cap Q_W(d).$$

This is a nonempty ample conditioning of A , and it is contained in D , because (2.68) leaves no member of K on that section. Now condition A only on the coordinates outside T . Its σ_z -section is C , while its opposite section, after deleting z , is the singleton $\{c\}$: there are no relative concepts on the opposite passive side, and $h = c^{\oplus z}$ is the only member of K in the face. This proves (2.76). The interval-exit path puts both d and $c = d^{\oplus W}$ in C . The identity $c^{\oplus z} = h \in K$ gives $c \in S_z$, while (2.68) gives $d^{\oplus z} \notin K$, and hence $d \notin S_z$. More generally, if $u \in C \cap S_z$, then $u^{\oplus z}$ belongs to $K \cap Q_T(d) = \{h\}$. Thus $u = c$, proving $C \cap S_z = \{c\}$.

Finally put $P = N_L(d)$. Every member of $Q_W(d)$ agrees with d on P , because $W \cap P = \emptyset$. This common P -trace is the missing trace defining the unique minimal carrier of d . Consequently C lies in that carrier. $\square$

The passive types in the preceding proposition are not arbitrary subsets of D . Relative Cohen–Macaulayness couples all of them through the fine numerator. The following elementary consequence is the first Helly-type constraint that we shall need.

Lemma 2.135 (Common owner when the active lower slice is empty). *Let $K \subsetneq A \subseteq Q_{Y \cup Z}$ be ample, put $D = A \setminus K$, and suppose that all members of D have the same restriction $\sigma \in Q_Z$. Suppose moreover that*

$$\{k|_Y : k \in K, k|_Z = \sigma\} = \emptyset.$$

For $z \in Z$, put

$$S_z = \{d \in D : d^{\oplus z} \in K\}, \quad Z_0 = \{z \in Z : S_z \neq \emptyset\}.$$

If $Z_0 \neq \emptyset$, then

$$\bigcap_{z \in Z_0} S_z \neq \emptyset. \tag{2.77}$$

Proof. For $Q \subseteq Y$, let $\mathcal{C}_Q$ be the fibers of the projection $D \rightarrow D|_Q$. The active lower slice is empty, so every such fiber is a relative projection block. Label a block $E \in \mathcal{C}_Q$ by

$$\lambda(E) = \{z \in Z_0 : E \cap S_z \neq \emptyset\},$$

and, for $R \subseteq Z_0$, put

$$n_Q(R) = |\{E \in \mathcal{C}_Q : \lambda(E) = R\}|, \quad h_{P,R} = \sum_{Q \subseteq P} (-1)^{|P \setminus Q|} n_Q(R).$$

The rank-selected supporting-cube identity for relative Yang modules [Bou26] expresses every $h_{P,R}$ as a sum of fine numerator coefficients of the full relative pair (A, K) . Both endpoints are ample, so that relative module is Cohen–Macaulay and all these coefficients are nonnegative. Hence

$$h_{P,R} \geq 0 \quad (P \subseteq Y, R \subseteq Z_0). \quad (2.78)$$

At $Q = \emptyset$, the unique projection block is D . It meets every S_z , by the definition of Z_0 , and therefore

$$n_{\emptyset}(Z_0) = h_{\emptyset, Z_0} = 1.$$

Boolean Möbius inversion and (2.78) now give

$$n_Y(Z_0) = \sum_{P \subseteq Y} h_{P, Z_0} \geq 1.$$

Since all members of D agree on Z , projection to Y is injective. Thus the blocks in $\mathcal{C}_Y$ are singletons, and a singleton has label Z_0 exactly when its concept belongs to every S_z . This proves (2.77). $\square$

We next localize the same argument to one minimal active carrier. Continue with the notation $L \subseteq B = L \dot{\cup} D \subseteq Q_Y$ of Proposition 2.133, but only assume that D is connected. Put

$$\mathcal{H}_Y = \text{Sh}(B) \setminus \text{Sh}(L).$$

If P is inclusion-minimal in $\mathcal{H}_Y$, then L misses one unique trace $u_P \in Q_P$, and its *minimal active carrier* is

$$E_P = \{d \in D : d|_P = u_P\}.$$

Proposition 2.136 (Common passive owner of a minimal active carrier). *Let $K \subsetneq A$ be ample, let $D = A \setminus K$ be connected, and use the preceding active-slice notation. For $z \in Z$, put*

$$S_z = \{d \in D : d^{\oplus z} \in K\}, \quad Z_P = \{z \in Z : E_P \cap S_z \neq \emptyset\}.$$

Then, whenever $Z_P \neq \emptyset$,

$$E_P \cap \bigcap_{z \in Z_P} S_z \neq \emptyset. \quad (2.79)$$

Thus all passive boundary types incident with one minimal active carrier have a common owner inside that carrier.

Proof. Condition both A and K on the active trace u_P . The relative concepts of the conditioned pair are exactly E_P . They agree on the passive assignment σ . Moreover, its lower slice at σ is empty, because u_P is not realized by L . Conditioning preserves ampleness, so Lemma 2.135 applies. Its nonempty passive boundary types are precisely

$$E_P \cap S_z \quad (z \in Z_P).$$

Their common intersection is (2.79). $\square$

Corollary 2.137 (A blocked active root transfers to carrier depth two). *Under the hypotheses of Proposition 2.133, let $d \in R$. Then the unique minimal active carrier E_P containing d contains a concept $o \neq d$ which belongs to at least two minimal active carriers.*

Proof. The one-concept colon criterion identifies $P = N_L(d)$ as the unique minimal relative support whose missing trace is realized by d . The blocking conclusion (2.72) supplies $z \in Z_P$ such that $d \notin S_z$. Choose a common owner

$$o \in E_P \cap \bigcap_{w \in Z_P} S_w$$

from Proposition 2.136. Then $o \neq d$.

Suppose that o belonged to only one minimal active carrier. The local carrier form of the one-concept criterion would give $L \cup \{o\}$ ample, with unique carrier E_P . Apply Proposition 2.133 to o . It gives a passive blocker S_w meeting E_P but not containing o . Hence $w \in Z_P$, contrary to the choice of the common owner. Thus o has minimal-carrier depth at least two. $\square$

The preceding owner statement records only which relative concepts have a neighbour across a passive coordinate. The whole opposite section is available, however, and ampleness makes it considerably more rigid. We record that extra structure before continuing the cardinality argument. For $u \in Q_Y$, let $\hat{u} \in Q_X$ denote its lift with passive restriction σ .

Lemma 2.138 (A passive coordinate gives two ample subintervals). *Use the notation of Proposition 2.133. For $z \in Z$, put*

$$M_z = \{u \in Q_Y : \hat{u}^{\oplus z} \in K\}. \quad (2.80)$$

Then $S_z = D \cap M_z$. Moreover, whenever the displayed families are nonempty, each of

$$L \cap M_z, \quad B \cap M_z, \quad L \cup M_z, \quad B \cup M_z \quad (2.81)$$

is ample, and

$$\begin{aligned} (B \cap M_z) \setminus (L \cap M_z) &= S_z, \\ (B \cup M_z) \setminus (L \cup M_z) &= D \setminus S_z. \end{aligned} \quad (2.82)$$

Thus a passive boundary type and its complement are not arbitrary subsets of D : they are the relative facets of two nested ample pairs on the same five active coordinates.

Proof. Condition K on the assignment σ on $Z \setminus \{z\}$, and then delete those fixed coordinates. The two z -sections of the resulting ample class are exactly L and M_z . Apply the two-layer shadow-intersection lemma, Lemma 7.22: its two sections, their union, and their intersection are ample whenever nonempty. This gives the two families involving L in (2.81).

The same conditioning of A has sections B and M_z . Indeed, every member of $A \setminus K = D$ has passive restriction σ , so the opposite section of A is the same as that of K . A second application of the two-layer lemma gives the two families involving B . Finally, $B = L \dot{\cup} D$, and hence elementary set algebra gives (2.82). The identity $S_z = D \cap M_z$ follows directly from the definitions. $\square$

Lemma 2.139 (Passive neighbours force a punctured cube). *Let $K \subsetneq A \subseteq Q_{Y \cup Z}$ be ample, put $D = A \setminus K$, and suppose that all members of D have the same restriction $\sigma \in Q_Z$. Put*

$$L = \{k|_Y : k \in K, k|_Z = \sigma\}, \quad S_z = \{d \in D : d^{\oplus z} \in K\} \quad (z \in Z).$$

For $d \in D$, define

$$Z(d) = \{z \in Z : d \in S_z\}, \quad t(d) = |Z(d)|.$$

Then

$$d^{\oplus T} \in K \quad \text{for every } d \in D \text{ and } \emptyset \neq T \subseteq Z(d). \quad (2.83)$$

Consequently,

$$|K| \geq |L| + \sum_{d \in D} (2^{t(d)} - 1). \quad (2.84)$$

If $L \neq \emptyset$, put

$$Z_0 = \{z \in Z : S_z \neq \emptyset\}.$$

For every $z \in Z_0$, there is an $a_z \in L$ such that the lift of a_z and its z -flip, $\hat{a}_z^{\oplus z}$, both belong to K . These collar vertices sharpen the count to

$$|K| \geq |L| + |Z_0| + \sum_{d \in D} (2^{t(d)} - 1). \quad (2.85)$$

Thus t passive neighbours of one relative concept cost the whole punctured t -cube, not merely t boundary vertices, and every nonempty passive direction has an additional collar cost.

Proof. Fix $d \in D$ with $Z(d) \neq \emptyset$. Condition K on the active word $d|_Y$ and on the passive assignment σ outside $Z(d)$, and delete the fixed coordinates. The resulting family $F_d \subseteq Q_{Z(d)}$ is nonempty and ample. Under the identification of a word with the set of coordinates on which it differs from $\sigma|_{Z(d)}$, one has

$$\emptyset \notin F_d, \quad \{z\} \in F_d \quad (z \in Z(d)).$$

Indeed, the first assertion is $d \notin K$, and the second is the definition of $Z(d)$.

By complement duality, $Q_{Z(d)} \setminus F_d$ is ample. It contains the empty word. If it contained a second word $T \neq \emptyset$, its one-inclusion graph would contain a geodesic from $\emptyset$ to T , because every ample family is isometric. The first edge of this geodesic would end at a singleton $\{z\}$, although every singleton belongs to F_d , a contradiction. Hence

$$Q_{Z(d)} \setminus F_d = \{\emptyset\},$$

which proves (2.83). There is nothing to prove when $Z(d) = \emptyset$.

The words in (2.83) belonging to two distinct concepts $d, d' \in D$ have distinct restrictions to Y , because d, d' have the same restriction to Z . They are therefore distinct. They are also disjoint from the lift of L , since their passive restriction differs from σ . For fixed d , there are $2^{t(d)} - 1$ of them. Summing these disjoint subsets of K proves (2.84).

Suppose that $L \neq \emptyset$, and fix $z \in Z_0$. Condition K on σ outside z , and delete the fixed passive coordinates. Its two z -sections are L and M_z , in the notation of Lemma 2.138. Both are nonempty: the first by hypothesis, and the second because $S_z \neq \emptyset$. The conditioned ample family is connected. Since active edges stay in one z -section, it must contain a z -edge. Equivalently, $L \cap M_z \neq \emptyset$; choose a_z in this intersection.

The opposite lift $\hat{a}_z^{\oplus z} \in K$ has active restriction in L , whereas every word counted in (2.84) has active restriction in D . It is therefore a new vertex. The chosen vertices for distinct coordinates z have different passive restrictions, so they are pairwise distinct. Adding these $|Z_0|$ vertices proves (2.85). $\square$

Lemma 2.140 (Every nonempty passive shore contains an active root). *Under the hypotheses of Proposition 2.133, suppose that (2.71) holds. Then $L \neq \emptyset$, and every nonempty S_z contains a member of*

$$R = \{d \in D : L \cup \{d\} \text{ is ample}\}.$$

Proof. Suppose first that $L = \emptyset$. Then every member of D belongs to R . Proposition 2.133 gives, for each $d \in D$, a nonempty passive type which excludes d . In particular the set Z_0 of nonempty passive types is nonempty. Lemma 2.135 supplies

$$o \in \bigcap_{z \in Z_0} S_z.$$

Applying the blocking conclusion of Proposition 2.133 to $o \in R$ produces $z \in Z_0$ with $o \notin S_z$, a contradiction. Thus $L \neq \emptyset$.

Fix z with $S_z \neq \emptyset$, and abbreviate $M = M_z$. The conditioned ample class with sections L, M has both sections nonempty. Its one-inclusion graph is connected, so it contains a z -edge. Hence

$$I := L \cap M \neq \emptyset. \quad (2.86)$$

By Lemma 2.138,

$$I \subsetneq (B \cap M) = I \dot{\cup} S_z$$

is a nested pair of ample families on at most five coordinates. Apply Theorem A.137 to obtain $p \in S_z$ for which $I \cup \{p\}$ is ample. The conditioned class with sections L, M is an ample expansion, $p \in M \setminus L$, and its section intersection is I . Lemma A.19 therefore shows that $L \cup \{p\}$ is ample. Thus $p \in R \cap S_z$, as required. $\square$

The preceding proof used only the first concept of a relative peeling of the shore. Keeping the whole peeling turns the shore into an actual ample intermediate family. This is the useful functorial form of the collar construction.

Lemma 2.141 (Passive-shore absorption and blocker-rank descent). *Under the hypotheses of Lemma 2.140, fix $z \in Z$ with $S_z \neq \emptyset$, and put*

$$G_z = L \cup S_z. \quad (2.87)$$

Then G_z is ample. More precisely, there is a saturated chain of ample families from L to B in which every member of S_z is added before every member of $D \setminus S_z$. Its first segment is

$$L = G_0 \subset G_1 \subset \cdots \subset G_{|S_z|} = G_z, \quad |G_i \setminus G_{i-1}| = 1, \quad (2.88)$$

whose added concepts are exactly the members of S_z .

Suppose in addition that $d \in R \setminus S_z$ has the peripheral blocker cell C , support W , and duplicated endpoint $c = d^{\oplus W}$ supplied by Lemma 2.134 for this same coordinate z . Then

$$C \cap G_z = \{c\}, \quad (G_z \cup \{d\}) \cap Q_W(d) = \{d, c\}. \quad (2.89)$$

Consequently, if $|W| \geq 2$, the extension $G_z \cup \{d\}$ is nonample, witnessed by the antipodal pair on the strictly smaller active face $Q_W(d)$. If $|W| = 1$, the named transverse blocker on $W \cup \{z\}$ has disappeared: its trace after absorption is the full one-cube. This last assertion does not exclude a different blocker for $G_z \cup \{d\}$.

Proof. Abbreviate $M = M_z$ and $I = L \cap M$. The proof of Lemma 2.140 gives $I \neq \emptyset$, and Lemma 2.138 gives the nested ample pair

$$I \subsetneq I \dot{\cup} S_z = B \cap M.$$

The five-coordinate relative-peeling theorem supplies an ordering $p_1, \dots, p_{|S_z|}$ of S_z for which every

$$I_i = I \cup \{p_1, \dots, p_i\}$$

is ample. Put $G_i = L \cup \{p_1, \dots, p_i\}$. We prove inductively that the two-layer family with sections G_i and M is ample. The case $i = 0$ is the conditioning of K used in the collar lemma. Moreover, $G_{i-1} \cap M = I_{i-1}$, while $I_{i-1} \cup \{p_i\} = I_i$ is ample. Lemma A.20 therefore duplicates p_i into the G_{i-1} -section and proves the induction step. Each G_i is a section of this ample two-layer family and hence is ample. This proves (2.88). If $G_z \neq B$, apply the five-coordinate relative-peeling theorem once more, now to $G_z \subsetneq B$, and append the resulting saturated chain. Its added concepts are exactly $B \setminus G_z = D \setminus S_z$. If $G_z = B$, the second segment is empty. This proves the asserted two-block ordering in both cases.

Now use the peripheral blocker cell. Since $C \subseteq D$, it is disjoint from L , while $C \cap S_z = \{c\}$. Hence $C \cap G_z = \{c\}$. Also $C = A \cap Q_W(d)$ and $G_z \subseteq B \subseteq A$, so $G_z \cap Q_W(d) = \{c\}$, which proves (2.89). When $|W| \geq 2$, apply Proposition 2.131 to the ample family G_z . When $|W| = 1$, the two displayed concepts are the two vertices of the one-cube, proving exactly the stated rank-one conclusion. $\square$

Complement duality gives the simultaneous upper-shore statement. Put

$$R^+ = \{d \in D : B \setminus \{d\} \text{ is ample}\}.$$

These are the active deletion roots of the conditioned upper slice.

Corollary 2.142 (The two directed root cycles forced by a terminal fibre). *In addition to the hypotheses of Proposition 2.133, suppose that*

$$A \setminus \{d\} \text{ is nonample for every } d \in D. \quad (2.90)$$

Then the following hold.

1. *Every nonempty S_z contains a member of R , and every nonempty $D \setminus S_z$ contains a member of R^+ . If both shores are nonempty, one saturated ample chain from L to B starts with a member of $R \cap S_z$, adds all of S_z before its complement, and ends with a member of $R^+ \cap (D \setminus S_z)$.*

2. *One has*

$$|R| \geq 2, \quad |R^+| \geq 2.$$

3. *There is a directed graph on R with positive out-degree at every vertex and no loop. An edge $d \rightarrow e$ is witnessed by a passive coordinate z such that*

$$d \notin S_z, \quad e \in S_z, \quad (2.91)$$

by an isolated peripheral blocker cell C_d in the unique minimal active carrier of d , and by a common passive owner $o \in S_z$ which is not in R . The shore also gives the ample gate $G_z = L \cup S_z$; absorbing it either lowers the named blocker from $W \cup \{z\}$ to W , or eliminates that blocker when $|W| = 1$. Consequently this graph contains a directed cycle of length at least two.

4. *Complementarily, there is such a directed cycle on R^+ , with every edge separated by an opposite shore $D \setminus S_z$ and carrying a common owner which is not in R^+ .*

Proof. The assertion for S_z is Lemma 2.140. Apply the same lemma to the complementary nested ample pair

$$Q_X \setminus A \subsetneq Q_X \setminus K.$$

Its relative concepts are again D , its active conditioned pair is

$$Q_Y \setminus B \subsetneq Q_Y \setminus L,$$

and its passive boundary type in coordinate z is $D \setminus S_z$. Indeed, $d^{\oplus z}$ belongs to A exactly when it belongs to K , because every member of $A \setminus K$ has passive restriction σ . Condition (2.90) is precisely the no-one-extension condition for this complementary pair. Its active roots are the members of R^+ . This proves item 1.

For the strengthened simultaneous statement, use the two-block chain from Lemma 2.141. Its first added concept is an active lower root and belongs to S_z ; its last added concept is an active upper root and belongs to $D \setminus S_z$.

Fix $d \in R$. Lemma 2.134 supplies a nonempty passive type S_z which excludes d , together with the isolated peripheral blocker cell in the unique minimal carrier of d . By item 1, S_z contains some $e \in R$, necessarily different from d ; direct the edge from d to one such e . Proposition 2.136 supplies a common owner o of all passive types incident with the unique minimal carrier of d . In particular $o \in S_z$, whereas $d \notin S_z$. Corollary 2.137 shows that o has minimal-carrier depth at least two, so $o \notin R$. Every vertex of the finite directed graph has an outgoing edge and there are no loops. It therefore contains a directed cycle of length at least two, and in particular R has at least two members. This proves items 2 and 3 on the lower shore.

Apply the same argument to the complementary pair. Minimal active carriers there are the upper carriers of the original active interval, and carrier depth one is exactly membership in R^+ . This gives the upper directed cycle, its non-root owners, and the remaining inequality in item 2. $\square$

The two root cycles can be coupled without choosing a common set of root vertices. The correct common objects are the ample gates.

Corollary 2.143 (Two-sided holonomy on the passive gates). *Under the hypotheses of Corollary 2.142, let*

$$\mathcal{S} = \{S_z : \emptyset \neq S_z \subsetneq D, z \in Z\},$$

where repeated shore sets are identified. Then $\mathcal{S} \neq \emptyset$. For every $S \in \mathcal{S}$, one may choose a two-block relative peeling of B over L , with S first, whose first and last relative concepts r_S, u_S satisfy

$$r_S \in R \cap S, \quad u_S \in R^+ \cap (D \setminus S).$$

There are maps

$$\lambda, \mu : \mathcal{S} \longrightarrow \mathcal{S}$$

such that

$$r_S \in S \setminus \lambda(S), \quad u_S \in \mu(S) \setminus S. \quad (2.92)$$

In particular, neither map has a fixed point, both maps contain a directed cycle, and every arrow is certified by a nonempty asymmetric difference of its two shores.

If $|\mathcal{S}| = 2$, say $\mathcal{S} = \{S, T\}$, then necessarily

$$r_S, u_T \in S \setminus T, \quad r_T, u_S \in T \setminus S.$$

Thus the two-shore case is one crossed lower–upper profile, rather than two unrelated root cycles.

Proof. The active root set R is nonempty. Applying Lemma 2.134 to any of its members produces a nonempty shore which excludes that member, and hence is proper. Thus $\mathcal{S} \neq \emptyset$.

For $S = S_z \in \mathcal{S}$, take the two-block chain from Lemma 2.141. Its first relative concept is an active lower root in S , and its last relative concept is an active upper root in $D \setminus S$; call them r_S, u_S .

Apply Lemma 2.134 to r_S . Its blocker shore is a nonempty S_w which excludes r_S . It is proper because it excludes a member of D , so its underlying shore set belongs to $\mathcal{S}$. Define $\lambda(S) = S_w$. This gives the first relation in (2.92).

Apply the complementary version of the same lemma to $u_S \in R^+$. It supplies a nonempty proper opposite shore $D \setminus S_v$ which excludes u_S . Equivalently $u_S \in S_v$, and S_v is itself nonempty and proper. Define $\mu(S) = S_v$. This gives the second relation. The displayed asymmetric differences exclude fixed points. Every self-map without a fixed point on a nonempty finite set contains a directed cycle of length at least two.

If $\mathcal{S} = \{S, T\}$, both fixed-point-free maps interchange the two members. Substituting this into (2.92) gives the final display. $\square$

The holonomy now has a quantitative cost. The following statement also identifies the exact zero-excess geometry, which will be the rigid branch of the remaining comparison theorem.

Corollary 2.144 (Exponential cost and equality form of gate holonomy). *Assume the hypotheses of Corollary 2.143, and let*

$$S_1, S_2, \dots, S_k$$

be a directed cycle of λ , indexed so that $\lambda(S_i) = S_{i+1}$ modulo k . Thus $k \geq 2$. For $d \in D$, let $t(d) = |\{z \in Z : d \in S_z\}|$, and put

$$Z_0 = \{z \in Z : S_z \neq \emptyset\}, \quad t_C(d) = |\{i : d \in S_i\}|.$$

Define the uncounted lower-fiber remainder

$$\eta = |K| - |L| - |Z_0| - \sum_{d \in D} (2^{t(d)} - 1). \quad (2.93)$$

Then $\eta \geq 0$, and

$$|K| - |L| \geq |Z_0| + \sum_{d \in D} (2^{t(d)} - 1) \geq 3k, \quad (2.94)$$

$$|K| - |L| - 3k = \eta + (|Z_0| - k) + \sum_{d \in D} (2^{t(d)} - 1 - t_C(d)) + \sum_{i=1}^k (|S_i| - 2). \quad (2.95)$$

Every summand on the right of (2.95) is nonnegative.

If the total bound in (2.94) is sharp, namely

$$|K| = |L| + 3k, \quad (2.96)$$

then the following conclusions hold.

1. *The cycle shores are pairwise disjoint and have order two. Every nonempty passive shore is one of them, and each cycle shore is realized by a unique passive coordinate.*
2. *Each S_i consists of one lower root $r_i \in R$ and one common owner $o_i \notin R$, and $r_i o_i$ is an active edge.*

3. If z_i is the unique passive coordinate realizing S_i , then $L \cap M_{z_i} = \{a_i\}$ and

$$M_{z_i} = \{a_i, r_i, o_i\}.$$

This is a three-vertex active path whose terminal edge is $r_i o_i$. Consequently K contains the attached three-edge path

$$\widehat{a}_i, \widehat{a}_i^{\oplus z_i}, r_i^{\oplus z_i}, o_i^{\oplus z_i},$$

in one of the two possible orders after $\widehat{a}_i^{\oplus z_i}$.

4. The upper family A contains the full square on the active-passive direction pair of $r_i o_i$ and z_i :

$$r_i, o_i \in D, \quad r_i^{\oplus z_i}, o_i^{\oplus z_i} \in K.$$

These k squares are pairwise vertex-disjoint.

5. Apart from the lift of L , the lower family K consists exactly of the $3k$ nonbase vertices in the attached paths.

Thus every departure from a system of pendant three-edge square-gate arms has a positive, explicitly charged cost in $K \setminus L$.

Proof. For every cycle shore S_i , the shore-root lemma gives a lower root in S_i . Moreover, S_i blocks the lower root of its predecessor S_{i-1} . The common-owner theorem and Corollary 2.137 therefore give a member of $S_i \setminus R$. Hence

$$|S_i| \geq 2, \quad \sum_{d \in D} t_C(d) = \sum_{i=1}^k |S_i| \geq 2k. \quad (2.97)$$

Moreover $t(d) \geq t_C(d)$, and $2^{t(d)} - 1 \geq t(d)$. The cycle contains k distinct shore sets, so Z_0 contains at least k coordinates. Lemma 2.139 and (2.97) prove (2.94), including $\eta \geq 0$. Subtracting $3k$ and inserting the middle incidence sum in the definition of η gives

$$\begin{aligned} |K| - |L| - 3k &= \eta + (|Z_0| - k) + \sum_{d \in D} (2^{t(d)} - 1 - t_C(d)) \\ &\quad + \sum_{i=1}^k (|S_i| - 2), \end{aligned}$$

which proves (2.95). Its summands are nonnegative by the two preceding inequalities.

Suppose now that (2.96) holds. Every inequality just used is then an equality. Thus $|S_i| = 2$ for every i , and $t(d) \leq 1$ for every $d \in D$, since $2^t - 1 = t$ only for $t \in \{0, 1\}$. Equality also gives $t(d) = t_C(d)$. It follows that the cycle shores are pairwise disjoint, that no other passive coordinate has a nonempty shore, and that no cycle shore is represented by a second passive coordinate. This proves item 1.

Choose a lower root $r_i \in R \cap S_i$. The common owner supplied from the predecessor belongs to $S_i \setminus R$; call it o_i . Since $|S_i| = 2$, these are its two members. The ample gate $G_{z_i} = L \cup \{r_i, o_i\}$ admits the two-block chain of Lemma 2.141. Its first addition is r_i , because $o_i \notin R$.

Suppose that r_i and o_i were not adjacent. Since $L \cup \{o_i\}$ is nonample, the one-concept colon criterion gives $y \in L$ which blocks o_i . Adding the non-neighbour r_i does not add a neighbour of o_i , so the same y blocks o_i over the ample family $L \cup \{r_i\}$. The criterion would make $L \cup \{r_i, o_i\} = G_{z_i}$ nonample, a contradiction. Hence $r_i o_i$ is an active edge, proving item 2.

Equality in the collar count gives $|L \cap M_{z_i}| = 1$; write this intersection as $\{a_i\}$. It also leaves no further member of M_{z_i} , since such a member would give another vertex of K in the z_i -layer. Hence

$$M_{z_i} = \{a_i, r_i, o_i\}.$$

This family is ample, and hence connected. It contains the edge $r_i o_i$, so its third vertex a_i is adjacent to exactly one of r_i, o_i . Together with the z_i -edge from $\hat{a}_i$ to $\hat{a}_i^{\oplus z_i}$, this is the attached three-edge path in item 3.

The definition of the shore gives $r_i^{\oplus z_i}, o_i^{\oplus z_i} \in K$, while $r_i, o_i \in D \subseteq A$. These are the four vertices of the claimed square. Different shores have disjoint active endpoints. Their flipped vertices also have distinct active restrictions, and no flipped vertex has passive restriction σ . The squares are therefore vertex-disjoint. This proves item 4. Finally, equality in the counting argument of Lemma 2.139 leaves no member of K beyond the lift of L and these $3k$ nonbase path vertices, proving item 5. $\square$

Lemma 2.145 (A shore outside the holonomy cycle pays its order plus one). *Retain the notation of Corollary 2.144, and put*

$$\varepsilon = |K| - |L| - 3k.$$

For every nonempty passive shore S_z , one has

$$\begin{cases} |S_z| \leq \varepsilon + 2, & \text{if } S_z \in \{S_1, \dots, S_k\}, \\ |S_z| \leq \varepsilon - 1, & \text{if } S_z \notin \{S_1, \dots, S_k\}. \end{cases} \quad (2.98)$$

Equivalently, a cycle shore costs at least $|S_z| - 2$ units of excess, whereas a shore outside the cycle costs at least $|S_z| + 1$.

Proof. If $S_z = S_i$ is a cycle shore, its summand $|S_i| - 2$ in (2.95) gives the first bound.

Suppose that S_z is not a cycle shore. The coordinate z is in Z_0 but is not one of the k coordinates needed to realize the distinct cycle shores. Hence its contribution gives $|Z_0| - k \geq 1$. For every $d \in S_z$, the noncycle incidence in coordinate z gives $t(d) \geq t_C(d) + 1$, and therefore

$$2^{t(d)} - 1 - t_C(d) \geq 1.$$

The corresponding terms for the $|S_z|$ distinct concepts, together with the extra coordinate term, contribute at least $|S_z| + 1$ to the right-hand side of (2.95). This proves the second bound. $\square$

Corollary 2.146 (A lower family of order at most nine has two-arm holonomy). *Under the hypotheses of Corollary 2.144, suppose that $|K| \leq 9$. Every directed cycle of λ has length two. For such a cycle S_1, S_2 , put*

$$\varepsilon = |K| - |L| - 6.$$

Then $0 \leq \varepsilon \leq 2$, and

$$\eta + (|Z_0| - 2) + \sum_{d \in D} (2^{t(d)} - 1 - t_C(d)) + \sum_{i=1}^2 (|S_i| - 2) = \varepsilon. \quad (2.99)$$

In particular, $|Z_0| \leq 4$ and $|\mathcal{S}| \leq 4$. If $|K| = 7$, then $|L| = 1$, $\varepsilon = 0$, and the complete pendant two-arm normal form of Corollary 2.144 holds.

Proof. Lemma 2.140 gives $|L| \geq 1$. A cycle of length k therefore gives

$$9 \geq |K| \geq |L| + 3k \geq 1 + 3k.$$

Since $k \geq 2$, one has $k = 2$. The same inequalities give $0 \leq \varepsilon \leq 2$, and (2.95) gives (2.99). Its first summand gives $|Z_0| \leq 4$, and every member of $\mathcal{S}$ is realized by a coordinate in Z_0 , so $|\mathcal{S}| \leq 4$. If $|K| = 7$, then the displayed chain of inequalities is sharp throughout, and the equality part of Corollary 2.144 applies. $\square$

Lemma 2.147 (A two-concept cycle shore is an active edge). *Under the hypotheses of Corollary 2.144, let S lie on a directed cycle of λ , and suppose that $|S| = 2$. Then*

$$S = \{r, o\}, \quad r \in R, \quad o \notin R,$$

and ro is an active edge.

Proof. The shore-root lemma gives $r \in R \cap S$. The predecessor of S on the λ -cycle is blocked by S , so the common-owner theorem and Corollary 2.137 give $o \in S \setminus R$. These are the two members of S .

The gate $L \cup S$ is ample, as is $L \cup \{r\}$, whereas $L \cup \{o\}$ is nonample. If r, o were nonadjacent, a blocking vertex in L for the latter one-concept extension would remain a blocking vertex after adding the non-neighbour r . The one-concept colon criterion would make $L \cup \{r, o\}$ nonample, a contradiction. $\square$

Corollary 2.148 (The one-unit holonomy anomalies form a trichotomy). *Use the notation of Corollary 2.146, and let S_1, S_2 be its directed two-cycle. If $\varepsilon = 1$, exactly one of the following alternatives holds:*

1. $\eta = 1$; the two shores are disjoint two-concept sets, and the two cycle shores are the only nonempty passive shores, each represented by one coordinate.
2. $\eta = 0$; the two shores have order two and meet in one concept. Their common concept belongs to $D \setminus R$: it is the common non-root owner of both shores.
3. $\eta = 0$; the two shores are disjoint, one has order two, and the other has order three.

In particular, an extra nonempty passive coordinate cannot occur at cost one.

If $\varepsilon = 2$ and $|Z_0| = 3$, then the third passive coordinate has a singleton shore disjoint from $S_1 \cup S_2$, while the two cycle shores are disjoint two-concept sets and $\eta = 0$. Thus a new passive direction has minimum total cost two.

Proof. For the two-cycle, the exact identity (2.99) is

$$\varepsilon = \eta + (|Z_0| - 2) + \sum_{d \in D} (2^{t(d)} - 1 - t_C(d)) + (|S_1| - 2) + (|S_2| - 2). \quad (2.100)$$

All summands are nonnegative integers.

Suppose first that $\varepsilon = 1$. If $|Z_0| > 2$, choose an additional nonempty shore coordinate w , and a concept $d \in S_w$. If $d \notin S_1 \cup S_2$, then its summand in the middle sum is at least one, in addition to $|Z_0| - 2 \geq 1$. If d belongs to a cycle shore, then $t(d) \geq t_C(d) + 1$ and its middle summand is at least two. Both conclusions contradict $\varepsilon = 1$. Hence $|Z_0| = 2$, and each cycle shore is represented by its unique coordinate.

With these two coordinates, one has $t(d) = t_C(d)$. The middle summand is zero unless $d \in S_1 \cap S_2$, in which case it is $2^2 - 1 - 2 = 1$. The final two summands count the concepts in the cycle shores

beyond their required root and non-root owner. Equation (2.100) therefore gives exactly the three listed alternatives.

In the overlap alternative, write $r_i \in R \cap S_i$ for the selected root. Since $\lambda(S_1) = S_2$, relation (2.92) gives $r_1 \notin S_2$; similarly $r_2 \notin S_1$. Each two-concept shore contains a root and a non-root owner. Their common concept is consequently the non-root member of both shores.

Finally suppose that $\varepsilon = 2$ and $|Z_0| = 3$. The extra coordinate already contributes one through $|Z_0| - 2$. The preceding argument shows that every concept in its shore contributes at least one more to the middle sum, and contributes at least two if it also lies in a cycle shore. Equality with total two therefore forces the extra shore to be a singleton disjoint from $S_1 \cup S_2$, and exhausts the entire budget. Thus $\eta = 0$, the cycle shores are disjoint and have order two, and no further anomaly occurs. $\square$

Proposition 2.149 (A shared-owner one-unit anomaly is impossible). *Under the hypotheses of Corollary 2.148, the second alternative of that corollary cannot occur. Thus, when $\varepsilon = 1$, the only possibilities are $\eta = 1$ or disjoint cycle shores of orders two and three.*

Proof. Suppose that

$$S_1 = \{r_1, o\}, \quad S_2 = \{r_2, o\},$$

where $r_1, r_2 \in R$ and $o \notin R$. Let z_i be the unique passive coordinate realizing S_i . By Lemma 2.147, $r_i o$ is an active edge; write its direction as f_i . The two directions are distinct, since r_1, r_2, o are three distinct cube vertices.

Here $\eta = 0$, so the forced-fiber count accounts for all of K . For each i , the collar intersection $L \cap M_{z_i}$ is therefore a singleton, say $\{a_i\}$, and

$$M_{z_i} = \{a_i, r_i, o\}.$$

This is an ample three-vertex family containing the edge $r_i o$; hence it is a path and a_i is adjacent to exactly one of r_i, o . The punctured passive fiber over o contributes

$$o^{\oplus z_1}, \quad o^{\oplus z_2}, \quad o^{\oplus \{z_1, z_2\}} \in K.$$

Together with the two collars and the two root lifts, these are the seven members of K outside the lift of L . Since $|K| \leq 9$, one has $|L| \leq 2$.

We determine the one-inclusion graph of K . First suppose that $a_1 = a_2 = a$, which includes the case $|L| = 1$. Translate o to the zero active word. If a is adjacent to o in both active paths, the vertices

$$\hat{a}, \hat{a}^{\oplus z_1}, o^{\oplus z_1}, o^{\oplus \{z_1, z_2\}}, o^{\oplus z_2}, \hat{a}^{\oplus z_2}$$

form a six-cycle. If a is adjacent to both r_1 and r_2 , then $a = o^{\oplus \{f_1, f_2\}}$, and inserting the two root lifts in the preceding display gives an eight-cycle. The two mixed alternatives are impossible: two adjacent cube vertices have no third common neighbour. Any remaining root lift, and the second vertex of L when present, is attached as a tree edge.

It remains that $|L| = 2$ and $a_1 \neq a_2$. The two vertices of the ample family L are adjacent. Again translate o to zero. If a_1 and a_2 were both adjacent to o , they would be at Hamming distance two; if both were adjacent to r_1, r_2 , respectively, their active words would both have even weight and could not be adjacent. After interchanging the indices, the only remaining possibility is

$$a_1 = o^{\oplus g}, \quad a_2 = o^{\oplus \{f_2, g\}}$$

for an active direction $g \notin \{f_1, f_2\}$. The collar paths, the edge $a_1 a_2$, and the mixed owner vertex then form an eight-cycle; the unused root lift is a pendant edge.

Thus in every case the one-inclusion graph of K is a cycle C_{2m} , with $m \in \{3, 4\}$, with possibly some trees attached. There are no further edges: two vertices in different passive layers already differ in the corresponding passive coordinates, while within one passive layer the active adjacencies are exactly those of the three-vertex paths displayed above. The only remaining possible adjacency is the edge of the two-vertex family L , which is one of the attached tree edges in the preceding description. It has no four-cycle. If t is the number of attached tree edges, then its isometric dimension is $m + t$: the cycle has m coordinate classes and every tree edge is a singleton coordinate class. Since no pair of coordinates supports a square, the only strongly shattered sets are the empty set and these $m + t$ singleton sets. Hence

$$|\text{SSh}(K)| = 1 + m + t < 2m + t = |K|.$$

This contradicts ampleness of K , which requires equality in the lower Sandwich bound. The shared-owner alternative is impossible. $\square$

Corollary 2.150 (Two-arm reduction for a small absolute obstruction). *Let H have minimum cardinality among nonempty cornerless ample families, suppose that $|H| \leq 39$, and use the coordinate-section notation*

$$A_b = \rho_e^b H, \quad B = A_0 \cap A_1, \quad D_b = A_b \setminus B$$

from Proposition 2.122. Then $|B| \leq 9$. If some D_b has order ten and is two-connected, the relative pair $B \subsetneq A_b$ is terminal under both one-concept addition and deletion, its lower holonomy consists of a two-cycle, and its excess satisfies (2.99). In particular, $|B| \geq 7$. The same conclusion holds for a two-connected order-eleven wing whenever $B \subsetneq A_b$ is terminal under one-concept addition.

Proof. The two relative wings satisfy $|D_0|, |D_1| \geq \rho_{\mathcal{D}_{\text{amp}}} \geq 10$. Consequently,

$$2|B| = |H| - |D_0| - |D_1| \leq 39 - 20,$$

and hence $|B| \leq 9$.

Suppose that $|D_b| = 10$. If $B \cup \{d\}$ were ample for some $d \in D_b$, then d would be a corner of this one-concept extension. Since $B \in \mathcal{D}_{\text{amp}}$, reversing a peeling of B would give $B \cup \{d\} \in \mathcal{D}_{\text{amp}}$. Moreover, $\text{Cor}(A_b) \subseteq B \subseteq B \cup \{d\}$. The pair

$$B \cup \{d\} \subsetneq A_b$$

would therefore be a trapped $\mathcal{D}_{\text{amp}}$ -interval with nine relative concepts, contrary to $\rho_{\mathcal{D}_{\text{amp}}} \geq 10$. Thus no one-concept addition is ample. No one-concept deletion $A_b \setminus \{d\}$ is ample either: the ample corner-deletion criterion would make $d \in D_b$ a corner of A_b , contrary to $\text{Cor}(A_b) \subseteq B$. The pair is terminal.

If D_b is two-connected, the passive-gate normalization and Corollary 2.146, with $K = B$, apply. They give the two-cycle, the displayed excess budget, and $|B| \geq |L| + 6 \geq 7$. For an order-eleven wing, terminality is assumed, and the same argument applies without change. $\square$

Proposition 2.151 (Pair-rich opposite wings have only three alternating support graphs). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, and put $B = A_0 \cap A_1$. Suppose that, for each $b \in \{0, 1\}$, there are two disjoint direction pairs

$$P_{b,1}, P_{b,2} \in \text{Sh}(A_b) \setminus \text{Sh}(B).$$

Then the four pairs are distinct. Moreover, either three of them are pairwise disjoint, or their two-coloured support graph is an alternating four-cycle, an alternating four-edge path, or two disjoint alternating two-edge paths.

Proof. The two-layer shadow-intersection identity gives

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B).$$

Consequently a pair acquired by one wing cannot also be acquired by the other. Together with disjointness inside each wing, this proves that the four pairs are distinct.

Colour a pair by the wing which acquires it. Each colour class is a matching of order two, so the four-edge graph has maximum degree at most two and every cycle is alternating. Suppose that it has no matching of order three. A component cannot be an isolated edge together with a three-edge path, a two-edge path together with two isolated edges, or any finer partition of four edges, because each of those graphs has a three-edge matching. Nor can it be a longer cycle, since there are only four edges. The only remaining possibilities are one alternating four-cycle, one alternating four-edge path, and two disjoint alternating two-edge paths. $\square$

Lemma 2.152 (A crossed pair freezes its common carrier). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, put $B = A_0 \cap A_1$, and denote the layer coordinate by e . Suppose that x, y, z are distinct directions whose singletons are shattered by B , and that

$$\{x, y\} \in \text{Sh}(A_0) \setminus \text{Sh}(B), \quad \{x, z\} \in \text{Sh}(A_1) \setminus \text{Sh}(B).$$

Then there are unique traces

$$q_{xy} \in Q_{\{x, y\}}, \quad q_{xz} \in Q_{\{x, z\}}$$

such that

$$B|_{\{x, y\}} = A_1|_{\{x, y\}} = Q_{\{x, y\}} \setminus \{q_{xy}\}, \tag{2.101}$$

$$B|_{\{x, z\}} = A_0|_{\{x, z\}} = Q_{\{x, z\}} \setminus \{q_{xz}\}. \tag{2.102}$$

In particular, $\{e, x, y\}$ and $\{e, x, z\}$ are minimal nonfaces of $\text{Sh}(H)$. Their unique omitted traces are respectively

$$(1, q_{xy}) \quad \text{and} \quad (0, q_{xz}),$$

where the first entry is the e -coordinate.

Moreover, every x -edge of B lies in the common slice

$$y = 1 - q_{xy}(y), \quad z = 1 - q_{xz}(z). \tag{2.103}$$

Consequently, if two distinguished occurrences of the direction x are different x -edges of B , an anchor path between them in the connected x -carrier of B uses neither y nor z . Every step of this path is therefore a square of B with support $\{x, t\}$ for some $t \notin \{x, y, z\}$.

Proof. Projection is a pc-minor operation and therefore preserves ampleness. The projection of B to $\{x, y\}$ shatters both singletons but not the pair. Its shattered-set complex consequently has three members, so the Sandwich equality makes the projection a three-vertex path in the square. It omits one unique trace, which we call q_{xy} .

The shadow-intersection identity gives

$$\{x, y\} \notin \text{Sh}(A_1).$$

The projection $A_1|_{\{x,y\}}$ contains $B|_{\{x,y\}}$, which already has three traces, and it cannot contain the fourth. This proves (2.101). The same argument with the two layers interchanged proves (2.102).

The $e = 0$ section realizes all four traces on $\{x, y\}$, while the $e = 1$ section realizes all except q_{xy} . Thus the projection of H to $\{e, x, y\}$ is the cube with the single trace $(1, q_{xy})$ removed. The pair $\{x, y\}$ is shattered by H , and the pairs $\{e, x\}$ and $\{e, y\}$ are shattered because B shatters x and y . Hence $\{e, x, y\}$ is a minimal nonface. The argument for $\{e, x, z\}$ is symmetric; its missing trace lies in the $e = 0$ layer.

Consider an x -edge of B . Its two endpoints have one common y -value. If that value were $q_{xy}(y)$, the endpoint whose x -value is $q_{xy}(x)$ would realize the omitted trace q_{xy} , contrary to (2.101). Its common y -value is therefore $1 - q_{xy}(y)$. Applying the same argument to q_{xz} fixes its z -value and proves (2.103).

The anchors of the x -edges of an ample family form an ample, hence connected, carrier. Equation (2.103) makes both y and z constant on this carrier. Neither direction can label an anchor-path edge. Finally, an anchor edge in direction t is exactly a full $\{x, t\}$ -square of B , which proves the last assertion. $\square$

Corollary 2.153 (One signed carrier object covers the three alternating graphs). *In the setting of Proposition 2.151, suppose in addition that every direction occurring in the four acquired pairs is shattered as a singleton by B . Form a bipartite graph J whose four vertices are the two acquired pairs of each wing, joining two opposite-wing vertices once for their common direction. If the four pairs have no matching of order three, then*

$$J \cong K_{2,2}, \quad J \cong P_4, \quad \text{or} \quad J \cong 2K_2.$$

For every edge of J , the common direction carries the two omitted trace labels and the conditioned carrier slice of Lemma 2.152. If its two distinguished arm edges do not coincide, their transport is a path of squares in B which avoids the two partner directions.

Thus the alternating four-cycle, four-edge path, and two disjoint two-edge paths are not three unrelated unpositioned exceptions: they are respectively the incidence shadows $K_{2,2}$, P_4 , and $2K_2$ of one signed carrier-transport problem.

Proof. Two acquired pairs belonging to the same wing are disjoint, so two opposite-wing pairs meet in at most one direction and J is a simple subgraph of $K_{2,2}$. The three support graphs in Proposition 2.151 have pair-incidence graphs $K_{2,2}$, P_4 , and $2K_2$, respectively. The carrier assertion is exactly Lemma 2.152, applied at each edge of J . $\square$

Proposition 2.154 (Opposite sharp two-arm systems have three alternating support graphs). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, and put $B = A_0 \cap A_1$. Suppose that, for each $b \in \{0, 1\}$, the terminal relative pair $B \subsetneq A_b$ satisfies the sharp two-arm conclusion of Corollary 2.144. For its two arms, let

$$P_{b,i} = \{f_{b,i}, z_{b,i}\} \quad (i \in \{1, 2\})$$

be the direction pair of the terminal active-passive square, where $f_{b,i}$ is the direction of the shore edge and $z_{b,i}$ is its passive direction. Then:

1. $P_{b,1}$ and $P_{b,2}$ are disjoint for each b ;
2. the four sets $P_{b,i}$ are pairwise distinct;

3. either three of the four pairs are pairwise disjoint, or the two-coloured graph with edge set $\{P_{b,i} : b \in \{0,1\}, i \in \{1,2\}\}$ is one of the following: an alternating four-cycle, an alternating four-edge path, or two disjoint alternating two-edge paths.

Thus, in the sharp shared-overlap branch, failure to produce three independent direction pairs has only three coordinate-incidence forms.

Proof. Fix b . The equality description writes B as its active slice L_b together with two pendant three-edge arms. Every edge of either arm is a bridge in the one-inclusion graph of B . In a partial cube, a bridge is the unique edge in its coordinate class: if a second edge had the same coordinate, convexity and connectivity of the two semicubes would give a path avoiding the first edge. The six arm edges therefore have six distinct directions. In particular $P_{b,1} \cap P_{b,2} = \emptyset$, which proves item 1.

The full $P_{b,i}$ -square belongs to A_b , so

$$P_{b,i} \in \text{Sh}(A_b).$$

On the other hand, the passive direction $z_{b,i}$ labels exactly one edge of B , namely the first edge of its pendant arm. An ample family which shatters $P_{b,i}$ contains a full $P_{b,i}$ -square, and such a square has two $z_{b,i}$ -edges. Hence

$$P_{b,i} \notin \text{Sh}(B).$$

Proposition 2.151 now applies to these two pairs in each wing. It proves simultaneously that the four pairs are distinct and that failure of a matching of order three has exactly the three forms in item 3. $\square$

Lemma 2.155 (Two pendant three-edge arms have a unique small core). *Let T be a tree. Suppose that T is obtained from a path C of order $\ell \in \{1, 2, 3\}$ by attaching two internally vertex-disjoint pendant paths of length three, each meeting C only at its initial vertex. Then C and the unordered pair of attached paths are uniquely determined by T among all such decompositions with core order ℓ .*

Proof. For $\ell = 1$, the tree is the seven-vertex path, its core is the unique central vertex, and its two components are the two arms.

For $\ell = 2$, write the core as the edge xy . If the two arms are attached at different endpoints, T is the eight-vertex path and xy is its unique central edge. If both are attached at, say, x , then x is the unique vertex of degree three. The three components of $T - x$ have orders 1, 3, 3; the one-vertex component contains y , and the two three-vertex components are the arms.

For $\ell = 3$, write the core as $x - y - z$. Up to reversing this path, there are four attachment patterns. Attaching at x, z gives the nine-vertex path, whose central three-vertex path is unique. Attaching twice at x gives one degree-three vertex with branch lengths 2, 3, 3; the length-two branch is the rest of the core. Attaching twice at y gives one degree-four vertex with branch lengths 1, 1, 3, 3; the two short branches complete the core. Finally, attaching at x, y gives one degree-three vertex, namely y , with branch lengths 1, 3, 4. The length-one branch is the vertex z , the length-three branch is the arm at y , and the first vertex on the length-four branch is x , after which its last three vertices form the other arm. In every case the decomposition is forced. These patterns exhaust the unordered pairs of attachment vertices, proving the lemma. $\square$

Corollary 2.156 (Opposite sharp two-arm wings do not fit below order ten). *In the setting of Proposition 2.154, one has*

$$|B| \geq 10.$$

Equivalently, if $|B| \leq 9$, the two opposite relative wings cannot both attain equality in the $3k$ holonomy bound.

Proof. Suppose that $|B| \leq 9$. For either wing, the sharp two-arm description writes the one-inclusion graph of B as the graph of its active slice L_b , with two pendant three-edge paths attached. It also gives

$$|L_b| = |B| - 6 \in \{1, 2, 3\}.$$

The ample family L_b is connected. Its graph is therefore a vertex, an edge, or a three-vertex path. The whole graph of B is consequently a tree of the form in Lemma 2.155.

Apply that lemma to the two wing decompositions of the same graph B . Their unordered pairs of arms coincide. The square-support pair attached to an arm consists of the coordinate labels of its attachment edge and its terminal edge, so the two wings have the same unordered pair of square supports. This contradicts item 2 of Proposition 2.154, which says that all four supports are distinct. Thus $|B| \geq 10$. $\square$

Lemma 2.157 (An ample one-vertex completion of a tree is a square fan). *Let B be ample, let $v \in B$, and suppose that the graph*

$$T = G(B) - v$$

is a nonempty tree. Write $d = \deg_{G(B)}(v)$. Define the square link Λ_v as follows. Its vertices are the d edge directions incident with v , and two such directions are adjacent when their square through v belongs to B . Then Λ_v is a tree. More precisely:

1. *every cube of B of dimension at least two is a square through v ;*
2. *B contains exactly $d - 1$ squares;*
3. *a direction not incident with v labels exactly one edge of $G(B)$, while a direction i incident with v labels $1 + \deg_{\Lambda_v}(i)$ edges.*

Thus the cycles created by the one added vertex form one connected tree of squares; no longer even cycle or disconnected collection of square repairs is possible.

Proof. Every square avoiding v would be a cycle in T , so every square contains v . A cube of dimension at least three has a square face avoiding each one of its vertices. Hence no such cube belongs to B , which proves item 1.

We shall use the elementary connected-carrier property of ample families. For a coordinate i , identify the two i -sections of B after deleting i . Their intersection is ample by the two-layer intersection lemma and is therefore connected. Its vertices correspond to the i -edges of B , and its edges correspond to squares joining two i -edges. Consequently all edges of one direction i are joined by a sequence of squares.

If i is not incident with v , this carrier consists of one edge: a square in a connecting sequence would contain v and would therefore have an i -edge incident with v . If i is incident with v , every other i -edge is the edge opposite the unique i -edge at v in a square through v . Such squares are indexed exactly by the neighbours of i in Λ_v . This proves item 3.

The graph Λ_v is a forest. Indeed, a cycle $i_1 i_2 \cdots i_s i_1$ in Λ_v would give in T the subdivision

$$v^{\oplus i_1}, v^{\oplus \{i_1, i_2\}}, v^{\oplus i_2}, \dots, v^{\oplus \{i_s, i_1\}}, v^{\oplus i_1}$$

of that cycle, contrary to the assumption that T is a tree.

Let r be the number of nonconstant coordinates of B , let $q = |E(\Lambda_v)|$, and put $n = |B|$. Item 3 gives

$$|E(G(B))| = r + 2q. \tag{2.104}$$

On the other hand, T has $n - 2$ edges and adjoining v adds d edges, so

$$|E(G(B))| = n - 2 + d.$$

Since B is ample, its shattered and strongly shattered sets coincide. By item 1 they are the empty set, the r singleton directions, and the q square supports through v . Therefore

$$n = 1 + r + q.$$

Substitution in (2.104) gives $q = d - 1$. A forest on d vertices with $d - 1$ edges is connected, so Λ_v is a tree and item 2 follows. $\square$

Lemma 2.158 (An ample two-vertex completion of a tree is a two-apex square complex). *Let B be ample, let $u, v \in B$ be distinct, and suppose that*

$$T = G(B) - \{u, v\}$$

is a nonempty tree. Put

$$d_u = \deg_{G(B)}(u), \quad d_v = \deg_{G(B)}(v), \quad \delta_{uv} = \begin{cases} 1, & uv \in E(G(B)), \\ 0, & uv \notin E(G(B)). \end{cases}$$

Then every cube of B of dimension at least two is a square meeting $\{u, v\}$, no two squares have the same direction support, and the number $q(B)$ of squares is

$$q(B) = d_u + d_v - \delta_{uv} - 2. \tag{2.105}$$

Thus two vertices added to a tree cannot create a higher-dimensional cube, a repeated square carrier, or an arbitrary collection of cycles: all cycles are uniquely supported squares incident with one of the two added vertices, and their number is determined by the two degrees.

Proof. Every square avoiding u, v would be a cycle in T , so every square meets $\{u, v\}$. A cube of dimension at least four has a square face avoiding any two prescribed vertices: after choosing two free directions, there are at least four parallel square faces and at most two of them meet $\{u, v\}$. Hence no such cube belongs to B .

Suppose that B contains a three-cube C . If C contains at most one of u, v , the square face opposite that vertex avoids both. If it contains both vertices and they are not antipodal in C , they agree in one of the three cube directions, and the opposite square face again avoids both. Finally, if u, v are antipodal in C , then $C - \{u, v\}$ contains the canonical six-cycle Q_3^{--} , which is a cycle in T . All alternatives contradict the tree hypothesis. Thus every nontrivial cube is a square meeting $\{u, v\}$.

Fix a direction pair S supporting a square. Successively intersecting the two coordinate sections in the directions of S shows that the family of anchors of the full S -squares is ample, by the two-layer intersection lemma. It is therefore connected. If two distinct S -squares existed, the first edge on a path between their anchors would join two opposite S -squares into a three-cube. This has just been excluded. Consequently every square support occurs exactly once.

Let r be the number of nonconstant directions of B , let $n = |B|$, and write $q = q(B)$. Ampleness and the preceding conclusions give

$$n = 1 + r + q. \tag{2.106}$$

We also use the edge-shadow identity

$$|E(G(B))| = r + 2q. \quad (2.107)$$

For completeness, in one direction i , the anchors of the i -edges form an ample carrier. Its shattered supports are exactly the sets $S \setminus \{i\}$ with $i \in S \in \text{Sh}(B)$. The Sandwich equality in that carrier therefore counts the i -edges by those supports. Summing over i proves the edge-shadow identity; here only singleton and pair supports occur, which gives (2.107).

The tree T has $n - 2$ vertices and $n - 3$ edges. Restoring u, v adds $d_u + d_v - \delta_{uv}$ edges, since the edge uv , when present, is counted twice by the two degrees. Hence

$$|E(G(B))| = n - 3 + d_u + d_v - \delta_{uv}.$$

Substituting (2.106) and (2.107), and cancelling r , gives

$$2q = q - 2 + d_u + d_v - \delta_{uv},$$

which is exactly (2.105). $\square$

Lemma 2.159 (A one-vertex completion of a unicyclic graph has a controlled square link). *Let B be ample, let $v \in B$, and suppose that*

$$T = G(B) - v$$

is connected and unicyclic. Assume that T contains at most one square, and let $q_0 \in \{0, 1\}$ be its number of squares. If $d = \deg_{G(B)}(v)$, define the square link Λ_v as in Lemma 2.157. Then:

1. B has no cube of dimension at least three, and every square support occurs exactly once;
2. B has exactly d squares, of which $d - q_0$ contain v ;
3. if $q_0 = 0$, then Λ_v is connected and unicyclic;
4. if $q_0 = 1$, then the unique cycle of T is that square and Λ_v is a tree.

Thus the cycle rank missing from the tree case appears exactly once: as the unique cycle of the square link when the base cycle is not already a square.

Proof. A three-cube avoiding v would place six squares in T . A three-cube containing v has three distinct square faces opposite v , again all contained in T . Both conclusions contradict the hypothesis on T . Every cube of larger dimension contains a three-cube, so B has no cube of dimension at least three.

As in the proof of Lemma 2.158, the anchors of all squares with one fixed direction support form an ample, hence connected, carrier. Two anchors would have an edge between consecutive anchors and therefore create a three-cube. Thus each square support occurs once.

Let r be the number of nonconstant directions, let q be the number of squares, and put $n = |B|$. The preceding paragraph and ampleness give

$$n = 1 + r + q, \quad |E(G(B))| = r + 2q.$$

The graph T has $n - 1$ vertices and, because it is unicyclic, $n - 1$ edges. Restoring v adds d edges, so

$$|E(G(B))| = n - 1 + d.$$

Substitution gives $q = d$. The edges of Λ_v are precisely the squares through v , and hence

$$|E(\Lambda_v)| = d - q_0. \quad (2.108)$$

A cycle of length s in Λ_v gives its canonical one-subdivision, a cycle of length $2s$, in T . Consequently Λ_v has cycle rank at most one. If $q_0 = 0$, then (2.108) gives d edges on d vertices. If it had c connected components, its cycle rank would be c ; hence $c = 1$, and it is connected and unicyclic.

If $q_0 = 1$, the unique cycle of T is its square. A cycle in Λ_v would instead give a subdivided cycle of length at least six, so no such cycle exists. Equation (2.108) now gives a forest with d vertices and $d - 1$ edges. It is therefore a tree. $\square$

Corollary 2.160 (The pure two-unit anomaly has two square apices). *Use the notation of Corollary 2.146. Suppose that $|K| \leq 9$, that the directed two-cycle has $\varepsilon = 2$, and that all of this excess lies in the uncounted remainder:*

$$\eta = 2, \quad |Z_0| = 2, \quad 2^{t(d)} - 1 = t_C(d) \quad (d \in D), \quad |S_1| = |S_2| = 2. \quad (2.109)$$

Then $|L| = 1$ and $|K| = 9$. After choosing one collar vertex for each passive direction, exactly two vertices $u, v \in K$ remain uncounted, and

$$G(K) - \{u, v\} = P_7,$$

the path obtained by attaching the two pendant three-edge arms to the one-vertex active slice. Consequently K has only uniquely supported squares, every such square meets $\{u, v\}$, and

$$q(K) = \deg_K(u) + \deg_K(v) - \mathbf{1}_{uv \in E(K)} - 2.$$

Let $P_i = \{z_i, f_i\}$ be the support of the terminal square of arm i , where z_i is its passive attachment direction and f_i its terminal direction. At most one of P_1, P_2 is shattered by K . Therefore at least one of the two terminal pairs is acquired by the upper family.

Proof. Equality in every term of (2.109), apart from $\eta = 2$, has the same consequences as the sharp part of Corollary 2.144: the two shores are disjoint active edges, each has one passive direction and one collar, and the counted lower vertices form two pendant three-edge arms attached to the lift of L . Moreover

$$|K| = |L| + 8.$$

The shore-root lemma gives $L \neq \emptyset$, while $|K| \leq 9$, so $|L| = 1$ and $|K| = 9$. The seven counted vertices therefore induce P_7 , and the two units of η are exactly the two remaining vertices u, v . Lemma 2.158 gives the asserted square structure and count.

It remains to compare the terminal supports. The direction of the attachment edge of arm i is z_i , and z_i occurs on no other edge of P_7 . Its two adjacent edges in the path have directions z_{3-i} and, say, g_i , neither of which is the active terminal direction f_i . If a P_i -square of K contained exactly one of u, v , deleting that apex from the square would leave in P_7 a two-edge path with directions z_i, f_i , a contradiction. A square cannot avoid both apices by the preceding lemma. Hence every P_i -square in K contains both u and v .

We next show that $f_1 \neq f_2$. Suppose instead that both terminal edges have direction f . The two terminal leaves lie in the same f -semicube: the path from the central vertex to either leaf crosses f only on its terminal edge. After deleting the two terminal f -edges, that semicube contains no other vertex of P_7 . Semicubes of a partial cube are convex and hence connected, so its two terminal leaves have a path through the two apices. The leaves belong to the same bipartition class. With only two apices available, they must therefore have a common apex and be at distance two.

After translating the active word of the central vertex to the empty set, the two leaves have respective direction words

$$z_1 \triangle g_1 \triangle f, \quad z_2 \triangle g_2 \triangle f.$$

The passive directions z_1, z_2 are distinct and the directions g_1, g_2 are active. Distance two consequently forces $g_1 = g_2$. The two active shore edges then have the same endpoint pair $\{g_1, g_1 \triangle f\}$, contrary to the disjointness of the cycle shores forced by (2.109). Thus $f_1 \neq f_2$, and the active-passive pairs P_1, P_2 are disjoint.

If squares on both supports existed and u, v were adjacent, the direction of uv would belong to both supports. If they were nonadjacent, they would be opposite in each square, and both supports would equal $\text{Diff}(u, v)$. Either conclusion is impossible. Thus at most one terminal pair is shattered by K . Both pairs are shattered by the upper family, which contains the corresponding terminal squares, so every pair not shattered by K is acquired there. $\square$

Corollary 2.161 (The two-unit budget has exactly seven profiles). *Use the notation of Corollary 2.144, let S_1, S_2 be a directed two-cycle, and assume that its excess is $\varepsilon = 2$. Up to interchanging S_1, S_2 , exactly one of the following alternatives holds:*

1. $\eta = 2$, and S_1, S_2 are disjoint two-concept shores;
2. $\eta = 1$, and S_1, S_2 are two-concept shores meeting in one concept;
3. $\eta = 1$, and S_1, S_2 are disjoint shores of orders two and three;
4. $\eta = 0$, and S_1, S_2 have orders two and three and meet in one concept;
5. $\eta = 0$, and S_1, S_2 are disjoint shores of orders two and four;
6. $\eta = 0$, and S_1, S_2 are disjoint three-concept shores;
7. $\eta = 0$, the cycle shores are disjoint two-concept sets, and a third passive coordinate has a singleton shore disjoint from their union.

In alternatives 1–6, $|Z_0| = 2$, each cycle shore is represented by exactly one passive coordinate, and there is no other nonempty passive shore. Alternative 7 is the rigid singleton-extra-shore profile. Consequently alternatives 1–3 are precisely the two-unit profiles with an uncounted lower vertex.

Proof. Write the exact excess identity as

$$2 = \eta + (|Z_0| - 2) + \sum_{d \in D} (2^{t(d)} - 1 - t_C(d)) + (|S_1| - 2) + (|S_2| - 2). \quad (2.110)$$

Every summand is a nonnegative integer. The two distinct cycle shores require two passive coordinates. Every further coordinate has a nonempty shore and therefore contributes at least one additional incidence to $\sum_d (t(d) - t_C(d))$. Since $2^t - 1 \geq t$, the second and third terms in (2.110) together are at least $2(|Z_0| - 2)$. Hence $|Z_0| \leq 3$. If $|Z_0| = 3$, equality in both contributions forces precisely the singleton-extra-shore profile of Corollary 2.148, which is alternative 7.

It remains that $|Z_0| = 2$. Each of the two distinct cycle shores is then represented by its unique passive coordinate, so $t(d) = t_C(d)$ for every $d \in D$. The middle sum in (2.110) is therefore

$$|S_1 \cap S_2| :$$

a concept belonging to one shore contributes zero, while a concept belonging to both contributes $2^2 - 1 - 2 = 1$. Put

$$a = |S_1 \cap S_2|, \quad x = |S_1| - 2, \quad y = |S_2| - 2.$$

Then

$$\eta + a + x + y = 2.$$

The shores have order at least two. If $\eta = 2$, all three remaining parameters vanish, giving alternative 1. If $\eta = 1$, the remaining unit is either $a = 1$ or, up to reversal, $x = 1$, giving alternatives 2 and 3. Finally let $\eta = 0$. The formal solution $a = 2, x = y = 0$ would make the two two-concept shores equal, contrary to their being distinct members of the directed cycle. The remaining solutions are $(a, x, y) = (1, 1, 0), (0, 2, 0), (0, 1, 1)$, up to reversal, which are exactly alternatives 4–6. $\square$

Lemma 2.162 (A disjoint passive shore cannot hide its terminal square). *Use the active–passive normalization of Lemma 2.139, and suppose that $L = \{a\}$. Let $z \in Z$ satisfy*

$$S_z \cap S_w = \emptyset \quad \text{for every } w \in Z \setminus \{z\}.$$

If xy is an active edge contained in S_z , with direction f , then

$$\{z, f\} \in \text{Sh}(A) \setminus \text{Sh}(K).$$

Thus the square over an edge of a shore separated from all the other shores is genuinely acquired by the upper family; arbitrary additional vertices of K cannot hide another square with the same support.

Proof. The definition of S_z gives

$$x, y, x^{\oplus z}, y^{\oplus z} \in A, \quad x, y \in D, \quad x^{\oplus z}, y^{\oplus z} \in K.$$

These four vertices form a full $\{z, f\}$ -square in A , so the pair is shattered there.

Suppose that K also shattered $\{z, f\}$. Since K is ample, it would contain a full square with this support. The anchors of all $\{z, f\}$ -squares of A , after the two directions are deleted, form the intersection of the four corresponding sections of A . Repeated application of the two-layer intersection lemma makes this anchor family ample and hence connected. It therefore contains a path from the anchor of the displayed split square to the anchor of a full square in K .

At the first anchor every passive coordinate other than z has value σ . The anchor of a full square in K cannot have this same passive restriction: its two vertices on the $z = \sigma(z)$ side would be distinct members of the one-concept slice L . Hence the anchor path has a first edge which changes some passive coordinate $w \neq z$. Immediately before this edge, consider the two vertices of the current square whose z -value agrees with σ . They have the full passive restriction σ , and therefore belong either to D or to the lift of L . Since they are distinct and L has one concept, at least one of them, say d , belongs to D .

The current square contains $d^{\oplus z}$, and the three-cube joining the two consecutive anchors contains $d^{\oplus w}$. Both vertices lie outside the common passive layer of D , so both belong to K . Thus

$$d \in S_z \cap S_w,$$

contrary to the shore-separation hypothesis. Hence K does not shatter $\{z, f\}$, as required. $\square$

Lemma 2.163 (At most one uncounted vertex separates the shore directions). *In the notation of Corollary 2.144, suppose that $L = \{a\}$, that the nonempty shores are pairwise disjoint, and that $\eta \leq 1$. If distinct shores S_z, S_w contain active edges of directions f_z, f_w , respectively, then*

$$f_z \neq f_w.$$

Consequently the two active-passive pairs $\{z, f_z\}$ and $\{w, f_w\}$ are disjoint.

Proof. Suppose that both shore edges had direction f . Their z - and w -flips are two f -edges of K , lying in the passive layers $\sigma \oplus z$ and $\sigma \oplus w$. The anchor family of the f -edges of K is ample and hence connected. Along an anchor path between these two edges, take the first step which changes the passive restriction. Before that step the passive restriction is $\sigma \oplus z$. The new f -edge lies either in the base layer σ , if the step removes z , or in a layer differing from σ in at least two coordinates.

The base layer contains only the lift of the one concept a , so it contains no edge. Because the shores are pairwise disjoint, the punctured fibres forced by Lemma 2.139 occur only in single-coordinate passive layers. The collar vertices do as well. Consequently both endpoints of an edge in a layer with at least two passive flips would have to belong to the uncounted remainder. This requires at least two uncounted vertices, contrary to $\eta \leq 1$. Thus the connected f -carrier cannot join the two displayed edges, a contradiction. $\square$

Corollary 2.164 (The feasible uncounted two-unit profiles are pair-rich). *In the setting of Corollary 2.161, assume in addition that $|K| \leq 9$. Alternatives 1–3 have the following outcomes:*

1. *alternative 1 has the two-apex P_7 normal form of Corollary 2.160, and both of its terminal pairs are acquired;*
2. *alternative 2 is impossible;*
3. *in alternative 3, $|L| = 1$, $|K| = 9$, and the terminal pair of each of its two cycle shores is acquired.*

Proof. In alternative 1, the two terminal pairs are disjoint by Corollary 2.160. Here L is a singleton and the two shores are disjoint, so Lemma 2.162 shows that both pairs are acquired. Consider alternative 2. The exact count gives

$$|K| = |L| + 8.$$

The shore-root lemma and $|K| \leq 9$ force $|L| = 1$ and $|K| = 9$. Write the shores as

$$S_1 = \{r_1, o\}, \quad S_2 = \{r_2, o\},$$

where r_1, r_2 are the two lower roots and o is their common non-root owner. Let z_i be the passive direction of S_i , let f_i be the direction of the active edge $r_i o$, and write $L = \{a\}$. Since $r_i \in R$, the two-concept family $\{a, r_i\}$ is ample and hence ar_i is an active edge. On the other hand, $o \notin R$, so a is not adjacent to o . Thus, after the single uncounted vertex v is deleted, the counted graph $T = G(K) - v$ is the shared-owner eight-cycle.

The graph T is square-free and unicyclic, so Lemma 2.159 makes the square link Λ_v connected and unicyclic. Its unique link cycle must have length four, and its subdivision uses all eight vertices of T . If its consecutive directions are i_1, i_2, i_3, i_4 , the cyclic direction sequence of the subdivided cycle is

$$i_2, i_1, i_3, i_2, i_4, i_3, i_1, i_4.$$

No direction occurs on opposite edges. In contrast, the provenance of the shared-owner eight-cycle gives the sequence

$$z_1, f_2, f_1, z_2, z_1, f_2, f_1, z_2,$$

in which every direction occurs on opposite edges. This contradiction excludes alternative 2.

Finally consider alternative 3, and let S_1 be its two-concept shore, and let S_2 be its three-concept shore. Again $|L| = 1$, $|K| = 9$, and the two shores are disjoint. The shore S_1 is an active edge by Lemma 2.147. The shore S_2 also contains an active edge. Indeed, it contains a common non-root owner o . The two-concept family $L \cup \{o\}$ is nonample, so o is not adjacent to the unique member of L . A path from o to L in the connected ample gate $L \cup S_2$ must therefore begin with an edge internal to S_2 .

Lemma 2.162 applied to these two shore edges gives one acquired pair from each shore. Their active directions are distinct by Lemma 2.163, while their passive directions are distinct and are disjoint from every active direction. The two acquired pairs are therefore disjoint, proving item 3. $\square$

Corollary 2.165 (The vertex-free two-unit profiles are separated by their passive carriers). *In the setting of Corollary 2.161, assume in addition that $|K| \leq 9$. The four vertex-free alternatives 4–7 have the following outcomes:*

1. *alternative 4, whose shores have orders two and three and meet in one concept, is impossible;*
2. *in each of alternatives 5–7, the two cycle shores supply two disjoint acquired direction pairs.*

Consequently every feasible two-unit wing is pair-rich: it has two disjoint terminal direction pairs acquired by the upper family.

Proof. In all four alternatives, $\eta = 0$ and the exact count gives $|K| = |L| + 8$. Hence $|L| = 1$, $|K| = 9$, and we write $L = \{a\}$. Equality in the collar and punctured-fiber counts describes all vertices of K : for every nonempty passive direction z , the family contains the collar edge

$$a a^{\oplus z},$$

and, for every active concept d , it contains $d^{\oplus T}$ for every nonempty $T \subseteq Z(d) = \{z : d \in S_z\}$. It contains no further vertex.

Suppose first that alternative 4 holds, and let $S_{z_1} \cap S_{z_2} = \{c\}$. The z_1 -edges of K are exactly

$$a a^{\oplus z_1}, \quad c^{\oplus z_2} c^{\oplus \{z_1, z_2\}}.$$

After contracting z_1 , their two anchors are a and $c^{\oplus z_2}$. They differ in the passive direction z_2 and also in at least one active direction, because $a \in L$ and $c \in D$. Thus the two anchors are nonadjacent. On the other hand, the anchor family of all edges in one direction is the intersection of the two corresponding sections of K ; it is ample and hence connected. This contradiction excludes alternative 4.

Now consider alternatives 5–7. Each cycle shore is disjoint from every other nonempty shore. Fix one of them, write it as S_z , and let $o \in S_z \setminus R$ be the common owner supplied by the predecessor on the holonomy cycle. The gate $L \cup S_z$ is ample and hence connected. The two-concept family $L \cup \{o\}$ is not ample, because o is not a lower root. Since $L = \{a\}$, the vertices a, o are not adjacent. A path from o to a in $G(L \cup S_z)$ therefore starts with an active edge xy having both endpoints in S_z ; let f be its direction.

Disjointness of the shores and the exact vertex description above show that the collar $a a^{\oplus z}$ is the only z -edge of K . Thus K cannot shatter $\{z, f\}$: by ampleness it would then contain a full

$\{z, f\}$ -square and hence two z -edges. In the upper family, both x, y and their z -flips are present, so this full square does exist there. The pair $\{z, f\}$ is therefore acquired. Applying the argument to each of the two cycle shores gives two acquired pairs. Their active directions are distinct by Lemma 2.163, applied with $\eta = 0$; their passive directions are distinct and disjoint from the active directions. Hence the pairs are disjoint, proving item 2. $\square$

Corollary 2.166 (Pair-rich comparisons in the two-unit row). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. If the terminal relative pair $B \subsetneq A_0$ has two-cycle holonomy and excess two, then it has two disjoint acquired direction pairs. If the opposite wing $B \subsetneq A_1$ is sharp, has the uncounted $\eta = 1$ one-unit form, or has any feasible two-unit profile, then either the two wings supply three pairwise disjoint acquired pairs or their four acquired pairs form one of the three alternating support graphs in Proposition 2.151.

At this stage the only apparent profile outside the pair-rich comparison is the one-unit extra-owner wing with disjoint shores of orders two and three. Proposition 2.172 below shows that an extra-owner wing opposite a two-unit wing is pair-rich as well.

Proof. Corollary 2.164 excludes alternative 2 and gives two disjoint acquired pairs in alternatives 1 and 3 of the two-unit classification. Alternative 4 is impossible, and Corollary 2.165 gives two disjoint acquired pairs in alternatives 5–7. This proves the first assertion.

A sharp wing has two such pairs by the proof of Proposition 2.154; the uncounted one-unit wing has them by Corollary 2.168; and a second two-unit wing has them by the preceding paragraph. Proposition 2.151 now gives the claimed matching or one of the three alternating graphs. The one-unit trichotomy and Proposition 2.149 leave only the extra-owner form outside these pair-rich cases. Its pair-poor subcase is localized by Corollary 2.170. $\square$

Lemma 2.167 (A small three–four arm decomposition is intrinsic except on a path). *Let G be a connected graph of order at most nine which is obtained from a path core C of order one or two by attaching the following two components at vertices of C :*

1. *a pendant path of length three;*
2. *a connected four-vertex graph F , attached by one edge at a distinguished vertex of F , where F is a tree or a four-cycle.*

Call the first component the ordinary arm. If G admits two such decompositions, then their ordinary arms coincide, except in the following two cases:

1. *$G = P_8$, the core has order one, and the two decompositions interchange the length-three and length-four sides of the central edge;*
2. *$G = P_9$, the core has order two, and the two decompositions interchange the length-three and length-four sides of the central three-vertex path.*

Moreover, if G also has a sharp two-arm decomposition as in Lemma 2.155, then the ordinary arm of every three–four decomposition is one of its two three-edge arms.

Proof. A connected ample graph of order four is either a four-vertex tree or a four-cycle. If F is a four-cycle, it is the unique nontrivial block of G . The block–cut tree then identifies F , the path

core, and the ordinary arm. Such a graph has no sharp two-arm decomposition, because the latter is a tree.

Suppose that F is a tree. There are four rooted possibilities for $(F, \text{attachment vertex})$: a four-vertex path rooted at an endpoint or at an internal vertex, and a three-leaf star rooted at its centre or at a leaf. In the internally rooted path case, the attachment vertex has degree three and the two branches inside F have orders one and two. In the centre-rooted star case it has degree four. In the leaf-rooted star case, the unique degree-three centre lies one edge beyond the attachment vertex and has two one-vertex branches away from the core. These three degree and branch-order profiles are different from one another and from the path core. They therefore identify the rooted copy of F ; the remaining three-vertex component is the ordinary arm.

It remains that F is a path rooted at an endpoint. Thus G is formed from its core by attaching paths of lengths three and four. For a one-vertex core this is P_8 , and either side of its central edge can be declared the ordinary arm. For a two-vertex core, attaching at different core endpoints gives P_9 , with the two choices stated in the lemma. Attaching both paths at the same endpoint gives a unique degree-three vertex with branch lengths 1, 3, 4, which distinguishes the ordinary arm. This proves the first assertion.

Finally, compare with a sharp decomposition. Its graph is a path core of order at most three with two length-three arms. The uniqueness proof of Lemma 2.155, or the same branch-length comparison, shows that any pendant three-vertex component in a three–four decomposition is one of those two arms. In the only path possibilities, this is the length-three side of the central edge or central core. Hence its attachment and terminal edges, and not merely its vertex set, coincide with a sharp arm. $\square$

Corollary 2.168 (The $\eta = 1$ anomaly is a tree of squares). *Use the notation of Corollary 2.148, suppose that $|K| \leq 9$, and assume its first alternative:*

$$\eta = 1,$$

with two disjoint two-concept cycle shores. Then $|L| \in \{1, 2\}$. After choosing one collar vertex for each passive direction, there is exactly one remaining, uncounted vertex $v \in K$, and $G(K) - v$ is obtained from the path $G(L)$ by attaching two pendant paths of length three. The square link of v is a tree. In particular, if $d = \deg_{G(K)}(v)$, then K contains exactly $d - 1$ squares, all through v .

For each arm, let z_i be the direction of its attachment edge and let f_i be the direction of its terminal edge. Then

$$P_i = \{z_i, f_i\} \notin \text{Sh}(K).$$

The two pairs P_1, P_2 are disjoint. They are precisely the supports of the two terminal squares acquired by the upper family.

Proof. The one-unit identity gives

$$|K| = |L| + 7,$$

so $|L| \leq 2$. The shore-root lemma gives $L \neq \emptyset$, and hence $|L| \in \{1, 2\}$.

For a shore $S_i = \{r_i, o_i\}$, the opposite active section M_{z_i} contains the edge $r_i o_i$ and at least one collar vertex in $L \cap M_{z_i}$. All but at most one vertex of M_{z_i} are accounted for by the lower slice, the collar count, and the punctured shore fibers. Since M_{z_i} is connected and bipartite, one may choose a collar vertex a_i adjacent to one endpoint of $r_i o_i$. Thus

$$a_i, r_i, o_i$$

induce a three-vertex path. After lifting to the z_i -layer and adjoining the collar edge from L , this is a pendant path of length three in K .

The two arms have disjoint nonbase vertices: their passive restrictions are respectively the z_1 - and z_2 -flips of the common passive word, and their shore endpoints have distinct active words. No edge joins the two arms, since such vertices differ in both z_1 and z_2 ; and an arm vertex is adjacent to the lower slice only along its collar edge. The six arm vertices together with L account for all but the single unit $\eta = 1$. Calling that remaining vertex v , we obtain the asserted tree $G(K) - v$. Lemma 2.157 gives the square-link conclusion.

Within $G(K) - v$, the passive direction z_i occurs only on the attachment edge of arm i . By the carrier description in Lemma 2.157, a square of K using z_i , if one exists, has this attachment edge as the side opposite v . Its other direction is therefore the direction of the next edge of the arm, not the terminal direction f_i . Hence $\{z_i, f_i\}$ is not strongly shattered by K , and ampleness gives $P_i \notin \text{Sh}(K)$.

The directions z_1, z_2 are distinct passive coordinates and do not occur on the active edges of either arm. If $f_1 = f_2$, connectedness of the f_1 -carrier would put both terminal edges opposite the f_1 -edge at v in two squares. The two corresponding length-two paths in $G(K) - v$, together with the path between the arms through L , would form a cycle, a contradiction. Thus $f_1 \neq f_2$, and the two pairs are disjoint. Finally, the four vertices over the shore edge $r_i o_i$ form the acquired terminal square in the upper family, as in the zero-excess case. $\square$

Lemma 2.169 (Comparison of one-unit two-arm decompositions). *Let G be an ample partial cube of order at most nine admitting an $\eta = 1$ decomposition from Corollary 2.168, with terminal support pairs P_1, P_2 .*

1. *If G also admits a sharp two-arm decomposition, one of its terminal support pairs is equal to P_1 or P_2 .*
2. *If G admits two $\eta = 1$ decompositions, then either their terminal-support systems have a common pair, three of their four pairs are pairwise disjoint, or G is the following exceptional graph: a four-cycle with a pendant path of length two attached at each vertex of one antipodal pair.*
3. *If G also admits a three-four decomposition from the extra-owner anomaly, then its ordinary-arm support is equal to one of P_1, P_2 , or it is disjoint from both of them.*

In the exceptional graph of item 2, write z_1, z_2 for the directions of the central square and f_1, f_2 for the terminal directions of the two pendant paths. The two decompositions use the two opposite remaining vertices of the square as their respective cores and extra vertices, and their four terminal supports are

$$\{z_i, f_j\} \quad (i, j \in \{1, 2\}). \quad (2.111)$$

Proof. Delete the extra vertex of an $\eta = 1$ decomposition. The remaining tree has one of only three branch profiles:

$$P_7, \quad P_8, \quad \text{a three-branch tree with branch lengths } 1, 3, 3.$$

Lemma 2.157 says that all cycles restored by the extra vertex form a connected square fan. We compare decompositions using the block-cut tree of this fan.

First suppose that G is a tree. Then the extra vertex is a leaf. The branch lengths identify both three-edge arms unless G itself is P_8 or P_9 . In the identified cases a second sharp, $\eta = 1$,

or three–four decomposition retains an entire arm and hence its terminal support. On P_8 or P_9 , distinct $\eta = 1$ decompositions move the extra leaf from one end to the other. Reading the first and third edge of each resulting arm gives three pairwise disjoint support pairs. The same reading shows that a sharp decomposition retains one arm, while an ordinary three–four arm either coincides with an $\eta = 1$ arm or has both of its support directions outside the two disjoint $\eta = 1$ supports.

Now suppose that G contains a cycle. Every nontrivial block is part of the square fan. If the fan has at least two squares, any second vertex whose deletion is a tree must belong to every square. The square-link tree is consequently a star, and that vertex is the common neighbour of the fan apex. Deleting either endpoint of this common fan edge leaves the same branches in the block–cut tree. Their lengths identify one whole arm, unless the other decomposition uses its ordinary arm outside both old arms; in the latter case its first and terminal directions are disjoint from both P_1 and P_2 . This proves the three asserted alternatives in the multi-square case.

If the two decompositions use the same fan apex, then they are already decompositions of the same deleted tree. The P_7, P_8 , and $(1, 3, 3)$ branch profiles above identify their arms directly, so their terminal-support systems agree. Thus the preceding common-neighbour argument covers every genuinely different apex.

It remains that the square fan is one four-cycle. Removing any candidate extra vertex turns this block into a three-edge path. Attach to its four vertices the components of the block–cut tree and compare their orders with the three base profiles displayed above. Unless one three-edge arm is unchanged, the only way that two removals both leave a P_7 core profile is to attach two-edge paths at two opposite vertices of the square and to use the other opposite pair as the two possible core–extra choices. This is the exceptional graph in item 2. If the central square directions are z_1, z_2 , each of the two core choices pairs each z_i with one of the two tail directions f_1, f_2 , and the choices interchange the tails. This gives exactly (2.111). For a sharp or three–four comparison, the same four block–cut positions leave an old arm unchanged or put the ordinary arm outside both old supports. This completes all three comparisons. $\square$

Corollary 2.170 (A pair-poor extra-owner wing has a rooted-claw overlap). *In the one-unit extra-owner alternative of Corollary 2.148, suppose that $|K| \leq 9$. If $|L| = 1$, then the two disjoint shores supply two disjoint acquired direction pairs. Consequently an extra-owner wing which is not pair-rich must satisfy*

$$|L| = 2, \quad |K| = 9.$$

In that case, write $G(L) = ab$, and choose the notation so that the ordinary three-edge arm is

$$ap_1p_2p_3.$$

Then $G(K)$ is obtained by adjoining a vertex c , three leaves q_1, q_2, q_3 adjacent to c , and one edge rc , where $r \in \{a, b\}$. Thus there are only two rooted graphs, according as $r = a$ or $r = b$.

Proof. The one-unit identity gives $|K| = |L| + 7$, so the only possible slice orders are one and two. Suppose that $L = \{a\}$. The two-concept shore is an active edge by Lemma 2.147. The three-concept shore contains a non-root owner o . Since $L \cup \{o\}$ is nonample, o is not adjacent to a ; connectivity of the ample shore gate therefore supplies an active edge internal to that shore.

The shores are disjoint, so Lemma 2.162 makes both active–passive pairs acquired. Their active directions are distinct by Lemma 2.163, applied with $\eta = 0$, and their passive directions are distinct. Thus the two pairs are disjoint. The order identity now gives the final assertion.

It remains to identify the graph when $|L| = 2$ and the wing is pair-poor. The proof of Lemma 2.167 writes $G(K)$ as the edge $G(L)$, an ordinary pendant path of length three, and a rooted four-vertex component F . The latter consists of its collar vertex and the three concepts of the larger shore.

The passive direction attaching F occurs on only the collar edge of K : the two shores are disjoint and $\eta = 0$, so the exact punctured-fibre count contains no further edge in that passive direction.

Suppose that two concepts of the larger shore were adjacent, in active direction f . Their passive translates in K , together with the corresponding two vertices in the upper family, give an acquired active–passive pair. This pair is disjoint from the ordinary-arm pair. Indeed, the two passive directions are distinct and are disjoint from the active directions, while the terminal edge of the ordinary pendant arm is a bridge of $G(K)$. A bridge in a partial cube is the unique edge of its direction, so its active direction cannot equal f . The wing would therefore be pair-rich, contrary to the hypothesis.

Hence the three shore concepts are pairwise nonadjacent in F . Since F is connected, each is adjacent to the collar vertex. Thus F is a claw rooted at its centre. There are no edges between the two attached components in the active–passive normal form, so the edge list in the statement is complete. $\square$

Lemma 2.171 (The two rooted-claw overlaps have rigid cuts). *Let $T^\parallel$ and $T^\times$ be the two trees with vertex set*

$$\{a, b, p_1, p_2, p_3, c, q_1, q_2, q_3\}$$

and common edge set

$$\{ab, ap_1, p_1p_2, p_2p_3, cq_1, cq_2, cq_3\},$$

where the final edge is ac in $T^\parallel$ and bc in $T^\times$. Call $ap_1p_2p_3$ the old arm. Then:

1. *if deleting two vertices leaves a path on seven vertices, viewed as two pendant paths of length three attached to its centre, then the tree is $T^\times$, the deleted vertices are two of q_1, q_2, q_3 , and one of the two arms is the old arm;*
2. *suppose that deleting one vertex leaves a connected graph obtained from a one-vertex core by attaching a pendant path of length three and a rooted four-vertex tree. If the three nonattachment vertices of the four-vertex tree contain an edge, then the pendant path is the old arm;*
3. *the only cut producing component orders $(3, 5)$ is the cut at a in $T^\times$, and its three-vertex component is $\{p_1, p_2, p_3\}$. The only cut producing component orders $(4, 4)$ is the cut at b in $T^\times$; one of its rooted four-vertex components is the claw on $\{c, q_1, q_2, q_3\}$, whose three nonattachment vertices are independent. Neither tree has a cut with component orders $(2, 3, 3)$.*

When either tree is embedded isometrically in a cube, the occurrence of the old arm in items 1–3 has the same attachment and terminal directions as in the original rooted-claw decomposition.

Proof. For item 1, the claw centre c cannot be deleted: its three leaves would be isolated, and only one further vertex may be removed. Thus at least two neighbours of c must be deleted in order that the remaining graph have maximum degree two. In $T^\parallel$, the neighbour a cannot be deleted, because the retained vertices in the old arm, at b , and at the claw would lie in different components. Hence two claw leaves must be deleted, but then a still has the three neighbours b, p_1, c . The remainder is not a path. In $T^\times$, the neighbour b likewise cannot be deleted without separating the old arm from the claw. Deleting two claw leaves is therefore necessary and sufficient, and the remaining path is

$$q_i cb ap_1 p_2 p_3.$$

Its two length-three arms meet at a , and one is the old arm.

For item 2, deletion of one vertex from a tree leaves a connected graph only when that vertex is a leaf. Checking the leaf orbits and then the component orders after deleting the proposed one-vertex core gives

tree	deleted leaf	core	outcome
$T^{\parallel}$	b	a	old arm and a centre-rooted claw
$T^{\times}$	p_3	b	a three-edge arm and a centre-rooted claw
$T^{\times}$	q_i	a	old arm and a rooted four-vertex tree.

These are all the possibilities: deleting p_3 or a claw leaf in $T^{\parallel}$ gives no cut into components of orders three and four; in the last row, the alternative core b attaches the three-vertex component at its middle and hence does not give a pendant path of length three. In the first two rows the three vertices of the claw away from its attachment are independent. The hypothesis of item 2 therefore leaves only the last row, where the pendant path is the old arm.

Finally, direct deletion of a vertex gives the following complete list of component orders. In $T^{\parallel}$, deleting a, p_1, p_2, c , or a leaf gives, respectively,

$$(1, 3, 4), \quad (2, 6), \quad (1, 7), \quad (1, 1, 1, 5), \quad (8).$$

Thus no vertex gives one of $(3, 5), (4, 4), (2, 3, 3)$. In $T^{\times}$, deleting a gives $(3, 5)$, deleting b gives $(4, 4)$, and every other deletion gives one of

$$(8), \quad (2, 6), \quad (1, 7), \quad (1, 1, 1, 5).$$

The components in the two relevant cuts are exactly those described in item 3. This proves the cut assertions. The final direction statement holds because the repeated arm has the same edges, and every edge of a tree partial cube is its own direction class. $\square$

Proposition 2.172 (A two-unit wing excludes a pair-poor extra-owner wing). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. If one terminal relative pair $B \subsetneq A_b$ has two-cycle holonomy and excess two, while the opposite wing has the one-unit extra-owner form, then the extra-owner wing is pair-rich. Consequently either the two wings supply three pairwise disjoint acquired direction pairs, or their four acquired pairs form one of the three alternating support graphs in Proposition 2.151.

Proof. Suppose for a contradiction that the extra-owner wing is pair-poor. Corollary 2.170 gives $|B| = 9$ and identifies $G(B)$ with one of the two rooted-claw trees in Lemma 2.171. Let Q be the acquired support determined by the attachment and terminal directions of its old arm. We compare this tree with the seven two-unit alternatives of Corollary 2.161.

In alternative 1, deleting the two uncounted vertices leaves the seven-vertex path with two length-three arms from Corollary 2.160. Item 1 of the rooted-claw cut lemma says that one of these arms is the old arm. Its acquired terminal support is therefore Q .

Alternative 2 is impossible because deleting its uncounted vertex leaves the six-cycle with two pendant terminal edges described in the proof of Corollary 2.164; an induced subgraph of the tree $G(B)$ cannot contain that cycle. In alternative 3, deleting the uncounted vertex leaves a one-vertex core, a three-edge arm, and a rooted four-vertex component. The three shore vertices in the latter contain an edge, as proved in the final paragraph of that same corollary. Item 2 of the rooted-claw cut lemma therefore makes the three-edge arm the old arm, so its acquired support is again Q .

Alternative 4 is impossible by Corollary 2.165. In the remaining vertex-free alternatives, deleting the one-vertex active slice leaves components of orders $(3, 5)$, $(4, 4)$, and $(2, 3, 3)$, respectively. Item 3 of the rooted-claw cut lemma shows that the first profile can occur only with its three-vertex component equal to the old arm; its acquired support is Q . In the second profile, the only possible cut has a rooted claw whose three shore vertices are independent, contrary to the internal shore edge used in the proof of Corollary 2.165. The third component profile does not occur in either rooted-claw tree.

Thus every feasible comparison makes the two-unit wing acquire Q . The extra-owner wing also acquires Q , whereas B does not shatter it. This contradicts the two-layer shadow-intersection identity

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B).$$

The extra-owner wing is therefore pair-rich. The two-unit wing is pair-rich by Corollaries 2.164 and 2.165; applying Proposition 2.151 gives the final alternative. $\square$

Proposition 2.173 (The extra-owner anomaly contradicts the opposite wing or forces three pairs). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. For each $b \in \{0, 1\}$, suppose that the terminal pair $B \subsetneq A_b$ has a two-cycle and is either sharp or has the one-unit extra-owner form: $\eta = 0$, with disjoint cycle shores of orders two and three. Then the two wings cannot both be sharp. If at least one has the extra-owner form, either the two wing structures are incompatible, or the acquired-support family contains three pairwise disjoint direction pairs.

Proof. The sharp-sharp case is Corollary 2.156. Consider an extra-owner wing. Its two-concept shore is an active edge by Lemma 2.147. Together with its unique passive coordinate it gives a full square in the upper wing. The passive coordinate labels exactly one edge of B , because $\eta = 0$, so this square support is not shattered by B . We call it the ordinary-arm support.

The opposite active section M_z of the three-concept shore has order four: it consists of the unique collar vertex and the three shore concepts. It is ample. Hence its graph is a four-vertex tree or a four-cycle. The graph of B is obtained from the active slice, of order

$$|B| - 7 \leq 2,$$

by attaching the ordinary three-edge arm and this rooted four-vertex component. It therefore has the form of Lemma 2.167.

If the other wing is sharp, the last assertion of that lemma makes the ordinary arm one of its two sharp arms. If the other wing also has the extra-owner form and the ordinary arms are intrinsic, the first assertion makes the two ordinary arms coincide. In either case their support pairs coincide. But each pair belongs to the shadow of its wing and not to $\text{Sh}(B)$, contradicting

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B)$$

from the two-layer shadow-intersection identity.

The only remaining case consists of two extra-owner decompositions of P_8 or P_9 . Write the consecutive edge directions of the path as $x_0, x_1, \dots, x_{q-2}$, where $q \in \{8, 9\}$. For P_8 , the two ordinary-arm supports are

$$\{x_0, x_2\}, \quad \{x_4, x_6\}.$$

The four-vertex branch of the first decomposition has passive direction x_3 and a shore edge of direction x_5 , so it also gives the full square support $\{x_3, x_5\}$. These three pairs are pairwise disjoint. For P_9 , the corresponding three supports are

$$\{x_0, x_2\}, \quad \{x_5, x_7\}, \quad \{x_4, x_6\},$$

again pairwise disjoint. Each is acquired by one of the two wings, because the relevant passive direction is a bridge of B and hence cannot belong to a square of B . This proves the proposition. $\square$

Proposition 2.174 (All one-unit opposite-wing comparisons reduce to one central square). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. Assume that both terminal pairs $B \subsetneq A_b$ have two-cycle holonomy and excess at most one. Then one of the following holds:

1. *the acquired-support family of the two wings contains three pairwise disjoint direction pairs;*
2. *B is the exceptional graph of Lemma 2.169: a central four-cycle with pendant paths of length two at two opposite vertices, and both wings have the $\eta = 1$ form based at the other two opposite vertices.*

In the second alternative the four terminal square supports form the complete bipartite graph

$$\{\{z_i, f_j\} : i, j \in \{1, 2\}\}$$

between the two central-square directions and the two tail directions. Thus, apart from this one graph, the entire two-connected one-unit frontier produces three independent direction pairs in one step.

Proof. By Proposition 2.149, a one-unit wing is either of $\eta = 1$ type or of extra-owner type. A zero-unit wing is sharp.

Two sharp wings are impossible by Corollary 2.156. If at least one wing has the extra-owner form and neither has the $\eta = 1$ form, Proposition 2.173 gives either a contradiction or three pairwise disjoint acquired supports.

It remains to compare an $\eta = 1$ wing with the opposite wing. Its two terminal pairs are acquired by that wing and are not shattered by B , by Corollary 2.168. The ordinary support of an extra-owner wing has the same property by the proof of Proposition 2.173; and the terminal pairs of a sharp wing do so by Proposition 2.154. The two-layer shadow identity

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B)$$

therefore forbids an acquired pair from occurring in both wings.

Apply Lemma 2.169. Against a sharp wing it forces such a repeated pair, a contradiction. Against an extra-owner wing it forces either a repeated pair or an ordinary support disjoint from both $\eta = 1$ supports; the latter gives three independent pairs. Against a second $\eta = 1$ wing it gives a repeated pair, three independent pairs, or the unique exceptional central-square graph. The repeated case is again forbidden by the shadow identity, and (2.111) gives the final support description in the exceptional case. $\square$

Lemma 2.175 (Passive layers can shield only nearby overlap corners). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless, put $B = A_0 \cap A_1$, and fix $b \in \{0, 1\}$. Suppose that there is a set of directions Z_b and a word $\sigma_b \in Q_{Z_b}$ such that every exclusive concept in $D_b = A_b \setminus B$ has restriction σ_b to Z_b . Then every corner v of B satisfies

$$d_{Q_{Z_b}}(v|_{Z_b}, \sigma_b) \leq 1. \quad (2.112)$$

If equality holds and z is the unique coordinate of Z_b on which v and σ_b differ, then the exclusive shield of v is forced:

$$v^{\oplus z} \in D_b. \quad (2.113)$$

Equivalently, a corner of the overlap at passive distance at least two from the exclusive layer of a wing makes the corresponding copy a corner of the full two-layer family, while a distance-one corner has only one possible exclusive neighbour.

Proof. Suppose that $v \in \text{Cor}(B)$ and that the distance in (2.112) is at least two. Every member of D_b differs from v in at least two coordinates of Z_b , so v has no neighbour in D_b . Let $Q = Q_B(v)$ be the unique maximal cube of B through v . Every horizontal edge of H incident with (v, b) is therefore an edge of $B \times \{b\}$, and its direction belongs to Q . The vertical edge through (v, b) is present because $v \in B$, and all these incident directions belong to the single cube

$$Q \times \{0, 1\}.$$

Thus (v, b) belongs to a unique maximal cube of H , contrary to cornerlessness.

Now suppose that the passive distance is one, with unique differing coordinate z . The same unique-maximal-cube argument shows that v must have a neighbour in D_b . Every member of D_b has passive word σ_b , so such a neighbour must change z . Changing any further coordinate would give Hamming distance at least two. The only possible neighbour is therefore $v^{\oplus z}$, which proves (2.113). $\square$

Proposition 2.176 (Connected pure two-apex incidence exposes a distant corner). *Let B be an ample family of order nine with two distinguished decompositions of the pure $\eta = 2$ form in Corollary 2.160. For decomposition $j \in \{0, 1\}$, write c_j for the central vertex of its P_7 , U_j for its two uncounted vertices, $Z_j = \{z_{j,1}, z_{j,2}\}$ for its passive directions, and*

$$\mathcal{P}_j = \{\{z_{j,1}, f_{j,1}\}, \{z_{j,2}, f_{j,2}\}\}$$

for its terminal-support system. Suppose that the four members of $\mathcal{P}_0 \cup \mathcal{P}_1$ are distinct, have no matching of order three, and form a connected support graph. Then, for some $j \in \{0, 1\}$, there is a vertex $v \in U_j$ such that

$$v \in \text{Cor}(B), \quad d_{Q_{Z_j}}(v|_{Z_j}, c_j|_{Z_j}) = 2.$$

Proof by a complete normal-form audit. Fix one distinguished decomposition and label its seven-vertex spine

$$P_7 = p_{-3}p_{-2}p_{-1}p_0p_1p_2p_3, \quad c_0 = p_0.$$

Every family in the statement is obtained by adjoining two vertices to this fixed path. The first has an arbitrary neighbourhood in P_7 ; the second has an arbitrary neighbourhood in the resulting eight-vertex graph. Thus there are exactly

$$2^7 2^8 = 32768$$

labelled candidates, with no assumption on the positions of the two new vertices.

The verifier `verify_damp_pure_eta_two_connected_incidence.py` performs the following direct tests. It computes the Djoković classes and retains exactly the isometric cube families satisfying the Sandwich equality. For every retained family it deletes every pair of vertices, rediscovers every induced P_7 with singleton centre, and checks that each of its two attachment directions occurs nowhere else on the path. It then forms the four terminal pairs for every two decompositions. Finally, it tests (2.176) directly.

For completeness, the corner test used in the last step is exact rather than a degree proxy. Lemma 2.158 says that every nontrivial cube is a uniquely supported square meeting the deleted pair. The verifier lists all such squares through v , adds every incident edge direction contained in no such square, and declares v a corner exactly when this list has one maximal member.

The audit retains 317 ample partial-cube completions, counted with their distinguished labelled P_7 . It finds respectively

support graph	C_4	P_5	$2P_3$
comparisons	52	32	42
without the corner in (2.176)	0	0	42

with the same harmless labelling multiplicities. Since Proposition 2.151 gives only C_4 , P_5 , and $2P_3$ when a matching of order three is absent, the connected cases are exactly the first two columns. The full record is `damp_pure_eta_two_connected_incidence.json`; its mathematical digest is

e12c192cb3e1ef387bc42655cd7486ad58b9eee3fd3085c32d3b75fcc3252e8.

□

Proposition 2.177 (Disconnected pure two-apex incidence has one path falsifier). *Keep the notation of Proposition 2.176, but suppose that the four terminal supports form $2P_3$. Define the forced passive-owner set*

$$O_j = \left\{ v^{\oplus z} : \begin{array}{l} v \in \text{Cor}(B), \ d_{Q_{Z_j}}(v|_{Z_j}, c_j|_{Z_j}) = 1, \\ z \text{ is the unique differing coordinate} \end{array} \right\}.$$

Then one of the following alternatives holds:

1.

$$O_0 \cap O_1 \neq \emptyset \quad \text{or} \quad (O_0 \cup O_1) \cap B \neq \emptyset;$$

2. $G(B) = P_9$. Up to reversing the path and interchanging the decompositions, write its vertices as $v_0 v_1 \cdots v_8$, with edge directions x_i on $v_{i-1} v_i$. The two P_7 's are $v_0 \cdots v_6$ and $v_2 \cdots v_8$, with central vertices v_3, v_5 . Their two inner arms have the common collar support

$$\{x_4, x_5\}.$$

Proof by the same complete normal-form audit. The verifier used in the preceding proposition continues the exact corner test through the 42 labelled $2P_3$ comparisons. For every corner at passive distance one, it computes the unique word obtained by correcting the differing passive coordinate. Thirty-six comparisons have either a word forced on both sides or a forced word already in B . After identifying repeated embedded comparisons, these are 20 types.

The remaining six labelled comparisons, or three embedded types, all have the path form in item 2. In both decompositions the shared directions of the two components reverse attachment and terminal roles. Reading the directions of the two overlapping P_7 's gives

	left arm	right arm
$v_0 \cdots v_6$	(x_3, x_2, x_1)	(x_4, x_5, x_6)
$v_2 \cdots v_8$	(x_5, x_4, x_3)	(x_6, x_7, x_8)

The collar pair is formed by the first two directions of an arm, so the right arm in the first row and the left arm in the second row both give $\{x_4, x_5\}$. No comparison remains unresolved. These further counts are recorded in the same JSON artifact and covered by the digest displayed in the preceding proof. $\square$

Corollary 2.178 (Two pure two-unit wings force three independent pairs). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless and ample, and put $B = A_0 \cap A_1$. Suppose that both relative pairs $B \subsetneq A_b$ have the pure $\eta = 2$ two-apex form. Then three of their four terminal acquired pairs are pairwise disjoint.

Proof. Both terminal pairs of either wing are acquired by Corollary 2.164. The two-layer shadow identity makes the two systems disjoint. If there is no matching of order three, Proposition 2.151 leaves C_4 , P_5 , and $2P_3$.

Suppose that one of the first two profiles occurs. Proposition 2.176 gives a decomposition j and a corner v satisfying (2.176). In the active–passive normalization for that wing, every exclusive concept has the common passive word $c_j|_{Z_j}$. Lemma 2.175 now contradicts cornerlessness of H .

It remains to exclude $2P_3$. Apply Proposition 2.177. In its first alternative, every forced word in O_j must belong to $D_j = A_j \setminus B$, by (2.113). A word in B cannot do so. Nor can one word belong to both O_0 and O_1 , because

$$D_0 \cap D_1 = \emptyset.$$

Thus the first alternative is impossible.

In the second alternative, the active–passive normal form makes the first two edges of every arm one side and one lift of a full collar square in the corresponding A_j . Hence (2) belongs to both $\text{Sh}(A_0)$ and $\text{Sh}(A_1)$. On the other hand, $B = P_9$ is a tree and shatters no direction pair. This contradicts

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B).$$

The no-matching assumption is therefore impossible, proving the corollary. $\square$

Proposition 2.179 (One pure two-apex wing resolves the sharp, uncounted, and two-unit rows). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless and ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. Assume that $B \subsetneq A_0$ has the pure $\eta = 2$ two-apex form. If $B \subsetneq A_1$ is sharp, has the uncounted $\eta = 1$ one-unit form, or has any feasible two-unit profile, then the acquired-support family of the two wings contains three pairwise disjoint direction pairs.

Proof by an exact-profile audit. We explain both the finite scope of the audit and how each machine outcome becomes a mathematical conclusion. Normalize the pure wing by fixing its counted spine

$$P_7 = p_{-3}p_{-2}p_{-1}p_0p_1p_2p_3.$$

As in Proposition 2.176, adjoining the two uncounted vertices with arbitrary neighbourhoods gives all $2^7 2^8$ labelled candidates. After the partial-cube and Sandwich tests, there are 158 distinct embedded ample overlaps. The audit rediscovers every pure decomposition of each overlap.

The opposite profiles are recognized without a graph catalogue. A sharp marker is a path core of order three with two pendant three-edge arms. An uncounted one-unit marker becomes a path core of order two with the same two arms after its unique uncounted vertex is deleted. For the two-unit row, the verifier applies the exact identity (2.110): it chooses the passive set, its common word, and the uncounted remainder, and requires the remaining vertices to be exactly the collar vertices and all nonempty punctured fibres of the shore-incidence sets. It also checks ampleness of every active gate. This recovers alternatives 1, 3, and 5–7 of Corollary 2.161; alternative 4 is already impossible by Corollary 2.165. The sharp and one-unit graph markers are deliberately over-inclusive, so in particular they contain every marker furnished by the corresponding structural theorems.

For each genuine marker the two terminal pairs are the acquired pairs proved in the sharp, one-unit, and two-unit results above. The audit also records an *acquired collar*: if z is passive and an active edge of direction g joins the active slice to a shore, the active–passive normal form supplies the full $\{z, g\}$ -square in the upper family. The pair is recorded only when B does not shatter it. Thus every recorded terminal or collar pair really belongs to $\text{Sh}(A_b) \setminus \text{Sh}(B)$.

The following table gives all 2268 candidate comparisons, with the harmless marker multiplicities retained. We abbreviate repeated terminal, terminal matching, distant corner, owner conflict, repeated collar, and collar-augmented matching by RT, TM, DC, OC, RC, and CM, respectively:

opposite profile	RT	TM	DC	OC	RC	CM
sharp	16	3	0	0	0	0
$\eta = 1$ one-unit	172	42	2	0	0	0
feasible two-unit	1188	592	118	95	24	16

There is no further outcome. Here is why the six columns suffice. A repeated acquired terminal or collar pair contradicts the two-layer shadow identity

$$\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B).$$

A distant overlap corner contradicts Lemma 2.175. At passive distance one, equation (2.113) forces the correcting word into the exclusive set $D_b = A_b \setminus B$. An owner conflict means either that this word already lies in B , or that the same word is forced into both D_0 and D_1 ; both alternatives are impossible because D_0, D_1, B are pairwise disjoint. The two matching columns give the desired three acquired pairs directly, before or after the acquired collars are added.

The verifier is `verify_damp_pure_eta_two_mixed_profiles.py`, and its complete output is `damp_pure_eta_two_mixed_profiles.json`. The mathematical digest is

f6fed025e0c67f1789358d287e116bc8d46cfbdca691c78b340a446cfb312ffd.

□

Proposition 2.180 (A two-unit wing resolves every opposite profile through two units). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless and ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. Assume that $B \subsetneq A_0$ has two-cycle holonomy and excess two, while $B \subsetneq A_1$ has two-cycle holonomy and excess at most two. Then the acquired-support family of the two wings contains three pairwise disjoint direction pairs.

Proof by the complete exact-profile audit. The classification and the two local impossibility results leave five feasible two-unit profiles. Alternative 2 of Corollary 2.164 is impossible, as is alternative 4 by Corollary 2.165. In each of alternatives 1, 3, and 5–7, the active slice is a singleton and the exact excess identity determines the entire counted overlap from its rooted active gates.

This gives a short exhaustive construction of the overlap universe. The pure profile is obtained by adjoining two arbitrary vertices to its P_7 spine. In alternative 3, one arbitrary vertex is adjoined to the wedge of a rooted three-vertex path and a rooted ample gate of order four. Alternatives 5 and 6 are respectively the wedges of rooted ample gates of orders three and five, and of two rooted ample gates of order four. Alternative 7 is the wedge of rooted paths of orders three, three, and two. Directly filtering all rooted graphs of orders four and five by partial-cube isometry and the Sandwich equality gives respectively

$$5 \quad \text{and} \quad 13$$

rooted ample gate types. Consequently this construction is exhaustive, without a catalogue of arbitrary nine-vertex graphs.

The accompanying exact-profile verifier constructs these normal forms and retains 148 distinct embedded ample overlaps. It then rediscovers 960 two-unit markers and every possible opposite sharp, uncounted one-unit, two-unit, or pair-rich extra-owner marker. The root/non-root orientation is checked explicitly: every cycle shore contains a lower root and a non-root owner, and a concept common to two cycle shores is a non-root in both. For an extra-owner marker, an edge inside its three-concept shore supplies the second acquired pair. If no such edge exists, the marker is pair-poor; the audit finds three such decompositions, all covered by the rooted-claw contradiction in Proposition 2.172.

Opposite wings which use the same marker repeat both acquired terminal supports and are immediately impossible by shadow intersection. For distinct markers, the audit makes 8395 comparisons. With the abbreviations from the preceding proof, their outcomes are

opposite row	RT	TM	DC	OC	RC	CM
two-unit	3131	1930	117	347	112	238
sharp	121	49	0	0	0	7
$\eta = 1$ one-unit	791	260	12	66	0	0
pair-rich extra-owner	718	334	0	94	0	68

and there is no unresolved comparison.

Each column has the proof meaning established above. Repeated acquired terminal or collar supports contradict $\text{Sh}(A_0) \cap \text{Sh}(A_1) = \text{Sh}(B)$. A distant corner contradicts passive-radius-one shielding. An owner conflict forces a word into B , or into both disjoint exclusive sets. The two matching columns give three independent acquired pairs directly. Thus every possible opposite profile has the conclusion of the proposition.

The verifier source and its complete output are, respectively,

```
verify_damp_two_unit_all_opposite_profiles.py,
damp_two_unit_all_opposite_profiles.json.
```

The mathematical digest is

272d55715bd51f23c8956fb0578ccc84c804eb5a54dd4995ca0d9e9484de1b81.

□

Corollary 2.181 (The central-square profile cannot be shielded). *Under the hypotheses of Proposition 2.174, suppose in addition that H is cornerless. Then the acquired-support family contains three pairwise disjoint direction pairs. In particular, the central-square $K_{2,2}$ alternative of that proposition cannot occur in a cornerless family.*

Proof. Suppose that the exceptional alternative occurs, and fix one of the two wings. In its $\eta = 1$ normal form, the active slice L is a singleton, say $\{c\}$, and the uncounted vertex v is the vertex of the central square opposite c . The two directions z_1, z_2 of that square are the two passive attachment directions of the wing. All relative concepts $D = A_b \setminus B$ have one common passive word, namely the passive word of c . Hence

$$v|_{\{z_1, z_2\}} = (c^{\oplus\{z_1, z_2\}})|_{\{z_1, z_2\}},$$

so v is at passive distance two from this word. The two pendant paths of the exceptional graph are attached at the other antipodal pair of vertices of the central square. Consequently v is a corner of B , contradicting Lemma 2.175. The exceptional alternative is impossible. $\square$

Lemma 2.182 (Forced wing words certify active coordinates). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless and ample, put $B = A_0 \cap A_1$, and fix one wing $D_b = A_b \setminus B$. Suppose that all members of D_b have one common word σ on a passive coordinate set Z . Let $\mathcal{S} \subseteq D_b$ be any collection of shore concepts whose opposite Z -fibres occur in B . For every corner v of B whose Z -word differs from σ in exactly one coordinate z , put $o_v = v^{\oplus z}$, and set

$$W_b = \mathcal{S} \cup \{o_v : v \in \text{Cor}(B), d_{Q_Z}(v|_Z, \sigma) = 1\}.$$

Then $W_b \subseteq D_b$. If the graph induced by D_b is connected, every coordinate which is nonconstant on W_b is active on that graph. Consequently, if the induced graph is two-connected, then

$$|D_b| \geq 2|\{x : |\{w|_x : w \in W_b\}| = 2\}|. \quad (2.114)$$

Proof. The shore concepts belong to D_b by definition. At passive distance one, Lemma 2.175 says that the unique concept which can shield the corner v is $v^{\oplus z}$, and cornerlessness forces that concept into D_b . Thus $W_b \subseteq D_b$.

If two members of W_b differ in coordinate x , a path between them inside the connected graph induced by D_b must use an x -edge. Hence x is active. Applying Lemma 2.130 gives (2.114). $\square$

Proposition 2.183 (The terminal two-connected matching row closes). *Let*

$$H = (A_0 \times \{0\}) \cup (A_1 \times \{1\})$$

be cornerless and ample, put $B = A_0 \cap A_1$, and suppose that $|B| \leq 9$. For $b \in \{0, 1\}$, suppose that the relative pair $B \subsetneq A_b$ is terminal under both one-concept addition and deletion, that the graph induced by $D_b = A_b \setminus B$ is two-connected, and that

$$|D_b| \in \{10, 11\}.$$

Then the two lower gate-holonomy cycles cannot both have excess at most two. In particular, no pair of terminal two-connected wings of orders ten or eleven occurs in the below-forty two-wing reduction.

Computer-assisted proof. By Corollary 2.146, both holonomy cycles have length two. The local comparison is already complete. If one wing has excess two, Proposition 2.180 gives either an immediate contradiction or three independent acquired pairs. If both excesses are at most one, Proposition 2.174 gives the same alternative or the central-square profile, and Corollary 2.181 excludes the latter. Thus it remains only to resolve the three-pair outcomes.

For each normal-form marker, form W_b from its actual shore concepts and the owners forced by every overlap corner at passive distance one. Corners at larger passive distance are already impossible. The complete two-unit normal-form universe of Proposition 2.180 contains 2886 matching comparisons. Passive distance and owner collisions exclude 302 of them. Among the remaining 2584, the ordered numbers of coordinates varying on (W_0, W_1) are

$(Y(W_0) , Y(W_1))$	(5, 5)	(6, 5)	(6, 6)
comparisons	504	108	1972.

Lemma 2.182 excludes the last two columns, because a two-connected wing using six active coordinates has at least twelve concepts.

The low-excess normal forms give the same dichotomy. The 76 ample one-vertex completions contain four distinct $\eta = 1$ versus $\eta = 1$ matchings and eight distinct $\eta = 1$ versus extra-owner matchings. Every positioned wing forces six active coordinates, except when the overlap is the eight-vertex path. The two extra-owner path decompositions behave identically: P_9 forces six coordinates, while P_8 is the sole five-coordinate residue.

It remains to check a fixed five-cube, not an ambient order interval. Up to cube automorphisms, the forced subsets of a wing are

$$\{0, 1, 2, 3, 12, 28\}, \quad \{0, 1, 2, 7, 15, 31\}, \quad \{0, 1, 2, 13, 29\} \subseteq Q_5.$$

They are, respectively, a square and a disjoint edge, two disjoint three-vertex paths, and the five-vertex P_8 equality pattern. Retaining their positions in B gives fourteen two-unit side states and six low-excess side states. For each state, the finite check adjoins every subset of the remaining Q_5 -vertices which gives a relative layer of order ten or eleven, and then tests induced two-connectivity, ampleness of $A_b = B \cup D_b$, and both terminality conditions. The results are

source	$ D_b = 10$		$ D_b = 11$	
	two-connected	ample upper	two-connected	ample upper
two-unit residue	504	238	3346	1330
low-excess P_8	744	198	4176	1026

Every ample upper family in the table has both an ample one-concept extension of B and a deletable concept in D_b . Hence none is terminal.

All tests use the exact cube adjacency relation and the Sandwich equality $|\text{Sh}(F)| = |F|$; no list of ambient families is used. The source files and compact certificates are

```

explore_damp_three_pair_global_compatibility.py
damp_three_pair_global_compatibility.json
verify_damp_low_excess_global_matching.py
damp_low_excess_global_matching.json
explore_damp_five_coordinate_matching_completions.py
damp_five_coordinate_matching_completions.json.

```


Their mathematical digests are, respectively,

ff2d10fdf95a608da5e7cdc0dca8cc9d973c4cfc345336f3b7278e40df8e3f92,
 760ff9566860cefa41b8a300dd9f99b03ecd08a79701a89803027f954e5afe08,
 fc485e9c7e6550c8478248effa90bf60b84719d42ea1579e2399b3d3f3cb5b10.

This exhausts every matching outcome and proves the proposition. $\square$

For a two-connected terminal layer of order ten or eleven, Lemma 2.130 supplies the five-active-coordinate hypothesis automatically. Corollary 2.142 therefore replaces all sixty-two residual graph types by two coupled directed cycles of roots and their carrier owners. Corollary 2.143 further replaces those two cycles by one two-sided holonomy system on ample intermediate gates. Corollary 2.144 adds a quantitative dichotomy: every shore overlap or additional owner pays a positive vertex cost, while zero excess is a system of pendant three-edge square-gate arms. At one unit, the shared-owner alternative is impossible, the extra-owner comparisons give three independent pairs or a contradiction, and the uncounted vertex is the apex of a connected tree of squares. Comparing the two wings first leaves two native support profiles: three independent acquired pairs, or the single central-square graph of Proposition 2.174, whose four supports form $K_{2,2}$. The latter profile is now excluded by Corollary 2.181: for either wing, its uncounted central-square vertex is a corner of B at passive distance two from every exclusive concept of that wing, so it cannot be shielded in H .

The remaining two-connected step is therefore no longer an open taxonomy of one- or two-unit anomalies. Every one-unit comparison gives three independent acquired pairs. The exact two-unit identity gives seven formal profiles; the shared-owner uncounted profile and the overlapping vertex-free profile are impossible, while each of the other five supplies two disjoint acquired pairs.

The three alternating support graphs $C_4, P_5, 2P_3$ were first merged by Corollary 2.153 into the incidence shadows $K_{2,2}, P_4, 2K_2$ of one signed carrier problem. Passive-radius shielding, forced distance-one owners, and acquired collar squares now resolve that problem completely. Proposition 2.180 excludes every alternating comparison of a two-unit wing with a sharp, one-unit, or two-unit opposite wing. Its audit includes the formerly separate neither-wing-pure and pair-rich extra-owner rows; the pair-poor extra-owner row is already impossible by the rooted-claw comparison. Thus every local two-connected profile through two units yields three independent acquired pairs.

Lemma 2.182 turns those pairs into positioned residual concepts. Proposition 2.183 then closes the global matching row: six varying coordinates contradict the open-ear bound, and the unique five-coordinate residue has no terminal completion. The octahedral support sphere of Remark 2.125 remains the geometry of the sharp twelve-concept example, but it is no longer a proof obligation for the orders-ten-and-eleven frontier. Proposition 2.129 also closes every terminal cut wing in that range. Thus all terminal wings through order eleven are two-connected and covered by the preceding comparison. The only remaining below-forty compatibility issue is the marked case of Lemma 2.186: one order-eleven wing may become a terminal order-ten core only after a single ample concept has been moved into the common overlap.

Proposition 2.184 (Nine-concept passive-carrier exclusion). *There is no trapped pair*

$$\emptyset \neq K \subsetneq A, \quad A, K \in \mathcal{D}_{\text{amp}}, \quad \text{Cor}(A) \subseteq K, \quad |A \setminus K| = 9.$$

Consequently,

$$10 \leq \rho_{\mathcal{D}_{\text{amp}}} \leq 12. \quad (2.115)$$

Computer-assisted proof. The arbitrary-upper relative linear-quotient theorem through eight facets [Bou26] already gives $\rho_{\mathcal{D}_{\text{amp}}} \geq 9$. Suppose that the displayed pair exists. Choose $C \in \mathcal{D}_{\text{amp}}$ inclusion-maximal subject to $K \subseteq C \subsetneq A$. The pair $C \subsetneq A$ is still trapped, so the lower bound nine forces $C = K$. Hence $K \cup \{c\}$ is not ample for every $c \in R := A \setminus K$: if it were, one could peel c first and contradict maximality. Also $A \setminus \{c\}$ is not ample, because c is not a corner of A . Thus every member of R is doubly blocked. Proposition 2.190 gives minimum residual degree two.

The graph induced by R is connected by Proposition 2.127.

Delete the coordinates constant on R . An exact cut-system classification tests all 160 connected bipartite graph types of order nine and minimum degree two. Ten admit an induced-cube embedding, and quotienting their coordinate cuts by graph automorphisms gives thirteen coordinate-coloured profiles. Applying the degree-two marker and cross-carrier rules of Lemma 2.126 excludes ten of them, leaving only the following active residual words:

profile	active words in Q_4
5.1	0, 2, 4, 6, 7, 8, 10, 11, 15
8.1	0, 2, 4, 6, 8, 10, 11, 14, 15
9.1	0, 2, 3, 4, 6, 7, 8, 10, 11

The classification and first reduction are reproduced by `enumerate_damp_nine_relative_graphs.py`, `enumerate_damp_nine_colored_embeddings.py`, and `verify_damp_nine_degree_two_carrier_frontier.py`. The last verifier keeps passive support parts symbolically through their inclusion preorder; its digest is

e5ea4fb30cb6611290a6555182a2af36dbd1339598d921ba1b04f37356b114ea.

It remains to exclude the three displayed profiles without conditioning away passive coordinates. Since A, K are ample, the relative support set $\mathcal{H}$ has exactly nine members. For each member, the final formula records its four-bit active part, its passive part, its position in the support order, and residual anchors for its lower and upper missing traces. It imposes support distinctness, carrier depth at least two at every residual vertex, the exact full-cube conclusion for every compatible minimal-maximal carrier pair, the degree-two isolated markers, and the degree-three dichotomy (A4g1c). These are only necessary conditions: in particular, the formula omits the adjacent-marker reversal law, so it is a relaxation of every genuine terminal interval.

Nine formal passive coordinates suffice. If $P_1, \dots, P_9$ are the actual passive parts, replace P_i by the down-set

$$\widehat{P}_i = \{j : P_j \subseteq P_i\}.$$

Then $\widehat{P}_i \subseteq \widehat{P}_j$ if and only if $P_i \subseteq P_j$. Thus this replacement faithfully preserves every passive inclusion and equality used in the formula.

The three structural formulas are unsatisfiable. CaDiCaL generated core proofs that were checked independently by `drat-trim`; the proof statistics are

profile	variables	clauses	resolution steps
5.1	6,366	71,937	9,796,884
8.1	6,615	76,490	6,854,743
9.1	6,627	74,647	7,636,141

All three checks use zero RAT lemmas. The generator, DIMACS manifest, proofs, and aggregate checker are

```
solve_damp_nine_passive_carrier.cnf.py,
damp_nine_passive_carrier.cnf.manifest.json,
damp_nine_passive_carrier.*.core.drat,
verify_damp_nine_passive_carrier_proofs.py.
```

The aggregate mathematical digest is

54b27131c083db9074d7c79ffcb303de8a8b36fe6d0c5de2ab6edd3c4f937d8b.

This excludes the last three profiles, proving that no trapped difference of order nine exists. The upper bound in (2.115) is Proposition 2.123. $\square$

Lemma 2.185 (Every sub-ten dismantlable-ample interval has a first concept). *Let $C \subsetneq H$ belong to $\mathcal{D}_{\text{amp}}$, and suppose that*

$$|H \setminus C| \leq 9.$$

Then some $h \in H \setminus C$ satisfies $C \cup \{h\} \in \mathcal{D}_{\text{amp}}$.

Proof. Suppose not, and choose $J \in \mathcal{D}_{\text{amp}}$ inclusion-minimal subject to $C \subsetneq J \subseteq H$. No corner of J lies outside C . Indeed, deleting such a corner j leaves a member of $\mathcal{D}_{\text{amp}}$. If $J \setminus \{j\} = C$, then $C \cup \{j\} = J$, contrary to the assumption; otherwise it contradicts the minimal choice of J . Consequently

$$\text{Cor}(J) \subseteq C, \quad 1 \leq |J \setminus C| \leq 9.$$

This is a trapped pair smaller than the lower bound $\rho_{\mathcal{D}_{\text{amp}}} \geq 10$ from Proposition 2.184, a contradiction. $\square$

Lemma 2.186 (Small trapped wings have at most one marked synchronization). *Let $B \subsetneq A$ be ample, suppose that*

$$\text{Cor}(A) \subseteq B, \quad |A \setminus B| \leq 11,$$

and assume that every ample family C with $B \subseteq C \subseteq A$ belongs to $\mathcal{D}_{\text{amp}}$. Then there is an ample family C with

$$B \subseteq C \subsetneq A, \quad |C \setminus B| \leq 1,$$

such that, on putting $R = A \setminus C$,

$$|R| \in \{10, 11\}, \quad C \cup \{r\} \text{ and } A \setminus \{r\} \text{ are nonample for every } r \in R. \quad (2.116)$$

Moreover, the graph induced by R is two-connected. In particular, a trapped wing of order ten is already terminal, whereas a trapped wing of order eleven is either terminal or consists of a terminal ten-concept core over one marked ample addition to B .

Proof. Choose C inclusion-maximal among the proper ample intermediate families between B and A . Such a family exists because $B \subsetneq A$. For every $r \in R = A \setminus C$, maximality makes $C \cup \{r\}$ nonample. If $A \setminus \{r\}$ were ample, the ample corner-deletion criterion would make r a corner of A . This is impossible because $r \notin C \supseteq B \supseteq \text{Cor}(A)$. Thus every member of R is doubly blocked.

The hypothesis puts both C and A in $\mathcal{D}_{\text{amp}}$, and $\text{Cor}(A) \subseteq B \subseteq C$. Hence $C \subsetneq A$ is a trapped pair in the definition of $\rho_{\mathcal{D}_{\text{amp}}}$. Proposition 2.184 gives

$$10 \leq |R| \leq |A \setminus B| \leq 11.$$

It follows that $|C \setminus B| = |A \setminus B| - |R| \leq 1$. Finally, Proposition 2.129 makes the residual graph two-connected. The last assertion is the specialization to the two possible initial orders. $\square$

Proposition 2.187 (A holonomy shore has a contracted split-square marker). *Let $K \subsetneq A$ belong to $\mathcal{D}_{\text{amp}}$, put $D = A \setminus K$, and suppose that*

$$K \cup \{d\} \text{ is nonample for every } d \in D, \quad |D| \leq 11.$$

Let z be constant on D , put

$$S_z = \{d \in D : d^{\oplus z} \in K\},$$

and suppose that $2 \leq |S_z| < |D|$. After contracting z , the relative concepts of the nested pair $\pi_z K \subsetneq \pi_z A$ are exactly the images of $D \setminus S_z$, and there is a concept $u \in D \setminus S_z$ whose image can be adjoined amply to $\pi_z K$. This root cannot lift to K . Consequently there is an active edge

$$uu^{\oplus f} \in E(D), \quad u \notin S_z, \quad u^{\oplus f} \in S_z, \quad (2.117)$$

whose z, f -square is split as

$$(u^{\oplus f})^{\oplus z} \in K, \quad u^{\oplus z} \notin A. \quad (2.118)$$

Proof. All members of D have the same z -value. Their projections remain distinct. A projected member already belongs to $\pi_z K$ exactly when its original concept lies in S_z . Hence

$$\pi_z A \setminus \pi_z K = (D \setminus S_z)|_{X \setminus \{z\}}.$$

Both contracted endpoints belong to $\mathcal{D}_{\text{amp}}$ by pc-minor closure, and

$$1 \leq |D \setminus S_z| \leq 11 - 2 = 9.$$

Lemma 2.185 supplies the asserted contracted root; denote its original lift by u .

The extension $K \cup \{u\}$ is nonample by hypothesis. Apply Proposition 2.191 with contracted coordinate z , lower family K , and upper family A . The other member $u^{\oplus z}$ of the z -fibre does not belong to A : it cannot belong to D , whose z -value is constant, and if it belonged to A it would belong to K , contrary to $u \notin S_z$. Thus u is the unique relative member of its fibre, as required by that proposition. Its wrong-side conclusion supplies an f for which

$$u^{\oplus f} \in D, \quad u^{\oplus \{z, f\}} \in K.$$

The second membership says exactly that $u^{\oplus f} \in S_z$, and also proves (2.118); this gives (2.117). $\square$

Corollary 2.188 (Lower gate holonomy is carried by split squares). *Assume in addition the hypotheses of Corollary 2.143, with $K, A \in \mathcal{D}_{\text{amp}}$ and $|D| \leq 11$. Every shore on a directed cycle of the map λ has at least two concepts and carries a split-square marker of the form (2.117)–(2.118). Thus the remaining two-connected obstruction is a cyclic system of oriented split squares, not a cyclic system of arbitrary boundary subsets.*

Proof. Let S lie on a λ -cycle and let T be its predecessor on that cycle. By construction, $S = \lambda(T)$ blocks the lower root r_T . The shore-root lemma gives a member of $R \cap S$, while the common-owner theorem gives a member of $S \setminus R$. These two concepts are distinct, so $|S| \geq 2$. Also $S \subsetneq D$, by the definition of $\mathcal{S}$. Proposition 2.187 now supplies the claimed marker. $\square$

Remark 2.189 (Why the relative threshold does not give the next absolute bound). The present calibration is

$$10 \leq \rho_{\mathcal{D}_{\text{amp}}} \leq 12.$$

In particular, the proposed estimate $\rho_{\mathcal{D}_{\text{amp}}} \geq 22$ is false. Proposition 2.122 remains a valid reduction, but by itself gives only order at least 22. Its coordinatewise form is stronger: it now excludes order 34 without an ambient-family census. Reaching the upper calibration $\rho_{\mathcal{D}_{\text{amp}}} = 12$ would make the same first-moment argument exclude every order below 40. The alternative bulk route is the uniform path-core repair theorem of Corollary 7.136.

The next proposition retains more of this compatibility. Rather than recording only the number of concepts in a trapped interval, it extracts a terminal block and charges separately for its cycles and for the cube corners that must be attached to the lower family.

Proposition 2.190 (Terminal block-curvature tax). *Let*

$$\emptyset \neq B \subsetneq A, \quad A, B \in \mathcal{D}_{\text{amp}}, \quad \text{Cor}(A) \subseteq B.$$

Choose $C \in \mathcal{D}_{\text{amp}}$ inclusion-maximal subject to $B \subseteq C \subsetneq A$, and put $D = A \setminus C$. Write $e(D)$ for the number of edges induced by D , and define

$$\tau(D) = e(D) - |D| + \max\{1, |\text{Cor}(D)|\}. \quad (2.119)$$

Then the graph induced by D has minimum degree at least two, every corner of the cube family D has a neighbor in C , and the relative Yang module for (A, C) is deletion-minimal Cohen–Macaulay: it is Cohen–Macaulay, whereas the module for $(A \setminus \{d\}, C)$ is not Cohen–Macaulay for any $d \in D$. Moreover,

$$\sum_{S \in \text{Sh}(A) \setminus \text{Sh}(B)} |S| \geq |A \setminus B| + \tau(D). \quad (2.120)$$

In particular, $\tau(D) \geq 1$.

Proof. No $d \in D$ satisfies the C -teaching condition. Indeed, that condition says that the one-concept extension $C \cup \{d\}$ is ample. It then belongs to $\mathcal{D}_{\text{amp}}$: first peel d , and then peel C . This contradicts the maximal choice of C .

Because A and C are ample, their relative Yang module is Cohen–Macaulay. On the other hand, $A \setminus \{d\}$ is not ample for any $d \in D$: otherwise d would be a corner of the ample family A , contrary to $\text{Cor}(A) \subseteq C$. The relative Cohen–Macaulay criterion over the ample lower family C proves the deletion-minimal assertion.

Suppose first that d is isolated in the graph induced by D . The ample family A is isometric. For every $c \in C$, the first edge of an isometric path from d to c therefore joins d to C in a coordinate on which d and c differ. Thus the C -neighbors of d meet $\text{Diff}(d, c)$ for every $c \in C$, which is exactly the C -teaching condition, a contradiction.

Suppose next that d has degree one in the graph induced by D . The relative leaf-deletion dichotomy [Bou26] says that either d satisfies the C -teaching condition, already excluded, or the relative module for $(A \setminus \{d\}, C)$ is Cohen–Macaulay. Since C is ample, the relative Cohen–Macaulay criterion makes $A \setminus \{d\}$ ample. Hence d is a corner of A , contrary to $d \notin C \supseteq \text{Cor}(A)$. This proves the minimum-degree assertion and, consequently, $e(D) \geq |D|$.

Let d be a corner of D , and let Q be the unique maximal cube of D containing it. If d had no neighbor in C , then every edge of A at d would already be an edge of D , and hence would lie in Q . Every cube of A through d would consequently lie in Q , making d a corner of A . This is again

impossible. Distinct corners of D give distinct boundary edges because their endpoints in D are distinct. Since the one-inclusion graph of A is connected, there is also at least one edge between the two nonempty sets C and D . If $e_\partial(C, D)$ denotes the number of these boundary edges, then

$$e_\partial(C, D) \geq \max\{1, |\text{Cor}(D)|\}. \quad (2.121)$$

For every ample cube family X , the edge-shadow identity gives

$$|E(X)| = \sum_{S \in \text{Sh}(X)} |S|.$$

Therefore

$$\begin{aligned} \sum_{S \in \text{Sh}(A) \setminus \text{Sh}(C)} |S| &= |E(A)| - |E(C)| \\ &= e(D) + e_\partial(C, D) \\ &\geq |D| + \tau(D). \end{aligned}$$

Finally, every support in $\text{Sh}(C) \setminus \text{Sh}(B)$ is nonempty, because $B \neq \emptyset$, and ampleness gives

$$|\text{Sh}(C) \setminus \text{Sh}(B)| = |C| - |B|.$$

Adding these earlier acquired supports proves (2.120). $\square$

Proposition 2.191 (Wrong-side witnesses localize on relative edges). *Let $C \subsetneq A \subseteq Q_E$, suppose that A is isometrically embedded, and fix a coordinate $e \in E$. Put*

$$D = A \setminus C, \quad \bar{A} = \pi_e A, \quad \bar{C} = \pi_e C.$$

For a cube family K and a vertex x , write $N_K(x) = \{f : x^{\{f\}} \in K\}$. Let $d \in D$, write $y = d|_{E \setminus \{e\}}$, and suppose that $y \in \bar{A} \setminus \bar{C}$ and that d is the unique member of D in its e -fibre. If y satisfies the $\bar{C}$ -teaching condition but d does not satisfy the C -teaching condition, then there is $c \in C$ such that, for every

$$f \in N_{\bar{C}}(y) \cap \text{Diff}(y, c|_{E \setminus \{e\}}),$$

one has

$$d^{\{f\}} \in D, \quad d^{\{e, f\}} \in C. \quad (2.122)$$

In particular, every witness supplied by the contracted teaching root but lost on lifting is the direction of an edge induced by D ; the opposite vertex of the corresponding e, f -square lies in C .

Proof. The hypotheses on the e -fibre imply that $d^{\{e\}} \notin A$. Indeed, the other lift cannot belong to C , since $y \notin \bar{C}$, and it cannot belong to D by uniqueness.

Because d fails the C -teaching condition, choose $c \in C$ such that

$$N_C(d) \cap \text{Diff}(d, c) = \emptyset.$$

The contracted root y teaches $\bar{C}$, so the set displayed in the statement is nonempty. Fix one of its members f . Since $y^{\{f\}} \in \bar{C}$, one of the two lifts $d^{\{f\}}$ and $d^{\{e, f\}}$ belongs to C . Moreover, $f \in \text{Diff}(d, c)$. The choice of c therefore excludes $d^{\{f\}} \in C$, and hence $d^{\{e, f\}} \in C$.

The two vertices d and $d^{\{e, f\}}$ are at Hamming distance two. Isometry of A supplies a two-edge path between them in A . Its intermediate vertex is either $d^{\{e\}}$ or $d^{\{f\}}$. The first is not in A , as observed above, so $d^{\{f\}} \in A$. It is not in C , and consequently it belongs to D . This proves (2.122) for every such f . $\square$

Remark 2.192 (The first two cyclic block shapes). The sole nine-concept relative graph not excluded by the current linear-quotient arguments is, up to signed coordinate permutation,

$$D_9 = \{0, 1, 2, 3, 32, 48, 52, 56, 60\} \subseteq Q_6.$$

It consists of two squares joined by a path of two edges. Thus $e(D_9) = 10$, and its six nonattachment square vertices are precisely its cube-family corners. If D_9 occurs as a terminal block, then

$$\tau(D_9) = 10 - 9 + 6 = 7, \quad \sum_{S \in \text{Sh}(A) \setminus \text{Sh}(B)} |S| \geq 16.$$

For comparison, the twelve-concept terminal block in Proposition 2.123 consists of three squares coupled cyclically. It has fifteen induced edges and six corners, so its curvature tax is 9; its twelve acquired supports have total rank 36, well above the lower bound $12 + 9 = 21$.

The gap in the last comparison is informative. Cycle rank and the number of boundary attachments are necessary data, but they still forget which square owns each boundary direction. Excluding D_9 , or realizing it, therefore requires a compatibility theorem for these side-coloured boundary owners rather than a further uncoloured moment inequality. Proposition 2.191 makes this requirement local. At the level of the relative graph, contracting either edge of the two-edge bridge identifies its endpoints and gives the eight-vertex barbell. The contracted relative pair may have fewer facets when another projected vertex already belongs to the contracted lower class, but it is nonempty and has at most eight facets. The relative linear-quotient theorem therefore supplies a contracted teaching root. Away from the contracted bridge fibre, such a root can fail to lift only by transferring every available witness along an internal edge of D_9 and completing the corresponding square into the lower class. Thus the remaining obstruction is a signed transfer along the block tree, not an unrestricted choice in the ambient cube.

3 Exact forbidden pc-minors

This section proves the exact basis criterion. The algebraic dictionary translates corner peelings into linear quotients and pc-minors into color-respecting specializations. The Yang-complex dictionary supplies the equivalent topological form. Pc-minor closure then reduces all proper pc-minors to the three elementary maps in each coordinate.

3.1 Alexander-dual formulation

The same problem has a concise algebraic form. This formulation is useful because it preserves both structures that matter here: a corner peeling is an ordering property of monomial generators, and a pc-minor is a color-respecting specialization rather than an arbitrary ideal minor.

Fix a field k . For an irredundant cube family $X \subseteq \{0, 1\}^E$, put

$$S_E = k[v_{e,0}, v_{e,1} : e \in E].$$

The *Yang complex* of X is the pure simplicial complex

$$\Delta_X = \langle F_x := \{v_{e,x_e} : e \in E\} : x \in X \rangle.$$

Its Alexander-dual ideal is the transversal squarefree ideal

$$J_X := (m_x : x \in X), \quad m_x := \prod_{e \in E} v_{e,1-x_e}. \quad (3.1)$$

Thus every minimal generator chooses exactly one variable from each color pair $\{v_{e,0}, v_{e,1}\}$.

Recall that an equigenerated monomial ideal $J = (u_1, \dots, u_t)$ has *linear quotients* in the displayed order if

$$(u_1, \dots, u_{j-1}) : u_j$$

is generated by variables for every $j > 1$. It has linear quotients if some ordering of its minimal generators has this property.

Proposition 3.1 (Algebraic pc-minor dictionary). *Let $X \subseteq \{0, 1\}^E$ be isometric in its given cube embedding. Then the following statements hold.*

1. $X \in \mathcal{D}_{\text{amp}}$ if and only if J_X has linear quotients.
2. Let $F, D \subseteq E$ be disjoint, let $f \in \{0, 1\}^F$ be realized by X , and set

$$K = \pi_D(X_f), \quad R = E \setminus (F \cup D).$$

Define the k -algebra specialization $\phi_{f,D} : S_E \rightarrow S_R$ by

$$\begin{array}{c|cc} & v_{e,f(e)} & v_{e,1-f(e)} \\ \hline e \in F & 0 & 1 \\ e \in D & 1 & 1 \end{array} \quad \text{and} \quad \phi_{f,D}(v_{e,b}) = v_{e,b} \quad (e \in R). \quad (3.2)$$

Then

$$J_K = \phi_{f,D}(J_X). \quad (3.3)$$

3. Consequently, $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ if and only if J_X has no linear-quotient order while the image under every proper specialization of the form (3.2) has linear quotients.

Proof. For an order $x_1, \dots, x_t$ of X , the monomial-colon formula gives

$$(m_{x_1}, \dots, m_{x_{j-1}}) : m_{x_j} = \left(\frac{m_{x_i}}{\gcd(m_{x_i}, m_{x_j})} : i < j \right). \quad (3.4)$$

The quotient belonging to x_i is the product of the literals indexed by the coordinates on which x_i and x_j differ. The right-hand side of (3.4) is generated by variables exactly when, for every $i < j$, there is $q < j$ such that x_q is a cube neighbor of x_j lying in the Boolean interval between x_i and x_j . Induction on Hamming distance shows that this is equivalent to every prefix $\{x_1, \dots, x_j\}$ being isometric. Proposition 4.2 of Chalopin–Chepoi–Moran–Warmuth makes this a prefix-building corner order; reversing it gives a corner peeling in our deletion convention [CCMW22, Proposition 4.2]. This proves item 1. Equivalently, it is the usual shellability–linear-quotients correspondence applied to Δ_X .

For item 2, consider a generator m_x . If $x|_F \neq f$, one of its factors is sent to zero. If $x|_F = f$, all its F -factors are sent to one; the two factors of every contracted color are also sent to one. The nonzero image is therefore

$$\phi_{f,D}(m_x) = \prod_{e \in R} v_{e,1-x_e} = m_{x|_R}.$$

Repeated residual concepts give repeated monomials and hence do not change the generated ideal. This proves (3.3). The normal form for pc-minors as a conditioning followed by coordinate contractions now gives item 3 from item 1. $\square$

Proposition 3.2 (Ball form of ampleness). *Let $\emptyset \neq X \subsetneq Q_E$, and fix a field k . Then the following are equivalent:*

1. X is ample;
2. Δ_X is a constructible simplicial ball of dimension $|E| - 1$;
3. Δ_X is a k -homology ball of dimension $|E| - 1$.

When these conditions hold,

$$\partial\Delta_X = \Delta_X \cap \Delta_{Q_E \setminus X}. \quad (3.5)$$

Moreover, X is corner-peelable if and only if this ball is shellable. Thus the corner-peelable ample families are exactly the cube families whose Yang complexes are shellable simplicial balls.

Proof. The ample–Cohen–Macaulay Yang dictionary says that X is ample if and only if Δ_X is Cohen–Macaulay over k [Bou26]. Because $\emptyset \neq X \subsetneq Q_E$, the Yang complex is a nonempty proper full-dimensional subcomplex of the boundary of the cross-polytope. Theorem 4.3 of Fløystad–Gabrielsen–Mafi applies to precisely such a subcomplex and makes it a constructible triangulated ball [FGM26, Theorem 4.3]. This proves that item 1 implies item 2. Conversely, a constructible simplicial ball is Cohen–Macaulay over every field, so the same Yang dictionary gives item 1. A simplicial ball is a k -homology ball, and a k -homology ball is Cohen–Macaulay over k ; the Yang dictionary therefore also proves the equivalence with item 3.

It remains to identify the boundary. Write $\Sigma_E = \Delta_{Q_E}$, the boundary of the $|E|$ -dimensional cross-polytope. The complementary-ball theorem for polarizations identifies $\Delta_{Q_E \setminus X}$ as the complementary full-dimensional ball in Σ_E , and the two balls meet in their common boundary [FGM26, Corollaries 8.5–8.6]. This proves (3.5). Finally, the shellability–corner-peeling correspondence [Bou26, CCMW22] gives the last assertion. $\square$

The specialization criterion in Proposition 3.1 appears to quantify over the whole pc-minor order. In fact, pc-minor closure reduces it to three substitutions per color. We record the resulting one-color parametrization explicitly.

For $e \in E$ and $b \in \{0, 1\}$, let $r_{e,b} : S_E \rightarrow S_{E \setminus \{e\}}$ and $c_e : S_E \rightarrow S_{E \setminus \{e\}}$ be the homomorphisms that fix all other variables and satisfy

$$\begin{aligned} r_{e,b}(v_{e,b}) &= 0, & r_{e,b}(v_{e,1-b}) &= 1, \\ c_e(v_{e,0}) &= 1, & c_e(v_{e,1}) &= 1. \end{aligned} \quad (3.6)$$

Thus $r_{e,b}$ retains the concepts on the side $x_e = b$, whereas c_e contracts color e . Repeated or divisible monomials are, as usual, removed when the image ideal is written with its minimal generators. We use the vacuous convention that an ideal with at most one minimal generator, including the zero ideal, has linear quotients; the zero colon is generated by the empty set of variables.

Let $\mathfrak{B}_d^{\text{col}}$ be the set of squarefree monomial ideals $J \subseteq k[v_{e,0}, v_{e,1} : e \in [d]]$ satisfying the following four conditions on J and its one-color images.

1. The minimal generators are distinct monomials of the form $\prod_{e \in [d]} v_{e,1-x_e}$, and both values of x_e occur among them for every e .
2. J has a d -linear resolution.
3. J has no linear-quotient order.

4. Each of the $3d$ ideals

$$r_{e,0}(J), \quad r_{e,1}(J), \quad c_e(J) \quad (e \in [d]) \quad (3.7)$$

has linear quotients.

The first condition lets us recover an irredundant cube family $X(J)$ from the exponent choices of the generators. The hyperoctahedral group $C_2 \wr S_d$ acts by permuting colors and interchanging the two variables within a color.

Theorem 3.3 (Colored-ideal parametrization). *Write $\text{Iso}(\mathcal{C})$ for the set of partial-cube isomorphism classes in a class $\mathcal{C}$. Then*

$$\text{Iso}(\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})) = \{[Q_n^-] : n \geq 3\} \dot{\cup} \coprod_{d \geq 1} \mathfrak{B}_d^{\text{col}} / (C_2 \wr S_d). \quad (3.8)$$

On the second summand the bijection is $[J] \mapsto [X(J)]$. Equivalently, for an irredundantly embedded ample family $X \subseteq Q_d$,

$$X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff \begin{cases} X \notin \mathcal{D}_{\text{amp}}, \\ \rho_e^0 X, \rho_e^1 X, \pi_e X \in \mathcal{D}_{\text{amp}} \text{ for every } e \in [d]. \end{cases} \quad (3.9)$$

In particular, membership in the ample part of the forbidden basis is decided from J and its $3d$ one-color images in (3.7); no additional proper pc-minor has to be inspected.

Within this parametrization, the cornerless basis members are exactly those $J \in \mathfrak{B}_d^{\text{col}}$ for which deletion of every minimal generator destroys d -linearity. The remaining ideals parametrize the cornered basis members.

Proof. Let $J \in \mathfrak{B}_d^{\text{col}}$. The first condition defines an irredundant family $X = X(J) \subseteq Q_d$. By the Yang-complex correspondence, the second condition is equivalent to ampleness of X ; in particular, X is isometric. Proposition 3.1 and the third condition give $X \notin \mathcal{D}_{\text{amp}}$.

The three maps belonging to e are precisely the Alexander-dual maps of the elementary pc-minors

$$\rho_e^0 X, \quad \rho_e^1 X, \quad \pi_e X.$$

The fourth condition and the same dictionary therefore put every elementary pc-minor of X in $\mathcal{D}_{\text{amp}}$. Every proper pc-minor of X is a pc-minor of at least one elementary pc-minor. Since $\mathcal{D}_{\text{amp}}$ is pc-minor closed, every proper pc-minor of X belongs to $\mathcal{D}_{\text{amp}}$, and hence $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$.

Conversely, let $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ have irredundant dimension d . Its ideal J_X satisfies the first condition. Ampleness gives the d -linear resolution, and $X \notin \mathcal{D}_{\text{amp}}$ rules out linear quotients. Every elementary pc-minor is proper and belongs to $\mathcal{D}_{\text{amp}}$, so all the ideals in (3.7) have linear quotients. Thus $J_X \in \mathfrak{B}_d^{\text{col}}$. The uniqueness of an irredundant partial-cube embedding up to relabelling and reorientation of its Θ -classes gives the quotient by $C_2 \wr S_d$. Proposition 2.8 supplies the first family in (3.8) and shows that the union is disjoint.

Finally, the ample corner-deletion criterion identifies a corner x with preservation of ampleness after deleting x . Under Alexander duality this is preservation of d -linearity after deleting m_x , which proves the last assertion. $\square$

Corollary 3.4 (Yang-ball parametrization). *Let $X \subseteq Q_d$ be irredundant and ample, put $\Delta = \Delta_X$, and write $V_e = \{v_{e,0}, v_{e,1}\}$ for the two vertices of color e . Then X is a forbidden pc-minor for $\mathcal{D}_{\text{amp}}$ if and only if*

1. Δ is a nonshellable constructible simplicial $(d-1)$ -ball;

2. for every $e \in [d]$ and $b \in \{0, 1\}$, the link $\text{lk}_\Delta(v_{e,b})$ is shellable; and
3. for every $e \in [d]$, the color-pair deletion $\Delta[V(\Delta) \setminus V_e]$ is shellable.

Consequently the full forbidden basis consists of the antipodally twice-punctured cubes Q_n^{--} , $n \geq 3$, and these color-minimal nonshellable Yang balls, modulo permutations and reorientations of the colors. No iterated link or additional pc-minor test is required.

Proof. The elementary Yang-complex identities are

$$\Delta_{\rho_e^b X} = \text{lk}_{\Delta_X}(v_{e,b}), \quad \Delta_{\pi_e X} = \Delta_X[V(\Delta_X) \setminus V_e]. \quad (3.10)$$

Indeed, taking the link selects precisely the facets whose concepts have e -value b and then removes that literal. Deleting both vertices of color e projects every facet to the remaining colors; repeated projected facets are immaterial, exactly as in contraction.

By Proposition 3.2, ampleness of X makes Δ a constructible ball, and $X \notin \mathcal{D}_{\text{amp}}$ is equivalent to nonshellability of that ball. The same proposition and (3.10) identify items 2 and 3 with corner peelability of all $3d$ elementary pc-minors. Theorem 3.3 now proves the equivalence. Proposition 2.8 supplies the nonample part of the final statement. $\square$

4 Core normal forms

Definition 4.1 (Critical cores and linear-extension words). Let $\mathfrak{C}_d^{\text{col}}$ be the set of squarefree monomial ideals $K \subseteq k[v_{e,0}, v_{e,1} : e \in [d]]$ with the following properties:

1. every minimal generator chooses exactly one variable from each color pair;
2. K has a d -linear resolution but has no linear-quotient order; and
3. $K \setminus u$ does not have a d -linear resolution for any minimal generator u of K .

Unlike the ideals in $\mathfrak{B}_d^{\text{col}}$, a member of $\mathfrak{C}_d^{\text{col}}$ need not use both variables of every color.

For any full-transversal ideal K , a *linear-extension word* over K is a sequence of distinct full-transversal monomials $\mathbf{u} = (u_1, \dots, u_t)$, none a generator of K , such that, on putting

$$K_0 = K, \quad K_i = K_{i-1} + (u_i) \quad (1 \leq i \leq t), \quad (4.1)$$

the colon $K_{i-1} : u_i$ is generated by variables for every i . The empty word is allowed.

Theorem 4.2 (Critical-core extension normal form). *The ideal family in Theorem 3.3 is exactly*

$$\mathfrak{B}_d^{\text{col}} = \left\{ K_t \left| \begin{array}{l} K \in \mathfrak{C}_d^{\text{col}}, (u_1, \dots, u_t) \text{ is a linear-extension word over } K, \\ K_t \text{ is irredundant and has no linear quotients,} \\ r_{e,0}(K_t), r_{e,1}(K_t), c_e(K_t) \text{ have linear quotients for every } e \in [d] \end{array} \right. \right\}. \quad (4.2)$$

Every K_i in (4.1) has a d -linear resolution. Moreover, the word has length zero exactly for the cornerless members of $\mathfrak{B}_d^{\text{col}}$; the cornered members have $t \geq 1$.

In simplicial language, every ample forbidden pc-minor is obtained by adjoining a sequence of shelling facets to a minimal Cohen–Macaulay, nonshellable Yang subcomplex. Pc-minimality is imposed only at the final complex by the $3d$ one-color tests.

Proof. First let $K \in \mathfrak{C}_d^{\text{col}}$ and let $(u_1, \dots, u_t)$ be a linear-extension word. Suppose inductively that K_{i-1} has a d -linear resolution. The exact sequence

$$0 \longrightarrow S/(K_{i-1} : u_i)(-d) \xrightarrow{\cdot u_i} S/K_{i-1} \longrightarrow S/K_i \longrightarrow 0 \quad (4.3)$$

and the fact that $K_{i-1} : u_i$ is generated by variables show that $\text{reg}(S/K_i) \leq d - 1$. Since K_i is generated in degree d , it has a d -linear resolution. Thus every K_i is linear. If the endpoint has the remaining properties displayed in (4.2), then it satisfies the four defining conditions of $\mathfrak{B}_d^{\text{col}}$.

Conversely, take $J \in \mathfrak{B}_d^{\text{col}}$. Starting from J , delete a minimal generator whenever the deletion still has a d -linear resolution. The process terminates at an ideal K for which no further deletion preserves d -linearity. Reverse the deleted generators and call them $u_1, \dots, u_t$. Every ideal K_i in the resulting chain has a d -linear resolution. Theorem 2.3 of Lu says that, after deleting any one minimal generator from a monomial ideal with a linear resolution, the colon of the deleted-generator ideal by that generator is generated by variables [Lu23, Theorem 2.3]. Applied to K_i and its generator u_i , it gives that each $K_{i-1} : u_i$ is generated by variables. Hence the reversed sequence is a linear-extension word.

It remains to check that K has no linear quotients. Otherwise take a linear-quotient order of K and append $u_1, \dots, u_t$. The defining colon condition makes the resulting order a linear-quotient order of J , contrary to $J \in \mathfrak{B}_d^{\text{col}}$. Thus $K \in \mathfrak{C}_d^{\text{col}}$, proving the reverse inclusion in (4.2).

The endpoint J is critical linear exactly when the deletion process stops immediately. By the ample corner-deletion criterion, this is equivalent to the corresponding concept class being cornerless. This proves the assertion about the word length. The final simplicial statement is the shelling–linear-extension translation; it is also the Yang-complex specialization of the theorem that every Cohen–Macaulay complex is shelled over a minimal Cohen–Macaulay subcomplex [DDL20, Theorem 2.1]. $\square$

5 Signed minimal nonfaces and one-concept extensions

Using the local extension test from Section 2.4, this section gives its algebraic and topological forms. It then encodes every ample family with a fixed shattered-set complex by the traces omitted on its minimal nonfaces and expresses corner peelings in the resulting trace table.

5.1 Algebraic and topological forms of one-concept additions

Proposition 5.1 (Algebraic and topological form of one-facet attachment). *Retain the hypotheses of Lemma 2.10, and suppose that $A = S \cup \{a\}$ is ample. Put $I = I_S(a)$, let*

$$F_a = \{v_{e,a_e} : e \in F\}$$

be the new Yang facet, and write 2^W for the simplex on a vertex set W . Then

$$J_S : m_a = (v_{e,a_e} : e \in I), \quad (5.1)$$

$$2^{F_a} \cap \Delta_S = \partial 2^{\{v_{e,a_e} : e \in I\}} * 2^{\{v_{e,a_e} : e \in F \setminus I\}}. \quad (5.2)$$

Equivalently, the intersection is the union of the ridges

$$F_a \setminus \{v_{e,a_e}\}, \quad e \in I.$$

If $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$, then $\emptyset \neq I \subsetneq F$, so the complex in (5.2) is a shellable $(|F| - 2)$ -ball.

Proof. The family A is isometric. For every $x \in S$, take an a - x geodesic in A . Its first edge has some direction $e \in I$. A hypercube geodesic uses each direction at most once, so $x_e \neq a_e$. Hence the monomial quotient

$$\frac{m_x}{\gcd(m_x, m_a)}$$

is divisible by v_{e,a_e} . Conversely, the neighbor $a^{\{e\}}$ belongs to S for every $e \in I$, and its quotient is precisely v_{e,a_e} . The monomial-colon formula proves (5.1).

A face $G \subseteq F_a$ belongs to Δ_S precisely when $G \subseteq F_x$ for some $x \in S$. By the preceding geodesic argument, x differs from a in some $e \in I$, and therefore $G \subseteq F_a \setminus \{v_{e,a_e}\}$. Conversely, this ridge is contained in the facet $F_{a^{\{e\}}}$ of Δ_S . This proves the ridge-union description, which is exactly the join in (5.2).

If S is an ample obstruction, it is nonempty, and connectedness of the isometric extension A gives $I \neq \emptyset$. If $I = F$, condition (2.1) gives $S = Q_F \setminus \{a\}$, which is corner-peelable. Thus $I \subsetneq F$. The displayed join is the join of a sphere with a nonempty simplex, hence a shellable ball of dimension $|F| - 2$. $\square$

5.2 Signed minimal nonfaces and first-loss orders

Definition 5.2 (Signed minimal-nonface presentation). Here a simplicial complex is a nonvoid downward-closed family, so it contains the empty face. Let $\Sigma \subseteq 2^F$ be such a complex, and let

$$\mathcal{N}(\Sigma) = \{I \subseteq F : I \notin \Sigma \text{ and every proper subset of } I \text{ belongs to } \Sigma\}$$

be its clutter of minimal nonfaces. A *signing* of $\mathcal{N}(\Sigma)$ is a choice $\sigma = (\sigma_I)_{I \in \mathcal{N}(\Sigma)}$, where $\sigma_I \in Q_I$. Its satisfying family is

$$H(\Sigma, \sigma) = \{x \in Q_F : x|_I \neq \sigma_I \text{ for every } I \in \mathcal{N}(\Sigma)\}. \quad (5.3)$$

We call the signing *admissible* when

$$|H(\Sigma, \sigma)| = |\Sigma|. \quad (5.4)$$

Thus (I, σ_I) forbids one Boolean trace on each minimal nonface. Admissibility is a single cardinality condition on this signed clutter; it contains no recursive minor or peeling test.

Theorem 5.3 (Parametrization by signed minimal nonfaces). *For every simplicial complex $\Sigma \subseteq 2^F$, write $H \upharpoonright I = \{x|_I : x \in H\}$. The map*

$$H \mapsto \sigma_H = (\sigma_I(H))_{I \in \mathcal{N}(\Sigma)}, \quad Q_I \setminus (H \upharpoonright I) = \{\sigma_I(H)\}, \quad (5.5)$$

is a bijection from the ample families $H \subseteq Q_F$ with $\text{Sh}(H) = \Sigma$ to the admissible signings of $\mathcal{N}(\Sigma)$. Its inverse is $\sigma \mapsto H(\Sigma, \sigma)$. In particular, the signed minimal nonfaces determine the entire ample family, not merely its shattered-set complex.

Proof. Let H be ample with shattered-set complex Σ , and take $I \in \mathcal{N}(\Sigma)$. Coordinate projection preserves ampleness [Law83, Theorem 2], and

$$\text{Sh}(H \upharpoonright I) = 2^I \setminus \{I\}.$$

The Sandwich equality for the ample family $H \upharpoonright I$ therefore gives $|H \upharpoonright I| = 2^{|I|} - 1$. Hence this projection omits a unique trace $\sigma_I(H)$.

Plainly $H \subseteq H(\Sigma, \sigma_H)$. The latter family does not shatter any member of $\mathcal{N}(\Sigma)$, and hence does not shatter any nonface of Σ . The upper Sandwich inequality gives

$$|H(\Sigma, \sigma_H)| \leq |\text{Sh}(H(\Sigma, \sigma_H))| \leq |\Sigma| = |H|.$$

Thus equality holds throughout and $H = H(\Sigma, \sigma_H)$; in particular the signing is admissible and is uniquely determined.

Conversely, let σ be admissible and put $H = H(\Sigma, \sigma)$. Every nonface of Σ contains a member of $\mathcal{N}(\Sigma)$, whose prescribed trace is absent from H . Consequently $\text{Sh}(H) \subseteq \Sigma$. Now

$$|\Sigma| = |H| \leq |\text{Sh}(H)| \leq |\Sigma|.$$

Equality in the Sandwich bound characterizes ample families, so H is ample and $\text{Sh}(H) = \Sigma$. Its omitted traces are the prescribed σ_I , proving that the two maps are inverse. $\square$

Coordinate expansions and their signed clauses. The cardinality test in Theorem 5.3 parametrizes all ample realizations at once. Fixing a contraction and a reduction gives a complementary constructive description: the remaining choices are independent labels on connected components. The following theorem identifies those labels with the new bits of the signed clauses. Its conclusion concerns ampleness. In the USO formulation, membership in $\mathcal{D}_{\text{amp}}$ asks whether the constructed support admits any acyclic USO; forbiddenness additionally asks for independent acyclic USOs on all proper pc-minors. A labeling alone supplies neither orientation nor those witnesses. Fixing a restriction and reduction representation map is a more constrained extension problem than choosing the support labels. The completion theory in the [representation-map writeup](#) treats that additional prescribed data. Its fixed-map criteria do not replace the existential conditions in the present support classification.

Theorem 5.4 (Component labels and expansion signs). *Let $\emptyset \neq C \subseteq A \subseteq Q_F$ be ample, let $e \notin F$, and let $K_1, \dots, K_k$ be the connected components of the graph induced by $A \setminus C$. For $b = (b_1, \dots, b_k) \in \{0, 1\}^k$, put*

$$X_b = (C \times \{0, 1\}) \cup \bigcup_{i=1}^k (K_i \times \{b_i\}).$$

Then X_b is ample, and

$$\text{Sh}(X_b) = \text{Sh}(A) \cup \{T \cup \{e\} : T \in \text{Sh}(C)\}. \quad (5.6)$$

Conversely, these are exactly the ample families on $F \cup \{e\}$ whose contraction in e is A and whose reduction in e is C . The labels are recovered uniquely.

Write $\mathcal{I} = \mathcal{N}(\text{Sh}(C)) \cap \text{Sh}(A)$. For $T \in \mathcal{I}$, let $\tau_T \in Q_T$ be the unique trace missing from $C|_T$. The nonempty fiber

$$A(T, \tau_T) = \{x \in A : x|_T = \tau_T\}$$

lies in a unique component $K_{i(T)}$. The signed minimal nonfaces of X_b are exactly

- 1. the signed minimal nonfaces of A , on supports not containing e ;*
- 2. for each $T \in \mathcal{I}$, the support $T \cup \{e\}$ with missing trace $(\tau_T, 1 - b_{i(T)})$.*

Every component occurs as some $i(T)$. Thus, with these supports, old clauses, and traces τ_T fixed, a choice of new-coordinate bits is admissible for the shadow in (5.6) if and only if its bits agree whenever $i(T) = i(U)$. There are exactly 2^k such choices. If $A = C$, the component list and $\mathcal{I}$ are empty and the construction is $A \times \{0, 1\}$.

Proof. We first prove ampleness directly by comparing shattered and strongly shattered supports. We use that complements and face sections of ample families are ample, and that nonempty ample families are connected [Law83, Theorem 2 and Lemma 1].

Let Q be any cube contained in A . The family $C \cap Q$ is ample inside Q , so its complement $Q \setminus C$, when nonempty, is connected. Consequently it belongs to one K_i . The entire cube Q therefore occurs in at least one layer of X_b . A support $T \subseteq F$ is shattered by X_b exactly when it is shattered by its contraction A ; since A is ample, the preceding cube argument shows that it is also strongly shattered by X_b .

A support $T \cup \{e\}$ is strongly shattered by X_b exactly when T is strongly shattered by C . If T is shattered by C , both layers already realize every T -trace. Conversely, suppose T is not shattered by C . If it is not shattered by A , it cannot give a shattered support of X_b . Otherwise choose a trace q missing from $C|_T$. The nonempty face section $A(T, q)$ is connected and avoids C , so it lies in one component K_i . Only layer b_i realizes q ; hence $T \cup \{e\}$ is not shattered. This proves (5.6) and equality with the strongly shattered supports, proving ampleness.

For the converse, let X have the stated contraction and reduction. Every member of C has both lifts, and every member of $A \setminus C$ has a unique lift. Two adjacent vertices of $A \setminus C$ cannot have different lift values: their two lifts would form an antipodal pair in a square face, with the other two vertices absent. Such a face section is not ample. Thus lift values are constant on each K_i , which gives $X = X_b$ and uniqueness.

The minimal nonfaces of the shadow in (5.6) which avoid e are exactly those of $\text{Sh}(A)$. A minimal nonface containing e has the form $T \cup \{e\}$, where T is shattered by A , is not shattered by C , and every proper subset of T is shattered by C . These are precisely the members of $\mathcal{I}$. The unique missing trace τ_T exists by Theorem 5.3. Its fiber in A is a nonempty connected face section outside C . Only its component's layer realizes that trace, giving the stated sign.

To see that every component occurs, take $x \in A \setminus C$. The signed-clause presentation of C supplies a minimal nonface T of $\text{Sh}(C)$ with $x|_T = \tau_T$. All proper subsets of T are shattered by both C and A . If T were not shattered by A , the two projections $C|_T \subseteq A|_T$ would both have $2^{|T|} - 1$ members. They would be equal, contradicting the presence of $x|_T$ in $A|_T$. Hence $T \in \mathcal{I}$ and $x \in A(T, \tau_T)$.

Finally, bits constant on each component's clause group produce X_b and are admissible by the construction. Conversely, any admissible signing with the displayed fixed data defines an ample family Y with that shadow. The old clauses imply that its contraction A' is contained in A . Every point in its reduction avoids each (T, τ_T) for $T \in \mathcal{I}$, because otherwise one of its two lifts is forbidden. Every minimal nonface of $\text{Sh}(C)$ outside $\mathcal{I}$ is a minimal nonface of $\text{Sh}(A)$; the two projections have the same unique missing trace, by the preceding cardinality argument. Thus the old clauses and these additional conditions imply all clauses defining C , so $C' \subseteq C$. Since

$$|Y| = |A'| + |C'| = |A| + |C|,$$

both inclusions are equalities. The converse already proved identifies $Y = X_b$ and forces equality of the bits within each clause group. $\square$

The theorem gives an intrinsic recursion for ample families: choose nested ample families on fewer coordinates and label their complementary components. Fixed coordinates may be added separately. Every active coordinate of a nonempty ample family has a nonempty reduction, by connectedness, so the converse supplies exhaustive recovery. This construction of the ambient ample class does not identify the forbidden labelings.

Corollary 5.5 (Constructing realizations in a fixed coordinate order). *Fix a simplicial complex $\Sigma \subseteq 2^F$ and an ordering $e_1, \dots, e_n$ of F . Start with the down-set*

$$D_\Sigma = \{x \in Q_F : \{f : x_f = 1\} \in \Sigma\}.$$

At stage j , form the contraction A and reduction C in e_j of the current family. Its unique lifts outside C are all in layer zero. Independently choose a label for every component of $A \setminus C$ and apply the construction of Theorem 5.4. If C is empty, e_j is constant and the choice is simply to retain or reflect that constant coordinate. After these n stages, the possible outputs are exactly all ample families with shadow Σ .

For the prescribed coordinate order, every output determines its entire sequence of intermediate families and component choices uniquely. In particular, the graph joining two realizations when one component is reflected in one coordinate is connected; reflection of a constant coordinate is included as a move.

Proof. Define $D_e(H)$ by moving every unique lift in coordinate e to layer zero and retaining every double lift. The theorem shows that $D_e(H)$ is ample with the same shadow and that its signed-clause presentation is obtained by setting the e -bit to one in every clause containing e . Other bits are unchanged. For a constant coordinate the same assertion holds for its singleton clause, and the operation is a coordinate reflection or the identity.

Consequently the D_e commute. Applying every one of them makes every clause the all-one trace, which gives exactly D_Σ . This is the signed-clause form of simultaneous shift normalization. A family for which all clause bits in coordinate e are one is downward closed in e : changing that coordinate from one to zero cannot create a forbidden trace. Thus all unique lifts at a stage involving an as-yet unprocessed coordinate lie in layer zero, as asserted.

Reversing the normalization steps, in the prescribed order, reconstructs any target realization by the component choices in the theorem. Each individual component may be reflected separately; all intervening families are ample with the same shadow. Conversely, every choice allowed by the recipe preserves those two properties. If H is the final output, the intermediate family after stage j is

$$D_{e_{j+1}} \cdots D_{e_n}(H).$$

Its signed clauses retain exactly the final bits in processed coordinates and have all-one bits in the remaining coordinates. These intermediate families are therefore uniquely determined, as are their recovered component labels. Reversing a path to D_Σ proves connectivity. $\square$

Proposition 5.6 (Alternative repairs in component labelings). *Let $C \subseteq S \subseteq Q_F$ be nonempty ample families, with C corner-peelable and S not corner-peelable. Choose distinct points $a_1, \dots, a_r$ outside S , where $r \geq 1$, and put $R = \{a_1, \dots, a_r\}$. Assume:*

1. $A = S \cup R$ is ample;
2. each $\{a_i\}$ is a connected component of $A \setminus C$;
3. every $S \cup \{a_i\}$ is corner-peelable.

In the expansion of Theorem 5.4, give all components contained in $S \setminus C$ label zero and give $\{a_i\}$ label ε_i . Then

$$X_\varepsilon \in \mathcal{D}_{\text{amp}} \iff \varepsilon_i = 0 \text{ for at least one } i.$$

If all $\varepsilon_i = 1$, the nonpeelable family S is a proper section of X_ε , so this negative expansion is not a forbidden minor.

Proof. Every family $S \cup U$, with $U \subseteq R$, is a component interpolant between C and A . It is ample: assign its selected components one label and the others the opposite label in Theorem 5.4, and take the appropriate section. Thus deleting any available a_i preserves ampleness and is a corner deletion. If $U \neq \emptyset$, delete all but one of its members and finish with the given peeling of $S \cup \{a_i\}$. In particular, A and every such $S \cup U$ are corner-peelable.

Suppose some $\varepsilon_i = 0$, and let U consist of these tips. Delete the unique lifts of the tips with label one. At each step the remaining family is an ample expansion over a component interpolant $S \cup U'$ and the same C , by the theorem, so each deletion is a corner deletion. The remainder is the peripheral expansion with base $S \cup U$ and site C . Both peel, and Proposition 2.6 completes its peeling. If instead all labels are one, the zero section is exactly S ; minor closure of corner peelability proves the negative assertion and also excludes pc-minimality. $\square$

Remark 5.7 (Ample sign choices and peelable sign choices differ). The fixed 291-concept cornerless family S has two independent repairing tips 91 and 157, with incident supports 102 and 390 in binary encoding. Let C be the union of their punctured incident four-cubes. The fixed replay verifies that $|C| = 26$, that C peels, and that $A = S \cup \{91, 157\}$ is ample and peels. Its complement of C has components of orders 265, 1, 1, the last two being the tips.

There are twelve new-coordinate clauses: ten belong to the first component and one to each tip. All eight component labelings give ample expansions of order 319. Precisely six peel. With the first component labelled zero, the labelings $(0, 1, 0)$ and $(0, 0, 1)$ peel, whereas their coordinatewise maximum $(0, 1, 1)$ does not. Thus the equality constraints describing admissible expansion signs do not describe the peelable subset. The negative examples have the proper nonpeelable section S , as the proposition requires; they do not test a criterion restricted to families whose proper minors all peel.

The independent checker `verify_component_expansion_signs.py` also tests all 2244 nested nonempty ample pairs through Q_3 and all 9656 new-clause bit assignments. Exactly 5384 assignments satisfy the component equalities and the asserted shadow formula. It also normalizes these small families in every coordinate order and replays the inverse component reflections. The fixed larger example is checked by full positive orders and a cornerless proper-section certificate, without a peeling search for negative answers. Inputs, orders, signs and scope are in `component_expansion_signs_verification.json`. These finite checks support the examples and the implementation; the universal claims use the preceding proofs.

Lemma 5.8 (Corners as unique facet traces). *Let $H \subseteq Q_F$ be ample, let $\Sigma = \text{Sh}(H)$. Then $x \in H$ is a corner if and only if there is a facet T of Σ such that $x|_T$ is realized by no other member of H .*

Proof. Deleting x loses a shattered face $D \in \Sigma$ exactly when $x|_D$ is realized by no other member of H . The ample corner-deletion criterion says that x is a corner exactly when $H \setminus \{x\}$ is ample. By the Sandwich inequalities, this happens exactly when deletion loses one shattered face: if none is lost, then $H \setminus \{x\}$ still has $|H|$ shattered faces and hence does not satisfy the Sandwich equality for its $|H| - 1$ members, while losing two or more faces would violate the upper Sandwich inequality $|H \setminus \{x\}| \leq |\text{Sh}(H \setminus \{x\})|$.

It remains to identify the possible unique face. If $x|_D$ is unique and $D \subseteq T \in \Sigma$, then $x|_T$ is also unique. If D is not a facet, choose a facet T with $D \subsetneq T$; there are then at least two uniquely realized faces, namely D and T . Therefore exactly one face is lost if and only if the lost face is a facet, proving the claim. $\square$

Proposition 5.9 (A simplicial maximal cube exposes a corner). *Let $X \subseteq Q_E$ be a nonempty cube family, and let $\Gamma(X)$ be the graph whose vertices are the inclusion-maximal cubes of X , with two*

such cubes adjacent when they intersect. If a maximal cube C is a simplicial vertex of $\Gamma(X)$, then C contains a vertex that belongs to no other maximal cube of X . In particular, X has a corner. Consequently, if X is cornerless, then $\Gamma(X)$ has no simplicial vertex; in particular, it is nonchordal and contains an induced cycle of length at least four.

Proof. Recall that a graph vertex is simplicial when its neighbors form a clique. First suppose that C has no neighbor in $\Gamma(X)$. Every vertex of C then belongs to no other maximal cube and is a corner.

Suppose next that C has neighbors $C_1, \dots, C_k$. The cubes in $\{C, C_1, \dots, C_k\}$ intersect pairwise: each C_i meets C , and the C_i meet one another because C is simplicial. Subcubes of a hypercube have the Helly property. Indeed, each subcube is described by fixing some coordinates, and pairwise intersection says precisely that none of these fixed-coordinate prescriptions conflict. Hence there is a vertex

$$z \in C \cap C_1 \cap \dots \cap C_k.$$

Let y be the antipode of z in C , obtained by reversing every coordinate that is free in C . For every i , the intersection $C \cap C_i$ is a proper subcube of C : otherwise $C \subseteq C_i$, contrary to the distinctness and maximality of C and C_i . Since $C \cap C_i$ is a proper subcube containing z , it fixes at least one free coordinate of C to its value at z and therefore does not contain y . Thus no neighbor C_i contains y , while a nonneighbor is disjoint from C and cannot contain y either. Hence C is the unique maximal cube containing y , so y is a corner.

It follows that the maximal-cube intersection graph of a cornerless family has no simplicial vertex. Every finite chordal graph has a simplicial vertex, whereas a nonchordal graph contains an induced cycle of length at least four. This proves the final assertions. $\square$

Lemma 5.10 (Square-free edges are bridges). *Let X be an ample family. Every cycle in the one-inclusion graph of X , regarded as an edge chain over $\mathbb{F}_2$, is a sum of boundaries of squares contained in X . Consequently, every edge of X that belongs to no square is a bridge.*

Proof. We prove the first assertion by induction on the number of active coordinates. It is immediate when there is at most one such coordinate. Fix an active coordinate e , and write X_0, X_1 for the two e -sections, with the e -coordinate deleted. Both sections are ample. Their intersection

$$I = X_0 \cap X_1$$

is also ample. Indeed, the contraction $X_0 \cup X_1$ is ample, and the two-layer shattering formula gives

$$|X_0| + |X_1| = |\text{Sh}(X_0 \cup X_1)| + |\text{Sh}(X_0) \cap \text{Sh}(X_1)|.$$

The first term is $|X_0 \cup X_1|$, so the second is $|I|$. Since

$$\text{Sh}(I) \subseteq \text{Sh}(X_0) \cap \text{Sh}(X_1)$$

and Sauer's inequality gives $|\text{Sh}(I)| \geq |I|$, equality holds and I is ample. It is therefore connected whenever it is nonempty.

Let Z be a cycle of X . Its e -edges occur in an even number. Pair them arbitrarily. For a paired pair with projected endpoints $u, v \in I$, choose a path from u to v in I . The two lifts of this path, together with the two paired e -edges, form a ladder. Its boundary is the sum of the boundaries of the squares corresponding to the edges of the chosen path. Adding all these ladder boundaries to Z cancels every e -edge. The remaining even edge chain is the disjoint sum of an even edge chain in

the lower section and one in the upper section. Decompose each into cycles and apply induction inside X_0 and X_1 . This expresses Z as a sum of square boundaries.

Now suppose that an edge a belongs to no square but is not a bridge. Then a lies on a graph cycle Z . In the square-boundary expression for Z , the coefficient of a is zero because no summand contains a , whereas its coefficient in Z is one. This contradiction proves that a is a bridge. $\square$

Lemma 5.11 (Local trace covers). *Let $X \subseteq Q_E$ be a nonempty cube family, let C be a maximal cube of X , and let $C_1, \dots, C_k$ be the other maximal cubes that meet C . Then*

$$\text{Cor}(X) \cap C = C \setminus \bigcup_{i=1}^k (C \cap C_i). \quad (5.7)$$

Moreover, for distinct i, j , one has

$$C_i \cap C_j \neq \emptyset \iff (C \cap C_i) \cap (C \cap C_j) \neq \emptyset. \quad (5.8)$$

Consequently, if X is cornerless, the nonempty proper subcubes $C \cap C_i$ cover C , and the graph induced by the neighbors of C in $\Gamma(X)$ is their intersection graph. After deleting repeated and redundant traces, one obtains an irredundant cover of C by proper subcubes.

Proof. A vertex $x \in C$ is not a corner precisely when it lies in some maximal cube other than C . Any such cube meets C and is one of the C_i , which proves (5.7).

If the two traces on the right-hand side of (5.8) meet, then C_i and C_j plainly meet. Conversely, if C_i and C_j meet, then the three subcubes C, C_i, C_j intersect pairwise. The hypercube Helly property used in Proposition 5.9 gives $C \cap C_i \cap C_j \neq \emptyset$. This proves the equivalence. Each trace is proper because C and C_i are distinct maximal cubes. When X is cornerless, (5.7) says that the traces cover C . Finally choose an inclusion-minimal subfamily of the distinct traces that still covers C ; every member of the resulting cover has a private vertex, so the cover is irredundant. $\square$

Proposition 5.12 (The rank-three local trace catalogue). *Up to coordinate permutations and reversals, the numbers of irredundant covers of Q_r by nonempty proper subcubes are as follows:*

r	proper subcubes	labelled covers	orbits	cover-size range
1	2	1	1	2
2	8	11	4	2–4
3	26	869	43	2–8

(5.9)

Consequently, Lemma 5.11 reduces the local cornerlessness condition for every family of VC dimension at most three to forty-eight fixed cover types.

Computer-assisted proof. A proper subcube of Q_r is encoded by a nonempty set of fixed coordinates and their prescribed bits, giving $3^r - 1$ possibilities. The verifier branches on the first uncovered cube vertex, retains a cover only when every selected subcube has a private vertex, and then takes the lexicographically least image under all $2^r r!$ cube automorphisms. The three rows of (5.9) are the resulting exhaustive counts.

The generator and output are

```
explore_damp_local_maximal_cube_covers.py,
damp_local_maximal_cube_covers.json,
```

and the output digest is

c570301788f52c46a536d0905c627cc05dfcf530625afc3124f6971d7613e931.

$\square$

Proposition 5.13 (Triangle-carrier trees in VC dimension three). *Let $H \subseteq Q_E$ be ample of VC dimension at most three, put $\Sigma = \text{Sh}(H)$, and fix a two-set $S \in \Sigma$. Define*

$$L_S = \{e \in E \setminus S : S \cup \{e\} \in \Sigma\}$$

and the S -cube carrier family by

$$\mathcal{K}_S(H) = \{y \in Q_{E \setminus S} : Q_S \times \{y\} \subseteq H\}. \quad (5.10)$$

After its constant coordinates are deleted, $\mathcal{K}_S(H)$ is a maximum VC-dimension-one class on L_S . Its one-inclusion graph is a tree with $|L_S| + 1$ vertices and exactly one edge in each direction $e \in L_S$.

The edge in direction e corresponds to the unique three-cube $C_{S \cup \{e\}}$ of H with support $S \cup \{e\}$, and its endpoints correspond to the two opposite S -facets of that cube. An endpoint is incident with another edge of the carrier tree if and only if the corresponding S -facet lies in another maximal three-cube of H . Consequently, both opposite S -facets of $C_{S \cup \{e\}}$ are covered by other maximal cubes if and only if the e -edge has no leaf endpoint in the carrier tree. In particular, if H is maximum of VC dimension three, then $L_S = E \setminus S$, so the carrier tree has $|E| - 1$ vertices and $|E| - 2$ edges.

Proof. For $A \subseteq E \setminus S$, the carrier family strongly shatters A if and only if H strongly shatters $S \cup A$. Likewise, if $\mathcal{K}_S(H)$ shatters A , then H shatters $S \cup A$: for every A -trace supplied by the carrier, the accompanying full S -cube supplies all traces on $S \cup A$. Since H is ample, these observations give

$$\text{SSH}(\mathcal{K}_S(H)) = \text{Sh}(\mathcal{K}_S(H)) = \{\emptyset\} \cup \{\{e\} : e \in L_S\}.$$

Thus the carrier is ample and has $1 + |L_S|$ members. A coordinate outside L_S is not shattered and is therefore constant on the carrier; delete all such coordinates.

Its one-inclusion graph is connected because every ample family is an isometric cube family. Choose a spanning tree. Every coordinate of L_S is shattered and hence occurs on an edge; moreover every path between the two coordinate semicubes must use an edge in that direction. The spanning tree therefore has at least one edge in each of the $|L_S|$ directions. It has exactly $|L_S|$ edges, so it uses every direction exactly once. There can be no further edge of the one-inclusion graph: if an additional e -edge joined u and v , the tree path from u to v would have to change only coordinate e . Because every tree-edge direction occurs at most once, that path would be the existing e -edge itself. Hence the one-inclusion graph equals the spanning tree and has exactly one edge of each direction.

By (5.10), an e -edge of the carrier is precisely a full cube of H with support $S \cup \{e\}$. Its endpoints are its two S -facets. Since the carrier has one e -edge, this three-cube is unique. A second carrier edge is incident with one of its endpoints precisely when the corresponding S -cube extends in a second direction, or equivalently lies in another maximal three-cube. The last assertion is now exactly the statement that neither endpoint of the e -edge is a leaf. $\square$

Proposition 5.14 (Reconstruction from anchored facet cubes). *Let $\mathcal{F} \subseteq 2^E$ be a nonempty antichain, let*

$$\Sigma = \{S \subseteq E : S \subseteq T \text{ for some } T \in \mathcal{F}\},$$

and, for each $T \in \mathcal{F}$, choose an affine cube $C_T \subseteq Q_E$ with support T . Put $H = \bigcup_{T \in \mathcal{F}} C_T$. Then

$$H \text{ is ample with } \text{Sh}(H) = \Sigma \iff |H| = |\Sigma|. \quad (5.11)$$

When these conditions hold, the cubes C_T , for $T \in \mathcal{F}$, are exactly the maximal cubes of H . Consequently,

$$H \text{ is cornerless} \iff |\{T \in \mathcal{F} : x \in C_T\}| \neq 1 \text{ for every } x \in Q_E. \quad (5.12)$$

Thus a fixed-rank minimum-order search is a static anchored-cube union problem: it requires one cardinality equality and forbids incidence multiplicity one.

Proof. Every face of Σ is strongly shattered by one of the selected cubes, so

$$|\Sigma| \leq |\text{SSh}(H)| \leq |H|.$$

If $|H| = |\Sigma|$, equality holds throughout. Equality in the lower Sandwich inequality makes H ample, and $\text{SSh}(H) = \Sigma$ then gives $\text{Sh}(H) = \Sigma$. The converse is the Sandwich equality.

It remains to identify the maximal cubes. For any $S \in \Sigma$, let $\mathcal{K}_S(H)$ be the carrier family defined as in (5.10). The argument in the proof of Proposition 5.13, without the rank-three specialization, gives

$$\text{Sh}(\mathcal{K}_S(H)) = \text{SSh}(\mathcal{K}_S(H)) = \text{link}_\Sigma(S).$$

If $S = T$ is a facet, this link consists only of the empty face, so the carrier has one member: there is a unique T -cube, namely C_T . If S is not a facet, its link contains a singleton. The carrier is then a nontrivial connected ample family, so every one of its vertices is incident with an edge along some carrier path. Equivalently, every S -cube of H extends to a larger cube and is not maximal. Hence the selected facet cubes are exactly the maximal cubes. A vertex is a corner precisely when it belongs to exactly one maximal cube, which proves (5.12). $\square$

Corollary 5.15 (Triangle cap at the current lower endpoint). *Let H be an irredundant ample family of VC dimension three and order 27, and put $d = |E|$. If f_2 is the number of triangular faces of $\text{Sh}(H)$, then*

$$f_2 \leq \begin{cases} 8, & d = 8, \\ 7, & d = 9, \\ 7, & d = 10. \end{cases} \quad (5.13)$$

In particular, each of the three candidate profiles in Theorem 7.159 is an anchored-cube union with at most eight maximal three-cubes.

Proof. Write f_1 for the number of two-element faces. Irredundancy and the Sandwich equality give

$$27 = 1 + d + f_1 + f_2.$$

The Kruskal–Katona theorem [Kru63, Kat68] says that the pair shadow of f_2 triples has at least as many pairs as the first f_2 triples in colexicographic order. For $f_2 = 8$ this minimum is 10, while for $f_2 = 9$ it is still 10; for $f_2 = 7$ it is 9, while for $f_2 = 8$ it is 10. Thus

d	$f_1 + f_2 = 26 - d$	first impossible f_2
8	18	9 (10 + 9 > 18),
9	17	8 (10 + 8 > 17),
10	16	8 (10 + 8 > 16).

This proves the displayed bounds. The final assertion follows from Proposition 5.14. $\square$

Lemma 5.16 (Four-coordinate compatibility). *Let Σ be a two-dimensional simplicial complex with complete one-skeleton such that every four-set contains at most two triangular faces, let $H \subseteq Q_F$ be ample with $\text{Sh}(H) = \Sigma$, and let $\sigma = (\sigma_I)_{I \in \mathcal{N}(\Sigma)}$ be its signing. Fix a four-set $U \subseteq F$. For each missing triple $I \subset U$, put*

$$E_U(I) = \{z \in Q_U : z|_I = \sigma_I\}.$$

Then the edges $E_U(I)$, indexed by the missing triples in U , have connected union. In particular:

1. *if U contains exactly two missing triples I, J , then $\sigma_I|_{I \cap J} = \sigma_J|_{I \cap J}$;*

2. if U contains exactly three missing triples and e belongs to all three, then their three omitted traces have the same value at e .

Proof. Let t be the number of triangular faces of Σ contained in U . There are $m = 4 - t$ missing triples in U . Projection preserves ampleness, and the shattered sets of $H|U$ are exactly $\Sigma[U] = \{D \in \Sigma : D \subseteq U\}$. Hence

$$|H|U| = |\Sigma[U]| = 1 + 4 + 6 + t = 11 + t.$$

The minimal nonfaces of $\Sigma[U]$ are precisely its m missing triples, with the same omitted traces as in H : projection cannot create a trace that was absent from $H|_I$, and both projections omit exactly one I -trace by ampleness. Applying Theorem 5.3 to the ample projection therefore identifies the complement of $H|U$ with $\bigcup_{I \in \mathcal{N}(\Sigma), I \subset U} E_U(I)$. Thus this union has

$$16 - (11 + t) = 5 - t = m + 1$$

vertices.

Every $E_U(I)$ is an edge in direction $U \setminus I$, so the m edges have distinct directions. A cycle in a hypercube uses every direction an even number of times; consequently these edges form a forest. A forest with m edges and $m + 1$ vertices is connected, proving the first claim.

Two cube edges in distinct directions meet exactly when their prescribed traces agree on the common coordinates. This gives item 1. For item 2, the three edges form a tree. Along each pair of incident edges the prescribed traces agree at their common coordinate e , so connectedness propagates one value of $\sigma_I(e)$ through all three edges. $\square$

Definition 5.17 (Coordinate-propagation graph). Under the hypotheses of Lemma 5.16, fix $e \in F$. The graph $P_e(\Sigma)$ has as vertices the minimal nonfaces $I \in \mathcal{N}(\Sigma)$ containing e . Join I and J if either

1. $|I \cup J| = 4$ and I, J are the only missing triples in $I \cup J$, or
2. $|I \cup J| = 4$ and $I \cup J$ contains exactly three missing triples, all containing e .

Proposition 5.18 (Rigidity from coordinate propagation). *Let Σ be a two-dimensional simplicial complex with complete one-skeleton such that every four-set contains at most two triangular faces. If $P_e(\Sigma)$ is connected whenever it is nonempty, then every ample family $H \subseteq Q_F$ with $\text{Sh}(H) = \Sigma$ is a cube translate of the canonical face family*

$$\{\mathbf{1}_D : D \in \Sigma\}.$$

Proof. Let σ be the signing of H . Each edge of $P_e(\Sigma)$ equates the two values of the corresponding omitted traces at e by Lemma 5.16. Connectivity therefore gives a bit h_e such that $\sigma_I(e) = h_e$ for every minimal nonface I containing e . If no minimal nonface contains e , choose h_e arbitrarily. With $h = (h_e)_{e \in F}$, we have $\sigma_I = h|_I$ for every $I \in \mathcal{N}(\Sigma)$. Let $g = h + \mathbf{1}_F$, with addition in Q_F . Translating Q_F by g changes every omitted trace to the all-one trace. The resulting satisfying family is

$$\{x \in Q_F : \text{supp}(x) \text{ contains no member of } \mathcal{N}(\Sigma)\} = \{\mathbf{1}_D : D \in \Sigma\},$$

which proves the claim. $\square$

Remark 5.19 (A three-coordinate example). Let $F = \{1, 2, 3\}$ and

$$\Sigma = \{D \subseteq F : \{1, 2\} \not\subseteq D\}.$$

Then $\mathcal{N}(\Sigma) = \{\{1, 2\}\}$. Choosing the omitted trace $\sigma_{\{1, 2\}} = (1, 1)$ gives

$$H(\Sigma, \sigma) = Q_F \setminus \{(1, 1, 0), (1, 1, 1)\}.$$

Both this family and Σ have six members, so the signing is admissible and the theorem makes H ample with shattered complex Σ . The same count works for any of the four choices of the omitted $\{1, 2\}$ -trace. Reorienting the single clause can therefore change four concepts, which is why clause reorientation alone is coarser than adjacency in the mutation graph below.

Definition 5.20 (Oriented one-hole cover). Let E be finite. An *oriented one-hole datum* is a pair (y, B) with $y \in Q_E$ and $B \subseteq E$, $|B| \geq 2$. Its punctured affine cube and distinguished antipode are

$$P(y, B) = \{y^U : \emptyset \neq U \subseteq B\}, \quad a(y, B) = y^B. \quad (5.14)$$

For a nonempty finite family $\mathcal{W}$ of such data, put

$$H(\mathcal{W}) = \bigcup_{(y, B) \in \mathcal{W}} P(y, B).$$

We call $\mathcal{W}$ an *admissible oriented one-hole cover* if

1. $y \notin H(\mathcal{W})$ for every $(y, B) \in \mathcal{W}$;
2. the distinguished antipodes cover the constructed family:

$$H(\mathcal{W}) = \{a(y, B) : (y, B) \in \mathcal{W}\}; \quad (5.15)$$

3. the Sandwich equality

$$|\text{Sh}(H(\mathcal{W}))| = |H(\mathcal{W})| \quad (5.16)$$

holds.

All three conditions are finite conditions on the displayed punctured cubes; none asks for a corner or a peeling order.

Theorem 5.21 (Intrinsic one-hole construction of cornerless ample families). *Every admissible oriented one-hole cover $\mathcal{W}$ constructs a nonempty cornerless ample family $H(\mathcal{W})$. Conversely, every nonempty cornerless ample family $H \subseteq Q_E$ is obtained in this way.*

More precisely, for $x \in H$ define its local cube-support complex by

$$\Lambda_H(x) = \{A \subseteq E : \{x^C : C \subseteq A\} \subseteq H\}. \quad (5.17)$$

Then the canonical recovered cover is

$$\mathcal{W}(H) = \{(x^B, B) : x \in H, B \in \mathcal{N}(\Lambda_H(x)), |B| \geq 2\}. \quad (5.18)$$

Thus oriented one-hole covers give an explicit, order-free, and canonically recoverable parametrization of all cornerless ample families. They do not by themselves characterize which such families are pc-minor-minimal outside $\mathcal{D}_{\text{amp}}$.

Proof. Let $\mathcal{W}$ be admissible and put $H = H(\mathcal{W})$. Condition (5.16) and the Sandwich inequalities make H ample. Fix $x \in H$. By (5.15), there is $(y, B) \in \mathcal{W}$ with $x = y^B$. Every proper subset $C \subsetneq B$ satisfies

$$x^C = y^{B \setminus C} \in P(y, B) \subseteq H,$$

whereas $x^B = y \notin H$. Hence B is a nontrivial minimal nonface of $\Lambda_H(x)$. Equivalently, the cubes of H through x cannot all lie in one maximal cube: such a cube would contain every direction of B and therefore also $x^B = y$. Thus every concept lies in at least two maximal cubes, so H is cornerless.

Conversely, let H be nonempty, ample, and cornerless. The complex $\Lambda_H(x)$ is not a simplex for any $x \in H$, because its maximal faces are precisely the supports of the maximal cubes through x . It therefore has a minimal nonface B of order at least two. For every proper $C \subsetneq B$, one has $x^C \in H$, while $x^B \notin H$. Putting $y = x^B$, we obtain

$$P(y, B) = \{x^{B \setminus U} : \emptyset \neq U \subseteq B\} \subseteq H, \quad a(y, B) = x.$$

Taking all such minimal nonfaces for all x gives $\mathcal{W}(H)$. Every punctured cube in this family lies in H , and every $x \in H$ occurs as its distinguished antipode. Hence $H(\mathcal{W}(H)) = H$; the missing-tip and antipode-cover conditions hold by construction, and the Sandwich equality follows from ampleness. This proves admissibility and the recovery formula. $\square$

The one-hole construction does not become a forbidden-minor construction by selecting its pc-minimal cornerless outputs. A cornerless ample class can have a corner in every nonempty proper pc-minor while still having a proper minor that is not corner-peelable. The following gluing statement isolates the distinction.

Proposition 5.22 (Cornerlessness can be minimal without forbiddenness). *Let $S \subseteq Q_F$ be an ample forbidden pc-minor whose only corner is 0, and let $P \subseteq Q_E$ be a nontrivial corner-peelable ample family whose only corner is 0. Assume that every proper root-preserving pc-minor of each family peels to the image of 0. The coordinate sets are disjoint and the embeddings irredundant. Set*

$$X = S \vee P = (S \times \{0\}^E) \cup (\{0\}^F \times P).$$

Then X is ample and cornerless, every nonempty proper pc-minor of X has a corner, and every non-corner-peelable pc-minor of X contains S as a pc-minor. In particular, S is the unique forbidden pc-minor of X , up to isomorphism, and X has no cornerless forbidden pc-minor.

Proof. The vertex-gluing formula in Proposition 2.18 proves ampleness. A nonzero vertex has exactly the same incident cubes as in its own piece, so it is not a corner. Both nontrivial pieces are connected and have a neighbor of 0. Two such neighbors, one from each piece, have no completing mixed square. Thus 0 is not a corner either. Contracting all coordinates of P recovers S , so X is not forbidden.

We describe all pc-minors, rather than only the elementary ones. A coordinate contraction acts within one of the two blocks and retains the common root. A convex restriction meeting both blocks away from that root must contain the root, since every path between the blocks passes through it. Its intersection with each block is a convex restriction of that block. A nonempty restriction omitting the root lies wholly in one block: it is then a proper pc-minor of S or a pc-minor of P . Both are corner-peelable, and every subsequent minor remains corner-peelable. Such a nonempty minor therefore has a corner.

All other minors retain the common root and have the form $S' \vee P'$, where S' and P' are root-preserving pc-minors of the respective pieces. Suppose first that S' is proper. It peels to 0 by

hypothesis. The family P' is corner-peelable, since P is and this property is pc-minor closed. Peel S' down to 0, then peel P' ordinarily. This gives a complete peeling of $S' \vee P'$. Therefore every nonpeelable minor must retain $S' = S$, and contains S by contracting the other block.

It remains to check corners when $S' = S$ and the minor is proper. Irredundancy forces P' to be proper. If $P' = \{0\}$, the minor is S , which has its corner 0. Otherwise P' peels to 0; the first vertex of such a peeling is a nonzero corner of P' , and remains a corner of the union. This proves the assertion about every nonempty proper minor. Finally, a forbidden minor containing S must be isomorphic to S , by its proper-minor minimality. Since S has a corner, no cornerless forbidden minor occurs. $\square$

Corollary 5.23 (A cornerless class whose forbidden minor has one corner). *There is an ample class $X \subseteq Q_{24}$ with 580 concepts, VC dimension three, and minimum graph degree four, such that X is cornerless and every nonempty proper pc-minor has a corner. Its unique forbidden pc-minor has 292 concepts and one corner. Thus pc-minimality among cornerless ample classes is strictly weaker than being a cornerless forbidden pc-minor, even among graph three-cores of VC dimension three.*

Proof. Let H be the 292-concept class of Theorem 5.75, and let G be the fixed 291-concept family used there. In its twelve-bit integer notation, let $\oplus$ denote coordinatewise addition modulo two and put

$$S = \{x \oplus 3959 : x \in H\}, \quad P = \{x \oplus 29 : x \in (G \cup \{91\}) \setminus \{61, 125, 93\}\}.$$

The cited theorem makes S ample and forbidden, with 0 as its only corner. Remark 2.50 shows that P is ample and ordinarily peelable, has 0 as its only corner, and has all proper root-preserving minors peelable to their roots. Both families have VC dimension three.

The additional condition on S is verified directly: all 24 elementary root-preserving minors have supplied corner peelings ending at 0. Root-preserving minor closure of peeling to the root gives the condition for every proper rooted minor. The checker `verify_damp_cornerless_minor_minimal_counterexample.py` reconstructs both families, recomputes their shadows and corners, and checks these 24 orders, the 24 rooted orders for P , its ordinary order, and the 36 ordinary proper-minor orders for S . In particular, the new rooted assertion does not rely on the ordinary orders ending at arbitrary vertices.

Use disjoint coordinate blocks and write the union as

$$X = S \cup \{2^{12}x : x \in P\}.$$

The preceding proposition applies. It gives $|X| = 292 + 289 - 1 = 580$ and all the assertions about its proper minors and forbidden minor. The union has dimension 24 and VC dimension three because its shattered supports lie in the two blocks. The checker verifies minimum degree four and cornerlessness directly. It also inspects all 72 elementary images: 48 have checked complete peelings and 24 project onto S , so do not peel, although each has a corner. These finite checks are recorded in `damp_cornerless_minor_minimal_counterexample_verification.json`. The assertion for all proper minors follows from the proposition's block decomposition, not from extrapolating the elementary corner checks. $\square$

This counterexample rules out selecting forbidden one-hole covers by testing only that their nonempty proper pc-minors have corners. It does not affect the minimum-cardinality argument: a globally smallest nonempty cornerless ample family is still forbidden. Every smaller ample family peels, since otherwise maximal corner deletion would leave a smaller nonempty cornerless ample family. Proper pc-minors are smaller. Minimality under pc-minors and global minimum cardinality are therefore different requirements.

Restrictions on small covers. The antipode convention can be compressed by allowing one punctured cube to protect several concepts. For $P_i = \mathbf{P}(y_i, L_i)$, write

$$I_i = \{y_i^T : T \subseteq L_i, |T| \geq 2\}, \quad r_i = |L_i|.$$

Consider a nonempty family of m distinct such pieces satisfying

$$H = \bigcup_{i=1}^m P_i = \bigcup_{i=1}^m I_i, \quad y_i \notin H \quad (1 \leq i \leq m). \quad (5.19)$$

If $|\text{Sh}(H)| = |H|$, then H is ample and cornerless: replace each (y_i, L_i) by all (y_i, T) with $T \subseteq L_i$, $|T| \geq 2$, and apply Theorem 5.21. The following restrictions use only (5.19).

For each $q \in L_i$, choose a piece $j \neq i$ with $y_i^{\{q\}} \in I_j$, and write $i \rightarrow j$, with label $q_{ij} = q$. Call j the chosen protector of that neighbour. If D_{ij} is the set of coordinates on which y_i, y_j differ, then

$$q_{ij} \in L_i \setminus L_j, \quad D_{ij} \setminus L_j = \{q_{ij}\}, \quad |D_{ij} \cap L_j| \geq 2. \quad (5.20)$$

Indeed, if $q \in L_j$, membership of $y_i^{\{q\}}$ in I_j would imply that y_i lies in the full carrier of P_j . Its absence from P_j would then force $y_i = y_j$, contradicting the distance-at-least-two condition defining I_j . Thus $q \notin L_j$, and the other assertions follow by comparing y_i and $y_i^{\{q\}}$ outside L_j . In particular, one piece cannot protect two neighbours of the same tip, so all chosen targets are distinct and $r_i \leq m - 1$.

Theorem 5.24 (Two supports of largest possible size). *Suppose (5.19) holds and $r_1 = r_2 = m - 1$. Put $a = q_{12}$, $b = q_{21}$. There is a set K of size $m - 2$ such that*

$$L_1 = K \dot{\cup} \{a\}, \quad L_2 = K \dot{\cup} \{b\}.$$

For $k \notin \{1, 2\}$, put $u_k = q_{1k}$, $v_k = q_{2k}$. Both lists enumerate K , so $\pi(u_k) = v_k$ defines a permutation of K . If $C = \{u \in K : \pi(u) \neq u\}$ and $c = |C|$, then $c \geq 2$ and

$$r_k \geq \begin{cases} c, & u_k \in C, \\ c + 2, & u_k \notin C. \end{cases} \quad \text{Moreover, } r_k \geq 3 \text{ for every } k \notin \{1, 2\}. \quad (5.21)$$

Proof. Both pieces choose every other piece as a protector. Applying (5.20) in both directions gives

$$D_{12} = \{a, b\} \dot{\cup} C_0, \quad \emptyset \neq C_0 \subseteq L_1 \cap L_2.$$

The common part is nonempty because $D_{12} \cap L_2$ contains b and has size at least two. For every third index k , the identity $D_{12} = D_{1k} \triangle D_{2k}$ gives

$$D_{12} \setminus L_k = \{u_k\} \triangle \{v_k\}. \quad (5.22)$$

The coordinate a differs from u_k , since piece 1 uses distinct labels for distinct targets; it also differs from $v_k \in L_2$, since $a \notin L_2$. Thus $a \in L_k$. The same argument with 1 and 2 interchanged gives $b \in L_k$.

We next show that $u_k \in L_2$. Otherwise $u_k \neq a$ and $D_{12} \setminus L_2 = \{a\}$ imply $u_k \notin D_{12}$. But $u_k \notin L_k$ and $u_k \neq v_k \in L_2$, contradicting (5.22) at coordinate u_k . Similarly $v_k \in L_1$. The $m - 2$ distinct labels u_k exhaust $L_1 \setminus \{a\}$ and belong to L_2 ; since $a \notin L_2$, this set is exactly $K = L_1 \cap L_2$. Interchanging the indices proves the assertion about the v_k 's and the two supports.

Since $a, b \in L_k$, equation (5.22) becomes

$$C_0 \setminus L_k = \{u_k\} \triangle \{v_k\}.$$

Both u_k, v_k are absent from L_k . If $u_k = v_k$, then $u_k \notin C_0$ and $C_0 \subseteq L_k$. If $u_k \neq v_k$, then $u_k, v_k \in C_0$ and $C_0 \setminus \{u_k, v_k\} \subseteq L_k$. As the u_k 's enumerate K , this proves that $C_0 = C$ is exactly the set of moved points of π . A permutation cannot move exactly one point, so $c \geq 2$. Counting these required coordinates together with a, b proves the two displayed rank bounds.

It remains to exclude $r_k = 2$. The bounds already proved force $c = 2$, $u_k \neq v_k$, $C = \{u_k, v_k\}$, and $L_k = \{a, b\}$. Translate all tips by y_1 , so that y_1 is the zero word, and identify words with their supports. Equation (5.20) for $1 \rightarrow k$ then forces $y_k = \{a, b, u_k\}$: its part outside L_k is $\{u_k\}$, and its part inside has at least two elements. But $y_2 = \{a, b, u_k, v_k\}$, so $y_k = y_2^{\{v_k\}} \in P_2$, contradicting the missing-tip condition in (5.19). $\square$

Corollary 5.25 (Capacity with two largest supports). *For a cover satisfying (5.19) with six pieces, if two supports have size five, then every other support has size at least three and*

$$\sum_{i=1}^6 |I_i| \geq 82.$$

In particular, the rank multiset $(5, 5, 4, 2, 2, 2)$ is impossible.

Proof. A rank- r piece has $f(r) = 2^r - r - 1$ protected concepts, and f is increasing for $r \geq 2$. In Theorem 5.24, $|K| = 4$, so $c \in \{2, 3, 4\}$. If $c = 2$, the two moved indices have rank at least three and the two fixed indices have rank at least four; the capacity is at least $2f(5) + 2f(3) + 2f(4) = 82$. If $c = 3$, three ranks are at least three and the remaining rank is at least five, giving $2f(5) + 3f(3) + f(5) = 90$. If $c = 4$, all four ranks are at least four, giving $2f(5) + 4f(4) = 96$. Each case proves the stated bound. $\square$

This is a bound on the sum of the protected-interior sizes, not on $|H|$: different interiors may overlap. Nor does it give a pc-minimality criterion for the constructed family. Its role is to exclude cover parameters using their signed neighbour incidences before examining any elementary image.

Proposition 5.26 (Elementary images of an oriented one-hole cover). *For $z \in Q_E$ and $C \subseteq E$, write*

$$Q(z, C) = \{z^U : U \subseteq C\}.$$

Let (y, B) be an oriented one-hole datum, fix $e \in E$, and let $\bar{y}$ be the word obtained by suppressing coordinate e . If $e \notin B$, then

$$\rho_e^i P(y, B) = \begin{cases} P(\bar{y}, B), & i = y_e, \\ \emptyset, & i \neq y_e, \end{cases} \quad \pi_e P(y, B) = P(\bar{y}, B). \quad (5.23)$$

If $e \in B$ and $C = B \setminus \{e\}$, then

$$\rho_e^{y_e} P(y, B) = P(\bar{y}, C), \quad \rho_e^{1-y_e} P(y, B) = Q(\bar{y}, C), \quad \pi_e P(y, B) = Q(\bar{y}, C). \quad (5.24)$$

Here $P(\bar{y}, C)$ has the same formula as in (5.14); it may have $|C| = 1$ and hence consist of one vertex.

Consequently, for every finite cover $\mathcal{W}$, each elementary restriction or contraction of $H(\mathcal{W})$ is the union of the corresponding images in (5.23)–(5.24). Thus it has a directly computed mixed cover by full affine cubes and oriented one-hole cubes, with the original missing-tip and support provenance retained.

Proof. If $e \notin B$, every vertex of $P(y, B)$ has the fixed e -value y_e , which gives (5.23). Suppose $e \in B$. On the side with value y_e , the flipped set U is a nonempty subset of $B \setminus \{e\}$, giving the first formula in (5.24). On the opposite side, U contains e , while $U \setminus \{e\}$ is arbitrary; after coordinate e is suppressed, this gives the full cube $Q(\bar{y}, C)$. Contraction is the union of the two restrictions and hence gives the same full cube. Finally, restrictions, contractions, and unions commute, proving the cover formula for $H(\mathcal{W})$. $\square$

Definition 5.27 (Punctured expansion profile). Let $H \subseteq Q_E$, fix $e \in E$, and put $\bar{E} = E \setminus \{e\}$. For $z \in \pi_e H$, $A \subseteq \bar{E}$, and $C \subseteq A$, let

$$\varepsilon_{H,e}(z, A; C) = \{i \in \{0, 1\} : \iota_e^i(z^C) \in H\}, \quad (5.25)$$

where ι_e^i inserts the value i in coordinate e . We call this a *punctured expansion profile* if

$$\varepsilon_{H,e}(z, A; A) = \emptyset, \quad \varepsilon_{H,e}(z, A; C) \neq \emptyset \quad (C \subsetneq A). \quad (5.26)$$

It is *direct* if one side contains the entire punctured cube, that is,

$$\bigcap_{C \subsetneq A} \varepsilon_{H,e}(z, A; C) \neq \emptyset, \quad (5.27)$$

and *switched* otherwise. For a switched profile write

$$\begin{aligned} P_A &= Q_A \setminus \{A\}, \\ D &= \{C \in P_A : \varepsilon_{H,e}(z, A; C) = \{0, 1\}\}, \\ Z_i &= \{C \in P_A : \varepsilon_{H,e}(z, A; C) = \{i\}\} \quad (i \in \{0, 1\}). \end{aligned} \quad (5.28)$$

Thus $P_A = D \dot{\cup} Z_0 \dot{\cup} Z_1$.

The distinction is necessary because contraction takes a union of two sections. A punctured cube in the union need not occur in either section: the two sections may cover different parts of it and change sides through their common fibres.

Theorem 5.28 (Canonical transport of local holes). *Let $H \subseteq Q_E$ be ample, let $e \in E$, and put $\bar{E} = E \setminus \{e\}$.*

1. *If $z \in \rho_e^i H$, then*

$$\mathcal{N}(\Lambda_{\rho_e^i H}(z)) = \{A \in \mathcal{N}(\Lambda_H(\iota_e^i(z))) : A \subseteq \bar{E}\}. \quad (5.29)$$

Hence restrictions transport the canonical one-hole data simply by discarding the supports that contain e .

2. *If $z \in \pi_e H$, then*

$$A \in \mathcal{N}(\Lambda_{\pi_e H}(z)) \iff \varepsilon_{H,e}(z, A; \cdot) \text{ is a punctured expansion profile.} \quad (5.30)$$

3. *For every switched profile, D, Z_0, Z_1 from (5.28) satisfy: $D, Z_0, Z_1 \neq \emptyset$; no edge of P_A joins Z_0 to Z_1 ; every component of $P_A \setminus D$ is contained in one Z_i ; and*

$$D \text{ is ample,} \quad \text{Sh}(D \cup Z_0) \cap \text{Sh}(D \cup Z_1) = \text{Sh}(D). \quad (5.31)$$

Conversely, suppose that $P_A = D \dot{\cup} Z_0 \dot{\cup} Z_1$, that D is ample, and that the shadow equality in (5.31) holds. Then the two-layer family

$$L(D, Z_0, Z_1) = ((D \cup Z_0) \times \{0\}) \cup ((D \cup Z_1) \times \{1\}) \quad (5.32)$$

is ample, its contraction is P_A , and A is a canonical local hole at the vertex $\emptyset$ of that contraction. If both Z_i are nonempty, this hole is switched. Thus (5.28)–(5.31) give an intrinsic local construction and recovery theorem for the contraction holes not visible in the independent images of the original one-hole pieces.

Proof. For $A \subseteq \bar{E}$, the A -cube through z lies in $\rho_e^i H$ exactly when the same cube through $\iota_e^i(z)$ lies in H . The local cube-support complex of the restriction is therefore the induced subcomplex of $\Lambda_H(\iota_e^i(z))$ on $\bar{E}$. Its minimal nonfaces are precisely the minimal nonfaces of the original complex that avoid e , proving (5.29).

An A -cube through z lies in $\pi_e H$ exactly when $\varepsilon_{H,e}(z, A; C) \neq \emptyset$ for every $C \subseteq A$. It is a minimal missing cube exactly when this holds for every proper C , but fails for $C = A$. This is (5.30).

Now fix a switched profile and abbreviate D, Z_0, Z_1 as in (5.28). Switchedness says that neither side occurs at every proper vertex, so both Z_0 and Z_1 are nonempty. Suppose that adjacent vertices $C \in Z_0$ and $C' \in Z_1$ existed. Their chosen lifts in H would differ in coordinate e and in the one coordinate of $C \triangle C'$. The only two intermediate vertices on a length-two cube path are the opposite lift of C and the opposite lift of C' , but both are absent by the definitions of Z_0, Z_1 . This contradicts the fact that the ample family H is an isometric cube subgraph. Hence no such edge exists, and every component of $P_A \setminus D$ is monochromatic. Since the punctured cube P_A is connected and both colours occur, D is nonempty.

Restrict H outside $A \cup \{e\}$ to the values prescribed by z . On the remaining coordinates the resulting ample family is exactly

$$L = ((D \cup Z_0) \times \{0\}) \cup ((D \cup Z_1) \times \{1\}),$$

and its contraction is P_A . For any two-layer family with sections L_0, L_1 , union $P = L_0 \cup L_1$, and intersection $D = L_0 \cap L_1$, one has the exact formula

$$\text{Sh}(L) = \text{Sh}(P) \dot{\cup} \{J \cup \{e\} : J \in \text{Sh}(L_0) \cap \text{Sh}(L_1)\}.$$

Moreover, $|L| = |P| + |D|$. Here L and $P = P_A$ are ample. Consequently

$$|D| = |\text{Sh}(L_0) \cap \text{Sh}(L_1)|.$$

Since $\text{Sh}(D) \subseteq \text{Sh}(L_0) \cap \text{Sh}(L_1)$, the Sandwich inequalities give

$$|D| \leq |\text{Sh}(D)| \leq |\text{Sh}(L_0) \cap \text{Sh}(L_1)| = |D|.$$

This proves that D is ample and establishes (5.31).

Conversely, construct $L(D, Z_0, Z_1)$ from the displayed partition. Its contraction is P_A , its order is $|P_A| + |D|$, and the same two-layer shadow formula, the ampleness of P_A, D , and (5.31) give

$$|\text{Sh}(L(D, Z_0, Z_1))| = |P_A| + |D| = |L(D, Z_0, Z_1)|.$$

The Sandwich equality makes $L(D, Z_0, Z_1)$ ample. The tip A is absent from its contraction while every proper $C \subsetneq A$ is present, so A is a minimal nonface of the local cube-support complex at $\emptyset$. Both exclusive parts being nonempty is exactly the switched condition. $\square$

Remark 5.29 (The first switched hole). Let $A = \{p, q\}$, put

$$D = \{\emptyset\}, \quad Z_0 = \{\{p\}\}, \quad Z_1 = \{\{q\}\}.$$

The family $L(D, Z_0, Z_1)$ is a four-vertex path. Its contraction is the punctured square $Q_{\{p,q\}} \setminus \{\{p, q\}\}$, but neither section contains that punctured square: the two arms occur on opposite sides and meet at the duplicated vertex $\emptyset$. Thus the missing square is a canonical one-hole certificate of the contraction and is not the image of a one-hole certificate supported away from the contracted coordinate.

Remark 5.30 (The switch phenomenon in Hall's obstruction). The distinction is not confined to the preceding path. A worker audit of all twelve contractions of Hall's 299-concept obstruction found 8,560 canonical local holes. Of these, 7,691 are direct and 869 are switched; every contraction contains switched holes. The same audit verified (5.29) for all 9,159 canonical holes in the twenty-four restrictions. The durable diagnostic is `damp_hall_contraction_hole_transport.json`, generated by `analyze_damp_hall_contraction_hole_transport.py`. These counts are not used in the proof.

Remark 5.31 (The remaining seed condition). Theorem 5.21 supplies the nonpeelable part of a seed structurally: the one-hole antipodes cover the family, so no first peeling step exists. To obtain the cornerless seeds of the forbidden basis, one must still characterize, in the same parameters, when all elementary restrictions and contractions are corner-peelable. Proposition 5.26 gives a useful piecewise image cover, but Theorem 5.28 shows that it is not the canonical local-hole data after contraction: switched holes arise by jointly assembling the two sections. The exact minor parameter must therefore retain the recovered punctured expansion profiles, including their ample separators and component colours. The missing theorem must make peelability follow from the global incidence and compatibility of those profiles. Encoding the image conditions by orders, acyclic trace-carrier schemes, or SAT would return to recognition and is not the desired next theorem.

Proposition 5.32 (Two-cone decomposition and the gluing dichotomy). *Let $H \subseteq Q_E$ be ample, let $e \in E$ be active, and write*

$$B_0 = \rho_e^0 H, \quad B_1 = \rho_e^1 H, \quad A = B_0 \cap B_1, \quad C = B_0 \cup B_1. \quad (5.33)$$

Then B_0, B_1, A, C are ample and

$$\Delta_H = (v_{e,0} * \Delta_{B_0}) \bigcup_{\Delta_A} (v_{e,1} * \Delta_{B_1}). \quad (5.34)$$

For nested ample families $A \subseteq B$, call an ordering $p_1, \dots, p_k$ of $B \setminus A$ an A -relative ample chain if

$$A \cup \{p_1, \dots, p_j\} \quad \text{is ample for every } 0 \leq j \leq k. \quad (5.35)$$

Equivalently, this is a relative shelling of (Δ_B, Δ_A) . If $A, B_{1-b} \in \mathcal{D}_{\text{amp}}$ and $A \subseteq B_b$ has an A -relative ample chain for one $b \in \{0, 1\}$, then $H \in \mathcal{D}_{\text{amp}}$.

Consequently, if $H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$, then every active coordinate has exactly one of the following two signatures.

1. Inherited-site signature: $A \notin \mathcal{D}_{\text{amp}}$.
2. Gluing-critical signature: $A \in \mathcal{D}_{\text{amp}}$, but neither relative pair (Δ_{B_0}, Δ_A) nor (Δ_{B_1}, Δ_A) is relatively shellable.

In the second case the three absolute Yang complexes $\Delta_{B_0}, \Delta_A, \Delta_{B_1}$ are shellable; the obstruction lies in their relative gluing rather than in the topology of an individual piece.

Proof. The two restrictions, their union, and their intersection are ample by Lemma 7.22. Every facet of Δ_H contains exactly one of $v_{e,0}, v_{e,1}$, so

$$\Delta_H = (v_{e,0} * \Delta_{B_0}) \cup (v_{e,1} * \Delta_{B_1}).$$

The two cones have distinct apex vertices. Their intersection is therefore $\Delta_{B_0} \cap \Delta_{B_1}$, which equals Δ_A by the signed trace identity (7.23). This proves (5.34).

Suppose, after interchanging the signs if necessary, that $p_1, \dots, p_k$ is an A -relative ample chain in B_1 . Starting from H , delete the upper concepts $(p_k, 1), \dots, (p_1, 1)$ in this order. Immediately before $(p_j, 1)$ is deleted, the upper section is $A \cup \{p_1, \dots, p_j\}$, and deleting p_j preserves its ampleness. The ample corner-deletion criterion therefore makes p_j a corner of that section. Since $p_j \notin A$, its lower mate is absent, so the same concept is a corner of the current two-layer family. After these deletions the remaining family is

$$(B_0 \times \{0\}) \cup (A \times \{1\}),$$

the peripheral expansion with base B_0 and site A . The exact peripheral-expansion test and $A, B_0 \in \mathcal{D}_{\text{amp}}$ give a peeling of this remainder. Reversing signs proves the symmetric assertion.

Now assume that H is a forbidden pc-minor. The sections B_0, B_1 and the contraction C are proper pc-minors and hence belong to $\mathcal{D}_{\text{amp}}$. If $A \notin \mathcal{D}_{\text{amp}}$, the first signature holds. Otherwise, the preceding implication shows that a relative shelling on either side would put H in $\mathcal{D}_{\text{amp}}$, a contradiction. This gives the second signature and proves the dichotomy. $\square$

Theorem 5.33 (Exact descent through a nonpeelable overlap). *Let $H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$, let e be active, and retain the notation*

$$B_b = \rho_e^b H, \quad A = B_0 \cap B_1 = H^{\{e\}}, \quad C = B_0 \cup B_1.$$

If $A \notin \mathcal{D}_{\text{amp}}$, then

$$A \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}.$$

Thus the inherited-site signature always exhibits a smaller member of the forbidden basis, not merely an unspecified nonpeelable subfamily. Notice that A is a reduction of H , not necessarily a pc-minor of H .

Consequently, repeated descent through nonpeelable overlaps terminates at an obstruction S for which

$$S^{\{q\}} \in \mathcal{D}_{\text{amp}} \quad \text{for every active coordinate } q.$$

Every coordinate of S is gluing-critical. At each inverse step the larger obstruction is recovered in the explicit form

$$H = (A \times \{0, 1\}) \cup (R_0 \times \{0\}) \cup (R_1 \times \{1\}), \quad R_b = B_b \setminus A, \quad (5.36)$$

where $R_0, R_1 \neq \emptyset$ and

$$A \cup R_0, \quad A \cup R_1, \quad A \cup R_0 \cup R_1 \quad \text{belong to } \mathcal{D}_{\text{amp}}.$$

The core A and the two signed repair wings R_0, R_1 are recovered canonically once e is chosen.

Proof. The ampleness of A is part of Lemma 7.22. We prove that all its elementary pc-minors belong to $\mathcal{D}_{\text{amp}}$. Fix a coordinate $q \neq e$ active in A . Restrictions commute with reduction:

$$(\rho_q^i H)^{\{e\}} = \rho_q^i A \quad (i \in \{0, 1\}).$$

The family $\rho_q^i H$ is a proper pc-minor of H , hence belongs to $\mathcal{D}_{\text{amp}}$; Proposition 2.5 puts $\rho_q^i A$ in $\mathcal{D}_{\text{amp}}$.

Contraction commutes as well, but here ampleness is essential. The two e -sections of $\pi_q H$ are $\pi_q B_0, \pi_q B_1$. Applying the signed trace identity (7.22) with the coordinates e, q suppressed gives

$$(\pi_q H)^{\{e\}} = \pi_q B_0 \cap \pi_q B_1 = \pi_q (B_0 \cap B_1) = \pi_q A.$$

Again $\pi_q H$ is a proper pc-minor in $\mathcal{D}_{\text{amp}}$, so its reduction $\pi_q A$ belongs to $\mathcal{D}_{\text{amp}}$. Therefore every elementary pc-minor of A is in $\mathcal{D}_{\text{amp}}$, and pc-minor closure handles every proper pc-minor. The assumed $A \notin \mathcal{D}_{\text{amp}}$ proves $A \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

Each descent removes one coordinate, so iteration terminates. At termination every overlap is peelable, and Proposition 5.32 makes every coordinate gluing-critical. Formula (5.36) is the section decomposition itself. The two repair wings are nonempty because an ample forbidden pc-minor is periphery-free by Proposition 2.52. Finally, the two sections and their union are proper pc-minors of H , proving the three displayed $\mathcal{D}_{\text{amp}}$ -memberships. $\square$

Theorem 5.34 (Reduction descent in the separator branch). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor, and let $\Gamma = \text{Cr}(H)$. For every $D \subseteq F$, a nonpeelable reduction H^D is an ample forbidden pc-minor using all coordinates of $F \setminus D$. In particular, if $H^{\{e\}}$ is nonpeelable, then e is a universal vertex of Γ , meaning that it is adjacent to every other vertex.*

Suppose that Γ is not two-connected, or equivalently that H has a cut vertex or a separating edge. Then exactly one of the following holds:

1. *Every reduction H^D with $D \neq \emptyset$ belongs to $\mathcal{D}_{\text{amp}}$.*
2. *There is a unique coordinate e such that*

$$H^D \notin \mathcal{D}_{\text{amp}} \iff D \in \{\emptyset, \{e\}\}. \quad (5.37)$$

In this case e is the unique universal vertex and the unique cut vertex of Γ ; the graph $\Gamma - e$ has exactly two components. The family H has a unique separating e -edge, and $H^{\{e\}}$ is an ample forbidden pc-minor with a cut vertex. Its two rooted blocks are the strong e -projections of the two canonical pieces at that edge.

Consequently every forbidden class with a cut vertex has only peelable nonempty-coordinate reductions. Throughout the separator branch, descent through nonpeelable overlaps has length at most one, and its terminal obstruction is uniquely determined.

Proof. Write $K = H^D \notin \mathcal{D}_{\text{amp}}$. Restrictions and contractions in a coordinate outside D commute with reduction along D . For contractions, iterate the ample signed-trace identity used in Theorem 5.33; all intermediate reductions are ample. Suppose a coordinate $q \in F \setminus D$ were inactive in K . The family $\pi_q H$ is a proper pc-minor of H , so reduction closure gives

$$\pi_q K = (\pi_q H)^D \in \mathcal{D}_{\text{amp}}.$$

The nonempty ample family K is connected. An inactive coordinate is therefore constant, making K isomorphic to $\pi_q K$, a contradiction. This proves the full-coordinate assertion. Every intermediate

projection along a subset of D is nonpeelable, since its further reduction is K . Repeated application of Theorem 5.33 now makes K a forbidden pc-minor.

If $D = \{e\}$, an edge of direction $q \neq e$ in K lifts to an $\{e, q\}$ -square in H . All such q are active by the preceding paragraph, so e is universal in Γ . More generally, if H^D is nonpeelable, then $H^{\{e\}}$ is nonpeelable for every $e \in D$, by reduction closure.

An ample forbidden class has VC dimension at least three, so Γ has at least three vertices. A graph of this order with two universal vertices is two-connected: after any single vertex is removed, a remaining universal vertex connects the rest. Thus Γ has at most one universal vertex under the hypothesis. It follows that either all nonempty-coordinate reductions peel, or their only nonpeelable member is $H^{\{e\}}$ for a unique e , proving (5.37). If Γ is disconnected it has no universal vertex, which also proves the assertion for a cut vertex of H .

In the remaining case, e is universal and Γ is connected. Removing any vertex other than e leaves a connected graph. Since Γ is not two-connected, e is its unique cut vertex. Put $K = H^{\{e\}}$. Its crossing graph uses all vertices of $\Gamma - e$, and every edge of $\text{Cr}(K)$ is an edge of $\Gamma - e$: a shattered pair in K lifts to a shattered triple in H . Thus $\text{Cr}(K)$ is disconnected, so K has a cut vertex by Proposition 2.33. Theorem 2.19 gives exactly two blocks in K . Their coordinate sets are precisely the two components of $\text{Cr}(K)$, as proved in the separator recovery proposition. The disconnected spanning supergraph $\Gamma - e$ consequently has exactly two components as well.

The graph Γ is connected, so H has no cut vertex. Proposition 2.33 now recovers a unique separating e -edge and its two canonical convex pieces H_1, H_2 . Normalize that edge to $\{0, \mathbf{1}_{\{e\}}\}$. Taking its reduction gives

$$K = H_1^{\{e\}} \vee H_2^{\{e\}},$$

with common root 0 and disjoint exclusive coordinate sets. Both factors are nontrivial: the full-coordinate assertion for K retains every coordinate assigned to each piece. The cut-vertex theorem therefore identifies these factors with the two rooted blocks of K . Equation (5.37) says that all further nonempty-coordinate reductions peel, proving the termination and uniqueness claims. $\square$

The universal-vertex condition locates the only possible descent coordinate; it does not decide whether that carrier peels. Both outcomes occur in the existing constructions. The 1158-concept terminal edge amalgam of Remark 6.39 has one universal crossing coordinate, but its 261-concept carrier peels. The one-layer tower over the 1158-concept terminal cut-vertex obstruction in Remark 6.37 has 2318 concepts; its layer coordinate is the unique universal crossing vertex, and its reduction is that nonpeelable cut-vertex obstruction. The fixed checks, including explicit orders transported from proper contractions when a carrier loses a coordinate, are recorded by `verify_damp_separator_strong_descent.py` in `damp_separator_strong_descent_verification.json`. This bounds and makes canonical an existing descent within the separator branch. It does not classify its rooted factors intrinsically or settle descent choices for cores with two-connected crossing graphs.

Proposition 5.35 (Square attachments leave nonpeelable carriers unchanged). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor, and let K be its canonical graph three-core. For every nonempty $D \subseteq F$,*

$$H^D \notin \mathcal{D}_{\text{amp}} \iff K^D \notin \mathcal{D}_{\text{amp}},$$

and in this case $H^D = K^D$ as embedded families. Moreover, no acquired coordinate pair in the square-attachment data meets a coordinate e for which $H^{\{e\}} \notin \mathcal{D}_{\text{amp}}$.

Consequently, if $\text{Cr}(K)$ is not two-connected, descent from H through nonpeelable reductions has length at most one and a unique terminal obstruction. When a descent exists, its result is the same cut-vertex obstruction obtained from K . This conclusion does not require a separator in H itself.

Proof. Theorem 2.53 and Theorem 2.60 reconstruct H from K by ample one-concept additions of support size two. Every intermediate family is forbidden and uses all of F . Consider one such addition $V = U \cup \{x\}$, with acquired pair I . Proposition 5.105 gives

$$V^D = \begin{cases} U^D \cup \{x|_{F \setminus D}\}, & D \subseteq I, \\ U^D, & D \not\subseteq I, \end{cases}$$

and the addition in the first case is ample, with acquired support $I \setminus D$. Therefore a peelable U^D always gives a peelable V^D : when the families differ, delete the new corner first. This includes the empty-family and singleton cases.

Suppose U^D is nonpeelable. If $D \subseteq I$, then $|I \setminus D| \leq 1$. But Theorem 5.34 makes U^D nonempty and active on every coordinate of $F \setminus D$. Its shattered complex already contains the empty set and every singleton, so none can be an acquired support. This contradiction proves $D \not\subseteq I$ and hence $V^D = U^D$. Thus the nonpeelable projections and their exact embedded families are unchanged at every attachment, proving the first assertion. For $D = \{e\}$, the same argument gives $e \notin I$ at every step, proving the assertion about acquired pairs.

The three-core has the same active coordinates as H . Apply Theorem 5.34 to K and use the just-proved equality of all nonpeelable projections. If every nonempty-coordinate projection of K peels, the same is true of H . Otherwise their only such nonpeelable projection is the same cut-vertex obstruction $K^{\{e\}} = H^{\{e\}}$, all of whose further nonempty-coordinate reductions peel. $\square$

For a concrete example where the extension matters, take the 2318-concept one-layer tower just described, in the integer coordinates of Remark 6.37. The word $32769 = 1 + 2^{15}$ completes the square at 0 in directions 0, 15. Its addition has degree two, so the tower is the three-core of the resulting 2319-concept obstruction. The new shattered pair joins the two components left after deleting the layer coordinate 26 from the crossing graph. That coordinate is still universal, so the new crossing graph is two-connected. The full obstruction consequently has neither a cut vertex nor a separating edge, yet its strong 26-projection is still the 1158-concept cut-vertex obstruction. The same verifier checks this square, the three-core, the crossing graph, the unchanged carrier, and all proper elementary-minor peelings of the enlarged family.

Theorem 5.36 (Replacing a coordinate by a block). *Let $H \subseteq Q_{G \cup \{z\}}$ be ample, and write*

$$B_i = \rho_z^i H, \quad C = B_0 \cup B_1, \quad A = B_0 \cap B_1.$$

Let E be a new nonempty coordinate set, disjoint from G , and put $m = |E|$. Define

$$W_E(H) = (B_0 \times \{0_E\}) \cup (B_1 \times \{1_E\}) \cup (C \times (Q_E \setminus \{0_E, 1_E\})). \quad (5.38)$$

For $m = 1$ this is H with its coordinate z renamed. For every $m \geq 1$, the following assertions hold.

1. *The family $W_E(H)$ is ample, and*

$$\text{Sh}(W_E(H)) = \{D \cup T : D \in \text{Sh}(C), T \subsetneq E\} \dot{\cup} \{D \cup E : D \in \text{Sh}(A)\}, \quad (5.39)$$

$$|W_E(H)| = (2^m - 1)|C| + |A|. \quad (5.40)$$

If $H \neq \emptyset$, its VC dimension increases by $m - 1$. In its signed minimal-nonface presentation, leave every clause avoiding z unchanged. In every other clause, replace z by all of E , repeating the forbidden sign at z on every new coordinate. These are exactly the signed minimal nonfaces of $W_E(H)$.

2. $W_E(H) \in \mathcal{D}_{\text{amp}}$ if and only if $H \in \mathcal{D}_{\text{amp}}$. If H is irredundantly embedded and z is active, then

$$W_E(H) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}).$$

3. If H is cornerless, then $W_E(H)$ is cornerless. If H is an irredundant ample forbidden pc-minor and $m \geq 2$, then $W_E(H)$ has minimum degree at least three and is its own graph three-core. Every coordinate in E is universal in its crossing graph, which is two-connected.

The marked construction has unique recovery: its two constant sections recover B_0, B_1 , and hence recover H . Equivalently, the marked signed clauses recover the original presentation by replacing E by z . No resolving tips or peeling orders are construction parameters.

Proof. We first determine the shadow, then lift corner deletions and check the proper minors. Empty families and empty sections are allowed in the shadow and peeling assertions; their shadows are empty.

Suppose first that $m \geq 2$. If $T \subsetneq E$, every word on T extends to a nonconstant word on E . Consequently a support $D \cup T$, with $D \subseteq G$, is shattered exactly when D is shattered by C . A support containing all of E is shattered exactly when its old part is shattered by both B_0 and B_1 . Identity (5.65), with its residual family $K = C$, identifies their common shadow with $\text{Sh}(A)$. If a section is empty, this identity is immediate. This proves (5.39); for $m = 1$ it is the usual two-layer formula. Its support count is $(2^m - 1)|C| + |A|$, also the vertex count in (5.38). The Sandwich equality therefore proves ampleness. The largest support has size

$$(m - 1) + \max\{\text{VC}(C), 1 + \text{VC}(A)\} = (m - 1) + \text{VC}(H),$$

with the second term omitted when A is empty.

The shadow formula also identifies its minimal nonfaces. Those avoiding z in the original shadow remain unchanged. Each old minimal nonface containing z becomes $(I \setminus \{z\}) \cup E$. Indeed, a nonface whose new-coordinate part is proper must already contain an old nonface of $\text{Sh}(C)$; a minimal remaining nonface must contain E , and deleting any element of its old part must put that part in $\text{Sh}(A)$. These are precisely the old minimality conditions for a nonface containing z . All nonconstant new-coordinate traces occur over C ; the two constant traces have exactly the original sections. Thus the unique missing trace repeats the old sign, proving the clause assertion using Theorem 5.3.

For the peeling assertion, first take $E = \{e, f\}$. Let (x, i) be a corner of H . If its mate $(x, 1 - i)$ belongs to H , the cube at that corner has all its old directions in the intersection A . Hence (x, ii) is a corner of $W_E(H)$; its deletion gives exactly $W_E(H \setminus \{(x, i)\})$. If its mate is absent, delete

$$(x, 01), \quad (x, 10), \quad (x, ii)$$

in that order. To verify these are corners, no old neighbor of $x \in B_i \setminus B_{1-i}$ can lie in $B_{1-i} \setminus B_i$. Otherwise the corresponding vertices of H , at Hamming distance two, would have neither possible intermediate vertex, contradicting isometry. Thus the old neighbor set of x in C equals its neighbor set in B_i . The corner cube of (x, i) supplies that entire old cube in B_i . Before each of the first two deletions its product with the remaining incident new edge is present. After those deletions, the old cube alone certifies the last corner. Again the remainder is exactly the replacement of the source after its corner deletion. Lifting every step of a peeling proves the forward implication. The reverse implication follows from

$$W_{\{e, f\}}(H)^{\{e\}} = H$$

with z renamed f , and reduction closure. Repeatedly replace one member of the new block by two coordinates. The constant sections remain B_0, B_1 , and every nonconstant section is C , so this

gives exactly $W_E(H)$ for every m . Induction proves the peeling equivalence, including empty and constant-coordinate intermediate families.

Now let H be an irredundant ample forbidden pc-minor. For a binary replacement, every restriction or contraction μ in an old coordinate $q \in G$ commutes with the construction: the formula uses only the two sections and their union. Thus $\mu W_{\{e,f\}}(H) = W_{\{e,f\}}(\mu H)$ peels by the preceding paragraph. A contraction in e or f is $C \square Q_1$. A restriction in either coordinate is a peripheral expansion of C along B_0 or B_1 . All three input families are proper pc-minors of H , so Proposition 2.6 makes these images peel. The replacement itself does not peel. This proves forbiddenness; iteration proves it for every block size. All coordinates remain active.

Conversely, suppose $W_E(H)$ is forbidden, with $m \geq 2$. The peeling equivalence makes H nonpeelable. Every proper old-coordinate image of H is a reduction of the corresponding proper image of $W_E(H)$, so it peels. The two constant sections B_i are proper face restrictions of $W_E(H)$, and C is a face restriction after contracting one new coordinate. They peel as well. These are all elementary pc-minors of H , proving the reverse implication. For $m = 1$ it is just coordinate renaming.

For cornerlessness, each vertex (x, t) of the replacement belongs to a full fibre on $E \setminus \{e\}$ for some $e \in E$. If $x \in A$, any e works. If $x \in B_i \setminus B_{1-i}$, choose e with $t_e = i$; such an e exists because the excluded constant word is absent. The reduction along that fibre is H , with z renamed e . A corner in a full fibre projects to a corner: every neighbor of the projected vertex lifts beside the given vertex, and the corner cube contains its product with the fibre. A corner of the replacement would therefore give a corner of H .

Finally, the degree of every new vertex is at least $\delta(H) + m - 1$. For an exclusive old word, the old neighbor set is unchanged by the isometry argument above, and its punctured cube row has degree at least $m - 1$. For an overlap word, a constant section retains its old neighbors and replaces its one z -edge by m edges; a nonconstant section has at least as many old neighbors. An ample forbidden family has minimum degree at least two, as shown in the proof of Theorem 2.53. Thus the replacement has minimum degree at least three. Each old coordinate is active in C , and $A \neq \emptyset$ since H is connected and z is active. The shadow formula now makes every new coordinate adjacent to all others in the crossing graph. Two universal vertices make a graph of this order two-connected. The recovery assertions follow directly from the constant sections and the signed clause rule. $\square$

Corollary 5.37 (Arbitrarily many maximal descent supports). *With the notation of Theorem 5.36, for every $D \subseteq G$ and $T \subseteq E$,*

$$W_E(H)^{D \cup T} = \begin{cases} W_{E \setminus T}(H^D), & T \subsetneq E, \\ H^{D \cup \{z\}}, & T = E. \end{cases} \quad (5.41)$$

Consequently its nonpeelable reductions are determined exactly by those of H : in the first case the projection is nonpeelable if and only if H^D is nonpeelable, and in the second case if and only if $H^{D \cup \{z\}}$ is nonpeelable.

In particular, suppose H is an ample forbidden pc-minor whose nonempty-coordinate reductions all peel. Then

$$\{U : W_E(H)^U \notin \mathcal{D}_{\text{amp}}\} = \{T : T \subsetneq E\}.$$

There are exactly m maximal descent supports, namely $E \setminus \{e\}$ for $e \in E$. Every maximal sequence of one-coordinate nonpeelable descents has length $m - 1$, and every endpoint is H with a coordinate renamed. Thus neither uniqueness of the maximal support nor a uniform bound on descent length extends to all forbidden three-cores with two-connected crossing graphs. This does not refute uniqueness of the terminal isomorphism type.

Proof. Reduction along $D \subseteq G$ commutes with the construction. The only point requiring justification is

$$(B_0 \cup B_1)^D = B_0^D \cup B_1^D.$$

If a full D -cube Q occurs in the union, apply (5.64) with residual family $K = Q$. One of $B_0 \cap Q, B_1 \cap Q$ shatters D , and hence contains the whole cube Q . This proves the nontrivial inclusion; the reverse inclusion is immediate. The argument also covers an empty section directly.

It remains to take the reduction in T . If $T \subsetneq E$, a constant surviving word has fibre B_i , and a nonconstant surviving word has fibre C . For $T = E$, a full fibre requires the old word to lie in A . Applying this description after the projection in D proves (5.41). The peeling equivalence in the theorem gives the exact status rule. Under the additional hypothesis, the first case is nonpeelable precisely when $D = \emptyset$, and the second case always peels. Maximal proper subsets of E have size $m - 1$; projection along any one of them leaves the original family on its remaining marked coordinate. This proves all the descent assertions. $\square$

Theorem 5.38 (Canonical reduction of repeated signed-clause columns). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor. Index its signed minimal nonfaces by $\mathcal{N}$, and associate to each coordinate $e \in F$ the column*

$$v_e(I) = \begin{cases} 0, & e \notin I, \\ 2\sigma_I(e) - 1, & e \in I, \end{cases} \quad I \in \mathcal{N}.$$

Declare $e \sim f$ if $v_e = v_f$ or $v_e = -v_f$. This partitions F into classes $E_1, \dots, E_r$. Reorient coordinates so that the columns within each class agree, choose representatives $e_j \in E_j$, and take the reduction

$$R = H^{F \setminus \{e_1, \dots, e_r\}} \quad (5.42)$$

of the reoriented family. Then R is an irredundant ample forbidden pc-minor whose signed-clause columns are pairwise distinct up to sign.

The family H is recovered from R by replacing e_j by a block of size $m_j = |E_j|$, for every j . The reduced family and these multiplicities are unique up to coordinate renaming and reorientation. Every sequence of compressions of equal columns up to sign gives this same result when no such pair remains.

Conversely, every irredundant ample forbidden R with pairwise distinct columns up to sign, together with arbitrary positive integers $m_1, \dots, m_r$, gives an ample forbidden class by these replacements. Thus the forbidden inputs with distinct columns, and the positive integer multiplicities on their coordinates, form an exact parametrization. The intrinsic description of those reduced forbidden inputs remains open.

Proof. Equality of columns up to sign is an equivalence relation. Reversing one coordinate negates its column, so the indicated normalization is possible. Consider a class E of size at least two after normalization. Every clause either avoids E or contains all of E , with a constant forbidden sign on that block. Let C be the set of old words satisfying the clauses that avoid E , and let B_0, B_1 be the two constant sections of H . Its nonconstant sections are all C . Also $B_0 \cup B_1 \subseteq C$.

Here ampleness forces the reverse inclusion. If $x \in C \setminus (B_0 \cup B_1)$, its face restriction in H would be exactly $Q_E \setminus \{0_E, 1_E\}$. This family has $2^{|E|} - 2$ concepts and shatters every proper subset of E , so its shadow has $2^{|E|} - 1$ members. It is not ample, contradicting closure of ampleness under face restriction. Hence $C = B_0 \cup B_1$. The section pattern is exactly (5.38).

Keeping one representative of E , the reduction along the other coordinates has sections B_0, B_1 , and is ample. Call it U . We have $H = W_E(U)$, so the peeling equivalence of Theorem 5.36 makes U

nonpeelable. The full-coordinate and descent assertion in Theorem 5.34, which requires no separator for that assertion, makes U an irredundant ample forbidden pc-minor. Its signed minimal nonfaces are obtained by replacing E by its representative in every clause.

This last operation changes neither the set of clause rows nor the entries in the remaining columns. The rows correspond bijectively: a support was a union of whole coordinate classes, so compressing a class cannot identify two different supports or change their inclusion relations. Consequently no two previously inequivalent columns become equivalent. The remaining equivalence classes are exactly the original classes with their compressed members removed. Induction gives (5.42), forbiddenness of R , and distinctness of its columns up to sign. Reductions commute, and choosing a different representative in a class merely renames its coordinate. Changing the normalization reverses that coordinate. This proves independence of all compression choices and orders.

Conversely, the block-replacement theorem preserves forbiddenness at every step. Its signed clause rule shows that replacements on different coordinates commute: each original column is simply repeated the chosen number of times. Since the input columns are distinct up to sign, the resulting equivalence classes are exactly these repeated blocks. Compression therefore recovers precisely R and the prescribed multiplicities, proving the converse and uniqueness assertions. $\square$

This reduction is canonical in the signed clause presentation and removes all freedom due to repetition of signed-clause columns. Its interaction with graph three-core reduction is settled by Theorem 5.39 below: the reductions need not commute, but their alternation is forced and canonical. Column reduction also differs from terminality under reduction. A repeated block supplies at least two nonpeelable one-coordinate projections, by (5.41). The verified 2318-concept tower has exactly one such projection, so its columns are already distinct up to sign although it still admits a nonpeelable reduction descent.

For an explicit control, use the fixed 291-concept cornerless obstruction and replace coordinate 0. Its contraction has 224 concepts and its reduction has 67. All its nonempty-coordinate reductions peel, because its VC dimension is three and those projections have VC dimension at most two [CCMW22, Proposition 4.11]. Blocks of sizes two and three therefore give cornerless forbidden three-cores of orders $3 \cdot 224 + 67 = 739$ and $7 \cdot 224 + 67 = 1635$, with respectively two and three maximal descent supports. In the binary case both singleton projections are nonpeelable but their joint projection peels; the nonpeelable supports need not be closed under union. The verifier `verify_damp_coordinate_block_replacement.py` checks all 832 replacements of ample inputs through Q_3 at block sizes two and three, including 1108 corner lifts and the signed clause and reduction formulas. It independently checks the fixed inputs and replays 117 elementary-minor orders, including all proper elementary minors of the two displayed obstructions. A further 36 positive carrier orders, together with cornerlessness of all proper-block projections, check their complete descent-support descriptions using reduction closure. The same checks recover the canonical signed-column reduction of every ample input through Q_3 , test every representative choice, and recover the 291-concept reduced input from both displayed replacements. The record is `damp_coordinate_block_replacement_verification.json`. The construction provides an intrinsic operation over every forbidden input, together with exact recovery of its marked outputs. It does not characterize the forbidden inputs that admit no such reduction.

Theorem 5.39 (Alternating graph and signed-column reductions). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor, and put $d = \text{VC}(H)$. Allow either of the following reductions whenever it is available:*

1. *delete a vertex of degree two;*

2. compress two signed minimal-nonface columns that agree up to sign, keeping one representative coordinate after reorientation.

Every maximal sequence of these reductions ends at the same ample forbidden family H_* , up to coordinate renaming and reorientation. It has minimum graph degree at least three and pairwise distinct signed minimal-nonface columns up to sign.

More precisely, group each maximal consecutive run of reductions of the same type into a phase. The phase sequence and its intermediate families are canonical: a graph phase takes the full graph three-core, and a column phase compresses all repeated columns. The types alternate. If there are q column phases and s graph phases, then

$$q \leq d - \text{VC}(H_*) \leq d - 3, \quad s \leq q + 1, \quad q + s \leq 2d - 5. \quad (5.43)$$

Every phase preserves the difference $\text{idim}(H) - \text{VC}(H)$.

Conversely, start from any irredundant ample forbidden family B of minimum degree at least three whose signed-clause columns are distinct up to sign. Apply a finite alternating sequence of the following two constructions, starting with either type, or apply neither:

1. nonempty admissible square attachment data of Definition 2.59;
2. coordinate-block replacements with positive integer multiplicities, at least one of which is greater than one.

All outputs are ample forbidden families. Their canonical phase reductions recover exactly the construction sequence, its square attachment data, its coordinate multiplicities, and its base B , up to the renamings allowed in the two constituent recovery theorems. Every ample forbidden family is obtained this way. In particular, a minimum-cardinality ample forbidden family is already fixed by both reductions.

Proof. The decisive fact is that the two types of reduction cannot be available simultaneously. If a forbidden family has repeated signed-clause columns, Theorem 5.38 expresses it as a nontrivial coordinate-block replacement of another ample forbidden family. Theorem 5.36 makes the output have minimum degree at least three. Thus any forbidden family with a degree-two vertex has no repeated columns.

Every allowed reduction preserves ample forbidden-minor status, by Theorems 2.53 and 5.38. It strictly reduces cardinality. This is immediate for a vertex deletion. For compression of a pair, write the family before compression as $W_{\{e,f\}}(U)$. Its excess over the compressed input U is

$$|W_{\{e,f\}}(U)| - |U| = 2|\pi_z U| > 0$$

by (5.40). Here z is the representative coordinate of U , which is nonempty and forbidden. Consequently every reduction sequence terminates.

Whenever a graph reduction is possible, no column reduction is possible until all degree-two vertices have been removed. The graph three-core theorem makes the resulting family independent of every deletion choice. Whenever a column reduction is possible, no graph reduction is possible as long as any repeated pair remains. The canonical column-reduction theorem makes the resulting family independent of the choices and order of all pair compressions. At the end of a phase, that type is exhausted; either the other type is available or both are exhausted. Induction over the strictly decreasing cardinalities proves that the phase sequence, its intermediate families, and its final family are canonical. A forbidden family has no vertex of degree zero or one, by the graph three-core theorem. Exhaustion of both types therefore gives the two asserted properties of H_* .

A graph phase changes neither isometric dimension nor VC dimension. A column phase with multiplicities $m_1, \dots, m_r$ decreases each of these dimensions by $\sum_j (m_j - 1)$, by the coordinate-block shadow formula. This sum is at least one. Hence their difference is invariant and $q \leq d - \text{VC}(H_*)$. Every ample forbidden class has VC dimension at least three [CCMW22, Proposition 4.11]. Alternation permits at most one more graph phase than column phases, proving (5.43).

We finally verify that the inverse description needs no additional compatibility conditions. A nonempty square attachment over a forbidden graph three-core has that core as its unique graph reduction, by Theorem 2.60. Its larger output has a degree-two vertex, so the mutual-exclusion argument gives distinct signed-clause columns. It can therefore serve as the input to the next coordinate-block phase, and canonical column compression will recover it exactly. A nontrivial block replacement of an input with distinct columns has that input as its canonical column reduction; its output has minimum degree at least three and can serve as the next square attachment base. Both statements hold at the initial base B , which has both required properties. Induction proves soundness and exact phase-by-phase recovery for every alternating construction sequence. Conversely, reversing the canonical reductions of H supplies precisely such a sequence and the already proved intrinsic attachment data and multiplicities. The endpoint satisfies both required base conditions. If a minimum-cardinality obstruction admitted either reduction, the strictly smaller forbidden output would contradict minimality. $\square$

The individual full reductions need not commute. Let A be the 579-concept square extension of the 577-concept three-core described after Theorem 2.114, in the coordinates of `damp_three_core_verification.json`. Replace coordinate 0 by $\{0, 24\}$, obtaining a 1603-concept family X . Then add the word 24578, which completes a square on $\{1, 14\}$ in the constant 00-section. The resulting 1604-concept obstruction Y has the forced phase sequence

$$Y_{1604} \xrightarrow{\text{graph}} X_{1603} \xrightarrow{\text{columns}} A_{579} \xrightarrow{\text{graph}} B_{577}, \quad (5.44)$$

with graph, column, and graph phases, respectively. The source has isometric dimension 25 and VC dimension 4; the final base has dimensions 24 and 3. Thus the dimension difference is 21, and three phases attain (5.43) at VC dimension four. On X , graph reduction is the identity, whereas column reduction gives A , whose graph reduction has 577 concepts. Applying the two full reductions once in opposite orders therefore gives different outputs, of orders 579 and 577.

The standalone verifier `verify_damp_alternating_reductions.py` checks the four ample families from actual cubes, replays 294 proper elementary-minor orders, and verifies nonpeelability by the original negative-state certificate or its unchanged reduction. Both column representatives and both graph-deletion priorities give the same final base. The recorded abstract square data reconstruct the two graph phases in either available dependency order. The report is `damp_alternating_reductions_verification.json`.

The theorem accounts for all square attachment and coordinate repetition layers, including their interaction. The remaining intrinsic construction problem lies in the final forbidden bases with minimum degree at least three and distinct signed-clause columns. These two conditions alone do not imply forbiddenness. For example, take the 42-concept down-set of words of Hamming weight at most three in Q_6 . Its shattered complex is that same down-set, so it is ample; deleting maximal words preserves this property and gives a peeling. Vertices of weight three have degree three, and all other vertices have degree six. Its minimal nonfaces are all four-element supports, each with missing trace 1111. For any two coordinates there is such a support containing one and omitting the other, so their columns cannot agree up to sign. The same verifier checks this positive control. Nor does being fixed by these two reductions imply that every nonempty-coordinate reduction peels: the preceding 2318-concept example remains a counterexample to that implication.

The earlier opposite-status clause pairs give stronger tests of this remaining problem. Take the intermediate families X'_j, Y'_j in Corollary 2.94 for $j = 0, 1$, of orders 577, 1158. All four are periphery-free and already fixed by both reductions. Their respective minimum-degree pairs are $(4, 3)$ and $(4, 4)$. Within each pair the full shadow is identical, only one four-coordinate forbidden trace changes, every proper pc-minor peels, and the two original families have opposite peeling status. Thus the reduced-base problem still retains the signed distinction in these existing examples. The independent replay `verify_damp_reduced_base_clause_pairs.py` checks actual cubes, all signed clauses, both fixed-point conditions, the two positive orders, the complete negative certificates, and all 300 proper-minor orders. Its report is `damp_reduced_base_clause_pairs_verification.json`. This is a verification of the remaining difficulty, not a new construction or an intrinsic admissibility criterion.

Remark 5.40 (What overlap descent does and does not classify). Theorem 5.33 separates two genuinely different parts of the requested parametrization. Its terminal objects are the *overlap-atomic obstructions*, namely those forbidden pc-minors whose one-coordinate reductions all belong to $\mathcal{D}_{\text{amp}}$. Every other obstruction is obtained from a smaller obstruction by one or more inverse steps of (5.36). The antipodal two-repair towers are explicit examples of such inverse steps.

This is a completeness decomposition, not yet a generative classification. In particular, the three displayed $\mathcal{D}_{\text{amp}}$ -memberships in the theorem are consequences for a recovered obstruction; using them as admissibility tests for arbitrary wings would be circular. Two intrinsic tasks remain: classify the overlap-atomic seeds by their signed Yang gluing data, and replace those three memberships by local attachment axioms that guarantee a valid inverse repair step.

Remark 5.41 (The candidate global Yang invariant). Proposition 5.32 identifies a smaller global object than a peeling order. By Theorem 5.33, it is needed only for the terminal overlap-atomic seeds. For every coordinate of such a seed, retain the paired relative Yang complexes

$$\mathfrak{R}_e(H) = ((\Delta_{B_0}, \Delta_A), (\Delta_{B_1}, \Delta_A)). \quad (5.45)$$

Each relative face module is fully Cohen–Macaulay and has 0/1 flag h -support $\text{Sh}(B_b) \setminus \text{Sh}(A)$, by Lemma A.12. Thus ordinary homology and the flag h -vector cannot detect the gluing-critical signature. What is missing is a finite signed incidence invariant describing how the two relative modules attach to their common Yang ball. Such an invariant must be extracted directly from the oriented one-hole and punctured expansion profiles and must force the absence or presence of a relative shelling without including a shelling order in its data.

This is a research reduction, not the requested intrinsic parametrization. In particular, replacing (5.45) by a search over relative shellings would merely rename the original problem. The expression *extension/holonomy class* is therefore only a specification for the missing invariant, not the name of an invariant already constructed.

The relative Yang theory nevertheless extracts a first, genuinely static part of the missing invariant. For $A \subseteq B \subseteq Q_F$, write

$$G(B, A) := Q_F[B \setminus A]$$

for the graph induced by the relative facets. Its edges are naturally labelled by their cube coordinates.

Proposition 5.42 (Cycle carriers at an overlap-atomic coordinate). *Let $H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be overlap-atomic, let e be active, and write*

$$B_b = \rho_e^b H, \quad A = B_0 \cap B_1, \quad R_b = B_b \setminus A.$$

Then both graphs $G(B_0, A) = Q[R_0]$ and $G(B_1, A) = Q[R_1]$ contain a cycle. In particular,

$$|R_0|, |R_1| \geq 4, \quad |H| = 2|A| + |R_0| + |R_1| \geq 2|A| + 8. \quad (5.46)$$

Proof. The overlap A and the two sections B_0, B_1 belong to $\mathcal{D}_{\text{amp}}$. In particular, the relative Yang module of each nested pair $A \subseteq B_b$ is Cohen–Macaulay. The forest theorem for relative Yang modules says that if $G(B_b, A)$ is a forest, then (Δ_{B_b}, Δ_A) has a relative shelling [Bou26]. Proposition 5.32 would then make H corner-peelable, a contradiction. Thus both relative-facet graphs contain cycles. A cube graph is bipartite, so each cycle has at least four vertices. The section decomposition gives

$$|H| = |B_0| + |B_1| = 2|A| + |R_0| + |R_1|,$$

which proves (5.46). $\square$

Remark 5.43 (The rooted signed cycle system). Proposition 5.42 identifies the first nonzero layer of a possible global invariant. It is useful to retain more than the two unlabelled cycle spaces. For a nested pair $A \subseteq B$, adjoin a root vertex $*$ to $G(B, A)$, and for every $x \in B \setminus A$ and every coordinate q with $x^{\{q\}} \in A$, add a root edge $x*$ labelled by the signed boundary incidence $(q, x^{\{q\}})$. Together with the coordinate labels on the internal edges, this gives a finite rooted signed graph

$$\widehat{G}(B, A).$$

It is recovered directly from the facet–ridge incidence of the relative Yang pair and involves no ordering. Thus an overlap-atomic coordinate canonically supplies the paired rooted cycle system

$$(\widehat{G}(B_0, A), \widehat{G}(B_1, A)).$$

The proposition proves that its two internal cycle spaces are nonzero. Cycle rank alone is not sufficient: shellable relative pairs can also have cyclic relative-facet graphs. The full signed rooted graph carries more information. Proposition 5.44 below shows that it is the complete first-syzygy presentation of the relative Alexander-dual module and hence determines relative shellability. Thus higher-dimensional Yang incidences are not needed to decide the individual relative problem. The remaining difficulty is to describe the nonshellable rooted presentations structurally, and then to describe the compatibility of the two rooted presentations at a gluing, without hiding a linear-quotient order in admissibility.

Retaining every concept label and boundary endpoint would still be too close to copying the two sections to serve as the final parameters. The presentation theorem instead directs attention to smaller labelled block, cut, or circuit data extracted from the graph. The root-cut criterion in the same proposition is the first such compression.

For the next statement, fix a field k , put

$$S_F = k[v_{q,0}, v_{q,1} : q \in F], \quad m_x = \prod_{q \in F} v_{q,1-x_q}, \quad J_X = (m_x : x \in X),$$

and, for $A \subseteq B \subseteq Q_F$, write

$$Q(B, A) = J_B / J_A.$$

The vertices of $\widehat{G}(B, A)$ are signed by their cube words. Thus an edge in coordinate q incident with x carries both its coordinate and the literal v_{q,x_q} .

Proposition 5.44 (Rooted presentation and root-cut criterion). *Let $A \subsetneq B \subseteq Q_F$ be ample and put $D = B \setminus A$, with $J_\emptyset = 0$. Let*

$$\varphi : \bigoplus_{x \in D} S_F \varepsilon_x \longrightarrow \mathcal{Q}(B, A), \quad \varepsilon_x \longmapsto \overline{m_x}.$$

Then

$$\begin{aligned} \ker \varphi = & \sum_{\substack{x \in D, q \in F \\ x^{q\} \in A}} S_F v_{q, x_q} \varepsilon_x \\ & + \sum_{\substack{\{x, y\} \in E(Q[D]) \\ y = x^{q\}}} S_F (v_{q, x_q} \varepsilon_x - v_{q, y_q} \varepsilon_y). \end{aligned} \quad (5.47)$$

Consequently, the signed rooted graph $\widehat{G}(B, A)$ determines the relative module with its distinguished generators and therefore determines all relative shelling orders of (Δ_B, Δ_A) .

More explicitly, for $x \in D$, let

$$L_A(x) = \{q \in F : x^{q\} \in A\}$$

be the coordinate labels of the root edges incident with x . Delete from $\widehat{G}(B, A)$ every internal or root edge whose coordinate label belongs to $L_A(x)$. Then

$$A \cup \{x\} \text{ is ample} \iff x \text{ and the root lie in different components of the resulting graph.} \quad (5.48)$$

Proof. Since A and B are ample, the relative Yang face module is Cohen–Macaulay. Its squarefree Alexander dual $\mathcal{Q}(B, A)$ therefore has an $|F|$ -linear resolution; in particular, it is linearly presented. The general monomial presentation has one annihilator relation for every pair $x \in D, a \in A$, and one pair relation for every two members of D . A relation of the first kind is linear exactly when x and a are cube neighbours, and a relation of the second kind is linear exactly when its two concepts are cube neighbours. Since the entire kernel is generated by its linear part, these are precisely the root and internal relations displayed in (5.47).

The presentation matrix is read verbatim from the signed rooted graph. An isomorphism of such graphs, including the coordinate and sign labels, therefore identifies the relative modules and their distinguished minimal generators. Linear quotients of those generators are the Alexander-dual form of a relative shelling, proving the comparison assertion.

It remains to prove the cut criterion. The monomial colon formula gives

$$J_A : m_x = \left(\prod_{q \in \text{Diff}(x, a)} v_{q, x_q} : a \in A \right). \quad (5.49)$$

Its linear monomials are exactly v_{q, x_q} for $q \in L_A(x)$. By the one-concept colon criterion, $A \cup \{x\}$ is ample precisely when every generator in (5.49) is divisible by one of these variables.

Suppose first that the edge-deleted graph contains an x -to-root path. Its internal edges and final root edge all have labels outside $L_A(x)$. The endpoint $a \in A$ of the final root edge consequently agrees with x on every coordinate in $L_A(x)$. The corresponding generator in (5.49) is divisible by none of the displayed linear variables, so $A \cup \{x\}$ is not ample.

Conversely, suppose that the colon is not generated by those variables. Then some $a \in A$ agrees with x on $L_A(x)$. Apply (5.47) to the annihilator relation indexed by x, a , in its signed multidegree μ . A relative generator divides μ exactly when its cube word lies in the Boolean interval $[x, a]$. After dividing each surviving basis vector by the unique monomial needed to reach μ , an internal edge relation becomes $e_z - e_w$, a root relation becomes e_z , and the annihilator relation indexed by x, a becomes e_x . Consequently e_x lies in the span of the edge differences and rooted unit vectors of this finite graph. In any component containing no rooted unit vector, the coordinate-sum functional vanishes on every edge difference but not on e_x . The component of x must therefore contain a root edge. This gives an x -to-root path whose labels all belong to $\text{Diff}(x, a)$, hence avoid $L_A(x)$. Thus x and the root remain connected after the prescribed edge deletion, proving (5.48). $\square$

Definition 5.45 (Root-locked and biroot-locked relative pairs). The nested ample pair $A \subsetneq B$ is *lower-root-locked* if, for every $x \in B \setminus A$, the root remains connected to x after deleting all edges labelled in $L_A(x)$. Equivalently, no one-concept ample extension of A lies in B .

Apply the same construction to the complementary nested pair

$$Q_F \setminus B \subsetneq Q_F \setminus A.$$

We call $A \subsetneq B$ *upper-root-locked* if that complementary pair is lower-root-locked, and *biroot-locked* if it is both lower- and upper-root-locked.

Corollary 5.46 (Static terminal-pair certificate). *Lower-root-locking is a finite, ordering-free certificate that (Δ_B, Δ_A) is not relatively shellable. Upper-root-locking is equivalent to*

$$B \setminus \{x\} \text{ being nonample for every } x \in B \setminus A.$$

Thus a biroot-locked pair admits neither a first nor a last relative shelling facet. Both assertions are decided entirely by labelled cut tests in the two rooted graphs.

Proof. The first assertion follows immediately from Proposition 5.44: a relative shelling must begin with an ample one-concept extension of A . For the second, observe that

$$B \setminus \{x\} \text{ is ample} \iff (Q_F \setminus B) \cup \{x\} \text{ is ample,}$$

and apply the same root-cut criterion to the complementary pair. $\square$

Remark 5.47 (What the rooted presentation accomplishes). The presentation theorem changes the proof contract in two ways. First, the signed rooted graph is not merely a cycle-count proxy: it is a complete invariant of each individual relative shelling problem. Second, lower- or upper-root-locking gives a direct nonshellability certificate without a shelling or carrier order. The remaining classification problem is now sharper. One must extract a smaller family of labelled cut blocks whose intrinsic assembly axioms force the required locking, describe how two such presentations attach to the same overlap, and recover those blocks from every overlap-atomic obstruction. The full rooted graph nearly records the original pair, while merely asking whether its presentation has linear quotients would still be circular. Neither is an admissible final parametrization.

Definition 5.48 (Relative one-hole block cover). Let $A \subseteq Q_F$, and let $\mathcal{W}$ be a finite family of oriented one-hole data (y, S) on F . Put

$$l(y, S) = \{y^T : T \subseteq S, |T| \geq 2\}, \quad B(A, \mathcal{W}) = A \cup \bigcup_{(y, S) \in \mathcal{W}} P(y, S).$$

We call $(A, \mathcal{W})$ an *admissible relative one-hole cover* if

1. A and $B(A, \mathcal{W})$ satisfy their Sandwich equalities;
2. every displayed tip is absent: $y \notin B(A, \mathcal{W})$ for $(y, S) \in \mathcal{W}$; and
3. $B(A, \mathcal{W}) \setminus A \neq \emptyset$; and
4. the interiors of the displayed punctured cubes protect every relative concept:

$$B(A, \mathcal{W}) \setminus A \subseteq \bigcup_{(y, S) \in \mathcal{W}} \mathsf{l}(y, S). \quad (5.50)$$

These are finite conditions on A and the displayed punctured cubes. In particular, admissibility contains no deletion, corner, shelling, linear-quotient, or rooted-connectivity predicate.

Theorem 5.49 (One-hole normal form for upper-root-locked pairs). *Every admissible relative one-hole cover constructs a nested ample pair*

$$A \subseteq B(A, \mathcal{W})$$

that is upper-root-locked. Conversely, every upper-root-locked nested ample pair $A \subsetneq B \subseteq Q_F$ is obtained in this way.

More precisely, put $D = B \setminus A$. Its canonical recovered block family is

$$\mathcal{W}_{\max}(B, A) = \left\{ (y, S) : \begin{array}{l} y \notin B, \mathsf{P}(y, S) \subseteq B, |S| \geq 2, \\ \mathsf{l}(y, S) \cap D \neq \emptyset, \text{ and } S \text{ is inclusion-maximal with } \mathsf{P}(y, S) \subseteq B \end{array} \right\}. \quad (5.51)$$

Consequently, upper-root-locked relative Yang presentations admit an explicit, descendant-free construction and recovery by punctured-cube blocks.

Proof. Let $(A, \mathcal{W})$ be admissible and put $B = B(A, \mathcal{W})$. The two Sandwich equalities make A and B ample. Fix $x \in B \setminus A$. By (5.50), some $(y, S) \in \mathcal{W}$ and $T \subseteq S$, $|T| \geq 2$, satisfy $x = y^T$. For every proper $C \subsetneq T$,

$$x^C = y^{T \Delta C} \in \mathsf{P}(y, S) \subseteq B, \quad x^T = y \notin B.$$

Thus T is a nontrivial minimal nonface of $\Lambda_B(x)$. The concept x is therefore contained in more than one maximal cube of B , so it is not a corner of the ample family B . Equivalently, $B \setminus \{x\}$ is nonample. This holds for every $x \in B \setminus A$, which is upper-root-locking.

Conversely, let $A \subsetneq B$ be upper-root-locked and ample. For each $x \in B \setminus A$, the family $B \setminus \{x\}$ is nonample. Hence x is not a corner of B , so its local cube-support complex $\Lambda_B(x)$ is not a simplex and has a minimal nonface S of order at least two. For every such S , put $y = x^S$. Minimality gives

$$\mathsf{P}(y, S) \subseteq B, \quad y \notin B, \quad x \in \mathsf{l}(y, S).$$

Extend each such S , with its tip y fixed, to an inclusion-maximal support S' satisfying $\mathsf{P}(y, S') \subseteq B$. Then $x \in \mathsf{l}(y, S')$, so the finite family $\mathcal{W}_{\max}(B, A)$ protects all of $B \setminus A$. Every one of its punctured cubes lies in B . It follows that

$$B(A, \mathcal{W}_{\max}(B, A)) = A \cup (B \setminus A) = B.$$

The missing-tip conditions hold by construction, while both Sandwich equalities follow from the ampleness of A and B . Hence the recovered cover is admissible and reconstructs the original pair. $\square$

Corollary 5.50 (Root cuts and one-hole blocks). *For nested ample $A \subsetneq B$, the following are equivalent:*

1. $A \subsetneq B$ is upper-root-locked;
2. every $x \in B \setminus A$ belongs to $\mathsf{l}(y, S)$ for some punctured cube $\mathsf{P}(y, S) \subseteq B$ with $y \notin B$ and $|S| \geq 2$;
3. in the complementary rooted presentation, every relative vertex remains joined to the root after deleting the labels incident with it at the root.

If $x = y^T \in \mathsf{l}(y, S)$ is protected as in (2), then $T \cap \{q : x^{\{q\}} \notin B\} = \emptyset$. The block therefore gives, through Proposition 5.44, a rooted path avoiding all deleted labels.

Proof. The equivalence of (1) and (2) is Theorem 5.49; the equivalence of (1) and (3) is the complementary root-cut criterion. For the final assertion, $x^{\{q\}} \in B$ for every $q \in T$, because $T \setminus \{q\}$ is a nonempty subset of S . Hence no coordinate of T is deleted in the complementary cut test. The root-cut criterion converts the corresponding nonlinear missing-tip relation into the asserted path. $\square$

Remark 5.51 (What has and has not been parametrized). Theorem 5.49 closes the individual upper-locking row: one need not classify arbitrary rooted graphs or use locking as an admissibility oracle. The primitive blocks may be taken to be the inclusion-maximal punctured cubes, one of which can protect many relative concepts, and they are canonically recovered. This is not yet the basis classification. The Hall audit shows that both wings at every tested coordinate have this normal form, but the missing theorem must still characterize which paired covers over a common overlap have all elementary pc-minors peelable. In particular, appending arbitrary shelling certificates for those images would violate the non-circularity gate.

Definition 5.52 (Internal and cross-layer protection). For $B \subseteq Q_F$, let

$$\text{Int}(B) = \{x \in B : \text{there are } y \notin B, S \subseteq F \text{ with } \mathsf{P}(y, S) \subseteq B \text{ and } x \in \mathsf{l}(y, S)\}. \quad (5.52)$$

For $B_0, B_1 \subseteq Q_F$, put $A = B_0 \cap B_1$. For $b \in \{0, 1\}$, let $\text{Cross}_b(B_0, B_1)$ consist of the words $a \in A$ for which there is a nonempty $T \subseteq F$ such that

$$\{a^U : U \subseteq T\} \subseteq B_b, \quad \{a^U : U \subsetneq T\} \subseteq B_{1-b}, \quad a^T \notin B_{1-b}. \quad (5.53)$$

Thus internal protection is supplied by an absent-tip punctured cube in one section. Cross-layer protection is supplied by a full T -cube on one side and the same cube punctured at a^T on the other.

Theorem 5.53 (Paired one-hole criterion for a cornerless gluing). *Let $B_0, B_1 \subseteq Q_F$, and form the two-layer family*

$$H = (B_0 \times \{0\}) \cup (B_1 \times \{1\}) \subseteq Q_{F \cup \{e\}}.$$

Suppose that H is nonempty and satisfies the Sandwich equality. Then H is cornerless if and only if

$$B_b = \text{Int}(B_b) \cup \text{Cross}_b(B_0, B_1) \quad (b \in \{0, 1\}). \quad (5.54)$$

All the sets in (5.54) are computed directly from full and punctured affine cubes in the two sections. Hence this is an explicit, ordering-free construction and recovery theorem for the nonpeelable root of a two-layer gluing.

Proof. The Sandwich equality makes H ample. Fix $x \in B_b$ and write $\tilde{x} = (x, b)$. By the local-hole characterization, $\tilde{x}$ is not a corner of H exactly when there are a missing tip $z \notin H$ and a support R , $|R| \geq 2$, such that

$$\tilde{x} = z^R, \quad P(z, R) \subseteq H. \quad (*)$$

Suppose first that $e \notin R$. The entire punctured cube in $(*)$ lies in layer b . Suppressing e gives a punctured cube in B_b whose protected interior contains x . Thus $x \in \text{Int}(B_b)$. Conversely, every block in (5.52) lifts within layer b and gives $(*)$.

Now suppose that $e \in R$, and put $T = R \setminus \{e\}$. Since $|R| \geq 2$, the set T is nonempty. The tip is

$$z = (x^T, 1 - b).$$

The layer- b section of $P(z, R)$ is the full T -cube through x , whereas its layer- $(1 - b)$ section is that cube punctured at x^T . The assertion $z \notin H$ says precisely that $x^T \notin B_{1-b}$. These are exactly the three conditions in (5.53). The converse follows by reversing the same construction: the full/punctured pair in (5.53) is $P((x^T, 1 - b), T \cup \{e\})$.

We have proved, for every $x \in B_b$,

$$(x, b) \text{ is not a corner of } H \iff x \in \text{Int}(B_b) \cup \text{Cross}_b(B_0, B_1).$$

The family H is cornerless exactly when this holds for both layers, which is (5.54). $\square$

Remark 5.54 (The remaining paired problem). Equation (5.54) is the missing nonpeelability compatibility law for the two sides: internal holes protect a section concept on their own, while full/punctured cross blocks protect the concepts exposed by the section split. It does not assert that the $3|F \cup \{e\}|$ elementary images are peelable. The intrinsic generative classification now requires attachment axioms on these same blocks whose transported direct and switched profiles construct peelable elementary images without storing or searching for their shellings.

Definition 5.55 (Latent punctured-face system). Let $H \subseteq Q_F$. Its *latent punctured-face system* is

$$\mathcal{L}(H) = \{(y, S) : y \in Q_F, S \subseteq F, |S| \geq 2, P(y, S) \subseteq H\}. \quad (5.55)$$

The tip y is allowed to belong to H . If it does, then $\{y\} \cup P(y, S)$ is a full S -cube; deleting y activates the punctured face. Thus $\mathcal{L}(H)$ records both the holes already present in H and the holes that can appear later during a peeling.

For an ordering $\mathbf{x} = (x_1, \dots, x_m)$ of H , put

$$\text{pos}_{\mathbf{x}}(y) = \begin{cases} 0, & y \notin H, \\ j, & y = x_j, \end{cases}$$

and let $b_{\mathbf{x}}(y, S)$ be the first member of $P(y, S)$ in this ordering.

Theorem 5.56 (Latent punctured-face word criterion). *Let $H \subseteq Q_F$ be ample. An ordering $\mathbf{x} = (x_1, \dots, x_m)$ is a corner peeling if and only if, for every $(y, S) \in \mathcal{L}(H)$,*

$$\text{pos}_{\mathbf{x}}(y) < \text{pos}_{\mathbf{x}}(b_{\mathbf{x}}(y, S)) \implies b_{\mathbf{x}}(y, S) = y^{\{q\}} \text{ for some } q \in S. \quad (5.56)$$

In words: if the tip has disappeared before any displayed vertex of a punctured cube, then the first displayed vertex must be a neighbor of the tip.

Proof. Suppose first that $\mathbf{x}$ is a corner peeling. Fix $(y, S) \in \mathcal{L}(H)$, let $x = b_{\mathbf{x}}(y, S) = y^T$, and suppose that y occurs before x , with a missing tip understood to occur at time zero. Immediately before x is deleted, the tip is absent and every member of $P(y, S)$ is still present. If $|T| \geq 2$, then $P(y, T)$ is still present. Equivalently, every proper T -flip of x is present while $x^T = y$ is absent. The local cube-support complex at x is therefore not a simplex, so x is not a corner, a contradiction. Hence $|T| = 1$, which is (5.56).

Conversely, suppose that the displayed rule holds. We prove inductively that x_j is a corner of $H_j = \{x_j, \dots, x_m\}$. The preceding deletions preserve ampleness by the induction hypothesis, so H_j is ample. If x_j were not a corner, its local cube-support complex would have a nonface $T \subseteq F$ of order at least two. Replacing T by a minimal nonface if necessary and putting $y = x_j^T$, we obtain

$$y \notin H_j, \quad P(y, T) \subseteq H_j.$$

Thus $(y, T) \in \mathcal{L}(H)$, the tip y is either absent from H or occurs before x_j , and x_j is the first member of $P(y, T)$. Since $x_j = y^T$ and $|T| \geq 2$, this contradicts (5.56). Therefore every x_j is a corner, and $\mathbf{x}$ is a corner peeling. $\square$

The word criterion admits a particularly small finite state space. Write $a(y, S) = y^S$ for the antipode of a latent block. Once the tip has disappeared, a corner deletion of the antipode forces one of its S -neighbors to have disappeared already.

Corollary 5.57 (Finite facet-protector criterion). *Let $H \subseteq Q_F$ be ample. For each $(y, S) \in \mathcal{L}(H)$, choose one of the following states:*

1. if $y \in H$, the escape state, contributing the arc

$$y^S \longrightarrow y;$$

2. a facet-protector state marked by a coordinate $q \in S$, contributing the arc

$$y^{S \setminus \{q\}} \longrightarrow y^S.$$

Then H is corner-peelable if and only if the states can be chosen so that the union of their arcs is acyclic.

Proof. Let $\mathbf{x}$ be a corner peeling and fix a latent block (y, S) . If y^S precedes the present tip y , choose the escape state. Otherwise the tip is absent or precedes the antipode $a = y^S$. Immediately before a is deleted, suppose that every neighbor $y^{S \setminus \{q\}}$, $q \in S$, is still present. Then every coordinate of S is incident with a in the remaining family. Because a is a corner, the unique maximal cube through a contains the cube generated by all its incident coordinates, including the S -cube. This would put the tip y in the remaining family, a contradiction. Hence some S -neighbor precedes a ; choose its coordinate as the facet-protector state. Every selected arc points forward in $\mathbf{x}$, so their union is acyclic.

Conversely, topologically order an acyclic selected-arc digraph and suppose that the word rule fails for a latent block (y, S) . Write its first displayed vertex as $x = y^T$, where $|T| \geq 2$. The subblock (y, T) is also latent, its antipode is x , and its tip is absent or precedes x . It cannot carry the escape state, whose arc $x \rightarrow y$ would reverse these positions. Its selected state is therefore a facet protector

$$y^{T \setminus \{q\}} \longrightarrow x \quad (q \in T).$$

The protector is another displayed vertex of the original block and precedes x , contradicting the choice of x . Thus the topological order satisfies (5.56), and Theorem 5.56 makes it a corner peeling. $\square$

Proposition 5.58 (Contraction transport of facet protectors). *Let $H \subseteq Q_{F \cup \{e\}}$ be ample and put $K = \pi_e H$. For $z \in K$, define its lift set*

$$L_e(z) = \{b \in \{0, 1\} : (z, b) \in H\}.$$

Then every edge $zz^{\{q\}}$ of K has a nonempty owner set

$$\Omega_e(z, q) = L_e(z) \cap L_e(z^{\{q\}}) \neq \emptyset. \quad (5.57)$$

Consequently, if $(y, S) \in \mathcal{L}(K)$ has antipode $z = y^S$, then every facet-protector state $q \in S$ lifts in each layer belonging to $\Omega_e(z, q)$. Contraction preserves the coordinate q ; its only additional state is the one- or two-element owner set $\Omega_e(z, q)$.

For a switched punctured expansion profile $P_S = D \dot{\cup} Z_0 \dot{\cup} Z_1$, written in the coordinates based at its antipode z , one has

$$\Omega_e(z^C, q) = \begin{cases} \{0, 1\}, & C, C \triangle \{q\} \in D, \\ \{b\}, & \{C, C \triangle \{q\}\} \not\subseteq D \text{ and its non-} D \text{ endpoints lie in } Z_b. \end{cases} \quad (5.58)$$

Thus the separator and component colours recover the transport of every facet-protector state without a peeling order.

Proof. Suppose that $L_e(z) \cap L_e(z^{\{q\}}) = \emptyset$. Both lift sets are nonempty because their projected vertices belong to K ; hence they are opposite singletons, say

$$(z, 0), (z^{\{q\}}, 1) \in H, \quad (z, 1), (z^{\{q\}}, 0) \notin H.$$

The two present vertices have Hamming distance two. Their only two possible intermediate cube vertices are precisely the two absent vertices displayed above, contradicting isometry of the ample family H . This proves (5.57). The protector of a latent block is the edge $z^{\{q\}}z$, so the first consequence follows immediately.

In a punctured expansion profile, the vertices of D have both lifts, whereas the vertices of Z_b have only their layer- b lift. The canonical profile theorem shows that no edge joins Z_0 to Z_1 . Intersecting the two endpoint lift sets now gives exactly (5.58). $\square$

Remark 5.59 (What remains after protector transport). Facet-protector transport is canonical and coordinate preserving. The remaining compatibility is smaller but still global: owner choices must agree when blocks share protector edges, and the resulting protector arcs must synchronize with escape arcs $y^S \rightarrow y$ of blocks whose tips are present. Selecting owners and testing the resulting digraph for acyclicity would again be recognition. The desired construction theorem must instead derive these choices from the incidence of the ample separators D , their monochromatic components Z_b , and the paired section attachments. This is the precise state-transport problem on which a signed holonomy invariant can now be tested.

Definition 5.60 (Colored Yang protector atlas). Let $H \subseteq Q_F$ be ample and write F_x for the Yang facet belonging to $x \in H$. For $(y, S) \in \mathcal{L}(H)$, define the *punctured cross-polytopal chart*

$$\mathcal{C}_H(y, S) = \langle F_{y^T} : \emptyset \neq T \subseteq S \rangle \subseteq \Delta_H.$$

It is the join of the simplex on the literals fixed outside S with the boundary of the S -cross-polytope after the relative interior of the facet indexed by $y|_S$ has been removed. Its distinguished *antipodal facet* is $A(y, S) = F_{y^S}$. For $q \in S$, its q -protector facet is

$$P_q(y, S) = F_{y^{S \setminus \{q\}}};$$

this is the facet of the chart sharing the q -ridge with $A(y, S)$. If $y \in H$, the additional facet $E(y, S) = F_y$ is called the *escape facet*. The family of these labelled charts and facets is the *colored Yang protector atlas* $\mathcal{A}(H)$.

Corollary 5.61 (Yang protector-atlas criterion). *Let $H \subseteq Q_F$ be ample. In each chart of $\mathcal{A}(H)$, choose either*

$$A(y, S) \prec E(y, S) \quad \text{when the escape facet is present,}$$

or

$$P_q(y, S) \prec A(y, S) \quad \text{for some } q \in S.$$

Then the Yang complex Δ_H is shellable if and only if these local precedence relations can be chosen so that their union is acyclic. In that case every linear extension is the facet order corresponding, under the shellability–corner-peeling dictionary, to a corner peeling of H .

Proof. Under the facet–concept bijection, the displayed relations are exactly the escape and facet-protector arcs of Corollary 5.57. Apply that corollary and the shellability–corner-peeling correspondence. $\square$

An elimination order is *distance-preserving* if every nonempty suffix induces an isometric subgraph of the original graph. A vertex v is *weakly 4-simplicial* if every two of its neighbors have distance at most two after v is deleted. A weakly 4-simplicial elimination order tests this condition in the graph remaining at each step.

Proposition 5.62 (Yang shellability is metric elimination). *Let $\emptyset \neq H \subseteq Q_F$ be isometric, and let $G_H = Q_F[H]$ be its concept graph. Then the following are equivalent:*

1. *the Yang complex Δ_H is shellable;*
2. *Δ_H is strongly shellable;*
3. *G_H has a distance-preserving elimination order; and*
4. *G_H has a weakly 4-simplicial elimination order.*

Every isometric family satisfying these conditions is itself ample and corner-peelable. More precisely, its distance-preserving elimination orders are exactly its corner-peeling orders, whose reversals give the corresponding Yang shellings.

Proof. All Yang facets have the same cardinality. Use the standard pairwise criterion for a pure shelling order. For earlier and later facets F_x, F_y , respectively, it supplies an earlier facet F_z such that

$$|F_y \setminus F_z| = 1, \quad F_x \cap F_y \subseteq F_z.$$

Suppose that F_z and F_y differ in color e . If $x_e = y_e$, then the e -literal belongs to $F_x \cap F_y$ but not to F_z , a contradiction. Hence $x_e \neq y_e$, and the new e -literal of F_z belongs to F_x . Therefore $F_z \subseteq F_x \cup F_y$, which is the additional condition for a strong shelling. Thus ordinary and strong shellability coincide for Yang complexes.

The codimension-one graph of Δ_H is G_H : two Yang facets meet in a ridge exactly when their concepts differ in one coordinate. Moreover, the facet distance between F_x and F_y is their Hamming distance. Since H is isometric, this equals the distance between x and y in G_H . Thus the codimension-one graph is harmonious. The graph characterization of pure strong shellability now

identifies item 2 with a distance-preserving elimination order [GSW19, Theorem 4.7]. Deleting a vertex preserves all distances exactly when that vertex is weakly 4-simplicial, so items 3 and 4 are equivalent [SZ25, Lemma 3.3 and Proposition 3.4]. To obtain the last assertion without assuming ampleness, reverse a distance-preserving elimination order, starting from its last singleton. Every next family is isometric in the cube. By Lemma 2.10, each addition preserves ampleness and adds a corner. Thus the original order is a corner peeling. Conversely, every suffix of a corner peeling is ample and therefore isometric. This proves the order-level assertion as well. $\square$

Proposition 5.63 (Metric criterion for a relative peeling). *Let $\emptyset \neq A \subseteq H \subseteq Q_F$ be ample, and list $H \setminus A$ as $x_1, \dots, x_k$. For a distance-two pair $u, v \in H$, put $C_H(u, v) = N_H(u) \cap N_H(v)$. The following conditions are equivalent:*

1. *Deleting $x_1, \dots, x_k$ is a relative corner peeling of H to A , with all of A retained.*
2. *Each $H \setminus \{x_1, \dots, x_j\}$, $0 \leq j \leq k$, is isometric in the original concept graph of H .*
3. *For every distance-two pair $u, v \in H$, either $C_H(u, v) \cap A \neq \emptyset$, or there are $z \in \{u, v\} \setminus A$ and $w \in C_H(u, v)$ such that z precedes w in the displayed deletion list.*

The target A need not be corner-peelable. A complete peeling of H with A as its final block exists exactly when such a relative peeling exists and A itself is corner-peelable.

Proof. A relative corner peeling leaves ample families, hence isometric families, proving $1 \Rightarrow 2$. Conversely, reverse a sequence in item 2, starting at the ample family A . At each addition the enlarged family is isometric in the cube. The metric extension criterion in Lemma 2.10 makes it ample and makes the added concept a corner. Reversing these additions proves $2 \Rightarrow 1$. This argument uses ampleness of A , but no peeling order for A .

We prove the relative version of the original-graph criterion [CDNS18, Corollary 2]. Assume item 2 and take u, v with $C_H(u, v) \cap A = \emptyset$. The pair cannot lie wholly in A , since A is isometric. Immediately before the first of its endpoints is deleted, an isometric residual graph still contains a common neighbor w . This neighbor is outside A and is deleted later than that endpoint, giving item 3.

Conversely, suppose item 3 and consider a first deletion x_j that fails to preserve distances. Its preceding graph is isometric in H . There must be two nonadjacent neighbors u, v of x_j with no common neighbor after the deletion: otherwise each two-edge segment through x_j on a shortest path could be replaced by a surviving two-edge segment, preserving every distance. The pair has distance two in H , all of $C_H(u, v)$ has been deleted by step j , and both endpoints remain. Neither alternative in item 3 can hold, a contradiction. This proves $3 \Rightarrow 2$.

Finally, concatenate a relative peeling with a peeling of A . Conversely, the two portions of a complete peeling with final block A give exactly those two orders. $\square$

Both target qualifications matter. In Q_3 , the deletion list 000, 111 leaves isometric graphs and satisfies item 3 for $A = Q_3 \setminus \{000, 111\}$, but its second deletion is not a corner deletion. The target is an isometric six-cycle, which is not ample. In the opposite direction, take the cornerless ample 291-concept seed G and its resolving extension $G \cup \{91\}$ from Remark 2.50. Deleting 91 is a relative peeling to G , although no complete peeling can have G as its final block.

The checker `verify_damp_relative_metric_peeling.py` verifies all 504 one-concept extension candidates over the 127 nonempty ample families in Q_3 , all 720 orders from Q_3 to a fixed edge, and both target qualifications above. Exactly 344 of the extensions and 176 of the relative orders pass their equivalent tests. Its report is `damp_relative_metric_peeling_verification.json`. This

gives an exact way to retain an ample core while expressing the exterior constraints in the original graph. Existence of the deletion list is still an ordering predicate, so it is not intrinsic admissibility for the generative classification.

The canonical overlap is a particularly restrictive choice of retained target. Its two wings cannot repair one another while the overlap is kept fixed. The next proposition makes this statement at the level of the original metric constraints, including partial deletion lists.

Proposition 5.64 (Metric separation across an ample overlap). *Let $H \subseteq Q_{F \cup \{e\}}$ be ample, with e active, and put*

$$B_b = \rho_e^b H, \quad A = B_0 \cap B_1, \quad C = B_0 \cup B_1, \quad D_b = B_b \setminus A.$$

These four families B_0, B_1, A, C are nonempty and ample. For $X \in \{B_0, B_1, C\}$, let $C_X(u, v)$ be the common-neighbor set of a distance-two pair in X , and define

$$\mathcal{T}_A(X) = \{(\{u, v\}, C_X(u, v)) : d_{G_X}(u, v) = 2, C_X(u, v) \cap A = \emptyset\}.$$

Then

$$\mathcal{T}_A(C) = \mathcal{T}_A(B_0) \dot{\cup} \mathcal{T}_A(B_1). \quad (5.59)$$

All vertices outside A in a member of $\mathcal{T}_A(B_b)$ belong to D_b . Thus these are independent constraint systems on the two wings, with their vertices in A retained permanently.

More precisely, let σ be any list of distinct vertices of $C \setminus A$, and let σ_b be its subsequence in D_b . Every successive deletion in σ preserves distances in C if and only if every successive deletion in each σ_b preserves distances in B_b . In particular, when σ lists all of $C \setminus A$, it is a relative corner peeling to A if and only if both σ_b are relative corner peelings of B_b to A . Every interleaving of two such wing orders is valid.

There is also an exact local description of new midpoints under contraction. Suppose $u, v \in B_b$ have distance two and $C_C(u, v) \setminus C_{B_b}(u, v) \neq \emptyset$. Then there are vertices w, z such that

$$C_{B_b}(u, v) = \{w\}, \quad C_C(u, v) = \{w, z\}, \quad \{u, v, w\} \subseteq A, \quad z \in D_{1-b}. \quad (5.60)$$

The four vertices $Q = \{u, v, w, z\}$ form a square, and the corresponding three-dimensional face of H is exactly

$$H \cap (Q \times \{0, 1\}) = (Q \times \{0, 1\}) \setminus \{(z, b)\}. \quad (5.61)$$

Conversely, each such punctured three-cube supplies the displayed new midpoint. Its old distance-two constraint already has a common neighbor in the retained overlap.

Proof. The ampleness assertions follow from Proposition 5.32; activity of e gives an e -edge and hence $A \neq \emptyset$. No edge of C joins D_0 to D_1 . Otherwise the exclusive lifts of its endpoints would have Hamming distance two in H , with both possible intermediate vertices absent, contradicting isometry.

If $u \in D_0$ and $v \in D_1$ have distance two in C , every common neighbor belongs to A : a neighbor outside A would give an edge between the wings. If both endpoints belong to A , its isometry supplies a common neighbor in A as well. Every remaining constraint therefore has an endpoint in exactly one wing, say D_b , and its other endpoint and all common neighbors lie in B_b . Its common-neighbor set is unchanged on passing to B_b . Conversely, a constraint of $\mathcal{T}_A(B_b)$ has an endpoint in D_b , since A is isometric, and cannot acquire a common neighbor in the other wing. This proves (5.59), including disjointness.

A sequence of deletions from an isometric graph preserves distances at every step exactly when no original distance-two pair loses all its common neighbors while both endpoints remain. Necessity

follows by looking at that pair; sufficiency follows at the first failed deletion by rerouting every two-edge passage through its deleted vertex, as in Proposition 5.63. While A is retained, every pair with a common neighbor in A satisfies this condition permanently. The other pairs split by (5.59). Deletions in one wing do not affect any vertex of a constraint in the other wing. This proves the assertion for partial lists. For full lists, Proposition 5.63 identifies the three metric conditions with the respective relative corner peelings.

Finally, a cube distance-two pair has just two possible common neighbors. The side B_b has at least one, so a new common neighbor means $C_{B_b}(u, v) = \{w\}$ and $C_C(u, v) = \{w, z\}$, with $z \in D_{1-b}$. Neither u nor v can lie in D_b , because each is adjacent to z ; thus both lie in A . Isometry of A then forces $w \in A$. The vertices u, v, w have both lifts, and z has only its $(1 - b)$ -lift. These are exactly the seven vertices in (5.61). Reading that face in the opposite direction proves the converse. $\square$

Corollary 5.65 (Contraction peelings must enter the overlap). *In the notation of Proposition 5.64, suppose that $H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$. The contraction C is peelable, but no complete corner peeling of C has A as its final block. If $A \in \mathcal{D}_{\text{amp}}$, the first deleted vertex of A occurs while both wings D_0, D_1 still have vertices remaining. If H is cornerless, then*

$$\text{Cor}(C) \subseteq A \setminus \text{Cor}(A). \quad (5.62)$$

Consequently every peeling of C enters A at its first deletion, and that deletion makes the remaining overlap nonample.

Proof. The contraction is a proper pc-minor and hence belongs to $\mathcal{D}_{\text{amp}}$. If $A \notin \mathcal{D}_{\text{amp}}$, it cannot be the final block of a complete peeling. If $A \in \mathcal{D}_{\text{amp}}$, the gluing-critical alternative of Proposition 5.32 says that neither B_b peels relatively to A . Suppose one wing were exhausted before the first deletion in A . Projecting that prefix to the exhausted wing, the partial-list assertion of Proposition 5.64 gives a distance-preserving deletion list from B_b to A . Proposition 5.63 makes it a relative corner peeling, a contradiction. This proves both assertions in this case.

For the last assertion, every cube $Q \subseteq C$ lies wholly in one side B_b . Indeed, $A \cap Q$ is ample, so its complement in Q is either empty or connected. In the nonempty case the absence of edges between the wings puts that complement in one D_b , and then $Q \subseteq B_b$; in the empty case both sides contain Q . If a corner x of C lay in D_b , its unique maximal cube would therefore lie in B_b . Its lift has no e -neighbor, and all its incident cubes in H lie in that lifted cube. It would be a corner of H , a contradiction. Now let $x \in A$ be a corner of C , and let its unique maximal cube Q lie in B_b . Every cube of B_{1-b} through x lies in Q and hence in A . Suppose x were a corner of A , with maximal cube R . Then all those cubes lie in R , and R has both lifts. The cube $R \times \{0, 1\}$ would contain every cube of H incident with $(x, 1 - b)$, making that lift a corner. This again contradicts cornerlessness of H , proving (5.62). The ample corner-deletion criterion makes $A \setminus \{x\}$ nonample. $\square$

The checker `verify_damp_overlap_metric_separation.py` verifies the constraint splitting and midpoint faces on all active splits in Q_3 , including partial and complete deletion lists. It also replays the 72 elementary-minor orders of the fixed 291- and 292-concept families from Theorem 5.75, and verifies the forced overlap entry at all 24 coordinate splits. For the cornerless source it also checks that every contraction corner is a noncorner of its overlap. Its report is `damp_overlap_metric_separation_verification.json`. The structural consequence is a restriction on the next construction: retaining the canonical overlap cannot produce the required contraction peelings. Their attachment rules must instead justify deletions inside that overlap, together with their effect on the two remaining wings. For a cornerless source the overlap already ceases to be ample at the first such step, so keeping an ample overlap throughout is also excluded.

The loss of overlap ampleness has an exact description in terms of shattered sets. On every ample contraction residual, the two section shadows intersect in precisely the overlap shadow. Consequently, all failure of ampleness in the lifted family is measured by the overlap.

Proposition 5.66 (Overlap and section deficits). *Retain the notation H, B_0, B_1, A, C of Proposition 5.64. For any cube family X , including the empty family, write*

$$\delta(X) = |\text{Sh}(X)| - |X|.$$

For an ample subfamily $K \subseteq C$, put

$$B_b^K = B_b \cap K, \quad A^K = A \cap K, \quad H[K] = \{(v, b) \in H : v \in K\},$$

where K may be empty. The section residuals and their overlap need not be ample. For every $J \subseteq F$, their signed traces satisfy

$$(B_0^K)|_J \cap (B_1^K)|_J = (A^K)|_J. \quad (5.63)$$

Moreover,

$$\text{Sh}(K) = \text{Sh}(B_0^K) \cup \text{Sh}(B_1^K), \quad (5.64)$$

$$\text{Sh}(A^K) = \text{Sh}(B_0^K) \cap \text{Sh}(B_1^K), \quad (5.65)$$

$$\delta(H[K]) = \delta(B_0^K) + \delta(B_1^K) = \delta(A^K). \quad (5.66)$$

In particular, $H[K]$ is ample if and only if A^K is ample.

Let x be a corner of K , put $K' = K \setminus \{x\}$, and let $m(x) = |\{b : x \in B_b\}|$. Define

$$L_b(x; K) = \text{Sh}(B_b^K) \setminus \text{Sh}(B_b^{K'}),$$

$$L_A(x; K) = \text{Sh}(A^K) \setminus \text{Sh}(A^{K'}).$$

Then

$$\begin{aligned} \delta(H[K']) - \delta(H[K]) &= m(x) - |L_0(x; K)| - |L_1(x; K)| \\ &= \begin{cases} 0, & x \notin A, \\ 1 - |L_A(x; K)|, & x \in A. \end{cases} \end{aligned} \quad (5.67)$$

An exterior deletion preserves both side deficits individually. An overlap deletion increases the deficit by at most one, and decreases it exactly when it loses at least two shattered supports of the overlap.

Proof. Suppose a trace on J is realized by $u \in B_0^K$ and $v \in B_1^K$. If either concept lies in A , the trace is already realized in A^K . Otherwise, u, v lie in opposite wings. Since K is ample, it is isometric; a shortest path in K from u to v flips no coordinate on which they agree, so every concept of the path has the same J -trace. There are no edges between the wings, by Proposition 5.64. The path must therefore pass through A^K , proving the nontrivial inclusion in (5.63). The reverse inclusion is immediate, and the assertion is also valid when K is empty. A support J shattered in both sections has all its traces in their intersection, so it is shattered in A^K . This proves (5.65).

Every cube of C lies wholly in one side. To see this, intersect such a cube Q with the ample family A . The complement of $A \cap Q$ in Q is ample and hence, if nonempty, connected. There are no edges between the two wings, so this complement lies in one wing and Q lies in that side. If the complement is empty, both sides contain Q .

Since K is ample, every support shattered by K is realized by a cube in K . That cube lies in one B_b^K , proving one inclusion in (5.64); the reverse inclusion follows from $B_b^K \subseteq K$. This includes $K = \emptyset$. For arbitrary two-layer families the shadow formula is

$$\text{Sh}(H[K]) = \text{Sh}(K) \dot{\cup} \{J \cup \{e\} : J \in \text{Sh}(B_0^K) \cap \text{Sh}(B_1^K)\}.$$

Use $|H[K]| = |K| + |A^K|$ and $|\text{Sh}(K)| = |K|$, together with (5.65), to obtain $\delta(H[K]) = \delta(A^K)$. Inclusion–exclusion in (5.64) also gives $\delta(B_0^K) + \delta(B_1^K) = \delta(A^K)$, proving (5.66). The ampleness equivalence follows because a family is ample exactly when its deficit is zero.

Deleting x removes $m(x)$ concepts from the two sections and loses exactly $|L_0(x; K)| + |L_1(x; K)|$ section supports. Taking the difference of the first equality in (5.66) gives the equality in (5.67). Both K and K' are ample, so the same balance identity identifies this difference with $\delta(A^{K'}) - \delta(A^K)$. If $x \notin A$, the overlap is unchanged, and so is one of the two sections. The sum of the side deficits is fixed; hence each side deficit is fixed. If $x \in A$, its deletion removes one overlap concept and exactly $|L_A(x; K)|$ overlap supports. This proves the second equality and all the final assertions. $\square$

Proposition 5.67 (The first nonample overlap has one-sided deficit). *In the same notation, let $x \in \text{Cor}(C) \cap (A \setminus \text{Cor}(A))$, and let Q be the unique maximal cube of C through x , with support I . There is a unique $b \in \{0, 1\}$ for which $Q \subseteq B_b$. The deletion of x then has the following effects:*

$$\text{Sh}(B_b \setminus \{x\}) = \text{Sh}(B_b) \setminus \{I\}, \quad (5.68)$$

$$\text{Sh}(B_{1-b} \setminus \{x\}) = \text{Sh}(B_{1-b}), \quad \text{Sh}(A \setminus \{x\}) = \text{Sh}(A), \quad (5.69)$$

$$\text{Sh}(H[C \setminus \{x\}]) = \text{Sh}(H) \setminus \{I\}. \quad (5.70)$$

In particular, $B_b \setminus \{x\}$ is ample, whereas

$$\delta(B_{1-b} \setminus \{x\}) = \delta(A \setminus \{x\}) = \delta(H[C \setminus \{x\}]) = 1.$$

The nonample section and overlap have the same conditioning-minimal nonample faces. Define

$$\mathcal{T}_x = \{T \subseteq F : |T| \geq 2, x^T \notin A, x^U \in A \text{ for every } U \subsetneq T\}.$$

This family is nonempty. For every $T \in \mathcal{T}_x$, let $Q_T(x) = \{x^U : U \subseteq T\}$. Then

$$(A \setminus \{x\}) \cap Q_T(x) = (B_{1-b} \setminus \{x\}) \cap Q_T(x) = Q_T(x) \setminus \{x, x^T\}, \quad x^T \in B_b \setminus A. \quad (5.71)$$

These are exactly the conditioning-minimal nonample faces of each displayed residual family. Thus their two missing vertices are the deleted overlap concept and a surviving concept of the opposite wing.

Proof. The cube containment proved above puts Q in at least one side. It cannot lie in both, because then it would be the unique maximal cube of A through x , making x an A -corner. Let b be its unique side. All cubes of B_b through x lie in Q , so x is a corner of B_b , with support I . Every cube of B_{1-b} through x also lies in Q and hence in A . Therefore the local cube-support complexes of A and B_{1-b} at x coincide, and x is a noncorner of both.

A corner deletion from an ample family loses its unique cube support, by Lemma 2.10 applied in reverse. A noncorner deletion loses no shattered support: losing any support would, by the upper Sandwich inequality, make the remaining family ample and hence make the deleted concept a corner. This proves (5.68) and (5.69). Since $C \setminus \{x\}$ loses I while the other side loses nothing, (5.64) implies $I \notin \text{Sh}(B_{1-b})$. The common section shadow consequently remains $\text{Sh}(A)$. The two-layer shadow formula now proves (5.70), and counting the deleted concepts gives all the deficit identities.

The supports in $\mathcal{T}_x$ are precisely the minimal nonfaces of order at least two in the common local cube-support complex at x . There is at least one such support because x is a noncorner. Every such support is contained in I ; hence its opposite vertex x^T lies in $Q \subseteq B_b$. It is absent from both A and B_{1-b} , which proves (5.71).

For completeness, let M be either original ample family A or B_{1-b} . A nonample conditioning of $M \setminus \{x\}$ must come from a face containing x , since other faces are unchanged conditionings of M . In that face, deleting x from an ample family fails exactly when its local cube-support complex at x has a minimal nonface of order at least two. Such a support is some $T \in \mathcal{T}_x$ and supplies the smaller face in (5.71). Minimality of the nonample conditioning forces equality with that face. Conversely, removing two antipodes from a cube of dimension at least two is nonample, and every proper nonempty conditioning is a punctured or full cube, hence ample. This proves the exact minimal-face assertion. $\square$

Remark 5.68 (Completing the corner cube need not repair the overlap). The preceding local description does not imply that adjoining all of Q to a lower ample family preserves ampleness. Here is an explicit failure even with a marked noncorner of the lower family. Encode the vertices of Q_4 by $v = \sum_{i=0}^3 v_i 2^i$, and put

$$C_0 = \{0, 1, 2, 3, 6, 7, 8, 9\}, \quad A_0 = \{0, 2, 6, 7, 8\}, \quad Q_0 = \{0, 1, 8, 9\}.$$

The graph of C_0 is a four-vertex path times an edge, and A_0 is the geodesic path 8, 0, 2, 6, 7, so both are ample. The union $A_0 \cup Q_0 = C_0 \setminus \{3\}$ is nonample: fixing coordinates 0 and 3 to 1 and 0, respectively, leaves the antipodal pair $\{1, 7\}$ on coordinates 1, 2. Now put

$$C = C_0 \times Q_2, \quad A = A_0 \times \{00, 10, 01\}, \quad x = (8, 00).$$

The families C, A are ample, of orders 32, 15. The vertex x is a corner of C and a noncorner of A , and its corner cube in C is $Q = Q_0 \times Q_2$. Nevertheless, the 00 section of $A \cup Q$ is the nonample family $A_0 \cup Q_0$, so $A \cup Q$ is nonample. Its order is 25, and its shattered shadow has order 28. This refutes the unconditional local cube-completion step; no forbidden-minor assertion is needed for this control.

The checker `verify_damp_contraction_section_deficits.py` verifies the signed trace and shadow identities on all ample residuals of the active Q_3 splits, and the deficit laws on their corner transitions. It checks all 24 first contraction corners of the fixed cornerless 291-concept family, including their positioned antipodal pairs, and a full saved 224-step contraction peeling. In that particular order the lifted deficit reaches 14 before returning to zero; this is not a claim about the minimum possible peak over all orders. The report `damp_contraction_section_deficits_verification.json` also verifies the cube-completion counterexample. The deficit identities record losses exactly, but their scalar values do not determine which losses are compatible. An exterior deletion can change which overlap concepts are corners of the contraction, but it cannot lower either side deficit. Intrinsic construction still requires compatible overlap losses to be derived from the positioned supports and the two section shadows.

Theorem 5.69 (Metric Yang obstruction class). *Let $H \subseteq Q_F$ be ample, put $m = |H| \geq 2$, and let*

$$\mathcal{O}(H) = \text{Ord}(2^H \setminus \{\emptyset, H\})$$

be the order sphere whose chambers are the orders of H . Let

$$\mathcal{D}_2(H) = \{\{u, v\} \subseteq H : d_{G_H}(u, v) = 2\}, \quad C_H(u, v) = N_{G_H}(u) \cap N_{G_H}(v).$$

For $D = \{u, v\} \in \mathcal{D}_2(H)$, let P_D be the poset on H with generating relations

$$w \triangleleft_D u, \quad w \triangleleft_D v \quad (w \in C_H(u, v)), \quad (5.72)$$

and define

$$\mathcal{R}_D^{\text{met}} = \text{Ord}(\mathcal{J}(P_D) \setminus \{\emptyset, H\}), \quad \mathcal{U}_{\text{met}}(H) = \bigcup_{D \in \mathcal{D}_2(H)} \mathcal{R}_D^{\text{met}}.$$

Then

$$H \text{ is nonpeelable} \iff \mathcal{U}_{\text{met}}(H) = \mathcal{O}(H) \iff \tilde{H}_{m-2}(\mathcal{U}_{\text{met}}(H); \mathbb{F}_2) \cong \mathbb{F}_2. \quad (5.73)$$

Consequently

$$\omega_{\text{met}}(H) = \dim_{\mathbb{F}_2} \tilde{H}_{m-2}(\mathcal{U}_{\text{met}}(H); \mathbb{F}_2) \in \{0, 1\}$$

is an intrinsic global invariant of the metric graph G_H , equal to one exactly when the Yang complex is nonshellable.

Proof. A static characterization of a distance-preserving elimination order $\prec$ says that, for every pair u, v at distance two, there is a common neighbor $w \in C_H(u, v)$ such that

$$u \prec w \quad \text{or} \quad v \prec w;$$

see [CDNS18, Corollary 2]. An order fails this condition at $D = \{u, v\}$ exactly when every common neighbor precedes both u and v . These are precisely the linear extensions of the poset P_D , hence precisely the chambers of $\mathcal{R}_D^{\text{met}}$. Proposition 5.62 therefore gives the first equivalence in (5.73).

The chamber-adjacency graph of the order sphere is connected. As in the proof of Theorem 5.81, a proper subcomplex of this sphere has zero top-dimensional homology over $\mathbb{F}_2$, whereas the full sphere has a one-dimensional fundamental class. This proves the second equivalence and the assertion about ω_{met} . $\square$

Remark 5.70 (What the metric class does and does not solve). The invariant ω_{met} is substantially more intrinsic than the latent-chart recognition theorem: it uses only distance-two pairs and their common-neighbor sets in the concept graph. It contains no corners, punctured faces, shelling orders, linear quotients, or descendant tests in its definition. It is nevertheless still an exact recognition invariant, not the requested generative parametrization. Computing whether its fundamental class survives is equivalent to deciding shellability.

The useful new target is now precise. One must describe the minimal supports of this metric fundamental class and prove how those supports are destroyed by restriction and contraction. A successful construction would recover an obstruction from finite distance-two interval data with local boundary and transport axioms. Merely requiring $\omega_{\text{met}}(H) = 1$ and its vanishing on proper pc-minors would again be circular and is not an admissibility condition.

Theorem 5.71 (Boolean-prefix cutset and exact pc-minor transport). *Let $H \subseteq Q_F$ be ample. For each $D = \{u, v\} \in \mathcal{D}_2(H)$, define the Boolean interval*

$$\mathcal{I}_D(H) = [C_H(u, v), H \setminus \{u, v\}] = \{P \subseteq H : C_H(u, v) \subseteq P, P \cap \{u, v\} = \emptyset\},$$

and put

$$\mathcal{U}_{\text{pre}}(H) = \bigcup_{D \in \mathcal{D}_2(H)} \mathcal{I}_D(H) \subseteq 2^H.$$

Then H is nonpeelable if and only if $\mathcal{U}_{\text{pre}}(H)$ meets every maximal chain of the Boolean lattice 2^H . Moreover, this interval system has the following exact pc-minor transport laws.

1. If $A = H_{e=b}$ is a coordinate restriction and $D \in \mathcal{D}_2(H)$ has $D \subseteq A$, then

$$\mathcal{I}_D(A) = \mathcal{I}_D(H) \cap 2^A.$$

Thus the bad-prefix system of A is obtained by retaining the intervals whose owners lie in A and restricting them to the Boolean sublattice $2^A \subseteq 2^H$.

2. Let $q: H \rightarrow \overline{H} = \pi_e H$ be a coordinate contraction. Pullback identifies $2^{\overline{H}}$ with the sublattice

$$\text{Sat}_q(H) = \{q^{-1}(P) : P \subseteq \overline{H}\} \subseteq 2^H$$

of unions of q -fibers. Under this identification, the interval owned by $\overline{D} = \{\overline{u}, \overline{v}\} \in \mathcal{D}_2(\overline{H})$ becomes

$$[q^{-1}C_{\overline{H}}(\overline{u}, \overline{v}), H \setminus q^{-1}\{\overline{u}, \overline{v}\}]_{\text{Sat}_q(H)}.$$

Hence contraction is represented without choosing an order: pass to the fiber-saturated Boolean sublattice and rebuild the finitely many owner intervals from the quotient metric.

Proof. Write a total order of H as $x_1, \dots, x_m$, and let $P_i = \{x_1, \dots, x_i\}$. The order fails the distance-two condition at $D = \{u, v\}$ precisely when every member of $C_H(u, v)$ precedes both u and v . If this happens, the prefix ending with the last member of $C_H(u, v)$ belongs to $\mathcal{I}_D(H)$. Conversely, if some prefix belongs to $\mathcal{I}_D(H)$, then every common neighbor precedes both endpoints, so the order fails at D . Orders of H are in bijection with maximal chains of 2^H . The static metric criterion used in Theorem 5.69 now proves the cutset assertion.

For a restriction, the common-neighbor identity in Proposition 5.79 gives $C_A(u, v) = C_H(u, v)$; intersecting the two endpoints of the Boolean interval with 2^A gives the displayed formula. For a contraction, pullback is a lattice isomorphism from $2^{\overline{H}}$ onto $\text{Sat}_q(H)$. Applying this isomorphism to the lower and upper endpoints of each quotient interval gives the second formula. $\square$

Proposition 5.72 (Unique-midpoint core and the square remainder). *Let V be finite and let $\mathcal{T}$ be a family of rooted-pair constraints $(w; \{u, v\})$ on three distinct members of V . Call a nonempty set $W \subseteq V$ a $\mathcal{T}$ -core if, for every $w \in W$, there is a constraint $(w; \{u, v\}) \in \mathcal{T}$ with $u, v \in W$. There is an order $\prec$ of V satisfying*

$$u \prec w \quad \text{or} \quad v \prec w \quad \text{for every } (w; \{u, v\}) \in \mathcal{T} \quad (\dagger)$$

if and only if no $\mathcal{T}$ -core exists.

This criterion is constructive and order-free. Starting with $W = V$, repeatedly delete any $w \in W$ for which no constraint rooted at w has both heads in W . The final set is the unique largest $\mathcal{T}$ -core; when it is empty, the deletion order satisfies $(\dagger)$.

Apply this to

$$\mathcal{T}_1(H) = \{(w; \{u, v\}) : \{u, v\} \in \mathcal{D}_2(H), C_H(u, v) = \{w\}\}.$$

If the $\mathcal{T}_1(H)$ -core is nonempty, then H is nonpeelable. If it is empty, the unique-midpoint constraints admit an order, and the only remaining metric conditions come from full squares. More precisely, a

total order of H is a distance-preserving elimination order if and only if it satisfies $(\dagger)$ for $\mathcal{T}_1(H)$ and, in every square of G_H , the first two vertices are adjacent.

The square conditions alone never obstruct elimination. For every cube vertex $z \in Q_F$, any total order of H that is nondecreasing in $d_{Q_F}(z, \cdot)$ has adjacent first two vertices in every square. Consequently, if $\mathcal{T}_1(H) = \emptyset$, then H is peelable.

Proof. Suppose first that W is a $\mathcal{T}$ -core, and take the first vertex w of W in an arbitrary total order. Some constraint $(w; \{u, v\})$ has both heads in W . Both heads follow w , so this constraint violates $(\dagger)$.

Conversely, perform the stated deletion procedure. A union of $\mathcal{T}$ -cores is again a $\mathcal{T}$ -core, so the set left at termination is the unique largest one. If the final set is empty, list the vertices in the order in which they were deleted. When w was deleted, every constraint rooted at w had a head already deleted. That head precedes w in the deletion order, proving $(\dagger)$.

For a unique common neighbor w , the metric condition for $D = \{u, v\}$ is exactly $(\dagger)$. On a square, the two diagonals give the two reciprocal two-root constraints. One of them fails exactly when the first two vertices of the square are the opposite diagonal that forms its common-neighbor set. Thus both diagonal constraints hold exactly when the first two square vertices are adjacent. Item 1 of Proposition 5.79 says that these are all possible distance-two constraints.

Finally, fix $z \in Q_F$. On any square there is a unique vertex nearest to z , the two vertices at the next distance are its two square neighbors, and the remaining vertex is farthest from z . Hence the first two square vertices in every distance-nonincreasing order are adjacent. If $\mathcal{T}_1(H)$ is empty, such an order satisfies all metric constraints. The static metric criterion completes the proof. $\square$

Proposition 5.73 (A one-sided criterion for empty unique-midpoint core). *For nested ample families $A \subseteq B$, let $\kappa_A(B)$ be the largest set $W \subseteq B \setminus A$ such that every $w \in W$ roots a constraint*

$$(w; \{u, v\}) \in \mathcal{T}_1(B) \quad \text{with} \quad u, v \in A \cup W.$$

The empty set is allowed. Equivalently, start with $W = B \setminus A$ and repeatedly delete a root having no such witness, keeping all of A present and keeping $\mathcal{T}_1(B)$ fixed.

Let $H \subseteq Q_F$ be ample, fix an active coordinate e , and write $B_i = \rho_e^i H$, $A = B_0 \cap B_1$, and $D_i = B_i \setminus A$. Suppose that $\kappa(B_0) = \kappa(B_1) = \emptyset$, where κ denotes the ordinary unique-midpoint core. If $\kappa_A(B_i) = \emptyset$ for either sign i , then $\kappa(H) = \emptyset$.

In particular, the conclusion holds if either B_i can be reduced to A by deleting corners outside A . Equivalently, it holds if either inclusion $A \subseteq B_i$ admits an A -relative ample chain as in (5.35).

Proof. Unions of sets satisfying the displayed witness condition again satisfy it. Every such set survives the stated pruning, and the set left at termination satisfies the condition. This proves existence, uniqueness, and the pruning description of $\kappa_A(B)$.

Suppose, after exchanging the signs if necessary, that $\kappa_A(B_0) = \emptyset$, and let K be any unique-midpoint core of H . A word $(w, 0)$ with $w \in D_0$ has no neighbor in direction e . Its witnessing constraint in K must therefore be horizontal. Its heads project into $A \cup \{v \in D_0 : (v, 0) \in K\}$. Thus

$$\{w \in D_0 : (w, 0) \in K\} \subseteq \kappa_A(B_0) = \emptyset.$$

Every remaining member of K in the zero section projects into A . A unique-midpoint constraint rooted at $(w, 0)$ and using e would have heads $(w, 1)$ and $(v, 0)$, where v is an old-coordinate neighbor of w . Uniqueness requires $(v, 1) \notin H$, so $v \in D_0$, whose copy cannot belong to K . Hence every constraint witnessing a member of the zero section of K is horizontal. That section would be a core

of B_0 , which is impossible. The zero section of K is therefore empty. Every witness for its one section is now horizontal as well, making that section a core of B_1 . Thus K is empty, as required.

Finally, suppose a corner-deletion sequence reduces B to A , and that W satisfies the witness condition defining $\kappa_A(B)$. At the first deletion of a member of a nonempty W , both heads of its witness are still present. Their other possible midpoint was absent from B , and is still absent, so the root cannot be a corner. This contradiction proves $\kappa_A(B) = \emptyset$. Reversing the deletion sequence gives the stated equivalence with a relative ample chain. $\square$

Corollary 5.74 (Nonempty cores force two large cyclic wings). *Let $H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be irredundantly embedded, and suppose $\kappa(H) \neq \emptyset$. For every active coordinate e , with sections, overlap, and wings as in Proposition 5.73, both $\kappa_A(B_0)$ and $\kappa_A(B_1)$ are nonempty. Neither inclusion $A \subseteq B_i$ admits an A -relative ample chain. Both induced wing graphs $Q[D_i]$ contain cycles, and*

$$|D_0|, |D_1| \geq 9, \quad |H| \geq 2|A| + 18. \quad (5.74)$$

These conclusions hold whether or not A is corner-peelable. If $A \notin \mathcal{D}_{\text{amp}}$, then additionally

$$|H| \geq 2m_{\text{amp}} + 18.$$

Proof. The two sections are proper pc-minors and hence corner-peelable, so their unique-midpoint cores are empty. Proposition 5.73 proves the first two assertions. An edge of H in direction e projects into the overlap, so A is nonempty. Proposition 5.32 makes it ample. Each relative Yang module of $A \subseteq B_i$ is Cohen–Macaulay by Lemma A.12. The forest and eight-facet linear-quotient theorems for relative Yang modules [Bou26] would give an A -relative ample chain if $Q[D_i]$ were a forest or if $|D_i| \leq 8$. Both alternatives are excluded. The identity $|H| = 2|A| + |D_0| + |D_1|$ gives (5.74). If A is nonpeelable, Theorem 5.33 makes it an ample forbidden pc-minor, so $|A| \geq m_{\text{amp}}$, proving the last bound. $\square$

One side suffices. For example, let B_0, B_1 be Hall’s two coordinate-zero sections, and put $A = B_0 \cap B_1$. The peripheral family $(B_0 \times \{0\}) \cup (A \times \{1\})$ has 224 concepts and empty core. Its zero section cannot even start a peeling down to A : all nine corners of B_0 lie in A , and $|\kappa_A(B_0)| = 52$. Its one section already equals A . Conversely, two nonempty sets $\kappa_A(B_i)$ at one coordinate do not force a nonempty global core: at coordinate zero of the 292-concept example of Theorem 5.75, their orders are 54 and 17, while the global core is empty. The fixed checker `verify_damp_relative_midpoint_wings.py` verifies these controls and both wing conditions at every coordinate of Hall’s family and its 300-concept extension at 373. These are necessary conditions on a possible terminal counterexample to layer existence, not a classification of such counterexamples.

Theorem 5.75 (An ample forbidden pc-minor with empty unique-midpoint core). *Let $G \subseteq Q_{12}$ be the certified 291-concept family stored as `base` in `damp_fixed_291_repair_creation.json`, with words encoded as $x = \sum_{i=0}^{11} x_i 2^i$. Put*

$$\begin{aligned} D &= \{96, 126, 394, 1922, 1934\}, \\ A &= \{553, 763, 2968, 4008, 4026\}, \\ \widehat{G} &= (G \setminus D) \cup A, \quad H = \widehat{G} \cup \{3959\}. \end{aligned}$$

Both $\widehat{G}$ and H are ample forbidden pc-minors. The family $\widehat{G}$ is cornerless and has 291 concepts; its largest unique-midpoint core has 76 concepts. The family H has 292 concepts, its only corner is 3959, and its largest unique-midpoint core is empty. The corner cube at 3959 has support $\{4, 7, 11\}$.

In particular, an ample forbidden pc-minor with empty unique-midpoint core need not peel, and it need not have a corner whose deletion leaves an empty unique-midpoint core.

Proof. This is a finite certificate proof. The complete two vertex sets, their source hashes, and the elementary-minor peelings are stored in `damp_empty_core_counterexample.json`. The checker `verify_damp_empty_core_counterexample.py` recomputes all shattered sets and all corners directly. It verifies

family	order	Sh	corners	largest core order
$\widehat{G}$	291	291	$\emptyset$	76
H	292	292	$\{3959\}$	0

The two Sandwich equalities prove ampleness. Cornerlessness makes $\widehat{G}$ nonpeelable. Every peeling of H would have to start at 3959, leaving $\widehat{G}$, so H is nonpeelable as well.

For completeness of the core calculation, the checker enumerates every distance-two endpoint pair and its common neighbors. It obtains exactly 766 unique-midpoint constraints in H , also checked by independently enumerating rooted neighbor pairs. The verification report stores a 292-word pruning order for this *fixed* constraint family: when a root is removed, each constraint rooted there already has a removed head. Proposition 5.72 proves that its core is empty. Repeating the deterministic calculation on $\widehat{G}$ leaves 76 vertices; one internal rooted-pair witness for each of them is recorded in `damp_empty_core_counterexample_verification.json`. Thus deleting the only corner activates a nonempty core.

Finally, every coordinate is active in both families. For each family, coordinate, and elementary restriction or contraction, the witness gives a complete corner peeling of the resulting image. The checker validates all 72 orders, testing the corner cube at every deletion and checking that the image is exhausted. Hence every elementary pc-minor of each family peels. Pc-minor closure of $\mathcal{D}_{\text{amp}}$, together with their nonpeelability, proves that both families are forbidden pc-minors. $\square$

Remark 5.76 (A compatible completion that remains nonpeelable). For the same G as in Theorem 5.75, put

$$T = \{186, 553, 680, 763, 898, 910, 1958, \\ 2184, 2592, 2968, 3137, 3959, 4008, 4026\}.$$

Every family $G \cup U$, $U \subseteq T$, is an ample forbidden pc-minor. Thus these fourteen additions do not repair G , even collectively. This asserts 2^{14} distinct embedded families, without an isomorphism count.

Here the finite negative certificate concerns only the largest family $C = G \cup T$, of order 305. It is ample and has empty unique-midpoint core. Assign a nine-bit unsigned rank r_x to each $x \in C$. For every distance-two pair u, v , impose

$$\bigvee_{\substack{x \in \{u, v\} \\ w \in C_C(u, v)}} r_x < r_w.$$

By Proposition 5.72, these conditions admit a solution exactly when C peels: a peeling supplies distinct ranks, and arbitrarily refining tied ranks preserves every satisfied strict inequality. The explicit comparator encoding has 17721 Boolean variables and 87217 clauses. Its stored DRAT refutation is independently checked by `drat-trim`. The checker `verify_damp_fixed_305_completion.py` regenerates the formula, checks the proof hashes, and validates complete peelings of all thirty-six elementary minors of both G and C . Its report is `damp_fixed_305_completion_verification.json`.

It remains to justify the simultaneous assertion about all subsets. The acquired supports of the fourteen individually ample additions are distinct. Each added tip is therefore a corner after any subset has been retained, by the union-shadow formula and the ample corner-deletion criterion. If

$G \cup U$ peeled, first deleting $T \setminus U$ from C would give a peeling of C , a contradiction. All intermediate families are consequently nonpeelable and ample; iterating Proposition 2.12 from the forbidden family G proves their pc-minimality. No subset enumeration is needed. This fixed conclusion does not prove a general prohibition on cooperating additions over a forbidden seed.

Remark 5.77 (Failure of the core-or-nonample criterion). The proposed equivalence between nonpeelability of an ample family and nonemptiness of its unique-midpoint core is false by Theorem 5.75. The counterexample also refutes the equivalent core-safe-corner assertion. It is ample, so all its pc-minors are ample; failure to repair its square constraints cannot produce a nonample antipodally punctured pc-minor. The sufficient implication from a nonempty core to nonpeelability in Proposition 5.72 remains valid.

The earlier bounded checks explain why that equivalence was plausible: it has no mismatch among the 5529 nonempty ample subfamilies of Q_4 , and Hall’s 299-concept family and the original 291-concept example have cores of orders 158 and 76. These checks do not extend to the new one-corner example. Its only corner has degree three, so it also satisfies the exclusions of Proposition 2.13 and Corollary 2.14. The example is not corner-descent terminal: deleting its corner leaves $\widehat{G}$, another ample forbidden pc-minor. It therefore does not settle the layer-existence or cornerless-seed questions for terminal obstructions.

Remark 5.78 (What the global invariant isolates). Theorem 5.71 is the Boolean-lattice shadow of nonshellability of the Yang complex: the metric intervals form a cutset for all chambers of its order sphere. It is finite, intrinsic, and its behavior under both pc-minor operations is explicit. However, “meets every maximal chain” is still equivalent to the forbidden ordering search, so it is not an admissibility axiom for the desired construction.

Proposition 5.72 removes that circularity for the entire incomplete-square part: its obstruction is the largest rooted-pair core, obtained by deterministic pruning. Consequently the remaining structural problem is narrower. One must parameterize how full squares couple otherwise prunable rooted-pair systems, and prove a minor-cancellation theorem for that coupling. This square-coupling datum, rather than another certificate that a shelling order does not exist, is the candidate global invariant that a complete generative classification must expose. Theorem 5.75 shows that this coupling creates an additional obstruction even in an ample forbidden pc-minor: the initial rooted-pair system is prunable, but deleting its sole corner activates a nonempty core.

Proposition 5.79 (Local alphabet and pc-minor span laws for metric supports). *Let $H \subseteq Q_F$ be isometric. For $\mathcal{S} \subseteq \mathcal{D}_2(H)$, put*

$$V(\mathcal{S}) = \bigcup_{\{u,v\} \in \mathcal{S}} (\{u,v\} \cup C_H(u,v)).$$

Call $\mathcal{S}$ order-infeasible if there is no total order $\prec$ of $V(\mathcal{S})$ such that, for every $\{u,v\} \in \mathcal{S}$, some $w \in C_H(u,v)$ satisfies

$$u \prec w \quad \text{or} \quad v \prec w.$$

An inclusion-minimal order-infeasible family is called a metric support. Then the following statements hold.

1. *Every $D = \{u,v\} \in \mathcal{D}_2(H)$ has either one or two common neighbors. Thus the metric failure cover has only two local types: a three-vertex constraint with one common neighbor and a four-vertex constraint supported on a square.*
2. *If $e \in F$, $b \in \{0,1\}$, and $H_{e=b} = \{x \in H : x_e = b\}$, then, after suppressing the constant coordinate e ,*

$$\mathcal{D}_2(H_{e=b}) = \{\{u,v\} \in \mathcal{D}_2(H) : u_e = v_e = b\}, \quad C_{H_{e=b}}(u,v) = C_H(u,v). \quad (*)$$

3. If H is ample, then H is nonpeelable if and only if its distance-two interval system contains a metric support.
4. Suppose that H is an ample forbidden pc-minor for corner peelability. Then every metric support is coordinate-spanning: for every effective coordinate e ,

$$V(\mathcal{S}) \cap H_{e=0} \neq \emptyset \quad \text{and} \quad V(\mathcal{S}) \cap H_{e=1} \neq \emptyset.$$

5. Under the hypotheses of item 4, put $\bar{H} = \pi_e H$ and $\bar{x} = \pi_e(x)$. For every effective coordinate e and every metric support $\mathcal{S}$, contraction causes at least one of the following three changes:

- (a) two vertices of $V(\mathcal{S})$ have the same projection;
- (b) some $\{u, v\} \in \mathcal{S}$ satisfies $u_e \neq v_e$, so that $\bar{u}$ and $\bar{v}$ are adjacent;
- (c) for some $\{u, v\} \in \mathcal{S}$ with $u_e = v_e$, the contraction creates a new common neighbor:

$$C_{\bar{H}}(\bar{u}, \bar{v}) \not\subseteq \pi_e(C_H(u, v)).$$

Proof. If $d_{G_H}(u, v) = 2$, isometry says that u and v differ in exactly two cube coordinates, say p and q . Their only possible common neighbors in the ambient cube are $u^{\{p\}}$ and $u^{\{q\}}$. At least one belongs to H , because a length-two geodesic exists in G_H . This proves item 1.

The coordinate halfspace $H_{e=b}$ is convex in G_H . Indeed, a shortest path between two of its vertices has Hamming length and therefore cannot flip the coordinate e . Hence distances inside the halfspace are unchanged. Moreover, a common neighbor of two distinct vertices with e -coordinate b also has e -coordinate b : otherwise both vertices would be obtained from that common neighbor by flipping only e , and would coincide. This proves (*).

By the static distance-two criterion used in Theorem 5.69, H is nonpeelable exactly when the full family $\mathcal{D}_2(H)$ is order-infeasible. Since this family is finite, it then contains an inclusion-minimal order-infeasible subfamily. Conversely, every order of H restricts to an order of $V(\mathcal{S})$, so an order-infeasible subfamily already prevents a peeling. This proves item 3.

Finally, let e be effective and suppose that $V(\mathcal{S}) \subseteq H_{e=b}$. By (*), the same constraints form an order-infeasible subfamily of the distance-two interval system of $H_{e=b}$. Item 3 makes this restriction nonpeelable. It is a proper pc-minor of H , contradicting pc-minimality. Therefore every metric support meets both coordinate halfspaces, proving item 4.

Suppose, finally, that none of the three changes in item 5 occurs. Projection is then injective on $V(\mathcal{S})$, every constrained pair remains at distance two, and

$$C_{\bar{H}}(\bar{u}, \bar{v}) = \pi_e(C_H(u, v)) \quad (\{u, v\} \in \mathcal{S}).$$

Indeed, projection preserves every old same-layer common neighbor, while the absence of item 5(c) excludes all new ones. Thus projection identifies the constraint system on $\mathcal{S}$ with an order-infeasible subsystem of $\mathcal{D}_2(\bar{H})$. Item 3 makes the proper contraction $\bar{H}$ nonpeelable, again contradicting pc-minimality. This proves item 5. $\square$

Remark 5.80 (Structural content and remaining contraction problem). The proposition replaces an arbitrary failure region by a finite family of ternary and quaternary interval constraints and proves two global minimality laws: no support can hide in a coordinate halfspace, and every contraction must hit every support by a collision, a distance drop, or a new common neighbor. These are necessary structural conditions, not yet admissibility for a generative construction, because order-infeasibility remains equivalent to the original shellability failure. The next missing functorial

statement is a cancellation theorem: one must give intrinsic attachment axioms under which the three local changes jointly destroy all minimal supports. The direct and switched punctured-expansion profiles retain precisely the owner data that the uncolored quotient interval system forgets.

Theorem 5.81 (Canonical Yang order-sphere obstruction). *Let $H \subseteq Q_F$ be ample, put $m = |H| \geq 2$, and let*

$$\mathcal{O}(H) = \text{Ord}(2^H \setminus \{\emptyset, H\})$$

be the order complex of the proper nonempty subsets of H . Thus $\mathcal{O}(H)$ is the barycentric subdivision of the boundary of an $(m-1)$ -simplex, and its chambers are the orders of H , represented by their chains of proper prefixes.

For a latent chart $B = (y, S)$, put

$$a_B = y^S, \quad p_{B,q} = y^{S \setminus \{q\}} \quad (q \in S),$$

and define a poset P_B on H by the following generating relations:

$$\begin{aligned} a_B \triangleleft_B p_{B,q} & \quad (q \in S), \quad y \notin H, \\ y \triangleleft_B a_B \triangleleft_B p_{B,q} & \quad (q \in S), \quad y \in H. \end{aligned} \tag{5.75}$$

Let $\mathcal{J}(P_B)$ be its lattice of order ideals, and define the failure region

$$\mathcal{R}_B = \text{Ord}(\mathcal{J}(P_B) \setminus \{\emptyset, H\}) \subseteq \mathcal{O}(H).$$

Finally put

$$\mathcal{U}(H) = \bigcup_{B \in \mathcal{L}(H)} \mathcal{R}_B.$$

Then

$$\Delta_H \text{ is nonshellable} \iff \mathcal{U}(H) = \mathcal{O}(H) \iff \tilde{H}_{m-2}(\mathcal{U}(H); \mathbb{F}_2) \cong \mathbb{F}_2. \tag{5.76}$$

If Δ_H is shellable, the last homology group is zero. Consequently

$$\omega_{\text{ord}}(H) = \dim_{\mathbb{F}_2} \tilde{H}_{m-2}(\mathcal{U}(H); \mathbb{F}_2) \in \{0, 1\}$$

is a canonical global invariant of the colored Yang protector atlas, equal to one exactly for nonpeelable H .

More precisely, let $\emptyset \neq \mathcal{I} \subseteq \mathcal{L}(H)$. Write $P_{\mathcal{I}}$ for the preorder generated by all relations in P_B , $B \in \mathcal{I}$, and let $\bar{P}_{\mathcal{I}}$ be the poset obtained by contracting the strongly connected classes of this preorder. If $k_{\mathcal{I}}$ is the number of these classes, then

$$\bigcap_{B \in \mathcal{I}} \mathcal{R}_B = \text{Ord}(\mathcal{J}(\bar{P}_{\mathcal{I}}) \setminus \{\emptyset, \bar{P}_{\mathcal{I}}\}). \tag{5.77}$$

This intersection is homotopy equivalent to

$$\begin{cases} S^{k_{\mathcal{I}}-2}, & \bar{P}_{\mathcal{I}} \text{ is an antichain,} \\ \{\text{pt}\}, & \bar{P}_{\mathcal{I}} \text{ is not an antichain,} \end{cases}$$

with $S^{-1} = \emptyset$. Hence the augmented Mayer–Vietoris spectral sequence of the canonical cover $(\mathcal{R}_B)_{B \in \mathcal{L}(H)}$ has

$$E_{p,q}^1 = \bigoplus_{\substack{\mathcal{I} \subseteq \mathcal{L}(H) \\ |\mathcal{I}|=p+1}} \tilde{H}_q \left(\bigcap_{B \in \mathcal{I}} \mathcal{R}_B; \mathbb{F}_2 \right) \implies \tilde{H}_{p+q}(\mathcal{U}(H); \mathbb{F}_2), \tag{5.78}$$

and its only nonzero E^1 -summands are the one-dimensional summands with $\overline{P}_{\mathcal{I}}$ an antichain and $q = k_{\mathcal{I}} - 2$.

Proof. An order $\mathbf{x} = (x_1, \dots, x_m)$ is a chamber of $\mathcal{R}_B$ precisely when every prefix of $\mathbf{x}$ is an order ideal of P_B , or equivalently when $\mathbf{x}$ is a linear extension of P_B . If $y \notin H$, this says that the antipode a_B precedes every protector $p_{B,q}$. If $y \in H$, it says that

$$y \prec a_B \prec p_{B,q} \quad \text{for every } q \in S.$$

These are exactly the two ways in which all allowed gate relations in the chart B point backwards. Corollary 5.61 therefore says that an order is a shelling order if and only if its chamber belongs to none of the $\mathcal{R}_B$. Thus Δ_H is nonshellable exactly when the failure regions contain every chamber. Since they are subcomplexes, they then contain every face, proving the first equivalence in (5.76).

The chamber-adjacency graph of $\mathcal{O}(H)$ is connected: adjacent chambers differ by interchanging two consecutive members of an order. Over $\mathbb{F}_2$, the boundary equation at their common ridge forces the two chamber coefficients of every top-dimensional cycle to agree. If a subcomplex of $\mathcal{O}(H)$ omits one chamber, give that chamber coefficient zero; connectivity then forces every coefficient to be zero. Hence a proper subcomplex has zero homology in dimension $m - 2$, whereas $\mathcal{O}(H) \cong S^{m-2}$ has one-dimensional top homology. This proves the second equivalence and the assertion about ω_{ord} .

A subset of H is an order ideal of every P_B , $B \in \mathcal{I}$, exactly when it is an order ideal of the generated preorder $P_{\mathcal{I}}$. Such an ideal is a union of strongly connected classes, and these unions are precisely the order ideals of $\overline{P}_{\mathcal{I}}$. This proves (5.77).

It remains to identify its homotopy type. If $\overline{P}_{\mathcal{I}}$ is an antichain on $k_{\mathcal{I}}$ members, its ideal lattice is Boolean, so its proper-part order complex is $S^{k_{\mathcal{I}}-2}$. Otherwise the atoms of its ideal lattice are the singleton minimal elements. Even the join of all these atoms is then a proper ideal. The atomic crosscut complex is consequently a full simplex, and the crosscut argument contracts the proper-part order complex. Finally, the standard augmented intersection double complex of a finite cover by subcomplexes has the E^1 -page displayed in (5.78): for each simplex of the union, the horizontal summands containing it form a simplex and hence an exact augmented chain complex. The intersection calculation gives the stated support of the E^1 -page. $\square$

Corollary 5.82 (Balanced circuit carrier). *If H is nonpeelable, there is a nonempty collection $\mathcal{I} \subseteq \mathcal{L}(H)$ such that*

1. *every generating relation of $P_{\mathcal{I}}$ lies on a directed cycle; equivalently, $\overline{P}_{\mathcal{I}}$ is an antichain; and*
- 2.

$$|\mathcal{I}| + k_{\mathcal{I}} = |H| + 1. \quad (5.79)$$

If $r_{\mathcal{I}}$ concepts are incident with a generating relation and these concepts form $c_{\mathcal{I}}$ nontrivial strongly connected components, then the balance equation is equivalently

$$|\mathcal{I}| = r_{\mathcal{I}} - c_{\mathcal{I}} + 1.$$

Such balanced circuit collections are necessary carriers of the global obstruction class; their existence alone is not asserted to be sufficient.

Proof. When H is nonpeelable, the top class in (5.76) survives to the abutment of (5.78). Some nonzero $E_{p,q}^1$ -summand of total degree $p + q = m - 2$ must therefore support it. Theorem 5.81 makes $\overline{P}_{\mathcal{I}}$ an antichain, $p = |\mathcal{I}| - 1$, and $q = k_{\mathcal{I}} - 2$. The total-degree identity is exactly (5.79). An antichain condensation has no edge between distinct strongly connected classes, so every generating relation lies within one class and hence on a directed cycle. Finally, the $m - r_{\mathcal{I}}$ isolated concepts and the $c_{\mathcal{I}}$ nontrivial components give $k_{\mathcal{I}} = m - r_{\mathcal{I}} + c_{\mathcal{I}}$, yielding the last formula. $\square$

Proposition 5.83 (The first cancellation page is a cut-nerve homology). *Let $\Pi(H)$ be the set of partitions of H . For $\pi \in \Pi(H)$, let $\mathcal{A}_\pi$ be the latent charts whose every generating failure relation has both endpoints in one part of π . Define $\mathcal{N}_\pi \subseteq 2^{\mathcal{A}_\pi}$ by*

$$\mathcal{N}_\pi = \{\mathcal{I} \subseteq \mathcal{A}_\pi : \text{the relations of } \mathcal{I} \text{ fail to make some part of } \pi \text{ strongly connected}\}.$$

This is a simplicial complex. If $|\pi| = k$, then the row $q = k - 2$ of the first differential in (5.78) decomposes as

$$(E_{\bullet, k-2}^1, d^1) \cong \bigoplus_{\substack{\pi \in \Pi(H) \\ |\pi| = k}} C_\bullet(2^{\mathcal{A}_\pi}, \mathcal{N}_\pi; \mathbb{F}_2). \quad (5.80)$$

Consequently,

$$E_{p, k-2}^2 \cong \bigoplus_{\substack{\pi \in \Pi(H) \\ |\pi| = k}} \tilde{H}_{p-1}(\mathcal{N}_\pi; \mathbb{F}_2). \quad (5.81)$$

The complexes on the right have a canonical cut-nerve model. For a part $C \in \pi$ and a nonempty proper subset $U \subsetneq C$, put

$$\mathcal{S}_\pi(C, U) = \{B \in \mathcal{A}_\pi : P_B \text{ has no generating arc from } U \text{ to } C \setminus U\}.$$

After discarding empty $\mathcal{S}_\pi(C, U)$, let $\mathcal{K}_\pi$ be the nerve of the family of simplices

$$\left(2^{\mathcal{S}_\pi(C, U)}\right)_{\substack{C \in \pi \\ \emptyset \neq U \subsetneq C}}.$$

Then

$$\mathcal{N}_\pi \simeq \mathcal{K}_\pi. \quad (5.82)$$

In particular, if H is nonpeelable, then for some partition π with $k = |\pi|$,

$$\tilde{H}_{|H|-k-1}(\mathcal{K}_\pi; \mathbb{F}_2) \neq 0. \quad (5.83)$$

Proof. A nonzero summand of $E_{p, k-2}^1$ is indexed by a chart collection $\mathcal{I}$ whose strongly connected quotient is an antichain of order k . Let π be its strongly connected partition. Every failure arc of $\mathcal{I}$ lies inside a part of π , and the arcs strongly connect every part. Conversely, these two conditions make the strongly connected partition exactly π . Thus the summands belonging to π are indexed by the faces of the simplex $2^{\mathcal{A}_\pi}$ that do not belong to $\mathcal{N}_\pi$.

Remove one chart from $\mathcal{I}$. If every part remains strongly connected, the strongly connected partition is still π . Both intersections in (5.77) are then the same Boolean sphere on the parts of π , so the inclusion induces the identity on its one-dimensional homology over $\mathbb{F}_2$. If a part splits, the new sphere has a different dimension and the component of d^1 in row $q = k - 2$ is zero. This is exactly the relative simplicial boundary of $(2^{\mathcal{A}_\pi}, \mathcal{N}_\pi)$, proving (5.80). Since the first member of the pair is a simplex, its relative homology is the shifted reduced homology of the second, which proves (5.81).

A directed graph on C is strongly connected exactly when every nonempty proper $U \subsetneq C$ has an arc leaving U . Therefore a chart collection belongs to $\mathcal{N}_\pi$ exactly when it is contained in $2^{\mathcal{S}_\pi(C, U)}$ for some displayed cut:

$$\mathcal{N}_\pi = \bigcup_{\substack{C \in \pi \\ \emptyset \neq U \subsetneq C}} 2^{\mathcal{S}_\pi(C, U)}.$$

Every nonempty intersection in this cover is again a simplex, on the intersection of the corresponding chart sets. The nerve lemma gives (5.82).

Finally, a surviving top class has total degree $m - 2$. It must have a nonzero representative on the second page, so (5.81) is nonzero for some $p + k - 2 = m - 2$, namely $p = m - k$. Apply (5.82) to obtain (5.83). $\square$

Remark 5.84 (Higher differentials are balanced partition splittings). The bidegree of the homological differential gives a sharp remaining contract. A possible nonzero map out of a spherical row has the form

$$d^r : E_{p,k-2}^r \longrightarrow E_{p-r,k+r-3}^r.$$

By Theorem 5.81, the target row can be supported only on strongly connected partitions with

$$k' = k + r - 1$$

parts. Thus an r -th differential can only record a net splitting into exactly $r - 1$ additional strongly connected classes. Dually, an incoming r -th differential can only originate from a partition with $k - r + 1$ parts. Hence a second-page class with no compatible balanced refinement or coarsening of these ranks is spectrally isolated and survives to the global obstruction class. The remaining generative problem is to read these balanced splitting maps, or their absence, from the direct and switched chart attachments.

Proposition 5.85 (A binary partition split has a canonical filling torsor). *Let π be a partition and let σ be obtained by replacing one part $C \in \pi$ by two nonempty parts C_0, C_1 . Put*

$$\mathbb{S}_\pi = \text{Ord}(2^\pi \setminus \{\emptyset, \pi\}), \quad \mathbb{S}_\sigma = \text{Ord}(2^\sigma \setminus \{\emptyset, \sigma\}).$$

Identifying a union of parts of π with the corresponding union of parts of σ embeds

$$\iota_{\pi,\sigma} : \mathbb{S}_\pi \hookrightarrow \mathbb{S}_\sigma.$$

For $b \in \{0, 1\}$, let $\mathbb{B}_b(\pi, \sigma)$ be the order complex of the nonempty proper subsets $\emptyset \neq A \subsetneq \sigma$ satisfying

$$C_{1-b} \in A \implies C_b \in A. \quad (5.84)$$

Then $\mathbb{B}_0(\pi, \sigma)$ and $\mathbb{B}_1(\pi, \sigma)$ are PL balls with

$$\mathbb{B}_0(\pi, \sigma) \cap \mathbb{B}_1(\pi, \sigma) = \iota_{\pi,\sigma}(\mathbb{S}_\pi), \quad \mathbb{B}_0(\pi, \sigma) \cup \mathbb{B}_1(\pi, \sigma) = \mathbb{S}_\sigma. \quad (5.85)$$

Thus the old sphere is the common equator of the refined sphere.

Over $\mathbb{F}_2$, let $h_b(\pi, \sigma)$ be the sum of the top-dimensional simplices of $\mathbb{B}_b(\pi, \sigma)$. The two chains are precisely the two top-dimensional fillings of the equatorial fundamental cycle:

$$\partial h_0(\pi, \sigma) = \partial h_1(\pi, \sigma) = [\mathbb{S}_\pi], \quad h_0(\pi, \sigma) + h_1(\pi, \sigma) = [\mathbb{S}_\sigma]. \quad (5.86)$$

Consequently the fillings form a canonical two-element $\mathbb{F}_2$ -torsor indexed by the two new parts.

Proof. The subsets satisfying (5.84) are the order ideals of the one-relation poset $C_b < C_{1-b}$, together with the unaffected parts of π . Their proper-part order complex is the closed half of the Coxeter sphere $\mathbb{S}_\sigma$ in which C_b precedes C_{1-b} ; it is a PL ball. A subset satisfies both implications exactly when it contains either both C_0, C_1 or neither, which is precisely the embedded Boolean lattice on the parts of π . Moreover, a chain of subsets cannot contain both a set containing C_0 but not C_1

and a set containing C_1 but not C_0 , because those two sets are incomparable at the first change of the split coordinates. Hence every simplex belongs to one of the two balls. This proves (5.85).

The boundary of the fundamental chain of either PL ball is the fundamental cycle of its common equator, giving the first identity in (5.86). Their sum is the fundamental cycle of the sphere obtained by gluing the two balls, giving the second identity. Finally, two top-dimensional chains with the displayed boundary differ by a top cycle of $\mathbb{S}_\sigma$. Its top cycle space over $\mathbb{F}_2$ is one-dimensional, so the two displayed chains are the only two fillings. $\square$

Proposition 5.86 (Iterated binary fillings factor through a permutation). *Let τ refine a partition π by replacing one part $C \in \pi$ with nonempty parts indexed by a finite set L . An oriented binary refinement tree for C is a rooted full binary tree whose leaves are L , together with a choice of first child at every internal node. Reading the first child before the second recursively induces a total order*

$$\lambda_1 \prec_T \lambda_2 \prec_T \cdots \prec_T \lambda_{|L|} \quad (5.87)$$

of L .

Starting from the parts of π , perform the binary splits prescribed by the tree and at each split take the hemisphere filling indexed by its first child. The resulting iterated filling region in $\mathbb{S}_\tau$ is

$$\mathbb{B}(T) = \text{Ord}(\mathcal{J}(P_T) \setminus \{\emptyset, \tau\}), \quad (5.88)$$

where P_T makes the members of L the chain in (5.87) and leaves all parts of $\pi \setminus \{C\}$ incomparable with that chain and with one another. In particular:

1. $\mathbb{B}(T)$ is a PL ball;
2. it depends only on the induced permutation of L , not on the binary refinement tree; and
3. every permutation of L is obtained.

Consequently all coherence changes between binary refinement trees factor through the natural symmetric-group action of $\mathfrak{S}_L$. Independent refinements of distinct parts commute, whereas interacting refinements of one part carry permutation-valued holonomy.

Proof. Proceed by induction on the refinement tree. A leaf contributes the one-element chain. At an internal node, the chosen hemisphere orders every leaf of the first child before every leaf of the second child. Thus it takes the ordinal sum of the two child posets. By induction both child posets are chains in their recursively read leaf orders, and their ordinal sum is the concatenated chain (5.87). Iterating the one-relation order-ideal description from Proposition 5.85 gives (5.88).

The linear-extension chambers of a poset consisting of one chain and otherwise incomparable elements form a closed convex region of the Coxeter complex, hence a PL ball. The poset P_T , and therefore its order-ideal complex, depends only on the final leaf order. Conversely, given any order of L , a binary comb whose successive first children are read in that order realizes it. Finally, trees on disjoint parts impose relations on disjoint leaf sets and their ordinal-sum operations commute. Trees refining the same part change only the induced order of its leaves, giving the stated symmetric-group action. $\square$

Remark 5.87 (The exact finite local state group). For a latent chart $B = (y, S)$, define its gate-label set by

$$G_B = \begin{cases} S, & y \notin H, \\ S \cup \{*\}, & y \in H, \end{cases}$$

where $q \in S$ labels the q -protector and $*$ labels the escape. The local gate states are the points of G_B . Proposition 5.86 shows that iterated binary fillings on this label set factor through the candidate coherence group $\mathfrak{S}(G_B)$, with gate alphabet the homogeneous space

$$G_B \cong \mathfrak{S}(G_B)/\mathfrak{S}(G_B \setminus \{g\})$$

for any fixed $g \in G_B$. Thus an absent rank- r chart has the natural $\mathfrak{S}_r/\mathfrak{S}_{r-1}$ alphabet, while a present rank- r chart has the $\mathfrak{S}_{r+1}/\mathfrak{S}_r$ alphabet. The binary owner switch is a transposition. This explains simultaneously why missing squares are binary and why ranks three and four cannot be compressed to a scalar parity before their permutation holonomy has been controlled. This identifies the smallest natural finite group containing the local binary moves; proving that the actual chart-cancellation maps factor through this action remains part of the coherence theorem.

Remark 5.88 (Owner transport is the local filling torsor). Proposition 5.85 supplies the missing topological meaning of the owner bit in Proposition 5.58. Once the protector coordinate q is fixed, splitting a projected class into its two lift layers creates exactly the two equatorial fillings h_0, h_1 . A singly owned protector edge permits only its owner's filling; a doubly owned separator edge permits both. Thus the $\mathbb{F}_2$ -torsor is canonical within each coordinate-labelled protector sector, while Proposition 5.86 supplies the ambient symmetric-group coherence in which transitions among sectors must land. What remains is to prove that the cancellation maps of the Yang failure cover factor through these local filling choices and that their compositions around partition-refinement diamonds give all, and only, the higher differentials. That coherence theorem would turn the present local torsors into the desired global holonomy invariant.

Remark 5.89 (Circuit existence is not the invariant). The balance condition in Corollary 5.82 is only the support condition on the first page of the spectral sequence. It is not sufficient even for the full square Q_2 . Take all four latent square charts, one for each present tip. Their failure relations form a strongly connected digraph on the four concepts, so

$$|\mathcal{I}| = 4 = |Q_2|, \quad k_{\mathcal{I}} = 1,$$

and the balance equation holds. Nevertheless Q_2 peels. The corresponding E^1 -generator is killed by a later differential.

The same warning appears at scale in the known obstructions. A worker diagnostic finds strongly connected generating subfamilies of 202 latent charts for Hall's order-299 seed and 201 charts for the certified order-291 seed. Padding them to 299 and 291 charts gives balanced carriers, but their mere existence does not distinguish the two seeds from shellable systems such as Q_2 . The durable artifact is `damp_yang_balanced_circuits.json`, generated by `analyze_damp_yang_balanced_circuits.py`. Thus the smaller invariant must encode survival of the canonical fundamental class—equivalently, the cancellation maps between circuit carriers—rather than cycle rank or strong connectivity alone.

Remark 5.90 (The candidate global invariant). Ordinary topology of Δ_H cannot detect the obstruction: every ample Yang complex in the present problem is already a constructible Cohen–Macaulay ball. Theorem 5.81 nevertheless gives an exact global invariant of its colored protector atlas. It is not yet the final generative answer. The order sphere has $m!$ chambers, and merely computing its top class is an order-free recognition oracle equivalent to shellability. The real gain is the spectral compression: every possible carrier of the fundamental class is a balanced collection of latent charts whose local failure relations close into directed circuits.

Associate local gate states to the charts and transport them through the color links and color-pair deletions. Proposition 5.58 shows that contraction preserves the protector coordinate q ; only the owner layer may change. Thus the natural smaller invariant is the monodromy of this finite-state

transport system on the balanced circuit carriers. If the transition relations can be proved to be canonical bijections, they define a holonomy representation into a finite permutation group. If, more strongly, the remaining owner states form two-element torsors, the representation reduces to a class

$$[\omega_H] \in H^1(\Gamma_H; \mathbb{F}_2),$$

whose value is obtained by multiplying transition signs around cycles.

Neither conclusion is currently automatic. Rank-three and rank-four charts have three and four coordinate-labelled protector states, present tips add escape states, and shared protector edges impose compatibility. Merely defining a global section to be an acyclic selection would restate the shelling search. The required structural theorem is therefore a *torsor-reduction theorem*: from the signed chart attachments alone, it must canonically quotient the gate states, prove functorial transport under all elementary pc-minors, and show that vanishing of the resulting holonomy constructs the minor shellings. Only after that theorem would $[\omega_H]$, or its finite-group analogue, be a genuine order-free global invariant rather than new notation for shellability.

Remark 5.91 (Why initially missing blocks do not suffice). The full square Q_2 has no missing-tip one-hole block. Nevertheless, after deleting 00, the punctured square $P(00, \{1, 2\})$ becomes active and prevents the antipode 11 from being the next corner. The latent word rule says exactly that if 00 is first, then the next vertex must be 01 or 10. Equivalently, the first two vertices of a peeling of Q_2 must be adjacent. Thus an invariant built only from the holes present in the initial family omits essential transition data even in dimension two. As a bounded regression, the static word rule and direct corner deletion agree on all 93,248 orders of all 127 nonempty ample families in Q_3 ; this computation is not used in the proof. The durable artifact is `damp_latent_punctured_word_q3.json`, generated by `verify_damp_latent_punctured_word.py`. The same implementation was checked on all 36 elementary images of Hall’s obstruction. It examined 104,839 latent blocks and found a facet protector for every nonescape state, as the corollary requires. The artifact is `damp_hall_latent_protectors.json`, generated on the worker by `analyze_damp_hall_latent_protectors.py`. Neither computation is used in the proof.

Remark 5.92 (Consequences for the holonomy program). Theorem 5.56 identifies the canonical local states that a global invariant must synchronize. They are coordinate-labelled rather than binary in general. For a present tip, an S -block has one escape state and $|S|$ facet-protector states; for an absent tip it has the $|S|$ protector states. Thus an absent square is binary, while absent blocks of ranks three and four have three and four states. Moreover, all latent blocks are needed: inclusion-maximal holes present at the root do not record the blocks activated by later deletions.

Corollary 5.57 is still a recognition theorem, not the desired generative classification. Choosing states and asking whether their precedence digraph is acyclic is another finite form of the shelling search. The remaining structural problem is to prove that, on the recovered paired one-hole presentations of overlap-atomic obstructions and their elementary images, these states factor through canonical block transitions. A binary $\mathbb{F}_2$ -holonomy can be complete only after proving a quotient theorem that collapses these coordinate states without losing compatibility; it cannot be assumed from the direct/switched dichotomy alone.

Remark 5.93 (Known Hall seeds are gluing-critical in every coordinate). A worker audit reconstructed the fixed 299-concept Hall obstruction, the certified 291-concept obstruction, and the 27 cornerless families in the radius-two Hall exchange atlas. Every one of their 348 coordinate overlaps $A = B_0 \cap B_1$ is an ample corner-peelable family of VC dimension two. Their orders are 67 in 339 cases, 65 in eight cases, and 59 in one case. Hence every tested coordinate has the gluing-critical signature. The same audit applied the two root-cut tests to both relative wings at every coordinate. All 696

wings are upper-root-locked, none is lower-root-locked, and the direct ampleness test agrees with the root-cut prediction in every case. More precisely, every wing has between two and fourteen possible first relative facets, but no possible last relative facet. Thus the known Hall mechanism is one-sided terminality, not biroot-locking.

The canonical relative one-hole recovery contains 164,762 minimal blocks over the 57,372 relative concepts. Their support orders are 2, 3, 4. At 49,404 concepts the least block has order two, at 3,288 it has order three, and at 4,680 it has order four; in particular, no tested wing is covered by punctured squares alone. Passing to inclusion-maximal blocks gives 86,226 canonical blocks. A deterministic greedy subcover uses 18,352 of them, between 26% and 40% as many blocks as relative concepts in each wing, and one order-four block protects as many as eleven relative concepts. The greedy counts are compression evidence only; the construction and recovery theorem does not depend on their optimality. The corresponding artifact is `damp_relative_one_hole_blocks.json`.

By contrast, if $K \notin \mathcal{D}_{\text{amp}}$, then the product $K \square Q_1$ has $B_0 = B_1 = A = C = K$ in the product coordinate and therefore has the inherited-site signature. The overlap signature thus separates the product adversary from all presently known Hall seeds, although this finite audit is not a proof that every Cartesian-prime seed is gluing-critical. The finite audit does not itself supply the desired construction: upper locking is equivalent to the absence of an ample-preserving deletion and therefore cannot simply be installed as an admissibility oracle. Its role is to identify the structural fact that the future cut-block axioms must force. The reproducible artifact is `damp_seed_overlap_shellability.json`, generated by `analyze_damp_seed_overlap_shellability.py`.

Proposition 5.94 (First-loss criterion for a corner peeling). *Let $\emptyset \neq H \subseteq Q_F$ be ample, put $\Sigma = \text{Sh}(H)$, and write $m = |H| = |\Sigma|$. For an ordering $\mathbf{x} = (x_1, \dots, x_m)$ of H , define*

$$\ell_{\mathbf{x}}(D, q) = \max\{i : x_i|_D = q\}, \quad \delta_{\mathbf{x}}(D) = \min_{q \in Q_D} \ell_{\mathbf{x}}(D, q) \quad (D \in \Sigma). \quad (5.89)$$

The maxima exist because H shatters D . Then $\mathbf{x}$ is a corner-peeling order if and only if

$$\delta_{\mathbf{x}} : \Sigma \longrightarrow [m] \quad \text{is a bijection.} \quad (5.90)$$

Thus corner peelability is an ordering property of the original trace table: no descendant family has to be constructed in order to state or check the criterion.

Proof. After deleting $x_1, \dots, x_k$, a support $D \in \Sigma$ is still shattered precisely when every D -trace has a representative whose index is greater than k . Equivalently,

$$D \in \text{Sh}(\{x_{k+1}, \dots, x_m\}) \iff \delta_{\mathbf{x}}(D) > k. \quad (5.91)$$

If $\delta_{\mathbf{x}}$ is bijective, the right-hand side holds for exactly $m - k$ supports. The suffix has $m - k$ concepts, so equality in the Sandwich bound makes every suffix ample. Successive deletion from an ample family preserves ampleness exactly when the deleted concept is a corner; hence $\mathbf{x}$ is a corner peeling.

Conversely, if $\mathbf{x}$ is a corner peeling, every suffix is ample. Taking cardinalities in (5.91) gives

$$|\{D \in \Sigma : \delta_{\mathbf{x}}(D) > k\}| = m - k \quad (0 \leq k \leq m).$$

It follows successively that exactly one support has each death time $1, \dots, m$, which is (5.90). $\square$

Definition 5.95 (Acyclic trace-carrier scheme). Let $H \subseteq Q_F$ be ample and put $\Sigma = \text{Sh}(H)$. For $D \in \Sigma$ and $q \in Q_D$, write

$$H(D, q) = \{x \in H : x|_D = q\}$$

for the corresponding trace fiber. Every such fiber is nonempty because D is shattered.

A trace-carrier scheme on H consists of

1. a bijection $\kappa : \Sigma \rightarrow H$; and
2. for every $D \in \Sigma$ and every $q \in Q_D \setminus \{\kappa(D)|_D\}$, a protector

$$p(D, q) \in H(D, q).$$

The *precedence digraph* $G(\kappa, p)$ has vertex set H and the following arcs, for every $D \in \Sigma$:

$$y \longrightarrow \kappa(D) \quad (y \in H(D, \kappa(D)|_D) \setminus \{\kappa(D)\}), \quad (5.92)$$

$$\kappa(D) \longrightarrow p(D, q) \quad (q \in Q_D \setminus \{\kappa(D)|_D\}). \quad (5.93)$$

The scheme is *acyclic* when $G(\kappa, p)$ has no directed cycle. Thus its data are a support–concept bijection and marked concepts in specified Boolean trace fibers; no total order of H is part of the definition.

Theorem 5.96 (Acyclic trace-carrier criterion). *Let $H \subseteq Q_F$ be ample. Then H has an ordering with bijective first-loss map if and only if H admits an acyclic trace-carrier scheme.*

Proof. Put $\Sigma = \text{Sh}(H)$ and $m = |H| = |\Sigma|$. Suppose first that $\mathbf{x} = (x_1, \dots, x_m)$ has bijective first-loss map $\delta_{\mathbf{x}}$. For $D \in \Sigma$, let

$$i_D = \delta_{\mathbf{x}}(D), \quad \kappa(D) = x_{i_D}.$$

The values i_D are $1, \dots, m$ without repetition, so κ is a bijection from Σ to H .

Let q_D be a trace whose last occurrence has index i_D . Its last representative is x_{i_D} , and hence $q_D = \kappa(D)|_D$. Every other member of $H(D, q_D)$ occurs before $\kappa(D)$. For each $q \neq q_D$, choose as $p(D, q)$ the last member of $H(D, q)$ in the order $\mathbf{x}$. Its index is strictly larger than i_D : it cannot be smaller by the definition of i_D , and equality would make the same concept carry two different D -traces. Consequently every arc in (5.92)–(5.93) points forward in $\mathbf{x}$. The resulting trace-carrier scheme is acyclic.

Conversely, let (κ, p) be an acyclic trace-carrier scheme and choose any topological ordering $\mathbf{x}$ of its precedence digraph. Fix $D \in \Sigma$, put $z = \kappa(D)$, and write $q_D = z|_D$. The arcs in (5.92) place every other member of $H(D, q_D)$ before z . Thus the last occurrence of the trace q_D is the position of z . For every $q \neq q_D$, the arc $z \rightarrow p(D, q)$ places a member of $H(D, q)$ strictly after z . Therefore

$$\delta_{\mathbf{x}}(D) = \text{pos}_{\mathbf{x}}(\kappa(D)).$$

Both $\kappa : \Sigma \rightarrow H$ and the position map $H \rightarrow [m]$ are bijections. Hence $\delta_{\mathbf{x}}$ is bijective, as required. $\square$

Remark 5.97 (What the trace-carrier criterion removes). For a specified scheme, all conditions are read directly from the trace table: fiber membership, bijectivity of κ , and acyclicity of a finite digraph. The criterion therefore removes the total-order variable. It does not yet give the desired generative classification of the obstructions, because one must still characterize those signed minimal-nonface presentations for which every trace-carrier scheme is cyclic. The remaining target is an intrinsic circuit or assembly description of that failure, rather than an enumeration of schemes.

Remark 5.98 (Protector choices cannot be saturated). One cannot make the precedence digraph canonical by replacing each chosen arc $\kappa(D) \rightarrow p(D, q)$ with arcs from $\kappa(D)$ to every member of $H(D, q)$. This stronger rule already fails for $H = Q_2$. Indeed, in any topological order of the saturated digraph, the carrier of a singleton support must occur after the other concept with the same trace and before both concepts with the opposite trace. It is therefore forced into the second position. The two singleton supports have distinct carriers under the bijection κ , so they cannot both occupy that position. Thus no saturated scheme exists, although Q_2 admits a trace-carrier scheme. The selected protector, rather than the whole competing fiber, is therefore essential incidence data.

Corollary 5.99 (Order-free Boolean recognition criterion for the ample forbidden basis). *Let σ be a signing of $\mathcal{N}(\Sigma)$, and put $H = H(\Sigma, \sigma)$. Suppose that $H \subseteq Q_F$ is irredundantly embedded. Then H is an ample forbidden pc-minor for $\mathcal{D}_{\text{amp}}$ if and only if all three conditions below hold.*

1. *The signing is admissible.*
2. *The family H admits no acyclic trace-carrier scheme.*
3. *Each of the $3|F|$ families $\rho_e^0 H, \rho_e^1 H, \pi_e H$, for $e \in F$, admits an acyclic trace-carrier scheme for its own shattered-set complex.*

Together with the nonample family $\{Q_n^{--} : n \geq 3\}$, these signed presentations, modulo coordinate permutations and reorientations, give the full forbidden basis. The conditions are static: they quantify over finite trace-fiber data of the original family and its explicit one-coordinate images, not over total orders, arbitrary proper pc-minors, or recursive peeling descendants.

Proof. Theorem 5.3 identifies admissibility with ampleness and fixes the shattered-set complex. Theorem 5.96, together with Proposition 5.94, identifies the second condition with failure of corner peelability and the third with corner peelability of every elementary pc-minor. Theorem 3.3 reduces pc-minimality to those elementary images and supplies the quotient by cube symmetries. Proposition 2.8 supplies the nonample part. $\square$

Remark 5.100 (Recognition is not a generative construction). Corollary 5.99 is an exact recognition criterion, but it does not solve the structural construction problem for the ample forbidden basis. Conditions 2 and 3 still ask whether specified families admit acyclic trace-carrier schemes. By Theorem 5.96, these are precisely the original corner-peelability questions with the total-order variable removed.

Encoding scheme existence as SAT, applying a fixed SAT or resolution procedure, or adjoining satisfying assignments and refutations as certificates does not change this point. Each is a finite decision or verification format for the same property, not structural data that explain why the root fails and all its elementary images succeed. We therefore do not regard such encodings or certificates as a parametrized construction of the obstructions.

Problem 5.101 (Intrinsic generative classification). *Write*

$$\mathfrak{B}_{\text{amp}} = \{H : H \in \mathcal{A} \setminus \mathcal{D}_{\text{amp}} \text{ and every proper pc-minor of } H \text{ belongs to } \mathcal{D}_{\text{amp}}\} / \cong.$$

Find a finite list of explicit combinatorial data types $(\mathcal{P}_\tau)_{\tau \in T}$, intrinsic admissibility predicates $\mathcal{A}_\tau$, and direct constructions $P \mapsto H_\tau(P)$ such that

$$\mathfrak{B}_{\text{amp}} = \{[H_\tau(P)] : P \in \mathcal{P}_\tau, \mathcal{A}_\tau(P)\}.$$

Here each individual parameter is finite, although its size and ambient coordinate set may vary. The displayed equality is required to hold for the following structural reasons.

1. *Explicit construction. The concepts and coordinates of $H_\tau(P)$ are obtained directly from the finite combinatorial data P . Constructing the output does not call a dismantlability, corner-peeling, or pc-minimality oracle.*
2. *Uniform soundness. Every P satisfying $\mathcal{A}_\tau(P)$ produces an ample forbidden pc-minor for $\mathcal{D}_{\text{amp}}$. Nonpeelability of the root and peelability of its proper pc-minors must follow uniformly from the structural form of P , rather than from running a recognition test on the output.*

3. Completeness and recovery. *Every ample forbidden pc-minor is isomorphic to some $H_\tau(P)$, and its parameter P is recovered from the obstruction, canonically or with explicitly controlled multiplicity, by a stated structural decomposition. In particular, completeness is not merely a proof that a suitable parameter exists.*
4. Non-circular admissibility. *The conditions $\mathcal{A}_\tau(P)$ are stated in native finite data such as signed incidences, supports, carriers, attachment maps, or prescribed assembly pieces. They do not invoke membership in $\mathcal{D}_{\text{amp}}$, the existence or nonexistence of a corner peeling, first-loss order, shelling order, linear-quotient order, or acyclic trace-carrier scheme; nor may they replace one of these tests by SAT/UNSAT of an equivalent formula or by an arbitrary proof certificate for the same assertion. In particular, one may not take P to be merely a lossless copy of $H_\tau(P)$ and hide the obstruction predicate in the definition of admissibility.*
5. Descendant-free verification. *Admissibility does not quantify over all proper pc-minors. The construction itself supplies a uniform structural reason why the required restrictions and contractions are peelable, for example through explicit parameter transformations or peelings prescribed directly from P .*

These restrictions concern the description of the parameters and the definition of admissibility. They do not prevent a soundness proof from deriving, uniformly from P , a canonical corner peeling of an elementary image. What is excluded is supplying such a peeling as auxiliary witness data, searching for one, or making its existence an admissibility condition. The nonample family $\{Q_n^{--} : n \geq 3\}$ is already explicit; the problem concerns the ample part of the basis.

We regard the classification as solved only when the construction, admissibility theorem, soundness theorem, and completeness-and-recovery theorem are all stated and proved. A parametrization of all ample or all cornerless ample families, a recognition criterion for the forbidden basis, or a construction proved only relative to an already classified seed is a component of such a solution, not the solution itself.

5.3 One-concept exchanges with fixed shattered sets

Proposition 5.102 (Exchange law for signed minimal nonfaces). *Let $H = H(\Sigma, \sigma)$ be the ample family of an admissible signing. For $a \in Q_F \setminus H$ and $c \in H$, put*

$$\mathcal{V}_H(a) = \{I \in \mathcal{N}(\Sigma) : a|_I = \sigma_I\}, \quad (5.94)$$

$$\mathcal{R}_H(a, c) = \left\{ I \in \mathcal{N}(\Sigma) : \begin{array}{l} a|_I \neq c|_I, \\ \{x \in H : x|_I = c|_I\} = \{c\} \end{array} \right\}. \quad (5.95)$$

Then

$$(H \setminus \{c\}) \cup \{a\} \text{ is ample with shattered complex } \Sigma \iff \mathcal{V}_H(a) = \mathcal{R}_H(a, c). \quad (5.96)$$

When these conditions hold, the new signing is

$$\sigma'_I = \begin{cases} c|_I, & I \in \mathcal{V}_H(a), \\ \sigma_I, & I \notin \mathcal{V}_H(a). \end{cases} \quad (5.97)$$

Moreover,

$$H \cup \{a\} \text{ is ample} \iff |\mathcal{V}_H(a)| = 1. \quad (5.98)$$

In the latter case, if $\mathcal{V}_H(a) = \{I\}$, then $\text{Sh}(H \cup \{a\}) = \Sigma \dot{\cup} \{I\}$.

Proof. For every $I \in \mathcal{N}(\Sigma)$, the trace set $H|I$ is $Q_I \setminus \{\sigma_I\}$. On replacing c by a , this projection still omits exactly one trace in precisely the following two cases: either $a|_I = \sigma_I$ and c is the unique old representative of the distinct trace $c|_I$, or $a|_I \neq \sigma_I$ and removal of c does not create a second missing trace. Simultaneously for all minimal nonfaces, these cases are exactly the equality $\mathcal{V}_H(a) = \mathcal{R}_H(a, c)$. They also give (5.97).

If the equality holds, the exchanged family omits one trace on every minimal nonface, so its shattered complex is contained in Σ . It has $|\Sigma|$ concepts. The Sandwich inequality therefore forces equality and ampleness. Conversely, an ample exchanged family with shadow Σ omits exactly one trace on every minimal nonface, so the preceding two-case analysis forces the displayed equality. This proves (5.96).

The set $\mathcal{V}_H(a)$ is nonempty because Theorem 5.3 says that H is exactly the common satisfying family. Adjoining a shatters every member of $\mathcal{V}_H(a)$. Hence, if $H \cup \{a\}$ is ample, its shadow has exactly one more member than Σ , and $|\mathcal{V}_H(a)| = 1$.

Conversely suppose $\mathcal{V}_H(a) = \{I_0\}$, and let a support $J \notin \Sigma$ become shattered after adjoining a . Since a single trace was added, $H|J = Q_J \setminus \{a|_J\}$. Choose a minimal nonface $I \subseteq J$. If $I \subsetneq J$, there is a trace on J that extends σ_I but differs from $a|_J$; it cannot be realized by H , a contradiction. Thus $J = I$, and necessarily $a|_I = \sigma_I$. Therefore $J = I_0$, and

$$\text{Sh}(H \cup \{a\}) = \Sigma \dot{\cup} \{I_0\}.$$

Its cardinality equals $|H| + 1$, so the Sandwich equality proves ampleness and completes the proof. $\square$

Proposition 5.103 (Fiber decomposition of the punctured-face parametrization). *Let $\mathcal{M}(S)$ be the minimal nonfaces of $\text{Sh}(S)$. By Theorem 5.3, every $I \in \mathcal{M}(S)$ has a unique omitted trace; denote it by σ_I . Define*

$$\mathcal{Y}_I(S) = \left\{ y \in Q_{F \setminus I} : \begin{array}{ll} (\sigma_I^U, y) \in S & \text{for every } \emptyset \neq U \subseteq I, \\ (\sigma_I, y^{\{e\}}) \notin S & \text{for every } e \in F \setminus I \end{array} \right\}. \quad (5.99)$$

Then

$$\mathcal{P}(S) = \bigcup_{I \in \mathcal{M}(S)} \{((\sigma_I, y), I) : y \in \mathcal{Y}_I(S)\}. \quad (5.100)$$

In particular, all punctured-face data with support I have the same missing trace on I . Their remaining datum is the outside assignment y : it locates a punctured I -cube and requires the missing tip to have no neighbor outside I .

Proof. Take $(a, I) \in \mathcal{P}(S)$. The punctured-cube condition realizes every I -trace except possibly $a|_I$, while $I \notin \text{Sh}(S)$ says that at least one trace is missing. Thus $a|_I$ is the unique missing trace σ_I . Put $y = a|_{F \setminus I}$. The punctured-cube condition is the first line of (5.99). Since $I = I_S(a)$, no coordinate outside I is incident with a , which is its second line.

Conversely, take I, σ_I, y on the right-hand side of (5.100) and put $a = (\sigma_I, y)$. The missing trace ensures $a \notin S$. The two lines in (5.99) say respectively that the punctured I -cube around a lies in S and that $I_S(a) = I$. Since I is a nonface, all conditions in (2.5) hold. The two constructions are inverse. $\square$

Proposition 5.104 (Adjacent square completions). *Let $G \subseteq Q_F$ be nonempty and ample. Suppose that $c, u \notin G$ are adjacent and that both individual additions are ample with the same acquired support I , where $|I| = 2$. Then $H = G \cup \{c, u\}$ is ample and*

$$H \in \mathcal{D}_{\text{amp}} \iff G \in \mathcal{D}_{\text{amp}}.$$

Proof. Write $I = \{p, q\}$ and $u = c^{\{f\}}$, so $f \notin I$. The two punctured squares in G , together with c , form the punctured $I \cup \{f\}$ -cube at u . This support is absent from $\text{Sh}(G \cup \{c\}) = \text{Sh}(G) \cup \{I\}$, so Lemma 2.10 makes H ample. If G peels, delete u, c in that order and then peel G .

For the converse, suppose that H peels and G does not, and choose such a counterexample with $|G|$ minimal. Put

$$C = Q_{I \cup \{f\}}(c), \quad x = c^{\{p,q\}}, \quad y = u^{\{p,q\}}, \quad W = \{c^{\{p\}}, c^{\{q\}}, u^{\{p\}}, u^{\{q\}}\}.$$

Thus $G \cap C = C \setminus \{c, u\}$. Neither added tip has a neighbor outside C . Moreover, a corner of H lying in C has no such neighbor: its incident cube already contains all directions of C , and an additional direction e would force the absent concept $c^{\{e\}}$.

Let z be the first deletion in a peeling of H . It cannot be c or u , because either deletion leaves a nonpeelable square completion of G , by Proposition 2.13. If $z \notin C$, its corner cube avoids both added tips, whose unique maximal cube is C . Thus z is also a corner of G . The ample family $G' = G \setminus \{z\}$ is nonpeelable, since otherwise deleting z first would peel G . Both punctured squares remain intact in G' , and both tips still acquire I . The peeling of $H \setminus \{z\}$ therefore contradicts the minimal choice of $|G|$.

If $z \in W$, interchange the two layers and the names p, q as necessary to write $z = c^{\{p\}}$. Since z has no outside neighbor, its incident $\{q, f\}$ -square lies in G , so it is a corner of G . Again $G' = G \setminus \{z\}$ is ample and nonpeelable. Adding u to G' completes its unchanged I -square, whose support is still unshattered because $G' \subseteq G$. The subsequent addition of c gives $H \setminus \{z\}$ and has incident support $\{q, f\}$. Both steps are ample and acquire supports of size two. Applying Proposition 2.13 twice contradicts the peelability of $H \setminus \{z\}$.

Only $z = x$ or $z = y$ remains. Exchanging the layer names handles the latter case, so assume $z = x$, and let w be the second deletion. The tips c, u retain every direction of C but their cube lacks x , so neither is a corner. If $w \notin C$, then w was already a corner of H : it is not adjacent to x , and deleting a nonneighbor cannot create a corner. Deleting w first leaves the entire corner cube of x intact. Swapping the first two deletions therefore gives a peeling starting outside C , which was just excluded.

Suppose instead that $w \in W$. It retains its neighbor in $\{c, u\}$, so any outside neighbor would still give a missing square through that tip. Hence w has no outside neighbor. It must also be adjacent to x , since otherwise all its cube directions remain incident while x is missing. Thus x, w are adjacent corners of the full cube, both without outside neighbors. Deleting either leaves the other as a corner of the opposite face. Swapping these two deletions gives a peeling starting in W , which was also excluded.

The only remaining choice is $w = y$. Its full I -square contains u , so an outside neighbor would require an outside neighbor of u . Therefore y , like x , has no neighbor outside C . The square automorphism interchanging c, x and fixing $c^{\{p\}}, c^{\{q\}}$, applied in both f -layers, interchanges the edges $\{c, u\}$ and $\{x, y\}$ and fixes W . It extends by the identity outside C , since neither exchanged edge has outside neighbors. Consequently $H \setminus \{x, y\} \cong G$, contradicting the remaining peeling. All first and second deletions have been accounted for. $\square$

The checker `verify_damp_square_pair_invariance.py` exercises these reductions on three ample bases with corners. Each has a proper restriction equal to the certified cornerless 291-concept family. It checks the two support-two additions after a common corner, both commutations, and the opposite-edge graph isomorphism. These fixed controls are nonminimal and require no peeling search.

Proposition 5.105 (Resolving additions and reductions). *Let $S \subseteq Q_F$ be nonempty and ample, and suppose that $H = S \cup \{a\}$ is an ample one-concept extension with acquired support I . For $D \subseteq F$, put $a_D = a|_{F \setminus D}$. Then*

$$H^D = \begin{cases} S^D \cup \{a_D\}, & D \subseteq I, \\ S^D, & D \not\subseteq I. \end{cases} \quad (5.101)$$

In the first case the added concept is new and has acquired support $I \setminus D$; if S^D is empty, the extension is a singleton and this support is empty.

Consequently, if $H \in \mathcal{D}_{\text{amp}}$ and $S^D \notin \mathcal{D}_{\text{amp}}$, then

$$D \subseteq I, \quad |I \setminus D| \geq 3. \quad (5.102)$$

In particular, a resolving addition with $|I| = 3$ forces every reduction of S on a nonempty coordinate set to belong to $\mathcal{D}_{\text{amp}}$.

Proof. The only full D -fibre that the addition can create is the fibre through a . If $D \subseteq I$, all its other concepts belong to the punctured I -cube in S , so that fibre becomes full. Its projection a_D was absent from S^D , because its old fibre lacked a . If instead $e \in D \setminus I$, the same fibre still lacks $a^{\{e\}}$, which is absent by the definition of $I = I_S(a)$. This proves (5.101), including $D = \emptyset$.

Assume $D \subseteq I$. For each $e \in I \setminus D$, the entire D -fibre through $a^{\{e\}}$ belongs to S , so e is incident with a_D in the extension. No $e \in F \setminus I$ can be incident: its fibre would require $a^{\{e\}} \in S$. The incident support is therefore exactly $I \setminus D$. Reductions preserve ampleness, so Lemma 2.10 identifies this as the acquired support whenever $S^D \neq \emptyset$. If S^D is empty, there are no incident directions, and the singleton acquires only the empty support.

Now suppose that H peels and S^D does not. To see directly that H^D peels, reverse a peeling of H to an ample prefix order, and order the full D -fibres by their completion times. The resulting prefixes are reductions of ample prefixes and hence are ample. Reversing this order peels H^D . Thus the two reductions cannot be equal, which forces $D \subseteq I$. Their difference is a resolving one-concept addition of support $I \setminus D$, and Proposition 2.13 gives $|I \setminus D| \geq 3$. For $|I| = 3$, this is impossible when $D \neq \emptyset$, proving the last assertion. $\square$

The prescribed-endpoint analogue of this projection argument is false, even when the root has a full fibre; see Proposition 2.108.

Corollary 5.106 (Deletion of adjacent degree-three corners). *Let $H \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor. If c, u are adjacent corners of degree three, then*

$$H \setminus \{c\}, \quad H \setminus \{u\}, \quad H \setminus \{c, u\} \quad \text{belong to } \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}.$$

In particular, a degree-three corner of a corner-descent-terminal ample forbidden pc-minor has no corner as a neighbor.

Proof. The adjacent corners have the same unique maximal cube C , because their edge belongs to a maximal cube containing both. Write P for its three-coordinate support, $u = c^{\{f\}}$, and $I = P \setminus \{f\}$. Neither corner has an outside neighbor. Deleting c leaves u as a corner of its I -face, so $G = H \setminus \{c, u\}$ is ample. Each individual addition to G acquires I . Proposition 5.104 makes G nonpeelable. Its embedding remains irredundant: the punctured cube $C \setminus \{c, u\}$ retains every direction of P , and a surviving cube vertex has the same outside coordinates as the two deleted concepts.

We check every elementary pc-minor μG , using $\mu H \in \mathcal{D}_{\text{amp}}$. Under a restriction in coordinate e , either both tips disappear, only one is retained, or both are retained. The single-tip case occurs

when $e = f$, and its acquired support is I . If both survive, their individual acquired supports are $I \setminus \{e\}$, and they remain adjacent in direction f . All these intermediate images are ample, being images of $G, G \cup \{c\}, G \cup \{u\}, H$.

For a contraction in $e \in I$, each tip is identified with a neighbor in G , so $\mu H = \mu G$. Contraction in f identifies the two tips with each other and adds one concept to $\pi_f G$, with acquired support I . A contraction outside P retains two distinct new tips: neither has a neighbor in that direction. Their individual acquired supports remain I , since contraction restricts shattered supports to those avoiding e and hence preserves the one-support gain (2.3).

Thus μH either equals μG , adds one concept with support at most two, or adds an adjacent pair with common individual support of size one or two. In the single-concept case, Proposition 2.13 makes μG peelable. For a pair with common support of size two, Proposition 5.104 does the same. For a pair with common support of size one, the two successive additions have supports of sizes one and two, so the one-concept invariance applies twice. The equal-image case is immediate. Every elementary pc-minor of G therefore peels, proving $G \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

The two single deletions from H are nonpeelable: a peeling of either could be prefixed by the deleted corner to peel H . They are ample one-concept extensions of the irredundant forbidden family G , so Proposition 2.12 makes both forbidden. Finally, adjacent corners necessarily share their maximal cube and hence their degree. A corner neighbor of a degree-three corner would therefore give an obstruction-preserving deletion, contradicting corner-descent terminality. $\square$

Proposition 5.107 (Same-shadow mutations). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be cornerless, let $(a, I) \in \mathcal{P}(S)$, and put $A = S \cup \{a\}$. Define the set of exposed tips of this punctured face by*

$$\mathcal{E}_S(a, I) = \{a^U : \emptyset \neq U \subseteq I, \quad a^{U \Delta \{e\}} \notin S \text{ for every } e \in F \setminus I\}. \quad (5.103)$$

Then

$$\text{Cor}(A) = \{a\} \dot{\cup} \mathcal{E}_S(a, I). \quad (5.104)$$

For every $c \in \mathcal{E}_S(a, I)$, the exchanged family

$$T_c = (S \setminus \{c\}) \cup \{a\} = A \setminus \{c\}$$

is ample and has the same shattered-set complex as S . Moreover,

$$(a, I) \in \mathcal{P}_{\text{peel}}(S) \iff T_c \in \mathcal{D}_{\text{amp}} \text{ for some } c \in \mathcal{E}_S(a, I). \quad (5.105)$$

No exposed tip is at Hamming distance one from a . If $|I| = 2$, then every exposed tip c satisfies $T_c \cong S$, and hence $(a, I) \in \mathcal{P}_{\text{obs}}(S)$. Consequently every peelable punctured-face datum over a cornerless forbidden family has

$$|I| \geq 3. \quad (5.106)$$

Proof. Lemma 2.10 makes a a corner of A , with unique maximal cube the completed I -cube. Let $c \in S$ be any other corner of A . Its unique maximal cube in A must contain a . Indeed, if it avoided a , it would remain a cube of S containing every cube of S through c , making c a corner of the cornerless family S . The maximal cube through c is therefore the completed I -cube. Thus $c = a^U$ for a nonempty $U \subseteq I$, and c has no neighbor in a direction outside I . This proves that the left-hand side of (5.104) is contained in the right-hand side.

Conversely, take $c \in \mathcal{E}_S(a, I)$. The completed I -cube contains every neighbor of c in a direction in I , and the definition of an exposed tip excludes every other direction. Hence this cube is the unique maximal cube of A through c , so c is a corner. This proves (5.104).

Deleting the corner c from the ample family A gives the ample family T_c . The unique maximal cube of c has support I , so the shattered-support form of corner deletion gives

$$\text{Sh}(T_c) = \text{Sh}(A) \setminus \{I\} = \text{Sh}(S),$$

where the second equality is (2.3). This proves the same-shadow assertion.

A nonempty ample family is corner-peelable precisely when some corner can be removed leaving a corner-peelable family. Removing a from A leaves the nonpeelable family S ; by (5.104), all other possible first vertices are exactly the exposed tips. This proves (5.105).

Suppose that $a^{\{e\}}$ were an exposed tip. In S , its incident directions would be exactly $I \setminus \{e\}$, and the punctured I -cube contains the entire cube on those directions. It would therefore be a corner of S , a contradiction. Hence exposed tips have Hamming distance at least two from a .

Finally let $I = \{p, q\}$. The only possible exposed tip is $c = a^{\{p, q\}}$. In A , the vertices a and c have the same two neighbors $a^{\{p\}}, a^{\{q\}}$ and no others. Interchanging a and c is consequently a graph automorphism of A , and it maps $T_c = A \setminus \{c\}$ to $S = A \setminus \{a\}$. Thus $T_c \cong S \notin \mathcal{D}_{\text{amp}}$. The exchange criterion and the parametrization dichotomy in Proposition 2.12 now give the last two assertions. $\square$

Remark 5.108 (Relation with oriented-matroid mutations). For a uniform oriented matroid, replacing a simplicial tope and its antipode by the two missing opposite tips is called a mutation, and the resulting mutation graph is a well-studied object [KM23, Section 3]. The operation in Proposition 5.107 is the one-sided ample analogue: it replaces one exposed tip of a completed cube by the opposite missing tip and preserves the shattered-set complex. We therefore call it a *same-shadow mutation*. Our exchange graph is more general than the oriented-matroid mutation graph: neither antipodality nor uniformity is assumed, and its relevant local ranks need not equal the VC dimension. The known connectivity theorems for rank-three oriented matroids consequently do not settle the shellability question here.

6 Layered extensions and irreducible families

We first give an exact trace formula for arbitrary marked layer systems. For the natural independent construction—one new concept in each selected layer over a common seed—we then determine ampleness, the signed presentation, and all inverse parameters. Allowing repair tips to persist through arbitrary layer families yields a labelled occurrence-cover normal form in which ampleness and the signed clauses separate exactly. Pc-minimality makes the singleton-occurrence specialization collapse to the two-extension lift over one new coordinate. We finish by reducing every ample nonpeelable family to a cornerless, periphery-free, Cartesian-prime family.

Proposition 6.1 (Trace transport for an arbitrary layer system). *Let E and F be disjoint coordinate sets. For each $t \in Q_E$, choose a section $H_t \subseteq Q_F$, and put*

$$X = \bigcup_{t \in Q_E} (H_t \times \{t\}) \subseteq Q_{F \cup E}.$$

For any cube family Y , support D , and trace $q \in Q_D$, write

$$Y[D, q] = \{y \in Y : y|_D = q\}.$$

For $L \subseteq E$ and $r \in Q_L$, define the projected union of the layers with L -trace r by

$$H_{L, r} = \bigcup_{\substack{t \in Q_E \\ t|_L = r}} H_t. \tag{6.1}$$

Then, for $D \subseteq F$, $q \in Q_D$, $L \subseteq E$, and $r \in Q_L$,

$$X[D \cup L, (q, r)] = \bigcup_{\substack{t \in Q_E \\ t|_L = r}} (H_t[D, q] \times \{t\}). \quad (6.2)$$

Consequently,

$$D \cup L \in \text{Sh}(X) \iff D \in \bigcap_{r \in Q_L} \text{Sh}(H_{L,r}). \quad (6.3)$$

Conversely, the marked layer set E recovers every parameter H_t as the projected t -section of X . Thus (6.2) gives an exact, canonically recoverable trace table for any finite number of binary assembly layers.

Proof. A vertex $(x, t) \in X$ has $(D \cup L)$ -trace (q, r) precisely when $t|_L = r$ and $x|_D = q$. Partitioning the corresponding trace fiber by the full layer word t proves (6.2). The support $D \cup L$ is shattered precisely when this fiber is nonempty for every $(q, r) \in Q_D \times Q_L$. For fixed r , that condition says exactly that $H_{L,r}$ shatters D , proving (6.3). The recovery statement is the definition of a section. $\square$

Proposition 6.2 (Independent repair layers). *Let $E \neq \emptyset$ and F be disjoint coordinate sets, let $S \subseteq Q_F$ be ample, and let $T \subseteq Q_E$. For every $t \in T$, choose a vertex $a_t \in Q_F \setminus S$, with the vertices $(a_t)_{t \in T}$ pairwise distinct, such that*

$$A_t = S \cup \{a_t\}$$

is ample. Write

$$\text{Sh}(A_t) \setminus \text{Sh}(S) = \{P_t\}, \quad P_t = \{f \in F : a_t^{\{f\}} \in S\}. \quad (6.4)$$

Place one repair tip in each selected layer and no repair tip in the other layers:

$$X_T = (S \times Q_E) \cup \{(a_t, t) : t \in T\}. \quad (6.5)$$

Then

$$X_T \text{ is ample} \iff t \mapsto P_t \text{ is injective on } T. \quad (6.6)$$

Assume henceforth that the supports are pairwise distinct, and put

$$\Sigma = \text{Sh}(S), \quad \Sigma^+ = \Sigma \dot{\cup} \{P_t : t \in T\}, \quad C_T = S \cup \{a_t : t \in T\}.$$

The shadow, the signed minimal nonfaces, and the inverse parameters of the assembly are all explicit. Namely,

$$\text{Sh}(X_T) = \{D \cup L : D \in \Sigma, L \subseteq E\} \dot{\cup} \{P_t : t \in T\}, \quad (6.7)$$

$$\mathcal{N}(\text{Sh}(X_T)) = \mathcal{N}(\Sigma^+) \dot{\cup} \{P_t \cup \{e\} : t \in T, e \in E\}. \quad (6.8)$$

For the corresponding unique missing traces,

$$\sigma_{X_T}(I) = \sigma_{C_T}(I) \quad (I \in \mathcal{N}(\Sigma^+)), \quad (6.9)$$

$$\sigma_{X_T}(P_t \cup \{e\}) = (a_t|_{P_t}, 1 - t_e) \quad (t \in T, e \in E). \quad (6.10)$$

In particular, these signed clauses reconstruct X_T . Conversely, the marked layer set E canonically recovers all the displayed parameters. If $H_v \subseteq Q_F$ is the projected v -section of X_T , then

$$S = \bigcap_{v \in Q_E} H_v, \quad T = \{v \in Q_E : H_v \neq S\}, \quad \{a_t\} = H_t \setminus S, \quad (6.11)$$

and each P_t is recovered by the local formula in (6.4).

Proof. For $U \subseteq T$, put $C_U = S \cup \{a_t : t \in U\}$. Since S is ample, it strongly shatters every member of Σ . The one-concept extension criterion shows that A_t strongly shatters P_t . If the supports P_t , $t \in U$, are distinct, then C_U therefore strongly shatters at least

$$|\Sigma| + |U| = |S| + |U| = |C_U|$$

distinct supports. The Sandwich inequalities force equality, so C_U is ample and

$$\text{Sh}(C_U) = \Sigma \dot{\cup} \{P_t : t \in U\}. \quad (6.12)$$

Apply Proposition 6.1 to the sections $H_t = A_t$ for $t \in T$ and $H_t = S$ otherwise. If $L = \emptyset$, their union is C_T , whose shadow is given by (6.12). If $L \neq \emptyset$, every union $H_{L,r}$ contains S , while a given new support P_t occurs only in the unique union with $r = t|_L$. Since Q_L has at least two words, the intersection of the shadows of the $H_{L,r}$ is exactly Σ . The shattering criterion (6.3) now proves (6.7). Its right-hand side has

$$2^{|E|}|\Sigma| + |T| = 2^{|E|}|S| + |T| = |X_T|$$

members, giving ampleness.

Conversely, suppose $P_t = P_u = P$ for distinct $t, u \in T$, and choose $e \in E$ with $t_e \neq u_e$. Each of the two unions of sections on the two sides of e contains an extension that shatters P . Hence $P \cup \{e\} \in \text{Sh}(X_T)$. The product $S \times Q_E$ already shatters all $D \cup L$ with $D \in \Sigma$ and $L \subseteq E$. Moreover, the old-coordinate projection C_T has at least $|C_T| = |S| + |T|$ shattered supports by Sauer's inequality, and hence at least $|T|$ old supports outside Σ . The support $P \cup \{e\}$ is an additional one. Therefore

$$|\text{Sh}(X_T)| \geq 2^{|E|}|S| + |T| + 1 = |X_T| + 1,$$

so X_T is not ample. This proves (6.6).

It remains to read the signed presentation from (6.7). Each P_t is a minimal nonface of Σ . A nonface using a layer coordinate is minimal precisely when it has the form $P_t \cup \{e\}$; nonfaces contained in F are exactly the minimal nonfaces of Σ^+ . This proves (6.8). Projecting to the old coordinates gives C_T , which proves (6.9). The trace $a_t|_{P_t}$ is missing from S . No other tip a_u supplies it: otherwise A_u would shatter P_t , contradicting

$$\text{Sh}(A_u) = \Sigma \dot{\cup} \{P_u\} \quad \text{and} \quad P_u \neq P_t.$$

Thus this trace occurs only in the full layer t , and the missing trace on $P_t \cup \{e\}$ is $(a_t|_{P_t}, 1 - t_e)$. This proves (6.10); the fixed-shadow signed-clutter theorem reconstructs X_T .

Finally, $E \neq \emptyset$, every tip occurs in exactly one section, and the tips are pairwise distinct. The intersection of all sections is therefore S , after which the remaining formulas in (6.11) and the local support formula are immediate. $\square$

Proposition 6.3 (Replicated repair layers). *Let $E \neq \emptyset$ and F be disjoint, let $S \subseteq Q_F$ be ample, and choose a finite set $\mathcal{A} \subseteq Q_F \setminus S$. Suppose that every $S \cup \{a\}$, $a \in \mathcal{A}$, is ample, and that their new supports*

$$\text{Sh}(S \cup \{a\}) \setminus \text{Sh}(S) = \{P_a\}$$

are pairwise distinct. For each $a \in \mathcal{A}$, choose a nonempty occurrence family $R_a \subseteq Q_E$, and put

$$X(\mathbf{R}) = (S \times Q_E) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times R_a). \quad (6.13)$$

Then its shattered-set complex is

$$\text{Sh}(X(\mathbf{R})) = \{D \cup L : D \in \text{Sh}(S), L \subseteq E\} \dot{\cup} \bigcup_{a \in \mathcal{A}} \{P_a \cup L : L \in \text{Sh}(R_a)\}. \quad (6.14)$$

Consequently,

$$X(\mathbf{R}) \text{ is ample} \iff R_a \text{ is ample for every } a \in \mathcal{A}. \quad (6.15)$$

Assume these equivalent conditions. Set

$$\Sigma = \text{Sh}(S), \quad \Sigma^+ = \Sigma \dot{\cup} \{P_a : a \in \mathcal{A}\}, \quad \Gamma_a = \text{Sh}(R_a), \quad C = S \cup \mathcal{A}.$$

Then the signed minimal nonfaces are

$$\mathcal{N}(\text{Sh}(X(\mathbf{R}))) = \mathcal{N}(\Sigma^+) \dot{\cup} \bigcup_{a \in \mathcal{A}} \{P_a \cup J : J \in \mathcal{N}(\Gamma_a)\}, \quad (6.16)$$

with signs

$$\sigma_{X(\mathbf{R})}(I) = \sigma_C(I) \quad (I \in \mathcal{N}(\Sigma^+)), \quad (6.17)$$

$$\sigma_{X(\mathbf{R})}(P_a \cup J) = (a|_{P_a}, \sigma_{R_a}(J)) \quad (a \in \mathcal{A}, J \in \mathcal{N}(\Gamma_a)). \quad (6.18)$$

Thus the old repair clause and the layer clause compose without any additional compatibility condition.

If every R_a is proper in Q_E , the marked layer set recovers the parameters canonically. For the projected sections $H_t \subseteq Q_F$,

$$S = \bigcap_{t \in Q_E} H_t, \quad \mathcal{A} = \left(\bigcup_{t \in Q_E} H_t \right) \setminus S, \quad R_a = \{t \in Q_E : a \in H_t\}, \quad (6.19)$$

and $P_a = \{f \in F : a^{\{f\}} \in S\}$. Without the properness assumption, the same formulas give the unique normalized presentation: every tip with $R_a = Q_E$ is absorbed into the common core.

Proof. For $U \subseteq \mathcal{A}$, the proof of Proposition 6.2 gives

$$\text{Sh}(S \cup U) = \text{Sh}(S) \dot{\cup} \{P_a : a \in U\}. \quad (6.20)$$

Indeed, these are $|S| + |U|$ distinct strongly shattered supports, so the Sandwich inequalities force equality.

For $L \subseteq E$ and $r \in Q_L$, the union of the sections with layer trace r is

$$H_{L,r} = S \cup \{a \in \mathcal{A} : R_a \cap \{t \in Q_E : t|_L = r\} \neq \emptyset\}.$$

Equation (6.20) therefore shows that $H_{L,r}$ shatters P_a exactly when R_a contains a word with L -trace r . Hence every $H_{L,r}$ shatters P_a exactly when R_a shatters L . Every one of these unions shatters all members of Σ , and their total union shatters no old support outside Σ^+ . Proposition 6.1 now gives (6.14).

The two sides of the Sandwich inequality differ by

$$|\text{Sh}(X(\mathbf{R}))| - |X(\mathbf{R})| = \sum_{a \in \mathcal{A}} (|\text{Sh}(R_a)| - |R_a|).$$

Every summand is nonnegative by Sauer's inequality. It vanishes exactly when R_a is ample, proving (6.15).

Now suppose all occurrence families are ample. A nonface contained in F is minimal exactly when it lies in $\mathcal{N}(\Sigma^+)$. If a minimal nonface uses a layer coordinate, its old part must be one of the minimal nonfaces P_a added to Σ , and its layer part must be a minimal nonface of Γ_a . Conversely, every $P_a \cup J$ with $J \in \mathcal{N}(\Gamma_a)$ is minimal: deleting an old coordinate puts its old part in Σ , while deleting a layer coordinate puts its layer part in Γ_a . This proves (6.16).

The old-coordinate projection of $X(\mathbf{R})$ is C , which gives (6.17). The trace $a|_{P_a}$ is missing from S and from every other repair tip, by the distinctness of the supports. It occurs in $X(\mathbf{R})$ precisely over the layer words in R_a . Thus a trace on $P_a \cup J$ is missing precisely when its old part is $a|_{P_a}$ and its layer part is the unique trace $\sigma_{R_a}(J)$ missing from R_a . This proves (6.18).

Finally, if all occurrence families are proper, every tip is absent from some section, whereas every member of S occurs in every section. This proves (6.19), and the local support formula follows from the one-concept extension criterion. In general,

$$\bigcap_{t \in Q_E} H_t = S \cup \{a \in \mathcal{A} : R_a = Q_E\}.$$

Absorbing these full rows into the core preserves ampleness and the distinct-support union formula (6.20); all remaining occurrence families are proper. The displayed intersection makes this normalized presentation unique. $\square$

Remark 6.4 (The first genuinely coupled parameters). The independent construction in Proposition 6.2 is the special case in which every R_a is a singleton. Proposition 6.3 replaces a search through layered cube families by two finite pieces of data: a repair tip labelled by P_a , and an ample occurrence family R_a on the layer cube. Overlap among the R_a is exactly the first coupling mechanism: several tips may occur in one section and one tip may persist through several sections. The remaining obstruction problem is no longer to guess which such systems are ample, but to characterize which ample labelled covers $(R_a)_{a \in \mathcal{A}}$ are pc-minimal and preserve the unavoidable trace circuits of the seed.

Corollary 6.5 (Layer pc-minors act on the occurrence cover). *Use the notation of Proposition 6.3. For $e \in E$ and $i \in \{0, 1\}$, let*

$$\rho_e^i R_a = \{t|_{E \setminus \{e\}} : t \in R_a, t_e = i\}, \quad \pi_e R_a = \{t|_{E \setminus \{e\}} : t \in R_a\}.$$

Then

$$\rho_e^i X(\mathbf{R}) = (S \times Q_{E \setminus \{e\}}) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times \rho_e^i R_a), \tag{6.21}$$

$$\pi_e X(\mathbf{R}) = (S \times Q_{E \setminus \{e\}}) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times \pi_e R_a). \tag{6.22}$$

Empty rows are discarded, and rows equal to $Q_{E \setminus \{e\}}$ are absorbed into the common core. After this normalization, both images are again replicated repair assemblies with the surviving labels P_a .

Thus restrictions and contractions in marked layer coordinates are computed entirely inside the finite labelled occurrence cover $(R_a, P_a)_{a \in \mathcal{A}}$. In particular, if $X(\mathbf{R})$ is a forbidden pc-minor, all $3|E|$ normalized covers in (6.21)–(6.22) are corner-peelable.

Proof. A word (x, t) survives the $e = i$ restriction precisely when $t_e = i$, and suppressing e replaces its layer word by $t|_{E \setminus \{e\}}$. This proves (6.21). Contraction takes the union of the two restricted sections, so it projects each R_a , proving (6.22). The normalization and label statements follow from the recovery part of Proposition 6.3. The final assertion is the elementary pc-minor condition for a forbidden pc-minor. $\square$

Proposition 6.6 (Corners and layer-minimality of a repair cover). *Use the notation and hypotheses of Proposition 6.3, assume that all occurrence families are ample and proper, and let*

$$C = S \cup \mathcal{A}, \quad Q_a = \{a^U : U \subseteq P_a\} \quad (a \in \mathcal{A}).$$

Then Q_a is the unique maximal cube of C through a . The maximal cubes of $X(\mathbf{R})$ through (a, t) are exactly

$$Q_a \times K, \quad K \text{ a maximal cube of } R_a \text{ through } t. \quad (6.23)$$

If S is cornerless, this gives the exact corner set

$$\text{Cor}(X(\mathbf{R})) = \bigcup_{a \in \mathcal{A}} (\{a\} \times \text{Cor}(R_a)). \quad (6.24)$$

Suppose in addition that S is nonpeelable, every one-concept repair $S \cup \{a\}$ belongs to $\mathcal{D}_{\text{amp}}$, and every R_a belongs to $\mathcal{D}_{\text{amp}}$. For a layer restriction or contraction

$$\nu \in \{\rho_e^0, \rho_e^1, \pi_e\}, \quad e \in E,$$

one then has

$$\nu X(\mathbf{R}) \in \mathcal{D}_{\text{amp}} \iff \nu R_a = Q_{E \setminus \{e\}} \text{ for some } a \in \mathcal{A}. \quad (6.25)$$

Consequently, all elementary pc-minors in marked layer coordinates are corner-peelable if and only if the occurrence rows own every signed facet of the layer cube:

$$\text{for every } e \in E \text{ and } i \in \{0, 1\}, \quad \{t \in Q_E : t_e = i\} \subseteq R_a \text{ for some } a \in \mathcal{A}. \quad (6.26)$$

Thus the entire layer-coordinate part of pc-minimality is a finite containment test on the labelled cover; it involves neither a corner order nor a trace-carrier scheme.

Proof. For every $U \subseteq \mathcal{A}$, equation (6.20) says that $S \cup U$ is ample. In particular, both C and $C \setminus \{a\}$ are ample, so a is a corner of C . The one-concept extension criterion identifies its unique maximal cube as Q_a .

Let L be the support of a cube of $X(\mathbf{R})$ through (a, t) in the layer coordinates. Its old-coordinate projection is a cube of C through a , hence is contained in Q_a . Moreover, all vertices with old word a show that the layer face through t on L lies in R_a . Conversely, whenever a layer cube $K \subseteq R_a$ contains t , the product $Q_a \times K$ lies in the assembly: its vertices with old word different from a belong to $S \times Q_E$, while its vertices with old word a are supplied by R_a . Maximality on both sides proves (6.23).

Now let $s \in S$, and let M be a maximal cube of S through s . The cube $M \times Q_E$ is maximal in the assembly. Indeed, a proper extension of it can use only old coordinates, and any new old word in that extension would have to occur in every layer. Such a word would be a repair tip with full occurrence row, contrary to properness. If S is cornerless, s lies in at least two maximal cubes of S , so (s, t) lies in at least two maximal cubes of the assembly. Together with (6.23), this proves (6.24).

Fix ν as in the statement and normalize the rows of νX as in Corollary 6.5. If no image row is full, strongly projecting all remaining layer coordinates gives exactly S . Since S is nonpeelable, Proposition 2.5 gives $\nu X \notin \mathcal{D}_{\text{amp}}$. This argument also covers a zero-dimensional remaining layer cube and requires no assumption on the corners of S .

Suppose instead that the set

$$\mathcal{A}_0 = \{a : \nu R_a = Q_{E \setminus \{e\}}\}$$

is nonempty. Absorb these rows into the new core $C_0 = S \cup \mathcal{A}_0$. Choose $a_0 \in \mathcal{A}_0$. Every other member of $\mathcal{A}_0$ is a corner of the current old-coordinate union, so these tips may be peeled until $S \cup \{a_0\}$ remains. The latter belongs to $\mathcal{D}_{\text{amp}}$, and hence $C_0 \in \mathcal{D}_{\text{amp}}$. Every remaining image row is a pc-minor of some R_a , so it belongs to $\mathcal{D}_{\text{amp}}$. Formula (6.23), applied successively, lifts a corner peeling of each proper row to the assembly. After all proper rows have been removed, the family is $C_0 \times Q_{E \setminus \{e\}}$, which belongs to $\mathcal{D}_{\text{amp}}$ by the product test. This proves (6.25).

For a restriction, the image row $\rho_e^i R_a$ is full exactly when the signed facet $\{t : t_e = i\}$ is contained in R_a . If this holds for either sign, then $\pi_e R_a$ is also full. Thus the $2|E|$ restriction conditions imply all $|E|$ contraction conditions, and (6.26) follows. $\square$

Lemma 6.7 (Compatible repairs do not collide). *Let $S \subseteq Q_F$ be ample, and let $\mathcal{A} \subseteq Q_F \setminus S$ be a set of one-concept ample extensions whose new shattered supports P_a , $a \in \mathcal{A}$, are pairwise distinct. Then no two members of $\mathcal{A}$ are adjacent in Q_F .*

More generally, let μ be a restriction or contraction in one old coordinate, put $S_\mu = \mu S$, and discard the tips which do not survive the restriction or whose image belongs to S_μ . The map $a \mapsto \mu a$ is injective on the remaining tips. Moreover, after any subset of these projected tips has been retained, every retained tip is a corner of the resulting old-coordinate family.

Proof. Suppose that $b = a^{\{f\}}$ for distinct $a, b \in \mathcal{A}$. Since neither tip belongs to S , the repair-support formula gives $f \notin P_a \cup P_b$. The family $C = S \cup \{a, b\}$ is ample by (6.20). Under contraction of f , the two tips merge into one word c , and $c \notin \pi_f S$; hence

$$|\pi_f C| = |\pi_f S| + 1.$$

On the other hand, both distinct supports $P_a, P_b \subseteq F \setminus \{f\}$ are shattered by $\pi_f C$ and not by $\pi_f S$. Indeed, the missing trace on P_a is $a|_{P_a}$, contraction does not change this trace, and the same argument applies to P_b . Both contracted families are ample, so the equality between cardinality and number of shattered supports makes a gain of at least two shattered supports incompatible with the displayed gain of one concept. This proves nonadjacency.

A restriction is injective on the tips which survive it. Two tips can have the same image under a contraction only when they are adjacent in the contracted coordinate, which has just been excluded. This proves the injectivity assertion. Finally, let $\mathcal{U}$ be any subset of the remaining tips. The families $S \cup \mathcal{U}$ and $S \cup (\mathcal{U} \setminus \{a\})$ are ample by (6.20); their images under μ are ample as well. Injectivity and the exclusion of images in S_μ show that the second image is obtained from the first by deleting precisely μa . The ample corner-deletion criterion therefore makes μa a corner, independently of the chosen subset. $\square$

Theorem 6.8 (Simple complete common-seed cover criterion). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be irredundantly embedded in Q_F . Let $\mathcal{A} \subseteq Q_F \setminus S$ be a finite set such that every $S \cup \{a\}$ belongs to $\mathcal{D}_{\text{amp}}$ and the new supports P_a are pairwise distinct. Let $E \neq \emptyset$, and assign to every $a \in \mathcal{A}$ a nonempty proper ample family $R_a \subsetneq Q_E$. Form $X(\mathbf{R})$ as in (6.13). Then*

$$X(\mathbf{R}) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff \begin{array}{l} R_a \in \mathcal{D}_{\text{amp}} \text{ for every } a \in \mathcal{A}, \\ \{t : t_e = i\} \subseteq R_a \text{ for some } a \in \mathcal{A} \quad (e \in E, i \in \{0, 1\}). \end{array} \quad (6.27)$$

Thus no condition in the old coordinates is needed: compatible repairs cannot collide under an elementary old-coordinate map, and their projected tips remain successively peelable. The two displayed conditions are finite and descendant-free, but the row condition $R_a \in \mathcal{D}_{\text{amp}}$ is still a recognition condition and not intrinsic structural data.

The normalized marked sections canonically recover $S, (a, P_a, R_a)_{a \in \mathcal{A}}$. Hence (6.27) is exact within the common-seed class over any fixed ample forbidden seed. It is not yet an intrinsic generative classification: the seed is assumed to be an obstruction, and peelability of the occurrence rows remains to be explained structurally. The terminal specialization below removes the latter defect.

Proof. Proposition 6.3 makes the assembly ample. Since every row is proper, its reduction along E is exactly the nonpeelable seed S . Proposition 2.5 therefore gives $X(\mathbf{R}) \notin \mathcal{D}_{\text{amp}}$, whether or not S has corners.

Suppose first that $X(\mathbf{R})$ is a forbidden pc-minor. Fixing all old coordinates to the word a realizes R_a as a proper pc-minor, so $R_a \in \mathcal{D}_{\text{amp}}$. Proposition 6.6 applied to every layer restriction gives the signed-facet condition in (6.27). This proves necessity.

Conversely, assume the two displayed conditions. The signed-facet condition and (6.25) put every elementary layer restriction and contraction in $\mathcal{D}_{\text{amp}}$. It remains to consider an old-coordinate map μ . Put $S_\mu = \mu S$, discard the tips which do not survive a restriction or whose image belongs to S_μ , and denote the remaining index set by $\mathcal{A}_\mu$. Lemma 6.7 gives

$$\mu X(\mathbf{R}) = (S_\mu \times Q_E) \cup \bigcup_{a \in \mathcal{A}_\mu} (\{\mu a\} \times R_a), \quad (6.28)$$

and says that every projected tip is a corner after any subset of the other projected tips has been removed.

Choose $a \in \mathcal{A}_\mu$. A corner peeling of R_a , lifted by the product argument in the proof of (6.23), removes its entire occurrence row and then its projected tip. The lemma permits this operation to be repeated in any order. Eventually only $S_\mu \times Q_E$ remains. Because the embedding of S is irredundant, S_μ is a proper pc-minor of the obstruction S , so it belongs to $\mathcal{D}_{\text{amp}}$; the product test finishes the peeling. Hence every old-coordinate elementary image lies in $\mathcal{D}_{\text{amp}}$.

All elementary pc-minors of $X(\mathbf{R})$ are therefore corner-peelable. Pc-minor closure proves sufficiency in (6.27). The trace-carrier reformulation follows from Theorem 5.96, and parameter recovery is the recovery statement of Proposition 6.3. $\square$

Proposition 6.9 (Unique-midpoint cores separate by layer). *Use the hypotheses and notation of Proposition 6.3, with all occurrence rows ample and proper. Write $\kappa(Y)$ for the largest unique-midpoint core of a family Y , allowing it to be empty, and put*

$$H_t = S \cup \{a \in \mathcal{A} : t \in R_a\} \quad (t \in Q_E).$$

If $\kappa(R_a) = \emptyset$ for every $a \in \mathcal{A}$, then

$$\kappa(X(\mathbf{R})) = \bigcup_{t \in Q_E} (\kappa(H_t) \times \{t\}). \quad (6.29)$$

In particular, if the rows and all the sections H_t are corner-peelable, then $X(\mathbf{R})$ has empty unique-midpoint core.

Proof. We first exclude occurrence words from a core, and then show that all remaining witnesses stay in one layer. The compatible tips are pairwise nonadjacent by Lemma 6.7. The cube Q_a of Proposition 6.6 is the unique maximal cube through a in $S \cup \mathcal{A}$, and $Q_a \setminus \{a\} \subseteq S$.

Consider a unique-midpoint constraint rooted at an occurrence word (a, t) . If its two incident directions are old coordinates, the two edges lie in $Q_a \times \{t\}$, so their opposite midpoint is present. If one direction is old and the other is a layer coordinate, the product description (6.23) again supplies that midpoint: the old neighbor belongs to S , which occurs in every layer. Thus every unique-midpoint constraint rooted at (a, t) uses two layer directions and is exactly a constraint of R_a , lifted at the word a .

Let K be any unique-midpoint core of $X(\mathbf{R})$. For fixed a , a nonempty set $K_a = \{t : (a, t) \in K\}$ would consequently be a unique-midpoint core of R_a . The hypothesis excludes this, so $K \subseteq S \times Q_E$. A witnessing constraint for $(s, t) \in K$ now has both heads in $S \times Q_E$. Two layer directions span a square in $\{s\} \times Q_E$. An old and a layer direction also span a full square in $S \times Q_E$. Hence a unique-midpoint witness with both heads in K must use two old directions and lie entirely in H_t . It follows that $\{s : (s, t) \in K\} \subseteq \kappa(H_t)$, proving one inclusion in (6.29).

Conversely, a horizontal distance-two pair has both possible midpoints in its own layer. Its unique-midpoint constraint in H_t therefore remains exactly the same constraint in $X(\mathbf{R})$. Each nonempty $\kappa(H_t) \times \{t\}$ is consequently a core of $X(\mathbf{R})$, and their union is a core as well. This proves the reverse inclusion. Finally, a corner-peelable family has empty unique-midpoint core by Proposition 5.72, which gives the last assertion. $\square$

Corollary 6.10 (Layer presentations force empty unique-midpoint core). *Every forbidden pc-minor in Theorem 6.8 has empty unique-midpoint core. More generally, if an ample forbidden pc-minor has a successful layer set, its unique-midpoint core is empty. Thus a nonempty unique-midpoint core excludes every successful layer set. The converse fails: empty core alone does not guarantee a successful layer set, even for an ample forbidden pc-minor.*

Proof. The common-seed criterion makes each occurrence row corner-peelable. For every $t \in Q_E$, the section H_t is obtained by fixing all layer coordinates. Since $E \neq \emptyset$ and the nonempty seed occurs in every layer, this is a proper pc-minor of the obstruction, and hence is corner-peelable. Proposition 6.9 applies. Theorem 6.27 places every successful layer presentation of an ample forbidden pc-minor in this common-seed class, proving the necessary condition.

For the failure of the converse, use the one-corner obstruction of Theorem 5.75. A successful layer presentation has at least two nonempty proper occurrence rows: one proper row cannot own both opposite facets of a layer coordinate. Each row peels, so it has a corner, which lifts to a distinct occurrence corner by (6.23). Such a presentation would therefore give at least two corners, contrary to the example's unique corner. $\square$

Corollary 6.11 (Terminal empty-core obstructions in unbounded dimension). *Let $G \subseteq Q_{12}$ be the certified 291-concept family stored as `base` in `damp_fixed_291_repair_creation.json`. For every integer $r \geq 1$, put*

$$X_r = (G \times Q_r) \cup (\{91\} \times (Q_r \setminus \{0^r\})) \cup (\{157\} \times (Q_r \setminus \{1^r\})).$$

Then X_r is a corner-descent-terminal ample forbidden pc-minor with empty unique-midpoint core, isometric dimension $12 + r$, $293 \cdot 2^r - 2$ concepts, and exactly $2r$ corners. For $r = 1$, both corner deletions have unique-midpoint core of order 76. In particular, terminality does not restore either the core-or-nonample criterion or its core-safe-corner form.

Proof. The stored family G is cornerless and forbidden. The two extensions at 91 and 157 have recorded complete corner peelings and acquire the distinct supports

$$P_{91} = \{1, 2, 5, 6\}, \quad P_{157} = \{1, 2, 7, 8\}.$$

Theorem 6.59 proves ampleness, forbidden status, and the order and dimension formulas for every r . The rows partition the signed layer facets between the two owners, so Theorem 6.22 proves terminality. Each punctured row has exactly r corners; since G is cornerless, the corner formula (6.24) gives exactly $2r$ corners of X_r . Corollary 6.10 proves that their unique-midpoint cores are empty.

When $r = 1$, deleting either occurrence corner empties its row. The two remaining horizontal sections are G , whose core has order 76, and a one-concept repair of G , whose core is empty. Discarding the empty row and applying (6.29) gives core order 76 for both deletions. Thus neither of the two corners of the terminal 584-concept example is core-safe.

The fixed checker `verify_damp_layer_midpoint_cores.py` validates the source hashes and both stored repair peelings. It checks the layer-core identity on the unrepaired product, on X_1, X_2 , and on all their corner deletions. Direct shattered-set counts verify these finite controls, and the report `damp_layer_midpoint_cores_verification.json` records complete static pruning orders and surviving-core witnesses. The uniform conclusion uses the preceding proofs, not extrapolation from the controls. $\square$

Theorem 6.12 (Universality of peelable occurrence rows). *Suppose that the cornerless obstruction $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ has three one-concept extensions*

$$S \cup \{a_0\}, \quad S \cup \{a_+\}, \quad S \cup \{a_-\}$$

in $\mathcal{D}_{\text{amp}}$, and that their three new shattered supports are pairwise distinct. Then every nonempty family $K \in \mathcal{D}_{\text{amp}}$ is isomorphic to a proper pc-minor of an explicitly constructed member

$$X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}.$$

More precisely, if $K \subseteq Q_D$, adjoin one layer coordinate z , put $E = D \dot{\cup} \{z\}$, choose any word $u \in Q_E$, and use the three occurrence rows

$$R_0 = K \times \{0\}, \quad R_+ = Q_E \setminus \{u\}, \quad R_- = Q_E \setminus \{\bar{u}\} \tag{6.30}$$

at a_0, a_+, a_- , respectively. The common-seed assembly determined by (6.30) is such an X_K , and fixing the old coordinates to a_0 and the coordinate z to zero recovers K .

Proof. The row R_0 is isomorphic to K , and the two other rows are singly punctured cubes. Hence all three rows are nonempty, proper, ample, and corner-peelable. It remains only to check the signed-facet condition in Theorem 6.8.

Fix $e \in E$. The facet $\{t \in Q_E : t_e = 1 - u_e\}$ avoids u , so it is contained in R_+ . The opposite facet $\{t : t_e = u_e\}$ avoids $\bar{u}$, so it is contained in R_- . Thus the two punctured rows own all $2|E|$ signed facets, independently of R_0 . The common-seed criterion gives $X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$.

Finally, no member of S , a_+ , or a_- has old-coordinate word a_0 . The section of X_K at that word is therefore exactly R_0 . Restricting further to $z = 0$ gives K . Since the construction has the nonempty old-coordinate factor $S \times Q_E$, this is a proper pc-minor. $\square$

Remark 6.13 (What shelling universality rules out). The verified Hall extension atlas supplies the required triple: in zero-based coordinate numbering one may take

$$a_0 = 183, \quad a_+ = 315, \quad a_- = 1836,$$

whose new support labels are, respectively,

$$\{2, 3, 6, 7\}, \quad \{2, 3, 8, 9\}, \quad \{4, 5, 10, 11\}.$$

Thus the conclusion of the theorem is unconditional. The underlying extension record is

$$\text{computations/hall_corner_exchange/source_H.json}.$$

Under the Yang dictionary, Theorem 6.12 says that an arbitrary shellable Yang complex of an ample family can occur as an iterated color-link inside the nonshellable Yang ball of a forbidden pc-minor. Consequently, the elementary-image part of a generative classification cannot be reduced to a special finite catalogue of easy shellable rows. Either the parameters must carry a structural construction of arbitrary corner-peelable ample families, or a global attachment theorem must make their shellability automatic without naming the rows individually. The theorem is a universality barrier, not the missing construction: using the condition $K \in \mathcal{D}_{\text{amp}}$ as admissibility would remain circular.

Theorem 6.14 (Universality within the reduced bases). *For every nonempty ample family K of isometric dimension n , there is an explicit ample family X_K with the following properties.*

1. $X_K \notin \mathcal{D}_{\text{amp}}$, and K is isomorphic to a proper coordinate face of X_K .
2. The minimum graph degree of X_K is at least four, its signed minimal-nonface columns are pairwise distinct up to sign, and its crossing graph is complete. In particular, it has no cut vertex or separating edge.
3. Its parameters are

$$\begin{aligned} |X_K| &= 301 \cdot 2^{n+4} + 2|K| + 1, \\ \text{idim}(X_K) &= n + 16, \quad \text{VC}(X_K) = n + 7. \end{aligned} \tag{6.31}$$

4. $X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ if and only if $K \in \mathcal{D}_{\text{amp}}$.

Consequently, every nonempty corner-peelable ample family is a proper coordinate face of a forbidden base already fixed by both reductions of Theorem 5.39. These bases can all be chosen with isometric dimension minus VC dimension equal to nine. There are infinitely many such bases, even with that difference fixed.

Proof. We first enlarge the input so that each of its coordinates has a distinct incidence pattern. We then insert it into the three-repair construction, whose two other rows ensure all required signed-facet coverage. The input is allowed to be nonpeelable throughout the construction; its peeling status will decide precisely the minimality of the output.

Embed $K \subseteq Q_D$ irredundantly, with $|D| = n$, and reverse coordinates so that $0 \in K$. Adjoin four distinct coordinates w, p, q, z , and put

$$A = D \dot{\cup} \{w\}, \quad P = \{p, q\}, \quad E = A \dot{\cup} P \dot{\cup} \{z\}, \quad m = |E| = n + 4.$$

Let $U = K \times Q_{\{w\}} \subseteq Q_A$ and form the vertex gluing

$$L = (U \times \{0_P\}) \cup (\{0_A\} \times Q_P) \subseteq Q_{A \cup P}. \quad (6.32)$$

Products and vertex gluing preserve ampleness, so L is ample. Moreover, $|L| = 2|K| + 3$. If $K \in \mathcal{D}_{\text{amp}}$, first delete the three nonzero vertices of the attached square, in decreasing Hamming weight, and then peel the product U . This proves $L \in \mathcal{D}_{\text{amp}}$ using Proposition 2.18 and the product test in Proposition 2.6.

Fix the 299-concept Hall obstruction $S \subseteq Q_F$, $|F| = 12$, and the three repairs a_0, a_+, a_- of Remark 6.13, taking F disjoint from E . Its shadow consists of all subsets of F of size at most three, and the three acquired supports P_0, P_+, P_- have size four. Define

$$\begin{aligned} R_0 &= L \times \{0_z\}, & R_+ &= Q_E \setminus \{0_E\}, & R_- &= Q_E \setminus \{1_E\}, \\ X_K &= (S \times Q_E) \cup (\{a_0\} \times R_0) \cup (\{a_+\} \times R_+) \cup (\{a_-\} \times R_-). \end{aligned} \quad (6.33)$$

The three rows are nonempty, proper, and ample. Proposition 6.3 makes X_K ample. Since all rows are proper, its full layer reduction is $X_K^E = S$. If X_K had a corner peeling, reverse it to an ample prefix order and order the complete E -fibres by the appearance of their last vertex. Every resulting prefix is a reduction of an ample prefix, hence is ample [CCMW22, Section 3.1]. This would give an ample prefix order of S , and therefore a peeling of S , a contradiction. Thus $X_K \notin \mathcal{D}_{\text{amp}}$, irrespective of the status of K . Fixing the old coordinates to a_0 and w, p, q, z to zero recovers exactly K . This is a proper coordinate face, since the entire nonempty product $S \times Q_E$ is also present.

If $K \in \mathcal{D}_{\text{amp}}$, then all three occurrence rows peel. The two punctured rows contain opposite facets in every layer direction, exactly as in the proof of Theorem 6.12. Theorem 6.8 now proves $X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$. Conversely, if X_K is forbidden, its proper coordinate face K must belong to $\mathcal{D}_{\text{amp}}$. This proves the equivalence in item 4.

We next verify both reduction stopping conditions directly. Every vertex of $S \times Q_E$ has at least $m \geq 4$ neighbors in its layer cube. A vertex at repair tip a_b has its four old-coordinate neighbors in $S \times Q_E$, one for each coordinate in P_b . Hence the minimum degree is at least four.

For column distinctness it suffices to distinguish the unsigned support-incidence columns. The old clauses of X_K have supports

$$\binom{F}{4} \setminus \{P_0, P_+, P_-\}.$$

Indeed, these are the minimal nonfaces after adding the three acquired supports to the seed shadow: a five-element set has five four-element subsets, so cannot become a new minimal nonface after just three additions. For distinct $f, g \in F$, there are $\binom{10}{3} = 120$ four-element supports containing f and omitting g ; removing three leaves at least one. These old rows distinguish all old coordinates from one another. Each old coordinate occurs in an old row, whereas no layer coordinate does, so they also distinguish old coordinates from layer coordinates.

It remains to distinguish the layer coordinates. The shadow of the vertex gluing L is the union of the shadows of its two factors. Every coordinate of A is active in U , including w even when $n = 0$. Thus $\{t, p\}$ and $\{t, q\}$ are minimal nonfaces of L for each $t \in A$; their missing traces are 11. The constant coordinate z supplies the additional minimal nonface $\{z\}$ of R_0 . By the exact clause formula (6.16), X_K therefore has clauses on

$$P_0 \cup \{t, p\}, \quad P_0 \cup \{t, q\} \quad (t \in A), \quad P_0 \cup \{z\}.$$

For distinct $t, t' \in A$, the first row for t distinguishes them. The row using w, p distinguishes p from q . The row using t, q distinguishes t from p , and the row using t, p distinguishes t from q . Finally

the last row distinguishes z from every other layer coordinate. All support columns are consequently distinct, which excludes equality of signed columns up to sign.

The product $S \times Q_E$ already shatters every coordinate pair, so the crossing graph is complete. Proposition 2.33 excludes cut vertices and separating edges. All $12 + m$ coordinates are active. The disjoint vertex count in (6.33) gives

$$|X_K| = 299 \cdot 2^m + (2|K| + 3) + 2(2^m - 1) = 301 \cdot 2^m + 2|K| + 1.$$

In the shadow formula (6.14), the seed product contributes maximum rank $3 + m$. Every proper ample row has VC dimension at most $m - 1$, so each repair contribution has rank at most $4 + (m - 1) = m + 3$. This proves (6.31). When K peels, items 2 and 4 put X_K among the reduced forbidden bases. Taking $K = Q_D$ for arbitrarily large $|D|$ gives pairwise nonisomorphic such bases with dimension difference nine, proving the last assertion. $\square$

The verifier `verify_damp_reduced_base_universality.py` replays all 36 elementary-minor orders of the fixed seed and the three repair orders. It checks the input and enlarged-row peelings for a singleton, an edge, and the ordinarily peelable 289-concept rooted factor. A cornerless 299-concept input checks the opposite direction: its output is nonpeelable but cannot be forbidden because that input is a proper coordinate face. The 4819- and 9637-concept outputs are materialized and checked from actual cubes, complete signed presentations, degrees, and recovered faces. The two larger outputs are checked through the exact formulas and compact inputs; no output-minor orders or output peeling search are claimed. The record is `damp_reduced_base_universality_verification.json`.

This theorem shows that the reduced-base problem retains the full peelability question even with complete crossing graph and fixed isometric dimension minus VC dimension. In particular, reduction does not leave only finitely many bases at each value of that difference, nor does it restrict proper coordinate faces to a smaller class of peelable ample families. It does not rule out the requested description by unrestricted finite structural parameters: the missing step is still intrinsic admissibility of those parameters.

Proposition 6.15 (Clause changes inside nonpeelable families). *Fix a finite coordinate set D and a simplicial complex $\Sigma \subseteq 2^D$ containing every singleton of D . For each ample $K \subseteq Q_D$ with $0 \in K$ and $\text{Sh}(K) = \Sigma$, use the same fixed coordinates and repairs in the construction of Theorem 6.14. All resulting X_K have the same shadow. Their inherited input clauses are exactly*

$$\sigma_{X_K}(P_0 \cup I) = (a_0|_{P_0}, \sigma_K(I)) \quad (I \in \mathcal{N}(\Sigma)); \quad (6.34)$$

every other output clause is independent of K . The map $K \mapsto X_K$ is injective and preserves both the number of changing clauses and the total number of changing sign bits between any two inputs. All outputs are nonpeelable; precisely those with peelable inputs are forbidden pc-minors.

In particular, there are two nonpeelable ample families with the same shadow whose presentations differ in one bit of one clause, such that exactly one is a forbidden pc-minor. Both have minimum graph degree at least four, distinct signed columns up to sign, and complete crossing graph. Thus an adjacent clause change can destroy pc-minimality while preserving ampleness, nonpeelability, and both reduction stopping conditions.

Proof. Retain $A = D \dot{\cup} \{w\}$, $P = \{p, q\}$, $E = A \dot{\cup} P \dot{\cup} \{z\}$, and L, R_0, R_+, R_- from the preceding theorem. The product $U = K \times Q_{\{w\}}$ has shadow $\{I \cup T : I \in \Sigma, T \subseteq \{w\}\}$. The two factors in the vertex gluing defining L use disjoint coordinates and meet at zero. A trace containing a nonzero coordinate from each factor is absent; traces supported in either factor are exactly its traces. Hence

$$\text{Sh}(L) = \text{Sh}(U) \cup 2^P.$$

Every coordinate of A is active in U . It follows that the minimal nonfaces of $R_0 = L \times \{0_z\}$ are precisely

$$\mathcal{N}(\Sigma) \dot{\cup} \{\{t, p\}, \{t, q\} : t \in A\} \dot{\cup} \{\{z\}\}.$$

On a support in $\mathcal{N}(\Sigma)$, the trace family is that of K : the extra square contributes only the already present zero trace. Its missing trace is therefore $\sigma_K(I)$. The missing trace on each displayed pair is 11, and that on $\{z\}$ is 1. These are all the clauses of R_0 .

The other two rows are fixed punctured cubes. Applying (6.14) shows that the output shadow depends only on Σ . Equations (6.16)–(6.18) now give (6.34); all remaining signs are fixed. Distinct input supports give distinct output supports, and adjoining the same old trace changes no sign-bit distance. This proves both preservation assertions. The proper coordinate face recovering K proves injectivity. Nonpeelability of every output and the exact forbiddenness equivalence are the preceding theorem.

For the last assertion, take the adjacent pair of order 1188 from Corollary 2.102. Call its peelable member K and its forbidden member G . Both contain zero on the same twenty-six active coordinates, and their only changing clause has support of bitmask 6531, with missing traces 2306 and 2307, respectively. Thus X_K and X_G differ in one bit of one clause. Both are nonpeelable and have all the stated degree, column, and crossing-graph properties. But X_K is forbidden, whereas X_G has the nonpeelable G as a proper coordinate face and is not pc-minimal. This proves the claimed failure of preservation. $\square$

The checker `verify_damp_clause_changes_minimality.py` replays the positive 1188-concept input order, all 78 elementary orders of the negative input, and the two ordinary factor orders. It checks the negative endpoint certificates from the unique corner and the two fixed overlaps with a missing root neighbor. Actual cube shadows and full signed presentations verify the input exchange. The enlarged rows have 2379 concepts; their clauses give the complete 1589-clause presentations of both outputs. Exactly one bit of one ten-coordinate clause changes. The outputs have dimension 42, VC dimension 33, and order 323 196 291 401; they are not materialized. Their nonpeelability and minimality statuses use the uniform theorem, not an output-minor enumeration. The record is `damp_clause_changes_minimality_verification.json`.

Thus pc-minimality can change between adjacent nonpeelable ample families. The full signed input variation survives in the outputs. Intrinsic admissibility of the reduced forbidden bases remains open, and the minimum-cardinality interval is unchanged.

Lemma 6.16 (Descent from the ambient cube). *Let $A \subsetneq Q_E$ be nonempty and ample. There is a sequence of corner deletions from Q_E to A if and only if $Q_E \setminus A \in \mathcal{D}_{\text{amp}}$.*

Proof. Put $B = Q_E \setminus A$, which is ample by complement duality. Suppose that $b_1, \dots, b_m$ are the successive deleted concepts in a descent from Q_E to A . Each residual $Q_E \setminus \{b_1, \dots, b_j\}$ is ample, so its complement $\{b_1, \dots, b_j\}$ is ample. Reversing these ample prefixes gives a corner peeling $b_m, \dots, b_1$ of B . Conversely, reverse a peeling of B to obtain ample prefixes $B_j = \{b_1, \dots, b_j\}$. All their complements are ample. The ample corner-deletion criterion therefore makes b_j a corner of $Q_E \setminus B_{j-1}$, giving the desired descent. Here $B_0 = \emptyset$, and the initial residual is the full cube. $\square$

Theorem 6.17 (Prescribed sections in complements of obstructions). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be irredundantly embedded in Q_F , with three resolving one-concept additions a_0, a_+, a_- having pairwise distinct acquired supports. Let $\emptyset \neq K \subsetneq Q_D$ belong to $\mathcal{D}_{\text{amp}}$, where $F \cap D = \emptyset$. Choose $u \in Q_D$, and define*

$$\begin{aligned} X_K = & (S \times Q_D) \cup (\{a_0\} \times K) \\ & \cup (\{a_+\} \times (Q_D \setminus \{u\})) \cup (\{a_-\} \times (Q_D \setminus \{\bar{u}\})). \end{aligned} \tag{6.35}$$

Then $X_K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$, and the section of $X_K^c = Q_{F \cup D} \setminus X_K$ obtained by fixing the old word to a_0 is exactly $Q_D \setminus K$. In particular, if $Q_D \setminus K \notin \mathcal{D}_{\text{amp}}$, then neither X_K nor X_K^c is corner-peelable, and X_K cannot be obtained from its ambient cube by corner deletions. Its order is

$$|X_K| = (|S| + 2)2^{|D|} + |K| - 2. \quad (6.36)$$

Proof. All three occurrence rows in (6.35) are nonempty, proper, ample, and corner-peelable. For every $e \in D$, the facet $t_e = 1 - u_e$ lies in the row at a_+ , and the facet $t_e = u_e$ lies in the row at a_- . Thus these two rows alone supply the signed-facet condition of Theorem 6.8, which proves that X_K is an ample forbidden pc-minor.

At the old word a_0 , only the row K occurs in X_K , so the complementary section is exactly $Q_D \setminus K$. This section is a proper pc-minor of the ample family X_K^c : the latter also contains (a_+, u) , outside that section. If X_K^c had a peeling, intersecting its reversed ample prefixes with this section would give an ample prefix order of $Q_D \setminus K$. Thus $Q_D \setminus K \notin \mathcal{D}_{\text{amp}}$ implies $X_K^c \notin \mathcal{D}_{\text{amp}}$. Lemma 6.16 then rules out the asserted cube descent. Finally, the four displayed parts of X_K have different old words, or old words in S , and are disjoint; counting them gives (6.36). $\square$

Corollary 6.18 (Forbidden minors with nonpeelable cube complements). *For every integer $r \geq 0$, there is an ample forbidden pc-minor X_r of isometric dimension $24 + r$ and order*

$$|X_r| = 1,236,693 \cdot 2^r - 2$$

whose cube complement is nonpeelable. In particular, ample forbidden pc-minors are not all obtainable by corner deletions from their ambient cubes.

Proof. Let $H \subseteq Q_{12}$ be the fixed 299-concept Hall obstruction used in Remark 6.13. Its complement $K_0 = Q_{12} \setminus H$ is corner-peelable: the shelling of the complementary Hall facets described in [CCMW22, Section 4.1] gives such a peeling. An independent complete order is supplied by the fixed check below. Use $S = H$, the three resolving additions in that remark, and $K = K_0 \times Q_r$ on a disjoint set of $12 + r$ coordinates in Theorem 6.17. The product test puts K in $\mathcal{D}_{\text{amp}}$, whereas its complement is $H \times Q_r$, which has the nonpeelable pc-minor H . Since $|K_0| = 4096 - 299 = 3797$, the order formula follows from (6.36). The full product $S \times Q_{12+r}$ inside X_r supplies edges in all $24 + r$ coordinates, proving the dimension claim. $\square$

Remark 6.19 (Verification and scope of the cube-descent obstruction). The fixed verifier `verify_damp_cube_descent_obstruction.py` checks the Hall source’s exact shattered-set complex and absence of corners, all 36 elementary-minor peeling orders, the three resolving additions and their distinct supports, and a 3797-concept peeling of its complement. It also checks both punctured-cube row orders, all 24 signed facets, and the complementary section identity. The report `damp_cube_descent_obstruction_verification.json` records hashes of these inputs and the checker’s source. For $r = 0$, the output has 1,236,691 concepts and its complement has 15,540,525; neither large family is enumerated. Their asserted properties follow from the uniform theorem and the verified inputs.

This closes the proposed exhaustive generation by descent from cubes. The examples need not be terminal under obstruction-preserving corner deletion: the row at a_0 can be peeled away while the other two rows continue to cover all signed facets. Intrinsic generation with other primitive data remains open, as does the complete classification. The minimum-cardinality bounds are unchanged.

Corollary 6.20 (Deletion test for a common-seed cover). *Let $X(\mathbf{R}) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ satisfy the hypotheses of Theorem 6.8. Fix $a \in \mathcal{A}$ and $t \in R_a$, and put*

$$R_a^- = R_a \setminus \{t\}.$$

In the modified cover, replace R_a by R_a^- , discarding the label a if $R_a^- = \emptyset$. Then

$$X(\mathbf{R}) \setminus \{(a, t)\} \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff \begin{array}{l} R_a^- = \emptyset \text{ or } R_a^- \in \mathcal{D}_{\text{amp}}, \\ \text{the modified rows own every signed facet of } Q_E. \end{array} \quad (6.37)$$

In particular, the deletion on the left is necessarily a corner deletion; this need not be imposed as a separate condition.

If S is cornerless, a common-seed cover admits no obstruction-preserving corner deletion if and only if, for every $a \in \mathcal{A}$ and $t \in R_a$, either $R_a^- \neq \emptyset$ and $R_a^- \notin \mathcal{D}_{\text{amp}}$, or deleting t destroys signed-facet ownership. This is an exact recognition test on the row data; the next result replaces it by a structural normal form.

Proof. After deleting (a, t) , the family remains a replicated repair assembly over the same seed, with the single row R_a replaced by R_a^- . Proposition 6.3 says that this family is ample exactly when the nonempty modified row is ample. Theorem 6.8, applied after discarding an empty row, now gives exactly the two conditions on the right of (6.37). When they hold, both the original and the deleted family are ample, so the ample corner-deletion criterion makes (a, t) a corner. The final characterization follows by negating the displayed equivalence, since for a cornerless seed all assembly corners lie in occurrence rows. $\square$

Lemma 6.21 (Signed-facet shell and residual core of a row). *Let $R \subsetneq Q_E$ be a proper cube family, and for $e \in E$, $i \in \{0, 1\}$, put*

$$F_{e,i} = \{t \in Q_E : t_e = i\}.$$

Define its set of full signed facets and the face opposite all of them by

$$\mathcal{O}(R) = \{(e, i) : F_{e,i} \subseteq R\}, \quad F_R = \bigcap_{(e,i) \in \mathcal{O}(R)} F_{e,1-i}, \quad (6.38)$$

where an empty intersection is Q_E , and put $B_R = R \cap F_R$. Then $\mathcal{O}(R)$ contains at most one of $(e, 0), (e, 1)$ for each e , and

$$R = \left(\bigcup_{(e,i) \in \mathcal{O}(R)} F_{e,i} \right) \dot{\cup} B_R = (Q_E \setminus F_R) \dot{\cup} B_R. \quad (6.39)$$

After the coordinates fixed on F_R are suppressed, one has

$$R \in \mathcal{A} \iff B_R \in \mathcal{A}, \quad R \in \mathcal{D}_{\text{amp}} \iff B_R \in \mathcal{D}_{\text{amp}}, \quad (6.40)$$

with the empty family allowed as a vacuous terminal row. Moreover, B_R contains no full signed facet of its residual cube.

Proof. If both $F_{e,0}$ and $F_{e,1}$ were contained in R , then $R = Q_E$, contrary to properness. The union of the facets in $\mathcal{O}(R)$ is the complement of their common opposite face F_R . This union lies in R , and every remaining member of R lies in F_R , proving (6.39).

After reorienting coordinates, write $K = \{e : (e, 0) \in \mathcal{O}(R)\}$ and $L = E \setminus K$. Then

$$R = ((Q_K \setminus \{\mathbf{1}_K\}) \times Q_L) \cup (\{\mathbf{1}_K\} \times B_R).$$

Choosing one coordinate of K at a time expresses this family as an iterated peripheral expansion: one section is a full cube and the other is the same construction with one fewer shell coordinate.

The exact peripheral-expansion test therefore gives both equivalences in (6.40), starting from B_R . The reverse implications also follow by restricting all coordinates of K to 1.

Finally, if B_R contained a full signed facet in a residual coordinate, then adjoining the already present shell $Q_E \setminus F_R$ would make the corresponding full signed facet of Q_E a subset of R . That facet would belong to $\mathcal{O}(R)$, contradicting that its coordinate is residual. $\square$

Theorem 6.22 (Terminal covers as signed-facet partitions). *Let $X(\mathbf{R}) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ satisfy the hypotheses of Theorem 6.8. For each $a \in \mathcal{A}$, put*

$$\mathcal{O}_a = \{(e, i) \in E \times \{0, 1\} : F_{e,i} \subseteq R_a\}.$$

If $X(\mathbf{R})$ is corner-descent terminal, then

1. *the sets $(\mathcal{O}_a)_{a \in \mathcal{A}}$ form a partition of $E \times \{0, 1\}$; and*
2. *every occurrence row is exactly its signed-facet shell:*

$$R_a = \bigcup_{(e,i) \in \mathcal{O}_a} F_{e,i}. \quad (6.41)$$

If S is cornerless, these two conditions are also sufficient. In particular, each block $\mathcal{O}_a$ is nonempty and contains at most one sign of any coordinate.

Conversely, retain cornerlessness and the seed and repair-tip hypotheses of Theorem 6.8. Let $(\mathcal{O}_a)_{a \in \mathcal{A}}$ be any partition of $E \times \{0, 1\}$ into nonempty blocks, no block containing both $(e, 0)$ and $(e, 1)$, and define the rows by (6.41). Then the resulting $X(\mathbf{R})$ is a corner-descent-terminal forbidden pc-minor. The marked sections recover every block canonically as the set of full signed facets of its row.

Proof. Assume first that $X(\mathbf{R})$ is corner-descent terminal. Apply Lemma 6.21 to a row R_a , and suppose that its residual core B_{R_a} is nonempty. Because $R_a \in \mathcal{D}_{\text{amp}}$, the lemma gives $B_{R_a} \in \mathcal{D}_{\text{amp}}$. Choose the first concept t in a corner peeling of this residual core. The same lemma, applied after deleting t , shows that $R_a \setminus \{t\} \in \mathcal{D}_{\text{amp}}$. Moreover, t lies in the opposite face F_{R_a} , hence in none of the full signed facets of R_a . Deleting it therefore leaves every signed-facet containment in the cover unchanged. Corollary 6.20 would then make $X(\mathbf{R}) \setminus \{(a, t)\}$ another forbidden pc-minor, contrary to terminality. Thus every residual core is empty, proving (6.41).

The signed-facet ownership condition in Theorem 6.8 says that the sets $\mathcal{O}_a$ cover $E \times \{0, 1\}$. Suppose that one signed facet $F_{e,i}$ belongs to both $\mathcal{O}_a$ and $\mathcal{O}_b$, with $a \neq b$. Choose a word $t \in R_a$ which has value i at e and, for every other $(f, j) \in \mathcal{O}_a$, has value $1 - j$ at f . Thus t lies in exactly one maximal facet of the union (6.41), namely $F_{e,i}$, and is a corner of R_a . Deleting t leaves a corner-peelable row: after the other shell facets are removed by Lemma 6.21, its residual row is a punctured cube in the coordinates not fixed by $\mathcal{O}_a$. All other facets of $\mathcal{O}_a$ remain full, while $F_{e,i}$ is still owned by R_b . The deletion corollary again produces a forbidden pc-minor, contradicting terminality. Hence the cover is disjoint and the $\mathcal{O}_a$ form a partition.

Conversely, assume that S is cornerless and start with such a partition. A row in (6.41) is the complement of the face opposite its block. Lemma 6.21, with empty residual core, makes it ample and corner-peelable. The partition owns every signed facet, so Theorem 6.8 gives $X(\mathbf{R}) \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

The maximal cubes of a row are exactly its full signed facets. Hence a row concept is a corner precisely when it lies in exactly one of them. Deleting such a concept destroys that signed facet in its row; the partition says that no other row owns it. The deletion criterion therefore shows that no corner deletion preserves membership in $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$. Thus the assembly is corner-descent terminal. Finally, $\mathcal{O}_a$ is read directly from R_a , proving canonical recovery. $\square$

Remark 6.23 (What the facet-partition theorem closes). For cornerless seeds, Theorem 6.22 removes both row peelability and the row-deletion test from the terminal common-seed parameters. The occurrence data are only a partition of the signed layer facets into consistent nonempty blocks; soundness and terminality follow uniformly from the facet shells. For two blocks, consistency forces the two blocks to choose opposite signs in every coordinate, recovering the antipodally punctured two-row towers.

This is a relative generative theorem, not the full classification. The seed is still assumed to be a cornerless forbidden pc-minor, and it is not yet intrinsic which one-concept extensions of the seed are admissible repair tips. Universal common-seed completeness is false by Theorem 2.66. The remaining intrinsic debts are seed classification, repair-tip admissibility, and a complete description of the additional terminal types. The structure of terminal occurrence rows within the common-seed class is settled.

Proposition 6.24 (Thickness of terminal facet partitions). *Use the construction in Theorem 6.22, put $r = |E|$, and write K_a for the coordinates represented in the owner block $\mathcal{O}_a$. Denote by $\lambda(e, i)$ the unique owner of (e, i) . For the two projected sections $H_i = \rho_e^i X$ of a layer coordinate e , one has*

$$|H_i \setminus H_{1-i}| = 2^{r-|K_{\lambda(e,i)}|} \quad (i = 0, 1). \quad (6.42)$$

Thus a layer coordinate has $|H_0 \triangle H_1| = 2$ if and only if there are exactly two repair tips. In that case every layer coordinate has this property. If there are at least three repair tips, then every coordinate of X , including every old coordinate, has two nonempty exclusive sections whose total order is at least three.

Proof. For $a = \lambda(e, i)$, the i -section of its row is the full $(r - 1)$ -cube. The opposite section misses the face fixing each coordinate in $K_a \setminus \{e\}$ opposite to its owned sign. That face has $2^{r-|K_a|}$ vertices. Apart from the two owners of e , all rows have equal e -sections. This proves (6.42). Both exclusive orders are one exactly when the two owners each represent every coordinate. They then exhaust the partition of the $2r$ signed facets, leaving no third owner. Conversely, a partition into two consistent blocks assigns one sign of every coordinate to each block.

For an old coordinate f , the seed is periphery-free by Proposition 2.52. Choose $s_i \in \rho_f^i S \setminus \rho_f^{1-i} S$, for $i = 0, 1$. The product core supplies 2^r occurrences of s_i in the i -section. These remain exclusive unless the opposite old word is a repair tip b_i . If it is a tip, its row fills all but $2^{r-|K_{b_i}|}$ of those occurrences. Thus each exclusive section is nonempty. If their total order were two, both selected contributions would be one. Since $r \geq 1$, both opposite words would have to be repair tips, and both owner supports would equal E . The two tips are distinct because their f -values differ. Their blocks would exhaust all signed facets, again excluding a third owner. $\square$

Remark 6.25 (A terminal obstruction with every coordinate thick). Use Hall's seed C_H and the three repair tips $a_0 = 183$, $a_+ = 315$, $a_- = 1836$ from Remark 6.13. On $E = \{p, q\}$, assign

$$\lambda(p, 0) = \lambda(q, 0) = a_0, \quad \lambda(p, 1) = a_+, \quad \lambda(q, 1) = a_-.$$

The rows are $R_0 = Q_E \setminus \{(1, 1)\}$, $R_+ = \{t : t_p = 1\}$, and $R_- = \{t : t_q = 1\}$. The facet-partition theorem gives a corner-descent-terminal ample forbidden pc-minor with

$$|X| = 4 \cdot 299 + 3 + 2 + 2 = 1203, \quad \text{idim}(X) = 14.$$

Proposition 6.24 proves that all fourteen coordinates are thick. Consequently thickness does not separate terminal obstructions inside this construction from those outside it.

Recovering layer coordinates from forbidden traces. The signed minimal nonfaces distinguish layer coordinates more directly than section sizes do. Write $\eta_a(e) = i$ when $(e, i) \in \mathcal{O}_a$, and put $C = S \cup \mathcal{A}$. The row R_a misses exactly the face with trace $1 - \eta_a$ on K_a . Its shattered complex therefore has the single minimal nonface K_a , with that missing trace. The replicated-repair formulas (6.16)–(6.18) give

$$\mathcal{N}(\text{Sh}(X)) = \mathcal{N}(\text{Sh}(C)) \dot{\cup} \{P_a \cup K_a : a \in \mathcal{A}\}, \quad (6.43)$$

$$\sigma_X(P_a \cup K_a) = (a|_{P_a}, 1 - \eta_a), \quad (6.44)$$

$$\sigma_X(J) = \sigma_C(J) \quad (J \in \mathcal{N}(\text{Sh}(C))).$$

These identities require only an ample seed and compatible one-concept extensions; the seed need not be an obstruction. Every layer coordinate occurs in exactly two members of this signed clause family, with opposite missing-trace values.

Definition 6.26 (Coordinates occurring twice with opposite signs). For a nonempty ample family $X \subseteq Q_V$, put $\mathcal{J} = \mathcal{N}(\text{Sh}(X))$, with missing traces $(\sigma_J)_{J \in \mathcal{J}}$, and define

$$\mathcal{B}(X) = \left\{ e \in V : \begin{array}{l} \exists J_0, J_1 \in \mathcal{J} : \\ \{J \in \mathcal{J} : e \in J\} = \{J_0, J_1\}, \\ J_0 \neq J_1, \quad \sigma_{J_0}(e) \neq \sigma_{J_1}(e) \end{array} \right\}.$$

For a nonempty $E \subseteq \mathcal{B}(X)$, put $F = V \setminus E$, $\mathcal{J}_E = \{J \in \mathcal{J} : J \cap E \neq \emptyset\}$, and

$$C_E = \{x \in Q_F : x|_J \neq \sigma_J \text{ for every } J \in \mathcal{J} \text{ with } J \subseteq F\}, \quad (6.45)$$

$$P_J = J \cap F, \quad K_J = J \cap E, \quad T_J = \{x \in C_E : x|_{P_J} = \sigma_J|_{P_J}\} \quad (J \in \mathcal{J}_E). \quad (6.46)$$

The family C_E is defined by the old clauses alone. The theorem below identifies it with the projection of X onto F .

Theorem 6.27 (Recovery from signed clauses and singleton trace fibres). *A marked nonempty ample family (X, E) , with $E \neq \emptyset$, has a common-seed presentation with a nonempty ample seed, distinct one-concept repair supports, and rows given by a partition of the signed E -facets if and only if $E \subseteq \mathcal{B}(X)$ and every T_J , $J \in \mathcal{J}_E$, is a singleton $\{a_J\}$, with these singletons pairwise distinct. In that case C_E is the projection of X onto F , and*

$$S_E = C_E \setminus \{a_J : J \in \mathcal{J}_E\} = X^E$$

is automatically nonempty and ample. Every $S_E \cup \{a_J\}$ is ample, its unique new shattered support is P_J , and these supports are pairwise distinct. Thus the recovery criterion uses only signed clauses and their trace fibres; seed and repair ampleness require no additional tests. The presentation is unique for the marked set E : its tips are a_J , its owner supports are K_J , and its owner signs are $1 - \sigma_J|_{K_J}$.

If X is an irredundantly embedded ample forbidden pc-minor, then S_E is itself a forbidden pc-minor and every $S_E \cup \{a_J\}$ is corner-peelable. If the recovered seed is also cornerless, this presentation belongs to the terminal common-seed obstruction class of Theorem 6.22.

Proof. First suppose such a presentation is given. Equations (6.43)–(6.44) show that $E \subseteq \mathcal{B}(X)$, that $C_E = C = S \cup \mathcal{A}$, and that its mixed clauses correspond bijectively to the repairs. For the clause associated with a , the missing trace on P_a is supplied in C only by a . Indeed, the union-shadow formula (6.20) shows that $C \setminus \{a\}$ is ample and does not shatter P_a ; it contains S , which already supplies every other trace on that minimal nonface. Hence $T_J = \{a\}$.

Conversely, assume the distinct-singleton condition and $E \subseteq \mathcal{B}(X)$. Write $C = C_E$, $S = C \setminus \{a_J : J \in \mathcal{J}_E\}$. We first identify the seed and its one-tip extensions as ample families, then show that their acquired supports are exactly the old clause parts.

Projection onto F preserves precisely the shattered sets contained in F . Its minimal nonfaces are consequently the members of $\mathcal{J}$ contained in F , with the same missing traces. This projection is ample, so Theorem 5.3 identifies it with C . In particular, C is nonempty and ample.

A word (x, t) satisfying all signed clauses of X must have $x \in C_E$. If $x \in S_E$, it matches none of the old parts of the mixed forbidden traces, so every $t \in Q_E$ is allowed. If $x = a_J$, the distinct-singleton condition says that the only mixed clause it can violate is J . Its allowed row is therefore

$$R_J = \{t \in Q_E : t|_{K_J} \neq \sigma_J|_{K_J}\} = \bigcup_{e \in K_J} \{t : t_e = 1 - \sigma_J(e)\}.$$

The same reconstruction theorem now gives

$$X = (S_E \times Q_E) \cup \bigcup_{J \in \mathcal{J}_E} (\{a_J\} \times R_J).$$

Every K_J is nonempty. Each $e \in E$ occurs in exactly two mixed clauses with opposite signs, so these displayed owner blocks partition all signed E -facets. Every row is proper, so $S = X^E$. Reductions of ample families are ample: if X^E shatters $D \subseteq F$, then X shatters $D \cup E$, hence contains a cube on that support; its reduction is a D -cube in X^E . The same argument proves

$$D \in \text{Sh}(S) \iff D \cup E \in \text{Sh}(X) \quad (D \subseteq F). \quad (6.47)$$

The seed cannot be empty. Otherwise X would not strongly shatter E , hence would not shatter it, so some $J \in \mathcal{J}$ would satisfy $J \subseteq E$. Then $P_J = \emptyset$ and $T_J = C$. But any $e \in E$ belongs to two distinct mixed clauses, whose distinct singleton fibres already give $|C| \geq 2$, a contradiction.

For each J , restrict X to a signed facet owned by J and strongly project the remaining layer coordinates. The row R_J becomes full. For any other row, its missing face still meets the chosen facet: its owner block does not contain that signed facet. Thus the resulting family is exactly $S \cup \{a_J\}$. Restrictions and reductions preserve ampleness, proving that each one-tip extension is ample.

By Lemma 2.10, write Q_J for its unique acquired support, a minimal nonface of $\text{Sh}(S)$. Since $T_J = \{a_J\}$, adjoining a_J adds exactly one word to the projection of S onto P_J . Both projected families are ample, so their shadows differ by one set. The one-tip shadow formula (2.3) therefore forces $Q_J \subseteq P_J$.

It remains to exclude a proper inclusion and coincident acquired supports. By (6.47), $Q_J \cup E$ is not shattered in X , so it contains some minimal nonface $I \in \mathcal{J}$. This cannot be an old clause: Q_J is shattered in $S \cup \{a_J\} \subseteq C$, and hence in the projection of X . Consequently $I \in \mathcal{J}_E$, and

$$Q_I \subseteq P_I \subseteq Q_J.$$

Minimality of Q_I, Q_J as nonfaces of $\text{Sh}(S)$ gives $Q_I = P_I = Q_J$. Every tip a_L with $Q_L = Q_J$ supplies the unique missing trace of S on this support. That trace is $a_I|_{P_I} = \sigma_I|_{P_I}$, so every such tip lies in T_I . Since $T_I = \{a_I\}$ and all tips are distinct, necessarily $L = I$. In particular $J = I$, which proves $Q_J = P_J$, and no other tip has that acquired support. This completes the presentation proof. All its data were forced by the clauses, proving uniqueness.

Finally, suppose X is a forbidden pc-minor. Each row R_J is a product of a singly punctured cube on K_J and a cube on $E \setminus K_J$, hence is corner-peelable. Equation (6.23) lifts these row peelings

to successive corner deletions in X , leaving $S_E \times Q_E$. If S_E were peelable, the product test (A4g1b) would finish a peeling of X , a contradiction. Thus S_E is nonpeelable. Every intermediate reduction is nonpeelable: otherwise its further reduction S_E would be peelable by Proposition 2.5. Repeated application of Theorem 5.33 therefore makes S_E a forbidden pc-minor, after suppressing any inactive coordinates.

For a repair a_J , choose a signed facet owned by its block. Restrict X to that facet and strongly project all the other layer coordinates. Its row becomes full. No other row becomes full, since no other block owns that signed facet. Thus the resulting family is exactly $S_E \cup \{a_J\}$. The initial restriction is a proper pc-minor of X , hence is peelable, and reduction closure makes $S_E \cup \{a_J\}$ peelable as well. There can be no inactive old coordinate in S_E : irredundancy of X would require a tip on its opposite side, whose one-tip extension would have the nonpeelable S_E as a proper restriction. If S_E is cornerless, the hypotheses of the terminal facet-partition theorem now all hold. $\square$

Theorem 6.28 (One-concept repairs of a recovered facet partition). *Let $X \subseteq Q_V$ be an irredundantly embedded ample forbidden pc-minor, and let E satisfy the singleton recovery criterion of Theorem 6.27. Retain its recovered seed S , tips a_J , supports P_J , and owner supports K_J . Then*

$$\{z \in Q_V \setminus X : X \cup \{z\} \in \mathcal{D}_{\text{amp}}\} = \{(a_J, \sigma_J|_E) : J \in \mathcal{J}_E, K_J = E\}. \quad (6.48)$$

The one-concept extension at the displayed word acquires the support $P_J \cup E = J$. Thus X has at most two one-concept extensions in $\mathcal{D}_{\text{amp}}$. It has two exactly when its recovered presentation has two repair tips and their rows are oppositely punctured layer cubes.

If the recovered seed $S = X^E$ itself has a nonempty layer set D satisfying the singleton recovery criterion, then both presentations have exactly two repair tips and flatten into one antipodally punctured two-repair presentation with layer set $E \cup D$ and seed S^D . In particular, $E \cup D$ satisfies the singleton recovery criterion for X .

Proof. Suppose $X \cup \{z\} \in \mathcal{D}_{\text{amp}}$. Its reduction along E must be peelable by Proposition 2.5, whereas $S = X^E$ is not. Adding z must therefore complete a full E -fibre. Since $E \neq \emptyset$, an old word outside the projection of X cannot acquire a full fibre from this single addition. An old word in S already has a full fibre. The only possibility is an old repair word a_J whose row is missing exactly one layer word. The complement of that row is a face of order $2^{|E|-|K_J|}$, so necessarily $K_J = E$ and $z = (a_J, \sigma_J|_E)$.

Conversely, add this word for a clause with $K_J = E$. The completed row joins the product seed, giving $(S \cup \{a_J\}) \times Q_E$. Peel each other occurrence row using (6.23); all these rows are products of punctured cubes with cubes. The recovered one-tip extension $S \cup \{a_J\}$ is peelable by Theorem 6.27, so the product test (A4g1b) finishes a peeling. The old-coordinate neighbours of the added word have support P_J , and every layer neighbour is present. The local extension test therefore identifies its acquired support as $P_J \cup E$.

For any fixed $e \in E$, only two owner blocks contain e . Hence at most two blocks can have support E . If two do, they exhaust all signed facets, so there are no other blocks; their signs are opposite in every coordinate. This proves the count and the two-row characterization.

For the last assertion, put $T = S^D$ and apply the result just proved to the recovered presentation of the obstruction S . The outer presentation of X supplies at least two distinct one-concept repairs of S , since every layer coordinate has two different owners. Therefore these are exactly the two possible repairs of S . Its inner presentation has the form

$$S = (T \times Q_D) \cup (\{a\} \times (Q_D \setminus \{u\})) \cup (\{b\} \times (Q_D \setminus \{\bar{u}\})),$$

and its two resolving concepts are (a, u) and $(b, \bar{u})$. These must be all the tips of the outer presentation. Its two rows are consequently $Q_E \setminus \{v\}$ and $Q_E \setminus \{\bar{v}\}$ for some $v \in Q_E$. Substituting the inner presentation into the outer one gives

$$\begin{aligned} X = & (T \times Q_{D \cup E}) \\ & \cup (\{a\} \times (Q_{D \cup E} \setminus \{(u, v)\})) \\ & \cup (\{b\} \times (Q_{D \cup E} \setminus \{(\bar{u}, \bar{v})\})). \end{aligned}$$

The inner repair supports are distinct, so this is a presentation of the kind recognized by Theorem 6.27. That theorem proves the asserted success of $D \cup E$. $\square$

Remark 6.29 (A terminal obstruction with no one-concept repair). Use the Hall seed and the three tips a_0, a_+, a_- from Remark 6.25. On $E = \{p, q, r\}$, assign their owner blocks as

$$\mathcal{O}_0 = \{(p, 0), (q, 0)\}, \quad \mathcal{O}_+ = \{(q, 1), (r, 0)\}, \quad \mathcal{O}_- = \{(p, 1), (r, 1)\}.$$

The facet-partition theorem gives a terminal ample forbidden pc-minor of order $8 \cdot 299 + 3 \cdot 6 = 2410$ and dimension 15. Every owner support is a proper subset of E , so Theorem 6.28 shows that no one-concept extension of this obstruction is corner-peelable. For comparison, the order-1203 example has exactly one such extension, obtained by completing the row of a_0 , and an antipodal two-repair tower has exactly two. The fixed verifier `verify_damp_facet_partition_repairs.py` checks all three cases and records explicit peelings of every positive extension. Hence a completeness argument cannot assume that a terminal obstruction itself has two resolving tips; that property belongs to the seed of a recovered presentation.

Remark 6.30 (Scope of common-seed factorization). For an ample forbidden pc-minor X , let $\mathcal{E}(X)$ consist of the nonempty subsets of $\mathcal{B}(X)$ satisfying the distinct-singleton trace-fibre condition in Theorem 6.27 and yielding a cornerless seed. The universal claim that $\mathcal{E}(X) \neq \emptyset$ for every cornered, corner-descent-terminal X is refuted by Theorem 2.66: that example has no nontrivial successful layer set at all. Even the restriction to two-connected terminal obstructions fails, by the edge amalgam in Proposition 6.38. The recovery theorem still gives an explicit map for every successful choice. Even when a successful choice exists, the largest recovered seed need not be cornerless: Proposition 6.36 gives two-connected terminal counterexamples. Theorem 6.32 and Corollary 6.34 below show that $\mathcal{E}(X)$ has at most one member: only the largest successful layer set can yield a cornerless seed. In particular, a cornered terminal obstruction with $\mathcal{B}(X) = \emptyset$ cannot have such a presentation. The mere existence of a coordinate in $\mathcal{B}(X)$ is only necessary; distinct singleton fibres and cornerlessness of the recovered seed must still be proved. Nonemptiness and ampleness of that seed, ampleness of each one-tip extension, and distinctness of the acquired supports follow from the singleton condition. Theorem 6.28 gives another direct falsifier: a cornered terminal obstruction with three distinct one-concept extensions in $\mathcal{D}_{\text{amp}}$ cannot have such a presentation. Theorem 6.33 sharpens this test: two resolving extensions already suffice if every coordinate is thick. This does not yet provide intrinsic admissibility for arbitrary forbidden seeds or their resolving repairs.

Theorem 6.31 (Forced enlargement of candidate layer sets). *Let $X \subseteq Q_V$ be nonempty and ample, and use the signed clauses and balanced coordinate set $\mathcal{B}(X)$ of Definition 6.26. Write T_J^E for the fibre in (6.46) when the marked set is E . For a nonempty family of words $T \subseteq Q_{V \setminus E}$, put*

$$\text{var}(T) = \{f \in V \setminus E : \exists x, y \in T \text{ with } x_f \neq y_f\}, \quad W(E) = \bigcup_{J \in \mathcal{J}_E} \text{var}(T_J^E).$$

Starting from any nonempty $E_0 \subseteq \mathcal{B}(X)$, perform these steps with $E = E_0$:

1. If $W(E) \not\subseteq \mathcal{B}(X)$, stop with failure.
2. If two distinct mixed clauses have intersecting fibres $T_I^E \cap T_J^E \neq \emptyset$, stop with failure.
3. If $W(E) = \emptyset$, return E . Otherwise replace E by $E \cup W(E)$ and repeat.

The procedure terminates after at most $|\mathcal{B}(X)| - |E_0| + 1$ passes. On success it returns the unique smallest superset of E_0 satisfying the singleton recovery criterion of Theorem 6.27. On failure, no superset of E_0 satisfies that criterion. The successful sets are closed under nonempty intersections.

The first failure has a certificate consisting of a mixed clause, two words in its fibre, and a coordinate outside $\mathcal{B}(X)$ on which they differ. The second has a certificate consisting of two mixed clauses and a word in both fibres. Thus existence of any nonempty layer set satisfying the singleton criterion is decided by starting the procedure at each of the $|\mathcal{B}(X)|$ one-coordinate sets. This assertion does not impose cornerlessness on the recovered seed.

Proof. For every $E \subseteq \mathcal{B}(X)$, the family C_E is the projection of X onto $V \setminus E$: its minimal forbidden supports are exactly the old clauses, and Theorem 5.3 identifies its words. If $J \in \mathcal{J}_E$, then $J \setminus E$ is a proper subset of a minimal nonface, hence is shattered by X . In particular, its prescribed trace occurs in C_E , so $T_J^E \neq \emptyset$.

Suppose $E \subseteq D \subseteq \mathcal{B}(X)$. Projecting a word of T_J^E onto $V \setminus D$ gives a word in C_D that still matches σ_J on $J \setminus D$. Therefore

$$\{x|_{V \setminus D} : x \in T_J^E\} \subseteq T_J^D \quad (J \in \mathcal{J}_E). \quad (6.49)$$

If D satisfies the singleton criterion, all coordinates on which two words in T_J^E differ must consequently belong to D . Thus $W(E) \subseteq D$. Furthermore, any word in two distinct fibres at E projects to a word in the corresponding two fibres at D , contrary to their being distinct singletons.

It follows inductively that every successful superset of E_0 contains every set produced by the procedure. Either failure therefore excludes every such superset, with the stated certificate. Otherwise each nonterminal pass adds at least one coordinate of $\mathcal{B}(X)$. At termination all fibres are nonempty with no varying coordinate, hence are singletons; the second test makes them distinct. The returned set is therefore successful and is contained in every successful superset of E_0 , proving existence, minimality, and uniqueness. Finally, any nonempty successful set contains some coordinate e , so the procedure started at $\{e\}$ must succeed. The converse is immediate from the success statement. For a nonempty intersection E_0 of successful sets, the procedure must return a superset of E_0 contained in each of those sets. It therefore returns E_0 itself, proving the intersection assertion. $\square$

Theorem 6.32 (A largest recovered layer set for an ample obstruction). *Let $X \subseteq Q_V$ be an irredundantly embedded ample forbidden pc-minor. Call a nonempty $E \subseteq \mathcal{B}(X)$ successful if it satisfies the singleton criterion of Theorem 6.27, without requiring its recovered seed to be cornerless. The successful sets are closed under union. Consequently, if any successful set exists, there is a unique largest one, given by*

$$E_{\max}(X) = \{e \in \mathcal{B}(X) : \text{the forced-enlargement procedure started at } \{e\} \text{ succeeds}\}. \quad (6.50)$$

Thus the one-coordinate starts of Theorem 6.31 recover this largest set directly. Its seed $X^{E_{\max}(X)}$ is determined by the marked family X , independently of a choice of presentation.

Proof. We first control how signed clauses change when a layer presentation is recovered. We then treat disjoint successful sets; reduction of their intersection will reduce the general case to that one.

Every coordinate of an irredundantly embedded ample forbidden pc-minor occurs with both missing signs among its minimal forbidden traces. Indeed, if a coordinate occurs with at most one sign, the signed-clause description in Theorem 5.3 makes one section contain the other. Both nonempty sections are proper pc-minors and hence peelable. The peripheral-expansion test (A4g1a) would make the whole family peelable, a contradiction.

Fix a successful E , put $F = V \setminus E$, and write its presentation with seed $S = X^E$, repair tips a , acquired supports P_a , and owner supports $K_a \subseteq E$. The seed is an irredundantly embedded ample forbidden pc-minor by Theorem 6.27. Each minimal nonface I of $\text{Sh}(S)$ supplies a distinct clause of X : it supplies I itself if it is not an acquired support, and supplies $I \cup K_a$ if $I = P_a$. In the first case I remains a minimal nonface after the selected supports are adjoined to the shadow. The clause formulas (6.16)–(6.18) prove the assertion and preserve all signs on I . Since every seed coordinate occurs with both signs, an old coordinate occurring in exactly two clauses of X must already occur in exactly two oppositely signed clauses of S . In particular,

$$\mathcal{B}(X) \setminus E \subseteq \mathcal{B}(S). \quad (6.51)$$

Suppose first that D is another successful set and $D \cap E = \emptyset$. Then $D \subseteq \mathcal{B}(S)$. Let I be a seed clause meeting D . Because $I \setminus D$ is a proper subset of a minimal nonface, some word x in the projection of S onto $F \setminus D$ has its prescribed missing trace. The projection of X along D contains $\{x\} \times Q_E$, supplied by the seed product. If the clause supplied by I is the unchanged old clause I , its recovery fibre for D therefore contains all these words, contradicting the singleton criterion. Hence $I = P_a$ for an outer repair. Its clause $P_a \cup K_a$ has at least $2^{|E \setminus K_a|}$ words in that fibre, obtained by fixing the missing trace on K_a . Thus $K_a = E$.

Choose any $f \in D$. Its two incident seed clauses consequently give two distinct owner blocks whose supports are all of E . They exhaust the signed E -facets. The outer presentation therefore has exactly two tips, say a, b , and antipodally punctured rows. Moreover, the only seed clauses meeting D are P_a, P_b . Every coordinate of D belongs to both, with opposite missing signs.

Consider the recovery fibres for these two clauses in the projection of S along D . Each is nonempty by the proper-subset shattering argument above. Appending the missing E -word of its respective outer row injects it into the corresponding singleton recovery fibre of X for D . Hence these seed fibres are singletons, say $\{x_a\}$ and $\{x_b\}$. They are distinct. Otherwise the restriction of S obtained by fixing $F \setminus D$ to their common word would be

$$Q_D \setminus \{u, \bar{u}\}.$$

Indeed, the old clauses hold at this projected word, and the only two clauses meeting D exclude precisely the two opposite words. For $|D| = 1$ this contradicts membership in the projection. For $|D| \geq 2$ this restriction is not ample: it has $2^{|D|} - 2$ words but shatters all $2^{|D|} - 1$ proper subsets of D . Restrictions of ample families are ample, giving the contradiction. Thus D satisfies the distinct-singleton criterion for S . The nested-recovery conclusion of Theorem 6.28 now shows that $E \cup D$ is successful for X .

For arbitrary successful E, D , containment makes the conclusion immediate. Otherwise put $I = E \cap D$, $E' = E \setminus I$, $D' = D \setminus I$, and $Y = X^I$. In the presentation marked by E , strongly projecting a row along I leaves the union of its owned facets whose coordinates lie in E' ; the row vanishes if it owns none of these facets. Discard those empty rows. The surviving tips remain distinct compatible repairs, and the surviving owner blocks partition all signed E' -facets. This gives a presentation of Y with layer set E' and seed X^E , so E' is successful. The presentation marked by D likewise proves success of D' in Y .

The family Y is irredundant: its seed X^E uses all its old coordinates, and the nonempty product with $Q_{E'}$ uses the others. It is nonpeelable because its further reduction X^E is nonpeelable. Reduction closure of $\mathcal{D}_{\text{amp}}$ and repeated application of Theorem 5.33 make Y an ample forbidden pc-minor; the same seed product keeps the remaining coordinates active at each intermediate projection. Apply the disjoint argument to E', D' in Y . It proves that D' is successful in $Y^{E'} = X^E$. Nested recovery in the original X then makes $E \cup D' = E \cup D$ successful.

Finally, the union of all successful sets is successful because V is finite. A coordinate belongs to that union exactly when it belongs to some successful set, which the forced-enlargement theorem decides by starting at its singleton. This proves (6.50) and uniqueness. $\square$

Theorem 6.33 (All successful layer choices and the thin coordinates). *Let $X \subseteq Q_V$ be an irredundantly embedded ample forbidden pc-minor. Write $\mathcal{L}(X)$ for its family of successful layer sets, without requiring a cornerless seed, and define its thin coordinates by*

$$\mathcal{T}(X) = \{e \in V : |\rho_e^0 X \setminus \rho_e^1 X| = |\rho_e^1 X \setminus \rho_e^0 X| = 1\}.$$

Then exactly one of the following alternatives holds.

1. *If $\mathcal{T}(X) \neq \emptyset$, then*

$$E_{\max}(X) = \mathcal{T}(X), \quad \mathcal{L}(X) = 2^{\mathcal{T}(X)} \setminus \{\emptyset\}. \quad (6.52)$$

The largest presentation has two repair tips and antipodally punctured rows. The obstruction X has exactly two one-concept extensions in $\mathcal{D}_{\text{amp}}$.

2. *If $\mathcal{T}(X) = \emptyset$, then either $\mathcal{L}(X) = \emptyset$, or $\mathcal{L}(X) = \{E_{\max}(X)\}$. In the latter case $|E_{\max}(X)| \geq 2$, its presentation has at least three repair tips, and X has at most one one-concept extension in $\mathcal{D}_{\text{amp}}$.*

In particular, a cornered terminal obstruction with every coordinate thick and two distinct one-concept extensions in $\mathcal{D}_{\text{amp}}$ has no common-seed presentation.

Proof. First consider any recovered presentation marked by E , with seed S , tips $\mathcal{A}$, and clauses $J_a = P_a \cup K_a$. For a nonempty $D \subsetneq E$, put $L = E \setminus D$. The recovery fibre of a clause meeting D has the exact form

$$T_{J_a}^D = \{a\} \times \{t \in Q_L : t|_{K_a \cap L} = \sigma_{J_a}|_{K_a \cap L}\} \quad (K_a \cap D \neq \emptyset). \quad (6.53)$$

Indeed, in the full old projection $C = S \cup \mathcal{A}$, only a has the missing trace on P_a , as proved in Theorem 6.27. Since $K_a \cap D \neq \emptyset$, projecting its row along D fills every L -word: choose a coordinate of $K_a \cap D$ opposite to its missing sign. The remaining clause condition fixes exactly the displayed coordinates of L , proving the formula. The fibres for distinct tips are disjoint. Thus D is successful if and only if

$$L \subseteq K_a \quad \text{for every } a \text{ with } K_a \cap D \neq \emptyset.$$

If this condition holds, choose a coordinate of D and let a, b be its two owners. Both owner supports contain all of L , so these two blocks exhaust the signed L -facets. Any other owner meeting D would also have to own a facet in each coordinate of L , which is impossible. No other owner can be supported inside L either. Hence there are exactly two owners. Conversely, with exactly two owners both supports equal E , and the displayed fibre formula makes every nonempty $D \subseteq E$ successful. We have proved: a recovered presentation admits a proper nonempty successful subset exactly when it has two owners, whenever $|E| \geq 2$.

A singleton $\{e\}$ is successful exactly when e is thin. One direction follows immediately from its two one-point rows; the other is the thin-coordinate recovery in Proposition 6.63. If a thin coordinate exists, the largest successful set exists by Theorem 6.32. Its singleton subset forces two owners by the preceding argument, unless the largest set is already a singleton, in which case its two signed facets give two owners directly. Every nonempty subset of that largest set is therefore successful, and its singleton members identify it with $\mathcal{T}(X)$.

If no thin coordinate exists but recovery is possible, the largest presentation cannot have two owners. It has at least three, so the fibre argument rules out every proper nonempty successful subset. Every successful set lies in the largest one, proving uniqueness and $|E_{\max}(X)| \geq 2$. Finally, Theorem 6.28 identifies the resolving additions with owner blocks supported on the entire layer set. There are exactly two such blocks in the first case and at most one when there are at least three owners. This proves the extension counts and the stated falsifier. $\square$

Corollary 6.34 (Cornerless recovery requires a maximal layer set). *If E satisfies the singleton recovery criterion and X^E is cornerless, then no proper superset of E satisfies that criterion. If X is an irredundantly embedded ample forbidden pc-minor and E is maximal among successful layer sets, its recovered seed X^E has no nonempty layer set satisfying the singleton criterion. In particular, it has no thin coordinate. That seed is an obstruction with at least two one-concept repairs in $\mathcal{D}_{\text{amp}}$, and can be cornered as in Remark 6.55. For such an obstruction X , a successful recovery with cornerless seed exists if and only if $E_{\max}(X)$ exists and $X^{E_{\max}(X)}$ is cornerless.*

Proof. Suppose a proper superset D also satisfies the criterion, and use its recovered presentation with seed S_D , tips a_J , owner supports $K_J = J \cap D$, and rows R_J . Put $L = D \setminus E$. Strongly projecting the rows along E gives

$$X^E = (S_D \times Q_L) \cup \bigcup_{J: K_J \cap L \neq \emptyset} (\{a_J\} \times \{t \in Q_L : t|_{K_J \cap L} \neq \sigma_J|_{K_J \cap L}\}).$$

Indeed, a row survives the reduction exactly when one of its owned facets uses a coordinate outside E ; the displayed remaining facets then describe it. Each surviving row is nonempty, proper, and a product of a singly punctured cube with a cube. At least one survives because $L \neq \emptyset$ and every signed L -facet has an owner. Such a row has a corner, which lifts to a corner of X^E by (6.23), a contradiction. For the second assertion, a successful layer set in the recovered seed would enlarge E by Theorem 6.28, contradicting maximality. Obstruction descent and the two distinct facet owners at any $e \in E$ give the remaining assertions. The seed has no thin coordinate because any such coordinate would give a successful singleton by Theorem 6.33. The last equivalence follows because Theorem 6.32 makes the maximal successful set unique. $\square$

Theorem 6.35 (Seed-corner deletions in a signed-facet assembly). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ and the compatible resolving tips $\mathcal{A}$ satisfy Theorem 6.8; the seed may have corners. Let $E \neq \emptyset$, let*

$$\lambda : E \times \{0, 1\} \longrightarrow \mathcal{A}$$

be a surjection with $\lambda(e, 0) \neq \lambda(e, 1)$, and put

$$R_a = \bigcup_{\lambda(e, i) = a} \{t \in Q_E : t_e = i\}, \quad X = (S \times Q_E) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times R_a).$$

Then X is an ample forbidden pc-minor. For each $s \in \text{Cor}(S)$, define

$$G_s = S \setminus \{s\}, \quad \mathcal{A}_s = \{a \in \mathcal{A} : a \text{ is not adjacent to } s \text{ in } Q_F, \quad G_s \cup \{a\} \in \mathcal{D}_{\text{amp}}\}.$$

Every obstruction-preserving corner deletion of X removes a seed concept (s, t) , and its exact criterion is

$$\begin{aligned} (s, t) \in \text{Cor}(X), \quad X \setminus \{(s, t)\} \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \\ \iff s \in \text{Cor}(S), \quad G_s \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}), \quad \lambda(e, t_e) \in \mathcal{A}_s \quad \text{for every } e \in E. \end{aligned} \quad (6.54)$$

Here $s \in S$ and $t \in Q_E$. For each fixed s with $G_s \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$, its allowable layer words form the possibly empty face

$$D_s = \prod_{e \in E} \{i \in \{0, 1\} : \lambda(e, i) \in \mathcal{A}_s\}. \quad (6.55)$$

Consequently X is corner-descent terminal if and only if

$$\begin{aligned} \text{for every } s \in \text{Cor}(S) \text{ with } G_s \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}), \\ \text{some } e \in E \text{ satisfies } \lambda(e, 0), \lambda(e, 1) \notin \mathcal{A}_s. \end{aligned} \quad (6.56)$$

In particular, every corner-descent-terminal seed gives terminal signed-facet assemblies, including when that seed has corners.

Proof. The common-seed criterion proves that X is forbidden: each facet shell is peelable and the rows own every signed facet. We first locate the seed corners and show that occurrence-row peelings survive their deletion. Reductions will then prove necessity, while projected row peelings prove sufficiency for all elementary pc-minors.

Every maximal seed cube through s , multiplied by Q_E , remains maximal in X , because all repair rows are proper. Thus a corner (s, t) requires $s \in \text{Cor}(S)$. All layer directions are incident at (s, t) . If a repair tip adjacent to s occurs at t , its old-coordinate direction would also have to belong to the corner cube. That cube would contain the tip in every layer, contradicting properness of its row. Conversely, if no adjacent repair occurs at t , the incident directions are precisely those of the corner cube of s in S , together with E , and their full product is present. We have proved

$$(s, t) \in \text{Cor}(X) \iff s \in \text{Cor}(S), \quad t \notin R_a \text{ for every repair } a \text{ adjacent to } s. \quad (6.57)$$

Fix such a corner and write $H = X \setminus \{(s, t)\}$. The family G_s is ample and nonpeelable: a peeling of G_s could be preceded by s to peel S . It still uses every old coordinate. Otherwise it would be a proper signed restriction of S and hence peelable. In particular, $G_s \times Q_E \subseteq H$ makes the embedding of H irredundant as well.

We will use the following observation about deleting occurrence rows from H . If the repair cube $Q_a = \{a^U : U \subseteq P_a\}$ contains s , then a is adjacent to s . Indeed, at distance at least two from a , all P_a -neighbors of s lie in the punctured cube contained in S . Since s is a corner of S , its corner cube would then contain the entire P_a -cube through s , including a , which is impossible. Thus whenever $s \in Q_a$, (6.57) gives $t \notin R_a$. Every cube $Q_a \times K$, with $K \subseteq R_a$, therefore avoids the deleted concept. The row peelings supplied by (6.23) remain valid in H , in any order and with any chosen rows retained. Applying a restriction or contraction to this sequence again gives corner deletions whenever its image changes: consecutive images are ample and differ by one concept.

Suppose now that H is forbidden. Its strong E -projection is exactly G_s , which is nonpeelable. Repeated inherited-overlap descent in Theorem 5.33 makes G_s a forbidden pc-minor. For each $e \in E$, put $a = \lambda(e, t_e)$. Its row contains t , so a is not adjacent to s by (6.57). Restricting H to the facet t_e and strongly projecting the remaining layer coordinates gives exactly $G_s \cup \{a\}$: only the owner row becomes full, and the s -fibre is missing t . The restriction is a proper pc-minor and hence peelable; reduction closure makes $G_s \cup \{a\}$ peelable. This proves necessity in (6.54).

Conversely, assume its right-hand side. Each adjacent tip is excluded from $\mathcal{A}_s$, so none of its owned facets contains t . Equation (6.57) makes (s, t) a corner; thus H is ample. Its nonpeelable reduction G_s makes H nonpeelable by Proposition 2.5.

Consider an elementary old-coordinate map μ . Peel every occurrence row in H , and project this sequence by μ , using the observation above. The remaining family is μZ , where

$$Z = (S \times Q_E) \setminus \{(s, t)\} = (G_s \times Q_E) \cup (\{s\} \times (Q_E \setminus \{t\})).$$

The base μG_s is peelable, since G_s is an irredundant forbidden pc-minor. If μ erases s or identifies it with a word of μG_s , then $\mu Z = (\mu G_s) \times Q_E$ peels. Otherwise $\mu S = (\mu G_s) \cup \{\mu s\}$ is an ample one-concept extension, so μs is a corner. Peel its punctured-cube occurrence row and then $(\mu G_s) \times Q_E$. This proves that μH peels in every old coordinate.

Every layer contraction fills the single deleted concept from its s -mate, and every layer restriction to the side opposite t is unchanged from X . These images therefore peel. For the remaining restriction $\nu = \rho_e^{t_e}$, put $a = \lambda(e, t_e) \in \mathcal{A}_s$. Peel all occurrence rows except a in H , and project the sequence by ν . Because the a -row becomes full, its final family is

$$((S \cup \{a\}) \times Q_{E \setminus \{e\}}) \setminus \{(s, t|_{E \setminus \{e\}})\}.$$

Nonadjacency of a, s leaves the corner cube of s unchanged in $S \cup \{a\}$. Its deletion is the peelable family $G_s \cup \{a\}$. If no layer coordinate remains, this is already the displayed final family. Otherwise peel the punctured layer row of s , then the product over $G_s \cup \{a\}$. All elementary pc-minors of H now peel, completing the sufficiency proof.

Finally, an occurrence-row corner in a facet shell lies in exactly one of its maximal facets. Deleting it destroys that facet's unique owner. The row-deletion criterion (6.37) therefore rules out every obstruction-preserving occurrence-row deletion. Formula (6.54) accounts for all remaining corners. Its choices of t_e are independent, giving the face (6.55). This face is empty exactly when some coordinate offers no owner in $\mathcal{A}_s$, proving (6.56). A terminal seed has no corner s with G_s forbidden, so the last assertion follows. $\square$

For a nonadjacent pair s, a in this theorem, deleting s leaves the punctured repair cube at a intact. Thus $G_s \cup \{a\}$ is automatically ample and acquires the same support P_a . The set $\mathcal{A}_s$ asks exactly which of these inherited ample repairs remain resolving. Formula (6.56) is therefore a complete terminality test on the signed-facet data once these seed-deletion and repair data are known. The necessary direction of Theorem 6.22 requires these facet partitions in every terminal common-seed cover, so the two results characterize terminality throughout that class, including cornered seeds. This is not intrinsic seed admissibility: it still uses $G_s \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$ and $G_s \cup \{a\} \in \mathcal{D}_{\text{amp}}$. The fixed verifier `verify_damp_seed_corner_facet_deletion.py` checks the corner formula on the 602- and 1202-concept Hall assemblies. For each of the two deletions of a copy of 373 from the first family, it records peelings of all 39 elementary images. For the cornered terminal 600-concept seed of the nested second family, it checks that each seed-corner deletion has Hall's obstruction as a proper layer restriction. These controls check both surviving deletions and terminal-seed inheritance. The next construction settles the global cornerless-recovery question negatively.

Proposition 6.36 (Largest layer recovery can retain a cut vertex). *Let $S \subseteq Q_F$ be an irredundantly embedded, corner-descent-terminal ample forbidden pc-minor with a cut vertex. Suppose that $a, b \notin S$ are distinct resolving tips: both $S \cup \{a\}$ and $S \cup \{b\}$ belong to $\mathcal{D}_{\text{amp}}$, and their acquired shattered supports are distinct. For a nonempty coordinate set E disjoint from F , put*

$$X_E = (S \times Q_E) \cup (\{a\} \times (Q_E \setminus \{0^E\})) \cup (\{b\} \times (Q_E \setminus \{1^E\})). \quad (6.58)$$

Then X_E is a two-connected, corner-descent-terminal ample forbidden pc-minor. Its unique largest successful layer set is exactly E , and its recovered seed is $X_E^E = S$. In particular, if S has corners, no successful layer choice for X_E gives a cornerless seed. Every X_E is already its graph three-core.

Proof. The two punctured rows are a signed-facet partition, with opposite facets assigned to different tips. The common-seed construction theorem and Theorem 6.35 show that X_E is ample, forbidden, and terminal. It is two-connected by Proposition 2.113. The displayed presentation also makes E a successful layer set and gives $X_E^E = S$.

By Theorem 6.32, there is a largest successful set D , and $E \subseteq D$. Suppose the inclusion is strict, and consider the facet-partition presentation marked by D , with seed $T = X_E^D$. Strongly projecting this presentation along E retains its product seed $T \times Q_{D \setminus E}$. An occurrence row becomes the union of its owned facets whose coordinates lie in $D \setminus E$; it vanishes if there are no such facets. Discard the vanishing rows. The remaining owner blocks still partition all signed facets of $Q_{D \setminus E}$, and each remaining row is nonempty and proper. There are at least two rows, since opposite facets have distinct owners. Their tips are distinct resolving tips of T , by the recovery theorem. Thus $S = X_E^E$ has a nontrivial replicated common-seed presentation with resolving tips. Proposition 2.113 would make S two-connected, contrary to its cut vertex. Hence $D = E$.

Any successful choice giving a cornerless seed must be the largest one, by Corollary 6.34. This proves the assertion when S is cornered. Finally, a degree-two vertex of an ample forbidden minor is a corner whose deletion preserves forbiddenness, by Theorem 2.53. Terminality excludes such a vertex; the same theorem excludes smaller degrees. Hence X_E has minimum degree at least three and equals its three-core. $\square$

Remark 6.37 (Two-connected terminal towers). Use $S = Y \vee C$ from Remark 2.67. Its two corners are in Y , and its copy of C uses coordinates $14, \dots, 25$. In that copy, take

$$a = 128 \cdot 2^{14}, \quad b = 290 \cdot 2^{14}.$$

These are resolving tips for S . Indeed, $S \cup \{a\} = Y \vee (C \cup \{128\})$ is ample. First peel $C \cup \{128\}$ to the common root, using its prescribed endpoint peeling, then peel Y ordinarily. The same argument applies to b . The acquired supports have bitmasks $390 \cdot 2^{14}$ and $480 \cdot 2^{14}$, respectively, so they are distinct. All hypotheses of Proposition 6.36 therefore hold.

For $|E| = r \geq 1$, the resulting family has

$$|X_E| = 1160 \cdot 2^r - 2, \quad \dim(X_E) = 26 + r, \quad \text{VCdim}(X_E) = 4 + r.$$

Its minimum degree is $r + 3$, and it has exactly $2^{r+1} + 2r$ corners. To see the corner count, neither new tip is adjacent to either seed corner: the tips and seed corners lie in the two disjoint coordinate blocks. Both seed corners therefore lift to all 2^r layers, by (6.57). Each punctured row has exactly r corners, namely the neighbors of its missing word. These row corners have degree $4 + (r - 1) = r + 3$, while every seed concept has degree at least $4 + r$. The shadow formula (6.14) gives the stated VC dimension.

Thus the first examples have 2318 and 4638 concepts, with six and twelve corners. Their largest recovered seed is the same 1158-concept terminal family, with two corners and a cut vertex. This refutes cornerlessness of the largest recovered seed even for two-connected terminal obstructions of minimum degree at least three.

The checker `verify_damp_cornered_maximal_seed.py` constructs both examples and replays all 165 elementary-minor peelings, as well as four resolving-extension peelings. It checks every initial corner deletion by exhibiting a proper minor whose reduction is either the nonpeelable 1158-concept seed or the cornerless 577-concept double. It also checks two-connectivity, exact shadows, and that

the only thin coordinates are the new layer coordinates, which independently identifies the largest layer set by Theorem 6.33. The report is `damp_cornered_maximal_seed_verification.json`.

The preceding family still admits layer recovery. The construction below shows that even the existence of such a recovery fails for two-connected, cornered terminal obstructions. Its separator is an edge shared by two coordinate blocks. The input conditions concern peeling to this entire edge, not just to one of its endpoints.

Proposition 6.38 (Terminal obstructions from an edge amalgam). *Let $C \subseteq Q_F$ be an irredundantly embedded, two-connected ample family which is corner-peelable and has 0 as its only corner. Choose $e \in F$ with $D = \{0, 0^{\{e\}}\} \subseteq C$. A peeling to D deletes corners outside D until precisely D remains. Suppose that*

1. *for $f \neq e$, the restriction $\rho_f^0 C$ and the contraction $\pi_f C$ peel to their copies of D ;*
2. *each of $\rho_e^0 C, \rho_e^1 C, \pi_e C$ peels to the singleton image of D ; and*
3. *there are distinct $a, b \notin C$ such that $C \cup \{a\}$ and $C \cup \{b\}$ are ample, peel to D , and acquire distinct shattered supports P_a, P_b .*

Take copies C_1, C_2 whose coordinate sets F_1, F_2 intersect only in e , identifying their edges D . Put zeros in the coordinates outside each copy. With two new coordinates p, q , form

$$X = C_2 \cup (C_1 \times (Q_{\{p,q\}} \setminus \{00\})) \cup \{(a_1, 10), (b_1, 01)\}, \quad (6.59)$$

where C_2 occupies the 00-section. Then X is a two-connected, corner-descent-terminal ample forbidden pc-minor, of order $4|C| + 2$ and dimension $2|F| + 1$. Its only corners are $(a_1, 10)$ and $(b_1, 01)$.

If $|F| \geq 4$ and e belongs to at least three minimal nonfaces of $\text{Sh}(C)$, no coordinate of X is balanced in the sense of Definition 6.26. Consequently X has no successful common-seed layer set, even with cornered recovered seeds allowed.

Proof. We first establish ampleness and ordinary peelings of the pieces. Then we check the three kinds of elementary minors, locate all corners, and exclude balanced coordinates. Write $L = Q_{\{p,q\}} \setminus \{00\}$ and, temporarily working with just the first old coordinate block, put

$$Y = (D \times \{00\}) \cup (C \times L) \cup \{(a, 10), (b, 01)\}.$$

Compatibility of the two acquired supports gives $\text{Sh}(C \cup \{a, b\}) = \text{Sh}(C) \cup \{P_a, P_b\}$. The shadows of the two sections in either new coordinate intersect in $\text{Sh}(C)$. A support using both new coordinates must already be shattered in the 00-section D , and that condition is sufficient since $D \times Q_{\{p,q\}} \subseteq Y$. It follows that $\text{Sh}(Y)$ is the disjoint union of

$$\text{Sh}(C) \cup \{P_a, P_b\}, \quad \{I \cup \{p\} : I \in \text{Sh}(C)\}, \quad \{I \cup \{q\} : I \in \text{Sh}(C)\}, \quad \{I \cup \{p, q\} : I \in \text{Sh}(D)\}.$$

Its cardinality is $3|C| + 4 = |Y|$, proving ampleness. In $X = Y \cup C_2$, a support using a coordinate of $(F_1 \setminus \{e\}) \cup \{p, q\}$ and a coordinate of $F_2 \setminus \{e\}$ cannot be shattered: their joint trace 11 is absent. Thus its shadow is precisely the union of the two block shadows, with intersection $\text{Sh}(D)$. Again shadow cardinality equals family cardinality, proving ampleness of X and its asserted order. Every indicated coordinate is active.

There is an ordinary peeling of Y : delete the two concepts in $D \times \{00\}$, then its two added tips, and then peel $C \times L$ using any ordinary peeling of C and the path peeling 01, 11, 10 of L . The first

two concepts lie on a full $D \times Q_{\{p,q\}}$ -cube and have no additional incident directions in their section. The tip corner cubes lie in their own sections. These observations justify the initial deletions as well as the subsequent product peeling.

We now construct peelings of all elementary minors of X . A corner outside the shared edge in either block remains a corner of the union: it has no additional neighbors in the other block. Hence a relative peeling of one block to the intersection can be followed by an ordinary peeling of the other.

For a coordinate $f \in F_2 \setminus \{e\}$, restriction to $f = 1$ discards Y and leaves a peelable restriction of C_2 . Restriction to zero or contraction first peels the changed C_2 to D , by item 1, and then peels Y . For $f \in F_1 \setminus \{e\}$, restriction to one similarly leaves a peelable restriction of Y . Under restriction to zero or contraction, write $C' = \mu C$. Delete any surviving images of the tips that lie outside $C' \times L$; these are corner deletions, being the changing images of the original tip deletions. The remaining family is $(D \times \{00\}) \cup (C' \times L)$. Use the peeling of C' to D from item 1, deleting each outside concept in the three layers in path-peeling order. Finally peel $D \times Q_{\{p,q\}}$ to $D \times \{00\}$, and follow with an ordinary peeling of C_2 .

For a new coordinate, restriction to one again discards C_2 and is a restriction of the peelable family Y . Restriction to zero leaves D in one layer and either $C \cup \{a\}$ or $C \cup \{b\}$ in the other. Item 3 peels the latter to its copy of D ; delete that copy and then peel C_2 . Under contraction, the two layers are $C \cup \{a\}$ and $C \cup \{b\}$, with the intersection edge in one of them. Delete the added tip in the layer not containing that edge, peel its remaining copy of C ordinarily, and peel the other layer to D using item 3. All corner cubes in the first copy of C lift across the two layers, since the other layer still contains all of C . Finish with C_2 .

For any of the three elementary operations in the shared coordinate e , the intersection becomes a singleton. Item 2 peels the image of C_2 to this singleton; the image of Y is peelable by pc-minor closure. This treats every elementary minor.

It remains to prove nonpeelability and terminality. Every vertex of C_2 outside D retains its old incident directions and is not a corner. At either vertex of D , two-connectivity of C_2 supplies an incident direction outside e , while p and q are incident from Y ; the required mixed squares are absent, so neither vertex is a corner. Consider $(v, 10)$ or $(v, 01)$ with $v \in C$. Its neighbor in the 11-section forces every old direction of a hypothetical corner cube to be supported in C . Thus v would be a corner of C , hence $v = 0$. But at zero the other new direction is also incident, and its full corner cube would require the corner cube of C at zero to lie inside the edge D . This is impossible since C is two-connected and zero has degree at least two. The same argument in the 11-section first forces $v = 0$ and then fails in the 00-section. The two tips are corners, each with its acquired support as its incident directions. This argument remains valid after either or both tips are deleted. Consequently the complete corner-deletion process is confined to these two tips and stops at a nonempty cornerless family. Therefore X is nonpeelable, and the elementary-minor checks make it forbidden.

After deleting $(a_1, 10)$, restrict to $q = 0$ and contract p . After deleting $(b_1, 01)$, interchange p, q . Both operations give $C_1 \cup C_2$, with intersection D . This family is cornerless: each copy has only zero as an old corner, and at zero there are incident exclusive directions in both blocks but no mixed square. It has $2|C| - 2$ concepts, so it is a proper nonpeelable minor of each corner deletion. Neither deletion is forbidden. Hence X is corner-descent terminal.

For two-connectivity, the Cartesian product $C \times L$ has no cut vertex. Indeed, after deleting one vertex, every remaining vertex can reach a layer $C \times \{t\}$ avoiding that vertex, using another vertex of C if necessary to pass the deleted layer position; that intact layer is connected. Each tip has at least two old neighbors: its acquired support cannot have size zero or one in an irredundant nonempty ample base. Each vertex of $D \times \{00\}$ has two neighbors in $C \times L$. Adding these vertices

therefore preserves two-connectivity of Y . Its union with the two-connected graph C_2 , along the two vertices of D , is two-connected as well.

Finally suppose the extra hypotheses hold, and put $m = |F|$. Each coordinate in $(F_1 \setminus \{e\}) \cup \{p, q\}$ belongs to $m - 1 \geq 3$ missing cross-pair supports, one for each coordinate in $F_2 \setminus \{e\}$. Each coordinate in the latter set belongs to $m + 1$ such supports. These are minimal nonfaces since all singleton supports are shattered. The three assumed minimal nonfaces through e in the second copy of C remain minimal nonfaces of X , because its projection to F_2 is exactly C_2 . Thus every coordinate belongs to at least three minimal nonfaces. No coordinate can be balanced, and Theorem 6.27 rules out every successful layer set. $\square$

Remark 6.39 (A fixed two-connected terminal obstruction without layers). Take the translated 289-concept family C used in Remark 2.67, choose the edge $D = \{0, 2\}$, and use $a = 128, b = 290$. Their acquired supports have bitmasks 390, 480. The input minor conditions require exactly $22 + 3 = 25$ relative peelings; the two repairs also peel to D . These 27 orders are explicitly recorded in `computations/round806_edge_amalgam_orders.json` and replayed by `verify_damp_terminal_edge_amalgam.py`. The coordinate of the edge belongs, for example, to the three minimal nonfaces with bitmasks 38, 70, 15; their omitted traces have bitmasks 32, 64, 8, respectively.

For an explicit embedding, shift the first copy's words by two bits, put the two tips at 514 and 1161, and embed the second copy by the coordinate map

$$1 \mapsto 3, \quad (0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11) \mapsto (14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24).$$

The resulting obstruction has 1158 concepts, dimension 25, VC dimension 4, minimum degree 4, and exactly the two displayed corners. Each corner deletion has the cornerless 576-concept edge amalgam as a proper pc-minor. The verifier checks all 75 elementary-minor peelings, the complete four-state negative certificate, two-connectivity, both terminality witnesses, and the exact shadow. It recovers all 1004 minimal signed clauses; every coordinate occurs in at least 24 of them. Hence there is no balanced coordinate and no common-seed layer presentation. The full report is `damp_terminal_edge_amalgam_verification.json`.

This refutes layer-existence completeness even for two-connected, cornered terminal graph three-cores. The edge-amalgam construction still assumes relative peelings of its input minors and repairs; it does not solve their intrinsic admissibility. Theorem 2.32 now gives exact decomposition and recovery throughout the separating-edge branch, using the five boundary peeling properties of its pieces. Their intrinsic admissibility, and the full generative classification beyond the known branches, remain open.

Proposition 6.40 (One-corner extensions and a two-neighbor falsifier). *Let $G \subseteq Q_F$ be an irredundantly embedded cornerless ample forbidden pc-minor. Suppose that $c, h \notin G$ are adjacent, both one-concept extensions are ample, and both acquire the same support I . Then $S = G \cup \{c\}$ is an ample forbidden pc-minor whose only corner is c . It has no successful layer set and no common-seed cover with nonempty proper rows satisfying Theorem 6.8.*

Suppose, more generally, that an irredundantly embedded ample forbidden pc-minor $S \subseteq Q_F$ has only one corner c , and that two distinct words $u, v \in Q_F \setminus S$ are adjacent to c and satisfy $S \cup \{u\}, S \cup \{v\} \in \mathcal{D}_{\text{amp}}$. For a new coordinate e , the family

$$X = ((S \cup \{u\}) \times \{0\}) \cup ((S \cup \{v\}) \times \{1\}) \tag{6.60}$$

is a corner-descent-terminal ample forbidden pc-minor with exactly two corners. Its only successful layer set is $\{e\}$, and its recovered seed is the cornered family S . Thus such S, u, v would give a counterexample with a one-corner seed. Proposition 6.36 already refutes general cornerless recovery using a two-corner seed.

Proof. For the first assertion, write $h = c^{\{f\}}$. The acquired-support formula gives $f \notin I$, and the punctured I -cubes at both c and h lie in G . The added word c is a corner of S , with completed cube Q_c on support I . Any other corner of S would have to have Q_c as its unique maximal cube: otherwise its maximal cube avoids c and makes it a corner of G . But every $x \in Q_c \setminus \{c\}$ has the neighbor $x^{\{f\}} \in G$, supplied by the punctured cube at h . This direction lies outside I , so Q_c cannot be its unique maximal cube. Hence c is the only corner of S . Deleting it leaves the nonpeelable G , so $S \notin \mathcal{D}_{\text{amp}}$; Proposition 2.12 makes it forbidden.

Every proper-row common-seed obstruction in Theorem 6.8 has at least two corners. There are at least two rows, since a proper row cannot contain both opposite facets of a layer coordinate. Each row is nonempty and peelable, and a corner of each row lifts to a distinct occurrence corner by (6.23). Successful layer presentations of an obstruction are such covers by Theorem 6.27. This proves both nonrepresentation assertions for S .

For the second assertion, write $u = c^{\{i\}}$ and $v = c^{\{j\}}$, with $i \neq j$. The word $c^{\{i,j\}}$ does not belong to S : otherwise isometry of S would require one of the missing words u, v on a length-two path to c . The support acquired by u therefore contains i and omits j ; the support acquired by v contains j and omits i . These supports are distinct. The common-seed theorem now makes X an ample forbidden pc-minor. Its two occurrence words $(u, 0), (v, 1)$ are corners. No seed word is a corner: only copies of c could be, and at each layer an adjacent repair occurs, which excludes them by (6.57). Deleting either occurrence corner leaves S as a proper layer restriction, so neither deletion is a forbidden pc-minor. Thus X is terminal.

The coordinate e is thin and hence successful. By Theorem 6.33, the largest presentation has two antipodally punctured rows on $E_{\max}(X)$. Each row has at least $|E_{\max}(X)|$ corners, namely the neighbors of its missing word. These lift to distinct corners of X . Since X has exactly two corners, $|E_{\max}(X)| = 1$, and therefore $E_{\max}(X) = \{e\}$, proving uniqueness and the cornered-seed assertion. Existence of the two resolving neighbors is an additional hypothesis here, not a consequence of the first assertion. $\square$

Corollary 6.41 (No resolving neighbor after a square completion). *Let $G \subseteq Q_F$ be nonempty, ample, and nonpeelable, and let $S = G \cup \{c\}$ be ample with acquired support I , where $|I| \leq 2$. No word adjacent to c is a resolving one-concept addition to S . Consequently, the one-corner construction in Proposition 6.40 can supply its two resolving neighbors only when $|I| \geq 3$.*

Proof. Suppose that $u = c^{\{f\}} \notin S$ and $H = S \cup \{u\}$ is ample. Then $f \notin I$, and the punctured-cube test gives its acquired support as $P = J \cup \{f\}$, where $J \subseteq I$: each $j \in J$ supplies $u^{\{f,j\}} = c^{\{j\}} \in G$. The family S is nonpeelable by Proposition 2.13 applied to G, S . The same proposition excludes a resolving addition when $|J| \leq 1$.

Otherwise $I = J$ has size two. Removing c leaves the punctured I -square at u intact and removes its only neighbor in direction f . Since $I \notin \text{Sh}(G)$, the one-concept test makes $G \cup \{u\}$ ample with acquired support I . Proposition 5.104 now makes H nonpeelable as well. $\square$

The fixed verifier `verify_damp_rank_two_additions.py` checks two nonpeelable controls over the certified 291-concept seed. Completing a square attached along a path changes the order from 293 to 294 and exposes the vertex opposite the addition; their complete deletion certificates have three and seven states. In the product of the seed with a punctured square, one and two adjacent completions give orders 874 and 875, with two- and four-state certificates. These controls contain the original seed as a proper pc-minor; they check the local mechanisms used above.

Proposition 6.42 (Two forced deletions at a rank-three neighbor). *Let $S \subseteq Q_F$ be ample and nonpeelable, with exactly one corner c , and let*

$$I = \{k \in F : c^{\{k\}} \in S\}, \quad |I| \geq 3.$$

Suppose that $u = c^{\{i\}} \notin S$ and that $H = S \cup \{u\}$ is ample with acquired support P of size three. Then $P = J \cup \{i\}$ for a two-element proper subset $J \subsetneq I$. Put

$$x = u^J, \quad y = c^J.$$

If H peels, every peeling starts by deleting x and then y . In particular, x has no neighbor outside the completed P -cube, and

$$\{y^{\{k\}} : k \in F \setminus I, y^{\{k\}} \in S\} = \{x\}. \quad (6.61)$$

Consequently, u is not resolving if there is another adjacent ample addition $v = c^{\{j\}} \neq u$ whose acquired support is $K \cup \{j\}$, with $J \subseteq K \subseteq I$.

Proof. The punctured-cube test for the ample addition u gives $i \in P$. For $k \in P \setminus \{i\}$, it also gives $u^{\{i,k\}} = c^{\{k\}} \in S$, so $k \in I$. This proves the asserted form of P . Since $u \notin S$, one has $i \notin I$. The corner cube at c is the full I -cube $Q_I(c)$.

We first locate every possible first deletion. The vertex c is not a corner of H : for $k \in I \setminus J$, its neighbors $c^{\{k\}}$ and u have the missing opposite vertex $u^{\{k\}}$. Any other old corner z of H must have its unique maximal cube contain u ; otherwise that cube would already make z a corner of S . Since u is a corner with cube $Q_P(u)$, this forces $z \in Q_P(u)$ and excludes every neighbor of z outside that cube. No vertex c^U , $U \subseteq J$, can satisfy this: its neighbor in any direction of $I \setminus J$ belongs to $Q_I(c)$, outside $Q_P(u)$.

Write $J = \{p, q\}$. On the other i -side of $Q_P(u)$, the remaining candidates are $u^{\{p\}}, u^{\{q\}}, x$, in addition to the new corner u . Each of $u^{\{p\}}, u^{\{q\}}$ has a neighbor outside $Q_P(u)$. Indeed, without such a neighbor its incident directions in S would span the full face of $Q_P(u)$ opposite u in the corresponding coordinate, making it a corner of S , different from c . Thus the only possible corners of H are u and x . Deleting u returns the nonpeelable S , so a peeling must start with x . In particular, x has no outside neighbor.

Removing a vertex can create a corner only at one of its neighbors: elsewhere the incident directions do not change, and deleting a point cannot fill a missing cube. The three neighbors of x are $u^{\{p\}}, u^{\{q\}}, y$. The first two retain their outside neighbors and their neighbor u . If an outside direction is $k \notin P$, the square joining these two directions would require $u^{\{k\}} \notin H$. Neither vertex can therefore become a corner after deleting x . The vertex u also ceases to be a corner: its incident directions remain P , but their cube now misses x . Hence y is the only possible second deletion.

At y , the full I -cube remains intact. Its i -neighbor was x , which has been removed. If another direction $k \notin I$ remained incident, the corner cube would have to contain $c^{\{k\}}$. Here $k \neq i$, so this vertex is absent even after adding u , by the definition of I . Thus y is a corner after deleting x exactly when (6.61) holds. This proves the forced prefix and the stated necessary condition.

Finally, a second addition v as in the proposition supplies $v^J \in S$, since $\emptyset \neq J \subseteq K$ and its punctured repair cube lies in S . This vertex is $y^{\{j\}}$. It is an outside- I neighbor of y distinct from $x = y^{\{i\}}$, contradicting (6.61). $\square$

Corollary 6.43 (A companion tip excludes rank-three resolving neighbors). *In the first construction of Proposition 6.40, every resolving addition adjacent to the unique corner c has acquired support of size at least four. If the initial support I has size three, each such resolving tip $u = c^{\{i\}}$ completes the entire $I \cup \{i\}$ -cube. Moreover, $G \cup \{u\}$ is itself an ample one-corner forbidden pc-minor acquiring I .*

Thus, at initial support three, the proposed two-neighbor falsifier requires three tips c, u, v with the same acquired support over G : their individual extensions are nonpeelable, while both $G \cup \{c, u\}$ and $G \cup \{c, v\}$ must peel.

Proof. The earlier square-completion result (Corollary 6.41) excludes $|I| \leq 2$, so assume $|I| \geq 3$. Write the companion tip as $h = c^{\{f\}}$. The two punctured I -cubes in G , together with c , form the punctured $I \cup \{f\}$ -cube at h . The one-concept test makes $S \cup \{h\}$ ample, with acquired support $I \cup \{f\}$: this support is not shattered by S , whose shadow is $\text{Sh}(G) \cup \{I\}$.

A resolving neighbor cannot have support of size at most two by Proposition 2.13. If it had size three, it would differ from h , whose support has size at least four. The last assertion of Proposition 6.42 would then exclude it, using h with $K = I$. This proves the lower bound four.

For any adjacent ample addition, the punctured-cube argument above gives $P_u \subseteq I \cup \{i\}$, regardless of its support size. If $|I| = 3$, the lower bound four therefore forces equality. Removing c leaves the entire punctured I -cube at u and removes precisely its neighbor in direction i . Since $I \notin \text{Sh}(G)$, the one-concept test makes $G \cup \{u\}$ ample with acquired support I . The adjacent same-support tips u, c satisfy the first part of Proposition 6.40, so this extension is one-corner and forbidden. Applying the same argument to a second resolving neighbor v proves the final assertion. $\square$

The fixed checker `verify_damp_rank_three_corner_neighbors.py` tests both stages of the forced-prefix argument. It attaches copies of the certified cornerless 291-concept family at selected vertices of two small ample families, leaving each with exactly one corner. In the first resulting family, the rank-three addition permits the forced deletions x, y ; in the second, a companion tip blocks y after x is deleted. Complete negative deletion certificates are checked, together with fifteen rank-three neighbors in the previously recorded one-corner examples. The two constructed controls have proper cornerless minors and are not forbidden witnesses. The result excludes support three in the companion construction; it does not exclude rank-three resolving neighbors of arbitrary one-corner families or settle the remaining support-four cooperation problem.

Corollary 6.44 (Reductions under adjacent pair completions). *Let $G \subseteq Q_F$ be a nonempty ample family. Suppose that $c, u, v \notin G$ are distinct tips whose individual extensions are ample with the same acquired support I , where $|I| = 3$. Suppose also that $u = c^{\{i\}}$, $v = c^{\{j\}}$, and*

$$G \cup \{c, u\}, \quad G \cup \{c, v\} \in \mathcal{D}_{\text{amp}}.$$

Then $G^D \in \mathcal{D}_{\text{amp}}$ for every nonempty coordinate set $D \subseteq F$.

Proof. Put $S = G \cup \{c\}$. The same-support hypothesis gives $i, j \notin I$ and $i \neq j$. The punctured cubes at c, u make the addition of u to S acquire $I \cup \{i\}$; the addition of v similarly acquires $I \cup \{j\}$. Both resulting families are peelable by assumption.

Let $D \neq \emptyset$ and $G^D \notin \mathcal{D}_{\text{amp}}$. Apply Proposition 5.105 first to G, S . Either $S^D = G^D$, or S^D adds one concept with support $I \setminus D$ of size at most two. In the second case Proposition 2.13 preserves nonpeelability. Hence $S^D \notin \mathcal{D}_{\text{amp}}$ in both cases. Applying (5.102) to both additions now gives

$$D \subseteq (I \cup \{i\}) \cap (I \cup \{j\}) = I, \quad 4 - |D| \geq 3.$$

Thus $D = \{k\}$ with $k \in I$. Put $A = G^{\{k\}}$. By (5.101), the projected tips c_k, u_k give individual ample additions to A with the same acquired support $I \setminus \{k\}$, of size two. They remain distinct and adjacent because $i \neq k$. Both have their k -mates in G , so

$$(G \cup \{c, u\})^{\{k\}} = A \cup \{c_k, u_k\} \in \mathcal{D}_{\text{amp}}.$$

Proposition 5.104 forces A to peel, contradicting the choice of D . Hence no nonempty D has a nonpeelable reduction. $\square$

The checker `verify_damp_repair_strong_projections.py` verifies (5.101) on every coordinate subset in six fixed controls, including the recorded rank-three tip, its two adjacent pair completions, and repaired terminal towers with one and two layer coordinates. It also extracts twelve overlap peelings of the certified 291-concept seed from three stored repair orders: the supports at tips 91, 157, 319 have empty intersection, so for each coordinate one repair leaves that reduction unchanged. Thus every reduction of this seed on a nonempty coordinate set is peelable. This seed meets the necessary projection condition. The next proposition nevertheless excludes its three-tip falsifier profile by using VC dimension and the positioned punctured cubes.

Proposition 6.45 (A third companion blocks a pair in VC dimension three). *Let $G \subseteq Q_F$ be a nonempty cornerless ample family with $\text{VCdim}(G) \leq 3$. Suppose that the individual extensions at three distinct missing tips c, u, v are ample and acquire the same support I , where $|I| = 3$. If $u = c^{\{i\}}$ and $v = c^{\{j\}}$, then*

$$G \cup \{c, u\} \notin \mathcal{D}_{\text{amp}}.$$

The same conclusion holds for $G \cup \{c, v\}$.

Proof. The two punctured cubes at c, u complete a four-dimensional cube when both tips are added. We show that the third punctured cube protects its entire c -side. After the first old vertex is deleted, the VC-dimension bound confines all further deletions to four vertices on the other side; they cannot reach the attached part of G .

Put $P = I \cup \{i\}$, $C = \{c^U : U \subseteq P\}$, and $H = G \cup \{c, u\}$. The acquired-support hypothesis implies $i, j \notin I$, $i \neq j$, and

$$C \setminus \{c, u\} \subseteq G, \quad \text{Sh}(H) = \text{Sh}(G) \cup \{I, P\}.$$

Indeed, the punctured cube at each tip has support I , and after adding c the punctured cube at u has support P . The one-concept test (Lemma 2.10) applies successively, since I is not shattered by G and no support containing I is shattered by G . Thus H is ample, its only shattered four-set is P , and neither added tip has a neighbor outside C .

Partition the vertices of C into

$$C_0 = \{c^U : U \subseteq I\}, \quad \{u\}, \quad W = \{u^{\{p\}} : p \in I\}, \quad B \cup \{z\},$$

where $B = \{u^{I \setminus \{p\}} : p \in I\}$ and $z = u^I$. Every $c^U \in C_0 \setminus \{c\}$ has the neighbor $v^U \in G \setminus C$, supplied by the punctured cube at v . Every $w = u^{\{p\}} \in W$ also has a neighbor outside C . Otherwise its incident directions in G would be exactly $P \setminus \{p\}$, whose entire cube through w lies in $C \setminus \{c, u\}$; this would make w a corner of G .

Any corner of H belonging to G has its corner cube containing c or u , since otherwise it was already a corner of G . It therefore lies in C ; all four directions of P are incident there, and the added tips have no further incident directions. Its corner cube is exactly C . Consequently every such corner has no neighbor outside C , and the preceding paragraph confines it to $B \cup \{z\}$. Deleting c first leaves $G \cup \{u\}$, whose only corner is u : each old vertex of the completed I -cube at u has its i -neighbor in G . Its only next deletion leaves the cornerless G . The same argument applies with c, u interchanged. Hence a successful peeling would have to start in $B \cup \{z\}$.

Such a first deletion has cube support P , so it removes precisely P from the shadow. The remaining ample family has VC dimension at most three, as does every family obtained by further

corner deletions. We prove that any further deletion must again lie in $B \cup \{z\}$ and have no neighbor outside C in H . This maintains all vertices outside C , as well as the twelve vertices of $C_0 \cup \{u\} \cup W$.

For the induction, assume that all deletions so far have the stated form. No vertex outside C can have become a corner: none was initially a corner, and deleting a nonneighbor cannot create a corner, because its incident directions remain unchanged. Each vertex of $C_0 \setminus \{c\}$ retains its three I -neighbors and its outside j -neighbor. The vertex c retains its three I -neighbors and u ; the vertex u retains W and c . Each has degree at least four, whereas a corner in an ample family of VC dimension at most three has degree at most three. Every $w = u^{\{p\}} \in W$ retains u and an outside neighbor $w^{\{k\}}$, with $k \notin P$. The square on these two edges would require $u^{\{k\}}$, which is absent from H . Thus w cannot be a corner either.

Each remaining vertex $b \in B$ retains its two neighbors in W and its mate in C_0 . If it has an outside neighbor, these four surviving neighbors prevent its deletion. Moreover, while z remains, every such b has four surviving neighbors even without an outside neighbor. Hence if the first deletion was a vertex of B , no second vertex of B can be deleted before z . At any subsequent deletion of z , it therefore still has its mate in C_0 and two neighbors in B . An outside neighbor would again give degree at least four. Thus every possible deletion of z , including a first deletion, has no outside neighbor. After z is deleted, the same conclusion for the remaining vertices of B follows from their three protected neighbors. This completes the induction.

The protected twelve vertices survive every deletion sequence that starts at an old vertex. Neither possible type of first deletion can therefore begin a complete peeling. Interchanging i, u with j, v proves the second assertion. $\square$

Corollary 6.46 (At most one resolving neighbor in VC dimension three). *Let $S \subseteq Q_F$ be an ample nonpeelable family with exactly one corner c , and put $I = \{k \in F : c^{\{k\}} \in S\}$. If $\text{VCdim}(S) \leq 3$, at most one missing neighbor u of c satisfies $S \cup \{u\} \in \mathcal{D}_{\text{amp}}$. Consequently the one-corner two-neighbor falsifier in Proposition 6.40 requires $\text{VCdim}(S) \geq 4$, whether or not its source comes from the companion construction.*

Without a VC-dimension assumption, if two resolving neighbors $u = c^{\{i\}}$, $v = c^{\{j\}}$ have acquired supports $J \cup \{i\}$, $K \cup \{j\}$ of size three, where $J, K \subseteq I$, then

$$J \cup K \subsetneq I. \quad (6.62)$$

In that companion construction, if $\text{VCdim}(G) \leq 3$, the specified companion h is the only possible resolving neighbor of c . If c has two distinct same-support companions over G , it has no resolving neighbor in $S = G \cup \{c\}$.

Proof. The predecessor $G = S \setminus \{c\}$ is ample and nonpeelable: otherwise deleting c first would peel S . Adding c to G acquires I . If $|I| \leq 2$, Corollary 6.41 excludes every resolving neighbor, so assume $|I| \geq 3$.

We first prove (6.62). The punctured-cube test gives $J, K \subseteq I$, each of size two, and distinct $i, j \notin I$. Proposition 6.42 excludes $J = K$ and makes every peeling of $H_u = S \cup \{u\}$ start at $x_u = u^J$ and $y_u = c^J$. Write $J = \{p, q\}$. After deleting x_u , the only corner is y_u . Its remaining neighbors, by (6.61), are

$$c^{\{p\}}, \quad c^{\{q\}}, \quad t_r = c^{J \cup \{r\}} \quad (r \in I \setminus J).$$

Deleting a nonneighbor cannot create a corner, so these are the only third-deletion candidates. For any $r \in I \setminus J$, the vertex $c^{\{p\}}$ retains the neighbors $c, c^{\{p,r\}}, u^{\{p\}}$. A cube spanning those three directions would contain $u^{\{r\}}$, which is absent because $r \notin J \cup \{i\}$. The argument for $c^{\{q\}}$ uses $c, c^{\{q,r\}}, u^{\{q\}}$ and the same missing word. Thus the third deletion must be some t_r .

Every t_r has an outside- I neighbor in S , since otherwise the full I -cube would make it a second corner of S . After deleting x_u, y_u , it retains all these outside neighbors and every direction in $I \setminus \{r\}$: x_u is not adjacent to t_r , and y_u was its neighbor along r . If it is the third corner deleted, choose any retained outside direction $k \notin I$. Its corner cube contains the face

$$\{t_r^U : U \subseteq (I \setminus \{r\}) \cup \{k\}\} \subseteq S. \quad (6.63)$$

The face avoids u , since its coordinate r is flipped from the value at c , whereas u agrees with c there.

If $r \in K$, then $y_v = c^K$ lies in this face, as does $y_v^{\{k\}}$. Applying (6.61) to v forces $k = j$. Choose $s \in I \setminus K$, which exists because $|I| \geq 3 > |K|$. The same face then contains $x_v^{\{s\}}$, where $x_v = v^K = y_v^{\{j\}}$. Indeed, $s \neq r$, so direction s belongs to its support. This gives a neighbor of x_v outside its completed $K \cup \{j\}$ -cube, contrary to its required first deletion. Hence every possible third deletion has $r \in I \setminus (J \cup K)$, proving (6.62).

Now impose $\text{VCdim}(S) \leq 3$. The full corner cube has support I , so $|I| = 3$. For any two putative resolving neighbors, write their supports as $J \cup \{i\}$, $K \cup \{j\}$, with $J, K \subseteq I$. Proposition 2.13 excludes supports of size at most two. Thus each has size three or four. If one has size three and the other size four, the last assertion of Proposition 6.42 excludes the former. If both have size three, that assertion gives $J \neq K$, whence $J \cup K = I$, contradicting (6.62).

The remaining case has both acquired supports of size four, so $J = K = I$. We show that the predecessor G is cornerless. Any corner created by deleting c must be a neighbor $c^{\{p\}}$, with $p \in I$, because c was the only corner of S . Such a vertex retains its two neighbors along $I \setminus \{p\}$, as well as $u^{\{p\}}$ and $v^{\{p\}}$, supplied by the two punctured repair cubes. These are four distinct neighbors in the ample G , whose VC dimension is at most three, so it cannot be a corner. Moreover, the individual extensions of G at c, u, v all acquire I . For u , removing c from its punctured repair cube removes its neighbor along i and leaves its entire punctured I -cube; the support I is not shattered by G , since it was the support removed with the corner c . The one-concept test therefore applies, and the same argument handles v . Proposition 6.45 now contradicts $G \cup \{c, u\} = H_u \in \mathcal{D}_{\text{amp}}$. This finishes the general assertion.

For the more specific companion conclusion, write $h = c^{\{f\}}$. Its common acquired support I over the nonempty ample G is nonempty. For every $p \in I$, the two punctured I -cubes contain the full face on $(I \setminus \{p\}) \cup \{f\}$ with coordinate p flipped from its value at c . Hence $|I| \leq \text{VCdim}(G) \leq 3$. Supports of size at most two are already excluded. At size three, Corollary 6.43 makes every resolving neighbor an individual addition to G acquiring I . If it differs from h , the three tips violate Proposition 6.45. Thus only h can resolve. Applying this conclusion to two companions leaves no possible resolving neighbor. $\square$

The fixed checker `verify_damp_vc_three_companion.py` attaches copies of the certified cornerless 291-concept family at the corners of two small ample families. The VC-three control has 1761 concepts in 77 coordinates; its complete negative deletion certificate has fifteen states and realizes the four possible old first deletions. The VC-four control permits the upper antipode to become a corner while retaining a neighbor outside the completed four-cube. It isolates the step that uses the VC bound. Both controls have proper nonpeelable minors, and the latter still does not peel: it is not a counterexample to a dimension-free nonpeelability claim. The checker `verify_damp_vc_three_one_corner.py` tests the additional argument for incomparable three-supports. In one fixed VC-three control, both additions permit their first two old deletions, but the third is blocked. In another, the third corner cube for one addition supplies the outside neighbor that blocks the other's second deletion. A VC-four control permits both three-deletion prefixes along a direction outside $J \cup K$. Each extension still has a proper cornerless nonpeelable

minor; these prefixes are not resolving examples. The general corollary excludes all one-corner sources of VC dimension at most three. The two-neighbor existence question remains open in higher VC dimension.

Remark 6.47 (A peelable class with a unique corner). The fixed family in `damp_one_corner_peelable_control.json` is an ample corner-peelable class with 294 concepts in twelve coordinates and VC dimension three. Its only corner is 1894, with support $\{0, 8, 11\}$, in the binary-word convention of the certificate. Thus having one corner does not imply nonpeelability.

The checker `verify_damp_one_corner_peelable_control.py` reconstructs this family from the stored owner colouring of the 232-concept Hall contraction and its 62-concept overlap. The two sections have orders 89 and 205. It verifies ampleness of the sections, overlap, and lift, checks the unique corner, and replays all 294 steps of the stored peeling, which begins 1894, 1638, 1634, 1632.

Every peeling of this nontrivial family must start at 1894, so none can end there. A corner-peelable ample family therefore cannot in general be peeled to an arbitrarily prescribed remaining vertex. VC dimension three is the smallest dimension where this can fail: the proof of [CCMW22, Proposition 4.11] constructs an ample prefix order starting at any chosen concept in VC dimension at most two. The checker also verifies that none of the nine missing neighbors of 1894 is an ample addition. This positive control does not supply the two resolving neighbors in Proposition 6.40.

Remark 6.48 (A fixed one-corner source and a disconnected placement fibre). For the certified 291-concept cornerless seed G , the support with integer mask $I = 2312$ has precisely five ample one-concept extensions, at

$$2184, \quad 2696, \quad 3720, \quad 3818, \quad 3822.$$

Use the convention $x = \sum_{f=0}^{11} x_f 2^f$. The adjacency graph on these tips consists of the path 2184–2696–3720 and the edge 3818–3822. They have the same trace on I , so the outside-placement fibre $\mathcal{Y}_I(G)$ from (5.99) is disconnected; it has five words but seven shattered supports, so it is not ample. Thus even for a cornerless forbidden seed one cannot assume that the possible placements of a single acquired support form an ample family.

Proposition 6.40 makes $S = G \cup \{2696\}$ a 292-concept forbidden pc-minor with exactly one corner, outside the proper-row common-seed cover class. Its only adjacent ample additions are 2568, 2184, 3720, and all three are nonpeelable. Their complete corner-deletion certificates have respectively three, five, and four states. Corollary 6.46 now gives a uniform reason that this source has no resolving neighbor: the seed has VC dimension three and 2696 has two companions. The fixed verifier `verify_damp_one_corner_neighbors.py` checks the complete placement fibre, the one-corner conclusion, and all adjacent ample additions at this source. The strengthened corollary also excludes two resolving neighbors for arbitrary one-corner sources of VC dimension three. Higher VC dimension and nonadjacent cooperative repairs remain outside that conclusion.

Proposition 6.49 (Separated cooperative additions). *For every integer $\ell \geq 0$, there is a cornerless ample family G_ℓ of order $597 + \ell$ and two outside words a_ℓ, b_ℓ such that all four families*

$$G_\ell, \quad G_\ell \cup \{a_\ell\}, \quad G_\ell \cup \{b_\ell\}, \quad G_\ell \cup \{a_\ell, b_\ell\}$$

are ample. The first three are nonpeelable, while the last is peelable. The two additions acquire disjoint four-coordinate supports, and their completed cubes have Hamming distance $8 + \ell$. Each of the first three families has Hall's cornerless obstruction as a proper pc-minor; in particular none is a forbidden pc-minor.

Proof. Use Hall's 299-concept family C_H , its resolving addition 183, and the certified peeling of $C_H \cup \{183\}$ ending at 4095, from the data in Remark 6.25. Translate every word by 4095, so that the last word becomes 0. Write C for the translated cornerless family and $v = 183 \oplus 4095$ for the translated repair tip, where $\oplus$ denotes bitwise addition modulo two. Its acquired support is $P = \{2, 3, 6, 7\}$. The word v has four nonzero coordinates outside P , so the completed P -cube has distance four from 0. Denote the translated peeling by $w_1, \dots, w_{300} = 0$.

Take two disjoint copies F_1, F_2 of the twelve-coordinate set, and a further set Z of ℓ coordinates. Put

$$T_\ell = \{1^k 0^{\ell-k} : 0 \leq k \leq \ell\} \subseteq Q_Z.$$

The desired family consists of two copies of C joined at their zero words by this path:

$$G_\ell = (C \times \{0\} \times \{0_Z\}) \cup (\{0\} \times C \times \{1_Z\}) \cup (\{(0, 0)\} \times T_\ell), \quad (6.64)$$

$$a_\ell = (v, 0, 0_Z), \quad b_\ell = (0, v, 1_Z). \quad (6.65)$$

For $\ell = 0$, the two endpoint words coincide and the copies meet at one vertex. For positive ℓ , the path has $\ell - 1$ internal vertices. In both cases $|G_\ell| = 597 + \ell$.

No pair of coordinates from different blocks F_1, F_2, Z is shattered. Between F_1 and F_2 the trace 11 is missing; between F_1 and Z it is again 11; between F_2 and Z the trace 10, with the F_2 -coordinate first, is missing. The Z -projection is a path and shatters only the empty set and its singletons. Consequently the full shattered complex is the union of the two block shadows and these ℓ singletons, with the empty set counted once. Its size is $597 + \ell$, proving ampleness by the Sandwich equality. The same calculation after replacing either or both copies of C by $C \cup \{v\}$ proves ampleness of the three extensions.

Every word in a copy of C retains its failure to be a corner: no cube on that copy's coordinates is added by the other copy or the path. An internal path vertex has two incident directions which do not span a square. Thus G_ℓ is cornerless. Restricting to $F_1 = 0, Z = 1_Z$ leaves the second copy of C , even after adding a_ℓ . Restricting to $F_2 = 0, Z = 0_Z$ leaves the first copy, even after adding b_ℓ . These are proper pc-minors and are nonpeelable, proving nonpeelability and failure of pc-minimality for the first three families.

After both additions, peel the first completed copy using $w_1, \dots, w_{299}$, leaving its attachment vertex. Every deleted word has exactly its old incident cubes, since it is not the attachment vertex. Do the same in the second completed copy. The remaining path T_ℓ peels from one endpoint; for $\ell = 0$ only the common vertex remains. This is an explicit peeling of the final family. The two added supports are the copies of P in F_1, F_2 . Their cubes are each at distance four from the respective attachment vertex, and all ℓ bridge coordinates differ. Their distance is therefore $4 + \ell + 4$, as asserted. $\square$

The verifier `verify_damp_separated_cooperative_repairs.py` checks the cases $\ell = 0, 3$, including the original peeling certificate, all four Sandwich equalities, cornerlessness, the proper Hall restrictions, and the combined peelings of lengths 599, 602. It computes the shattered sets block by block using the missing mixed pairs, without an ambient-cube enumeration. This construction rules out geometric separation of completed cubes as a substitute for pc-minimality in a repair-inheritance argument. Whether such cooperation can occur over a cornerless forbidden pc-minor remains open.

Theorem 6.50 (Terminality of iterated common-seed covers). *Let $G \subseteq Q_F$ be an irredundantly embedded ample forbidden pc-minor, and let*

$$S = (G \times Q_D) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times R_a), \quad D \neq \emptyset,$$

be a common-seed cover satisfying Theorem 6.8, with S itself forbidden. Thus the inner tips resolve G , their acquired supports are distinct, and their occurrence rows are nonempty and proper. The inner rows need not be facet shells or have disjoint facet ownership. Let $E \neq \emptyset$ be disjoint from $F \cup D$, and let $X \subseteq Q_{F \cup D \cup E}$ be a second common-seed cover over S , satisfying the same theorem with nonempty proper outer rows.

If X is a corner-descent-terminal forbidden pc-minor, then both stages have exactly two repair tips and antipodally punctured rows. More precisely, for some distinct $a, b \in \mathcal{A}$, $u \in Q_D$, and $v \in Q_E$,

$$\begin{aligned} S &= (G \times Q_D) \cup (\{a\} \times (Q_D \setminus \{u\})) \cup (\{b\} \times (Q_D \setminus \{\bar{u}\})), \\ X &= (G \times Q_{D \cup E}) \cup (\{a\} \times (Q_{D \cup E} \setminus \{(u, v)\})) \\ &\quad \cup (\{b\} \times (Q_{D \cup E} \setminus \{(\bar{u}, \bar{v})\})). \end{aligned} \tag{6.66}$$

In particular, $D \cup E$ is successful for X . Consequently, the seed recovered from the largest layer set of a terminal obstruction admits no further common-seed cover satisfying the displayed hypotheses, even with arbitrary proper occurrence rows. More generally, a terminal output of two or more successive covers of this kind is a single antipodal two-repair tower over the initial seed.

Proof. We first determine all resolving additions to the inner cover, without assuming that its rows are facet shells. Put

$$\mathcal{A}_* = \{a \in \mathcal{A} : R_a = Q_D \setminus \{u_a\} \text{ for some } u_a \in Q_D\}.$$

Then

$$\{z \in Q_{F \cup D} \setminus S : S \cup \{z\} \in \mathcal{D}_{\text{amp}}\} = \{(a, u_a) : a \in \mathcal{A}_*\}. \tag{6.67}$$

Indeed, $S^D = G \notin \mathcal{D}_{\text{amp}}$, so a resolving addition must change this reduction and hence complete a full D -fibre. An old word outside the projection cannot acquire a full fibre from one addition, since $D \neq \emptyset$. The only possible additions are therefore the displayed missing words. Conversely, completing such a row gives the product $(G \cup \{a\}) \times Q_D$, together with the other occurrence rows. These other rows peel by the common-seed criterion, and their peelings lift by (6.23). The remaining product peels because $G \cup \{a\} \in \mathcal{D}_{\text{amp}}$. The added word's incident directions are exactly $P_a \cup D$, where P_a is the support acquired over G : every layer neighbor is present, and compatible old tips are nonadjacent. The local extension test gives this new support, and these supports remain pairwise distinct.

Index the chosen outer tips by a subset $\mathcal{B} \subseteq \mathcal{A}_*$, and write $T_a \subsetneq Q_E$ for the outer row of (a, u_a) . Substitution expresses X over the original seed G , with one row for every $a \in \mathcal{A}$:

$$X = (G \times Q_{D \cup E}) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times U_a), \quad U_a = \begin{cases} (R_a \times Q_E) \cup (\{u_a\} \times T_a), & a \in \mathcal{B}, \\ R_a \times Q_E, & a \notin \mathcal{B}. \end{cases} \tag{6.68}$$

Every U_a is nonempty and proper: an unselected row retains a missing D -word, and a selected row misses $\{u_a\} \times (Q_E \setminus T_a)$. Each is ample, being the restriction of the ample X obtained by fixing its F -word to a . Hence this is another common-seed cover meeting the hypotheses of Theorem 6.8.

Terminality of X and Theorem 6.22 force the rows U_a to be a signed-facet partition on $D \cup E$. Their reductions along E are exactly R_a , because every outer row is proper. Reduction of a facet shell retains precisely its owned facets whose coordinates lie in D . Every R_a is nonempty, so each owner block retains at least one such facet. The retained blocks partition all signed D -facets. Thus the original inner rows R_a themselves form a facet partition, and D is successful for S .

Terminality also makes the original outer rows a facet partition, so E is successful for X . The nested-recovery conclusion of Theorem 6.28 now applies. It forces two owners at each stage and gives (6.66) and success of $D \cup E$. If E was the largest successful set of X , this strict enlargement is impossible, proving the canonical-seed assertion. For a longer chain, apply the two-stage result to the last two covers, then regard their flattened output as the outer cover over the preceding stage. Repeating this operation proves the assertion by induction on the number of stages. $\square$

Remark 6.51 (A redundant path row and its outer lift). Take Hall's seed $G = C_H$, its three certified repairs $a = 183, b = 315, d = 1836$, and $D = \{p, q, r\}$. In Boolean-word notation, set

$$R_a = Q_D \setminus \{000\}, \quad R_b = Q_D \setminus \{111\}, \quad R_d = \{000, 100, 110, 111\}.$$

These rows give a 2410-concept ample forbidden pc-minor S . The path row owns no signed facet, while the other two rows partition all signed facets. The two resolving additions to S are $(a, 000)$ and $(b, 111)$. Every layer coordinate belongs to four minimal forbidden supports: the two punctured-row clauses and two path-row clauses. Old coordinates belong to 165 or 167 clauses. Thus $\mathcal{B}(S) = \emptyset$. The only obstruction-preserving corner deletions remove the two endpoints of the path row. Each resulting three-word path has one constant coordinate and one missing trace on the other two coordinates; every layer coordinate then belongs to three clauses. Their balanced coordinate sets are again empty, so neither predecessor has layer recovery.

Use the two resolving additions as opposite owners on a new coordinate e . The resulting 4822-concept outer lift has $\mathcal{B}(X) = \{e\}$ and unique layer recovery to S . Formula (6.68) identifies its extra row over G as $R_d \times Q_{\{e\}}$. Peel these eight occurrences. Every intermediate family is a forbidden pc-minor by the common-seed cover criterion, and the endpoint is the terminal 4814-concept antipodal tower over G on four layers. The fixed verifier `verify_damp_iterated_common_seed_covers.py` checks the complete inner repair list with two explicit peelings, all eight inner corners, both preserving predecessors, the flattening identity, and all eight outer corner deletions. Here the iterated-cover exclusion applies even though neither preserving corner predecessor admits layer recovery.

Theorem 6.52 (Repair growth after extending a recovered obstruction). *Let $H \subseteq Q_V$ be an irredundantly embedded ample forbidden pc-minor with a successful nonempty layer set E , and put $F = V \setminus E$. Suppose that $c \notin H$ and $S = H \cup \{c\}$ is ample and nonpeelable. For an ample family $Y \subseteq Q_V$, write*

$$\mathcal{R}(Y) = \{a \in Q_V \setminus Y : Y \cup \{a\} \in \mathcal{D}_{\text{amp}}\}.$$

Then

$$\mathcal{R}(H) \subseteq \mathcal{R}(S), \quad |\mathcal{R}(S) \setminus \mathcal{R}(H)| \leq 1, \quad |\mathcal{R}(S)| \leq 3. \quad (6.69)$$

Every member of $\mathcal{R}(H)$ is nonadjacent to c and retains its acquired support over S . If a new resolving tip a exists, then $a|_F = c|_F$, and the missing part of that fibre in H is the edge with endpoints $a|_E, c|_E$. In particular, a is adjacent to c .

No signed-facet assembly over S , using compatible resolving tips, is corner-descent terminal. Consequently, if a terminal facet assembly has a cornered seed S , every obstruction-preserving corner deletion $S \setminus \{s\}$ has no successful layer presentation.

Proof. Proposition 2.12 makes S a forbidden pc-minor. Recover the facet-partition presentation of H , and write $K = H^E$ for its nonpeelable seed. By Theorem 6.28, the one-concept additions that complete an entire E -fibre are exactly $\mathcal{R}(H)$; each has an acquired support containing E . Since S is nonpeelable, adding c completes no such fibre, and hence $S^E = K$. Its acquired support Q_c cannot contain E : the new corner cube on Q_c would otherwise complete the E -fibre through c .

For $a \in \mathcal{R}(H)$, its acquired support Q_a therefore differs from Q_c . The compatible-repair union formula (6.20) makes $H \cup \{a, c\}$ ample and preserves both acquired supports. Deleting c leaves the peelable family $H \cup \{a\}$, so the union peels as well. This proves $\mathcal{R}(H) \subseteq \mathcal{R}(S)$. Lemma 6.7 makes a, c nonadjacent.

Now take $a \in \mathcal{R}(S) \setminus \mathcal{R}(H)$. The strong E -projection of S is still the nonpeelable family K , whereas that of $S \cup \{a\}$ is peelable by Proposition 2.5. Thus adding a completes an E -fibre. If its old word differed from $c|_F$, the same fibre would already be completed in $H \cup \{a\}$, implying $a \in \mathcal{R}(H)$, a contradiction. The two additions therefore share their old word, and the original fibre has exactly the two missing words $c|_E, a|_E$.

For an old word in the projection of H , this missing set is the face opposite a recovered owner block; a two-word face is an edge. For an old word outside that projection, the entire Q_E is missing, so two missing words force $|E| = 1$ and again form an edge. There is at most one other missing word in the fibre through c , proving the second inequality and adjacency. The previous repair theorem gives $|\mathcal{R}(H)| \leq 2$, proving the last inequality.

Finally, c is a corner of S , and deleting it returns the forbidden family H . Among any compatible chosen resolving tips of S , exactly the inherited ones belong to $\mathcal{A}_c$ in Theorem 6.35: all inherited tips are nonadjacent to c and resolve H , while any new tip is adjacent to c . At most one owner is therefore outside $\mathcal{A}_c$. Opposite facets have distinct owners, so they cannot both be outside that set. The exact deletion criterion supplies an obstruction-preserving copy of c in every facet assembly over S , proving nonterminality. A nonpeelable corner deletion of S still uses every coordinate, as proved in Theorem 6.35. Thus applying this conclusion with $H = S \setminus \{c\}$ proves the final assertion. $\square$

Remark 6.53 (A sharp three-repair example without layer recovery). Let $K = C_H$ be Hall's 299-concept obstruction, and use its three compatible repairs $a = 183, b = 315, d = 1836$ from Remark 6.25. On one new coordinate p , put

$$H = ((K \cup \{b\}) \times \{0\}) \cup ((K \cup \{a\}) \times \{1\}), \quad c = (d, 0), \quad S = H \cup \{c\}.$$

The family H is the terminal 600-concept two-repair lift. The family S has 601 concepts and is an ample forbidden pc-minor: its three singleton occurrence rows are peelable and own both signed p -facets. Completing any one of these rows gives a peelable extension, while any resolving addition must change the nonpeelable strong p -projection K . Hence

$$\mathcal{R}(H) = \{(a, 0), (b, 1)\}, \quad \mathcal{R}(S) = \{(a, 0), (b, 1), (d, 1)\}.$$

The new repair $(d, 1)$ is adjacent to c , attaining the bound in (6.69). Three resolving tips preclude any successful layer set for S , by Theorem 6.28. Equivalently, every old coordinate occurs in 165 minimal forbidden traces and p occurs in three, so $\mathcal{B}(S) = \emptyset$. Nevertheless every facet assembly over S is nonterminal by the new theorem.

For example, put $u = (d, 1)$, $v = (b, 1)$, and form the 1204-concept opposite-layer lift

$$X = ((S \cup \{u\}) \times \{0\}) \cup ((S \cup \{v\}) \times \{1\})$$

on a second coordinate q . Relative to deleting c , the owner u is ineligible and v is inherited. Thus deleting $(c, 1)$ preserves obstruction status. In the remaining family, the Hall repair rows own the disjoint facet blocks

$$a : \{(p, 1)\}, \quad b : \{(p, 0), (q, 1)\}, \quad d : \{(q, 0)\}.$$

The result is a terminal 1203-concept Hall assembly. Before deletion, p occurs in three minimal forbidden traces and q in two, while every old coordinate occurs in 165. Thus $\{q\}$ is the unique

successful layer set of X , with recovered seed S . The deletion makes p balanced as well, allowing recovery of the cornerless Hall seed on layers $\{p, q\}$. The fixed verifier `verify_damp_layered_predecessor_repair_growth.py` records all five resolving-extension peelings for H, S , checks the three-repair list and clause degrees, and verifies this deletion and its terminal facet partition.

Proposition 6.54 (Removing a seed extension throughout the layer cube). *Let $G \subseteq Q_F$ be an irredundantly embedded cornerless ample forbidden pc-minor. Suppose that $c \notin G$, that $S = G \cup \{c\}$ is ample and nonpeelable, and that $\mathcal{A} \subseteq Q_F \setminus S$ is a set of tips with $G \cup \{a\} \in \mathcal{D}_{\text{amp}}$ for every $a \in \mathcal{A}$. Assume that the one-concept acquired supports P_c and $(P_a)_{a \in \mathcal{A}}$ over G are pairwise distinct. Let $(R_a)_{a \in \mathcal{A}}$ be rows given by a partition of the signed facets of a nonempty layer cube Q_E , as in Theorem 6.22, and put*

$$Y = (G \times Q_E) \cup \bigcup_{a \in \mathcal{A}} (\{a\} \times R_a), \quad X = Y \cup (\{c\} \times Q_E).$$

Then X is an ample forbidden pc-minor, and there is a sequence of $2^{|E|}$ obstruction-preserving corner deletions removing precisely $\{c\} \times Q_E$ and ending at Y . In particular, X is not corner-descent terminal, although its marked layer set E recovers the cornered seed S .

Proof. The facet-partition theorem makes Y a terminal ample forbidden pc-minor. Choose a corner peeling of the cube Q_E , and denote its successive suffix families by $T_0 = Q_E, T_1, \dots, T_m = \emptyset$, where $m = 2^{|E|}$. Every T_i is ample. The compatible-repair union formula (6.20) and Proposition 6.3 make

$$Y_i = Y \cup (\{c\} \times T_i) \quad (0 \leq i \leq m)$$

ample. Their reductions along E are G for $i > 0$ and S for $i = 0$. All are therefore nonpeelable by Proposition 2.5.

Starting with $Y_m = Y$, traverse these families in reverse order. Each step is an ample one-concept extension in the same ambient cube, whose coordinates are already active in Y . The extension remains nonpeelable, so Proposition 2.12 makes it another forbidden pc-minor. Thus every Y_i is a forbidden pc-minor. Successive families differ by one concept and are ample, so traversing them from $Y_0 = X$ to $Y_m = Y$ gives the asserted corner deletions. Finally, all original rows R_a are proper, whence $X^E = S$. The compatible-repair union formula says that the acquired supports over S are still the distinct P_a , so this is the marked presentation recognized by Theorem 6.27. $\square$

Remark 6.55 (Maximal recovery need not give a cornerless seed). Let $H = C_H$ be Hall's 299-concept obstruction, and put $S = H \cup \{373\}$. Its added support is $P_c = \{1, 2, 7, 9\}$. The two corners of S are 373 and 1011; deleting either leaves a nonempty cornerless family. Hence S is nonpeelable and, by Proposition 2.12, is itself a forbidden pc-minor. It is not terminal because deleting 373 recovers H .

Take the Hall repairs $a = 183$, $b = 315$ from Remark 6.25, whose acquired supports are distinct from P_c , and form

$$X = ((S \cup \{b\}) \times \{0\}) \cup ((S \cup \{a\}) \times \{1\}).$$

Proposition 6.54 makes X a 602-concept ample forbidden pc-minor. Its signed minimal nonfaces are all old four-sets except the three acquired supports, together with the two mixed repair clauses. Thus an old coordinate occurs in 164 clauses if it belongs to the added support of 373, and in 165 otherwise; the new layer coordinate occurs twice with opposite signs. Consequently that coordinate is the only member of $\mathcal{B}(X)$. Its singleton layer set is successful and maximal, and its recovered seed is the cornered obstruction S . Deleting the two layer copies of 373, in the order supplied by the proposition, reaches the terminal 600-concept Hall lift.

The fixed diagnostic `explore_damp_cornered_hall_repair_signatures.py` checks the four nonpeelable Hall extensions at words 373, 1105, 1360, 1525. Each has nineteen ample one-concept extensions: twelve peelable and seven nonpeelable. The twelve resolving tips, with their acquired supports, are exactly those of H . The record contains explicit positive peelings and complete corner-deletion certificates for the negative decisions. Thus every facet-partition assembly using one of these four cornered seeds satisfies the proposition and is nonterminal. This rules out that source of counterexamples; it does not show that every cornered recovered seed permits the same descent. Proposition 6.36 supplies terminal counterexamples using a different, two-corner seed.

Remark 6.56 (Union closure requires the obstruction hypothesis). Union closure in Theorem 6.32 does not hold for arbitrary ample families. Using the convention $x = \sum_{i=0}^3 x_i 2^i$, consider

$$X = \{0, 1, 2, 3, 7, 8\} \subseteq Q_{\{0,1,2,3\}}.$$

It consists of a square and two pendant edges. Its shattered sets are the empty set, the four singletons, and $\{0, 1\}$, so it is ample. Its minimal forbidden traces, written as support and missing word in the same integer convention, are

$$(12, 12), \quad (10, 10), \quad (9, 9), \quad (6, 4), \quad (5, 4).$$

Thus $\mathcal{B}(X) = \{0, 1\}$. Marking coordinate 0 gives fibres $T_9 = \{8\}, T_5 = \{6\}$; marking coordinate 1 gives $T_{10} = \{8\}, T_6 = \{5\}$. Both choices are successful. For their union, however,

$$T_9 = T_{10} = \{8\}, \quad T_5 = T_6 = \{4\},$$

so recovery fails by a collision. This ample example is peelable. For obstruction inputs, in contrast, the union theorem permits all successful one-coordinate closures to be united and recovers the unique largest layer set. Neither universal layer existence nor cornerlessness of the largest recovered seed holds: Theorem 2.66 and Proposition 6.36 give the respective counterexamples.

The fixed verifier `verify_damp_obstruction_layer_unions.py` checks the full recovery families of six Hall constructions. Antipodal towers on two and three layers have respectively three and seven successful sets; the three-layer check includes overlapping incomparable choices. It also checks the order-1203 three-owner example and the 1204-to-1203 descent in Remark 6.53, including the cornered seed of the larger input. The cyclic order-2410 example in Remark 6.29 supplies the case with no resolving addition. The verifier checks the sublayer fibre formula, all thin coordinates, and the zero-, one-, and two-extension cases of Theorem 6.33. These finite regressions accompany the proof and retain the six-word example as a control for its hypothesis.

Remark 6.57 (Unique layer recovery for the three-repair example). For the order-1203 family in Remark 6.25, one has $\mathcal{B}(X) = \{p, q\}$, and the only successful layer set in Theorem 6.27 is $\{p, q\}$. Indeed, the Hall shadow consists of all subsets of F of order at most three. Adjoining the three repairs adds exactly their three four-element supports. No five-element set has all its four-subsets among these three additions, so the old minimal nonfaces are exactly $\binom{F}{4} \setminus \{P_0, P_+, P_-\}$. The mixed clauses replace the three removed clauses while retaining their old supports. Each old coordinate therefore occurs in exactly $\binom{11}{3} = 165$ clauses, whereas p, q each occur twice with opposite signs. If only p is marked, the old part of the clause $P_+ \cup \{p\}$ has the two solutions $(a_+, 0)$ and $(a_+, 1)$ in $C_{\{p\}}$, so its trace fibre is not a singleton. Marking only q fails in the same way at a_- . Marking both coordinates recovers the Hall seed and the displayed three repairs. In Theorem 6.31, starting at $\{p\}$ therefore forces q , and starting at $\{q\}$ forces p ; both procedures stop at $\{p, q\}$. The diagnostic `explore_damp_forced_layer_closure.py` records these trajectories and the two failure certificates on fixed small examples. The fixed witness verifier `verify_damp_terminal_clause_recovery.py`

checks these traces and all fourteen section differences directly; its output also contains explicit corner peelings of the three seed extensions.

Remark 6.58 (The reduced completeness target). Recall that an ample forbidden pc-minor is *corner-descent terminal* if deleting none of its corners gives another forbidden pc-minor. Starting from any ample obstruction and repeatedly making an obstruction-preserving corner deletion must terminate. Reversing the deletions expresses the original family as a finite sequence of the punctured-face extensions from Proposition 2.12, starting at a corner-descent-terminal obstruction.

The terminal objects are not all cornerless: Corollary 6.69 gives two-extension lifts with exactly two corners and no preserving deletion. They do not all admit common-seed presentations either: Theorem 2.66 constructs a terminal cut-vertex obstruction, while every nontrivial common-seed assembly with resolving tips is two-connected. Proposition 6.38 further constructs two-connected terminal obstructions with no successful layer set. Thus splitting off cut vertices does not make the common-seed construction exhaustive; shared-edge assemblies must also be accounted for.

Inside the common-seed class, Corollary 6.20 gives the exact deletion criterion. The thin/thick dichotomy in Corollary 6.64 remains useful: a thin coordinate canonically exposes a two-sided fragile kernel; otherwise every coordinate has symmetric section difference at least three. The new terminal cut-vertex example lies outside all successful layer recoveries, and hence has no thin coordinate. Its two resolving extensions also yield the terminal towers of Proposition 6.36, whose largest recovered seed is cornered. Thus the seed class cannot be restricted to cornerless obstructions even after terminality and largest-layer recovery. Intrinsic admissibility of the recovered seeds, rooted factors, and their repairing extensions remains necessary for the full generative classification.

Theorem 6.59 (Antipodally punctured two-repair towers). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be cornerless and irredundantly embedded in Q_F . Let $a, b \in Q_F \setminus S$ be distinct, suppose that $S \cup \{a\}, S \cup \{b\} \in \mathcal{D}_{\text{amp}}$, and suppose that their new supports P_a, P_b are distinct. Let $E \neq \emptyset$ be disjoint from F , and let $R_a, R_b \subsetneq Q_E$ be nonempty ample families. Then*

$$X = (S \times Q_E) \cup (\{a\} \times R_a) \cup (\{b\} \times R_b) \quad \text{belongs to } \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \quad (6.70)$$

if and only if there is a word $u \in Q_E$ such that, after possibly interchanging a and b ,

$$R_a = Q_E \setminus \{u\}, \quad R_b = Q_E \setminus \{\bar{u}\}. \quad (6.71)$$

In that case

$$|X| = 2^{|E|}(|S| + 2) - 2, \quad \text{idim}(X) = |F| + |E|. \quad (6.72)$$

The normalized marked presentation recovers S, a, b and the antipodal pair $\{u, \bar{u}\}$. Modulo layer-coordinate permutations and reorientations, the only new parameter is $|E|$. For $|E| = 1$, the construction is the opposite-layer two-extension lift.

Proof. Suppose first that X is a forbidden pc-minor. Restricting all old coordinates to the word a , respectively b , realizes R_a and R_b as proper pc-minors of X . Hence both rows belong to $\mathcal{D}_{\text{amp}}$. Proposition 6.6 and pc-minimality imply that the two rows own every signed facet of Q_E . For each $e \in E$, a proper row cannot contain both opposite e -facets. Consequently there is a word $v \in Q_E$ such that

$$\{t : t_e = v_e\} \subseteq R_a, \quad \{t : t_e = 1 - v_e\} \subseteq R_b \quad (e \in E).$$

The union of the facets on the left is $Q_E \setminus \{\bar{v}\}$. Since R_a is proper, it equals this punctured cube. Similarly $R_b = Q_E \setminus \{v\}$, which is (6.71) with $u = \bar{v}$.

Conversely, suppose (6.71) holds. A punctured cube belongs to $\mathcal{D}_{\text{amp}}$, as follows inductively from the peripheral-expansion test. Proposition 6.3 therefore makes X ample, and Proposition 6.6

makes it non-corner-peelable: all its corners lie in the two proper rows, and after any rowwise corner deletions one either gets stuck in a row or reaches the cornerless product $S \times Q_E$.

Every elementary layer restriction makes exactly one of the two rows full and leaves the other row empty when $|E| = 1$, or punctured when $|E| > 1$. Every layer contraction makes both rows full. Equation (6.25) therefore puts all these layer images in $\mathcal{D}_{\text{amp}}$.

It remains to consider an elementary restriction or contraction μ in an old coordinate. Put $S' = \mu(S)$. This is a proper pc-minor of the obstruction S , so $S' \in \mathcal{D}_{\text{amp}}$. By Lemma 6.7, the two repair tips cannot merge outside S' , and every surviving outside tip is a corner after the other has been removed. Each such tip retains its punctured occurrence row. The product description (6.23) lifts a peeling of its punctured row. Peeling all surviving rows leaves $S' \times Q_E$, which belongs to $\mathcal{D}_{\text{amp}}$. Thus all old-coordinate elementary images are corner-peelable as well. Pc-minor closure now proves $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

The two rows each have $2^{|E|} - 1$ members, giving the order formula. The product core realizes every old and layer direction, giving the isometric dimension. Recovery follows from (6.19); the complements of the two recovered rows are the two antipodal words. Finally, the automorphism group of Q_E is transitive on its words, so all choices of u are equivalent in the marked layer cube. $\square$

Remark 6.60 (The first Hall towers). For Hall's 299-concept obstruction, the repair tips

$$a = 183, \quad b = 315$$

are corner-peelable one-concept extensions with respective support labels in the verifier's zero-based coordinate numbering

$$P_a = \{2, 3, 6, 7\}, \quad P_b = \{2, 3, 8, 9\}.$$

They therefore satisfy Theorem 6.59. Layer dimensions one, two, and three give explicit forbidden pc-minors of orders

$$600, \quad 1202, \quad 2406,$$

and isometric dimensions 13, 14, 15, respectively. More generally, the order in layer dimension $r \geq 1$ is $301 \cdot 2^r - 2$. These are the first members of one tower; every pair of Hall repair tips with distinct support labels gives another.

Lemma 6.61 (Exact ampleness test for a pair of one-concept extensions). *Let $S \subseteq Q_F$ be ample, let $a, b \in Q_F \setminus S$ be distinct, and suppose that*

$$A = S \cup \{a\}, \quad B = S \cup \{b\}$$

are ample. Each extension has a unique new shattered coordinate set; write

$$\text{Sh}(A) \setminus \text{Sh}(S) = \{P_a\}, \quad \text{Sh}(B) \setminus \text{Sh}(S) = \{P_b\}. \quad (6.73)$$

These supports are local: explicitly,

$$P_a = \{f \in F : a^{\{f\}} \in S\}, \quad P_b = \{f \in F : b^{\{f\}} \in S\}. \quad (6.74)$$

Put $C = S \cup \{a, b\}$, and, for a new coordinate e , put

$$X = (B \times \{0\}) \cup (A \times \{1\}).$$

Then

$$X \text{ is ample} \iff P_a \neq P_b. \quad (6.75)$$

When these equivalent conditions hold, C is ample and

$$\text{Sh}(C) = \text{Sh}(S) \dot{\cup} \{P_a, P_b\}, \quad (6.76)$$

while

$$\text{Sh}(A) \cap \text{Sh}(B) = \text{Sh}(S). \quad (6.77)$$

Proof. Lemma 2.10, applied to a and b , proves the uniqueness and local formulas in (6.73)–(6.74). In particular, P_a and P_b are strongly shattered in A and B , respectively.

Suppose first that $P_a \neq P_b$. The family C strongly shatters every member of $\text{Sh}(S)$, as well as P_a and P_b . Hence

$$|\text{SSh}(C)| \geq |S| + 2 = |C|.$$

The lower Sandwich inequality gives the reverse inequality. Equality in that inequality characterizes ample families, so C is ample. Since $|\text{Sh}(C)| = |C| = |S| + 2$, the shattered supports already exhibited are all of them, proving (6.76). The distinctness of the two new supports also gives (6.77).

A set of old coordinates is shattered by X exactly when it is shattered by C , while $D \cup \{e\}$ is shattered exactly when D is shattered by both A and B . Therefore

$$\text{Sh}(X) = \text{Sh}(C) \dot{\cup} \{D \cup \{e\} : D \in \text{Sh}(A) \cap \text{Sh}(B)\}. \quad (6.78)$$

Using the ampleness of C, S and (6.77), we obtain

$$|\text{Sh}(X)| = |C| + |S| = 2|S| + 2 = |X|,$$

so X is ample.

If $P_a = P_b$, then the intersection in (6.78) contains $\text{Sh}(S) \cup \{P_a\}$. Sauer's inequality gives $|\text{Sh}(C)| \geq |C|$, and consequently

$$|\text{Sh}(X)| \geq |C| + |S| + 1 = |X| + 1.$$

Thus X is not ample, proving (6.75). $\square$

Proposition 6.62 (Signed trace transport and recovery for a two-repair lift). *Use the notation and hypotheses of Lemma 6.61, assume that $P_a \neq P_b$, and put*

$$\Sigma = \text{Sh}(S), \quad \Sigma^+ = \Sigma \dot{\cup} \{P_a, P_b\}.$$

Then the shattered-set complex of the lift is

$$\text{Sh}(X) = \Sigma^+ \dot{\cup} \{D \cup \{e\} : D \in \Sigma\}. \quad (6.79)$$

Its entire trace table is obtained from the two sections as follows:

$$X[D, q] = (B[D, q] \times \{0\}) \dot{\cup} (A[D, q] \times \{1\}) \quad (D \in \Sigma^+, q \in Q_D), \quad (6.80)$$

$$X[D \cup \{e\}, (q, 0)] = B[D, q] \times \{0\},$$

$$X[D \cup \{e\}, (q, 1)] = A[D, q] \times \{1\} \quad (D \in \Sigma, q \in Q_D). \quad (6.81)$$

Let $\mathcal{N}(\Gamma)$ denote the minimal nonfaces of a simplicial complex Γ , and let σ_Y denote the unique missing trace on each minimal nonface of an ample family Y . Then

$$\mathcal{N}(\text{Sh}(X)) = \mathcal{N}(\Sigma^+) \dot{\cup} \{P_a \cup \{e\}, P_b \cup \{e\}\}, \quad (6.82)$$

and its signing is

$$\sigma_X(I) = \sigma_C(I) \quad (I \in \mathcal{N}(\Sigma^+)), \quad (6.83)$$

$$\sigma_X(P_a \cup \{e\}) = (a|_{P_a}, 0), \quad \sigma_X(P_b \cup \{e\}) = (b|_{P_b}, 1). \quad (6.84)$$

In particular, the signed presentation in (6.82)–(6.84) reconstructs X .

Conversely, the marked pair (X, e) recovers all the assembly data, up to reversing e . Indeed, after projecting the two e -sections to Q_F ,

$$B = \rho_e^0 X, \quad A = \rho_e^1 X, \quad S = A \cap B, \quad \{a\} = A \setminus B, \quad \{b\} = B \setminus A, \quad (6.85)$$

and

$$P_a = \{f \in F : a^{\{f\}} \in S\}, \quad P_b = \{f \in F : b^{\{f\}} \in S\}. \quad (6.86)$$

Proof. Equation (6.76) gives $\text{Sh}(C) = \Sigma^+$, and (6.77) gives $\text{Sh}(A) \cap \text{Sh}(B) = \Sigma$. Substitution in (6.78) proves (6.79). The two trace formulas follow directly from whether the traced support contains the layer coordinate e .

A set not containing e is a minimal nonface of $\text{Sh}(X)$ precisely when it is a minimal nonface of Σ^+ . Now consider $D \cup \{e\}$. It is a nonface precisely when $D \notin \Sigma$. It is minimal precisely when $D \in \mathcal{N}(\Sigma)$ and deletion of e leaves a face of Σ^+ . Since P_a and P_b are the two distinct minimal nonfaces added to Σ , this happens exactly for $D \in \{P_a, P_b\}$. This proves (6.82).

On an old minimal nonface, the union of the projected traces of the two sections is exactly C , which proves (6.83). The trace $a|_{P_a}$ is the unique trace on P_a missing from S . It is also missing from B : otherwise adding b would shatter P_a , contrary to $\text{Sh}(A) \cap \text{Sh}(B) = \Sigma$. The upper section contains that trace through a , so the unique trace of $P_a \cup \{e\}$ missing from X is $(a|_{P_a}, 0)$. The argument for P_b in the lower section is symmetric. This proves (6.84). The fixed-shadow signed-clutter theorem now reconstructs X from these clauses.

Finally, the two sections of the displayed lift are B and A . Their common concepts are exactly S , and their two differences are the singleton repair tips. This proves (6.85); the local formula from Lemma 6.61 gives (6.86). Reversing e merely interchanges (a, A, P_a) with (b, B, P_b) . $\square$

Proposition 6.63 (A thin coordinate recovers a two-sided fragile kernel). *Let $X \subseteq Q_{F \cup \{e\}}$ be ample and not corner-peelable. Project its two e -sections to Q_F , and write*

$$H_i = \rho_e^i X \quad (i \in \{0, 1\}), \quad K = H_0 \cap H_1.$$

Suppose that the two section differences are singletons, say

$$H_0 \setminus K = \{b\}, \quad H_1 \setminus K = \{a\}. \quad (6.87)$$

Then K , $H_0 = K \cup \{b\}$, $H_1 = K \cup \{a\}$, and $C = K \cup \{a, b\}$ are ample, whereas $K \notin \mathcal{D}_{\text{amp}}$. Moreover, the two one-concept extensions acquire distinct shattered supports:

$$\text{Sh}(K \cup \{a\}) \setminus \text{Sh}(K) = \{P_a\}, \quad \text{Sh}(K \cup \{b\}) \setminus \text{Sh}(K) = \{P_b\}, \quad P_a \neq P_b. \quad (6.88)$$

The family is exactly the two-extension lift

$$X = ((K \cup \{b\}) \times \{0\}) \cup ((K \cup \{a\}) \times \{1\}). \quad (6.89)$$

Thus the marked family (X, e) recovers K, a, b, P_a, P_b , up to reversing e , by intersections, differences, and the local formulas in (6.86).

If, in addition, $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$, then

$$K \cup \{a\}, \quad K \cup \{b\}, \quad K \cup \{a, b\} \quad \text{all belong to } \mathcal{D}_{\text{amp}}. \quad (6.90)$$

In particular, the thin coordinate exposes a finite, intrinsic two-sided fragile kernel. In this obstruction case, $K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ as well.

Proof. A set of old coordinates is shattered by X exactly when it is shattered by $C = H_0 \cup H_1$. A set $D \cup \{e\}$ is shattered exactly when D is shattered by both sections. Hence

$$\text{Sh}(X) = \text{Sh}(C) \dot{\cup} \{D \cup \{e\} : D \in \text{Sh}(H_0) \cap \text{Sh}(H_1)\}. \quad (6.91)$$

Since every support shattered by K is shattered by both sections, Sauer's inequality and the ampleness of X give

$$\begin{aligned} 2|K| + 2 &= |X| = |\text{Sh}(X)| \\ &\geq |C| + |\text{Sh}(K)| \geq (|K| + 2) + |K| = 2|K| + 2. \end{aligned}$$

Equality holds throughout. It follows that C and K are ample and that

$$\text{Sh}(H_0) \cap \text{Sh}(H_1) = \text{Sh}(K). \quad (6.92)$$

The sections H_0, H_1 are signed restrictions of the ample family X , so they are ample as well. Each one-concept extension of K therefore acquires exactly one shattered support. These supports are distinct by (6.92), proving (6.88). The section decomposition gives (6.89) and the stated recovery formulas.

It remains to prove that $K \notin \mathcal{D}_{\text{amp}}$. Deleting $(a, 1)$ from X leaves the peripheral expansion

$$((K \cup \{b\}) \times \{0\}) \cup (K \times \{1\}),$$

which is ample because both its base and its duplicated site are ample. Thus $(a, 1)$ is a corner. After deleting it, $(b, 0)$ is a corner, and deleting both exclusive tips leaves $K \square Q_1$. If $K \in \mathcal{D}_{\text{amp}}$, the product equivalence would make this last family corner-peelable, and the two preceding corner deletions would give a corner peeling of X , a contradiction. Hence $K \notin \mathcal{D}_{\text{amp}}$.

Finally, if $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$, then H_0, H_1 , and $C = X/e$ are proper pc-minors of X . Pc-minimality gives (6.90). Since $K = X^{\{e\}} \notin \mathcal{D}_{\text{amp}}$, Theorem 5.33 gives $K \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$. $\square$

Corollary 6.64 (Thin-thick terminal dichotomy). *Let $O \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be corner-descent terminal and suppose that O has a corner. For every coordinate e , put*

$$H_i(e) = \rho_e^i O, \quad D_i(e) = H_i(e) \setminus H_{1-i}(e) \quad (i \in \{0, 1\}).$$

Exactly one of the following alternatives holds.

1. Thin fragile-kernel branch. *Some coordinate e satisfies $|D_0(e)| = |D_1(e)| = 1$. Proposition 6.63 canonically recovers an ample forbidden pc-minor K and the peelable diamond (6.90). The two exclusive concepts are corners of O . Deleting either one leaves a family outside $\mathcal{D}_{\text{amp}}$, but terminality says that neither deletion is a forbidden pc-minor.*

2. Thick-section branch. *For every coordinate e ,*

$$|D_0(e)| \geq 1, \quad |D_1(e)| \geq 1, \quad |D_0(e)| + |D_1(e)| \geq 3. \quad (6.93)$$

Proof. Proposition 2.52 implies that neither section of O is contained in the other. Thus $D_0(e)$ and $D_1(e)$ are nonempty for every e . If both have order one for some coordinate, Proposition 6.63 applies. Its proof shows that the two exclusive concepts are corners and that deleting either one gives a peripheral expansion with duplicated site $K \notin \mathcal{D}_{\text{amp}}$; the exact peripheral-expansion test therefore puts both deletions outside $\mathcal{D}_{\text{amp}}$. Corner-descent terminality says that they are not forbidden pc-minors. If there is no such coordinate, two positive integers $|D_0(e)|, |D_1(e)|$ cannot have sum two, which gives (6.93). $\square$

Corollary 6.65 (Protector-incidence transport alphabet). *For a trace-carrier scheme on X , every precedence arc is read from (6.80)–(6.81). More explicitly:*

1. *an old-support carrier has same-trace predecessors in the union of the two corresponding section fibers, and each of its protector choices lies in such a two-section union;*
2. *if the carrier of $D \cup \{e\}$ lies in layer i , all of its same-trace predecessors lie in layer i , while for every $q \in Q_D$ it has a protector demand in the opposite-layer fiber*

$$X[D \cup \{e\}, (q, 1 - i)].$$

Consequently every directed trace circuit of the lift is a finite word in these explicitly listed old-support, same-layer, and cross-layer incidence types. Its number of cross-layer arcs is even.

Proof. The carrier and protector arcs are defined solely by membership in the relevant trace fibers. The two assertions therefore follow from (6.80) and (6.81). Traversing a closed directed walk changes layers an even number of times. $\square$

Theorem 6.66 (Two-extension lift). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be irredundantly embedded in Q_F . Let $a, b \in Q_F \setminus S$ be distinct, and put*

$$A = S \cup \{a\}, \quad B = S \cup \{b\}, \quad C = S \cup \{a, b\}.$$

Assume that $A, B \in \mathcal{D}_{\text{amp}}$ and that their new shattered supports P_a, P_b from (6.73) are distinct. For a corner z of an ample family Z , denote its unique maximal cube by $Q_Z(z)$. Suppose also that

$$\begin{aligned} Q_B(z) &\not\subseteq A \quad \text{for every } z \in S \text{ that is a corner of } B, \\ Q_A(z) &\not\subseteq B \quad \text{for every } z \in S \text{ that is a corner of } A. \end{aligned} \tag{6.94}$$

Then, for a new coordinate e , the family

$$X = (B \times \{0\}) \cup (A \times \{1\}) \tag{6.95}$$

is an ample forbidden pc-minor for $\mathcal{D}_{\text{amp}}$. Its only corners are $(b, 0)$ and $(a, 1)$.

Proof. Lemma 6.61 shows that C and X are ample.

The concepts $(b, 0)$ and $(a, 1)$ have no e -mates. They are corners of X because b and a are the new corners of the ample one-concept extensions B and A . Now let $z \in S$. The old directions incident to $(z, 0)$ are exactly those incident to z in B , and the direction e is also incident. Consequently $(z, 0)$ is a corner of X exactly when z is a corner of B and its corner cube $Q_B(z)$ is contained in the opposite section A . The symmetric statement holds for $(z, 1)$. Condition (6.94) therefore leaves exactly the two displayed corners.

Deleting $(a, 1)$ leaves

$$(B \times \{0\}) \cup (S \times \{1\}),$$

which is non-corner-peelable by Proposition 2.6, because $S \notin \mathcal{D}_{\text{amp}}$. The same argument after reversing e shows that deleting $(b, 0)$ is non-corner-peelable. Since these are the only corners, $X \notin \mathcal{D}_{\text{amp}}$.

It remains to prove pc-minimality. The two signed e -restrictions are A and B . The e -contraction is C , which belongs to $\mathcal{D}_{\text{amp}}$: the concept a is a corner of the ample family C , and deleting it leaves $B \in \mathcal{D}_{\text{amp}}$.

Let μ be a restriction or contraction in an old coordinate. Put $S' = \mu(S)$, $A' = \mu(A)$, and $B' = \mu(B)$, suppressing repeated projected concepts. The family $\mu(X)$ is ample because ampleness is pc-minor closed. The family S' is a proper pc-minor of the obstruction S , so $S' \in \mathcal{D}_{\text{amp}}$. Each of A' and B' is either S' or an ample one-concept extension of S' , because it is a pc-minor of A or B . If neither section has a new concept, then $\mu(X) = S' \sqcup Q_1 \in \mathcal{D}_{\text{amp}}$. If just one section has a new concept, deleting it leaves $S' \sqcup Q_1$, which is ample; the ample corner-deletion criterion therefore makes that concept a corner of $\mu(X)$. Peel it and then peel $S' \sqcup Q_1$. If the sections have distinct new concepts, deleting either one leaves a peripheral expansion whose base is the other ample one-concept extension and whose site is S' . Both the base and the site belong to $\mathcal{D}_{\text{amp}}$, so the exact expansion test places this deletion in $\mathcal{D}_{\text{amp}}$. The deleted concept is therefore a corner of $\mu(X)$; peeling it leaves a family already in $\mathcal{D}_{\text{amp}}$. Finally, if their new concepts coincide, then $\mu(X) = D \sqcup Q_1$, where $D = A' = B'$ is a pc-minor of $A \in \mathcal{D}_{\text{amp}}$, and hence this product also lies in $\mathcal{D}_{\text{amp}}$.

Thus every elementary pc-minor of X is corner-peelable, and pc-minor closure gives the same conclusion for every proper pc-minor. Hence $X \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$. $\square$

Corollary 6.67 (Complete multipartite extension graph). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be cornerless and irredundantly embedded in Q_F . Define its set of peelable extension vertices by*

$$\mathcal{R}(S) = \{a \in Q_F \setminus S : S \cup \{a\} \in \mathcal{D}_{\text{amp}}\}.$$

For $a \in \mathcal{R}(S)$, let $P(a)$ be the unique member of

$$\text{Sh}(S \cup \{a\}) \setminus \text{Sh}(S).$$

By Lemma 6.61, this label is the local neighbour-direction signature

$$P(a) = \{f \in F : a^{\{f\}} \in S\}. \quad (6.96)$$

and a minimal nonface of $\text{Sh}(S)$. For distinct $a, b \in \mathcal{R}(S)$, form the opposite-layer lift

$$X_{a,b} = ((S \cup \{b\}) \times \{0\}) \cup ((S \cup \{a\}) \times \{1\}). \quad (6.97)$$

Then

$$X_{a,b} \text{ is ample} \iff P(a) \neq P(b). \quad (6.98)$$

Whenever these conditions hold,

$$X_{a,b} \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$$

and $X_{a,b}$ has exactly the two corners $(b, 0)$ and $(a, 1)$. Consequently, the graph on $\mathcal{R}(S)$ in which a, b are adjacent precisely when their lift is ample is the complete multipartite graph whose parts are the nonempty fibers of P . Every edge of this graph gives a forbidden pc-minor without any pc-minor test. In particular, the compatibility graph is computed from the Hamming neighbours of the extension vertices, without computing shattered sets or proper pc-minors.

Proof. Lemma 6.61 proves (6.98). Suppose that the supports are distinct. We verify that the reflection hypothesis in Theorem 6.66 is automatic.

Let $z \in S$ be a corner of $S \cup \{b\}$. If its unique maximal cube did not contain b , that cube would lie in S . Every cube of S through z is also a cube of $S \cup \{b\}$ through z , and hence is contained in this unique maximal cube. Thus z would be a corner of S , contrary to the hypothesis. The corner cube therefore contains b , which does not belong to $S \cup \{a\}$. It is not contained in the opposite one-concept extension. The symmetric argument applies to old corners of $S \cup \{a\}$.

Theorem 6.66 now gives the asserted obstruction and corner statements. The adjacency condition depends only on inequality of the support labels, which is exactly the complete multipartite description. $\square$

Corollary 6.68 (Independent repair assemblies stop at two layers). *Let $S \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$ be cornerless and irredundantly embedded in Q_F . Use the construction of Proposition 6.2, with $E \neq \emptyset$, $T \subseteq Q_E$, and $a_t \in \mathcal{R}(S)$ for every $t \in T$. Assume that the tips and their supports are pairwise distinct, so that X_T is ample. Then*

$$X_T \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \iff |E| = 1 \text{ and } T = Q_E. \quad (6.99)$$

Thus the two-extension lift is the only forbidden pc-minor in the entire class of independent one-tip-per-layer assemblies over a common seed. In particular, placing three tips in three layers of a square cannot produce a forbidden pc-minor, and filling all four layers cannot produce a new one.

Proof. If $|E| = 1$ and $T = Q_E$, the family is the two-extension lift associated with its two tips. Their support labels are distinct, so Corollary 6.67 places X_T in $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

Suppose next that $T \neq Q_E$, and choose $v \in Q_E \setminus T$. Restricting all layer coordinates to v gives exactly S . This is a proper pc-minor of X_T outside $\mathcal{D}_{\text{amp}}$, so X_T is not pc-minimal outside $\mathcal{D}_{\text{amp}}$.

It remains to consider $T = Q_E$ and $|E| \geq 2$. Choose $e \in E$ and fix all coordinates in $E \setminus \{e\}$. The resulting proper pc-minor has two sections $S \cup \{a_t\}$ and $S \cup \{a_u\}$, where t, u are the endpoints of the selected layer edge. Their support labels are distinct, so this restriction is itself a forbidden two-extension lift by Corollary 6.67. Again X_T is not pc-minimal. These three cases prove (6.99). $\square$

Corollary 6.69 (Two-extension lifts refute direct corner descent). *Every family X produced by Theorem 6.66 is a counterexample to the corner-deletion assertion: neither of its corners c satisfies*

$$X \setminus \{c\} \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}). \quad (6.100)$$

Thus direct corner descent fails even for forbidden endpoints with exactly two corners.

Proof. Theorem 6.66 says that the only corners of X are $(b, 0)$ and $(a, 1)$. Deleting the upper corner gives

$$X \setminus \{(a, 1)\} = (B \times \{0\}) \cup (S \times \{1\}).$$

This family is not corner-peelable by the exact peripheral-expansion test, because $S \notin \mathcal{D}_{\text{amp}}$. Its restriction to the upper semicube is S , a proper pc-minor outside $\mathcal{D}_{\text{amp}}$; hence the deletion is not pc-minimal outside $\mathcal{D}_{\text{amp}}$, and therefore does not belong to $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$. The same argument after reversing the two layers applies to the deletion of $(b, 0)$. This proves (6.100). $\square$

Theorem 6.70 (Reduction to an irreducible cornerless family). *Every finite ample non-corner-peelable cube family X can be reduced to an ample, cornerless, periphery-free, Cartesian-prime, non-corner-peelable family H by a finite sequence of the following operations:*

1. delete a corner;

2. if the current family is a peripheral expansion

$$(B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B,$$

replace it by a nonempty non-corner-peelable member of $\{B, S\}$;

3. if the current family is $A \square B$ with both factors nontrivial, replace it by a non-corner-peelable factor.

Every intermediate family selected by this procedure is ample and non-corner-peelable. In ideal language, the terminal family is a periphery-free Cartesian-prime member of $\mathfrak{C}_r^{\text{col}}$ for some $r \leq d$. Thus cornerless prime cores are the starting objects that cannot be reduced further by these three operations.

Proof. Deleting a corner preserves ampleness. It also preserves failure of a complete corner peeling: otherwise the deleted corner followed by a peeling of the remainder would peel the original family. Therefore we may delete corners until a cornerless ample nonpeelable family is reached.

If that family has a peripheral coordinate, write it in the form in item 2. Its nonempty projected sections are ample. The same expansion test in Proposition 2.6 shows, when $S \neq \emptyset$, that at least one of B, S is non-corner-peelable. If $S = \emptyset$, deleting the redundant coordinate identifies the family with B , which is therefore non-corner-peelable. Select a nonempty non-corner-peelable section. If the family has a nontrivial Cartesian factorization, the factors are ample, and the same proposition shows that at least one factor is non-corner-peelable; select one. Either selection strictly decreases the isometric dimension. Starting from the selected family, delete corners again and repeat.

The process terminates because the pair consisting of isometric dimension and order, ordered lexicographically, strictly decreases: a corner deletion decreases the order without increasing dimension, whereas a section or factor selection decreases the dimension. At termination the family is cornerless and has neither a peripheral coordinate nor a nontrivial Cartesian factor. It is therefore periphery-free and Cartesian-prime. Cornerlessness says that deletion of every generator destroys linearity, so its ideal belongs to $\mathfrak{C}_r^{\text{col}}$. $\square$

7 Examples and the minimum-order problem

The preceding recognition criteria characterize the full forbidden basis, while the construction theorems describe specified subclasses. We now record representative members and study the independent extremal question of the least ample member.

7.1 The Hall obstruction and its first exchanges

The theorem of Chalopin, Chepoi, Moran, and Warmuth produces a maximum VC-dimension-three family with no corner. Cornerlessness proves that the family is not in $\mathcal{D}_{\text{amp}}$, but does not by itself prove pc-minor minimality. The following proposition records the additional finite check needed here and the first neighboring obstructions it reveals. Integers in the statement encode their twelve-bit binary expansions in the coordinate convention of [CCMW22, Theorem 4.5].

Proposition 7.1 (The Hall obstruction and four first exchanges). *Let $C_H \subseteq Q_{12}$ be the 299-concept maximum VC-dimension-three class of [CCMW22, Theorem 4.5]. Then*

$$C_H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}.$$

It has exactly sixteen ample one-concept extensions. Twelve are corner-peelable and have four corners. The other four are nonpeelable, have exactly two corners, and are described by

<i>adjoined concept a</i>	<i>other corner c</i>	
373	1011	(7.1)
1105	192	
1360	3120	
1525	252	

For every row, both

$$C_H \cup \{a\} \quad \text{and} \quad (C_H \setminus \{c\}) \cup \{a\}$$

belong to $\text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}}) \cap \mathcal{A}$; the second family is cornerless, has order 299, and has the same shattered-set complex as C_H . The four second families are pairwise nonisomorphic and are not isomorphic to C_H .

Computer-assisted proof. The cited theorem proves that C_H is maximum of VC dimension three and has no corner, hence is ample and not corner-peelable. Exact evaluation of its twenty-four signed restrictions and twelve contractions gives a complete corner peeling of every elementary pc-minor. The criterion in Theorem 3.3 therefore proves $C_H \in \text{Obs}_{\text{pc}}(\mathcal{D}_{\text{amp}})$.

For each of the $2^{12} - 299 = 3797$ outside concepts, the finite check first tests the one-concept isometry condition and then the Sandwich equality, the complete corner set, and corner peelability. This gives the split $16 = 12 + 4$ and the four pairs in (7.1). Proposition 2.12 makes each nonpeelable union $C_H \cup \{a\}$ a forbidden pc-minor. Its only possible first peeling steps are its two corners. Deleting a returns the nonpeelable family C_H , so deleting the other corner c also leaves a nonpeelable ample family. The same-shadow criterion in Proposition 5.107 proves that this deletion preserves the shattered-set complex. Direct corner evaluation makes the resulting family cornerless. Finally, the thirty-six elementary minors of each exchanged family are corner-peelable, and exact canonicalization under coordinate permutations and independent coordinate reversals distinguishes the five cornerless types. $\square$

Proposition 7.2 (The carrier paths in Hall’s class). *For each of the sixty-six coordinate pairs $S \subseteq [12]$, the carrier tree $\mathcal{K}_S(C_H)$ from Proposition 5.13 is a path on eleven vertices. It has ten edges, of which eight are internal. Among the 220 maximal three-cubes of C_H , exactly 36 are internal edges in one of their three carrier paths, 60 are internal edges in two, and 124 are internal edges in all three. In particular, every maximal cube of C_H is covered by two opposite square facets belonging to other maximal cubes.*

Computer-assisted proof. The exact verifier reconstructs the unique three-cube on every triple support from the fixed list of the 299 concepts. For each pair S , it forms the carrier graph whose vertices are the S -cube anchors and whose e -edge is the cube with support $S \cup \{e\}$. All sixty-six graphs have eleven vertices, ten edges, and degree sequence

$$1, 1, 2, 2, 2, 2, 2, 2, 2, 2,$$

so each is a path. Counting, for every triple, in how many of its three carrier paths its labeled edge has no leaf endpoint gives the asserted histogram. Proposition 5.13 then identifies such an internal edge with an opposite-facet cover.

The durable verifier and output are

```
verify_damp_hall_carrier_trees.py,
damp_hall_carrier_trees.json,
```

and the mathematical-profile digest is

65c031beb26e738618488674026fd5308367c99c513773edf2793e09c2ba7dcf.

□

Remark 7.3 (What the Hall neighborhood shows). The obstruction C_H is not isolated. Repeating the same-shadow exchange through distance two gives twenty-seven nonisomorphic cornerless obstructions of order 299 and thirty-two nonisomorphic two-corner obstructions of order 300. Continuing one further one-concept layer yields 454 nonisomorphic forbidden pc-minors in dimension twelve, of orders 299, 300, and 301. These exact finite atlases are useful evidence about the geometry of the basis, but the paper uses them neither as a classification nor as a substitute for the symbolic parametrizations above.

Proposition 7.4 (A maximum fibre inside every ample family). *Let $H \subsetneq Q_E$ be a nonempty ample family, and let $z \notin E$. Write $\overline{H} = \{\overline{h} : h \in H\}$ and form the antipodal double*

$$\widehat{H} = (\{0\} \times H) \cup (\{1\} \times \overline{H}) \subseteq Q_{\{z\} \cup E}.$$

There are a set $R \subseteq E$ and a word $f \in Q_R$ such that, after deleting the fixed coordinates R ,

$$H_f = H_{\overline{f}} =: A, \tag{7.2}$$

the common family $A \subseteq Q_{E \setminus R}$ is maximum, and

$$(\{0\} \times A) \cup (\{1\} \times \overline{A})$$

is minimal nonample. In particular, A is the tope set on one side of a distinguished element of a simple uniform oriented matroid, and hence is a pseudogeometric maximum class.

The set R can be taken empty precisely when $\widehat{H}$ is itself minimal nonample. Put $r = |R|$. If R is nonempty, then

$$(r+1)|A| \leq |H|. \tag{7.3}$$

Equality holds only if, for every $a \in A$, the section

$$P_a = \{g \in Q_R : (g, a) \in H\}$$

is a geodesic path from f to $\overline{f}$, and every section over a word outside A is empty. In fact all these paths are the same: equality holds if and only if

$$H = A \times P \tag{7.4}$$

for one antipodal geodesic path $P \subseteq Q_R$.

Proof. Put $\Sigma = \text{Sh}(H)$. Since H is proper, Σ has a minimal nonface I . Every proper subset of I is shattered by H . The projection $H|_I$ is ample, so it has exactly $2^{|I|} - 1$ traces and therefore omits a unique word $q \in Q_I$. Its antipodal image omits $\overline{q}$, and $q \neq \overline{q}$. Consequently $(H \cup \overline{H})|_I = Q_I$, so $\widehat{H}$ shatters I .

For every $S \in \Sigma$, both S and $S \cup \{z\}$ are shattered by $\widehat{H}$: the two z -layers each realize every trace on S . These $2|\Sigma|$ sets do not include I . Since H is ample,

$$|\text{Sh}(\widehat{H})| \geq 2|\Sigma| + 1 = 2|H| + 1 = |\widehat{H}| + 1.$$

Thus $\widehat{H}$ is nonample.

Choose a conditioning of $\widehat{H}$ that is nonample and has as few unfixed coordinates as possible. It is minimal nonample. The coordinate z remains unfixed: fixing it produces a conditioning of H or of $\overline{H}$, which is ample because ampleness is closed under conditionings. Hence the chosen conditioning fixes a word f on some $R \subseteq E$ and has the two z -layers

$$H_f \quad \text{and} \quad \overline{H_{\bar{f}}}.$$

The classification of minimal nonample families [Bou26] says that such a family is antipodally closed and that each nonempty one-coordinate conditioning is maximum. Antipodal closure identifies the displayed layers, giving $H_f = H_{\bar{f}} = A$, and the $z = 0$ conditioning shows that A is maximum. The same classification identifies the minimal nonample double as a uniform-oriented-matroid tope set, so its side A is pseudogeometric. The assertion about $R = \emptyset$ follows from the choice of the conditioning.

For each $a \in A$, the section P_a is a nonempty conditioning of the ample family H , and hence is itself ample and isometric. It contains the antipodal words f and $\bar{f}$. Their Hamming distance is r , so P_a contains a shortest path of length r and therefore at least $r + 1$ vertices. The sections over different words of $Q_{E \setminus R}$ are disjoint, and summing over $a \in A$ proves (7.3). Equality leaves no nonempty section outside A and forces every P_a to consist of exactly the vertices of one shortest antipodal path. Such a path flips every coordinate of R exactly once.

It remains to use ampleness. For every $S \in \text{Sh}(A)$, the family H shatters S and $S \cup \{e\}$ for every $e \in R$: the endpoint layers f and $\bar{f}$ both contain all of A . These are $(r+1)|\text{Sh}(A)| = (r+1)|A| = |H|$ distinct shattered sets, so in the equality case they exhaust $\text{Sh}(H)$. If two paths P_a and P_b were different, their coordinate orders would reverse some pair $\{e, g\} \subseteq R$. One path realizes three traces on that pair, including the trace obtained by flipping e but not g , while the other realizes the complementary intermediate trace obtained by flipping g but not e . Together with their common antipodal endpoints, their union therefore shatters $\{e, g\}$. This is an additional shattered set, a contradiction. Hence all P_a equal one path P , which proves (7.4). Conversely, the product $A \times P$ is ample and attains equality. $\square$

Corollary 7.5 (Maximum-core dichotomy below Hall's order). *Let H be cornerless and ample with $|H| < 299$. Then either H is itself a pseudogeometric maximum class, or H contains two opposite equal conditionings whose common value is a pseudogeometric maximum class A with*

$$|A| \leq \left\lfloor \frac{298}{|R| + 1} \right\rfloor \leq 149.$$

Proof. Apply Proposition 7.4. If its set R is empty, then $A = H$. Otherwise (7.3) gives the displayed bound. $\square$

Corollary 7.6 (Maximum core in a least cornerless family). *Let H have minimum cardinality among all nonempty cornerless ample families, and suppose $|H| < 299$. Then either H is a pseudogeometric maximum class, or Proposition 7.4 gives a set R with $r = |R| \geq 2$ and a pseudogeometric maximum class A such that*

$$(r + 1)|A| + 1 \leq |H|, \quad |A| \leq \left\lfloor \frac{297}{r + 1} \right\rfloor \leq 99. \quad (7.5)$$

Proof. Assume $R \neq \emptyset$. Equality in (7.3) would give $H = A \times P$. If A were cornerless, it would be a smaller nonempty cornerless ample family. Otherwise A has a corner, and its product with either endpoint of the path P is a corner of $A \times P$. Both alternatives contradict the choice of H . The

inequality is therefore strict. If $r = 1$, there are only the two opposite layers, so $H = A \times Q_1$ and equality is automatic; hence $r \geq 2$. Finally, $|H| \leq 298$ gives $(r + 1)|A| \leq 297$, and therefore the displayed bound. $\square$

Proposition 7.7 (Geodesic-disagreement tax). *Use the maximum core A , the coordinate set R , and the sections P_a from Proposition 7.4, and put*

$$\Delta(H; A, R) = |H| - (|R| + 1)|A|.$$

Let $G_R(H)$ be the graph on R in which eg is an edge exactly when the two-coordinate projection $H|_{\{e,g\}}$ is the full square. Then

$$|E(G_R(H))| \leq \Delta(H; A, R). \quad (7.6)$$

Moreover,

$$\Delta(H; A, R) = |H \setminus (Q_R \times A)| + \sum_{a \in A} (|P_a| - (|R| + 1)). \quad (7.7)$$

In particular, if shortest antipodal paths chosen inside P_a and P_b flip e and g in opposite orders, then $eg \in E(G_R(H))$ and this disagreement spends one unit of the excess in (7.6).

Proof. As in the proof of Proposition 7.4, H shatters

$$S \quad \text{and} \quad S \cup \{e\} \quad (S \in \text{Sh}(A), e \in R).$$

These are $(|R| + 1)|A|$ distinct supports and each uses at most one coordinate of R . For every edge eg of $G_R(H)$, the additional support $\{e, g\}$ is shattered. Ampleness now gives

$$|H| = |\text{Sh}(H)| \geq (|R| + 1)|A| + |E(G_R(H))|,$$

which is (7.6). The sections over the words in A are disjoint, each has at least $|R| + 1$ vertices, and all remaining vertices of H lie over words outside A . Partitioning H by these sections proves (7.7).

Finally, orient the two coordinates from the common endpoint f . One of the two paths contains the intermediate trace 10 on $\{e, g\}$ and the other contains 01; their common antipodal endpoints supply 00 and 11. Hence the projection is the full square, as claimed. $\square$

Lemma 7.8 (Everything below a least cornerless family is dismantlable). *Let H have minimum cardinality among all nonempty cornerless ample families. If K is ample and $|K| < |H|$, then $K \in \mathcal{D}_{\text{amp}}$.*

Proof. Suppose that $K \notin \mathcal{D}_{\text{amp}}$, and delete corners successively for as long as possible. Every remaining family is ample. The process cannot delete all of K , since the reverse order would then be a corner peeling of K . It therefore stops at a nonempty cornerless ample family K' . But $|K'| \leq |K| < |H|$, contrary to the choice of H . $\square$

Proposition 7.9 (Every coordinate shift resolves a least obstruction). *Let H have minimum cardinality among all nonempty cornerless ample families, and let e be an active coordinate of H . Write its two e -sections as B_0, B_1 , and put*

$$A = B_0 \cap B_1, \quad C = B_0 \cup B_1.$$

For either orientation of e , replace the two sections by A on the lower side and C on the upper side. The resulting coordinate shift

$$U_e(H) = (A \times \{0\}) \cup (C \times \{1\}) \quad (7.8)$$

is ample, has the same order and the same shattered-set complex as H , and belongs to $\mathcal{D}_{\text{amp}}$.

Consequently a least cornerless ample family is shift-critical in every active coordinate: every one-coordinate move to the standard polarization is already corner-peelable. In the Yang-complex language, every paired coordinate shift of its strongly nonshellable Yang ball is shellable.

Proof. The two-layer shadow-intersection lemma makes A and C ample. Since e is active and H is connected, both sections are nonempty and $A \neq \emptyset$; hence

$$|A| < |H|, \quad |C| = |H| - |A| < |H|.$$

Lemma 7.8 therefore gives $A, C \in \mathcal{D}_{\text{amp}}$.

The family in (7.8) is the peripheral expansion with lower section $A \subseteq C$. It is ample by the two-layer shadow formula: its supports avoiding e are indexed by $\text{Sh}(C)$, while those containing e are indexed by $\text{Sh}(A)$. Their total number is $|C| + |A| = |U_e(H)|$.

To peel the shifted family, first take any corner peeling of A in its lower copy. If a is the next corner of the current lower section and Q is its unique maximal cube there, then $Q \times \{0, 1\}$ is present, because the upper section contains the lower one. It contains every edge of the shifted family incident with $(a, 0)$, so $(a, 0)$ is a corner. After the lower copy has been deleted, peel the remaining upper copy of C . Thus $U_e(H) \in \mathcal{D}_{\text{amp}}$.

The equality of the shattered complexes is the paired coordinate-shift theorem; it also follows directly from the two-layer identities

$$\text{Sh}(C) = \text{Sh}(B_0) \cup \text{Sh}(B_1), \quad \text{Sh}(A) = \text{Sh}(B_0) \cap \text{Sh}(B_1).$$

Interchanging the two signs of e gives the other orientation. The Yang-complex assertion is the shellability–corner-peeling dictionary. $\square$

Corollary 7.10 (Every overlap corner has two exclusive shields). *Keep the notation of Proposition 7.9. If a is a corner of A , then for each $b \in \{0, 1\}$ there is a coordinate f_b such that*

$$a^{\{f_b\}} \in B_b \setminus A. \quad (7.9)$$

The two shield directions are distinct. Thus every corner of the contraction overlap is owned on both sides by two different exclusive wing concepts.

Proof. Let Q be the unique maximal cube of A through a . Suppose that a has no neighbour in $B_b \setminus A$. Then its incident directions in B_b are exactly the directions of Q . The other copy of every vertex of Q belongs to H , because $Q \subseteq A$. Hence all directions incident with (a, b) generate the single cube $Q \times \{0, 1\}$, making (a, b) a corner of H , a contradiction. This proves (7.9) on both sides. If $f_0 = f_1$, the same neighbour of a would belong to both sections and hence to A , contrary to the displayed inclusions. $\square$

Lemma 7.11 (Repeated ownership forces a punctured cube). *Let A be an isometric cube family, let $z \notin A$, and suppose that the distinct neighbours*

$$a_i = z^{\{f_i\}} \in A \quad (i \in I)$$

are corners of A . Then the face through z on $F = \{f_i : i \in I\}$ satisfies

$$Q_F(z) \setminus \{z\} \subseteq A. \quad (7.10)$$

In particular,

$$2^{|I|} - 1 \leq |A|. \quad (7.11)$$

Consequently, if $|H| < 48$ in Corollary 7.10, no one exclusive wing concept can shield four corners of A .

Proof. For distinct $i, j \in I$, the two vertices a_i, a_j have Hamming distance two. Isometry of A supplies a two-edge path between them. Their only common cube neighbours are z and $z^{\{f_i, f_j\}}$; the first is absent from A , so the second belongs to A .

Fix $i \in I$. The corner a_i is therefore incident in every direction f_j , $j \neq i$. All these directions lie in its unique maximal cube, which contains

$$z^{\{f_i\} \Delta J} \quad (J \subseteq F \setminus \{f_i\}).$$

As i varies, these are all nonempty vertices of $Q_F(z)$, proving (7.10) and the cardinality bound.

For the last assertion, Proposition 2.184 shows that the two exclusive wings in any active coordinate have at least ten concepts each. Hence

$$|A| \leq \frac{|H| - 20}{2} < 14.$$

Four corners with one owner would force $|A| \geq 2^4 - 1 = 15$, a contradiction. $\square$

Proposition 7.12 (A fibre corner forces a transverse puncture). *Let $H \subseteq Q_{R \dot{\cup} X}$ be ample and cornerless, let $g \in H|_R$, and put*

$$H_g = \{a \in Q_X : (g, a) \in H\}.$$

If a is a corner of H_g , then there is a set $J \subseteq I_H((g, a))$ such that

$$|J| \geq 2, \quad J \cap R \neq \emptyset, \quad (g, a)^U \in H \ (U \subsetneq J), \quad (g, a)^J \notin H. \quad (7.12)$$

Thus the punctured cube rooted at (g, a) is not contained in the fibre: at least one of its directions is a projection direction.

Proof. Put $v = (g, a)$. Since H is cornerless, the directions incident with v do not generate a cube of H . Choose an inclusion-minimal set $J \subseteq I_H(v)$ for which $v^J \notin H$. Every singleton incident with v is present, so $|J| \geq 2$, and minimality gives $v^U \in H$ for every proper subset $U \subsetneq J$.

It remains to prove that J uses a coordinate of R . If $J \subseteq X$, then $J \subseteq I_{H_g}(a)$. Because a is a corner of H_g , all directions incident with a belong to its unique maximal cube. That cube would contain a^J , and hence $v^J = (g, a^J)$ would belong to H , a contradiction. Therefore $J \cap R \neq \emptyset$. $\square$

Proposition 7.13 (A column corner forces a core-transverse puncture). *Let $H \subseteq Q_{R \dot{\cup} X}$ be ample and cornerless, let $a \in H|_X$, and put*

$$P_a = \{g \in Q_R : (g, a) \in H\}.$$

If g is a corner of P_a , then there is a set $J \subseteq I_H((g, a))$ such that

$$|J| \geq 2, \quad J \cap X \neq \emptyset, \quad (g, a)^U \in H \ (U \subsetneq J), \quad (g, a)^J \notin H. \quad (7.13)$$

Thus a punctured cube rooted at a column corner must use a core direction.

Proof. Put $v = (g, a)$. Since H is cornerless, the directions incident with v do not generate a cube of H . Choose an inclusion-minimal $J \subseteq I_H(v)$ for which $v^J \notin H$. As in Proposition 7.12, one has $|J| \geq 2$ and $v^U \in H$ for every proper subset $U \subsetneq J$.

If $J \subseteq R$, then $J \subseteq I_{P_a}(g)$. Because g is a corner of P_a , all those directions lie in the unique maximal cube of P_a through g . That cube contains g^J , so $v^J = (g^J, a) \in H$, a contradiction. Hence $J \cap X \neq \emptyset$. $\square$

Corollary 7.14 (Fibrewise transverse-puncture cover). *Let H have minimum cardinality among all nonempty cornerless ample families, and use the nonempty repair set R supplied by Proposition 7.4. Put $W = H|_R$. Every nonempty fibre H_g belongs to $\mathcal{D}_{\text{amp}}$. In particular, for every $g \in W$ there are a corner a_g of H_g and a set J_g satisfying (7.12). The resulting rooted punctures have pairwise distinct repair roots g . If $|H| < 48$, then*

$$2 \leq |J_g| \leq 5 \quad (g \in W).$$

Proof. Every nonempty fibre H_g is ample, because it is a conditioning of H . It is strictly smaller than H : the two opposite repair fibres from Proposition 7.4 are nonempty and $R \neq \emptyset$. Lemma 7.8 therefore gives $H_g \in \mathcal{D}_{\text{amp}}$.

Choose the first member a_g of a corner peeling of H_g and apply Proposition 7.12. Distinct words g give distinct roots. Finally, the punctured J_g -face contains $2^{|J_g|} - 1$ vertices of H . If $|H| < 48$, this excludes $|J_g| \geq 6$. $\square$

Corollary 7.15 (Bidirectional transverse-puncture cover). *Let $H \subseteq Q_{R \dot{\cup} X}$ have minimum cardinality among all nonempty cornerless ample families, and suppose that both projections*

$$W = H|_R, \quad Z = H|_X$$

have at least two members. Then every nonempty row H_g and every nonempty column P_a belongs to $\mathcal{D}_{\text{amp}}$. One may therefore choose

- for every $g \in W$, a row corner a_g and a rooted puncture J_g with $J_g \cap R \neq \emptyset$;
- for every $a \in Z$, a column corner g_a and a rooted puncture L_a with $L_a \cap X \neq \emptyset$.

The roots within the first family have distinct repair words, and the roots within the second have distinct core words. If $|H| < 48$, every puncture in both families has rank between two and five.

Proof. Rows and columns are conditionings of H , hence are ample. The hypothesis $|W|, |Z| \geq 2$ makes every nonempty one strictly smaller than H . Lemma 7.8 therefore puts all of them in $\mathcal{D}_{\text{amp}}$.

Choose a corner in each row and column. Proposition 7.12 gives the row-indexed punctures, while Proposition 7.13 gives the column-indexed punctures. Their stated root distinctions follow from their indices. Finally, each punctured J -face contains $2^{|J|} - 1$ vertices of H , so $|H| < 48$ excludes $|J| \geq 6$ in either family. $\square$

Proposition 7.16 (Section corners force a mixed defect or an incidence cycle). *Let $H \subseteq Q_{R \dot{\cup} X}$ be ample and cornerless, put $W = H|_R$ and $Z = H|_X$, and suppose that every nonempty row H_g and column P_a has a corner. Choose one row corner $a_g \in H_g$ for every $g \in W$ and one column corner $g_a \in P_a$ for every $a \in Z$. Then at least one of the following holds.*

1. *Some cell $(g, a) \in H$ is both selected: $a = a_g$ and $g = g_a$. At this cell there is a rooted puncture J satisfying*

$$J \cap R \neq \emptyset, \quad J \cap X \neq \emptyset. \quad (7.14)$$

2. *The incidence graph*

$$\mathcal{I}(H) = (W \dot{\cup} Z, \{ga : (g, a) \in H\}) \quad (7.15)$$

contains an even cycle of length at least four whose edges alternate between selected row corners and selected column corners.

In particular, a least cornerless ample family in the path-spine regime always contains either a mixed transverse puncture at a common section corner or an alternating cycle in its row-column incidence graph.

Proof. Direct the selected incidence edge from $g \in W$ to $a_g \in Z$, and direct the selected incidence edge from $a \in Z$ to $g_a \in W$. Every vertex of the resulting finite directed bipartite graph has out-degree one. Following outgoing edges therefore eventually enters a directed cycle. Its length is even. A directed two-cycle is exactly an incidence edge (g, a) selected from both endpoints, giving item 1. Every longer directed cycle has distinct vertices and, after orientations are forgotten, gives the alternating incidence cycle in item 2.

It remains to verify the mixed puncture assertion in item 1. Put $v = (g, a)$. Cornerlessness of H gives an inclusion-minimal $J \subseteq I_H(v)$ such that $v^J \notin H$; as before, $|J| \geq 2$ and $v^U \in H$ for every proper $U \subsetneq J$. If $J \subseteq X$, then the unique maximal cube through the row corner a in H_g contains a^J , a contradiction. If $J \subseteq R$, the unique maximal cube through the column corner g in P_a gives the same contradiction. Hence J meets both coordinate blocks, proving (7.14). $\square$

Conjecture 7.17 (Mutual section-corner principle). *Let $H \subseteq Q_{R \dot{\cup} X}$ be ample, with R and X nonempty. Suppose that every nonempty row H_g and every nonempty column P_a belongs to $\mathcal{D}_{\text{amp}}$. Then some concept $(g, a) \in H$ is a corner in both of its sections: a is a corner of H_g and g is a corner of P_a .*

For a least cornerless ample family, the section hypothesis holds by Lemma 7.8. The conjecture would therefore force item 1 of Proposition 7.16, and hence a genuinely mixed rooted puncture, without analyzing longer alternating cycles. The dismantlability hypothesis cannot simply be omitted: if C is cornerless and ample, then $C \times Q_1$ is ample but has no mutual section corner for the split $Q_1 \mid C$, because its two C -rows are cornerless. Exact tests verify the stated conjecture for every ample family in Q_4 , for the induced-six-cycle witness described in the campaign ledger, and for every coordinate split of the order-291 Hall obstruction. These tests are not used in any proof.

Lemma 7.18 (Fixed-support punctures have disjoint carriers). *Let J be a coordinate set and let $v_1, \dots, v_s \in H$ be distinct roots such that*

$$v_i^U \in H \quad (U \subsetneq J), \quad v_i^J \notin H \quad (1 \leq i \leq s).$$

Then the parallel J -faces $Q_J(v_1), \dots, Q_J(v_s)$ are pairwise disjoint. Consequently

$$s(2^{|J|} - 1) \leq |H|. \quad (7.16)$$

Proof. Two parallel J -faces are either equal or disjoint. Suppose that $Q_J(v_i) = Q_J(v_j)$. The puncture rooted at v_i contains every vertex of this face except v_i^J , whereas the puncture rooted at v_j omits v_j^J . If the two omitted vertices were different, each would be required to be both present and absent. Hence $v_i^J = v_j^J$, and flipping J gives $v_i = v_j$. Distinct roots therefore have disjoint faces. Each face contributes $2^{|J|} - 1$ distinct vertices of H , which proves (7.16). $\square$

Lemma 7.19 (Projected cube carriers conserve support). *Let $H \subseteq Q_{R \dot{\cup} X}$ be ample, let $\emptyset \neq K \subseteq X$ be strongly shattered by H , and define the K -cube carrier and its repair projection by*

$$\mathcal{K}_K(H) = \{y \in Q_{(R \dot{\cup} X) \setminus K} : Q_K \times \{y\} \subseteq H\}, \quad W_K = \mathcal{K}_K(H)|_R. \quad (7.17)$$

Then $\mathcal{K}_K(H)$ and W_K are nonempty ample families. Moreover,

$$\text{Sh}(W_K) = \{T \subseteq R : K \dot{\cup} T \in \text{Sh}(H)\}. \quad (7.18)$$

Consequently, if m pairwise distinct repair words occur as projections of K -cube anchors, then H shatters at least m supports whose intersection with X is exactly K .

Proof. For $A \subseteq (R \dot{\cup} X) \setminus K$, the carrier $\mathcal{K}_K(H)$ strongly shatters A if and only if H strongly shatters $K \dot{\cup} A$. Indeed, a full A -cube of carrier anchors and the full K -cube over each of its vertices together form a full $(K \dot{\cup} A)$ -cube of H , and the converse follows by contracting the K -directions of such a cube.

If the carrier merely shatters A , then for every trace on A it contains an anchor of a full K -cube. Those cubes supply all traces on $K \dot{\cup} A$, so H shatters that union. Since H is ample, the union is strongly shattered, and the preceding equivalence shows that A is strongly shattered by the carrier. Thus the shattered and strongly shattered sets of $\mathcal{K}_K(H)$ coincide, so the carrier is ample. It is nonempty because K is strongly shattered. Projections are pc-minors, hence preserve ampleness, and therefore W_K is ample as well.

The trace of the carrier on a set $T \subseteq R$ is the same as the trace of its projection W_K on T . Thus $T \in \text{Sh}(W_K)$ implies that the carrier shatters T , and the preceding argument gives $K \dot{\cup} T \in \text{Sh}(H)$. Conversely, if H shatters $K \dot{\cup} T$, ampleness makes that support strongly shattered. Contracting the K -directions of the resulting cube gives a full T -cube in $\mathcal{K}_K(H)$, and hence T is shattered by W_K . This proves (7.18).

Finally, if m distinct repair words occur among the projected anchors, then $|W_K| \geq m$. Ampleness gives $|\text{Sh}(W_K)| = |W_K|$, and (7.18) supplies the asserted m supports. Their core part is exactly K because every $T \in \text{Sh}(W_K)$ is contained in R . $\square$

Corollary 7.20 (Every core carrier below a least obstruction is dismantlable). *Let $H \subseteq Q_{R \dot{\cup} X}$ have minimum cardinality among all nonempty cornerless ample families, and suppose that H uses at least two core words. For a core word $a \in Q_X$, put*

$$P_a = \{g \in Q_R : (g, a) \in H\}.$$

Every nonempty column P_a belongs to $\mathcal{D}_{\text{amp}}$. For every nonempty strongly shattered $K \subseteq X$, the carrier $\mathcal{K}_K(H)$ and its repair projection W_K in (7.17) also belong to $\mathcal{D}_{\text{amp}}$.

Proof. The column P_a is a conditioning of H , so it is ample. It is strictly smaller than H because H contains a vertex with a different core word. Lemma 7.8 gives $P_a \in \mathcal{D}_{\text{amp}}$.

Lemma 7.19 makes the carrier and its projection ample. The full K -cubes over distinct carrier anchors are disjoint, and hence

$$2^{|K|} |\mathcal{K}_K(H)| \leq |H|.$$

Thus both $\mathcal{K}_K(H)$ and W_K have order strictly below $|H|$. Another application of Lemma 7.8 proves the claim. $\square$

Proposition 7.21 (Anchored coface ascent). *Let $H \subseteq Q_E$ have minimum cardinality among all nonempty cornerless ample families, and put $\Sigma = \text{Sh}(H)$. For a nonempty $K \in \Sigma$, a positioned K -cube is a full K -cube $Q_K(y) \subseteq H$, where $y \in Q_{E \setminus K}$ is its anchor. The following statements hold.*

First, the positioned K -cubes can be matched bijectively with the cofaces of K in Σ : there is a bijection

$$\psi_K : \{Q_K(y) \subseteq H\} \longrightarrow \{L \in \Sigma : K \subseteq L\} \quad (7.19)$$

such that $Q_K(y)$ is contained in a positioned $\psi_K(Q_K(y))$ -cube. Exactly one value of ψ_K is K ; all other values are strict cofaces of K .

Second, let $\mathcal{A}$ be any finite multiset of anchored faces $(K, Q_K(y))$, with $K \neq \emptyset$. Each member may be enlarged, by repeatedly replacing it with a containing positioned coface, so that the following normal form holds:

$$\text{for every final support } L, \text{ all members labelled by } L \text{ occupy one positioned } L\text{-cube.} \quad (7.20)$$

In particular, either the members of $\mathcal{A}$ acquire pairwise distinct support labels, or at least two of them end in one common positioned cube. Thus collisions between distinct cube positions never form a terminal obstruction: they ascend to strict cofaces. The only terminal multiplicity is a same-positioned-cube packet.

Proof. Fix $K \neq \emptyset$. Regard K as one coordinate block and $E \setminus K$ as the other. Lemma 7.19 makes the full carrier

$$\mathcal{K}_K(H) = \{y \in Q_{E \setminus K} : Q_K(y) \subseteq H\}$$

ample and gives the full link identity

$$\text{Sh}(\mathcal{K}_K(H)) = \{S \subseteq E \setminus K : K \dot{\cup} S \in \Sigma\}. \quad (7.21)$$

Indeed, both implications are proved before projection in that lemma: a full S -cube of carrier anchors is the same thing as a full $K \dot{\cup} S$ -cube of H .

The disjoint positioned K -cubes show that $2^{|K|}|\mathcal{K}_K(H)| \leq |H|$. Hence the carrier is smaller than H , and minimum-cardinality makes it a member of $\mathcal{D}_{\text{amp}}$, as in Corollary 7.20. Choose a corner peeling

$$F_1 = \mathcal{K}_K(H) \supset F_2 \supset \cdots \supset F_s = \{y_s\}, \quad F_{i+1} = F_i \setminus \{y_i\}.$$

For $i < s$, let S_i be the support of the unique maximal cube of F_i through y_i , and put $S_s = \emptyset$. Deleting the corner y_i removes exactly the shattered support S_i . Therefore the S_i are distinct and, by ampleness, exhaust $\text{Sh}(\mathcal{K}_K(H))$. Moreover the positioned S_i -cube of carrier anchors through y_i gives a positioned $K \dot{\cup} S_i$ -cube of H containing $Q_K(y_i)$. Together with (7.21), the assignment

$$Q_K(y_i) \longmapsto K \dot{\cup} S_i$$

is the required bijection. Only S_s is empty, which proves the strict-coface assertion.

We now prove the normal form. Start with the given multiset. Whenever some support K labels members occupying at least two distinct positioned K -cubes, choose one member at each occupied position and apply the preceding bijection to those distinct positions. Replace each chosen member by the containing positioned coface supplied by the bijection. The new support labels are pairwise distinct; at most one chosen member keeps the label K , while every other chosen member moves to a strict coface.

Use

$$\Phi = \sum_{(L, Q_L(z)) \in \mathcal{A}} |L|$$

as a potential, counted with multiplicity. Every such operation strictly increases Φ , and $\Phi \leq |E||\mathcal{A}|$. Hence the procedure terminates. At termination no support is represented at two different positions, which is precisely (7.20). If no final position contains two members, then every support labels at most one member and the labels are pairwise distinct. Otherwise the repeated members form the asserted same-positioned-cube packet. $\square$

Lemma 7.22 (Compatible signed traces in an ample two-layer family). *Let x be a coordinate and let*

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

be ample, with both sections nonempty. Then P , Q , $P \cup Q$, and $P \cap Q$ are ample. More precisely, for every set T of coordinates different from x ,

$$\pi_T(P) \cap \pi_T(Q) = \pi_T(P \cap Q). \quad (7.22)$$

Equivalently, their Yang complexes satisfy the signed identity

$$\Delta_P \cap \Delta_Q = \Delta_{P \cap Q}. \quad (7.23)$$

Consequently,

$$\text{Sh}(P) \cap \text{Sh}(Q) = \text{Sh}(P \cap Q). \quad (7.24)$$

Moreover,

$$\text{Sh}(Y) = \text{Sh}(P \cup Q) \dot{\cup} \{S \cup \{x\} : S \in \text{Sh}(P \cap Q)\}. \quad (7.25)$$

Furthermore,

$$\text{Sh}(P \cup Q) = \text{Sh}(P) \cup \text{Sh}(Q). \quad (7.26)$$

Proof. Fix a trace $\alpha \in \pi_T(P) \cap \pi_T(Q)$, and choose $p \in P$ and $q \in Q$ which both realize α . The ample family Y is isometric. A hypercube geodesic in Y from $(p, 0)$ to $(q, 1)$ crosses the x -coordinate in an edge

$$(r, 0)(r, 1), \quad r \in P \cap Q.$$

The endpoints p and q agree on every coordinate of T , so a hypercube geodesic between them changes no coordinate of T . Hence $r|_T = \alpha$. This proves the nontrivial inclusion in (7.22); the reverse inclusion is immediate. Faces of a Yang complex are precisely signed traces, so (7.23) is equivalent to the trace identity. Taking all traces on a coordinate support gives (7.24).

The two sections and their union are restrictions and a contraction of Y , so they are ample. Direct inspection of the two sections gives

$$\text{Sh}(Y) = \text{Sh}(P \cup Q) \dot{\cup} \{S \cup \{x\} : S \in \text{Sh}(P) \cap \text{Sh}(Q)\}.$$

Using the Sandwich equality for Y and $P \cup Q$ now gives

$$|\text{Sh}(P) \cap \text{Sh}(Q)| = |P| + |Q| - |P \cup Q| = |P \cap Q|.$$

On the other hand, the Sandwich inequality gives $|\text{Sh}(P \cap Q)| \geq |P \cap Q|$. Equality holds throughout. Thus $P \cap Q$ is ample, and substitution gives (7.25). Finally, $\text{Sh}(P) \cup \text{Sh}(Q)$ is contained in $\text{Sh}(P \cup Q)$, while

$$|\text{Sh}(P) \cup \text{Sh}(Q)| = |P| + |Q| - |P \cap Q| = |P \cup Q| = |\text{Sh}(P \cup Q)|.$$

This proves (7.26). □

Corollary 7.23 (Monochromatic upper stars outside the overlap). *In the setting of Lemma 7.22, put $U = P \cup Q$ and $K = P \cap Q$. If*

$$S \in \text{Sh}(U) \setminus \text{Sh}(K),$$

then exactly one of P, Q shatters S . The same section shatters every $T \in \text{Sh}(U)$ with $S \subseteq T$.

In particular, if U is a maximum class of VC dimension three on d coordinates and a pair S is not shattered by K , then all $d - 2$ triples containing S are shattered by one common section.

Proof. Equation (7.26) puts S in at least one section shadow, whereas (7.24) prevents it from lying in both. Suppose, say, that P shatters S . If Q shattered a set $T \supseteq S$, downward closure of the shattering complex would put S in $\text{Sh}(Q)$, a contradiction. Thus every such T is shattered by P , again by (7.26). A maximum VC-dimension-three class shatters every triple, which gives the final assertion. $\square$

Proposition 7.24 (The bichromatic core must be nonample). *Let $U \subseteq Q_E$ be a maximum class of VC dimension three whose cubical complex is pure, and let*

$$Y = (P \times \{0\}) \cup (Q \times \{1\}) \subseteq Q_{E \cup \{e\}}$$

be ample. Put $K = P \cap Q$, and suppose that $\text{VCdim}(K) \leq 2$. For every triple $T \subseteq E$, let C_T be the unique T -cube of $U = P \cup Q$. The cube C_T has one of the two owners P, Q , according as

$$T \in \text{Sh}(P) \setminus \text{Sh}(Q) \quad \text{or} \quad T \in \text{Sh}(Q) \setminus \text{Sh}(P).$$

Define the bichromatic core

$$M = \{v \in U : v \text{ belongs to triple cubes of both owners}\}.$$

If M is nonempty and ample and $M \subsetneq K$, then Y has a corner.

Consequently, if U has a corner and Y is cornerless, then M is nonempty and nonample.

Proof. The two-layer shadow identities imply that every triple belongs to exactly one section shadow, since K has VC dimension at most two. Maximum classes have one cube on each top-dimensional support, which defines the asserted owner. A vertex incident with cubes of both owners belongs to both sections, and hence

$$M \subseteq K.$$

Suppose first that M is nonempty and ample and that $M \subsetneq K$. The relative VC-two peeling lemma, Lemma A.16, supplies a corner $v \in K \setminus M$. All maximal three-cubes of U through v have one owner; interchange the layers so that this owner is P . Let D be the unique maximal cube of K through v .

We claim that the upper copy $(v, 1)$ lies in the unique maximal cube $D \times \{0, 1\}$ of Y . Indeed, any cube of Q through v is a cube of U . Purity extends it to a maximal three-cube of U through v , whose owner is P . Every vertex of the original Q -cube therefore lies in both P and Q , so that cube is contained in K , and then in D . The same is automatic for every vertical cube through $(v, 1)$. Thus $(v, 1)$ is a corner of Y .

Now assume that U has a corner u and that Y is cornerless. The unique maximal cube of U through u has one owner, so $u \notin M$. If $u \notin K$, its sole copy in the owner layer lies in a unique maximal cube of Y , contrary to cornerlessness. Hence $u \in K \setminus M$, and in particular $M \subsetneq K$. If M were empty, any corner of the nonempty ample VC-two family K would be monochromatic, and the preceding unique-maximal-cube argument would again give a corner of Y . Thus $M \neq \emptyset$. The first part now shows that M cannot be ample. $\square$

Proposition 7.25 (Exact relative-corner criterion). *In the setting of Proposition 7.24, write $v^P = (v, 0)$ and $v^Q = (v, 1)$. If $v \notin M$, let $o(v) \in \{P, Q\}$ be the common owner of all maximal three-cubes of U through v , and let $\bar{o}(v)$ be the other owner. Then*

$$\text{Cor}(Y) = \{u^{o(u)} : u \in \text{Cor}(U) \setminus K\} \dot{\cup} \{v^{\bar{o}(v)} : v \in \text{Cor}(K) \setminus M\}.$$

Consequently,

$$Y \text{ is cornerless} \iff \text{Cor}(U) \subseteq K \text{ and } \text{Cor}(K) \subseteq M.$$

Proof. First let $v \notin K$. Then $v \notin M$, so all maximal three-cubes of U through v have owner $o(v)$, and only the copy $v^{o(v)}$ belongs to Y . Purity of U identifies the maximal cubes of Y through this copy with the maximal three-cubes of U through v . Hence $v^{o(v)}$ is a corner of Y exactly when v is a corner of U .

Now let $v \in K$. Every maximal cube D of K through v gives a vertical maximal cube $D \times \{0, 1\}$ of Y . If v belongs to at least two maximal cubes of K , these vertical cubes already prevent both copies of v from being corners.

It remains to suppose that v is a corner of K , with unique maximal cube D . If $v \in M$, then a P -owned and a Q -owned maximal three-cube of U pass through v . Each copy of v therefore lies in both a horizontal maximal cube of its own layer and the vertical cube $D \times \{0, 1\}$, so neither copy is a corner.

Finally suppose that $v \notin M$, and interchange the layers if necessary so that $o(v) = P$. Every cube of Q through v extends in U to a P -owned maximal three-cube. Its vertices consequently belong to both sections, so the cube lies in K , and then in D . Thus v^Q lies only in the vertical maximal cube $D \times \{0, 1\}$ and is a corner. The copy v^P also lies in a horizontal P -owned maximal three-cube and is not a corner. These three cases prove the identity and its stated consequence. $\square$

Proposition 7.26 (Relative leaf-charge identity). *Let $K \subseteq Q_E$ be ample of VC dimension at most two, and suppose that every maximal cube of K is a square. For $i \in E$, define the singleton carrier $T_i(K)$ as follows: its vertices are the i -edges of K , and two such edges are adjacent when their union is a square of K . Every nonempty $T_i(K)$ is a tree.*

For an i -edge e , an endpoint $x \in e$, and a set $M \subseteq K$, let $h_i(x)$ be the number of maximal squares of K that contain x but whose support does not contain i . Then

$$2|\text{Cor}(K) \setminus M| = \sum_{i \in E} \sum_{e \text{ leaf of } T_i(K)} |\{x \in e \setminus M : h_i(x) = 0\}|. \quad (7.27)$$

In particular, $\text{Cor}(K) \subseteq M$ exactly when every endpoint outside M of every carrier leaf has a maximal-square shield not using the carrier coordinate.

Proof. The carrier of a strongly shattered singleton in an ample family is ample. After contracting its i -edges, its one-inclusion graph has VC dimension at most one and is connected. An ample family of VC dimension at most one has a tree as its one-inclusion graph. This is precisely $T_i(K)$.

Fix an i -edge e and $x \in e$. Every maximal square through x whose support contains i contains the whole edge e , and corresponds to one edge of $T_i(K)$ incident with e . Hence the number of maximal squares of K through x is

$$\deg_{T_i(K)}(e) + h_i(x).$$

If e is a leaf, the endpoint x is therefore a corner of K exactly when $h_i(x) = 0$. Conversely, if x is a corner and its unique maximal square has support $\{i, j\}$, then the i -edge and the j -edge of that square through x are carrier leaves and have no off-carrier shield at x . Thus every corner outside M is counted twice on the right of (7.27), once for each direction of its unique square, and no other endpoint is counted. $\square$

Proposition 7.27 (Owner-deficit identity). *In the setting of Proposition 7.24, let A_P be the set of vertices of U incident only with P -owned maximal cubes, and define A_Q symmetrically. Let t_P and t_Q be the numbers of triples owned by P and Q , respectively, and put*

$$\ell_P = |\{S \in \text{Sh}(P) \setminus \text{Sh}(K) : |S| \leq 2\}|,$$

with the analogous definition of ℓ_Q . Then

$$|A_P| - t_P = |A_P \cap K| + \ell_P, \quad |A_Q| - t_Q = |A_Q \cap K| + \ell_Q. \quad (7.28)$$

In particular, both owner deficits are nonnegative. If Y is cornerless, then

$$|A_P| - t_P \geq |A_P \cap \text{Cor}(U)|, \quad |A_Q| - t_Q \geq |A_Q \cap \text{Cor}(U)|. \quad (7.29)$$

Consequently, an owner with zero deficit cannot own a corner of U .

Proof. Purity implies that every vertex of U is incident with a maximal three-cube. Every vertex in a P -owned cube belongs to P , every vertex in a Q -owned cube belongs to Q , and every vertex incident with both owners belongs to K . It follows that

$$P \setminus K = A_P \setminus K, \quad Q \setminus K = A_Q \setminus K.$$

The nested families $K \subseteq P$ are both ample, so their Sandwich equalities give

$$|A_P \setminus K| = |P \setminus K| = |\text{Sh}(P) \setminus \text{Sh}(K)|.$$

Because K has VC dimension at most two, the three-element supports on the right are precisely the t_P triples owned by P ; the remaining supports are the ℓ_P lower-dimensional ones. Therefore

$$|A_P| - |A_P \cap K| = t_P + \ell_P,$$

which is the first identity in (7.28). The second is symmetric.

If Y is cornerless, the proof of Proposition 7.24 shows that every corner of U belongs to K . Such a corner is incident with one maximal cube and hence belongs to the corresponding one of A_P, A_Q . Dropping the nonnegative terms ℓ_P, ℓ_Q from (7.28) proves (7.29). $\square$

Corollary 7.28 (Forced-core shadow bound). *In the setting of Proposition 7.24, suppose that Y is cornerless and put*

$$F = M \cup \text{Cor}(U).$$

Then

$$F \subseteq K, \quad |K| \geq |\text{Sh}(F)|, \quad |Y| \geq |U| + |\text{Sh}(F)|. \quad (7.30)$$

Thus a target order for Y imposes one owner-only shadow bound before the overlap is chosen.

Proof. The bichromatic core lies in K , and the proof of Proposition 7.24 puts every corner of U in K . Hence $F \subseteq K$. Since K is ample, the Sandwich equality and monotonicity of shattering give

$$|K| = |\text{Sh}(K)| \geq |\text{Sh}(F)|.$$

Finally, the two-layer decomposition gives $|Y| = |U| + |K|$. $\square$

Corollary 7.29 (Cubical-completion bound). *For $F \subseteq U$, define*

$$\kappa_U(F) = \min\{|Z| : F \subseteq Z \subseteq U \text{ and every } S \in \text{Sh}(F) \text{ is strongly shattered by } Z\}.$$

In the setting of Corollary 7.28, every cornerless lift satisfies

$$|K| \geq \kappa_U(F) \quad \text{and} \quad |Y| \geq |U| + \kappa_U(F).$$

Equivalently, Z must contain a positioned S -cube for every support S shattered by the forced core. This can be strictly stronger than merely counting the supports in $\text{Sh}(F)$.

Proof. Because $F \subseteq K$, monotonicity gives $\text{Sh}(F) \subseteq \text{Sh}(K)$. Ampleness identifies shattering and strong shattering, so K strongly shatters every support in $\text{Sh}(F)$. Thus K is feasible in the minimization defining $\kappa_U(F)$, and the first inequality follows. The second again uses $|Y| = |U| + |K|$. $\square$

Corollary 7.30 (Relative-corner completion bound). *For $M \subseteq F \subseteq U$, define*

$$\lambda_U(F, M) = \min\{|Z| : F \subseteq Z \subseteq U, Z \text{ is ample, } \text{VCdim}(Z) \leq 2, \text{Cor}(Z) \subseteq M\},$$

with value ∞ if no such Z exists. In the setting of Corollary 7.28, every cornerless lift satisfies

$$|K| \geq \lambda_U(F, M) \quad \text{and} \quad |Y| \geq |U| + \lambda_U(F, M).$$

Proof. The overlap K contains F , is ample of VC dimension at most two, and Proposition 7.25 gives $\text{Cor}(K) \subseteq M$. Thus K is feasible in the minimization defining $\lambda_U(F, M)$. $\square$

Corollary 7.31 (Corner-cube cover obstruction). *In the setting of Proposition 7.24, suppose that Y is cornerless. If u is a corner of U and C is the unique maximal cube of U containing u , then*

$$u \in K \quad \text{and} \quad C \setminus (M \cup \{u\}) \neq \emptyset.$$

Thus every contraction-corner cube has a vertex distinct from its corner that is incident only with maximal cubes of its own owner.

Proof. The proof of Proposition 7.24 shows that $u \in K$. If every other vertex of C belonged to M , then $M \subseteq K$ would give $C \subseteq K$. The support of C would then be shattered by K , contrary to $\text{VCdim}(K) \leq 2$. $\square$

Remark 7.32. Proposition 7.24 and Corollary 7.31 can be checked before the overlap K is chosen. They therefore eliminate complete owner colourings, simultaneously for every possible overlap order, rather than merely eliminating one cardinality layer of the reverse-expansion search.

Corollary 7.33 (Monochromatic components of the overlap-defect graph). *In the setting of Corollary 7.23, suppose that U is maximum of VC dimension three on d coordinates. Form a graph M_K on the coordinates by declaring ij to be an edge when $\{i, j\} \notin \text{Sh}(K)$. Every nontrivial connected component of M_K has one owner: all its edges, and all triples containing any of those edges, belong to the same one of the two section shadows.*

If K shatters every singleton, let m be the number of edges of M_K and let c be the number of triples shattered by K . Then

$$|K| = 1 + d + \binom{d}{2} - m + c. \tag{7.31}$$

In particular, an overlap smaller than the maximum VC-two order satisfies $m \geq c + 1$. Every one of its common triples is supported on a triangle of the complement graph $\overline{M_K}$.

Proof. Two incident missing pairs have a three-element union T . The monochromatic-upper-star property assigns T the owner of either pair, so the two owners must agree. Propagation along a path in M_K proves the component statement, and the same upper-star property assigns every triple containing a component edge to that component's owner.

The shadow of K is a down-set contained in the rank-three shadow of U . When all singletons are present, it therefore consists of the empty set, the d singletons, the $\binom{d}{2} - m$ common pairs, and the c common triples. Since K is ample, the Sandwich equality gives (7.31). Comparing this identity with $1 + d + \binom{d}{2}$ gives $m \geq c + 1$ in the stated range. Finally, downward closure forbids a common triple from containing an edge missing from the common shadow. $\square$

Proposition 7.34 (Carrier-subtree and one-cut principles). *Let $U \subseteq Q_E$ be an ample family of VC dimension at most three and let $P \subseteq U$ be ample. For a pair $R \in \text{Sh}(P)$, the edges of the R -cube carrier of U whose triple supports belong to $\text{Sh}(P)$ form a connected subtree of the carrier of U . If there is no such triple, the carrier of P consists of one vertex.*

Now suppose that U is maximum of VC dimension three, every pair carrier of U is a path, and

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

is ample with $P \cup Q = U$. If, for some pair S ,

$$\text{Sh}(P \cap Q) = \{R \subseteq E : |R| \leq 2\} \setminus \{S\}, \quad (7.32)$$

then the following assertions hold.

1. *All triples containing S belong to one section shadow.*
2. *Every other pair-carrier path is cut into at most two monochromatic edge intervals, one belonging to P and one to Q .*
3. *If both colours occur, their unique common pair cube is the carrier vertex at the cut. If the path is monochromatic, the section with no edge chooses one carrier vertex, and this is again the unique common pair cube.*
4. *If Y is cornerless and v is a corner of U with unique maximal triple cube C_T , then at least two of the three pair facets of C_T containing v belong to $P \cap Q$. Equivalently, at least two of the three corresponding pair carriers choose the vertex v .*
5. *If Y is cornerless and v, w are two corners of U whose unique triple supports T_v, T_w meet in one coordinate, then T_v and T_w have the same owner.*
6. *More generally, if a missing common pair S' meets the unique triple support T_v of a corner v in one coordinate, then S' and T_v have the same owner.*

Thus a lift satisfying (7.32) is specified at rank three by one colour per triple, at most one cut per pair-carrier, and one vertex choice only on a monochromatic carrier. No independent choice of the vertices of P and Q remains.

Proof. The R -cube carrier $\mathcal{K}_R(P)$ is an ample subfamily of $\mathcal{K}_R(U)$. Its one-inclusion graph is connected. An edge labelled e belongs to this subfamily exactly when P contains the unique $(R \cup \{e\})$ -cube of U , or equivalently when $R \cup \{e\} \in \text{Sh}(P)$. Since the carrier of U is a tree, these edges and their incident vertices form a connected subtree. With no edge, ampleness and the Sandwich equality leave one carrier vertex.

Apply Lemma 7.22. Under (7.32), every triple of U belongs to exactly one of the two section shadows. The first assertion is Corollary 7.23. Fix $R \neq S$. Both sections shatter R , and the preceding paragraph says that the edges of either colour form a connected subpath. The two edge sets partition the carrier path, so they are two intervals with at most one cut. If both are nonempty, the corresponding vertex subpaths meet exactly at the cut vertex. A full R -cube lies in $P \cap Q$ precisely when its carrier vertex belongs to both subpaths, so that cut vertex is the unique common R -cube. If, say, every edge belongs to P , then the carrier of P is the full path, whereas the edge-free carrier of Q is one vertex; their intersection is that vertex. This proves the first three assertions and the rank-three parametrization.

It remains to prove the corner assertions. The unique triple cube C_T is owned by one section, say P . The copy of v in that layer belongs to one horizontal maximal cube, so cornerlessness requires a vertical maximal cube through it. Such a cube is the product with the new coordinate of a common pair facet of C_T through v . This vertical cube also puts the opposite-layer copy of v in Y . That copy belongs to no horizontal triple cube, because C_T is the only maximal cube of U containing v and is not owned by Q . A second vertical pair cube is therefore required to keep the opposite-layer copy from being a corner. Hence at least two of the three pair facets through v are common.

For the two-corner assertion, a common pair cube through the corner v is a leaf of its carrier path. It cannot be the cut vertex of two nonempty colour intervals, because such a cut is internal. Consequently the path is monochromatic, with the same owner as T_v . Among any two pair facets of T_v , one contains the coordinate in $T_v \cap T_w$; the same holds at w . If the two corner triples had opposite owners, choose such a common pair at each corner. Their union is a triple and would belong to both section shadows by the monochromatic-upper-star property, a contradiction. Thus the owners agree.

The same argument proves the more general statement. Of the two common pair facets forced through v , at least one contains the coordinate in $S' \cap T_v$. Its carrier is monochromatic with the owner of T_v , while Corollary 7.23 makes the entire upper star of S' monochromatic with the owner of S' . The union of the two pairs is a triple, so the two owners must coincide. $\square$

Corollary 7.35 (Two-corner trace obstruction). *Assume the one-pair-deficit hypotheses of Proposition 7.34, and write the missing pair as $S = \{a, b\}$. Suppose that U has corners v_0, v_1 with unique triple supports T_0, T_1 such that*

$$T_0 \cap S = T_1 \cap S = \{a\} \quad \text{and} \quad v_0(b) \neq v_1(b).$$

Then the lift Y has a corner.

Proof. Suppose that Y is cornerless and put $K = P \cap Q$. At least two of the three pair facets through v_i are common, by Proposition 7.34. Since $a \in T_i$, at least one of these two pair facets contains a . Its common pair cube contains both v_i and $v_i \triangle \{a\}$. Thus the trace of K on S contains both labels having b -coordinate $v_i(b)$. The two values $v_0(b)$ and $v_1(b)$ are different, so together the two corners force all four labels on S . This says that K shatters S , contrary to (7.32). $\square$

Remark 7.36 (The first Hall shield profile closes without search). In the coordinate-zero projection of Hall's class, the two corners are the projected vertices 1600 and 1894. Their unique triple supports are $\{0, 5, 6\}$ and $\{0, 7, 8\}$, while their values on coordinate 1 are 0 and 1, respectively. Taking $S = \{0, 1\}$ in Corollary 7.35 excludes every one-pair-deficit lift with this missing pair. In particular, the apparent 3×3 choices of two shields at the two corners are not nine separate cases: the same two-line trace argument eliminates all of them.

Proposition 7.37 (Exterior-component normal form for a one-pair deficit). *Let $U \subseteq Q_E$ be maximum of VC dimension three, and write C_T for its unique triple cube with support $T \in \binom{E}{3}$. Let $K \subseteq U$ be ample with*

$$\text{Sh}(K) = \{R \subseteq E : |R| \leq 2\} \setminus \{S\}, \quad S \in \binom{E}{2}.$$

Form a graph on the triple supports by joining T, T' whenever $C_T \cap C_{T'}$ contains a vertex outside K , and additionally join all triples containing S . Let $\mathcal{B}$ be its set of connected components, let B_S

be the component containing the triples above S , and put

$$V_B = \left(\bigcup_{T \in B} C_T \right) \setminus K, \quad \omega(B) = |V_B| - |B| - \mathbf{1}_{B=B_S}.$$

Then the sets V_B partition $U \setminus K$ and

$$\sum_{B \in \mathcal{B}} \omega(B) = 0. \quad (7.33)$$

For $\mathcal{R} \subseteq \mathcal{B}$ with $B_S \in \mathcal{R}$, define

$$P_{\mathcal{R}} = K \cup \bigcup_{B \in \mathcal{R}} \bigcup_{T \in B} C_T, \quad Q_{\mathcal{R}} = K \cup \bigcup_{B \notin \mathcal{R}} \bigcup_{T \in B} C_T.$$

The following are equivalent:

1. $P_{\mathcal{R}}$ and $Q_{\mathcal{R}}$ are ample;
2. $\sum_{B \in \mathcal{R}} \omega(B) = 0$.

Under these conditions, $P_{\mathcal{R}} \cup Q_{\mathcal{R}} = U$, $P_{\mathcal{R}} \cap Q_{\mathcal{R}} = K$, and the corresponding two-layer lift is ample. Conversely, every ample two-layer lift with contraction U and overlap K is obtained in this way, up to interchanging its two layers.

For $v \in U$, let $\nu_K(v)$ be the number of maximal pair cubes of K containing v , and let $\rho_{\mathcal{R}}(v)$ and $\rho_{\overline{\mathcal{R}}}(v)$ count the triple cubes through v whose supports lie in $\mathcal{R}$ and its complement. The lift is cornerless if and only if

$$\nu_K(v) + \rho_{\mathcal{R}}(v) \neq 1, \quad \nu_K(v) + \rho_{\overline{\mathcal{R}}}(v) \neq 1 \quad (v \in U). \quad (7.34)$$

Equivalently, the lift is cornerless if and only if

1. every corner of U belongs to at least two maximal pair cubes of K ; and
2. every corner of K is incident with triple cubes from both sides of the component partition.

If $\mathcal{B}$ has one component, no cornerless lift exists.

Proof. If $v \notin K$, all triple cubes containing v belong to the same graph component. The triple cubes cover U , so the sets V_B are nonempty, pairwise disjoint, and cover $U \setminus K$. Since

$$|U \setminus K| = \left(1 + |E| + \binom{|E|}{2} + \binom{|E|}{3} \right) - \left(1 + |E| + \binom{|E|}{2} - 1 \right) = 1 + \binom{|E|}{3},$$

their orders give (7.33).

The definitions and the exterior-incidence edges give $P_{\mathcal{R}} \cup Q_{\mathcal{R}} = U$ and $P_{\mathcal{R}} \cap Q_{\mathcal{R}} = K$. The selected facet cubes strongly shatter exactly

$$\text{Sh}(K) \cup \{S\} \cup \{T : T \in B, B \in \mathcal{R}\}$$

on the $P_{\mathcal{R}}$ side, and exactly

$$\text{Sh}(K) \cup \{T : T \in B, B \notin \mathcal{R}\}$$

on the other side. Here all triples above S lie in $B_S \subseteq \mathcal{R}$, so the full S -carrier belongs to $P_{\mathcal{R}}$, whereas $Q_{\mathcal{R}}$ does not shatter S . Therefore

$$|P_{\mathcal{R}}| - |\text{SSh}(P_{\mathcal{R}})| = \sum_{B \in \mathcal{R}} \omega(B).$$

The corresponding difference for $Q_{\mathcal{R}}$ is the negative of this sum by (7.33). Each difference is nonnegative by the lower Sandwich inequality. Thus either one vanishes if and only if both vanish, and Proposition 5.14 gives the stated ampleness equivalence. The horizontal and vertical strongly shattered supports of the lift have total order $|U| + |K|$. This is also the order of the lift, so equality in the lower Sandwich inequality proves that the lift is ample.

Conversely, consider any ample two-layer lift with contraction U and overlap K . Lemma 7.22 shows that its sections are ample and that exactly one of them shatters S . Every triple support is shattered by at least one section, so its unique triple cube in U lies in that section; it cannot lie in both, because then its support would be shattered by K . Triple cubes meeting at a vertex outside K must lie in the same section, and the section that shatters S contains every triple cube above S . After interchanging the two layers if necessary, its triple supports are therefore the union of a set $\mathcal{R}$ of graph components with $B_S \in \mathcal{R}$. Hence its sections are exactly $P_{\mathcal{R}}$ and $Q_{\mathcal{R}}$.

The maximal cubes of the lift are precisely the horizontal triple cubes and the products of the new-coordinate edge with the maximal pair cubes of K . Thus the two copies of v have maximal-cube incidence multiplicities $\nu_K(v) + \rho_{\mathcal{R}}(v)$ and $\nu_K(v) + \rho_{\overline{\mathcal{R}}}(v)$; multiplicity zero means that the copy is absent. Proposition 5.14 now proves (7.34).

At a corner of U , exactly one of the two ρ -terms is one and the other is zero. The two inequalities therefore force $\nu_K(v) \geq 2$. At a corner of K , one has $\nu_K(v) = 1$, so both ρ -terms must be positive. Finally, these two conditions are also sufficient. Indeed, a vertex outside K that is not a corner of U belongs to at least two triple cubes, all on the same side, while a vertex in K that is not a corner of K belongs to at least two maximal pair cubes. Thus every vertex not covered by the two conditions already satisfies (7.34). If $\mathcal{B}$ has one component, one section is U and the other is K . The latter has a corner by the VC-two corner-peeling theorem, and its copy in the lift belongs only to the corresponding vertical maximal cube. $\square$

Proposition 7.38 (Exterior-component normal form for an arbitrary overlap). *Let $U \subseteq Q_E$ be maximum of VC dimension three, let C_T be its unique triple cube of support $T \in \binom{E}{3}$, and let $K \subseteq U$ be a nonempty ample family. Put $\Sigma = \text{Sh}(K)$ and*

$$\mathcal{N} = \binom{E}{3} \setminus \Sigma, \quad \mathcal{D} = \{A \subseteq E : |A| \leq 2 \text{ and } A \notin \Sigma\}.$$

Form a graph on $\mathcal{N}$ by joining two triple supports when their cubes meet at a vertex outside K , and, for every $A \in \mathcal{D}$, joining all triple supports that contain A . Let $\mathcal{B}$ be the set of connected components. For $B \in \mathcal{B}$, define

$$V_B = \left(\bigcup_{T \in B} C_T \right) \setminus K, \quad \mathcal{D}_B = \{A \in \mathcal{D} : A \subseteq T \text{ for some } T \in B\},$$

and

$$\omega_K(B) = |V_B| - |B| - |\mathcal{D}_B|. \quad (7.35)$$

Then the V_B partition $U \setminus K$, the $\mathcal{D}_B$ partition $\mathcal{D}$, and every component is balanced individually:

$$\omega_K(B) = 0 \quad (B \in \mathcal{B}). \quad (7.36)$$

For $\mathcal{R} \subseteq \mathcal{B}$, put

$$P_{\mathcal{R}} = K \cup \bigcup_{B \in \mathcal{R}} \bigcup_{T \in B} C_T, \quad Q_{\mathcal{R}} = K \cup \bigcup_{B \notin \mathcal{R}} \bigcup_{T \in B} C_T.$$

Then $P_{\mathcal{R}}$ and $Q_{\mathcal{R}}$ are both ample for every $\mathcal{R} \subseteq \mathcal{B}$, and the corresponding two-layer family is ample. Every ample two-layer lift with contraction U and overlap K arises from such a component cut, up to interchanging the layers.

For $v \in U$, let $\nu_K(v)$ be the number of maximal cubes of K through v , and let $\rho_{\mathcal{R}}(v)$ and $\rho_{\overline{\mathcal{R}}}(v)$ count the noncommon triple cubes through v on the two sides of the component cut. The corresponding lift is cornerless if and only if

$$\nu_K(v) + \rho_{\mathcal{R}}(v) \neq 1, \quad \nu_K(v) + \rho_{\overline{\mathcal{R}}}(v) \neq 1 \quad (v \in U). \quad (7.37)$$

Equivalently, every corner of U belongs to at least two maximal cubes of K , and, for every corner v of K , the component set

$$\Gamma_K(v) = \{B \in \mathcal{B} : v \in C_T \text{ for some } T \in B\}$$

meets both sides of the cut.

Proof. Every support in $\mathcal{D}$ lies below at least one triple support. All triples above it belong to $\mathcal{N}$, by downward closure of Σ , and the second kind of graph edge puts them in one component. Thus the $\mathcal{D}_B$ partition $\mathcal{D}$. Every vertex of the maximum class U lies in a triple cube. If a vertex outside K belongs to cubes with two noncommon supports, the first kind of graph edge joins those supports. It follows that the sets V_B partition $U \setminus K$.

Both U and K are ample. Since the supports missing from Σ are exactly $\mathcal{N} \dot{\cup} \mathcal{D}$, the two partitions first give

$$\sum_{B \in \mathcal{B}} |V_B| = |U| - |K| = |\text{Sh}(U)| - |\Sigma| = |\mathcal{N}| + |\mathcal{D}| = \sum_{B \in \mathcal{B}} (|B| + |\mathcal{D}_B|),$$

and hence $\sum_B \omega_K(B) = 0$.

Fix one component B . The families $P_{\{B\}}$ and $Q_{\{B\}}$ strongly shatter, respectively, at least the supports

$$\Sigma \dot{\cup} B \dot{\cup} \mathcal{D}_B \quad \text{and} \quad \Sigma \dot{\cup} (\mathcal{N} \setminus B) \dot{\cup} (\mathcal{D} \setminus \mathcal{D}_B).$$

Their vertex sets are $K \dot{\cup} V_B$ and $K \dot{\cup} \bigcup_{B' \neq B} V_{B'}$. The lower Sandwich inequality therefore gives

$$\omega_K(B) \geq 0 \quad \text{and} \quad -\omega_K(B) \geq 0.$$

Thus (7.36) holds component by component. This two-sided argument is the point at which ampleness of the overlap eliminates every possible cancellation between exterior components.

The exterior-incidence edges give $P_{\mathcal{R}} \cup Q_{\mathcal{R}} = U$ and $P_{\mathcal{R}} \cap Q_{\mathcal{R}} = K$. The positioned cubes contained in $P_{\mathcal{R}}$ strongly shatter at least

$$\Sigma \cup \bigcup_{B \in \mathcal{R}} B \cup \bigcup_{B \in \mathcal{R}} \mathcal{D}_B,$$

and the complementary family of supports is strongly shattered by $Q_{\mathcal{R}}$. Componentwise balance gives

$$|P_{\mathcal{R}}| = |\Sigma| + \sum_{B \in \mathcal{R}} (|B| + |\mathcal{D}_B|),$$

and the complementary equality holds for $Q_{\mathcal{R}}$. Equality in the lower Sandwich inequality now proves that both families are ample. The corresponding two-layer family has order $|U| + |K|$ and strongly shatters the disjoint support families

$$\text{Sh}(U) \quad \text{and} \quad \{A \cup \{e_0\} : A \in \Sigma\},$$

where e_0 is the new coordinate. These have total order $|U| + |K|$, so the lower Sandwich equality proves that the lift is ample as well.

Conversely, let two ample sections have union U and intersection K . The exact shadow intersection and union identities put every triple in Σ in both sections and every triple in $\mathcal{N}$ in exactly one. Every missing lower support $A \in \mathcal{D}$ and its entire upper star have one owner. Triple cubes meeting outside K also have one owner. Thus the owner classes are unions of the components in $\mathcal{B}$, and the two sections are the displayed $P_{\mathcal{R}}, Q_{\mathcal{R}}$.

Finally, the maximal cubes of the lift are the horizontal noncommon triple cubes and the products of the new-coordinate edge with the maximal cubes of K . Hence the two copies of v have maximal-cube incidence multiplicities shown in (7.37); multiplicity zero means that the copy is absent. At a corner of U the condition reduces to $\nu_K(v) \geq 2$, and at a corner of K it requires a noncommon triple from each side. Every remaining present vertex already has incidence at least two, proving the last equivalence. $\square$

Corollary 7.39 (Property-B criterion for arbitrary overlaps). *In the setting of Proposition 7.38, form the hypergraph $\mathcal{H}_K$ on $\mathcal{B}$ whose hyperedges are the sets $\Gamma_K(v)$ over the corners v of K . There exists a cornerless ample two-layer lift with contraction U and overlap K if and only if*

1. *every corner of U belongs to at least two maximal cubes of K ; and*
2. *$\mathcal{H}_K$ has Property B, that is, its vertices admit a two-colouring with no monochromatic hyperedge.*

No weight or section-ameness condition remains to be checked. If the lift is a minimum-cardinality ample non-corner-peelable family, then K is corner-peelable and has a corner; consequently the colouring uses at least two exterior components.

Proof. Every component cut gives ample sections by the proposition. Its exact relative-corner criterion says that the two copies of a corner of K are shielded precisely when its hyperedge meets both colours, while a corner of U is shielded precisely when it lies in at least two maximal cubes of K . This proves the equivalence. In the minimum-cardinality case, K is an ample family of smaller order. It must therefore be corner-peelable, since otherwise it would itself be a smaller ample non-corner-peelable family. Its first vertex in a corner peeling is a corner, so $\mathcal{H}_K$ has a nonempty hyperedge; Property B then forces both colours. $\square$

Proposition 7.40 (All-rank corner localization in Hall projections). *Let U be an eleven-coordinate contraction of Hall's maximum class, let $K \subseteq U$ be a non-singleton ample family, and let v be a corner of K . Write C for its unique maximal cube and A for the support of C ; thus $1 \leq |A| \leq 3$. Form a graph $L_U(A, v)$ on the triple supports T for which $v \in C_T$, omitting $T = A$ when $|A| = 3$. Join T, T' when, for some $e \notin A$, both cubes contain the neighbour $v \triangle \{e\}$. Then*

$$|\Gamma_K(v)| \leq \text{cc}(L_U(A, v)) \leq \begin{cases} 1, & |A| = 1, \\ 2, & |A| \in \{2, 3\}. \end{cases} \quad (7.38)$$

Consequently a cornerless two-layer lift has no corner of K supported by a maximal edge. Every corner supported by a maximal square or a common triple cube is one of the listed two-component incidences, and its two local components have opposite owners. In particular, every hyperedge of $\mathcal{H}_K$ has order two, and the owner constraints form a bipartite graph on the exterior components.

Across all twelve contractions, the audit covers 14,784 positioned edge–vertex incidences, 26,400 square–vertex incidences, and 15,840 triple–cube–vertex incidences. The respective numbers with two local components are 0, 480, and 184; no local graph has more than the bound in (7.38). In the coordinate-zero contraction there are forty binary square incidences and fifteen binary triple incidences.

Proof with a finite local audit. Because C is the unique maximal cube of K through v , every neighbour $v \triangle \{e\} \in U$ with $e \notin A$ lies outside K . Therefore all triple cubes containing such a neighbour belong to one exterior component. Each component of $L_U(A, v)$ is consequently contained in one member of $\Gamma_K(v)$, which proves the first inequality in (7.38). The arbitrary-overlap normal form, Proposition 7.38, then shows that cornerlessness requires at least two members of $\Gamma_K(v)$.

The remaining statement is a fixed local computation, independent of the order of K . The verifier constructs every positioned edge, square, and triple cube in every Hall projection, forms $L_U(A, v)$ for every incidence, and computes its connected components directly. It records the complete distributions and exceptional support lists in `damp_hall_all_corner_localization.json`. The mathematical digest is

c32cc3bbdc908fe80c5886762d64f3c58b2e69d9dea58a21a730fdeeb9760ddf.

The rank-one bound rules out maximal-edge corners. In ranks two and three, Property B rules out local component count at most one and forces the two components at every remaining corner to have opposite colours. Hence the corner hypergraph is a graph, and its Property-B colourings are exactly its bipartite two-colourings, unique up to reversal on each connected component. $\square$

Proposition 7.41 (Incidence separators and signed holonomy). *Let U be a Hall projection on coordinate set E . Form the bipartite incidence graph $\mathfrak{I}_U$ with one class $\binom{E}{3}$ and the other class*

$$U \dot{\cup} \binom{E}{1} \dot{\cup} \binom{E}{2}.$$

A triple support T is adjacent to $x \in U$ when $x \in C_T$, and it is adjacent to a singleton or pair A when $A \subseteq T$. For an ample overlap K with shadow Σ , delete from $\mathfrak{I}_U$

$$\{T \in \binom{E}{3} : T \in \Sigma\} \dot{\cup} K \dot{\cup} \{A \in \binom{E}{1} \cup \binom{E}{2} : A \in \Sigma\}. \quad (7.39)$$

The connected components of the surviving triple nodes are exactly the exterior components $\mathcal{B}$ of Proposition 7.38. Moreover, the deletion set in (7.39) has order

$$|K| + |\Sigma \setminus \{\emptyset\}| = 2|K| - 1. \quad (7.40)$$

Suppose now that every corner of K is one of the two-component incidences in Proposition 7.40. Give all paths in the residual incidence graph label zero, and, for each corner, add one label-one edge between representatives of its two local classes. Then the corner-component graph is bipartite if and only if this signed graph is balanced, that is, every cycle contains an even number of label-one edges. Equivalently, the affine equations

$$z_T + z_{T'} = 0 \quad \text{on residual incidence paths,} \quad z_T + z_{T'} = 1 \quad \text{at overlap corners}$$

have a solution over $\mathbb{F}_2$.

For the coordinate-zero Hall projection, the minimum unit vertex-separator widths in $\mathfrak{J}_U$ between the two local classes of the fifty-five possible binary corners have distribution

$$13^{(10)}, \quad 14^{(43)}, \quad 21^{(2)}. \quad (7.41)$$

Here terminal triple nodes are undeletable, while every other triple, projection-vertex, singleton, and pair node has unit capacity. If one additionally imposes the cube-closure condition that the eight vertices of a terminal triple cube cannot all be deleted while its terminal triple node survives, the distribution becomes

$$13^{(10)}, \quad 14^{(42)}, \quad 16^{(1)}, \quad 22^{(2)}. \quad (7.42)$$

Proof. A surviving projection-vertex node joins precisely the noncommon triple cubes that meet at a vertex outside K . A surviving singleton or pair node joins precisely the noncommon upper star of a support missing from Σ . Taking the two-section of the residual bipartite graph therefore gives the two kinds of edges in the definition of $\mathcal{B}$, proving the first claim. The three parts of the deletion set are disjoint and together contain one node for every member of K and one node for every nonempty member of Σ . Since K is ample, $|K| = |\Sigma|$, which proves (7.40).

A zero-labelled connected component must have one owner. A corner asks its two incident components to have opposite owners, and hence contributes the displayed affine equation of label one. Such an affine system on a graph is solvable exactly when the sum of its edge labels on every cycle is zero: necessity follows by summing the vertex equations around the cycle, and sufficiency follows by propagating a value from one root in each connected component. This proves the signed-holonomy criterion.

The last statement is a fixed max-flow/min-cut computation in the ambient incidence graph and is independent of K and of its order. The verifier splits every nonterminal node into an in-node and an out-node of capacity one, computes a maximum flow for each of the fifty-five terminal pairs, and checks the corresponding cut directly. It imposes terminal-cube closure by branching on a vertex retained from every terminal cube completed by a candidate cut. The complete cuts are recorded in `damp_hall_binary_separator_cuts.json`; their mathematical digest is

d3daab1128456dc68c5f356f93f696c1618275373cc7a963f74dada147195827.

□

Remark 7.42 (The remaining Hall separator inequality). Fix the coordinate-zero Hall contraction. Componentwise balance makes section ampleness automatic, and the proposition reduces every overlap corner to one of fifty-five binary local incidences. Thus the common-triple branch is the following single problem: can the deletion set (7.39) of an ample overlap K of order at most 58 double-shield both contraction corners and leave a balanced signed incidence graph? By Menger's theorem, each binary corner requires one of the separator packages in (7.41). The missing step is an isoperimetric transfer from these ambient blocker packages to the coupled identity $|K| = |\Sigma|$: prove that double shielding and balanced holonomy force $|\Sigma| \geq 59$. This formulation ranges over all overlap orders at once and contains neither section variables nor a cardinality-by-cardinality census.

Proposition 7.43 (A 298-concept ample obstruction). *There is an ample forbidden pc-minor for corner peelability with 298 concepts. Consequently,*

$$m_{\text{amp}} \leq 298.$$

Proof. Use the coordinate-zero contraction U of Hall's class. There is an ample overlap K of order 66 whose shadow is the full rank-two shadow except for the pair $\{6, 9\}$. Its exterior graph has two components. The distinguished component contains 95 triple cubes and 96 exterior vertices, while the other contains 70 triple cubes and 70 exterior vertices. Thus both component weights are zero. Assigning the two components to opposite sections gives section orders

$$|P| = 66 + 96 = 162, \quad |Q| = 66 + 70 = 136.$$

The two contraction corners 1600 and 1894 each lie in three maximal pair cubes of K . The three corners of K , namely

$$412, \quad 1025, \quad 2039,$$

are each incident with both exterior components. The arbitrary-overlap Property-B criterion therefore shows that

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

is ample and cornerless, and $|Y| = 162 + 136 = 298$.

The exact section lists are recorded in `damp_hall_overlap_component_c0_cnf.json`; their mathematical digest is

c9a32cff523db600553d42ef6a288d986be49bd2e6aa7df78d50625d03b575d3.

The independent verifier `verify_damp_hall_order298_obstruction.py` reconstructs Y from those lists, recomputes its shattered sets and corners, and constructs a complete corner peeling for each of its 24 restrictions and 12 contractions. Its mathematical digest is

bfa75ccdb16a493fa3fc805fb5516d80f3a060edaa14051b57935f927d472772.

Every proper pc-minor factors through one of these elementary minors, and corner peelability is pc-minor closed. Hence every proper pc-minor of Y is corner-peelable, whereas Y itself has no first corner. Thus Y is an ample forbidden pc-minor. $\square$

Corollary 7.44 (Owner-optimality of the 298-concept lift). *Fix the coordinate-zero Hall contraction and the ownership of its maximal triple cubes used in Proposition 7.43. Every cornerless ample two-layer lift realizing this ownership has at least 298 concepts.*

Proof. For this ownership, the bichromatic core has 59 vertices. Adjoining the two contraction corners gives a forced core F of order 61, and a direct shadow calculation gives

$$|\text{Sh}(F)| = 66.$$

The contraction has order 232, so Corollary 7.28 gives

$$|Y| \geq 232 + 66 = 298.$$

The verifier `verify_damp_hall_bichromatic_core.py` performs the shadow calculation directly from the owner list; its mathematical digest is

81fa19a62fb7e653593621f72048b59e961336f7be360c39371e6a19d8f2bf29.

Proposition 7.43 attains equality. $\square$

Proposition 7.45 (A 291-concept ample obstruction). *There is an ample forbidden pc-minor for corner peelability with 291 concepts. Consequently,*

$$m_{\text{amp}} \leq 291.$$

Proof. Again let U be the coordinate-zero contraction of Hall's class. There is an ample overlap $K \subseteq U$ of order 59 whose shadow contains every singleton and all pairs except

$$\begin{aligned} &\{3, 7\}, \{4, 7\}, \{3, 8\}, \{4, 8\}, \\ &\{5, 9\}, \{6, 9\}, \{5, 10\}, \{6, 10\}. \end{aligned}$$

Thus the missing-pair graph is the disjoint union of the two four-cycles $K_{\{3,4\},\{7,8\}}$ and $K_{\{5,6\},\{9,10\}}$. Assigning 139 of the 165 maximal three-cubes of U to the first section gives sections P, Q of orders

$$|P| = 206, \quad |Q| = 85.$$

Their intersection is K , both sections are ample, and their union is U . The two corners of U belong to K , while the three corners 412, 1025, 2039 of K all belong to the bichromatic core. The exact relative-corner criterion therefore shows that the lift

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

is ample and cornerless, with $|Y| = 206 + 85 = 291$.

The overlap, owner set, and reconstructed sections are recorded in `damp_hall_carrier_defect_min8.json`. The generating formula uses one variable for the missing status of each pair and one choice among the eleven vertices of its Hall carrier path. Hence its single threshold constraint covers all full-singleton, square-pure overlaps in this carrier model with at least eight missing pairs; it is not an order-by-order enumeration. The stored witness is checked directly by `solve_damp_hall_carrier_defect_cnf.py`; its mathematical digest is

f2ea815743a1d96b8ac3cdc00efb0ab6dd80777483dd38ebe91c972df4184ffa.

The two four-cycles are coupled rather than independent. The complete 3^{11} -conditioning audit compares the mixed cores of the certified orders 298, 297, 295, 291, whose orders are respectively 59, 56, 52, 42. All four have exactly the same unique conditioning-minimal nonample trace. After fixing

$$x_0 = 1, \quad x_3 = x_4 = 0, \quad x_9 = x_{10} = 1,$$

and leaving $\{1, 2, 5, 6, 7, 8\}$ free, this trace is the antipodal pair $\{0^6, 1^6\}$. Thus one coherent rank-six antipodal bridge persists while the missing-pair graph grows from one edge to two edges, then four, then eight. In the last stage it spans the two defect cycles and carries zero relative-corner charge. The audit is recorded in `damp_hall_mixed_core_minimal_nonample_traces.json`, with mathematical digest

b584060c83d61722762f0c9f785f7ff5a6d8ee0a62c99c60ca1407694c38e9e1.

The companion comparison `analyze_damp_hall_bridge_defect_chain.py` verifies that the missing pair sets are strictly nested, with orders 1, 2, 4, 8, and have successive shapes

$$K_2, \quad P_3, \quad P_3 \dot{\cup} P_3, \quad C_4 \dot{\cup} C_4.$$

The corresponding mixed-core orders are 59, 56, 52, 42. Its artifact `damp_hall_bridge_preserving_defect_chain.json` has mathematical digest

`acfb63e3dc1c42196847a47315a7b98e90c3a753836cfe3d6c9d67d35d1faa7.`

The independent verifier `verify_damp_hall_carrier_defect_witness.py` reconstructs Y from the overlap and owner list, recomputes ampleness and cornerlessness, and constructs a complete corner peeling for each of its 24 restrictions and 12 contractions. The verification artifact `damp_hall_carrier_defect_min8_verification.json` has mathematical digest

`681a0bc8ba1584ad7732783da3526ae5e6eb11271f7216caa1fe6ee4254b0f56.`

Thus every elementary pc-minor of Y is corner-peelable. Every proper pc-minor factors through an elementary one, whereas Y itself has no corner, so Y is an ample forbidden pc-minor. $\square$

Proposition 7.46 (Exact capacity of the rank-six Hall bridge). *Let U be the coordinate-zero Hall contraction, and let*

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

be a cornerless ample lift with overlap $K = P \cap Q$. Suppose that K shatters all singletons, has VC dimension at most two, and is square-pure. Let M be the bichromatic core, and condition on

$$x_0 = 1, \quad x_3 = x_4 = 0, \quad x_9 = x_{10} = 1.$$

If the resulting trace of M on the free coordinates $\{1, 2, 5, 6, 7, 8\}$ is exactly $\{0^6, 1^6\}$, then K misses at most eight pairs. Consequently

$$|Y| \geq 291.$$

The bound is sharp.

Proof. Let m be the number of pairs not shattered by K . Ampleness and the full singleton shadow give

$$|K| = 1 + 11 + \binom{11}{2} - m = 67 - m, \quad |Y| = |U| + |K| = 299 - m.$$

For completeness, square-purity also implies that every shattered pair has exactly one maximal square. Indeed, summing $|E(T_i(K))| = |V(T_i(K))| - 1$ over the eleven singleton carrier trees and using the cubical Euler identity gives that the number of maximal squares is the number of shattered pairs.

The formula in `solve_damp_hall_carrier_defect_cnf.py` is therefore exact for this class. It chooses the unique positioned square for every shattered pair and the owner of every maximal three-cube of U . Proposition 7.34 gives the one-transition condition on every Hall pair carrier. The three Sandwich equalities impose ampleness of K, P, Q , and Proposition 7.25 imposes cornerlessness. Finally, unit clauses prescribe the displayed trace of M .

With $m \geq 9$, the resulting CNF has 15,032 variables and 80,469 clauses and is unsatisfiable. CaDiCaL generated a 440,994-line DRAT proof. The independent checker `drat-trim` returned VERIFIED; its backward core uses 9,696 original clauses, 32,496 lemmas, and 1,792,976 resolution steps. The formula, proof, checker hashes, and verification statistics are recorded in `damp_hall_rank6_bridge_capacity_proof_verification.json`, whose mathematical digest is

`db9ba53d639a8e1b87fa80e5d5f2133c51ebb08f99cedd224fcba96049cbd799.`

Thus $m \leq 8$, and the displayed order identity gives $|Y| \geq 291$. The construction in Proposition 7.45 has the prescribed rank-six trace and $m = 8$, so equality is attained. $\square$

Corollary 7.47 (Capacity of the observed Hall bridge types). *In the full-singleton square-pure Hall carrier model, suppose that the mixed core has one of the five exact conditioning-minimal antipodal traces appearing in the forced-shadow-61, forced-shadow-55, forced-shadow-46, or certified order-298 owner cores. Then $m \leq 8$, and hence*

$$|Y| \geq 291.$$

Proof. Besides the common rank-six trace of Proposition 7.46, the complete conditioning audit gives two adjacent rank-four traces for the forced-shadow-61 core, one rank-five trace for the forced-shadow-55 core, and one rank-four trace for the forced-shadow-46 core. Prescribing each trace separately and imposing $m \geq 9$ gives four further unsatisfiable carrier formulas. Independent **drat-trim** checks return VERIFIED for all four proof logs. Together with the rank-six check, the aggregate verification covers all five observed bridge signatures. It is recorded in `damp_hall_observed_bridge_capacity_proof_verification.json`, with mathematical digest

a2174b2e8107e2912eae634dbc916381202fba62ea3561c89b66edf15d20a4f0.

□

Remark 7.48. Corollary 7.47 is a finite classification of the bridge signatures that occur in the present Hall owner-core artifacts. It does not assert that every possible Hall owner colouring has one of these five traces. The next theorem handles the complementary owner colourings directly, without making such an assertion.

Theorem 7.49 (Exact optimum in the Hall carrier model). *Let U be the coordinate-zero Hall contraction, and let*

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

be a cornerless ample lift. If $K = P \cap Q$ shatters all singletons, has VC dimension at most two, and is square-pure, then

$$|Y| \geq 291.$$

This bound is sharp.

Proof. As in Proposition 7.46, write m for the number of missing pairs. It is enough to exclude $m \geq 9$. Proposition 7.24 shows that the mixed core M is nonample.

Partition the owner colourings according to whether M has one of the five exact antipodal traces in Corollary 7.47. The five independently checked formulas cited there exclude $m \geq 9$ in the first part of the partition. In the residual part, compile the nonampleness of M by the strict Sandwich inequality

$$|\text{Sh}(M)| \geq |M| + 1$$

and forbid all five exact traces. This residual formula has 19,229 variables and 150,552 clauses and is unsatisfiable. Lingeling generated the DRAT proof, and an independent **drat-trim** check returned VERIFIED. Its backward core contains 21,951 original clauses and 131,531 lemmas, using 25,223,694 resolution steps.

The five profile certificates and the residual certificate form an exhaustive logical partition of the exact carrier formula. Their hashes, checker statistics, and the sharp $m = 8$ witness are collected in `damp_hall_full_carrier_capacity_proof_verification.json`, with mathematical digest

5758b4f9df00f6e627cdcb172e32ee15e0e812b735e2d616f1b67bb6d53b7993.

Thus $m \leq 8$, and $|Y| = 299 - m \geq 291$. Sharpness follows from Proposition 7.45. □

Lemma 7.50 (Cubical normal form in VC dimension two). *Let K be a nonempty ample family of VC dimension at most two. Every shattered pair supports exactly one square of K . Moreover, K has a maximal edge of support $\{i\}$ if and only if $\{i\}$ is shattered and i is isolated in the graph of shattered pairs; in that case the maximal i -edge is unique. Finally, K has a maximal vertex if and only if $|K| = 1$.*

Proof. Fix a shattered pair S . Apply Proposition 5.13 to K and S . Since K shatters no triple, the set L_S in that proposition is empty. The S -carrier therefore consists of one vertex, which is exactly one square of support S .

Now fix a shattered singleton $\{i\}$, and form the i -carrier by contracting each full i -edge to a vertex. The argument in the proof of Proposition 5.13, with $S = \{i\}$, shows that this carrier is an ample VC-dimension-one family: its shattered coordinates are precisely the j for which $\{i, j\}$ is shattered by K . Hence its one-inclusion graph is a tree, with one vertex for each i -edge of K and one edge for each square whose support contains i .

An i -edge of K is maximal precisely when the corresponding carrier vertex is isolated. A tree has an isolated vertex precisely when it consists of that single vertex. Thus a maximal i -edge exists precisely when no shattered pair contains i , and in that case it is unique. Finally, the one-inclusion graph of an ample family is connected. A maximal zero-cube is an isolated vertex of that graph, so it can occur precisely when K is a singleton. $\square$

Theorem 7.51 (Exact Hall optimum for every VC-two overlap). *Let U be the coordinate-zero Hall contraction and let*

$$Y = (P \times \{0\}) \cup (Q \times \{1\})$$

be a cornerless ample lift. If $K = P \cap Q$ has VC dimension at most two, then

$$|Y| \geq 291.$$

This bound is sharp.

Computer-assisted proof. Theorem 7.49 handles the case in which K shatters every singleton and every maximal cube of K is a square. By Lemma 7.50, every other VC-two overlap either misses a singleton or has an isolated shattered coordinate with one maximal edge. These alternatives form an exact structural partition, not a partition by order.

The all-profile formula in `solve_damp_hall_overlap65_cnf.py` chooses K , every triple owner, and both sections simultaneously. It imposes the three Sandwich equalities, the exact relative-corner criterion, the nonampleness of the bichromatic core, and the cubical normal form of Lemma 7.50. Requiring one of the two exceptional alternatives and $|K| \leq 58$, equivalently $|Y| \leq 290$, gives a formula with 26,406 variables and 250,636 clauses. It is unsatisfiable. Lingeling generated a DRAT proof, and the independent checker `drat-trim` returned VERIFIED. Its backward core uses 13,448 original clauses and 3,102 lemmas, with 262,558 resolution steps.

The formula, proof, hashes, and checker statistics are recorded in `damp_hall_vc2_normal_form_capacity_proof_verification.json`, whose mathematical digest is

6d0ef7268aa38ca55a2a19ce16f9f551f95420cb050fe3a39e3547649ff6f0b0.

Thus no exceptional VC-two overlap has order at most 58. Together with Theorem 7.49, this gives $|K| \geq 59$, and hence $|Y| = |U| + |K| \geq 232 + 59 = 291$. The construction in Proposition 7.45 attains equality. $\square$

Remark 7.52. Theorem 7.51 closes every Hall lift whose overlap has VC dimension at most two, including missing-singleton and non-pure overlaps. A smaller lift over this Hall contraction must therefore have a common triple cube, equivalently an overlap of VC dimension three. The theorem does not normalize an arbitrary least obstruction to a Hall two-layer lift.

The remaining VC-three overlaps are excluded by a direct search over all two-layer lifts, using the facet-cube description of ample families. The same search applies to many further bases near Hall’s class.

Theorem 7.53 (Hall lifts with arbitrary overlaps; computer-assisted). *Let B be one of the following ample families:*

1. *the contraction of Hall’s class C_H at any one of its twelve coordinates;*
2. *the contraction of C_H at any two of its coordinates;*
3. *the contraction, at any one coordinate, of any of the twenty-seven cornerless order-299 classes of Remark 7.3.*

If $Y = (P \times \{0\}) \cup (Q \times \{1\})$ is a cornerless ample family with $P \cup Q = B$, then $|Y| \geq 291$. For the coordinate-zero contraction of C_H , the bound is attained by Proposition 7.45.

Computer-assisted proof. Every listed base uses all of its coordinates and has a corner; these finite properties are checked directly by `scripts/verify_hall_lift_audit_inputs.py`. Since the projection of Y is B , all base coordinates remain active in Y . The added coordinate is active as well: otherwise Y is a copy of B in a single layer and inherits a corner, contrary to the hypothesis. Thus forcing every singleton into the encoded shattering complex loses no possible cornerless lift.

Every vertex of such a lift Y has the form $(x, 0)$ or $(x, 1)$ with $x \in B$. By Lemma 7.100, Y is the union of one cube per facet of $\text{Sh}(Y)$, every such cube lies inside $B \times \{0, 1\}$, and cornerlessness says that every vertex of Y lies in at least two of these cubes. The formula has a variable for each coordinate set, saying that it belongs to a down-closed complex Σ . It has one cube-choice variable for each facet of Σ and each cube inside $B \times \{0, 1\}$, and a covering variable for each of the $2|B|$ candidate vertices. It requires that the union shatters no set outside Σ , that every covered vertex is covered twice, that for every $x \in B$ at least one of $(x, 0), (x, 1)$ is covered, and that at most 290 vertices are covered. By the converse half of Lemma 7.100, its models are exactly the cornerless ample lifts of B of order at most 290.

The $12 + 66 + 324 = 402$ labeled cases, comprising 272 distinct formulas, are generated by `explore_damp_lift_cornerless_sat.py` with `--max-order=290`. For every one of them, CaDiCaL 2.1.2 reports UNSAT, and its DRAT proof is accepted by `drat-trim`. The same formula for the coordinate-zero contraction with bound 291 is satisfiable. Its decoded model is a cornerless ample family of order 291, which recovers the equality case independently. The fresh replay retains all 272 formulas and checked proofs, together with input maps, stage logs and tool identities, in the split archive `evidence/hall_lift_optimum/replay-20260924/`. Its manifest records the part and archive hashes; the atlas reconciliation records every payload hash and the coverage of all 402 labeled bases. $\square$

Remark 7.54 (Scope of the lift bound). Theorem 7.53 removes the overlap hypothesis of Theorem 7.51 and extends the bound from one base to 402 bases. It is still a statement about two-layer lifts of these bases. Cornerless ample families below order 291 exist (Proposition 7.55 below), but none of them is a one-coordinate lift of any of these bases. Each single contraction of C_H has exactly two corners. Its double contractions have two, three, or four corners, and exactly six of them have

two: those over the coordinate pairs $\{0, 1\}, \{2, 3\}, \dots, \{10, 11\}$, which form a perfect matching of the coordinates. In the other direction, Proposition 7.101 requires at least eight coordinates for any cornerless ample family.

The same step can be iterated: contract a cornerless family at one coordinate and search for a cornerless ample lift of least order of the contraction. Starting from the 291-concept lift of the coordinate-zero contraction, three further steps, each contracting at coordinate zero, give cornerless ample families of orders 283, 275 and 267. None of these lifts contains the previous family.

Proposition 7.55 (A 267-concept ample obstruction; computer-assisted). *There is an ample forbidden pc-minor for corner peelability with 267 concepts on twelve coordinates, of VC dimension three. Consequently*

$$m_{\text{amp}} \leq 267.$$

Computer-assisted proof. The family $Y_{267} \subseteq Q_{12}$ is listed in `evidence/order267_obstruction/record.json`, with SHA-256 digest

45eebc6ef09bf97ffa814295b621bb09d9788f87258031ffc25c5181f879573c.

It was found by the search just described, but its properties are checked directly, without a SAT solver, by `./ample-corners/scripts/verify_ample_corners_data.py`. First, $|\text{Sh}(Y_{267})| = |\text{SSh}(Y_{267})| = |Y_{267}| = 267$, so Y_{267} is ample by the Sandwich equality, and all twelve coordinates are active. Second, no vertex lies in a unique maximal cube, so $Y_{267} \notin \mathcal{D}_{\text{amp}}$. Third, each of the 36 elementary minors $\rho_e^0 Y_{267}$, $\rho_e^1 Y_{267}$ and $\pi_e Y_{267}$, with inactive coordinates removed, is ample and is emptied by repeatedly deleting a corner, so it lies in $\mathcal{D}_{\text{amp}}$. The contractions all have 208 concepts, and the restrictions have between 77 and 190. By the recognition criterion (1.1), Y_{267} is an ample forbidden pc-minor. $\square$

A single contract-and-lift step does not go below 267 from the families found so far. For every single and double contraction of the families of order at most 275 found in this search, 573 distinct bases in all, CaDiCaL reports that no cornerless ample lift of order at most 266 exists. These runs kept no proofs, and we make no claim based on them.

Remark 7.56 (A relative-corner obstruction below the shadow threshold). The owner-only shadow bound is not uniformly 66. An exact all-owner formula finds a colouring with

$$|M| = 44, \quad |F| = 46, \quad |\text{Sh}(F)| = 61.$$

For this fixed colouring, there is an ample overlap K of order 62 whose three corners consist of two vertices of M and one vertex outside M . The induced sections are ample, and their lift has order 294 and exactly one corner. Thus neither raw shadow size nor ample completion alone controls cornerlessness.

Remark 6.47 verifies that this lift is corner-peelable, although no peeling can leave its unique corner last.

The relative-corner criterion gives the exact obstruction. Since an ample VC-dimension-two family in Q_{11} has order at most 67, one formula with upper bound 67 covers every possible overlap. For the colouring above, this formula is unsatisfiable when $\text{Cor}(K) \subseteq M$. Its DRAT certificate was independently checked by `drat-trim`; the reduced certificate contains 10,685 original clauses and 550,770 proof lines. Together with the order-62 witness, this proves that the minimum number of corners of K outside M is exactly one.

The corner defect is visible along a nested sequence

$$K_{62} \subset K_{65} \subset K_{66} \subset K_{67}.$$

At every stage the overlap and both induced sections are ample, while the unique corner outside M moves along

$$1894 \xrightarrow{7} 2022 \xrightarrow{4} 2038 \xrightarrow{3} 2046.$$

Its unique maximal-square support is, respectively,

$$\{0, 8\}, \quad \{0, 7\}, \quad \{0, 4\}, \quad \{0, 3\}.$$

Thus repairing one endpoint exposes the next one along a singleton carrier. The last overlap is maximum of VC dimension two. This nested chain is verified by `analyze_damp_hall_relative_corner_transport.py`; the corresponding artifact has mathematical digest

$$5f25fec458bd3750d63cd5c3c8678cc7387b3980e635614d31cb479d96c37a0b.$$

The witness, proof-verification record, and generating program are, respectively,

- `damp_hall_forced_core_relative_defect1_f61_o67.json`,
- `damp_hall_forced_core_relative_completion_f61_o67_proof_verification.json`, and
- `analyze_damp_hall_forced_core_completion.py`.

Their mathematical and verification digests are recorded in the accompanying campaign ledger. This computation excludes one complete owner-colouring class at every overlap order; it does not establish the corresponding all-owner bound.

Proposition 7.57 (Two-ended core-corner surcharge). *Use the maximum core A , the repair set R , and the sections P_a of Proposition 7.4, and put $r = |R|$. If H is cornerless, then*

$$|P_a| \geq r + 2 \quad \text{for every } a \in \text{corn}(A). \quad (7.43)$$

Consequently

$$\Delta(H; A, R) \geq |\text{corn}(A)|. \quad (7.44)$$

If $r \geq 3$, then the stronger bounds

$$|P_a| \geq r + 3 \quad \text{for every } a \in \text{corn}(A), \quad \Delta(H; A, R) \geq 2|\text{corn}(A)| \quad (7.45)$$

hold. Both charges are independent of the number of repair coordinates.

Proof. Fix a corner a of A and let $Q_A(a)$ be its unique maximal cube, with support S . The copies of $Q_A(a)$ in the opposite fibres H_f and $H_{\bar{f}}$ give two vertices of the S -cube carrier of H . That carrier is ample and isometric, by the carrier argument in the proof of Proposition 5.13. The two carrier vertices differ precisely on the coordinates of R . A shortest path between them therefore has length r and changes no old coordinate. In other words, it supplies an antipodal geodesic $P \subseteq Q_R$ such that

$$Q_A(a) \subseteq H_g \quad (g \in P). \quad (7.46)$$

In particular, $P \subseteq P_a$ and $|P| = r + 1$.

Suppose that $|P_a| = r + 1$. Then $P_a = P$ is itself this geodesic. At the copy (f, a) , the repair section has exactly one incident direction, namely the first direction of P . The old incident directions are exactly the directions of $Q_A(a)$, because a is a corner of A . Equation (7.46) says that all these directions together generate the cube $Q_A(a)$ times the first edge of P . The local corner criterion therefore makes (f, a) a corner of H , a contradiction. This proves (7.43).

Now suppose that $r \geq 3$ and $|P_a| = r + 2$. Since the carrier geodesic P already has $r + 1$ vertices, write $P_a = P \cup \{w\}$. If w is not adjacent to f , then the first edge of P is again the only repair edge at (f, a) , and the preceding corner argument applies unchanged. Thus cornerlessness forces $d_{Q_R}(w, f) = 1$. Applying the same argument at the other endpoint forces $d_{Q_R}(w, \bar{f}) = 1$. The triangle inequality would then give

$$r = d_{Q_R}(f, \bar{f}) \leq 2,$$

contrary to $r \geq 3$. Hence $|P_a| \geq r + 3$.

Finally sum the excess $|P_a| - (r + 1)$ over the core corners and use the exact decomposition (7.7). All other terms in that decomposition are nonnegative. This proves (7.44), and, when $r \geq 3$, summing the two units supplied by each core corner proves (7.45). $\square$

Proposition 7.58 (VC-one path-core propagation surcharge). *Use the maximum core A , repair set R , and sections P_a from Proposition 7.4. Suppose that A has VC dimension one and uses $n \geq 2$ active coordinates. Thus its one-inclusion graph is a path*

$$a_0 a_1 \cdots a_n.$$

If H is cornerless and $r = |R| \geq 3$, then

$$\Delta(H; A, R) \geq \begin{cases} 6, & n = 2, \\ 7, & n \geq 3. \end{cases} \quad (7.47)$$

Proof. For an endpoint $a \in \{a_0, a_n\}$, let b be its unique neighbour in A and put

$$C = P_a \cap P_b.$$

Fixing all core coordinates except the coordinate of the edge ab produces an ample two-layer conditioning of H with sections P_a and P_b . Lemma 7.22 therefore makes C ample.

By Proposition 7.57, the section excess

$$e_a = |P_a| - (r + 1)$$

is at least two. Suppose first that equality holds. Since P_a is ample and contains two antipodes, its shadow already contains the empty set and all r singletons. The equality $|P_a| = r + 3$ leaves exactly two further shattered supports. No support of size at least three can occur, because it and its three two-element subsets would already give four further supports. Hence $\text{VCdim}(P_a) \leq 2$.

If $C \subsetneq P_a$, relative peeling in VC dimension two (Lemma A.16) supplies a corner $w \in P_a \setminus C$. If $C = P_a$, take any corner w of P_a . Let K be the unique maximal cube of P_a through w . In the first case the core edge ab is absent over w ; in the second case $K \times ab$ is present. Because a is an endpoint of the core path, these are all incident core directions belonging to A . Consequently, unless H_w contains a core neighbour of a outside A , all directions incident with (w, a) generate respectively K or $K \times ab$, making (w, a) a corner of H . Cornerlessness therefore forces at least one vertex of H over w whose core word lies outside A .

The same conclusion holds when $e_a = 3$. Indeed, in that case the shadow of P_a has only three members beyond the empty set and the r singletons. A shattered support of size at least three would

bring with it that support and at least three two-element subsets, so $\text{VCdim}(P_a) \leq 2$ again, and the preceding relative-peeling argument applies unchanged. We have therefore proved the following alternative independently at the two path endpoints:

$$e_a \geq 4 \quad \text{or} \quad \text{an external core vertex adjacent to } a \text{ is present.}$$

The two endpoint sections already contribute four units to $\Delta(H; A, R)$. The preceding argument in particular recovers the preliminary bounds $\Delta \geq 5$ for $n = 2$ and $\Delta \geq 6$ for $n \geq 3$: when $n \geq 3$, an external core word cannot be adjacent to both a_0 and a_n , whereas for $n = 2$ the unique missing word of the core square is adjacent to both endpoints. We now exclude equality in each preliminary bound.

Suppose first that $n = 2$ and $\Delta = 5$. Put

$$L = P_{a_0}, \quad M = P_{a_1}, \quad N = P_{a_2}.$$

The excess decomposition and the endpoint alternatives force

$$|L| = |N| = r + 3, \quad |M| = r + 1,$$

and exactly one vertex of H has core word outside A . The two endpoint corners exposed by relative peeling must both be hidden by that vertex. Thus the outside core word is the fourth word of the core square and, for some repair word w ,

$$w \in (L \cap N) \setminus M. \tag{7.48}$$

Let

$$s = |\text{Sh}(L) \cap \text{Sh}(N)|.$$

Both endpoint sections contain the two repair antipodes, so their shadows contain the empty set and all r singletons; hence $s \geq r + 1$. Partition the shattered supports of H according to their intersection with the two core coordinates. Supports using neither core coordinate contain

$$\text{Sh}(L) \cup \text{Sh}(N),$$

of size $2(r + 3) - s$. Supports using only the first core coordinate contain a copy of $\text{Sh}(L) \cap \text{Sh}(N)$: on its two sides the projected repair families are L and $M \cup N$, because $w \in L$. The second core coordinate gives another such copy, using $L \cup M$ and N , because $w \in N$. Finally the two-coordinate support itself is shattered, since all four core sections are nonempty. Therefore

$$|\text{Sh}(H)| \geq 2(r + 3) - s + 2s + 1 = 2r + 7 + s \geq 3r + 8 = |H|.$$

Ampleness forces equality throughout, and in particular $s = r + 1$.

On the other hand, $L \cap M$ is ample by Lemma 7.22; it contains the two repair antipodes and is contained in the $(r + 1)$ -vertex family M . Isometry therefore gives $L \cap M = M$. Similarly $M \cap N = M$, so $M \subseteq L \cap N$. Together with (7.48), this gives $|L \cap N| \geq r + 2$. The Sandwich inequality now yields

$$r + 2 \leq |\text{Sh}(L \cap N)| \leq |\text{Sh}(L) \cap \text{Sh}(N)| = r + 1,$$

a contradiction. Hence $\Delta \geq 6$ when $n = 2$.

Now suppose that $n \geq 3$ and $\Delta = 6$. The two endpoint alternatives show that both endpoint excesses equal two, every interior section has excess zero, and there are exactly two external-core

vertices, one adjacent to each endpoint. Indeed, an endpoint excess of three still requires an external vertex, an excess of at least four already spends two additional units, and external neighbours of the two endpoints are distinct.

Write $P_i = P_{a_i}$. Every adjacent intersection $C_i = P_{i-1} \cap P_i$ is ample and contains the two repair antipodes. Since each interior section has order $r + 1$, isometry gives

$$P_1 = P_2 = \cdots = P_{n-1} =: P, \quad P \subseteq P_0 \cap P_n.$$

The relative endpoint corners lie at words

$$w_0 \in P_0 \setminus P, \quad w_n \in P_n \setminus P.$$

Write their external core neighbours as

$$b_0 = a_0^{\{x_j\}} \quad (2 \leq j \leq n), \quad b_n = a_n^{\{x_k\}} \quad (1 \leq k \leq n-1).$$

Let

$$t = |(P_0 \cap P_n) \setminus P|.$$

Every word counted by t occurs at both endpoints and at no internal path vertex. Isometry of its core fibre forces at least $n - 1$ external core vertices between a_0 and a_n . These vertices are disjoint for different repair words, so

$$2 \geq (n - 1)t.$$

For the residual shadow count needed below, put

$$W = H|_R = P_0 \cup P_n$$

and remove from $\text{Sh}(H)$ the disjoint family

$$\mathcal{B} = \text{Sh}(W) \dot{\cup} \bigcup_{i=1}^n \{T \cup \{x_i\} : T \in \text{Sh}(P)\}.$$

The projection W and the common carrier P are ample, and every displayed support is shattered by H . Since $|W| = r + 5 - t$ and $|H| = (n + 1)(r + 1) + 6$, the residual family $\Omega = \text{Sh}(H) \setminus \mathcal{B}$ has

$$|\Omega| = |H| - |W| - n|P| = t + 2.$$

Each external endpoint neighbour forces two kinds of residual supports. For the left endpoint, the four core traces supplied by a_0, a_{j-1}, a_j, b_0 shatter the pure core pair $\{x_1, x_j\}$. Project H onto $R \cup \{x_j\}$ and let D_j be the intersection of its two x_j -sections. The two-layer shadow lemma makes D_j ample, and $P \subseteq D_j$. Moreover $w_0 \in D_j \setminus P$. Hence we may choose

$$T_0 \in \text{Sh}(D_j) \setminus \text{Sh}(P).$$

Then $T_0 \cup \{x_j\} \in \Omega$. Notice that T_0 contains at least two repair coordinates, because P contains the repair antipodes and hence shatters the empty set and every repair singleton. Symmetrically, the right endpoint forces the pure core pair $\{x_k, x_n\}$ and a support $T_n \cup \{x_k\} \in \Omega$ with a nonempty repair part.

If $t = 0$, these four named supports have at least three distinct values. Indeed, if the two pure core pairs are distinct, either mixed support is a third value. If they coincide, then necessarily $j = n$

and $k = 1$, so the two mixed supports use the distinct core coordinates x_n and x_1 . This contradicts $|\Omega| = 2$.

It remains to consider $t > 0$. Equation (7.1) forces $n = 3$ and $t = 1$. The unique common word outside P must be both w_0 and w_n , and its core fibre must use the two external vertices as the internal vertices of a shortest a_0 – a_3 path. Hence b_0 and b_n are adjacent and $j \neq k$. The original core path and this second antipodal geodesic have coordinate orders $(1, 2, 3)$ and, respectively, $(2, 3, 1)$, $(3, 2, 1)$, or $(3, 1, 2)$. The latter order has at least two inversions relative to the former, and every inverted pair is shattered by the union of the two geodesics. Thus Ω contains at least two distinct pure core pairs. The two mixed supports above are also distinct, because their unique core coordinates are $x_j \neq x_k$, and neither is a pure core pair. Hence $|\Omega| \geq 4$, contradicting $|\Omega| = t + 2 = 3$.

Thus equality is impossible for $n \geq 3$, and $\Delta \geq 7$. $\square$

Proposition 7.59 (Path-spine Euler identity). *Keep the VC-one path core $a_0 a_1 \cdots a_n$ and its edge coordinates $x_1, \dots, x_n$ from Proposition 7.58. Put*

$$P_i = P_{a_i}, \quad C_i = P_{i-1} \cap P_i \quad (1 \leq i \leq n),$$

let

$$W = H|_R, \quad U = \bigcup_{i=0}^n P_i,$$

and let

$$V = |H \setminus (Q_R \times A)|.$$

For $g \in U$, let $c(g)$ be the number of connected components of

$$\{i \in \{0, \dots, n\} : g \in P_i\}$$

in the path on $\{0, \dots, n\}$, and put

$$\Phi = \sum_{g \in U} (c(g) - 1).$$

Finally define the disjoint spine family

$$\mathcal{B} = \text{Sh}(W) \dot{\cup} \bigcup_{i=1}^n \{T \cup \{x_i\} : T \in \text{Sh}(C_i)\} \subseteq \text{Sh}(H) \quad (7.49)$$

and its residual shadow

$$\Omega = \text{Sh}(H) \setminus \mathcal{B}.$$

Then

$$|\Omega| = \Phi + V - |W \setminus U|. \quad (7.50)$$

In particular, both terms on the right are nonnegative: Φ counts disconnected reappearances of repair words along the core path, while $V - |W \setminus U|$ counts external-core multiplicity after subtracting the one vertex needed to introduce each repair word outside U .

Proof. The projection W is ample. Each C_i is ample by applying Lemma 7.22 to the conditioning that fixes all core coordinates except x_i . That conditioned family also shows that every $T \cup \{x_i\}$ with $T \in \text{Sh}(C_i)$ is shattered by H . The supports in (7.49) are pairwise distinct, so

$$|\mathcal{B}| = |W| + \sum_{i=1}^n |C_i|.$$

Since H is ample,

$$|\Omega| = |H| - |W| - \sum_{i=1}^n |C_i| = V + \sum_{i=0}^n |P_i| - |W| - \sum_{i=1}^n |C_i|. \quad (7.51)$$

Fix $g \in U$. If it belongs to $m(g)$ of the sections and to both sections at $b(g)$ adjacent path edges, then its contribution to $\sum_i |P_i| - \sum_i |C_i|$ is $m(g) - b(g)$. The induced subgraph on its section indices is a disjoint union of $c(g)$ paths, and hence $m(g) - b(g) = c(g)$. Summing over g gives

$$\sum_{i=0}^n |P_i| - \sum_{i=1}^n |C_i| = \sum_{g \in U} c(g) = |U| + \Phi.$$

Substitution in (7.51), together with $U \subseteq W$, proves (7.50).

Finally, every word of $W \setminus U$ supports at least one vertex of H whose core word lies outside A . These vertices are distinct for distinct repair words, so $V \geq |W \setminus U|$. This proves the final assertion. $\square$

Proposition 7.60 (The support-interval exact sequence). *Keep the notation of Proposition 7.59, and put*

$$\mathcal{L}_R = \{\emptyset\} \cup \binom{R}{1}, \quad \mathcal{S} = \bigcup_{j=0}^n \text{Sh}(P_j), \quad \mathcal{T} = \mathcal{S} \setminus \mathcal{L}_R.$$

For $T \in \mathcal{T}$, let

$$I_T = \{j : T \in \text{Sh}(P_j)\}, \quad m_T = |I_T|,$$

and let c_T be the number of connected components of I_T in the section path. Finally, put

$$q_{\text{ext}} = |\text{Sh}(W) \setminus \mathcal{S}|, \quad \Sigma_0 = \text{Sh}(A) * \mathcal{L}_R, \quad \Xi = \mathcal{B} \setminus \Sigma_0.$$

Then

$$\sum_{j=0}^n (|P_j| - (r+1)) = \sum_{T \in \mathcal{T}} m_T, \quad (7.52)$$

$$|\Xi| = \sum_{T \in \mathcal{T}} (m_T - c_T + 1) + q_{\text{ext}}, \quad (7.53)$$

$$\Delta(H; A, R) = |\Xi| + |\Omega|, \quad (7.54)$$

$$|\Omega| = V + \sum_{T \in \mathcal{T}} (c_T - 1) - q_{\text{ext}}. \quad (7.55)$$

There is also an exact linear form of these identities. Fix a field k . For each $T \in \mathcal{T}$, choose one component of I_T and one index j_T in that component. Let $C_0(T)$ have basis $e_{j,T}$, $j \in I_T$, and let $C_1(T)$ have one basis vector u_T , together with one basis vector $u_{i,T}$ for every path edge $a_{i-1}a_i$ whose two ends belong to I_T . The map

$$\partial_T u_T = e_{j_T, T}, \quad \partial_T u_{i,T} = e_{i,T} - e_{i-1,T} \quad (7.56)$$

is injective and gives an exact sequence

$$0 \longrightarrow C_1(T) \xrightarrow{\partial_T} C_0(T) \longrightarrow k^{c_T-1} \longrightarrow 0. \quad (7.57)$$

The basis of $C_0(T)$ records the occurrences of T in the section shadows. The basis of $C_1(T)$ records exactly the spine supports

$$T, \quad T \cup \{x_i\} \quad (T \in \text{Sh}(C_i)).$$

Consequently a successful adjacent duplication, which transports one support T across one section edge, changes the occurrence vector by a member of $\text{im } \partial_T$. Successful acquisitions are therefore relations in the support-interval complex, not an additional cardinality charge. The only supportwise cokernel is the one-dimensional defect of every component of I_T after the distinguished component.

Proof. Every P_j , every C_i , and W contains the two repair antipodes. Their shadows therefore contain the common repair baseline $\mathcal{L}_R$, of order $r + 1$. Since each P_j is ample, counting every nonbaseline repair support in every section gives (7.52).

The two-layer shadow-intersection identity gives

$$\text{Sh}(C_i) = \text{Sh}(P_{i-1}) \cap \text{Sh}(P_i).$$

For a fixed $T \in \mathcal{T}$, the number of path edges whose two ends belong to I_T is $m_T - c_T$. Thus the carrier blocks of $\mathcal{B} \setminus \Sigma_0$ contain exactly $m_T - c_T$ supports of the form $T \cup \{x_i\}$. The pure repair block contains T itself for every $T \in \mathcal{T}$, together with the q_{ext} supports that occur in $\text{Sh}(W)$ but in no section shadow. The blocks are disjoint, proving (7.53).

The baseline Σ_0 has order $(n + 1)(r + 1)$. Since H is ample and

$$\text{Sh}(H) = \Sigma_0 \dot{\cup} \Xi \dot{\cup} \Omega,$$

subtracting the baseline proves (7.54). On the other hand, the geodesic excess decomposition (7.7) and (7.52) give

$$\Delta(H; A, R) = V + \sum_{T \in \mathcal{T}} m_T.$$

Subtracting (7.53) proves (7.55).

For the linear statement, adjoin a new root vertex to the component of the path induced by I_T that contains j_T , using one edge from the root to j_T . Keep all internal edges of the induced path and no other edges. The resulting graph is a forest with $m_T + 1$ vertices, $m_T - c_T + 1$ edges, and c_T components. After deleting the row of the new root from its oriented incidence matrix, one obtains exactly (7.56). The columns are linearly independent, and their cokernel has dimension

$$m_T - (m_T - c_T + 1) = c_T - 1.$$

This proves (7.57). Finally, Lemma A.51 says that a successful duplication across $a_{i-1}a_i$ transports one existing support T between the two corresponding section shadows. Its change in $C_0(T)$ is therefore, up to orientation, $e_{i,T} - e_{i-1,T} = \partial_T u_{i,T}$, proving the last assertion. $\square$

Proposition 7.61 (Projected carriers double support fragmentation). *Keep the notation of Proposition 7.60. For $T \in \mathcal{T}$, define the carrier of all positioned T -cubes and its core projection by*

$$\mathcal{K}_T(H) = \{y \in Q_{(R \dot{\cup} X) \setminus T} : Q_T \times \{y\} \subseteq H\}, \quad Z_T = \mathcal{K}_T(H)|_X.$$

Write the components of I_T as

$$[\ell_1, r_1], \dots, [\ell_{c_T}, r_{c_T}], \quad r_h + 1 < \ell_{h+1},$$

and put

$$\lambda_T = \sum_{h=1}^{c_T-1} (\ell_{h+1} - r_h - 1).$$

Finally let

$$\Omega[T] = \{S \in \Omega : S \cap R = T\}.$$

Then Z_T is ample,

$$Z_T \cap A = \{a_j : j \in I_T\}, \quad |Z_T \setminus A| \geq \lambda_T \geq c_T - 1, \quad (7.58)$$

and

$$|\Omega[T]| = |Z_T \setminus A| + c_T - 1 \geq \lambda_T + c_T - 1 \geq 2(c_T - 1). \quad (7.59)$$

Consequently

$$|\Omega| \geq 2 \sum_{T \in \mathcal{T}} (c_T - 1), \quad (7.60)$$

$$V - q_{\text{ext}} \geq \sum_{T \in \mathcal{T}} (c_T - 1). \quad (7.61)$$

Thus changing the positioned cube that realizes T between two section intervals creates no additional multiplicity problem: all positions are already assembled by the ample projected carrier Z_T , and each extra interval component has two residual units before any ranks are used.

Proof. The support T is shattered by some P_j , hence by H , and ampleness makes it strongly shattered. Apply Lemma 7.19 with the roles of R and X interchanged. It makes $\mathcal{K}_T(H)$ and Z_T ample and gives

$$\text{Sh}(Z_T) = \{K \subseteq X : T \dot{\cup} K \in \text{Sh}(H)\}. \quad (7.62)$$

A path word a_j belongs to Z_T exactly when some positioned full T -cube occurs in the repair section P_j , which is equivalent to $T \in \text{Sh}(P_j)$. This proves the first equality in (7.58).

The ample family Z_T is isometric. Consider two consecutive components of I_T , ending at $p = r_h$ and beginning at $q = \ell_{h+1}$. A shortest Z_T -path from a_p to a_q has length $q - p$ and lies in their interval subcube. None of its $q - p - 1$ internal vertices belongs to A , because the internal path vertices $a_{p+1}, \dots, a_{q-1}$ are absent from Z_T . The interval subcubes belonging to different gaps have disjoint interiors. Summing their internal vertices gives

$$|Z_T \setminus A| \geq \sum_{h=1}^{c_T-1} (\ell_{h+1} - r_h - 1) = \lambda_T.$$

Every gap is nonempty, so $\lambda_T \geq c_T - 1$, completing (7.58).

It remains to count supports. Equation (7.62) and ampleness of Z_T show that H has exactly $|Z_T|$ shattered supports whose exact repair part is T . Among them, the path-spine family contains T itself and precisely the $m_T - c_T$ supports

$$T \cup \{x_i\} \quad (T \in \text{Sh}(C_i))$$

coming from the internal edges of I_T . Every other support with exact repair part T belongs to $\Omega[T]$. Since $|Z_T \cap A| = m_T$, subtraction gives

$$|\Omega[T]| = |Z_T| - 1 - (m_T - c_T) = |Z_T \setminus A| + c_T - 1.$$

Together with (7.58), this proves (7.59).

The families $\Omega[T]$, $T \in \mathcal{T}$, are pairwise disjoint subsets of Ω . Summing their lower bounds proves (7.60). Finally compare this inequality with the exact residual balance (7.55) and cancel $\sum_T (c_T - 1)$. The result is (7.61). $\square$

Proposition 7.62 (Punctured antipodal-section surcharge). *Let $P \subseteq Q_R$ be ample, let $f, \bar{f} \in P$ be antipodes, and let $T \subsetneq R$ have $|T| = k \geq 2$. Suppose that*

$$f^U \in P \quad (U \subsetneq T), \quad f^T \notin P. \quad (7.63)$$

Then

$$|P| \geq |R| + 2^k. \quad (7.64)$$

More precisely, for every $t \in T$ there is a coordinate $c_t \in R \setminus T$ such that P shatters $\{t, c_t\}$, and these k pairs are distinct.

Proof. Translate the repair cube so that $f = 0_R$. The family P is isometric. Fix $t \in T$ and consider a shortest path in P from $T \setminus \{t\}$ to R , where words are identified with their supports. The coordinates changed by this path are exactly $(R \setminus T) \cup \{t\}$. Its first edge cannot have direction t , because the resulting vertex would be the absent word T . Hence that edge has a direction $c_t \in R \setminus T$, and

$$(T \setminus \{t\}) \cup \{c_t\} \in P.$$

The four words

$$\emptyset, \quad \{t\}, \quad (T \setminus \{t\}) \cup \{c_t\}, \quad R$$

realize the four traces on $\{t, c_t\}$. Thus this pair is shattered. The pairs obtained for distinct t are distinct, since their coordinate in T is different and their other coordinate lies outside T .

It remains to count shattered supports. The empty set and the $|R|$ singletons are shattered because P contains two antipodes. Every proper subset of T is strongly shattered by the punctured face in (7.63), while T itself is shattered because $\bar{f}$ supplies its one missing trace. Hence P shatters all $2^k - k - 1$ subsets of T of order at least two. The k pairs $\{t, c_t\}$ are distinct from these supports. Therefore

$$|\text{Sh}(P)| \geq (|R| + 1) + (2^k - k - 1) + k = |R| + 2^k.$$

Ampleness gives $|P| = |\text{Sh}(P)|$, proving the result. $\square$

Proposition 7.63 (Opposite-puncture projection signature). *Let $P \subseteq Q_R$ be ample, let $f, \bar{f} \in P$ be antipodes, and let $T \subsetneq R$ have $|T| \geq 2$. Suppose that*

$$\begin{aligned} f^U &\in P \quad (U \subsetneq T), \quad f^T \notin P, \\ \bar{f}^U &\in P \quad (U \subsetneq T), \quad \bar{f}^T \notin P. \end{aligned} \quad (7.65)$$

For every $c \in R \setminus T$, the projection $P|_{T \cup \{c\}}$ is exactly one of

$$Q_{T \cup \{c\}}, \quad Q_{T \cup \{c\}} \setminus \{(f|_T, \bar{f}_c)\}, \quad Q_{T \cup \{c\}} \setminus \{(\bar{f}|_T, f_c)\}. \quad (7.66)$$

In the first case $T \cup \{c\}$ is shattered by P . In particular, $|R \setminus T| \geq 2$. Moreover, writing $k = |T|$, one has

$$|P| \geq |R| + 2^{k+1} - k - 2. \quad (7.67)$$

One also has the independent shadow bound

$$|P| \geq 2^k + (2^k - 1)(|R| - k). \quad (7.68)$$

Proof. Fix $c \in R \setminus T$. The two punctured faces in (7.65) realize every trace on $T \cup \{c\}$ except possibly the antipodal pair

$$(\bar{f}|_T, f_c), \quad (f|_T, \bar{f}_c).$$

Thus the projection contains the twice-punctured cube obtained by deleting these two traces. Projections of ample families are ample. The twice-punctured cube is not ample, whereas adjoining either missing trace gives a singly punctured cube and adjoining both gives the full cube. These are all possibilities, proving (7.66). The full-cube case shatters its entire coordinate set.

If $R \setminus T = \{c\}$, the projection is P itself. Both exceptional traces are absent by (7.65), so P would be the nonample twice-punctured cube, a contradiction. Hence $|R \setminus T| \geq 2$.

For the cardinality bound, put $C = R \setminus T$ and translate so that $f|_C = 0_C$. Let

$$F_0 = \{g \in Q_C : (f|_T, g) \in P\}, \quad F_1 = \{g \in Q_C : (\bar{f}|_T, g) \in P\}.$$

These are nonempty ample conditionings of P . The two punctured faces in (7.65) contribute $2(2^k - 1)$ vertices, and among them the only member counted by F_0 is 0_C , while the only member counted by F_1 is 1_C .

For every $c \in C$, the projection trichotomy says that c is active in at least one of F_0, F_1 . Assign it to one such fiber and write $C = C_0 \dot{\cup} C_1$ for the resulting partition. Since an ample family is isometric and hence connected, a family active on s coordinates has at least $s + 1$ vertices. Therefore

$$|F_0| \geq |C_0| + 1, \quad |F_1| \geq |C_1| + 1.$$

Apart from their displayed base vertices, these two fibers contribute at least $|C_0| + |C_1| = |C|$ further vertices, disjoint from the two punctured faces. Consequently

$$|P| \geq 2(2^k - 1) + |C| = |R| + 2^{k+1} - k - 2,$$

as required.

For the shadow bound, every subset of T is shattered: the two punctured faces supply all proper traces, and either antipodal face supplies the trace missing from the other. More strongly, fix $c \in C$ and a proper subset $S \subsetneq T$. The lower punctured face, on which $c = f_c$, realizes every trace on S : any prescribed S -trace has an extension to T different from the one missing lower tip, because $T \setminus S \neq \emptyset$. The upper punctured face gives the same conclusion with $c = \bar{f}_c$. Together the two faces realize every trace on $S \cup \{c\}$, so this support is shattered. The supports wholly contained in T give 2^k distinct members of the shadow, and, independently for each $c \in C$, the proper subsets $S \subsetneq T$ give another $2^k - 1$. Consequently

$$|\text{Sh}(P)| \geq 2^k + (2^k - 1)|C| = 2^k + (2^k - 1)(|R| - k).$$

Ampleness gives the claimed order bound. □

Corollary 7.64 (Bichromatic collisions lie in an ordered two-cube collar). *Keep the hypotheses of Proposition 7.63, and suppose that no $T \cup \{c\}$ is shattered. Translate so that $f = 0_R$. Partition $C = R \setminus T$ as $L \dot{\cup} U$, where $c \in L$ when $P|_{T \cup \{c\}}$ contains $(1_T, 0_c)$ and omits $(0_T, 1_c)$, and $c \in U$ in the opposite case. For every $c \in L$ and $d \in U$, either P shatters $\{c, d\}$ or every $g \in P$ satisfies $g_c \leq g_d$. Consequently, if no such pair is shattered, then*

$$P|_C \subseteq (\{0_L\} \times Q_U) \cup (Q_L \times \{1_U\}). \quad (7.69)$$

Proof. Fix $c \in L$ and $d \in U$. The two antipodal carrier words give the traces 00 and 11 on $\{c, d\}$. Since $c \in L$, there is a carrier word with T -trace 1_T and c -trace zero. The defining omission for $d \in U$ forces its d -trace to be one, so the trace 01 is also present. If 10 occurs, the pair is shattered. Otherwise every carrier word avoids 10, which is exactly the implication $g_c \leq g_d$.

Suppose that no bichromatic pair is shattered. If a word in $P|_C$ has a one in some coordinate of L , all its coordinates in U are one by the preceding implication, and it belongs to $Q_L \times \{1_U\}$. If it has no one in L , it belongs to $\{0_L\} \times Q_U$. This proves (7.69). $\square$

Proposition 7.65 (Shadow-tight opposite collisions are serial peripheral). *Keep the hypotheses of Proposition 7.63, translate so that $f = 0_R$, and put $C = R \setminus T$. Suppose that*

$$T \cup \{c\} \notin \text{Sh}(P) \quad (c \in C), \quad \{c, d\} \notin \text{Sh}(P) \quad (c, d \in C, c \neq d). \quad (7.70)$$

Then there are an ordering $c_1, \dots, c_m$ of C , where $m = |C|$, and an index h with $1 \leq h \leq m - 1$ such that, on writing

$$v_j = 1_{\{c_1, \dots, c_j\}} \in Q_C \quad (0 \leq j \leq m),$$

the family P is exactly

$$\begin{aligned} P = & \bigcup_{j < h} ((Q_T \setminus \{1_T\}) \times \{v_j\}) \\ & \cup (Q_T \times \{v_h\}) \cup \bigcup_{j > h} ((Q_T \setminus \{0_T\}) \times \{v_j\}). \end{aligned} \quad (7.71)$$

In the notation of Corollary 7.64, one has

$$U = \{c_1, \dots, c_h\}, \quad L = \{c_{h+1}, \dots, c_m\}.$$

Contracting every coordinate of $C \setminus \{c_h, c_{h+1}\}$ sends P to the canonical collar

$$\mathcal{K}_T = ((Q_T \setminus \{1_T\}) \times \{00\}) \cup (Q_T \times \{10\}) \cup ((Q_T \setminus \{0_T\}) \times \{11\}). \quad (7.72)$$

In particular, both orientation classes are nonempty, the outside projection is a coordinate-simple path, and $P \in \mathcal{D}_{\text{amp}}$. Thus every opposite collision either shatters one of the supports excluded in (7.70), or its entire carrier has the explicit corner-peelable normal form (7.71).

Proof. The supports already shown to be shattered in the proof of Proposition 7.63 are precisely

$$\Sigma_0 = 2^T \cup \{S \cup \{c\} : S \subsetneq T, c \in C\}. \quad (7.73)$$

Every support outside Σ_0 either contains two coordinates of C , or contains $T \cup \{c\}$ for some $c \in C$. By downward closure and (7.70), none is shattered. Therefore $\text{Sh}(P) = \Sigma_0$. Its minimal nonfaces are exactly the pairs $\{c, d\} \subseteq C$ and the sets $T \cup \{c\}$, $c \in C$.

The antipodes realize the traces 00 and 11 on every outside pair. Theorem 5.3 therefore says that the unique omitted trace on each $\{c, d\}$ is either 01 or 10. Regard the corresponding forbidden trace as one of the implications $x_c \leq x_d$ and $x_d \leq x_c$, and let $F \subseteq Q_C$ be their common satisfying family. For a trace $\alpha \in Q_T \setminus \{0_T, 1_T\}$, none of the signed clauses on $T \cup \{c\}$ is active. The same signed-minimal-nonface theorem gives

$$\{z \in Q_C : (\alpha, z) \in P\} = F. \quad (7.74)$$

Both punctured faces show that $0_C, 1_C \in F$, and this conditioning of the ample family P is ample.

The implications orient every pair of coordinates of C . Collapse the strong components of the resulting tournament. Coordinates in one strong component have equal values in every member of F , while the condensation is an acyclic tournament and hence a total order. If it has q components, its satisfying family has exactly $q + 1$ threshold words. On the other hand, F contains two antipodes, so it shatters every singleton of C . Ampleness therefore gives

$$q + 1 = |F| = |\text{Sh}(F)| \geq m + 1.$$

Since $q \leq m$, equality holds and every strong component is a singleton. After relabelling the coordinates, F is the coordinate-simple path

$$F = \{v_0, v_1, \dots, v_m\}. \quad (7.75)$$

For $\beta \in Q_T$, write $P_\beta = \{z \in Q_C : (\beta, z) \in P\}$. Use the partition $C = L \dot{\cup} U$ from Corollary 7.64. The signed clause on $T \cup \{c\}$ forces $c = 0$ in the 0_T -fiber when $c \in L$, and forces $c = 1$ in the 1_T -fiber when $c \in U$. It imposes no other restriction. Consequently

$$P_{0_T} = \{v_j : c = 0 \text{ on } v_j \text{ for every } c \in L\}, \quad P_{1_T} = \{v_j : c = 1 \text{ on } v_j \text{ for every } c \in U\}, \quad (7.76)$$

while every other T -fiber is the full path F . The missing tips $(1_T, 0_C)$ and $(0_T, 1_C)$ show respectively that U and L are nonempty.

Let p be the first position occupied by a member of L , and let q be the last position occupied by a member of U in the order $c_1, \dots, c_m$. Equations (7.74) and (7.76) give

$$|P| = (2^{|T|} - 2)(m + 1) + p + (m - q + 1).$$

On the other hand, (7.73) and ampleness give

$$|P| = |\Sigma_0| = 2^{|T|} + (2^{|T|} - 1)m.$$

Thus $p - q = 1$. Since L and U partition C , every member of U occurs before every member of L . Put $h = q = p - 1$. The fiber description now becomes exactly (7.71). Its projection to $T \cup \{c_h, c_{h+1}\}$ has outside traces 00, 10, 11; the three corresponding T -sections are respectively $Q_T \setminus \{1_T\}$, Q_T , and $Q_T \setminus \{0_T\}$. This proves (7.72).

It remains to verify the final dismantlability assertion. Delete the layers $v_0, v_1, \dots, v_{h-1}$ successively from the left. At each step the coordinate changed by the first path edge is peripheral: its smaller section is a fixed copy of $Q_T \setminus \{1_T\}$, contained in the next section. Delete the layers $v_m, v_{m-1}, \dots, v_{h+1}$ symmetrically from the right; the duplicated site is now a fixed copy of $Q_T \setminus \{0_T\}$. The remaining layer is Q_T . The cube and the two singly punctured cubes are corner-peelable, so repeated application of the exact peripheral-expansion test (A4g1a) proves $P \in \mathcal{D}_{\text{amp}}$. $\square$

Corollary 7.66 (Relative peeling above the canonical low-rank collar). *Let $2 \leq |T| \leq 3$, let $\mathcal{K}_T$ be the family in (7.72), and let*

$$\mathcal{K}_T \subseteq B \subseteq Q_{T \cup \{u, \ell\}}$$

be ample. For every $s \in \mathcal{K}_T$, the family B has a corner peeling in which every member of $B \setminus \mathcal{K}_T$ precedes every member of $\mathcal{K}_T$, and s is last.

Proof. The ambient cube has $|T| + 2 \leq 5$ coordinates. Apply the five-coordinate relative-peeling theorem, Theorem A.137, to the nested ample pair $\mathcal{K}_T \subseteq B$; ampleness of $\mathcal{K}_T$ also follows from Proposition 7.65 or directly from its peripheral construction. $\square$

Proposition 7.67 (The canonical collar has exactly two bad extensions). *Let $|T| \geq 2$, write the last two coordinates of $\mathcal{K}_T$ in the order used in (7.72), and put*

$$b_0 = (0_T, 01), \quad b_1 = (1_T, 01).$$

For every $x \in Q_{T \cup \{u, \ell\}} \setminus \mathcal{K}_T$,

$$\mathcal{K}_T \cup \{x\} \text{ is nonample} \iff x \in \{b_0, b_1\}. \quad (7.77)$$

More precisely, for each $t \in T$, the two bad tips have the antipodal blockers

$$(1_T^{\{t\}}, 00) \longleftrightarrow b_1 \text{ on } \{t, \ell\}, \quad (0_T^{\{t\}}, 11) \longleftrightarrow b_0 \text{ on } \{t, u\}. \quad (7.78)$$

Thus every other missing word is an allowable first step in a relative peeling above $\mathcal{K}_T$.

Proof. Each pair in (7.78) is antipodal in the indicated square, and that square meets the corresponding one-concept extension in exactly those two vertices. Proposition 2.131 shows that the two extensions are nonample.

Conversely, suppose that adjoining x is nonample. The same proposition gives a blocker $y \in \mathcal{K}_T$ and a set J of at least two coordinates such that the J -face through y contains no other collar vertex and $x = y^J$. If a coordinate of J were incident with y in $\mathcal{K}_T$, the corresponding neighbor would be a second collar vertex in that face. Hence

$$J \subseteq (T \cup \{u, \ell\}) \setminus I_{\mathcal{K}_T}(y). \quad (7.79)$$

The nonincident directions can be read directly from the three layers in (7.72). In the 00-layer, a word has at least two nonincident directions only when it is $1_T^{\{t\}}$ for some $t \in T$; its two nonincident directions are then t and ℓ . Every other word in that layer has only ℓ nonincident. Symmetrically, in the 11-layer only the words $0_T^{\{t\}}$ have two nonincident directions, namely t and u . Every word in the middle 10-layer has at most one nonincident direction. Since $|J| \geq 2$, inclusion (7.79) is therefore an equality in one of the two displayed cases. Flipping those two directions gives respectively b_1 and b_0 , completing the proof. $\square$

Corollary 7.68 (No ample interval gap immediately above the canonical collar). *Let $|T| \geq 2$ and $\mathcal{K}_T \subsetneq B \subseteq Q_{T \cup \{u, \ell\}}$ be ample. Then there is an $x \in B \setminus \mathcal{K}_T$ such that $\mathcal{K}_T \cup \{x\}$ is ample.*

Proof. Otherwise Proposition 7.67 gives $B \setminus \mathcal{K}_T \subseteq \{b_0, b_1\}$. A one-tip extension is nonample by the same proposition. If both tips occur, conditioning $u = 0$ and $\ell = 1$ leaves the antipodal pair $\{0_T, 1_T\}$, which is nonample because $|T| \geq 2$. Both alternatives contradict ampleness of B . $\square$

Corollary 7.69 (High-rank opposite collisions exceed the bulk threshold). *Keep the path-spine notation of Proposition 7.59, with $n \geq 2$. If one of $W, C_1, \dots, C_n$ contains opposite punctured faces with the same repair support T and $|T| \geq 4$, then $|H| \geq 46$.*

Proof. Put $k = |T|$. Proposition 7.63 gives $r - k = |R \setminus T| \geq 2$ and says that the exceptional carrier has order at least

$$r + 2^{k+1} - k - 2 = (r + 1) + (2^{k+1} - k - 3).$$

Every other member of the list $W, C_1, \dots, C_n$ is ample and contains the two repair antipodes, so it has at least $r + 1$ vertices. The disjoint path-spine shadow decomposition therefore gives

$$|H| \geq (n + 1)(r + 1) + 2^{k+1} - k - 3.$$

The stronger shadow estimate (7.68) says that the exceptional carrier has tax at least

$$2^k - k - 1 + (2^k - 2)(r - k)$$

above its $r + 1$ antipodal-geodesic baseline. Hence

$$|H| \geq (n + 1)(r + 1) + 2^k - k - 1 + (2^k - 2)(r - k).$$

Using $n \geq 2$ and $r \geq k + 2$, the right-hand side is at least

$$3(k + 3) + 2^k - k - 1 + 2(2^k - 2) = 3 \cdot 2^k + 2k + 4.$$

This equals 60 at $k = 4$ and is increasing for $k \geq 4$. $\square$

Corollary 7.70 (An adjacent-carrier collision pays twice). *Keep the path-spine notation of Proposition 7.59, with $n \geq 2$, and suppose that some C_i contains opposite punctured faces with common support T , where $k = |T| \geq 2$. Put*

$$\tau_k(r) = 2^k - k - 1 + (2^k - 2)(r - k). \quad (7.80)$$

Then

$$|H| \geq (n + 1)(r + 1) + 2\tau_k(r). \quad (7.81)$$

Consequently, if $|H| < 46$, no C_i contains a collision of rank at least three. A rank-two collision in a C_i has $4 \leq r \leq 6$ and satisfies

$$\begin{array}{c|ccc} r & 4 & 5 & 6 \\ \hline n & \leq 6 & \leq 4 & = 2. \end{array} \quad (7.82)$$

Thus every rank-three collision below order 46 lies in the pure repair projection W , not in an adjacent path carrier.

Proof. The inclusion $C_i \subseteq W$ puts the same two punctured faces in both ample carriers. By (7.68), each carrier has order at least $(r + 1) + \tau_k(r)$. Every other member of the disjoint path-spine list $W, C_1, \dots, C_n$ has order at least $r + 1$. Summing their shadows in (7.49) proves (7.81).

For $k = 3$, one has $r \geq 5$ and $\tau_3(r) = 6r - 14$. Hence

$$|H| \geq 3(r + 1) + 2(6r - 14) = 15r - 25 \geq 50.$$

The lower bound increases with k , so every $k \geq 3$ is excluded below 46. For $k = 2$, one has $r \geq 4$ and $\tau_2(r) = 2r - 3$, giving

$$|H| \geq (n + 1)(r + 1) + 4r - 6.$$

Substitution of $r = 4, 5, 6$ gives the displayed bounds on n , while $r = 7$ already gives $3 \cdot 8 + 22 = 46$. $\square$

Corollary 7.71 (A projection collision forces a global puncture cover). *Keep the path-spine notation of Proposition 7.59, suppose that H has minimum cardinality among all nonempty cornerless ample families, and suppose that W contains opposite punctured faces with common repair support T of order $k \geq 2$. Then $W \in \mathcal{D}_{\text{amp}}$, every nonempty fibre H_g belongs to $\mathcal{D}_{\text{amp}}$, and H contains at least*

$$2^k + (2^k - 1)(r - k) \quad (7.83)$$

rooted transverse punctures whose repair roots are pairwise distinct. If $|H| < 48$, every one of these punctures has rank between two and five. Moreover, a fixed support of rank j can own at most

$$\left\lfloor \frac{47}{2^j - 1} \right\rfloor = \begin{cases} 15, & j = 2, \\ 6, & j = 3, \\ 3, & j = 4, \\ 1, & j = 5 \end{cases} \quad (7.84)$$

of these roots.

Proof. Proposition 7.63 gives

$$|W| \geq 2^k + (2^k - 1)(r - k).$$

The projection W is ample. It is strictly smaller than H , because the endpoint repair fibre contains the path core with at least three concepts. Lemma 7.8 gives $W \in \mathcal{D}_{\text{amp}}$, while Corollary 7.14 gives the same conclusion for every nonempty H_g . That corollary also supplies one rooted transverse puncture over every word of W , with distinct repair roots, and bounds their ranks by five when $|H| < 48$. Finally apply Lemma 7.18 to all roots that choose the same support J and use $|H| \leq 47$. $\square$

Proposition 7.72 (Cross-support conservation for the fibrewise punctures). *Keep the path-spine notation of Proposition 7.59, and let the rooted transverse punctures $(g, a_g), J_g$ be chosen as in Corollary 7.14. Write*

$$\Omega_{\geq 2} = \{S \in \Omega : |S \cap X| \geq 2\}.$$

For $K \subseteq X$, put

$$m_K = |\{g \in W : J_g \cap X = K\}|. \quad (7.85)$$

For $1 \leq i \leq n$, split the singleton-core owners into the spine-aligned and off-carrier counts

$$\begin{aligned} m_i^{\parallel} &= |\{g \in C_i : J_g \cap X = \{x_i\}\}|, \\ m_i^{\perp} &= |\{g \in W \setminus C_i : J_g \cap X = \{x_i\}\}|. \end{aligned} \quad (7.86)$$

Then

$$|\Omega_{\geq 2}| \geq \sum_{\substack{K \subseteq X \\ |K| \geq 2}} m_K \quad (7.87)$$

and, more precisely,

$$|\Omega| \geq \sum_{\substack{K \subseteq X \\ |K| \geq 2}} m_K + \sum_{i=1}^n m_i^{\perp}. \quad (7.88)$$

In particular,

$$|\Omega| \geq \sum_{\substack{K \subseteq X \\ |K| \geq 2}} m_K + \sum_{i=1}^n \max\{0, m_{\{x_i\}} - |C_i|\}. \quad (7.89)$$

Equivalently, the only puncture anchors that can be absorbed without a distinct residual support have no core direction, or are spine-aligned singleton-core owners:

$$m_{\emptyset} + \sum_{i=1}^n m_i^{\parallel} \geq |W| - |\Omega|. \quad (7.90)$$

Consequently,

$$m_\emptyset + \sum_{i=1}^n \min\{m_{\{x_i\}}, |C_i|\} \geq |W| - |\Omega|. \quad (7.91)$$

Thus punctures with two or more core directions admit no cross-support reuse at all: each such repair root pays a different member of the residual shadow, even when many roots use the same core support.

Proof. Fix a nonempty $K \subseteq X$ and a repair word counted by m_K . Write $J_g = S_g \dot{\cup} K$, where $S_g = J_g \cap R$ is nonempty. Every subset of K is then a proper subset of J_g . The puncture relations (7.12) show that the full K -cube through (g, a_g) lies in H . Hence g occurs in the repair projection W_K of the K -cube carrier. The repair roots are pairwise distinct, so $|W_K| \geq m_K$.

Lemma 7.19 now gives at least m_K shattered supports with exact core part K . For distinct sets K , these collections are disjoint. If $|K| \geq 2$, every one of them belongs to $\Omega_{\geq 2}$, which proves (7.87).

Suppose next that $K = \{x_i\}$. Among the supports supplied by the projected carrier, the only ones that can belong to the spine family (7.49) are

$$T \cup \{x_i\}, \quad T \in \text{Sh}(C_i).$$

There are $|\text{Sh}(C_i)| = |C_i|$ of these. Consequently at least $\max\{0, m_{\{x_i\}} - |C_i|\}$ of the supplied supports belong to Ω . Their exact core parts distinguish different indices i and also distinguish them from the supports counted in (7.87). Summing proves (7.89).

The same argument retains the positions of the roots. Every member of C_i anchors the path edge $a_{i-1}a_i$, and hence belongs to the repair projection $W_{\{x_i\}}$ of the x_i -cube carrier. All $m_i^\perp$ off-carrier roots are further, distinct members of that projection. Therefore

$$|W_{\{x_i\}}| - |C_i| \geq m_i^\perp.$$

Because both nested families are ample, the same difference counts the supports in $\text{Sh}(W_{\{x_i\}}) \setminus \text{Sh}(C_i)$. After adjoining x_i , all those supports belong to Ω . These families are disjoint for different i , and disjoint from the high-core families. This proves (7.88).

Finally the owner classes partition W , so

$$|W| = m_\emptyset + \sum_{i=1}^n (m_i^\parallel + m_i^\perp) + \sum_{|K| \geq 2} m_K.$$

Subtracting the right-hand side of (7.88) from this identity gives (7.90). The bound $m_i^\parallel \leq \min\{m_{\{x_i\}}, |C_i|\}$ then gives (7.91). $\square$

Proposition 7.73 (The column cover gives the dual support-capacity law). *Keep the path-spine notation of Proposition 7.59, and choose the column-indexed punctures $(g_a, a), L_a$ from Corollary 7.15. For $S \subseteq R$, put*

$$n_S = |\{a \in Z : L_a \cap R = S\}|, \quad b_S = \mathbf{1}_{S \in \text{Sh}(W)} + \sum_{i=1}^n \mathbf{1}_{S \in \text{Sh}(C_i)}. \quad (7.92)$$

Then

$$|\Omega| \geq \sum_{\emptyset \neq S \subseteq R} \max\{0, n_S - b_S\}. \quad (7.93)$$

Equivalently, the column owners that can be absorbed by the path-spine shadow satisfy

$$n_\emptyset + \sum_{\emptyset \neq S \subseteq R} \min\{n_S, b_S\} \geq |Z| - |\Omega|. \quad (7.94)$$

Together with Proposition 7.72, this gives the bidirectional bound

$$|\Omega| \geq \max \left\{ \sum_{\substack{K \subseteq X \\ |K| \geq 2}} m_K + \sum_{i=1}^n m_i^\perp, \quad \sum_{\emptyset \neq S \subseteq R} \max\{0, n_S - b_S\} \right\}. \quad (7.95)$$

Thus a residual support may pay one row demand and one column demand at the same Boolean-matrix entry. Absorption by the spine is recorded by b_S , while every remaining row-column reuse is localized to the exact pair (S, K) of that residual support.

Proof. Fix a nonempty $S \subseteq R$ with $n_S > 0$, and define the repair-cube carrier and its core projection by

$$\mathcal{K}_S(H) = \{y \in Q_{(R \dot{\cup} X) \setminus S} : Q_S \times \{y\} \subseteq H\}, \quad Z_S = \mathcal{K}_S(H)|_X.$$

If a is counted by n_S , write $L_a = S \dot{\cup} K_a$. Proposition 7.13 gives $K_a \neq \emptyset$. Hence every subset of S , including S itself, is a proper subset of L_a . The puncture relations show that the full S -cube through (g_a, a) lies in H , so $a \in Z_S$. Distinct column indices give distinct words a , and therefore

$$|Z_S| \geq n_S. \quad (7.96)$$

Apply Lemma 7.19 after interchanging R and X . It gives the exact row-column identity

$$\text{Sh}(Z_S) = \{K \subseteq X : S \dot{\cup} K \in \text{Sh}(H)\}. \quad (7.97)$$

Thus the number of shattered supports of H whose exact repair part is S equals $|\text{Sh}(Z_S)| = |Z_S|$.

The members of the spine family (7.49) with exact repair part S are precisely S itself when $S \in \text{Sh}(W)$ and the supports $S \cup \{x_i\}$ for which $S \in \text{Sh}(C_i)$. Their number is exactly b_S . Consequently the residual shadow has exactly $|Z_S| - b_S$ members with repair part S , and (7.96) makes this at least $\max\{0, n_S - b_S\}$. Supports with distinct exact repair parts are different, so summing over nonempty S proves (7.93).

The owner classes partition Z , giving

$$|Z| = n_\emptyset + \sum_{\emptyset \neq S \subseteq R} n_S.$$

Subtracting the right-hand side of (7.93) proves (7.94). Finally combine (7.93) with (7.88) to obtain (7.95). $\square$

Corollary 7.74 (Forced row-column collisions in the residual shadow). *In the setting of Proposition 7.73, put*

$$\begin{aligned} \rho &= \sum_{\substack{K \subseteq X \\ |K| \geq 2}} m_K + \sum_{i=1}^n m_i^\perp, \\ \kappa &= \sum_{\emptyset \neq S \subseteq R} \max\{0, n_S - b_S\}. \end{aligned}$$

There are subsets $\mathcal{R}, \mathcal{C} \subseteq \Omega$ with

$$|\mathcal{R}| = \rho, \quad |\mathcal{C}| = \kappa,$$

such that every member of $\mathcal{R}$ pays one row-owner demand with its exact core part and every member of $\mathcal{C}$ pays one column-owner demand with its exact repair part. Consequently

$$|\mathcal{R} \cap \mathcal{C}| \geq \max\{0, \rho + \kappa - |\Omega|\}. \quad (7.98)$$

Every member of the intersection is a shattered support $S \dot{\cup} K$ that is simultaneously selected by the projected K -carrier and the projected S -carrier. Hence all unavoidable reuse of the two transverse puncture covers is localized in common full $(S \dot{\cup} K)$ -cubes of H .

Proof. The proof of Proposition 7.72 selects, for each $|K| \geq 2$, at least m_K residual supports with exact core part K , and for $K = \{x_i\}$ at least $m_i^\perp$ members of the relative carrier shadow

$$\{T \cup \{x_i\} : T \in \text{Sh}(W_{\{x_i\}}) \setminus \text{Sh}(C_i)\}.$$

These groups have different exact core parts, so their selected members form a set $\mathcal{R}$ of order ρ .

Likewise, the proof of Proposition 7.73 supplies, for each nonempty S , at least $\max\{0, n_S - b_S\}$ residual supports with exact repair part S . The groups for different S are disjoint, and selected members form a set $\mathcal{C}$ of order κ . Both sets lie in Ω , so inclusion–exclusion gives (7.98).

If $S \dot{\cup} K \in \mathcal{R} \cap \mathcal{C}$, ampleness of H makes this support strongly shattered. A full cube on that support is simultaneously an S -cube in the K -carrier and a K -cube in the S -carrier, which proves the final assertion. $\square$

Lemma 7.75 (The shadow-neutral defects reduce to square tips). *Let v, J be a rooted puncture in H :*

$$v^U \in H \quad (U \subsetneq J), \quad v^J \notin H, \quad |J| \geq 2.$$

For every two-set $L \subseteq J$, the vertex $u = v^{J \setminus L}$ is the root of a punctured L -square:

$$u, u^{\{p\}}, u^{\{q\}} \in H, \quad u^L \notin H \quad (L = \{p, q\}). \quad (7.99)$$

In particular, keep the path-spine setting of Proposition 7.72 and write $v = (g, a)$.

1. *A repair-only puncture contains a repair-only punctured square in the column P_a .*
2. *Suppose that $J = S \dot{\cup} \{x_i\}$ with $\emptyset \neq S \subseteq R$, and choose $s \in S$. The punctured square on $\{s, x_i\}$ has repair root $h = g^{S \setminus \{s\}}$, and h belongs to the repair projection $W_{\{x_i\}}$ of the x_i -cube carrier. In the full carrier, its anchor y is present and the s -neighbour $y^{\{s\}}$ is absent. Either $h \in C_i$, so this is a spine-aligned rank-two defect, or H has a residual shattered support with exact core part $\{x_i\}$.*

Consequently the local shadow-neutral transport problem has only two terminal defect types: a punctured repair square, or a punctured $\{s, x_i\}$ -square rooted over C_i . Both are boundary data inside smaller dismantlable carriers, by Corollary 7.20. Each square is retained together with the parent puncture (v, J) from which it was obtained.

Proof. If $A \subsetneq L$, then $(J \setminus L) \cup A$ is a proper subset of J . Hence

$$u^A = v^{(J \setminus L) \cup A} \in H.$$

On the other hand, $u^L = v^J \notin H$. This proves (7.99) and immediately gives the first special case.

For the second, take $L = \{s, x_i\}$. The root u has repair word $h = g^{S \setminus \{s\}}$ and the same core word as v . The two vertices u and $u^{\{x_i\}}$ form an x_i -edge of H , so $h \in W_{\{x_i\}}$. If y denotes the word obtained from u by deleting its x_i -coordinate, then $y \in \mathcal{K}_{\{x_i\}}(H)$, whereas $y^{\{s\}} \notin \mathcal{K}_{\{x_i\}}(H)$ because the latter anchor would require the absent vertex $u^{\{s, x_i\}}$. If $h \notin C_i$, then $C_i \subsetneq W_{\{x_i\}}$. The family C_i is ample by Proposition 7.59, while $W_{\{x_i\}}$ is ample by Lemma 7.19; therefore

$$\text{Sh}(C_i) \subsetneq \text{Sh}(W_{\{x_i\}}).$$

Choose T in the difference. Lemma 7.19 makes $T \cup \{x_i\}$ a shattered support of H , while its absence from $\text{Sh}(C_i)$ keeps it out of the spine family (7.49). Thus it belongs to Ω and has exact core part $\{x_i\}$. The alternative $h \in C_i$ is precisely the asserted spine-aligned square. $\square$

Remark 7.76 (The terminal squares retain their provenance). The map $(v, J, L) \mapsto v^{J \setminus L}$ need not be injective when the parent supports J differ. Thus the square-tip reduction does not replace the distinct-root statement in Corollary 7.14 by a new distinct-root statement. Its output is a family of provenance-tagged square tips. The remaining transport theorem must either separate coincident tips by their carrier provenance, charging a new residual support, or show that the coincidence survives a contraction. This is precisely the synchronization issue; forgetting the parent punctures would discard the global multiplicity information.

Proposition 7.77 (Isometric path-fibre gap tax). *Keep the notation of Proposition 7.59. For $g \in U$, write the connected components of*

$$I(g) = \{i : g \in P_i\}$$

as index intervals

$$[\ell_1, r_1], \dots, [\ell_c, r_c], \quad r_s + 1 < \ell_{s+1},$$

and put

$$\lambda(g) = \sum_{s=1}^{c-1} (\ell_{s+1} - r_s - 1), \quad \Lambda = \sum_{g \in U} \lambda(g).$$

Then

$$V \geq |W \setminus U| + \Lambda, \quad \Lambda \geq \Phi. \quad (7.100)$$

Consequently the residual shadow satisfies the doubled fragmentation bound

$$|\Omega| \geq \Phi + \Lambda \geq 2\Phi. \quad (7.101)$$

More precisely, every gap $r_s + 1, \dots, \ell_{s+1} - 1$ forces that many distinct external core concepts over g , and the concepts forced by different gaps are disjoint.

Proof. Fix $g \in U$ and two consecutive components of $I(g)$. Put $p = r_s$ and $q = \ell_{s+1}$. The core fibre

$$H_g = \{z \in Q_X : (g, z) \in H\}$$

is a nonempty conditioning of the ample family H , and hence is isometric. It contains a_p and a_q , whose Hamming distance is $q - p$. A shortest H_g -path between them therefore has $q - p - 1$ internal vertices and changes exactly the coordinates $x_{p+1}, \dots, x_q$.

No internal vertex of this path belongs to A . Indeed, the only members of the core path A in the interval subcube between a_p and a_q are $a_p, a_{p+1}, \dots, a_q$, while none of the intermediate concepts

$a_{p+1}, \dots, a_{q-1}$ belongs to H_g by the choice of consecutive components. Thus all $q - p - 1$ internal vertices are external core concepts over g .

The open coordinate intervals

$$\{x_{r_s+1}, \dots, x_{\ell_s+1}\}$$

for different gaps are disjoint. Their interval subcubes can meet only at a common endpoint belonging to A , so their internal vertices are disjoint. Summing the forced internal vertices over the gaps proves that at least $\lambda(g)$ external core concepts lie over g .

For a repair word in $W \setminus U$, every vertex over it has core word outside A , so at least one external concept is present. Summing over all repair words gives the first inequality in (7.100). Every one of the $c(g) - 1$ gaps has positive length, so $\lambda(g) \geq c(g) - 1$; summing gives $\Lambda \geq \Phi$. Finally, Proposition 7.59 gives

$$|\Omega| = \Phi + V - |W \setminus U| \geq \Phi + \Lambda \geq 2\Phi,$$

which proves (7.101). $\square$

Proposition 7.78 (Transverse path-spine decomposition). *Keep the notation of Proposition 7.59. Orient the path coordinates so that*

$$a_i = a_0^{\{x_1, \dots, x_i\}} \quad (0 \leq i \leq n).$$

For $1 \leq i \leq n$ and $b \in \{0, 1\}$, put

$$B_i^b = \{g \in Q_R : \text{some } (g, z) \in H \text{ has } z_{x_i} = b\}, \quad D_i = B_i^0 \cap B_i^1.$$

Then D_i is ample, $C_i \subseteq D_i$, and the residual shadow in (7.50) decomposes as

$$\Omega = \bigcup_{i=1}^n \{T \cup \{x_i\} : T \in \text{Sh}(D_i) \setminus \text{Sh}(C_i)\} \dot{\cup} \Omega_{\geq 2}, \quad (7.102)$$

where every member of $\Omega_{\geq 2}$ contains at least two coordinates of the core path. Consequently

$$\Phi + V - |W \setminus U| = \sum_{i=1}^n |D_i \setminus C_i| + |\Omega_{\geq 2}|. \quad (7.103)$$

Moreover, let $g \in P_0$ and suppose that H_g contains a core neighbour $a_0^{\{x_j\}}$ outside A . Then $j \geq 2$, and either

$$g \in D_j \setminus C_j$$

or $\{x_1, x_j\} \in \Omega_{\geq 2}$. Symmetrically, an external core neighbour $a_n^{\{x_j\}}$ of a_n either belongs over a word in $D_j \setminus C_j$ or forces $\{x_j, x_n\} \in \Omega_{\geq 2}$.

Proof. Project H onto $R \cup \{x_i\}$. This projection is ample and its two x_i -sections are B_i^0 and B_i^1 . Lemma 7.22 makes $D_i = B_i^0 \cap B_i^1$ ample and shows that the shattered supports of the projection containing x_i are exactly

$$\{T \cup \{x_i\} : T \in \text{Sh}(D_i)\}.$$

Every repair word supporting both adjacent path concepts a_{i-1}, a_i lies in D_i , so $C_i \subseteq D_i$. The family C_i is ample by the same two-layer lemma applied after fixing the other core coordinates. Therefore the one-core-coordinate members of Ω are exactly the first family on the right of (7.102); all remaining members use at least two core coordinates. The union is disjoint. Taking cardinalities, using ampleness of C_i, D_i , and applying (7.50) gives (7.103).

For the final assertion, the only neighbour of a_0 in the core path is $a_1 = a_0^{\{x_1\}}$, so an external core neighbour has $j \geq 2$. The two words a_0 and $a_0^{\{x_j\}}$ over g put g in D_j . If $g \notin C_j$, the first alternative holds. If $g \in C_j$, then H_g also contains a_{j-1} and a_j . Their traces on $\{x_1, x_j\}$, together with the traces of $a_0, a_0^{\{x_j\}}$, are respectively all four binary words. Hence H shatters $\{x_1, x_j\}$. This pair is not in the spine family (7.49), so it belongs to $\Omega_{\geq 2}$. The proof at a_n is the same after reversing the path: the four words $a_{j-1}, a_j, a_n^{\{x_j\}}, a_n$ realize all traces on $\{x_j, x_n\}$. $\square$

Proposition 7.79 (Endpoint trapped-wing-or-escape dichotomy). *Keep the path-spine notation of Proposition 7.59, and suppose that H has minimum cardinality among all nonempty cornerless ample families. At the left endpoint, at least one of the following two alternatives holds:*

1. $C_1 \subsetneq P_0$, every corner of P_0 belongs to C_1 , and

$$|P_0 \setminus C_1| \geq \rho_{\mathcal{D}_{\text{amp}}} \geq 10; \quad (7.104)$$

2. there are $g \in P_0$ and $j \in \{2, \dots, n\}$ such that $(g, a_0^{\{x_j\}}) \in H$.

The symmetric alternatives hold at the right endpoint, with C_n, P_n , and an external neighbour $a_n^{\{x_j\}}$, $1 \leq j \leq n-1$.

Consequently, if $\Delta(H; A, R) < 13$, then both endpoints have external core neighbours. If both endpoints pay the trapped-wing tax, then $\Delta(H; A, R) \geq 20$.

Proof. The families P_0 and $C_1 = P_0 \cap P_1$ are ample. They are proper pc-minors of H , and the minimum-cardinality hypothesis therefore puts both of them in $\mathcal{D}_{\text{amp}}$. The two opposite repair fibres of H belong to every P_i , so C_1 contains two repair antipodes. Isometry gives

$$|C_1| \geq r + 1. \quad (7.105)$$

Suppose first that $C_1 \subsetneq P_0$ and $\text{Cor}(P_0) \subseteq C_1$. The pair $C_1 \subsetneq P_0$ occurs in the minimum defining $\rho_{\mathcal{D}_{\text{amp}}}$, and hence

$$|P_0 \setminus C_1| \geq \rho_{\mathcal{D}_{\text{amp}}} \geq 10.$$

This is the first alternative.

Otherwise choose a corner g of P_0 outside C_1 when the inclusion is proper, and choose any corner g of P_0 when $C_1 = P_0$. Let K be the unique maximal repair cube of P_0 through g . In the first case the core edge $a_0 a_1$ is absent over g , while in the second case the incident cube in that direction is $K \times a_0 a_1$. Because a_0 is an endpoint of the core path, these account for every incident core direction whose other endpoint lies in A . If no core neighbour of a_0 outside A occurred over g , the local corner criterion would therefore make (g, a_0) a corner of H , a contradiction. Such a neighbour must have the form $a_0^{\{x_j\}}$ with $j \geq 2$, proving the second alternative. Reversing the path proves the assertion at a_n .

It remains to account for the excess. In a trapped alternative, (7.105) gives

$$|P_0| - (r + 1) \geq |P_0 \setminus C_1| \geq 10.$$

In an escape alternative, Proposition 7.57 gives endpoint-section excess at least two, and the escaping concept contributes at least one vertex to $H \setminus (Q_R \times A)$. Thus two trapped endpoints cost at least 20, while one trapped and one escaping endpoint cost at least $10 + 2 + 1 = 13$ in the exact decomposition (7.7). Therefore excess below 13 forces an escape at each endpoint. $\square$

Proposition 7.80 (Exact two-coordinate repair calculus). *Let u, v be two coordinates, let A be ample, and let $B, C \subseteq A$ be nonempty ample families with $B \cup C = A$. Define*

$$H_{00} = H_{11} = A, \quad H_{01} = B, \quad H_{10} = C. \quad (7.106)$$

Then

$$\text{Sh}(H) = \text{Sh}(A) \dot{\cup} \{S \cup \{u\} : S \in \text{Sh}(A)\} \dot{\cup} \{S \cup \{v\} : S \in \text{Sh}(A)\} \dot{\cup} \{S \cup \{u, v\} : S \in \text{Sh}(B) \cap \text{Sh}(C)\}. \quad (7.107)$$

Consequently, H is ample if and only if

$$B \cap C \text{ is ample} \quad \text{and} \quad \text{Sh}(B) \cap \text{Sh}(C) = \text{Sh}(B \cap C). \quad (7.108)$$

Equivalently,

$$\text{Sh}(B) \cup \text{Sh}(C) = \text{Sh}(A). \quad (7.109)$$

In that case

$$|H| = 3|A| + |B \cap C|. \quad (7.110)$$

Suppose now that H is ample. If a is a corner of A , denote its unique maximal cube by $Q_A(a)$ and put

$$K_a = \begin{cases} B, & a \in B \setminus C, \\ C, & a \in C \setminus B, \\ B \cap C, & a \in B \cap C. \end{cases}$$

Then both endpoint copies $(00, a)$ and $(11, a)$ are corners of H if and only if

$$Q_A(a) \subseteq K_a. \quad (7.111)$$

Proof. Let S use only coordinates of A . Since the two endpoint fibres in (7.106) are both A , the sets S , $S \cup \{u\}$, and $S \cup \{v\}$ are shattered by H exactly when S is shattered by A . The set $S \cup \{u, v\}$ is shattered exactly when all four fibres realize every trace on S ; because $B, C \subseteq A$, this is equivalent to $S \in \text{Sh}(B) \cap \text{Sh}(C)$. This proves (7.107).

Since A, B, C are ample and $B \cup C = A$,

$$|H| = 2|A| + |B| + |C| = 3|A| + |B \cap C|.$$

Comparing this identity with (7.107) shows that H is ample exactly when

$$|\text{Sh}(B) \cap \text{Sh}(C)| = |B \cap C|.$$

Now

$$\text{Sh}(B \cap C) \subseteq \text{Sh}(B) \cap \text{Sh}(C),$$

whereas the Sandwich inequality gives $|\text{Sh}(B \cap C)| \geq |B \cap C|$. Equality therefore holds throughout, which proves (7.108). Inclusion–exclusion then gives

$$|\text{Sh}(B) \cup \text{Sh}(C)| = |B| + |C| - |B \cap C| = |A|.$$

The union on the left is contained in $\text{Sh}(A)$, which also has size $|A|$, proving (7.109). Conversely, that shadow cover and inclusion–exclusion give $|\text{Sh}(B) \cap \text{Sh}(C)| = |B \cap C|$, so the Sandwich inequality gives (7.108). Finally, (7.110) has already been established.

It remains to inspect a core corner. At either endpoint copy of a , the old-coordinate edge directions generate $Q_A(a)$. If $a \in B \setminus C$, there is exactly one incident repair direction and the cube generated by all incident directions is present precisely when the corresponding second copy of $Q_A(a)$ lies in B , that is, precisely when $Q_A(a) \subseteq B$. The case $a \in C \setminus B$ is symmetric. If $a \in B \cap C$, both repair directions are incident and they generate the full repair square; its product with $Q_A(a)$ is present precisely when $Q_A(a) \subseteq B \cap C$. The local corner criterion now proves (7.111) at both endpoints. $\square$

Proposition 7.81 (Unrestricted two-coordinate shadow identity). *Let u, v be two coordinates and suppose that an ample family H has fibres*

$$H_{00} = H_{11} = A, \quad H_{01} = B, \quad H_{10} = C, \quad (7.112)$$

where A is nonempty and $A \subseteq B \cup C$. Put $K = B \cup C$. Then

$$D := B \cap C \subseteq A, \quad D \text{ is ample}, \quad (7.113)$$

and

$$\text{Sh}(A \cup B) \cap \text{Sh}(A \cup C) = \text{Sh}(A), \quad (7.114)$$

$$\text{Sh}(A) \cap \text{Sh}(B) \cap \text{Sh}(C) = \text{Sh}(D). \quad (7.115)$$

Consequently

$$\begin{aligned} \text{Sh}(H) &= \text{Sh}(K) \\ &\quad \dot{\cup} \{S \cup \{u\}, S \cup \{v\} : S \in \text{Sh}(A)\} \\ &\quad \dot{\cup} \{S \cup \{u, v\} : S \in \text{Sh}(D)\}, \end{aligned} \quad (7.116)$$

and

$$|H| = |K| + 2|A| + |D|. \quad (7.117)$$

Thus ampleness forces the external parts of the two middle fibres to be disjoint; it does not merely constrain their cardinalities.

Proof. The families $A, B, C, K, A \cup B$, and $A \cup C$ are conditionings or contractions of H , and hence are ample. For a set S of old coordinates, direct inspection of the four fibres gives the disjoint decomposition

$$\text{Sh}(H) = \text{Sh}(K) \dot{\cup} \{S \cup \{u\}, S \cup \{v\} : S \in I\} \dot{\cup} \{S \cup \{u, v\} : S \in J\},$$

where

$$I = \text{Sh}(A \cup B) \cap \text{Sh}(A \cup C), \quad J = \text{Sh}(A) \cap \text{Sh}(B) \cap \text{Sh}(C).$$

Let $W = B \cap C$ and $D_0 = A \cap W$. Because $(A \cup B) \cup (A \cup C) = K$ and all three unions are ample, inclusion–exclusion gives

$$\begin{aligned} |I| &= |A \cup B| + |A \cup C| - |\text{Sh}(A \cup B) \cup \text{Sh}(A \cup C)| \\ &\geq |A \cup B| + |A \cup C| - |K| = |A \cup W| = |A| + |W| - |D_0|. \end{aligned}$$

Moreover, $\text{Sh}(D_0) \subseteq J$, so the Sandwich inequality gives $|J| \geq |D_0|$.

On the other hand, ampleness of H and K and the ordinary inclusion–exclusion identity $|B| + |C| = |K| + |W|$ give

$$2|I| + |J| = |H| - |K| = 2|A| + |W|.$$

Combining the last three displays yields

$$2|A| + |W| \geq 2(|A| + |W| - |D_0|) + |D_0| = 2|A| + 2|W| - |D_0|.$$

Thus $|W| \leq |D_0|$. Since $D_0 \subseteq W$, equality holds and $W = D_0 \subseteq A$. We may therefore write $D = W$. Now $\text{Sh}(A) \subseteq I$ and $\text{Sh}(D) \subseteq J$, while

$$2|I| + |J| = 2|A| + |D|.$$

The Sandwich inequality for D forces equality in both inclusions: $I = \text{Sh}(A)$, $J = \text{Sh}(D)$, and $|\text{Sh}(D)| = |D|$. Hence D is ample. Substitution gives (7.116); the order identity follows either from that formula or directly from the four fibres. $\square$

Theorem 7.82 (Two-coordinate repair over a relatively peelable core). *In Proposition 7.81, suppose that A has a corner and that every nonempty proper ample subfamily $D' \subsetneq A$ omits some corner of A . Then H has a corner.*

The hypothesis holds, in particular, if either

1. $\text{VCdim}(A) \leq 2$, or
2. *after deleting constant coordinates, A uses at most four coordinates.*

Thus no cornerless ample family can have such a core separated by two repair coordinates, regardless of whether the repair is internal or external.

Proof. By Proposition 7.81, the overlap $D = B \cap C$ is an ample subfamily of A . We first choose a corner a of A such that either $D = A$ or $a \notin D$. If D is a nonempty proper subfamily, this is exactly the relative-corner hypothesis. If D is empty or $D = A$, use any corner of A . Denote the unique maximal cube of A through a by $Q_A(a)$.

Suppose first that $D = A$. Every vertex of $Q_A(a)$ belongs to all four fibres in (7.112). Hence $Q_{\{u,v\}} \times Q_A(a)$ lies in H . At the endpoint copy $(00, a)$, the old incident directions are exactly those of $Q_A(a)$, because a is a corner of A , and both repair directions are incident. All incident directions therefore generate the displayed cube, and the local corner criterion makes $(00, a)$ a corner of H .

It remains to consider $a \notin D$. The carrier family of the copies of $Q_A(a)$ in H is ample and isometric, by the carrier argument in the proof of Proposition 5.13. It contains the anchors of $Q_A(a)$ in the 00- and 11-fibres. Those two anchors differ precisely in u and v , so a shortest carrier path between them starts with a u - or a v -edge. Equivalently, the copy of $Q_A(a)$ in the 00-fibre extends to a cube in one repair direction.

Since $A \subseteq B \cup C$ and $a \notin B \cap C$, the concept a belongs to exactly one middle fibre. Thus $(00, a)$ has exactly one repair neighbor, and the carrier path says that this repair direction together with all old incident directions spans the extension of $Q_A(a)$. There are no other old directions because a is a corner of A . Hence all incident directions at $(00, a)$ generate a single cube, so $(00, a)$ is a corner of H .

It remains to verify the two sufficient conditions. For VC dimension at most two, translate a member of a nonempty $D' \subseteq A$ to the empty set and use the relative extension theorem of Mészáros and Rónyai [MR14, Theorem 4.1 and Lemma 4.1]. The last concept in a valid one-concept construction from D' to A is a corner of A outside D' . Taking D' to be a singleton also shows that A has a corner. For at most four active coordinates, the same conclusions follow from Theorem A.18. $\square$

Proposition 7.83 (Maximum-core corner shields). *In Proposition 7.80, suppose that A is maximum of VC dimension k , put $D = B \cap C$, and assume that H is ample. Then every maximal cube of A has dimension k , and for each k -set S there is exactly one S -cube in A . Consequently, every corner $a \in A \setminus D$ satisfies (7.111).*

If H is cornerless, it follows that

$$\text{corn}(A) \subseteq D, \quad Q_A(a) \setminus D \neq \emptyset \quad (a \in \text{corn}(A)). \quad (7.118)$$

Moreover, every corner cube $Q_A(a)$ is contained wholly in exactly one of B, C . Thus $A \setminus D$ meets every corner cube although it contains no core corner.

If, in addition, H has minimum cardinality among nonempty cornerless ample families and $|H| < 299$, then D , B , and C are nonempty proper subfamilies of A . In particular, the internal two-coordinate branch is a genuine two-sided cover by smaller ample families, not a one-sided repair with $B = A$ or $C = A$.

Proof. The complete-cube and reduction theorems for maximum classes say that A is the union of a complete collection of k -cubes and that reduction in the directions of a t -cube is maximum of VC dimension $k - t$ [RRB14, Section 2]. The vertex of the reduction representing that t -cube therefore lies in a $(k - t)$ -cube; lifting it shows that the original cube lies in a k -cube. Thus every maximal cube has dimension k . For a k -set S , the same simultaneous reduction has VC dimension zero, so it has one concept. Equivalently, A has exactly one S -cube.

Let $a \in \text{corn}(A) \setminus D$, and let S support its unique maximal cube. By (7.109), one of B, C shatters S . That family is ample, so it contains an S -cube; uniqueness forces this cube to be $Q_A(a)$. Since a belongs to exactly one of B, C , the family containing the cube is precisely K_a . Equation (7.111) now gives two corners of H . Hence cornerlessness forces $\text{corn}(A) \subseteq D$. For a corner in D , the same shadow-cover argument puts its unique cube in at least one of B, C . Equation (7.111) says that it cannot lie in both, proving (7.118) and the asserted monochromaticity.

Now assume that H is a least cornerless family. The family D is nonempty by (7.108). If $D = A$, then $B = C = A$ and $H = A \times Q_2$, the equality product already excluded in Corollary 7.6. Thus $D \subsetneq A$. Suppose, say, that $B = A$. Then $C = D$. The smaller nonempty ample family D cannot be cornerless, so choose a corner a of D . At its C -layer copy, both repair directions and exactly the old directions of $Q_D(a)$ are incident, and the cube generated by them is $Q_2 \times Q_D(a) \subseteq H$. This copy of a is a corner of H , a contradiction. Therefore $B \subsetneq A$, and symmetry gives $C \subsetneq A$. $\square$

Corollary 7.84 (Compatible corner for a VC-two core). *Under the hypotheses of Proposition 7.80, suppose in addition that A is maximum of VC dimension at most two and that H is ample. Then H has a corner. Consequently, the internal two-coordinate branch of a least cornerless ample family cannot have a maximum core of VC dimension at most two.*

Proof. Put $D = B \cap C$. This family is ample by (7.108), and it is nonempty because otherwise the empty support would belong to both $\text{Sh}(B)$ and $\text{Sh}(C)$ but not to $\text{Sh}(D)$. Translate a concept of D to the empty set. This preserves inclusion, ampleness, maximumness, VC dimension, cubes, and corners, and ensures that $\emptyset \in D \subseteq A$.

Mészáros and Rónyai proved that every ample family of VC dimension at most two containing the empty set belongs to their building class, and, more strongly, that whenever two such families are nested, the smaller one can be extended to the larger one by valid building steps [MR14, Theorem 4.1 and Lemma 4.1]. Each step adds one concept: a step of type A attaches it by one new coordinate edge, whereas a step of type B completes a square and attaches it by the two edges at the new vertex.

If $D = A$, take the concept added last in a valid construction of A from the singleton $\{\emptyset\}$; if A itself is a singleton, take its only concept. In either case this gives a corner a of A and $Q_A(a) \subseteq D = K_a$, so (7.111) applies.

Suppose now that $D \subsetneq A$, and take the concept a added by the last step of a valid relative construction from D to A . Thus $a \notin D$, and a is a corner. Let S be the support of its unique maximal cube $Q_A(a)$. If A has VC dimension one, the last step is of type A and its new coordinate occurs on no other edge of A ; hence $Q_A(a)$ is the unique S -cube. If A has VC dimension two, the last step cannot be of type A: after that last step its new coordinate would belong to no square, whereas a maximum VC-two family shatters every pair of active coordinates. The step is therefore of type B. It completes the square $Q_A(a)$, and this is the unique S -square in A . The last uniqueness also follows from Proposition 5.13, because the carrier of an S -cube in a VC-two family has no active coordinate. The case of VC dimension zero cannot occur when $D \subsetneq A$.

By (7.109), at least one of B and C shatters S . Since that family is ample, it contains an S -cube, necessarily the unique cube $Q_A(a)$. On the other hand, $a \notin D$ and $B \cup C = A$, so a belongs to exactly one of B, C ; only that family can contain $Q_A(a)$. Consequently $Q_A(a) \subseteq K_a$, and (7.111) again gives two corners of H . A least cornerless family cannot contain such an internal repair, which proves the final assertion. $\square$

Corollary 7.85 (Finite-rank remainder of the two-coordinate branch). *Let H be as in Corollary 7.6 and let $r = 2$. After the constant coordinates of A are deleted,*

$$\text{VCdim}(A) \geq 3 \quad \text{and} \quad 5 \leq \text{idim}(A) \leq 8.$$

More precisely, if $k = \text{VCdim}(A)$ and $n = \text{idim}(A)$, then

$$k = 3 \implies n \leq 8, \quad k = 4 \implies n \leq 7, \quad k \geq 5 \implies n \leq 6.$$

Proof. The VC-dimension bound follows from Theorem 7.82. The four-coordinate alternative of the same theorem gives the lower bound $\text{idim}(A) \geq 5$. If the active dimension n were at least nine, maximumness and $\text{VCdim}(A) \geq 3$ would give

$$|A| = \sum_{i=0}^{\text{VCdim}(A)} \binom{n}{i} \geq 1 + 9 + \binom{9}{2} + \binom{9}{3} = 130,$$

contrary to $|A| \leq 99$ in (7.5). For $k = 4$, the first excluded value is $\sum_{i=0}^4 \binom{8}{i} = 163$, and for $k \geq 5$ it is already $\sum_{i=0}^5 \binom{7}{i} = 120$. These give the two sharper bounds. $\square$

Proposition 7.86 (Core-repair shadow budget at order twenty-seven). *Let H have minimum cardinality among nonempty cornerless ample families, and suppose $|H| = 27$. Write $d = \text{idim}(H)$, and use the core A and repair set R from Corollary 7.6. Then H is not maximum, A is a pseudogeometric maximum VC-one path, and, with*

$$r = |R|, \quad n = d - r,$$

the path has exactly $n + 1$ vertices.

Let X be the n active coordinates of A . The shattered-set complex of H contains every pair xe with $x \in X$ and $e \in R$. If q is the number of its other two-element faces and t is the number of its three-element faces, then

$$27 = (n + 1)(r + 1) + q + t, \quad q \geq 1, \quad t \geq 1. \quad (7.119)$$

Consequently the only numerical possibilities are

$$\begin{array}{c|c|c|c} d & n & r & q+t \\ \hline 8 & 5, 4, 3, 2 & 3, 4, 5, 6 & 3, 2, 3, 6 \\ 9 & 2 & 7 & 3. \end{array} \quad (7.120)$$

In particular, the isometric-dimension-ten profile and the maximum VC-two core are impossible.

Proof. A maximum class of order 27 would have $27 = \sum_{i=0}^k \binom{n}{i}$. The only solution is $k = 1, n = 26$, and a maximum VC-one class is a tree and has a corner. Thus H is not maximum. Corollary 7.6 gives

$$|A| \leq \left\lfloor \frac{26}{r+1} \right\rfloor.$$

The class A is not a full cube: otherwise $\text{Susp}(A)$ would also be a full cube and hence ample, contrary to Proposition 7.4. If $r = 2$, then $|A| \leq 8$. A proper maximum class of VC dimension at least three has at least $\sum_{i=0}^3 \binom{4}{i} = 15$ concepts, so $\text{VCdim}(A) \leq 2$. Theorem 7.82 then gives a corner of H , a contradiction. Hence $r \geq 3$.

Now $|A| \leq 6$. A proper maximum VC-two class already has $1 + 3 + \binom{3}{2} = 7$ concepts, while VC dimension zero gives a singleton full cube and is excluded by the suspension condition. Hence A has VC dimension one.

The family A is maximum on all $n = d - r$ coordinates left after the conditioning; these are not auxiliary constant coordinates. A maximum VC-one family on n coordinates has $n + 1$ concepts and, in the pseudogeometric case, its one-inclusion graph is a path. Its shattered sets are the empty set and the n coordinate singletons. The proof of Proposition 7.7 shows that H shatters S and $S \cup \{e\}$ for every $S \in \text{Sh}(A)$ and $e \in R$. Thus it contains all nr cross-pairs between X and R .

The preliminary facet-excess bound gives order at least 34 in VC dimension at least four, so the endpoint has VC dimension three. Relative peeling and Proposition 7.158 give $d \geq 8$, whereas Corollary 7.155 gives $d \leq 10$. Its Sandwich equality now gives

$$27 = 1 + d + nr + q + t = (n + 1)(r + 1) + q + t.$$

There is at least one triangle because the VC dimension is three. The complete bipartite graph on $X \dot{\cup} R$ is triangle-free, so the pair shadow of that triangle contains at least one noncross pair; hence $q \geq 1$. Therefore $(r + 1)(d - r + 1) \leq 25$. For $d = 10$ the left side is at least 27 for $2 \leq r \leq 8$. For $d = 9$ only $r = 7$ remains, with value 24. For $d = 8$, the values at $r = 3, 4, 5, 6$ are 24, 25, 24, 21. Subtracting from 27 proves (7.120). $\square$

Proposition 7.87 (Small-excess bipartite shadows). *Let H be an irredundant ample family of VC dimension three and order 27. Suppose that, for a partition $E = X \dot{\cup} R$, every cross-pair xe with $x \in X$ and $e \in R$ belongs to $\text{Sh}(H)$. No cornerless such family exists in any of the following cases, or after exchanging the two blocks:*

$$(|X|, |R|, q + t) = (4, 4, 2), \quad (5, 3, 3), \quad (7, 2, 3),$$

where q and t have the meaning of Proposition 7.86.

Computer-assisted proof. For fixed block sizes and excess, the pair shadow of every chosen triangle forces its noncross pairs. After these pairs are inserted, the verifier chooses the remaining noncross pairs and quotients the resulting facet antichains by permutations within X and within R . It obtains two block-coloured types for $(4, 4, 2)$, nine for $(5, 3, 3)$, and seven for $(7, 2, 3)$. The two balanced types become isomorphic if the blocks are exchanged.

For each of these eighteen support systems, Proposition 5.14 reduces realization to the anchors of one affine cube on every facet support. The exact CNF chooses one anchor per facet, requires the union to have order 27, and forbids incidence one at every covered vertex. It also includes the carrier cardinality equalities, which follow from ampleness and improve propagation. The formulas have between 29,155 and 60,248 variables and between 67,857 and 138,340 clauses. CaDiCaL returns UNSAT on all eighteen formulas.

The deterministic verifier and its full list of facet representatives, formula sizes, and CNF digests are

```
verify_damp_small_excess_bipartite_shadows.py,  
damp_small_excess_bipartite_shadows.json.
```

The mathematical-profile digest is

```
e3383a65257458ea5d27bfdc27bef2e9f61beaeac9e295945de59e29a8a2b764.
```

Thus none of the listed support systems has a cornerless ample anchored realization. $\square$

Corollary 7.88 (Single oriented residual shadow at order twenty-seven). *Under the hypotheses of Proposition 7.86, one has $d = 8$ and*

$$(n, r, q + t) = (2, 6, 6).$$

Thus the maximum core is a three-vertex path on the two-coordinate block, the repair set is the six-coordinate block, and the shattered complex contains $K_{2,6}$ with exactly six further pair or triangle faces.

Proof. Proposition 7.87 excludes the dimension-nine row and the three middle dimension-eight rows of (7.120). The displayed row is the only remaining possibility. $\square$

Proposition 7.89 (The oriented $K_{2,6}$ endpoint is impossible). *Let H be an irredundant ample family of VC dimension three and order 27. Suppose that its coordinates split as $X \dot{\cup} R$ with $|X| = 2$ and $|R| = 6$, that its shattered complex contains every cross-pair of $K_{X,R}$ and exactly six further pair or triangle faces, and that two antipodal R -fibres of H are the same maximum VC-one family on X . Then H has a corner.*

Computer-assisted proof. Coordinate reversals normalize the two antipodal repair words to 0^6 and 1^6 and their common three-vertex path to $Q_X \setminus \{10\}$. The bounded support generator first chooses the triangles, inserts their forced noncross pair shadows, and then chooses the remaining noncross pairs. A coloured incidence-graph certificate quotients by permutations within X and within R . This gives 321 support systems; it is a census of the single oriented shadow layer in the hypotheses, not of cube families or of another cardinality.

For each support system, Proposition 5.14 gives an exact anchored-cube CNF. It chooses one affine cube on every facet support, requires their union to have order 27, forbids incidence one, and imposes the carrier-cardinality equalities forced by ampleness. Eight unit clauses fix the two normalized endpoint fibres to $Q_X \setminus \{10\}$. The formulas have between 29,123 and 30,046 variables and between 64,909 and 68,784 clauses. CaDiCaL returns UNSAT on all 321 formulas.

The support generator, verifier, and merged certificate are

```
explore_damp_k26_residual_structure.py,  
verify_damp_k26_residual_anchors.py,  
damp_k26_oriented_maximum_core.json.
```

The mathematical-profile digest is

21129e2baf496b277be92b60b369acd132dd05a50f3211bca5314ee9549ffb16.

The CNF is equivalent to a cornerless ample anchored realization with the prescribed maximum core, so unsatisfiability proves the claim. $\square$

Corollary 7.90 (Exclusion of order twenty-seven). *The least order of a nonempty cornerless ample family is not 27.*

Proof. If a least family had order 27, Corollary 7.88 would put it under Proposition 7.89, which supplies a corner. $\square$

In the two-coordinate case of Corollary 7.6, isometry first gives $A \subseteq H_{01} \cup H_{10}$: a shortest path between the two copies of $(a, 00)$ and $(a, 11)$ must use one of the intermediate copies of a . Proposition 7.80 handles the internal-repair branch in which both intermediate fibres lie in A . It reduces that entire branch to one explicit statement: under the compatible shadow cover (7.108), find a corner a of A for which (7.111) holds. Corollary 7.84 proves this in all dimensions through VC dimension two. Thus the remaining internal branch has VC dimension at least three and active dimension at most eight. Proposition 7.83 sharpens its obstruction: the proper ample overlap $D = B \cap C$ must contain every core corner, while the complement $A \setminus D$ meets every corner cube and each such cube is assigned to exactly one cover. It is therefore a bounded-rank relative-corner problem rather than a cardinality-by-cardinality search. The remaining external-repair branch is the case in which an intermediate fibre contains a word outside A ; those words already contribute strict geodesic-bundle excess and should be charged to the core cubes whose corners they repair.

The compatible-corner statement has a direct bounded check on the canonical maximum VC-dimension-two core

$$A = \{x \in Q_4 : |x| \leq 2\}.$$

Among its 426 ample subfamilies there are 2,169 ordered pairs (B, C) satisfying the hypotheses of Proposition 7.80. Every pair has at least three of the six core corners satisfying (7.111); the resulting repaired family has at least eight corners. Moreover, repeatedly deleting the least available corner completely peels all 2,169 families, whose orders range from 33 to 44. Thus this test produces neither a cornerless repair nor a cornerless descendant. The exhaustive profile is generated by `forbidden_pc_minor_campaign/explore_damp_two_coordinate_maximum_core_repairs.py` and has mathematical digest

b1d4d540fa0d0852f6910b8baad08c468c15d37dd992025e2c639d82f03f865c

This finite result is evidence for the compatible-corner lemma, not its proof in arbitrary dimension.

A second audit removes the choice of core. There are 832 proper labelled maximum families in Q_4 , forming eight orbits under cube automorphisms. Across one representative of each orbit, the script `forbidden_pc_minor_campaign/verify_damp_two_cover_compatible_corner_q4.py` checks all 10,820 compatible ordered ample covers. Every cover has a surviving core corner, and the same deterministic rule completely peels every repaired family. There are no cornerless repairs or cornerless descendants. The mathematical digest is

ff73f017733c75767269a7f81f276a26c0b9ff3a2daa0f4cfd65eef6a353b41e

This broader check also includes the punctured-cube maximum VC-dimension-three core; it remains a finite-dimensional falsifier rather than a proof of the proposed lemma.

The corollary is a dimension-free reduction, but it does not by itself preserve cornerlessness: a corner of the maximum fibre A may be covered in H by a cube using one of the conditioned coordinates in R . The remaining nonmaximum problem is therefore a labelled repair problem. One must show that these repairs either leave a proper cornerless pc-minor or require at least 299 concepts. This is the point at which the minimal-nonample classification supplies a genuine maximum core without silently assuming that the original family was maximum. Moreover, the near-extremal objects are now concrete: their sections over A are short antipodal paths, equality is a literal product with one common path, and the first excess measures deviations from that common path. Compatibility of these deviations across the cubes of A is the next dimension-free object to control.

7.2 Codimension-two tree wedges

The preceding example lives in twelve coordinates, but a maximum class of codimension two already has a rigid local form. Its complement is a tree, and its corners are read directly from the wedges of that tree. Besides controlling every five-coordinate projection of a maximum VC-dimension-three family, the formula will identify the only pseudogeometric maximum VC-two core on four active coordinates.

Proposition 7.91 (Codimension-two tree-wedge formula). *Let Z be a coordinate set of order $d \geq 3$, let $A \subseteq Q_Z$ be maximum of VC dimension $d - 2$, and put $D = Q_Z \setminus A$. The one-inclusion graph T_D is a tree on $d + 1$ vertices with exactly one edge in each coordinate of Z . Moreover, there is a bijection*

$$\text{corn}(A) \longleftrightarrow \{(v, \{e, f\}) : \text{the } e\text{- and } f\text{-edges of } T_D \text{ meet at } v\}.$$

Under this bijection, the corner corresponding to $(v, \{e, f\})$ is $v^{\{e, f\}}$, and its unique maximal cube has support $Z \setminus \{e, f\}$. Consequently,

$$|\text{corn}(A)| = \sum_{v \in D} \binom{\deg_{T_D}(v)}{2} \geq d - 1. \quad (7.121)$$

Equality holds if and only if T_D is a path; otherwise A has at least d corners.

For $X \subseteq Q_Z$, let $\bar{X} = \{\bar{x} : x \in X\}$, where $\bar{x}$ is the antipode of x in Q_Z . Then

$$|A \cap \bar{A}| = 2^d - 2d - 2 + |D \cap \bar{D}| = \begin{cases} 2^d - 2d, & T_D \text{ is a path,} \\ 2^d - 2d - 2, & T_D \text{ is not a path.} \end{cases} \quad (7.122)$$

In particular, by the affine-oriented-matroid criterion of Gärtner and Welzl [GW94, Theorem 51], A is the tope set of a simple affine uniform oriented matroid if and only if T_D is a path.

Proof. Complementation takes a maximum class of VC dimension three on five coordinates—and, in general, a maximum class of VC dimension $d - 2$ on d coordinates—to a maximum class of VC dimension one [CCMW22, Theorem 3.5]. Thus D is ample, has six vertices when $d = 5$, and has $d + 1$ vertices in general. It shatters every singleton, and therefore contains an edge in each coordinate direction. No direction occurs twice: in a one-dimensional ample family every edge is maximal, while distinct maximal cubes have distinct supports [CCMW22, Corollary 3.3]. Since an ample family is isometric and hence connected, these d edges form the asserted tree.

For $e \in Z$ and $x \in Q_Z$, remove the unique e -edge from T_D and let $H_e(x)$ be the component on which coordinate e has value x_e . For distinct $e, f \in Z$, the trace $x|_{\{e, f\}}$ is realized by D exactly when

$$H_e(x) \cap H_f(x) \neq \emptyset. \quad (7.123)$$

The restriction of D to $\{e, f\}$ realizes three of the four binary patterns. Hence the missing pattern fixes the unique cube of A with support $Z \setminus \{e, f\}$. It follows from (7.123) that the $(d-2)$ -cubes of A through x are indexed by the pairs $\{e, f\}$ for which $H_e(x)$ and $H_f(x)$ are disjoint.

If $x \in A$, the d subtrees $H_e(x)$, $e \in Z$, have empty total intersection, since a vertex in their intersection would be a member of D equal to x . Subtrees of a tree have the Helly property, so at least one pair is disjoint. Thus every vertex of A belongs to one of the displayed $(d-2)$ -cubes.

Suppose first that the e - and f -edges meet at v , and set $x = v^{\{e, f\}}$. The two halfspaces $H_e(x)$ and $H_f(x)$ are the two branches directed away from v , so they are disjoint. Every other pair intersects: if neither coordinate is e or f , both halfspaces contain v ; and if the pair is $\{e, g\}$ with $g \neq f$, both halfspaces contain the neighbor of v across the e -edge. The case $\{f, g\}$ is symmetric. Moreover, the e -neighbor of x is the neighbor of v across the f -edge of T_D , and the f -neighbor of x is the neighbor of v across the e -edge. Both belong to D . All remaining neighbors of x lie in the displayed $(Z \setminus \{e, f\})$ -cube. Every cube of A through x is therefore contained in that cube, so x is a corner with the stated unique maximal cube.

Conversely, suppose that $H_e(x)$ and $H_f(x)$ are the unique disjoint pair. If the e - and f -edges were not incident, choose an edge g strictly between them in T_D . Disjointness forces $H_e(x)$ and $H_f(x)$ to be the two outward branches. Neither component of $T_D - g$ meets both branches, contradicting the required intersections of $H_g(x)$ with each of them. Hence the two edges meet at a vertex v . For every $g \notin \{e, f\}$, the component $H_g(x)$ must meet both outward branches and therefore contains v . Thus $x_g = v_g$, while $x_e \neq v_e$ and $x_f \neq v_f$, proving $x = v^{\{e, f\}}$. If two pairs of halfspaces are disjoint, the corresponding distinct $(d-2)$ -cubes both contain x , so x is not a corner. Together with the preceding paragraph and the Helly argument, this proves the claimed bijection.

Counting incident pairs gives the equality in (7.121). Since every vertex of the tree has positive degree,

$$\sum_{v \in D} \binom{\deg_{T_D}(v)}{2} \geq \sum_{v \in D} (\deg_{T_D}(v) - 1) = 2d - (d+1) = d-1.$$

Equality holds exactly when every degree is at most two, that is, when the tree is a path. Every nonpath therefore has at least d wedges.

Finally,

$$|A \cap \bar{A}| = 2^d - |D \cup \bar{D}| = 2^d - 2d - 2 + |D \cap \bar{D}|.$$

If D contains u and its antipode, isometry gives a path of length d between them. This path uses all d edges of T_D , so T_D is a path and $u, \bar{u}$ are its endpoints. Conversely, the endpoints of a path T_D differ in every coordinate, because its edges have distinct coordinate labels. These are the only antipodal vertices in the path, so $|D \cap \bar{D}| = 2$ in the path case and is zero otherwise. This proves (7.122). The final equivalence is exactly the cited criterion, whose required antipodal-core cardinality is

$$2 \sum_{i=0}^{d-3} \binom{d-1}{i} = 2^d - 2d.$$

□

Corollary 7.92 (Five-coordinate repair obligation). *Let $H \subseteq Q_E$ be a cornerless maximum family of VC dimension three. For every five-set $Z \subseteq E$, the projection $A = H|_Z$ has at least four corners. If A is not the tope set of a simple affine uniform oriented matroid, it has at least five.*

More precisely, let p be one of these projected corners, let S be the support of its unique maximal cube in A , and let $x \in H$ be the unique lift of p in the S -cube of H . Then x lies in a second maximal

cube of H , with support R , such that

$$R \cap Z \subseteq S \quad \text{and} \quad R \setminus Z \neq \emptyset. \quad (7.124)$$

Suppose in addition that $d = |E| \geq 6$, and let $b_5(H)$ be the number of five-sets $Z \subseteq E$ for which $H|_Z$ is not an affine uniform oriented-matroid tope set. For $x \in H$, let t_x be the number of maximal cubes of H containing x . Then

$$\binom{d-4}{2} \sum_{x \in H} t_x(t_x - 1) \geq 4 \binom{d}{5} + b_5(H). \quad (7.125)$$

More precisely, if $\mathcal{I}(H)$ is the set of triples (S, x, R) such that the distinct S - and R -cubes of H both contain x , then the left-hand side may be replaced by

$$\sum_{(S, x, R) \in \mathcal{I}(H)} \binom{d - |S \cup R|}{2}. \quad (7.126)$$

Proof. Restrictions of maximum families are maximum, so the numerical assertions follow from Proposition 7.91. There is a unique S -cube of H , and its projection is the unique S -cube of A ; hence it contains a unique lift x of p . Cornerlessness supplies another maximal cube B through x ; write R for its support. The projection of B is a cube of A through p , so it is contained in the unique maximal cube through p . This gives $R \cap Z \subseteq S$. If $R \subseteq Z$, then $R \subseteq S$. Since both cubes contain x , the cube B is contained in the global S -cube, contradicting its maximality. Thus $R \setminus Z \neq \emptyset$.

Choose one such support R for every projected corner. This assigns each local corner to a triple $(S, x, R) \in \mathcal{I}(H)$. For a fixed triple, an eligible five-set has the form $Z = S \cup W$, where

$$W \in \binom{E \setminus (S \cup R)}{2}.$$

Thus (S, x, R) receives at most $\binom{d - |S \cup R|}{2}$ assignments. Summing these capacities and using the lower bounds of four for every Z , with one additional unit for each of the $b_5(H)$ non-pseudogeometric projections, proves (7.126). Maximality gives $R \not\subseteq S$, so $|S \cup R| \geq 4$. Moreover, $|\mathcal{I}(H)| \leq \sum_x t_x(t_x - 1)$. Replacing every summand by $\binom{d-4}{2}$ and then using this inequality gives (7.125). $\square$

The preceding capacity bound deliberately forgets which projected corners are repaired together. The next description retains that information and turns all five-coordinate corner obligations into collisions in a family of eight-colourings.

Proposition 7.93 (Tip-colouring identity). *Let $H \subseteq Q_E$ be maximum of VC dimension three, with $d = |E| \geq 5$. For each $S \in \binom{E}{3}$, let C_S be the unique three-cube of H with support S . If $e \in E \setminus S$, then $H|_{S \cup \{e\}}$ is a four-cube with one trace $q_{S,e}$ deleted. Define the e -tip of C_S to be the unique vertex $\tau_S(e) \in C_S$ satisfying*

$$\tau_S(e)|_S = q_{S,e}|_S.$$

For $x \in C_S$, put

$$A_S(x) = \{e \in E \setminus S : \tau_S(e) = x\}, \quad a_S(x) = |A_S(x)|.$$

Then the eight sets $A_S(x)$, $x \in C_S$, partition $E \setminus S$. Moreover, for distinct $e, f \in E \setminus S$, the projection $H|_{S \cup \{e, f\}}$ has a corner whose unique maximal cube has support S if and only if

$$\tau_S(e) = \tau_S(f).$$

Consequently,

$$\sum_{Z \in \binom{E}{5}} |\text{corn}(H|_Z)| = \sum_{S \in \binom{E}{3}} \sum_{x \in C_S} \binom{a_S(x)}{2}, \quad \sum_{x \in C_S} a_S(x) = d - 3. \quad (7.127)$$

Let

$$\epsilon_5(H) = \sum_{Z \in \binom{E}{5}} \left(\sum_{v \in T_Z} \binom{\deg_{T_Z}(v)}{2} - 4 \right),$$

where T_Z is the complement tree of $H|_Z$. If H is cornerless, then

$$4 \binom{d}{5} + \epsilon_5(H) = \sum_{S \in \binom{E}{3}} \sum_{x \in C_S} \binom{a_S(x)}{2} \leq \binom{d}{3} \binom{d-4}{2}. \quad (7.128)$$

In particular, no cornerless maximum VC-dimension-three family exists on five or six coordinates.

Proof. The projection to $S \cup \{e\}$ is maximum of VC dimension three and therefore has fifteen vertices. Its unique missing trace lies on the side opposite the projected S -cube, so its restriction to S selects a unique vertex of the global cube C_S . This proves that $\tau_S(e)$ is well-defined. Every outside coordinate has one tip, which gives the partition and the second identity in (7.127).

Write $Z = S \cup \{e, f\}$. In the punctured four-cube $H|_{S \cup \{e\}}$, the vertex of the projected S -cube whose S -trace is $q_{S,e}|_S$ is the unique corner supported by S . Thus, if a vertex p of $H|_Z$ is a corner supported by S , its projections after deleting f and after deleting e show that its unique lift in C_S equals both $\tau_S(e)$ and $\tau_S(f)$.

Conversely, suppose $\tau_S(e) = \tau_S(f) = x$, and put $p = x|_Z$. After deleting either outside coordinate, p is a corner supported by S . If a cube of $H|_Z$ through p used direction e or direction f , projecting away the other outside coordinate would give a cube through one of these four-coordinate corners that is not contained in its unique S -cube. This is impossible. Every cube through p therefore lies in the projected S -cube, so p is a corner supported by S . We have proved the claimed equivalence. Counting the pairs in each fibre gives the first identity in (7.127).

Proposition 7.91 identifies the left-hand side of that identity with $4 \binom{d}{5} + \epsilon_5(H)$. If some fibre has size $d - 3$, then its vertex x is the tip for every direction outside S . The four-coordinate corner characterization just used shows that no cube through x has a direction outside S ; hence the global S -cube is the unique maximal cube through x , and x is a corner. Thus, in a cornerless family every part of the partition of $E \setminus S$ has size at most $d - 4$. Among nontrivial partitions of a $(d - 3)$ -element set, the convex sum $\sum_i \binom{a_i}{2}$ is maximized by the partition $(d - 4) + 1$. Summing over the $\binom{d}{3}$ supports proves (7.128). For $d = 5$ its right-hand side is zero while its left-hand side is at least four; for $d = 6$, the two sides are at least 24 and at most 20, respectively. $\square$

The collision identity is the second member of a whole hierarchy. This hierarchy is useful because a lower-dimensional corner theorem can be fed directly into the next ambient dimension, without enumerating the classes in that dimension.

Proposition 7.94 (Projection-corner moment identity). *Keep the notation of Proposition 7.93. For every integer r with $4 \leq r \leq d$,*

$$\sum_{Z \in \binom{E}{r}} |\text{corn}(H|_Z)| = \sum_{S \in \binom{E}{3}} \sum_{x \in C_S} \binom{a_S(x)}{r-3}. \quad (7.129)$$

In particular, the full tip fibres $a_S(x) = d - 3$ are in bijection with the corners of H .

For $r \geq 4$, let c_r denote the minimum number of corners among maximum VC-dimension-three families on r coordinates. If H is cornerless and $5 \leq r < d$, then

$$c_r \binom{d}{r} \leq \binom{d}{3} \binom{d-4}{r-3}, \quad \text{or equivalently} \quad c_r \leq \binom{r}{3} \frac{d-r}{d-3}. \quad (7.130)$$

If H is cornerless, then for $r = d - 1$ the left-hand side of (7.129) counts exactly the near-full fibres $a_S(x) = d - 4$. More precisely, a corner of $H|_{E \setminus \{e\}}$ supported by S corresponds to the unique fibre

$$A_S(x) = E \setminus (S \cup \{e\}). \quad (7.131)$$

Thus any lower bound on corners in dimension $d - 1$ becomes, in one step, a labelled near-full-fibre obligation in dimension d .

Proof. Fix $Z \in \binom{E}{r}$ and $S \in \binom{Z}{3}$. The projection $H|_Z$ is again maximum of VC dimension three, and its unique S -cube is the projection of C_S . A vertex p of that cube has a unique lift $x \in C_S$.

We claim that p is a corner of $H|_Z$ supported by S if and only if

$$\tau_S(e) = x \quad \text{for every } e \in Z \setminus S. \quad (7.132)$$

Suppose first that p is such a corner. If $\tau_S(e) \neq x$ for some $e \in Z \setminus S$, then the trace that agrees with p on S and has the opposite e -value is present in $H|_{S \cup \{e\}}$. Projections commute, so it has a lift $y \in H|_Z$. The vertices p and y agree on S . Since the maximum class $H|_Z$ is ample, its one-inclusion graph is isometric; a shortest p, y -path therefore begins with an edge at p in a direction outside S . A maximal cube containing this edge is not contained in the S -cube, contradicting that p is a corner supported by S . Hence $\tau_S(e) = x$ for every such e .

Conversely, assume (7.132). A cube of $H|_Z$ through p that is not contained in the S -cube uses some direction $e \in Z \setminus S$. Its projection to $S \cup \{e\}$ would then give a cube through the e -tip x that is not contained in the S -cube, contradicting the four-coordinate tip characterization. This proves the claim.

For fixed S and x , condition (7.132) holds for precisely $\binom{a_S(x)}{r-3}$ choices of Z . Every corner of a maximum VC-dimension-three projection lies in a unique maximal three-cube, so summing over S and x proves (7.129). Taking $r = d$ gives the assertion about full fibres.

Now suppose that H is cornerless. Then every fibre has size at most $d - 4$. For $r \geq 5$, superadditivity of $a \mapsto \binom{a}{r-3}$ on the nonnegative integers shows that, among nontrivial partitions of $d - 3$, the sum of these binomial moments is maximized by the partition $(d - 4) + 1$. Hence

$$\sum_{S,x} \binom{a_S(x)}{r-3} \leq \binom{d}{3} \binom{d-4}{r-3}.$$

Every r -coordinate projection has at least c_r corners. Combining this observation with (7.129) gives the first inequality in (7.130); cancelling $\binom{d}{r}$ and simplifying the binomial coefficients gives the second.

Finally take $r = d - 1$. Since all fibres have size at most $d - 4$, the binomial coefficient $\binom{a_S(x)}{d-4}$ is one exactly for a near-full fibre and zero otherwise. Such a fibre omits a unique coordinate $e \in E \setminus S$, and the already proved corner criterion identifies its vertex with a corner of $H|_{E \setminus \{e\}}$. This is precisely (7.131). $\square$

For $r = 5$, Proposition 7.91 gives $c_5 = 4$. Substituting this value into (7.130) forces $d \geq 7$, recovering the simultaneous exclusion of dimensions five and six. Higher values of r retain strictly more of the projection structure: in particular, the case $r = d - 1$ records not merely how many projected corners exist, but which deleted coordinate owns each one.

Corollary 7.95 (Exact six-coordinate corner number). *Every maximum VC-dimension-three family on six coordinates has at least four corners, and this bound is attained. Thus $c_6 = 4$.*

Computer-assisted proof. The six five-coordinate projections contribute at least $6c_5 = 24$ corners. There are $\binom{6}{3} = 20$ tip partitions, each partitioning three labels. A partition with no full part contributes at most one to the second binomial moment, whereas a full part contributes three and corresponds to a global corner. If c is the number of global corners, then

$$24 \leq \sum_{S,x} \binom{a_S(x)}{2} \leq 20 + 2c,$$

and hence $c \geq 2$. This illustrates the intended use of (7.129): a corner theorem in one dimension becomes a simultaneous constraint on all larger dimensions.

For the sharp bound, Proposition 7.96 below is compiled with one corner marker for every source-cube vertex and a cardinality constraint allowing at most three markers. A vertex belonging to no second cube forces its marker, so any family with at most three corners satisfies the formula. Cube translation, the first outside tip, and its remaining coordinate orbit are normalized. The deterministic formula has 3,935 variables and 16,617 clauses. CaDiCaL versions 1.9.5 and 2.1.2 both return UNSAT. The latter proof is accepted by `drat-trim`; its backward core contains 6,528 original clauses and 24,836 lemmas and uses 2,317,233 resolution steps. The formula and proof digests are

e5c70eb2330bb0d221451aa67a087164c6ed0ec6a17fbb704ed93a96ba9ecef4,
dec117586483eb7eb3c293566fa44cb3ff1c103fc21f39b17457cc7eb7515503.

Finally, every six-coordinate projection of Hall's class has exactly four corners, as independently checked by `analyze_damp_projection_corner_moments.py`. Hence the lower bound is attained. $\square$

The tip maps admit an exact description using only the traces missing on four coordinates. Besides shortening the computational formulation, this separates the maximum-class compatibility conditions from cornerlessness.

Proposition 7.96 (Local missing-trace characterization). *Let E have $d \geq 4$ elements. For each $A \in \binom{E}{4}$, choose a word $q_A \in Q_A$, and define*

$$H(q) = \{x \in Q_E : x|_A \neq q_A \text{ for every } A \in \binom{E}{4}\}.$$

For $S \in \binom{E}{3}$, let $C_S(q)$ be the three-cube with support S whose fixed coordinates are

$$x_e = 1 - q_{S \cup \{e\}}(e), \quad e \in E \setminus S. \quad (7.133)$$

Then $H(q)$ is maximum of VC dimension three, and the q_A are its missing four-traces, if and only if

$$\text{for all } S \in \binom{E}{3}, A \in \binom{E}{4} \text{ with } |A \setminus S| \geq 2, \quad q_A(e) = q_{S \cup \{e\}}(e) \text{ for some } e \in A \setminus S. \quad (7.134)$$

When these conditions hold,

$$H(q) = \bigcup_{S \in \binom{E}{3}} C_S(q). \quad (7.135)$$

More explicitly, for $u \in Q_S$, let $x_{S,u} \in C_S(q)$ be the vertex with $x_{S,u}|_S = u$. The maximum class $H(q)$ is cornerless if and only if, for every pair (S, u) , there is a triple $T \neq S$ such that $x_{S,u} \in C_T(q)$. In missing-trace coordinates this last condition says that, for every $h \in E \setminus T$,

$$\begin{cases} u(h) = 1 - q_{T \cup \{h\}}(h), & h \in S, \\ q_{S \cup \{h\}}(h) = q_{T \cup \{h\}}(h), & h \notin S. \end{cases} \quad (7.136)$$

Thus the existence of a cornerless maximum VC-dimension-three class is an exact Boolean compatibility problem on the missing four-traces; no cardinality constraint or enumeration of concept classes is required.

Proof. Assume first that (7.134) holds. We claim that $C_S(q) \subseteq H(q)$ for every S . Fix $A \in \binom{E}{4}$. If $A = S \cup \{e\}$, then every vertex of $C_S(q)$ differs from q_A at e by (7.133). Otherwise $|A \setminus S| \geq 2$, and (7.134) supplies $e \in A \setminus S$ with $q_A(e) = q_{S \cup \{e\}}(e)$. The fixed value of $C_S(q)$ at e is its complement, so again no vertex of the cube realizes q_A . This proves the claim.

The family $H(q)$ omits a trace on every four-set, so its VC dimension is at most three. On the other hand, the cubes $C_S(q) \subseteq H(q)$ strongly shatter every set of size at most three. The Sandwich inequality and Sauer's inequality therefore give

$$\sum_{i=0}^3 \binom{d}{i} \leq |H(q)| \leq \sum_{i=0}^3 \binom{d}{i}.$$

Thus $H(q)$ is maximum. The same argument applied to the union of the cubes proves (7.135).

Conversely, let H be maximum of VC dimension three and let q_A be the unique trace missing on A . A word outside H cannot avoid all the q_A , because adjoining it would preserve VC dimension at most three and violate Sauer's bound. Hence $H = H(q)$. Its unique S -cube has the anchor in (7.133). Since this cube avoids q_A , one of its fixed coordinates in $A \setminus S$ differs from q_A whenever $|A \setminus S| \geq 2$; rewriting that difference gives (7.134).

Finally, (7.135) lists all maximal cubes: each has dimension three and distinct maximal cubes have distinct supports. Therefore $x_{S,u}$ is a corner precisely when it belongs to no $C_T(q)$ with $T \neq S$. Comparing the fixed coordinates of the two anchored cubes gives (7.136), completing the proof. $\square$

Corollary 7.97 (Seven-coordinate maximum classes). *Every maximum VC-dimension-three family on seven coordinates has a corner.*

Computer-assisted proof. Proposition 7.96 gives an exact Boolean formulation with $4\binom{7}{4} = 140$ mathematical missing-trace bits. We normalize a cube translation, the first outside tip, and the order of the remaining outside coordinates. Tseitin variables record the choice of a second cube through every source-cube vertex and the redundant five-coordinate tip-collision inequalities from Proposition 7.93. The resulting deterministic formula has 10,374 variables and 68,677 clauses.

Lingeling and CaDiCaL independently return UNSAT. Lingeling's DRAT proof is independently accepted by `drat-trim`; the backward core contains 5,281 original clauses and 17,677 lemmas, and uses 474,870 resolution steps. The durable files are

```
solve_damp_cornerless_maximum_vc3_missing_traces.cnf.py,
damp_cornerless_maximum_vc3_missing_traces_d7.cnf,
damp_cornerless_maximum_vc3_missing_traces_d7.drat,
damp_cornerless_maximum_vc3_missing_traces_d7.cnf.json,
damp_cornerless_maximum_vc3_missing_traces_d7_proof_verification.json.
```


The SHA-256 digests of the DIMACS formula and DRAT proof are, respectively,

```
900eb029ad9947f89982d45d474dd7cfa62bfcde56c5e3c79c34c07210b54ef0,
f3effdb6192b724325e07b6a1b99471d8294a999dab63fb8ab1a0732a0bedd5b.
```

Since the clauses are equivalent to maximum-class compatibility and cornerlessness, unsatisfiability proves the claim. $\square$

Corollary 7.98 (Eight-coordinate maximum classes). *Every maximum VC-dimension-three family on eight coordinates has a corner.*

Computer-assisted proof. We use the same exact missing-trace formulation as in Corollary 7.97. In dimension eight it has $4\binom{8}{4} = 280$ mathematical missing-trace bits. After the repair and tip-collision witnesses are introduced, the deterministic formula has 26,824 variables and 230,952 clauses. CaDiCaL and Kissat 4.0.4 independently return UNSAT.

Kissat’s binary DRAT trace was written directly in gzip-compressed form and independently checked by streaming its decompression to `drat-trim`. The checker retains 28,741 original clauses and 710,850 lemmas in the backward core, and uses 27,205,283 resolution steps. The durable files are

```
damp_cornerless_maximum_vc3_missing_traces_d8.cnf,
damp_cornerless_maximum_vc3_missing_traces_d8.cnf.json,
damp_cornerless_maximum_vc3_missing_traces_d8_kissat.proof.gz,
damp_cornerless_maximum_vc3_missing_traces_d8_proof_verification.json.
```

The SHA-256 digests of the DIMACS formula and compressed proof are, respectively,

```
c254620be1f44e24332b1e75f3e30882d52c9472d2c0250f50de952f1ebddced,
d5d4d2e66f240a777c3b28d8bd9b1e67d924a000a764c215cfdefe661616c816.
```

The independent verification record has mathematical digest

```
d9c94c7387253dff8a2be493bce0d7b9d805dcb3c1815c8c2739bc47838f7d40.
```

Proposition 7.96 identifies the formula’s unsatisfiability with the asserted corner conclusion. $\square$

The same question makes sense in every VC dimension. By Corollary 7.6, a least nonempty cornerless ample family of order below 299 is either a pseudogeometric maximum class or has a maximum core of order at most 99. For the first alternative, only finitely many pairs (d, n) are relevant. An ample family of VC dimension at most two is corner-peelable, and a maximum family of VC dimension $n - 1$ on n coordinates is a cube with one vertex deleted, which also peels. Hence $d \geq 3$, $n \geq d + 2$, and

$$\Phi_d(n) = \sum_{i=0}^d \binom{n}{i} < 299.$$

For every maximum family H of VC dimension d , each maximal cube has dimension d . Suppose that a k -cube with $k < d$ and support A is maximal at x . The reduction H^A is maximum of VC dimension $d - k \geq 1$, hence a connected family with at least two vertices, so $x|_{E \setminus A}$ has a neighbour in H^A . This neighbour extends the cube, contradicting maximality. Consequently, by [CCMW22, Lemma 4.1], a vertex of H is a corner exactly when it lies in one d -cube of H .

Proposition 7.99 (Further maximum classes with corners; computer-assisted). *Every maximum family of VC dimension d on n coordinates has a corner when*

$$(d, n) \in \{(4, 6), (4, 7), (4, 8), (5, 7), (5, 8), (6, 8)\}.$$

Computer-assisted proof. Choose one d -cube for every d -set of coordinates. Such a choice determines the union U of the chosen cubes. A maximum family of VC dimension d contains exactly one d -cube with each support, and these cubes cover it: their union strongly shatters every set of size at most d , hence has at least $\Phi_d(n)$ vertices. Conversely, if U shatters no $(d + 1)$ -set, then its shattered and strongly shattered sets coincide, so U is ample, and by the Sandwich equality it is maximum. By the preceding paragraph, a cornerless maximum family is therefore exactly such a choice in which U shatters no $(d + 1)$ -set and every vertex of U lies in at least two chosen cubes. These conditions are encoded in propositional logic with one variable per cube choice, one covering variable per vertex, a sequential counter for double coverage, and one auxiliary variable for each $(d + 1)$ -set and missing trace.

The formula is invariant under the hyperoctahedral group acting on the coordinates. We add lexicographic-leader constraints $v \leq_{\text{lex}} g(v)$ on the cube-choice variables for the generators g : the adjacent transpositions and the single-coordinate reflections. The lexicographically least member of an orbit of solutions satisfies these constraints, so they preserve satisfiability.

For each listed pair, CaDiCaL 2.1.2 reports UNSAT, and its DRAT proof is accepted by `drat-trim`. Without the double-coverage constraint, each formula is satisfiable, and the solver’s model is a maximum family. The formulas are generated by `explore_damp_group_cornerless_sat.py` with options `none --vc=d --vc-encoding --lex`. The compressed formulas and proofs, the checker logs, and SHA-256 digests are in `evidence/cornerless_maximum_small/`. For $(d, n) = (4, 8)$ the 1.16 GB proof is not archived. Its digest and checker log are recorded there; `drat-trim` retains 3,010,081 lemmas in its backward core. To recheck it, one regenerates the formula, compares its digest, reruns the solver and checks the fresh proof; a fresh proof need not coincide byte for byte with the original. Their sizes are as follows.

d	n	$\Phi_d(n)$	variables	clauses
4	6	57	2 571	5 299
4	7	99	12 205	27 252
4	8	163	49 858	118 878
5	7	120	6 519	13 232
5	8	219	34 752	75 548
6	8	247	16 386	32 958

For $d = 3$ the same encoding returns UNSAT for $n = 8$, in agreement with Corollary 7.98, and for $n = 7$ its DRAT proof is again accepted by `drat-trim`. $\square$

The same cube-choice description holds for every ample family, with the facets of its shattered complex in place of the d -sets.

Lemma 7.100 (Facet cubes of an ample family). *Let $C \subseteq Q_E$ be a nonempty ample family, and call the inclusion-maximal members of $\text{Sh}(C)$ its facets. For every facet S , the family C contains exactly one cube with support S , the facet cube of S . The maximal cubes of C are exactly its facet cubes. Consequently C is the union of its facet cubes, and a vertex of C is a corner if and only if it lies in exactly one facet cube.*

Conversely, let Σ be a down-closed family of subsets of E , and choose one cube with support S for every inclusion-maximal $S \in \Sigma$. If the union U of the chosen cubes shatters no set outside Σ , then U is ample and $\text{Sh}(U) = \Sigma$.

Proof. For $T \subseteq E$, the placements of the T -cubes of C form the reduction C^T . It is ample and shatters exactly the sets U with $T \cup U \in \text{Sh}(C)$. If S is a facet, then C^S shatters only the empty set, so $|C^S| = |\text{Sh}(C^S)| = 1$. Now let a cube with support T be maximal in C . Its placement has no neighbour in C^T , since a neighbour would extend the cube. Every nonempty ample family is connected, so C^T is this one point. Hence C^T shatters only the empty set, and T is a facet. Every vertex lies in a maximal cube, so the union statement follows. The corner statement is then [CCMW22, Lemma 4.1].

For the converse, every member of Σ is a subset of a chosen cube's support, so it is strongly shattered by U , and $\Sigma \subseteq \text{SSh}(U)$. By hypothesis $\text{Sh}(U) \subseteq \Sigma$. The Sandwich inequalities give

$$|\Sigma| \leq |\text{SSh}(U)| \leq |U| \leq |\text{Sh}(U)| \leq |\Sigma|,$$

so equality holds throughout, U is ample, and $\text{Sh}(U) = \Sigma$. $\square$

Proposition 7.101 (Ample families on seven coordinates have corners; computer-assisted). *Every nonempty ample family on at most seven coordinates has a corner. Consequently every nonempty cornerless ample family uses at least eight coordinates.*

Computer-assisted proof. A family on at most six coordinates has at most 64 vertices, and the only one with 64 is a full cube, which has corners. By Theorem 7.159 and its proof, every nonempty cornerless ample family has at least 64 vertices, so the claim holds on at most six coordinates. A family on seven coordinates with a constant coordinate lives on six coordinates, so we may assume all seven coordinates vary.

Let C be a nonempty cornerless ample family on seven coordinates and let d be its VC dimension. Ample families of VC dimension at most two admit corner peelings [CCMW22, Proposition 4.11], so $3 \leq d \leq 7$. By Lemma 7.100, C is the union of its facet cubes, and every vertex lies in at least two of them. The formula for fixed d has one variable for each subset of coordinates, saying that the subset belongs to a down-closed complex Σ containing all singletons, together with facet indicators. It has one cube-choice variable per facet, a covering variable per vertex, and a sequential counter demanding double coverage. It also requires that some set of size d lies in Σ , that none of size $d + 1$ does, and that the union shatters no set outside Σ . By the converse part of Lemma 7.100, the satisfying assignments are exactly the cornerless ample families of VC dimension d , with Σ their shattered complex. Lexicographic-leader constraints for the adjacent transpositions and the coordinate reflections are added, as in Proposition 7.99.

For each $d = 3, \dots, 7$, CaDiCaL 2.1.2 reports UNSAT, and `drat-trim` accepts its DRAT proof. The backward cores contain 1,299,213, 7,972,306, 4,607,445, 72,680, and 1 lemmas, respectively. The formulas have 58,531 variables and at most 148,463 clauses and are generated by `explore_damp_cornerless_ample_sat.py` with options `7 --lex --vc-exact=d`. Formula and proof digests, the checker logs, and the proofs for $d = 6, 7$ are in `evidence/cornerless_ample_n7/`. The proofs for $d = 3, 4, 5$ are between 0.5 and 3.9 GB and are not archived. They are rechecked by regenerating each formula, comparing its digest, rerunning the solver and checking the fresh proof. The same formula without the double-coverage constraint is satisfiable on small instances, and its models were checked independently to be ample. With Σ fixed to the sets of size at most three, the encoding reproduces the maximum-class results above. $\square$

Lemma 7.102 (Four-coordinate pseudogeometric VC-two core). *Let $A \subseteq Q_E$ be maximum of VC dimension two and use all four coordinates of E . Then A has at least three corners. If A is pseudogeometric, its complement is a five-vertex path and it has exactly three corners. Consequently, up to a cube automorphism, the only pseudogeometric possibility is*

$$A_4^{\text{pg}} = Q_4 \setminus \{0000, 1000, 1100, 1110, 1111\}. \quad (7.137)$$

Proof. Apply Proposition 7.91 with $d = 4$. It identifies the corners with the two-edge paths of the complement tree and gives at least $d - 1 = 3$ of them. The same proposition and the Gärtner–Welzl criterion say that pseudogeometricity is equivalent to the complement tree being a path. A path whose four edges have distinct coordinate labels is carried by a cube translation and a coordinate permutation to

$$0000, 1000, 1100, 1110, 1111.$$

Taking its complement proves the displayed normal form, and the path has exactly three two-edge subpaths. $\square$

Proposition 7.103 (Bulk reduction below order forty-eight). *Let H have minimum cardinality among all nonempty cornerless ample families, and suppose that $|H| < 48$. Then H is not maximum. For the maximum core A and repair set R supplied by Corollary 7.6, after deleting the constant coordinates of A , one has*

$$r := |R| \geq 3, \quad 1 \leq \text{VCdim}(A) \leq 2.$$

More precisely, exactly the following two structural regimes remain.

1. *If $\text{VCdim}(A) = 1$, then A is a path on $n + 1$ vertices for some $2 \leq n \leq 9$. If $n = 2$, then $3 \leq r \leq 12$; if $n \geq 3$, then*

$$(n + 1)(r + 1) \leq 40. \tag{7.138}$$

2. *If $\text{VCdim}(A) = 2$, then $A \cong Q_3 \setminus \{v\}$ and $r \in \{3, 4\}$.*

Consequently, excluding these two multi-coordinate repair regimes would raise the lower bound for the least ample forbidden pc-minor directly to 48 without examining the intermediate cardinalities separately.

Proof. Suppose first that H is maximum, and put $k = \text{VCdim}(H)$ and $d = \text{idim}(H)$. Maximum families of VC dimension at most two have a corner by Lemma A.16. For $k = 3$, the inequality

$$\sum_{i=0}^3 \binom{d}{i} < 48$$

forces $d \leq 6$. The cases $d \leq 4$ are full or once-punctured cubes, the five-coordinate case has at least four corners by Proposition 7.91, and the six-coordinate case has at least four corners by Corollary 7.95. For $k = 4$, order less than 48 forces $d \leq 5$, again leaving only a full or once-punctured cube. Finally, if $k \geq 5$, then either $d = k$ and H is a full cube, or

$$|H| \geq \sum_{i=0}^k \binom{k+1}{i} = 2^{k+1} - 1 \geq 63.$$

Every case contradicts cornerlessness. Thus H is not maximum.

Corollary 7.6 now gives a proper pseudogeometric maximum core A and $r \geq 2$ with

$$(r + 1)|A| + 1 \leq |H| \leq 47. \tag{7.139}$$

If $\text{VCdim}(A) \geq 3$, then, after its constant coordinates are deleted, properness and maximumness give

$$|A| \geq 1 + 4 + \binom{4}{2} + \binom{4}{3} = 15.$$

Together with (7.139), this forces $r = 2$, $|A| = 15$, and $A \cong Q_4 \setminus \{z\}$. Write B, C for the two middle repair fibres and put $K = B \cup C$, $D = B \cap C$. Proposition 7.81 gives

$$D \subseteq A, \quad D \text{ ample}, \quad |H| = |K| + 2|A| + |D|.$$

Both middle fibres are nonempty. Otherwise the other one contains all of A ; any core corner then has an endpoint copy whose unique maximal core cube extends in the single incident repair direction, making that copy a corner of H .

Since $A \subseteq K \subseteq Q_4$, either $K = A$ or $K = Q_4$. In the first case the repair is internal. Proposition 7.83 makes D a proper ample subfamily of A containing every corner of A , contrary to Theorem A.18.

Suppose that $K = Q_4$. The order identity gives $|D| \leq 1$. The empty support belongs to the three nonempty shadows of A, B, C , so (7.115) makes D nonempty; hence D is a singleton. Moreover

$$\text{Sh}(B) \cap \text{Sh}(C) = \{\emptyset\}.$$

Indeed, the double-link identity excludes every proper nonempty support, because A shatters all proper subsets of the four-coordinate set; the full support cannot be shattered by both B and C , since that would make both families equal to Q_4 . As B, C are ample, inclusion–exclusion now gives

$$|\text{Sh}(B) \cup \text{Sh}(C)| = |B| + |C| - 1 = |K| + |D| - 1 = 16.$$

Thus one of B, C shatters the full coordinate set and is Q_4 ; the other has only the empty shattered support and is the singleton D . Assume $B = Q_4$. Choose a corner a of A outside D , which is possible because $A = Q_4^-$ has four corners. At either endpoint repair copy of a , the only incident repair direction leads into B , and the unique maximal core cube through a is contained in B . The local corner criterion exposes both endpoint copies of a , a contradiction. The case $C = Q_4$ is symmetric.

Thus the core has VC dimension at most two. A maximum core of VC dimension zero is a singleton and has no active coordinate; it is the full zero-dimensional cube, again contradicting properness. Theorem 7.82 excludes $r = 2$, so $r \geq 3$.

Let first $\text{VCdim}(A) = 1$, and let n be its active dimension. Pseudogeometric maximumness makes its one-inclusion graph a path on $n + 1$ vertices, and properness gives $n \geq 2$. Proposition 7.58 gives

$$|H| \geq \begin{cases} 3(r+1) + 6, & n = 2, \\ (n+1)(r+1) + 7, & n \geq 3. \end{cases}$$

Combining this with $|H| \leq 47$ gives $r \leq 12$ in the first case and (7.138) in the second. Since $r \geq 3$, the latter inequality also gives $n \leq 9$.

Finally let $\text{VCdim}(A) = 2$ and let n again be its active dimension. The two-ended surcharge (7.45) gives $\Delta(H; A, R) \geq 2|\text{corn}(A)| \geq 2$. If $n \geq 5$, then $|A| \geq 1 + 5 + \binom{5}{2} = 16$, already contradicting (7.139). If $n = 4$, then $|A| = 11$, and Lemma 7.102 together with the two-ended surcharge gives

$$|H| \geq 4 \cdot 11 + 2 \cdot 3 = 50,$$

again a contradiction. Thus $n = 3$, and the only maximum VC-two family is a three-cube with one vertex deleted. It has three corners, so the same surcharge gives

$$|H| \geq 7(r+1) + 6.$$

The inequality $|H| \leq 47$ now forces $r \leq 4$, completing the reduction. $\square$

Proposition 7.104 (Bulk reduction through order sixty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families, and suppose that $|H| \leq 67$. Then H is not maximum. For the maximum core A and repair set R supplied by Corollary 7.6, after deleting the constant coordinates of A , one has*

$$r := |R| \geq 3, \quad 1 \leq \text{VCdim}(A) \leq 2.$$

More precisely, exactly the following profiles remain:

1. *If $\text{VCdim}(A) = 1$, then A is a path on $n + 1$ vertices for some $2 \leq n \leq 14$. If $n = 2$, then $3 \leq r \leq 19$; if $n \geq 3$, then*

$$(n + 1)(r + 1) \leq 60. \quad (7.140)$$

2. *If $\text{VCdim}(A) = 2$ and A uses three coordinates, then $A \cong Q_3 \setminus \{v\}$ and $3 \leq r \leq 7$.*

3. *If $\text{VCdim}(A) = 2$ and A uses four coordinates, then A is cube-isomorphic to the pseudogeometric core A_4^{pg} in (7.137), and $r \in \{3, 4\}$.*

There are no further core isomorphism types through order 67.

Proof. Suppose first that H is maximum, and put $k = \text{VCdim}(H)$ and $d = \text{idim}(H)$. For $k \leq 2$, relative VC-two peeling gives a corner. If $k = 3$, then $\sum_{i=0}^3 \binom{d}{i} \leq 67$ forces $d \leq 7$; the cases through four coordinates are full or once-punctured cubes, the cases $d = 5, 6$ have corners by Proposition 7.91 and Corollary 7.95, and the case $d = 7$ has a corner by Corollary 7.97. If $k = 4$, then $d \leq 6$. The cases $d \leq 5$ are again full or once-punctured cubes, while the case $d = 6$ is codimension two and has at least five corners by Proposition 7.91. If $k = 5$, only the full five-cube and the once-punctured six-cube have order at most 67, and both have corners. Finally, if $k \geq 6$, either $d = k$ and H is a full cube, or

$$|H| \geq \sum_{i=0}^k \binom{k+1}{i} = 2^{k+1} - 1 \geq 127.$$

Thus H cannot be maximum.

Corollary 7.6 now gives a proper pseudogeometric maximum core A and a repair set of order $r \geq 2$ with

$$(r + 1)|A| + 1 \leq |H| \leq 67. \quad (7.141)$$

The case $r = 2$ is impossible in one step. If $\text{VCdim}(A) \leq 2$, apply Theorem 7.82. If its VC dimension is at least three and its active dimension is at most four, apply the four-coordinate alternative of the same theorem. If instead its active dimension is at least five, then

$$|A| \geq 1 + 5 + \binom{5}{2} + \binom{5}{3} = 26,$$

so $3|A| + 1 \geq 79$, contrary to (7.141). Hence $r \geq 3$.

Suppose that $\text{VCdim}(A) \geq 3$. If the active dimension is at least five, then $4|A| + 1 \geq 105$, again impossible. The only remaining proper maximum core is $Q_4 \setminus \{v\}$, of order fifteen. It has four corners, so the two-ended surcharge (7.45) gives

$$|H| \geq 4 \cdot 15 + 2 \cdot 4 = 68,$$

another contradiction. Thus the core has VC dimension one or two.

In VC dimension one, pseudogeometric maximumness makes A a path on $n + 1$ vertices, with $n \geq 2$. Proposition 7.58 gives

$$|H| \geq \begin{cases} 3(r+1) + 6, & n = 2, \\ (n+1)(r+1) + 7, & n \geq 3. \end{cases}$$

Together with $|H| \leq 67$ this gives $r \leq 19$ in the first case and (7.140) in the second. Since $r \geq 3$, the latter inequality gives $n \leq 14$.

Finally suppose that the core has VC dimension two, and let n be its active dimension. If $n \geq 5$, then $|A| \geq 16$. Relative VC-two peeling gives at least two core corners. Indeed, apply Lemma A.16 with lower family $\{v\}$ to obtain a corner $q \neq v$, and apply it again with lower family $\{q\}$ to obtain a second corner distinct from q . The two-ended surcharge therefore strengthens the core budget to $|H| \geq 4|A| + 4 \geq 68$, impossible. For $n = 3$ the core is $Q_3 \setminus \{v\}$; its three corners and the two-ended surcharge give

$$|H| \geq 7(r+1) + 6,$$

and hence $r \leq 7$. For $n = 4$, Lemma 7.102 gives both the displayed pseudogeometric normal form and three corners. Thus

$$|H| \geq 11(r+1) + 6,$$

which leaves exactly $r = 3, 4$. □

Proposition 7.105 (Path-core exclusion through order sixty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 67$, and let the maximum core supplied by Corollary 7.6 have VC dimension one. Then H has a corner.*

Proof. Use the path-section system and baseline shadow Σ_0 from Corollary A.67. The support-interval exact sequence, the two-sweep reduction, and Corollary A.58 cancel every successful acquisition and place every remaining neutral or failure demand in the original excess $\text{Sh}(H) \setminus \Sigma_0$. Corollaries A.67 and A.68 reduce a saturated cornerless state to a same-position anchored packet or to packet-free equality.

If a packet ascends to support L , some facet T of $\text{Sh}(H)$ contains L . The uniform facet-rank bound gives

$$|H| \geq 2^{|T|+1} + 4 \geq 68$$

when $|L| \geq 5$; here we use Corollary 7.153. Hence every packet has rank three or four. The mixed rank-three profile separates by Corollary A.74; the pure profile either ascends or has a maximal repair three-cube owner by Corollary A.75, and the latter has strict excess by Propositions A.76 and A.82. Rank four is closed by Corollary A.92. In the packet-free case, Corollary A.71 makes the whole shattered complex two-dimensional and exposes a corner. All alternatives contradict cornerlessness. □

Corollary 7.106 (VC-two core frontier through order sixty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families and suppose that $|H| \leq 67$. Then its maximum core has VC dimension two and is one of exactly the following normalized types:*

core	$ A $	repair rank
$Q_3 \setminus \{v\}$	7	$3 \leq r \leq 7,$
A_4^{pg}	11	$r \in \{3, 4\}.$

Proof. Apply Proposition 7.104 and exclude its path row by Proposition 7.105. □

Proposition 7.107 (Two-ended facet-orbit quotient). *Keep the hypotheses and notation of Corollary 7.106, and write the two equal antipodal core fibres as*

$$H_{0_R} = H_{1_R} = A.$$

Choose a corner a of the normalized core whose unique maximal core cube is a square, and denote its support by S . At each of the two endpoint copies of this square, choose a maximal cube of H that contains it. The two supports have the form

$$S \dot{\cup} U, \quad S \dot{\cup} V, \quad (7.142)$$

where

$$1 \leq |U|, |V| \leq 2, \quad U \not\subseteq V, \quad V \not\subseteq U. \quad (7.143)$$

After permuting the repair coordinates, the ordered pair (U, V) is therefore exactly one of

type	U	V
1/1	$\{1\}$	$\{2\}$,
1/2	$\{1\}$	$\{2, 3\}$,
2/1	$\{1, 2\}$	$\{3\}$,
2/2, intersecting	$\{1, 2\}$	$\{1, 3\}$,
2/2, disjoint	$\{1, 2\}$	$\{3, 4\}$.

(7.144)

The last row occurs only when $r \geq 4$. Thus the entire VC-two frontier through order 67 reduces to four ordered extension types in repair rank three and five in every larger repair rank. If the two antipodal repair endpoints are left unoriented, the 1/2 and 2/1 rows are equivalent. Hence there are only three genuine types in repair rank three and four in every larger repair rank; the reduction does not branch on the cardinality of H .

Proof. The chosen square in the endpoint fibre is contained in a maximal cube of H . Its support contains S and contains no further core coordinate, because restriction to that endpoint fibre would otherwise give a core cube properly containing the unique maximal core cube through a . Consequently its support is $S \dot{\cup} U$ for some $U \subseteq R$; the same argument at the opposite endpoint gives $S \dot{\cup} V$.

The least cornerless family H is periphery-free. If either displayed maximal cube had rank at least five, the uniform facet-rank bound, Corollary 7.153, would give

$$|H| \geq 2^6 + 4 = 68,$$

contrary to $|H| \leq 67$. Since $|S| = 2$, this proves $|U|, |V| \leq 2$.

The two repair supports are nonempty. Indeed, if $U = \emptyset$, the core square at the first endpoint would itself be a maximal cube. The copy of that square at the opposite endpoint lies in some maximal cube. Its support either properly contains S , contrary to the antichain property of maximal-cube supports, or equals S , contrary to the fact that distinct maximal cubes of an ample family have distinct supports. The same argument excludes $V = \emptyset$.

The supports in (7.142) belong to distinct maximal cubes: their repair projections contain the two endpoints of Q_R , whose distance is $r \geq 3$, whereas either cube uses at most two repair coordinates. Maximal-cube supports are an antichain, and distinct maximal cubes of an ample family cannot have the same support. Hence the two supports, and therefore U and V , are incomparable.

It remains only to take orbits under permutations of R . Normalize U to $\{1\}$ or $\{1, 2\}$. If $|U| = 1$, incomparability makes V disjoint from U , leaving the first two rows. If $|U| = 2$ and $|V| = 1$, the

unique element of V lies outside U . If both sets have order two, their intersection has order one or zero, giving the final two rows. These are all the stabilizer orbits listed in (7.144).

Finally, complement every repair coordinate and leave every core coordinate fixed. This cube automorphism exchanges the two antipodal repair fibres, preserves their common core A , and replaces (U, V) by (V, U) . It therefore identifies the 1/2 and 2/1 rows. The other three rows are fixed as types by endpoint exchange, proving the unoriented count. $\square$

Proposition 7.108 (Core-coordinate wing squeeze). *Keep the hypotheses and notation of Corollary 7.106. Let X be the active coordinate set of the maximum VC-dimension-two core A , put $n = |X|$, and write $m = |H|$. For every $x \in X$,*

$$\deg_{\text{Sh}(H)}(x) \leq \left\lfloor \frac{m-20}{2} \right\rfloor. \quad (7.145)$$

In particular,

$$m \geq 20 + 2n(r+1). \quad (7.146)$$

Choose a square core corner and the two endpoint extensions $S \dot{\cup} U$ and $S \dot{\cup} V$ from Proposition 7.107. Let

$$t = |\{W \in \{U, V\} : |W| = 2\}|.$$

Then

$$m \geq 20 + 2n(r+1) + 4t. \quad (7.147)$$

Consequently, below order 64, the initial 26 unoriented endpoint orbits reduce analytically to the following 14:

core	r	possible endpoint types	
$Q_3 \setminus \{v\}$	3	1/1, 1/2, 2/2 intersecting	
$Q_3 \setminus \{v\}$	4	1/1, 1/2, 2/2 intersecting or disjoint	
$Q_3 \setminus \{v\}$	5	1/1, 1/2	
$Q_3 \setminus \{v\}$	6	1/1	(7.148)
A_4^{pg}	3	1/1, 1/2, 2/2 intersecting	
A_4^{pg}	4	1/1.	

There is no repair-rank-seven row. At this stage the unresolved frontier consists of one parametric 1/1 branch at repair ranks four, five, and six, rather than unrelated cardinality cases. The next proposition removes the rank-six endpoint of this branch without a finite search.

Proof. Fix an active coordinate e of H . Write its two sections as A_e^0, A_e^1 and put $B_e = A_e^0 \cap A_e^1$. The two-layer shadow identity makes B_e ample and gives

$$|B_e| = \deg_{\text{Sh}(H)}(e).$$

All three families have smaller order than H , so Lemma 7.8 puts them in $\mathcal{D}_{\text{amp}}$. Since H is periphery-free, both inclusions $B_e \subsetneq A_e^b$ are proper. No corner of either upper section lies outside B_e , for such a corner would lift to a corner of H . Proposition 2.184 therefore gives

$$|A_e^b \setminus B_e| \geq 10 \quad (b \in \{0, 1\}).$$

Hence

$$m = |A_e^0 \setminus B_e| + |A_e^1 \setminus B_e| + 2|B_e| \geq 20 + 2 \deg_{\text{Sh}(H)}(e),$$

which proves (7.145).

Both normalized cores are maximum of VC dimension two, so $\text{Sh}(A) = \binom{X}{\leq 2}$. The equal antipodal core fibres force

$$\binom{X}{\leq 2} * \left(\{\emptyset\} \cup \binom{R}{1} \right) \subseteq \text{Sh}(H).$$

For a fixed $x \in X$, exactly n members of $\binom{X}{\leq 2}$ contain x . The forced join therefore contributes $n(r+1)$ faces through x . Substitution in the degree cap proves (7.146).

Now fix $x \in S$. If $W \in \{U, V\}$ has order two, the facet support $S \dot{\cup} W$ is shattered. Downward closure supplies the two faces

$$\{x\} \dot{\cup} W, \quad S \dot{\cup} W.$$

Neither belongs to the forced join, because each contains two repair coordinates. Distinct two-element members of $\{U, V\}$ give distinct faces; if both have order two, their incomparability in (7.143) makes them distinct. Thus

$$\deg_{\text{Sh}(H)}(x) \geq n(r+1) + 2t.$$

The coordinate-degree cap now gives (7.147).

For $Q_3 \setminus \{v\}$ one has $n = 3$, so the lower bounds for $t = 0, 1, 2$ are $6r + 26, 6r + 30, 6r + 34$. For A_4^{pg} one has $n = 4$, and they are $8r + 28, 8r + 32, 8r + 36$. Requiring these quantities to be at most 63, and remembering that the disjoint 2/2 type needs at least four repair coordinates, gives exactly the table in (7.148). Its row counts are $3 + 4 + 2 + 1 + 3 + 1 = 14$. The only rows in that table not already covered by the stored proof certificates are the 1/1 rows with $Q_3 \setminus \{v\}$ and $r \in \{4, 5, 6\}$, proving the final assertion. $\square$

Corollary 7.109 (The order-sixty-four–sixty-five endpoint slab). *Let H have minimum cardinality among all nonempty cornerless ample families and suppose that $64 \leq |H| \leq 65$. After exchanging the two repair endpoints, its core, repair rank, and endpoint type belong to one of the following seventeen orbits, grouped into six rows:*

core	r	available degree slots	possible endpoint types
$Q_3 \setminus \{v\}$	3	10	1/1, 1/2, 2/2 intersecting
$Q_3 \setminus \{v\}$	4	7	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	5	4	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	6	1	1/1
A_4^{pg}	3	6	1/1, 1/2, 2/2 intersecting
A_4^{pg}	4	2	1/1, 1/2.

In particular, orders 64 and 65 form one structural slab: the core-coordinate degree allowance and the endpoint-orbit list do not depend on which of the two cardinalities occurs.

Proof. The strengthened core frontier and the two-ended facet-orbit quotient apply through order 65. For both possible orders,

$$\left\lfloor \frac{|H| - 20}{2} \right\rfloor = 22.$$

After the forced join is removed, the number of available supports through one core coordinate is therefore

$$22 - n(r+1),$$

where $n = 3$ for $Q_3 \setminus \{v\}$ and $n = 4$ for A_4^{pg} . This gives the third column.

The lower bound (7.147) is $6r + 26 + 4t$ for the three-coordinate core and $8r + 28 + 4t$ for the four-coordinate core, where t counts endpoint repair supports of order two. Retain the types whose bound is at most 65, and remember that a disjoint 2/2 type needs at least four repair coordinates. The six row counts are 3, 4, 4, 1, 3, 2, which sum to seventeen. $\square$

Corollary 7.110 (The order-sixty-six–sixty-seven endpoint slab). *Let H have minimum cardinality among all nonempty cornerless ample families and suppose that $66 \leq |H| \leq 67$. After exchanging the two repair endpoints, its core, repair rank, and endpoint type belong to one of the following fifteen orbits, grouped into five rows:*

core	r	available degree slots	possible endpoint types
$Q_3 \setminus \{v\}$	4	8	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	5	5	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	6	2	1/1, 1/2
A_4^{pg}	3	7	1/1, 1/2, 2/2 intersecting
A_4^{pg}	4	3	1/1, 1/2.

Thus orders 66 and 67 form one structural slab. Relative to the preceding slab, the three repair-rank-three punctured-core orbits disappear, and the only new endpoint orbit is the 1/2 type for $Q_3 \setminus \{v\}$ at repair rank six.

Proof. Corollary 7.106 and Proposition 7.107 apply through order 67. For either possible order,

$$\left\lfloor \frac{|H| - 20}{2} \right\rfloor = 23.$$

The number of degree slots beyond the forced join is therefore $23 - n(r + 1)$, with $n = 3$ for $Q_3 \setminus \{v\}$ and $n = 4$ for A_4^{pg} . This gives the third column.

The endpoint-rank charge (7.147) retains the listed types. It removes the repair-rank-three punctured-core row because that profile uses only $3 + 3 = 6$ active coordinates and hence has order at most $2^6 = 64$. It excludes repair rank seven because its three-coordinate-core baseline is $20 + 2 \cdot 3 \cdot 8 = 68$, and it excludes both 2/2 types in the repair-rank-six row because either would cost at least $20 + 2 \cdot 3 \cdot 7 + 8 = 70$. The five row counts are 4, 4, 2, 3, 2, which sum to fifteen. $\square$

Corollary 7.111 (The unified order-sixty-four–sixty-seven endpoint slab). *Let H have minimum cardinality among all nonempty cornerless ample families and suppose that $64 \leq |H| \leq 67$. After exchanging the two repair endpoints, its core, repair rank, and endpoint type belong to one of the following eighteen orbits:*

core	r	possible endpoint types
$Q_3 \setminus \{v\}$	3	1/1, 1/2, 2/2 intersecting
$Q_3 \setminus \{v\}$	4	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	5	1/1, 1/2, 2/2 intersecting or disjoint
$Q_3 \setminus \{v\}$	6	1/1, 1/2
A_4^{pg}	3	1/1, 1/2, 2/2 intersecting
A_4^{pg}	4	1/1, 1/2.

Thus all four orders form one structural slab with six core–rank rows and eighteen endpoint orbits. In particular, a certificate for this common list can move the lower bound from 64 directly to 68.

Proof. Apply Corollary 7.109 when $|H| \leq 65$, and Corollary 7.110 otherwise. The displayed table is the union of their orbit lists. It has row counts 3, 4, 4, 2, 3, 2, whose sum is eighteen. Among the five shared rows, the only type present in just one subrange is the rank-six 1/2 type. The entire repair-rank-three punctured-core row can occur only at order 64, since it uses six active coordinates. Hence the union is exhaustive throughout the stated range. $\square$

Lemma 7.112 (Core-carrier defect ledger). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| < 64$, and suppose that its maximum core is*

$$A = Q_3 \setminus \{v\}$$

on the active coordinate set X , with a repair set R of order $r \in \{4, 5, 6\}$. Put $\Sigma = \text{Sh}(H)$. For each $S \in \binom{X}{2}$, let $C_S \subseteq Q_R$ be the repair projection of the full positioned S -cube carrier, and put

$$q_S = \left| \text{Sh}(C_S) \setminus \left(\{\emptyset\} \cup \binom{R}{1} \right) \right|.$$

For $x \in X$, also put

$$p_x = |\{T \subseteq R : |T| \geq 2, \{x\} \dot{\cup} T \in \Sigma\}|, \quad h = \mathbf{1}_{\{x \in \Sigma\}}.$$

Define the defect allowance

$$\delta(H, r) = \left\lfloor \frac{|H| - 20}{2} \right\rfloor - 3(r + 1).$$

Then $0 \leq \delta(H, r) \leq 3(6 - r)$. If $S, S' \in \binom{X}{2}$ are the two core squares containing x , then

$$h + q_S + q_{S'} + p_x \leq \delta(H, r). \quad (7.149)$$

Consequently,

$$3h + 2 \sum_{S \in \binom{X}{2}} q_S + \sum_{x \in X} p_x \leq 3\delta(H, r) \leq 9(6 - r). \quad (7.150)$$

For later use, the allowance at the four odd order thresholds is

upper bound on $ H $	57	59	61	63
$r = 4$	3	4	5	6
$r = 5$	0	1	2	3

For every $S \in \binom{X}{2}$ and $x \in S$, one also has

$$q_S \leq p_x, \quad h + 2q_S + q_{S'} \leq \delta(H, r), \quad (7.151)$$

where S' is the other core square containing x .

If an endpoint extension of the core square S uses a two-element repair support T , then T contributes one unit to q_S and one unit to p_x for each $x \in S$. If $C_S, C_{S'}$ are antipodal geodesics and their path orders disagree on a repair pair T , then T contributes one unit to p_x , where $\{x\} = S \cap S'$. Thus at repair rank five all carrier defects, missing-core concepts, endpoint two-supports, and carrier-order disagreements must share a total defect budget of nine. At repair rank six every quantity in the two displays vanishes.

Proof. The forced join in the proof of Proposition 7.108 contributes exactly $3(r + 1)$ shattered supports through each $x \in X$. The coordinate degree cap therefore leaves at most

$$\left\lfloor \frac{|H| - 20}{2} \right\rfloor - 3(r + 1) = \delta(H, r)$$

further supports through x . This quantity is nonnegative because the forced join is present. Since $|H| \leq 63$, it is at most $3(6 - r)$.

The support-conservation identity for a projected cube carrier, Lemma 7.19, says that every

$$T \in \text{Sh}(C_S) \setminus \left(\{\emptyset\} \cup \binom{R}{1} \right)$$

supplies the shattered support $S \dot{\cup} T$ of H . For the two squares S, S' containing x , these give $q_S + q_{S'}$ distinct nonbaseline supports through x . The p_x supports have core part exactly $\{x\}$, while X , when it is shattered, has core part X . These three groups are pairwise disjoint and none belongs to the forced join. Counting them inside the available $\delta(H, r)$ slots proves (7.149).

Sum that inequality over $x \in X$. Each square S contains two core coordinates, so its term q_S occurs twice; h occurs three times. This proves (7.150).

For each support T counted by q_S , support conservation puts $S \dot{\cup} T$ in Σ . Downward closure then puts $\{x\} \dot{\cup} T$ in Σ for either $x \in S$. Distinct supports T remain distinct, so $q_S \leq p_x$. Substitution in (7.149) gives the second inequality in (7.151).

Finally, if a maximal endpoint cube has support $S \dot{\cup} T$ with $|T| = 2$, support conservation puts T in the acquired shadow of C_S . Downward closure also puts $\{x\} \dot{\cup} T$ in Σ for both $x \in S$. This proves the endpoint charge.

Now suppose that $C_S, C_{S'}$ are antipodal geodesics and their orders disagree on $T = \{e, f\}$. Their union realizes both orders of e, f and therefore shatters T . Both full square carriers contain the common core direction $x \in S \cap S'$. Whichever carrier supplies a given trace on T also supplies that trace on both x -sides. Hence $\{x\} \dot{\cup} T \in \Sigma$, so T is counted by p_x . When $r = 6$, the right side of (7.150) is zero, giving the final assertion. $\square$

Lemma 7.113 (Zero-carrier endpoint charge). *In the setting of Lemma 7.112, suppose that H is cornerless and that*

$$q_S = 0 \quad \text{for every } S \in \binom{X}{2}.$$

Then

$$\sum_{x \in X} p_x \geq \begin{cases} 3, & r = 4, \\ 2, & r = 5. \end{cases} \quad (7.152)$$

In particular, the zero-carrier branch is controlled by the bounded defect intervals

	$h = 0$	$h = 1$
$r = 4$	$3 \leq \sum_x p_x \leq 18$	$3 \leq \sum_x p_x \leq 15$
$r = 5$	$2 \leq \sum_x p_x \leq 9$	$2 \leq \sum_x p_x \leq 6$.

Proof. Translate the core so that its missing word is 111. Label its three maximal squares S_0, S_1, S_2 , and let a_i be the corner of the core belonging only to S_i . Thus a_i is the characteristic word of S_i . For $i \neq j$, the squares S_i, S_j share one core coordinate, say x_{ij} , whereas

$$a_i \triangle a_j = X \setminus \{x_{ij}\}. \quad (7.153)$$

Since $q_{S_i} = 0$, support conservation shows that the repair projection of the full S_i -carrier has shadow

$$\{\emptyset\} \cup \binom{R}{1}.$$

It is ample and contains $0_R, 1_R$, so it is an antipodal geodesic P_i . Write π_i for the order in which its repair coordinates are crossed. The repair section

$$F_i = \{g \in Q_R : (g, a_i) \in H\}$$

contains P_i . At either endpoint of P_i , the two core directions in S_i together with the incident path direction already span the product of the unique core square through a_i and the endpoint path edge. Since H is cornerless and there is no further core direction at a_i , the section F_i contains both

$$\{j_i\}, \quad R \setminus \{k_i\},$$

where j_i is not the first and k_i is not the last coordinate in the order π_i . If $E_i \subseteq \binom{R}{2}$ is the pair shadow of F_i , it therefore contains the two endpoint stars

$$\begin{aligned} L_i &= \{\{e, j_i\} : e <_{\pi_i} j_i\}, \\ R_i &= \{\{k_i, e\} : k_i <_{\pi_i} e\}. \end{aligned} \quad (7.154)$$

We record how these pair shadows are charged to the defect ledger. Let I_{ij} be the inversion set of the two orders π_i, π_j . Lemma 7.112 puts every member of I_{ij} among the pairs counted by $p_{x_{ij}}$. On the other hand, if $T \in E_i \cap E_j$, then each of the two repair sections realizes every trace on T . Equation (7.153) consequently puts $\{x\} \dot{\cup} T$ in the shadow of H for both $x \in X \setminus \{x_{ij}\}$. Hence, on defining

$$D_x = I_{ij} \cup \bigcup_{\substack{k < \ell \\ x \neq x_{k\ell}}} (E_k \cap E_\ell), \quad x = x_{ij}, \quad (7.155)$$

we have

$$\sum_{x \in X} p_x \geq \sum_{x \in X} |D_x|. \quad (7.156)$$

Suppose first that $r = 5$. If the three orders are not all equal, their three pairwise Kendall distances have sum at least two. The inversion sets in (7.155) then already give the required lower bound. If the three orders are equal, assume that the sets E_0, E_1, E_2 are pairwise disjoint. Number positions in the common order from 0 to 4, and write j_i, k_i also for the positions of these coordinates. Pairwise disjointness of the stars in (7.154) forces the j_i to be distinct, the k_i to be distinct, and

$$j_i \leq k_\ell \quad (i \neq \ell).$$

After relabelling, let $j_0 < j_1 < j_2$. Then $j_2 \geq 3$, so the two distinct values k_0, k_1 must both belong to $\{3\}$, a contradiction. Some E_i, E_j therefore share a pair. That pair belongs to the two sets D_x with $x \neq x_{ij}$, and again the total charge is at least two.

It remains to prove the rank-four bound. If the sum of the three Kendall distances is at least three, there is nothing to prove. We treat the only two smaller possibilities explicitly. For one fixed order 0123, the nine possible unions $L_i \cup R_i$ are displayed below; we abbreviate $\{a, b\}$ to ab .

$$\begin{array}{c|ccc}
& k=0 & k=1 & k=2 \\
\hline
j=1 & 01, 02, 03 & 01, 12, 13 & 01, 23 \\
j=2 & 01, 02, 03, 12 & 02, 12, 13 & 02, 12, 23 \\
j=3 & 01, 02, 03, 13, 23 & 03, 12, 13, 23 & 03, 13, 23.
\end{array} \tag{7.157}$$

If all three orders agree, three entries of this table contribute at least three to the right side of (7.156). Here is a short check that does not conceal a case. A total charge at most two would allow at most one pair of entries to meet, and their intersection would have order one; the third entry would have to be disjoint from both. The unique two-edge entry is $\{01, 23\}$. Any two two-edge entries therefore meet in two edges. If the disjoint third entry is $\{01, 23\}$, the table has only one entry disjoint from it, namely $\{02, 12, 13\}$, so the other two entries coincide in at least three edges. If the disjoint third entry has at least three edges, its complement has at most three edges. The other two entries would then both have to be the unique two-edge entry in order to meet in only one edge and fit inside that complement; they instead meet in two edges. Thus the charge is at least three.

Finally, if the sum of the Kendall distances is two, two orders agree and the third differs from them by one adjacent transposition. Reversal of the common order identifies the first and last transpositions. Relabel the coordinates in (7.157) by the transposition and compare the nine entries. The resulting minima are

$$\begin{array}{c|ccc}
\text{third order} & 1023 & 0213 & 0132 \\
\hline
\min \sum_x |D_x| & 3 & 5 & 3.
\end{array}$$

This covers every order triple with total inversion charge below three and proves the rank-four assertion. The upper endpoints in the final table of the statement are exactly (7.150) with $q_S = 0$. $\square$

Lemma 7.114 (Rank-five allowance-one normal form). *In the setting of Lemma 7.112, suppose that the repair rank is five and that $\delta(H, 5) \leq 1$. Then*

$$\delta(H, 5) = 1, \quad h = 0, \quad q_S = 0 \quad \left(S \in \binom{X}{2} \right), \quad 2 \leq \sum_{x \in X} p_x \leq 3, \quad p_x \leq 1 \quad (x \in X).$$

Consequently, up to permuting the three core coordinates, the vector $(p_x)_{x \in X}$ is either $(1, 1, 0)$ or $(1, 1, 1)$.

Proof. Suppose that $q_S > 0$ for some core square S , and choose $x \in S$. Equation (7.151) gives $p_x \geq q_S \geq 1$. The coordinate defect inequality at x would give

$$2 \leq q_S + p_x \leq h + q_S + q_{S'} + p_x \leq \delta(H, 5) \leq 1,$$

where S' is the other core square containing x , a contradiction. Thus every q_S vanishes.

Lemma 7.113 now gives $\sum_x p_x \geq 2$. Since $\delta(H, 5)$ is a nonnegative integer, the total defect inequality first forces $\delta(H, 5) = 1$ and then gives

$$3h + \sum_x p_x \leq 3\delta(H, 5) \leq 3,$$

so $h = 0$ and $\sum_x p_x \leq 3$. Finally, the coordinate defect inequality reduces to $p_x \leq 1$ for every x . The two listed vectors are the only possibilities. $\square$

Lemma 7.115 (Rank-five allowance-one carrier orders). *Under the hypotheses of Lemma 7.114, label the three maximal core squares S_0, S_1, S_2 , and let π_i be the repair-coordinate order of the antipodal geodesic carrier of S_i . If x_{ij} is the core coordinate common to S_i and S_j , then*

$$d_K(\pi_i, \pi_j) \leq p_{x_{ij}} \leq 1,$$

where d_K denotes Kendall distance. Consequently the three orders assume at most two values; if two values occur, they differ by one adjacent transposition.

In the normalized 1/1 endpoint branch, let the prescribed near and far repair coordinates of S_0 be 0 and 1. After permuting the other three repair coordinates, one may take

$$\pi_0 = (0, 2, 3, 4, 1).$$

There are then only thirteen labelled possibilities for (π_1, π_2) : the pair (π_0, π_0) , and, for each of the four orders τ obtained from π_0 by one adjacent transposition, the three pairs

$$(\pi_0, \tau), \quad (\tau, \pi_0), \quad (\tau, \tau).$$

Proof. The allowance-one normal form gives $q_{S_i} = 0$ for every i , so the proof of Lemma 7.113 identifies each carrier with an antipodal geodesic. The inversion set of (π_i, π_j) injects into the repair supports counted by $p_{x_{ij}}$, by Lemma 7.112. Its order is the Kendall distance, proving the displayed inequalities.

The graph of permutations joined by adjacent transpositions is bipartite, and in particular triangle-free. Three permutations at pairwise distance at most one therefore assume at most two values, and two distinct values are adjacent. In the fixed endpoint representative, the selected maximal cube at 0_R makes coordinate 0 the first edge of the S_0 carrier, while the selected maximal cube at 1_R makes coordinate 1 its last edge. The remaining coordinates are symmetric, giving the stated normalization of π_0 . Relative to this order, each of π_1, π_2 is either π_0 or an adjacent order; triangle-freeness forces the same adjacent order whenever both differ from π_0 . This gives $1 + 4 \cdot 3 = 13$ labelled pairs. $\square$

Lemma 7.116 (Rank-five allowance-two normal form). *In the setting of Lemma 7.112, suppose that the repair rank is five and that $\delta(H, 5) = 2$. Then exactly one of the following holds.*

1. $h = 1$, every q_S vanishes, and, up to permuting X , $(p_x)_{x \in X}$ is $(1, 1, 0)$ or $(1, 1, 1)$.
2. $h = 0$, every q_S vanishes, and, up to permuting X , $(p_x)_{x \in X}$ belongs to

$$\{(0, 0, 2), (0, 1, 1), (0, 1, 2), (0, 2, 2), (1, 1, 1), (1, 1, 2), (1, 2, 2), (2, 2, 2)\}.$$

3. $h = 0$, there is a unique square S with $q_S = 1$, all other $q_{S'}$ vanish, and

$$p_x = 1 \quad (x \in S), \quad p_z \in \{0, 1, 2\} \quad (X \setminus S = \{z\}).$$

Moreover, the unique repair support counted by q_S is the unique such support counted by each p_x , $x \in S$.

Thus allowance two has thirteen numerical defect shapes modulo core symmetry.

Proof. Equation (7.151) implies $q_S \leq 1$ for every S : if $q_S \geq 2$, then for either $x \in S$ the coordinate ledger spends at least $q_S + p_x \geq 4$. At most one square can have $q_S = 1$. Indeed, any two core squares meet in a coordinate x , and two nonzero carrier defects together with $p_x \geq 1$ would spend at least three units at x .

If $h = 1$, even one carrier defect would, at either coordinate of its square, spend at least $h + q_S + p_x = 3$ units. Hence all q_S vanish. The zero-carrier endpoint charge gives $\sum_x p_x \geq 2$, while the total and coordinate ledgers give $\sum_x p_x \leq 3$ and $p_x \leq 1$. This is the first case.

Suppose next that $h = 0$. If every q_S vanishes, the endpoint charge and the two ledgers give

$$2 \leq \sum_x p_x \leq 6, \quad 0 \leq p_x \leq 2.$$

The eight displayed triples are exactly the integer solutions modulo permutation. Finally, suppose that $q_S = 1$. For $x \in S$, one has $p_x \geq q_S = 1$, while the coordinate ledger gives $q_S + p_x \leq 2$; hence $p_x = 1$. The third coordinate satisfies only $0 \leq p_z \leq 2$. Each support counted by q_S is counted by both p_x , and all three counts are one, proving the final assertion. $\square$

Lemma 7.117 (Rank-five allowance-two carrier orders). *In either of the first two cases of Lemma 7.116, label the three maximal core squares S_0, S_1, S_2 , and let π_i be the repair-coordinate order of the carrier of S_i . If x_{ij} is the core coordinate common to S_i and S_j , then*

$$d_K(\pi_i, \pi_j) \leq p_{x_{ij}} \leq 2 - h. \quad (7.158)$$

After normalizing the designated carrier order to

$$\pi_0 = (0, 2, 3, 4, 1),$$

there are thirteen labelled possibilities for (π_1, π_2) when $h = 1$, and 106 when $h = 0$. In the latter case, the sorted triple of pairwise Kendall distances has one of the following five profiles, with the indicated multiplicity:

distance profile	000	011	022	112	222
number of labelled pairs	1	12	27	36	30.

(7.159)

Proof. Since $q_{S_i} = 0$, the repair projection of the full S_i -cube carrier is ample, contains the two antipodal endpoint words, and has shadow $\{\emptyset\} \cup \binom{R}{1}$. It is therefore an antipodal geodesic and has a well-defined order π_i . The inversion injection from Lemma 7.112 gives the first inequality in (7.158). The coordinate defect ledger gives $h + p_{x_{ij}} \leq 2$, proving the second.

The endpoint normalization fixes the first and last directions of π_0 , and a permutation of the remaining three directions gives the displayed representative. When $h = 1$, all three pairwise distances are at most one. As in Lemma 7.115, the triangle-free permutation graph leaves the base pair and three labelled pairs for each of the four adjacent transpositions, hence thirteen pairs.

It remains to count the radius-two case. Write $s_1, \dots, s_4$ for the adjacent transpositions relative to π_0 . The ball of Kendall radius two consists of the identity, the four s_i , and the following nine elements of length two:

$$s_1 s_3, s_1 s_4, s_2 s_4, \quad s_i s_{i+1}, s_{i+1} s_i \quad (1 \leq i \leq 3).$$

Each of the three pairwise distances is at most two, and their sum is even by permutation parity. Hence the five profiles in (7.159) are exhaustive. The profiles 011 and 022 arise by making two of the three orders equal, and therefore occur $3 \cdot 4 = 12$ and $3 \cdot 9 = 27$ times. Among the length-two

elements, the three products of commuting generators have two neighbours of length one, whereas each of the other six has one; there are consequently twelve incidences between the first and second spheres. The profile 112 comprises the twelve ordered pairs of distinct length-one elements and the $2 \cdot 12$ orientations of these incidences, for a total of 36. Finally, within the second sphere the three commuting products have degree four at Kendall distance two, and the other six have degree three. Thus profile 222 occurs $3 \cdot 4 + 6 \cdot 3 = 30$ times. Adding the identity pair gives the table, whose entries sum to 106. $\square$

Lemma 7.118 (One-defect antipodal carrier). *Let R be a five-element set, and let $C \subseteq Q_R$ be ample with $0_R, 1_R \in C$. Suppose that, for some $e, f \in R$,*

$$\text{Sh}(C) = \{\emptyset\} \cup \binom{R}{1} \cup \{\{e, f\}\}.$$

Then C is the union of exactly two antipodal geodesics. The two geodesics agree outside one square, traverse that square in the two opposite orders ef and fe , and the directions e, f are consecutive in either path order. Equivalently, C is one antipodal geodesic together with the missing fourth vertex of a square on two consecutive path directions. The square may occur in any of the four positions along the path.

Proof. Ampleness gives $|C| = |\text{Sh}(C)| = 7$. The family C is isometric and contains two vertices at distance five, so it contains an antipodal geodesic P of order six. Write z for the unique member of $C \setminus P$.

In an ample family, every shattered support is the support of a positioned cube. Hence C contains an $\{e, f\}$ -square. The geodesic P realizes three of the four traces on $\{e, f\}$, so this square contains z and three vertices of P . A monotone antipodal path contains three vertices of one positioned $\{e, f\}$ -square only when its e - and f -edges are consecutive. The vertex z is then the fourth vertex of that square. Replacing the two-edge ef -arc of P by the fe -arc through z gives a second antipodal geodesic. Every antipodal geodesic in C must use one of these two arcs, so these are the only two. A five-edge path has four consecutive pairs, giving the last assertion. $\square$

Corollary 7.119 (One-defect three-carrier synchronization). *In case 3 of Lemma 7.116, let S be the unique square with $q_S = 1$, and let T be its unique extra repair support. Write π, π' for the two geodesic orders in the carrier of S . Then T consists of two consecutive directions of π , and π' is obtained by transposing them. Each of the other two square carriers has order π or π' . If the two clean carriers have common core coordinate z and $p_z = 0$, then their orders are equal. After normalizing the designated endpoint carrier (choosing one of its two orders when it is defective), there are 32 labelled three-carrier patterns: sixteen when the designated carrier is defective, and eight when either one of the other two carriers is defective.*

Proof. The first assertion is Lemma 7.118. Let S' be a clean square and $x \in S \cap S'$. The allowance-two normal form says that $p_x = 1$ and that its unique support is T . Compare the geodesic carrier of S' first with the π -geodesic contained in the carrier of S , and then with the π' -geodesic. The inversion argument in Lemma 7.112 applies to either contained geodesic: every inversion belongs to the singleton-core defect family at x . Thus both inversion sets are contained in $\{T\}$. Since π, π' differ by the single adjacent transposition T , the only order having this property is π or π' . This proves the assertion for both clean carriers. Finally, an inversion between the two clean orders would be counted by p_z , so $p_z = 0$ forces equality. The square ear has four possible adjacent positions. If the designated carrier is defective, each clean carrier independently chooses one of its two geodesics, giving $4 \cdot 2 \cdot 2 = 16$ patterns. If one of the other carriers is defective, the remaining clean carrier has two choices, giving $4 \cdot 2 = 8$ patterns for either defective-carrier position. $\square$

Corollary 7.120 (Rank-five punctured-core exclusion through order fifty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 57$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order five. Then H has a corner.*

Proof. The threshold table in Lemma 7.112 gives $\delta(H, 5) \leq 0$. Since the defect allowance is nonnegative, it vanishes. Equations (7.149) and (7.150) then force

$$h = 0, \quad q_S = 0 \quad (S \in \binom{X}{2}), \quad p_x = 0 \quad (x \in X).$$

The zero-carrier endpoint charge, Lemma 7.113, instead gives $\sum_x p_x \geq 2$. This contradiction proves the claim. $\square$

Lemma 7.121 (Three-section endpoint packing on six coordinates). *Let*

$$P = (p_0, p_1, \dots, p_6), \quad p_t = \{0, \dots, t-1\},$$

be the standard antipodal geodesic in $Q_{[6]}$, and let F_1, F_2, F_3 be families containing P . For each i , let G_i be the graph on $[6] = \{0, \dots, 5\}$ whose edges are the two-element supports shattered by F_i . Suppose that the three edge sets are pairwise disjoint and that, for every i ,

$$\{j_i\} \in F_i \quad \text{for some } 1 \leq j_i \leq 5, \quad [6] \setminus \{k_i\} \in F_i \quad \text{for some } 0 \leq k_i \leq 4.$$

Then

$$\sum_{i=1}^3 |E(G_i)| \geq 11. \tag{7.160}$$

Moreover, none of the graphs G_i contains a triangle.

Proof. We use two elementary consequences of the fixed path order. First,

$$\begin{aligned} L_j &:= \{hj : 0 \leq h < j\} \subseteq E(G_i) \quad \text{if } \{j\} \in F_i, \\ R_k &:= \{kh : k < h \leq 5\} \subseteq E(G_i) \quad \text{if } [6] \setminus \{k\} \in F_i. \end{aligned} \tag{7.161}$$

Indeed, the displayed word supplies the 01-trace on each listed pair, while P supplies the other three traces.

Second, the path order is an umbrella-free order for every G_i : if $a < b < c$ and $ac \in E(G_i)$, then

$$ab \in E(G_i) \quad \text{or} \quad bc \in E(G_i). \tag{7.162}$$

To see this, choose $w \in F_i$ with $w_a = 0, w_c = 1$, which exists because ac is shattered and P already realizes the other three traces. If $w_b = 0$, then w supplies the missing trace on bc ; if $w_b = 1$, it supplies the missing trace on ab .

Pairwise edge-disjointness first implies that the j_i are distinct and that the k_i are distinct. It also gives

$$j_i \leq k_h \quad (i \neq h), \tag{7.163}$$

because $k_h < j_i$ would put the edge $k_h j_i$ in both R_{k_h} and L_{j_i} . Relabel the families so that $j_1 < j_2 < j_3$. Then $j_s \geq s$. The two k -values belonging to families other than F_3 are distinct, at least j_3 , and at most four. Hence $j_3 = 3$, and consequently

$$(j_1, j_2, j_3) = (1, 2, 3), \quad \{k_1, k_2, k_3\} = \{2, 3, 4\}, \quad k_3 = 2.$$

There are only two possible pairings of the remaining values. The forced edges from (7.161) are

	(j_1, k_1)		(j_2, k_2)		(j_3, k_3)
I	$(1, 3) : (01, 34, 35)$		$(2, 4) : (02, 12, 45)$		$(3, 2) : (03, 13, 23, 24, 25)$
II	$(1, 4) : (01, 45)$		$(2, 3) : (02, 12, 34, 35)$		$(3, 2) : (03, 13, 23, 24, 25)$

(7.164)

In either row these are eleven distinct edges, proving (7.160). They comprise every edge of K_6 except

$$M = \{04, 05, 14, 15\}. \quad (7.165)$$

Thus an additional edge of any G_i must belong to M .

It remains to exclude a triangle. In row I, the only possible triangles are $01h$, with $h \in \{4, 5\}$, in G_1 , and $a45$, with $a \in \{0, 1\}$, in G_2 ; the forced edges of G_3 together with M contain no triangle. A triangle $01h$ contradicts (7.162) on the long edge $0h$ with intermediate vertex 2: the edge 02 belongs to G_2 , while $2h$ belongs to G_3 , so edge-disjointness excludes both from G_1 . Similarly, a triangle $a45$ contradicts the umbrella-free condition on $a5$ with intermediate vertex 3: the edge $a3$ belongs to G_3 , while 35 belongs to G_1 .

In row II, only G_1 can contain a triangle, and it must again be of the form $01h$ or $a45$. The same two umbrella arguments apply, now using $02 \in E(G_2)$, $2h \in E(G_3)$, $a3 \in E(G_3)$, and $35 \in E(G_2)$. This excludes every triangle. $\square$

Proposition 7.122 (Rank-six punctured-core exclusion). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| < 64$, and suppose that the maximum core from Corollary 7.6 is $A = Q_3 \setminus \{v\}$ with a repair set R of order six. Then H has a corner.*

Proof. Write X for the three core coordinates, put $\Sigma = \text{Sh}(H)$, and translate the repair coordinates so that the two prescribed core fibres occur over 0_R and 1_R . The forced join in the proof of Proposition 7.108 contributes exactly $3(6+1) = 21$ faces through every $x \in X$. On the other hand, (7.145) and $|H| \leq 63$ give

$$\deg_{\Sigma}(x) \leq 21.$$

Consequently equality holds, and no support through a core coordinate lies outside the forced join

$$\binom{X}{\leq 2} * \left(\{\emptyset\} \cup \binom{R}{1} \right). \quad (7.166)$$

In particular, H has no concept whose core trace is v , since one such concept together with the two endpoint fibres would shatter X . Also, no shattered support can contain both a core coordinate and two repair coordinates.

The three maximal squares of A have a common repair carrier path. Indeed, for such a square support S , Lemma 7.19 and (7.166) show that the repair projection of its full S -cube carrier is ample and has shadow

$$\{\emptyset\} \cup \binom{R}{1}.$$

It contains $0_R, 1_R$, so it has seven vertices and is an antipodal geodesic. Two maximal core squares share a core edge. If their carrier geodesics ordered some repair pair in opposite ways, their union would shatter that pair in both sections of the shared core edge. The union of the pair with the core-edge coordinate would then be shattered by H , contrary to (7.166). Thus the three carriers are one path P . Permute R so that

$$P = (\emptyset, \{0\}, \{0, 1\}, \dots, R). \quad (7.167)$$

Every core word of A lies in a maximal core square, and hence every repair section

$$F_a = \{g \in Q_R : (g, a) \in H\}$$

contains P . Each F_a is ample because it is a conditioning of H .

If $a \neq b$ and a repair support T of order at least two were shattered by both F_a and F_b , choose a core coordinate x on which a and b differ. The two sections would realize every trace on T on the two opposite x -sides, so $T \cup \{x\}$ would be shattered. Hence

$$(\text{Sh}(F_a) \setminus \text{Sh}(P)) \cap (\text{Sh}(F_b) \setminus \text{Sh}(P)) = \emptyset \quad (a \neq b). \quad (7.168)$$

Let a_1, a_2, a_3 be the three corners of A . At the copy of a_i over the first endpoint of P , the two incident core directions and the first repair direction span the product of its unique core square with the first path edge. There is no incident core direction leading outside A . Cornerlessness therefore supplies another repair neighbour $\{j_i\}$, where $1 \leq j_i \leq 5$. The same argument at the opposite endpoint supplies $R \setminus \{k_i\}$, where $0 \leq k_i \leq 4$. The pair-shadow graphs of the three corner sections therefore satisfy Lemma 7.121. In particular, their union has at least eleven edges and none of them contains a triangle.

Suppose first that H shatters a pure repair support of order at least three. By downward closure it shatters a repair triple T , and ampleness supplies a full T -cube in one fixed core section F_b . The trace b belongs to A , by the first paragraph. If b is a core corner, its pair-shadow graph contains the triangle $\binom{T}{2}$, contrary to the endpoint-packing lemma. If b is not a core corner, the T -cube contributes at least four vertices outside P : an antipodal geodesic meets an affine three-cube in a contiguous subpath of at most four vertices. The eleven distinct pair supports in the corner sections contribute at least eleven further section vertices, because $P \subseteq F_a$ and both families are ample. Since no section outside A is nonempty,

$$|H| = 7|P| + \sum_{a \in A} |F_a \setminus P| \geq 49 + 11 + 4 = 64,$$

again a contradiction.

It remains that no pure repair triple is shattered. Then every F_a has VC dimension at most two. Each corner section F_{a_i} properly contains P , so Lemma A.16, applied to $P \subsetneq F_{a_i}$, supplies a corner $w \in F_{a_i} \setminus P$. The word w is not a prefix in the path order. Hence there are $e < f$ with $w_e = 0, w_f = 1$, and $P \cup \{w\}$ shatters $\{e, f\}$. If w occurred in any other core section F_b , that repair pair would be shattered in both sections, contrary to (7.168). There is also no section over the missing core trace v . Thus the concept (w, a_i) has no incident core edge. Its unique maximal repair cube in F_{a_i} is consequently its unique maximal cube in H , so (w, a_i) is a corner of H . This final contradiction proves the claim. $\square$

Proposition 7.123 (VC-two core exclusion through order fifty-four). *Let H have minimum cardinality among all nonempty cornerless ample families. If*

$$48 \leq |H| \leq 54,$$

then H has a corner.

Computer-assisted proof. Corollary 7.106 reduces H to a VC-two maximum core. For $A = Q_3 \setminus \{v\}$, the two-ended surcharge gives

$$|H| \geq 7(r+1) + 6,$$

so $r \leq 5$. For the pseudogeometric eleven-concept core A_4^{pg} , it gives

$$|H| \geq 11(r + 1) + 6,$$

so $r = 3$. Thus there are four core–repair profiles, not seven separate cardinality cases:

$$(Q_3 \setminus \{v\}, r), \quad r \in \{3, 4, 5\}, \quad (A_4^{\text{pg}}, 3).$$

For each profile we use the exact static facet-cube model of Proposition 5.14. The variables choose an antichain of facet supports, one anchored affine cube for each chosen support, their union K , and one order selector. The clauses enforce the two prescribed antipodal core fibres, the equality

$$|K| = \left| \bigcup_{T \in \mathcal{F}} 2^T \right|,$$

and incidence multiplicity different from one at every vertex. These conditions are equivalent to ampleness and cornerlessness; the formulas contain neither a recursive minor test nor a peeling search. Since $|H| < 64$, the strict facet-excess inequality restricts every facet to rank at most four.

The rank-three punctured-cube formula is already unsatisfiable without fixing an endpoint-extension orbit. In every other row, choose a square core corner. Proposition 7.107 gives three unoriented endpoint orbits when $r = 3$ and four when $r \geq 4$. The archived certificate keeps the two endpoint orientations separate and therefore contains the redundant pair 1/2, 2/1. One formula for each ordered orbit fixes the two corresponding maximal cubes. The resulting formulas are

core	r	orders	formulas	(variables, clauses)
$Q_3 \setminus \{v\}$	3	48–54	1	(3,440, 32,685)
$Q_3 \setminus \{v\}$	4	48–54	5	(8,024, 112,468–112,470)
$Q_3 \setminus \{v\}$	5	48–54	5	(18,524, 380,182–380,184)
A_4^{pg}	3	50–54	4	(8,024, 112,493–112,495).

Thus fifteen formulas cover every admissible profile and every order in the interval simultaneously. Only twelve are distinct modulo the full endpoint-exchange symmetry; retaining the three duplicates makes the certificate independent of this final quotient.

All fifteen formulas are unsatisfiable. An independently compiled **drat-trim** at source commit 2e3b2dc0ecf938addbd779d42877b6ed69d9a985 verifies every proof; the checker binary has SHA-256 digest

7f24ee5af22745ac2dbe3cf86cc4d61ef684694d3dce46875bb1f2bc6c8a421f.

For the largest verification core, the formula digest is

5b400716b5241d00f5dc4ddc636226bf42c9d3314c1a235c7dd944a1a3cffd05,

and the proof digest is

d5c61c2e31887ce4a5d2126c562f20b29714cd2b98609b8448f1b959d42120af.

The checker retains 59,722 of the 380,184 original clauses and 1,229,938 of the 3,249,406 proof lemmas, using 669,133,373 resolution steps and no RAT lemma. The generators, manifests, per-profile verification records, and aggregate certificate are stored under the prefixes **damp_vctwo_core_o48_54** and **damp_vctwo_below55**. The aggregate mathematical digest is

c1903eaf24e3f9cbb979eca762f5ee57c7ea0e338b820001b89eb365832d0839.

Hence no listed profile exists, proving the claim. $\square$

Proposition 7.124 (Rank-four punctured-core exclusion through order fifty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 57$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order four. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit quotient and the stored branch certificates leave only the 1/1 endpoint type. Translate and permute the repair coordinates so that its two endpoint supports are the fixed representative $\{0\}, \{1\}$. For every core coordinate, the forced join contributes fifteen shattered supports. Since $|H| \leq 57$, the coordinate-degree bound gives

$$\deg_{\text{Sh}(H)}(x) \leq \left\lfloor \frac{57 - 20}{2} \right\rfloor = 18.$$

Thus the core-coordinate extra-face cap is three. Equivalently, the defect allowance of Lemma 7.112 satisfies $\delta(H, 4) \leq 3$.

We compile the exact static facet-cube model under this cap. The formula contains every facet support of rank at most four and every possible anchor, enforces the two antipodal core fibres and the normalized endpoint cubes, and imposes ampleness and cornerlessness through exact facet-cube reconstruction. The strict facet-excess inequality makes the rank-four truncation exhaustive below order 64. No cardinality selector occurs. The three coordinate inequalities and their weighted sum

$$3h + 2 \sum_S q_S + \sum_x p_x \leq 9$$

are encoded explicitly.

There are two formulas, according as the missing core word is or is not realized. Each has 7,116 variables and 98,787 clauses. Their CNF SHA-256 digests are

1396bc9ad064f8cdce82e77577e82986bfff2b8359a6fe0f16426b23353b1984,
ccf75c091f423c0b8405e714eeb725fe6de757334fad3b5ca6cb9d612e96ca29.

Kissat returns UNSAT and emits proof traces with digests

27836a554d7c75cdb654e46fb02ca8554b3a4df5140d3564fe6de6d80550fd40,
597b2a593b8532ef850649eaf9cc9831cbbac46c7922b4a1370a3945f02d605e.

Independently compiled `drat-trim` verifies both. For $h = 0$, it retains 17,540 original clauses and 389,356 lemmas and uses 52,899,189 resolution steps. For $h = 1$, the corresponding numbers are 16,881, 216,279, and 27,061,423. The aggregate verification record is `damp_vctwo_q3minus_r4_deg3_below58_verification.json`; its mathematical digest is

1b133d3afa01d2d0700eb3716b9c9ddb301c43e21eca0baf3597ed3e378e3221.

The two missing-core states exhaust the normalized 1/1 branch, proving the claim. $\square$

Proposition 7.125 (Rank-four punctured-core exclusion through order fifty-nine). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 59$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order four. Then H has a corner.*

Computer-assisted proof. As in Proposition 7.124, the stored endpoint-orbit certificates reduce the problem to the normalized 1/1 endpoint type. The forced join contributes fifteen shattered supports through each core coordinate, while the coordinate-degree bound gives

$$\deg_{\text{Sh}(H)}(x) \leq \left\lfloor \frac{59 - 20}{2} \right\rfloor = 19.$$

Thus the core-coordinate extra-face cap is four, or equivalently $\delta(H, 4) \leq 4$.

We use the same exact facet-cube reconstruction as in the preceding proposition, now with the three coordinate inequalities at allowance four and the weighted inequality

$$3h + 2 \sum_S q_S + \sum_x p_x \leq 12.$$

The formula has no cardinality selector. The strict facet-excess inequality again makes the rank-four facet truncation exhaustive below order 64. Splitting only by $h = 0$ and $h = 1$ gives two formulas, each with 7,116 variables and 98,787 clauses. Their CNF SHA-256 digests are

caf7a41381f2f4cd952f4bfc405af32e1d76465291610067922c467d7920ef80,
0806415174e1ef646c0323b496d70ae75cd48988ddf19aa787571523592876d2.

Kissat returns UNSAT and emits proof traces with digests

d43895e356786ccad41437149e30d25da9bdd712e2ddc96aacba24fd269dc5f5,
ad897495967cc214d5ee55c597af094dc9aebb99c144412718ffa72aecf3f257.

Independently compiled `drat-trim` verifies both traces. For $h = 0$, it retains 21,814 original clauses and 2,347,194 proof lemmas and uses 423,344,899 resolution steps. For $h = 1$, the corresponding numbers are 18,740, 759,210, and 110,604,115. The aggregate verification record is `damp_vctwo_q3minus_r4_deg4_below60_verification.json`; its mathematical digest is

5f68d1cadde08559f6761767bcd6048e324d217161b444d4e0d0cd47151803c3.

The two values of h exhaust the normalized branch, proving the claim. $\square$

Proposition 7.126 (Rank-five punctured-core exclusion through order fifty-nine). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 59$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order five. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit quotient and the stored branch certificates leave only the $1/1$ endpoint type. Its defect allowance satisfies $\delta(H, 5) \leq 1$. Lemma 7.114 therefore gives

$$h = 0, \quad q_S = 0 \quad (S \in \binom{X}{2}), \quad \sum_x p_x \in \{2, 3\}.$$

After fixing the endpoint representative, Lemma 7.115 normalizes one carrier order and leaves thirteen labelled order pairs for the other two carriers.

We add this order normal form to the exact static facet-cube reconstruction as redundant clauses. For each core square and repair word, an auxiliary literal is equivalent to the conjunction saying that all four concepts of the positioned core square occur. Unit clauses fix the normalized first carrier, and one of thirteen selectors fixes the other two geodesics. The resulting formulas split only by singleton-core mass two or three. They contain no cardinality selector, and each has 17,757 variables and 333,455 clauses. Their CNF SHA-256 digests are

953b947aeb1f6c24d843948bdb35534b476639145effb7ba92c59d2da45c0f8d,
25d69aab0165358079bbc25cc3ae0963975cc75335832657025e2d01feef7984.

Kissat returns UNSAT on both formulas and emits proof traces with digests

a8ebf575f9780163cbeb6015642d082623f12b043459749cbe0a0275a039d984,
9c6c4557797b643ce04bc4c51b65c9c46783f53db63be4fdb87f9744b2e6a1c8.

Independently compiled `drat-trim` verifies both. For singleton mass two it retains 44,743 original clauses and 507,751 proof lemmas and uses 96,352,829 resolution steps. For singleton mass three the corresponding numbers are 46,505, 746,828, and 143,769,041. The aggregate verification record is `damp_vctwo_q3minus_r5_deg1_ordercuts_below60_verification.json`; its mathematical digest is

51548309160088e68fe4ed9450cfb60ac1d68b6115fd6737e8e9176df02e37eb.

The two mass values exhaust the allowance-one normal form, proving the claim. $\square$

Corollary 7.127 (VC-two core exclusion through order fifty-seven). *Let H have minimum cardinality among all nonempty cornerless ample families. If*

$$48 \leq |H| \leq 57,$$

then H has a corner.

Proof. Orders through 54 are excluded by Proposition 7.123. For orders 55, 56, 57, the core-coordinate wing squeeze and the stored endpoint certificates leave only the 1/1 punctured-cube core at repair ranks four, five, and six. Proposition 7.124 excludes rank four, Corollary 7.120 excludes rank five, and Proposition 7.122 excludes rank six. $\square$

Corollary 7.128 (VC-two core exclusion through order fifty-nine). *Let H have minimum cardinality among all nonempty cornerless ample families. If*

$$48 \leq |H| \leq 59,$$

then H has a corner.

Proof. Orders through 57 are excluded by Corollary 7.127. At orders 58, 59, the core-coordinate wing squeeze and the stored endpoint certificates leave only the 1/1 punctured-cube core at repair ranks four, five, and six. Propositions 7.125 and 7.126 exclude ranks four and five, while Proposition 7.122 excludes rank six. $\square$

Proposition 7.129 (Rank-four punctured-core exclusion through order sixty-one). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 61$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order four. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit certificates again reduce the problem to the normalized 1/1 endpoint type. Its forced join contributes fifteen shattered supports through each core coordinate, while the trapped-wing degree bound gives

$$\deg_{\text{Sh}(H)}(x) \leq \left\lfloor \frac{61 - 20}{2} \right\rfloor = 20.$$

Thus the core-coordinate extra-face cap is five, or equivalently $\delta(H, 4) \leq 5$.

We use the exact facet-cube reconstruction with the three coordinate inequalities and the weighted defect inequality

$$3h + 2 \sum_S q_S + \sum_x p_x \leq 15.$$

The strict facet-excess bound makes the rank-four facet truncation exhaustive below order 64. Splitting only by $h = 0$ and $h = 1$ gives two cardinality-free formulas, each with 7,116 variables and 98,787 clauses. Their CNF SHA-256 digests are

0ccd9f2f256a74751d22e7ef59562826221b7eddb8922b88fb90547dc2cd5acf,
7b2f7513be88239d6a461abf694dffe91be0d112ac98fe1363efd8623c220ac9.

Kissat returns UNSAT and emits proof traces with digests

00296b53291b81912d77ca423a8a4bafc8d137ecfbe5886b36e7df2c9d276434,
5dff75ce553a2d321d70a7643064d71f48081cf9bb92693166e3c871f8b3ba90.

Independently compiled `drat-trim` verifies both traces. For $h = 0$, it retains 24,992 original clauses and 12,952,612 proof lemmas and uses 2,403,191,122 resolution steps. For $h = 1$, the corresponding numbers are 22,906, 4,277,360, and 802,767,916. The aggregate verification record is `damp_vctwo_q3minus_r4_deg5_below62_verification.json`; its mathematical digest is

1c1dc2f041aaa1f3bb7e576fe5aad72b46384ba7d7dea1ea7e3bf9c308392cf7.

The two values of h exhaust the normalized branch. □

Proposition 7.130 (Rank-five punctured-core exclusion through order sixty-one). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 61$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order five. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit certificates reduce the problem to the normalized 1/1 endpoint type. The forced join contributes eighteen shattered supports through each core coordinate, while the trapped-wing degree bound gives

$$\deg_{\text{Sh}(H)}(x) \leq \left\lfloor \frac{61 - 20}{2} \right\rfloor = 20.$$

Thus the core-coordinate extra-face cap is two, or equivalently $\delta(H, 5) \leq 2$. The cases with smaller allowance are contained in the allowance-two formula, so it suffices to use Lemma 7.116.

We compile the same exact static facet-cube reconstruction as above, with maximum facet rank four and no cardinality selector, and add the carrier-order normal forms from Lemma 7.117 and Corollary 7.119. The structural split is as follows.

state	further subdivision	formulas
$h = 0, \sum_S q_S = 0$	002, 011, 012, 022, 111, 112; 122 at distances 000, 011, 022, 112; 222 at distances 000, 011, 022, 112, 222	15
$h = 1, \sum_S q_S = 0$	all compatible clean-carrier profiles	1
$h = 0, \sum_S q_S = 1$	all synchronized square-ear profiles	1

Here a three-digit word in the first row is the sorted singleton-defect profile, and a word following “at distances” is the sorted triple of pairwise Kendall distances. The omitted combination 122/222 is empty: three pairwise distances equal to two force every corresponding singleton defect to be two.

An independent coverage audit enumerates the finite quotients rather than trusting the formula split. In the clean $h = 0$ state, the 106 normalized order pairs and 23 labelled singleton profiles give exactly 422 compatible order–profile pairs, all covered by the fifteen cells. In the clean $h = 1$ state, thirteen order pairs and four labelled singleton profiles give 28 compatible pairs. In the remaining state, the four positions of the square ear give 32 distinct labelled three-carrier patterns: sixteen with the designated carrier defective and eight for each of the other two defective-carrier positions. The audit finds no uncovered case.

The seventeen formulas have between 17,231 and 18,826 variables and between 330,074 and 344,517 clauses. Kissat returns UNSAT on every formula. Independently compiled `drat-trim` verifies all seventeen proof traces, using between 10,017,433 and 710,424,249 resolution steps per trace and

4,427,395,811 steps in total. The aggregate verification record is `damp_vctwo_q3minus_r5_deg2_below62_verification.json`; its mathematical digest is

b821dd601f63ac7e6cacc063be8257a9ed6e8bb0c114364e9e96bba0055dd183.

Thus every allowance-two normal form is impossible, proving the claim. $\square$

Corollary 7.131 (VC-two core exclusion through order sixty-one). *Let H have minimum cardinality among all nonempty cornerless ample families. If*

$$48 \leq |H| \leq 61,$$

then H has a corner.

Proof. Orders through 59 are excluded by Corollary 7.128. At orders 60, 61, the core-coordinate wing squeeze and the stored endpoint certificates leave only the 1/1 punctured-cube core at repair ranks four, five, and six. Propositions 7.129 and 7.130 exclude ranks four and five, while Proposition 7.122 excludes rank six. $\square$

Proposition 7.132 (Rank-four punctured-core exclusion through order sixty-three). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 63$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order four. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit certificates and the core-coordinate wing squeeze leave the normalized 1/1 endpoint type with extra-face allowance six. The carrier audit reconstructs all antipodal ample carriers and retains 50 carrier-defect states, 23,511 carrier patterns, and 349,503 compatible carrier-singleton-profile pairs. Quotienting by exchange of the two nondesignated carriers and grouping by total defect gives seventeen certificate cells. Carrier compatibility itself removes six nominal cells: totals 7, 8, 9 do not occur when $h = 0$, and totals 5, 6, 7 do not occur when $h = 1$.

The seventeen formulas have no cardinality selector. They have between 7,564 and 63,744 variables and between 100,451 and 265,984 clauses. Kissat returns UNSAT on every formula, and independently compiled `drat-trim` verifies every trace. The aggregate checker independently reconstructs the carrier quotient, finds no uncovered or multiply covered state, and records 1,728,418,557 resolution steps. Its verification record is `damp_vctwo_q3minus_r4_deg6_below64_verification.json`, with mathematical digest

03308601e7ce9fa9f6ab858f2ded838b90cdd8ac2c42ebba271fdeb60aed6e4a.

Thus the allowance-six normal form is impossible. $\square$

Proposition 7.133 (Rank-five punctured-core exclusion through order sixty-three). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| \leq 63$, and suppose that its maximum core is $Q_3 \setminus \{v\}$ with a repair set of order five. Then H has a corner.*

Computer-assisted proof. The endpoint-orbit certificates reduce again to the normalized 1/1 type, now with extra-face allowance three. The carrier-order normal form is split by the missing-core indicator, the three carrier defects, the sorted singleton-defect profile, and, only for the five resistant profiles, the sorted triple of pairwise Kendall distances. This gives sixty-four certificate cells and no cardinality selector.

An independent reconstruction finds 1,191 carrier patterns and 5,810 compatible carrier-profile pairs. Every pair lies in exactly one of the sixty-four cells. All sixty-four individual formulas are

UNSAT, and independently compiled **drat-trim** verifies all sixty-four traces. For the final aggregate, one additional unprofiled $h = 1, q = 000$ proof replaces the eight cells in that row. The resulting coverage certificate therefore uses fifty-seven proof records while still covering all sixty-four cells. The formulas represented in the aggregate have between 17,310 and 24,140 variables and between 330,104 and 375,007 clauses. The checker records 25,761,684,586 resolution steps in total.

The aggregate verification record is `damp_vctwo_q3minus_r5_deg3_below64_verification.json`. Its mathematical digest is

60daf3f6e845b3b858a5013291797fd6016b2c85f3fe56560fe51e8e50caae8e.

The independently audited coverage map has no uncovered or multiply covered pair, so every allowance-three normal form is impossible. $\square$

Corollary 7.134 (VC-two core exclusion through order sixty-three). *Let H have minimum cardinality among all nonempty cornerless ample families. If*

$$48 \leq |H| \leq 63,$$

then H has a corner.

Proof. Corollary 7.131 handles orders through 61. At orders 62, 63, the endpoint quotient and the core-coordinate wing squeeze leave only the punctured-cube core at repair ranks four, five, and six. Propositions 7.132 and 7.133 exclude the first two ranks, while Proposition 7.122 excludes rank six. $\square$

Proposition 7.135 (Punctured-cube core exclusion below order forty-eight). *Let X be a three-coordinate set, let $A = Q_X \setminus \{v\}$, and let R be disjoint from X with $r = |R| \in \{3, 4\}$. Suppose that $H \subseteq Q_{R \cup X}$ is ample and that two antipodal repair fibres satisfy*

$$H_f = H_{\bar{f}} = A.$$

If $|H| \leq 47$, then H has a corner.

Computer-assisted proof. Translate and permute coordinates so that $f = 0_R$ and $A = Q_X \setminus \{1_X\}$. Put $\Sigma = \text{Sh}(H)$. The two equal antipodal fibres force the join subcomplex

$$\Sigma_0 = \binom{X}{\leq 2} * (\{\emptyset\} \cup \binom{R}{1}) \subseteq \Sigma, \quad |\Sigma_0| = 7(r+1). \quad (7.169)$$

Indeed, both fibres realize all traces on every member of $\binom{X}{\leq 2}$, and their repair bits are opposite in every coordinate. Thus the available face excess is at most 19 for $r = 3$ and at most 12 for $r = 4$.

We use the static model of Proposition 5.14. Choose an antichain $\mathcal{F}$ of facet supports, one anchored affine T -cube C_T for each $T \in \mathcal{F}$, and put $K = \bigcup_{T \in \mathcal{F}} C_T$. The model enforces

$$|K| = \left| \bigcup_{T \in \mathcal{F}} 2^T \right|, \quad |\{T \in \mathcal{F} : x \in C_T\}| \neq 1 \quad (x \in Q_{R \cup X}), \quad (7.170)$$

together with $K_f = K_{\bar{f}} = A$ and $|K| \leq 47$. By Proposition 5.14, these conditions are equivalent to ampleness and cornerlessness; no recursive minor or peeling test occurs in the formula.

The face budget makes the facet-rank truncation exhaustive. For $r = 3$ the ambient dimension is six, and a rank-six facet already has 64 vertices, so it suffices to allow ranks at most five. For $r = 4$, a rank-six facet has old-repair type $3 + 3$ or $2 + 4$ and contributes at least

$$\min \left\{ 2^6 - \left(1 + 3 + \binom{3}{2} \right) (1 + 3), 2^6 - \left(1 + 2 + \binom{2}{2} \right) (1 + 4) \right\} = 36$$

faces outside Σ_0 , exceeding the twelve-face budget. Hence ranks at most five suffice in both cases. The resulting two global formulas include every facet support and every anchor of the permitted ranks. One order selector ranges over all feasible orders at once. Their sizes are

r	variables	clauses
3	34,541	215,670
4	60,777	558,329.

Lingeling returns UNSAT on both formulas and emits DRAT proofs. An independently compiled `drat-trim` checker returns VERIFIED on both. Its exact backward-checking statistics are

r	core clauses	core lemmas	proof lemmas	resolution steps
3	26,800	630,456	1,401,951	114,807,062
4	81,446	300,032	910,206	87,424,450.

Neither core uses a RAT lemma. The formula hashes are

462ff85630445e4fa2d9677cf196e1e414ba1357765ab30ad30f89c51e8d00bb,
bf850214e54f198586d8bffb5c52e2e3951b37410f06dbb0f8ea34927ae38841b,

and the corresponding proof hashes are

9f359404780c4cbc79d0d7fc7570e3cb5481c2af76b9a76a9d1f93fd9e12a913,
3f5107c6a9d47e61add8cb49c8c857e042cf1d41d3010c6c119120ebcc405eec.

The generator, formulas, proofs, and independent verification record are stored under the common prefix `damp_q3minus_core_below48`. Unsatisfiability proves the claim. $\square$

Corollary 7.136 (Path-only reduction below order forty-eight). *Let H have minimum cardinality among all nonempty cornerless ample families, and suppose that $|H| < 48$. Then the maximum core supplied by Corollary 7.6 has VC dimension one. It is a path on $n + 1$ vertices, where $2 \leq n \leq 9$; its repair set has $r \geq 3$, with $r \leq 12$ when $n = 2$, and*

$$(n + 1)(r + 1) \leq 40 \quad (n \geq 3).$$

Proof. Apply Proposition 7.103. Its VC-dimension-two alternative has core $Q_3 \setminus \{v\}$ and $r \in \{3, 4\}$, contrary to Proposition 7.135. The stated path bounds are the remaining alternative of the bulk reduction. $\square$

Proposition 7.137 (Relative-wing first-moment bound). *Let H have minimum cardinality among all nonempty cornerless ample families, suppose that $|H| < 48$, and keep the path-core notation of Corollary 7.136. Put*

$$m = |H|, \quad d = n + r, \quad q = \left\lfloor \frac{m - 20}{2} \right\rfloor,$$

and write $\Sigma = \text{Sh}(H)$. For $e \in R \dot{\cup} X$, let

$$\deg_{\Sigma}(e) = |\{S \in \Sigma : e \in S\}|.$$

Then

$$\deg_{\Sigma}(e) \leq q \quad (e \in R \dot{\cup} X), \quad (7.171)$$

$$n + 1 \leq q, \quad r + 1 \leq q, \quad (7.172)$$

$$2m - d - 2 \leq dq. \quad (7.173)$$

Consequently $m \geq 35$. Thus one compatibility calculation for all coordinate wings excludes every order from 28 through 34 simultaneously.

Proof. Fix an active coordinate e . As in Proposition 2.122, write its two sections as

$$A_b = \rho_e^b H \quad (b \in \{0, 1\}), \quad B = A_0 \cap A_1, \quad D_b = A_b \setminus B.$$

The pairs $B \subseteq A_b$ are trapped intervals in $\mathcal{D}_{\text{amp}}$. Proposition 2.184 gives $|D_b| \geq 10$ for both b .

The two-layer shadow-intersection identity says that the members of Σ containing e are exactly

$$\{T \cup \{e\} : T \in \text{Sh}(B)\}.$$

Since B is ample, their number is $|B|$. Hence

$$m = |D_0| + |D_1| + 2|B| \geq 20 + 2 \deg_{\Sigma}(e),$$

which proves (7.171).

Let $x_1, \dots, x_n$ be the coordinates of the core path. Its two antipodal repair fibres force the following join skeleton in the shattered complex:

$$\Sigma_0 = \{\emptyset\} \cup \{\{e\} : e \in R \dot{\cup} X\} \cup \{\{t, x_i\} : t \in R, 1 \leq i \leq n\} \subseteq \Sigma. \quad (7.174)$$

Indeed, the core path shatters every core singleton. Every adjacent path carrier contains the two repair antipodes, and therefore shatters every repair singleton; the two copies of the corresponding core edge then shatter every pair $\{t, x_i\}$. In particular, a repair coordinate belongs to at least $n + 1$ members of Σ , and a core coordinate to at least $r + 1$. Combining this with (7.171) proves (7.172).

The family Σ_0 has $1 + d + nr = (n + 1)(r + 1)$ members and first support moment

$$\sum_{S \in \Sigma_0} |S| = d + 2nr.$$

Every member of $\Sigma \setminus \Sigma_0$ has order at least two, because Σ_0 already contains the empty set and every singleton. Consequently

$$\begin{aligned} \sum_{e \in R \dot{\cup} X} \deg_{\Sigma}(e) &= \sum_{S \in \Sigma} |S| \\ &\geq d + 2nr + 2(m - (1 + d + nr)) = 2m - d - 2. \end{aligned}$$

The coordinatewise upper bound (7.171) gives the opposite estimate $\sum_e \deg_{\Sigma}(e) \leq dq$, proving (7.173).

It remains only to read these uniform inequalities. The already proved exclusion through order 27 gives $m \geq 28$.

If $28 \leq m \leq 29$, then $q \leq 4$, so the baseline degree bounds give $n, r \leq 3$ and $d \leq 6$. On the other hand, (7.173) gives $2m - 2 \leq 5d$, and hence $d \geq 11$, a contradiction.

If $30 \leq m \leq 31$, then $q \leq 5$, so $d \leq 8$, whereas the first moment gives $2m - 2 \leq 6d$ and hence $d \geq 10$. If $32 \leq m \leq 33$, then $q \leq 6$. The first moment gives $d \geq 9$, while $n, r \leq 5$. Since $n \geq 2$, $r \geq 3$, and $n + r \geq 9$, one has $n \geq 4$, and the smallest possible path baseline together with the seven-unit propagation surcharge is

$$(n + 1)(r + 1) + 7 \geq 5 \cdot 6 + 7 = 37 > m,$$

a contradiction.

Finally suppose $m = 34$. Then $q = 7$, and the first moment gives

$$66 \leq 8(n + r),$$

so $n + r \geq 9$. If $n = 2$, the six-unit path surcharge gives $34 \geq 3(r + 1) + 6$, while $r + 1 \leq 7$; hence $n + r \leq 8$, a contradiction. If $n \geq 3$, then $n, r \geq 3$, and the seven-unit surcharge gives $(n + 1)(r + 1) \leq 27$. But $n + r \geq 9$ implies $(n + 1)(r + 1) \geq 4 \cdot 7 = 28$. Therefore $m \geq 35$. $\square$

Proposition 7.138 (Rank-weighted relative-wing inequality). *Let H have minimum cardinality among all nonempty cornerless ample families, delete its constant coordinates, and put*

$$E = \text{idim}(H), \quad d = |E|, \quad \Sigma = \text{Sh}(H), \quad m = |H|.$$

For $j \in \{1, 2\}$, write

$$M_j = \sum_{S \in \Sigma} |S|^j.$$

For every $e \in E$, write its sections as A_e^0, A_e^1 , put $B_e = A_e^0 \cap A_e^1$, and, for $b \in \{0, 1\}$, choose $C_e^b \in \mathcal{D}_{\text{amp}}$ inclusion-maximal subject to

$$B_e \subseteq C_e^b \subsetneq A_e^b.$$

Set $D_e^b = A_e^b \setminus C_e^b$, and define $\tau(D_e^b)$ by (2.119). Then

$$(d + 3)M_1 - 2M_2 \geq dm + \sum_{e \in E} \sum_{b=0}^1 \tau(D_e^b) \geq d(m + 2). \quad (7.175)$$

Equivalently, the first coordinatewise wing bound has a second-moment refinement that remembers the ranks and terminal block curvature of every coloured acquired-support family.

Proof. Fix now $e \in E$. Write the two sections of H as A_e^0, A_e^1 , put $B_e = A_e^0 \cap A_e^1$, and define the two coloured acquired-support families

$$\mathcal{W}_e^b = \text{Sh}(A_e^b) \setminus \text{Sh}(B_e) \quad (b \in \{0, 1\}).$$

As in Proposition 2.122, both pairs $B_e \subsetneq A_e^b$ are trapped intervals in $\mathcal{D}_{\text{amp}}$. Applying Proposition 2.190 to all $2d$ pairs gives

$$\sum_{e \in E} \sum_{b=0}^1 \sum_{S \in \mathcal{W}_e^b} |S| \geq \sum_{e \in E} \sum_{b=0}^1 (|A_e^b \setminus B_e| + \tau(D_e^b)). \quad (7.176)$$

The right-hand wing count is

$$\sum_{e,b} |A_e^b \setminus B_e| = dm - 2M_1. \quad (7.177)$$

Indeed, for fixed e it equals $m - 2|B_e|$, and $|B_e|$ is the number of members of Σ containing e .

For the rank-weighted left-hand side, the two-layer shadow identities say that the union and intersection of the two section shadows are, respectively, the deletion and the link of e in Σ . A face $S \in \Sigma$ of order s contributes $s(d - s)$ over all deletions and $s(s - 1)$ over all links. Hence

$$\sum_{e,b} \sum_{S \in \mathcal{W}_e^b} |S| = \sum_{S \in \Sigma} |S|(d + 1 - 2|S|) = (d + 1)M_1 - 2M_2. \quad (7.178)$$

Substituting (7.177) and (7.178) into (7.176) gives

$$(d + 1)M_1 - 2M_2 \geq dm - 2M_1 + \sum_{e,b} \tau(D_e^b),$$

which gives the first inequality in (7.175). The second follows from $\tau(D_e^b) \geq 1$ for all $2d$ terminal blocks. $\square$

Proposition 7.139 (The four order-thirty-two equality shadows). *Under the hypotheses of Proposition 7.137, suppose that $m = 32$. Then $d = 8$ and*

$$(n, r) \in \{(2, 6), (3, 5), (4, 4), (5, 3)\}. \quad (7.179)$$

The shattered complex has VC dimension three and contains either one or two three-element faces. Let G be its graph of two-element faces and let $M = K_8 \setminus G$. Thus M has no edge between the core and repair blocks. Up to permuting coordinates within the two blocks and, when appropriate, interchanging the blocks, exactly the following four types are possible:

$ \Sigma \cap \binom{E}{3} $	(n, r)	M	three-element faces
1	(3, 5)	$P_3 \dot{\cup} P_5$	the centre of P_3 and two distance-two degree-two vertices of P_5
2	(2, 6)	$K_2 \dot{\cup} C_6$	the two alternating triples of C_6
2	(4, 4)	$C_4 \dot{\cup} P_4$	the two C_4 diagonals, paired with the two internal P_4 vertices
2	(4, 4)	$C_4 \dot{\cup} K_{1,3}$	the two C_4 diagonals, both joined to the claw centre.

(7.180)

The first row also represents the block-swapped split (5, 3).

Proof. At $m = 32$ one has $q = 7$. Equation (7.173) gives $62 \leq 8d$, so $d \geq 8$. On the other hand, (7.172) gives $n, r \leq 6$. If $n = 2$, the path surcharge gives $32 \geq 3(r + 1) + 6$, and hence $r \leq 6$. If $n \geq 3$, it gives

$$(n + 1)(r + 1) \leq 25.$$

For $n, r \geq 3$, a sum $n + r \geq 9$ would make the left-hand side at least $4 \cdot 7 = 28$. Thus $d \leq 8$ in every case. This proves $d = 8$ and the four splits in (7.179).

Every ample family of VC dimension at most two is dismantlable, so cornerlessness gives $\text{VCdim}(H) \geq 3$. Besides the empty set and the eight singletons, Σ has twenty-three faces. Put

$$h = \sum_{\substack{S \in \Sigma \\ |S| \geq 3}} (|S| - 2).$$

The coordinate-degree bound gives

$$54 + h = \sum_{S \in \Sigma} |S| = \sum_e \deg_{\Sigma}(e) \leq 56.$$

Thus $1 \leq h \leq 2$. A four-element face would bring its four three-element subsets and contribute at least six to h . Hence there is no face of order at least four, and h is exactly the number $t \in \{1, 2\}$ of three-element faces. It follows that G has $23 - t$ edges, so M has $5 + t$ edges.

For a coordinate v , let t_v be the number of three-element faces containing it. Since $\deg_G(v) = 7 - \deg_M(v)$,

$$\deg_{\Sigma}(v) = 1 + \deg_G(v) + t_v = 8 - \deg_M(v) + t_v \leq 7,$$

and therefore

$$\deg_M(v) \geq 1 + t_v. \tag{7.181}$$

If $t = 1$, the two sides of (7.181) have total sums 12 and 11, respectively. Thus one vertex has one extra missing-pair incidence and every other vertex has equality. The triangle is an independent set in M . Applying the handshaking lemma separately in the two blocks leaves only block orders three and five: the smaller block contains one triangle coordinate and has degrees $(2, 1, 1)$, while the larger contains two and has degrees $(2, 2, 2, 1, 1)$, with the extra incidence on a nontriangle vertex. The independence condition makes the two graphs respectively P_3 and P_5 , in the positions stated in the first row. For completeness, the other three block splits fail before anchors are considered: a triangle vertex in a two-vertex block would require degree two, three triangle vertices in the six-vertex block would require six incident missing edges although only five remain there, and the parity and independence inequalities give the analogous contradiction in the $4 + 4$ split.

If $t = 2$, both sides of (7.181) have total sum 14. Hence equality holds at every vertex. In the $2 + 6$ split, the two-vertex block cannot contain a triangle coordinate. The two triangles are disjoint in the other block: if they shared a vertex, that vertex would require three missing neighbours but would have at most two coordinates with which it shares neither triangle. All six degrees in that block are therefore two, and triangle independence forces the alternating C_6 in the second row.

The $3 + 5$ split is impossible. A triangle uses at most one coordinate of the three-vertex block, because a used coordinate has missing degree two and hence is adjacent in M to both other coordinates. Parity forces exactly one of the two triangles to meet that block. The other triangle then lies wholly in the five-vertex block; its three independent vertices require at least six incident missing edges, although that block has only five. The $5 + 3$ case is symmetric.

Finally consider the $4 + 4$ split. Parity and the same independence argument force total triangle incidences four and two in the two blocks, up to exchange. In the first block, every coordinate occurs once; all missing degrees are two, giving C_4 , and the two triangle pairs are its diagonals. In the second block, either two coordinates occur once, giving degree sequence $(2, 2, 1, 1)$ and hence P_4 , or one coordinate occurs twice, giving $(3, 1, 1, 1)$ and hence $K_{1,3}$. These are the last two rows and complete the classification. $\square$

Proposition 7.140 (Exclusion of orders thirty-two and thirty-three). *No minimum-cardinality nonempty cornerless ample family has order 32 or 33.*

Proof. Suppose first that $m = 33$. As in the preceding proof, $\text{VCdim}(H) \geq 3$, so at least one member of $\Sigma \setminus \Sigma_0$ has order at least three. The first-moment estimate therefore improves by one:

$$2m - d - 1 \leq dq.$$

Here $q = 7$, so $65 \leq 8d$ and $d \geq 9$. But (7.172) gives $n, r \leq 6$. If $n = 2$, the path surcharge and $m = 33$ give $r \leq 6$, whence $d \leq 8$. If $n \geq 3$, the same surcharge gives $(n + 1)(r + 1) \leq 26$, whereas $n, r \geq 3$ and $d \geq 9$ force this product to be at least 28. Both alternatives are impossible.

It remains to exclude $m = 32$. Proposition 7.139 leaves the four fixed shattered complexes in (7.180). For each one, apply the anchored-facet reconstruction of Proposition 5.14: choose one affine cube on every facet support, require their union to have order 32, and forbid a covered vertex from lying in exactly one chosen cube. These conditions are equivalent to ampleness and cornerlessness.

The four deterministic Boolean formulas have respectively

type	variables	clauses	core clauses
$P_3 \dot{\cup} P_5$	10,061	139,815	49,006
$K_2 \dot{\cup} C_6$	9,824	132,040	23,718
$C_4 \dot{\cup} P_4$	9,782	128,655	21,243
$C_4 \dot{\cup} K_{1,3}$	9,095	112,632	16,023.

CaDiCaL 2.1.2 returns UNSAT on all four formulas. For each formula, **drat-trim** extracts a subformula and independently verifies its DRAT proof. The original formula hashes are

```
de88895ed588caeb03d46423ea2018c4a61de90eb63a382f260231e032c39fb1,
3bbdbdd8be2db7cd7d190c218621b0ace4cee64f4728ce4654ab5492e6a557e9,
59bb2568d027448bd3f3db3d4eff68c1f439a9ea796f9e3fd8df66d34080497e,
ab9f9d2438c4733e05ee3943f4e5cc714139ddffdea85349718ffe7ec4bcb863,
```

and the corresponding core-proof hashes are

```
58369ac1d61e09a4bf705186c7ec246d5dd5782d490fa30a93bd20ce5fd8a1d2,
4f007c122d1980b8389248ee5a4301890d96b162822a3085d3cf338859fe60b0,
a33913af549b646c44aeb0241ea86779bb4a33cdc476580166e6d817f9783bed,
603e5c9d789a90c925b1e89a877dcec08c6193bf700895e067ebe34a59862bb7.
```

The compiler, formulas, extracted cores, proofs, and verifier are stored under the common prefix **damp_order32_wing_moment**. The manifest and verification digests are, respectively,

```
8ee1871622a5af2e8c83a41bd99ac344f2e191acbcc0cc68b0b5abbe45d5505a,
6ac2a3e3f1ac9f7041c8f177ef19cd21c1e221d76736deb466c4a3a7513fa08d.
```

Thus none of the four exact equality shadows has a cornerless anchored realization, excluding order 32. $\square$

The tip-colouring identity is stronger than the repair-capacity bound in one essential respect: it records the exact fibres in which projected corners occur. The coarse second-moment inequality by itself is too loose to force a new dimension bound; even Hall's example has substantial slack in that proxy. The active problem is therefore to constrain the compatible system of missing traces q_A , or equivalently of tip maps $\tau_S : E \setminus S \rightarrow C_S$. Equation (7.128) excludes dimensions five and six in one argument, and Corollary 7.97 settles dimension seven. The compatibility problem remains in dimensions eight through eleven.

7.3 A concrete two-extension obstruction

The symbolic two-extension theorem also produces examples that are not cornerless. For Hall's family, take

$$a = 183, \quad b = 315.$$

The two one-concept extensions are corner-peelable, and their new shattered supports are

$$P_a = \{2, 3, 6, 7\}, \quad P_b = \{2, 3, 8, 9\}.$$

They are distinct, so Corollary 6.67 gives an ample forbidden pc-minor

$$X = ((C_H \cup \{315\}) \times \{0\}) \cup ((C_H \cup \{183\}) \times \{1\}) \subseteq Q_{13}.$$

It has order 600 and exactly the two corners $(315, 0)$ and $(183, 1)$. Deleting either corner leaves a nonpeelable peripheral expansion containing C_H as a proper pc-minor, which explains concretely why direct corner descent fails.

Remark 7.141 (Greedy shelling is not the missing invariant). The same example rules out a tempting strengthening of the Yang route. Its two restrictions in the new coordinate are the corner-peelable one-concept repairs

$$C_H \cup \{315\}, \quad C_H \cup \{183\}.$$

In the first restriction, 315 is a corner whose deletion leaves the nonpeelable family C_H ; the second restriction has the symmetric bad corner 183. Thus even a proper pc-minor of a forbidden family need not have the property that every available first peeling step is safe. Equivalently, its shellable Yang ball need not be greedily shellable in the sense required by this deletion process.

A worker audit tested every initial corner of all 39 elementary pc-minors of the order-600 family. Among 432 corner-minor pairs, exactly the two just described are unsafe. The computation is only a regression check; the preceding symbolic argument is the reason for the failure. The artifact is `damp_two_repair_minor_corner_safety.json`, with mathematical digest

fff90aec268a0b9c85619ae1470e724f2f2ead7965c479431aa9a7f3f421cf5b.

7.4 The minimum-order problem

The connection with partial shellings also suggests a construction route that does not enumerate the intervening orders. Bognini and Sentinelli call a proper pure subcomplex of a simplex skeleton *austere* when every missing facet contains at least two minimal nonfaces [BS26]. Since a possible next shelling facet contains a unique minimal nonface, austerity is a static certificate that a partial shelling is stuck. For a partial shelling of a cross-polytope, the correspondence of Chalopin, Chepoi, Moran, and Warmuth identifies the missing facets with an ample cube family; the partial shelling is stuck exactly when this complementary family has no corner [CCMW22, Corollary 4.4]. Thus transversal austerity is the shelling form of cornerlessness.

This observation does not make the echo construction from [BS26] a Yang complex: an echo facet may use both vertices of one pair and omit another pair, whereas a cross-polytope facet selects exactly one vertex from every pair. Their explicit quiet two-complex nevertheless suggests a natural paired template. It has eight vertices, complete one-skeleton, twenty-one triangles, and hence fifty-eight faces including the empty face. Let Γ denote this complex. Since its one-skeleton is complete, every missing triple is a minimal nonface; quietness ensures that every four-set contains a missing triple, so there are no larger minimal nonfaces. Thus the minimal nonfaces are exactly the thirty-five missing triples.

The signed-trace constraints in fact make this displayed template rigid. This also serves as a worked application of Proposition 5.18. In the facet list below, copied from [BS26], we abbreviate a triple such as $\{1, 2, 4\}$ to 124:

$$148, 348, 234, 347, 368, 356, 168, 357, 236, 147, 167, 156, 267, 257, \\ 245, 125, 458, 128, 278, 123, 456.$$

For a coordinate e , the following row lists the edges of a spanning tree of $P_e(\Gamma)$.

e	spanning-tree edges
1	124–134, 124–145, 134–137, 134–138, 145–146, 145–158, 137–135, 137–157, 138–136, 157–127, 136–126, 127–178
2	124–127, 124–247, 124–248, 127–126, 247–237, 248–238, 248–258, 126–256, 126–268, 237–235, 256–246
3	134–137, 134–138, 137–136, 137–367, 138–238, 136–135, 367–237, 238–378, 135–235, 235–345, 345–346, 345–358
4	124–134, 124–145, 124–248, 145–146, 248–247, 248–478, 146–467, 146–468, 247–246, 247–457, 468–346, 457–345
5	135–235, 235–256, 235–345, 256–567, 345–358, 345–457, 567–157, 358–568, 457–145, 568–158, 158–258, 258–578
6	126–136, 126–256, 126–268, 256–246, 256–567, 268–678, 246–346, 567–367, 346–467, 346–468, 467–146, 468–568
7	127–137, 127–157, 127–178, 127–237, 157–567, 178–478, 178–678, 237–247, 237–367, 567–457, 567–467, 478–378, 478–578
8	138–238, 238–248, 238–268, 248–258, 268–678, 258–158, 258–578, 678–178, 158–568, 578–358, 578–378, 178–478, 568–468

Each listed pair satisfies one of the two conditions in Definition 5.17: this can be checked by inspecting the displayed facet list, since its complement is the set of missing triples. The table therefore proves that all eight propagation graphs are connected. Proposition 5.18 now shows that every ample $X \subseteq Q_8$ with $\text{Sh}(X) = \Gamma$ is a cube translate of the canonical face family of Γ . The canonical family has corners: the characteristic vector of any triangular facet lies only in the corresponding maximal cube. Consequently no such X is cornerless. Thus the displayed quiet complex does not improve the upper bound. In fact the same method, followed by a small residual audit, excludes the entire quiet family from [BS26].

Proposition 7.142 (Exclusion of the quiet fifty-eight-face templates). *None of the fifty-nine quiet two-complexes on eight vertices with complete one-skeleton and twenty-one triangles enumerated in [BS26] is the shattered-set complex of a cornerless ample family.*

Finite audit after coordinate propagation. We first reproduce the source census. Fix a Steiner system $S(3, 4, 8)$. Its fourteen blocks partition the fifty-six triples. If x, y, z blocks contain respectively two, one, and no chosen triangles, then quietness and the total of twenty-one triangles give

$$2x + y = 21, \quad x + y + z = 14,$$

so $x - z = 7$ and at least seven blocks contain two triangles. The affine group of the Steiner system has fourteen orbits on seven-subsets of its blocks. For one representative of each orbit, we choose two triangles in the seven marked blocks, impose the total of twenty-one triangles, complete one-skeleton, and quietness on every four-set, and quotient the surviving choices by all permutations of the eight vertices. Exhausting the fourteen marked problems gives fifty-nine isomorphism classes, agreeing with [BS26].

For forty-seven of these complexes every nonempty graph $P_e(\Sigma)$ is connected, so Proposition 5.18 excludes them. Consider one of the remaining twelve. If c_e is the number of components of $P_e(\Sigma)$, the four-coordinate lemma forces the e -bit of the omitted traces to be constant on each component. Translation in coordinate e complements all these component bits, so we normalize one of them to zero. Every admissible signing therefore occurs among exactly

$$2^r, \quad r = \sum_{e=1}^8 (c_e - 1),$$

component signings. The residual audit is summarized below.

r	complexes	signings	ample signings	least corners
1	1	2	2	21
2	3	12	10	18
3	4	32	28	15
4	3	48	36	15
11	1	2048	176	13
total	12	2142	252	13

For completeness, the audit evaluates a component signing σ by forming the common satisfying family $H(\Sigma, \sigma)$. By Theorem 5.3, it is ample with shattered-set complex Σ exactly when it has fifty-eight members. For each such signing we count the members that uniquely realize a trace on a triangular facet; by Lemma 5.8, these are exactly its corners. The table exhausts all 2142 normalized component signings. All 252 normalized ample signings produce families with at least thirteen corners, which excludes the remaining twelve complexes. $\square$

The exact basis theorem is a recognition theorem and does not determine the least order of an ample member of the basis. Hall's family gives order 299. We next obtain a conceptual lower bound from the fibers over a maximal shattered set. The argument is uniform in the ambient dimension; the earlier finite exclusions at particular orders remain independent campaign regressions and are not used here.

The minimum-order question admits a uniform conceptual treatment. The point is to count how many outside coordinates can fail section containment in the fibers over one maximal shattered set. This produces a bound in arbitrary dimension, rather than a separate argument at each order.

Lemma 7.143 (Unmatched coordinates in a rooted fiber). *Let $C \subseteq Q_U$ be an isometric family containing 0, with*

$$|C| \geq 2.$$

Define

$$D(C) = \{e \in U : \text{there is } z \in C \text{ with } z_e = 1 \text{ and } z^{\{e\}} \notin C\}.$$

Then

$$|D(C)| \leq |C| - 2.$$

If equality holds and $s = |C|$, then there are distinct coordinates $e_1, \dots, e_{s-1}$ such that

$$C = \{0, \mathbf{1}_{\{e_1\}}, \mathbf{1}_{\{e_1, e_2\}}, \dots, \mathbf{1}_{\{e_1, \dots, e_{s-1}\}}\}$$

and $D(C) = \{e_1, \dots, e_{s-2}\}$.

Proof. Root a spanning tree of C at 0 as follows. For every nonzero $z \in C$, choose as its parent a neighbor whose support has cardinality

$$|\text{supp}(z)| - 1.$$

Such a neighbor exists because C is isometric: a shortest path in C from 0 to z has length $|\text{supp}(z)|$. The chosen parent edges form a tree, and each root-to-vertex path in this tree is a hypercube geodesic.

Fix $e \in D(C)$, and choose a witnessing vertex z . The tree path from 0 to z crosses coordinate e exactly once. Let v_e be the endpoint of that edge farther from the root. This edge is not the last edge of the path, since otherwise the preceding vertex would be $z^{\{e\}} \in C$, contrary to the choice of z . Thus v_e is a nonroot, nonleaf vertex of the rooted tree. Distinct coordinates give distinct vertices v_e , because every nonroot vertex has a unique parent edge. A rooted tree with at least two vertices has a nonroot leaf, so it has at most $|C| - 2$ nonroot, nonleaf vertices. The injection $e \mapsto v_e$ proves the assertion.

Suppose equality holds. Then the injection uses every nonroot, nonleaf vertex, and the tree has exactly one nonroot leaf. A finite rooted tree with only one nonroot leaf is a path. Its edge labels $e_1, \dots, e_{s-1}$ are distinct because the path is a hypercube geodesic. The vertices of C are therefore exactly the successive prefixes displayed in the statement. Each e_i , for $i < s - 1$, is unmatched at the final vertex, whereas the only vertex with e_{s-1} -value one has its mate at the preceding vertex. This identifies $D(C)$. $\square$

Theorem 7.144 (Facet-excess inequality). *Let $H \subseteq Q_F$ be ample and cornerless, let T be a facet of $\text{Sh}(H)$, and put $U = F \setminus T$. If $p_T(H)$ is the number of nonperipheral coordinates in U , then*

$$p_T(H) \leq |H| - 2^{|T|+1}. \quad (7.182)$$

In particular, if H is periphery-free, then

$$|H| \geq 2^{|T|+1} + |F| - |T|. \quad (7.183)$$

Proof. Since H is ample, it strongly shatters T . Thus there is a word $y \in Q_U$ such that

$$\{(q, y) : q \in Q_T\} \subseteq H.$$

Translate the coordinates in U so that $y = 0$. For $q \in Q_T$, let

$$C_q = \{z \in Q_U : (q, z) \in H\}.$$

Every C_q contains 0. It is the projection of an intersection of coordinate semicubes of H , hence is an ample pc-minor and, in particular, isometric. Moreover,

$$|C_q| \geq 2.$$

Indeed, if $(q, 0)$ were the unique member with T -trace q , then the facet T would be uniquely realized by that member, and Lemma 5.8 would make it a corner.

Let $e \in U$ be nonperipheral. In the orientation fixed by the common base word, the containment

$$\rho_e^1 H \subseteq \rho_e^0 H$$

fails. Hence some $z \in C_q$ has $z_e = 1$ but $z^{\{e\}} \notin C_q$, so $e \in D(C_q)$. Therefore

$$p_T(H) \leq \left| \bigcup_{q \in Q_T} D(C_q) \right| \leq \sum_{q \in Q_T} |D(C_q)| \leq \sum_{q \in Q_T} (|C_q| - 2) = |H| - 2^{|T|+1},$$

where the third inequality is Lemma 7.143. If H is periphery-free, then all $|U| = |F| - |T|$ outside coordinates are counted by $p_T(H)$, which gives (7.183). $\square$

Lemma 7.145 (Facets have at least two outside coordinates). *Under the hypotheses of Theorem 7.144, one has*

$$|F \setminus T| \geq 2.$$

Proof. Put $U = F \setminus T$, and use the rooted fibers $C_q \subseteq Q_U$ from the proof of Theorem 7.144. If U were empty, the strongly shattered T -cube would be all of Q_F , so $H = Q_F$; this contradicts cornerlessness. If $|U| = 1$, then every fiber contains 0 and has at least two vertices, and hence every fiber equals Q_U . Thus $H = Q_F$ again. Equivalently, F is shattered, contrary to the maximality of the proper face T in $\text{Sh}(H)$. Therefore $|U| \geq 2$. $\square$

Equality in (7.183) is itself incompatible with ampleness.

Corollary 7.146 (Strict facet-excess inequality). *Under the hypotheses of Theorem 7.144, if H is periphery-free, then*

$$|H| \geq 2^{|T|+1} + |F| - |T| + 1.$$

Proof. Put $U = F \setminus T$ and $k = |U|$. If $k = 0$, the weak inequality would give $|H| \geq 2^{|T|+1}$, although $H \subseteq Q_T$ has at most $2^{|T|}$ members. Thus $k \geq 1$. Suppose equality holds in (7.183). With the notation from the proof of Theorem 7.144,

$$k = \left| \bigcup_{q \in Q_T} D(C_q) \right| = \sum_{q \in Q_T} |D(C_q)| = \sum_{q \in Q_T} (|C_q| - 2).$$

Hence the sets $D(C_q)$ are pairwise disjoint and partition U , and every fiber satisfies equality in Lemma 7.143. That lemma makes each C_q a rooted geodesic path. Write $t(q)$ for the label of its last edge. Then

$$\text{supp}(C_q) = D(C_q) \dot{\cup} \{t(q)\}.$$

Because the defect sets partition U , the terminal coordinate $t(q)$ belongs to $D(C_{q'})$ for a unique trace q' , and necessarily $q' \neq q$. Choose q with $|C_q| \geq 3$, set $a = t(q)$, and choose $b \in D(C_q)$. Such a fiber exists: if every fiber had two vertices, then all defect sets would be empty, contrary to $k \geq 1$. The equality description in Lemma 7.143 then ensures that $D(C_q) \neq \emptyset$.

Since $b \in D(C_q)$ and $a = t(q)$, the path C_q contains the zero trace, the singleton trace on b , and the trace with both a and b equal to one. The fiber $C_{q'}$ has $a \in D(C_{q'})$, so its terminal coordinate is not a . Moreover $b \notin D(C_{q'})$, because the defect sets are disjoint; if b occurs on this path, it is therefore its terminal coordinate and occurs after a . Thus the path contains a vertex with a -value one and b -value zero. Together with the common base trace 00, these vertices realize all four traces on $\{a, b\}$. Thus $\{a, b\}$ is shattered by H .

For each $e \in U$, let

$$P_e = \{q \in Q_T : \text{some } z \in C_q \text{ has } z_e = 1\}.$$

The sets $S \cup \{e\}$, with $S \in \text{Sh}(P_e)$, are shattered by H : the value-zero e -section contains the common T -cube, and the value-one section projects onto P_e . Conversely this criterion is exact. Since

$$|P_e| = \#\{q : e \in \text{supp}(C_q)\},$$

the fiber identity above gives

$$\sum_{e \in U} |P_e| = \sum_{q \in Q_T} |\text{supp}(C_q)| = \sum_{q \in Q_T} (|D(C_q)| + 1) = k + 2^{|T|}.$$

The Sandwich inequality for the families $P_e \subseteq Q_T$ now yields

$$\begin{aligned} |\text{Sh}(H)| &\geq 2^{|T|} + \sum_{e \in U} |\text{Sh}(P_e)| + 1 \\ &\geq 2^{|T|} + \sum_{e \in U} |P_e| + 1 = 2^{|T|+1} + k + 1 > |H|. \end{aligned}$$

Here the final 1 is the shattered pair $\{a, b\}$, and the counted families are disjoint because their intersections with U have sizes zero, one, and two. This contradicts the Sandwich equality for H . $\square$

The preceding proof has a stable first layer. Besides excluding equality, it excludes one additional vertex uniformly; this is the point at which the minimum-order argument ceases to be an order-by-order calculation.

Lemma 7.147 (First-slack rooted fibers). *Let $C_1, \dots, C_m \subseteq Q_U$ be ample families containing 0, with $|C_i| \geq 2$. Put*

$$D_i = D(C_i), \quad \delta_i = |C_i| - 2 - |D_i|,$$

and suppose that

$$\bigcup_{i=1}^m D_i = U, \quad \sum_{i=1}^m (|C_i| - 2) = |U| + 1. \quad (7.184)$$

Let $C = \bigcup_i C_i$, and for any family $A \subseteq Q_U$ write

$$h(A) = |\{R \in \text{Sh}(A) : |R| \geq 2\}|.$$

Then

$$h(C) > \sum_{i=1}^m h(C_i). \quad (7.185)$$

Proof. Write

$$\mu = \sum_i |D_i| - \left| \bigcup_i D_i \right|.$$

The defect bound and (7.184) give

$$\mu + \sum_i \delta_i = 1.$$

Thus there are two cases.

Suppose first that $\mu = 1$. Then every $\delta_i = 0$, so Lemma 7.143 makes every C_i a rooted geodesic path. Exactly one coordinate, say x , occurs twice among the defect sets, and every other coordinate occurs once. There is a coordinate $a \neq x$: otherwise two occurrences of the sole coordinate as a defect would require a path with two distinct edge labels in a one-dimensional cube. Let P be the path with $a \in D(P)$, and let t be its terminal coordinate. Let P' be a path with $t \in D(P')$, which exists by defect coverage. The path P realizes 00, 10, 11 on $\{a, t\}$. Since a is not the repeated coordinate, it does not belong to $D(P')$. Thus it is either absent from P' or is the terminal coordinate of P' , after t . In both cases P' realizes 01. Hence $\{a, t\}$ is shattered by C . No individual path shatters a pair, proving (7.185).

Now suppose that $\mu = 0$. The sets D_i partition U , and there is a unique index j with $\delta_j = 1$; all other fibers are rooted geodesic paths. Set $A = C_j$. Since A is ample,

$$|A| = 1 + |\text{supp}(A)| + h(A). \quad (7.186)$$

Combining this identity with $|A| = |D_j| + 3$ gives

$$|\text{supp}(A) \setminus D_j| + h(A) = 2.$$

The first summand is nonzero. Otherwise $h(A) = 2$; downward closure says that the two higher shattered sets are coordinate pairs. Since A is ample, these are precisely the two edges of its crossing graph. That graph is a forest, so the crossing-forest theorem [KM02, Theorem 5.5] makes the one-inclusion graph of A a C_4 -cactoid. Its recursive construction has a peripheral coordinate whose peripheral side does not contain the root. This coordinate is not a member of D_j , a contradiction.

Choose $a \in \text{supp}(A) \setminus D_j$, and let $P \neq A$ be the path fiber with $a \in D(P)$. Such a fiber exists because the defect sets partition U . Let t be the terminal coordinate of P , and let B be the unique fiber with $t \in D(B)$. The path P realizes 00, 10, 11 on $\{a, t\}$. Since $a \notin D(B)$, the fiber B realizes 01: if B is a path, then a is absent or terminal after t ; if $B = A$, take a witness for $t \in D(A)$, and, when that witness has a -value one, use its a -mate, which belongs to A because $a \notin D(A)$. Thus C shatters $\{a, t\}$.

If $h(A) = 0$, choose distinct coordinates $a, b \in \text{supp}(A) \setminus D_j$. No individual fiber shatters a pair, whereas the preceding paragraph gives a pair shattered by C . This proves (7.185) in this case.

If $h(A) = 1$, let $\{a, b\}$ be the unique higher shattered set of A , and let a_0 be the unique member of $\text{supp}(A) \setminus D_j$. The coordinate a_0 is not one of a, b . Indeed, the crossing graph of A has one edge, so the crossing-forest theorem again makes its one-inclusion graph a C_4 -cactoid with one square block. Root its block tree at the block containing 0. If a leaf block away from the root is a bridge, its coordinate is nondefect and lies outside $\{a, b\}$. Otherwise the unique square is the leaf farthest from the root. Both of its directions are then nondefect: every vertex on its side away from the attachment has its mate in the square. This contradicts the uniqueness of a_0 . If the block tree has only one block, A is the square and gives the same contradiction. Thus the first alternative occurs, and its nondefect bridge coordinate must be $a_0 \notin \{a, b\}$. Applying the preceding paragraph with a_0 therefore produces a pair different from $\{a, b\}$. It is a new higher shattered set of C , which proves (7.185). $\square$

Lemma 7.148 (Exit structure of a low-slack fiber). *Let $C \subseteq Q_U$ be ample, contain 0, and have $|C| \geq 2$. Write*

$$E(C) = \text{supp}(C) \setminus D(C), \quad h(C) = |\{R \in \text{Sh}(C) : |R| \geq 2\}|,$$

and suppose that $\delta(C) = |C| - 2 - |D(C)| \leq 2$. Then:

1. $E(C) \neq \emptyset$, no set of cardinality at least three is shattered by C , and the crossing graph of C is a forest;
2. for every $a \in D(C)$, there is a $t \in E(C)$ whose value-one peripheral side is contained in the value-one a -semicube. Thus C realizes 00, 10, 11, but not 01, on the ordered pair (a, t) ;
3. if $|E(C)| = 1$, every pair shattered by C is contained in $D(C)$.

Moreover,

$$\delta(C) = |E(C)| + h(C) - 1. \quad (7.187)$$

Proof. Ampleness and the definition of the support give

$$|C| = 1 + |\text{supp}(C)| + h(C).$$

Since $D(C) \subseteq \text{supp}(C)$, this proves (7.187).

We first show that $E(C)$ is nonempty. Otherwise $h(C) = \delta(C) + 1 \leq 3$. A shattered triple would contribute its three pairs and the triple itself to $h(C)$, so all higher shattered sets are pairs. These pairs are precisely the edges of the crossing graph. If that graph is a forest, the crossing-forest theorem [KM02, Theorem 5.5] makes the one-inclusion graph of C a C_4 -cactoid. Root its block-cut tree at the block containing 0. An outer edge of a leaf K_2 -block, or a new direction in a leaf square of a leaf C_4 -tree, has a peripheral side that avoids the root. Its direction belongs to $E(C)$, a contradiction.

If the crossing graph is not a forest, its at most three edges form one triangle; every other component is an isolated vertex. Crossing-graph components are the crossing graphs of the blocks, by [KM02, Theorem 3.4] applied blockwise. The triangular block is one of Q_3, Q_3^- , and C_6 , by [KM02, Proposition 5.2]. The cube Q_3 has four higher shattered sets, and C_6 is not ample, so the block is Q_3^- . Every other block is a K_2 . Root the block-cut tree at the block containing 0 and choose a leaf block away from the root, allowing the root block itself when it is the only block. A K_2 -leaf has an outer peripheral edge direction avoiding the root. For a Q_3^- -leaf, translate its attachment vertex (or the root in the one-block case) to zero and write $w \neq 0$ for the omitted cube vertex. For every coordinate $t \in \text{supp}(w)$, each present vertex with t -value one has its t -mate. The corresponding side avoids the attachment vertex, so it remains peripheral in all of C , and $t \in E(C)$. Both cases contradict $E(C) = \emptyset$.

Equation (7.187) now gives $h(C) \leq 2$. Thus no triple is shattered, the crossing graph is a forest, and C is a C_4 -cactoid by the cited theorem.

We use the peripheral-split lemma for median graphs [KM02, p. 244, paragraph preceding Theorem 5.4]: inside either side of any split there is a peripheral split whose peripheral side is contained in that side. Let $a \in D(C)$, and choose an unmatched vertex z in the value-one a -semicube. That semicube contains an edge. Indeed, on a geodesic from 0 to z , the first vertex after the a -edge cannot be z , since its predecessor would be $z^{\{a\}} \in C$. Apply the lemma to the value-one a -semicube and let t be the resulting peripheral direction. Its peripheral side is contained in that semicube and avoids the root. Thus $t \neq a$ and $t \in E(C)$: the direction cannot be a , since $a \in D(C)$ says precisely that the value-one a -side is not peripheral. The root and the two endpoints of a boundary t -edge realize 00, 10, 11 on (a, t) , whereas containment excludes 01. This proves item 2.

Finally, suppose that the unique member t of $E(C)$ crossed a coordinate a . Then $a \in D(C)$, so item 2 supplies an $s \in E(C)$ whose peripheral side is contained in the value-one a -semicube. Uniqueness gives $s = t$. But crossing of a and t makes the value-one t -semicube meet both a -semicubes, so it cannot be contained in the value-one a -semicube. This contradiction proves item 3. $\square$

Lemma 7.149 (Second-slack rooted fibers). *Let $C_1, \dots, C_m \subseteq Q_U$ be ample families containing 0, with $|C_i| \geq 2$ and $|U| \geq 4$. Suppose that*

$$\bigcup_{i=1}^m D(C_i) = U, \quad \sum_{i=1}^m (|C_i| - 2) = |U| + 2. \quad (7.188)$$

Then, with $C = \bigcup_i C_i$,

$$h(C) > \sum_{i=1}^m h(C_i). \quad (7.189)$$

Proof. Put $D_i = D(C_i)$, $E_i = E(C_i)$, $h_i = h(C_i)$, and

$$\mu = \sum_i |D_i| - |U|.$$

The defect bound and (7.188) give

$$\mu + \sum_i \delta(C_i) = 2. \quad (7.190)$$

In particular every fiber satisfies the hypotheses of Lemma 7.148, and

$$\sum_i |E_i| + \sum_i h_i = m + 2 - \mu. \quad (7.191)$$

We first record the common routing step. Suppose that a coordinate $a \in D_i$ occurs in no other defect set. By Lemma 7.148, choose $t \in E_i$ for which C_i realizes 00, 10, 11 on (a, t) . Defect coverage gives an index $j \neq i$ with $t \in D_j$. Choose a witness $z \in C_j$ with $z_t = 1$ and $z^{\{t\}} \notin C_j$. If $z_a = 0$, then z realizes the missing trace 01. If $z_a = 1$, uniqueness of the defect occurrence gives $a \notin D_j$, so $z^{\{a\}} \in C_j$ and realizes 01. Thus $\{a, t\}$ is shattered by C . We call it the pair routed from a .

There are three cases in (7.190). If $\mu = 2$, all fibers have $\delta = 0$, hence are rooted geodesic paths and have $h_i = 0$. The $|U| + 2$ defect occurrences leave a coordinate that occurs only once, because $|U| \geq 4$. Its routed pair proves (7.189).

Suppose that $\mu = 1$. Exactly one coordinate is repeated among the defect sets, and at least $|U| - 1 \geq 3$ coordinates occur only once. Moreover $\sum_i h_i \leq \sum_i \delta(C_i) = 1$. If the sum is zero, any unique defect coordinate gives the required routed pair. If it is one, let R be the unique pair shattered inside a fiber. Choose a unique defect coordinate outside the two members of R . Its routed pair is different from R , so C shatters at least two pairs.

It remains that $\mu = 0$. The defect sets partition U . If $\sum_i h_i \leq 1$, choose a defect coordinate outside the unique internal pair when that pair exists; there is such a coordinate because $|U| \geq 4$. Its routed pair is new. Finally suppose that $\sum_i h_i = 2$. Equation (7.191) and the nonemptiness of every E_i imply $|E_i| = 1$ for all i . Hence every pair shattered inside C_i is contained in D_i , by Lemma 7.148. The two internal pairs are distinct, because the sets D_i are disjoint. On the other hand, a routed pair from $a \in D_i$ has its other coordinate in some D_j , $j \neq i$; it is therefore not internal to any fiber. Thus the union shatters the two internal pairs and an additional routed pair. This proves (7.189) in every case. $\square$

Lemma 7.150 (Structure at fiber slack three). *Let $C \subseteq Q_U$ be ample, contain 0, have $|C| \geq 2$, and satisfy $\delta(C) = 3$. Then $E(C) \neq \emptyset$, no set of cardinality at least three is shattered by C , and the crossing graph of C is either a forest or consists of one triangle and isolated vertices. In the forest case, every $a \in D(C)$ has a $t \in E(C)$ whose value-one peripheral side is contained in the value-one a -semicube.*

In the triangular case, $h(C) = 3$, $|E(C)| = 1$, and the unique nontrivial block is Q_3^- ; all other blocks are K_2 's. Root the block-cut tree at the root. Either its terminal block is a K_2 , in which case some $a \in D(C)$ and the unique $t \in E(C)$ have the containment above, or its terminal block is Q_3^- . In the latter case, if a, b, t are its three directions, then t is the unique member of $E(C)$ and the block realizes every trace on $\{a, b, t\}$ except the singleton trace on t .

Proof. The potential identity (7.187) remains valid and gives

$$|E(C)| + h(C) = 4. \quad (7.192)$$

We claim that $E(C) \neq \emptyset$. Otherwise $h(C) = 4$. A shattered triple is strongly shattered by ampleness and gives a Q_3 -block; its block-cut tree has a peripheral side avoiding the root. If no triple is shattered, the crossing graph has four edges. When it is triangle-free, [KM02, Theorem 5.1] makes C a cube-free median graph, and the peripheral-split lemma supplies a peripheral side avoiding the root. When it contains a triangle, its fourth edge either forms a separate crossing component, in which case a terminal block is peripheral, or is pendant at the triangle. Any additional isolated crossing-graph vertex is a bridge block, whose block-cut tree has an outward peripheral leaf. Thus only a two-connected block with the four-vertex paw as crossing graph remains. Its shattered-set complex has the three minimal nonfaces

$$\{a, b, c\}, \quad \{b, t\}, \quad \{c, t\}.$$

The signed-minimal-nonface equality of Theorem 5.3 has, modulo cube translation and interchanging b, c , two solutions. Their satisfying families are

$$A = \{6, 7, 9, 10, 11, 12, 13, 14, 15\} \quad \text{and} \quad B = \{2, 3, 9, 10, 11, 12, 13, 14, 15\}.$$

For clarity, this two-orbit calculation is just inclusion-exclusion for the three forbidden subcubes: translate the forbidden trace on $\{a, b, c\}$ to 000; the equality $|C| = 9$ leaves six pairs of forbidden traces on $\{b, t\}$ and $\{c, t\}$, which form the two orbits above under flipping t and interchanging b, c . The following table lists, for each possible root, a nonempty set of nondefect directions; the entries are written as four-bit masks in the order a, b, c, t .

C	A -root	6	7	9	10	11	12	13	14	15
$E(C)$		6	7	9	10	11	12	13	14	15

C	B -root	2	3	9	10	11	12	13	14	15
$E(C)$		2	3	9	10	11	8	9	10	11

Thus the pendant-triangle case also contradicts $E(C) = \emptyset$.

Equation (7.192) gives $h(C) \leq 3$, so no triple is shattered. The crossing graph has at most three edges and is therefore a forest unless its nontrivial component is a triangle. In the forest case C is a median C_4 -cactoid, and the nested peripheral-split argument from Lemma 7.148 proves the asserted containment.

Suppose that the crossing graph contains a triangle. Then $h(C) = 3$, so (7.192) gives $E(C) = \{t\}$. The triangular block is Q_3^- , since Q_3 shatters a triple and C_6 is not ample; all other blocks are bridges. Every leaf block away from the root supplies an outward peripheral direction, and different blocks use different directions. Hence the rooted block-cut tree has a unique terminal block and is a path.

If the terminal block is a bridge of direction t , let x be its attachment vertex and y its outer endpoint. We have $x \neq 0$: if $x = 0$, the Q_3^- -block on the other side has a peripheral direction

avoiding 0, distinct from t . Choose $a \in \text{supp}(x)$. The value-one t -side is $\{y\}$, lies in the value-one a -semicube, and makes a a defect coordinate. This gives the first alternative.

It remains that the terminal block is Q_3^- . Translate its attachment vertex, or the root in the one-block case, to zero, and let $w \neq 0$ be the omitted cube vertex. For every coordinate in $\text{supp}(w)$, the corresponding value-one side is peripheral and avoids the attachment. It remains peripheral in all of C , so uniqueness of $E(C)$ forces $w = \{t\}$. The block therefore realizes exactly the other seven traces on its directions a, b, t , as claimed. $\square$

Lemma 7.151 (Third-slack rooted fibers). *Let $C_1, \dots, C_m \subseteq Q_U$ be ample families containing 0, with $|C_i| \geq 2$ and $|U| \geq 4$. Suppose that*

$$\bigcup_{i=1}^m D(C_i) = U, \quad \sum_{i=1}^m (|C_i| - 2) = |U| + 3. \quad (7.193)$$

Then, with $C = \bigcup_i C_i$,

$$h(C) > \sum_{i=1}^m h(C_i). \quad (7.194)$$

Proof. Write $D_i = D(C_i)$, $E_i = E(C_i)$, $h_i = h(C_i)$, and

$$\mu = \sum_i |D_i| - |U|.$$

The defect bound and (7.193) give

$$\mu + \sum_i \delta(C_i) = 3. \quad (7.195)$$

Call a coordinate *unique* if it belongs to exactly one defect set. At most μ coordinates are nonunique, so there are at least $|U| - \mu$ unique coordinates. For every unique $a \in D_i$ in a fiber of slack at most two, Lemma 7.148 gives an exposed pair $\{a, t\}$, with $t \in E_i$, on which C_i realizes all traces except 01. Defect coverage supplies a fiber with t as a defect; because a is unique, downward flips in coordinate a produce the missing trace exactly as in the common routing step of Lemma 7.149. Thus the union shatters this pair. Routed pairs from distinct unique coordinates are distinct except possibly when two coordinates route to each other, so k unique coordinates produce at least $\lceil k/2 \rceil$ distinct pairs.

If $\mu = 3$, all fibers are paths and one unique coordinate suffices. If $\mu = 2$, then $\sum_i \delta(C_i) = 1$ and $\sum_i h_i \leq 1$. When an internal pair exists, its fiber has a unique exit and the pair is contained in its defect set by Lemma 7.148. A routed pair spans two defect sets and is therefore new. If $\mu = 1$, there are at least three unique coordinates and $\sum_i h_i \leq 2$. When the latter sum is at most one, the two distinct routed pairs force a new one. When it equals two, equality in the potential identities makes every exit set a singleton. All internal pairs are contained in individual defect sets, whereas every routed pair spans two of them.

It remains that $\mu = 0$, so the defect sets partition U and $\sum_i \delta(C_i) = 3$. First suppose that every fiber has slack at most two, and put $H = \sum_i h_i$. The potential identity gives

$$\sum_i (|E_i| - 1) = 3 - H. \quad (7.196)$$

If $H \leq 1$, the at least four unique coordinates yield two routed pairs. If $H = 3$, (7.196) makes every E_i a singleton, so every routed pair is new. Finally, if $H = 2$, (7.196) gives total exit surplus one. Thus one fiber has two exits and the others have one. The exceptional fiber has at most one internal

pair, since otherwise its slack would be at least three; all other internal pairs lie within their defect sets. Of the two distinct routed pairs, at most one can therefore be internal.

We are left with one fiber A of slack three and path fibers of slack zero. If the crossing graph of A is a forest and $D(A) \neq \emptyset$, Lemma 7.150 gives an exposed pair from a defect of A , and defect coverage completes it in another fiber. This pair is not internal because its peripheral containment omits trace 01. If $D(A) = \emptyset$, then $E(A) = \text{supp}(A)$. A forest with $h(A) \geq 2$ has at least $h(A) + 1$ nonisolated vertices, contradicting $|E(A)| + h(A) = 4$. Hence $h(A) \leq 1$, while the four path defects yield at least two routed pairs.

Finally suppose that the crossing graph of A contains a triangle. In the bridge-terminal alternative of Lemma 7.150, its exposed pair is completed as before. In the Q_3^- -terminal alternative, let a, b, t be the three block directions. The family A realizes all traces except the singleton t , and $a, b \in D(A)$. The unique exit t is a defect of a path fiber P . Choose a t -defect witness in P . Since the defect sets partition U , neither a nor b is a defect of P ; successively flipping either coordinate that has value one produces a vertex of P whose trace on $\{a, b, t\}$ is the singleton t . The union therefore shatters this triple. The three pairs were the only internal higher faces, so (7.194) follows. This exhausts (7.195). $\square$

Corollary 7.152 (Second strict facet-excess inequality). *Under the hypotheses of Theorem 7.144, if H is periphery-free, then*

$$|H| \geq 2^{|T|+1} + |F| - |T| + 2.$$

Proof. Put $U = F \setminus T$, and suppose that equality held in the first strict bound, so that

$$|H| = 2^{|T|+1} + |U| + 1.$$

Use the rooted fibers C_q from the proof of Theorem 7.144. They satisfy (7.184). For $e \in U$, let

$$P_e = \{q \in Q_T : e \in \text{supp}(C_q)\}.$$

Because each C_q is ample,

$$|C_q| = 1 + |\text{supp}(C_q)| + h(C_q).$$

Every set with exactly one outside coordinate that is counted by $\text{Sh}(P_e)$, and every higher set shattered by $C = \bigcup_q C_q$, is shattered by H . These collections are disjoint. Lemma 7.147 and the Sandwich inequality therefore give

$$\begin{aligned} |\text{Sh}(H)| &\geq 2^{|T|} + \sum_{e \in U} |\text{Sh}(P_e)| + h(C) \\ &\geq 2^{|T|} + \sum_q |\text{supp}(C_q)| + h(C) \\ &> 2^{|T|} + \sum_q (|C_q| - 1) = |H|, \end{aligned}$$

contradicting ampleness. $\square$

Corollary 7.153 (Uniform facet-rank bound). *Under the hypotheses of Theorem 7.144, if H is periphery-free, then*

$$|H| \geq 2^{|T|+1} + 4.$$

In particular, a facet of rank at least five forces $|H| \geq 68$.

Proof. Lemma 7.145 and Corollary 7.152 give

$$|H| \geq 2^{|T|+1} + |F \setminus T| + 2 \geq 2^{|T|+1} + 4.$$

□

Corollary 7.154 (Third strict facet-excess inequality). *Under the hypotheses of Theorem 7.144, suppose that H is periphery-free and $|F \setminus T| \geq 4$. Then*

$$|H| \geq 2^{|T|+1} + |F| - |T| + 3.$$

Proof. Put $U = F \setminus T$, and suppose equality held in the second strict bound. The rooted fibers C_q from the proof of Theorem 7.144 then satisfy (7.188). Lemma 7.149 gives

$$h\left(\bigcup_q C_q\right) > \sum_q h(C_q).$$

Using the one-coordinate incidence families P_e from the proof of Corollary 7.152, the same disjoint Sandwich count gives

$$\begin{aligned} |\text{Sh}(H)| &\geq 2^{|T|} + \sum_{e \in U} |\text{Sh}(P_e)| + h\left(\bigcup_q C_q\right) \\ &> 2^{|T|} + \sum_q (|C_q| - 1) = |H|, \end{aligned}$$

contradicting ampleness. □

Corollary 7.155 (Fourth strict facet-excess inequality). *Under the hypotheses of Theorem 7.144, suppose that H is periphery-free and $|F \setminus T| \geq 4$. Then*

$$|H| \geq 2^{|T|+1} + |F| - |T| + 4.$$

Proof. Put $U = F \setminus T$, and suppose equality held in the third strict bound. The rooted fibers from Theorem 7.144 satisfy (7.193), so Lemma 7.151 gives a strict gain in their union's higher shattered sets. Adding this gain to the one-coordinate incidence-family Sandwich count, exactly as in Corollary 7.154, gives $|\text{Sh}(H)| > |H|$, contradicting ampleness. □

The low-dimensional relative-peeling theorem proved in Appendix A also eliminates two ambient dimensions at once.

Corollary 7.156 (Ample families in at most six coordinates). *Every nonempty ample family of isometric dimension at most six can be corner-peeled to any prescribed concept.*

Proof. The assertion in dimensions at most five follows from Theorem A.137, applied with S the prescribed singleton. In dimension six, orient an active coordinate so that the desired endpoint lies in the upper section, and write

$$H = (B \times \{0\}) \cup (M \times \{1\}) \subseteq Q_{E \cup \{e\}}, \quad |E| = 5,$$

for the family. If one section is empty, the other uses at most five coordinates and is already covered. Otherwise the intersection $S = B \cap M$ is nonempty: the one-inclusion graph of an ample family is

connected, so a path between the two sections contains an e -edge. Moreover, S is ample. Indeed, if a support is shattered by S , then adjoining e gives a shattered support of H ; ampleness makes it strongly shattered, and the cube crossing e makes the original support strongly shattered in S . The Sandwich equality follows.

Apply Theorem A.137 to $S \subseteq B$, ending at any chosen $s \in S$. In H , the same order deletes the lower copy of $B \setminus S$: such a vertex has no e -mate, so its corner cube is unchanged. It then deletes the lower copy of S : each old-coordinate corner cube lies in S and is duplicated in M , so adjoining the e -direction completes the corner cube in H . The remaining family is the upper copy of M , which uses at most five coordinates. By the first sentence it peels to the prescribed upper endpoint. $\square$

The next identity records the contribution of indexed fibers to the shattered sets of a family. It also supports the finite four-coordinate calculation used in Proposition 7.86.

Lemma 7.157 (Indexed fiber shadow decomposition). *Let T and U be disjoint coordinate sets, let $H \subseteq Q_{T \cup U}$, and write*

$$C_q = \{z \in Q_U : (q, z) \in H\} \quad (q \in Q_T).$$

Assume that $0 \in C_q$ for every $q \in Q_T$. For $R \subseteq U$ and $\alpha \in Q_R$, put

$$A_{R,\alpha} = \{q \in Q_T : \alpha \in \pi_R(C_q)\}, \quad \mathcal{K}_R = \bigcap_{\alpha \in Q_R} \text{Sh}(A_{R,\alpha}).$$

Then

$$\text{Sh}(H) = \{S \cup R : R \subseteq U, S \in \mathcal{K}_R\}. \quad (7.197)$$

Suppose in addition that every C_q is ample. Put

$$C = \bigcup_{q \in Q_T} C_q, \quad P_e = \{q \in Q_T : e \in \text{supp}(C_q)\} \quad (e \in U),$$

and, for a cube family A , let

$$h(A) = |\{R \in \text{Sh}(A) : |R| \geq 2\}|.$$

Then the exact shadow gap is

$$\begin{aligned} |\text{Sh}(H)| - |H| &= \sum_{e \in U} (|\text{Sh}(P_e)| - |P_e|) + h(C) - \sum_{q \in Q_T} h(C_q) \\ &\quad + \sum_{\substack{R \subseteq U \\ |R| \geq 2}} (|\mathcal{K}_R| - \mathbf{1}_{\{R \in \text{Sh}(C)\}}). \end{aligned} \quad (7.198)$$

The first sum and the last sum in (7.198) are nonnegative. No sign is asserted for the middle, unindexed union gap.

Proof. Fix $S \subseteq T$ and $R \subseteq U$. The set $S \cup R$ is shattered by H exactly when, for every $\alpha \in Q_R$, the fibers that realize α project onto all of Q_S . This is equivalent to $S \in \text{Sh}(A_{R,\alpha})$ for every α , and proves (7.197).

The common root gives

$$\mathcal{K}_\emptyset = 2^T, \quad \mathcal{K}_{\{e\}} = \text{Sh}(P_e).$$

Since every fiber is ample,

$$|H| = \sum_q |C_q| = 2^{|T|} + \sum_{e \in U} |P_e| + \sum_q h(C_q).$$

Moreover, $\mathcal{K}_R$ is nonempty exactly when every trace on R occurs in at least one fiber, which is equivalent to $R \in \text{Sh}(C)$. When it is nonempty, it contains $\emptyset$. Hence

$$\sum_{\substack{R \subseteq U \\ |R| \geq 2}} |\mathcal{K}_R| = h(C) + \sum_{\substack{R \subseteq U \\ |R| \geq 2}} (|\mathcal{K}_R| - \mathbf{1}_{\{R \in \text{Sh}(C)\}}).$$

Substitution in (7.197) gives (7.198). The Sandwich inequality makes every summand in the first sum nonnegative, and the preceding nonemptiness criterion makes every summand in the last sum nonnegative. $\square$

Proposition 7.158 (Complete indexed total-excess-eleven classification in four coordinates). *Let $|T| = 3$, let $|U| = 4$, and let*

$$H = \bigcup_{q \in Q_T} (\{q\} \times C_q) \subseteq Q_{T \cup U},$$

where every $C_q \subseteq Q_U$ is ample, contains 0, and has at least two members. Suppose that

$$\bigcup_{q \in Q_T} D(C_q) = U, \quad \sum_{q \in Q_T} (|C_q| - 2) = 11.$$

If H is ample, then it has at least four corners. In particular, no such H is cornerless.

Computer-assisted proof. Direct enumeration gives all 2,720 rooted ample subfamilies of Q_4 having order between two and thirteen. The exact union-state program checks all fifty-two partitions of total excess eleven among the eight fibers. For a fixed covered-defect mask and union family it retains the largest internal higher-face count, which is exactly the value that minimizes the unindexed gap. Among the full-defect terminal states it finds 7,293 states with nonpositive gap, including 148 with negative gap.

One common S_4 -action reduces these to 423 union rows. For every row, the verifier expands every fiber decomposition whose realized internal count makes the gap nonpositive; thus the negative-gap rows do not lose nonextremal decompositions. This gives 179,937 fiber multisets in 76,274 common-coordinate orbits. Every orbit has a fiber type of multiplicity one. Place the least such type at $0 \in Q_T$, quotient the remaining placements by the S_3 -stabilizer of 0, and give every representative eight times its stabilizer-orbit weight. Because the marked fiber occurs once, this is an exact transversal for the full forty-eight-element automorphism action on Q_T .

The resulting audit represents all 1,313,602,248 labelled assignments. Exactly 217,368 are ample, and every ample assignment has at least four corners. For each profile, the minimum indexed shadow is also checked by a direct seven-coordinate shattered-set count. On the 156 profiles having at most 168 labelled assignments, full-labelled enumeration independently agrees with the quotient calculation, including the shadow, ampleness, corner, and peripheral-coordinate data.

The reproducible programs and outputs are

```
explore_damp_q4_seventh_slack_nonpositive.py,
damp_q4_seventh_slack_nonpositive.json,
verify_damp_q4_seventh_slack_indexed.py,
damp_q4_seventh_slack_indexed_certificate.json.
```

The two output digests are respectively

```
c176842bba8798943cdb3d51de47874a204a0d108ba917c972d5f8c91bc9d5b0,
1e4ac11fe91392103e057cfbf490167d059aefb6c96c3f8ef7195edfbcfbfeae.
```

Positive unindexed gap is incompatible with ampleness by (7.198), while the exhaustive indexed audit covers every nonpositive-gap decomposition. This proves the claim. $\square$

Theorem 7.159 (Order bounds). *The least order of an ample forbidden pc-minor satisfies*

$$64 \leq m_{\text{amp}} \leq 267.$$

Proof. Proposition 7.55 gives the upper bound. For the lower bound, suppose that an ample forbidden pc-minor O has order less than 64. Delete corners from O as long as possible. The ample corner-deletion criterion preserves ampleness, and the process cannot reach the empty family: such a sequence would be a corner peeling of O . It therefore leaves a nonempty cornerless ample family of order at most $|O|$.

Choose H of minimum cardinality among all nonempty cornerless ample families. The preceding construction ensures that $|H| \leq |O| < 64$. There are two exhaustive ranges.

If $|H| < 48$, Corollary 7.136 gives a maximum core of VC dimension one. The path-core exclusion of Proposition 7.105 then gives a corner of H , contrary to its choice. We use that proposition only in the range $|H| < 48$.

If $48 \leq |H| \leq 63$, Corollary 7.134 again gives a corner of H , a contradiction. Both ranges are impossible, so no ample forbidden pc-minor has order below 64. $\square$

8 Conclusion

The nonample part of the forbidden pc-minor basis of $\mathcal{D}_{\text{amp}}$ is the family of antipodally twice-punctured cubes. The ample part has exact recognition criteria in cubical, algebraic, topological, and signed-set language. A complete intrinsic generative classification of the ample members remains open.

The canonical three-core reduction and square attachment theorem settle construction and recovery of every extension over a given forbidden three-core. Cut vertices and separating edges give further exact decompositions, and down-set completion constructs additional examples from the same rooted factors. Replicated repairs give a separate family of constructions; over a cornerless seed, terminal layer data are precisely partitions of the signed layer facets. In each case, the unresolved primitive inputs are explicit: forbidden cores, rooted factors, resolving additions, or relative peeling properties of pieces.

In the USO formulation, rooted factors obstruct a prescribed sink and separator pieces obstruct attainable boundary out-maps. The exact gluing laws in Proposition 2.20 and Theorem 2.30 specify which boundary data must be compatible. Protected missing cubes give the uniform cycle certificate of Proposition 2.9. These descriptions keep existence of an acyclic map separate from the behaviour of one chosen representation map.

Two tasks remain. These primitive properties need to follow from structural conditions on generative parameters, and the constructions need to recover all forbidden cores. The terminal cut-vertex and two-connected examples outside the common-seed class prevent reducing coverage to that class. The infinite family with nonpeelable cube complements also excludes generation solely by corner deletions from cubes. Moreover, a three-core shadow can force corners in every ample realization (Corollary 2.56), so changing signs alone cannot normalize all primitive inputs to cornerless families. These restrictions leave the intrinsic classification problem open without requiring a bounded list of examples or parameters.

The independent extremal question currently has the bounds $64 \leq m_{\text{amp}} \leq 267$. Table 1 summarizes the construction and recovery results separately from these bounds and from the remaining primitive-input problem.

A Relative peeling in five coordinates

This appendix proves the low-dimensional interval theorem used in Corollary 7.156. For nested ample families $S \subseteq B$, a *relative peeling* means a corner peeling of B in which every member of $B \setminus S$ is deleted before every member of S . The proof is symbolic; the bounded five-coordinate computations retained in the campaign archive are independent regression checks.

Lemma A.1 (One-concept augmentation as a colon test). *Let $B \subseteq Q_E$ be ample, let $x \in Q_E \setminus B$, and put*

$$m_x = \prod_{e \in E} v_{e, 1-x_e}, \quad \delta(x, y) = \{e \in E : x_e \neq y_e\}, \quad z_e = v_{e, x_e}. \quad (\text{A.1})$$

Then the following are equivalent.

1. $B \cup \{x\}$ is ample.
2. The one-generator colon $(J_B : m_x)$ is generated by variables.
3. For every $y \in B$, there is an $e \in \delta(x, y)$ such that $x^{\{e\}} \in B$.

When they hold, the colon is exactly

$$(J_B : m_x) = (z_e : x^{\{e\}} \in B). \quad (\text{A.2})$$

More generally, suppose that $B \subseteq B_0 \subseteq Q_E$ are ample and put $A = B_0 \setminus B$. If $x \in A$ cannot be adjoined amply to B , then there is a blocking vertex $y \in B$ for which

$$\delta(x, y) \cap \{e : x^{\{e\}} \in B\} = \emptyset. \quad (\text{A.3})$$

Every B_0 -geodesic from x to such a y starts with an edge from x to another member of A . Consequently, if no member of A can be adjoined amply to B , then the subgraph induced by A has no isolated vertex.

Proof. The monomial-colon formula gives

$$(J_B : m_x) = \left(\frac{m_y}{\gcd(m_y, m_x)} : y \in B \right) = \left(\prod_{e \in \delta(x, y)} z_e : y \in B \right). \quad (\text{A.4})$$

A variable z_e belongs to a minimal monomial generating set on the right only if $x^{\{e\}} \in B$. It follows that this colon is generated by variables exactly when every displayed monomial is divisible by a variable coming from a cube neighbor of x in B . This is precisely item 3, and it also proves (A.2).

If $B \cup \{x\}$ is ample, it is isometric. A geodesic from x to any $y \in B$ starts with a neighbor $x^{\{e\}} \in B$ for some $e \in \delta(x, y)$, proving item 3. Conversely, suppose the colon is generated by variables. The exact sequence

$$0 \longrightarrow J_B \longrightarrow J_B + (m_x) \longrightarrow (S_E / (J_B : m_x))(-|E|) \longrightarrow 0 \quad (\text{A.5})$$

has an $|E|$ -linear module on the left, by ampleness of B , and an $|E|$ -linear module on the right, by the Koszul resolution of a variable-generated ideal. Its Tor sequence therefore makes $J_B + (m_x) = J_{B \cup \{x\}}$ $|E|$ -linear. The Yang-ideal criterion implies that $B \cup \{x\}$ is ample.

If augmentation fails, the negation of item 3 supplies y satisfying (A.3). Since B_0 is ample, it is isometric. The first edge of a B_0 -geodesic from x to y flips some coordinate in $\delta(x, y)$. Its other endpoint cannot belong to B , by (A.3), and hence belongs to A . The last assertion follows by applying this argument to every $x \in A$. $\square$

Lemma A.2 (A conditioned augmentation lifts or has a transverse blocker). *Let E and F be disjoint coordinate sets, let $H \subseteq Q_{E \dot{\cup} F}$ be ample, fix $\alpha \in Q_F$, and put*

$$K = \{u \in Q_E : (u, \alpha) \in H\}.$$

Suppose that $x \notin K$ and $K \cup \{x\}$ is ample, and write $v = (x, \alpha)$. Then either $H \cup \{v\}$ is ample, or there is a set $J \subseteq E \dot{\cup} F$ such that

$$|J| \geq 2, \quad J \cap F \neq \emptyset, \quad (H \cup \{v\}) \cap Q_J(v) = \{v, v^J\}. \quad (\text{A.6})$$

Here $Q_J(v)$ denotes the J -face through v . Thus a conditioning-valid one-concept extension can fail globally only through an antipodal blocker using a coordinate that was fixed in the conditioning.

Proof. If $H \cup \{v\}$ is nonample, Proposition 2.131 supplies a set J of order at least two for which the last equality in (A.6) holds. If $J \cap F$ were empty, the whole J -face would lie in the F -conditioning α . Deleting the fixed coordinates would then give an antipodal-pair conditioning of $K \cup \{x\}$, contradicting its ampleness. Hence $J \cap F \neq \emptyset$. $\square$

Lemma A.3 (A blocked puncture forces a complete bipartite two-shadow). *Let $H \subseteq Q_E$ be ample, let $v \in Q_E \setminus H$, and let $T, J \subseteq E$ satisfy $|T| \geq 2$, $|J| \geq 2$,*

$$v^U \in H \quad (\emptyset \neq U \subseteq T), \quad (H \cup \{v\}) \cap Q_J(v) = \{v, v^J\}. \quad (\text{A.7})$$

Then $T \cap J = \emptyset$,

$$v^{J \cup \{t\}} \in H \quad (t \in T), \quad (\text{A.8})$$

and H shatters every pair

$$\{t, e\}, \quad t \in T, \quad e \in J. \quad (\text{A.9})$$

Thus a punctured T -cube and a transverse antipodal blocker do not merely give one owner: they force the complete bipartite graph $K_{T,J}$ into the two-dimensional shattered shadow.

Proof. For $t \in T$, the vertex $v^{\{t\}}$ belongs to H . If $t \in J$, this vertex would be a member of $Q_J(v)$ different from the two vertices allowed in (A.7). Hence $T \cap J$ is empty.

Put $y = v^J \in H$ and fix $t \in T$. The vertices y and $v^{\{t\}}$ differ exactly on $J \dot{\cup} \{t\}$. For every $e \in J$, the neighbour $y^{\{e\}} = v^{J \setminus \{e\}}$ is excluded by (A.7). Since H is ample, it is isometric. An H -geodesic from y to $v^{\{t\}}$ must therefore start in direction t , proving (A.8).

Now fix $e \in J$, and choose $u \in T \setminus \{t\}$. Relative to the trace of v on $\{t, e\}$, the four vertices

$$v^{\{u\}}, \quad v^{\{t\}}, \quad v^J, \quad v^{J \cup \{t\}}$$

have traces 00, 10, 01, 11, respectively. They all belong to H , so $\{t, e\}$ is shattered. This proves (A.9). $\square$

Corollary A.4 (Antipodal blockers in a relative-peeling gap). *Let $S \subsetneq B \subseteq Q_E$ be ample, and suppose that $S \cup \{x\}$ is nonample for every $x \in B \setminus S$. For each such x there are $J_x \subseteq E$, $|J_x| \geq 2$, and $y_x = x^{J_x} \in S$ such that the J_x -face through x contains no other member of $S \cup \{x\}$. Moreover, every B -geodesic from x to y_x lies in that face, has all its internal vertices in $B \setminus S$, and has length $|J_x|$.*

Proof. Apply Proposition 2.131 to $S \cup \{x\}$. Since B is ample, it is isometric. A shortest B -path between the two resulting antipodes flips exactly the coordinates of J_x and therefore remains in their face. No internal vertex belongs to S , by the exact two-point intersection in (2.68). $\square$

Lemma A.5 (Relative peeling and saturated ample intervals). *Let $\emptyset \neq S \subseteq B$ be ample families. The following conditions are equivalent.*

1. *The family B has a relative corner peeling that deletes exactly $B \setminus S$ and retains all of S .*
2. *There is a chain of ample families*

$$S = C_0 \subset C_1 \subset \cdots \subset C_k = B, \quad |C_i \setminus C_{i-1}| = 1 \quad (1 \leq i \leq k).$$

Proof. If the asserted peeling exists, stop it when S remains and reverse the deleted prefix. Every intermediate family is ample because corner deletion preserves ampleness. This gives the displayed chain. Conversely, reverse such a chain. At each step one vertex is deleted from an ample family and the remaining family is ample. The ample corner-deletion criterion makes that vertex a corner, so the reversed chain is the required peeling prefix. $\square$

Remark A.6 (The ample-interval conjecture). Requiring the equivalent conditions in Lemma A.5 for every nested ample pair is equivalent, by finite induction, to the interval conjecture of Moran and Warmuth [MW16, Conjecture 2]: whenever two ample families differ by at least two concepts, their inclusion interval contains a third ample family. The Hall class refutes that conjecture [CCMW22, Remark 4.6]. Thus the relative-peeling theorems below are genuinely low-coordinate results; they cannot be continued by a dimension-free assertion for arbitrary ample families. By Proposition 5.63, the chain is equivalently a sequence of isometric one-concept extensions from S to B . No peeling of the retained lower family is part of this relative assertion.

Proposition A.7 (Simultaneous shift normalization of ample intervals). *Let $\emptyset \neq S \subsetneq B \subseteq Q_E$ be ample. For a coordinate $e \in E$, define the up-shift $U_e(X)$ by its two projected sections:*

$$U_e(X)_0 = X_0 \cap X_1, \quad U_e(X)_1 = X_0 \cup X_1.$$

Then $U_e(S) \subsetneq U_e(B)$, both shifted families are ample, and

$$\text{Sh}(U_e(S)) = \text{Sh}(S), \quad \text{Sh}(U_e(B)) = \text{Sh}(B).$$

After shifting both endpoints once in every coordinate, in any order, the pair becomes

$$\mathcal{H}_{\text{Sh}(S)}^\uparrow \subsetneq \mathcal{H}_{\text{Sh}(B)}^\uparrow, \quad \mathcal{H}_\Sigma^\uparrow := \{x \in Q_E : \{e : x_e = 0\} \in \Sigma\}. \quad (\text{A.10})$$

The terminal pair admits a saturated chain of ample families and hence a relative corner peeling. Consequently, if $S \subsetneq B$ has no relative peeling, then along every complete simultaneous-shift sequence there is a first coordinate shift that changes a nonpeelable interval into a peelable one. More specifically, if $S \subsetneq B$ has no ample one-concept intermediate family, then along every such sequence there is a first shift after which an ample one-concept intermediate family exists.

Proof. On each fiber over $E \setminus \{e\}$, the up-shift acts as follows:

$$\emptyset \mapsto \emptyset, \quad \{0\} \mapsto \{1\}, \quad \{1\} \mapsto \{1\}, \quad \{0, 1\} \mapsto \{0, 1\}.$$

This map is monotone under inclusion, so $U_e(S) \subseteq U_e(B)$; the inclusion remains proper because each shift preserves cardinality. The Yang paired-shift theorem [Bou26] says that an up-shift of an ample family is ample and preserves its shattered complex. Its coordinate shifts commute, and shifting every coordinate gives the standard representatives in (A.10). Since $S \subseteq B$, one has $\text{Sh}(S) \subseteq \text{Sh}(B)$, so these standard representatives are nested.

It remains to peel the standard pair. Order the faces in $\text{Sh}(B) \setminus \text{Sh}(S)$ so that every proper subface occurs first. Adjoin the corresponding concepts, whose zero sets are those faces, in that order. Every intermediate zero-set family is a simplicial complex and therefore an ample down-set. Lemma A.5 converts this saturated ample chain into a relative peeling. If the original interval is not peelable, the finite shift sequence starts nonpeelable and ends peelable; its first peelable term has the asserted predecessor. The same argument, using the first member of the terminal saturated chain, proves the final assertion about one-concept intermediate families. $\square$

Proposition A.8 (One-coordinate inverse-shift criterion). *Let $X \subseteq Y \subseteq Q_{F \cup \{e\}}$ be ample. Write X_0, X_1 for the two e -sections and put*

$$A = X_0 \cap X_1, \quad C = X_0 \cup X_1, \quad \widehat{X} = U_e(X). \quad (\text{A.11})$$

Suppose that $p = (z, c) \in U_e(Y) \setminus \widehat{X}$ and that $\widehat{X} \cup \{p\}$ is ample.

If $c = 0$, then some concept in the e -fiber over z belongs to $Y \setminus X$ and can be adjoined amply to X .

If $c = 1$, then $z \notin C$, the family $C \cup \{z\}$ is ample, and, for

$$N_C(z) = \{i \in F : z^{\{i\}} \in C\}, \quad (\text{A.12})$$

every available lift $q_b = (z, b) \in Y$ satisfies

$$X \cup \{q_b\} \text{ is ample} \iff z^{\{i\}} \in X_b \text{ for every } i \in N_C(z). \quad (\text{A.13})$$

More precisely, in the residual literal ring put

$$I_b = (J_{X_b} : m_z) \quad (b \in \{0, 1\}).$$

After extension to the full literal ring,

$$(J_X : m_{q_b}) = I_b + v_{e,b} I_{1-b}, \quad I_0 + I_1 = (J_C : m_z) = (v_{i,z_i} : i \in N_C(z)). \quad (\text{A.14})$$

Consequently, if $i \in N_C(z)$ but $z^{\{i\}} \notin X_b$, then $z^{\{i\}} \in X_{1-b} \setminus X_b$ and $v_{e,b} v_{i,z_i}$ is a minimal quadratic generator of $(J_X : m_{q_b})$.

Proof. First suppose $c = 0$. If $z \in C$, then $z \notin A$, so exactly one lift of z belongs to X . Since the lower lift belongs to $U_e(Y)$, both lifts belong to Y . The lower section of $\widehat{X} \cup \{p\}$ is $A \cup \{z\}$, which is ample. Duplicating z into the missing section therefore preserves ampleness, by Lemma A.20.

Suppose instead that $z \notin C$. Both lifts of z belong to Y . For $w \in C$, apply the one-concept augmentation criterion to p and the upper lift $(w, 1) \in \widehat{X}$. The e -neighbor of p is absent, so there is an $i \in \delta(z, w)$ with $(z^{\{i\}}, 0) \in \widehat{X}$, or equivalently $z^{\{i\}} \in A$. Both lifts of this neighbor belong to X . The same criterion now shows that either lift of z can be adjoined amply to X .

Now suppose $c = 1$. Since the upper section of $\widehat{X}$ is C , one has $z \notin C$. The upper section of $\widehat{X} \cup \{p\}$ is $C \cup \{z\}$, so this family is ample. Assume first that every neighbor in (A.12) belongs to X_b . Given $(w, d) \in X$, the augmentation criterion in $C \cup \{z\}$ supplies an $i \in \delta(z, w)$ with $z^{\{i\}} \in C$. By hypothesis, $(z^{\{i\}}, b) \in X$, so the same criterion proves that $X \cup \{q_b\}$ is ample.

Conversely, if $z^{\{i\}} \in C \setminus X_b$, then it belongs to $X_{1-b} \setminus X_b$. The concepts q_b and $(z^{\{i\}}, 1-b) \in X$ differ exactly in coordinates e, i , but neither of the two intermediate cube vertices belongs to X . This is a blocking vertex for q_b , so $X \cup \{q_b\}$ is not ample. This proves (A.13).

Finally, sort the generators in the monomial-colon formula according to their e -section. Generators from section b give I_b , whereas those from section $1-b$ acquire the factor $v_{e,b}$. This proves

the first equality in (A.14). Since $J_C = J_{X_0} + J_{X_1}$, monomial colons give $(J_C : m_z) = I_0 + I_1$. The augmentation criterion for $C \cup \{z\}$ gives the displayed variable generators. If the i -neighbor is missing from X_b , then $v_{i,z_i} \notin I_b$ but $v_{i,z_i} \in I_{1-b}$. Neither a unit nor either of the two variables dividing $v_{e,b}v_{i,z_i}$ belongs to the full colon, so this quadratic generator is minimal. $\square$

Corollary A.9 (Minimal repair sets give split corners). *Let $S \subsetneq B \subseteq Q_E$ be ample and suppose that no $q \in B \setminus S$ makes $S \cup \{q\}$ ample. Fix an orientation of every coordinate and, for $T \subseteq E$, let U_T denote the product of the corresponding commuting up-shifts. Choose T inclusion minimal such that $U_T(S) \subsetneq U_T(B)$ has an ample one-concept intermediate family. Then $T \neq \emptyset$.*

For every $e \in T$, some first augmentation after the e -shift has the upper form $p_e = (z_e, 1)$. Immediately before that shift, put

$$X = U_{T \setminus \{e\}}(S), \quad Y = U_{T \setminus \{e\}}(B).$$

Then $C = X_0 \cup X_1$ satisfies $C \cup \{z_e\}$ ample. Its neighbor set $N_e = N_C(z_e)$ is an inclusion-minimal member of

$$\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S), \tag{A.15}$$

and $e \notin N_e$. For every sign b for which $(z_e, b) \in Y$, at least one neighbor $z_e^{\{i\}}$, $i \in N_e$, lies exclusively in the opposite section $X_{1-b} \setminus X_b$, and the corresponding colon has the minimal quadratic generator $v_{e,b}v_{i,(z_e)_i}$. Moreover, at least one of

$$(z_e, 1 - b), \quad (z_e^{\{i\}}, b) \tag{A.16}$$

belongs to $Y \setminus X$. Thus every such quadratic generator points to an adjacent concept in the relative layer, in direction e or i . If both signs are available, N_e has an exclusive neighbor on each side. In particular, if $\mathcal{K}$ is supported only in ranks two and three, then

$$|N_e| \in \{2, 3\} \quad (e \in T). \tag{A.17}$$

Proof. The fully shifted pair is the canonical nested down-set pair from Proposition A.7, so it has a one-concept intermediate family. The unshifted pair has none, proving that a nonempty minimal T exists. Fix $e \in T$. By minimality, the pair $X \subsetneq Y$ has no ample one-concept intermediate, whereas its e -shift does. A lower first augmentation would lift to X by Proposition A.8, a contradiction. Thus it has the asserted upper form, and the same proposition gives the section split and the minimal quadratic colon generator for every available sign. The available lift and its opposite-section blocking neighbor differ exactly in coordinates e, i . Since Y is ample, it is isometric, so a length-two Y -geodesic contains one of the two intermediate vertices in (A.16). Neither belongs to X , by construction, proving the relative-layer assertion.

It remains to identify the new support. The family $C \cup \{z_e\}$ is ample and deleting z_e leaves the ample family C , so z_e is a corner. Its incident directions are exactly N_e , and its unique maximal cube contains every z_e^R with $\emptyset \neq R \subseteq N_e$. Hence $C \cup \{z_e\}$ shatters N_e . The family C does not realize the pattern $z_e|_{N_e}$: otherwise an isometric path from z_e to such a realization would have to start in a direction outside N_e . Thus N_e is the unique new shattered support. Coordinate shifts preserve the two endpoint shattered complexes, so $N_e \in \mathcal{K}$. Since adjoining one concept adds only N_e , downward closure makes it inclusion-minimal in $\mathcal{K}$. This also proves $e \notin N_e$ and (A.17). $\square$

Remark A.10 (A split corner in three coordinates). Let $F = \{a, b\}$, put $z = 00$, and take

$$X_0 = \{10, 11\}, \quad X_1 = \{01, 11\}.$$

The lift $X \subseteq Q_{F \cup \{e\}}$ is a four-vertex path and is ample. Its shifted sections are $A = \{11\} \subset C = \{10, 01, 11\}$. The upper concept $p = (00, 1)$ can be adjoined to $U_e(X)$, because $C \cup \{00\} = Q_F$. The lower lift $q = (00, 0)$, however, cannot be adjoined to X : the concept $(01, 1) \in X$ differs from q in coordinates b, e , and neither intermediate vertex belongs to X . Here $N_C(00) = \{a, b\}$, but the b -neighbor 01 lies exclusively in X_1 . Thus $U_e(X \cup \{q\}) = U_e(X) \cup \{p\}$ is ample although $X \cup \{q\}$ is not.

Remark A.11 (What the shift reduction does and does not prove). Example A.10 rules out a pointwise inverse-shift theorem even in three coordinates, while the Hall example rules out unrestricted lifting of a complete relative peeling. The proved gain is the necessary normal form in Corollary A.9: a minimum five-coordinate gap would carry, after partial standardization, a constellation of rank-two or rank-three split corners with specified quadratic colon generators. Ruling out that constellation using the complementary signed blockers would replace the relative-ear argument below. It is a second sufficient proof route, not an equivalence that has already been established.

Lemma A.12 (Relative Cohen–Macaulay form of an ample interval). *Let $\emptyset \neq S \subseteq B \subseteq Q_E$ be ample, and let*

$$\Psi(B, S) = (\Delta_B, \Delta_S)$$

be the corresponding relative Yang complex. Then $\Psi(B, S)$ is a fully Cohen–Macaulay relative complex of dimension $|E| - 1$. Its balanced flag h -numbers are

$$h_R(\Psi(B, S)) = \begin{cases} 1, & R \in \text{Sh}(B) \setminus \text{Sh}(S), \\ 0, & \text{otherwise,} \end{cases} \quad (R \subseteq E). \quad (\text{A.18})$$

Moreover, a relative shelling of $\Psi(B, S)$ is exactly an ordering $p_1, \dots, p_k$ of $B \setminus S$ for which every family

$$S_i = S \cup \{p_1, \dots, p_i\} \quad (0 \leq i \leq k)$$

is ample.

Proof. The Yang complexes $\Delta_S \subseteq \Delta_B$ are Cohen–Macaulay of the same dimension, by the ampleness–linear-resolution dictionary and Alexander duality. The relative face module occurs in the exact sequence

$$0 \longrightarrow M[\Psi(B, S)] \longrightarrow k[\Delta_B] \longrightarrow k[\Delta_S] \longrightarrow 0.$$

The depth lemma therefore makes the relative module Cohen–Macaulay as well. This is the fully Cohen–Macaulay presentation in the terminology of [CKS19].

For $X \in \{S, B\}$, the flag f -number on a color set R counts the faces of Δ_X using exactly the colors in R . Such a face is a pattern realized by the projection of X to R , and hence

$$f_R(\Delta_X) = |\pi_R(X)| = |\{T \subseteq R : T \in \text{Sh}(X)\}|. \quad (\text{A.19})$$

Here the second equality uses minor closure of ampleness: the projection is ample, and its shattered supports are precisely the shattered supports of X contained in R . Boolean Möbius inversion gives

$$h_R(\Delta_X) = \mathbf{1}_{R \in \text{Sh}(X)}.$$

Subtracting the two flag h -vectors proves (A.18).

Finally, the relative shelling condition for adjoining the facet of p_i says precisely that $(J_{S_{i-1}} : m_{p_i})$ is generated by variables. By Lemma A.1, this is equivalent to ampleness of S_i . Applying this at every step proves the last assertion. $\square$

Lemma A.13 (Balanced Artinian model of an ample interval). *For $X \subseteq Q_E$ ample, put*

$$\theta_e = v_{e,0} + v_{e,1} \quad (e \in E).$$

Then $\theta = (\theta_e : e \in E)$ is a linear system of parameters for $k[\Delta_X]$, and

$$\frac{k[\Delta_X]}{(\theta)} \cong \frac{k[x_e : e \in E]}{(x_e^2 : e \in E; x_R : R \notin \text{Sh}(X))}, \quad x_R := \prod_{e \in R} x_e. \quad (\text{A.20})$$

In particular, the residue classes x_R , $R \in \text{Sh}(X)$, form a vector-space basis.

For a nested ample pair $S \subseteq B$, quotienting the relative face module by θ gives the monomial module

$$\frac{M[\Psi(B, S)]}{(\theta)M[\Psi(B, S)]} \cong \text{span}_k\{x_R : R \in \text{Sh}(B) \setminus \text{Sh}(S)\}. \quad (\text{A.21})$$

Its nonzero multiplication maps are exactly

$$x_e x_R = x_{R \cup \{e\}} \quad \text{when } e \notin R \text{ and } R \cup \{e\} \in \text{Sh}(B) \setminus \text{Sh}(S). \quad (\text{A.22})$$

Proof. Every facet of the balanced complex Δ_X contains exactly one vertex of each color. The coefficient matrix of the color sums on any facet is therefore the identity matrix, so the color sums form a linear system of parameters. Since Δ_X is Cohen–Macaulay, they form a regular sequence.

Modulo the color sums, write x_e for the image of $v_{e,0}$; then the image of $v_{e,1}$ is $-x_e$. The nonface $\{v_{e,0}, v_{e,1}\}$ gives $x_e^2 = 0$. More generally, a signed face on a color set R maps, up to sign, to x_R . Such a signed face is absent from Δ_X for at least one sign pattern exactly when R is not shattered by X . These are all the Stanley–Reisner relations after the substitution, proving (A.20) and its basis statement.

Apply the same regular sequence to the exact sequence

$$0 \longrightarrow M[\Psi(B, S)] \longrightarrow k[\Delta_B] \longrightarrow k[\Delta_S] \longrightarrow 0.$$

Lemma A.12 makes all three modules Cohen–Macaulay of the same dimension, so successive quotienting by the regular sequence preserves exactness. The resulting map sends x_R to itself when $R \in \text{Sh}(S)$ and to zero otherwise. Its kernel is therefore precisely the span in (A.21). Since shattered-set families are order ideals, multiplication has exactly the form (A.22). $\square$

Lemma A.14 (The five-color Artinian graph). *Suppose that the support difference $\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$ in Lemma A.13 is supported only in ranks two and three, and that $|E| = 5$. Put*

$$G_2 = \mathcal{K} \cap \binom{E}{2}, \quad G_3 = \{E \setminus K : K \in \mathcal{K} \cap \binom{E}{3}\}. \quad (\text{A.23})$$

The multiplication graph of the Artinian relative module has bipartition $G_2 \dot{\cup} G_3$, with an edge between $R \in G_2$ and $Q \in G_3$ exactly when $R \cap Q = \emptyset$. That edge records multiplication by the unique variable x_e , where

$$\{e\} = E \setminus (R \cup Q). \quad (\text{A.24})$$

Thus this multiplication graph is a subgraph of the bipartite double cover of the Petersen graph $KG(5, 2)$. It has no four-cycle, and the complementary ample interval has the same colored graph with its two parts exchanged.

Proof. By (A.22), a nonzero multiplication from the basis element x_R in degree two to the basis element x_K in degree three occurs exactly when $K = R \cup \{e\}$. Writing $Q = E \setminus K$, this is equivalent to $R \cap Q = \emptyset$, and the remaining color is the one in (A.24). This is precisely the stated subgraph of the bipartite double cover of $KG(5, 2)$.

For completeness, a four-cycle would give distinct two-sets R_1, R_2 and Q_1, Q_2 such that every R_i is disjoint from every Q_j . Then both R_i 's lie in $E \setminus (Q_1 \cup Q_2)$, which has size at most two and therefore cannot contain two distinct two-sets. Finally, support complementation interchanges G_2 and G_3 , while disjointness and the remaining edge color are symmetric. $\square$

Lemma A.15 (The support defect of a one-concept augmentation). *Let $\emptyset \neq S \subsetneq B \subseteq Q_E$ be ample, put*

$$\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S),$$

and, for $p \in B \setminus S$, put

$$\mathcal{H}_S(p) = \text{Sh}(S \cup \{p\}) \setminus \text{Sh}(S). \quad (\text{A.25})$$

Then $\emptyset \neq \mathcal{H}_S(p) \subseteq \mathcal{K}$, and the following conditions are equivalent:

1. $S \cup \{p\}$ is ample;
2. $|\mathcal{H}_S(p)| = 1$;
3. $(J_S : m_p)$ is generated by variables;
4. the intersection of the facet F_p with Δ_S is pure of dimension $|E| - 2$.

For every $R \subseteq E$,

$$|\pi_R(B) \setminus \pi_R(S)| = |\{K \in \mathcal{K} : K \subseteq R\}|. \quad (\text{A.26})$$

Consequently, for $R \in \mathcal{K}$,

$$R \in \mathcal{H}_S(p) \iff \{K \in \mathcal{K} : K \subseteq R\} = \{R\} \quad \text{and} \quad p|_R \notin \pi_R(S). \quad (\text{A.27})$$

Proof. Sauer's inequality and ampleness of S give

$$|\text{Sh}(S \cup \{p\})| \geq |S| + 1 = |\text{Sh}(S)| + 1,$$

so $\mathcal{H}_S(p) \neq \emptyset$. Monotonicity of shattering and $S \cup \{p\} \subseteq B$ give $\mathcal{H}_S(p) \subseteq \mathcal{K}$. Equality in the displayed Sauer bound is precisely ampleness of $S \cup \{p\}$, proving the equivalence of the first two conditions. The equivalence with the colon condition is Lemma A.1. Under the shellability–linear–quotients dictionary, the same colon condition says that $F_p \cap \Delta_S$ is a union of ridges of F_p , which is the fourth condition.

Subtracting the two flag f -identities (A.19) gives

$$|\pi_R(B)| - |\pi_R(S)| = \sum_{K \subseteq R} h_K(\Psi(B, S)).$$

Lemma A.12 turns the right-hand side into the count in (A.26).

If $R \in \mathcal{H}_S(p)$, then $S \cup \{p\}$ shatters R , so B shatters R and $R \in \mathcal{K}$. The projection $\pi_R(S)$ has all patterns except $p|_R$. The left-hand side of (A.26) is therefore one, which is equivalent to the first condition on the right of (A.27). Conversely, that condition makes $p|_R$ the unique pattern of $\pi_R(B)$ absent from $\pi_R(S)$. Adjoining p fills it and shatters R , proving the reverse implication. $\square$

Lemma A.16 (Relative peeling in VC dimension two). *Let $\emptyset \neq S \subseteq B$ be ample families. If B has VC dimension at most two, then B has a corner peeling in which every vertex of $B \setminus S$ precedes every vertex of S .*

Proof. Every support in $\text{Sh}(B) \setminus \text{Sh}(S)$ has size at most two. Theorem 2.57 therefore reduces B to S without deleting any concept of S . Choose $s \in S$. The same theorem applied to $\{s\} \subseteq S$ reduces S to $\{s\}$, since S also has VC dimension at most two. Concatenate these two sequences and finish by deleting s . This gives the required full peeling, and even allows any prescribed last concept in S . $\square$

Lemma A.17 (Nested ample families in three coordinates). *Let $\emptyset \neq S \subseteq B \subseteq Q_E$ be ample, and suppose that B uses at most three coordinates. For every $s \in S$, the family B has a corner peeling in which every vertex of $B \setminus S$ precedes every vertex of S , and s is last.*

Proof. It is enough to prove that whenever $S \subsetneq B$, some corner of B lies outside S . Deleting that corner preserves ampleness by the ample corner-deletion criterion, so iteration first reduces B to S . Applying the same assertion to the nested pair $\{s\} \subseteq S$ then peels S down to s .

For at most two active coordinates the assertion follows by inspecting a vertex, an edge, a three-vertex tree, and a square. Suppose that B uses all three coordinates. Up to a symmetry of Q_3 , the possibilities for B and its set $\text{Cor}(B)$ of corners are

$ B $	B	$\text{Cor}(B)$	
4	$\{000, 001, 010, 100\}$	$\{001, 010, 100\}$	(A.28)
4	$\{000, 001, 011, 100\}$	$\{011, 100\}$	
5	$\{000, 001, 010, 011, 100\}$	$\{001, 010, 011, 100\}$	
6	$\{000, 001, 010, 011, 100, 101\}$	$\{010, 011, 100, 101\}$	
7	$Q_3 \setminus \{111\}$	$\{011, 101, 110\}$	
8	Q_3	Q_3	

Here is a short completeness check. Full support shatters the empty set and the three singleton coordinate sets. If $|B| = k < 8$, ampleness says that exactly $k - 4$ coordinate pairs are shattered, and each is strongly shattered, hence represented by a square in B . For $k = 4$, the isometric graph of B is a three-edge tree, giving the star and the path. For $k = 5$, the unique square has one pendant vertex in the third direction. For $k = 6$, the two squares share their common-coordinate edge. The cases $k = 7, 8$ are the punctured and full cube. These are exactly the six rows of (A.28).

An ample family is an isometric subgraph of its cube. In the first row, an isometric subfamily containing the three leaves also contains their centre. In the second row, the two endpoints force the whole path. In the third row, the pendant corner forces its attachment vertex. In the fourth row, shortest paths between 010, 100 and between 011, 101 force the two vertices of the common edge. In the fifth row, shortest paths among 011, 101, 110 force 001, 010, 100. The only proper subfamily that could then contain all three corners is the resulting six-cycle. That cycle shatters the empty set, all three singletons, and all three coordinate pairs, so it has at least seven shattered sets but only six vertices and is not ample. In the last row every vertex is a corner. Thus no proper ample $S \subsetneq B$ contains every corner of B , as required. $\square$

Theorem A.18 (Nested ample families in four coordinates). *Let $\emptyset \neq S \subseteq B \subseteq Q_E$ be ample, and suppose that B uses at most four coordinates. For every $s \in S$, the family B has a corner peeling in which every vertex of $B \setminus S$ precedes every vertex of S , and s is last.*

Proof. As in Lemma A.17, it is enough to show that a proper ample subclass $S \subsetneq B$ omits a corner of B . We may assume that B uses all four coordinates. If its VC dimension is at most two, this follows from Lemma A.16. If its VC dimension is four, then $B = Q_4$ and every vertex is a corner. It remains to consider VC dimension three.

Choose a strongly shattered triple and orient the remaining coordinate so that its cube is in the lower layer. Then

$$B = (Q_3 \times \{0\}) \cup (U \times \{1\}), \quad (\text{A.29})$$

where U is a nonempty proper ample subfamily of Q_3 . It is nonempty because B uses the fourth coordinate, and it is proper because B has VC dimension three. The maximal cubes in (A.29) give

$$\text{Cor}(B) = ((Q_3 \setminus U) \times \{0\}) \dot{\cup} (\text{Cor}(U) \times \{1\}). \quad (\text{A.30})$$

The possible sites U , up to cube symmetry, are obtained from the support-zero, support-one, and support-two cases together with the six support-three types in (A.28). The corner formula gives the following short table; the last column is obtained by projecting the displayed corner set.

U	$ U $	$ \text{Cor}(B) $	$ \text{Sh}(\text{Cor}(B)) $
K_1	1	8	9
K_2	2	8	10
P_3	3	7	10
Q_2	4	8	12
$K_{1,3}$	4	7	11
P_4	4	6	10
Q_2 with one pendant vertex	5	7	10
two squares sharing an edge	6	6	8
Q_3^-	7	4	5

(A.31)

Put $C = \text{Cor}(B)$, and suppose first that the last entry of the table is not in force. If $C \subseteq S$, then ampleness and monotonicity of shattering give

$$|S| = |\text{Sh}(S)| \geq |\text{Sh}(C)| \geq 8. \quad (\text{A.32})$$

Let $A = Q_4 \setminus B$ and $T = Q_4 \setminus S$. Complements of ample cube families are ample [Law83, BCDK06], so $A \subsetneq T$ are ample, and (A.32) gives $|T| \leq 8$. If T uses at most three coordinates, Lemma A.17 applies. If it uses all four, then it cannot have VC dimension three: a shattered triple and the remaining active singleton already give at least nine shattered sets. Lemma A.16 therefore applies instead.

In either case, peel $T \setminus A$ down to A , and let x be the last vertex removed before A remains. Then $A \cup \{x\}$ is ample. Its complement is $B \setminus \{x\}$, which is consequently ample. The ample corner-deletion criterion makes x a corner of B . But

$$x \in T \setminus A = B \setminus S,$$

contradicting $C \subseteq S$.

It remains to treat $U = Q_3^-$, for which $B = Q_4^-$. Normalize the missing vertex to 1111. The four corners of B are its weight-three vertices. If an isometric subfamily contains them, then a shortest path between each pair cannot use the missing vertex and forces all six weight-two vertices. Let K be the resulting ten-vertex family of all weight-two and weight-three vertices. It shatters the empty set, the four singletons, and the six coordinate pairs.

Suppose that an ample $S \subseteq B$ contains K . If it also contains the zero vertex, then every coordinate triple is shattered, so $|\text{Sh}(S)| \geq 15$; a proper subfamily of B has order at most fourteen, a contradiction. Otherwise let k be the number of weight-one vertices in S . Each such vertex shatters the unique coordinate triple avoiding its support, and these triples are distinct. Hence

$$|\text{Sh}(S)| \geq 11 + k > |K| + k = |S|,$$

again contradicting ampleness. Thus no proper ample subclass of Q_4^- contains all four corners.

We have proved in every case that a proper ample $S \subsetneq B$ omits a corner of B . Delete such a corner and iterate; corner deletion preserves ampleness. This peels $B \setminus S$ first. Applying the same assertion to $\{s\} \subseteq S$ completes the peeling with s last. $\square$

Lemma A.19 (Augmentation across an expansion separator). *Let*

$$H = (C \times \{0\}) \cup (Y \times \{1\}) \subseteq Q_{E \cup \{r\}}$$

be ample, put $S = C \cap Y$, and let $p \in Y \setminus C$. If $S \cup \{p\}$ is ample, then $C \cup \{p\}$ is ample.

Proof. Put $Z = C \cup Y$. This is the contraction of H in coordinate r , so it is ample and hence isometric. Moreover, no edge of Z joins $C \setminus S$ to $Y \setminus S$. Indeed, if $c \in C \setminus S$ and $y \in Y \setminus S$ were adjacent, then $(c, 0)$ and $(y, 1)$ would have Hamming distance two in H . The only two possible intermediate cube vertices are $(y, 0)$ and $(c, 1)$, neither of which belongs to H , contradicting isometry.

We verify item 3 of the one-concept augmentation criterion, Lemma A.1. Fix $c \in C$. If $c \in S$, that criterion applied to $S \cup \{p\}$ gives a coordinate $e \in \delta(p, c)$ for which $p^{\{e\}} \in S \subseteq C$. Suppose instead that $c \in C \setminus S$. Choose a Z -geodesic from p to c . The preceding separation forces this geodesic to meet S ; let s be one of its vertices in S . Since s lies on a cube geodesic from p to c ,

$$\delta(p, s) \subseteq \delta(p, c).$$

Applying the augmentation criterion to $s \in S$ gives $e \in \delta(p, s)$ such that $p^{\{e\}} \in S \subseteq C$. Thus every $c \in C$ satisfies the criterion, and $C \cup \{p\}$ is ample. $\square$

Lemma A.20 (Duplicating an augmenting concept across an expansion). *Let*

$$H = (C \times \{0\}) \cup (Y \times \{1\}) \subseteq Q_{E \cup \{r\}}$$

be ample, put $I = C \cap Y$, and let $p \in Y \setminus C$. If $I \cup \{p\}$ is ample, then

$$H \cup \{(p, 0)\} = ((C \cup \{p\}) \times \{0\}) \cup (Y \times \{1\}) \tag{A.33}$$

is ample.

Proof. Put $Z = C \cup Y$. The two sections C, Y and the contraction Z are ample. The two-section shattering formula and ampleness of H give

$$\text{Sh}(C) \cap \text{Sh}(Y) = \text{Sh}(I), \tag{A.34}$$

and also show that I is ample. For completeness, the count is

$$|C| + |Y| = |Z| + |I| = |\text{Sh}(Z)| + |\text{Sh}(C) \cap \text{Sh}(Y)|.$$

Since $\text{Sh}(I)$ is contained in the intersection and Sauer's inequality gives $|\text{Sh}(I)| \geq |I|$, equality is forced throughout.

The ample singleton augmentation $I \cup \{p\}$ adds one shattered support, say R . Because it is contained in Y , we have $R \in \text{Sh}(Y)$. Equation (A.34) gives $R \notin \text{Sh}(C)$. Lemma A.19 shows that $C \cup \{p\}$ is ample. Its unique new shattered support must also be R , by monotonicity from $I \cup \{p\} \subseteq C \cup \{p\}$.

In (A.33), the union of the two sections remains Z , whereas their intersection changes from I to $I \cup \{p\}$. The two-section formula therefore adds exactly the support $R \cup \{r\}$ and no other shattered support. The family has gained one concept, so it is ample. $\square$

Lemma A.21 (Section synchronization or a nine-concept cycle). *Let*

$$H = (C \times \{0\}) \cup (Y \times \{1\}) \subseteq Q_{E \cup \{r\}}$$

be ample, put $S = C \cap Y$, and suppose that

$$\emptyset \neq S \subsetneq Y.$$

Then at least one of the following holds:

1. *some $p \in Y \setminus S$ satisfies*

$$S \cup \{p\} \text{ is ample,} \quad C \cup \{p\} \text{ is ample.} \quad (\text{A.35})$$

For every such p , both

$$(C \times \{0\}) \cup ((S \cup \{p\}) \times \{1\}) \quad \text{and} \quad H \cup \{(p, 0)\} \quad (\text{A.36})$$

are ample;

2. *put $D = Y \setminus S$, and let $c(D)$ be the number of connected components of the graph induced by D . Then $|D| \geq 9$, every component of that graph contains a cycle, and*

$$\sum_{T \in \text{Sh}(Y) \setminus \text{Sh}(S)} |T| \geq |D| + c(D) \geq 10. \quad (\text{A.37})$$

Proof. The sections Y and S are ample by Lemma 7.22. Suppose first that some $p \in Y \setminus S$ makes $S \cup \{p\}$ ample. Lemma A.19 then makes $C \cup \{p\}$ ample, proving (A.35).

For the first family in (A.36), the two-layer trace formula gives the supports not containing r as $\text{Sh}(C \cup \{p\})$. The supports containing r are indexed by

$$\text{Sh}(C) \cap \text{Sh}(S \cup \{p\}).$$

The original ample family H satisfies

$$\text{Sh}(C) \cap \text{Sh}(Y) = \text{Sh}(S)$$

by Lemma 7.22. Since $S \subseteq S \cup \{p\} \subseteq Y$, the last intersection is exactly $\text{Sh}(S)$. The first family therefore has

$$|\text{Sh}(C \cup \{p\})| + |\text{Sh}(S)| = |C| + 1 + |S|$$

shattered supports, equal to its number of concepts. The upper Sandwich equality proves ampleness. The second family in (A.36) is ample by Lemma A.20.

It remains to suppose that no $p \in Y \setminus S$ makes $S \cup \{p\}$ ample. The relative Yang complex of the nested ample pair $S \subsetneq Y$ is Cohen–Macaulay by the relative Cohen–Macaulay criterion of [Bou26].

The arbitrary-upper linear-quotient theorem through eight relative facets therefore gives $|Y \setminus S| \geq 9$. Put $D = Y \setminus S$.

The linear presentation of the Alexander-dual relative module splits over the connected components of the graph induced by D ; each component summand consequently has a linear resolution. If one component were a forest, the relative forest theorem of the same paper, applied to that summand, would supply a concept satisfying the S -teaching condition. Adjoining that concept to S would be ample, contrary to the assumption. Thus every component contains a cycle, and hence the graph induced by D has at least $|D|$ edges.

Every component of D has an edge to S , because the ample family Y is connected and $S \neq \emptyset$. The edge-shadow identity for the two nested ample families now gives

$$\sum_{T \in \text{Sh}(Y) \setminus \text{Sh}(S)} |T| = |E(Y)| - |E(S)| = |E(D)| + |E(D, S)| \geq |D| + c(D).$$

Together with $|D| \geq 9$, this proves (A.37). $\square$

Theorem A.22 (Complete section synchronization or a terminal cyclic remainder). *Keep the notation of Lemma A.21. There are an ample family Z with*

$$S \subseteq Z \subseteq Y$$

and an ordering $p_1, \dots, p_q$ of $Z \setminus S$ such that, on putting $Z_h = S \cup \{p_1, \dots, p_h\}$, every Z_h and every family

$$H_h = ((C \cup Z_h) \times \{0\}) \cup (Y \times \{1\}) \quad (\text{A.38})$$

is ample. Moreover, exactly one of the following conclusions holds:

1. $Z = Y$, so all concepts of $Y \setminus S$ may be duplicated across the r -edge, one at a time, while preserving ameness;
2. put $D = Y \setminus Z$. Then $|D| \geq 9$, every component of the graph induced by D contains a cycle, and

$$\sum_{T \in \text{Sh}(Y) \setminus \text{Sh}(Z)} |T| \geq |D| + c(D) \geq 10. \quad (\text{A.39})$$

Proof. Starting with $Z_0 = S$, repeatedly choose $p_{h+1} \in Y \setminus Z_h$ for which $Z_h \cup \{p_{h+1}\}$ is ample, and stop when no such concept exists. Every intermediate Z_h is ample by construction. The base family in (A.38) is H . If H_h is ample, its two sections are $C \cup Z_h$ and Y , with intersection Z_h . Lemma A.20 shows that adjoining the lower copy of p_{h+1} gives the ample family H_{h+1} . This proves the progressive synchronization assertion.

If the process reaches Y , the first alternative holds. Otherwise apply Lemma A.21 to the ample family H_q , whose section intersection is $Z = Z_q$. Maximality of the construction excludes the synchronization alternative of that lemma. Its cyclic alternative gives all assertions in the second conclusion. $\square$

Corollary A.23 (An adjacent path transition synchronizes or owns a terminal cycle). *Keep the path-spine notation of Proposition 7.59. For every $1 \leq i \leq n$ and each $j \in \{i-1, i\}$, at least one of the following holds:*

1. writing k for the other member of $\{i-1, i\}$, all concepts of $P_j \setminus C_i$ can be ordered so that duplicating their missing copies into P_k , in that order, preserves ameness at every step;

2. after such duplications have synchronized a proper ample intermediate family Z , with $C_i \subseteq Z \subsetneq P_j$, the remainder $D = P_j \setminus Z$ has at least nine concepts, every component of its induced graph contains a cycle, and

$$\sum_{T \in \text{Sh}(P_j) \setminus \text{Sh}(Z)} |T| \geq |D| + c(D) \geq 10. \quad (\text{A.40})$$

Proof. Condition all core coordinates except x_i to the values shared by a_{i-1} and a_i . The resulting ample two-layer family has sections P_{i-1}, P_i and overlap C_i . Apply Theorem A.22 in the chosen orientation. $\square$

Corollary A.24 (A noncyclic transition exposes an exclusive section corner). *For every oriented incidence (i, j) in the path spine for which $P_j \setminus C_i \neq \emptyset$, either the terminal cyclic alternative of Corollary A.23 occurs, or there is a concept*

$$p_{i,j} \in P_j \setminus C_i \quad (\text{A.41})$$

that is a corner of the ample section P_j .

Proof. In the complete-synchronization alternative of Theorem A.22, let $p_{i,j}$ be the last concept in the chain from C_i to P_j . Deleting it from P_j leaves the preceding ample member of the chain. The ample corner-deletion criterion therefore makes $p_{i,j}$ a corner of P_j , and it lies outside C_i . $\square$

Proposition A.25 (Acquired section shadows obey path-boundary conservation). *Keep the path-spine notation of Proposition 7.59, and put*

$$\mathcal{S} = \bigcup_{j=0}^n \text{Sh}(P_j).$$

For $T \in \mathcal{S}$, let $c_{\text{sh}}(T)$ be the number of connected components of

$$I_{\text{sh}}(T) = \{j : T \in \text{Sh}(P_j)\}$$

in the section path. Then

$$\begin{aligned} & \sum_{i=1}^n (|P_{i-1} \setminus C_i| + |P_i \setminus C_i|) \\ & \leq 2 \sum_{T \in \mathcal{S}} c_{\text{sh}}(T) = 2(|U| + \Phi) \leq 2(|W| + |\Omega|). \end{aligned} \quad (\text{A.42})$$

Thus all oriented section wings are controlled at once: repeated acquisition of one repair support is possible only through a new interval of section indices, and the path-spine Euler term pays for every additional interval.

Proof. For every path edge, the ample two-layer shadow-intersection identity gives

$$\text{Sh}(C_i) = \text{Sh}(P_{i-1}) \cap \text{Sh}(P_i).$$

Since all three families are ample, the left side of (A.42) is therefore

$$\sum_{i=1}^n (|\text{Sh}(P_{i-1}) \setminus \text{Sh}(C_i)| + |\text{Sh}(P_i) \setminus \text{Sh}(C_i)|).$$

Fix $T \in \mathcal{S}$. Its contribution to this sum is the number of path edges having exactly one endpoint in $I_{\text{sh}}(T)$. A union of $c_{\text{sh}}(T)$ nonempty path intervals has at most $2c_{\text{sh}}(T)$ boundary edges. Summing proves the first inequality.

Counting the vertices and internal edges of every set $I_{\text{sh}}(T)$ gives

$$\begin{aligned} \sum_{T \in \mathcal{S}} c_{\text{sh}}(T) &= \sum_{j=0}^n |\text{Sh}(P_j)| - \sum_{i=1}^n |\text{Sh}(C_i)| \\ &= \sum_{j=0}^n |P_j| - \sum_{i=1}^n |C_i| = |U| + \Phi, \end{aligned}$$

where the last equality is the word-level interval count in the proof of Proposition 7.59. Finally, that proposition gives the exact cancellation

$$|W| + |\Omega| = |U| + \Phi + V,$$

and $V \geq 0$. This proves the final inequality. $\square$

Proposition A.26 (Boundary-corner roots are distinct up to fragmentation). *Let $\mathcal{A}$ be any set of oriented incidences for which the corner alternative in Corollary A.24 holds, and choose one corner $p_{i,j}$ for each $(i,j) \in \mathcal{A}$. Let Z be the set of distinct chosen repair words. Then*

$$|\mathcal{A}| \leq 2(|Z| + \Phi), \quad |Z| \geq \frac{|\mathcal{A}|}{2} - \Phi. \quad (\text{A.43})$$

Thus repeated use of one repair word is already paid for by the path-spine fragmentation term.

Proof. Fix $p \in Z$. An incidence selecting p is an oriented boundary edge of the subgraph induced by

$$I(p) = \{h : p \in P_h\}$$

in the core path. A union of $c(p)$ nonempty path intervals has at most $2c(p)$ such boundary incidences. Hence

$$|\mathcal{A}| \leq 2 \sum_{p \in Z} c(p) = 2|Z| + 2 \sum_{p \in Z} (c(p) - 1) \leq 2(|Z| + \Phi),$$

which proves both displayed inequalities. $\square$

Proposition A.27 (The only neutral boundary-corner defect is an inward square tip). *Fix $1 \leq i \leq n$, let $j \in \{i-1, i\}$, and choose any*

$$p \in \text{Cor}(P_j) \setminus C_i.$$

Write $v = (p, a_j)$. Since H is cornerless, Proposition 7.13 gives a rooted puncture (v, J) . Put

$$S = J \cap R, \quad K = J \cap X.$$

Then $K \neq \emptyset$ and $x_i \notin K$. Moreover, at least one of the following holds:

1. *the puncture contains a concept of H over a core word outside A , with repair root determined by p ;*
2. *$|K| \geq 2$, $S \neq \emptyset$, and $K \in \Omega_{\geq 2}$;*

3. $K = \{x_\ell\}$, where x_ℓ is the other path direction incident with a_j . For every $s \in S$, the puncture has a provenance-tagged terminal square on $\{s, x_\ell\}$ whose repair root belongs to C_ℓ .

In particular, after charging external-core concepts and residual supports, all uncharged boundary-corner occurrences reduce to inward, spine-aligned repair-core square tips. No higher-rank neutral defect remains.

Proof. The transition direction x_i is not incident with v , because $p \notin C_i$. Since J consists of directions incident with v , we have $x_i \notin K$; core transversality gives $K \neq \emptyset$.

Suppose first that $S = \emptyset$. Then $|K| = |J| \geq 2$, and every single flip $v^{\{x\}}$, $x \in K$, belongs to H . At most one direction in K is the other core-path edge at a_j . Some such neighbor therefore lies over a core word outside A , proving the first alternative.

Suppose next that $S \neq \emptyset$ and $|K| \geq 2$. The set K is a proper subset of J , so the puncture relations contain the full K -cube through v . Hence H strongly shatters K . This support has at least two core coordinates and therefore belongs to $\Omega_{\geq 2}$, proving the second alternative.

It remains that $K = \{x_\ell\}$. If $a_j^{\{x_\ell\}} \notin A$, the present neighbor $v^{\{x_\ell\}}$ proves the first alternative. Otherwise x_ℓ is a path direction incident with a_j . It differs from the absent transition direction x_i , so it is the other incident path edge. In particular, the repair word p belongs to both path sections across that edge and hence to C_ℓ .

For $s \in S$, apply the square-tip reduction of Lemma 7.75 to $L = \{s, x_\ell\}$. Its repair root is $p^{S \setminus \{s\}}$. Both vertices in the x_ℓ -direction are proper flips of J , so this repair root also belongs to C_ℓ . The retained parent puncture supplies the asserted provenance tag. $\square$

Proposition A.28 (Closed neutral transfers localize in one repair cube). *For every occurrence of the third alternative in Proposition A.27, retain the parent puncture and write its support as*

$$J = S \dot{\cup} \{x_\ell\}.$$

Then $S \neq \emptyset$, and the opposite repair tip

$$q = p^S \tag{A.44}$$

belongs to $P_j \setminus C_\ell$. Thus the neutral defect transfers the chosen boundary occurrence at one side of P_j to a boundary occurrence at its other side.

For every interior section P_j , make disjoint copies of its two sets of boundary corners:

$$\mathcal{L}_j = \{(L, p) : p \in \text{Cor}(P_j) \cap (P_j \setminus C_j)\}, \quad \mathcal{R}_j = \{(R, p) : p \in \text{Cor}(P_j) \cap (P_j \setminus C_{j+1})\}.$$

For each retained neutral parent puncture rooted at one of these occurrences, draw an arrow to the opposite occurrence of the word in (A.44), provided that this opposite word is again a corner with a retained neutral parent puncture. Every directed cycle in the resulting neutral-transfer digraph stays in one $\mathcal{L}_j \dot{\cup} \mathcal{R}_j$, has even length, and all of its repair words lie in one positioned maximal cube Q of P_j . If T is the support of Q and $t = |T|$, then $t \geq 2$, and

$$Q \not\subseteq P_{j-1}, \quad Q \not\subseteq P_{j+1}. \tag{A.45}$$

Choose any arrow in each orientation on such a cycle, and let S_+ be the repair label of the arrow from $\mathcal{L}_j$ to $\mathcal{R}_j$ and S_- the label of the arrow in the reverse direction. Put $k_+ = |S_+|$ and $k_- = |S_-|$. Then

$$\Delta(H; A, R) \geq 2^t - t - 1 + \max\{0, 2^{k_+} - k_+ - 2\} + \max\{0, 2^{k_-} - k_- - 2\}. \tag{A.46}$$

A directed two-cycle has a stronger symmetric description. More precisely, if $p \in P_j \setminus C_j$ and $q \in P_j \setminus C_{j+1}$ form such a two-cycle, then for one nonempty $S \subseteq R$,

$$q = p^S, \quad p = q^S, \quad (\text{A.47})$$

and

$$\begin{aligned} \{p^U : U \subseteq S\} &\subseteq P_j, \\ \{p^U : U \subsetneq S\} &\subseteq C_{j+1}, \quad p^S \notin C_{j+1}, \\ \{q^U : U \subsetneq S\} &\subseteq C_j, \quad q^S \notin C_j. \end{aligned} \quad (\text{A.48})$$

The two chosen concepts p, q are opposite vertices of one face in their common positioned maximal cube Q .

If $k = |S|$, this local cycle gives the path excess bound

$$\Delta(H; A, R) \geq 2^t - t - 1 + 2 \max\{0, 2^k - k - 2\}. \quad (\text{A.49})$$

For $k \geq 2$, this is at least

$$3 \cdot 2^k - 3k - 5.$$

Consequently a neutral two-cycle in a family of order below 46 has $k \leq 3$. In the rank-one case the two oppositely oriented punctured squares on $\{s, x_j\}$ and $\{s, x_{j+1}\}$ still pay at least one non-singleton repair support.

Proof. In the neutral alternative one has $J = S \dot{\cup} \{x_\ell\}$ and $|J| \geq 2$, so $S \neq \emptyset$. The parent puncture contains the vertex obtained by the proper flip S , namely (p^S, a_j) , but omits its x_ℓ -neighbour. This is exactly the assertion that $q = p^S$ lies in $P_j \setminus C_\ell$.

Every arrow stays at the same section and exchanges its two incident path edges. Thus the neutral-transfer digraph is the disjoint union of the bipartite digraphs on $\mathcal{L}_j \dot{\cup} \mathcal{R}_j$, and every directed cycle is even.

Consider an arrow from a corner p to a corner $q = p^S$. The parent puncture supplies the full S -face through p in P_j . This face is contained in the unique maximal cube through p , and that maximal cube also contains q . Since q is a corner, the same cube is its unique maximal cube. Following the arrows around a cycle therefore shows that all of its corners have one common positioned maximal cube Q . The cycle contains a vertex of $\mathcal{L}_j$ and a vertex of $\mathcal{R}_j$, so Q contains a word absent from P_{j-1} and a word absent from P_{j+1} . This proves (A.45).

If $t = 1$, the two endpoints of Q have no repair neighbor outside Q : such a neighbor would lie in a second maximal cube through that endpoint. Connectedness would then force $P_j = Q$, which is impossible because P_j contains two repair antipodes and $r \geq 3$. Hence $t \geq 2$.

The projection W contains the full T -cube Q , so beyond the empty support and the r repair singletons it has at least $2^t - t - 1$ supports contained in T . An arrow from $\mathcal{L}_j$ to $\mathcal{R}_j$, with label S_+ , gives a punctured S_+ -face in C_{j+1} ; an arrow in the reverse orientation, with label S_- , gives a punctured S_- -face in C_j . Each such carrier strongly shatters every proper subset of its arrow label and hence has at least $\max\{0, 2^{k_\pm} - k_\pm - 2\}$ supports beyond its empty support and repair singletons. The three contributions lie in different blocks of the disjoint path-spine shadow family. This proves (A.46).

Suppose now that the cycle has two arrows. Write S_p and S_q for their repair labels. The arrows give $q = p^{S_p}$ and $p = q^{S_q}$, and therefore $S_p = S_q = \text{Diff}(p, q) = S$. The proper-flip relations of the first parent puncture give the full S -face in P_j and the punctured S -face rooted at p in C_{j+1} .

The second parent gives the oppositely punctured face rooted at q in C_j . These are precisely the inclusions and omissions in (A.48). Substituting $S_+ = S_- = S$ in (A.46) gives (A.49). Since $t \geq k$, its right side is at least $3 \cdot 2^k - 3k - 5$ when $k \geq 2$. If $k \geq 4$, then $r \geq k$, $n \geq 2$, and

$$|H| \geq (n+1)(r+1) + 3 \cdot 2^k - 3k - 5 \geq 46,$$

with equality in the last lower bound at $k = 4$. This proves the final assertions. $\square$

Proposition A.29 (Section excess pays the whole neutral-transfer forest). *Fix an interior path section P_j . Extend its neutral-transfer digraph as follows. For every retained neutral boundary-corner occurrence, draw the arrow to its opposite repair tip $q = p^S$, whether or not q is again a retained neutral corner. If it is not, adjoin q as a terminal vertex; terminal vertices with the same repair word are identified. Let κ_j be the number of weak components of this extended digraph. Then*

$$\kappa_j \leq |P_j| - (r+1). \quad (\text{A.50})$$

Consequently, over all interior sections,

$$\sum_{j=1}^{n-1} \kappa_j \leq \sum_{j=1}^{n-1} (|P_j| - (r+1)) \leq \Delta(H; A, R). \quad (\text{A.51})$$

Thus one additive budget pays every closed component and every acyclic terminal component simultaneously. The bound is independent of the repair labels, the ranks of terminal maximal cubes, and any reuse of a terminal support by several arrows in the same component.

Proof. Every nonterminal vertex has out-degree one. Hence each weak component contains either one directed cycle or one terminal vertex. For a weak component D , let $W(D)$ be the set of underlying repair words represented by its occurrence and terminal vertices. Every component contains an arrow $p \rightarrow p^S$, where $S \neq \emptyset$, and therefore

$$|W(D)| \geq 2. \quad (\text{A.52})$$

Choose antipodal repair geodesics

$$L \subseteq C_j, \quad R' \subseteq C_{j+1},$$

and let ν be the number of vertices exclusive to either one of these two geodesics. Both have $r+1$ vertices, so

$$|L \cup R'| = r+1 + \nu. \quad (\text{A.53})$$

Consider an arrow whose source lies in $\mathcal{L}_j$. The neutral parent puncture uses the other path direction x_{j+1} , whereas the source is absent across x_j . Its source and target therefore satisfy

$$p \in C_{j+1} \setminus C_j, \quad q \notin C_{j+1}.$$

If one of its two words belongs to $L \cup R'$, that word belongs to the symmetric difference of the two geodesics. More precisely, when both words belong to the union,

$$p \in R' \setminus L, \quad q \in L \setminus R'.$$

The reverse orientation gives the symmetric conclusion. In particular, no word of $L \cap R'$ occurs in the neutral digraph.

We count incidences between weak components and their underlying repair words. A word outside $L \cup R'$ belongs to at most two weak components, one through each of its left and right boundary occurrences. If it is adjoined as a terminal word, the occurrence on the target side is absent; all terminal copies of the word are identified, so the same bound still holds. A word in $L \triangle R'$ has only the boundary occurrence on the side of the geodesic which omits it and hence belongs to at most one weak component. Therefore

$$\begin{aligned} 2\kappa_j &\leq \sum_D |W(D)| \\ &\leq 2|P_j \setminus (L \cup R')| + |L \triangle R'| \\ &= 2(|P_j| - |L \cup R'|) + 2\nu \\ &= 2(|P_j| - (r + 1)), \end{aligned}$$

where the last equality uses (A.53). This proves (A.50) while retaining the possible left–right duplication of one repair word.

Finally, the path-section excess partition (A.64) gives

$$\Delta(H; A, R) = V + \sum_{h=0}^n (|P_h| - (r + 1)).$$

All terms are nonnegative, so summing the sectionwise bound over the interior indices proves (A.51). $\square$

Corollary A.30 (Equality in the neutral-forest budget is two-word). *If equality holds in (A.50), then every weak component of the extended neutral-transfer digraph uses exactly two underlying repair words. Every word of $P_j \setminus (L \cup R')$ belongs to exactly two components, and every word of $L \triangle R'$ belongs to exactly one. In particular, every directed cycle in a section attaining equality is a directed two-cycle.*

More precisely, form the bipartite incidence graph $\mathfrak{N}_j$ between the weak components and the underlying repair words which they use. Then every component vertex has degree two, every word vertex has degree one or two, and

$$\mathfrak{N}_j \text{ is a disjoint union of paths and cycles.} \tag{A.54}$$

Its path endpoints are exactly the words in $L \triangle R'$; every word on a cyclic component lies outside $L \cup R'$.

Proof. Equality at the two ends of the incidence count in the preceding proof forces equality at every intermediate step. Thus $|W(D)| = 2$ for each component, every outside word attains its multiplicity bound two, and every exclusive geodesic word attains its multiplicity bound one. A directed cycle with only two underlying words alternates between those words and the two occurrence sides, so its out-degree-one rule closes after the two opposite arrows. It is a directed two-cycle. The asserted degrees in $\mathfrak{N}_j$ are the same equality conditions. A finite graph of maximum degree two is a disjoint union of paths and cycles. Its degree-one word vertices are precisely the exclusive geodesic words, while all degree-two word vertices lie outside the two geodesics. This proves the last assertions. $\square$

Corollary A.31 (Only a two-word component can save a second side demand). *In the extended neutral-transfer digraph of one interior section, mark at most one initial neutral boundary-corner demand in each of $\mathcal{L}_j$ and $\mathcal{R}_j$. Suppose that the two marked vertices lie in the same weak component $\mathcal{C}$. Then exactly one of the following holds:*

1. $\mathcal{C}$ uses at least three underlying repair words. In this case

$$\kappa_j + 1 \leq |P_j| - (r + 1), \quad (\text{A.55})$$

so this component pays both marked demands;

2. $\mathcal{C}$ uses exactly two underlying repair words and is the directed two-cycle whose vertices are the two marked demands.

Proof. If $\mathcal{C}$ contains at least three underlying repair words, then every other weak component contains at least two. The component–word incidence count in the proof of Proposition A.29 gives

$$2\kappa_j + 1 \leq 2(|P_j| - (r + 1)).$$

Integrality proves the displayed strict-slack alternative. Suppose instead that $\mathcal{C}$ contains exactly two repair words p, q . An arrow changes occurrence side and flips the fixed nonempty set $\text{Diff}(p, q)$. Hence the possible nonterminal pairs are

$$\{(L, p), (R, q)\}, \quad \{(R, p), (L, q)\},$$

and no arrow joins the two pairs. Since the two marked neutral sources belong to opposite occurrence sides and lie in the same weak component, they are the two vertices of one displayed pair. Each has its prescribed outgoing arrow to the other, so the component is their directed two-cycle. $\square$

Corollary A.32 (Only a directed two-cycle is a tight coalescence). *Keep the marked two-sided setting of Corollary A.31. If the common component is not the directed two-cycle whose two vertices are the two marked demands, then*

$$\kappa_j + 1 \leq |P_j| - (r + 1). \quad (\text{A.56})$$

Thus every other common component already pays both initial side demands. The only coalescence that can attain the one-component bound is precisely the opposite-puncture two-cycle described in Proposition A.28.

Proof. This is the first alternative of Corollary A.31; its second alternative is precisely the excluded directed two-cycle. $\square$

Proposition A.33 (A neutral transfer to a noncorner exposes a split cube). *Let p be a boundary corner of P_j with a neutral parent puncture, let $q = p^S$ be its opposite repair tip as in (A.44), and let Q be the unique maximal cube of P_j through p . Write T for the support of Q . If q is not a corner of P_j , then q lies in a second maximal cube Q' whose support T' is incomparable with T .*

If $|T'| \geq 2$, then

$$|P_j| - (r + 1) \geq 2^{|T|} - |T|. \quad (\text{A.57})$$

If no such additional support tax is available, one may choose $T' = \{s\}$ with $s \notin T$: the neutral transfer terminates at a rank-one cube attached to the maximal repair cube Q . This is the only a priori shadow-neutral split-corner configuration; Proposition A.34 below shows that the path-core carrier geometry excludes it.

Proof. The parent puncture contains the full S -face through p , so $q = p^S$ belongs to Q . Since q is not a corner, it lies in a maximal cube $Q' \neq Q$. Parallel cubes with the same support are equal or disjoint, so their common vertex gives $T' \neq T$. If $T' \subsetneq T$, then Q' is contained in the full T -cube Q ,

contrary to maximality of Q' . The reverse containment would contradict maximality of Q . Hence T and T' are incomparable.

The full cube Q makes every subset of T a member of $\text{Sh}(P_j)$. If $|T'| \geq 2$, then T' is neither a repair singleton nor a subset of T , so it supplies one further member of the shadow. Together with the empty support and all r repair singletons, ampleness gives

$$|P_j| \geq (r+1) + (2^{|T|} - |T| - 1) + 1,$$

which is (A.57). If this extra-support alternative is absent, every second maximal cube through q has singleton support. Incomparability then puts its unique direction outside T , giving the stated attached rank-one cube. $\square$

Proposition A.34 (A neutral transfer cannot terminate in a rank-one attachment). *Keep the path-core notation. Let p be a boundary corner of P_j with a neutral parent puncture of support $S \dot{\cup} \{x_\ell\}$, and let*

$$q = p^S \in P_j \setminus C_\ell$$

be its opposite repair tip. Suppose that q is not a corner of P_j , let Q be the unique maximal cube through p , and let $Q' \neq Q$ be a maximal cube through q . Then

$$|\text{supp}(Q')| \geq 2.$$

Consequently the rank-one alternative in Proposition A.33 never occurs in a path-core repair. Every neutral transfer reaching a noncorner pays the additional-support bound

$$|P_j| - (r+1) \geq 2^{|\text{supp}(Q)|} - |\text{supp}(Q)|.$$

Proof. Suppose for a contradiction that Q' is the edge

$$e = qz, \quad z = q^{\{s\}},$$

of repair direction s . Since this edge is an inclusion-maximal cube, it belongs to no square of P_j . Lemma 5.10 makes e a bridge.

It is also the only s -edge of P_j . Indeed, if $q'z'$ were a second s -edge, choose paths from q to q' and from z to z' inside the two s -sections of P_j . These sections are nonempty ample families and hence connected. The two paths together with e and $q'z'$ would contain a cycle through e , contradicting that e is a bridge.

The adjacent carrier C_ℓ contains the two fixed repair antipodes $f, \bar{f}$. It is ample and therefore connected. Since the two antipodes differ in coordinate s , every path between them in C_ℓ contains an s -edge. The only such edge in P_j is e , so $e \subseteq C_\ell$. In particular $q \in C_\ell$, contrary to the choice of the opposite tip. Hence Q' cannot have rank one. The final inequality is now the nonsingleton case of Proposition A.33. $\square$

Proposition A.35 (A rank-one closed transfer has length at least six). *Keep the hypotheses of Proposition A.28, and suppose that every arrow of a directed neutral-transfer cycle in P_j has a singleton repair label. If the cycle is not a two-cycle, then its length is at least six and*

$$|C_j| \geq r+2, \quad |C_{j+1}| \geq r+2. \tag{A.58}$$

Consequently, if T is the support of its common positioned maximal cube and $t = |T|$, then

$$\Delta(H; A, R) \geq 2^t - t + 1. \tag{A.59}$$

Thus a four-cycle of rank-one square tips is impossible; every longer rank-one cycle pays one additional concept in each adjacent carrier, besides the maximal-cube tax.

Proof. Write the cycle as

$$p_0, p_1, \dots, p_{2m-1}, p_0,$$

where the even vertices belong to $\mathcal{L}_j$, the odd vertices belong to $\mathcal{R}_j$, and the arrow from p_h to p_{h+1} flips the repair coordinate s_h . Indices are read modulo $2m$. The repair words p_h are distinct. Indeed, a repetition in the same vertex class would repeat a vertex of the directed cycle, while a repetition in opposite classes would give an odd closed walk in the bipartite repair cube.

The carrier C_{j+1} contains every even vertex and omits every odd vertex. Fix consecutive even vertices p_{2h} and p_{2h+2} . The two coordinates s_{2h} and s_{2h+1} are distinct, since otherwise these even vertices would be equal. They are therefore at Hamming distance two, with p_{2h+1} as one of their two common cube neighbours. Since C_{j+1} is ample and hence isometric, and since $p_{2h+1} \notin C_{j+1}$, the other common neighbour, say z_h , belongs to C_{j+1} .

If $m = 2$, the two odd cycle vertices are exactly the two common neighbours of the two even vertices. Both are absent from C_{j+1} , contradicting isometry. Thus every non-two-cycle has $m \geq 3$.

Suppose for a contradiction that $|C_{j+1}| = r + 1$. The two fixed repair antipodes force the empty support and all r repair singletons into $\text{Sh}(C_{j+1})$. Ampleness shows that these are its entire shadow. The edge-shadow identity gives exactly r edges, while isometry gives connectedness. Hence its one-inclusion graph is a tree. Moreover, the geodesic between the repair antipodes has r edges, so it uses the whole tree; the graph is a path.

The length-two paths

$$p_{2h}, z_h, p_{2h+2} \quad (0 \leq h < m)$$

concatenate to a closed walk in this path. Every closed walk in a tree has an immediate backtrack. No backtrack can occur at a vertex z_h , because the adjacent even vertices are distinct. A backtrack at p_{2h} would give $z_{h-1} = z_h$; that common vertex would then be adjacent to the three distinct vertices $p_{2h-2}, p_{2h}, p_{2h+2}$, contradicting that the tree is a path. Therefore $|C_{j+1}| \geq r + 2$. Interchanging the two sides of the neutral cycle gives $|C_j| \geq r + 2$, proving (A.58).

Finally, the disjoint path-spine decomposition gives the exact excess identity

$$\Delta(H; A, R) = |W| - (r + 1) + \sum_{i=1}^n (|C_i| - (r + 1)) + |\Omega|.$$

The full T -cube in W gives $|W| - (r + 1) \geq 2^t - t - 1$, the two displayed carrier surcharges contribute two more, and every other term is nonnegative. This proves (A.59). $\square$

Lemma A.36 (Boolean-chain divergence is bounded by order disagreement). *Let*

$$\pi = (a_1, \dots, a_r), \quad \sigma = (b_1, \dots, b_r)$$

be two orders of the same coordinate set R , and let

$$A_i = \{a_1, \dots, a_i\}, \quad B_i = \{b_1, \dots, b_i\} \quad (0 \leq i \leq r)$$

be their Boolean chains. Put

$$\nu(\pi, \sigma) = |\{i : A_i \neq B_i\}|.$$

Then

$$\nu(\pi, \sigma) \leq \text{inv}(\pi, \sigma), \tag{A.60}$$

where $\text{inv}(\pi, \sigma)$ is the number of coordinate pairs that occur in opposite orders in π and σ . Consequently, every matching between the exclusive occurrences of the two chains has at most $\text{inv}(\pi, \sigma)$ edges. The pairs in this matching need not be cube edges.

Proof. List the common prefix ranks as

$$0 = i_0 < i_1 < \cdots < i_m = r, \quad A_{i_h} = B_{i_h}.$$

The coordinates split into consecutive common blocks

$$R_h = A_{i_h} \setminus A_{i_{h-1}} = B_{i_h} \setminus B_{i_{h-1}}, \quad \ell_h = |R_h|.$$

There is no common prefix rank strictly between i_{h-1} and i_h . Thus the two orders restricted to R_h have $\ell_h - 1$ distinct nonterminal prefix occurrences. Summing over the blocks gives

$$\nu(\pi, \sigma) = \sum_{h=1}^m (\ell_h - 1). \quad (\text{A.61})$$

We claim that the two orders of R_h have at least $\ell_h - 1$ inversions. Transform the first order into the second by a shortest sequence of adjacent transpositions. Its length is the inversion number of the two restricted orders. If it had fewer than $\ell_h - 1$ steps, one of the $\ell_h - 1$ boundaries between consecutive positions would never be used by a transposition. No coordinate could then cross that boundary, so the set of coordinates in the positions to its left would be the same in both orders. This would give a common prefix rank strictly between i_{h-1} and i_h , a contradiction.

Coordinates in different blocks occur in the same relative order under π and σ , so no inversion crosses two blocks. The total inversion number is therefore the sum of the block inversion numbers. Together with (A.61), this proves (A.60).

Each chain has exactly one occurrence of every cardinality. Hence $A_i = B_k$ implies $i = k$, and each chain has exactly $\nu(\pi, \sigma)$ exclusive occurrences. A matching between the two exclusive sets has at most this many edges, proving the final assertion. $\square$

Proposition A.37 (Carrier excess and order disagreements pay every closed transfer). *Fix an interior path section P_j , and put*

$$\epsilon_j = |C_j| - (r + 1), \quad \epsilon_{j+1} = |C_{j+1}| - (r + 1).$$

Choose antipodal repair geodesics $L \subseteq C_j$ and $R' \subseteq C_{j+1}$. Let $\nu(L, R')$ be their number of exclusive vertices on either path, and let $d(L, R')$ be the number of repair coordinate pairs that they flip in opposite orders.

For every collection $\mathcal{Z}$ of pairwise vertex-disjoint directed cycles in the neutral-transfer digraph of P_j ,

$$|\mathcal{Z}| \leq \epsilon_j + \epsilon_{j+1} + \nu(L, R') \leq \epsilon_j + \epsilon_{j+1} + d(L, R') \leq \Delta(H; A, R). \quad (\text{A.62})$$

In particular, every functional subdigraph has at most $\Delta(H; A, R)$ directed cycles. This pays all same-section closed transfers simultaneously, independently of their arrow labels, lengths, common maximal-cube supports, or overlap inside one positioned cube.

Proof. Every ample repair carrier contains the two fixed repair antipodes and is isometric. It therefore contains an antipodal repair geodesic and has at least $r + 1$ concepts, so the displayed excesses are nonnegative.

Every directed neutral cycle alternates between $\mathcal{L}_j$ and $\mathcal{R}_j$. Select from each member of $\mathcal{Z}$ one arrow directed from $\mathcal{L}_j$ to $\mathcal{R}_j$. If this arrow has root p and target q , the neutral parent at p gives

$$p \in C_{j+1} \setminus C_j, \quad q \in C_j \setminus C_{j+1}.$$

Indeed, the empty proper repair flip keeps p in the opposite carrier, whereas membership in $\mathcal{L}_j$ excludes it from C_j ; the corresponding statements at the target follow because q is a retained neutral occurrence in $\mathcal{R}_j$. Since the cycles in $\mathcal{Z}$ are vertex-disjoint, the selected pairs (p, q) form a matching between the two carrier differences.

Charge a selected pair having an endpoint outside $L \cup R'$ to one such endpoint. The matching property makes these charges distinct, and there are exactly $\epsilon_j + \epsilon_{j+1}$ carrier vertices outside the two geodesics. Every uncharged pair has

$$p \in R' \setminus L, \quad q \in L \setminus R',$$

so the uncharged pairs form a matching between the exclusive occurrences of the two Boolean chains. Lemma A.36 bounds their number first by $\nu(L, R')$ and then by $d(L, R')$. This proves the first two inequalities in (A.62).

Every coordinate pair ordered differently on L and R' is shattered by their union and hence by the ample repair projection W . Thus $|W| - (r + 1) \geq d(L, R')$. The path-spine shadow partition gives the exact identity

$$\Delta(H; A, R) = |W| - (r + 1) + \sum_{i=1}^n (|C_i| - (r + 1)) + |\Omega|.$$

All summands are nonnegative, and the two carrier summands indexed by $j, j + 1$ are $\epsilon_j, \epsilon_{j+1}$. This proves the last inequality. $\square$

Corollary A.38 (Closed components in distinct sections pay independently). *Let $\mathcal{J} \subseteq \{1, \dots, n-1\}$, and suppose that for every $j \in \mathcal{J}$ the neutral-transfer digraph in P_j contains a directed cycle whose common positioned maximal cube has support T_j , where $t_j = |T_j|$. Then*

$$\Delta(H; A, R) \geq V + \sum_{j \in \mathcal{J}} (2^{t_j} - t_j - 1) \geq V + |\mathcal{J}|. \quad (\text{A.63})$$

In particular, overlap among the repair supports T_j at different sections causes no double counting: the corresponding concepts lie over different core words.

Proof. The concepts of H over the path and outside it form a disjoint partition, so

$$\Delta(H; A, R) = V + \sum_{h=0}^n (|P_h| - (r + 1)). \quad (\text{A.64})$$

Every P_h is ample and contains two repair antipodes; hence it shatters the empty set and every repair singleton and has order at least $r + 1$. For $j \in \mathcal{J}$, the full T_j -cube adds all subsets of T_j to $\text{Sh}(P_j)$, and therefore

$$|P_j| - (r + 1) \geq 2^{t_j} - t_j - 1.$$

Summing these inequalities in (A.64) proves the first bound. The second follows from $t_j \geq 2$, proved in Proposition A.28. $\square$

Proposition A.39 (Parallel closed-component cubes pay by coordinate degree). *Fix a path section P_j , a repair support $T \subseteq R$ of order $t \geq 2$, and let $q_{j,T} \geq 1$ be the number of distinct positioned T -cubes in P_j that occur as common maximal cubes of directed neutral-transfer cycles. Then*

$$|P_j| - (r + 1) \geq 2^t - t - 1 + (q_{j,T} - 1)2^{t-1}. \quad (\text{A.65})$$

More generally, if the closed components in P_j use d_j distinct maximal-cube supports and $q_{j,T}$ positioned cubes of support T , then

$$|P_j| - (r + 1) \geq \max \left\{ d_j, \max_T (2^{|T|} - |T| - 1 + (q_{j,T} - 1)2^{|T|-1}) \right\}. \quad (\text{A.66})$$

Thus neither different supports nor different positioned cubes with one support can hide their local concept cost. Proposition A.37 separately pays all remaining closed-component multiplicity, including several cycles in one positioned maximal cube.

Proof. Choose $e \in T$. Distinct positioned T -cubes are disjoint, and each contains exactly 2^{t-1} edges of direction e . Hence

$$|E_e(P_j)| \geq q_{j,T} 2^{t-1}. \quad (\text{A.67})$$

The coordinatewise edge-shadow identity for the ample family P_j gives

$$|E_e(P_j)| = |\{S \in \text{Sh}(P_j) : e \in S\}|.$$

One positioned full T -cube already supplies the 2^{t-1} subsets of T that contain e . Therefore (A.67) forces at least $(q_{j,T} - 1)2^{t-1}$ further shattered supports outside 2^T . Those supports are distinct from the empty support, all repair singletons, and the $2^t - t - 1$ nonsingleton subsets of T . Ampleness now proves (A.65).

For the more general assertion, every distinct maximal-cube support has order at least two and is itself a different member of $\text{Sh}(P_j)$ outside the repair baseline. This gives the lower bound d_j . Applying (A.65) separately to each support and taking the maximum gives the other term in (A.66). $\square$

Corollary A.40 (Two disjoint cycles in one cube force rank at least three). *Suppose that a functional subdigraph of the neutral-transfer digraph in P_j has two vertex-disjoint directed cycles with the same common positioned maximal cube Q , of support T . Then $|T| \geq 3$, and*

$$|P_j| - (r + 1) \geq 4. \quad (\text{A.68})$$

This rank restriction is a local geometric refinement of the general closed-transfer packing theorem.

Proof. If $|T| = 2$, the two cycles use at least four distinct vertices and hence use every vertex of Q . Every one of these vertices is a corner of P_j whose unique maximal cube is Q . Consequently no vertex of Q has a repair neighbour outside Q . Since P_j is connected, this forces $P_j = Q$. But P_j contains two repair antipodes at distance $r \geq 3$, a contradiction. Therefore $|T| \geq 3$. The full T -cube then adds at least $2^3 - 3 - 1 = 4$ shattered supports beyond the empty support and repair singletons, proving the displayed bound. $\square$

Proposition A.41 (The bare three-section cube collar is peripheral). *Let T be a coordinate set, let $A, B \subseteq Q_T$ be nonempty ample families, and introduce two further coordinates x, y . Then*

$$\mathcal{L}(A, B) = (A \times \{00\}) \cup (Q_T \times \{10\}) \cup (B \times \{11\}) \subseteq Q_{T \cup \{x, y\}} \quad (\text{A.69})$$

is ample. If $A, B \in \mathcal{D}_{\text{amp}}$, then $\mathcal{L}(A, B) \in \mathcal{D}_{\text{amp}}$.

Consequently, conditioning a same-cube neutral-transfer configuration to its common repair cube and the three consecutive core sections cannot by itself produce a smaller forbidden pc-minor: in a least-cardinality cornerless family, both outer conditionings are smaller dismantlable ample families, so the bare local collar is dismantlable. Any contraction theorem that uses such a local collar must retain parent-puncture provenance or other attachments outside the three sections.

Proof. First form

$$C = (A \times \{0\}) \cup (Q_T \times \{1\}) \subseteq Q_{T \cup \{x\}}.$$

After reversing the x -orientation, this is a peripheral expansion of the cube Q_T along its ample subfamily A . It is therefore ample; if $A \in \mathcal{D}_{\text{amp}}$, the exact peripheral-expansion criterion puts C in $\mathcal{D}_{\text{amp}}$.

Identify B with the subfamily $B \times \{1\} \subseteq C$. With respect to the new coordinate y , the family in (A.69) is

$$(C \times \{0\}) \cup ((B \times \{1\}) \times \{1\}),$$

another peripheral expansion, now of C along the copy of B . The same exact criterion proves ampleness and, when $B, C \in \mathcal{D}_{\text{amp}}$, dismantlability. The final assertion follows because conditionings and pc-minors of an ample family are ample, while every smaller ample family belongs to $\mathcal{D}_{\text{amp}}$ by Lemma 7.8. $\square$

Proposition A.42 (A fixed repair support has bounded closed-component capacity). *Keep the hypotheses of Proposition A.28, and fix a repair support $T \subseteq R$ of order $t \geq 2$. Let c_T be the number of distinct pairs (j, Q) such that the neutral-transfer digraph in P_j has a directed cycle with common positioned maximal cube Q of support T . Let q_T be the number of distinct positioned T -cubes among these pairs. If $c_T > 0$, then*

$$|U| \geq 2^t q_T, \quad V \geq 2^{t-1}(c_T - q_T), \quad (\text{A.70})$$

and hence

$$c_T \leq \frac{|U| + V}{2^{t-1}} - 1 \leq \frac{|W| + |\Omega|}{2^{t-1}} - 1. \quad (\text{A.71})$$

Thus repeated closed components with one repair support cannot be hidden by reusing the same local picture: a new positioned cube costs 2^t path repair words, while reappearance of an old positioned cube forces full T -cubes over core words outside the path. Several cycles with the same pair (j, Q) count only once in this estimate.

Proof. Fix one positioned repair T -cube $Q \subseteq Q_R$, and let c_Q be the number of section indices j for which (j, Q) is counted by c_T . Consider its core anchor family

$$A_Q = \{a \in Q_X : Q \times \{a\} \subseteq H\}.$$

The family of anchors of all full T -cubes of H is ample by the cube-carrier theorem. Fixing the repair coordinates in $R \setminus T$ to the position of Q shows that A_Q is an ample conditioning, and is therefore connected.

If a cycle using Q is centered at a_j , then $Q \subseteq P_j$, whereas (A.45) gives $Q \not\subseteq P_{j-1}$ and $Q \not\subseteq P_{j+1}$. Hence a_j is an isolated component of $A_Q \cap A$. Distinct counted pairs have distinct section indices, so the latter intersection has at least c_Q components.

Put $B_Q = A_Q \setminus A$. If $A_Q \cap A$ has at least two components, connectivity of A_Q gives an edge from every component to B_Q . A core word outside the coordinate-simple path A has at most two neighbors on A : two such neighbors have path distance at most two, distance one is excluded by bipartiteness of the cube, and three path neighbors are therefore impossible. Consequently

$$|B_Q| \geq \frac{c_Q - 1}{2};$$

the same inequality is trivial when $c_Q = 1$. Every anchor in B_Q supports all 2^t concepts of Q , and all of them are counted by V . Thus Q forces at least $2^{t-1}(c_Q - 1)$ external concepts.

Distinct positioned T -cubes are disjoint, both over the path and over external anchors. Summing over the q_T cubes gives

$$|U| \geq 2^t q_T, \quad V \geq 2^{t-1} \sum_Q (c_Q - 1) = 2^{t-1} (c_T - q_T),$$

which proves (A.70). Adding the two estimates and using $q_T \geq 1$ gives the first inequality in (A.71). The second follows from the path-spine identity

$$|W| + |\Omega| = |U| + \Phi + V \geq |U| + V.$$

□

Proposition A.43 (Global conservation of terminal transition charges). *For each of the $2n$ oriented adjacent-section incidences in Corollary A.23, perform progressive synchronization and let $\mathcal{T}$ be the incidences that stop with a terminal cyclic remainder. For $(i, j) \in \mathcal{T}$, denote that remainder by $D_{i,j}$ and put*

$$\chi_{i,j} = |D_{i,j}| + c(D_{i,j}).$$

Then

$$\sum_{(i,j) \in \mathcal{T}} \chi_{i,j} \leq 2 \sum_{\ell=0}^n |E(P_\ell)| \leq 2 \sum_{T \in \text{Sh}(H)} |T \cap R|. \quad (\text{A.72})$$

In particular, if $b = |\mathcal{T}|$, then

$$5b \leq \sum_{T \in \text{Sh}(H)} |T \cap R|. \quad (\text{A.73})$$

Thus cyclic transition failures cannot be reused without limit along the path: each failure consumes ten units of section-edge rank, and every section can pay for at most its two incident transitions.

Proof. For a terminal incidence (i, j) , equation (A.40) charges $\chi_{i,j}$ to the rank sum of a subfamily of $\text{Sh}(P_j)$. A fixed section P_j is incident with at most two path edges. Consequently, after summing all terminal incidences, every member of $\text{Sh}(P_j)$ has been charged at most twice. The edge-shadow identity gives

$$\sum_{(i,j) \in \mathcal{T}} \chi_{i,j} \leq 2 \sum_{j=0}^n \sum_{T \in \text{Sh}(P_j)} |T| = 2 \sum_{j=0}^n |E(P_j)|.$$

The edges counted by $E(P_j)$ are precisely the repair-coordinate edges of H whose core word is a_j . These sets are disjoint for distinct j , so their union is contained in the set of all repair-coordinate edges of H . A second application of the edge-shadow identity, resolved by coordinate, counts the latter edges as

$$\sum_{T \in \text{Sh}(H)} |T \cap R|.$$

This proves (A.72). Each terminal remainder has $\chi_{i,j} \geq 10$, and division by two yields (A.73). □

Corollary A.44 (The adjacent-carrier extension-or-owner trichotomy). *Keep the path-spine notation and fix an oriented incidence (i, j) , with $j \in \{i-1, i\}$. Run the progressive synchronization in Corollary A.23, and try its proposed duplications in the full family, in the same order. Put $\hat{H}_0 = H$, and after h successful duplications denote the resulting ample family by $\hat{H}_h$. One of the following occurs:*

1. the whole exclusive side is duplicated into the other path section by the sequence of ample one-concept extensions $\widehat{H}_h$;
2. a first global extension fails, and it has an antipodal blocker $y = v^J \in \widehat{H}_h$, where v is the attempted next concept and

$$x_i \notin J, \quad J \cap (X \setminus \{x_i\}) \neq \emptyset; \quad (\text{A.74})$$

in particular, the failure is owned by a core direction other than the transition direction x_i ;

3. every proposed global extension succeeds until the local process stops at a terminal cyclic remainder D , which satisfies (A.40).

This trichotomy is independent of the orders of H and of the core path.

Proof. At any stage before a terminal remainder is reached, condition all core coordinates in $X \setminus \{x_i\}$ to the values common to a_{i-1} and a_i . The conditioned family, with the previously duplicated concepts included, is exactly the corresponding ample family in (A.38). Apply Lemma A.2 to $\widehat{H}_h$, with

$$E = R \dot{\cup} \{x_i\}, \quad F = X \setminus \{x_i\}.$$

If an extension fails, its blocker satisfies $J \cap (X \setminus \{x_i\}) \neq \emptyset$ by Lemma A.2. The concept being duplicated has its original copy across the x_i -edge in the current ample family. If $x_i \in J$, that copy would be a third member of $Q_J(v) \cap (\widehat{H}_h \cup \{v\})$, contrary to (A.6). Hence $x_i \notin J$, proving (A.74). If none fails, the local progressive process either duplicates the whole side or reaches the terminal cyclic alternative of Theorem A.22. These are the other two conclusions. $\square$

Proposition A.45 (Transverse owners move strictly toward an endpoint). *Suppose that the second alternative of Corollary A.44 occurs while duplicating a concept into the section over a_k , where $k \in \{i-1, i\}$. Let v be the missing copy and let $y = v^J \in \widehat{H}_h$ be its antipodal blocker in the current ample extension state. Then either the core word of y lies outside the path $A = \{a_0, \dots, a_n\}$, or it is a_h for an index satisfying*

$$h \leq i-2 \quad \text{if } k = i-1, \quad h \geq i+1 \quad \text{if } k = i. \quad (\text{A.75})$$

Thus an on-spine owner jumps strictly toward the endpoint on the target side of the transition. In particular, successive on-spine owner transfers cannot form a directed cycle and have length at most n .

Proof. Put $K = J \cap X$. Equation (A.74) gives $K \neq \emptyset$ and $x_i \notin K$. The core word of v is a_k , so the core word of its blocker is a_k^K . If this word lies outside A , the first alternative holds.

Otherwise $a_k^K = a_h$ for some h . Along the coordinate-simple path,

$$K = \text{Diff}(a_k, a_h) = \{x_{\min\{h,k\}+1}, \dots, x_{\max\{h,k\}}\}.$$

This interval is nonempty and omits x_i . If $k = i-1$, it must lie strictly to the left of x_i , so $h \leq i-2$. If $k = i$, it must lie strictly to the right, so $h \geq i+1$. The index therefore moves strictly toward the corresponding endpoint at every on-spine transfer, which excludes a directed cycle and bounds the length of a transfer chain. $\square$

Proposition A.46 (Every transverse owner pays the residual shadow). *Keep the failed-extension notation of Proposition A.45. Write $\widehat{H}$ for the current ample extension state immediately before the failed augmentation and write the blocker as $y = (g, b) \in \widehat{H}$. Define the path sections $\widehat{P}_j$, adjacent carriers $\widehat{C}_i$, full transverse carriers $\widehat{D}_i$, and residual shadow $\widehat{\Omega}$ for $\widehat{H}$ as in Propositions 7.59 and 7.78. Then*

$$y^{\{x_i\}} \in \widehat{H}, \quad g \in \widehat{D}_i. \quad (\text{A.76})$$

Moreover, at least one of the following residual payments exists:

1. if $g \notin \widehat{C}_i$, there is a support in $\widehat{\Omega}$ with exact core part $\{x_i\}$;
2. if $g \in \widehat{C}_i$, there is an $\ell \neq i$ such that $\{x_i, x_\ell\} \in \widehat{\Omega}_{\geq 2}$.

Thus every failed lift in the adjacent-carrier trichotomy produces a member of the residual shadow of its current extension state. In particular, a failure on the first proposed duplication pays the original residual shadow Ω of H .

Proof. Let v be the missing duplicated concept and put $w = v^{\{x_i\}}$. The original copy w belongs to the current ample family $\widehat{H}$, while $y = v^J$ is the blocker and $x_i \notin J$. Hence

$$\text{Diff}(y, w) = J \dot{\cup} \{x_i\}.$$

For every $e \in J$, the neighbor $y^{\{e\}}$ lies in the J -face through v and is different from its two allowed members v, y . The blocker identity therefore gives $y^{\{e\}} \notin \widehat{H}$. An isometric $\widehat{H}$ -geodesic from y to w must start in a direction belonging to $J \dot{\cup} \{x_i\}$. All directions in J are absent at y , so its first edge has direction x_i . This proves (A.76).

If $g \notin \widehat{C}_i$, then $g \in \widehat{D}_i \setminus \widehat{C}_i$. Both carriers are ample, so $\text{Sh}(\widehat{D}_i) \setminus \text{Sh}(\widehat{C}_i) \neq \emptyset$. Every support in this difference, after adjoining x_i , belongs to $\widehat{\Omega}$ by (7.102); this proves the first alternative.

It remains to suppose that $g \in \widehat{C}_i$. The fibre $\widehat{H}_g$ then contains the path edge $a_{i-1}a_i$ as well as the parallel edge $bb^{\{x_i\}}$ from (A.76). The word b is neither a_{i-1} nor a_i : it differs from the target core word of v on the nonempty set $J \cap X$, which omits x_i . Therefore some coordinate $x_\ell \neq x_i$ has a value on b opposite to its common value on a_{i-1} and a_i . The two parallel x_i -edges then realize all four traces on $\{x_i, x_\ell\}$. This pair is shattered by $\widehat{H}$, and it lies in $\widehat{\Omega}_{\geq 2}$ because the spine family contains no support with two core coordinates. $\square$

Proposition A.47 (Residual shadows transport backwards through synchronization). *Keep the path-spine notation for an ample family H . Let $j, k \in \{i-1, i\}$ be distinct, let $p \in P_j \setminus P_k$, and suppose that adjoining the missing copy $v = (p, a_k)$ gives an ample family*

$$H^+ = H \cup \{v\}.$$

Let Ω^+ be the residual shadow of H^+ . Then there is an injection

$$\iota_v : \Omega^+ \longrightarrow \Omega \quad (\text{A.77})$$

that fixes every support in $\Omega^+ \cap \Omega$, with at most one exception. More precisely, let $s(v) \in \{0, 1\}$ record whether a_k has a second path neighbor $a_h \neq a_j$ for which $p \in P_h$. Then

$$|\Omega^+| = |\Omega| - s(v). \quad (\text{A.78})$$

If the unique new shattered support of H^+ belongs to Ω^+ , it is sent by ι_v to an old residual support that entered the new path-spine family; all other members are fixed.

Proof. The repair projection W , its union U , and the external-core count V do not change: the repair word p was already present over a_j , and the new concept lies over the path core. The section P_k gains p . Consequently the adjacent carrier C_i gains p , and the only other carrier that can change is the one on the second path edge at a_k ; it gains p exactly when $s(v) = 1$. All changed carriers are conditionings of the ample family H^+ , hence are ample. Since each has gained one concept, its shadow has gained exactly one support.

Let $\mathcal{B}$ and $\mathcal{B}^+$ be the path-spine families from (7.49). The carrier supports are distinguished by their core coordinates, so the preceding paragraph gives

$$\mathcal{B} \subseteq \mathcal{B}^+, \quad |\mathcal{B}^+ \setminus \mathcal{B}| = 1 + s(v). \quad (\text{A.79})$$

On the other hand, the ample one-concept extension has a unique new shattered support; call it R_v . Thus

$$\text{Sh}(H^+) = \text{Sh}(H) \dot{\cup} \{R_v\}.$$

Subtracting (A.79) from this identity proves (A.78).

If $R_v \notin \Omega^+$, then every member of Ω^+ already belongs to Ω , so inclusion is the desired injection. Suppose that $R_v \in \Omega^+$. Then none of the $1 + s(v)$ new spine supports is R_v . Every one therefore belonged to $\text{Sh}(H)$, and, because it did not belong to $\mathcal{B}$, it belonged to Ω . Choose one such displaced support Q_v , send R_v to Q_v , and fix every common residual support. The support Q_v is absent from Ω^+ , so this map is injective and has the asserted form. $\square$

Corollary A.48 (All later residual payments return to the original shadow). *Let*

$$H = H_0 \subset H_1 \subset \cdots \subset H_q$$

be any sequence of successful adjacent-section duplications of the kind in Corollary A.44, with every H_h ample. If Ω_h is the residual shadow of H_h , then the maps in Proposition A.47 compose to an injection

$$\Omega_q \hookrightarrow \Omega_0. \quad (\text{A.80})$$

Consequently every residual support paid by a later failed lift has a well-defined representative in the original residual shadow of H .

Proof. Apply Proposition A.47 to each inclusion $H_{h-1} \subset H_h$ and compose the resulting injections in reverse chronological order. $\square$

Proposition A.49 (The two progressive synchronizations commute). *Fix an interior path section P_j . Suppose that the complete- synchronization alternative of Corollary A.23 holds for P_j at both adjacent path edges. Retain orders*

$$p_1, \dots, p_a \quad \text{of } P_j \setminus C_j, \quad q_1, \dots, q_b \quad \text{of } P_j \setminus C_{j+1}$$

supplied by that corollary. Duplicate the p 's into the section over a_{j-1} and the q 's into the section over a_{j+1} , in any shuffle that preserves the two displayed orders.

At every stage, the next proposed duplication is ample in the corresponding two-layer conditioning. Consequently, exactly one of the following occurs:

1. *a first duplication fails in the full family; it has a transverse owner and produces a residual payment which transports injectively to the residual shadow of the original family;*

2. every duplication succeeds in the full family. The resulting ample extension H^* has path sections

$$P_{j-1}^* = P_{j-1} \cup P_j, \quad P_j^* = P_j, \quad P_{j+1}^* = P_{j+1} \cup P_j, \quad (\text{A.81})$$

and its residual shadow injects into the residual shadow of H .

In the second outcome, both adjacent two-layer conditionings are peripheral toward their outer section.

Proof. Consider a proposed duplication of p_h across the edge $a_{j-1}a_j$. Condition every core coordinate except x_j to the common values of these two core words. Every copy previously added over a_{j+1} is absent from this conditioning, because a_{j+1} differs from a_j in x_{j+1} , which has just been fixed. The conditioned family is therefore exactly the h -th stage of the progressive synchronization order for the left edge, together with the earlier p -prefix. Corollary A.23 says that adjoining the next copy is ample in this conditioning. The symmetric argument applies to every proposed q_h -duplication. Hence the two local orders may be shuffled arbitrarily.

Run the shuffled sequence in the full family. At a first failed duplication, apply Lemma A.2 to the current ample extension state and the corresponding two-layer conditioning. The other-side additions vanish from that conditioning by the preceding paragraph. The argument in Corollary A.44 therefore applies verbatim: the blocker is transverse, and the original copy across the transition edge excludes that edge from the blocker support. Proposition A.46 supplies a residual payment in the current extension state. Corollary A.48 transports it through all earlier successful duplications to the original residual shadow.

If no duplication fails, every intermediate family and the final family H^* are ample. The left target section has acquired exactly $P_j \setminus C_j$; since $C_j = P_{j-1} \cap P_j$, it is

$$P_{j-1} \cup (P_j \setminus C_j) = P_{j-1} \cup P_j.$$

The right identity is symmetric, proving (A.81). Backward residual transport along the full successful sequence gives the asserted injection. Finally, P_j is contained in each outer union in (A.81), which is precisely the peripheral alternative in the corresponding two-layer conditioning. $\square$

Corollary A.50 (A tight two-cycle hands off to two peripheral transitions). *Suppose that the two marked initial demands in an interior section form the tight directed two-cycle of Corollary A.32. Retain the two complete-synchronization orders for which these marked corners are the last terms. Then either a shuffled synchronization produces a transverse residual payment, or there is an ample extension satisfying (A.81). Thus a tight two-cycle creates no further terminal geometry: its only unpaid outcome is a double-peripheral conditioned handoff.*

Proof. Apply Proposition A.49, using a shuffle in which all nonfinal terms precede the two marked final terms. Its two alternatives are exactly the stated outcomes. $\square$

Lemma A.51 (An adjacent duplication transports one section-shadow boundary). *Keep the path-spine notation, let j, k be the two ends of the path edge $a_{i-1}a_i$, and suppose that*

$$p \in P_j \setminus P_k, \quad H^+ = H \cup \{(p, a_k)\}$$

is ample. Put $C_i = P_j \cap P_k$. Then $P_k \cup \{p\}$ and $C_i \cup \{p\}$ are ample one-concept extensions, and there is a unique repair support T such that

$$\begin{aligned} \text{Sh}(P_k \cup \{p\}) &= \text{Sh}(P_k) \dot{\cup} \{T\}, \\ \text{Sh}(C_i \cup \{p\}) &= \text{Sh}(C_i) \dot{\cup} \{T\}. \end{aligned} \quad (\text{A.82})$$

Moreover,

$$T \in \text{Sh}(P_j) \setminus \text{Sh}(P_k). \quad (\text{A.83})$$

Thus a successful concept duplication carries exactly one existing section-shadow support across the same path edge; it creates no new repair support.

Proof. The two displayed families are conditionings of the ample family H^+ , so they are ample. Each has one more concept than its counterpart before the duplication and consequently acquires one shattered support. Call the new supports T_k and T_C , respectively.

Apply the two-layer shadow-intersection identity before and after the duplication. Since the source section does not change, it gives

$$\text{Sh}(C_i) = \text{Sh}(P_j) \cap \text{Sh}(P_k), \quad \text{Sh}(C_i \cup \{p\}) = \text{Sh}(P_j) \cap \text{Sh}(P_k \cup \{p\}).$$

The second intersection is obtained from the first by testing only the one new support T_k . It must grow, because $C_i \cup \{p\}$ is an ample one-concept extension. Hence $T_k \in \text{Sh}(P_j)$ and $T_C = T_k$. Taking $T = T_k$ proves both assertions. $\square$

Proposition A.52 (Weighted path boundaries are Lyapunov functions). *For a tuple of nonempty ample path sections $\mathbf{P} = (P_0, \dots, P_n)$, define its directed word-boundary potential by*

$$\Lambda_0(\mathbf{P}) = \sum_{i=1}^n (i|P_i \setminus P_{i-1}| + (n-i+1)|P_{i-1} \setminus P_i|) \quad (\text{A.84})$$

and its rank-weighted shadow-boundary potential by

$$\begin{aligned} \Lambda_1(\mathbf{P}) = \sum_{i=1}^n \Big(& i \sum_{T \in \text{Sh}(P_i) \setminus \text{Sh}(P_{i-1})} |T| \\ & + (n-i+1) \sum_{T \in \text{Sh}(P_{i-1}) \setminus \text{Sh}(P_i)} |T| \Big). \end{aligned} \quad (\text{A.85})$$

If one concept is duplicated successfully from one path section into its neighbour, then Λ_0 decreases by at least one. If T is the support in Lemma A.51, then Λ_1 decreases by at least $|T|$. More precisely, each decrease is either the displayed minimum or the annihilation value obtained when the transported boundary meets the oppositely oriented boundary at the next edge.

Put

$$\mathcal{E}(\mathbf{P}) = \sum_{j=0}^n |E(P_j)|.$$

Along every sequence of successful adjacent-section duplications,

$$\mathcal{E}(\mathbf{P}) + \Lambda_1(\mathbf{P}) \quad \text{is nonincreasing.} \quad (\text{A.86})$$

In particular, every such sequence has length at most $\Lambda_0(\mathbf{P})$ at its initial state. The initial potentials obey

$$\Lambda_0(\mathbf{P}) \leq 2n(|W| + |\Omega|), \quad (\text{A.87})$$

$$\Lambda_1(\mathbf{P}) \leq 2n \left(\sum_{j=0}^n |E(P_j)| - \sum_{i=1}^n |E(C_i)| \right). \quad (\text{A.88})$$

Thus no successful handoff can circulate: at concept level and at rank-weighted shadow level every moved boundary travels strictly toward an endpoint or annihilates with an opposite boundary.

Proof. Consider first a duplication from P_i into P_{i-1} . It removes the directed boundary occurrence of weight i . If $i > 1$ and the duplicated word is absent from P_{i-2} , it creates the same directed occurrence at the preceding edge, now of weight $i - 1$. The net decrease is one. If the word is present in P_{i-2} , the duplication instead removes the oppositely directed occurrence at that edge; the decrease is larger. At $i = 1$ the occurrence simply leaves the path. The argument for a duplication from P_{i-1} into P_i is symmetric, with weights $n - i + 1$ and $n - i$. This proves the assertion for Λ_0 .

Lemma A.51 says that the same operation is performed on exactly one section-shadow support T . Since the target section was nonempty, it already shattered the empty set, so $T \neq \emptyset$. Multiplying the preceding calculation by $|T|$ proves the assertion for Λ_1 . The edge-shadow identity gives

$$|E(P_k \cup \{p\})| - |E(P_k)| = |T|.$$

No other section changes. Hence the increase of $\mathcal{E}$ is at most the simultaneous decrease of Λ_1 , proving (A.86). Since every successful duplication decreases the nonnegative integer Λ_0 , the length bound follows.

Every coefficient in (A.84) is at most n . Proposition A.25 therefore gives

$$\Lambda_0(\mathbf{P}) \leq n \sum_{i=1}^n |P_{i-1} \triangle P_i| \leq 2n(|W| + |\Omega|),$$

which is (A.87).

For the last bound, fix $T \in \bigcup_j \text{Sh}(P_j)$. If its section-index set has $c_{\text{sh}}(T)$ components, it has at most $2c_{\text{sh}}(T)$ boundary edges. The rank-weighted version of the interval count in Proposition A.25 is

$$\sum_T |T| c_{\text{sh}}(T) = \sum_{j=0}^n |E(P_j)| - \sum_{i=1}^n |E(C_i)|.$$

Multiplying the boundary count by the maximum weight n and summing over T proves (A.88). $\square$

Corollary A.53 (Boundary descent pays every acquired marked copy). *Consider any sequence of successful adjacent-section duplications, and mark any collection of directed boundary demands that disappear during the sequence. Charge a marked demand to the first duplication that adjoins its missing copy. Then*

$$\#\{\text{disappearing marked demands}\} \leq \Lambda_0(\mathbf{P}^{\text{in}}) - \Lambda_0(\mathbf{P}^{\text{out}}). \quad (\text{A.89})$$

In particular, two different marked demands can be resolved by the same duplication only when two oppositely oriented boundaries annihilate there; that step decreases Λ_0 by $n + 2$, and so pays both demands with strict slack.

Proof. A duplication always removes the directed boundary across which it is performed. It can remove at most one further marked boundary, namely the oppositely directed boundary at the other edge incident with its target section. If it removes no such second boundary, the proof of Proposition A.52 gives a decrease of at least one. If it removes the second boundary, the two removed weights sum to

$$i + (n - i + 2) = n + 2 \quad \text{or} \quad (n - i + 1) + (i + 1) = n + 2,$$

according to the direction of transport. This is at least two. Assigning each marked demand to its first resolving duplication and summing these stepwise inequalities proves (A.89). $\square$

Proposition A.54 (All original tight two-cycles reduce in two sweeps). *In the original path-spine state, mark every interior section whose two selected initial demands form the tight directed two-cycle of Corollary A.32; let $\mathcal{J}$ be the set of their indices. All members of $\mathcal{J}$ can be treated together as follows.*

First process the leftward complete-synchronization orders in increasing order of $j \in \mathcal{J}$. Then process the unresolved rightward incidences in decreasing order. At a rightward incidence use a fresh progressive synchronization in the current ample two-layer conditioning, and stop as soon as the marked missing copy is acquired. For every $j \in \mathcal{J}$, one of the following occurs:

- 1. one of the required lifts fails globally and supplies a transverse residual payment, which transports to the original residual shadow;*
- 2. a fresh rightward synchronization stops before the marked copy and supplies a terminal cyclic block;*
- 3. both marked missing copies are acquired by successful ample extensions, so the original tight two-cycle disappears.*

Every successful duplication in the two sweeps is charged by the strict descent in Proposition A.52. Consequently all successful work in the simultaneous reduction is finite and satisfies the original-state edge-shadow bound (A.86). No iterated handoff cycle, owner-repetition alternative, or cardinality-by-cardinality closure is needed.

Proof. For the leftward sweep, an operation at an earlier centre $k < j$ changes only the section over a_{k-1} ; in particular, it changes neither section over the edge $a_{j-1}a_j$. After conditioning to that edge, all earlier additions therefore disappear. The retained leftward order for the original tight two-cycle is still locally valid. Trying its duplications in the full current family either gives the transverse failure in Corollary A.44, with backward residual transport, or acquires the marked left copy.

After the leftward sweep, consider the indices in decreasing order. An already processed larger centre changes only a section strictly to the right of the current edge. The leftward sweep may have enlarged the two current sections, but they remain conditionings of the current ample full family and their intersection is ample. Apply Theorem A.22 afresh in the rightward orientation. If the marked copy has already been acquired, no operation is needed. Otherwise, either the progressive process reaches that copy, a proposed full lift first fails and gives the residual outcome, or the process stops earlier with the stated terminal cyclic block. These are exactly the three alternatives in the proposition.

All full families retained during the construction are ample, and every successful step is an adjacent-section duplication. Proposition A.52 therefore applies to their complete concatenation. Corollary A.53 pays all marked copies acquired in the third alternative, including the case in which one addition resolves marked demands from two neighbouring centres. This gives the final assertions. $\square$

Proposition A.55 (Dynamic terminal blocks obey original-state conservation). *Start from a path-spine state with section tuple $\mathbf{P}^{\text{in}}$, perform any sequence of successful adjacent-section duplications, and process each oriented path incidence at most once. The incidences may be processed in different intermediate ample states. Let $\mathcal{T}$ be those that stop with a terminal cyclic remainder, and retain the notation*

$$\chi_{i,j} = |D_{i,j}| + c(D_{i,j})$$

from Proposition A.43. If $\mathbf{P}^{\text{out}}$ is the final section tuple, then

$$\sum_{(i,j) \in \mathcal{T}} \chi_{i,j} \leq 2\mathcal{E}(\mathbf{P}^{\text{out}}) \leq 2(\mathcal{E}(\mathbf{P}^{\text{in}}) + \Lambda_1(\mathbf{P}^{\text{in}})). \quad (\text{A.90})$$

Thus terminal blocks created during the two-sweep reduction require no new state-dependent budget: their complete rank charge returns to the initial section-edge rank together with the initial shadow-boundary potential.

Proof. At the state in which (i, j) stops, equation (A.40) charges $\chi_{i,j}$ to the rank sum of a subfamily of $\text{Sh}(P_j)$ in that state. Sections only grow during the duplication sequence, and every intermediate section remains ample. Consequently its shattered complex is contained in the shattered complex of the final section P_j^{out} .

A fixed section is the source of at most its two oriented path incidences. Therefore every support in $\text{Sh}(P_j^{\text{out}})$ is counted at most twice after summing all terminal charges based at that section. The edge-shadow identity gives

$$\sum_{(i,j) \in \mathcal{T}} \chi_{i,j} \leq 2 \sum_{j=0}^n \sum_{T \in \text{Sh}(P_j^{\text{out}})} |T| = 2\mathcal{E}(\mathbf{P}^{\text{out}}).$$

Finally apply the Lyapunov inequality (A.86) to the complete successful duplication sequence and discard the nonnegative terminal value of Λ_1 . This proves the second inequality in (A.90). $\square$

Proposition A.56 (Successful acquisitions form a spacetime support forest). *Start from an initial path-section tuple $\mathbf{P}^{\text{in}}$, perform a sequence of successful adjacent-section duplications, and let $\mathbf{P}^{\text{out}}$ be the final tuple. Fix a nonbaseline repair support T . Write*

$$I_T^{\text{in}} = \{j : T \in \text{Sh}(P_j^{\text{in}})\}, \quad I_T^{\text{out}} = \{j : T \in \text{Sh}(P_j^{\text{out}})\},$$

and denote their orders and numbers of path components by $m_T^{\text{in}}, c_T^{\text{in}}$ and $m_T^{\text{out}}, c_T^{\text{out}}$, respectively. Let q_T be the number of successful duplications whose transported repair support is T , and let μ_T be the number of those steps at which the newly added section index joins two components of the current T -index set. Then

$$m_T^{\text{out}} = m_T^{\text{in}} + q_T, \quad (\text{A.91})$$

$$c_T^{\text{out}} = c_T^{\text{in}} - \mu_T, \quad (\text{A.92})$$

$$(m_T^{\text{out}} - c_T^{\text{out}} + 1) - (m_T^{\text{in}} - c_T^{\text{in}} + 1) = q_T + \mu_T. \quad (\text{A.93})$$

More precisely, start with the rooted support forest from (7.56). A nonmerging acquisition adjoins one occurrence vertex and its processed path edge. A merging acquisition adjoins one occurrence vertex and both incident path edges. The resulting graph is the rooted support forest of I_T^{out} ; in particular, no spacetime support cycle is created.

Consequently, if a demand family $\mathcal{D}^{\text{out}}$ injects into the final nonbaseline occurrence set

$$\mathfrak{D}^{\text{out}} = \{(j, T) : T \notin \mathcal{L}_R, j \in I_T^{\text{out}}\},$$

then it and the failed lifts admit a joint injection into the disjoint union of

$$\Xi, \quad \Omega_0, \quad \mathcal{A} = \{\text{successful support-acquisition steps}\}.$$

Equivalently,

$$|\mathcal{D}^{\text{out}}| + |\mathcal{F}| \leq \Delta(H; A, R) + |\mathcal{A}|. \quad (\text{A.94})$$

Thus the only resource in a final-state terminal payment that is not already present in the original excess is the successful acquisition itself.

Proof. At a successful duplication, Lemma A.51 supplies one unique repair support T that is present in the source-section shadow and absent from the target-section shadow. The step adds the target index to the current set I_T , so every T -acquisition increases its order by one. This proves (A.91).

The new index is adjacent to its source index. If its other path neighbour is absent from I_T , it extends one component and leaves the component number unchanged. If the other neighbour is present, the two neighbours belong to different components of the path with the target index deleted; adjoining the target joins those components and decreases their number by one. These are the only cases, proving (A.92). Subtraction gives (A.93).

The same argument proves the forest description. In the first case one adjoins a leaf and one edge. In the second, one adjoins a degree-two vertex between two previously different tree components. Neither operation creates a cycle. The final graph has one root edge in a chosen component and every internal edge of the path induced by I_T^{out} , exactly as in the proof of Proposition 7.60.

It remains to count the available columns without charging a component merger twice. For fixed T ,

$$m_T^{\text{out}} = (m_T^{\text{in}} - c_T^{\text{in}} + 1) + q_T + (c_T^{\text{in}} - 1).$$

The first term is the number of original rooted and internal-edge supports in the exact-repair part of Ξ . Use the q_T acquisition steps as formal columns for the second term. Use the $c_T^{\text{in}} - 1$ original fragmentation generators for the last term. Proposition A.64 injects all these fragmentation generators, jointly with all failures, into Ω_0 . Therefore any injection of $\mathcal{D}^{\text{out}}$ into the final occurrence set lifts to the asserted joint injection. Summing over T , and applying (7.54), proves (A.94). $\square$

Proposition A.57 (A dynamic terminal block has an unmarked acquisition when it needs one). *Keep the two-sweep reduction of Proposition A.54. Suppose that the original tight two-cycle centred at the interior section P_j^{in} ends in the terminal cyclic alternative in a later section P_j^{cur} . Put*

$$e_j^{\text{in}} = |P_j^{\text{in}}| - (r + 1),$$

and let a_j be the number of successful duplications before that terminal state whose target is P_j . Then

$$e_j^{\text{in}} \geq 1, \quad e_j^{\text{in}} + a_j \geq 9. \quad (\text{A.95})$$

Consequently exactly one of the following holds:

1. $e_j^{\text{in}} \geq 2$, so the two sides of the original tight cycle can be assigned two distinct original nonbaseline occurrences in P_j^{in} ;
2. $e_j^{\text{in}} = 1$, and at least one of the acquisitions into P_j is unmarked, meaning that it does not first acquire either selected missing copy of an original two-sided demand.

The unmarked acquisition in the second alternative may be chosen injectively over all dynamic terminal sections.

Proof. The original directed two-cycle has a common positioned T -cube with $|T| \geq 2$ by Proposition A.28. Hence Corollary A.38, applied in the single section j , gives

$$e_j^{\text{in}} \geq 2^{|T|} - |T| - 1 \geq 1.$$

At the later terminal state, write Z for the last synchronized subfamily and $D = P_j^{\text{cur}} \setminus Z$ for the terminal remainder. The family Z contains the current adjacent carrier, which contains the two fixed repair antipodes. It is isometric and therefore has at least $r + 1$ concepts. The terminal theorem gives $|D| \geq 9$, so

$$|P_j^{\text{cur}}| - (r + 1) = |Z| + |D| - (r + 1) \geq 9.$$

Every successful duplication into P_j adds one concept and no operation deletes a concept. Thus

$$|P_j^{\text{cur}}| = |P_j^{\text{in}}| + a_j,$$

which proves (A.95).

If $e_j^{\text{in}} \geq 2$, ampleness identifies the original nonbaseline section-shadow supports with exactly e_j^{in} distinct occurrence labels, proving the first alternative. Suppose instead that $e_j^{\text{in}} = 1$. The displayed inequality gives $a_j \geq 8$. Only the two oriented incidences directed into P_j can have a selected missing copy whose acquisition step targets P_j , and each incidence has at most one such marked copy. Therefore at most two of these a_j steps are marked; at least six are unmarked. Choose one.

Finally, an acquisition step has one target section. Dynamic terminal cycles with different centres use different target sections P_j , so the choices made in the second alternative are automatically distinct. $\square$

Corollary A.58 (Terminal remainders add no residual-shadow debt). *In the two-sweep assembly, every initial terminal incidence and every dynamic terminal outcome can be paid by original section-shadow occurrence columns and by successful-acquisition columns that are not committed to a marked acquisition. These choices can be made disjointly from the fragmentation and failure payments in Ω_0 .*

Proof. If a progressive synchronization is terminal already in the initial state, its terminal remainder D is disjoint from an ample synchronized family containing the two repair antipodes. Hence its base section has at least $|D| \geq 9$ original nonbaseline occurrence columns. There are at most two oriented incidences based at one section, so choose distinct columns for all such initial terminal demands and for the selected demand on the other side, if present.

For a dynamic terminal outcome of an original tight two-cycle, the two selected side demands had coalesced into one neutral component. This is the only weak component based at P_j that contains a selected initial demand; components containing no selected demand need not be included in the assembly demand family. One original occurrence column pays the selected component by Proposition A.29. Proposition A.57 pays the possible second-side deficit either with another original occurrence or with an unmarked acquisition column. The final sentence of that proposition makes these latter choices distinct. Original occurrence columns lift through Proposition A.65, while acquisition columns are the additional forest columns in Proposition A.56. Neither type uses a member of the residual shadow already assigned to a failure or fragmentation generator. $\square$

Proposition A.59 (Residual collisions have a bounded genealogy). *Let*

$$H_0 \subset H_1 \subset \cdots \subset H_q$$

be a sequence of successful adjacent-section duplications, let Ω_h be the residual shadow of H_h , and use the backward injections of Proposition A.47. Interspersed with the successful steps, process each oriented path incidence at most once. Let $\mathcal{F}$ be a collection of failed lifts, and for a failure across the path edge x_i choose one of the residual supports supplied by Proposition A.46. Thus its chosen support $R \in \Omega_h$ satisfies $x_i \in R$.

For a family of supports $\mathcal{A}$, put

$$M_X(\mathcal{A}) = \sum_{R \in \mathcal{A}} |R \cap X|.$$

At a successful step $H_{h-1} \subset H_h$, call the backward injection exceptional if the unique new shattered support N_h belongs to Ω_h and is therefore sent to an old residual support displaced by the enlarged path-spine family. If $\mathcal{X}$ is the set of exceptional steps, then

$$|\mathcal{F}| \leq 2M_X(\Omega_0) + 2 \sum_{h \in \mathcal{X}} |N_h \cap X|. \quad (\text{A.96})$$

Moreover,

$$\sum_{h \in \mathcal{X}} |N_h \cap X| \leq |E_X(H_q)| - |E_X(H_0)| \leq \Lambda_0(\mathbf{P}^{\text{in}}) - \Lambda_0(\mathbf{P}^{\text{out}}) + 2V, \quad (\text{A.97})$$

where E_X denotes the set of edges whose direction lies in X , and V is the number of original concepts over core words outside the path. Consequently,

$$|\mathcal{F}| \leq 2M_X(\Omega_0) + 2\Lambda_0(\mathbf{P}^{\text{in}}) + 4V. \quad (\text{A.98})$$

Thus failures at different extension times cannot reuse one original residual support arbitrarily: all extra multiplicity is localized to new residual-support segments, and their total core rank is paid by boundary descent and the fixed external-core incidence budget.

Proof. Draw one node (h, R) for every $R \in \Omega_h$ and join it backwards to $(h-1, \iota_h(R))$. Every node has one backward edge, and injectivity of ι_h gives at most one forward edge at every node. The resulting graph is therefore a disjoint union of paths rooted at the members of Ω_0 . On every nonexceptional backward edge the support label is unchanged. An exceptional step introduces at most one new label, namely N_h . After compressing consecutive nodes with the same label, the paths therefore consist of the $|\Omega_0|$ initial segments and at most one further segment for each $h \in \mathcal{X}$.

Attach every failed lift to the node carrying its chosen residual support at the state in which the failure occurs. On a segment with fixed label R , a failure across x_i is possible only when $x_i \in R$. There are only two oriented incidences of the path edge x_i , and no incidence is processed twice. Hence at most $2|R \cap X|$ failures attach to that segment. Summing first over the initial segments and then over the exceptional segments proves (A.96).

Every successful ample one-concept extension has a different unique new shattered support. The coordinatewise edge-shadow identity therefore gives

$$|E_X(H_q)| - |E_X(H_0)| = \sum_{h=1}^q |N_h^{\text{new}} \cap X|,$$

where N_h^{new} denotes the unique new support at step h . The exceptional supports form a subfamily of these new supports, proving the first inequality in (A.97).

It remains to count the new core edges geometrically. A duplicated path concept creates one edge to its source section and possibly the edge to its other path neighbour. The proof of Proposition

A.52 shows that the corresponding decrease of Λ_0 is at least the number of these new path edges: it is one when a boundary moves, and $n + 2$ when two boundaries annihilate. Summing pays every new core edge whose two ends lie over the path.

Every remaining new core edge joins the duplicated path concept to an original concept whose core word lies outside the path. A core word outside a coordinate-simple cube path has at most two neighbours on that path: any two such neighbours have path distance two, while bipartiteness excludes consecutive path vertices, and three path neighbours are impossible. There are V original external concepts, so at most $2V$ such edges are created. This proves the second inequality in (A.97). Substitute it into (A.96) and discard the nonnegative final potential to obtain (A.98). $\square$

Proposition A.60 (Failed lifts have distinct original boundary ancestors). *Keep the dynamic setting of Proposition A.59, and stop processing an oriented incidence at its first failed global lift. Then the failures inject into the directed word-boundary occurrences of the initial section tuple. They also inject into the directed section-shadow boundary occurrences of the initial tuple. Consequently,*

$$|\mathcal{F}| \leq \sum_{i=1}^n |P_{i-1}^{\text{in}} \triangle P_i^{\text{in}}| \quad (\text{A.99})$$

$$= \sum_{i=1}^n |\text{Sh}(P_{i-1}^{\text{in}}) \triangle \text{Sh}(P_i^{\text{in}})| \leq 2(|W| + |\Omega_0|). \quad (\text{A.100})$$

More precisely, every failure carries two compatible labels: a distinct initial directed boundary ancestor and a residual-support lineage from Proposition A.59. Thus two failures transported to the same original residual support are already separated by their original boundary labels.

Proof. A directed word-boundary occurrence is a word p present on one end of a path edge and absent on the other, together with its orientation from the present end toward the missing end. Under a successful adjacent duplication, exactly one of three things happens to the occurrence used by that duplication: it leaves the path at an endpoint, it annihilates with the oppositely oriented occurrence at the other edge of the target section, or it is transported to that other edge. In the last case the support word and orientation are unchanged and its endpoint-distance weight decreases by one. No operation creates two descendants of one occurrence. Tracing backwards therefore injects every directed boundary occurrence in every intermediate state into a unique initial occurrence.

A failed lift reserves the occurrence across which it was attempted. That occurrence cannot later be transported: transport across the same oriented edge would require continuing the very incidence that was stopped at its first failure. An operation from the other edge of the missing section can only annihilate it. Hence two failures cannot reserve occurrences with the same initial ancestor. This proves (A.99).

There is a parallel shadow statement. Although the full lift fails, the proposed duplication is ample in its two-layer conditioning. Its target section and its section overlap are ample one-concept extensions. The proof of Lemma A.51, applied inside that conditioning, gives one common new support T already present in the source section. Thus the failure reserves the directed section-shadow boundary occurrence labelled by T . Successful global duplications transport or annihilate these shadow occurrences by Lemma A.51, again without branching. After a failure, moving the reserved occurrence across the same oriented edge would require resuming the stopped incidence. The same backward-ancestor argument therefore gives an injection into the initial shadow-boundary occurrences.

For each edge, the two-layer shadow-intersection identity and ampleness give

$$\begin{aligned} |P_{i-1} \triangle P_i| &= |P_{i-1}| + |P_i| - 2|C_i| \\ &= |\text{Sh}(P_{i-1})| + |\text{Sh}(P_i)| - 2|\text{Sh}(C_i)| \\ &= |\text{Sh}(P_{i-1}) \triangle \text{Sh}(P_i)|. \end{aligned}$$

Finally apply Proposition A.25 to the initial tuple to obtain the last inequality in (A.100). Pairing this boundary ancestor with the residual genealogy gives the final assertion. $\square$

Proposition A.61 (The residual label of a failed lift is canonical). *Keep the notation of a failed adjacent-section lift across the path edge $a_{i-1}a_i$. Thus a repair word p is present in the source section, absent from the target section, and adjoining its missing copy is ample in the corresponding two-layer conditioning but not in the full current family $\widehat{H}$. Let T be the unique repair support acquired by the target section in the locally ample extension. Then*

$$L(p, i) := T \cup \{x_i\} \in \widehat{\Omega}. \quad (\text{A.101})$$

In particular, every failed lift has a canonical residual payment whose core part is exactly its transition coordinate. No choice of a transverse owner or of a support in a shadow difference is required.

Proof. Write P_s, P_t for the source and target sections and $C_i = P_s \cap P_t$. The locally ample extension makes both $P_t \cup \{p\}$ and $C_i \cup \{p\}$ ample one-concept extensions. The proof of Lemma A.51, which uses only this two-layer conditioning, gives one common new repair support T , with

$$T \in \text{Sh}(P_s) \setminus \text{Sh}(P_t), \quad T \notin \text{Sh}(C_i).$$

In the two-layer extension, the repair projection does not change and the section-shadow intersection gains exactly T . Hence the unique new strongly shattered support of that conditioning is $T \cup \{x_i\}$.

Suppose that $T \cup \{x_i\}$ were not shattered by $\widehat{H}$. The family obtained by adjoining the missing copy would then strongly shatter all $|\widehat{H}|$ supports of the ample family $\widehat{H}$, together with the new support $T \cup \{x_i\}$. Its number of strongly shattered supports would equal its number of concepts. Equality in the lower Sandwich inequality would make the full extension ample, contrary to the definition of the failed lift. Thus $T \cup \{x_i\} \in \text{Sh}(\widehat{H})$.

The support is not in the path-spine family: it has exact core part $\{x_i\}$, while its repair part T is absent from $\text{Sh}(C_i)$. It therefore belongs to $\widehat{\Omega}$, proving (A.101). $\square$

Proposition A.62 (Failed lifts inject into the original residual shadow). *Keep the dynamic two-sweep setting, and let $\mathcal{F}$ be the failed lifts. Give each failure its canonical support $L(p, i)$ from Proposition A.61. These supports are pairwise distinct over the complete history, and every one of them belongs to the original residual shadow Ω_0 . Consequently the canonical label map is an injection*

$$\mathcal{F} \hookrightarrow \Omega_0, \quad |\mathcal{F}| \leq |\Omega_0|. \quad (\text{A.102})$$

This removes the residual core moment, the word-boundary potential, and all multiplicity loss from (A.98).

Proof. Fix a repair support T and a path edge x_i . At any state in which $T \cup \{x_i\}$ is the canonical support of a failure, T lies in the shadow of exactly the source section across that edge. Section shadows only grow. Hence this orientation can never reverse: a later acquisition may erase the boundary, but cannot recreate it with the opposite orientation. The same oriented incidence is

stopped at its first failure and is not processed again. Therefore $T \cup \{x_i\}$ labels at most one failure over the whole history.

Suppose first that such a label already belongs to $\text{Sh}(H_0)$. At the later failure one has $T \notin \text{Sh}(C_i)$. Since carrier shadows only grow, this was also true initially. The label has no other core coordinate, so it does not belong to any other block of the initial path-spine family. Consequently it lies in Ω_0 .

It remains to exclude a canonical label absent from $\text{Sh}(H_0)$. Consider the first state in which $T \cup \{x_i\}$ is shattered. Every change before a failure is an ample one-concept extension, so this label is the unique new shattered support of one earlier successful duplication. Let v be the concept added at that birth step, and let x_k be the path coordinate across which it was duplicated. The original copy of v is an x_k -neighbour already present before the step. Lemma 2.10 says that the unique new support consists of all directions incident with v before its addition. Hence

$$x_k \in T \cup \{x_i\}.$$

The set T contains only repair coordinates, so $k = i$.

In the locally ample target-section extension, the repair directions incident with v are exactly T . The full $T \cup \{x_i\}$ -cube through v therefore makes the adjacent carrier C_i acquire T at this birth step. Thus $T \cup \{x_i\}$ enters the path-spine family at its first appearance and, because carrier shadows only grow, remains there forever. This contradicts Proposition A.61, which places the same support in the residual shadow at the later failure. Every canonical label must consequently belong to $\text{Sh}(H_0)$, and the first part of the proof puts it in Ω_0 . Together with pairwise distinctness, this proves (A.102). $\square$

Proposition A.63 (Gap edges reserve the fragmentation payment). *Keep the initial path-spine state and the dynamic two-sweep setting of Proposition A.62. Fix $T \in \mathcal{T}$, and write the components of its initial section-index set as*

$$I_T = [\ell_1, r_1] \dot{\cup} \cdots \dot{\cup} [\ell_{c_T}, r_{c_T}].$$

For $1 \leq h < c_T$, put

$$G_{T,h} = \{r_h + 1, \dots, \ell_{h+1}\}, \quad d_{T,h} = |G_{T,h}| = \ell_{h+1} - r_h,$$

and define

$$\Gamma_{T,h} = \{T \cup \{x_i\} : i \in G_{T,h}\}, \quad \Gamma_T = \bigcup_{h=1}^{c_T-1} \Gamma_{T,h}. \quad (\text{A.103})$$

Then

$$\Gamma_T \subseteq \Omega[T], \quad |\Gamma_T| = \sum_{h=1}^{c_T-1} d_{T,h} = \lambda_T + c_T - 1. \quad (\text{A.104})$$

Let $\mathcal{F}_T$ be the failed lifts whose acquired repair support is T . Trace their section-shadow boundaries back to the initial state as in Proposition A.60. Let $f_{T,h}$ be the number of failures descended from the two boundary occurrences facing the h -th internal gap, and put

$$b_T = |\{h : d_{T,h} = 2 \text{ and } f_{T,h} = 2\}|. \quad (\text{A.105})$$

Then $0 \leq f_{T,h} \leq 2$ and

$$|\mathcal{F}_T| + c_T - 1 \leq |\Omega[T]| + b_T. \quad (\text{A.106})$$

In particular, canonical failures and the $c_T - 1$ fragmentation generators admit a joint injection into $\Omega[T]$ unless some one-section gap has one failed descendant from each of its two boundary occurrences. No longer gap can obstruct this joint payment.

Proof. Fix an internal gap between a_{r_h} and $a_{\ell_{h+1}}$. The ample projected carrier Z_T is isometric and contains both endpoints. Every shortest path between them uses each of the directions

$$x_{r_h+1}, \dots, x_{\ell_{h+1}}$$

exactly once. Hence Z_T has an x_i -edge for every $i \in G_{T,h}$, so $\{x_i\} \in \text{Sh}(Z_T)$. The support conservation identity (7.62) gives $T \cup \{x_i\} \in \text{Sh}(H)$.

The two endpoints of the path edge $a_{i-1}a_i$ cannot both belong to I_T , because the edge lies between distinct components. The two-layer shadow-intersection identity therefore gives $T \notin \text{Sh}(C_i)$. Thus $T \cup \{x_i\}$ is absent from the path-spine family and belongs to $\Omega[T]$. Different gap edges have different core coordinates, which proves the disjoint inclusion in (A.104). Finally,

$$d_{T,h} = \ell_{h+1} - r_h = (\ell_{h+1} - r_h - 1) + 1;$$

summing proves the cardinality formula.

Every failure with acquired support T descends from a distinct initial directed T -boundary occurrence. A boundary occurrence facing an internal gap can move only through that gap: it either disappears at an endpoint, annihilates with the occurrence coming from the other side, or stops at its first failure. It never branches. Hence at most the two boundary occurrences of one gap contribute failures, proving $f_{T,h} \leq 2$. Failures descended from an outer boundary of I_T have canonical labels outside Γ_T ; these labels are pairwise distinct members of $\Omega[T]$ by Proposition A.62.

For each internal gap,

$$f_{T,h} + 1 \leq d_{T,h} + \mathbf{1}_{\{d_{T,h}=2, f_{T,h}=2\}}. \quad (\text{A.107})$$

Indeed $d_{T,h} \geq 2$, while $f_{T,h} \leq 2$; the right-hand correction is needed only when both inequalities are equalities. Sum (A.107) over the $c_T - 1$ gaps, add the failures descended from outer boundaries, and use the disjoint reserved family Γ_T together with their canonical labels. This gives (A.106). $\square$

Proposition A.64 (Tight support gaps reserve distinct core squares). *Keep the initial path-spine state and the dynamic two-sweep setting of Proposition A.63. Let $\mathcal{H}$ be the set of pairs (T, h) counted by b_T . Thus the h -th gap of I_T has one missing section and both of its directed boundary descendants end in failure. If that missing section is a_j , put*

$$Q_{T,h} = \{x_j, x_{j+1}\}. \quad (\text{A.108})$$

Then the map

$$(T, h) \longmapsto Q_{T,h} \quad (\text{A.109})$$

is an injection into the original residual shadow Ω_0 . Its image is disjoint from every exact-repair block

$$\Omega_0[T] = \{S \in \Omega_0 : S \cap R = T\},$$

$T \in \mathcal{T}$. Consequently all canonical failures and all support fragmentation generators fit simultaneously in the original residual shadow:

$$|\mathcal{F}| + \sum_{T \in \mathcal{T}} (c_T - 1) \leq |\Omega_0|. \quad (\text{A.110})$$

This is independent of the repair positions and antipodal blockers of the failed lifts.

Proof. Fix $(T, h) \in \mathcal{H}$. Write the two components adjacent to its gap as

$$[\ell_h, r_h] \quad \text{and} \quad [\ell_{h+1}, r_{h+1}].$$

Since the gap has edge length two, $\ell_{h+1} = r_h + 2$. Put $j = r_h + 1$, so that

$$a_{j-1}, a_{j+1} \in Z_T, \quad a_j \notin Z_T$$

by (7.58). The endpoints have Hamming distance two. The two possible middle vertices of a two-edge cube path between them are a_j and

$$q_j = a_j^{\{x_j, x_{j+1}\}}.$$

The projected carrier Z_T is ample and therefore isometric. Since it omits a_j , it must contain q_j .

Membership $q_j \in Z_T$ means that some full positioned T -cube of the original family has core word q_j ; in particular, some concept of H_0 projects to q_j . On the coordinate pair $\{x_j, x_{j+1}\}$, the three path vertices a_{j-1}, a_j, a_{j+1} realize three consecutive traces, while q_j realizes the fourth. The path core A occurs as a repair fibre of H_0 , so these four traces all occur in H_0 . Hence

$$Q_{T,h} = \{x_j, x_{j+1}\} \in \text{Sh}(H_0).$$

This pure two-core support does not belong to $\text{Sh}(W)$, and every other block of the path-spine family $\mathcal{B}$ has exact core part $\{x_i\}$. Therefore $Q_{T,h} \in \Omega_0$.

It remains to prove distinctness. A tight gap with missing section a_j initially has its two directed T -boundaries on the incidences directed into P_j , one across x_j and one across x_{j+1} . Transporting either boundary would make P_j acquire T . Since both neighbouring sections already shatter T , that acquisition would annihilate both boundaries; neither could then have a failed descendant. Thus, when both descendants fail, they fail on precisely those two oriented incidences. The two-sweep protocol processes each oriented incidence at most once and stops it at its first failure.

It follows that two pairs in $\mathcal{H}$ with the same missing index j use the same two failed lifts. A failed lift has one unique acquired repair support, so the two pairs have the same support T , and then they are the same gap. Distinct pairs in $\mathcal{H}$ consequently have distinct missing indices. Since the edge directions $x_1, \dots, x_n$ of the isometric path A are pairwise distinct, their labels in (A.108) are distinct. This proves (A.109).

Every acquired section support is nonbaseline and hence belongs to $\mathcal{T}$. The exact-repair blocks $\Omega_0[T]$ are pairwise disjoint, and every member of them has nonempty repair part T , whereas the labels in (A.109) have empty repair part. Summing (A.106) over $T \in \mathcal{T}$, and then paying its total correction $\sum_T b_T = |\mathcal{H}|$ with these distinct pure core-square labels, gives (A.110). $\square$

Proposition A.65 (The support-interval complex separates occurrence and failure payments). *Keep the initial path-spine state and the two-sweep failure family $\mathcal{F}$. Put*

$$\mathfrak{D} = \{(j, T) : T \in \mathcal{T}, j \in I_T\}.$$

Let $\mathcal{D}$ be any finite demand family admitting an injection $\eta : \mathcal{D} \hookrightarrow \mathfrak{D}$. Then there is an injection

$$\mathcal{D} \dot{\cup} \mathcal{F} \hookrightarrow \Xi \dot{\cup} \Omega_0. \tag{A.111}$$

In particular,

$$|\mathcal{D}| + |\mathcal{F}| \leq |\Xi| + |\Omega_0| = \Delta(H; A, R). \tag{A.112}$$

Proof. For $T \in \mathcal{T}$, let d_T be the number of demands whose η -label has second coordinate T . Since η is injective,

$$d_T \leq m_T.$$

The exact-repair part of Ξ has precisely $m_T - c_T + 1$ members: the pure repair support T , which is the chosen root column of ∂_T , and the supports $T \cup \{x_i\}$ on the internal edges of the components of I_T . Assign

$$\min\{d_T, m_T - c_T + 1\}$$

of the T -labelled demands injectively to these supports. The number left is at most

$$\max\{0, d_T - (m_T - c_T + 1)\} \leq c_T - 1.$$

Proposition A.64 gives one joint injection of all failures and all $\sum_T (c_T - 1)$ formal fragmentation generators into Ω_0 . Use the generators belonging to T for the remaining T -labelled demands. Different exact-repair parts give disjoint supports in Ξ , and the fragmentation injection is disjoint from the failure labels. This proves (A.111). Finally apply (7.54) to obtain (A.112). $\square$

Corollary A.66 (Neutral forests and failed lifts share one exact budget). *For each interior section P_j , let κ_j be the number of weak components of its extended neutral-transfer digraph from Proposition A.29. Then*

$$\sum_{j=1}^{n-1} \kappa_j + |\mathcal{F}| \leq \Delta(H; A, R). \quad (\text{A.113})$$

This is a joint injection statement, not the sum of the separate neutral and failure estimates.

Proof. Because P_j is ample and contains the two fixed repair antipodes, its baseline shadow is exactly $\mathcal{L}_R = \{\emptyset\} \cup \binom{R}{1}$, of order $r + 1$. Consequently

$$|\{T \in \mathcal{T} : j \in I_T\}| = |P_j| - (r + 1).$$

Proposition A.29 gives $\kappa_j \leq |P_j| - (r + 1)$. For each j , label its neutral weak components injectively by distinct supports $T \in \mathcal{T} \cap \text{Sh}(P_j)$. Section indices distinguish equal supports chosen in different sections, so these labels give an injection of all neutral components into $\mathfrak{D}$. Apply Proposition A.65. $\square$

Corollary A.67 (The joint path budget has an anchored normal form). *Keep the hypotheses and notation of Proposition A.65. For every demand $d \in \mathcal{D}$, write $\eta(d) = (j, T)$, and choose a positioned full T -cube in the section P_j . For every failed lift, use its canonical support $T \cup \{x_i\}$ from Proposition A.61 and choose a positioned full cube with that support in the original family. Then these anchored demands admit one of the following two alternatives:*

1. *they acquire pairwise distinct coface labels in $\text{Sh}(H) \setminus \Sigma_0$;*
2. *at least two of them ascend to the same support and the same positioned cube of H .*

This is a normal form for the chosen occurrence labels. The positioned cube chosen for a neutral demand need not contain the puncture from which that demand arose. Consequently this statement alone does not localize the geometry of a neutral demand; the provenance-preserving statement for the closed neutral components is given below. If $|H| < 64$, the support L of such a packet satisfies

$$2 \leq |L| \leq 4. \quad (\text{A.114})$$

Thus the internal face flags of every unresolved packet live in one fixed four-cube, independently of the path length and repair dimension.

Proof. Every $T \in \mathcal{T}$ has at least two repair coordinates. Ampleness of P_j makes T strongly shattered there, so the required positioned T -cube exists and lifts to H over the core word a_j . The acquired repair support of a failed lift also belongs to $\mathcal{T}$. Its canonical support is a member of the original shattered complex by Proposition A.62; ampleness of the original family supplies the chosen positioned cube.

Apply Proposition 7.21 simultaneously to this multiset of anchored faces. Every initial support contains at least two repair coordinates, and coface ascent never removes a coordinate. Hence every final label lies outside $\Sigma_0 = \text{Sh}(A) * \mathcal{L}_R$, whose members have at most one repair coordinate. The two alternatives in the anchored-coface normal form are therefore exactly the alternatives displayed above.

Finally, every final support still contains the initial repair support T , so its order is at least two. Extend L to a facet M of $\text{Sh}(H)$. The facet-excess inequality, Theorem 7.144, gives

$$|H| \geq 2^{|M|+1} \geq 2^{|L|+1}.$$

If $|H| < 64$, this excludes $|L| \geq 5$ and proves (A.114). $\square$

Corollary A.68 (Neutral cycles and canonical failure labels have an anchored normal form). *Suppose that $|H| < 64$. For every interior section P_j , let $\mathcal{Z}_j$ be the set of directed cycles in its neutral-transfer digraph. Keep also the failed-lift family $\mathcal{F}$ from the two-sweep construction. One can associate with every member of*

$$\mathcal{Z} = \bigcup_{j=1}^{n-1} \mathcal{Z}_j \quad \text{and with every member of } \mathcal{F}$$

a positioned full face of H so that the associated faces are pairwise distinct and every one of their supports contains at least two repair coordinates. The face of a neutral cycle lies in its actual common maximal repair cube. The support of a failure face is its canonical residual label; the position of that face is an auxiliary realization and need not lie at the failed path edge. After anchored coface ascent, exactly one of the following holds:

1. *all members of $\mathcal{Z} \dot{\cup} \mathcal{F}$ have pairwise distinct labels in $\text{Sh}(H) \setminus \Sigma_0$;*
2. *at least two members occupy one positioned cube whose support L satisfies*

$$3 \leq |L| \leq 4.$$

Thus a neutral-cycle/canonical-failure packet below order 64 has rank three or four. In particular, the rank-two packet in the occurrence-label normal form is an artefact of using arbitrary section occurrences and is not an active obstruction.

Proof. Fix a positioned maximal repair cube Q of a section P_j , let T be its support, and put $t = |T|$. Proposition A.28 gives $t \geq 2$. We first assign the cycles based in Q distinct owner words. Each cycle contains at least two distinct repair words, different cycles are disjoint as boundary occurrences, and a repair word has at most its left and right occurrence. Hence every collection of k cycles collectively contains at least k distinct repair words. Hall's theorem therefore assigns to each cycle a distinct repair word occurring on that cycle.

We next match these owner words to distinct faces of Q of dimension at least two, with every word lying on its assigned face. For $t \geq 4$, use the incidence graph between the vertices and the two-faces of Q . Its left degree is $\binom{t}{2}$, its right degree is four, and $\binom{t}{2} \geq 4$. Counting the incidence

edges from an arbitrary vertex set proves Hall's condition. For $t = 3$, use the six two-faces together with Q itself. A set of at most four vertices has at least four incident faces, whereas a set of at least five vertices meets every two-face and hence has all seven faces as neighbours. There cannot be eight cycles based in Q : equality would use both boundary occurrences of every vertex of Q . All these vertices are corners whose unique maximal cube is Q , so connectedness would force $P_j = Q$; but each adjacent carrier is a nonempty subset of P_j , contradicting the use of the corresponding boundary occurrence at every vertex of Q . Thus Hall's condition again applies to the owner words. Finally, if $t = 2$, Corollary A.40 gives at most one cycle, which we assign to Q itself. Lift every assigned face to H over the core word a_j .

These neutral faces are pairwise distinct. Within one Q , they are distinct by the face matching. Every owner word is a corner of P_j whose unique maximal cube is Q . Hence a face through that owner cannot also lie in a different positioned maximal cube of the same section. Faces chosen in different sections have different core anchors.

For a failed lift, use a positioned full cube on its canonical support $T \cup \{x_i\}$. Proposition A.62 makes these supports pairwise distinct. They cannot equal a neutral support, since they contain one core coordinate, whereas every neutral support chosen above is contained in R . This cube realizes the canonical support for the coface matching; it is not asserted to contain the failed puncture. Thus all associated faces are pairwise distinct, as claimed.

Apply Proposition 7.21 to these faces. Every final support contains its initial support and therefore has at least two repair coordinates, so it lies outside $\Sigma_0 = \text{Sh}(A) * \mathcal{L}_R$. Suppose that a terminal packet had support L of order two. The initial support K of each member of the packet satisfies $K \subseteq L$ and $|K| \geq 2$, hence $K = L$. The containing positioned L -cube is then the initial face itself. Two members of the packet would therefore have had the same initial anchored face, contrary to the preceding paragraph. Hence every packet has rank at least three. The upper bound four follows from the facet-excess argument in (A.114). $\square$

Lemma A.69 (Root-flexible cube-support matching). *Let $F \in \mathcal{D}_{\text{amp}}$, and let $r \in F$. There is a bijection*

$$\mu_r : F \longrightarrow \text{Sh}(F) \tag{A.115}$$

such that

$$\mu_r(r) = \emptyset, \quad Q_{\mu_r(y)}(y) \subseteq F \quad (y \in F).$$

The bijection need not be induced by a corner peeling ending at r .

Proof. Take a corner peeling

$$F = F_1 \supset F_2 \supset \cdots \supset F_s = \{y_s\}, \quad F_{i+1} = F_i \setminus \{y_i\}.$$

For $i < s$, let S_i be the support of the unique maximal cube through y_i in F_i , and put $S_s = \emptyset$. Corner deletion removes exactly the shattered support S_i . Thus

$$M = \{(y_i, S_i) : 1 \leq i \leq s\}$$

is a perfect matching in the bipartite incidence graph whose left side is F , whose right side is $\text{Sh}(F)$, and where yS is an edge precisely when $Q_S(y) \subseteq F$.

Delete the right vertex $\emptyset$ and its incident matching edge. The resulting matching saturates all nonempty shattered supports and leaves y_s unmatched. We claim that every left vertex is reachable from y_s by an alternating path. This follows by descending induction on the peeling index. Suppose

that every y_j with $j > i$ is reachable. The family F_i is connected and has at least two concepts, so the unique maximal cube

$$C_i = Q_{S_i}(y_i) \subseteq F_i$$

contains a concept $z_i \neq y_i$. Its peeling index is larger than i , so it is reachable by induction. The edge $z_i S_i$ is not in the matching, whereas $S_i y_i$ is. Appending these two edges reaches y_i , proving the claim.

In particular, there is an alternating path from y_s to the prescribed concept r . Switching the matching along this path leaves r unmatched and still saturates every nonempty support. Add the edge $(r, \emptyset)$. The resulting perfect matching is the bijection μ_r in (A.115). $\square$

Proposition A.70 (Carriers ascend strictly to facets). *Let $H \subseteq Q_E$ have minimum cardinality among all nonempty cornerless ample families, and put $\Sigma = \text{Sh}(H)$. Let $K \in \Sigma$ be nonempty. Every proper collection of positioned K -cubes can be matched to pairwise distinct strict cofaces of K , with each cube contained in a positioned cube on its assigned coface.*

Consequently, let $\mathcal{A}$ be a finite multiset of anchored faces whose supports have order at least two. By repeatedly replacing a face with a containing positioned coface, one reaches one of the following two alternatives:

1. *at least two members occupy the same positioned cube;*
2. *all members have pairwise distinct support labels, and every label is a facet of Σ .*

Proof. Fix K . The full carrier

$$\mathcal{K}_K(H) = \{y \in Q_{E \setminus K} : Q_K(y) \subseteq H\}$$

is ample by Lemma 7.19. Its concepts are the positioned K -cubes of H , and the disjoint positioned cubes give

$$2^{|K|} |\mathcal{K}_K(H)| \leq |H|.$$

Since $K \neq \emptyset$, the carrier is smaller than H , so minimum-cardinality puts it in $\mathcal{D}_{\text{amp}}$.

Let $\mathcal{Q}$ be a proper collection of its concepts and choose $y_* \in \mathcal{K}_K(H) \setminus \mathcal{Q}$. Apply Lemma A.69 to the carrier with prescribed root y_* . For every $y \in \mathcal{Q}$, its matched support S_y is nonempty, the supports S_y are pairwise distinct, and the full S_y -cube of carrier anchors through y lifts to a full $K \dot{\cup} S_y$ -cube of H containing $Q_K(y)$. This gives the required matching to strict cofaces.

We now iterate this strict matching. At a support K , first stop if two members occupy the same positioned K -cube. Otherwise the members labelled by K occupy a set $\mathcal{Q}$ of distinct positions. If $\mathcal{Q}$ is proper, move all of them to distinct strict cofaces by the first part. If every position is occupied, use the bijection in Proposition 7.21: at most one member keeps K , and all the others move strictly upward. Unless K is already a facet, the carrier has at least two concepts, so this operation strictly increases the sum of the support orders. A remaining single member at K then occupies a proper set of positions and moves strictly at the next step.

The sum of the support orders is bounded by $|E||\mathcal{A}|$. Hence the process terminates. If it does not stop with a same-position packet, every terminal label is a facet; labels are pairwise distinct because a repeated terminal label at distinct positions would still move, whereas repetition at one position is the first alternative. This proves the dichotomy. $\square$

Corollary A.71 (Packet-free saturation exposes a corner). *Keep the path-core notation, and write*

$$\Sigma_0 = \text{Sh}(A) * (\{\emptyset\} \cup \binom{R}{1}), \quad \Delta = |\text{Sh}(H) \setminus \Sigma_0|.$$

Suppose that a saturated anchored demand family consists of exactly Δ faces, every initial support contains at least two repair coordinates, and strict facet ascent produces no same-position packet. Then H has a corner. In particular, the packet-free equality branch of the path-wide joint budget is impossible.

Proof. Proposition A.70 assigns the Δ demands pairwise distinct facet labels. Every label remains outside Σ_0 , because coface ascent preserves the two initial repair coordinates. There are exactly Δ supports outside Σ_0 , so the labels exhaust $\Sigma \setminus \Sigma_0$. Consequently every support outside the baseline is a facet and contains at least two repair coordinates.

Suppose that $L \in \Sigma \setminus \Sigma_0$ has order at least three. Choose two repair coordinates $s, t \in L \cap R$. Downward closure gives $\{s, t\} \in \Sigma \setminus \Sigma_0$. Both $\{s, t\}$ and L would then be facets, contrary to $\{s, t\} \subsetneq L$. Hence every nonbaseline support is a two-element subset of R .

The maximum VC-one core satisfies $\text{Sh}(A) = \{\emptyset\} \cup \binom{X}{1}$. Thus every member of Σ_0 also has order at most two: besides the empty set and the singletons, its only members are the mixed pairs $\{x, t\}$, $x \in X$, $t \in R$. We have proved that $\text{VCdim}(H) \leq 2$. Apply Lemma A.16 to the nested ample pair $\{h\} \subseteq H$, for any $h \in H$. It supplies a corner peeling of H , and in particular a corner, contradicting the choice of H . $\square$

Lemma A.72 (Rank-three packets have two local Hall obstructions). *Let Q_L be a positioned full cube of H , where $|L| = 3$, and let $\mathcal{A}$ be a family of pairwise distinct full faces of Q_L , each with support $K \subseteq L$ of order at least two. Put*

$$c = \begin{cases} 1, & Q_L \in \mathcal{A}, \\ 0, & Q_L \notin \mathcal{A}, \end{cases}$$

and let d be the number of two-subsets $K \in \binom{L}{2}$ for which both opposite positioned K -faces of Q_L belong to $\mathcal{A}$. The faces in $\mathcal{A}$ can be assigned pairwise distinct supports, using for a K -face only K or L , if and only if

$$c + d \leq 1. \tag{A.116}$$

Consequently every packet that cannot be separated inside its terminal three-cube contains one of exactly two minimal configurations:

1. *the full L -cube together with the two opposite K -faces for one $K \in \binom{L}{2}$;*
2. *the two opposite K -faces and the two opposite K' -faces for distinct $K, K' \in \binom{L}{2}$.*

Proof. For each two-subset K , at most two members of $\mathcal{A}$ have support K , namely the two opposite K -faces of Q_L . Assign one such face, when present, the label K . Every second face with that support must use L , and Q_L itself must also use L . Thus the displayed assignment succeeds exactly when the number $c + d$ of faces forced to use L is at most one.

The same count is also necessary. A full L -face has only the label L among the permitted supports. Two opposite K -faces have the common two-element label set $\{K, L\}$, so at least one of them must use L . The families forced to use L are disjoint, and only one face can receive that label. This proves (A.116). If the inequality fails, then either $c = 1$ and some direction is doubled, which contains the first configuration, or $c = 0$ and at least two directions are doubled, which contains the second. $\square$

Corollary A.73 (Core profiles of rank-three packets). *Apply Lemma A.72 to the initial faces of a packet from Corollary A.68.*

1. If $L \subseteq R$, every member is a neutral-cycle face, and the two configurations in Lemma A.72 are the only minimal local obstructions.
2. If $L = T \dot{\cup} \{x_i\}$, where $T \in \binom{R}{2}$, every neutral-cycle face has support T , every failure face has support L , and at most one failure occurs. The only minimal local obstruction is therefore one canonical failure representative together with the two opposite T -faces of its positioned L -cube.

These cases exhaust the rank-three packets.

Proof. Every initial face contains at least two repair coordinates. A neutral face has support contained in R , while a failure face has canonical support consisting of at least two repair coordinates and exactly one core coordinate. If $L \subseteq R$, no failure support can be contained in L , proving the first assertion.

Otherwise the three-element support L has exactly two repair coordinates and one core coordinate, so write it as displayed. A neutral initial support contained in L must then equal T , and a failure initial support contained in L must equal L . Canonical failure supports are pairwise distinct, so there is at most one such failure. The second minimal configuration of Lemma A.72 needs two different doubled two-supports and is impossible. Its first configuration has precisely the form asserted. No third core profile is possible because every initial support has at least two repair coordinates. $\square$

Corollary A.74 (Mixed rank-three packets separate). *In the anchored normal form of Corollary A.68, every rank-three packet whose support contains a core coordinate can be assigned pairwise distinct labels using supports carried by its terminal cube. Consequently every locally deficient rank-three packet has pure repair support and contains one of the two neutral-cycle configurations in Lemma A.72.*

Proof. By Corollary A.73, a locally deficient mixed packet would have support

$$L = T \dot{\cup} \{x_i\}, \quad T \in \binom{R}{2},$$

and would consist of one failure representative of support L together with the two opposite T -faces of its positioned L -cube. The failure occurred across the path edge of direction x_i , and Proposition A.61 gives

$$T \notin \text{Sh}(C_i). \tag{A.117}$$

Each of the two T -faces is a neutral-cycle face and hence, by the construction in Corollary A.68, lies over a path-core word. Write those words as a_h and a_k . The faces are opposite in one positioned L -cube, so their core anchors differ exactly in x_i . A coordinate-simple cube path flips every path direction once; two of its vertices differ in only x_i precisely when they are the two endpoints of its x_i -edge. Therefore $\{a_h, a_k\} = \{a_{i-1}, a_i\}$.

The two opposite faces of a positioned L -cube have the same repair anchor outside T . Thus one positioned full T -cube belongs to both sections P_{i-1} and P_i , and hence to their intersection C_i . This gives $T \in \text{Sh}(C_i)$, contradicting (A.117). The mixed minimal obstruction is impossible. Lemma A.72 now assigns distinct local labels to every mixed packet, while Corollary A.73 leaves only the two stated pure-repair configurations. $\square$

Corollary A.75 (A pure rank-three packet ascends or owns a maximal three-cube). *Let a locally deficient rank-three packet from Corollary A.74 have pure repair support L . All of its initial faces lie in one path section P_j , and its terminal positioned L -cube Q_L is contained in one common positioned*

maximal repair cube Q of P_j . If the support of Q properly contains L , the packet has a common strict coface of rank four. Consequently a deficient alternating component can stop at rank three only when $Q = Q_L$ is itself a maximal repair three-cube of P_j .

Proof. Every member of a pure-support packet is a neutral-cycle face. Such a face was lifted over the path word of its owner section. Since all faces belong to the same positioned repair cube Q_L , their core anchors agree, and they therefore lie in one section P_j .

Each chosen neutral face contains its assigned owner word p , which is a corner of P_j . Its unique positioned maximal cube $Q(p)$ contains that face. It also contains the terminal cube Q_L : both are cubes through p , and every cube through a corner is contained in its unique maximal cube. If p and p' are owners of two packet faces, then $Q_L \subseteq Q(p) \cap Q(p')$. In particular $p \in Q(p')$, so uniqueness of the maximal cube through p gives $Q(p) = Q(p')$. This proves the common-owner assertion.

Write M for the support of Q . If $s \in M \setminus L$, the positioned $L \cup \{s\}$ -face of Q containing Q_L is a full rank-four cube and contains every member of the packet. It supplies the claimed common strict coface. If no rank-four ascent is available, then $M = L$; maximality of Q gives $Q = Q_L$, as required. $\square$

Proposition A.76 (An overloaded maximal repair three-cube pays an exterior tax). *Let a positioned maximal repair three-cube Q of an interior section P_j be the common owner of c directed neutral cycles. Then*

$$\Delta(H; A, R) \geq 4 + \max\{0, 2c - 8\}. \quad (\text{A.118})$$

In the rank-three motif branch $c \geq 3$, and

$$\begin{aligned} c = 3 &\implies \Delta(H; A, R) \geq c + 1, \\ c = 4 &\implies \Delta(H; A, R) \geq c + 1, \\ c \geq 5 &\implies \Delta(H; A, R) \geq 2c - 4 \geq c + 1. \end{aligned}$$

Thus every maximal-three-cube rank-three motif supplies at least one local support unit beyond its cycle population.

Proof. The full three-cube Q contributes its three two-subsets and its three-element support beyond the empty support and the repair singletons. Hence

$$|P_j| - (r + 1) \geq 4.$$

The c cycles are disjoint as vertices of the left–right occurrence digraph and have length at least two, so they use at least $2c$ boundary occurrences. The cube Q has eight repair words, and each word has at most its left and right occurrence. Consequently at least $\max\{0, 2c - 8\}$ repair words have both occurrences on these cycles. Every such word p is a corner of P_j whose unique maximal repair cube is Q , and

$$p \notin C_j \cup C_{j+1}.$$

Consider the concept (p, a_j) of H . It has no core neighbour in either adjacent path section. If it had no core neighbour over a word outside the path, then every cube of H through it would lie in the section P_j and hence in Q ; it would be a corner of H , contrary to the standing hypothesis. Thus every doubled boundary word forces a concept of H over a core word outside the path. Distinct repair words force distinct concepts, and therefore

$$V \geq \max\{0, 2c - 8\}.$$

There is one extra point when $c = 4$. If none of the eight minimum cycle occurrences shared an underlying repair word, they would use all eight vertices of Q . Every vertex of Q would then be a corner of P_j whose unique maximal cube is Q . No repair edge would leave Q , and connectedness would force $P_j = Q$. Since P_j contains the two fixed repair antipodes, this would force $r = 3$ and make Q the full repair cube. Both antipodes belong to each adjacent carrier, so neither has a left or right boundary occurrence, contradicting the use of all eight words. Therefore some word occurs on both sides, and the preceding ambient-corner argument gives

$$c = 4 \implies V \geq 1. \quad (\text{A.119})$$

Substitute this and the section bound into the exact path-section partition

$$\Delta(H; A, R) = V + \sum_{h=0}^n (|P_h| - (r+1))$$

to obtain (A.118). If $c = 3$, the initial cube contribution four is already $c + 1$. If $c = 4$, combine the four cube supports with (A.119). If $c \geq 5$, the remaining inequalities are immediate. Either local Hall motif uses at least three cycles, so all possible cycle populations have at least one unit of strict local tax. $\square$

Corollary A.77 (A saturated maximal-three-cube motif has three two-cycles). *Suppose that the neutral-forest bound is tight in the section P_j :*

$$\kappa_j = |P_j| - (r+1).$$

If a positioned maximal repair three-cube Q of P_j owns a locally deficient rank-three packet, then it owns exactly three directed neutral cycles. They are directed two-cycles and use six distinct repair words of Q .

Proof. Corollary A.30 makes every directed neutral cycle in P_j a two-cycle. In the notation of (A.48), one endpoint belongs to $C_j \setminus C_{j+1}$ and the other to $C_{j+1} \setminus C_j$. Each endpoint consequently has only one boundary occurrence, so two distinct two-cycles cannot share an underlying repair word. Since Q has eight vertices, it owns at most four such cycles.

A locally deficient rank-three motif uses at least three cycles. If there were four, their eight distinct words would be all vertices of Q . Every vertex would then be a corner of P_j with unique maximal cube Q , so connectedness would force $P_j = Q$. As in the proof of Proposition A.76, the two fixed repair antipodes would then lie in both adjacent carriers and could not be boundary occurrences. This is a contradiction. Hence exactly three cycles occur, and they use six distinct words. $\square$

Lemma A.78 (Three opposite punctures force a short common pair). *Let Q be a three-cube and let $D_-, D_+ \subseteq Q$ be ample families such that*

$$|D_-| = |D_+| = 5, \quad D_- \cup D_+ = Q, \quad |D_- \cap D_+| = 2. \quad (\text{A.120})$$

Suppose that the three words of $D_- \setminus D_+$ can be paired with the three words of $D_+ \setminus D_-$ so that every pair (x, y) , with $S = \text{Diff}(x, y)$, satisfies

$$\{x^U : U \subsetneq S\} \subseteq D_-, \quad \{y^U : U \subsetneq S\} \subseteq D_+. \quad (\text{A.121})$$

Then, after a cube automorphism and possibly interchanging the two sides, there are directions a, b, e such that

$$D_- = \{0, a, b, ab, e\}, \quad D_+ = \{e, ae, be, abe, z\}, \quad z \in \{0, a, b\}. \quad (\text{A.122})$$

Here a word denotes the set of directions in which it differs from 0. In particular, the two common words are e, z , at distance one or two. Consequently, if the two common words are opposite in Q , at most two disjoint pairs can satisfy (A.121).

Proof. An ample five-vertex subfamily of a three-cube uses all three coordinates: five vertices cannot lie in one facet. It therefore shatters the empty set, all three singletons, and exactly one pair. Thus it is a full square with one vertex attached to a vertex of that square. Normalize D_- as

$$D_- = \{0, a, b, ab, e\},$$

where $\{0, a, b, ab\}$ is its square and e is attached at 0. Since the two families cover Q , the three vertices outside D_- , namely ae, be, abe , belong to D_+ . The remaining two vertices of D_+ form the intersection $I = D_- \cap D_+$.

There are only ten possible pairs $I \in \binom{D_-}{2}$, and the ample condition on D_+ has a direct square-pendant interpretation. The complete list is

I	D_+ is ample	a three-pair matching satisfies (A.121)
$\{0, a\}, \{0, b\}, \{0, ab\}, \{a, b\}$	no	no
$\{0, e\}$	yes	yes
$\{a, ab\}, \{b, ab\}$	yes	no
$\{a, e\}, \{b, e\}$	yes	yes
$\{ab, e\}$	yes	no.

For completeness, the three successful matchings are respectively

$$\begin{aligned} I = \{0, e\} : \quad & a \leftrightarrow ae, \quad b \leftrightarrow be, \quad ab \leftrightarrow abe, \\ I = \{a, e\} : \quad & 0 \leftrightarrow ae, \quad b \leftrightarrow be, \quad ab \leftrightarrow abe, \\ I = \{b, e\} : \quad & 0 \leftrightarrow be, \quad a \leftrightarrow ae, \quad ab \leftrightarrow abe. \end{aligned}$$

Every failed ample row is checked by one proper flip: in each of $I = \{a, ab\}, \{b, ab\}, \{ab, e\}$, the exclusive word 0 has no admissible partner in $\{ae, be, abe\}$. For example, pairing it with ae requires both proper intermediate words a, e on the two prescribed sides, and the analogous obstruction holds for be and abe . In each of the four nonample rows, the displayed five-set shatters at least two of the three coordinate pairs, in addition to the empty set and the three singletons, so its shadow has order at least six. This proves both the list and (A.122). In the last row the common words ab, e are opposite, which proves the final assertion. $\square$

Proposition A.79 (Three neutral two-cycles force six section units). *Let Q be a positioned maximal repair three-cube in an interior section P_j . If Q owns three directed neutral two-cycles whose six underlying repair words are distinct, then*

$$|P_j| - (r + 1) \geq 6. \quad (\text{A.123})$$

In particular, neither the four nonbaseline supports of Q nor those four supports together with one additional section support can realize this three-cycle profile.

Proof. Write T for the support of Q , so $|T| = 3$, and suppose first for a contradiction that

$$|P_j| - (r + 1) = 4. \quad (\text{A.124})$$

The section P_j contains the two fixed repair antipodes. It therefore shatters the empty set and all r repair singletons. The full cube Q supplies the three pairs in $\binom{T}{2}$ and T itself. These are four further supports, so ampleness and (A.124) give the exact identity

$$\text{Sh}(P_j) = \{\emptyset\} \cup \binom{R}{1} \cup \binom{T}{2} \cup \{T\}. \quad (\text{A.125})$$

We first extract the graph forced by this equality. For every $s \in R \setminus T$, no square of P_j uses s . Hence every s -edge is a bridge by Lemma 5.10. There is exactly one such edge. Indeed, there is at least one because the repair antipodes differ in s . If there were two, paths between their same-side endpoints inside the two convex s -semicubes would form a cycle containing both edges, contradicting that they are bridges.

Thus P_j has exactly $r - 3$ edges whose directions lie outside T . On the other hand, (A.124) gives

$$|P_j \setminus Q| = (r + 5) - 8 = r - 3.$$

Deleting the $r - 3$ bridge edges produces $r - 2$ connected components. There are only the cube Q and $r - 3$ vertices outside it, so equality forces every outside component to be a single vertex. Consequently the one-inclusion graph of P_j is obtained from Q by attaching a tree of $r - 3$ bridge edges and no other edges.

Every one of these bridges separates the two repair antipodes, since they have opposite values in its direction. After contracting Q , every edge of the attachment tree therefore lies on the path between the two antipodes. The tree is that path. In P_j , it consists of two possibly empty pendant paths attached to vertices $u, v \in Q$. A geodesic between the repair antipodes uses all $r - 3$ bridge directions and has total length r ; hence its segment in Q has length three. Thus u and v are opposite vertices of Q .

Both adjacent carriers C_j and C_{j+1} are connected and contain the repair antipodes. They must therefore contain both pendant paths and the vertices u, v . The endpoints of each neutral two-cycle belong to opposite exclusive carrier sides by (A.48). In particular, none of the six cycle words is u or v . They are distinct by hypothesis, so they are precisely the other six vertices of Q , three on each exclusive side. Put

$$D_- = C_j \cap Q, \quad D_+ = C_{j+1} \cap Q.$$

Repeated coordinate restrictions preserve ampleness, and therefore D_- and D_+ are ample five-vertex subfamilies of Q satisfying

$$D_- \cup D_+ = Q, \quad D_- \cap D_+ = \{u, v\}. \quad (\text{A.126})$$

The two words of every neutral two-cycle satisfy (A.121) by Proposition A.28. Since u, v are opposite, the final assertion of Lemma A.78 permits at most two such disjoint pairs, contradicting the three cycles. Thus (A.124) is impossible.

It remains to exclude section excess five. In that case the shadow in (A.125) has exactly one further member S . Downward closure forces S to be a two-set not contained in T . Moreover P_j contains exactly one positioned S -square: the full S -cube carrier is ample, and its shadow is the link of S , which here consists only of the empty set.

Every direction outside $T \cup S$ belongs to no square. Its unique edge is a bridge, by the argument above, and both adjacent carriers contain that bridge because they contain the repair antipodes.

Contract all these bridge directions. Contraction preserves ampleness, the full cube Q , the two adjacent carriers, and the six neutral-cycle words. None of those words is incident with a contracted bridge, since it is a corner whose unique maximal cube is Q . We obtain one of two reduced configurations.

Suppose first that $S = \{t, e\}$, where $t \in T$ and $e \notin T$. The reduced family has ten vertices: the eight vertices of Q and two others. Its unique S -square must therefore share a t -edge with Q , and its other two vertices are precisely the vertices outside Q . Exactly one repair antipode u belongs to Q . The other antipode belongs to the opposite edge of this square, so the shared t -edge does not contain u : its two fixed $T \setminus \{t\}$ coordinates are opposite to those of u . The two vertices of the shared edge lie in both Q and the S -square and hence are not corners. The word u lies in both adjacent carriers. Thus at most $8 - 2 - 1 = 5$ vertices of Q can be distinct exclusive carrier corners, whereas the three cycles require six. This is impossible.

Suppose next that $S = \{e, f\} \subseteq R \setminus T$. The reduced family has eleven vertices. Its unique S -square shares one vertex v with Q and contributes the other three; otherwise the order would be at least twelve. The two repair antipodes show that the vertex $u \in Q$ belonging to one antipode and the attachment vertex v are opposite in Q . The square is the only connection from Q to the other repair antipode. Hence both adjacent carriers contain u and v . The six distinct cycle words are consequently exactly the other six vertices of Q , three exclusive to each carrier. Their intersections with Q are ample five-vertex families satisfying (A.120). Since their two common words are opposite, Lemma A.78 permits only two opposite-puncture pairs, a contradiction.

Both possible forms of S are excluded. The section excess is neither four nor five, and (A.123) follows. $\square$

Proposition A.80 (A three-cycle attachment is short or costs seven). *Let Q be a positioned maximal repair three-cube in an interior section P_j , and suppose that it owns three directed neutral two-cycles on six distinct repair words. Let u, v be the two vertices of Q not used by those cycles. Then at least one of the following alternatives holds:*

1.

$$|P_j| - (r + 1) \geq 7; \tag{A.127}$$

2. *both adjacent carriers contain u, v , and their restrictions to Q have the short common-pair normal form (A.122). In particular, $\text{dist}_Q(u, v) \in \{1, 2\}$.*

Consequently, if the section excess is six, the second alternative is the only remaining maximal-three-cube profile.

Proof. The six cycle words are corners of P_j whose unique maximal cube is Q . Hence no edge of P_j leaves Q at any of those words.

Suppose first that one adjacent carrier, say C_j , omits one of u, v . The carrier is ample, contains the two fixed repair antipodes, and is therefore isometric. Choose an antipodal repair geodesic $G \subseteq C_j$, so $|G| = r + 1$. Its intersection with Q has at most one vertex. Indeed, a coordinate-simple cube geodesic meets a fixed subcube in a contiguous subpath: for every coordinate fixed by Q , the times at which its value agrees with the fixed value form an initial or a terminal interval, and their intersection is an interval. If this subpath were nontrivial, each of its endpoints would be u or v . For an internal endpoint of G , this follows because its next edge leaves Q ; for an endpoint of G , it follows because a repair antipode belongs to both adjacent carriers and hence is not one of the six exclusive cycle words. The two endpoints of a nontrivial subpath are distinct, so C_j would contain both u, v , contrary to the choice of C_j . Therefore

$$|P_j| \geq |Q \cup G| \geq 8 + (r + 1) - 1 = r + 8,$$

which is (A.127).

It remains that both carriers contain u, v . Each cycle contributes one word exclusively to C_j and its mate exclusively to C_{j+1} . Thus

$$D_- = C_j \cap Q, \quad D_+ = C_{j+1} \cap Q$$

are ample five-vertex restrictions, cover Q , and intersect exactly in $\{u, v\}$. Their three exclusive pairs satisfy (A.121) by Proposition A.28. Lemma A.78 gives the asserted short common-pair normal form and the distance conclusion. $\square$

Lemma A.81 (Two-square attachment antipode dichotomy). *Let $F \subseteq Q_R$ be ample and contain two antipodal words α, β . Suppose that $T \subseteq R$ has order three, that $Q \subseteq F$ is a positioned full T -cube, and that*

$$\text{Sh}(F) = \{\emptyset\} \cup \binom{R}{1} \cup \binom{T}{2} \cup \{T, S_1, S_2\}, \quad (\text{A.128})$$

where S_1, S_2 are distinct pairs not contained in T and

$$R = T \cup S_1 \cup S_2. \quad (\text{A.129})$$

If at least six vertices of Q are corners of F whose unique maximal cube is Q , put

$$N = Q \setminus \text{Cor}(F).$$

Then $1 \leq |N| \leq 2$, and the following alternatives hold.

1. If exactly one of α, β , say α , belongs to Q , then $\alpha \notin N$. If moreover $|N| = 1$, its unique word is opposite to α in Q .
2. If neither antipode belongs to Q , then N consists of two opposite words of Q .

Proof. For $i \in \{1, 2\}$, the full S_i -cube carrier is ample and its shadow is the link of S_i in $\text{Sh}(F)$. That link contains only the empty set. Consequently there is a unique positioned S_i -square; denote it by A_i . The same carrier argument shows that the positioned T -cube is unique. For $U \in \binom{T}{2}$, the link of U consists of the empty set and the singleton $T \setminus U$. Its carrier therefore has two vertices, supplied by the two opposite U -faces of Q . Thus those six faces and A_1, A_2 are all the squares of F .

Every coordinate in R belongs to a shattered pair. Its full edge-carrier is therefore a connected ample family with at least two vertices. No carrier vertex is isolated, so every edge of F lies in a square. The only squares are the six faces of Q and the two squares A_1, A_2 . Since F is connected,

$$F = Q \cup A_1 \cup A_2, \quad (\text{A.130})$$

and the intersection graph of these three cube blocks is connected. Moreover,

$$N = Q \cap (A_1 \cup A_2), \quad 1 \leq |N| \leq 2. \quad (\text{A.131})$$

The set $R \setminus T$ is nonempty, so Q contains at most one of the two antipodes.

Suppose first that $\alpha \in Q$. We claim that $\alpha \notin A_1 \cup A_2$. Otherwise, after interchanging the squares, assume $\alpha \in A_1$. The other antipode cannot lie in Q or A_1 , and hence belongs to A_2 . The two squares are disjoint: if $z \in A_1 \cap A_2$, then

$$R = \text{Diff}(\alpha, \beta) \subseteq S_1 \cup S_2,$$

whereas two pairs not contained in the three-set T cover at most two coordinates of T . Connectivity of the block-intersection graph now forces both squares to meet Q . Their intersections with Q are disjoint and together have at most two vertices by (A.131). Thus each intersection is a singleton, so S_1, S_2 are disjoint from T . If $y \in A_2 \cap Q$, then

$$\text{Diff}(\alpha, \beta) \subseteq T \cup S_2.$$

The distinct pure pairs S_1, S_2 leave a coordinate of S_1 outside $T \cup S_2$, another contradiction. This proves $\alpha \notin N$.

Assume in addition that $N = \{w\}$. The two squares cannot both meet Q : both intersections would be the singleton w , so both supports would be pure, and whichever square contains β would give $\text{Diff}(\alpha, \beta) \subseteq T \cup S_i$, missing a coordinate of the other pure pair. Hence, after relabelling,

$$Q \cap A_1 = \{w\}, \quad Q \cap A_2 = \emptyset, \quad A_1 \cap A_2 \neq \emptyset.$$

In particular, $S_1 \cap T = \emptyset$. Put

$$k = |S_2 \cap T|, \quad h = |S_1 \cap S_2|.$$

Here $k, h \in \{0, 1\}$, and

$$|R| = 7 - k - h.$$

Since a nonempty intersection of two positioned cubes is a cube on the common support, $|A_1 \cap A_2| = 2^h$. On the other hand, ampleness and (A.128) give $|F| = |R| + 7$, while inclusion-exclusion in (A) gives

$$|F| = 8 + 4 + 4 - 1 - 2^h.$$

Therefore

$$2^h = 1 + k + h,$$

which forces $k = 0$. Both square supports are pure. The antipode β belongs to A_2 , since membership in A_1 would miss a coordinate of $S_2 \setminus S_1$. Choose $z \in A_1 \cap A_2$. The words w, z, β agree on T , whereas α and β differ on all three coordinates of T . Thus w is opposite to α in Q .

Finally suppose that neither antipode belongs to Q . They cannot belong to the same square, so, after relabelling, $\alpha \in A_1$ and $\beta \in A_2$. The squares are disjoint, for a word in their intersection would again imply $R \subseteq S_1 \cup S_2$. Their block-intersection graph can therefore be connected only if both squares meet Q . The two intersections are nonempty and disjoint, and their union is N , which has order at most two. Hence

$$Q \cap A_1 = \{x\}, \quad Q \cap A_2 = \{y\},$$

and both square supports are disjoint from T . The pairs α, x and β, y agree on T . Since the antipodes differ on all three T -coordinates, x, y are opposite in Q . This proves the second alternative. $\square$

Proposition A.82 (Three neutral two-cycles force seven section units). *Let Q be a positioned maximal repair three-cube in an interior section P_j . If Q owns three directed neutral two-cycles whose six underlying repair words are distinct, then*

$$|P_j| - (r + 1) \geq 7. \tag{A.132}$$

Proof. Proposition A.79 gives a lower bound of six. Suppose for a contradiction that equality holds. Besides the empty set and the repair singletons, the four supports $\binom{T}{2} \cup \{T\}$ of Q leave room for exactly two further shattered supports. Downward closure makes them distinct pairs S_1, S_2 not contained in T .

If a repair coordinate lies outside $T \cup S_1 \cup S_2$, it belongs to no shattered pair. Its edge-carrier is a singleton, so there is one edge in that direction and the edge is a bridge. Contract all such directions. Each contraction lowers both the order and the dimension by one and preserves the section excess, the owner cube, the adjacent carriers, and the six cycle words. None of those six words is incident with a contracted edge, because its unique maximal cube is Q . We obtain an ample family F satisfying (A.128) and (A.129). The six cycle words remain corners of F with unique maximal cube Q .

Let u, v be the two unused words of Q . Equality excludes the first alternative of Proposition A.80. Hence both adjacent carriers contain u, v , and

$$\text{dist}_Q(u, v) \in \{1, 2\}. \quad (\text{A.133})$$

Every repair antipode belongs to both carriers, so an antipode in Q must be one of u, v .

Apply Lemma A.81 and write $N = Q \setminus \text{Cor}(F)$. If one antipode α lies in Q , then α is a corner. When $|N| = 2$, the cube has only six corners, one of which is α ; there are not six further corners available for the cycle words. When $|N| = 1$, the cycle words are exactly the six vertices outside $\{\alpha\} \cup N$. Thus $\{u, v\} = \{\alpha\} \cup N$, and the lemma makes u, v opposite, contradicting (A.133).

If neither antipode lies in Q , the lemma makes N a pair of opposite words. The six cycle words are precisely $Q \setminus N$, so $\{u, v\} = N$, again contradicting (A.133). These cases are exhaustive because Q contains at most one repair antipode. Section excess six is impossible, proving (A.132). $\square$

Corollary A.83 (A saturated maximal three-cube has four further components). *Under the hypotheses of Corollary A.77, one has*

$$|P_j| - (r + 1) = \kappa_j \geq 7. \quad (\text{A.134})$$

Thus besides the three two-cycles owned by Q , the section contains at least four further weak components, all using two underlying repair words. The six-unit short-common-pair profile does not belong to the global equality branch.

Proof. The three cycles are directed two-cycles on six distinct words by Corollary A.77. Proposition A.82 gives the lower bound in the display, while section saturation gives its equality on the left. Three of the κ_j components are the cycles owned by Q , so at least four others remain. Corollary A.30 makes every component in a saturated section a two-word component. $\square$

Lemma A.84 (Coface matching is controlled by upper ideals). *Let Q_L be a positioned cube, let $\mathcal{P} \subseteq 2^L$ be an upper ideal, and let $\mathcal{A}$ be a family of pairwise distinct full faces of Q_L whose supports belong to $\mathcal{P}$. For $K \in \mathcal{P}$, write*

$$m(K) = |\{A \in \mathcal{A} : \text{supp}(A) = K\}|.$$

Each face of support K may use any label $J \in \mathcal{P}$ with $K \subseteq J$. The faces admit pairwise distinct labels if and only if

$$\sum_{K \in \mathcal{U}} m(K) \leq |\mathcal{U}| \quad \text{for every upper ideal } \mathcal{U} \subseteq \mathcal{P}. \quad (\text{A.135})$$

Proof. Necessity follows by considering all faces whose supports belong to $\mathcal{U}$. Every allowed label for such a face remains in $\mathcal{U}$, because $\mathcal{U}$ is an upper ideal.

Conversely, let $\mathcal{B} \subseteq \mathcal{A}$, and let $\mathcal{U}$ be the upward closure in $\mathcal{P}$ of the supports of the faces in $\mathcal{B}$. This is exactly the set of labels available to at least one member of $\mathcal{B}$. Hence

$$|\mathcal{B}| \leq \sum_{K \in \mathcal{U}} m(K) \leq |\mathcal{U}|.$$

Hall's theorem now gives a matching of all faces to distinct labels. □

Corollary A.85 (Weighted-graph criterion inside a four-cube). *Let $|L| = 4$ and $\mathcal{P} = \{K \subseteq L : |K| \geq 2\}$. In the notation of Lemma A.84, put*

$$q = m(L), \quad t_v = m(L \setminus \{v\}), \quad p_{uv} = m(L \setminus \{u, v\})$$

for $v \in L$ and $\{u, v\} \in \binom{L}{2}$. Then the faces can be assigned distinct coface labels if and only if

$$q - 1 + \sum_{v \in W} (t_v - 1) + \sum_{\{u, v\} \in \binom{W}{2}} (p_{uv} - 1)_+ \leq 0 \quad (W \subseteq L). \quad (\text{A.136})$$

Thus local rank-four deficiency is a positive induced-subgraph weight, not a list of positioned face configurations.

Proof. Index the four triple supports by their missing elements. Index a pair support $L \setminus \{u, v\}$ by the edge $\{u, v\}$; its two triple cofaces are indexed by u, v . Every nonempty upper ideal contains L . It is therefore specified by a vertex set $W \subseteq L$ of triple supports and an arbitrary edge set $E \subseteq \binom{W}{2}$ of pair supports. Its demand minus its order is

$$q - 1 + \sum_{v \in W} (t_v - 1) + \sum_{\{u, v\} \in E} (p_{uv} - 1).$$

For fixed W , this expression is maximized by taking precisely the edges with $p_{uv} > 1$. Lemma A.84 now gives (A.136). □

Proposition A.86 (An overloaded maximal repair four-cube pays an exterior tax). *Let a positioned maximal repair four-cube Q of an interior section P_j be the common owner of c directed neutral cycles. Then*

$$\Delta(H; A, R) \geq 11 + \max\{0, 2c - 16\}. \quad (\text{A.137})$$

In particular, if $c \geq 3$, then

$$\Delta(H; A, R) \geq c + 1. \quad (\text{A.138})$$

Proof. The four-cube contributes all six repair pairs, all four repair triples, and its four-element support beyond the empty support and the repair singletons. Thus

$$|P_j| - (r + 1) \geq 11.$$

The c cycles are disjoint as boundary occurrences and use at least $2c$ occurrences. A repair word of Q has at most its left and right occurrence, so at least $\max\{0, 2c - 16\}$ words occur on both boundary sides. Every such word is a corner of P_j whose unique maximal repair cube is Q , and it belongs to neither adjacent carrier. As in the proof of Proposition A.76, ambient cornerlessness

forces a distinct concept over a core word outside the path for every doubled word. Hence the external-concept term in the path-section partition satisfies

$$V \geq \max\{0, 2c - 16\}.$$

Together with the eleven section supports, this proves (A.137).

For $3 \leq c \leq 10$, the initial eleven units already give $\Delta(H; A, R) \geq c + 1$. For $c \geq 11$, the right-hand side of (A.137) is $2c - 5 \geq c + 1$. This proves the strict-tax assertion. $\square$

Corollary A.87 (Pure rank-four packets have strict excess). *Let $|H| < 64$, and let a locally deficient packet from Corollary A.68 occupy a positioned four-cube whose support lies in R . Then all packet members are neutral cycles in one section and have one common positioned maximal repair four-cube owner. This owner supplies at least one unit of strict excess beyond its cycle population. Consequently a pure rank-four packet does not belong to the global equality branch.*

Proof. Every packet member is a neutral-cycle face, because a canonical failure support contains a core coordinate. Since the terminal cube has pure repair support, all its faces have the same core anchor and lie in one section P_j .

Each initial face contains its assigned owner word, which is a corner of P_j . Its unique maximal repair cube contains both that face and the terminal four-cube. The argument in the proof of Corollary A.75 shows that all packet members have the same maximal owner Q . If its support had order at least five, the facet-excess inequality would give

$$|H| \geq 2^6 = 64,$$

contrary to $|H| < 64$. Hence Q is the terminal four-cube itself.

Any family of at most two distinct faces has distinct coface labels: use their own supports when these differ, and use the common support and the full four-support when they agree. A deficient packet therefore contains at least three cycles. Apply Proposition A.86 to all cycles owned by Q . Equation (A.138) gives the claimed strict unit. $\square$

Corollary A.88 (Two-repair two-core rank-four packets separate). *Let a rank-four packet from Corollary A.68 have support*

$$L = T \dot{\cup} \{x_i, x_k\}, \quad T \in \binom{R}{2}, \quad x_i, x_k \in X. \quad (\text{A.139})$$

Then its members can be assigned pairwise distinct coface labels inside the terminal four-cube.

Proof. Every neutral face has support T . A failure face has support $T \cup \{x_i\}$ or $T \cup \{x_k\}$, and canonical failure supports are distinct, so at most one of each kind occurs. The neutral T -faces are indexed by core anchors in the square on $\{x_i, x_k\}$. They arise over vertices of the coordinate-simple path A , with every other core coordinate fixed by the terminal cube. The intersection of A with this positioned core square is a path of order at most three. Thus at most three neutral faces occur.

The four available labels are

$$T, \quad T \cup \{x_i\}, \quad T \cup \{x_k\}, \quad L.$$

An explicit matching fails only if all three possible neutral faces and both failure faces occur, giving five demands for these four labels. Assume this profile.

After interchanging i, k , the three core anchors are consecutive path vertices

$$a_{i-1}, a_i, a_{i+1}, \quad k = i + 1.$$

Indeed, they agree outside $\{x_i, x_k\}$, so no other path direction can occur between them. The three neutral faces belong to one positioned terminal cube and therefore have the same repair anchor outside T . The corresponding positioned T -cube lies in all three sections P_{i-1}, P_i, P_{i+1} , and hence

$$T \in \text{Sh}(C_i) \cap \text{Sh}(C_{i+1}).$$

On the other hand, the two failure supports are $T \cup \{x_i\}$ and $T \cup \{x_{i+1}\}$. Proposition A.61 says that their acquired repair support T is absent from C_i and C_{i+1} , respectively. This contradiction eliminates the only deficient profile. $\square$

Corollary A.89 (The remaining rank-four packet is a three-arm potential). *Let a rank-four packet have support*

$$L = B \dot{\cup} \{x_i\}, \quad B \in \binom{R}{3}. \quad (\text{A.140})$$

For $s \in B$, put $K_s = B \setminus \{s\}$, and define

$$\begin{aligned} p_s &= \#\{\text{neutral faces of support } K_s\}, \\ b &= \#\{\text{neutral faces of support } B\}, \\ f_s &= \#\{\text{failure faces of support } K_s \cup \{x_i\}\}, \\ g &= \#\{\text{failure faces of support } L\}. \end{aligned}$$

Then

$$0 \leq p_s \leq 4, \quad 0 \leq b \leq 2, \quad f_s, g \in \{0, 1\}. \quad (\text{A.141})$$

Canonical carrier absence also gives

$$f_s = 1 \implies p_s \leq 2, \quad g = 1 \implies b \leq 1. \quad (\text{A.142})$$

The packet can be assigned distinct coface labels inside its terminal cube if and only if

$$\Phi_4 := g + b - 2 + \sum_{s \in B} (p_s + f_s - 2)_+ \leq 0. \quad (\text{A.143})$$

Consequently every locally deficient rank-four packet below order 64 has the support profile (A.140) and satisfies $\Phi_4 > 0$. This is the sole remaining rank-four profile.

Proof. The support types follow from provenance. A neutral face has pure repair support of order at least two, hence support K_s or B . A failure support has exactly one core coordinate, hence is $K_s \cup \{x_i\}$ or L . The numbers in (A.141) are the numbers of parallel faces of the relevant supports; failure multiplicities are at most one because canonical failure supports are distinct.

If $f_s = 1$, then $K_s \notin \text{Sh}(C_i)$ by Proposition A.61. For each of the two values of the remaining repair coordinate s , at most one of the two core-side K_s -faces can therefore occur. Hence $p_s \leq 2$. The same argument for the two opposite B -faces gives $b \leq 1$ when $g = 1$. This proves (A.142).

It remains to apply Lemma A.84. The admissible support poset has bottom elements K_s , middle elements B and $K_s \cup \{x_i\}$, and top element L . An upper ideal containing a nonempty set $D \subseteq B$ of

bottom arms must contain B, L and, for every $s \in D$, the mixed triple $K_s \cup \{x_i\}$. Its maximum demand minus order is

$$g + b - 2 + \sum_{s \in D} (p_s + f_s - 2). \quad (\text{A.144})$$

Optional mixed triples have multiplicity at most one and cannot increase this excess. An ideal containing no bottom arm has nonpositive excess: the only potentially positive middle contribution is $b - 1$, and $g = 1$ forces $b \leq 1$.

The carrier bounds give $g + b - 2 \leq 0$. Every nonempty arm set D has excess at most Φ_4 . Hence $\Phi_4 \leq 0$ makes all upper-ideal excesses nonpositive. If $\Phi_4 > 0$, at least one arm has positive value; taking exactly all positive arms gives a nonempty upper ideal whose excess is Φ_4 . The upper-ideal criterion proves (A.143). Pure rank-four packets have strict excess by Corollary A.87, and the two-repair two-core profile separates by Corollary A.88. Since every initial support has at least two repair coordinates, these cases and (A.140) exhaust rank four. $\square$

Lemma A.90 (Two occupied core sides synchronize the repair cube). *Let Q_L be a positioned full cube of H , where*

$$L = B \dot{\cup} \{x_i\}, \quad B \subseteq R.$$

Suppose that Q_L contains a neutral-cycle face on each of its two x_i -sides. Then those two faces lie over the consecutive path words a_{i-1}, a_i , and the positioned full B -cube given by either x_i -side belongs to the carrier $C_i = P_{i-1} \cap P_i$. In particular,

$$B \in \text{Sh}(C_i). \quad (\text{A.145})$$

Proof. A neutral-cycle face lies over the path word of its owner section. Let the two path words supplied by the two x_i -sides of Q_L be a_h, a_k . The positioned cube fixes every core coordinate other than x_i , so a_h and a_k differ exactly in x_i . A coordinate-simple path flips x_i only on the edge $a_{i-1}a_i$. Hence

$$\{a_h, a_k\} = \{a_{i-1}, a_i\}.$$

The two x_i -sides of Q_L are full positioned B -cubes with the same repair anchor. They therefore give one positioned full B -cube in both P_{i-1} and P_i , hence in C_i . This proves (A.145). $\square$

Proposition A.91 (Rank-four three-arm deficiency descends to rank three). *Keep the notation of Corollary A.89, and suppose that $\Phi_4 > 0$. Then the initial neutral faces contain, on one x_i -side of the terminal cube, a locally deficient pure-repair rank-three packet in the positioned B -cube. Thus a three-repair one-core packet has no rank-four-irreducible Hall obstruction.*

Proof. Orient the two x_i -sides of the terminal cube by $\epsilon \in \{0, 1\}$. Let $p_{s,\epsilon}$ be the number of neutral K_s -faces on side ϵ , and let $b_\epsilon \in \{0, 1\}$ record whether the neutral B -face on that side occurs. Thus

$$p_s = p_{s,0} + p_{s,1}, \quad b = b_0 + b_1.$$

For $s \in B$, call side ϵ a *loaded s -row* if

$$p_{s,\epsilon} = 2 \quad \text{and} \quad (f_s = 1 \text{ or } p_{s,1-\epsilon} > 0). \quad (\text{A.146})$$

Let $\rho_{s,\epsilon}$ be its indicator, and put

$$q_\epsilon = b_\epsilon + \sum_{s \in B} \rho_{s,\epsilon}.$$

Finally, let d count the coordinates $s \in B$ for which $f_s = 1$ and

$$p_{s,0} = p_{s,1} = 1.$$

For such an s , canonical carrier absence prevents the two neutral faces from having the same fixed value of s ; they are the two diagonal K_s -faces, one on each core side.

The four possible K_s -faces form a 2-by-2 array, indexed by the x_i -side and the fixed value of s . Reading this array gives the exact identity

$$(p_s + f_s - 2)_+ = \rho_{s,0} + \rho_{s,1} + \mathbf{1}_{\{f_s=1, p_{s,0}=p_{s,1}=1\}}. \quad (\text{A.147})$$

Indeed, if $f_s = 0$, three selected faces give one loaded row and four give two, while at most two give neither. If $f_s = 1$, the carrier bound gives $p_s \leq 2$; two selected faces form either one row or the diagonal pair just described. Summing (A.147) yields

$$\Phi_4 = g + q_0 + q_1 + d - 2. \quad (\text{A.148})$$

If $q_\epsilon \geq 2$ for some side ϵ , choose any two of the events counted by q_ϵ . A b_ϵ -event is the full B -face, while a loaded-row event consists of the two opposite K_s -faces for one s . Two events therefore contain either the full B -cube together with one opposite pair, or two opposite pairs with distinct supports. These are exactly the two minimal rank-three Hall obstructions in Lemma A.72. All their faces are neutral and lie on one core side, so this is the required pure-repair rank-three subpacket.

It remains to rule out $q_0, q_1 \leq 1$. Suppose first that $g = 1$. Positivity in (A.148) gives $q_0 + q_1 + d \geq 2$. If $d > 0$, a diagonal arm supplies a neutral face on each core side. If $d = 0$, then $q_0 = q_1 = 1$, and again each core side contains a neutral face. Lemma A.90 gives $B \in \text{Sh}(C_i)$. But $g = 1$ is a canonical failure of support $B \cup \{x_i\}$, so Proposition A.61 gives $B \notin \text{Sh}(C_i)$, a contradiction.

Finally suppose that $g = 0$. Positivity gives $q_0 + q_1 + d \geq 3$, and hence $d > 0$. Choose a diagonal arm s . The same synchronization lemma gives $B \in \text{Sh}(C_i)$, and downward closure gives $K_s \in \text{Sh}(C_i)$. On the other hand, the diagonal arm has $f_s = 1$, whose canonical failure identity gives $K_s \notin \text{Sh}(C_i)$. This final contradiction proves that some q_ϵ is at least two, and the first part of the proof then supplies the rank-three subpacket. $\square$

Corollary A.92 (Rank four has no new packet obstruction). *Below order 64, every locally deficient rank-four packet either is strictly paid by the pure-repair four-cube tax or contains one of the already classified pure-repair rank-three packets. The two-repair two-core profile and every rank-four-irreducible three-repair one-core profile separate. Consequently rank four contributes no new equality profile beyond the rank-three row.*

Proof. Pure-repair packets have strict excess by Corollary A.87. Two-repair two-core packets separate by Corollary A.88. The only other support profile is $B \cup \{x_i\}$. If it is locally deficient, then $\Phi_4 > 0$ by Corollary A.89, and Proposition A.91 finds a proper rank-three deficient subpacket unless the configuration is already contradictory. This exhausts the rank-four support profiles. $\square$

Proposition A.93 (Repair-position carriers conserve strict cofaces). *Keep the notation of Proposition 7.61. Besides the core projection Z_T , project the full T -cube carrier to the coordinates outside T in the repair block:*

$$Y_T = \mathcal{K}_T(H)|_{R \setminus T}.$$

Then Y_T is ample and

$$\text{Sh}(Y_T) = \{S \subseteq R \setminus T : T \dot{\cup} S \in \text{Sh}(H)\}. \quad (\text{A.149})$$

Consequently, if two positioned full T -cubes of H have distinct repair anchors outside T , then for every coordinate s on which those anchors differ,

$$T \cup \{s\} \in \text{Sh}(W) \setminus \mathcal{L}_R \subseteq \Xi. \quad (\text{A.150})$$

Proof. Apply Lemma 7.19 with T as the cube support and $R \setminus T$ as the projection block. This gives ampleness of Y_T and (A.149). If two anchors differ on s , an isometric Y_T -geodesic between them has an s -edge. Hence $\{s\} \in \text{Sh}(Y_T)$, and the displayed identity gives $T \cup \{s\} \in \text{Sh}(H)$. This support is purely in R , so it lies in $\text{Sh}(W)$; it has order at least three because $|T| \geq 2$, and therefore is outside $\mathcal{L}_R$ and belongs to Ξ . $\square$

Corollary A.94 (A tight two-sided failure is a coface or one word hole). *Fix a one-section gap of I_T and suppose that one failed lift descends from each of its two initial boundary occurrences. For a failure F , let p_F be its attempted repair word and put*

$$u_F = p_F|_{R \setminus T}.$$

Then $u_F \in Y_T$. For the two failures F_-, F_+ , exactly one of the following holds:

1. the anchors u_{F_-} and u_{F_+} differ, and every coordinate in their difference supplies the strict pure-repair coface in (A.150);
2. $p_{F_-} = p_{F_+}$. Thus the two failure lineages terminate at the same missing concept in the middle section: one repair word is present on both sides of the gap and absent in its centre.

Hence the position-sensitive part of a tight hole has only one residual profile: a common missing repair word. Distinct positions already produce a strict-coface candidate in Ξ .

Proof. At a failed lift, adjoining p_F to the target section and to the adjacent carrier is locally ample with unique new repair support T . The one-concept extension test gives

$$p_F^U \in C_i \quad (\emptyset \neq U \subseteq T).$$

The source section also contains p_F . It therefore contains the full positioned T -cube through p_F , proving $u_F \in Y_T$.

If the two anchors differ, Proposition A.93 gives the first alternative. Suppose that they agree. The two repair words then belong to the same positioned T -face. Both failed lifts target the unique middle section of the tight gap. At the time of either attempt, that section contains every other vertex of the corresponding T -face. Section families grow throughout the two sweeps. If the two attempted words were different, the earlier punctured face would already contain the word attempted later, contradicting its absence at the later attempt. Thus the attempted words are equal, proving the second alternative. $\square$

Corollary A.95 (Transverse blockers supply pure-repair cross-supports). *Consider a failed adjacent-section lift with acquired repair support T , attempted concept v , and antipodal blocker v^J . Put*

$$S = J \cap R, \quad K = J \cap X.$$

Then $|T| \geq 2$, $T \cap S = \emptyset$, and

$$\{t, s\} \in \text{Sh}(H) \quad (t \in T, s \in S). \quad (\text{A.151})$$

If the blocker has exactly one core direction, then $S \neq \emptyset$, so the failure forces at least $|T|$ nonbaseline pure-repair supports. In particular, a tight two-sided hole whose two blockers lie over the two opposite neighbouring path sections exposes such a support family on each side.

Proof. Every repair section contains the two fixed repair antipodes, so it already shatters the empty set and every repair singleton. The support newly acquired by a locally ample section extension is therefore nonbaseline, and $|T| \geq 2$. The local one-concept extension test gives

$$v^U \in H \quad (\emptyset \neq U \subseteq T).$$

Apply Lemma A.3. It gives $T \cap J = \emptyset$ and shatters $\{t, e\}$ for every $t \in T$ and $e \in J$. Restricting e to S proves (A.151). Finally the blocker has order at least two and has nonempty core part by Corollary A.44; if that core part is a singleton, its repair part S must be nonempty. Distinct pairs $\{t, s\}$ have order two and hence lie outside the common repair baseline. $\square$

Remark A.96 (The remaining path-wide proof contract). Corollary A.44 separates the local problem from the global one. A transition that stops locally pays a cyclic block governed by Proposition A.43. A transition that fails only after lifting to H has a transverse core owner, and Proposition A.45 makes every on-spine transfer move strictly toward an endpoint. Proposition A.46 already assigns a current-state residual support to every failed lift. Thus the remaining theorem need not construct local peeling states, exclude owner cycles, or compare different extension states: Corollary A.48 transports every later payment back to the original residual shadow. Proposition A.25 controls repeated section acquisitions by path intervals.

The neutral part now has one uniform accounting theorem. Proposition A.29 adjoins every opposite tip as a terminal vertex and counts incidences between weak components and their underlying repair words. Every component uses at least two words. A word outside two chosen antipodal carrier geodesics can belong to at most its left and right components, while an exclusive geodesic word belongs to at most one. The two multiplicity bounds cancel exactly to give

$$\kappa_j \leq |P_j| - (r + 1).$$

Summing over the interior sections pays every cyclic and acyclic neutral component directly from the additive path-section excess. In particular, terminal rank, terminal-support reuse, arrow labels, and repeated cycles in one positioned cube create no separate multiplicity problem. The closed-cycle packing, rank-one surcharge, sectionwise cube tax, positioned-cube degree bound, fixed-support capacity, and common-cube rank restriction above remain valid quantitative refinements, but they are no longer active proof-debt rows.

Corollary A.30 also classifies the only way this bound can be tight. Every weak component then uses two repair words, and the component–word incidence graph is a disjoint union of paths and cycles. Its path endpoints are exactly the exclusive words of the two carrier geodesics; its cyclic word vertices lie outside both. In particular, every closed neutral component in the global equality branch is already a directed two-cycle. Longer neutral cycles and higher-word acyclic components create strict section slack and are not part of the remaining saturation problem.

There is also no hidden singleton exception. Lemma 5.10 makes a maximal singleton attachment a bridge and the only edge in its repair direction. Since the target carrier contains the two repair antipodes, it must use that direction and therefore must contain the bridge, contradicting the definition of the opposite tip. This is Proposition A.34. It is a structural strengthening, although the forest estimate already makes the rank of an acyclic terminal irrelevant.

The path-wide accounting can consequently be assembled without a terminal classification. For the $2n$ oriented adjacent-section incidences, compare the selected initial boundary-corner demands

with the terminal cyclic charges, transverse-owner payments, and neutral weak components, all in the original state. By Corollaries A.31 and A.32, a component containing the two selected demands from opposite sides of one section is already paid twice unless those demands form a directed two-cycle. Proposition A.28 identifies its two parent punctures exactly: they have one common repair label and opposite missing tips in the two adjacent carriers.

The directed two-cycle is no longer an unclassified terminal. Proposition A.49 and Corollary A.50 show that it either produces a residual support which transports to the original shadow, or has an ample extension in which its middle section is contained in both neighbouring sections.

The handoff itself is now globally controlled, rather than left as an iteration problem. Lemma A.51 shows that each successful concept duplication carries exactly one existing repair support across the same section edge. Proposition A.52 weights a boundary by its remaining distance to the endpoint toward which it moves. Both the word potential and the rank-weighted shadow potential decrease strictly; the latter decrease pays the simultaneous increase of the section-edge rank. Corollary A.53 also handles the only possible collision: if one duplication resolves marked demands from both neighbouring centres, two opposite boundaries annihilate and the potential has more, not less, slack.

Proposition A.54 therefore treats all tight two-cycles visible in the original state in two directed sweeps. Every one is either erased with an explicit boundary-descent payment, replaced by a terminal cyclic block, or replaced by a transverse residual payment transported to the original shadow. There is no remaining infinite closure, owner-repetition, or endpoint-confluence branch.

The remaining assembly row is narrower still. Proposition A.55 combines every terminal block created before or during the sweeps into the original section-edge rank plus the original shadow-boundary potential. Successful acquisitions are already paid by Corollary A.53. Only the transverse failures remain. Proposition A.61 identifies their residual payments exactly: a failure across x_i carries the support $T \cup \{x_i\}$, where T is its transported section-shadow boundary. Proposition A.62 shows that these labels are pairwise distinct over the complete history and all belong to Ω_0 . Thus transverse failures inject directly into the original residual shadow. The residual core moment, Λ_0 , and every multiplicity factor in Proposition A.59 are no longer active proof debt.

The dynamic collision problem is therefore closed. Proposition 7.60 also removes a second apparent overlap. For each repair support T , successful acquisitions are columns of the forest boundary map ∂_T . They identify section occurrences along one interval and are not an additional cardinality charge. The original excess has the exact static partition

$$\Delta(H; A, R) = |\Xi| + |\Omega_0|,$$

where Ξ consists of one pure-support root, the internal carrier edges of every section-support interval, and the repair supports seen only in W . The only cokernel has one generator for every interval component after the distinguished one; equation (7.55) records exactly how those generators, external concepts, and repair supports seen only in W balance the residual shadow.

Thus boundary descent is no longer a separate active row. Proposition 7.61 assembles all positioned T -cubes in one ample projected carrier Z_T . If the section-index set of T has c_T components, then the residual shadow with exact repair part T has at least $2(c_T - 1)$ members. One unit comes from the cokernel component in (7.57); the other is forced by an external core vertex on an isometric carrier path between consecutive components.

Proposition A.63 makes those two units explicit. A gap of edge length d reserves all d labels

$$T \cup \{x_i\}$$

on its path edges. At most two failure lineages can enter one gap. Hence the failure labels and one fragmentation generator fit in the reserved family whenever $d \geq 3$, and also when $d = 2$ unless both

lineages fail. The former supportwise complementarity problem is therefore reduced to one normal form: a single missing section between two T -intervals, with a failed descendant from each side.

The exceptional tight hole has a simpler payment that forgets the repair positions altogether. Its projected carrier Z_T contains the two path vertices on either side of the missing section and is isometric. It must therefore use the fourth vertex of their core square. Together with the three consecutive path vertices, this external carrier vertex shatters the pure core pair $\{x_j, x_{j+1}\}$. Two different exceptional holes cannot have the same missing section, because that would reuse the same two failed oriented incidences with two different unique acquired supports. Proposition A.64 consequently pays all correction terms b_T by distinct original residual supports and proves the global complementarity inequality

$$|\mathcal{F}| + \sum_{T \in \mathcal{T}} (c_T - 1) \leq |\Omega_0|.$$

The repair-position and blocker results remain useful local refinements, but they are no longer required for this payment.

Thus the failure–fragmentation row is closed without a support-incidence Hall theorem. The occurrence row is closed as well. Proposition A.65 embeds every chosen original-state occurrence demand jointly with all failures in $\Xi \dot{\cup} \Omega_0$. Proposition A.56 adds exactly one formal column per successful acquisition, and Corollary A.58 shows that dynamic terminal remainders use only unmarked columns of this kind. Hence no local collision between neutral components, terminal remainders, successful acquisitions, fragmentation generators, or failures remains.

This conclusion is a capacity theorem, not yet a contradiction. An injection of the selected demands into the excess budget proves that the payments can coexist; it does not prove that cornerlessness creates more payments than the budget contains. The independent fixed-spine argument now supplies the lower endpoint 40, while the path-core target remains 46.

The arbitrary occurrence-label normal form in Corollary A.67 is only a capacity statement: its chosen section cube need not contain the puncture that created the demand. Corollary A.68 supplies the geometric replacement for the closed neutral components and retains the canonical residual labels of the failures. It anchors each cycle in its actual owner cube and represents each failure by an auxiliary cube on its canonical support. The full positioned-cube carrier then moves every collision between different positions strictly upward in the support poset. Below order 48, every terminal packet consequently has rank three or four; rank two occurs only in the occurrence proxy. Acyclic neutral components require no such packet analysis, because the forest accounting charges them at their terminal outcomes.

The final contract is therefore an exact matching statement for these anchored faces. Lemma A.72 reduces the rank-three branch to two minimal local motifs: a full cube with one opposite pair of square faces, or two opposite pairs in different directions. Corollary A.74 excludes the mixed core–repair instance of the first motif, because it would put the failure’s repair support back into the path-edge carrier from which it is canonically absent. Thus both surviving motifs are pure-repair neutral-cycle configurations. Corollary A.75 then shows that either the alternating component already reaches rank four or the terminal cube is a maximal repair three-cube in its common owner section. It remains to show that the two motifs in this maximal-cube normal form supply strict surplus or expose a corner. Proposition A.76 already gives strict tax for every possible cycle population, including the apparent four-cycle equality case. Thus no isolated rank-three equality profile remains; in a globally saturated component, its spare unit must be consumed by another alternating demand. The equality skeleton makes this handoff considerably more rigid: Corollary A.77 shows that a maximal-three-cube motif in a saturated section consists of exactly three directed two-cycles on six distinct cube words. All other cycle populations create strict section slack.

Proposition A.79 now excludes both the four-unit cube profile and its one-support extension. In a saturated section the three cycles therefore coexist with at least three further two-word components; the former one- and two-attachment profiles do not occur. Proposition A.80 reduces section excess six to two unused cube words that belong to both adjacent carriers and have distance one or two.

The two-square attachment lemma closes that residue. At exact excess six, the shadow has only two pair supports beyond the four supports of the owner three-cube. After contracting singleton bridge directions, edge-carrier connectedness makes the section the union of exactly three maximal blocks: the owner cube and the two positioned squares. The intersection nerve and the two repair antipodes then force the following dichotomy. If one antipode lies in the owner cube, it is a cube corner; when only one attachment word remains, that word is opposite to the antipode. If neither antipode lies in the cube, the two attachment words are opposite. With six distinct cycle corners, either there are too few corners outside the common antipode or the two unused words are therefore opposite. Both conclusions contradict the short common-pair normal form. Proposition A.82 consequently gives

$$|P_j| - (r + 1) \geq 7.$$

In a saturated section the three cycles coexist with at least four further two-word components.

Thus the last rank-three equality profile is closed by one cube-block nerve argument, not by the fixed-rank realization list that suggested it. Every deficient anchored packet below order 48 now either reaches rank four or already creates strict excess.

The rank-four row also has an exact order-free compression. Coface matching is equivalent to the upper-ideal inequalities (A.135); in a pure four-cube these are the induced-subgraph weights in (A.136). A pure-repair packet has a common maximal repair four-cube owner, whose eleven nonbaseline supports give strict excess. A packet with two repair and two core coordinates separates: its only Hall-deficient profile would put the same repair two-cube in two path-edge carriers from which the two canonical failure labels require it to be absent.

The final mixed row initially appears as one three-arm potential. Its terminal support is $B \dot{\cup} \{x_i\}$, with three repair coordinates and one core coordinate, and it is deficient exactly when

$$\Phi_4 = g + b - 2 + \sum_{s \in B} (p_s + f_s - 2)_+ > 0.$$

This potential has a second, geometric decomposition. Its positive units are loaded rows on the two core sides together with diagonal failure arms. Two loaded events on one side are exactly a deficient pure-repair rank-three subpacket. If neither side contains two events, positivity forces neutral faces on both sides of the x_i -edge. The terminal cube then puts the full B -cube in C_i , contradicting the canonical carrier absence of either the top failure or a diagonal arm. Proposition A.91 therefore shows that every positive Φ_4 descends to the already closed rank-three row. Corollary A.92 closes rank four without a face-placement list.

The packet-free equality case is closed by strict facet ascent. Every proper carrier has a corner peeling, whose cube-support matching can be rerooted by alternating-path exchange at any prescribed carrier position. This matching flexibility holds even when no peeling ends at that position. A proper set of positioned faces can consequently be moved entirely to strict cofaces, with no carrier-order threshold. In the absence of a packet, iteration ends with distinct facet labels. Equality exhausts the nonbaseline shadow with these facets. Downward closure then forbids a support of rank at least three: any two of its repair coordinates would form a proper nonbaseline subface that is also a facet. The whole shattered complex has VC dimension at most two, and relative VC-two peeling exposes a corner. Thus packet-free equality is impossible, simultaneously for all orders $40, \dots, 47$.

Together with the rank-three and rank-four closures, this completes the path-core repair theorem below order 48. No cardinality or path-profile census enters the final step.

The shift-criticality theorem and the two-sided shield-capacity lemma remain independent structural checks on a least obstruction; they are not needed for the completed lower bound.

Lemma A.97 (Four-coordinate section synchronization). *Let $|E| \leq 4$, let $C, Y \subseteq Q_E$, and suppose that*

$$H = (C \times \{0\}) \cup (Y \times \{1\}) \subseteq Q_{E \cup \{r\}} \quad (\text{A.152})$$

is ample. Put $S = C \cap Y$. If $S \neq \emptyset$ and $Y \setminus C \neq \emptyset$, then some $p \in Y \setminus C$ satisfies

$$S \cup \{p\} \text{ is ample,} \quad C \cup \{p\} \text{ is ample.} \quad (\text{A.153})$$

For every such p , the subfamily

$$(C \times \{0\}) \cup ((S \cup \{p\}) \times \{1\}) \quad (\text{A.154})$$

is ample.

Proof. The coordinate sections and their intersection are ample. Theorem A.18, applied to the proper inclusion $S \subsetneq Y$, gives a concept $p \in Y \setminus S = Y \setminus C$ such that $S \cup \{p\}$ is ample. Lemma A.19 then shows that $C \cup \{p\}$ is ample as well. This proves (A.153) without a finite classification of the section pairs.

It remains to justify the final assertion without computation. Both singleton augmentations in (A.153) add exactly one new shattered support; call them R_S and R_C , respectively. Since $S \cup \{p\} \subseteq Y$, we have $R_S \in \text{Sh}(Y)$. Ampleness of H , applied to supports containing r , gives

$$\text{Sh}(C) \cap \text{Sh}(Y) = \text{Sh}(S). \quad (\text{A.155})$$

Thus $R_S \notin \text{Sh}(C)$. On the other hand, $S \cup \{p\} \subseteq C \cup \{p\}$, so monotonicity of shattering forces $R_S = R_C$. The two-section formula now gives both required equalities for (A.154): supports not containing r add the common support R_S , while supports containing r remain exactly $\text{Sh}(S)$. Hence the sublift is ample. $\square$

Lemma A.98 (A peripheral upper endpoint lowers the interval dimension). *Suppose that every proper nested pair of nonempty ample families using at most $d - 1$ coordinates has an ample one-concept intermediate family. Let $\emptyset \neq S \subsetneq B \subseteq Q_E$ be ample, with $|E| = d$. If B has a peripheral coordinate, then some $p \in B \setminus S$ makes $S \cup \{p\}$ ample.*

Proof. Orient a peripheral coordinate r so that

$$B = (C \times \{0\}) \cup (Y \times \{1\}), \quad Y \subseteq C,$$

and write

$$S = (A \times \{0\}) \cup (D \times \{1\}), \quad A \subseteq C, \quad D \subseteq Y.$$

If $A = \emptyset$, choose any $p \in D$. Adding its lower copy is a peripheral expansion of D along the singleton $\{p\}$, and is ample. Thus assume $A \neq \emptyset$.

First suppose that $D \setminus A \neq \emptyset$. Connectivity of S gives $I = A \cap D \neq \emptyset$. The lower-dimensional hypothesis applied to $I \subsetneq D$ supplies $p \in D \setminus I = D \setminus A$ for which $I \cup \{p\}$ is ample. Lemma A.20, applied to S , then shows that adjoining the lower copy of p to S is ample. Since $p \in D \subseteq Y \subseteq C$, that copy belongs to B .

It remains to suppose that $D \subseteq A$. If $A \subsetneq C$, the lower-dimensional hypothesis supplies $p \in C \setminus A$ with $A \cup \{p\}$ ample. The sections $D \subseteq A \cup \{p\}$ are nested, so adding the lower copy of p preserves ampleness. Finally, if $A = C$, properness of $S \subset B$ gives $D \subsetneq Y$. Apply the lower-dimensional hypothesis to $D \subsetneq Y$, or choose any $p \in Y$ when $D = \emptyset$, and add the upper copy. Again the resulting sections are nested ample families. This exhausts the cases. $\square$

Lemma A.99 (Sparse crossing graphs force a peripheral coordinate). *Let $A \subseteq Q_5$ be ample and use all five coordinates. If the crossing graph of its five Θ -classes has at most four edges, then A has a peripheral coordinate.*

Proof. Let G be the one-inclusion graph of A , and write $G^\sharp$ for its crossing graph. By hypothesis, $|E(G^\sharp)| \leq 4$.

If $G^\sharp$ is a forest, then G is a C_4 -cactoid by [KM02, Theorem 5.5]. A leaf block is a K_2 or a C_4 -tree. In the former case its outer edge class is peripheral. In the latter, root the tree of edge-glued squares at the unique cut vertex through which the rest of G is attached. A farthest leaf square has a new Θ -class whose outer two-vertex side avoids the cut vertex and is peripheral in all of G . The same conclusion is immediate if G itself is a single block.

Suppose that $G^\sharp$ contains a cycle. A connected graph on five vertices with a cycle has at least five edges, so $G^\sharp$ is disconnected. It has only one cyclic component: two cyclic components would use at least six vertices. The components of the crossing graph are the crossing graphs of the blocks of G , with a K_1 component corresponding to a bridge block; this follows from [KM02, Theorem 3.4] applied blockwise. The block-cut tree has at least two leaf blocks, and at most one of them has cyclic crossing graph. Choose a leaf block whose crossing graph is a tree. By [KM02, Theorem 5.4], it is a K_2 or a C_4 -tree. The preceding leaf-block argument again supplies a peripheral Θ -class of G . Thus the full-support family always has a peripheral coordinate. $\square$

Corollary A.100 (Small five-coordinate endpoints are peripheral). *If $A \subseteq Q_5$ is ample and $|A| \leq 10$, then either A is contained in a coordinate hyperplane or A has a peripheral coordinate.*

Proof. Suppose that A uses all five coordinates. A crossing pair is a shattered coordinate pair. Write $G^\sharp$ for the crossing graph of the one-inclusion graph of A . Then

$$|A| = |\text{Sh}(A)| \geq 1 + 5 + |E(G^\sharp)|.$$

Thus $|E(G^\sharp)| \leq 4$, and Lemma A.99 applies. $\square$

Lemma A.101 (Two-sided normal form of a minimal ample-interval gap). *Fix a coordinate set E , and suppose that some proper nested ample pair has no ample one-concept intermediate family. Choose such a pair $S \subsetneq B \subseteq Q_E$ minimizing $|U|$, where $U = B \setminus S$, and put*

$$A = Q_E \setminus B, \quad T = Q_E \setminus S. \tag{A.156}$$

Then there is no ample family strictly between S and B . The complementary pair $A \subsetneq T$ has the same property. In particular, for every $p \in U$, none of

$$S \cup \{p\}, \quad B \setminus \{p\}, \quad A \cup \{p\}, \quad T \setminus \{p\} \tag{A.157}$$

is ample. Equivalently, every corner of B belongs to S , and every corner of T belongs to A .

Moreover, every $p \in U$ has two blocking vertices $s_p \in S$ and $a_p \in A$. A B -geodesic from p to s_p and a T -geodesic from p to a_p both start with an edge whose other endpoint lies in U . Finally, $\Psi(B, S) = (\Delta_B, \Delta_S)$ is fully Cohen–Macaulay but loses this property when any one of its relative facets is filled into Δ_S .

Proof. If an ample C satisfied $S \subsetneq C \subsetneq B$, then the pair $S \subset C$ has smaller difference. Minimality would give an ample augmentation of S by a member of $C \setminus S$, contrary to the choice of $S \subset B$. Thus no such C exists, and the first two failures in (A.157) follow. Set complementation in Q_E reverses the interval and preserves ampleness, proving the corresponding assertions for $A \subset T$ and the other two failures. The ample corner-deletion criterion says that deleting a vertex preserves ampleness exactly when that vertex is a corner. Applying it to B and T gives the two corner containments.

Apply the blocking-vertex assertion of Lemma A.1 first to $S \subset B$ and then to $A \subset T$. It supplies s_p, a_p and the two asserted first edges. Lemma A.12 makes $\Psi(B, S)$ fully Cohen–Macaulay. Filling the relative facet of p replaces S by $S \cup \{p\}$, which is not ample and hence does not have a Cohen–Macaulay Yang complex. This proves the final claim. $\square$

Proposition A.102 (Central relative-CM profile in five coordinates). *If the five-coordinate ample-interval theorem fails, then it has a minimal counterexample $S \subsetneq B \subseteq Q_5$ with the following properties. With A, T, U as in (A.156), all four endpoint families S, B, A, T use all five coordinates and have no peripheral coordinate. Writing*

$$m_X = |\text{Sh}(X) \cap \binom{E}{2}|, \quad t_X = |\text{Sh}(X) \cap \binom{E}{3}|,$$

one has

$$5 \leq m_B \leq 10, qquad 1 \leq t_B \leq 5, qquad 12 \leq |B| = 6 + m_B + t_B \leq 21, \quad (\text{A.158})$$

and the same bounds hold for T . Consequently

$$11 \leq |S|, |A| \leq 20. \quad (\text{A.159})$$

The new shattered supports

$$\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$$

have only sizes two and three. If

$$a = |\mathcal{K} \cap \binom{E}{2}|, \quad b = |\mathcal{K} \cap \binom{E}{3}|,$$

then

$$|U| = a + b, qquad h(\Psi(B, S)) = (0, 0, a, b, 0, 0). \quad (\text{A.160})$$

Every nonzero balanced flag h -entry is equal to one. For the complementary interval $A \subset T$, the new two-supports are the complements of the members of $\mathcal{K} \cap \binom{E}{3}$, and the new three-supports are the complements of the members of $\mathcal{K} \cap \binom{E}{2}$.

Proof. Choose the counterexample by Lemma A.101. The four-coordinate theorem makes every coordinate active in the two upper endpoints B and T . The upper endpoint B cannot have a peripheral coordinate, by Lemma A.98; applying the same argument to the complementary counterexample $A \subset T$ shows that T has none either. Set complementation reverses the inclusions of the two coordinate sections, so it preserves the existence of a peripheral coordinate. Hence S and A are periphery-free as well; if either omitted a coordinate, its complement would have nested sections in that coordinate. Thus all four endpoints have full support.

For an ample $X \subseteq Q_E$, complement duality gives

$$R \in \text{Sh}(Q_E \setminus X) \iff E \setminus R \notin \text{Sh}(X). \quad (\text{A.161})$$

Since A uses every coordinate, all singleton supports are shattered by A ; therefore no four-subset is shattered by B . Neither B nor A is the full cube, so the five-set is not shattered either. Thus

$$|B| = |\text{Sh}(B)| = 1 + 5 + m_B + t_B.$$

If $m_B \leq 4$, Lemma A.99 makes B peripheral. Hence $m_B \geq 5$. Applying the same lemma to A , and using (A.161), gives

$$5 \leq m_A = 10 - t_B,$$

so $t_B \leq 5$. If $t_B = 0$, then B has VC dimension at most two, and Lemma A.16 supplies the forbidden augmentation. Hence $t_B \geq 1$, proving (A.158). Repeating the argument for $A \subset T$ gives the same bounds for T , and complementation then gives (A.159).

All four endpoints have full support, so their shattered-set systems agree in ranks zero and one. The preceding argument excludes ranks four and five for B , and therefore $\mathcal{K}$ is supported in ranks two and three. Ampleness gives

$$|U| = |B| - |S| = |\text{Sh}(B)| - |\text{Sh}(S)| = a + b.$$

Equation (A.160) and the flag statement now follow from Lemma A.12. Applying (A.161) to both ends of the interval interchanges new supports with their set complements, proving the last assertion. $\square$

Lemma A.103 (Paired defect cylinders force two neighbours in the intermediate layer). *Let $S \subsetneq B \subseteq Q_E$ be a minimum five-coordinate gap in the normal form of Proposition A.102. Put*

$$U = B \setminus S, \quad A = Q_E \setminus B, \quad T = Q_E \setminus S,$$

and write

$$\mathcal{K}_i = \mathcal{K} \cap \binom{E}{i}, \quad \mathcal{K}^* = \{E \setminus R : R \in \mathcal{K}\}, \quad \mathcal{H}_A(p) = \text{Sh}(A \cup \{p\}) \setminus \text{Sh}(A).$$

Define the eligible supports

$$\mathcal{E}(\mathcal{K}) = \mathcal{K}_2 \dot{\cup} \{R \in \mathcal{K}_3 : \text{no member of } \mathcal{K}_2 \text{ is contained in } R\}. \quad (\text{A.162})$$

Then, for every $p \in U$,

$$\mathcal{H}_S(p) \subseteq \mathcal{E}(\mathcal{K}), \quad \mathcal{H}_A(p) \subseteq \mathcal{E}(\mathcal{K}^*), \quad |\mathcal{H}_S(p)|, |\mathcal{H}_A(p)| \geq 2. \quad (\text{A.163})$$

Each $R \in \mathcal{E}(\mathcal{K})$ has a unique pattern $\sigma_R \in Q_R \setminus \pi_R(S)$. The cylinder

$$C_R = \{x \in Q_E : x|_R = \sigma_R\} \quad (\text{A.164})$$

is contained in T , and $R \in \mathcal{H}_S(p)$ exactly when $p \in C_R$. Dually, every $Q \in \mathcal{E}(\mathcal{K}^*)$ defines a cylinder $D_Q \subseteq B$, and $Q \in \mathcal{H}_A(p)$ exactly when $p \in D_Q$. Hence

$$p^{\{e\}} \in U \quad \text{for every } e \in E \setminus (R \cup Q), \quad R \in \mathcal{H}_S(p), \quad Q \in \mathcal{H}_A(p). \quad (\text{A.165})$$

Let $N_U(p) = \{e \in E : p^{\{e\}} \in U\}$. It is nonempty. If $N_U(p) = \{e\}$, then the other four coordinates can be labelled a, b, c, d so that

$$\begin{aligned} \mathcal{H}_S(p) &= \{\{a, b, c\}, \{a, b, d\}\}, \\ \mathcal{H}_A(p) &= \{\{a, c, d\}, \{b, c, d\}\}. \end{aligned} \quad (\text{A.166})$$

In particular,

$$\{e, a\}, \{e, b\} \in \mathcal{K}_2, \quad \{e, c\}, \{e, d\} \in \{E \setminus R : R \in \mathcal{K}_3\}. \quad (\text{A.167})$$

Moreover,

$$N_S(p) = \{a, b\}, \quad N_A(p) = \{c, d\}, \quad N_U(p) = \{e\}, \quad (\text{A.168})$$

and B contains the two squares through p on $\{e, a\}$ and $\{e, b\}$, while T contains the two squares through p on $\{e, c\}$ and $\{e, d\}$. This singleton alternative is impossible. Consequently,

$$|N_U(p)| \geq 2 \quad \text{for every } p \in U. \quad (\text{A.169})$$

Proof. We first convert the two one-concept failures into intersecting signed cylinders. We then show that a unique neighbor in U forces the four triple supports displayed in (A.166). Finally, a minimal blocking vertex and an isometric geodesic in S rule out that configuration.

The central profile supports $\mathcal{K}$ only in ranks two and three. Equation (A.27) therefore says exactly that a support contributing to $\mathcal{H}_S(p)$ is either a member of $\mathcal{K}_2$, or a member of $\mathcal{K}_3$ containing no member of $\mathcal{K}_2$. This is the first inclusion in (A.163); the complementary interval gives the second. None of the four one-concept families in (A.157) is ample, so Lemma A.15 gives both lower bounds.

For $R \in \mathcal{E}(\mathcal{K})$, the projection-defect count (A.26) is one. Since B shatters R , the sole pattern absent from $\pi_R(S)$ is σ_R . This proves the cylinder statement, and the dual argument gives $D_Q \subseteq B$. If both cylinders contain p , their intersection is the subcube through p whose free coordinates are $E \setminus (R \cup Q)$. It lies in $B \cap T = U$, proving (A.165).

We use the following elementary five-set observation. If $R \in \mathcal{E}(\mathcal{K})$, $Q \in \mathcal{E}(\mathcal{K}^*)$, and $R \cup Q = E$, then

$$Q = E \setminus R. \quad (\text{A.170})$$

Indeed, two two-sets cannot cover E . If their sizes are two and three, equality forces complementation. If both have size three, then $E \setminus Q \in \mathcal{K}_2$ and $E \setminus Q \subseteq R$, contradicting the eligibility of R .

Choose two distinct members of each defect family in (A.163). A fixed R has only one set complement, so one of the four cross pairs is not complementary. By (A.170), its union is not E , and (A.165) proves $N_U(p) \neq \emptyset$.

Suppose now that $N_U(p) = \{e\}$, and put $H = E \setminus \{e\}$. For any two chosen lower supports R_1, R_2 and upper supports Q_1, Q_2 , equation (A.165) gives

$$H \subseteq R_i \cup Q_j \quad (1 \leq i, j \leq 2). \quad (\text{A.171})$$

None of the four supports can be a two-set. Suppose, for example, that R_1 is a two-set and put $C = H \setminus R_1$. A two-set Q_j satisfying (A.171) must equal C . Then $E \setminus Q_j = R_1 \cup \{e\} \in \mathcal{K}_3$, while the covering law forces R_2 to contain R_1 ; it can neither be the same edge nor an eligible triple. Thus both Q_j 's are triples. A triple through e that covers H with R_1 is uniquely $E \setminus R_1$; if one Q_j has this form, the covering law again forces R_2 to contain R_1 . The only remaining two triples are $C \cup \{r\}$, one for each $r \in R_1$. Their complements are the two pairs $\{e, r\}$, and covering both with R_2 once more forces $R_1 \subseteq R_2$, contradicting distinctness or eligibility. The dual argument excludes a two-set among the Q_j 's.

Nor can one of the four three-sets contain e . If, say, $e \in R_i$, put $C = H \setminus R_i$, a two-set. A two-set Q_j covering H with R_i must equal $C = E \setminus R_i$, so at most one of the two distinct Q_j 's has this form. Every covering triple contains C , and its complementary pair lies inside R_i . Since such a triple is eligible for $\mathcal{K}^*$, that complementary pair belongs to $\mathcal{K}_2$, contradicting the eligibility of R_i . Hence a second upper support cannot exist. The dual argument handles the Q_j 's.

The four supports are therefore three-subsets of the four-set H . Write the two lower supports as $H \setminus \{c\}$ and $H \setminus \{d\}$, and denote the remaining elements by a, b . The covering law forces each upper support to contain c, d ; since the two are distinct, they are $H \setminus \{a\}$ and $H \setminus \{b\}$. This is (A.166). No further lower support is possible: it would have to contain a, b ; the edge $\{a, b\}$ is excluded by eligibility of the two displayed lower triples, a triple through e contains one of the forced edges $\{e, a\}, \{e, b\}$, and the two remaining triples are already listed. The dual argument excludes further upper supports.

Taking complements of the two upper triples gives $\{e, a\}, \{e, b\} \in \mathcal{K}_2$. Taking complements of the two lower triples gives the two asserted edges of the complementary graph, proving (A.167).

Finally, the cylinder of $H \setminus \{c\}$ is the square through p with free coordinates c, e , and it lies in T ; the other lower cylinder gives the d, e -square. The two upper cylinders give the a, e - and

b, e -squares in B . Since the only U -neighbor of p is its e -neighbor, its a, b -neighbors belong to S , and its c, d -neighbors belong to A . This proves (A.168) and the final assertion.

It remains to rule out this normal form. By Lemma A.1, there is a blocking vertex $s \in S$ for p . Choose one for which

$$D = \delta(p, s)$$

is inclusion-minimal. The blocking condition and (A.168) give $D \cap \{a, b\} = \emptyset$. Because B is isometric and $p, s \in B$, some first step of a B -geodesic from p to s flips a coordinate of D . The c - and d -neighbours of p belong to A , whereas its e -neighbour belongs to U . Thus $e \in D$.

The two T -squares in the conclusion contain $p^{\{e, c\}}$ and $p^{\{e, d\}}$, so neither vertex belongs to S . Also $p^{\{e\}} \in U$, not in S . Hence

$$D = \{c, d, e\}, \quad s = p^{\{c, d, e\}}. \quad (\text{A.172})$$

Minimality of D says that no vertex p^F , with $F \subsetneq D$, belongs to S : such a vertex would itself be a blocking vertex with a smaller difference set, because $F \cap N_S(p) = \emptyset$.

Now $p^{\{a\}} \in S$. Consider an S -geodesic from this vertex to s , which exists because S is ample and hence isometric. The coordinate a cannot be removed before all of c, d, e have been added: otherwise the geodesic would contain a vertex $p^F \in S$ for some proper subset $F \subsetneq D$. It follows that $p^{\{a, c, d, e\}} \in S$. The four vertices

$$p^{\{b\}}, \quad p^{\{a\}}, \quad p^{\{c, d, e\}}, \quad p^{\{a, c, d, e\}}$$

realize all four traces on $\{e, a\}$. Thus $\{e, a\} \in \text{Sh}(S)$, contradicting $\{e, a\} \in \mathcal{K}_2 = \text{Sh}(B) \setminus \text{Sh}(S)$ from (A.167). The singleton alternative is impossible, proving (A.169). $\square$

Corollary A.104 (Minimum degree of the intermediate layer). *In the setting of Lemma A.103, the subgraph of Q_E induced by $U = B \setminus S$ has minimum degree at least two. Consequently, each of its connected components contains an even cycle. Moreover,*

$$4 \leq |U| \leq 10,$$

so a shortest such cycle has length 4, 6, 8, or 10.

Proof. The degree assertion is (A.169). Every finite graph of minimum degree at least two contains a cycle, and every subgraph of a hypercube is bipartite. Proposition A.102 gives $|S| \geq 11$ and $|B| \leq 21$, so $|U| \leq 10$. This proves the remaining assertions. $\square$

Lemma A.105 (Persistence of defect labels along an intermediate edge). *Retain the setting of Lemma A.103, and let $p, q = p^{\{e\}} \in U$. For $X = S$ and for $X = A$, one has*

$$\{R \in \mathcal{H}_X(p) : e \notin R\} = \{R \in \mathcal{H}_X(q) : e \notin R\} = \mathcal{H}_X(p) \cap \mathcal{H}_X(q). \quad (\text{A.173})$$

Consequently, along a cycle of the subgraph induced by U , membership of a fixed support R in the defect family can change only across an edge whose color belongs to R . Across such an edge, R cannot occur at both endpoints. In particular, the number of changes encountered while traversing the cycle is even.

Proof. If $e \notin R$, then $p|_R = q|_R$. The characterization (A.27) therefore gives $R \in \mathcal{H}_X(p)$ if and only if $R \in \mathcal{H}_X(q)$. If $e \in R$, the two traces on R are different. An eligible support has a unique pattern missing from the projection of X , so it cannot belong to both defect families. This proves (A.173). The cycle assertions follow by reading these two alternatives around the closed walk. $\square$

Lemma A.106 (Convex signed defect carriers). *Retain the setting and notation of Lemma A.103. For $R \in \mathcal{E}(\mathcal{K})$ and $Q \in \mathcal{E}(\mathcal{K}^*)$, put*

$$\Gamma_R^- = B \cap C_R = U \cap C_R, \quad \Gamma_Q^+ = T \cap D_Q = U \cap D_Q. \quad (\text{A.174})$$

After the coordinates fixed by the corresponding cylinder are suppressed, each nonempty carrier is ample. Moreover, its one-inclusion graph is a convex subgraph of the graph induced by U . Finally,

$$R \in \mathcal{H}_S(p) \iff p \in \Gamma_R^-, \quad Q \in \mathcal{H}_A(p) \iff p \in \Gamma_Q^+. \quad (\text{A.175})$$

Proof. The cylinder C_R is disjoint from S , so $B \cap C_R = (S \dot{\cup} U) \cap C_R = U \cap C_R$. Fixing the pattern σ_R on R is a sequence of restrictions of the ample family B . Restrictions preserve ampleness, proving the first assertion for Γ_R^- . The upper assertion follows in the same way from $D_Q \cap A = \emptyset$ and the ample family T .

Let $x, y \in \Gamma_R^-$. Since B is isometric, every B -geodesic from x to y has their Hamming distance. Such a geodesic cannot flip a coordinate of R , since the two endpoints agree on R ; hence it stays in C_R , and therefore in Γ_R^- . In particular, a geodesic of Hamming length lies in U . Conversely, every U -geodesic between x and y has at least their Hamming distance, and equality is attained inside Γ_R^- . A shortest one cannot flip a coordinate of R , so it also stays in the carrier. Thus the carrier is convex in the graph induced by U . The proof for Γ_Q^+ is identical, using T . The equivalences in (A.175) are the two cylinder characterizations from Lemma A.103. $\square$

Corollary A.107 (Interval law on a shortest relative cycle). *Let C be a shortest cycle in the graph induced by U , of length $2k$. After a cyclic shift and a relabelling of its active coordinates, the edge-color word of C is*

$$e_1, e_2, \dots, e_k, e_1, e_2, \dots, e_k, \quad 2 \leq k \leq 5. \quad (\text{A.176})$$

The intersection of C with any signed defect carrier is either the whole cycle or a single cyclic interval. If the carrier has support R , a proper interval has at most $5 - |R|$ edges, and the colors of its two boundary edges belong to R .

If $k \geq 3$, the whole-cycle alternative is impossible. Consequently, on a shortest ten-cycle every pair carrier contains at most four consecutive cycle vertices, every triple carrier contains at most three, and the two boundary-edge colors of each carrier are distinct.

Proof. A shortest cycle in any graph is isometric: a path shorter than the shorter of its two cycle arcs would create a shorter cycle. Hence every arc of C with at most k edges is a hypercube geodesic and uses each coordinate at most once. The first k edge colors are therefore distinct. The other half is another geodesic between the same two vertices and uses the same k colors. Applying the distinctness statement to every cyclic block of k edges gives $e_{i+k} = e_i$, proving (A.176).

By Lemma A.106, a carrier is convex in the graph induced by U . Its intersection with the isometric cycle is therefore geodesically convex in that cycle, hence is the whole cycle or one interval. The endpoints of a proper interval agree on the $|R|$ fixed coordinates, so their Hamming distance, which equals the length of the interval, is at most $5 - |R|$. If a boundary edge had a color outside R , its two endpoints would have the same trace on R , contrary to maximality of the interval.

Suppose that a carrier contains C and $k \geq 3$. The canonical cycle in (A.176) shatters any pair of its active coordinates. The carrier is ample, so it strongly shatters that pair and contains a square. This square lies in U , contradicting the choice of a cycle of length greater than four as shortest. Thus every carrier interval is proper when $k \geq 3$. On a ten-cycle, two occurrences of the same edge color are separated by five edges. An interval bounded by them has more than $5 - |R|$ edges for $|R| \in \{2, 3\}$, so the boundary colors are distinct. $\square$

Corollary A.108 (Endpoint profile forced by a shortest ten-cycle). *If a shortest cycle in U has length ten, then it contains every vertex of U , and*

$$\begin{array}{c|cccc} X & S & A & B & T \\ \hline |X| & 11 & 11 & 21 & 21 \\ m_X & 5 & 5 & 10 & 10 \\ t_X & 0 & 0 & 5 & 5. \end{array} \quad (\text{A.177})$$

In particular,

$$|\mathcal{K}_2| = |\mathcal{K}_3| = 5. \quad (\text{A.178})$$

The lower and upper defect carriers give two covers of this ten-cycle, each with multiplicity at least two at every vertex. Their traces on the cycle are the intervals described in Corollary A.107. If a lower carrier of support R and an upper carrier of support Q meet, then

$$|R \cup Q| \geq 4. \quad (\text{A.179})$$

When equality holds, the sole color outside $R \cup Q$ is the color of one of the two cycle edges incident with every common vertex.

Proof. The cycle has ten distinct vertices, while Corollary A.104 gives $|U| \leq 10$; hence U is exactly the cycle. Its canonical word uses all five coordinates and realizes all four patterns on every coordinate pair. Thus $m_B = m_T = 10$. Write t_B, t_T for the numbers of shattered triples. The central profile gives

$$|B| = 16 + t_B, \quad |T| = 16 + t_T, \quad t_B, t_T \leq 5.$$

Since $S = Q_E \setminus T$ and $|B| - |S| = |U| = 10$, one obtains $t_B + t_T = 10$. Therefore $t_B = t_T = 5$, and complement duality gives

$$m_S = m_A = 5, \quad t_S = t_A = 0.$$

This proves (A.177). Taking the differences of the two shattered-set systems gives (A.178).

The multiplicity-two assertion is (A.163), and the interval description is Corollary A.107. If two opposite carriers meet at p , their defining cylinders have intersection equal to the full subcube through p on the coordinates outside $R \cup Q$, and that subcube lies in U . The ten-cycle contains no square, proving (A.179). If the union has size four, the intersection is an edge of U ; since U is the cycle, it must be one of the two cycle edges at each of its vertices. $\square$

Theorem A.109 (Five-color circular-carrier exclusion). *Let C be the canonical edge-colored ten-cycle in (A.176). There is no support system $\mathcal{K} \subseteq \binom{E}{2} \dot{\cup} \binom{E}{3}$, with $|\mathcal{K}_2| = |\mathcal{K}_3| = 5$, having the following data. For every support in $\mathcal{E}(\mathcal{K})$, choose one trace pattern on C ; do the same for every support in $\mathcal{E}(\mathcal{K}^*)$. Require every chosen trace fiber to be a nonempty convex interval, every cycle vertex to lie in at least two chosen fibers on each side, and every intersecting lower–upper pair of fibers to have supports whose union has size at least four.*

Proof. Write

$$X = \mathcal{K}_2, \quad Z = \{E \setminus R : R \in \mathcal{K}_3\}.$$

Thus X and Z are five-element families of two-subsets of E . Join $x \in X$ to $z \in Z$ when $x \cap z = \emptyset$, and let X_0 and Z_0 be the isolated vertices in the two parts. By (A.162), the lower eligible supports are

$$X \dot{\cup} \{E \setminus z : z \in Z_0\},$$

and the upper eligible supports are

$$Z \dot{\cup} \{E \setminus x : x \in X_0\}.$$

We call the pair supports mandatory and the displayed triple supports extra.

Every cycle vertex lies in an extra carrier on at least one side. Indeed, otherwise two mandatory lower carriers and two mandatory upper carriers contain that vertex. The cross-union condition says that each of the two lower pairs is disjoint from each of the two upper pairs, which gives a four-cycle in the bipartite disjointness graph. This is impossible by Lemma A.14. Every triple trace fiber on the canonical ten-cycle has at most three vertices, by Corollary A.107. Hence

$$10 \leq 3(|X_0| + |Z_0|). \quad (\text{A.180})$$

We next note that $|X_0|, |Z_0| \leq 2$. To see this, three distinct edges of K_5 have at most four common edge transversals, where an edge transversal means an edge meeting all three. If two of the three edges are disjoint, their four cross edges are the only possibilities. Otherwise choose two that meet at c . If the third also contains c , only the four edges through c can meet all three; if it does not, at most two of those four edges and the edge joining the two other endpoints of the chosen pair can do so. If X_0 contained three members, every member of the five-element family Z would be such a transversal, a contradiction. The argument with X and Z interchanged is identical. Equation (A.180) now gives

$$|X_0| = |Z_0| = 2. \quad (\text{A.181})$$

For $x \in X_0$, consider its nonempty mandatory lower pair fiber. It cannot meet any mandatory upper pair fiber, because x meets every member of Z . Both upper labels at each of its vertices must therefore be the two extra carriers. In particular, the pair fiber is contained in the upper triple fiber with support $E \setminus x$. These two trace patterns together fix every coordinate, so their fibers meet in at most one cube vertex. The mandatory pair fiber is therefore a singleton. Dually, the mandatory upper pair fiber of every $z \in Z_0$ is a singleton.

Index the colors in their cyclic order in (A.176). A pair trace fiber on C is a singleton exactly when its two colors are consecutive in this order. Thus X_0 and Z_0 are two-edge subsets of the color pentagon. Every edge in X_0 meets every edge in Z_0 . Two nonadjacent pentagon edges have only one common pentagon-edge transversal, so the two members of X_0 are adjacent. The only pentagon edges meeting both are those two edges themselves. Therefore $X_0 = Z_0$. After a cyclic relabelling, and writing ij for $\{i, j\}$,

$$X_0 = Z_0 = \{01, 04\}. \quad (\text{A.182})$$

The five edges of K_5 meeting both 01 and 04 are 01, 02, 03, 04, 14. Since every member of X and of Z must meet both isolated edges, it follows that

$$X = Z = \{01, 02, 03, 04, 14\}. \quad (\text{A.183})$$

It remains only to read the forced trace intervals. Write the vertices of C as $v_0, \dots, v_9$, with the edge $v_i v_{i+1}$ colored $i \pmod{5}$. The singleton fibers of 01 are $\{v_1\}$ and $\{v_6\}$, while those of 04 are $\{v_0\}$ and $\{v_5\}$. By the antipodal symmetry, suppose the lower 01-fiber is $\{v_1\}$. It is contained in both upper extra fibers, so the latter are forced to be

$$\{v_0, v_1, v_2\} \quad \text{on } 234, \quad \{v_9, v_0, v_1\} \quad \text{on } 123. \quad (\text{A.184})$$

Their common vertices force the lower 04-fiber to be $\{v_0\}$.

The upper 01-fiber cannot be $\{v_1\}$: the two corresponding lower extra fibers would then equal the intervals in (A.184), whereas lower and upper fibers with the same triple support may not meet. Hence the upper 01-fiber is $\{v_6\}$, and the same containment argument forces the lower extra fibers to be

$$\{v_5, v_6, v_7\} \quad \text{on } 234, \quad \{v_4, v_5, v_6\} \quad \text{on } 123, \quad (\text{A.185})$$

and the upper 04-fiber to be $\{v_5\}$.

The vertex v_3 lies in none of the four extra fibers and none of the four isolated mandatory fibers. Its two lower and two upper labels must therefore be chosen from the pair supports 02, 03, 14. Every chosen lower pair must be disjoint from every chosen upper pair, again producing a four-cycle in the bipartite disjointness graph. This final contradiction proves the theorem. $\square$

Corollary A.110 (Exclusion of the ten-cycle branch). *In a minimum five-coordinate gap, a shortest cycle in the intermediate layer has length 4, 6, or 8.*

Proof. A shortest ten-cycle would have the exact endpoint and carrier profile of Corollary A.108, contradicting Theorem A.109. $\square$

Lemma A.111 (Exhaustion by a shortest eight-cycle). *If a minimum five-coordinate gap has a shortest intermediate-layer cycle C of length eight, then $U = V(C)$. Moreover, the vertices of U agree on the unique coordinate not used by the edges of C .*

Proof. Corollary A.104 gives $|U| \leq 10$, so at most two vertices of U lie outside C . No such vertex can have two neighbours on C . Indeed, those two cycle vertices would be at Hamming distance two. Since C is isometric by Corollary A.107, their two-edge path on C , together with the outside vertex, would give a four-cycle in U , contrary to the choice of C .

One outside vertex would consequently have degree at most one in U , contradicting (A.169). If there were two, each would have to be adjacent to the other and to a distinct cycle vertex. They would give a three-edge path between two vertices of C . The distance of those endpoints along the isometric cycle is odd and at most three, so the outside path and a shortest cycle arc would form a four- or six-cycle in U , again a contradiction. Thus $U = V(C)$.

The repeated color word (A.176) uses exactly four coordinates when $|C| = 8$. No edge of $C = U$ flips the remaining coordinate, so all vertices of U agree there. $\square$

Lemma A.112 (Elimination of an unused coordinate). *Let $S \subsetneq B \subseteq Q_E$ be a minimum five-coordinate gap, and suppose that all vertices of $U = B \setminus S$ agree in coordinate r . Orient r so that U lies in its zero-section, and write the two sections of the upper and lower endpoints as*

$$B = (M \times \{0\}) \cup (Y \times \{1\}), \quad S = (L \times \{0\}) \cup (Y \times \{1\}), \quad M \setminus L = U. \quad (\text{A.186})$$

Then

$$U \cap Y = \emptyset. \quad (\text{A.187})$$

Moreover, no member of $\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$ contains r .

Proof. Put $I_M = M \cap Y$ and $I_L = L \cap Y$. The two-section shattering calculation used in Lemma A.20 shows that both intersections are ample. Suppose that $I_L \subsetneq I_M$. If $I_L = \emptyset$, choose any $p \in I_M$; otherwise Theorem A.18 gives $p \in I_M \setminus I_L$ for which $I_L \cup \{p\}$ is ample. In either case, Lemma A.20, applied to the lower endpoint S , shows that adjoining the missing zero-lift of p preserves ampleness. This contradicts the definition of a minimum gap. Thus $I_L = I_M$, which is (A.187).

For an ample two-section family, the supports containing r are obtained from the intersection of the shattered sets of its two sections. Equation (A.34) applied to the two endpoints, together with $I_L = I_M$, gives

$$\text{Sh}(M) \cap \text{Sh}(Y) = \text{Sh}(I_M) = \text{Sh}(I_L) = \text{Sh}(L) \cap \text{Sh}(Y).$$

Thus the supports containing r are the same at the two endpoints. $\square$

Theorem A.113 (Exclusion of the eight-cycle branch). *In a minimum five-coordinate gap, a shortest cycle in the intermediate layer has length 4 or 6.*

Proof. Suppose that a shortest cycle has length eight, and use the section notation of (A.186). Put $\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$. Lemma A.112 says that no member of $\mathcal{K}$ contains r .

Write $a = |\mathcal{K}_2|$ and $b = |\mathcal{K}_3|$. The central profile and $|U| = 8$ give $a + b = 8$. All these supports use the four active cycle colors, so $a \leq 6$ and $b \leq 4$; hence

$$a \geq 4, \quad b \geq 2. \quad (\text{A.188})$$

Let

$$X = \mathcal{K}_2, \quad Z = \{E \setminus R : R \in \mathcal{K}_3\}.$$

Each member of Z has the form $\{r, i\}$, where i is an active color. As in the proof of Theorem A.109, join $x \in X$ to $z \in Z$ when they are disjoint. A lower eligible triple would correspond to an isolated $z = \{r, i\}$, which would mean that every pair in X contains i . Only three active pairs contain a fixed color, whereas $a \geq 4$. Thus every lower eligible support is a pair.

An upper eligible triple corresponds to an isolated $x \in X$. Such an x must contain the active color of every member of Z . Since $b \geq 2$, there is at most one such x , and hence at most one upper triple carrier. Consider a cycle vertex outside that carrier, or any cycle vertex if the carrier does not exist. Its two lower labels and two upper labels are pair supports. The cross-cylinder law makes each chosen lower pair disjoint from each chosen upper pair, producing a four-cycle in the bipartite disjointness graph. This contradicts Lemma A.14. Therefore the sole upper triple carrier would have to contain the entire eight-cycle. This is impossible by Corollary A.107. The eight-cycle branch is excluded. $\square$

Lemma A.114 (Small-ear normal form around a shortest six-cycle). *Suppose that a minimum five-coordinate gap has a shortest intermediate-layer cycle C of length six, and put $W = U \setminus V(C)$. Then the graph induced by U is connected. If $W \neq \emptyset$, its induced graph is one of*

$$P_3, \quad P_4, \quad K_{1,3}. \quad (\text{A.189})$$

No vertex of W has two neighbours on C , and every leaf of the tree induced by W has exactly one neighbour on C .

More generally, suppose that $w, w' \in W$ have respective cycle neighbours u, u' , and let t be their distance in the tree induced by W . Then

$$d_C(u, u') + t \geq 4, \quad d_C(u, u') \equiv t \pmod{2}. \quad (\text{A.190})$$

Consequently, if $W = P_3$, only its endpoints attach to C , and their cycle neighbours are at distance two. If $W = K_{1,3}$, only its leaves attach to C , at the three alternating cycle vertices.

Proof. By (A.169), every component of the graph induced by U has minimum degree at least two. Hence each contains a cycle. A component disjoint from C would contain at least six vertices, giving $|U| \geq 12$, contrary to Corollary A.104. Thus U is connected and $|W| \leq 4$.

No vertex outside C has two cycle neighbours: as in the proof of Lemma A.111, isometry of C would turn those two edges into a four-cycle. The graph induced by W is a forest, because it has at most four vertices and U has no four-cycle. Each of its leaves must therefore use its remaining incident edge to attach to C , and the attachment is unique.

A component of this forest cannot have one vertex, by minimum degree. Nor can it have two vertices. In the latter case both vertices attach to C , giving a three-edge path between two cycle vertices. Parity and the absence of a four-cycle force those endpoints to be antipodal on C . The path is then a cube geodesic of length three and uses the three colors of C . It lies in the same three-cube as C . But a canonical six-cycle in a three-cube omits an antipodal pair, whereas the two internal vertices of the path are adjacent. This is impossible. Thus every forest component has at least three vertices. Since $|W| \leq 4$, there is only one, and the three possible trees are exactly those in (A.189).

For the attachment law, the path through W and its two attachment edges has length $t + 2$. Together with a shortest u, u' -arc of C , it forms a cycle in U . Its length is even and at least six, which gives both assertions in (A.190).

If $W = P_3$, an attachment at its middle vertex would have to be antipodal on C to each of the two leaf attachments. The two leaves would then attach to the same cycle vertex and form a four-cycle through the middle vertex. Hence only the leaves attach, and (A.190) puts their cycle neighbours at distance two. The same argument excludes an attachment at the center of $K_{1,3}$. Its three leaf attachments are pairwise at cycle distance two, so they are the three alternating vertices of C . $\square$

Corollary A.115 (Elimination of the bare and three-vertex-ear six-cycles). *In the setting of Lemma A.114, the outside graph W is either P_4 or $K_{1,3}$. In particular,*

$$|U| = 10. \tag{A.191}$$

Proof. Put $\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$. Suppose first that $W = \emptyset$. Then $U = C$ uses three coordinates and is constant in the other two. Applying Lemma A.112 to both unused coordinates shows that no support in $\mathcal{K}$ contains either of them. The central profile supports $\mathcal{K}$ only in ranks two and three, so there are at most $\binom{3}{2} + \binom{3}{3} = 4$ possible supports. This contradicts $|\mathcal{K}| = |U| = 6$.

Now suppose that $W = P_3$. Its two leaf attachments are separated by a two-edge arc of C . The path through W , together with that arc, is another shortest six-cycle. If the two arc colors are e_1 and e_2 , its canonical repeated color word makes the four-edge ear word

$$f, e_1, e_2, f \tag{A.192}$$

up to reversal. The color f is not the third color of C : if it were, all three vertices of W would lie in the three-cube of C , which has only the two vertices omitted by its canonical six-cycle. Thus U uses exactly four coordinates and is constant in the fifth, say r .

Lemma A.112 implies that every member of $\mathcal{K}$ avoids r . Write $a = |\mathcal{K}_2|$ and $b = |\mathcal{K}_3|$. Since $a + b = |U| = 9$, while there are only six pairs and four triples on the four active colors,

$$a \geq 5, \quad b \geq 3. \tag{A.193}$$

As before, let $X = \mathcal{K}_2$ and $Z = \{E \setminus R : R \in \mathcal{K}_3\}$. Each member of Z is $\{r, i\}$ for an active color i . No member of Z is isolated in the bipartite disjointness graph: isolation would require all at

least five pairs in X to contain the same color, but only three active pairs do. No member of X is isolated either, because it would have to contain the at least three distinct active colors indexed by Z . Hence both eligible-support families contain only pairs. At any vertex of U , two lower and two upper defect labels therefore give a four-cycle in the bipartite disjointness graph, contradicting Lemma A.14. Thus $W \neq P_3$, and the small-ear normal form proves the assertion. $\square$

Lemma A.116 (A ten-vertex intermediate layer uses all five colors). *In a minimum five-coordinate gap with $|U| = 10$, the graph induced by U uses all five coordinates.*

Proof. Suppose that every vertex of U agrees in coordinate r . By Lemma A.112, every support in $\mathcal{K} = \text{Sh}(B) \setminus \text{Sh}(S)$ avoids r . There are exactly $\binom{4}{2} + \binom{4}{3} = 10$ rank-two and rank-three supports on the other four colors. Since the central profile gives $|\mathcal{K}| = |U| = 10$, all of them belong to $\mathcal{K}$.

Every lower eligible support is consequently a pair: each active triple contains an active pair in $\mathcal{K}$. After complementation, every upper eligible support is also a pair. Two lower and two upper labels at any vertex of U would give a four-cycle in the bipartite disjointness graph, contrary to Lemma A.14. Thus no coordinate is unused by U . $\square$

Proposition A.117 (Five-color bowtie normal form for the six-cycle branch). *If a minimum five-coordinate gap has a shortest cycle of length six, then, after a cube symmetry, its intermediate layer is*

$$U = \{\emptyset, 0, 01, 012, 12, 2\} \cup \{\emptyset, 0, 04, 034, 34, 3\}. \quad (\text{A.194})$$

Thus U is the union of a canonical six-cycle on colors $\{0, 1, 2\}$ and a canonical six-cycle on colors $\{0, 3, 4\}$, sharing exactly the edge $\emptyset 0$. Moreover,

$$|\mathcal{K}_2| = |\mathcal{K}_3| = 5. \quad (\text{A.195})$$

Proof. By Corollary A.115, the outside tree is P_4 or $K_{1,3}$, and $|U| = 10$.

Suppose first that it is a claw. Its three leaves attach to alternating vertices of the base cycle. For each pair of leaves, the path through the center and the two-edge arc between their attachments form a canonical six-cycle. Reading its repeated color word shows that the three leaf-to-cycle edges have one common color f , while the three center-to-leaf edges have the three base-cycle colors. The common color f is different from every base color, because each of the three two-edge arcs omits a different base color. Hence U uses only four coordinates, contradicting Lemma A.116.

Now let the outside tree be the path P_4 , and let u, v be the cycle neighbours of its leaves. The attachment law gives $d_C(u, v) \in \{1, 3\}$. If the distance is three, the five-edge path between the antipodal vertices u, v flips the three base colors an odd number of times and all other colors an even number of times. Its two excess flips therefore use one color twice. That color cannot be a base color, for then all four internal vertices would lie in the three-cube of C , outside a six-cycle that omits only two vertices. Thus the path, and hence U , uses only four colors, again contradicting Lemma A.116.

It remains that u, v are adjacent, say by an edge of color 0. Together with that edge, the five-edge path is a canonical six-cycle. Its path word is

$$f, g, 0, f, g \quad (\text{A.196})$$

up to reversal and interchange of f, g . If both f, g were base colors, the four path vertices would again have to lie among the two vertices of the base three-cube outside C . If exactly one were a base color,

U would use only four coordinates. Therefore f, g are the two colors outside the base three-cube. Relabel them as 3, 4; the resulting vertex set is exactly (A.194). A direct Hamming-distance check also shows that the internal path vertices have no further neighbours on the base cycle.

Finally, write m_B, t_B, m_T, t_T for the numbers of shattered pairs and triples at the two upper endpoints. The central profile gives

$$|B| = 6 + m_B + t_B, \quad |T| = 6 + m_T + t_T.$$

Since $|B| - |S| = 10$ and $T = Q_E \setminus S$, their sum is 42. The four support counts therefore sum to 30, their maximum possible value. Hence $m_B = m_T = 10$ and $t_B = t_T = 5$. Complement duality gives five new pairs and five new triples, proving (A.195). $\square$

Lemma A.118 (Trace fibers on the five-color bowtie). *Let U be the bowtie in (A.194). Every trace fiber on a three-element support has at most three vertices.*

Now let $x_1, x_2 \in \binom{E}{2}$ be distinct and intersecting. For each i , choose a singleton trace fiber F_i on x_i and a trace fiber H_i on $E \setminus x_i$. If

$$F_1 \cup F_2 \subseteq H_1 \cap H_2,$$

then, up to interchanging x_1, x_2 , the complete list of possibilities is the following. The last column records $H_1 \cup H_2$.

$\{x_1, x_2\}$	$F_1 \cup F_2$	$H_1 \cup H_2$
$\{01, 02\}$	$\{12, 012\}$	$\{01, 012, 12, 2\}$
$\{01, 12\}$	$\{12, 2\}$	$\{\emptyset, 012, 12, 2\}$
$\{02, 12\}$	$\{012, 01\}$	$\{0, 01, 012, 12\}$
$\{03, 04\}$	$\{034, 34\}$	$\{04, 034, 34, 3\}$
$\{03, 34\}$	$\{034, 04\}$	$\{0, 04, 034, 34\}$
$\{04, 34\}$	$\{34, 3\}$	$\{\emptyset, 3, 34, 034\}$.

(A.197)

In particular, x_1, x_2 are two edges of one of the color triangles $\{0, 1, 2\}$ and $\{0, 3, 4\}$. Both singleton fibers and both complementary triple fibers lie in the corresponding six-cycle.

Proof. The bowtie contains no square. A trace fiber on a three-element support lies in a two-dimensional subcube, so four vertices in such a fiber would form a square. This proves the first assertion.

For the second assertion, the pair supports admitting a singleton trace fiber are

$$01, 02, 12, 03, 04, 34.$$

Their singleton fibers are, respectively,

$$\{12\}, \quad \{012\}, \quad \{01\} \text{ or } \{2\}, \quad \{034\}, \quad \{34\}, \quad \{3\} \text{ or } \{04\}.$$

Two complementary triple fibers H_1, H_2 containing both singleton vertices force those vertices to agree outside $x_1 \cap x_2$. Thus they are the endpoints of an edge of that common color. Comparing the displayed singleton fibers leaves exactly the six rows of (A.197). Reading the two complementary projections gives the last column. $\square$

Theorem A.119 (Five-color bowtie-carrier exclusion). *Let U be the bowtie in (A.194). There is no support system $\mathcal{K} \subseteq \binom{E}{2} \cup \binom{E}{3}$, with*

$$|\mathcal{K}_2| = |\mathcal{K}_3| = 5,$$

having the following data. For every support in $\mathcal{E}(\mathcal{K})$, choose one nonempty trace fiber on U ; do the same for every support in $\mathcal{E}(\mathcal{K}^*)$. Require every vertex of U to lie in at least two chosen fibers on each side, and require every intersecting lower–upper pair of fibers to have supports whose union has size at least four.

Proof. Put

$$X = \mathcal{K}_2, \quad Z = \{E \setminus R : R \in \mathcal{K}_3\},$$

and join $x \in X$ to $z \in Z$ when $x \cap z = \emptyset$. Let X_0, Z_0 be the isolated vertices in the two parts. As in the proof of Theorem A.109, the mandatory supports are the pairs in X and Z , while the extra supports are

$$\{E \setminus z : z \in Z_0\} \quad \text{and} \quad \{E \setminus x : x \in X_0\}.$$

Every vertex of U lies in an extra fiber on at least one side. Otherwise its two lower and two upper labels are pair supports, and the cross-union condition makes each lower pair disjoint from each upper pair. This is a four-cycle in the bipartite disjointness graph, contrary to Lemma A.14. Lemma A.118 bounds every extra fiber by three vertices, whence

$$10 \leq 3(|X_0| + |Z_0|).$$

The common-transversal argument in the proof of Theorem A.109 gives $|X_0|, |Z_0| \leq 2$: five edges cannot all meet three prescribed distinct edges of K_5 . Consequently

$$|X_0| = |Z_0| = 2.$$

Write $X_0 = \{x_1, x_2\}$. The nonempty lower pair fiber on x_i cannot meet a mandatory upper pair fiber, since x_i meets every member of Z . At each of its vertices, the two required upper labels must therefore be the two extra upper fibers. In particular, it is contained in the extra fiber on $E \setminus x_i$, so fixing the two complementary supports makes it a singleton. Both singleton fibers are contained in both extra upper fibers.

There are five members of Z , all meeting both x_1, x_2 . The two edges x_1, x_2 cannot be disjoint, because only their four cross edges meet both. They are therefore intersecting, and Lemma A.118 says that X_0 is a two-edge wedge in one of the two color triangles. Its two isolated pair fibers and the two upper extra fibers lie in the associated six-cycle. The dual argument gives the same conclusion for Z_0 , its upper pair fibers, and the two lower extra fibers.

Every member of X_0 meets every member of Z_0 . If the two wedges lie in the same color triangle, all four extra fibers and all four isolated mandatory fibers lie in the same six-cycle. Any vertex of the other cycle outside the shared edge lies in none of them. If the wedges lie in different triangles, the cross-intersection condition forces

$$X_0 = \{01, 02\}, \quad Z_0 = \{03, 04\},$$

or the reverse. The first and fourth rows of (A.197) then show that neither endpoint of the shared edge $\emptyset 0$ lies in an extra or isolated mandatory fiber.

In either case choose a vertex p outside all extra fibers and all isolated mandatory fibers. Its two lower labels are distinct pairs in $X \setminus X_0$, and its two upper labels are distinct pairs in $Z \setminus Z_0$. The cross-union condition makes every chosen lower pair disjoint from every chosen upper pair, producing a four-cycle in the bipartite disjointness graph. This final contradiction proves the theorem. $\square$

Corollary A.120 (Exclusion of the six-cycle branch). *In a minimum five-coordinate gap, a shortest cycle in the intermediate layer cannot have length six.*

Proof. Proposition A.117 gives the bowtie and its five-pair/five-triple support split. Its lower and upper defect carriers satisfy the hypotheses of Theorem A.119 by Lemma A.103 and the cross-cylinder law (A.165). $\square$

Corollary A.121 (Four-cycle reduction). *Every minimum five-coordinate gap has a four-cycle in its intermediate layer.*

Proof. Corollary A.104 supplies a shortest cycle of length 4, 6, 8, or 10. The other three lengths are excluded by Corollaries A.120, A.113, and A.110. $\square$

Lemma A.122 (The intermediate layer contains no three-cube). *In a minimum five-coordinate gap, the graph induced by U contains no three-dimensional subcube.*

Proof. Suppose that a three-face F of Q_E is contained in U . Since $8 \leq |U| \leq 10$, there are at most two vertices outside F . If $U = F$, then U uses only three coordinates. Applying Lemma A.112 to the other two coordinates leaves at most $\binom{3}{2} + \binom{3}{3} = 4$ possible supports in $\mathcal{K}$, contrary to $|\mathcal{K}| = |U| = 8$.

If there is exactly one vertex outside F , it has at most one neighbour in F , contradicting the minimum-degree conclusion of Lemma A.103. Suppose finally that there are two outside vertices. Each has at most one neighbour in F , so minimum degree forces the two outside vertices to be adjacent and each to have a neighbour in F . A vertex adjacent to F differs from it in one of the two coordinates fixed on F . The two outside vertices must differ in the same fixed coordinate; otherwise they differ in both fixed coordinates and cannot be adjacent. Hence U uses only the three coordinates of F and one further coordinate. This contradicts Lemma A.116. $\square$

Lemma A.123 (A degree-two vertex in the square branch). *In a minimum five-coordinate gap, the graph induced by U has a vertex of degree exactly two.*

Proof. Fix a square $C \subseteq U$, with active colors a, b , and translate so that its three inactive coordinates are zero. Suppose for a contradiction that every vertex of U has degree at least three. For each inactive color i , let $S_i \subseteq C$ be the set of square vertices whose i -neighbour belongs to U . Put

$$m = \sum_i |S_i|, \quad r = |\{i : S_i \neq \emptyset\}|.$$

Every vertex of C already has its two square neighbours and therefore needs an inactive neighbour. Hence

$$\bigcup_i S_i = C, \quad m \geq 4.$$

No S_i equals C , since that would give a three-cube in U , contrary to Lemma A.122. Thus every nonempty S_i has size at most three, and in particular $r \geq 2$.

Let W_1 be the set of the m recorded inactive neighbours, and put $W_2 = U \setminus (C \cup W_1)$, with $t = |W_2|$. Since $|U| \leq 10$,

$$t \leq 6 - m.$$

Write e_{11} for the number of edges within W_1 and e_{12} for the number between W_1 and W_2 . Edges within W_1 join lifts in the same inactive direction. A proper nonempty induced subgraph of the square on S_i has at most $|S_i| - 1$ edges, so

$$e_{11} \leq m - r.$$

A vertex of W_2 has at most two neighbours in W_1 : to have any, its inactive support must have size two, and deleting either inactive color gives the only two possibilities. Hence $e_{12} \leq 2t$.

Every vertex of W_1 has exactly one neighbour in C and, by the degree assumption, at least two neighbours in $W_1 \cup W_2$. Counting its incident edges gives

$$2m \leq 2e_{11} + e_{12} \leq 2(m - r) + 2t,$$

and therefore $r \leq t$. Combining the inequalities yields

$$2 \leq r \leq t \leq 6 - m \leq 2.$$

Thus $r = t = 2$ and $m = 4$, with equality throughout the edge count. Let i, j be the two inactive colors used in W_1 . Each of the two vertices of W_2 then has two neighbours in W_1 , so its inactive support is $\{i, j\}$, and its square trace belongs to $S_i \cap S_j$. The two vertices have distinct square traces; hence $|S_i \cap S_j| \geq 2$. Since $|S_i| + |S_j| = 4$, this forces

$$S_i = S_j$$

with both sets of size two. Their union cannot be all four vertices of C , a final contradiction. The minimum-degree lemma already gives degree at least two, so some vertex has degree exactly two. $\square$

Corollary A.124 (Opposite carriers meet in at most a square). *If a lower carrier of support R and an upper carrier of support Q meet, then*

$$|R \cup Q| \geq 3.$$

If equality holds, their defining cylinders meet in a square contained in U ; if $|R \cup Q| = 4$, they meet in an edge contained in U .

Proof. The intersection of the two defining cylinders is the subcube through any common vertex on the coordinates in $E \setminus (R \cup Q)$. As in the proof of (A.165), the lower cylinder lies in T , the upper cylinder lies in B , and their intersection is therefore contained in $B \cap T = U$. Lemma A.122 bounds its dimension by two. $\square$

Lemma A.125 (Degree-two vertices require a triple carrier). *Let $p \in U$, and suppose that its two neighbours in U have edge colors a, b . Then p lies in an eligible triple carrier on at least one of the lower and upper sides.*

Proof. Suppose instead that every lower and upper defect label at p is a pair support. Choose distinct lower labels x_1, x_2 and distinct upper labels z_1, z_2 , which is possible by (A.163). For every i, j , the intersection of their cylinders contains the subcube through p on $E \setminus (x_i \cup z_j)$. Every free coordinate is therefore the color of a neighbour of p . On putting $I = E \setminus \{a, b\}$, we obtain

$$I \subseteq x_i \cup z_j \quad (1 \leq i, j \leq 2).$$

Fix x_1, x_2 . Each z_j must contain

$$(I \setminus x_1) \cup (I \setminus x_2) = I \setminus (x_1 \cap x_2).$$

Two distinct two-sets z_1, z_2 can both contain this set only if it has size at most one. Thus $x_1 \cap x_2$ contains at least two elements of the three-set I . Since x_1, x_2 are themselves two-sets, this forces $x_1 = x_2$, a contradiction. $\square$

Lemma A.126 (Signed split normal form at a degree-two vertex). *Let $p \in U$ have two neighbours in U . After possibly exchanging the complementary gaps $S \subseteq B$ and $A \subseteq T$, the five colors can be labelled a, b, c, d, e so that*

$$N_U(p) = \{a, b\}, \quad N_S(p) = \{c\}, \quad N_A(p) = \{d, e\}. \quad (\text{A.198})$$

Every support in $\mathcal{H}_S(p)$ contains c , while

$$\mathcal{H}_A(p) \subseteq \{\{a, d, e\}, \{b, d, e\}, \{c, d, e\}\}, \quad |\mathcal{H}_A(p)| \geq 2. \quad (\text{A.199})$$

Put $X = \mathcal{K}_2$, and let X_0 be the set of members of X isolated from $\{E \setminus R : R \in \mathcal{K}_3\}$ in the bipartite disjointness graph. Then at least two of

$$\{a, b\}, \quad \{a, c\}, \quad \{b, c\} \quad (\text{A.200})$$

belong to X_0 .

Proof. For $R \in \mathcal{H}_S(p)$, every coordinate in $N_S(p)$ belongs to R . Indeed, if $f \in N_S(p) \setminus R$, then p and $p^{\{f\}}$ have the same trace on R , whereas the former realizes the unique pattern absent from S and the latter belongs to S . This is impossible. Dually, every member of $\mathcal{H}_A(p)$ contains $N_A(p)$.

The three colors outside $N_U(p)$ partition into $N_S(p)$ and $N_A(p)$. Neither part has size three. For if, say, $|N_S(p)| = 3$, then a rank-two or rank-three support containing it must be that unique three-set, contradicting $|\mathcal{H}_S(p)| \geq 2$. The dual argument applies to $N_A(p)$. Their sizes are consequently one and two, giving (A.198) after exchanging the two sides if necessary.

Every upper label contains $\{d, e\}$. If the pair $\{d, e\}$ were an upper label, it would be an upper eligible pair. No upper triple containing it would then be eligible, by (A.162); since there is no second pair containing both d, e , this would contradict the two-label lower bound. Thus all upper labels are triples, and the only possibilities are those in (A.199).

An upper eligible triple Q has the form $E \setminus x$ for an isolated pair $x \in X_0$. Complementing the three triples in (A.199) gives, respectively, $\{b, c\}, \{a, c\}, \{a, b\}$. At least two upper labels occur, proving (A.200). $\square$

Lemma A.127 (Marked complementary vertices). *Retain the notation of Lemma A.126. If $x \in X_0$, then the mandatory lower pair carrier and its complementary upper triple carrier meet in exactly one vertex:*

$$\Gamma_x^- \cap \Gamma_{E \setminus x}^+ = \{v_x\} \subseteq U. \quad (\text{A.201})$$

Thus every isolated pair support has a canonically marked relative vertex, determined by the two missing trace patterns.

Proof. Every pair in $\mathcal{K}_2$ is eligible. The assumption $x \in X_0$ says exactly that the complementary triple $E \setminus x$ is eligible for $\mathcal{K}^*$. Hence the two carriers in (A.201) are defined. Their cylinders fix complementary sets of coordinates, so their two trace patterns have a unique common extension $v_x \in Q_E$. The lower cylinder is contained in T , while the upper cylinder is contained in B . Consequently their common vertex belongs to $B \cap T = U$, and the carrier definitions give the displayed equality. $\square$

Lemma A.128 (Blocker face forced by the local upper triples). *Use the signed split normal form (A.198), and put*

$$I = \{i \in \{a, b, c\} : \{i, d, e\} \in \mathcal{H}_A(p)\}, \quad J = \{a, b, c\} \setminus I. \quad (\text{A.202})$$

Then $2 \leq |I| \leq 3$, and the restriction of A to the three-face through p on $\{a, b, c\}$ is a nonempty ample family satisfying

$$\emptyset \neq A \cap \{p^F : F \subseteq \{a, b, c\}\} \subseteq \{p^I, p^{I \cup J}\}. \quad (\text{A.203})$$

When $|I| = 2$, the right-hand side is an edge; when $|I| = 3$, it is the singleton $\{p^{\{a, b, c\}}\}$.

More precisely, set $z_F = \prod_{f \in F} v_{f, p_f}$. There is an $M \in \{I, I \cup J\}$ such that

$$(J_A : m_p) = (v_{d, p_d}, v_{e, p_e}, z_M). \quad (\text{A.204})$$

If $|I| = 3$, then necessarily $M = \{a, b, c\}$, the vertex $p^{\{a, b, c\}}$ is the unique blocker for adjoining p to A , and A contains the full d, e -square based at that vertex. In this case the four vertices

$$p, \quad p^{\{a\}}, \quad p^{\{b\}}, \quad p^{\{a, b\}} \quad (\text{A.205})$$

all belong to U .

Proof. The bounds on $|I|$ are (A.199). For every $i \in I$, the upper defect cylinder with support $\{i, d, e\}$ contains p , is disjoint from A , and is free on the other two coordinates. Thus a vertex of A that agrees with p in coordinates d, e must differ from p in every coordinate of I . This proves the containment in (A.203).

The family on the left of (A.203) is a restriction of the ample family A , and is therefore ample. It is nonempty because a blocking vertex for adjoining p to A differs from p in none of the coordinates in $N_A(p) = \{d, e\}$. This also proves the point-versus-edge description.

The monomial-colon formula (A.4) now gives (A.204). Indeed, the two neighbours $p^{\{d\}}, p^{\{e\}} \in A$ supply the two variables. Every other vertex outside the displayed three-face contributes a monomial divisible by one of them. Inside the face there are only the one or two vertices in (A.203); their unique inclusion-minimal difference support is some $M \in \{I, I \cup J\}$.

Suppose finally that $|I| = 3$. The blocker face is then the singleton $p^{\{a, b, c\}}$. Since $p^{\{d\}}, p^{\{e\}} \in A$ and A is isometric, the other intermediate vertex on a geodesic between them is $p^{\{d, e\}} \in A$. These three vertices and $p^{\{a, b, c\}}$ realize all four patterns on $\{d, e\}$, so A shatters that pair. Ampleness makes shattering strong. The zero-pattern vertex of a full d, e -square must lie in the blocker face, and hence is $p^{\{a, b, c\}}$. This proves the asserted square in A .

The upper cylinder with support $\{c, d, e\}$ contains the full a, b -square through p in B . Its first three displayed vertices belong to U by (A.198). If $p^{\{a, b\}}$ belonged to S , then the restriction of S to the face where d, e agree with p would contain both $p^{\{c\}}$ and $p^{\{a, b\}}$. Within that face, every neighbour of the latter vertex is one of $p^{\{a\}}, p^{\{b\}}, p^{\{a, b, c\}}$, none of which belongs to S . This would disconnect a nonempty restriction of the ample, hence isometric, family S . Therefore $p^{\{a, b\}} \notin S$; it lies in B , so it belongs to U , proving (A.205). $\square$

Lemma A.129 (Two upper labels force a peripheral contraction). *In the setting of Lemma A.128, suppose that $|I| = 2$. Write $I = \{r, s\}$ and $J = \{t\}$. Then*

$$p^{\{r, s\}} \in A \quad \text{and} \quad (J_A : m_p) = (v_{d, p_d}, v_{e, p_e}, z_{\{r, s\}}). \quad (\text{A.206})$$

Translate p to the empty set, identify vertices with their difference supports from p , and contract coordinate t in A . The resulting ample family $C = \pi_t A \subseteq Q_{\{r, s, d, e\}}$ satisfies

$$\begin{aligned} \{rs, d, e, de, rsd, rse, rsde\} &\subseteq C \\ &\subseteq \{rs, d, e, de, rsd, rse, rsde, rd, sd, re, se, rde, sde\}. \end{aligned} \quad (\text{A.207})$$

Moreover, the $t = p_t$ section of A , after suppressing t , contains the seven vertices on the left-hand side. At least one of d, e is a peripheral coordinate of C .

Proof. By Lemma A.128, the part of A agreeing with p in coordinates d, e is a nonempty subset of $\{p^{\{r,s\}}, p^{\{r,s,t\}}\}$. Suppose that it is the singleton $\{p^{\{r,s,t\}}\}$. The vertices $p^{\{d\}}$ and $p^{\{e\}}$ belong to A , and isometry of A , applied to this pair, also gives $p^{\{d,e\}} \in A$. These three vertices together with $p^{\{r,s,t\}}$ show that A shatters $\{d, e\}$. Ampleness makes this shattering strong. The base of the resulting full d, e -square must agree with p in d, e , so the singleton description forces that base to be $p^{\{r,s,t\}}$.

That square realizes the four traces on $\{t, d, e\}$ with $t \neq p_t$. The vertices $p^{\{d\}}, p^{\{e\}}, p^{\{d,e\}}$ realize the three nonconstant traces with $t = p_t$, while no vertex of A realizes the trace of p : any such vertex would lie in the displayed singleton and would instead differ in t . Thus adjoining p to A newly shatters $\{t, d, e\}$. This would be a third member of $\mathcal{H}_A(p)$, contrary to $I = \{r, s\}$. Hence $p^{\{r,s\}} \in A$, and the colon formula in Lemma A.128 has its inclusion-minimal third generator at $\{r, s\}$. This proves (A.206).

We now use the translated set notation in the statement. The two upper defect cylinders have supports rde and sde . They exclude both lifts of each of $\emptyset, r, s$ from A , so $\emptyset, r, s \notin C$. The vertices d, e, de, rs belong to C . Let $X = \rho_t^{p_t} A$. This ample section contains d, e, de, rs , so it shatters $\{d, e\}$. Strong shattering in X forces a full d, e -square. Its zero-pattern vertex is rs , because the blocker-face description leaves no other vertex of X agreeing with p in d, e . Hence $rsd, rse, rsde \in X$, proving the section assertion and the corresponding containment in C . These seven forced vertices and the three excluded vertices leave exactly the six optional vertices displayed in (A.207).

It remains to prove the peripheral assertion. Put

$$\alpha = [rd \in C], \quad \beta = [sd \in C], \quad \gamma = [re \in C], \quad \delta = [se \in C], \quad \epsilon = [rde \in C], \quad \phi = [sde \in C].$$

Here brackets denote truth values. Since the upper defect on rde makes C shatter $\{r, d\}$, ampleness of C supplies a full r, d -square. The missing vertices $\emptyset, s$ rule out the two squares with $e = 0$; the two remaining bases give

$$\begin{aligned} \alpha \vee \beta, \quad \gamma \vee \delta, \\ (\gamma \wedge \epsilon) \vee (\delta \wedge \phi), \quad (\alpha \wedge \epsilon) \vee (\beta \wedge \phi). \end{aligned} \tag{A.208}$$

Here the first line follows from isometry between d or e and rs . The final disjunction is obtained in the same way as the preceding one, using strong shattering of $\{r, e\}$. Isometry of C , applied respectively to the pairs rd, re and sd, se , gives

$$\alpha \wedge \gamma \implies \epsilon, \quad \beta \wedge \delta \implies \phi. \tag{A.209}$$

Reading the two sections from (A.207), coordinate d is peripheral precisely when

$$\gamma \implies \epsilon \quad \text{and} \quad \delta \implies \phi, \tag{A.210}$$

and coordinate e is peripheral precisely when

$$\alpha \implies \epsilon \quad \text{and} \quad \beta \implies \phi. \tag{A.211}$$

The same section check gives

$$\begin{aligned} r \text{ is peripheral} &\iff \alpha \wedge \gamma \wedge \epsilon, \\ s \text{ is peripheral} &\iff \beta \wedge \delta \wedge \phi. \end{aligned} \tag{A.212}$$

Suppose that d is not peripheral. If $\gamma \wedge \neg \epsilon$ holds, then (A.208) forces $\delta \wedge \phi$. Failure of peripheralness for e would give either $\alpha \wedge \neg \epsilon$, contradicting the first implication in (A.209), or $\beta \wedge \neg \phi$, contradicting ϕ . If instead $\delta \wedge \neg \phi$ holds, then (A.208) forces $\gamma \wedge \epsilon$. Nonperipheralness of e would now give either $\alpha \wedge \neg \epsilon$, immediately impossible, or $\beta \wedge \neg \phi$, contrary to the second implication in (A.209). Therefore d or e is peripheral. $\square$

Lemma A.130 (Carrier compatibility in the two-label core). *Let $C \subseteq Q_{\{r,s,d,e\}}$ be an ample family satisfying (A.207), and put*

$$K = \{rs, d, e, de, rsd, rse, rsde\}.$$

Let $D \subseteq C$ be a nonempty isometric family. Orient a peripheral coordinate q of C so that its $q = 0$ section is contained in its $q = 1$ section. Call this orientation D -compatible if

$$x \in D, \quad q \notin x \implies x \cup \{q\} \in D. \quad (\text{A.213})$$

If some connected component of $C \setminus D$ is disjoint from K , then C has a D -compatible peripheral coordinate.

Proof. Use the six optional-vertex indicators $\alpha, \beta, \gamma, \delta, \epsilon, \phi$ from the proof of Lemma A.129. The optional vertices induce two disjoint three-vertex paths,

$$rd - rde - re \quad \text{and} \quad sd - sde - se. \quad (\text{A.214})$$

Let P be a component of $C \setminus D$ disjoint from K . After interchanging r, s , assume that $P \subseteq \{sd, sde, se\}$. For $x \in \{d, e, de\}$, if $sx \in P$, then both x and rsx belong to D : they are core vertices adjacent to sx , and otherwise would belong to the same component of $C \setminus D$. The two vertices x, rsx have precisely rx, sx as their common cube neighbours. Isometry of D and $sx \notin D$ therefore force

$$sx \in P \implies x, rx, rsx \in D. \quad (\text{A.215})$$

Up to interchanging d, e , the connected subpath P has four possible shapes. We treat them in decreasing order.

The whole path $P = \{sd, sde, se\}$. Formula (A.215) puts

$$d, rd, rsd, \quad de, rde, rsde, \quad e, re, rse$$

in D . Thus r is peripheral by (A.212). Every $r = 0$ vertex of D has its r -mate in the displayed list; the three optional s -vertices lie in P , not in D . Hence r is D -compatible.

The edge $P = \{sd, sde\}$. The opposite-path rule gives

$$d, rd, rsd, de, rde, rsde \in D. \quad (\text{A.216})$$

Consequently e is peripheral by (A.211). All its compatibility conditions are already supplied by (A.216), except possibly

$$rs \in D \implies rse \in D. \quad (\text{A.217})$$

Suppose this implication fails. If $se \in C$, then $se \in D$, because it is adjacent to the component P . But $rs, se \in D$ have Hamming distance two, and their two common neighbours are $s \notin C$ and $rse \notin D$, contradicting isometry. Hence $se \notin C$. The ladder laws (A.208) now force $re, rde \in C$, so d is peripheral. Its D -compatibility follows from (A.216). Explicitly, the possible lower-side pairs are

$$rs \rightarrow rsd, \quad e \rightarrow de, \quad rse \rightarrow rsde, \quad re \rightarrow rde, \quad se \rightarrow sde;$$

the first four required mates are in D whenever their sources are, and $se \notin C$.

The middle vertex $P = \{sde\}$. We have $de, rsde \in D$, and their isometry forces $rde \in D$. If sd or se is present in C , then it belongs to D , since it is adjacent to P . Because $rde, sde \in C$, both d and e are peripheral by (A.210) and (A.211).

The only possible compatibility failures for d are

$$rs \in D, rsd \notin D, \quad \text{or} \quad se \in D; \quad (\text{A.218})$$

for e , they are

$$rs \in D, rse \notin D, \quad \text{or} \quad sd \in D. \quad (\text{A.219})$$

All other required mates are $de, rde, rsde$. The vertices sd, se cannot both lie in D , because their common neighbours are $s \notin C$ and $sde \notin D$. If $sd \in D$, isometry between sd and $rsde$ forces $rsd \in D$, so d is compatible. Symmetrically, if $se \in D$, then $rse \in D$, so e is compatible. If neither lies in D , the only possible failures in (A.218)–(A.219) require $rs \in D$. Isometry between rs and $rsde$ then puts rsd or rse in D , making d or e , respectively, compatible.

The endpoint $P = \{sd\}$. The opposite-path rule gives $d, rd, rsd \in D$. If $sde \in C$, then $sde \in D$, because it is adjacent to P . The ladder laws give

$$re \vee se, \quad re \implies rde, \quad se \implies sde. \quad (\text{A.220})$$

Thus d is peripheral. Its compatibility cannot fail at $e \rightarrow de$: the vertices $d, e \in D$ would force $de \in D$. It cannot fail at $re \rightarrow rde$: the vertices $rd, re \in D$ would force $rde \in D$. Nor can it fail at $se \rightarrow sde$, by (A.220) and the component boundary. The only possible failure is therefore

$$rse \in D, \quad rsde \notin D. \quad (\text{A.221})$$

Assume (A.221). Isometry between $rsd, rse \in D$ forces $rs \in D$. If $sde \in C$, then $sde \in D$, but rsd, sde would have neither of their common neighbours $sd, rsde$ in D . Hence $sde \notin C$, and (A.220) gives

$$se \notin C, \quad re, rde \in C.$$

Coordinate r is consequently peripheral. It is also compatible. Indeed, $d \rightarrow rd$ is already contained in D . If $e \in D$, isometry between e, rse forces $re \in D$, since $se \notin C$. Finally suppose $de \in D$. A D -geodesic from de to rse must start at rse through re , because $se \notin C$ and $rsde \notin D$. Its next vertex is e or rde . In the former case, isometry between $rd, re \in D$ forces $rde \in D$; in the latter case this is immediate. Thus every $r = 0$ carrier vertex has its r -mate in D . This completes all four path shapes. $\square$

Corollary A.131 (The two-label branch is impossible in a minimum gap). *In the minimum-gap setting of Proposition A.102 and Lemma A.128, one has $|I| = 3$. Equivalently, after the signed normalization,*

$$\mathcal{H}_A(p) = \{ade, bde, cde\}. \quad (\text{A.222})$$

Proof. Suppose that $|I| = 2$, use $I = \{r, s\}$, $J = \{t\}$, and translate p to the empty set as in Lemma A.129. Let

$$X = \rho_t^{p_t} A, \quad Y = \rho_t^{1-p_t} A, \quad C = X \cup Y = \pi_t A, \quad D = X \cap Y. \quad (\text{A.223})$$

The section assertion of that lemma gives $K \subseteq X$. The two-section shattering formula for an ample expansion, used in Lemma A.20, shows that D is ample, hence isometric. It is nonempty whenever both signed sections are nonempty, because otherwise the two sections would be disconnected in the one-inclusion graph of A .

All four endpoints of a minimum gap are periphery-free by Proposition A.102. In particular, t is not peripheral in A , so $X \setminus Y$ and $Y \setminus X$ are both nonempty. Moreover, no edge of C joins these two

sets: otherwise its two signed lifts in A would be at Hamming distance two with neither possible intermediate vertex present, contradicting isometry of A . Hence each connected component of $C \setminus D$ lies wholly in one of the two differences. A component meeting $Y \setminus X$ is disjoint from $K \subseteq X$.

Lemma A.130 now supplies a D -compatible peripheral coordinate q of C . This coordinate is peripheral in A . Indeed, if a vertex on its smaller side belongs to D , compatibility puts its q -mate in D . If it lies in $C \setminus D$, its mate either lies in D , or the edge between them stays in one component of $C \setminus D$ and therefore in the same signed section. In every case each lift of the smaller-side vertex has its q -mate in the same t -section. This contradicts periphery-freeness of A . Therefore $|I| = 3$. $\square$

Corollary A.132 (Marked-square profile with three upper labels). *In the setting of Lemma A.128, suppose that $|I| = 3$. Then*

$$\begin{aligned}\Gamma_{\{c,d,e\}}^+ &= \{p, p^{\{a\}}, p^{\{b\}}, p^{\{a,b\}}\}, \\ \Gamma_{\{a,d,e\}}^+ \cap \Gamma_{\{c,d,e\}}^+ &= \{p, p^{\{b\}}\}, \\ \Gamma_{\{b,d,e\}}^+ \cap \Gamma_{\{c,d,e\}}^+ &= \{p, p^{\{a\}}\}.\end{aligned}\tag{A.224}$$

The marked vertices of the three complementary isolated pairs satisfy

$$v_{\{a,b\}} \in \{p^{\{a\}}, p^{\{b\}}, p^{\{a,b\}}\}, \quad v_{\{a,c\}} \in \{p, p^{\{a\}}, p^{\{a,c\}}\}, \quad v_{\{b,c\}} \in \{p, p^{\{b\}}, p^{\{b,c\}}\}.\tag{A.225}$$

In particular, the mandatory a, b -pair carrier meets the forced intermediate square in the single vertex $v_{\{a,b\}} \neq p$.

Proof. The three upper cylinders through p are free, respectively, on $\{b, c\}$, $\{a, c\}$, and $\{a, b\}$. The last cylinder is the square in (A.205), so its upper carrier is that entire square. Intersecting it with either of the other two cylinders fixes, respectively, the a - or the b -coordinate at its value in p , giving the two displayed edges. All their vertices belong to U , proving (A.224).

For an isolated pair x , its marked vertex lies in the upper cylinder on $E \setminus x$, by Lemma A.127. Reading the three corresponding squares and deleting the vertex $p^{\{c\}} \in S$ gives the three containments in (A.225). Finally, $\{a, b\} \notin \mathcal{H}_S(p)$, because every lower defect support at p contains c . The carrier-label equivalence (A.175) therefore excludes $v_{\{a,b\}} = p$, while (A.201) makes its intersection with the square unique. $\square$

Lemma A.133 (Antipodal four-section rigidity). *Let $F = \{a, b, c\}$, write abc for the all-one vertex of Q_F , and let $X, Y, Z \subseteq Q_F$. In the coordinate order (d, e) , suppose that*

$$\mathcal{A} = (\{abc\} \times \{00\}) \dot{\cup} (X \times \{10\}) \dot{\cup} (Y \times \{01\}) \dot{\cup} (Z \times \{11\}) \subseteq Q_{F \cup \{d,e\}}\tag{A.226}$$

is ample and

$$\emptyset, abc \in X \cap Y \cap Z.\tag{A.227}$$

If $\mathcal{A}$ has no peripheral coordinate, then, after permuting a, b, c and possibly interchanging d, e ,

$$\begin{aligned}X &= Q_F \setminus \{b, c, bc\}, \\ Y &= Q_F \setminus \{a, c, ac\}, \\ Z &= Q_F \setminus \{c, ac, bc\}.\end{aligned}\tag{A.228}$$

Consequently, $\mathcal{A}$ is a maximum class of VC dimension two. In particular, it has order sixteen.

Proof. We first encode each of the three nontrivial sections by one interval in a six-cycle. The six vertices of Q_F other than $\emptyset, abc$ induce the cycle

$$a - ab - b - bc - c - ac - a. \quad (\text{A.229})$$

Each of X, Y, Z is ample, because coordinate face restrictions preserve ampleness. We shall use the following elementary description of such a section. If an ample $P \subseteq Q_F$ contains $\emptyset$ and abc , then

$$H_P := Q_F \setminus P \quad (\text{A.230})$$

is either empty or a cyclic interval of (A.229) having at most four vertices. Indeed, isometry supplies a three-edge geodesic from $\emptyset$ to abc . Its two middle vertices are adjacent on the cycle, so H_P is contained in the complementary four-vertex path. Since P contains the two antipodes, it shatters all three singletons. If $H_P \neq \emptyset$, it does not shatter F . Moreover, a pair is not shattered precisely when H_P contains the two endpoints of an edge of the corresponding color in the middle cycle. The three edges in any four-vertex subpath of the cycle have distinct colors, so the number of unshattered pairs is exactly $|E(H_P)|$. Ampleness therefore gives

$$8 - |H_P| = |\text{Sh}(P)| = 4 + 3 - |E(H_P)|,$$

and hence

$$|E(H_P)| = |H_P| - 1.$$

A subset of a path has this many induced edges exactly when it is an interval. Conversely, the same count shows that the complement of each listed interval is ample. Thus the interval description is exact.

Put $H_X = Q_F \setminus X$, and define H_Y, H_Z similarly. For an arbitrary subset H of the middle cycle, let

$$\mathcal{N}(H) = \{R \subseteq F : R \notin \text{Sh}(Q_F \setminus H)\}.$$

This set is read directly from the cycle: it is empty for $H = \emptyset$; otherwise it consists of F , together with the pair $F \setminus \{i\}$ for every cycle edge of color i whose two endpoints lie in H .

We next record the exact ampleness equation for (A.226). A support $R \subseteq F$ is shattered by $\mathcal{A}$ precisely when it is shattered by $X \cup Y \cup Z$. The support $R \cup \{d\}$ is shattered precisely when R is shattered by both Y and $X \cup Z$, and the analogous condition for $R \cup \{e\}$ uses X and $Y \cup Z$. Finally, among the supports containing both d, e , only $\{d, e\}$ is shattered, because the 00-section is the singleton $\{abc\}$. Since $X \cup Z = Q_F \setminus (H_X \cap H_Z)$, and similarly for the other unions, the equality

$$|\text{Sh}(\mathcal{A})| = |\mathcal{A}|$$

is equivalent to

$$\begin{aligned} & |\mathcal{N}(H_X \cap H_Y \cap H_Z)| + |\mathcal{N}(H_Y) \cup \mathcal{N}(H_X \cap H_Z)| \\ & + |\mathcal{N}(H_X) \cup \mathcal{N}(H_Y \cap H_Z)| = |H_X| + |H_Y| + |H_Z|. \end{aligned} \quad (\text{A.231})$$

Nonperipherality of d and e gives two simple interval noncontainments. If $H_Z \subseteq H_Y$, then $Y \subseteq Z$, so the $d = 0$ section Y is contained in the $d = 1$ section $X \cup Z$. Thus d would be peripheral. The analogous argument for e shows that nonperipherality forces

$$H_Z \not\subseteq H_Y, \quad H_Z \not\subseteq H_X. \quad (\text{A.232})$$

For completeness, the short interval calculation from (A.231) and (A.232) is displayed below. We write, for example, bc for the vertex $\{b, c\}$ of Q_F . Up to a permutation of F and interchange of

X, Y , these are all the possibilities. The last column lists an old coordinate whose zero-section is contained in its one-section in $\mathcal{A}$.

H_X	H_Y	H_Z	peripheral old coordinate
$\{c, bc\}$	$\{b, bc\}$	$\{b, c, bc\}$	a
$\{c, bc\}$	$\{c, ac\}$	$\{c, ac, bc\}$	a, b
$\{c, ac, bc\}$	$\{b, bc\}$	$\{b, c, bc\}$	a
$\{b, c, bc\}$	$\{c, ac\}$	$\{c, ac, bc\}$	a
$\{c, ac, bc\}$	$\{b, bc\}$	$\{b, c, ac, bc\}$	a
$\{b, c, bc\}$	$\{c, ac\}$	$\{b, c, ac, bc\}$	a
$\{c, ac, bc\}$	$\{b, ab, bc\}$	$\{b, c, bc\}$	a
$\{b, c, bc\}$	$\{a, c, ac\}$	$\{c, ac, bc\}$	none
$\{c, ac, bc\}$	$\{b, c, bc\}$	$\{b, c, ac, bc\}$	a

(A.233)

Here is a direct way to check exhaustiveness. Each H is one of the empty interval and the six intervals of each length 1, 2, 3, 4 in (A.229). For each possible length of H_Z , place one vertex of $H_Z \setminus H_X$ and one of $H_Z \setminus H_Y$, as required by (A.232). The rule defining $\mathcal{N}$ then determines the three summands on the left of (A.231). Equality leaves exactly the nine rows above. Thus the table checks only interval endpoints on the displayed six-cycle; no endpoint family is being classified.

For $f \in F$, the section of $\mathcal{A}$ with $f = 0$ is contained in the section with $f = 1$ exactly when, in each of X, Y, Z , every vertex with $f = 0$ has its f -mate. This verifies the last column of (A.233) directly. Since $\mathcal{A}$ is periphery-free, only the eighth row remains. Taking complements gives exactly (A.228).

Each of X, Y, Z has five vertices, so

$$|\mathcal{A}| = 1 + 5 + 5 + 5 = 16.$$

The section X shatters $\{b, c\}$, Y shatters $\{a, c\}$, and Z shatters $\{a, b\}$. The endpoints $\emptyset, abc$ in all three sections also show that every pair containing one of d, e is shattered, and the vertices in the four d, e -sections shatter $\{d, e\}$. Hence $\mathcal{A}$ shatters all ten coordinate pairs. Together with the empty support and the five singletons, these are sixteen shattered supports. Ampleness and $|\mathcal{A}| = 16$ leave no shattered triple. Thus $\mathcal{A}$ is maximum of VC dimension two. $\square$

Corollary A.134 (The three-label branch is impossible). *The three-label profile in Corollary A.132 cannot occur in a minimum five-coordinate gap.*

Proof. Translate the selected vertex p to $\emptyset$, and use the coordinates a, b, c, d, e from (A.198). The blocker-face conclusion of Lemma A.128 says that the four d, e -sections of A , with $F = \{a, b, c\}$, have the form (A.226): the 00-section is the singleton $\{abc\}$, while the other three sections all contain $\emptyset$ and abc . The endpoint A is ample and periphery-free by Proposition A.102. Lemma A.133 therefore makes A maximum of VC dimension two.

Complement duality (A.161) now makes $B = Q_E \setminus A$ maximum of VC dimension two as well: a support R is shattered by B exactly when $E \setminus R$ is not shattered by A , which here is equivalent to $|R| \leq 2$. Lemma A.16 supplies an ample one-concept intermediate family between S and B , contrary to the definition of a minimum gap. $\square$

Remark A.135. Theorem A.109 is a purely signed set-system statement on five colors. By Corollary A.108, it rules out the ten-cycle branch of a minimum gap without reconstructing either endpoint. The bowtie-carrier argument in Theorem A.119 similarly excludes the six-cycle branch. Corollary A.121 therefore leaves only the four-cycle branch. A carrier may contain that whole cycle,

so the interval argument requires a different replacement. Lemmas A.127 and A.128 initially reduce that replacement to two signed local profiles. The distinction is the number of upper triple cylinders through the selected vertex, not the total number of isolated pair supports: a globally isolated pair need not have its complementary cylinder pass through that vertex. The two-label profile is now excluded by Corollary A.131. Its contraction consists of a seven-vertex core and two optional three-vertex paths. Any nonconstant signed expansion has a core-free excess component in one of those paths; isometry of the duplicated carrier then forces a compatible peripheral semicube, which remains peripheral after lifting. Thus every minimum gap would have all three local upper triples. Lemma A.133 then identifies its periphery-free complementary endpoint with the pentagonal maximum VC-two class, and Corollary A.134 excludes it by relative VC-two peeling. This closes the four-cycle branch without choosing between the adjacent and opposite positions of the marked pair carrier.

Remark A.136 (The exact symbolic replacement for the bounded audit). The replacement is now entirely symbolic. The two-sided defect cylinders give minimum degree two in the intermediate layer. Convex carrier intervals exclude cycles of lengths ten, eight, and six, leaving a square and a degree-two vertex. The signed blocker at that vertex has two or three upper labels. Carrier compatibility excludes the two-label case. In the three-label case, the complementary endpoint splits into four antipodal Q_3 -sections. The holes of each section are short intervals of the middle six-cycle (A.229); the single ampleness budget (A.231) and periphery-freeness force the pentagonal maximum VC-two class. Relative VC-two peeling gives the final contradiction.

The balanced Artinian graph and the relative Yang-ball formulation remain useful interpretations of the reduction, but neither is needed as an additional hypothesis. In particular, the earlier order-through-twenty-one files are now regression checks only; they are not part of the proof of the five-coordinate theorem.

Theorem A.137 (Relative peeling in five coordinates). *Let $\emptyset \neq S_0 \subseteq B \subseteq Q_E$ be ample, and suppose that B uses at most five coordinates. For every $s \in S_0$, the family B has a corner peeling in which every vertex of $B \setminus S_0$ precedes every vertex of S_0 , and s is last.*

Proof. It is enough to prove that whenever $S_0 \subsetneq B$, some $p \in B \setminus S_0$ makes $S_0 \cup \{p\}$ ample. Otherwise choose a minimum counterexample as in Lemma A.101. Proposition A.102 puts it in the central five-coordinate profile. Corollary A.121 gives a square in its intermediate layer, and Lemma A.123 supplies a degree-two vertex on the square. Lemmas A.126 and A.128 give the signed blocker normal form at that vertex. Corollary A.131 excludes two upper labels, while Corollary A.134 excludes three. These are the only possibilities, so the minimum counterexample does not exist.

We may consequently adjoin one ample concept repeatedly until B is reached. Reversing this chain peels $B \setminus S_0$ first. Applying the same assertion to $\{s\} \subseteq S_0$ then completes the peeling with s last. $\square$

Remark A.138 (Why the five-coordinate proof is conceptual). Proposition A.102 still explains why ranks two and three are the only relevant relative h -supports. The proof does not classify the endpoint families in that order range. Instead it follows one shortest cycle in the intermediate layer, uses signed carrier convexity to eliminate the three longer cycle lengths, and reduces the remaining square to one degree-two vertex. The two-label profile is controlled by two three-vertex paths; the three-label profile is controlled by three intervals in the fixed six-cycle (A.229). Thus both finite-looking steps are consequences of constant-size structural lemmas, and the order-through-twenty-one verification is only an independent regression check.

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